*TM 1-1520-237-23-5

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

CHAPTER 5
MAINTENANCE INSTRUCTIONS

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

HEADQUARTERS, DEPARTMENT OF THE ARMY

17 April 2006

WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

WARNING

FIRST AID - refer to FM 4-25-11.

AC POWER - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

CARBON MONOXIDE - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

CONSUMABLE MATERIALS - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

SAFETY PINS - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

ELECTRICAL GROUNDING OF THE HELICOPTER - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

HIGH VOLTAGE - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

HYDRAULIC SYSTEMS - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

USE OF FIRE EXTINGUISHER - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

MAIN ROTOR PYLON SLIDING COVER - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

TECHNICAL ACETONE - is extremely flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

ALKALINE AQUEOUS CLEANER - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

CLEANING COMPOUND SOLVENT - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

ELECTRON - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

ISOPROPYL ALCOHOL - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

DS-108 - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/ goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

LOCKWIRE - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

PURPOSE OF A WARNING - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE NO. 3	HEADQUARTERS DEPARTMENT OF THE ARMY WASHINGTON, D.C., 20 June 2008

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

CHAPTER 5 MAINTENANCE INSTRUCTIONS

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5. Remove old pages and insert new pages as indicated below.

Remove Pages	Insert Pages
a and b	a and b
A and B	A and B
i through vi	i through vi

6. Replace the following work packages with their revised version.

Work Package Number

WP 0182 00
WP 0192 00

Work Package Number

WP 0194 00
WP 0195 00
WP 0196 00
WP 0198 00
WP 0200 00
WP 0204 00
WP 0205 00
WP 0209 00
WP 0210 00
WP 0212 00
WP 0215 00
WP 0217 00
WP 0218 00
WP 0219 00
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WP 0227 00
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WP 0233 00
WP 0240 00
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WP 0248 00
WP 0250 00
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WP 0261 00
WP 0262 00
WP 0263 00
WP 0265 00
WP 0266 00
WP 0268 00
WP 0285 00
WP 0286 00
WP 0287 00
WP 0296 00
WP 0303 00
WP 0305 00
WP 0312 00
WP 0317 00
WP 0324 00
WP 0325 00
WP 0327 00
WP 0333 00

Work Package Number

WP 0334 00
WP 0335 00
WP 0338 00
WP 0341 00
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WP 0351 00
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WP 0354 00
WP 0355 00
WP 0358 00
WP 0359 00
WP 0360 00
WP 0363 00
WP 0364 00
WP 0365 00
WP 0370 00
WP 0372 00
WP 0375 00

7. Add the following new work packages.

Work Package Number

WP 0261 01

By Order of the Secretary of the Army:

GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:

Joyce E. Morrow

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
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CHANGE
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TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

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Remove Pages	Insert Pages
None	a and b
A and B	A and B

6. Replace the following work packages with their revised version.

Work Package Number

WP 0175 00
WP 0183 00
WP 0184 00
WP 0188 00

Work Package Number

WP 0200 00
WP 0220 00
WP 0221 00
WP 0222 00
WP 0226 00
WP 0231 00
WP 0232 00
WP 0245 00
WP 0332 00
WP 0337 00
WP 0353 00
WP 0354 00
WP 0358 00
WP 0369 00

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General, United States Army
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TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE

FOR
ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L HELICOPTERS

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Remove pages	Insert pages
A and B	A and B
i through vi	i through vi
Chapter 5 Title and Blank	Chapter 5 Title and Blank
Cover	Cover

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Work Package Number

WP 0175 00
WP 0181 00
WP 0184 00
WP 0187 00
WP 0188 00

Work Package Number

WP 0194 00
WP 0204 00
WP 0216 00
WP 0226 00
WP 0231 00
WP 0232 00
WP 0245 00
WP 0332 00
WP 0333 00
WP 0338 00
WP 0342 00
WP 0353 00
WP 0360 00

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
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Dates of issue for original and changed pages / work packages are:

Original	0	17 April 2006
Change	1	21 March 2007
Change	2	15 December 2007
Change	3	20 June 2008

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 12 AND TOTAL NUMBER OF WORK PACKAGES IS 205 CONSISTING OF THE FOLLOWING:

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Chapter 5 Title Page	1
WP 0173 00 (2 pgs)	0
WP 0174 00 (2 pgs)	0
WP 0175 00 (4 pgs)	2
WP 0176 00 (4 pgs)	0
WP 0177 00 (2 pgs)	0
WP 0178 00 (4 pgs)	0
WP 0179 00 (4 pgs)	0
WP 0180 00 (2 pgs)	0
WP 0181 00 (6 pgs)	1
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WP 0239 00 (2 pgs)	0
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WP 0241 00 (4 pgs)	3
WP 0242 00 (6 pgs)	0
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WP 0244 00 (4 pgs)	0
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WP 0247 00 (14 pgs)	3
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* Zero in this column indicates an original page or work package.

LIST OF EFFECTIVE PAGES/WORK PACKAGES - CONTINUED

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WASHINGTON, D.C., 17 April 2006

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

Army Model Helicopters UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of this manual, directly to: Commander, US Army Aviation and Missile Command, ATTN:AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also send in your comments electronically to our E-mail address: 2028@redstone.army.mil or by fax 256-842-6546/DSN 788-6546. For the World Wide Web use: https://amcom2028.redstone.army.mil. Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028.

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CHAPTER 5

MAINTENANCE INSTRUCTIONS

FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L

HELICOPTERS

UNIT LEVEL

AIRFRAME

CORROSION INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
WP 0189 00
WP 0212 00

NOSE DOOR LATCH CORROSION INSPECTION

1. Visually inspect latch for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. Remove corrosion from latch (WP 0189 00).
2. Raise latch. Loosen jamnut and inspect fit of latch hook and jamnut on threaded stud. **IF LATCH HOOK OR JAMNUT IS LOOSE OR WOBBLES ON THREAD, REPLACE LATCH (WP 0189 00).**
3. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

COCKPIT DOOR CORROSION INSPECTION

1. Visually inspect latch for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting.
2. If corrosion is present, remove rod end from hinge and remove corrosion (WP 0212 00).
3. Open and close door to check for proper operation. **IF ROD END BEARING IS STICKING OR RATCHETY, REPLACE ROD END (WP 0212 00).**

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

COCKPIT DOOR INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Foam Padding
Spring Scale, AAA-S-133

Material/Parts
Wire, Nonelectrical, Item 351, WP 1803 00

Personnel Required
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0198 00
WP 0199 00
WP 1803 00

COCKPIT DOOR JETTISON CHECK

1. Position foam padding to prevent damage to cockpit door during jettison check.
2. Have assistants outside helicopter to guide jettisoned cockpit door onto foam padding.
3. Make sure cockpit door is closed and latched.
4. Connect spring scale to cockpit door jettison handle.
5. Pull jettison handle with spring scale. **FORCE SHALL BE BETWEEN 27 TO 33 LBS.**
6. If force is not 27 to 33 lbs, reset jettison handle and adjust hinge bolts.
7. Repeat pull test until force is between 27 to 33 lbs.
8. **REPEAT HANDLE PULL A MINIMUM OF 4 TIMES.**
9. Pull jettison handle, manually push or kick door from helicopter.
10. Inspect cockpit door and jettison mechanism for damage. **NO DAMAGE ALLOWED.**
11. Install cockpit door (WP 0198 00).
12. Open and close cockpit door a few times to check for proper operation. Adjust door if required (WP 0199 00).
13. Use nonelectrical wire, Item 351, WP 1803 00 (breakaway wire), lockwire jettison handle to clip on lower front edge of window frame.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

WINDOW AND WINDSHIELD INSPECTIONS

This WP supersedes WP 0175 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23-10
WP 0224 00
WP 0225 00
WP 0226 00
WP 0227 00
WP 0228 00
WP 0237 00
WP 0245 00

UPPER WINDOW INSPECTION

NOTE

If plastic window has minor scratches, hazing, crazing, damage by airborne abrasives or surface imperfections polish window per TM 1-1502-204-23-10.

1. Inspect upper window for cracks. **NO CRACKS ALLOWED.** Replace window if cracks are found (WP 0224 00).
2. Inspect upper window for crazing. **CRAZING SHALL NOT INTERFERE WITH PILOT'S VISIBILITY.** Replace window if crazing interferes with pilot's visibility (WP 0224 00).
3. Inspect upper window for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP.** Replace window if scratches exceed limit (WP 0224 00).
4. Inspect upper window for chips and nicks. **CHIPS AND NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN A 6 INCH DIAMETER AREA.** Replace window if damage exceeds limits (WP 0224 00).
5. Inspect upper window mounting holes for chips. **CHIPS SHALL NOT EXCEED 0.063-INCH DEEP OR 0.090-INCH LONG EXTENDING FROM EDGE OF MOUNTING HOLES. DAMAGE MUST NOT BE ON MORE THAN SIX HOLES AND NO MORE THAN TWO OUT OF FIVE CONSECUTIVE HOLES.** Replace window if damage exceeds limits (WP 0224 00).
6. Inspect upper window panel edge for chips. **CHIPS SHALL NOT EXCEED 1/2-INCH LONG OR 0.13-INCH WIDE THROUGH FULL DEPTH OF PANEL ALONG EDGE. NO MORE THAN TWO CHIPS ARE ALLOWED IN ANY 6 INCH LENGTH.** Replace window if damage exceeds limits (WP 0224 00).
7. Replace upper windshield if:
   a. **DAMAGE PREVENTS SECURE ATTACHMENT (WP 0224 00).**
   b. **DAMAGE EXCEEDING ABOVE LIMITS IS FOUND ON EITHER THE INNER OR OUTER SURFACE OF PANEL (WP 0224 00).**
   c. **DAMAGE INTERFERES WITH PILOT'S VISIBILITY (WP 0224 00).**

LOWER WINDOW INSPECTION

NOTE

If plastic window has minor scratches, hazing, crazing, damage by airborne abrasives or surface imperfections polish window per TM 1-1500-204-23-10.

1. Inspect lower window for cracks. **NO CRACKS ALLOWED.** Replace window if cracks are found (WP 0225 00).
2. Inspect lower window for crazing. **CRAZING SHALL NOT INTERFERE WITH PILOT'S VISIBILITY.** Replace window if crazing interferes with pilot's visibility (WP 0225 00).
3. Inspect lower window for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP.** Replace window if scratches exceed limit (WP 0225 00).
4. Inspect lower window for chips and nicks. **CHIPS AND NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN A 6 INCH DIAMETER AREA.** Replace window if damage exceeds limits (WP 0225 00).
5. Inspect lower window mounting holes for chips. **CHIPS SHALL NOT EXCEED 0.063-INCH DEEP OR 0.090-INCH LONG EXTENDING FROM EDGE OF MOUNTING HOLES. DAMAGE MUST NOT BE ON MORE THAN SIX HOLES AND NO MORE THAN TWO OUT OF FIVE CONSECUTIVE HOLES.** Replace window if damage exceeds limits (WP 0225 00).
6. Inspect lower window panel edge for chips. **CHIPS SHALL NOT EXCEED 1/2-INCH LONG OR 0.13-INCH WIDE THROUGH FULL DEPTH OF PANEL ALONG EDGE. NO MORE THAN TWO CHIPS ARE ALLOWED IN ANY 6 INCH LENGTH.** Replace window if damage exceeds limits (WP 0225 00).
7. Replace lower windshield if:
   a. **DAMAGE PREVENTS SECURE ATTACHMENT (WP 0225 00).**
   b. **DAMAGE EXCEEDING ABOVE LIMITS IS FOUND ON EITHER THE INNER OR OUTER SURFACE OF PANEL (WP 0225 00).**
   c. **DAMAGE INTERFERES WITH PILOT'S VISIBILITY (WP 0225 00).**
8. Check conductive tape for cuts and looseness. **REPAIR TAPE IF CUTS DO NOT SEVER IT.** Replace tape if severed or loose (WP 0225 00).

OUTBOARD WINDSHIELD INSPECTION

NOTE

If plastic window has minor scratches, hazing, crazing, damage by airborne abrasives or surface imperfections polish window per TM 1–1500-204-23-10.

1. Inspect outboard windshield for cracks. **NO CRACKS ALLOWED.** Replace windshield if cracks are found (WP 0226 00).
2. Inspect outboard windshield for erosion. **EROSION SHALL NOT INTERFERE WITH PILOT'S VISIBILITY.** Replace windshield if erosion interferes with pilot's visibility (WP 0226 00).
3. Inspect outboard windshield for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.005-INCH DEEP.** Replace windshield if scratches exceed limit (WP 0226 00).
4. Inspect outboard windshield for chips and nicks. **CHIPS AND NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN A 6 INCH DIAMETER AREA.** Replace windshield if damage exceeds limits (WP 0226 00).

OUTBOARD WINDSHIELD INSPECTION - Continued

5. Inspect outboard windshield for delamination of reinforced plastic edging. **DELAMINATION SHALL NOT CAUSE REINFORCED PLASTIC EDGING TO DISBOND FROM GLASS PANEL.** Replace windshield if reinforced plastic edging is disbonded from glass panel (WP 0226 00).
6. Replace outboard windshield if:
   a. **DAMAGE PREVENTS SECURE ATTACHMENT (WP 0226 00).**
   b. **DAMAGE EXCEEDING ABOVE LIMITS IS FOUND ON EITHER THE INNER OR OUTER SURFACE OF PANEL (WP 0226 00).**
   c. **DAMAGE INTERFERES WITH PILOT'S VISIBILITY (WP 0226 00).**
   d. **DEICING IS NOT OPERATIONAL DUE TO CRACKS OR OTHER DAMAGE (WP 0226 00).**
   e. **DEICING IS NOT OPERATIONAL DUE TO CRACKS OR OTHER DAMAGE (WP 0226 00).**

CENTER WINDSHIELD INSPECTION

NOTE

If plastic window has minor scratches, hazing, crazing, damage by airborne abrasives or surface imperfections polish window per TM 1-1500-204-23-10.

1. **W/O HCW** Inspect plastic center windshield as follows:
   a. Inspect plastic center windshield for cracks. **NO CRACKS ALLOWED.** Replace window if cracks are found (WP 0228 00)
   b. Inspect plastic center windshield for crazing. **CRAZING SHALL NOT INTERFERE WITH PILOT'S VISIBILITY.** Replace window if crazing interferes with pilot's visibility (WP 0228 00)
   c. Inspect plastic center windshield for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP.** Replace window if scratches exceed limit (WP 0228 00).
   d. Inspect plastic center windshield for chips and nicks. **CHIPS AND NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN A 6 INCH DIAMETER AREA.** Replace window if damage exceeds limits (WP 0228 00).
   e. Inspect plastic center windshield mounting holes for chips. **CHIPS SHALL NOT EXCEED 0.063-INCH DEEP OR 0.090-INCH LONG EXTENDING FROM EDGE OF MOUNTING HOLES. DAMAGE MUST NOT BE ON MORE THAN SIX HOLES AND NO MORE THAN TWO OUT OF FIVE CONSECUTIVE HOLES.** Replace window if damage exceeds limits (WP 0228 00).
   f. Inspect plastic center windshield panel edge for chips. **CHIPS SHALL NOT EXCEED 1/2-INCH LONG OR 0.13-INCH WIDE THROUGH FULL DEPTH OF PANEL ALONG EDGE. NO MORE THAN TWO CHIPS ARE ALLOWED IN ANY 6 INCH LENGTH.** Replace window if damage exceeds limits (WP 0228 00).
   g. Replace plastic center windshield if:
      (1) **DAMAGE PREVENTS SECURE ATTACHMENTS (WP 0228 00).**
      (2) **DAMAGE EXCEEDING ABOVE LIMITS IS FOUND ON EITHER THE INNER OR OUTER SURFACE OF PANEL (WP 0228 00).**
      (3) **DAMAGE INTERFERES WITH PILOT'S VISIBILITY (WP 0228 00).**
2. **HCW** Inspect glass center heated windshield as follows:
   a. Inspect glass center windshield for cracks. **NO CRACKS ALLOWED.** Replace windshield if cracks are found (WP 0227 00).

CENTER WINDSHIELD INSPECTION - Continued

b. Inspect glass center windshield for erosion. **EROSION SHALL NOT INTERFERE WITH PILOT'S VISIBILITY.** Replace windshield if erosion interferes with pilot's visibility (WP 0227 00).

c. Inspect glass center windshield for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.005-INCH DEEP.** Replace windshield if scratches exceed limit (WP 0227 00).

d. Inspect glass center windshield for chips and nicks. **CHIPS AND NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH MORE THAN FOUR CHIPS OR NICKS WITHIN A 6 INCH DIAMETER AREA.** Replace windshield if damage exceeds limits (WP 0227 00).

e. Inspect glass center windshield for delamination of reinforced plastic edging. **DELAMINATION SHALL NOT CAUSE REINFORCED PLASTIC EDGING TO DISBOND FROM GLASS PANEL.** Replace windshield if reinforced plastic edging is disbonded from glass panel (WP 0227 00).

f. Replace glass center windshield if:

(1) **DAMAGE PREVENTS SECURE ATTACHMENT (WP 0227 00).**

(2) **DAMAGE EXCEEDING ABOVE LIMITS IS FOUND ON EITHER THE INNER OR OUTER SURFACE OF PANEL (WP 0227 00).**

(3) **DAMAGE INTERFERES WITH PILOT'S VISIBILITY (WP 0227 00).**

(4) **DEICING IS NOT OPERATIONAL DUE TO CRACKS OR OTHER DAMAGE (WP 0227 00).**

GUNNER WINDOW INSPECTION

1. Inspect gunners window for cracks. **NO CRACKS ALLOWED.** Replace window (WP 0237 00).
2. If mission requirements are effected by reduced vision from crazing and or scratches, replace window if unable to restore visibility by polishing (WP 0237 00).

TROOP/CARGO WINDOW INSPECTION

1. Inspect troop cargo windows for cracks. **NO CRACKS ALLOWED.** Replace window (WP 0245 00).
2. If mission requirements are effected by reduced vision from crazing and or scratches replace window if unable to restore visibility by polishing (WP 0245 00).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

PILOT'S/COPILOT'S SEAT INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Spring Scale, AAA-S-133

Material/Parts
Rubber Sheet, Solid, Item 256, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0231 00
WP 0232 00
WP 1803 00

SHOULDER HARNESS INSPECTION

1. Sit in seat and adjust seat full up and full rear.
2. Check shoulder inertia reel mounting hardware for cracks and security. **NO CRACKS ALLOWED.** Replace cracked hardware (WP 0231 00).
3. Check inertia reel surface for smoothness and freedom from sharp edges or burrs that may snag webbing.
4. Check operation of control cable and inertia reel.
5. Check shoulder harness webbing (TM 1-1500-204-23).

LAP BELT INSPECTION

WARNING

Pilot and copilot seat lap belts must be compatible with the restraint buckles. Mismatched lap belt fittings and restraint buckles may degrade the integrity of the belt fittings attachment into the buckle. If installing a lap belt, confirm the lap belt fitting is correct for the restraint buckle.

1. Sit in seat and adjust full up and full rear.
2. Check lap belt anchor fittings for cracks, snags, and sharp edges. **NO CRACKS ALLOWED.** Replace cracked fittings (WP 0231 00). Round off edges and remove snags with file on serviceable fittings.
3. Check lap belt inertia reel mounting hardware for cracks and security. **NO CRACKS ALLOWED.** Replace cracked hardware (WP 0231 00).
4. Check lap belt webbing (TM 1-1500-204-23).

CROTCH BELT, BUCKLE, AND PAD INSPECTION

WARNING

Pilot and copilot seat lap belts must be compatible with the restraint buckles. Mismatched lap belt fittings and restraint buckles may degrade the integrity of the belt fittings attachment into the buckle. If installing a lap belt, confirm the lap belt fitting is correct for the restraint buckle.

1. Sit in seat and adjust full up and full rear.
2. Check lap belt anchor fittings for cracks, snags, and sharp edges. **NO CRACKS ALLOWED.** Replace cracked fittings (WP 0231 00). Round off edges and remove snags with file on serviceable fittings.
3. Check crotch belt mounting hardware for cracks and security. **NO CRACKS ALLOWED.** Replace cracked hardware (WP 0231 00).
4. Check rotary buckle for cleanness and smooth operation. Blow dirt from buckle mechanisms with dry compressed air.
5. Insert shoulder and lap belt harness fittings in rotary buckle. **ENGAGEMENT MUST BE POSITIVE AND SECURE.** Replace rotary buckle failing engagement test (WP 0231 00).
6. Attach spring scale to fluke on rotary buckle and check rotary buckle release tension. **TENSION SHALL BE NOT LESS THAN 5 OR MORE THAN 45 INCH- POUNDS. ALL HARNESS FITTINGS SHALL SEPARATE FROM ROTARY BUCKLE INSTANTLY AND CLEANLY AS RELEASE ROTATES APPROXIMATELY 1/8 OF A TURN IN EITHER DIRECTION.** Replace rotary buckle failing either tension or rotation test (WP 0231 00).
7. Check lap belt webbing (TM 1-1500-204-23).

PILOT’S/COPILOT’S SEAT, D-3801-1 AND D-3177 CLEVIS SCREW INSPECTION (SEAT SERIAL NOS 0009 THROUGH 1773)

NOTE

Helicopters with Norton or Simula seats need no further inspection or adjustment.

1. Inspect for ARA seats. Identify from nameplate on back of seat.
2. Check for clevis screw looseness by observing if the screw head has backed away from full contact with the column clevis. Any gap indicates looseness.

NOTE

Whenever a seat is removed to correct a loose clevis screw condition, lockwire clevis screw on the other side, also.

3. Remove seat (WP 0232 00).
4. Tilt the seat forward to gain access to the column bottom and clevis screws.
5. To tighten, engage screw slots on both columns and tighten one screw against to the other until the screws bottom out, then turn until both screw slots are parallel to the track position (perpendicular to the column).

PILOT'S/COPILOT'S SEAT, D-3801-1 AND D-3177 CLEVIS SCREW INSPECTION (SEAT SERIAL NOS 0009 THROUGH 1773) - Continued

NOTE

There are two configurations of isolators. Seats with serial Nos 0009 through 0719 have a thicker rubber body isolator. Seats with serial Nos 0720 and subsequent have a thin bodied rubber isolator.

6. If inspection shows chafing of rail guide by column clevis, add rubber separator. Make washer from solid rubber sheet, Item 256, WP 1803 00, and assemble it between outer clevis leg and isolator.
7. Place 2 pieces of nonelectrical wire, Item 354, WP 1803 00, 12 inches long, between isolator and inside of outer leg of column clevis, parallel to each on opposite sides of clevis screw shaft. If necessary, force in thicker rubber body isolator to get nonelectrical wire in place.
8. Twist two nonelectrical wires together at front and back, capturing screw shaft (serial Nos 0009 through 0719) or isolator bushing (serial Nos 0720 and subsequent). Twist full length of nonelectrical wire up to screw shaft or isolator bushing.
9. Verify position of screwheads, perpendicular to column. Bring twisted wires forward so that lockwire seats into screw slot.
10. Join forward and aft nonelectrical wires. Twist all nonelectrical wires together so wire is in screw slot tightly.
11. Trim length of four wire wrap to 0.5-inch.
12. On seats with serial Nos 0009 through 0719, loop wire ends outward, back, and slightly upward. On seats with serial Nos 0720 and subsequent, loop wire ends inward, back, and slightly upward.
13. Reinstall seat (WP 0232 00).
14. Make sure nonelectrical wire loops do not contact any surface when seat is operated in tilt back mode.

PILOT'S/COPILOT'S SEAT, D-3801-1 AND D-3177 SHOCK ABSORBER CLEVIS INSPECTION

1. Slide seat fully back.

WARNING

To prevent injury to personnel, do not pull forward on control lever unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and control lever is pulled forward, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes will be seriously injured.

2. Sit in seat and adjust it fully up.
3. Remove back cushion (WP 0232 00).
4. Grasping top of seat bucket, rock seat vigorously back and forth, and from side to side. **NO VISIBLE MOVEMENT OF CLEVISES OR RETAINING NUTS IS ALLOWED (Figure 1, Detail A).**

PILOT'S/COPILOT'S SEAT, D-3801-1 AND D-3177 SHOCK ABSORBER CLEVIS INSPECTION - Continued

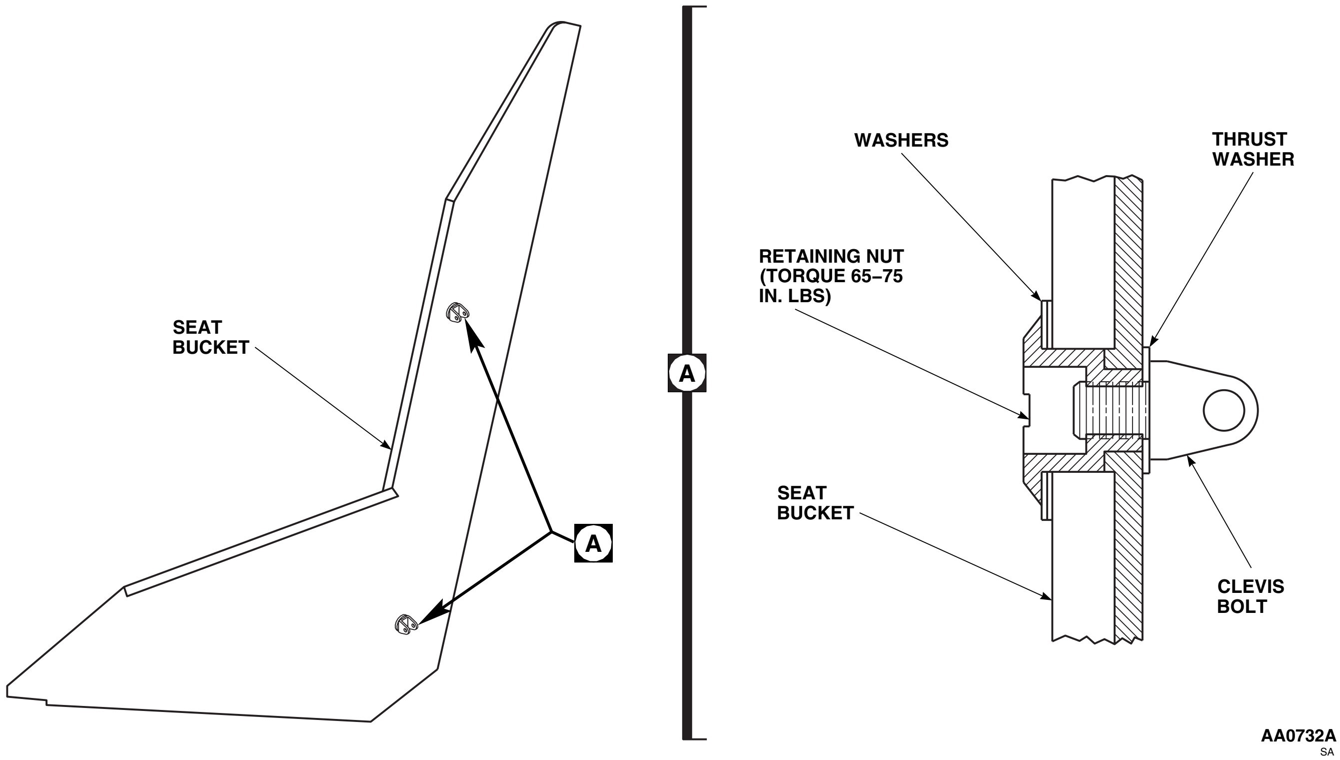

Figure 1. Shock Absorber Clevis.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TROOP/GUNNER'S SEAT INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Spring Scale, AAA-S-133
Torque Wrench, 0 - 30 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0233 00
WP 0234 00
WP 0235 00

TROOP SEAT RESTRAINT SYSTEM INSPECTION

1. Check restraint system webbing (TM 1-1500-204-23).
2. Check buckle for cleanness and smooth operation. Blow dirt out of buckle mechanism with dry compressed air.
3. Insert shoulder and lap belt harness fittings in buckle. **ENGAGEMENT MUST BE POSITIVE AND SECURE.** Replace buckles which fail engagement test (WP 0233 00).
4. Check buckle release tension with harness fittings installed, as follows:
   a. Modify socket for torque wrench to fit buckle release handle.
   b. Measure buckle release tension. **TENSION SHALL BE NOT LESS THAN 5 INCH-POUNDS OR MORE THAN 15 INCH-POUNDS. HARNESS FITTINGS MUST SEPARATE FROM BUCKLE INSTANTLY AND CLEANLY AS RELEASE HANDLE ROTATES THROUGH 30 DEGREES TO 45 DEGREES.**
   c. Replace rotary buckle failing either tension or rotation test (WP 0234 00).

GUNNER'S SEAT RESTRAINT SYSTEM INSPECTION

1. Check restraint seat anchor fittings for cracks, snags, and sharp edges. **NO CRACKS ALLOWED.** Replace cracked fittings (WP 0234 00). Radius edges and remove snags on serviceable fittings.
2. Check shoulder and lap belt inertia reel mounting hardware for cracks and security. **NO CRACKS ALLOWED.** Replace cracked hardware (WP 0234 00).
3. Pull webbing off shoulder and lap belt inertia reels in sequence and inspect (TM 1-1500-204-23).
4. Check buckle for cleanness and smooth operation. Blow dirt out of buckle mechanism with dry compressed air.
5. Insert shoulder and lap belt harness fittings in buckle. **ENGAGEMENT MUST BE POSITIVE AND SECURE.** Replace buckles failing engagement test (WP 0234 00).
6. Check buckle release tension with harness fittings installed, as follows:
   a. Modify socket for torque wrench to fit buckle release handle.
   b. Measure buckle release tension. **TENSION SHALL BE NOT LESS THAN 5 INCH-POUNDS OR MORE THAN 15 INCH-POUNDS. HARNESS FITTINGS MUST SEPARATE FROM BUCKLE INSTANTLY AND CLEANLY AS RELEASE HANDLE ROTATES THROUGH 30 DEGREES TO 45 DEGREES.**

GUNNER'S SEAT RESTRAINT SYSTEM INSPECTION - Continued

c. Replace rotary buckle failing either tension or rotation test (WP 0235 00).

7. Using a spring scale, measure shoulder harness and lap belt inertia reel tension against webbing. **AVERAGE TENSION BETWEEN MAXIMUM WEBBING EXTENSION AND RETRACTION TO WITHIN THE LAST 2 INCHES OF TRAVEL IS 2 TO 9 POUNDS FOR ALL THREE INERTIA REELS.** Replace reels failing tension test (WP 0234 00).

8. Give shoulder harness a quick pull. **INERTIA REEL SHALL LOCK.** Release tension. Reel shall smoothly retract or release webbing as required.

9. To check left and right lap belt inertia reel operation, do this:

   a. Pull several inches of webbing from reel.

   b. Put reel control in MANUAL-LOCK. Tug on webbing. **REEL SHALL NOT RELEASE ANY MORE THAN 1/2-INCH OF WEBBING** (compensating for webbing slip and stretch).

   c. Release tension. **REEL MUST RETRACT WEBBING BUT LOCK WHEN WEBBING IS PULLED.**

   d. Put reel control in AUTO-LOCK. Pull out entire webbing length.

   e. Release lap belt. **INERTIA REEL MUST FULLY RETRACT ENTIRE WEBBING LENGTH.**

   f. Give webbing a quick hard pull. **INERTIA REEL SHALL LOCK AND HOLD UNTIL TENSION IS RELEASED.**

   g. Give webbing a quick hard pull to lock reel. Maintain tension on lock. Put reel control in MANUAL-LOCK, then put reel control in AUTO-LOCK.

10. Replace reels that fail any part of preceding inspection (WP 0234 00).

TROOP SEAT CABLE INSPECTION

1. Check strands for breakage. **NO MORE THAN THREE BROKEN STRANDS ALLOWED.** If more than three broken strands are found replace cable (WP 0233 00).

2. Inspect cables for kinks. **NO KINKS ALLOWED.** If kinks are found replace cable (WP 0233 00).

3. Inspect sheathing surrounding cable. **CRACKS THROUGH SHEATHING DOWN TO CABLE ARE NOT ALLOWED.** If cracks through sheathing down to cable are found replace cable (WP 0233 00).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

GUNNER'S WINDOW INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 1-1500-344-23
TM 55-1500-345-23
WP 0237 00
WP 0381 00
WP 0396 00
WP 0397 00
WP 0398 00

LOWER TRACK INSPECTION

1. Turn off all electrical power.
2. Remove gunner's windows (WP 0237 00).
3. Visually inspect lower track for cracks, **NO CRACKS ALLOWED.** Replace cracked track (WP 0396 00 and TM 1-1500-204-23).
4. Inspect front wear strips for wear and security (Figure 1, Detail A and Detail B). **WEAR SHALL NOT BE MORE THAN 0.010-INCH DEEP.** If wear is more than 0.010-inch deep or wear strip is loose or missing, replace wear strip (WP 0397 00).
5. Inspect rear wear strips between STA 265.0 and STA 270.0, if installed, for wear, (Figure 1, Detail C). **WEAR SHALL NOT BE MORE THAN 0.010-INCH DEEP.** If wear is more than 0.010-inch deep, or wear strip is loose or missing, replace wear strip (WP 0397 00).
6. Inspect lower track for wear, (Figure 1, Detail B). **WEAR SHALL NOT BE MORE THAN 0.025-INCH DEEP, WITH NO LIMIT TO LENGTH OR WIDTH.** If wear is located between STA 265.0 and STA 270.0, repair track and install locally-made wear strips (WP 0397 00). If wear exceeds 0.025-inch at any location, replace track (WP 0396 00 and TM 1-1500-204-23).
7. Inspect areas of track around rivets for corrosion. Remove and treat corrosion (TM 1-1500-344-23).
8. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

UPPER TRACK INSPECTION

1. Visually inspect upper track for cracks. **NO CRACKS ALLOWED.** Replace cracked track (WP 0398 00).
2. Inspect track lip surfaces for wear by rollers. **MINIMUM MATERIAL THICKNESS ALLOWED IS 0.040-INCH.** Replace track if limit is exceeded (Figure 2, Detail A).
3. Inspect wear strips, if installed, for wear and security.**WEAR SHALL NOT BE MORE THAN 0.010-INCH DEEP.** If wear is more than 0.010-inch deep or wear strip is loose or missing, replace wear strip(WP 0397 00).

UPPER TRACK INSPECTION - Continued

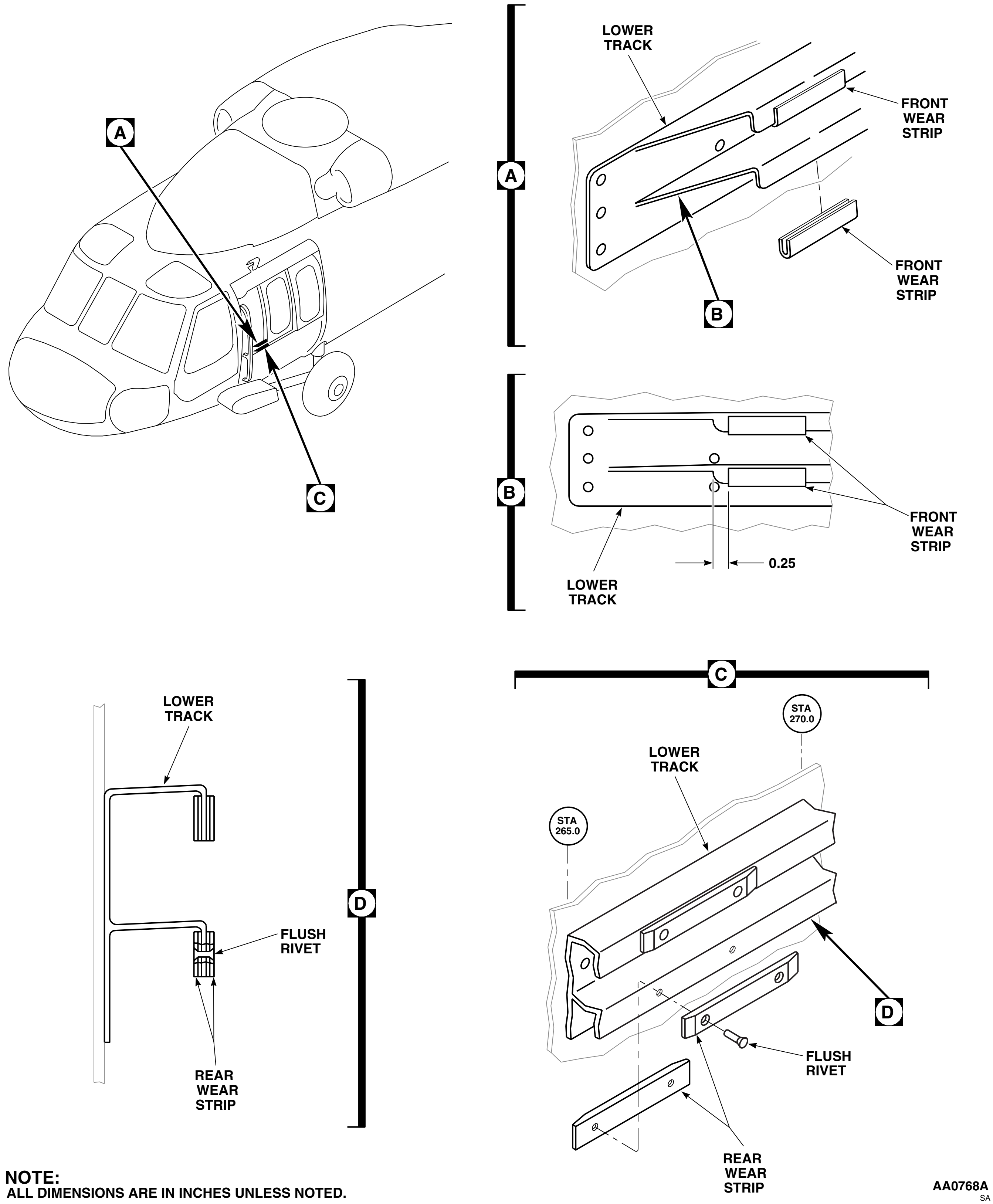

Figure 1. Lower Track Wear Strips.

UPPER TRACK INSPECTION - Continued

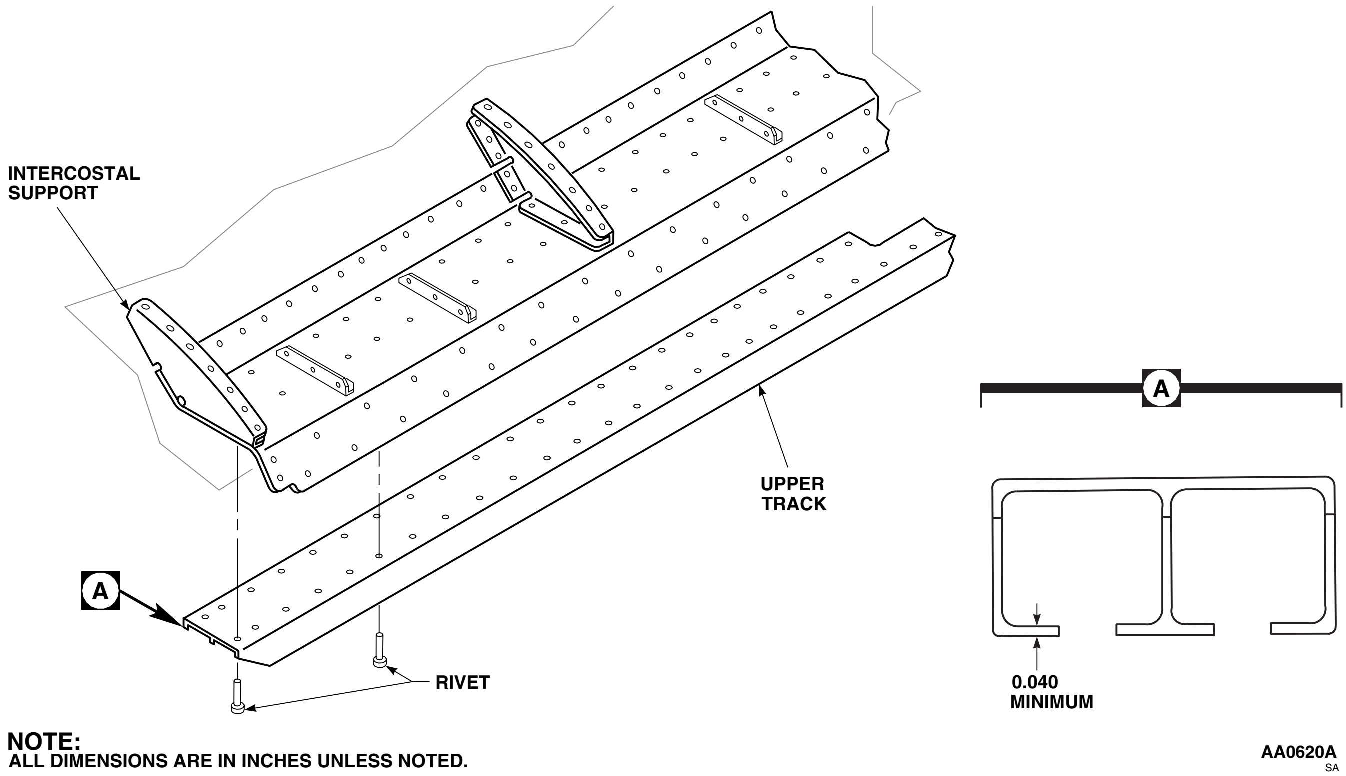

Figure 2. Replace Upper Track.

4. If wear spots are present and track material thickness is within limit, repair track by installing locally fabricated wear strips (WP 0397 00).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TROOP/CARGO DOOR INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Padding
Spring Scale, AAA-S-133

Material/Parts
Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00

Personnel Required
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0245 00
WP 0246 00
WP 1803 00

LOWER TRACK INSPECTION

1. Inspect track for visual cracks and for pieces broken off. **NO CRACKS OR BROKEN PIECES ALLOWED.** Replace lower track if cracks or broken off pieces are found (WP 0246 00).
2. Inspect wear strips for wear. Replace wear strips that are worn to less than 0.010-inch.
3. Inspect track under wear strips that are completely worn through. Measure thickness of track at worn areas (Figure 1, Detail C). **REPLACE TRACK IF THICKNESS IS LESS THAN 0.080-INCH (WP 0246 00).**
4. Inspect attaching screws for damaged threads, damaged or deformed heads. Inspect washers for deformation or damage. **NO DAMAGED THREADS, DEFORMED SCREWHEADS, OR WASHERS ALLOWED (WP 0246 00).**

WINDOW JETTISON CHECK

1. Position foam padding to prevent damage to troop/cargo door window during jettison check.
2. Have assistants outside helicopter to guide jettisoned troop/cargo door window onto padding.

NOTE

Turning emergency release lever will jettison both windows.

3. If only one window need be removed, secure other window in place with pressure sensitive adhesive tape, Item 334, WP 1803 00.
4. To gain access to emergency release lever, pull guard from cover (if installed).
5. Connect spring scale to troop/cargo door window jettison handle.
6. Pull jettison handle with spring scale. **FORCE SHALL NOT EXCEED 30 LBS.**

NOTE

Force required to jettison window from helicopter shall not be so great that an average size person cannot jettison troop/cargo door window.

7. Manually push base of window outward from helicopter. **WINDOW SHALL FALL AWAY FROM HELICOPTER.**

WINDOW JETTISON CHECK - Continued

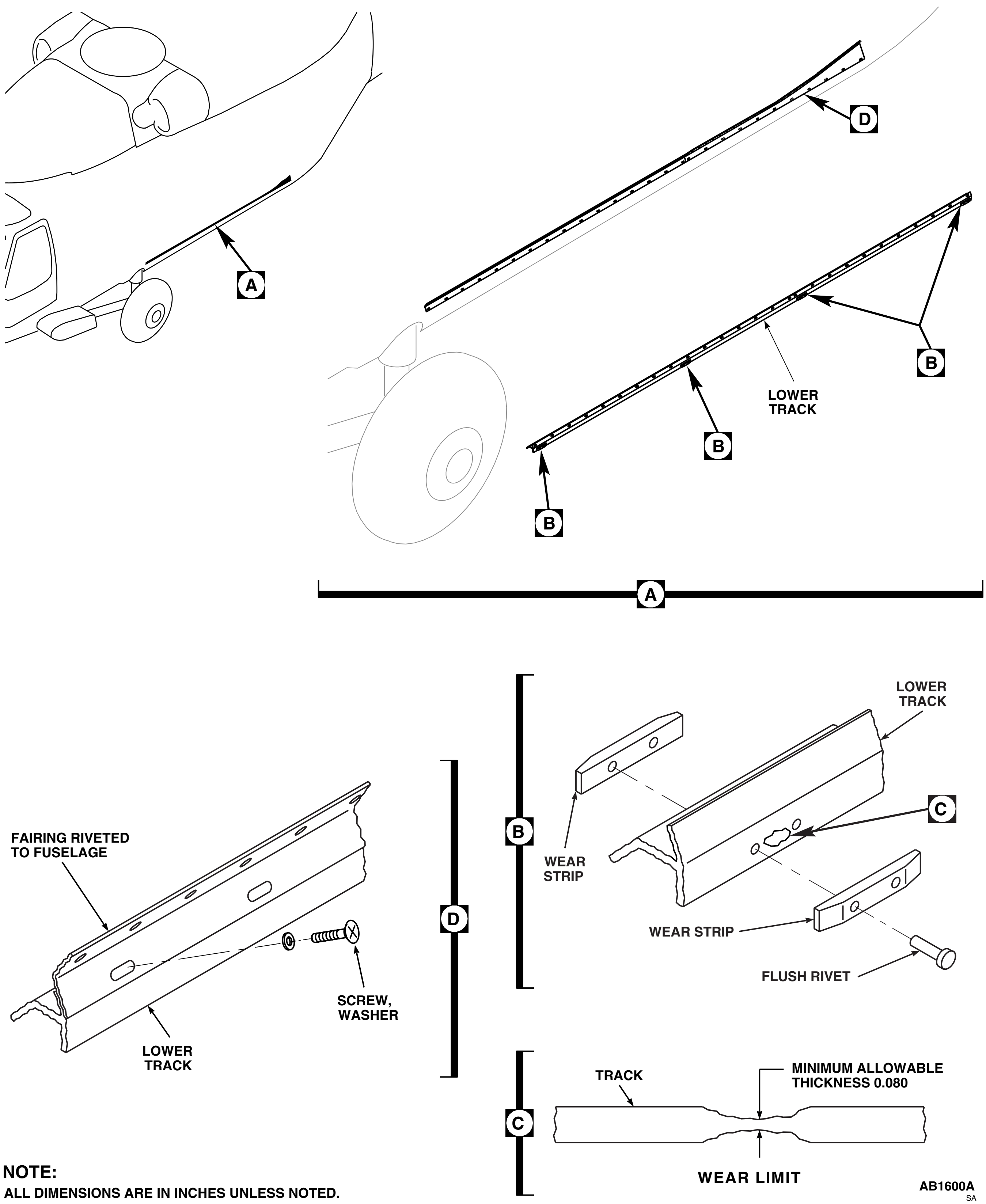

Figure 1. Troop/Cargo Door Inspections.

WINDOW JETTISON CHECK - Continued

8. Inspect troop/cargo door window and jettison mechanism for damage. **NO DAMAGE ALLOWED.**
9. Reinstall troop/cargo door window (WP 0245 00).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CABIN FLOOR INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
WP 0402 00

TIEDOWN RINGS AND FLOOR PANS INSPECTIONS

1. Turn off all electrical power.
2. Visually inspect cargo tiedown rings and floor pans for corrosion (TM 1-1500-344-23). Corrosion is indicated by pitting. **PITTING SHALL NOT COVER MORE THAN 25% OF WELD AREA.**
3. Remove corrosion from tiedown rings and floor pans (WP 0402 00).
4. Visually inspect floor attachment screws for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting.
5. Replace corroded screws or remove corrosion (WP 0402 00).
6. If corrosion is present, cut a slot in floor shims to drain water away from screws (WP 0402 00).
7. Visually inspect fasteners, skin under stringers and around bulkheads, and floor attachment fittings for corrosion.
8. Inspect drain holes for obstructions.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

WIRE STRIKE PROTECTION SYSTEM INSPECTIONS

This WP supersedes WP 0181 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0271 00
WP 0272 00
WP 0273 00
WP 0274 00
WP 0275 00
WP 0276 00
WP 0277 00
WP 0278 00
WP 0279 00
WP 0280 00
WP 0281 00
WP 0282 00
WP 0283 00
WP 0284 00

CUTTER INSPECTION

NOTE

The following inspection applies to all wire strike protection components.

1. Inspect cutters for defects (Figure 2).
2. Visually check the following cutters for evidence of wire strike (Figure 1, Sheet 1 and 2):
   a. Wire Strike Upper Cutter.
   b. Wire Strike Pitot Cutter.
   c. Wire Strike Pitot Cutter Guide Channel.
   d. Wire Strike Windshield Wiper Drive Post Deflector.
   e. Wire Strike Main Landing Gear Cutter.
   f. Wire Strike Main Landing Gear Cutter Clamp.
   g. Wire Strike Main Landing Gear Joint Deflector.
   h. Wire Strike Tail Landing Gear Deflector.
   i. Wire Strike Tail Landing Gear Deflector Clamp.
   j. Wire Strike Step Deflector.
   k. Wire Strike Step Extension.
   l. Wire Strike Door Latch Deflector.
   m. Wire Strike Windshield Center Post Deflector Insert.
   n. Wire Strike Windshield Center Post Deflector Channel.
3. **REPLACE THE FOLLOWING CUTTERS IF EVIDENCE OF WIRE STRIKE IS SEEN:**

CUTTER INSPECTION - Continued

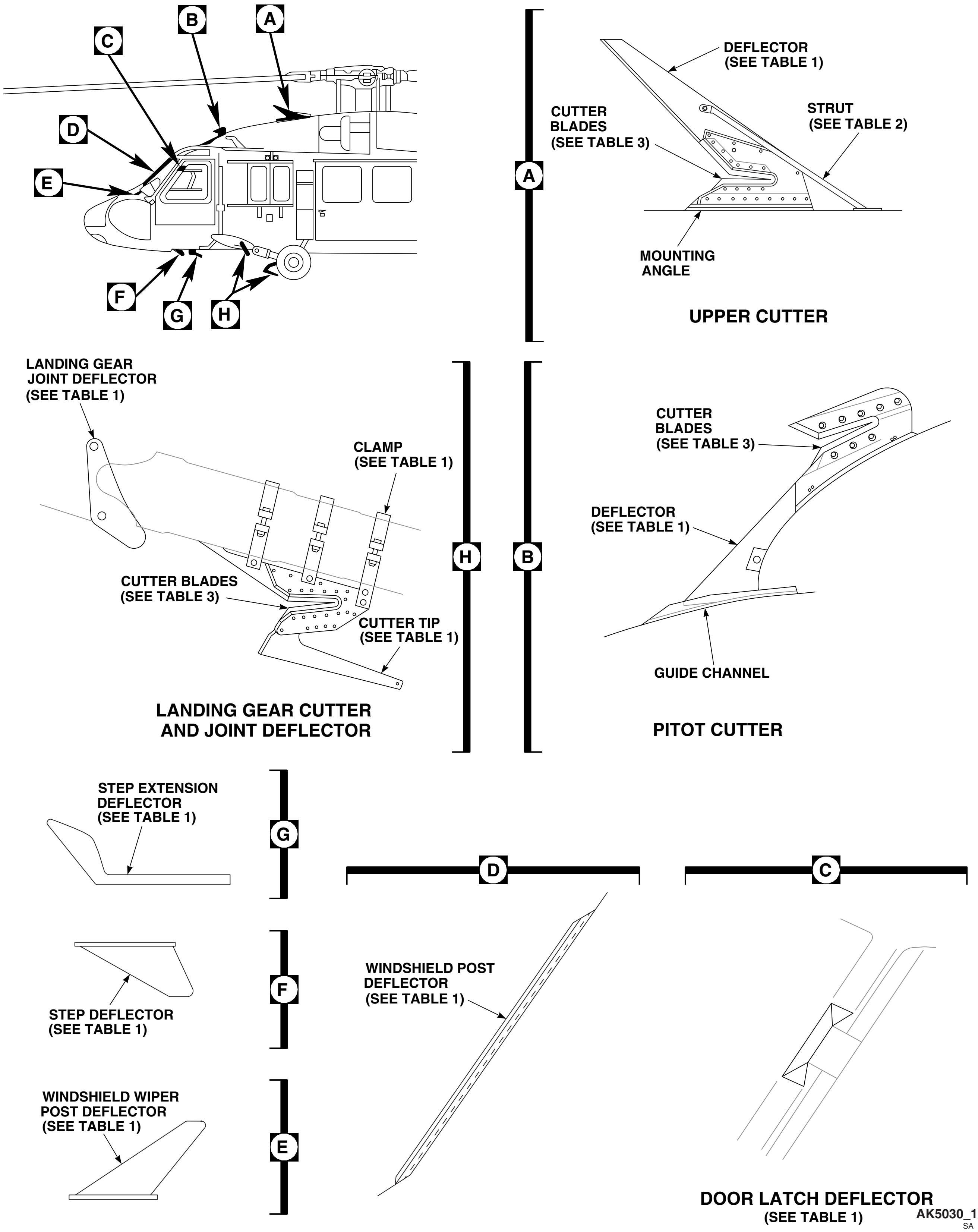

Figure 1. Repair Wire Strike Protection System. (Sheet 1 of 3)

CUTTER INSPECTION - Continued

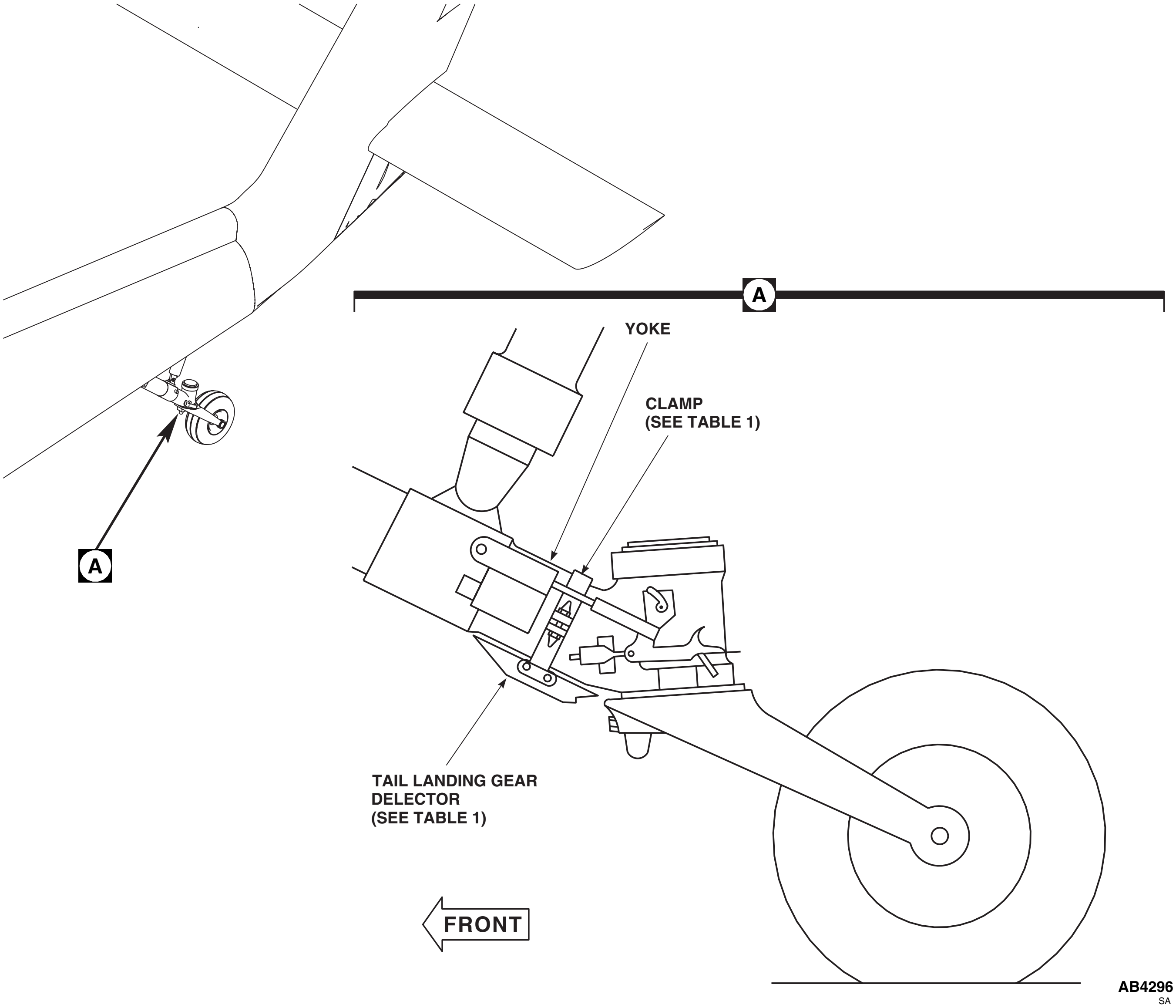

Figure 1. Repair Wire Strike Protection System. (Sheet 2 of 3)

a. Wire Strike Upper Cutter (WP 0271 00).

b. Wire Strike Pitot Cutter (WP 0272 00).

c. Wire Strike Pitot Cutter Guide Channel (WP 0273 00).

d. Wire Strike Windshield Wiper Drive Post Deflector (WP 0274 00).

e. Wire Strike Main Landing Gear Cutter (WP 0275 00).

f. Wire Strike Main Landing Gear Cutter Clamp (WP 0276 00).

g. Wire Strike Main Landing Gear Joint Deflector (WP 0277 00).

h. Wire Strike Tail Landing Gear Deflector (WP 0278 00).

CUTTER INSPECTION - Continued

TABLE 1

DEFECT	ACTION
SCRATCHES, NICKS, GOUGES TO A MAXIMUM OF 0.010-IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND ALUMINUM WOOL. BRUSH ALODINE, PRIME AND PAINT.
PAINT DETERIORATION.	CLEAN AFFECTED AREA USING ALUMINUM WOOL. BRUSH ALODINE, PRIME AND PAINT.
LIGHT CORROSION TO A MAXIMUM OF 0.010-IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND ALUMINUM WOOL. BRUSH ALODINE, PRIME AND PAINT.
MODERATE TO HEAVY CORROSION / WHITE POWDER / MATERIAL BULGING.	REPLACE.
CORRODED / MISSING / INCORRECT / BROKEN HARDWARE.	REPLACE HARDWARE.

TABLE 2

DEFECT	ACTION
SCRATCHES, NICKS, GOUGES TO A MAXIMUM OF 0.005-IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND STEEL WOOL. PRIME AND PAINT.
PAINT DETERIORATION.	CLEAN AFFECTED AREA USING STEEL WOOL. PRIME AND PAINT.
LIGHT CORROSION TO A MAXIMUM OF 0.005-IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND STEEL WOOL. PRIME AND PAINT.
MODERATE TO HEAVY CORROSION / PITTING.	REPLACE.
CORRODED / MISSING / INCORRECT / BROKEN HARDWARE.	REPLACE HARDWARE.

TABLE 3

DEFECT	ACTION
MISSING, PEELING RUBBER.	REMOVE LOOSE RUBBER AND REPLACE WITH SEALANT.
ANY SCRATCHES, NICKS, GOUGES, CHIPPING, ABRASION.	REPLACE.
ANY CORROSION.	REPLACE.
CORRODED / MISSING / INCORRECT / BROKEN HARDWARE.	REPLACE HARDWARE.

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Figure 1. Repair Wire Strike Protection System. (Sheet 3 of 3)

CUTTER INSPECTION - Continued

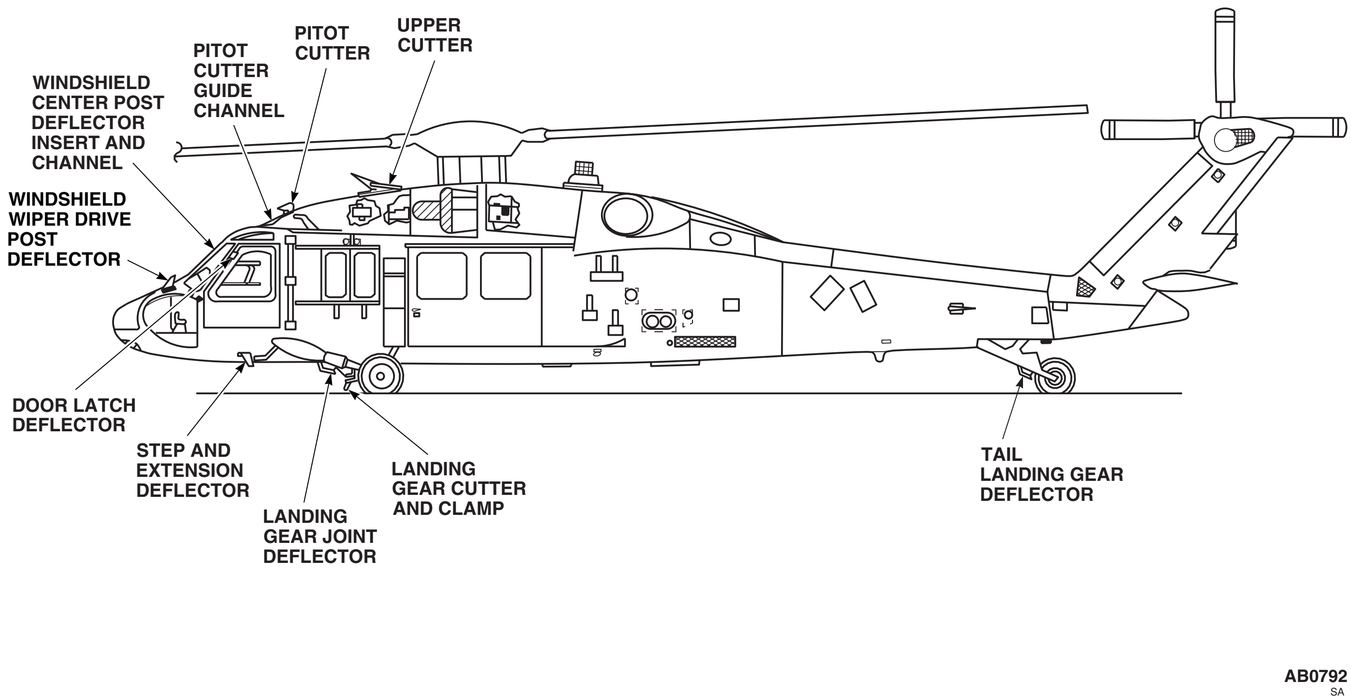


Figure 2. Wire Strike Protection System Inspections.

i. Wire Strike Tail Landing Gear Deflector Clamp (WP 0279 00).

j. Wire Strike Step Deflector (WP 0280 00).

k. Wire Strike Step Extension (WP 0281 00).

l. Wire Strike Door Latch Deflector (WP 0282 00).

m. Wire Strike Windshield Center Post Deflector Insert (WP 0283 00).

n. Wire Strike Windshield Center Post Deflector Channel (WP 0284 00).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

MAIN ROTOR PYLON INSPECTIONS

This WP supersedes WP 0182 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Caliper, Vernier, GGG-C-111
Caliper Set, Micrometer, Outside, GGG-C-105
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive Primer, Item 23, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Grommet, Nonmetallic, Item 139, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 0297 00
WP 0298 00
WP 0299 00
WP 0303 00
WP 0308 00
WP 0377 00
WP 1803 00

MAIN ROTOR SLIDING COVER INSPECTION

1. Visually inspect latch for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. Remove corrosion from latch.
2. Raise latch, loosen jamnut, and inspect fit of latch hook and jamnut on threaded stud. **IF LATCH HOOK OR JAMNUT IS LOOSE OR WOBBLES ON THREAD, REPLACE LATCH (WP 0297 00).**
3. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
4. Visually inspect roller assemblies for the following:
   a. Inspect for disbonding of nylon rollers from bushings.
   b. Inspect for interference between rivets, MS20470AD5-6, and horizontal main pylon roller.
5. Repair roller assemblies as follows:
   a. Replace damaged sliding cover rollers (WP 0298 00 or WP 0299 00)
   b. If interference occurs between rivets, MS20470AD5-6, and forward horizontal main pylon roller, replace two rivets, MS20470AD5-6, with countersunk rivets, NAS1097AD5 (WP 0299 00).
6. Inspect pin receivers and support structure for cracks. **NO CRACKS ALLOWED.** Repair cracks (WP 0377 00).
7. Inspect pin receiver holes for wear using vernier caliper and outside micrometer caliper from caliper set. **WEAR SHALL NOT EXCEED 0.60-INCH.**
8. **UH-60A 86-24516 - SUBQ UH-60L EH-60A 86-24561 - SUBQ** Inspect forward locating pin support for wear or damage to nylon edging grommet.
9. Replace pin support edging grommet as follows:
   a. Remove damage edging grommet using nonmetallic scraper.

MAIN ROTOR SLIDING COVER INSPECTION - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Remove adhesive residue from pin support and clean bonding surfaces thoroughly, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply thin coat of adhesive primer, Item 23, WP 1803 00, to bonding surfaces of pin support.

d. Let primer cure for at least 1 hour at room temperature.

e. Cut piece of nonmetallic grommet, Item 139, WP 1803 00, to fit edge of pin support.

f. Mix adhesive, Item 11, WP 1803 00, and apply to bonding surfaces of pin support and nonmetallic grommet.

g. Immediately install nonmetallic grommet on edge of pin support and press in place.

APU ACCESS DOOR INSPECTION

1. With access door closed and catches locked, try to move door horizontally. **MOVEMENT AT HINGE LINE OF 0.060-INCH OR MORE IS NOT ALLOWED. IF LIMIT IS EXCEEDED, REPLACE HINGES (WP 0303 00).**
2. Inspect flush catches. **IF CATCH OPENS WITH MINOR DOOR MOVEMENT, OR CATCH SPRING IS BROKEN, REPLACE FLUSH CATCH (WP 0303 00).**
3. Inspect door hinges for cracks and wear. **WEAR SHALL NOT EXCEED 10% OF MATERIAL THICKNESS, NO CRACKS ALLOWED. IF LIMITS ARE EXCEEDED, REPLACE HINGE (WP 0303 00).**
4. Inspect hinges, catches, and catch fittings for corrosion. **CORROSION REMOVAL SHALL NOT EXCEED 10% OF MATERIAL THICKNESS (WP 0303 00).**

OIL COOLER COMPARTMENT ACCESS DOOR INSPECTION

1. With access door closed and catches locked, try to move door horizontally. **MOVEMENT AT HINGE LINE OF 0.060-INCH OR MORE IS NOT ALLOWED. IF LIMIT IS EXCEEDED, REPLACE HINGES (WP 0308 00).**
2. Inspect flush catches. **IF CATCH OPENS WITH MINOR DOOR MOVEMENT, OR CATCH SPRING IS BROKEN, REPLACE FLUSH CATCH (WP 0308 00).**
3. Inspect door hinges for cracks and wear. **WEAR SHALL NOT EXCEED 10% OF MATERIAL THICKNESS, NO CRACKS ALLOWED. IF LIMITS ARE EXCEEDED, REPLACE HINGE (WP 0308 00).**
4. Inspect hinges, catches, and catch fittings for corrosion. **CORROSION REMOVAL SHALL NOT EXCEED 10% OF MATERIAL THICKNESS (WP 0308 00).**

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

ENGINE COWLING INSPECTIONS

This WP supersedes WP 0183 00, dated 17 April 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer, 0 - 1 inch

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
WP 0316 00
WP 0378 00

LATCH INSPECTION

1. Open engine cowling.
2. Lift latch handle away from surface of cowling.
3. **W/O MWO 50-64** Inspect inner and outer sides of latch handle for installation of spacer on center pivot shanks (Figure 1).
4. **W/O MWO 50-64** Inspect spring and spacer for proper movement and security.
5. Inspect flush head rivet forming the center pivot for properly bucked end. **BUCKED END OF RIVET MUST HAVE A MINIMUM DIAMETER OF 0.170-INCH. FLUSH HEADS OF RIVET MUST FILL 75% OF COUNTERSUNK WASHER.**
6. Inspect lockpin tubes for kinks, distortion or cracks (no cracks allowed, replace immediately if cracks are found). Replace kinked, or distorted lockpin tubes (WP 0316 00).
7. Inspect lockpin and lockpin fittings for cracks and galling. **REPLACE IF CRACKS ARE FOUND OR GALLING IS DEEPER THAN 0.020-INCH (WP 0316 00).**
8. Inspect latch handle release for ease of movement.
9. Visually inspect latch handle assembly for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. **PITS SHALL NOT EXCEED 0.015-INCH DEEP.** Remove corrosion that does not exceed this limit.
10. Inspect latch handle and handle support for wear at bolt holes and slotted holes. **REPLACE IF WEAR EXCEEDS 10% OF MATERIAL THICKNESS (WP 0316 00).**
11. Inspect handle support for cracks across bolt hole lugs. **CRACKS SHALL NOT EXTEND ACROSS BOLT HOLE LUG.** Replace handle support if crack separates lug from handle support (WP 0316 00).
12. Inspect chrome-plated handle guide track. **WEAR UP TO 0.025-INCH IS ALLOWED IF OPERATION IS NOT IMPAIRED.**
13. Inspect cowling hinges on airframe and adjacent structure for cracks, wear, and security (WP 0378 00).
14. Inspect latch, hinges, and latch fittings for nicks, scratches, and corrosion. **NICKS AND SCRATCHES SHALL NOT EXCEED 10% OF MATERIAL THICKNESS. CORROSION REMOVAL SHALL NOT EXCEED 10% OF MATERIAL THICKNESS (WP 0316 00).**

LATCH INSPECTION - Continued

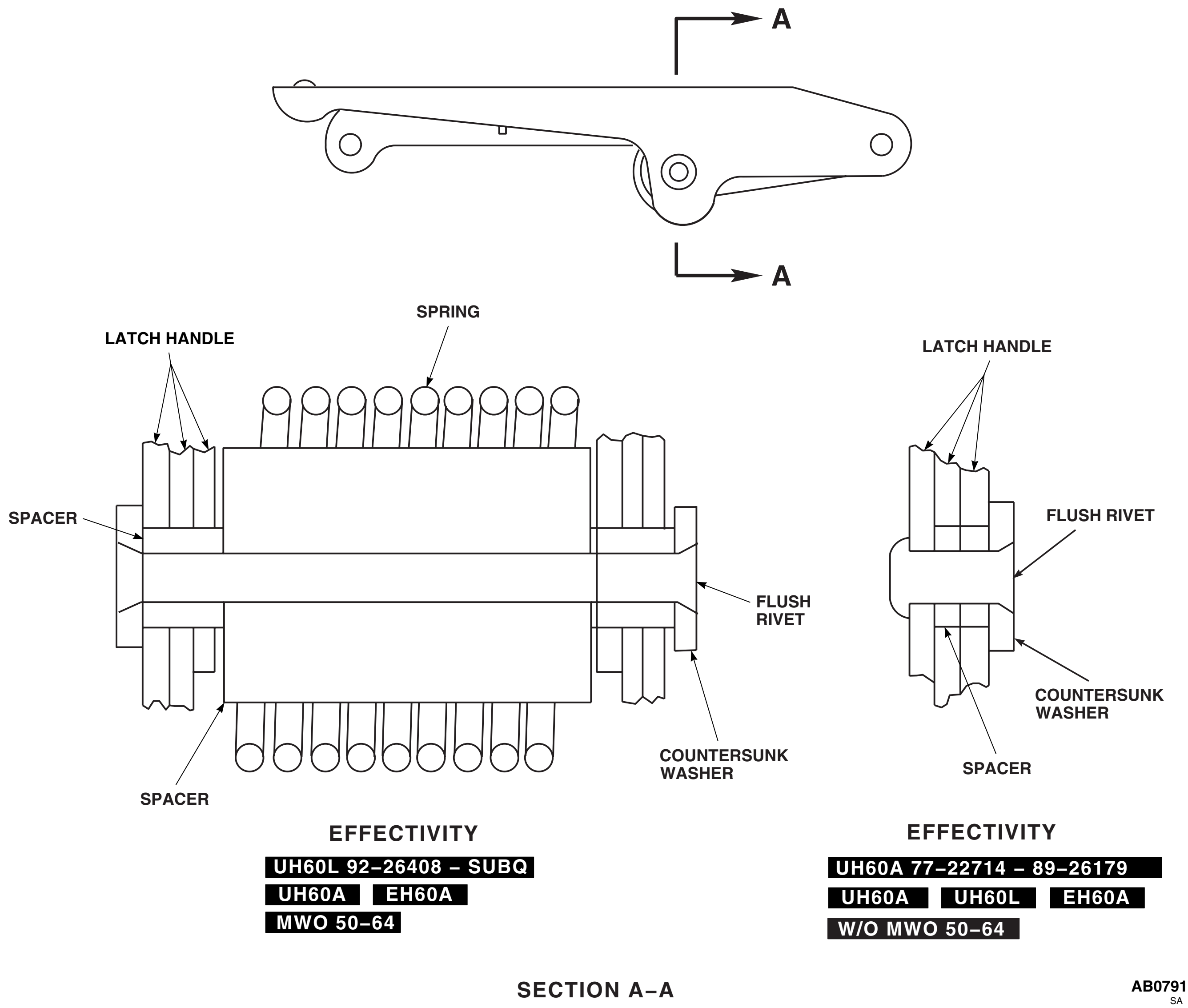


Figure 1. Engine Cowling Latch Inspection.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TRANSITION SECTION INSPECTIONS

This WP supersedes WP 0184 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-253-724-13&P
WP 0233 00
WP 0258 00
WP 0349 00
WP 1289 00
WP 1291 00
WP 1312 00
WP 1315 00

TRANSITION SECTION-TO-TAIL CONE BOLTS TORQUE CHECK

1. **UH-60A UH-60L** Do this:
   a. Remove troop seats (WP 0233 00).
   b. Remove soundproofing panel from left or right stowage compartment (WP 0258 00).
   c. Unlock cargo netting in stowage compartment.
2. **EH-60A** Remove the following ECS equipment:
   a. Plenum (WP 1289 00).
   b. Evaporator pallet (WP 1315 00) or condenser pallet (WP 1312 00).
   c. Supply ducting (WP 1291 00).
3. **HH-60A HH-60L** On Medevac helicopters do the following:
   a. Open avionics compartment access door (WP 0349 00).
   b. Purge OBOGS system (TM 1-1520-253-724-13&P).
   c. Remove oxygen concentrator (TM 1-1520-253-724-13&P).
4. Check torque on transition section to tail cone bolts in the sequence shown in Figure 1. **TORQUE SHALL NOT BE LESS THAN 190 INCH-POUNDS. RETORQUE NUTS THAT FAIL TORQUE CHECK TO 190 - 210 INCH-POUNDS.**
5. Check bolt for proper thread extension beyond nut. **BOLT SHALL EXTEND AT LEAST 0.084-INCHES BEYOND NUT.** If bolt extends too far one additional washer may be added to keep nut from bottoming out on shank of bolt.

TRANSITION SECTION-TO-TAIL CONE BOLTS TORQUE CHECK - Continued

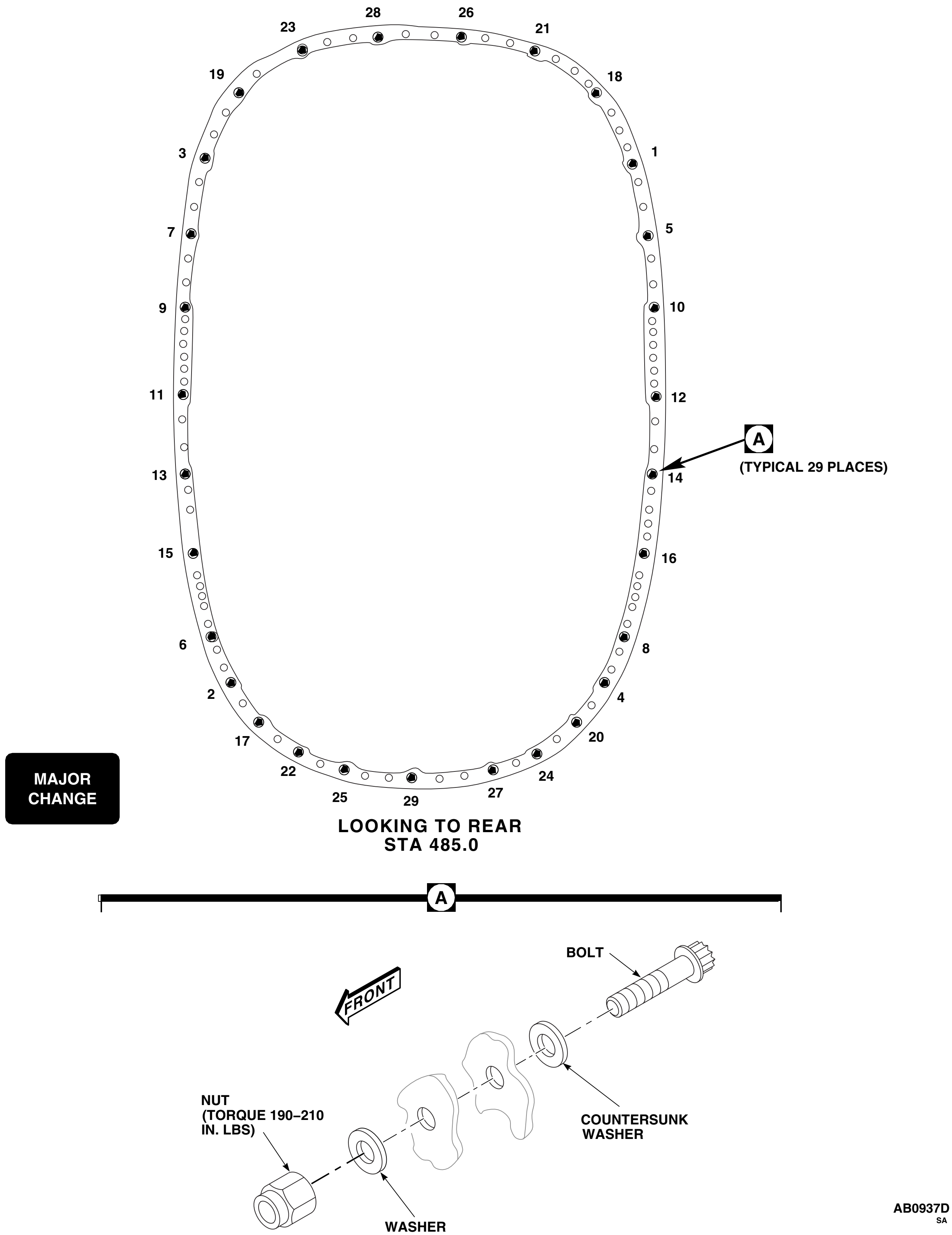


Figure 1. Transition Section-To-Tail Cone Bolts Torque Check Sequence.

TRANSITION SECTION-TO-TAIL CONE BOLTS TORQUE CHECK - Continued

NOTE

Nuts may be on front side of stackup for easy access.

6. Make sure area is clean and free of foreign material.
7. **UH-60A UH-60L** Do this:
   a. Hook cargo netting in rear section stowage compartment.
   b. Install soundproofing panel (WP 0258 00).
   c. Install rear troop seats (WP 0233 00).
8. **EH-60A** Install the following ECS equipment:
   a. Plenum (WP 1289 00).
   b. Evaporator pallet (WP 1315 00) or condenser pallet (WP 1312 00).
   c. Supply ducting (WP 1291 00).
9. **HH-60A HH-60L** On Medevac helicopters do the following:
   a. Install oxygen concentrator (TM 1-1520-253-724-13&P).
   b. Perform leak check of OBOGS system (TM 1-1520-253-724-13&P).
   c. Close avionics compartment access door (WP 0349 00).
10. If bolts required retorquing, check DA Form 2408-13-1 to make sure transition section to tail cone bolts torque check is shown due 40 flight-hours.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TAIL ROTOR DRIVE SHAFT COVER INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer, 0 - 1 inch

Material/Parts
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lubricating Oil, Helicopter, Item 187, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Polyurethane Coating, Item 232, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

QUARTER-TURN FASTENER AND HINGE INSPECTION

1. Visually inspect quarter-turn fasteners and pin type hinges for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. Remove corrosion from hinges.
2. **INOPERATIVE OR CORRODED QUARTER TURN FASTENERS SHALL BE REPLACED.**
3. Remove corrosion manually, using scouring pad, Item 205, WP 1803 00. **PITS ON HINGES SHALL NOT EXCEED 0.005-INCH DEEP AFTER CLEANUP.**
4. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, and 1 part water.
5. Apply compound solution to corroded areas of fasteners and hinges.
6. Rinse thoroughly with clean water.

CAUTION

Paint on bearing surfaces can cause improper operation of hinges and fasteners. Keep paint off of bearing surfaces.

7. Touch up paint with epoxy primer coating kit, Item 120, WP 1803 00, and polyurethane coating, Item 232, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
8. Coat hinge pins with helicopter lubricating oil, Item 187, WP 1803 00.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TAIL ROTOR PYLON INSPECTIONS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA101

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-265-23
WP 0337 00
WP 1681 00

TAIL ROTOR GEAR BOX FITTING INSPECTION

1. Inspect tail rotor gear box fitting for surface cracks around lower barrel nut holes, using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
2. Inspect skin for cracks, especially in area surrounding tail rotor gear box fitting. **NO CRACKS ALLOWED.** If skin cracks are noted, inspect inside surfaces of fitting where attached to front spar cap angles, using inspection mirror and flashlight through access hole in front spar web. **NO CRACKS ALLOWED.** A one-time flight to depot is allowed.

LOWER STEP INSPECTION

1. Turn off all electrical power.
2. Fold pylon 90 degrees (WP 1681 00).
3. Turn step towards right side of helicopter and lower step from pylon.
4. Inspect step for proper operation, if step does not operate properly, inspect as follows:
   a. Inspect spring for security as follows:
      (1) **IF SPRING HAS DISCONNECTED, REINSTALL SPRING.**
      (2) **IF SPRING IS BROKEN, REPLACE SPRING.**
   b. Inspect pin for looseness. **IF LOOSE, REPLACE STEP ASSEMBLY.**
   c. On step assembly, 70207-06052-041,inspect cotter pin for security. **REPLACE MISSING COTTER PIN.**
5. Using fluorescent penetrant inspection kit, visually inspect step assembly for cracks. **NO CRACKS ALLOWED.** If crack is found, replace step assembly (WP 0337 00 and TM 1-1520-265-23).
6. Inspect tube for wear at lower support point. **IF ANODIZED COATING IS WORN TO BASE METAL, REPLACE STEP ASSEMBLY.**

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

STABILATOR INSPECTIONS

This WP supersedes WP 0187 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer, 0 - 1 inch
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Corrosion Preventive Compound, Item 105, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
TM 55-1500-345-23
WP 0328 00
WP 0338 00
WP 0339 00
WP 0340 00
WP 0342 00
WP 0343 00
WP 0345 00
WP 0370 00
WP 0381 00
WP 0421 00
WP 0422 00
WP 0423 00
WP 0427 00
WP 0428 00
WP 0430 00
WP 1803 00

STABILATOR ACTUATOR ATTACH FITTING BEARING INSPECTION

1. Check spherical bearing for radial play. **IF RADIAL PLAY IS MORE THAN 0.010-INCH, REPLACE FITTING.**

WARNING

CSI

- Bonding of liners in center lobe of stabilator actuator attach fitting is a critical characteristic. Proper bonding must be verified.
- Damage to fitting will occur if feeler gage is improperly used. Liner may become dis-bonded. Use care when using feeler gage to check for liner edge separation.

2. Visually inspect bonding of liners in center lobe of fitting. Probe for edge separation using 0.0015-inch feeler gage. **IF VOIDS EXCEED 5% OF JOINT LINE LENGTH, REPLACE FITTING.**

STABILATOR SPAR INSPECTION

1. Inspect top stabilator surface at front spar, both left and right sides, for skin cracks and/or loose rivets. Repair cracks (WP 0427 00 or WP 0428 00) or replace stabilator (WP 0338 00).
2. **UH-60A 84-23935 - SUBQ UH-60L EH-60A** Remove screws, washers, and leading edge inspection plate cover.
3. **UH-60A 84-23935 - SUBQ UH-60L EH-60A** Using a flashlight and inspection mirror, check top internal spar cap and web for cracks and/or loose rivets. Replace stabilator with cracks, or loose rivets (WP 0338 00).

ELECTROSTATIC DISCHARGER INSPECTION

1. Inspect electrostatic discharger for cracks in rubber coating. If rubber coating is cracked, check for broken or frayed wires in cracked area. **NO BROKEN OR FRAYED WIRES ALLOWED.** Replace electrostatic discharger if broken or frayed wires are found (WP 0339 00).
2. If no broken or frayed wires are found, remove electrostatic discharger for repair to rubber coating (WP 0339 00).

STABILATOR FREE PLAY INSPECTION

1. With stabilator at 0 degrees, check stabilator up and down free play at left and right tip caps. **TOTAL VERTICAL MOVEMENT AT EITHER SIDE SHALL NOT BE MORE THAN 5/16-INCH.**
2. Check up and down free play at stabilator trailing edge center directly behind actuator. **TOTAL VERTICAL MOVEMENT SHALL NOT BE MORE THAN 1/4-INCH (Figure 1, Sheet 1, Detail A).**
3. If free movement is greater than specified, do this:
   a. Check endplay and sideplay in actuator assembly.
   b. Check for loose actuator fittings or loose stabilator hinge fittings
   c. Check for loose or worn pivot bolts, bearings, and bushings.
   d. Check elastomeric bearings.

STABILATOR ACTUATOR SHAFT PLAY INSPECTION

NOTE

Ensure stabilator is in neutral or 0 degree position before starting inspection.

1. Measure endplay as axial motion of clevis and relative to actuator housing (Figure 1, Sheet 1, Detail B). **IF END-PLAY OF EITHER CLEVIS, UNDER REVERSING LOAD OF 20 POUNDS, IS MORE THAN 0.025-INCH, REPLACE ACTUATOR ASSEMBLY (WP 0342 00).**
2. Measure sideplay as radial motion of dual actuator housing relative to fixed clevis ends (Figure 1, Sheet 1, Detail B). **IF TOTAL SIDEPLAY, UNDER REVERSING LOAD OF 10 POUNDS, IS MORE THAN 0.025-INCH, REPLACE ACTUATOR ASSEMBLY (WP 0342 00).**
3. If endplay is within limits. Refer to, STABILATOR UPPER ACTUATOR ATTACH FITTING INSPECTION and STABILATOR LOWER ACTUATOR ATTACH FITTING INSPECTION, in this work package.

STABILATOR UPPER ACTUATOR ATTACH FITTING INSPECTION

1. Visually inspect pylon actuator attach fitting for looseness and wear.
2. If pylon actuator attach fitting is loose, **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
3. If fitting shows wear, refer to WP 0423 00 for repair limits and procedures.
4. Remove bolt from actuator pylon fitting (WP 0338 00).

STABILATOR UPPER ACTUATOR ATTACH FITTING INSPECTION - Continued

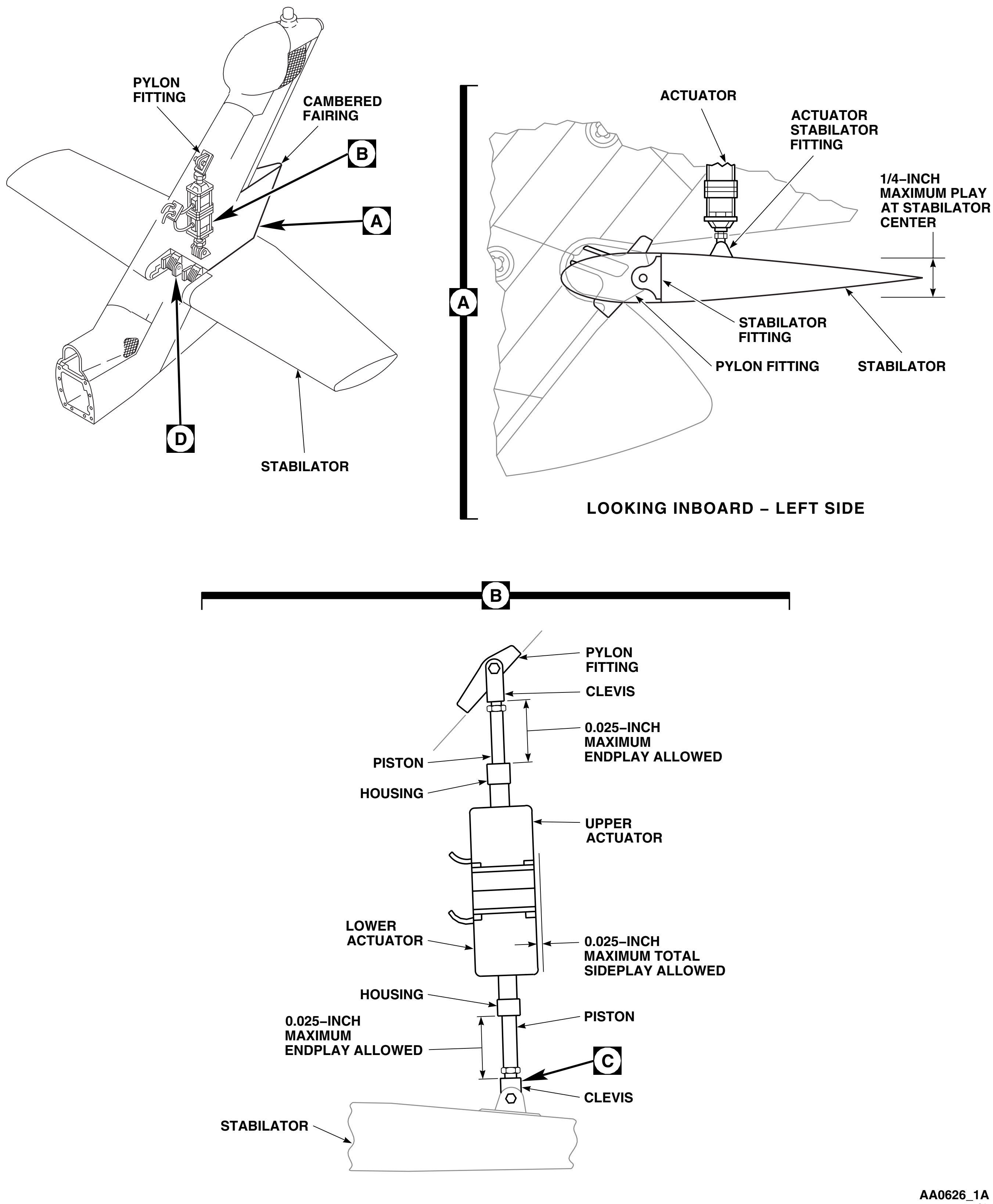
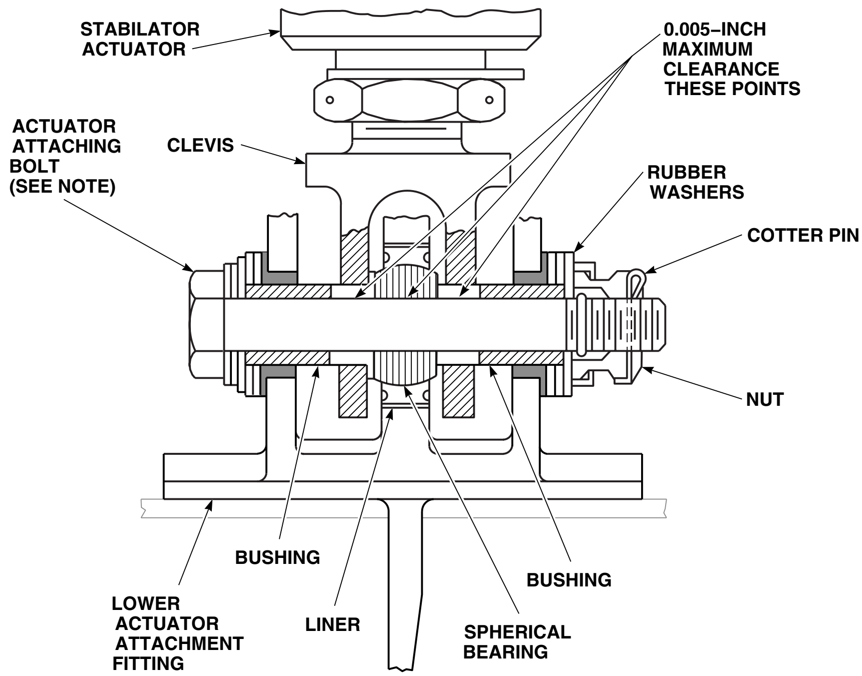


Figure 1. Inspect Stabilator and Actuator Attach Fittings. (Sheet 1 of 4)

STABILATOR UPPER ACTUATOR ATTACH FITTING INSPECTION - Continued



EFFECTIVITY

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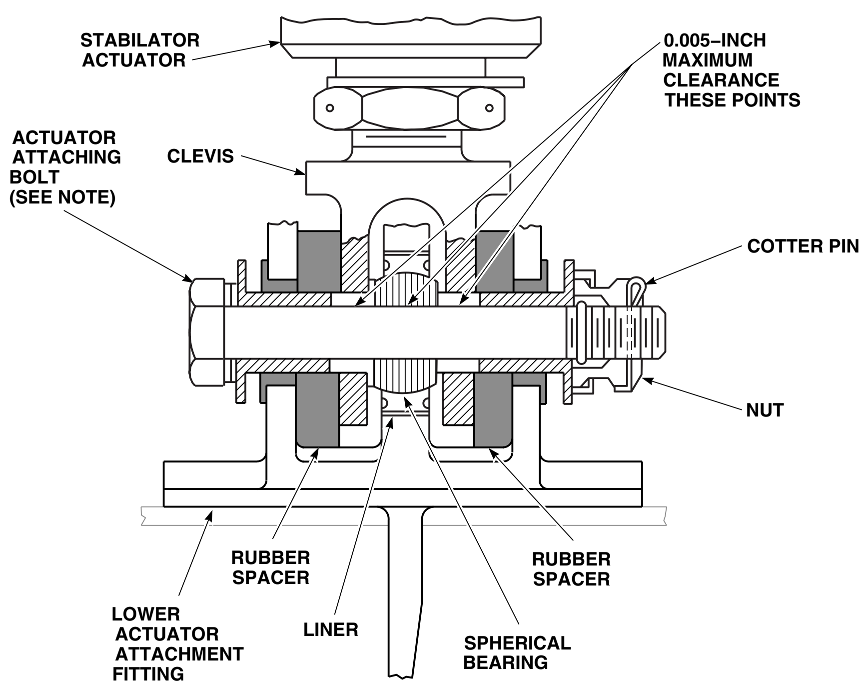

EFFECTIVITY

UH–60A 79–24274 – SUBQ

UH–60L **EH–60A**

NOTE:

BOLTHEADS MAY BE INSTALLED IN EITHER DIRECTION.

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Figure 1. Inspect Stabilator and Actuator Attach Fittings. (Sheet 2 of 4)

STABILATOR UPPER ACTUATOR ATTACH FITTING INSPECTION - Continued

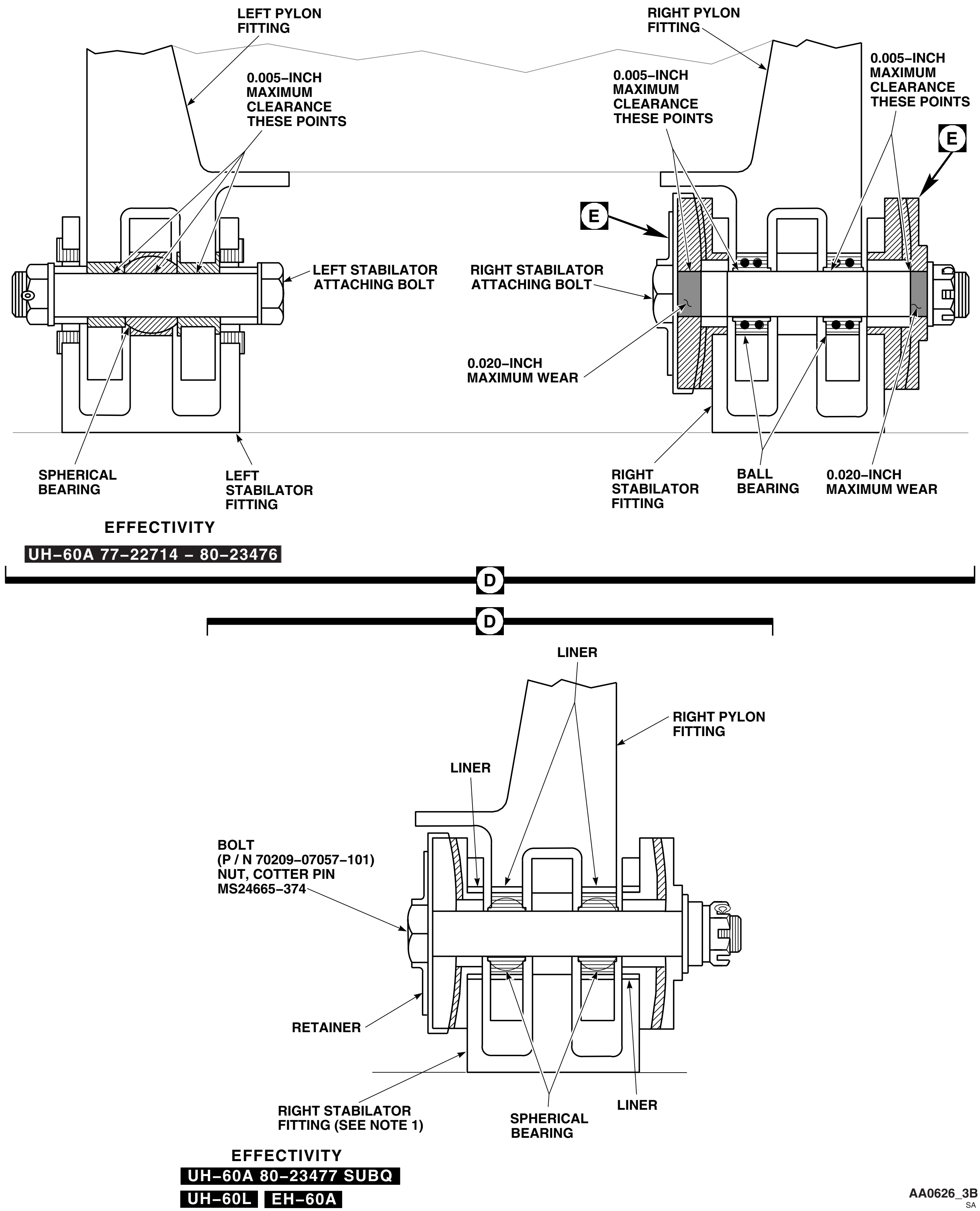


Figure 1. Inspect Stabilator and Actuator Attach Fittings. (Sheet 3 of 4)

STABILATOR UPPER ACTUATOR ATTACH FITTING INSPECTION - Continued

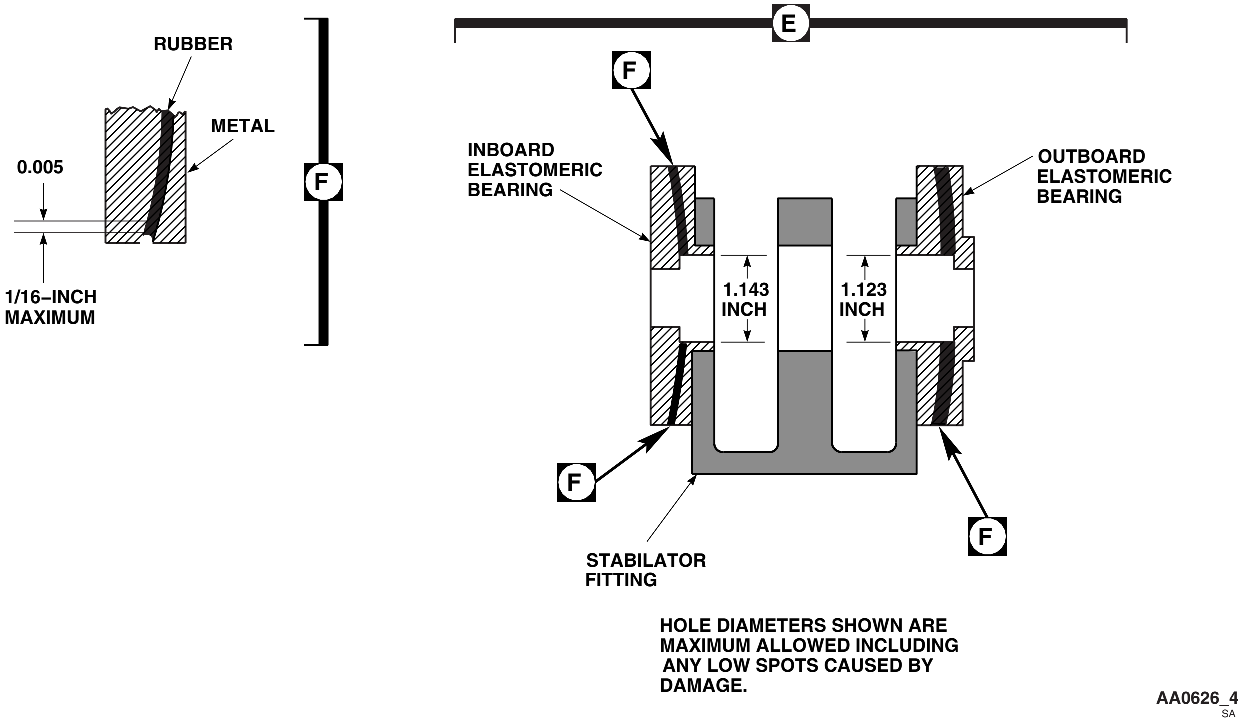


Figure 1. Inspect Stabilator and Actuator Attach Fittings. (Sheet 4 of 4)

WARNING

CSI

Bolthead shoulder and bolt shank of stabilator actuator attach bolts are critical surfaces. **DIMENSION LIMITS MUST NOT BE EXCEEDED.**

5. Check clearance between pylon fitting bushings and pivot bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH, CHECK BOLT AS FOLLOWS:**
   a. Replace bolt if cadmium-plating is worn through to steel.
   b. Measure diameter of bolt. **IF DIMENSION IS LESS THAN 0.370-INCH, REPLACE BOLT (WP 0338 00).**
   c. If bolt diameter is within limits, replace upper fitting (WP 0423 00).
6. Check clearance between upper actuator clevis bushing bore and pivot bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH, AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE UPPER ACTUATOR CLEVIS (WP 0343 00).**
7. Remove bolt from stabilator lower attach fitting (WP 0338 00) and check bolt.

STABILATOR LOWER ACTUATOR ATTACH FITTING INSPECTION

WARNING

CSI

Bolthead shoulder and bolt shank of stabilator actuator attach bolts are critical surfaces. **DIMENSION LIMITS MUST NOT BE EXCEEDED.**

1. Replace bolt if outside diameter and shoulder show signs of nicks, dents, scratches, and tool or scuff marks (WP 0338 00).
2. Measure diameter of bolt. **IF DIMENSION IS LESS THAN 0.370-INCH, REPLACE BOLT (WP 0338 00).**
3. Check stabilator fitting spherical bearing for radial play. **IF RADIAL PLAY IS MORE THAN 0.012-INCH, REPLACE BEARING (WP 0430 00).**

WARNING

CSI

- Bonding of liners in center lobe of stabilator actuator attach fitting is a critical characteristic. Proper bonding must be verified.
- Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.

4. Visually inspect bonding of liners in center lobe of fitting. Probe for edge separation using 0.0015-inch feeler gage. **IF VOIDS EXCEED 5% OF JOINT LINE LENGTH, REPLACE STABILATOR (WP 0338 00).**
5. Check clearance between lower actuator clevis bushing bore and pivot bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE LOWER ACTUATOR CLEVIS (WP 0343 00).**
6. Check clearance between stabilator attach fitting bearing/bushing bore and pivot bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE STABILATOR (WP 0338 00).**

LEFT STABILATOR-TO-PYLON ATTACH FITTING INSPECTION

1. Check pylon fitting for looseness. If loose, replace pylon (WP 0328 00).
2. Check stabilator fitting for looseness. If loose, tighten bolts (WP 0340 00).
3. Remove stabilator (WP 0338 00) and check left bolt.

WARNING

CSI

Bolthead shoulder and bolt shank of stabilator attach bolts are critical surfaces. Dimension limits must not be exceeded.

4. **REPLACE BOLT IF CADMIUM-PLATING IS WORN THROUGH TO STEEL (WP 0338 00).**
5. Measure diameter of bolt. **IF DIMENSION IS LESS THAN 0.6215-INCH, REPLACE BOLT (WP 0338 00).**

LEFT STABILATOR-TO-PYLON ATTACH FITTING INSPECTION - Continued

6. Check stabilator left attach fitting spherical bearing for radial play. **IF RADIAL PLAY IS MORE THAN 0.012-INCH, REPLACE FITTING (WP 0340 00) (Figure 1, Sheet 3, Detail D).**
7. Check clearance between stabilator spherical bearing bore and attaching bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH, AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE FITTING (WP 0340 00).**
8. Check clearance between left side pylon fitting bushing bore and attach bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH, AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE PYLON FITTING BUSHINGS (WP 0421 00).**

RIGHT STABILATOR-TO-PYLON ATTACH FITTING INSPECTION

1. Check pylon fitting for looseness (Figure 1, Sheet 3, Detail D). If loose, replace pylon (WP 0328 00).
2. Check stabilator fitting for looseness. If loose, tighten attach bolts (WP 0338 00).
3. Remove stabilator (WP 0338 00).
4. **UH-60A 77-22714 - 80-23476** Check ball bearings for radial play. **IF BALL BEARING RADIAL PLAY IS MORE THAN 0.001-INCH IN EITHER BEARING, REPLACE BEARING (WP 0422 00).**
5. **UH-60A 80-23477 - SUBQ UH-60L EH-60A** Check spherical bearing radial play. **IF RADIAL PLAY IS MORE THAN 0.012-INCH IN EITHER BEARING, REPLACE BEARING (WP 0422 00).**
6. Check clearance between pylon fitting bearing bore and attachment bolt (Figure 1, Sheet 3, Detail D). **IF CLEARANCE IS MORE THAN 0.005-INCH, AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE BEARING (WP 0422 00).**
7. Check clearance between stabilator fitting elastomeric bearing bore and attachment bolt. **IF CLEARANCE IS MORE THAN 0.005-INCH, AND BOLT DIAMETER IS WITHIN LIMITS, REPLACE FITTING (WP 0340 00).**

RIGHT ATTACH BOLT INSPECTION/REPAIR

WARNING

CSI

The bolthead shoulder and unthreaded shank of right attachment bolt are critical surfaces that must not be damaged. Use extreme care when removing and installing. Use protective covering when removed.

1. Measure attachment bolt diameter throughout shank. Except in areas shaded in Figure 1, Sheet 3, Detail D, and areas discussed in step 6. **MINIMUM BOLT DIAMETER ALLOWED IS 0.7465-INCH. IF BOLT DIAMETER, WHERE REQUIRED, IS LESS THAN 0.7465-INCH, REPLACE BOLT (WP 0338 00).**
2. Visually inspect bolt for scuffs and scratches. **NICKS, DENTS, SCRATCHES AND SCUFF MARKS, NOT MORE THAN 0.003-INCH WIDE, ARE ALLOWED.** Damage blending is permitted within limits of step 1.

NOTE

Right stabilator attach bolt, 70209-07057-101, may become discolored (reddish-brown) while in service. Surface discoloration to this bolt is not a cause for replacement.

3. Visually inspect bolthead and threaded portion of shank for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. **PITS SHALL NOT EXCEED 0.010-INCH DIAMETER.**

RIGHT ATTACH BOLT INSPECTION/REPAIR - Continued

4. Remove corrosion from noncritical areas of bolt manually, using scouring pad, Item 205, WP 1803 00.
5. Coat bolt with corrosion preventive compound, Item 105, WP 1803 00.

ELASTOMERIC BEARING INSPECTION

1. Measure large ID of inboard bearing (Figure 1, Sheet 4, Detail E). **IF DIAMETER IS MORE THAN 1.145-INCHES, INCLUDING ANY LOW SPOTS CAUSED BY DAMAGE, REPLACE ELASTOMERIC BEARING (AVIM) (WP 0370 00).**
2. Measure large ID of outboard bearing. **IF DIAMETER IS MORE THAN 1.123-INCHES, INCLUDING ANY LOW SPOTS CAUSED BY DAMAGE, REPLACE ELASTOMERIC BEARING (AVIM) (WP 0370 00).**
3. Check elastomeric bearings for cracking or separation. Measure gaps between rubber and metal with 0.015-inch feeler gage (Figure 1, Sheet 4, Detail F). **IF CRACKS OR SEPARATION BETWEEN METAL AND RUBBER IS MORE THAN 1/16-INCH DEEP, REPLACE ELASTOMERIC BEARING (AVIM) (WP 0370 00).**
4. Visually inspect stabilator hinge fitting for disbonding between mating surface of bearing and fitting. Up to 50% disbonding of mating surface is allowed. If bonding is 50% or greater, repair disbond per (WP 0370 00).

STABILATOR ACTUATOR ATTACH FITTINGS INSPECTION

WARNING

CSI

Nicks, dents, scratches and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.

1. Check stabilator actuator attach fittings for nicks, dents, scratches, and tool or scuff marks in lug area. **NONE ALLOWED.**
2. Check fittings for cracks in lugs, base, and attachment holes. **IF PYLON ACTUATOR ATTACH FITTING IS CRACKED, REPLACE FITTING (WP 0423 00). IF STABILATOR ACTUATOR ATTACH FITTING IS CRACKED, REPLACE STABILATOR (WP 0338 00). IF RIGHT OR LEFT PYLON ATTACH FITTING IS CRACKED, REPLACE PYLON (WP 0328 00). IF RIGHT OR LEFT STABILATOR ATTACH FITTING IS CRACKED, REPLACE FITTING (WP 0340 00).**
3. Check fittings for nicks, scratches, gouges, and tool marks. **IF DAMAGE IS NOT MORE THAN 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.500-INCH LONG, REPAIR BY BLENDING OUT DAMAGE TO 0.008-INCH MAXIMUM DEPTH, WITH MINIMUM BLEND RADIUS OF 3/8-INCH.**
4. **DAMAGE GREATER THAN 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.500-INCH LONG, REQUIRES ENGINEERING EVALUATION FOR REPAIR OR REPLACEMENT.**

STABILATOR ACTUATOR GROUNDING STRAP INSPECTION

1. Check grounding strap for cracks or tears in shrink tubing. **NO CRACKS OR TEARS ALLOWED.** Replace strap if damage is found (WP 0345 00).
2. Check for disbonding of shrink tubing at strap end fittings. **NO DISBONDING ALLOWED.** Replace strap if disbonding is found (WP 0345 00).
3. Inspect chemical conversion coated mating surfaces of grounding strap, grounding bracket, and rear spar (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR

This WP supersedes WP 0188 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Preventive Compound, Item 375, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
TM 11-1520-237-23
WP 1803 00
WP 0189 00
WP 0193 00
WP 0194 00

REMOVE

1. Turn off all electrical power.
2. **HH-60A HH-60L** Remove FLIR nose door fairing (WP 0194 00).
3. Unlatch and open nose door.
4. Remove radar warning system nose receiver, antennas, and associated wiring from nose door (TM 11-1520-237-23).
5. Support door in open position.
6. **UH-60A 77-22714 - 80-23415** Remove pin and washers from front end of strut support bracket on nose door (Figure 1, Detail A).
7. **UH-60A 80-23416 - SUBQ EH-60A UH-60L** Remove pin, spacers and washers from support bracket (Figure 1).
8. Lower door to closed position.
9. Visually inspect bolts holding hinge halves together for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. If corrosion is present and bolt is seized in hinge, do not use pneumatic tools to remove bolt. Use penetrating oil to loosen and tap bolt out. Replace bolt.
10. Remove bolts, washers, and nut holding hinge halves together. Remove door.
11. Remove door lock assembly from door (WP 0193 00).

INSTALL

1. Turn off all electrical power.
2. Place nose door in position on helicopter.
3. **QA** Line up hinge halves. Coat bolts with corrosion preventive compound, Item 375, WP 1803 00, and install bolts, washers, nuts, and cotter pins (Figure 1). Coat hinge halves with corrosion preventive compound, Item 375, WP 1803 00.
4. Support door in open position.

INSTALL - Continued

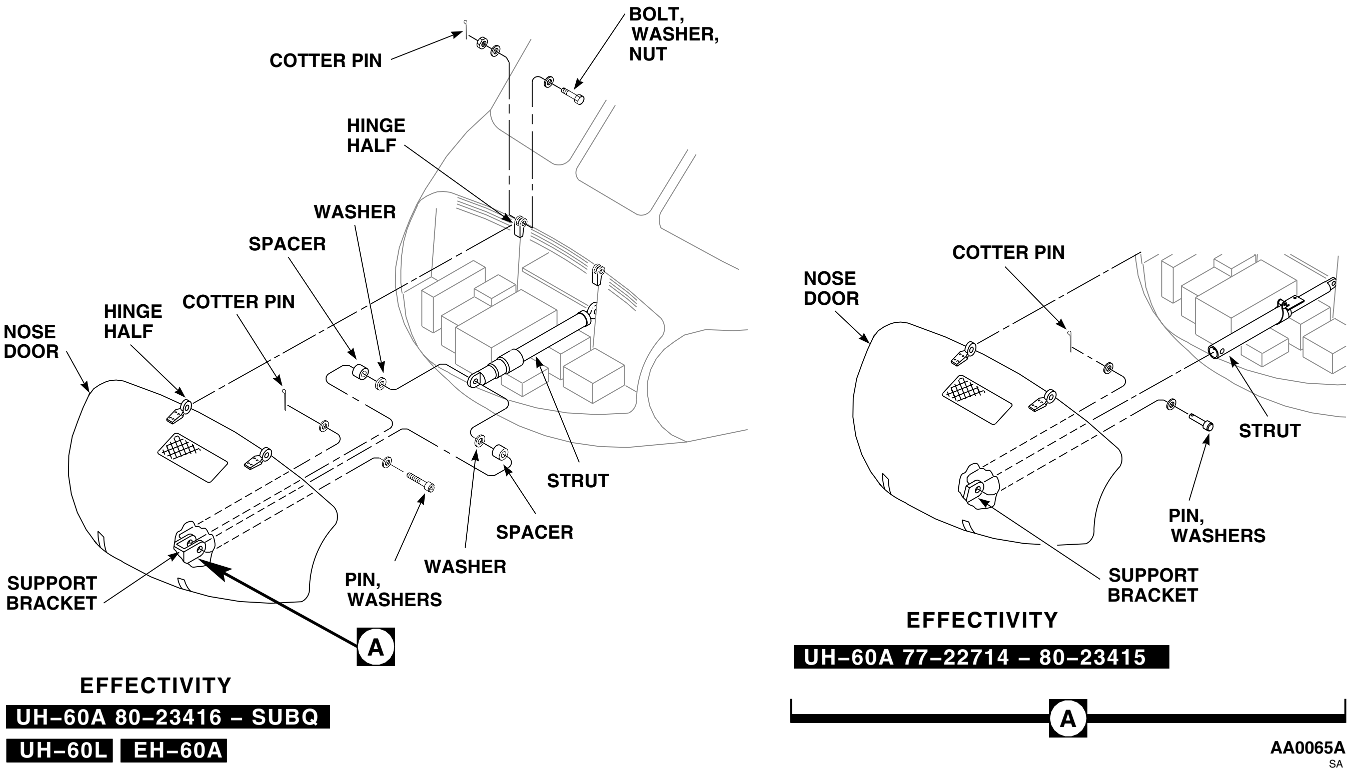


Figure 1. Replace Nose Door.

5. **QA** **UH-60A 80-23416 - SUBQ EH-60A UH-60L** Install end of strut in support bracket. Install washer and spacer each side of strut end. Install pin and washers. Install cotter pin.

6. **QA** **UH-60A 77-22714 - 80-23415** Install front end of strut to support bracket on nose door with pin and washers (Figure 1, Detail A). Install cotter pin.

7. Close door. See if door is centered within door opening.

8. If centering is required, adjust door by loosening screws and nuts that hold upper serrated hinge fittings, and reposition door as necessary.

NOTE

To keep locksets uniform when replacing nose door, install lock removed from old door.

9. Install door lock assembly (WP 0193 00).

10. Check door for adjustment by doing this:

    a. Press down on door.

    b. Pull outward on latch. If latch hooks can be pulled out from bracket, adjust latch (WP 0189 00).

11. Install radar warning system nose receiver, antennas, and associated wiring on nose door (TM 11-1520-237-23).

12. Make sure area clean and free of foreign material before closing up.

13. **HH-60A HH-60L** Install FLIR nose door fairing (WP 0192 00).

INSTALL - Continued

14. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR ADJUSTMENT

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Adhesive, Item 11, WP 1803 00
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Plastic Sheet, Item 221, WP 1803 00
Polyurethane Coating, Item 232, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55 1500-345-23
WP 0381 00
WP 1803 00

REPLACE, ADJUST, AND REPAIR LATCH

Remove

1. Turn off all electrical power.
2. Unlatch and open nose door. Lock strut in open position.
3. Remove screws, washers, and nuts, and remove latch from nose door (Figure 1, Sheet 1, Detail A).

Repair

1. To remove corrosion from latch, do this:
   a. Loosen jamnut and remove latch hook from threaded stud. Remove jamnut from stud (Figure 1, Sheet 2, Detail B).
   b. Remove corrosion manually from latch, using scouring pad, Item 205, WP 1803 00. **PITS ON NON-THREADED AREAS OF LATCH SHALL NOT EXCEED 0.005-INCH AFTER REWORK.**
   c. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
   d. Apply compound solution to corroded areas of latch. Rinse thoroughly with clean water.
   e. Touch up paint with 2 coats of epoxy primer coating kit, Item 120, WP 1803 00, and top coat of polyurethane coating, Item 232, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
   f. Install jamnut on threaded stud. Install latch hook on stud.
2. To replace secondary lock, do this:
   a. Loosen jamnut and remove latch hook from threaded stud. Remove jamnut from stud.
   b. Remove rivets attaching secondary lock to plate. Remove lock.
   c. Position replacement secondary lock on plate and attach with rivets.
   d. Install jamnut on threaded stud. Install latch hook on threaded stud.
   e. Install and adjust latch on door.

REPLACE, ADJUST, AND REPAIR LATCH - Continued

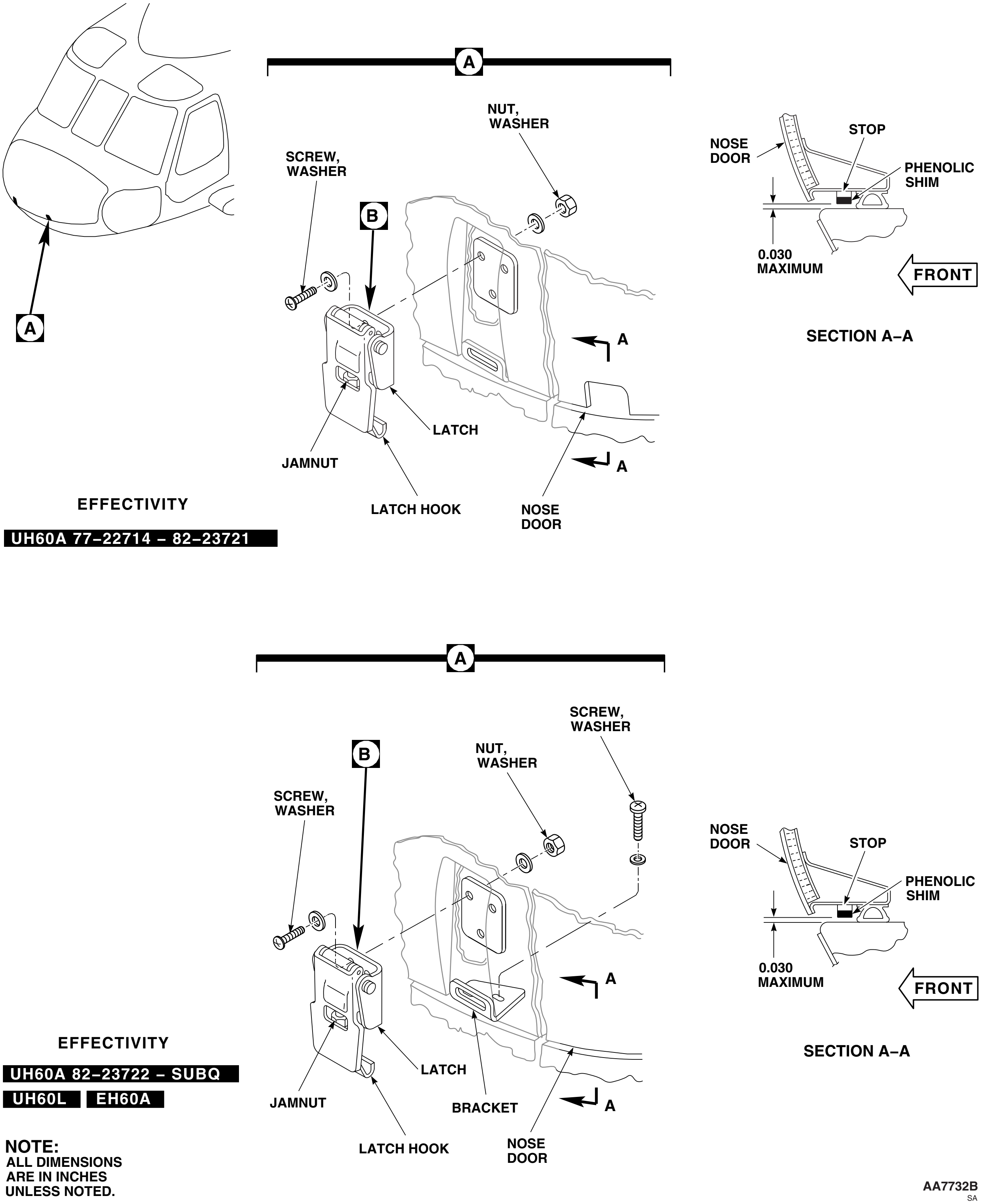


Figure 1. Replace/Repair Latch. (Sheet 1 of 2)

REPLACE, ADJUST, AND REPAIR LATCH - Continued

NOTE:
TIGHTEN 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

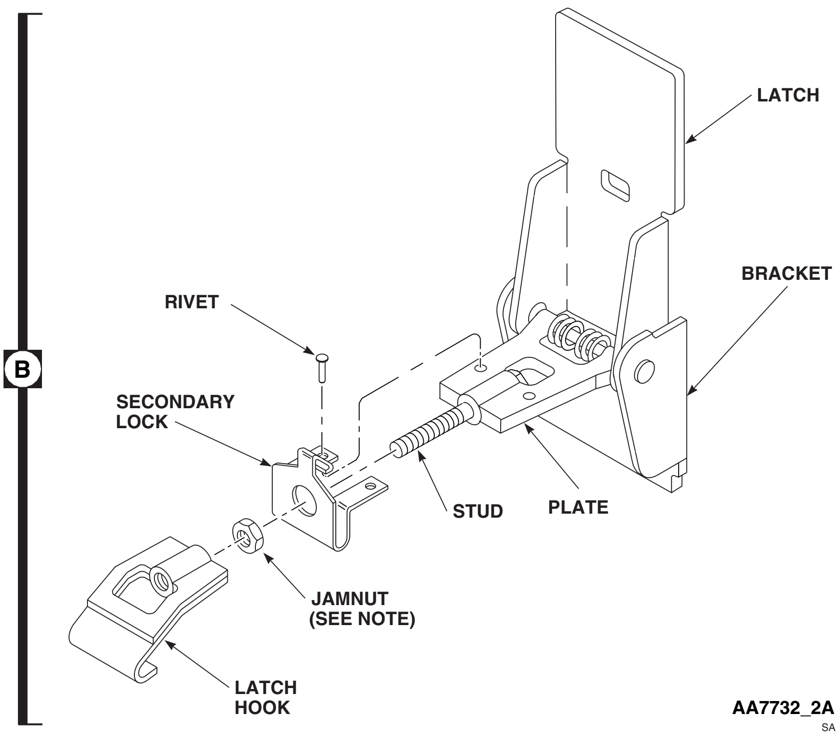


Figure 1. Replace/Repair Latch. (Sheet 2 of 2)

Install

1. Turn off all electrical power.
2. **QA** Fasten latch to nose door with screws, washers, and nuts (Figure 1, Sheet 1, Detail A).
3. **QA** **UH60A 82-23722 - SUBQ UH60L EH60A** Loosen two latch bracket screws and move bracket to center latch hook in bracket slot as required. Tighten bracket screws.
4. Close nose door and fasten latches.

Adjust

1. Open latch, loosen jamnut on latch hook (Figure 1, Sheet 2).
2. **QA** Verify that all four stops are installed.
3. Turn latch hook to adjust latch. Close latch. Door will draw down on seal.
4. Adjust latches until one stop on each side of door contacts sill with latch closed.
5. Check gaps between other two stops and sill. **MAXIMUM GAP ALLOWED IS 0.030-INCH.**
6. Open latch.
7. **QA TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
8. Close latch, recheck clearance.
9. If the stop to sill gap still exceeds limit, do this:
   a. Make shim from plastic sheet, Item 221, WP 1803 00.

REPLACE, ADJUST, AND REPAIR LATCH - Continued

b. Bond shim to stop with adhesive, Item 11, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR STRUT

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Unlatch and open nose door.
3. Install temporary nose door support and support door in open position while removing front end of strut from support bracket (Figure 1, Detail A).
4. **UH60A 77-22714 - 80-23415** Remove pin and washers from front end of strut support bracket on nose door.
5. Remove bolt, washers, and nut securing rear end of strut to support fitting on compartment bulkhead. Remove strut and spacers from compartment.
6. **UH60A 80-23416 - SUBQ UH60L EH60A** Remove pin, cotter pin, and washers from front end of strut. Remove washers and spacers from support bracket on nose door.
7. Remove bolt, washer, cotter pin, and nut securing rear end of strut to support fitting on compartment bulkhead. Remove strut and washer from compartment.

INSPECT

UH60A 80-23416 - SUBQ UH60L EH60A Measure play between rod end bearing and race. **IF PLAY IS 0.012 - INCH OR MORE, REPLACE ROD END.**

INSTALL

1. Turn off all electrical power.
2. **QA** **UH60A 77-22714 - 80-23415** Place a spacer on each side of support fitting on compartment bulkhead. Install rear end of strut on support fitting with bolt, washers, and nut. Install cotter pin (Figure 1, Detail A).
3. Attach front end of strut to support bracket on nose door with pin and washers. Install cotter pin.
4. **QA** **UH60A 80-23416 - SUBQ UH60L EH60A** Install rear end of strut on left side of support fitting. Place a washer between strut and fitting. Attach strut to fitting with bolt, washers, and nut. **TORQUE NUT TO 42 - 47 INCH-POUNDS.**
5. **QA** Install front end of strut in support bracket clevis on door. Install washer and spacer each side of strut end. Install pin and washers. Install cotter pin.
6. Remove temporary nose door support.
7. Make sure area is clean and free of foreign material.

INSTALL - Continued

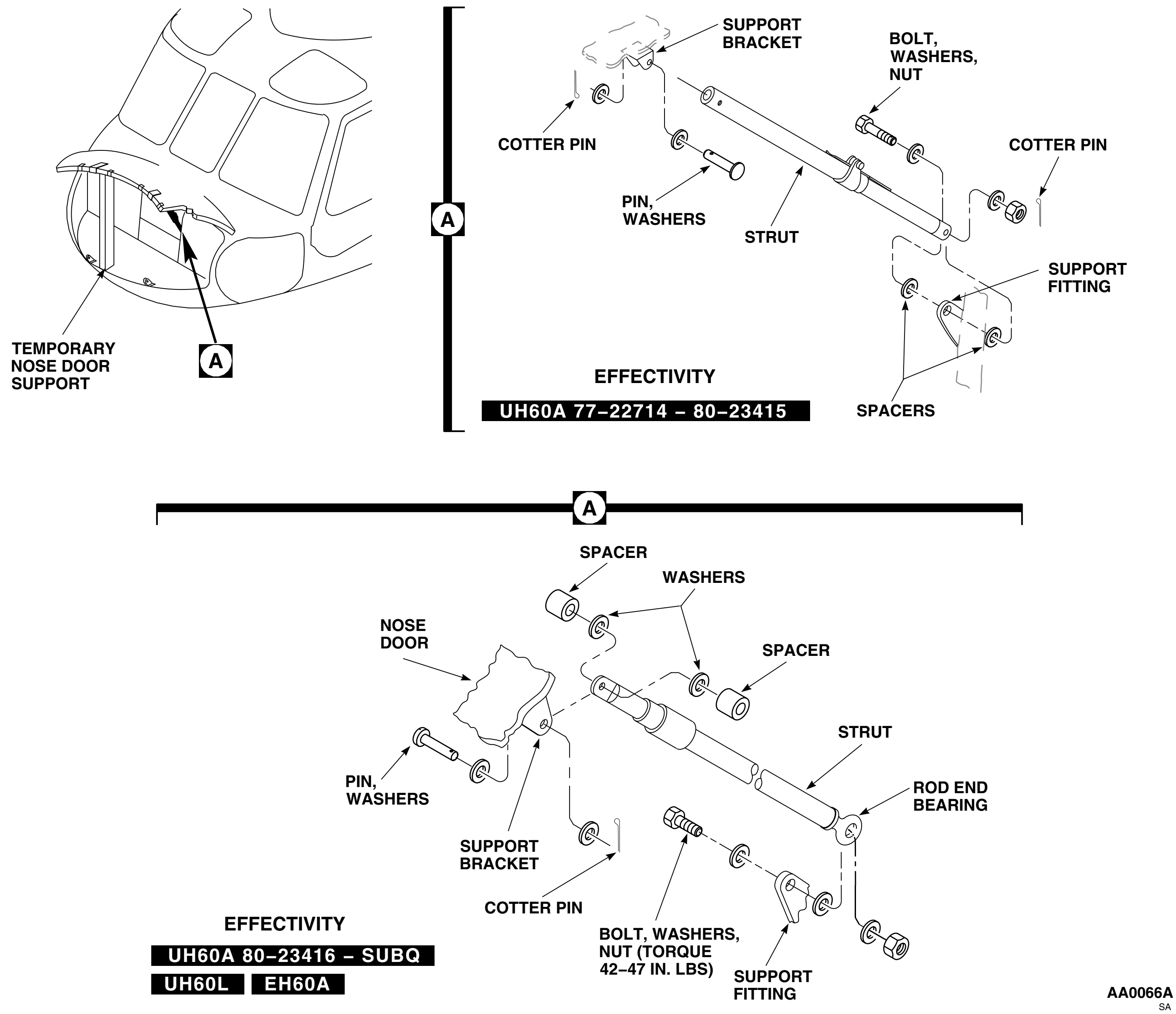


Figure 1. Replace Nose Door Strut.

8. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR STRUT CATCH ASSEMBLY

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Unlatch and open nose door.
3. Install temporary nose door support to support door in open position while removing catch assembly from strut assembly (Figure 1, Detail A).
4. Remove screw, washer, and nut securing catch assembly to clevis on strut.

INSTALL

1. Turn off all electrical power.
2. **QA** Bolt catch assembly to clevis with screw, washers, and nut (Figure 1, Detail A).
3. Check that catch assembly works properly.
4. Remove temporary nose door support.
5. Make sure area is clean and free of foreign material.
6. Close door and fasten latches.

INSTALL - Continued

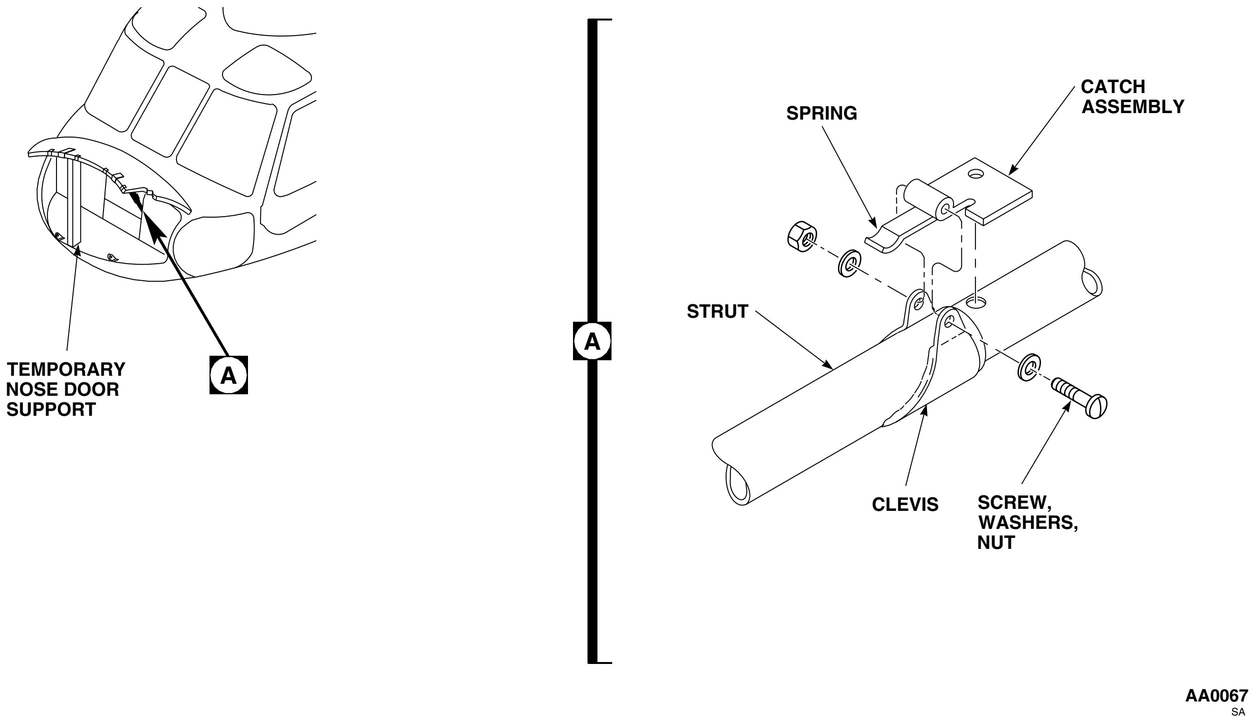


Figure 1. Replace Strut Catch Assembly.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR HINGE FITTING

This WP supersedes WP 0192 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper,

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Abrasive, Item 77, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Unlatch and open nose door.
3. Support door in open position while replacing hinge fitting.

NOTE

Removal of left and right nose door hinge fitting is the same.

4. Remove bolt, washers, and nut holding hinge fittings together (Figure 1).
5. Remove screws, washers, and nuts holding hinge fitting to door (Figure 1, Detail A). Remove hinge fitting from door.
6. Remove serrated plate and shims.
7. Using nonmetallic scraper, remove seal(s), if installed.

INSPECT

NOTE

- This inspection applies only to fittings 70207-01012-104, 70207-01012-106, and 70207-01015-101.
- Pay particular attention to pivot pin hole areas.

1. Inspect hinge fitting using a magnifying glass, paying particular attention to pivot pin hole area for cracks. **NO CRACKS ALLOWED.**
2. If cracks are found, replace hinge fitting.

INSPECT - Continued

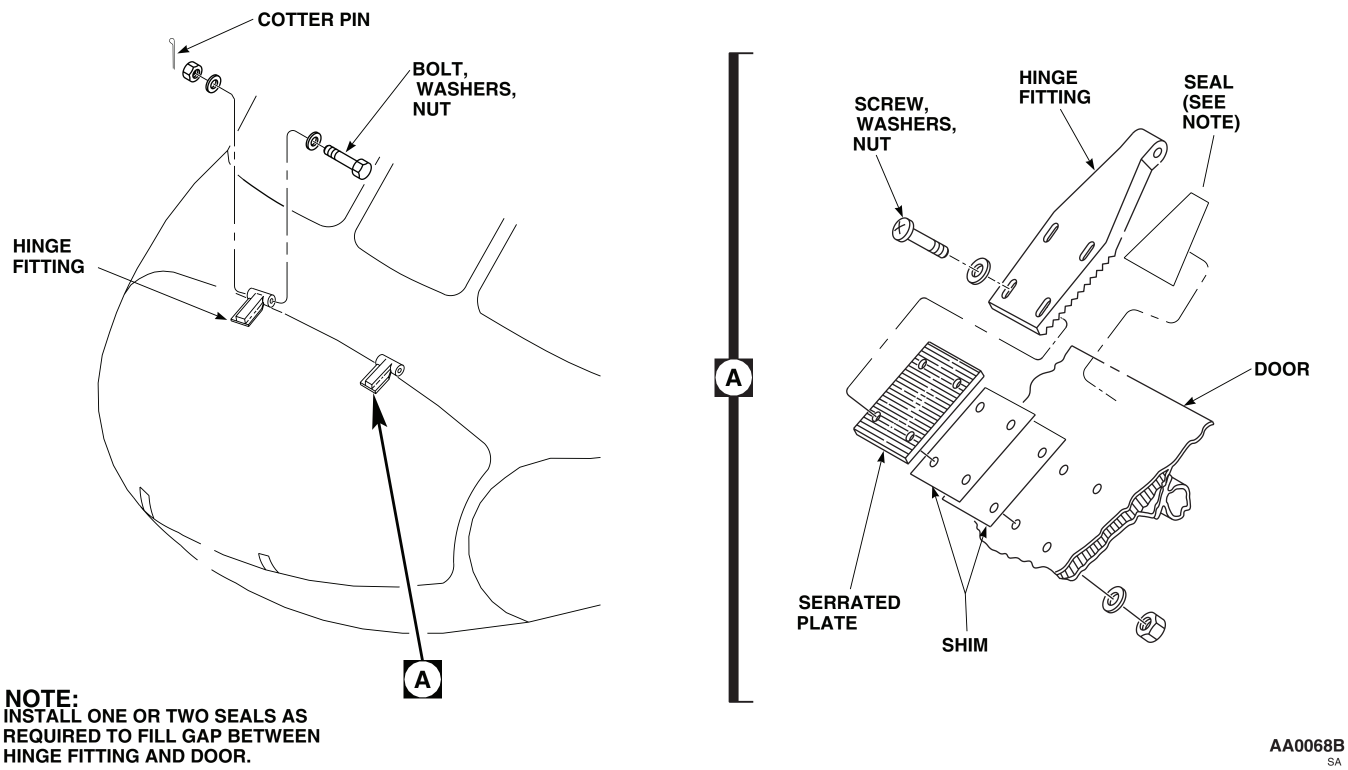


Figure 1. Replace Hinge Fitting.

INSTALL

1. Turn off all electrical power.

NOTE

Installation of left and right nose door hinge fitting is the same.

2. **QA** Bolt hinge fittings together. Install cotter pin (Figure 1).
3. Position door on nose of helicopter and temporarily install serrated plates and hinge fittings (Figure 1, Detail A). Add or remove shims to line up door with contour of nose. No more than two shims may be used.
4. Replace seal(s) as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Wipe seal(s) and door clean using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

INSTALL - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

NOTE

Do not sand through paint or primer.

b. Lightly scuff mating surface on door with abrasive cloth, Item 77, WP 1803 00. Wipe door clean using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply a thin, uniform coat of adhesive, Item 14, WP 1803 00, to seal(s) and to mating surface.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Wipe hinge parts clean using a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
6. Apply sealing compound, Item 273, WP 1803 00, to door hinge screwholes and screws.
7. **QA** Install hinge fittings, serrated plates, and shims on door with screws, washers, and nuts.
8. Remove door support and close door. See if door is centered within door opening.
9. If centering is required, adjust door by loosening screws and nuts which hold upper serrated hinge fitting. Reposition door as necessary. Retighten screws.
10. Wipe off extra sealing compound with clean machinery wiping towel, Item 344, WP 1803 00.
11. Make sure area is clean and free of foreign material.
12. Apply a fillet of sealing compound, Item 273, WP 1803 00, to perimeter of hinge fitting and seal.
13. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
14. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR LOCK

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

NOTE

Lock cylinders are available only in sets. When replacing nose door lock cylinder, replace lock cylinders on all doors.

1. Turn off all electrical power.
2. Unlatch and open nose door.
3. Remove nut from back of lock cylinder. Remove latch cam and lockwasher from lock (Figure 1, Detail A).
4. Remove nut securing lock cylinder to nose door. Push lock cylinder out of door.
5. **UH60A 77-22714 - 80-23508** If catch is damaged, remove it from nose of helicopter by removing attachment bolts, washers, and nuts. Remove shims.
6. **UH60A 80-23509 - SUBQ UH60L EH60A** If catch is damaged, remove it from nose of helicopter by removing attachment bolts and washers. Remove shims.

INSTALL

1. Turn off all electrical power.
2. Insert lock cylinder through front of nose door and secure with nut (Figure 1, Detail A).

NOTE

- **UH60A 77-22714 - 80-23508** Lock cylinder is lined up by flat side of hole in door. Install with dust cover up.
- **UH60A 80-23509 - SUBQ UH60L EH60A** Line up square end of lock square with sill.

3. **QA** Install lockwasher and latch cam on back of lock cylinder and secure with nut.
4. **QA** **UH60A 77-22714 - 80-23508** If catch is removed, install it and shims on nose door sill by installing bolts, washers, and nuts.
5. **QA** **UH60A 80-23509 - SUBQ UH60L EH60A** If catch is removed, install it and shims on nose door sill by installing bolts and washers. Adjust catch from side to side, if necessary, to line it up with latch cam when lock cylinder is locked.
6. If necessary, remove or install shims to make sure latch cam engages latch without forcing when locking nose door.
7. Add a washer, under boltheads, for each shim removed, to take up bolt shank length.

INSTALL - Continued

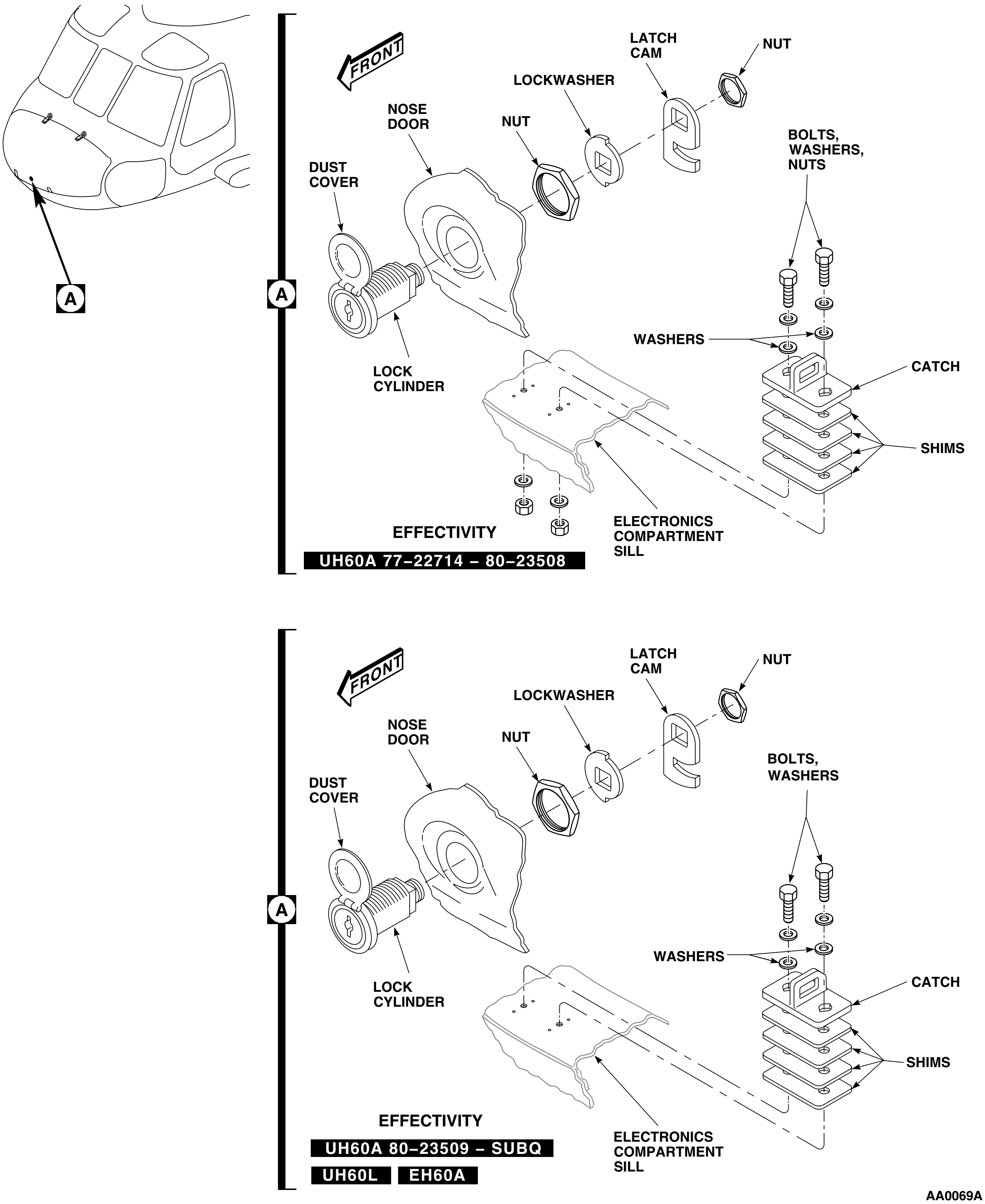


Figure 1. Replace Door Lock.

INSTALL - Continued

8. Make sure area is clean and free of foreign material.
9. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR COVER AND SCREEN

This WP supersedes WP 0194 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Seal S7020-11001-106
Seal S7020-11001-108
Seal S7020-11001-109
Seal S7020-11001-110

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REPLACE NOSE DOOR COVER AND SCREEN

Remove

1. Turn off all electrical power.
2. Remove screws and washers holding cover and screen on nose door (Figure 1, Detail A).

NOTE

Remove cover when air temperature above 32°F(0°C) makes it desirable to allow air flow into nose area.

3. Store cover as required.

Install

1. Turn off all electrical power.

NOTE

Install cover when air temperature below 32°F(0°C) makes it desirable to block air flow into nose area.

2. **QA** Install cover over screen. Install cover and screen on nose door with screws and washers (Figure 1, Detail A).

REPLACEMENT OF FLIR NOSE DOOR FAIRING HH-60A HH-60L

Remove

1. Turn off all electrical power.
2. Unlatch FLIR nose door fairing.

REPLACEMENT OF FLIR NOSE DOOR FAIRING HH-60A HH-60L - Continued

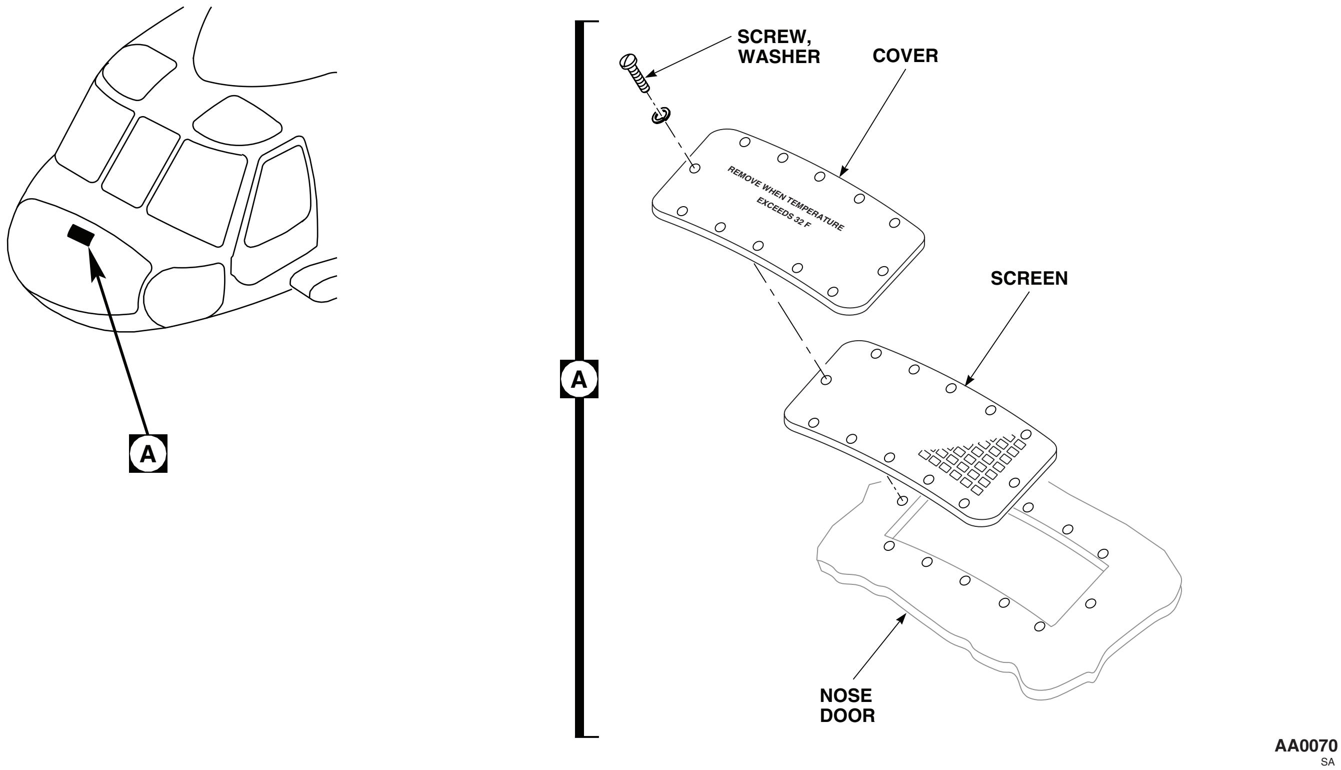


Figure 1. Replace Nose Door Cover and Screen.

3. Open nose door.
4. Remove screws, washers, and nuts securing FLIR nose door fairing, and if required FLIR nose door fairing hinge mount, to nose door (Figure 2, Detail A and Detail B). Remove FLIR nose door fairing from nose door.

Inspect/Repair

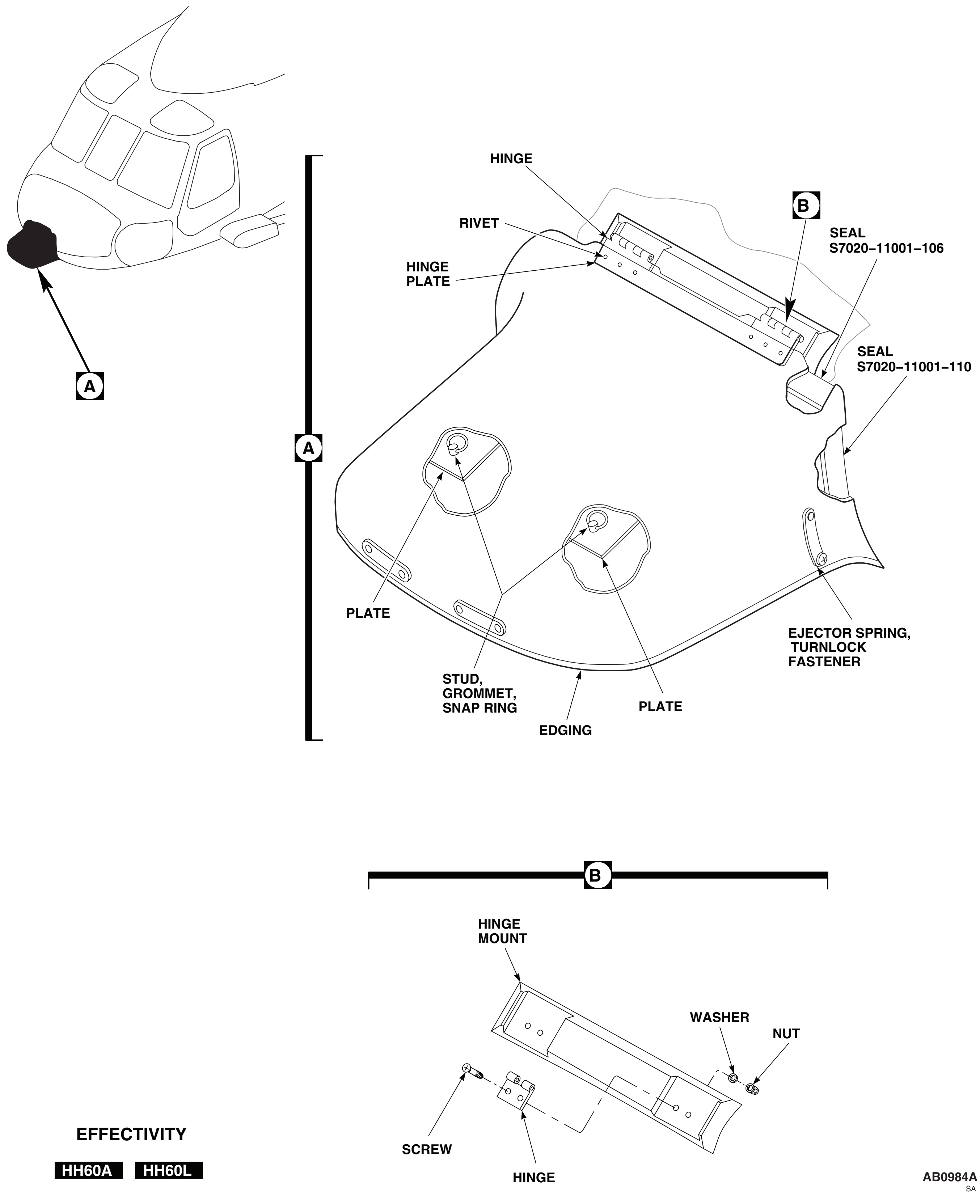
1. Visually inspect hinges, hinge strap, hinge mount, turnlock fasteners, ejector springs, and turnlock receptacles for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked item or door assembly, as required.
2. Remove corrosion and blend out wear, using scouring pad, Item 205, WP 1803 00. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. If seal(s) require replacement do this:
   a. Remove seals and old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 2, Detail A).

REPLACEMENT OF FLIR NOSE DOOR FAIRING HH-60A HH-60L - Continued



EFFECTIVITY

HH60A HH60L

Figure 2. Replace Flir Nose Door Fairing.

REPLACEMENT OF FLIR NOSE DOOR FAIRING HH-60A HH-60L - Continued

b. Thoroughly clean bonding surfaces of seal and FLIR nose door fairing, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply even coat of adhesive, Item 14, WP 1803 00, to both bonding surfaces.

d. Let adhesive dry for about 10 minutes until tacky.

e. Install seals on door and press firmly in place.

4. If edging requires replacement do this:

WARNING

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a. Remove edging, and old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 2, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Thoroughly clean bonding surfaces of FLIR lower nose fairing, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply even coat of adhesive, Item 14, WP 1803 00, to both bonding surfaces.

d. Let adhesive dry for about 10 minutes until tacky.

e. Install edging on fairing and press firmly in place.

5. If hinge requires replacement do this:

a. Remove rivets securing hinge plate and hinge to FLIR nose door fairing.

b. Remove hinge plate and hinge from FLIR nose door fairing, using nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Remove old adhesive from hinge plate and FLIR nose door fairing, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

d. Apply even coat of adhesive, Item 11, WP 1803 00, to all bonding surfaces.

e. Let adhesive dry for about 10 minutes until tacky.

f. Position new hinge and hinge plate on FLIR nose door fairing, and secure using rivets.

REPLACEMENT OF FLIR NOSE DOOR FAIRING HH-60A HH-60L - Continued

6. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. If hinge mount requires replacement do this:
   a. Remove old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   b. Thoroughly clean bonding surfaces of hinge mount and nose door, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   c. Apply even coat of adhesive, Item 11, WP 1803 00, to both bonding surfaces.
   d. Let adhesive dry for about 10 minutes until tacky.
   e. Install hinge mount on door and press firmly in place.
8. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

Install

1. **QA** Position FLIR nose door fairing on nose door. Secure fairing to nose door using screws, washers, and nuts (Figure 2, Detail A).
2. Make sure area is clean and free of foreign objects.
3. Close and secure nose door.
4. Make sure area is clean and free of foreign objects.
5. Close and fasten FLIR nose door fairing.

REPLACEMENT OF FLIR LOWER NOSE FAIRING HH-60A HH-60L

Remove

1. Turn off all electrical power.
2. Unlatch FLIR nose door fairing.
3. Open nose door.

CAUTION

To prevent damage to FLIR lower nose fairing and glide slope antenna, support fairing while removing screws and washers.

4. While supporting FLIR lower nose fairing, remove screws and washers securing FLIR lower nose fairing to helicopter (Figure 3, Detail A).
5. Carefully lower fairing until access to coaxial connector, P622R, is obtained. Disconnect coaxial connector, P622R, from glide slope antenna.

REPLACEMENT OF FLIR LOWER NOSE FAIRING HH-60A HH-60L - Continued

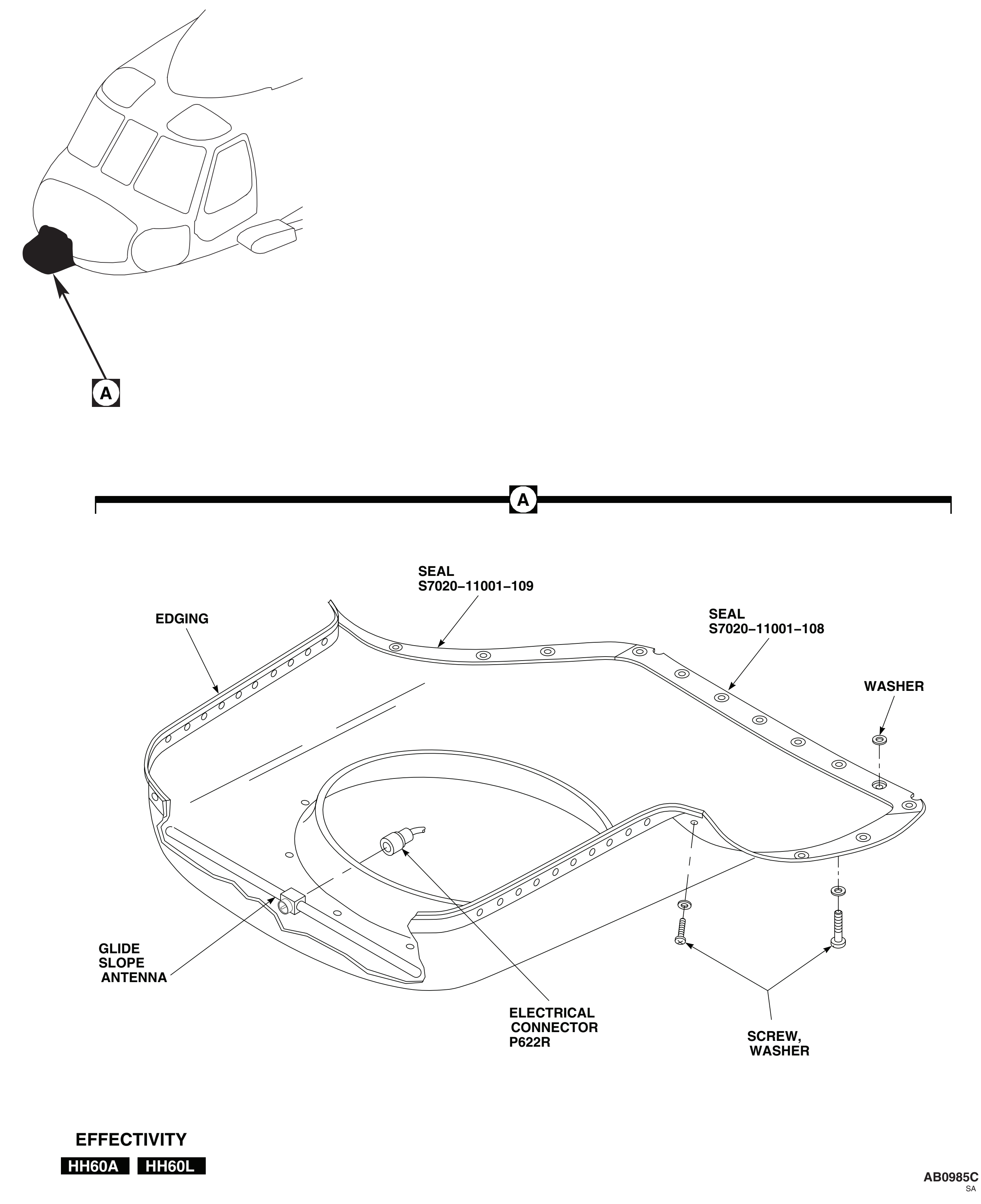

Figure 3. Replace Flir Lower Nose Fairing.

REPLACEMENT OF FLIR LOWER NOSE FAIRING HH-60A HH-60L - Continued

Inspect/Repair

1. Visually inspect FLIR lower nose fairing for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked item or fairing assembly, as required.
2. Remove corrosion and blend out wear, using scouring pad, Item 205, WP 1803 00. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. If seal(s) require replacement do this:
   a. Remove seals, washers, and old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 3, Detail A).
   b. Thoroughly clean bonding surfaces of seal and FLIR lower nose fairing, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   c. Apply even coat of adhesive, Item 14, WP 1803 00, to both bonding surfaces.
   d. Let adhesive dry for about 10 minutes until tacky.
   e. Install seal on fairing and press firmly in place.
   f. Open seal 0.050-inch diameter at fairing mount holes and bond washers to inside of fairing, centered with holes, using adhesive.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. If edging requires replacement do this:
   a. Remove edging, and old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 3, Detail A).
   b. Thoroughly clean bonding surfaces of FLIR lower nose fairing, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   c. Apply even coat of adhesive, Item 14, WP 1803 00, to both bonding surfaces.
   d. Let adhesive dry for about 10 minutes until tacky.
   e. Install edging on fairing and press firmly in place.
5. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPLACEMENT OF FLIR LOWER NOSE FAIRING HH-60A HH-60L - Continued

Install

CAUTION

To prevent damage to FLIR lower nose fairing and glide slope antenna, support fairing while connecting coaxial connector and securing fairing to helicopter.

1. **QA** Position and support FLIR lower nose fairing in approximate installed position and connect coaxial connector, P622R, to glide slope antenna. Secure fairing to nose door using screws, and washers (Figure 3, Detail A).
2. Make sure area is clean and free of foreign objects.
3. Close and secure nose door.
4. Make sure area is clean and free of foreign objects.
5. Close and fasten FLIR nose door fairing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR SEALS

This WP supersedes WP 0195 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 9, WP 1803 00
Adhesive, Item 8, WP 1803 00
Adhesive, Item 14, WP 1803 00
Paper, Abrasive, Item 208, WP 1803 00
Plastic Sheet, Item 220, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Stirring Stick, Beverage, Item 300, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0188 00
WP 1803 00

REPLACE NOSE DOOR SEALS

Remove

1. Turn off all electrical power.
2. Remove nose door (WP 0188 00).
3. Remove old seal from nose door, using a nonmetallic scraper (Figure 1, Sheet 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Remove remaining adhesive, using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper.

Install

1. Turn off all electrical power.

CAUTION

Do not cut into tube section of seal. Cut only the flange.

2. Fit new seal to door with spliced joints where shown (Figure 1, Sheet 1, Detail A and Detail B). To keep seal from bunching up or stretching around corners, make 30 degree to 40 degree cutouts on inner radius flange and slit outer radius flange (Figure 1, Sheet 1, Detail C).
3. Slit flange of seal (Figure 1, Sheet 2, Detail D) to prevent disbonding in bolt recess area.

REPLACE NOSE DOOR SEALS - Continued

4. Roughen all adhesive contact surfaces of new seal with abrasive paper, Item 208, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Wipe all adhesive contact areas with a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Wipe inside seal ends and splice reinforcing tube with a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
7. Apply a light coat of adhesive, Item 9, WP 1803 00, to inside of seal ends and outside surface of splice reinforcing tube.
8. Immediately install seal splice reinforcing tube in ends of seal and close up gap.
9. Line up seal joints on a flat surface covered with plastic sheet, Item 220, WP 1803 00, and hold for 10 to 15 minutes while adhesive sets up.
10. Roughen seal splice reinforcing strip and seal joint surface with abrasive paper, Item 208, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

11. Wipe roughened areas with a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
12. Apply adhesive, Item 9, WP 1803 00, to inside of reinforcing strip and seal splice. Install reinforcing strip over splice. Work out any gaps or blisters.
13. Mix door seal bonding adhesive, Item 8, WP 1803 00, following manufacturer's instructions.
14. Apply an even coat of adhesive, Item 8, WP 1803 00, to door and seal joining surfaces. Allow adhesive to air dry for 30 minutes or until tacky.

REPLACE NOSE DOOR SEALS - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

15. Attach seal to door. Work out gaps, trapped air and strained areas all the way around seal installation. Remove extra adhesive with a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
16. Mix and apply a bead of adhesive, Item 9, WP 1803 00, around the inner and outer edges of the door seal installation. Shape seal bead as shown with a beverage stirring stick, Item 300, WP 1803 00, and allow to cure at least 24 hours.

REPLACE NOSE DOOR RAIN GUTTER SEAL

Remove

1. Turn off all electrical power.
2. Remove nose door (WP 0188 00).
3. Remove old rain gutter seal from airframe, using a nonmetallic scraper (Figure 2, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Remove remaining adhesive, using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper.

Install

1. Turn off all electrical power.
2. Apply adhesive, Item 14, WP 1803 00, uniformally to rain gutter seal and airframe. Allow adhesive to dry for 10 minutes or until adhesive is tacky.
3. Position rain gutter seal on airframe making sure rain gutter seal is in direct contact with airframe (Figure 2, Detail A). Allow adhesive to cure for 24 hours at room temperature.
4. Fillet seal all edges of rain gutter seal using sealing compound, Item 273, WP 1803 00. Fill all voids and make sure there are no irregularities present.
5. Install nose door (WP 0188 00).

REPLACE NOSE DOOR RAIN GUTTER SEAL - Continued

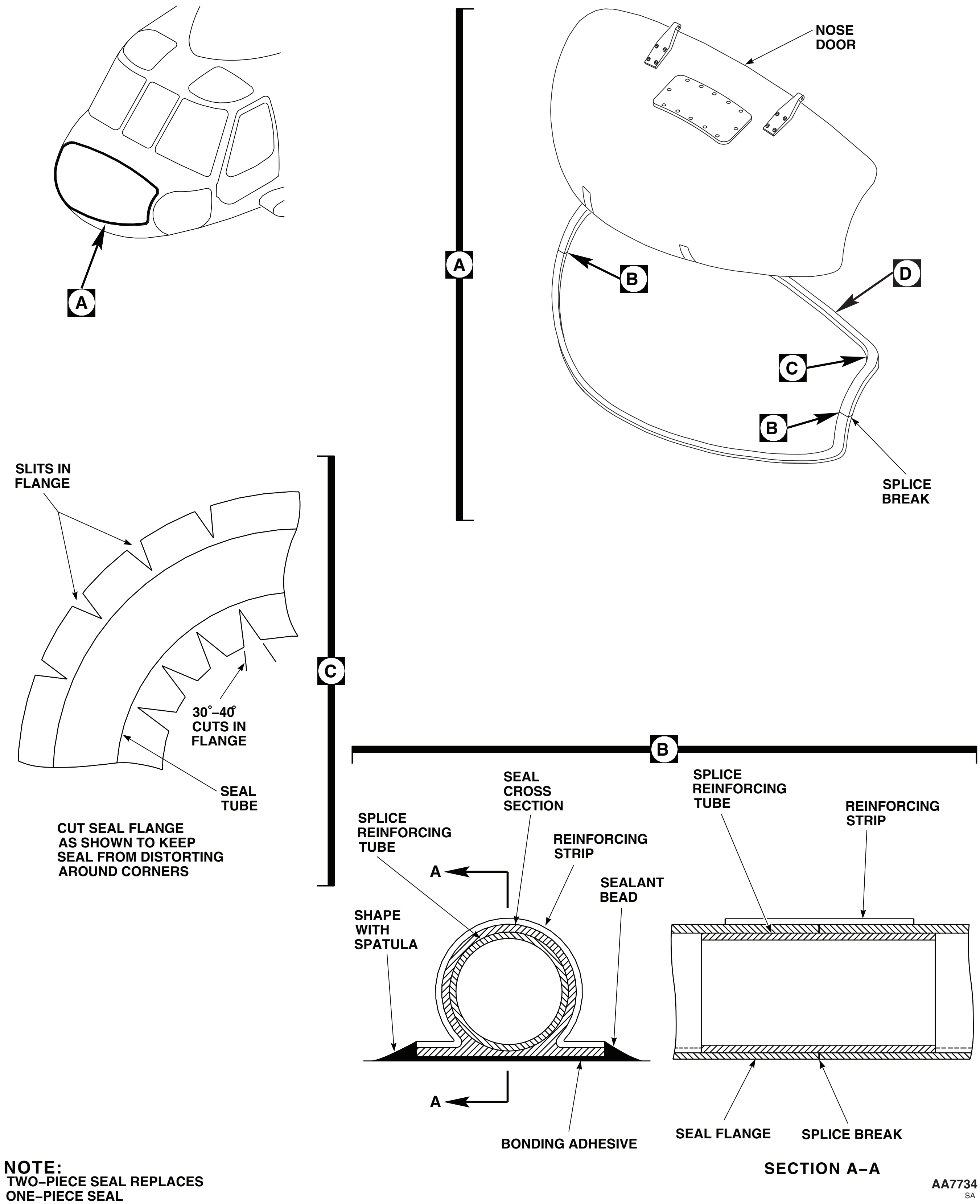

Figure 1. Replace Nose Door Seals. (Sheet 1 of 2)

REPLACE NOSE DOOR RAIN GUTTER SEAL - Continued

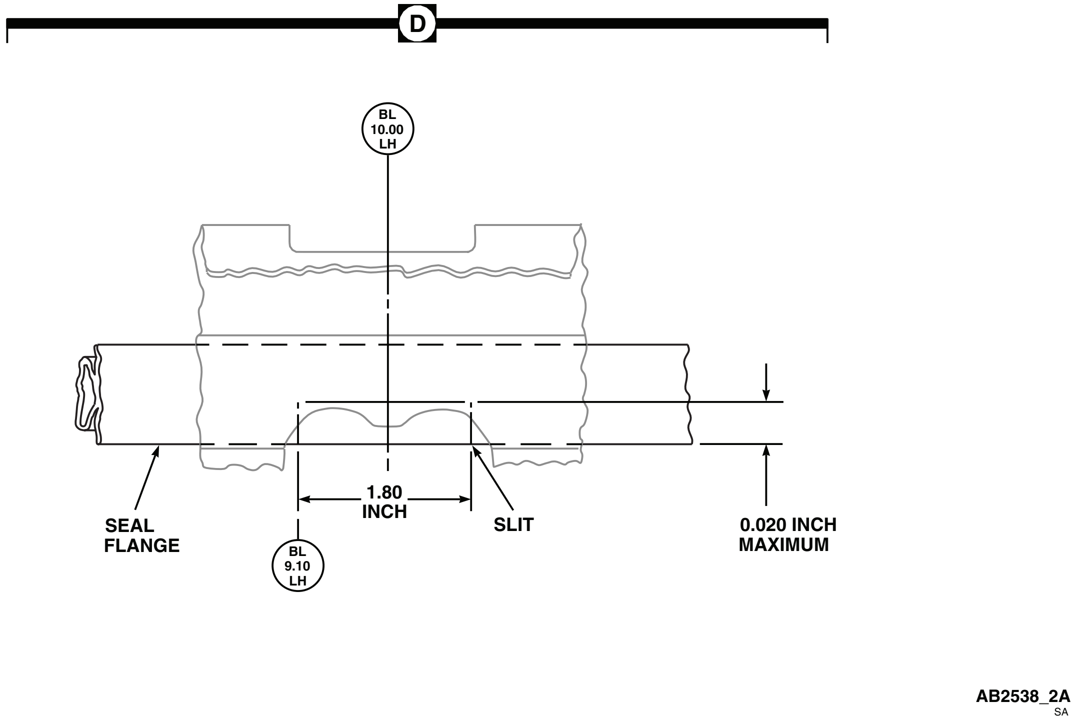

Figure 1. Replace Nose Door Seals. (Sheet 2 of 2)

REPLACE NOSE DOOR RAIN GUTTER SEAL - Continued

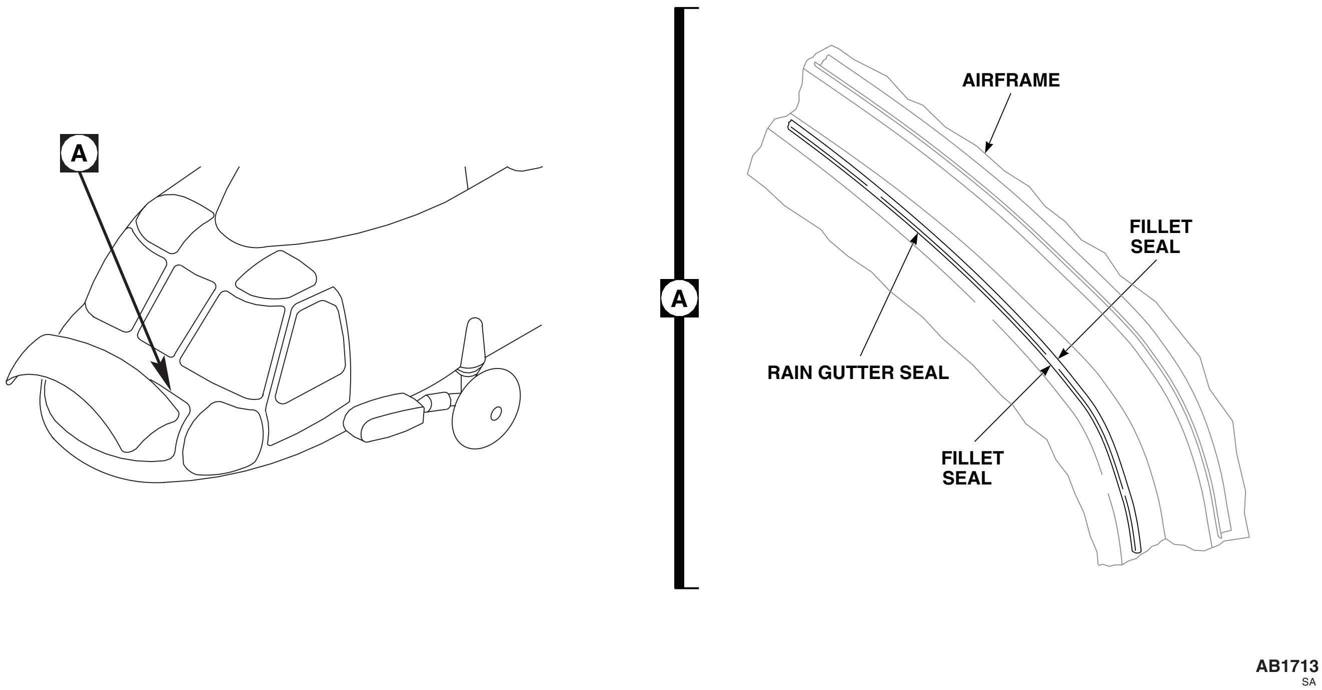

Figure 2. Replace Nose Door Rain Gutter Seal.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR STOPS

This WP supersedes WP 0196 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive Primer, Item 23, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Remove loose stop and old adhesive, using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper (Figure 1, Detail A and Detail B).

INSTALL

1. Turn off all electrical power.
2. Apply a light coat of adhesive primer, Item 23, WP 1803 00, to bonding surface of door. Allow adhesive primer coat to dry for at least 1 hour.
3. Using adhesive, Item 11, WP 1803 00, mix 100 parts by weight of part A to 22 parts by weight of part B. Apply a light coat of adhesive to door and stop.
4. **QA** Immediately install stop on door (Figure 1, Detail A and Detail B). Allow adhesive to dry at room temperature for at least 24 hours or for 2 hours at 110°F(43°C) to 130°F(54°C).
5. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

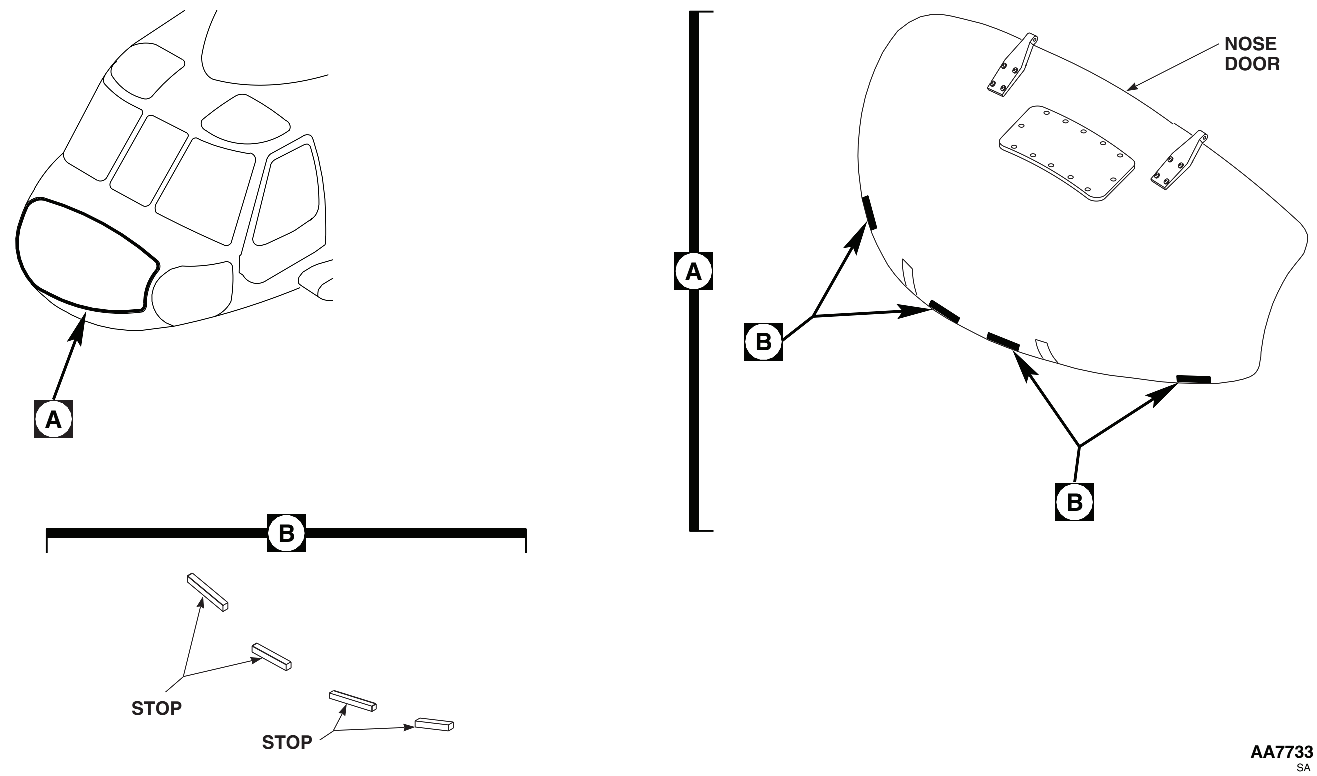

Figure 1. Replace Door Stops.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE DOOR DRAIN HOSES

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

NOTE

UH60A 82-23680 - SUBQ UH60L EH60A Are equipped with drain hoses.

1. Turn off all electrical power.
2. Open nose door. Engage support strut catch.
3. Loosen hose clamps. Remove screw, washers and nut on support clamp. Remove hose (Figure 1, Detail A).

INSTALL

1. Turn off all electrical power.
2. Loosely assemble support clamp and hose to door. Fit hose to upper and lower drain fittings and cut to correct length (Figure 1, Detail A).
3. Install end clamps on hoses, and tighten support clamp.
4. Release support strut catch and close nose door.

INSTALL - Continued

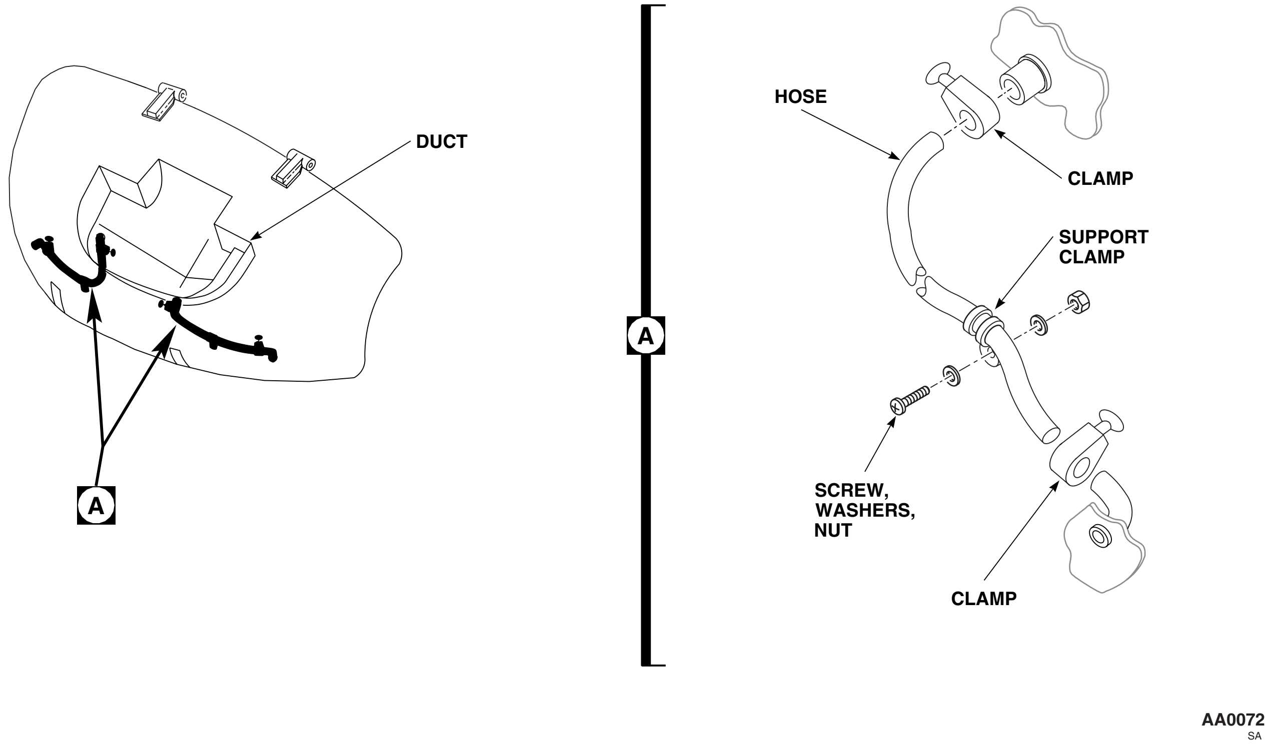

Figure 1. Replace Nose Door Drain Hoses.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR

This WP supersedes WP 0198 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Gasket, 70217-01000-142
Gasket, 70217-01000-143
Gasket, 70217-01000-144
Grease, General Purpose, Item 136, WP 1803 00
Sealing Compound, Item 282, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0174 00
WP 0199 00
WP 0214 00
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

NOTE

Removal of left and right cockpit door is the same.

2. Open cockpit door.
3. Support door. Pull jettison handle to disengage door hinges, and jettison pin. Remove door from helicopter (Figure 1, Detail A).
4. Check condition of gaskets under door hinge assemblies. If damaged, remove from hinge.
5. If door is being replaced, remove door lock assembly from door (WP 0214 00).

INSTALL

1. Turn off all electrical power.

NOTE

- To keep locksets uniform when replacing cockpit door, install lock removed from old door.
- Installation of left and right cockpit door is the same.

2. If not installed, install door lock assembly (WP 0214 00).

INSTALL - Continued

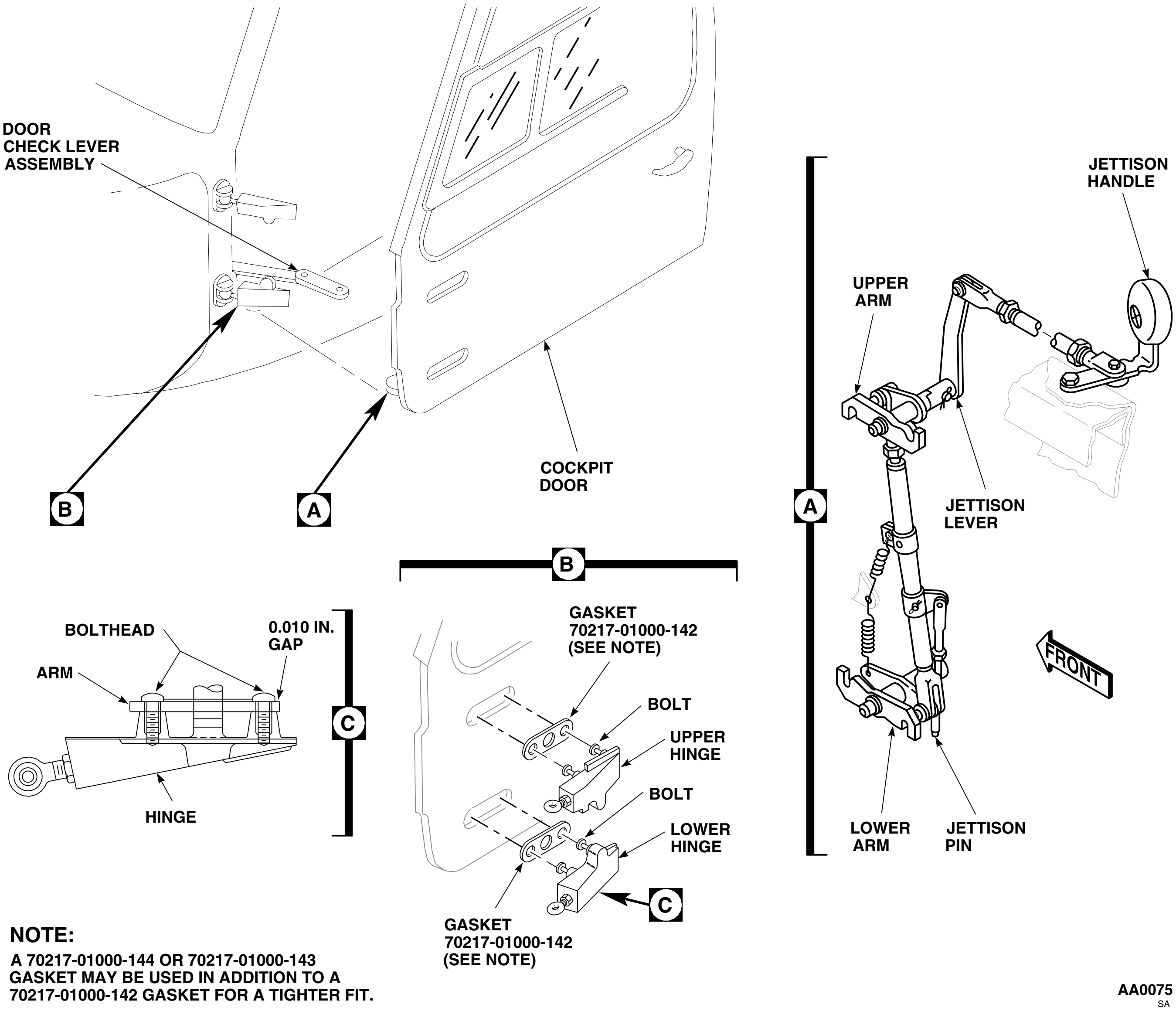

Figure 1. Replace Cockpit Door.

NOTE

An additional gasket under hinge assemblies may be used for a tighter fit.

3. If removed, place replacement gaskets, 70217-01000-142, 70217-01000-143, or 70217-01000-144 (in combination, if required) under hinge assemblies (Figure 1, Detail B).
4. Have assistant pull jettison handle to hold arm assemblies in open position (Figure 1, Detail A).

INSTALL - Continued

NOTE

It may be necessary to push on jettison lever to engage arm assemblies with hinge bolts.

5. Place door in position and allow jettison handle to retract. Arm assemblies will engage with bolts of hinge assemblies.
6. If door edge touches frame, trim edge until edge is clear of frame when door is fully closed.
7. Touch up paint on trimmed edge (WP 0381 00 and TM 55-1500-345-23).
8. After arms are installed in hinge bolts, check for 0.010-inch gap between bolthead and arm (Figure 1, Detail C).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. If necessary to adjust gap, remove bolts. Clean bolts and tapped hole using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and dry.
10. Coat bolt threads with sealing compound, Item 282, WP 1803 00, and install in hinge.
11. Apply thin coat of general purpose grease, Item 136, WP 1803 00, between boltheads and arm.
12. If duct and outlet assembly are installed, remove them, by removing screws and washers, and bolt and washer. Push down on duct to disengage hidden retaining latch.
13. Remove screws and cover from door.
14. Reinstall jettison pin through door check lever.
15. With assistant helping, make functional check of door jettison mechanism.
16. **QA** Reinstall door, including jettison pin through door check lever.
17. Open and close door a few times to check for proper operation. Adjust door if required (WP 0199 00).
18. Perform cockpit door jettison check (WP 0174 00).
19. **QA** Install cover on door with washers and screws.
20. **QA** Install duct and outlet assembly on door by sliding up from bottom to engage hidden retaining latch. Install screws and washers, and bolt and washer.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR ADJUSTMENT

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Shim, 70217-01005-107

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0174 00
WP 0198 00
WP 0212 00

ADJUST HINGES

NOTE

- Adjustment of right and left cockpit doors is the same.
- Adjusting both hinge rod ends equally in same direction in or out will move door front and rear.
- Adjusting hinge rod ends equally in opposite direction to each other will move rear end of door up or down to line up latches with striker plates.
- To move hinge end of door in to compress seal against frame requires peeling shims under door hinge fittings (WP 0212 00).

1. Remove cockpit door (WP 0198 00).
2. Remove headed pin, cotter pin, and washer from hinge to be adjusted (Figure 1, Detail A).
3. Loosen jamnut and adjust rod end to lengthen or shorten rod end as required.
4. Remove hinge fittings (WP 0212 00).

NOTE

A maximum of 2 new shims may be used under each hinge fitting.

5. Add or remove shims to adjust door.
6. Install hinge fittings (WP 0212 00).
7. Reinstall headed pin. Do not install cotter pin yet.
8. Open and close door to check adjustment. Continue to adjust door until latchbolt engages and releases properly at striker plate.
9. Once adjusted, tighten jamnut on rod end. **TIGHTEN 1/6 TO 1/3 TURN BEYOND POINT WHERE A SHARP RISE IN TORQUE IS FELT.** Install washer and cotter pin.

ADJUST HINGES - Continued

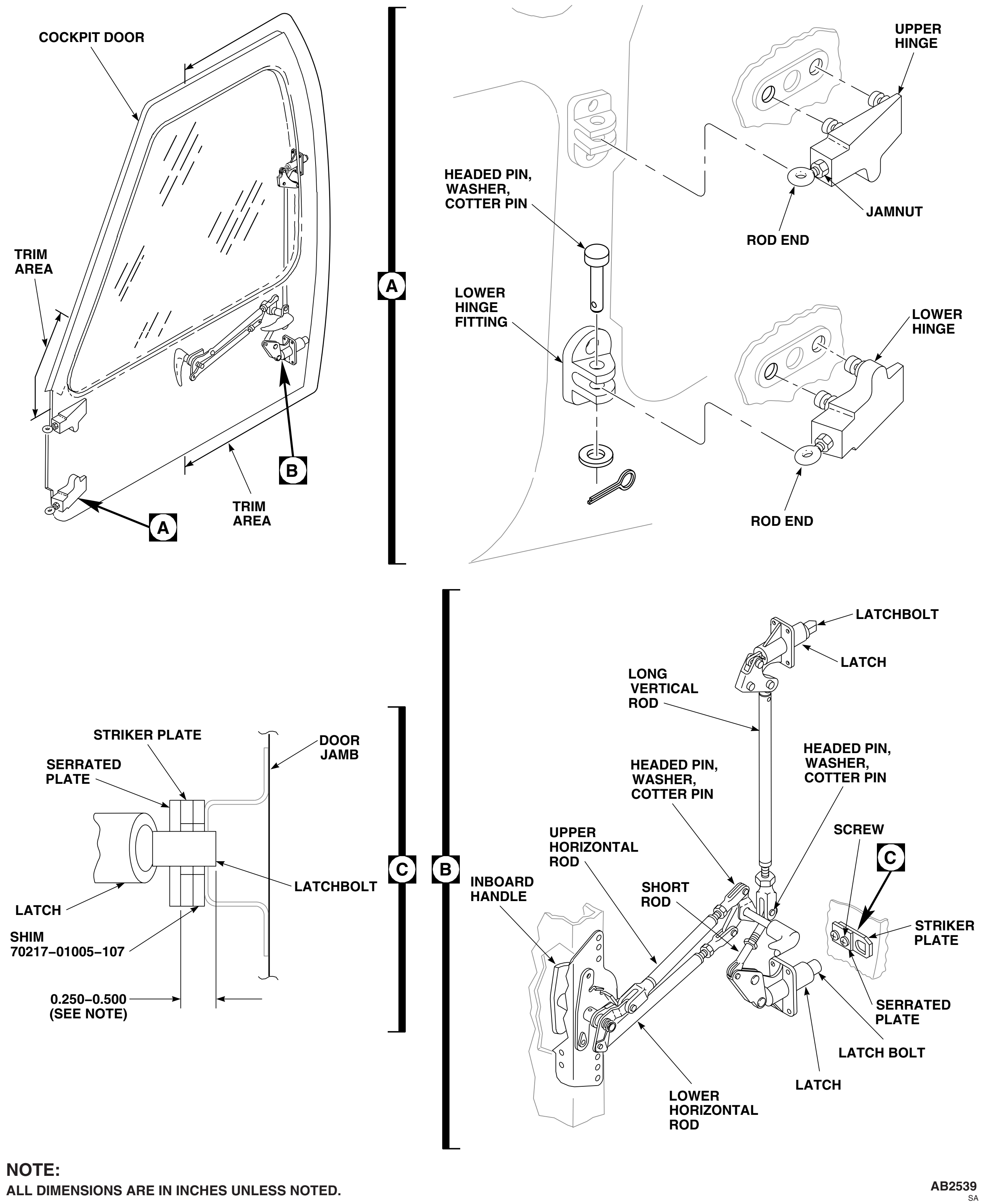

Figure 1. Adjust Hinges, Latches, Striker Plates.

ADJUST HINGES - Continued

10. Install cockpit door (WP 0198 00).

ADJUST LATCHES

1. Open cockpit door.
2. Operate inside/outside handle and check latchbolt action. Latchbolt must be flush with latch with door handle in full open position. **LATCHBOLT TRAVEL MUST BE AT LEAST 0.600-INCH BETWEEN OPEN AND LOCK POSITIONS.** Handle must turn freely into LOCK (over center) position without preloading linkage (Figure 1, Detail B).
3. Adjust long vertical rod and short rod to keep both latchbolts same measurement from latch.
4. Adjust lower horizontal rod to position latchbolt flush with latch when handle is in full open position. Vertical and short rods may have to be adjusted at same time.
5. Adjust upper horizontal rod with handle in locked position.
6. Remove headed pin from rod to be adjusted at adjustable end. Loosen jamnut and adjust rod end to lengthen or shorten rod. Install headed pin.
7. Operate handle and check latchbolt action.
8. Continue adjustment until latchbolt and lock action meet requirements.
9. **QA TIGHTEN JAMNUTS 1/6 TO 1/3 TURN BEYOND POINT WHERE A SHARP RISE IN TORQUE IS FELT.** Install washers and cotter pins.
10. Shut cockpit door and check latchbolt engagement in striker plate.
11. Perform cockpit door jettison check (WP 0174 00).

ADJUST STRIKER PLATES

1. Open door.
2. Close door and make sure striker plate is positioned properly. Make sure door latches and locks without too much force, yet compresses seal enough to prevent leakage.
3. Loosen screws, move striker plate in or out under serrated plate as necessary, and tighten screws.
4. Check latch engagement in striker plate (Figure 1, Detail C).
5. Latchbolt should extend into striker plate 0.250 to 0.500-inch. If necessary add shim, 70217-01005-107, under striker plate.
6. Perform cockpit door jettison check (WP 0174 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR CHECK COMPONENTS

This WP supersedes WP 0200 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Crew Door Post Doubler, Locally Made , Item 276, WP 1805 00
Gloves
Goggles/Face Shield

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0174 00
WP 0381 00
WP 1803 00
WP 1805 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door.
3. Remove bolt, washers, and nut attaching lever assembly to fitting (Figure 1, Detail A).
4. If replacing lever, pull jettison pin to release lever assembly from door.
5. Remove cotter pins and washers attaching spring to fitting. Pull spring out of fitting.
6. If worn, press out bushings in fitting.
7. If fitting needs replacement, remove as follows (Figure 1, Detail B and Detail C):

NOTE

Do not drill out both swage lock pins fastening the fitting body to the frame at the same time. There is a backing plate behind the door post that is required for installation of the new fitting.

a. Drill a 0.50-inch hole on inboard side of door post As shown in Figure 1, Detail C. Drill out rivets fastening flange of fitting to frame.
b. Drill out 3 rivets fastening flange of fitting to frame.
c. Remove one swage lock pin fastening the fitting body to door frame. Insert suitable tool into the 0.50-inch hole to secure the backing plate to door post.
d. Remove other swage lock pin fastening the fitting body to door frame
e. Remove fitting from door frame.
f. Locally manufacture door post doubler. Seal, prime, and paint doubler (WP 0381 00).

REMOVE - Continued

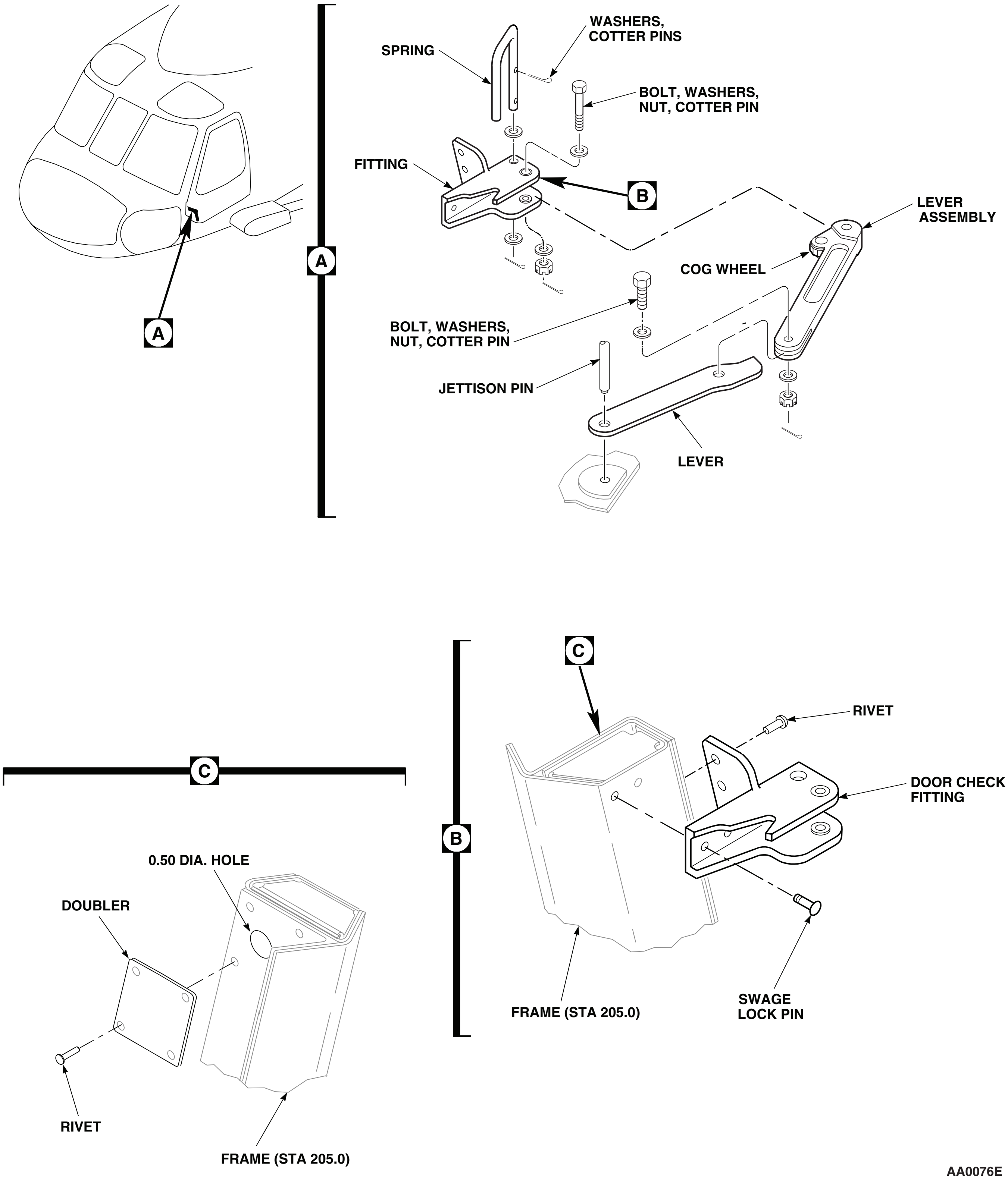

Figure 1. Replace Cockpit Door Check Components.

REMOVE - Continued

g. Using crew door post doubler, Item 276, WP 1805 00, as guide, drill 4 each holes in inboard door frame. As shown in Figure 1, Detail C.

INSTALL

1. Turn off all electrical power.
2. If fitting has been removed from frame, install it as follows (Figure 1, Detail B):

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Thoroughly clean mating surfaces of fitting and frame using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

b. Wipe dry with clean cheesecloth cloth, Item 85, WP 1803 00.

c. Apply sealing compound, Item 273, WP 1803 00, or epoxy primer coating kit, Item 120, WP 1803 00, to all mating surfaces and fasteners.

d. Install fitting while coating is still wet.

CAUTION

- Backing plate must be reinstalled or damage to door frame may result.
- Use only HL20PB86 or equivelant fasteners or damage to the door frame may result.

e. **QA** Install 2 HL20PB86 fasteners through the 0.50-inch in door post and secure backing plate and fitting to door frame so that collars are now on outboard side of fitting (TM 1-1500-204-23).

f. **QA** Install 3 NAS1738B5-X blind fasteners and secure flange of fitting to door frame (TM 1-1500-204-23).

g. Fair coating squeeze out into fillet where possible.

h. Apply sealing compound, Item 273, WP 1803 00, or epoxy primer coating kit, Item 120, WP 1803 00, to back of crew door post doubler, Item 276, WP 1805 00. Install crew door post doubler, Item 276, WP 1805 00, on door frame using 4 NAS 1738B5-X fasteners. (TM 1-1500-204-23).

i. Touch up paint as required (WP 0381 00).

3. If removed, coat OD of replacement bushings, with primer coating, Item 243, WP 1803 00, and press into fitting while primer is wet.
4. **QA** Install a washer on leg of spring that has holes in it. Insert that leg of spring into forward hole of fitting. Install remaining washer and secure with cotter pin (Figure 1, Detail A).
5. Attach lever assembly to fitting with bolt, washers, and nut. Install cotter pin.
6. If lever was replaced:

a. Remove screws, washers, and bolt, washer, and duct and outlet assembly from door by pushing down on duct to disengage hidden retaining latch.

b. Remove bolts and cover from door.

c. Connect release lever assembly to door by inserting jettison pin.

INSTALL - Continued

d. **QA** Install cover on door with washers and bolts.

e. Install duct and outlet assembly on door by sliding up from bottom to engage hidden retaining latch. Install screws and washers, and bolt and washer.

NOTE

Make sure cog wheel on lever assembly will engage spring when door is open.

7. Open and close cockpit door a few times to check for correct operation.
8. Perform cockpit door jettison check (WP 0174 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR JETTISON COMPONENTS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit SC 5180-99-B01
Airframe Repairer's Toolkit SC 5180-99-B02
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Gasket, 70217-01000-142
Gasket, 70217-01000-143
Gasket, 70217-01000-144

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0174 00
WP 0198 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and washers, and bolt and washer. Remove duct and outlet assembly from door by pushing down on duct to release hidden retaining latch.
3. Remove screws and cover from inside of door.
4. With assistant holding door, pull jettison handle. If door does not jettison, reach through door opening and disengage arms and jettison pin by pulling up on tube assembly (Figure 1, Detail A).
5. Remove cockpit door from helicopter.
6. Remove headed pin and washer attaching control rod to jettison handle.
7. Remove headed pin and washer attaching control rod to jettison lever. Remove control rod.
8. Remove tension/return springs from tube assembly.
9. Remove bolt, washer, and nuts securing upper end of tube assembly to upper arm.
10. Remove bolt, washer, and nut securing lower end of tube assembly to lower arm. Remove tube assembly.
11. Remove headed pin and washer attaching jettison pin to tube assembly.
12. Remove headed pin attaching jettison lever to arm pin A. Remove jettison lever.
13. Remove retaining ring holding arm pin A in door. Remove arm pin and upper arm.
14. Remove headed pin attaching sleeve to arm pin B. Remove sleeve.
15. Remove retaining ring holding arm pin B in door. Remove arm pin and lower arm.
16. Remove bolt, spacer, washer, and nut and remove jettison handle.

INSTALL

1. Turn off all electrical power.
2. **QA** Install jettison handle with bolt, spacer, washer, and nut (Figure 1, Detail A). **TORQUE NUT TO 95 - 105 INCH-POUNDS.** Install cotter pin.

INSTALL - Continued

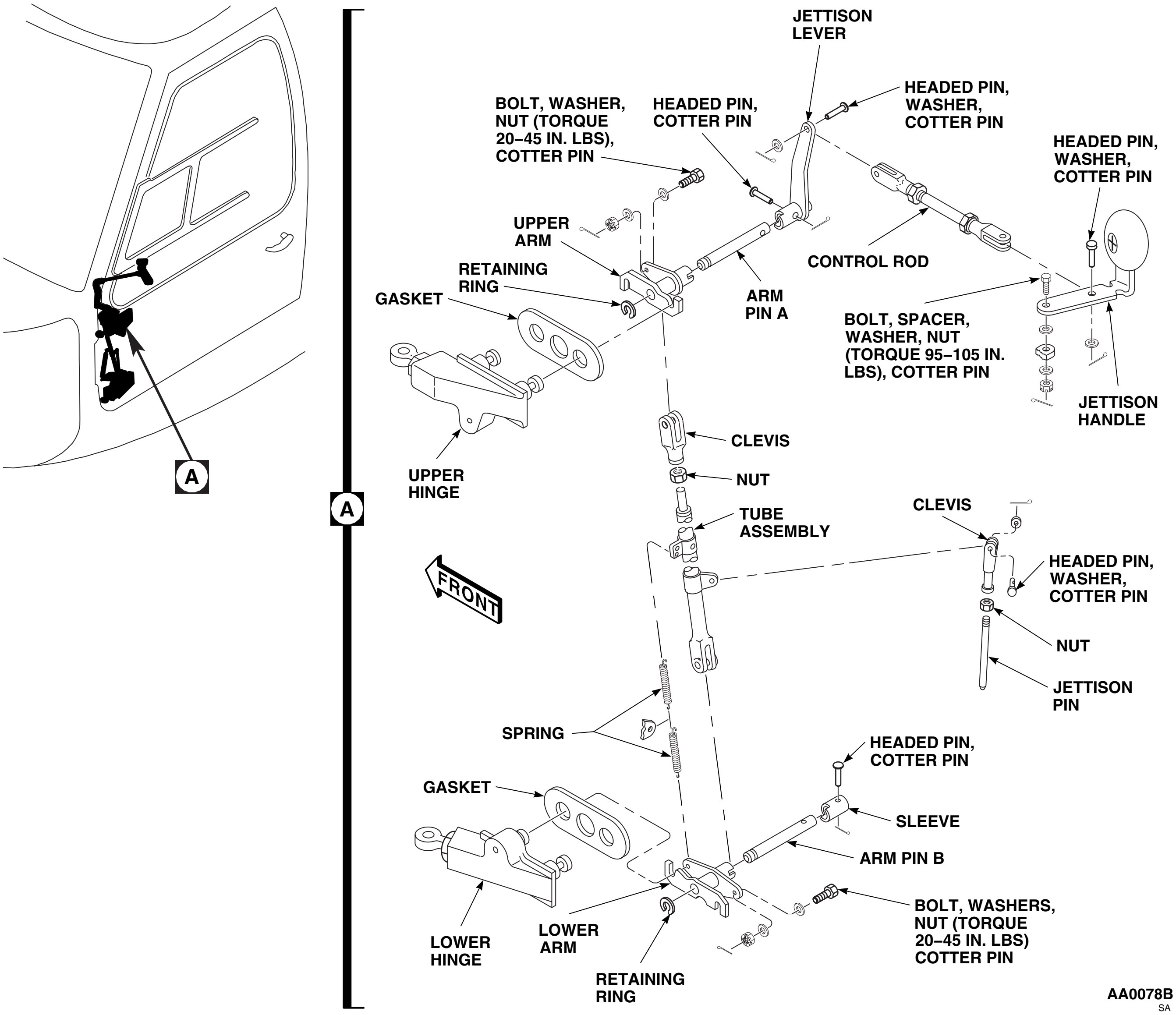

Figure 1. Replace Cockpit Door Jettison Components.

3. **QA** Install arm pin B in door and secure with retaining ring.
4. **QA** Install lower arm and sleeve on arm pin, fasten sleeve on arm pin with headed pin. Install cotter pin.
5. **QA** Install arm pin A in door and secure with retaining ring.
6. **QA** Install upper arm and jettison lever on arm pin A and fasten lever on arm pin A with headed pin. Install cotter pin.
7. **QA** Install jettison pin in fitting on bottom of door and attach jettison pin clevis to tube assembly with headed pin and washer. Install cotter pin.
8. **QA** Connect lower end of tube assembly to lower arm with bolt, washers, and nut. **TORQUE NUT TO 20 - 45 INCH-POUNDS.** Install cotter pin.

INSTALL - Continued

9. **QA** Connect upper end of tube assembly to upper arm with bolt, washers, and nut. **TORQUE NUT TO 20 - 45 INCH-POUNDS.** Install cotter pin.
10. **QA** Attach tension/return springs to tube assembly and to lower arm.
11. **QA** Attach control rod to jettison lever with headed pin and washer. Install cotter pin.
12. **QA** Attach control rod to jettison handle with headed pin and washer. Install cotter pin.
13. Adjust jettison components, as follows:
    a. Loosen jamnuts on control rod and turn rod to increase or decrease length until jettison handle is in full forward position. Tighten jamnuts.

NOTE

An additional gasket under hinge assemblies may be used for a tighter fit.

    b. If gaskets were removed, place replacement gaskets, 70217-01000-142, 70217-01000-143, or 70217-01000-144 (in combination, if required), under hinge assemblies.
    c. Have assistant pull jettison handle to jettison position to hold upper and lower arms in jettison position.
    d. Place door in position on helicopter and place hinges in door, move jettison handle forward to engage arms with bolts of hinge assembly.

NOTE

It may be necessary to push on jettison lever to engage arms.

    e. Check gap between arms and head of hinge bolt for 0.010-inch gap. If necessary, adjust bolt (WP 0198 00).
    f. If necessary, remove headed pin and washer connecting tube assembly to upper arm, loosen jamnut, and adjust clevis to put both arms in the engaged position. Tighten jamnut and connect tube assembly to upper arm with headed pin and washer. Install cotter pin.
14. Test jettison mechanism for operation and full engagement of arms and of jettison pin in door check lever. Adjust if necessary.
15. Perform cockpit door jettison check (WP 0174 00).
16. **QA** Install cover on inside of door with screws and washers.
17. **QA** Install duct and outlet assembly in door by sliding up from bottom to engage hidden retaining latch. Install bolt and washer, and screws and washers.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR RETAINER AND CLIP

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0174 00
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door.
3. Remove screws, washers, and nuts attaching plate and shims to door. Remove plate and shims (Figure 1, Detail A).
4. Check condition of clip riveted to canopy. If damaged, replace clip as follows:
   a. Carefully drill out and remove four rivets that secure clip to nose section, using No. 11 drill and 5/32-inch drive pin punch (Figure 1).
   b. Remove clip and clean up rivet debris.
   c. Open pilot holes in clip using No. 11 drill. Deburr holes.
   d. Position clip on canopy structure and hold in place using three 3/16-inch diameter clecos.
   e. Check open/close function of door for clearance of plate in clip slot.
   f. Install blind rivet in remaining hole.
   g. Remove one cleco from diagonal corner hole and install rivet.
   h. Remove remaining clecos and install rivets.
   i. Seal all gaps between clip and structure with sealing compound, Item 273, WP 1803 00.

INSTALL

1. Turn off all electrical power.
2. **QA** Position retainer plate and shims on door. Install short screw, washer, and nut in forward hole, with washer and nut against plate. Install remaining screws, washers, and nuts with screwheads against plate (Figure 1, Detail A).
3. Adjust plate by doing this:
   a. Close door.

INSTALL - Continued

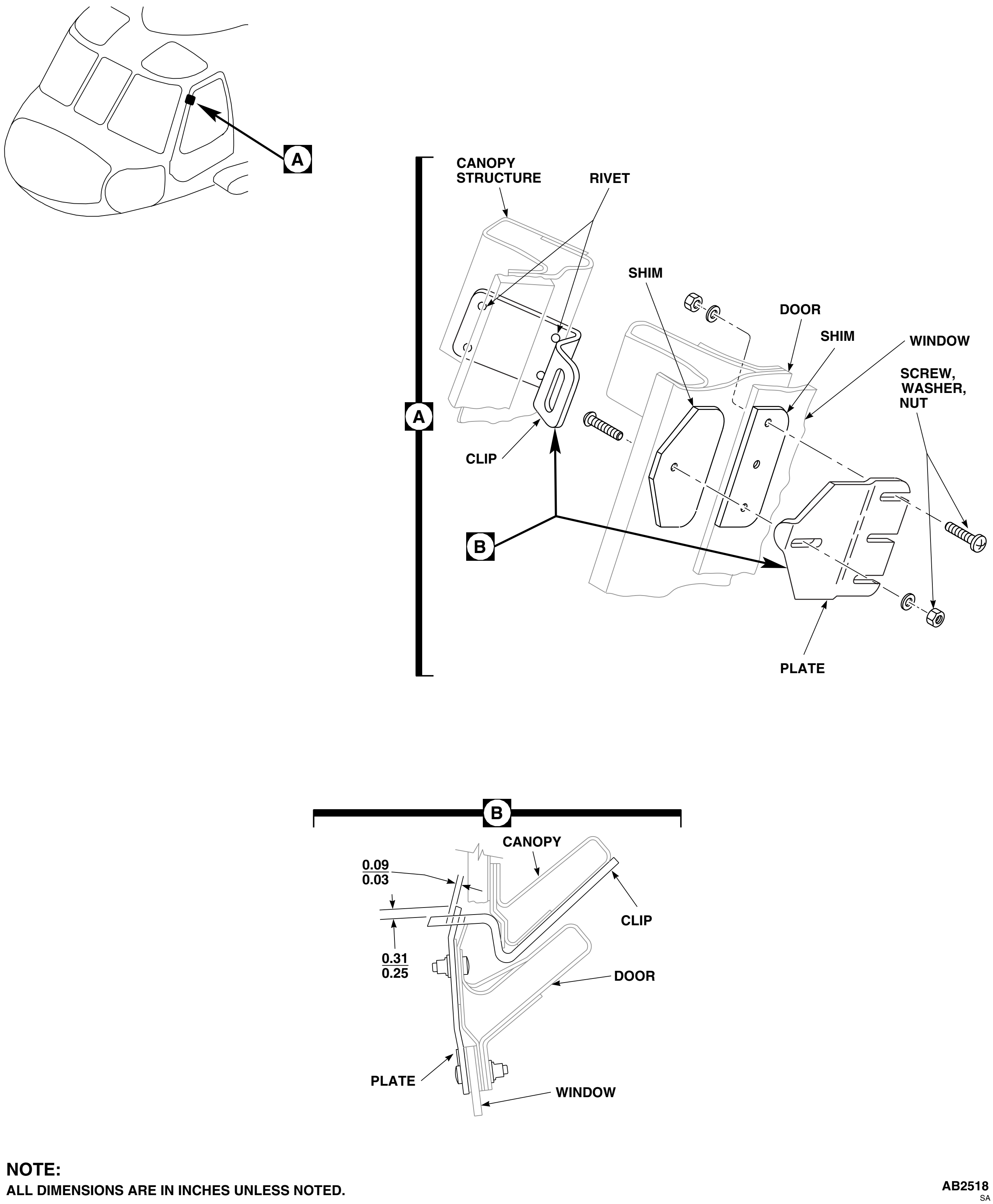

Figure 1. Replace Cockpit Door Retainer and Clip.

INSTALL - Continued

b. Check clearance between outer edge of plate and slot in clip.

c. Peel shims as necessary to make clearance 0.03 to 0.09-inch (Figure 1, Detail B).

d. Mount plate loosely with screws.

e. Close door.

f. Adjust plate until tip is 0.25 to 0.31-inch beyond clip.

g. Tighten screws.

4. Open and close cockpit door a few times to check operation.
5. Perform cockpit door jettison check (WP 0174 00).
6. Touch up paint in repaired area (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR STATIONARY WINDOW

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Remover, Paint, Item 252, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Tape, Insulation, Electrical, Item 318, WP 1803 00
Wire, Nonelectrical, Item 351, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0202 00
WP 0204 00
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove vent window (WP 0204 00).
3. Remove screws, washers, and nuts attaching window and lockwire clip to door frame (Figure 1, Detail A).
4. Loosen screw, washer, and nut attaching upper door retainer to door. Do not remove from door (WP 0202 00).
5. Remove window and lockwire clip from door.
6. Remove seal tape from door and clean flanges with paint remover, Item 252, WP 1803 00. Wipe dry.

INSTALL

1. Turn off all electrical power.
2. Trial fit window in frame. Trim window, if necessary, to seat in window frame (Figure 1, Detail A).
3. If holes in window and door frame do not line up, do this:
   a. Mark holes in frame that do not line up.
   b. Fill holes and touch up paint area (WP 0381 00 and TM 55-1500-345-23).
   c. Using replacement window as template, mark new holes.
   d. Using 7/32-inch diameter drill bit, drill new holes through frame.
4. Put electrical insulation tape, Item 318, WP 1803 00, on flanges of replacement window, with sticky side against flange. Trim tape as necessary.
5. Coat tape with a thin coating (about 0.003-inch) of sealing compound, Item 273, WP 1803 00.

INSTALL - Continued

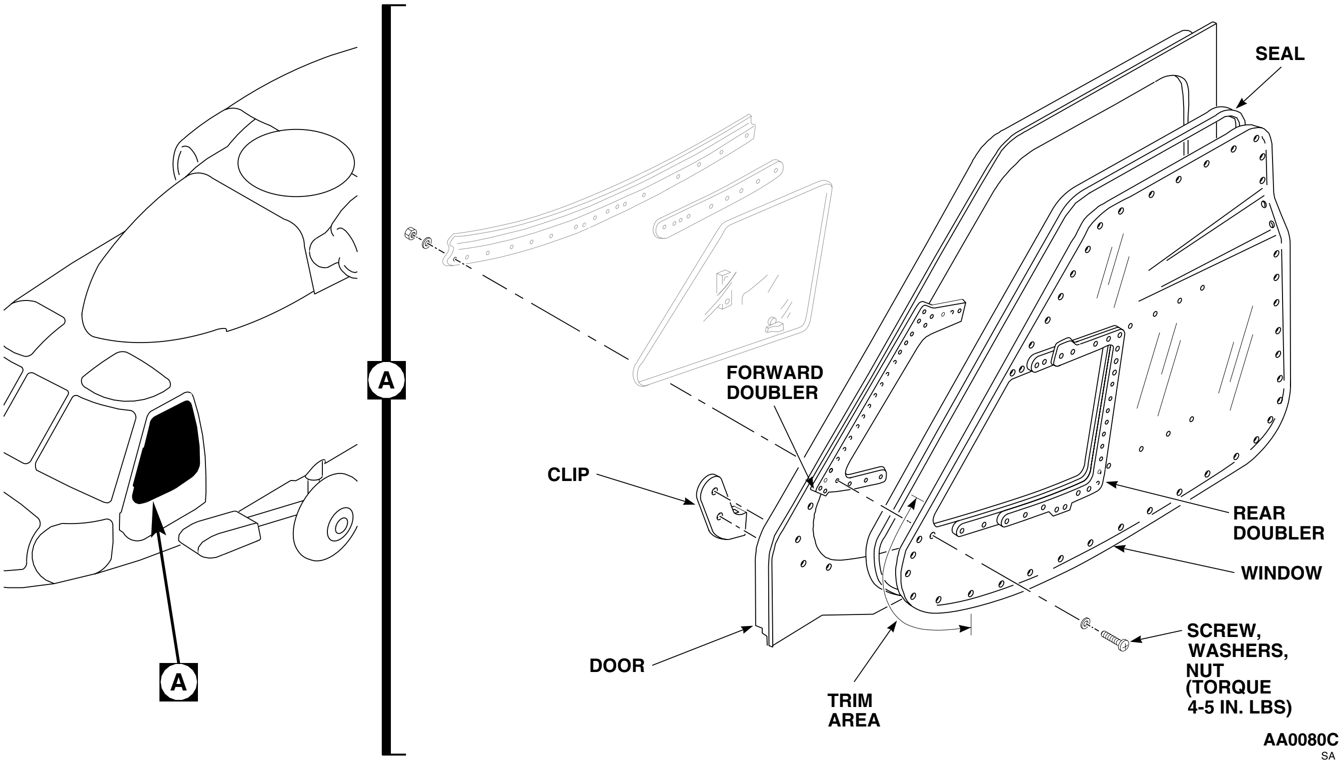

Figure 1. Replace Cockpit Door Stationary Window.

CAUTION

Overtightening of attaching screws will crack window. Do not overtighten attaching screws.

6. **QA** Position window on door. Place lockwire clip on lower forward edge of door window frame. Install screws, washers, and nuts. Tighten screws until window touches tape. **TORQUE NUTS TO 4 - 5 INCH-POUNDS.**
7. Tighten screws, washers, and nuts attaching upper door retainer to door.
8. Fill any gaps between window and frame with sealing compound, Item 273, WP 1803 00. Smooth out sealing compound until surface is flush with window and frame.
9. Install vent window (WP 0204 00).
10. With an assistant inside, spray window with water and check for leaks, especially around window edge. Seal any leaks with sealing compound, Item 273, WP 1803 00.
11. **QA** Using nonelectrical wire, Item 351, WP 1803 00 (breakaway wire), lockwire jettison handle to clip on lower forward edge of door window frame.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR VENT WINDOW

This WP supersedes WP 0204 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 40, WP 1803 00
Cloth, Abrasive, Item 84, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0378 00
WP 1647 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door. Move vent window to open position.
3. UH-60A 79-23283 - SUBQ Remove screws, washers, and nuts attaching upper vent window retainer and radius block to window (Figure 1, Sheet 1, Detail B).
4. UH-60A 79-23283 - SUBQ Remove screws, washers, and nuts attaching lower vent window retainer and radius block to window (Figure 1, Sheet 1, Detail C).
5. Remove screws, sealing washers, washers and nuts attaching vent window lower track to window (Figure 1, Sheet 1, Detail D). Remove track.
6. Lift vent window and pull inboard slightly to remove window from upper track.
7. Remove screws, sealing washers, washers, and nuts attaching upper track to window. Remove track.

INSPECT

1. Check condition of sheet metal doubler. If damaged, repair (WP 0378 00) or replace doubler.
2. Check condition of vent window seals. If damaged or dried out, replace seals.

REPAIR

1. Remove damaged seals from tracks or doubler using a nonmetallic scraper.

REPAIR - Continued

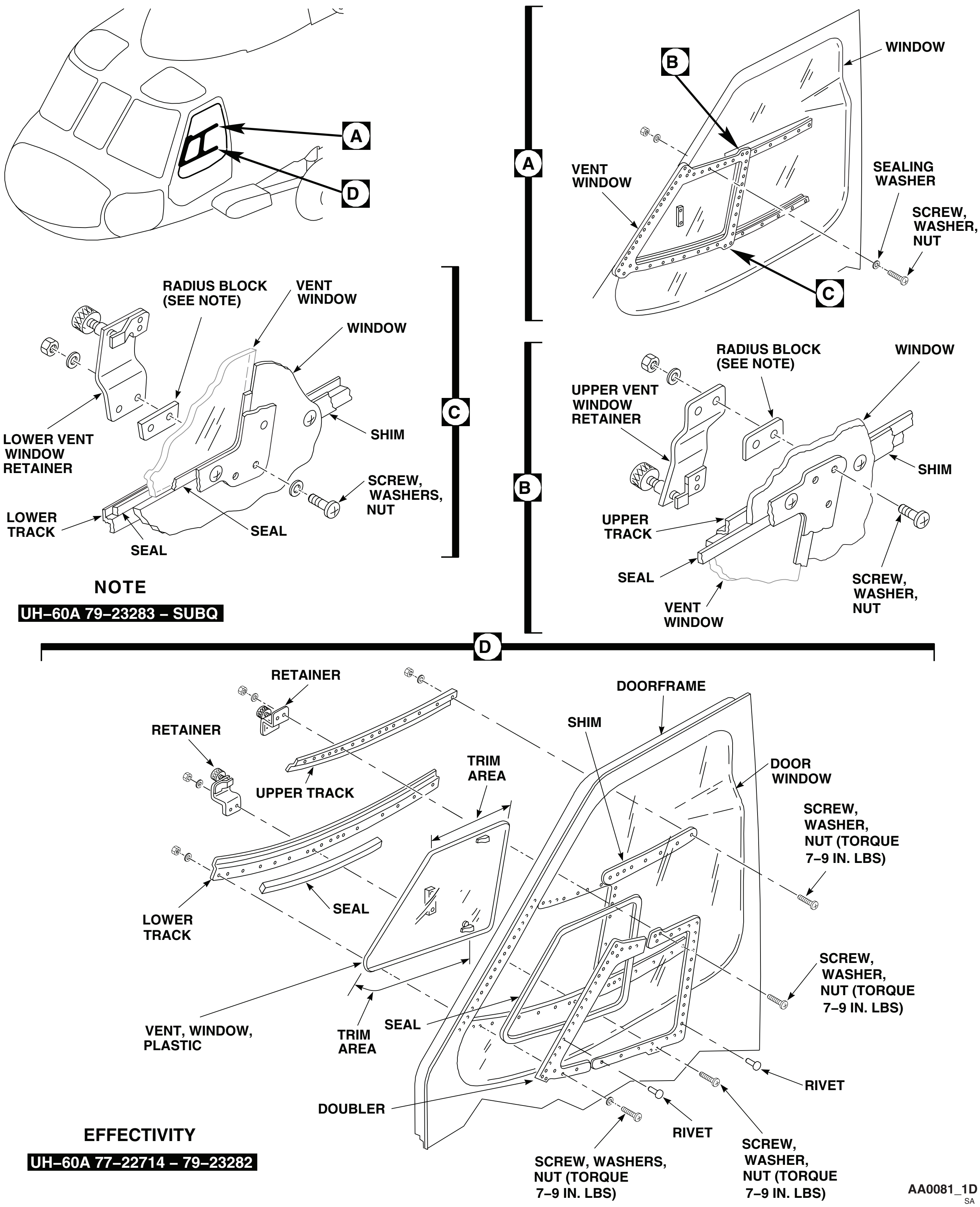

Figure 1. Replace Cockpit Door Vent Window. (Sheet 1 of 2)

REPAIR - Continued

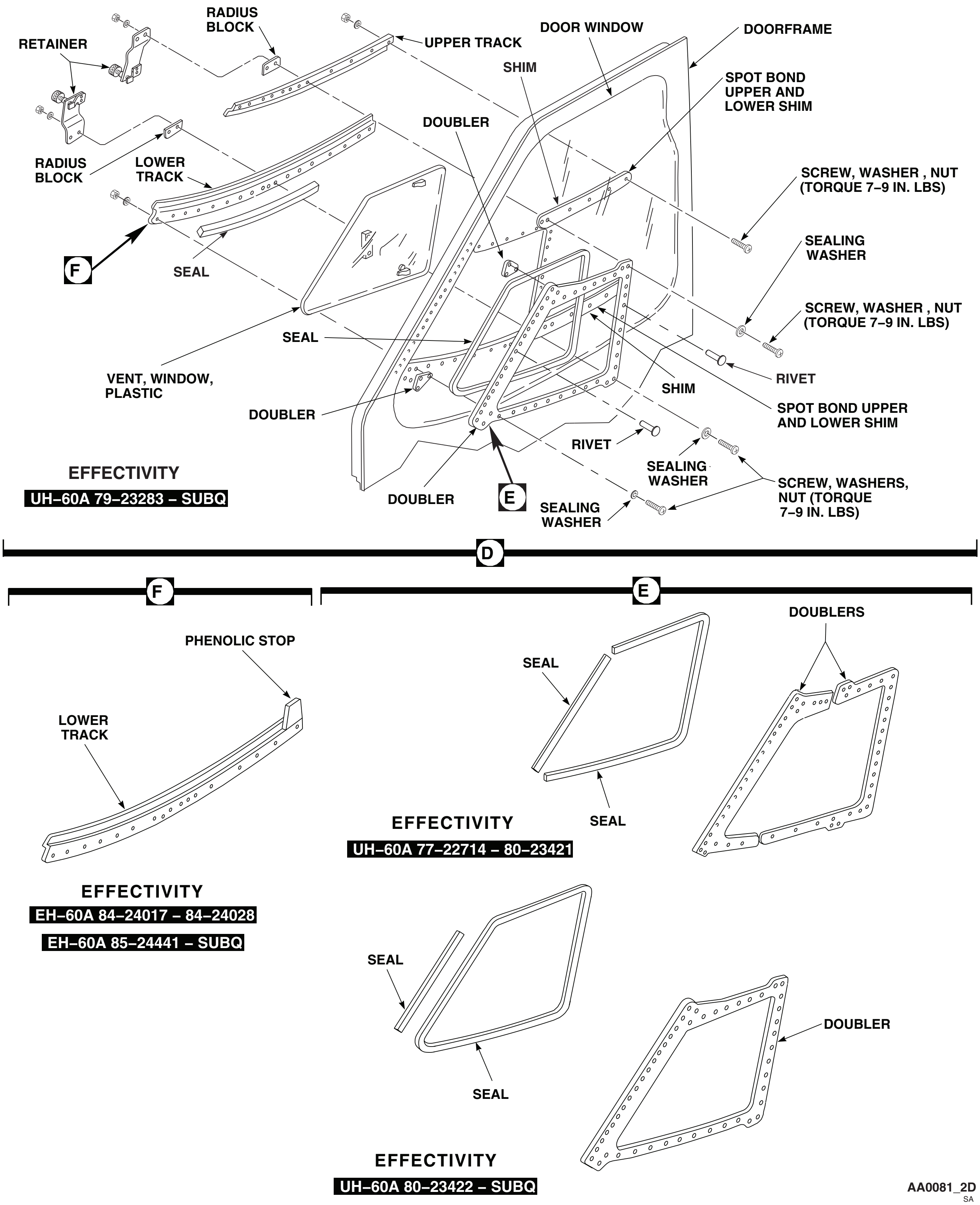

Figure 1. Replace Cockpit Door Vent Window. (Sheet 2 of 2)

REPAIR - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Plastic windows are damaged by contact with adhesives or adhesive solvents. Do not spill adhesive compounds or solvents on windows.

2. Remove old adhesive using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Trim new seals to correct size.
4. Roughen replacement seals, tracks, and doubler contact surfaces using abrasive cloth, Item 84, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

NOTE

Oil from fingers will weaken adhesive bond. Avoid touching adhesive contact surfaces after cleanup or after application of adhesive compound.

5. Remove debris from roughened parts with clean cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe parts clean and dry.
6. Apply a thin, even coat of adhesive, Item 14, WP 1803 00, to track, doubler, and seal bonding surfaces. Allow adhesive to air dry for 5 to 10 minutes or until tacky.

NOTE

Do not stretch seal when installing around corners of doubler.

7. Align and attach seals to bonding surfaces. Press seals firmly in place. Work out wrinkles or trapped air. Allow seal to cure 24 hours at 68°F(20°C).
8. Clean and polish window as required (WP 1647 00).

INSTALL

1. Turn off all electrical power.

CAUTION

To prevent cracking window, do not overtighten attaching nuts.

2. Spot bond plastic shims to window with adhesive, Item 40, WP 1803 00 (Figure 1, Sheet 2, Detail D).
3. Apply sealing compound, Item 273, WP 1803 00, under sealing washers on cockpit door window screws.

INSTALL - Continued

4. **QA** **UH-60A 79-23283 - SUBQ** Position upper track in place. Install retainer, radius block, screws with sealing washers under screwheads, washers, and nuts. **TORQUE NUTS TO 7 - 9 INCH POUNDS.**
5. Place vent window into upper track in rear (open) position.
6. **QA** **UH-60A 79-23283 - SUBQ** Install lower track retainer and radius block, screws with sealing washers under screwheads, washer, and nuts. **TORQUE NUTS TO 7 - 9 INCH POUNDS.**
7. Slide vent window open and closed to check operation.
8. If window binds in track, trim top and bottom of window as required, or in lieu of trimming vent window, drill out 0.210-inch diameter screw holes in upper track and retainer, as required, to a maximum of 0.250-inch diameter, so track can be adjusted to eliminate vent window binding (Figure 1, Sheet 1, Detail D).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR JETTISONABLE WINDOW

This WP supersedes WP 0205 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 21, WP 1803 00
Brush, Acid Swabbing, Item 57, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Cloth, Flannel, Item 88, WP 1803 00
Fastener Tape, Pile, Item 125, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 267, WP 1803 00
Stirring Stick, Beverage, Item 300, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0202 00
WP 0381 00
WP 1803 00

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW

Remove

1. Turn off all electrical power.
2. Open cockpit door, have assistant hold door during removal of window.

CAUTION

Extreme care must be taken during removal to prevent damage to jettisonable window.

3. Pull emergency strap inboard to disengage jettisonable window from seal.
4. Push jettisonable window outboard, through window opening, and remove from helicopter.
5. Close cockpit door.

Inspect/Repair

1. Check condition of jettisonable window seal. If damaged or dried out, replace seal as follows:
   a. Starting at twelve o'clock position, pull seal from frame (Figure 1, Detail A).

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW - Continued

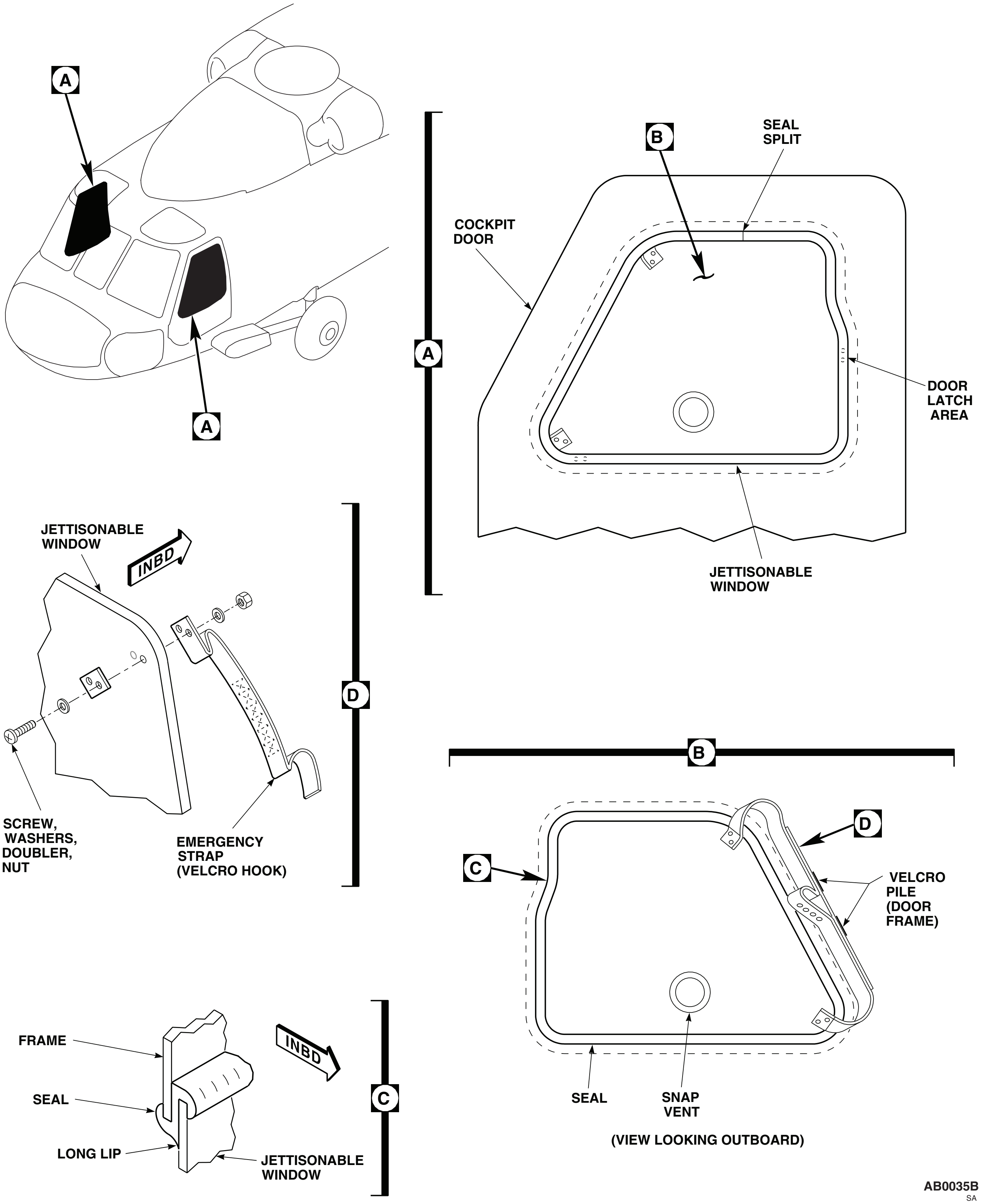

Figure 1. Replace Cockpit Door Jettisonable Window.

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW - Continued

WARNING

Improper installation of the window seal may result in the window being blown off the helicopter and subsequent damage to the helicopter and/or injury to personnel.

NOTE

- Long lip of seal must rest against window when window is installed.
- Make sure split in replacement seal is at twelve o'clock position on frame.

b. While making sure long lip of seal will rest on window, position split in seal at twelve o'clock, push seal over frame, continue applying seal around entire frame (Figure 1, Detail C).

2. Check condition of emergency pull strap. If damaged replace strap as follows:

a. Remove screws, washers, doubler, and nut attaching strap to jettisonable window (Figure 1, Detail D).

b. Position replacement strap on window, install screws, washers, doubler, and nut (Figure 1, Detail D).

3. Check condition of pile fastener tape on door frame. If damaged repair or replace as follows:

a. If pile fastener tape is not damaged but separated from door frame repair as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

(1) Clean exposed door frame surface using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

(2) Apply adhesive, Item 21, WP 1803 00, to door frame with acid swabbing brush, Item 57, WP 1803 00.

(3) Install loose pile fastener tape against door frame using light hand pressure.

b. If pile fastener tape is damaged replace as follows:

(1) Remove pile fastener tape from door frame using nonmetallic scraper.

WARNING

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(2) Remove adhesive residue from door frame using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

(3) Lightly abrade surface of door frame where pile fastener tape is located using abrasive cloth, Item 83, WP 1803 00.

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW - Continued

WARNING

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(4) Wipe surface with cheesecloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

(5) Apply adhesive, Item 21, WP 1803 00, using acid swabbing brush, Item 57, WP 1803 00, to door frame surface where pile fastener tape will be located.

(6) Install pile fastener tape, Item 125, WP 1803 00, on door frame using light hand pressure.

Install

1. Turn off all electrical power.
2. Open cockpit door, have assistant hold door during installation of window.

WARNING

Improper installation of the window seal may result in the window being blown off the helicopter and subsequent damage to the helicopter and/or personnel.

CAUTION

- To prevent damage to jettisonable window extreme care must be taken during installation.
- Make sure window is installed into seal on the inboard side of the door frame.

NOTE

- Make sure long lip of seal rests on window (Figure 1, Detail C).
- Make sure split in seal is at twelve o'clock position.
- Do not use lubricant when installing jettisonable window. Powdery residue present on seal is adequate for installation.

3. Starting in area of door latch insert window from inboard side of door into seal. Flex and manipulate window as required to properly engage seal. A beverage stirring stick, Item 300, WP 1803 00, may be used to lift seal over window (Figure 1, Detail A).
4. **QA** Check seal for any area that may have lifted from jettisonable window during installation. If seal has lifted push seal towards center of jettisonable window until seal has fully engaged window.
5. Make sure area of interface of jettisonable window and seal is clean. Apply a 3/16-inch bead of sealing compound, Item 273, WP 1803 00, around entire perimeter of jettisonable window and seal. Allow 3 to 4 hours for sealant to dry.

NOTE

If window frame is to be painted, apply sealant after frame has been painted.

6. Make sure area of interface of window frame and seal is clean. Apply a 3/16-inch bead of sealing compound, Item 273, WP 1803 00, around entire perimeter of window frame and seal. Allow 3 to 4 hours for sealant to dry.

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW - Continued

7. Close cockpit door.

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW ASSEMBLY FRAME

Remove

1. Turn off all electrical power.
2. Open cockpit door.
3. If installed, remove jettisonable window.
4. Remove upper door retainer (WP 0202 00).
5. Remove nuts, screws, and washers securing jettisonable window frame and clip to door.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Clean door flange and jettisonable window assembly using a clean flannel cloth, Item 88, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean flannel cloth.

Install

1. Turn off all electrical power.
2. Using pressure sensitive adhesive tape, Item 334, WP 1803 00, tape off door flange and jettisonable window assembly.
3. Apply sealing compound, Item 267, WP 1803 00, to the door flange.
4. **QA** Position jettisonable window assembly on the door. During the work life of the sealing compound, install jettisonable window assembly and clip with screws, washers, and nuts (Figure 2, Detail C).
5. Install upper door retainer (WP 0202 00).
6. Fillet seal area around and upper door retainer with sealing compound, Item 267, WP 1803 00, to a smooth surface.
7. Touch up paint in repaired area (WP 0381 00 and TM 55-1500-345-23).
8. Remove adhesive tape.
9. Install jettisonable window. Refer to, INSTALL, in this work package.

REPLACEMENT OF COCKPIT DOOR JETTISONABLE WINDOW ASSEMBLY FRAME - Continued

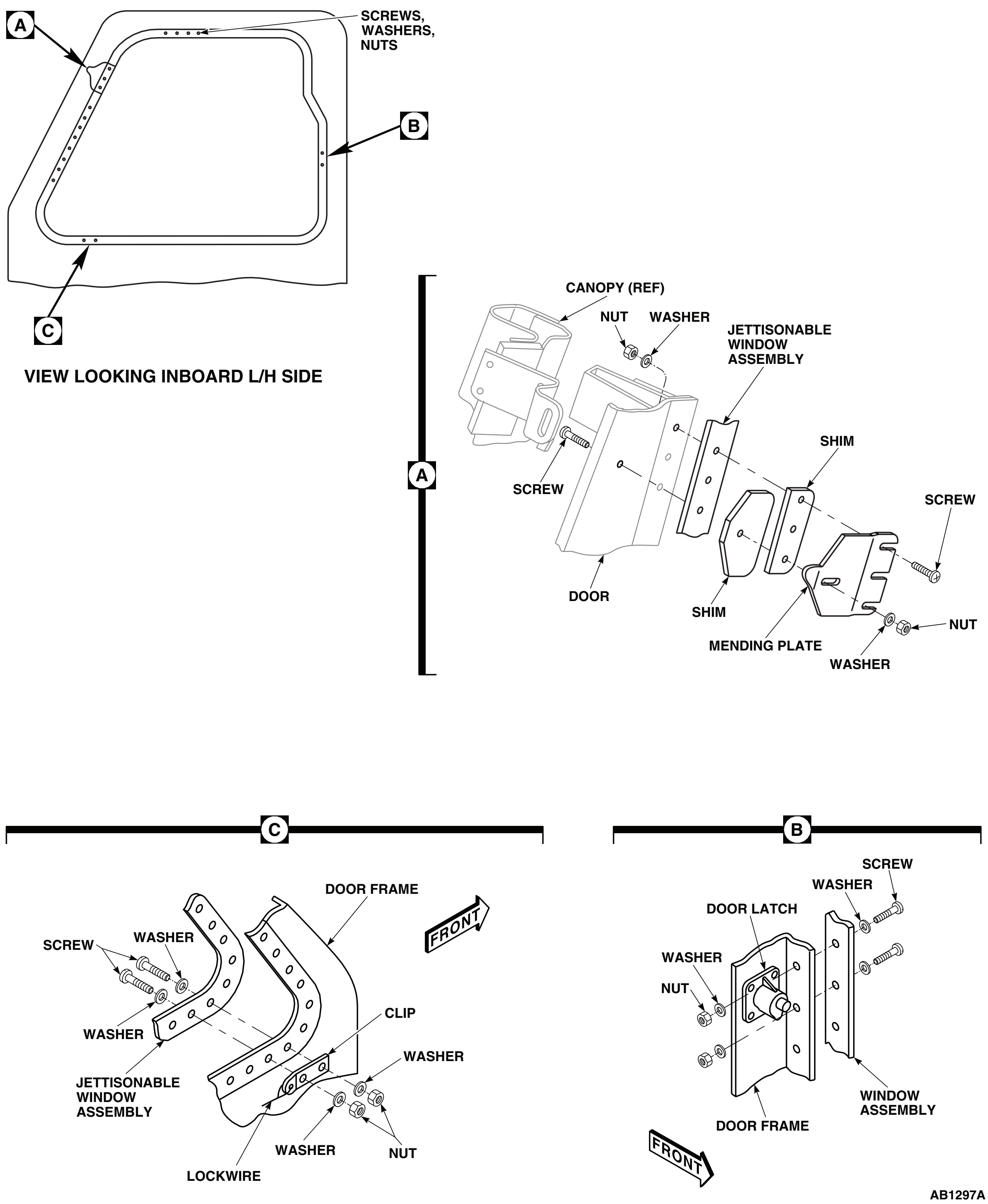


Figure 2. Installation Cockpit Door Jettisonable Window Assembly Frame.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR FRONT CRANK MECHANISM

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0199 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door.
3. Drive out rollpin attaching inside door handle shaft to front crank (Figure 1, Detail A).
4. Unhook handle return spring from cam and clip on lower rod.
5. Pull handle and shaft out of door. Remove spring and washers from between door and front crank.
6. Push crank outboard to get at clevis attaching points.
7. Remove headed pins attaching upper and lower rods to front crank.
8. Remove front crank from door.
9. Remove screws, washers, and nut attaching cam to doubler. Remove cam.
10. Drive out rollpin and remove handle from shaft.
11. Drill out rivets and remove doubler from door.
12. Drill out rivets and remove bearing from doubler (TM 1-1500-204-23).

INSTALL

1. Turn off all electrical power.
2. Rivet bearing to doubler (Figure 1, Detail A).
3. **QA** Rivet doubler to door (TM 1-1500-204-23).
4. **QA** Position cam in door. Attach cam to door and doubler with screw and washer, and screw, washer, and nut.
5. Place front crank in door. Connect upper and lower rods to front crank with headed pins, washers, and cotter pins.
6. Install front crank boss in doubler bearing. Place spring and washers in door behind crank.
7. Attach handle to shaft with rollpin.
8. Install handle and shaft through bearing, front crank, washers, and spring.
9. **QA** Attach shaft to front crank with rollpin.

INSTALL - Continued

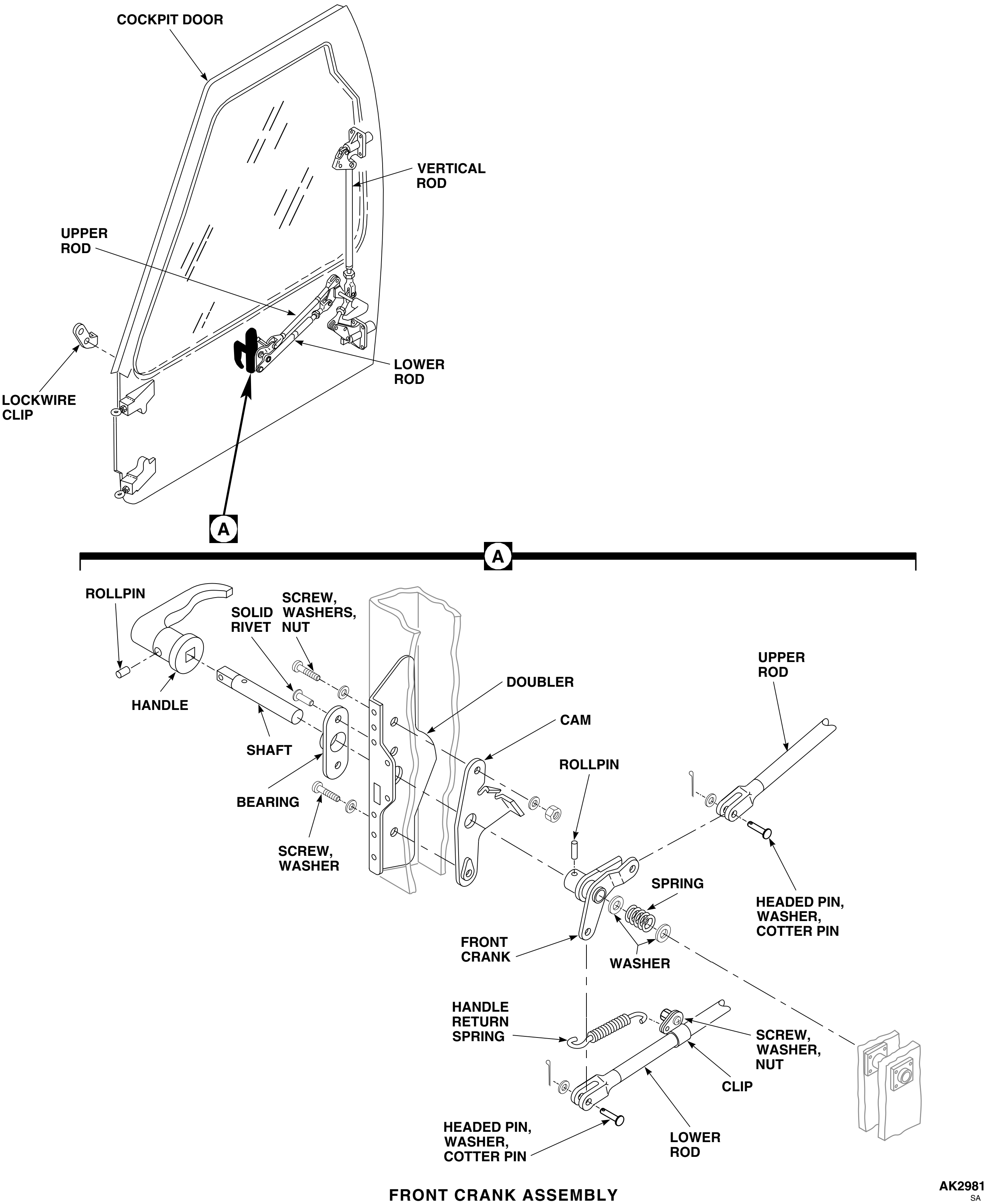


Figure 1. Replace Cockpit Door Front Crank Mechanism.

INSTALL - Continued

10. **QA** Hook handle return spring to cam and clip on lower rod.
11. Open and close door a few times to check for correct operation.
12. Adjust door as required (WP 0199 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR LATCHES

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Sealing Compound, Item 281, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0199 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door.
3. Remove headed pins, washers, and cotter pins. Remove vertical rod from upper bellcrank and remove bellcrank from latchbolt housing and latchbolt (Figure 1, Detail A).
4. Remove headed pins, washers, and cotter pins. Remove link from lower bellcrank, and bellcrank from latchbolt housing and latchbolt (Figure 1, Detail C).
5. Push latchbolt into housing, turn bolt to line up slotted end with slot in housing, and remove latchbolt and spring from housing (Figure 1, Detail B).
6. **UH60A 77-22714 - 78-23006** Drill out blind rivets and blind bolts and remove latch housing from door (TM 1-1500-204-23).
7. **UH60A 78-23007 - SUBQ UH60L EH60A** Remove bolts and screw from latch housing. Remove latch housing from door.
8. Remove drilled metal from inside door through drain hole in bottom of door.

INSTALL

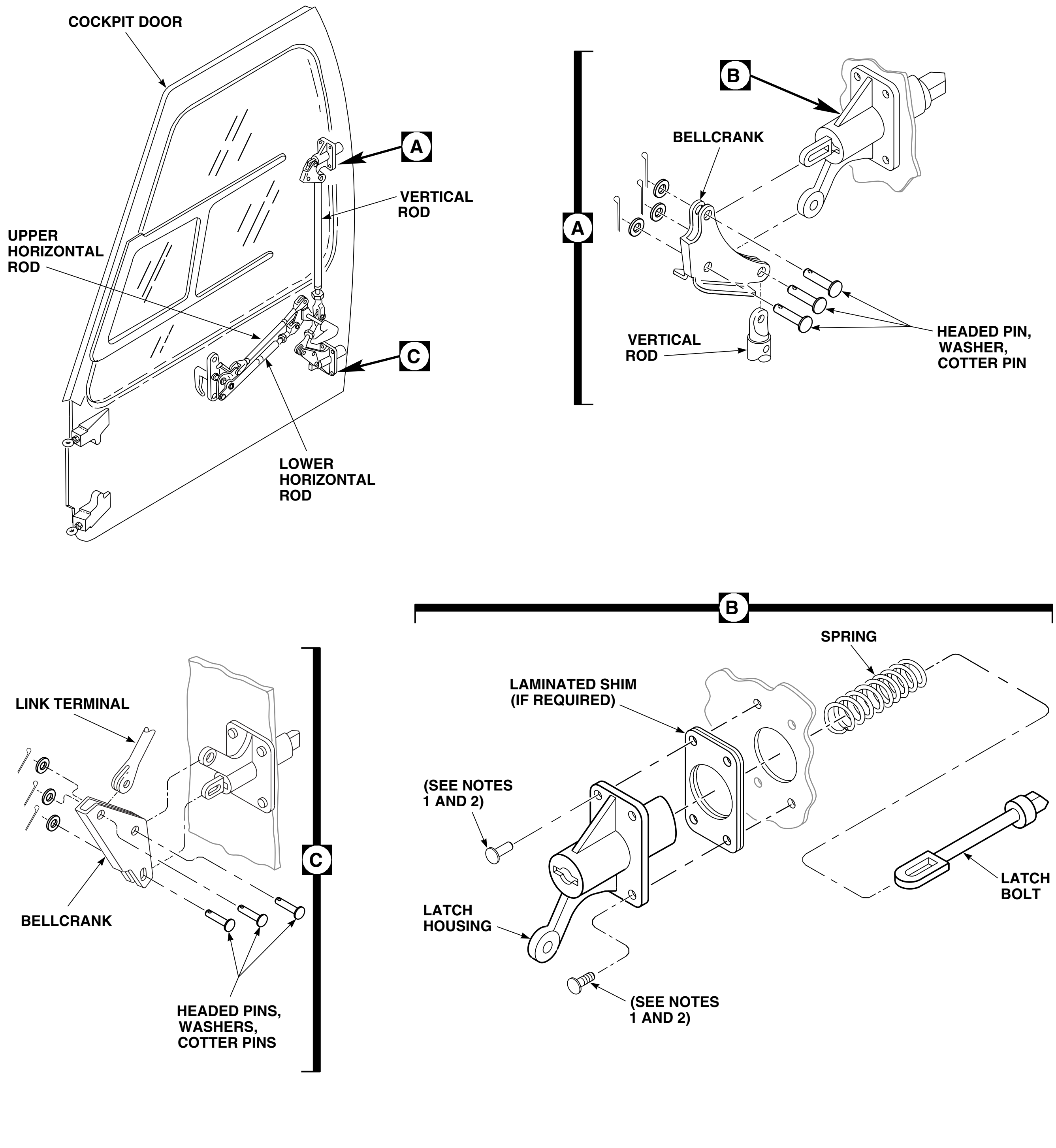
1. Turn off all electrical power.
2. Install latch housing in door frame (Figure 1, Detail B). Position upper latch housing with bellcrank arm down. Position lower latch housing with bellcrank arm up.

NOTE

If required for proper latch bolt engagement with striker plate, laminated shim may be used (Figure 1, Detail B).

3. **QA** **UH60A 77-22714 - 78-23006** Install blind rivets on inboard side of housing and blind bolts on outboard side of housing (TM 1-1500-204-23).

INSTALL - Continued



NOTES:

1. UH60A 77−22714 − 78−23006
   BLIND RIVETS AND BLIND BOLTS.
2. UH60A 78−23007 − SUBQ
   BOLTS, WASHERS, AND SCREW.

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SA

Figure 1. Replace Cockpit Door Latches.

INSTALL - Continued

4. **QA** **UH60A 78-23007 - SUBQ UH60L EH60A** Install two bolts and two washers on inboard side of housing. Install one bolt and washer to bottom outboard side and one screw in upper outboard side of housing. Apply sealing compound, Item 281, WP 1803 00, to threads of bolts and screws.
5. Install spring and latchbolt in latch housing. Push latchbolt slotted end through slot in housing and turn 90 degrees. Face beveled side of latchbolt inboard.
6. **QA** Install lower bellcrank on lower latch housing arm and on latchbolt slotted pin hole. Install headed pins, washers, and cotter pins (Figure 1, Detail C).
7. **QA** Install link in lower bellcrank with headed pin, washer, and cotter pin.
8. **QA** Install upper bellcrank on upper latch housing arm and on latchbolt, with slotted pin hole on latchbolt. Install headed pins, washers, and cotter pins (Figure 1, Detail A).
9. **QA** Install vertical rod in upper bellcrank with headed pin, washer, and cotter pin.
10. Close door and check that latchbolts engage striker plates correctly. Adjust as necessary (WP 0199 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR REAR CRANK MECHANISM

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0199 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door.
3. Remove screws, washers, lockwashers, and nuts attaching support to inside of door (Figure 1, Detail A). Remove support from door.
4. Remove screws, washers, and nuts attaching door handle to door. Remove handle.
5. Remove headed pin connecting vertical rod, link, and bellcrank.
6. Remove headed pins attaching upper and lower rods to rear crank and bellcrank.
7. Remove retaining ring and washer from rear crank shaft.
8. Remove rear crank and bellcrank from door. Slide inboard until clear of door structure.
9. Remove washer from rear crank and slide bellcrank off rear crank shaft.

INSTALL

1. Turn off all electrical power.
2. Slide rear crank shaft into bellcrank and install washer on rear crankshaft (Figure 1, Detail A).
3. **QA** Install rear crank in door structure and install washer and retaining ring.
4. **QA** Connect upper rod to bellcrank and lower rod to rear crank with headed pins, washers, and cotter pins.
5. **QA** Connect vertical rod and link to bellcrank with headed pin, washer, and cotter pin.
6. **QA** Install door handle on rear crank shaft and fasten to door with screws, washers, and nuts.
7. **QA** Place support on rear crank and inside of door. Fasten with screws, washers, lockwashers, and nuts.
8. Open and close door a few times to check for proper operation.
9. Adjust latches as required (WP 0199 00).

INSTALL - Continued

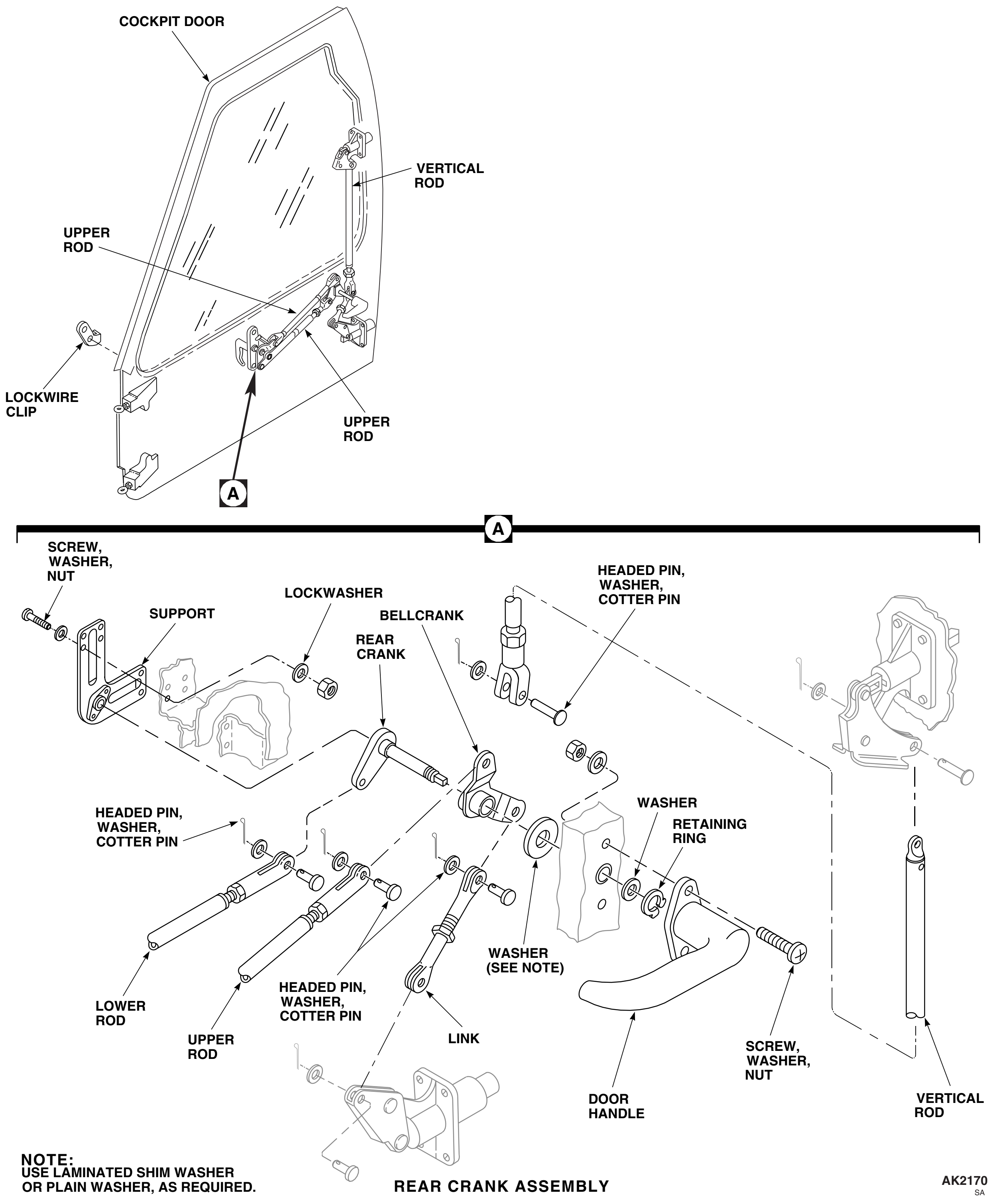


Figure 1. Replace Cockpit Door Rear Crank Mechanism.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR STRIKER PLATE

This WP supersedes WP 0209 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Corrosion Preventive Compound, Item 107, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0199 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open cockpit door.
3. Remove screws and remove serrated plate, striker plate, and shim from door jamb (Figure 1, Detail A).

INSTALL

1. Turn off all electrical power.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean off any oil or grease from serrated plate, striker plate, and shim using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Apply sealing compound, Item 273, WP 1803 00, to smooth side of striker plate. Do not get sealing compound on serrated side of plate.
4. Press striker plate and shim together with fingers (Figure 1, Detail A). Remove any sealing compound that squeezes out with clean machinery wiping towel, Item 344, WP 1803 00.
5. Apply corrosion preventive compound, Item 107, WP 1803 00, to serrated sides of striker plate and serrated plate.
6. **QA** Install shim, striker plate, and serrated plate on door jamb with screws.
7. Adjust as required (WP 0199 00).

INSTALL - Continued

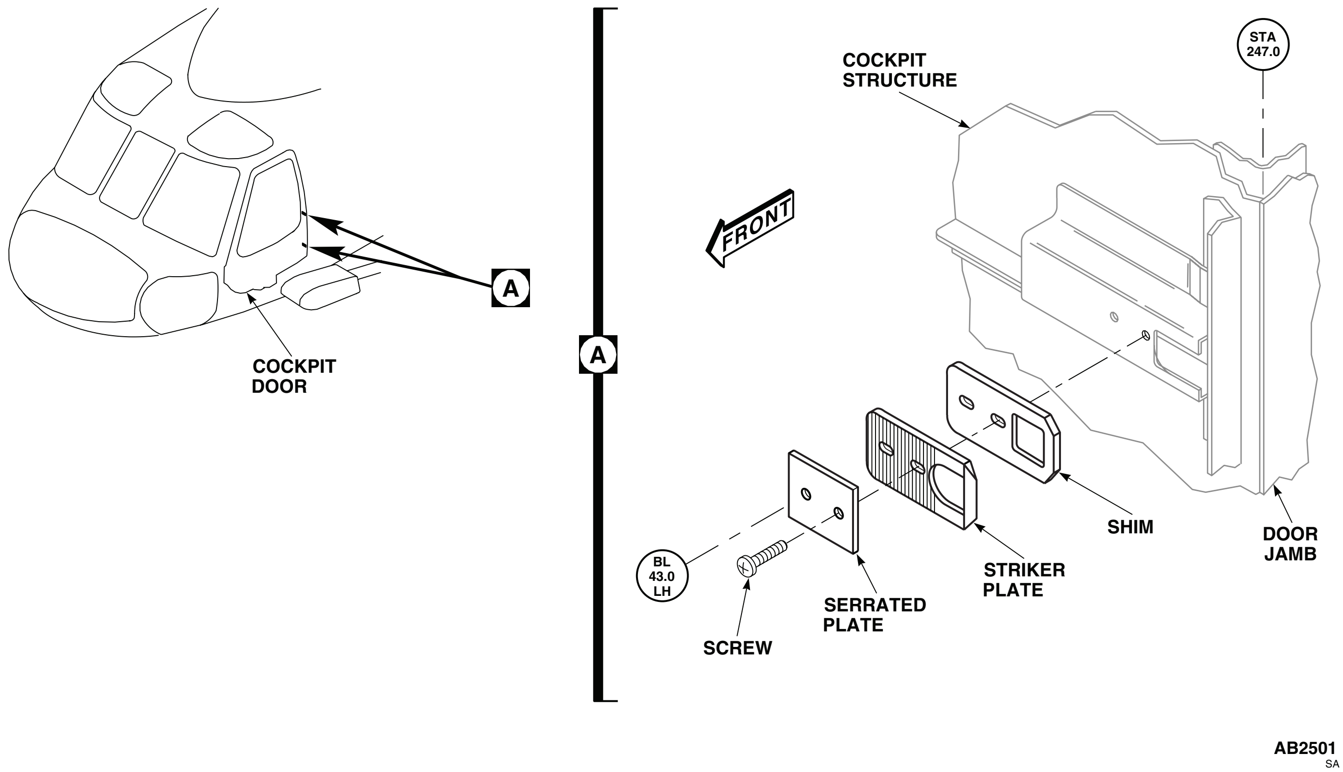


Figure 1. Replace Cockpit Door Striker Plate.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR SEAL

This WP supersedes WP 0210 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 36, WP 1803 00
Adhesive, Item 41, WP 1803 00
Adhesive, Item 47, WP 1803 00
Cloth, Abrasive, Item 84, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0198 00
WP 0199 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove cockpit door (WP 0198 00).
3. Remove damaged seal from door using a nonmetallic scraper (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to window will result if technical acetone contacts it. Do not allow technical acetone to contact window.

4. Remove adhesive from door seal flange using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Scrape off remaining adhesive using a nonmetallic scraper.

REMOVE - Continued

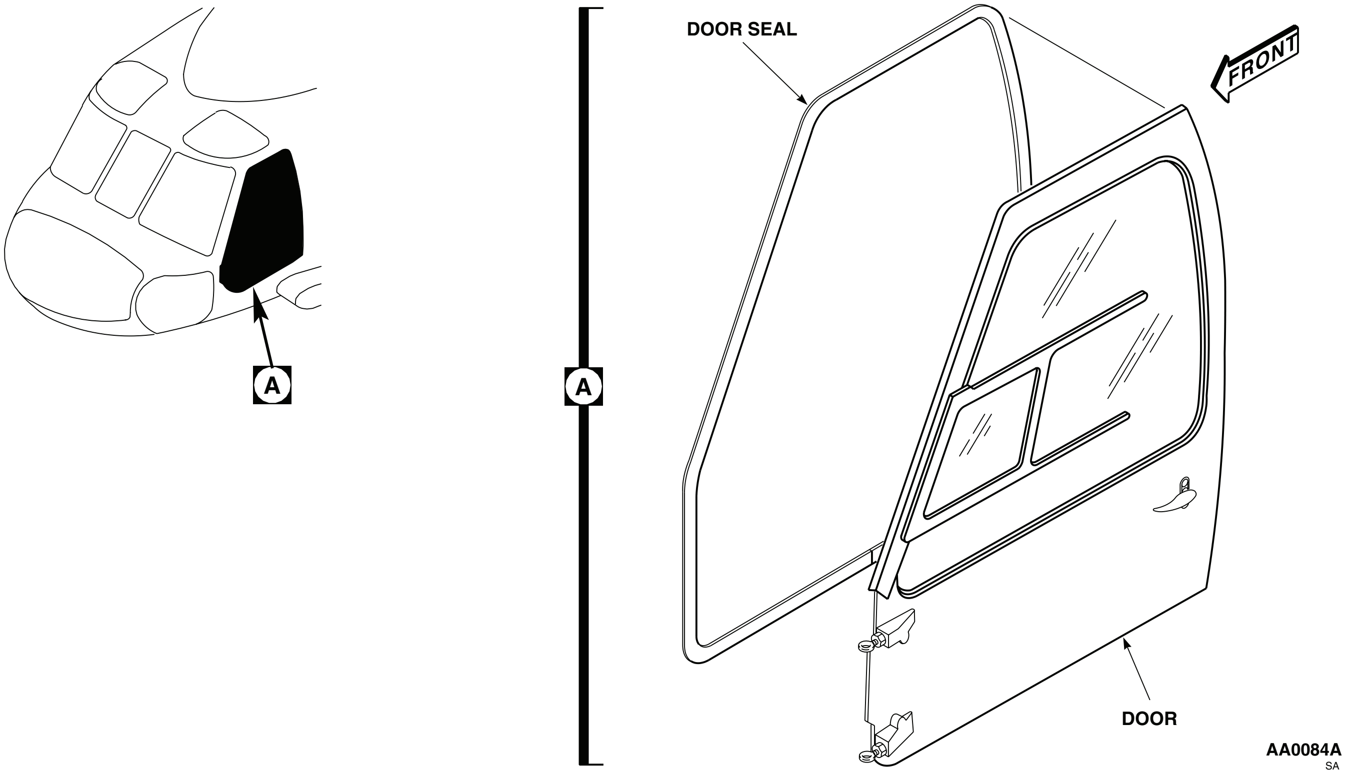


Figure 1. Replace Cockpit Door Seal.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean any adhesive residue from flange using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry using clean machinery towel.

INSTALL

1. Turn off all electrical power.
2. Roughen mating surface of replacement seal lightly, using abrasive cloth, Item 84, WP 1803 00, or scouring pad, Item 205, WP 1803 00.

INSTALL - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Wipe seal clean using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean machinery towel.
4. Apply a thin, even coat of adhesive, Item 47, WP 1803 00, to door seal flange and mating surface of replacement seal.
5. For alternate adhesive application:
   a. Apply adhesive, Item 36, WP 1803 00, to door seal flange and mating surface of seal.
   b. Allow adhesive to become visibly dry.
   c. Apply adhesive, Item 41, WP 1803 00, to flange and seal.
6. Starting at bottom, center of door, install seal to door flange area (Figure 1, Detail A). Do not stretch seal while installing, especially around curved parts of door.
7. Trim seal to fit and join at bottom of door.
8. Apply a fillet of sealing compound, Item 273, WP 1803 00, to seal joining area.
9. Install cockpit door (WP 0198 00).
10. Spray water on door and check for leaks. Adjust door as required (WP 0199 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR CONDUCTIVE SEALS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Set, GGG-D-751

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23

REMOVE

1. Turn off all electrical power.
2. To remove conductive seals from edge of door, remove screws (Figure 1, Detail A).
3. To remove conductive seal from door jamb, do this (Figure 1, Detail B):
   a. Drill out and remove rivets (TM 1-1500-204-23).
   b. Remove plates, conductive seals and angles from door jamb.

INSTALL

1. Turn off all electrical power.
2. To install conductive seals to edge of door do this (Figure 1, Detail B):
   a. Replace conductive seal with screws.
   b. Adjust seal by loosening screws, sliding seal.
3. Install angles, conductive seals, and plates on door jamb with rivets (Figure 1, Detail A).

INSTALL - Continued

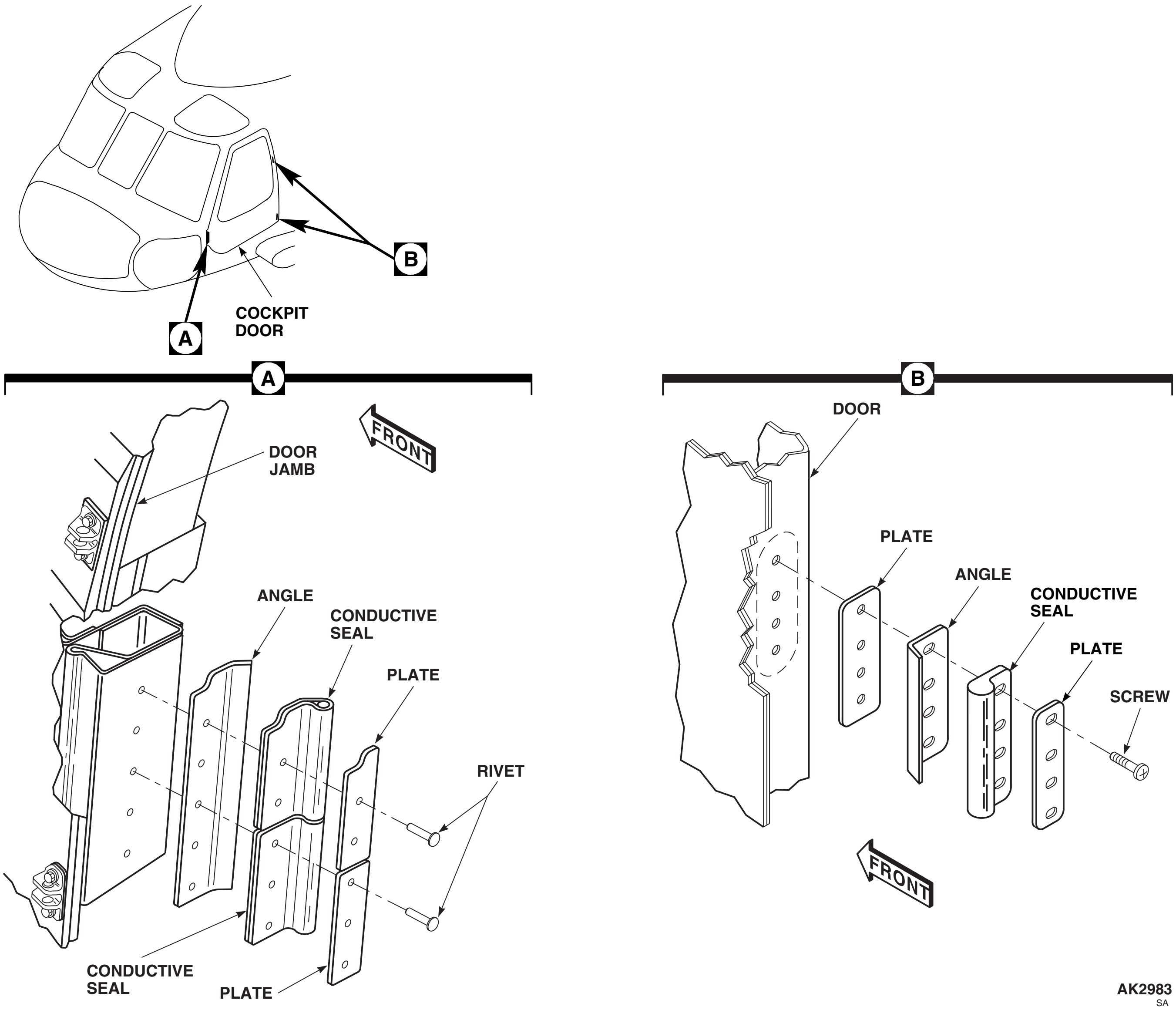


Figure 1. Replace Cockpit Door Conductive Seals.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR HINGE FITTINGS

This WP supersedes WP 0212 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit SC 5180-99-B01
Drift
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Reamer Set, GGG-R-00180
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00
Isopropyl, Alcohol Technical, Item 170, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0174 00
WP 1803 00

REMOVE

NOTE

- Remove one hinge fitting at a time so as not to disturb door adjustment.
- Removal and installation procedures are the same for upper and lower hinge fitting.

1. Turn off all electrical power.
2. Using assistant inside cockpit, make sure door is closed and latched.
3. Remove headed pin, washer, and cotter pin attaching hinge assembly to hinge fitting (Figure 1).
4. Using a nonmetallic scraper, remove sealing compound from hinge fitting.
5. **UH-60A 77-22714 - 87-26004 EH-60A** Remove bolts, washers, and nuts attaching hinge fitting and shim(s) to cockpit structure (Figure 1, Detail A). Keep shim(s) for reinstallation.
6. **UH-60A 87-26005 - SUBQ UH-60L** Remove bolts, washers, and nuts attaching hinge fitting and shim(s) to cockpit structure (Figure 1, Detail A). Keep shim(s) for reinstallation.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Using a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, clean sealing compound, Item 273, WP 1803 00, from cockpit structure, shim(s), serrated plate, hinge fitting, wedge-shaped washer, and bolts.

REMOVE - Continued

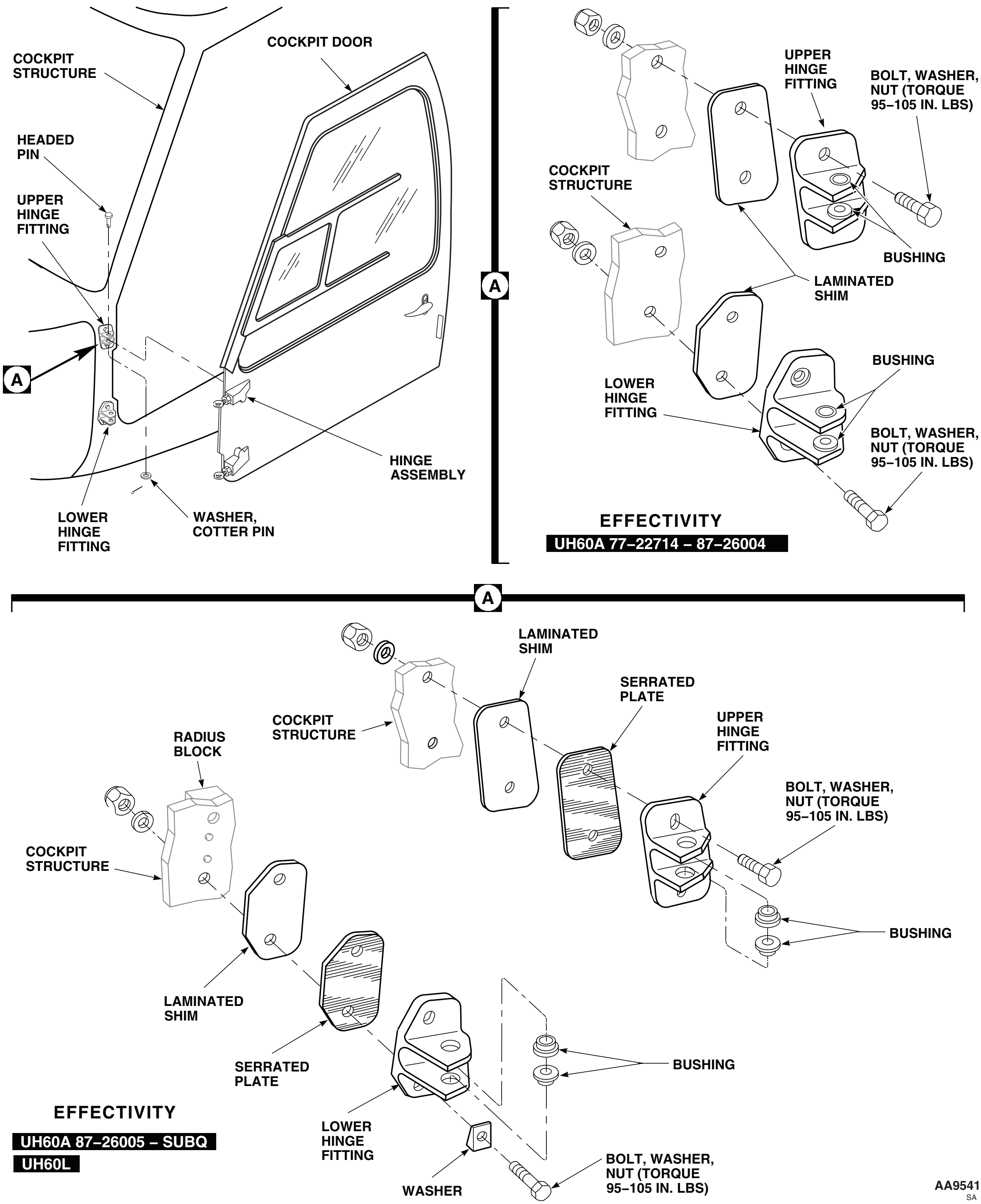


Figure 1. Replace Cockpit Door Hinge Components.

REMOVE - Continued

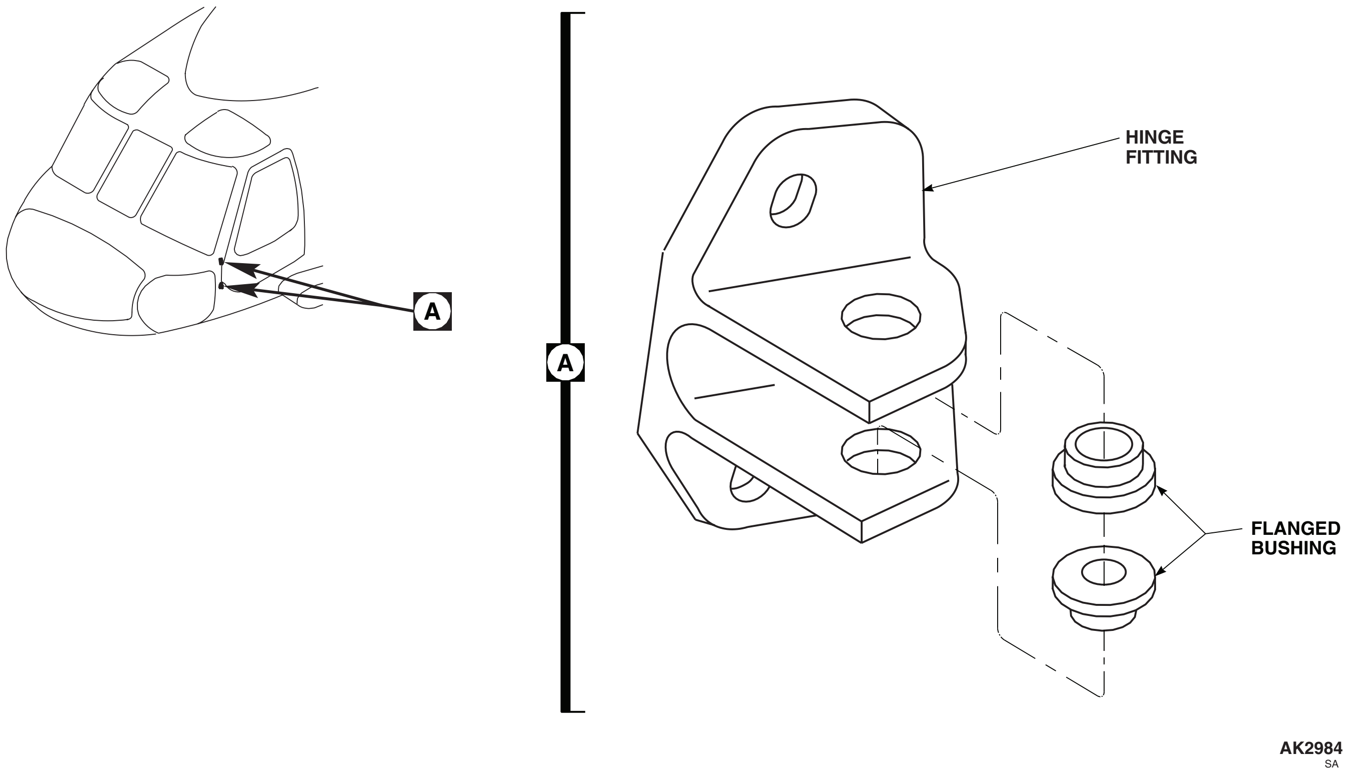


Figure 2. Replace Cockpit Door Hinge Fitting Bushings.

REPAIR

1. Turn off all electrical power.
2. Using an arbor press and drift, press bushing from hinge fitting (Figure 2, Detail A).

WARNING

Personal injury may occur if carbon dioxide or chilled parts are handled without gloves. Wear protective gloves when handling carbon dioxide or chilled parts.

3. Place new bushing in a container of carbon dioxide, Item 64, WP 1803 00, for about 15 minutes.
4. Remove bushing from carbon dioxide. Apply a thin coat of primer coating, Item 243, WP 1803 00, to bushing OD.
5. Using an arbor press and drift, press bushing into hinge fitting.

INSTALL

1. Turn off all electrical power.

NOTE

When applying sealing compound to shank of bolt, make sure no sealing compound gets on the threads.

2. Apply a light coat (approximately 0.005-inch) of sealing compound, Item 273, WP 1803 00, on both faces of shim(s) and boltshank.

INSTALL - Continued

3. Clean immediate surface with cloth moistened with technical isopropyl alcohol, Item 170, WP 1803 00. Allow to dry completely.
4. **UH-60A 77-22714 - 87-26004 EH-60A** Position shim(s) on hinge fitting and place hinge fitting on cockpit structure. Line up holes in fitting and structure. Install bolt in hinge fitting (Figure 1, Detail A).
5. **UH-60A 87-26005 - SUBQ UH-60L** Position shim(s) on serrated plate, place serrated plate on hinge fitting, line up holes, and place on cockpit structure (Figure 1, Detail A). Install bolt in hinge fitting with wedge-shaped washer under bolthead on bottom bolt of lower hinge fitting.
6. **QA** Install washers and nuts on bolts. **TORQUE NUT TO 95 - 105 INCH-POUNDS.**
7. Apply a continuous bead of sealing compound, Item 273, WP 1803 00, around hinge fitting.
8. **QA** Connect hinge assembly to hinge fitting with headed pin, washer, and cotter pin.
9. Open and close door to check for proper operation.
10. Perform cockpit door jettison check (WP 0174 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR HINGE ROD END

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Polyurethane Coating, Item 232, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0174 00
WP 0198 00
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove cockpit door (WP 0198 00).
3. Remove cotter pin, headed pin, and washer from hinge fitting and remove hinge (Figure 1, Detail A).
4. Measure distance between hole center of rod end and face of hinge (Figure 1, Detail B). Record dimension.
5. Loosen jamnut on rod end.
6. Remove rod end from hinge. Remove jamnut from rod end (Figure 1, Detail C).

REPAIR

1. Remove corrosion manually, using scouring pad, Item 205, WP 1803 00.
2. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.

CAUTION

Corrosion removal and treatment compound solution can damage rod end bearing. Do not allow solution to contact bearing.

3. Mask rod end bearing to prevent contact with corrosion removal and treatment compound solution, using pressure sensitive adhesive tape, Item 334, WP 1803 00.
4. Apply corrosion removal and treatment compound solution to corroded areas of rod end.
5. Rinse thoroughly with clean water.
6. Remove adhesive tape from bearing.
7. Touch up paint with two coats of epoxy primer coating kit, Item 120, WP 1803 00, and top coat of polyurethane coating, Item 232, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

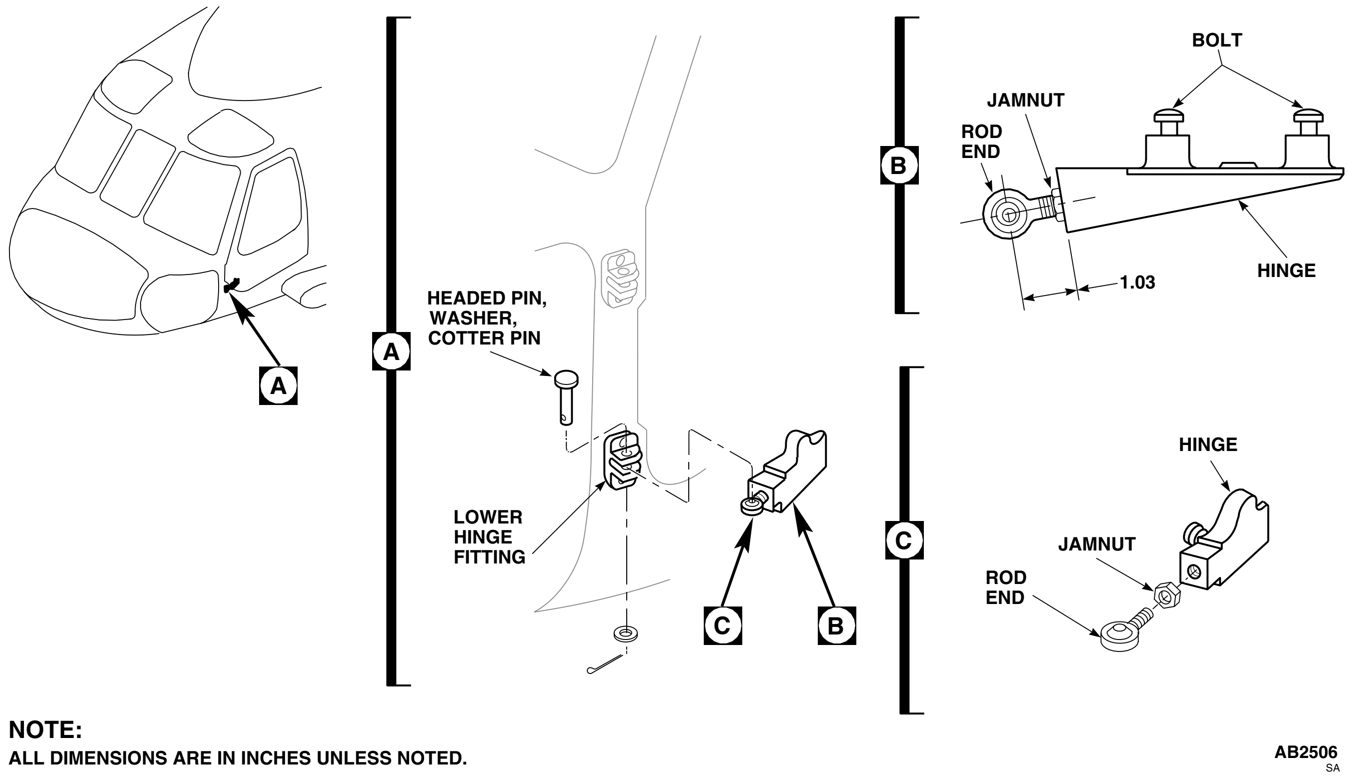


Figure 1. Replace Cockpit Door Hinge Rod End.

INSTALL

1. Turn off all electrical power.
2. Install rod end in hinge as follows (Figure 1, Detail C):
   a. Install jamnut on rod end.
   b. Install rod end in hinge.
   c. Adjust rod end length to dimension recorded. **IF DIMENSION WAS NOT RECORDED, SET ROD END LENGTH TO 1.03-INCH FROM CENTER OF ROD END TO FACE OF HINGE (FIGURE 1, DETAIL B).**
   d. **QA TIGHTEN JAMNUT 1/6 - 1/3 TURN BEYOND POINT WHERE A SHARP RISE IN TORQUE IS FELT.**
3. **QA** Install hinge on hinge fitting (Figure 1, Detail A). Fasten with headed pin and cotter pin.
4. Install cockpit door (WP 0198 00).
5. If duct and outlet assembly is installed, remove screws, bolts, and washers. Push down on duct to disengage hidden retaining latch and remove duct assembly from door.
6. Remove screws and washers, and remove cover from inside of door.
7. Perform cockpit door jettison check (WP 0174 00).
8. **QA** Install cover on door with screws and washers.

INSTALL - Continued

9. **QA** Install duct and outlet assembly by sliding up to engage hidden retaining latch and installing screws, bolt, and washers.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR LOCK

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0383 00

NOTE

Lock cylinders are available only in sets. When replacing cockpit door cylinder, replace lock cylinders on all doors.

REMOVE

1. Turn off all electrical power.
2. Unlock and open cockpit door.
3. Remove screws and washers attaching support to inside of door (WP 0383 00). Remove support.
4. Remove screws, washers, and nuts attaching handle to door. Remove handle (Figure 1, Detail A).
5. Remove snapring and washers fastening handle to escutcheon. Remove escutcheon from handle (Figure 1, Detail B).
6. Gently tap lock case with rubber mallet to remove it from handle.
7. Remove 2 pins holding shell in lock case. Remove shell from lock case.
8. Using thin bladed screwdriver, depress retaining ring into cylinder groove and remove cylinder from shell.

INSTALL

1. Turn off all electrical power.
2. **QA** Compress retaining ring, insert cylinder into shell, and snap into place (Figure 1, Detail B).
3. Insert shell into lock case and install 2 pins. Pins must be flush with surface of lock case.
4. Install lock case in handle.
5. Install washer on handle and handle in escutcheon.
6. **QA** Install small washers and snapring on handle.
7. **QA** Install handle on door with screws, washers, and nuts.
8. Make sure area is clean and free of foreign material before closing up.
9. Place support on door and install attaching screws, and washers (Figure 1, Detail A).
10. Close and open door a few times to check operation of handle.

INSTALL - Continued

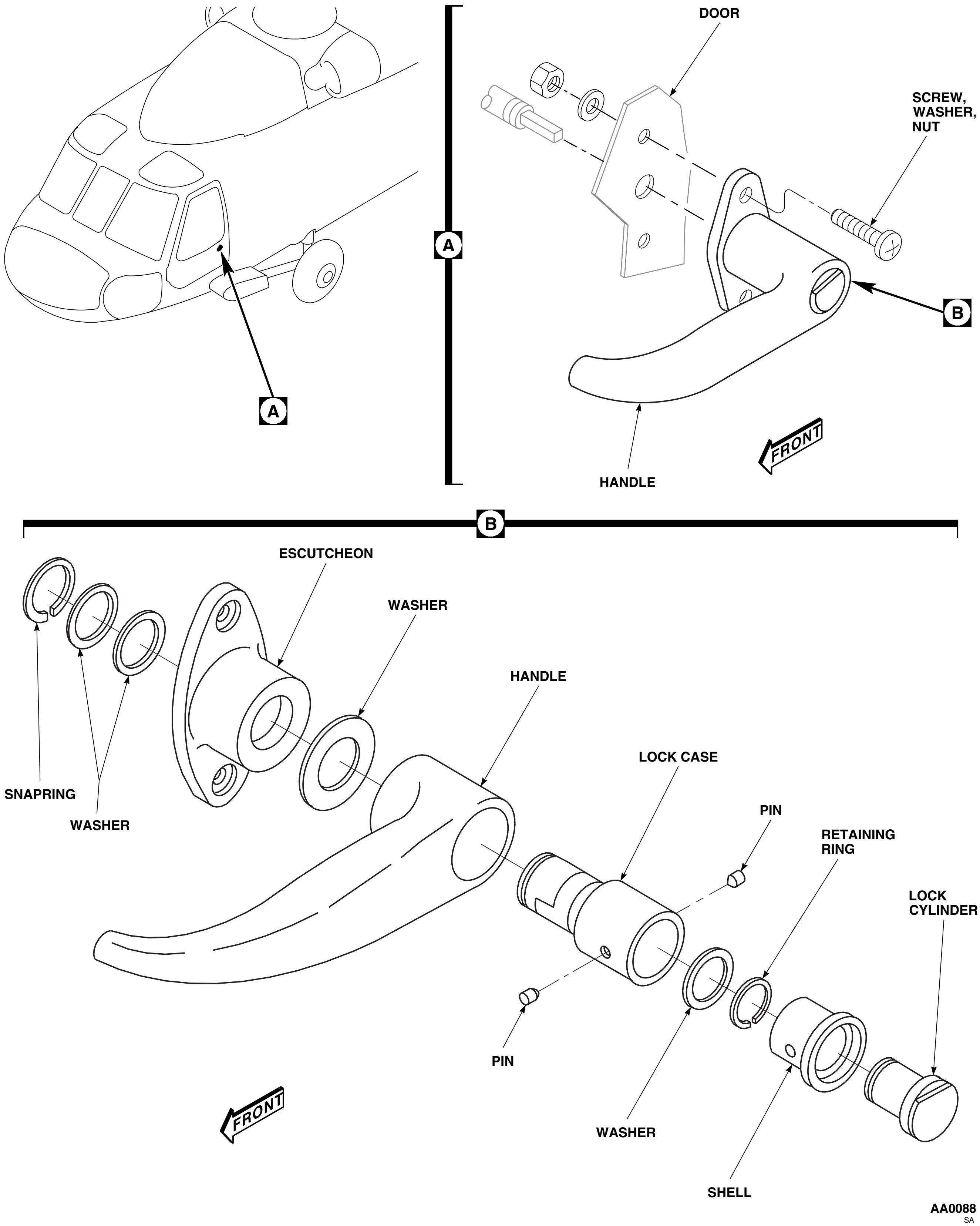


Figure 1. Replace Cockpit Door Lock.

INSTALL - Continued

11. Close and lock cockpit door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GLARE SHIELD

This WP supersedes WP 0215 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Abrasive, Item 75, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Strap, Tiedown, Electrical, Item 307, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0379 00
WP 0790 00
WP 1589 00
WP 1703 00
WP 1803 00

REPLACE GLARE SHIELD

Remove

1. Turn off all electrical power.
2. Disconnect pilot's and copilot's master warning panel electrical connector, P130 pilot, P131 copilot (Figure 1, Detail A).
3. **MWO 50-82** Disconnect pilot's forward air bag module electrical connector, P6 and copilot's forward air bag module electrical connector, P8, (Figure 1, Detail A).
4. Carefully unfasten and remove glare shield from instrument panel enough to disconnect main glare shield electrical connector. Disconnect connector, P162, and remove shield from helicopter.
5. Remove screws and washers securing instrument light assemblies to instrument panel.
6. Remove pilot's and copilot's master warning panel (WP 0790 00).
7. Cut tiedown straps attaching wiring harness to glare shield. Remove harness.
8. Remove forward air bag modules (WP 1589 00).

Install

1. Turn off all electrical power.
2. **QA** Attach wiring harness to glare shield using electrical tiedown straps, Item 307, WP 1803 00.
3. **QA** Install light assemblies on glare shield with screws and washers. Make sure ground wire is installed under screwhead (Figure 1, Detail A).
4. Install pilot's and copilot's master warning panel on glare shield with screws and washers (WP 0790 00).
5. Inspect and treat electrical connector (WP 1703 00).

REPLACE GLARE SHIELD - Continued

NOTE

Make sure master warning panel electrical connectors are hanging over top of instrument panel before installing glare shield in place.

6. **QA** Place glare shield in cockpit in position and connect main glare shield electrical connector, P162.
7. **QA** Attach glare shield to instrument panel with screws and washers.
8. Inspect and treat electrical connectors (WP 1703 00).
9. **QA** Connect electrical connector, P130, to pilot's and, P131, to copilot's master warning panel.
10. **MWO 50-82** Install forward air bag modules (WP 1589 00).
11. **MWO 50-82** Inspect and treat electrical connectors (WP 1703 00).
12. **QA** **MWO 50-82** Connect pilot's forward air bag module electrical connector, P6 and copilot's forward air bag module electrical connector, P8, (Figure 1, Detail A).

REPAIR STRUCTURAL DAMAGE

Repair structure of glare shield using procedures in WP 0379 00.

REPLACE FOAM PADDING AND RUBBER CHANNEL

NOTE

Replace rubber channel using same procedure as in replacing foam padding.

1. Remove loose or damaged foam padding using a nonmetallic scraper (Figure 2).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean remaining adhesive from front edge of glare shield using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Wipe edge of glare shield dry with clean machinery wiping towel, Item 344, WP 1803 00.

NOTE

If required, padding may be spliced in transition areas, joints to be bonded with adhesive, Item 14, WP 1803 00.

4. Lightly scuff inside surfaces of foam pad using abrasive cloth, Item 75, WP 1803 00.
5. Apply an even coat of adhesive, Item 14, WP 1803 00, to inside surfaces of foam padding and front edge of glare shield.
6. Allow adhesive to air-dry for 5 to 10 minutes until tacky.
7. Install foam padding to front edge of glare shield. Make sure surfaces of foam padding and glare shield are in direct contact.

REPLACE FOAM PADDING AND RUBBER CHANNEL - Continued

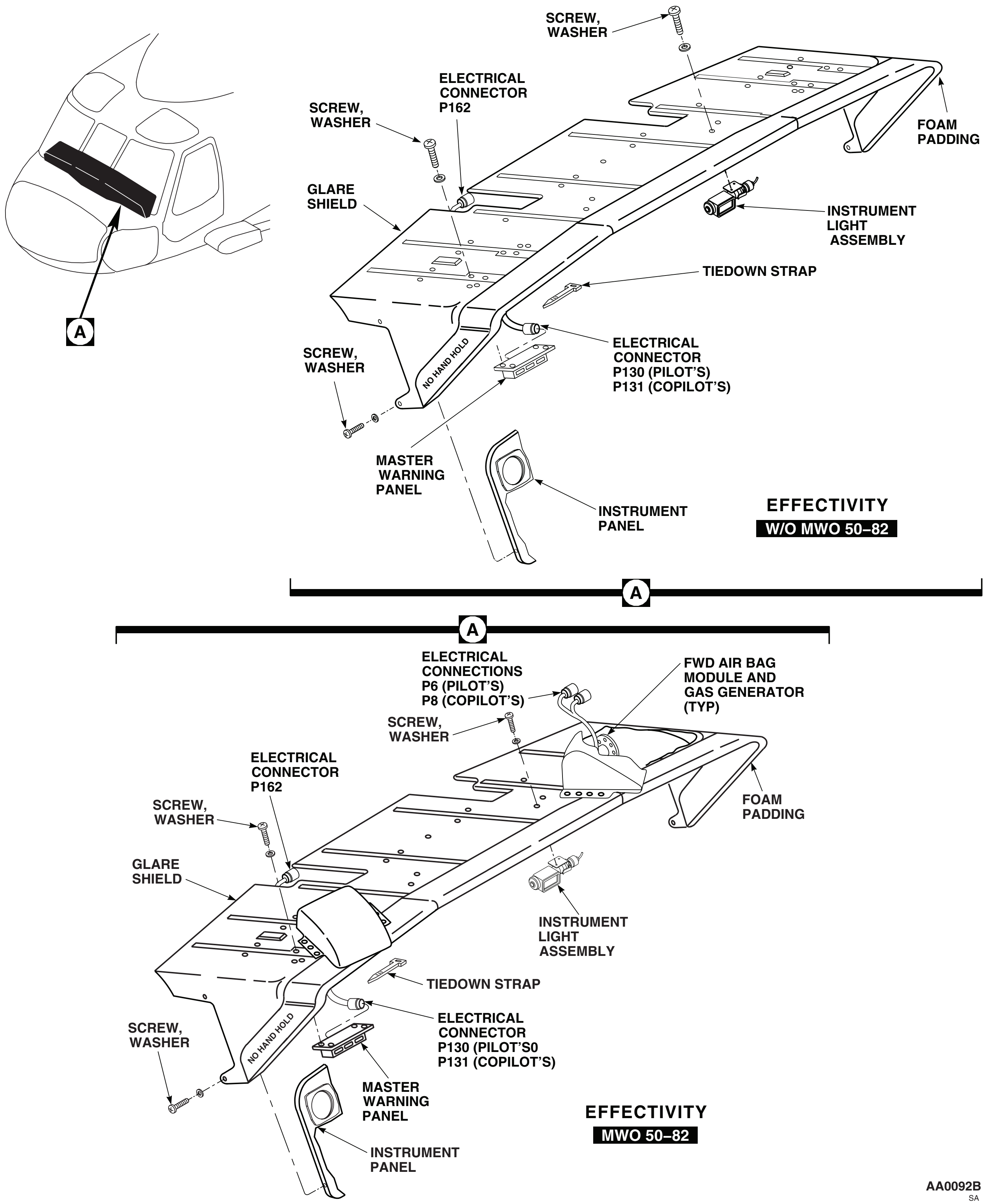


Figure 1. Replace Glare Shield.

REPLACE FOAM PADDING AND RUBBER CHANNEL - Continued

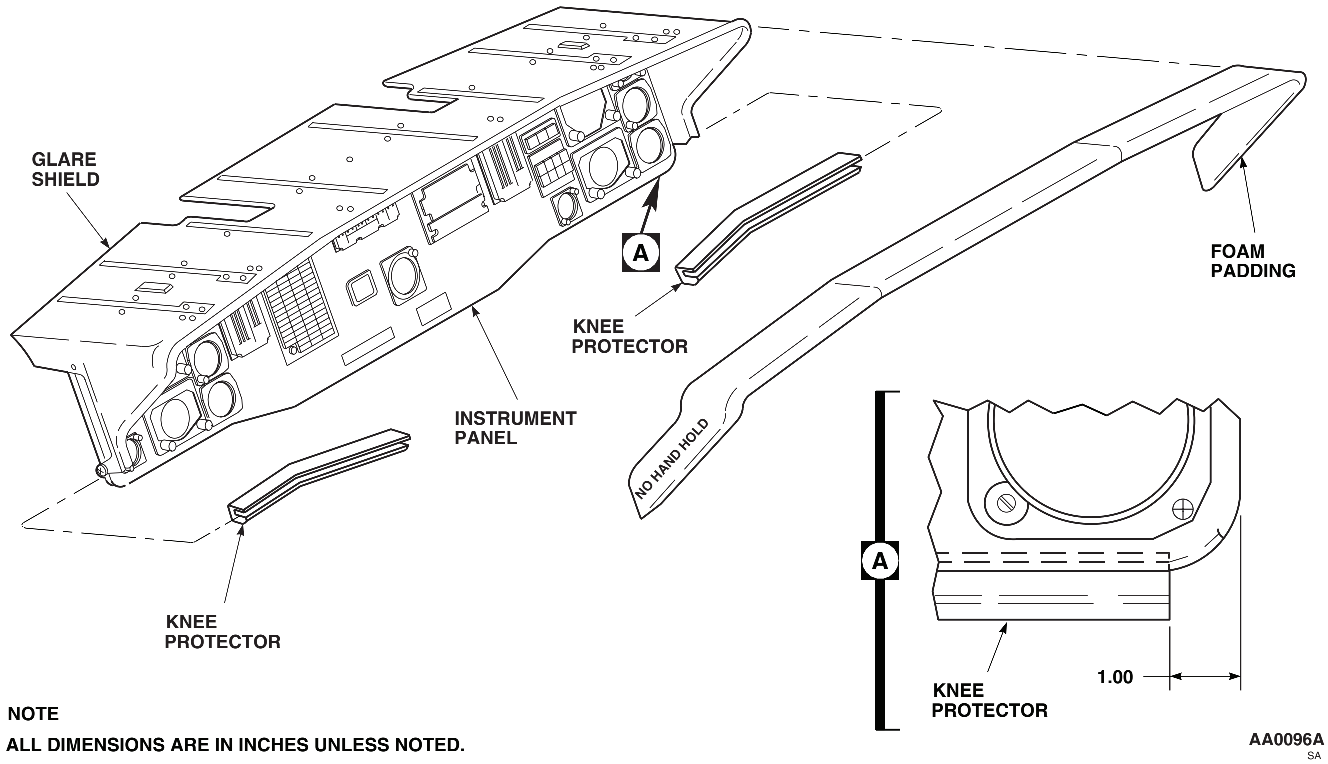


Figure 2. Repair Glare Shield.

8. Apply a bead of sealing compound, Item 273, WP 1803 00, to where top and side edges of foam padding contact glare shield.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

INSTRUMENT PANEL

This WP supersedes WP 0216 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Digital Low-Resistance Ohmmeter, DLRO24700
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Corrosion Resistant Coating, Item 111, WP 1803 00

Personnel Required
Airframe Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 11-1520-237-23
TM 11-1520-249-23
TM 55-1500-345-23
WP 0215 00
WP 0378 00
WP 0381 00
WP 0767 00
WP 0770 00
WP 0771 00
WP 0772 00
WP 0776 00
WP 0782 00
WP 0785 00
WP 0794 00
WP 0799 00
WP 0800 00
WP 0923 00
WP 0934 00
WP 0935 00
WP 1221 00
WP 1223 00
WP 1240 00
WP 1448 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove glare shield (WP 0215 00).
3. **UH-60A 77-22714 - 78-22968** Remove screws and washers holding right and left kick panels to bottom of instrument panel and clips attached to structure (Figure 1, Sheet 2, Detail C). Remove kick panels.
4. Disconnect these instruments and remove from instrument panel:
   a. Airspeed Indicators (WP 0771 00).
   b. Barometric Altimeters (WP 0772 00).
   c. Vertical Speed Indicators (WP 0770 00).
   d. Pilot's/Copilot's Display Units (WP 0782 00).
   e. Clocks (WP 0794 00).
   f. Caution/Advisory Panel (WP 0785 00).
   g. Central Display Unit (WP 0776 00).
   h. Blade Deice Control Panel (WP 1223 00).
   i. Blade Deice Test Panel (WP 1240 00).

REMOVE - Continued

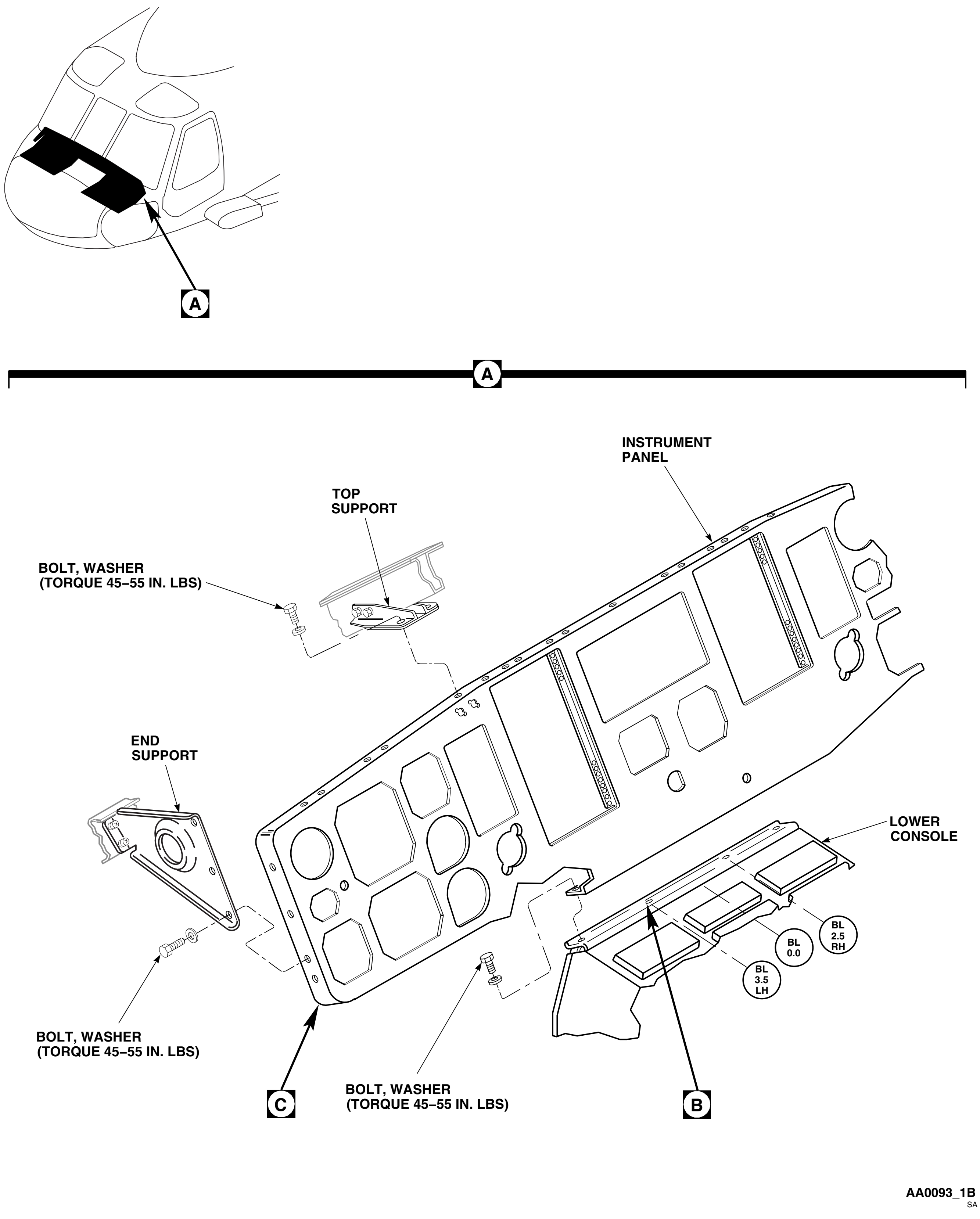


Figure 1. Replace Instrument Panel. (Sheet 1 of 2)

REMOVE - Continued

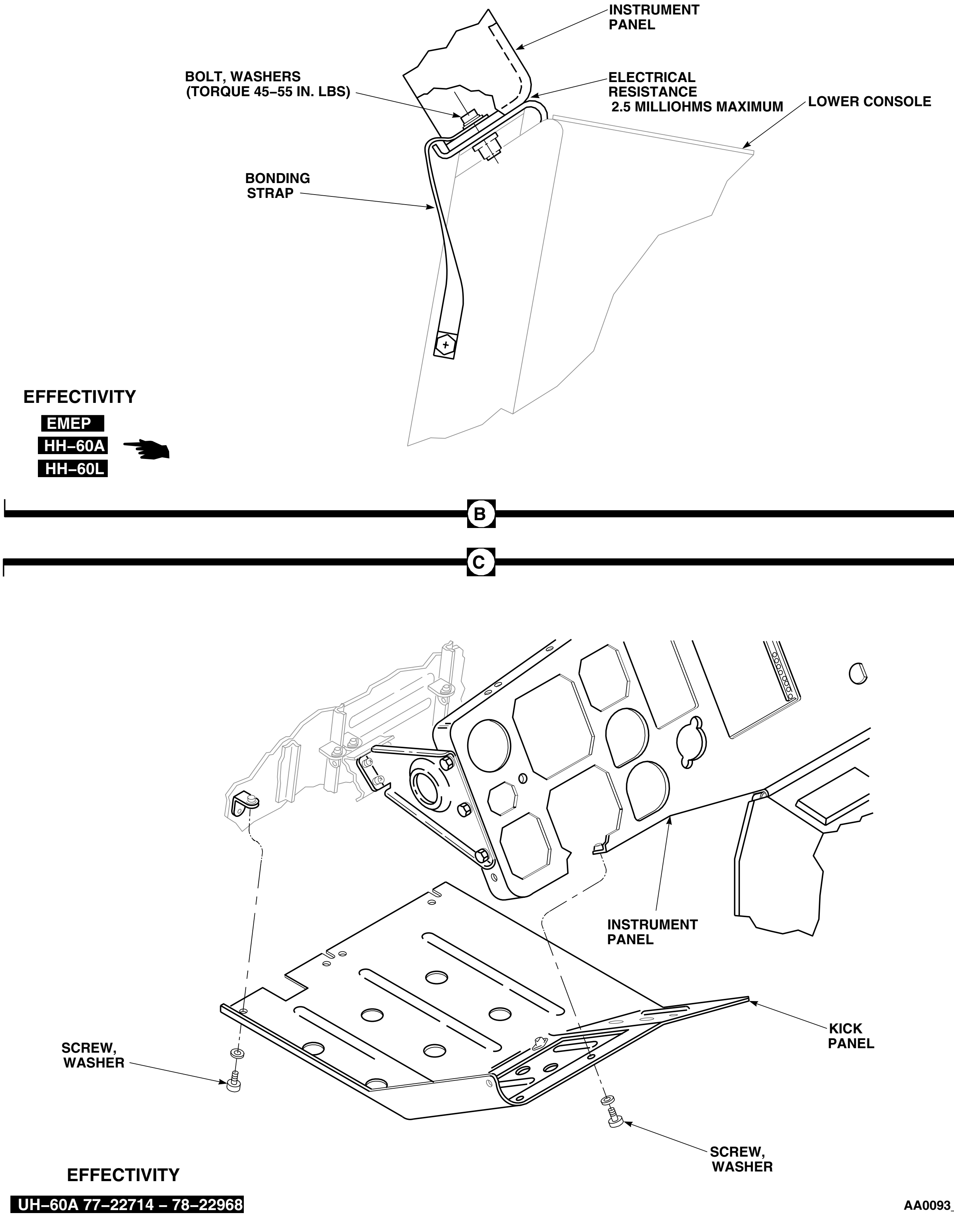


Figure 1. Replace Instrument Panel. (Sheet 2 of 2)

REMOVE - Continued

j. Icing Rate Meter (WP 1221 00).

k. Keylock Ignition Switch (WP 0923 00).

l. **EH-60A** Flares Switch (TM 11-1520-249-23).

m. **EH-60A** Bearing Distance Heading Indicator (WP 0767 00).

n. **HH-60A HH-60L** Indicator Light Switch (WP 0800 00).

o. **HH-60A HH-60L** Lower Console Auxiliary Dimming Control (WP 0935 00).

p. **HH-60A HH-60L** Master Warning and Radar Altimeter Dimming Control (WP 0934 00).

q. **HH-60A HH-60L** Multifunction Display (WP 0799 00).

r. **MWO 50-74** Auxiliary Fuel Management Panel (WP 1448 00).

5. Disconnect these instruments and remove from instrument panel (TM 11-1520-237-23):

a. Stabilator Indicator.

b. HSI/VSI Mode Selectors.

c. Vertical Situation Indicators.

d. Horizontal Situation Indicators.

e. Radar Altimeter Receiver/Transmitter Indicator.

f. Radar Signals Indicator.

g. Height Indicator (Radar Altimeter).

h. **EH-60A** Systems Select Panel (TM 11-1520-249-23).

i. **EH-60A** ECM Antenna Switch (TM 11-1520-249-23).

j. **EH-60A** ASE Status Panel (TM 11-1520-249-23).

k. **HH-60A HH-60L** Stormscope.

l. **HH-60A HH-60L** Remote Display Unit.

m. **HH-60A HH-60L** BRG 1/DIST Switch.

6. Remove bolts and washers holding instrument panel to end supports and top supports (Figure 1, Sheet 1, Detail A).

7. **EMEP HH-60A HH-60L** Remove bolts, washers, and bonding straps, if installed, holding instrument panel to front of lower console (Figure 1, Sheet 2, Detail B).

8. Remove instrument panel.

INSTALL

1. Turn off all electrical power.

2. **QA** **W/O EMEP** Position instrument panel and install bolts and washers securing it to end supports, top supports, and lower console (Figure 1, Sheet 1, Detail A). **TORQUE BOLTS TO 45 - 55 INCH-POUNDS.**

3. **EMEP HH-60A HH-60L** Install instrument panel as follows:

a. Inspect chemical conversion coated surfaces where bonding straps mate with instrument panel (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

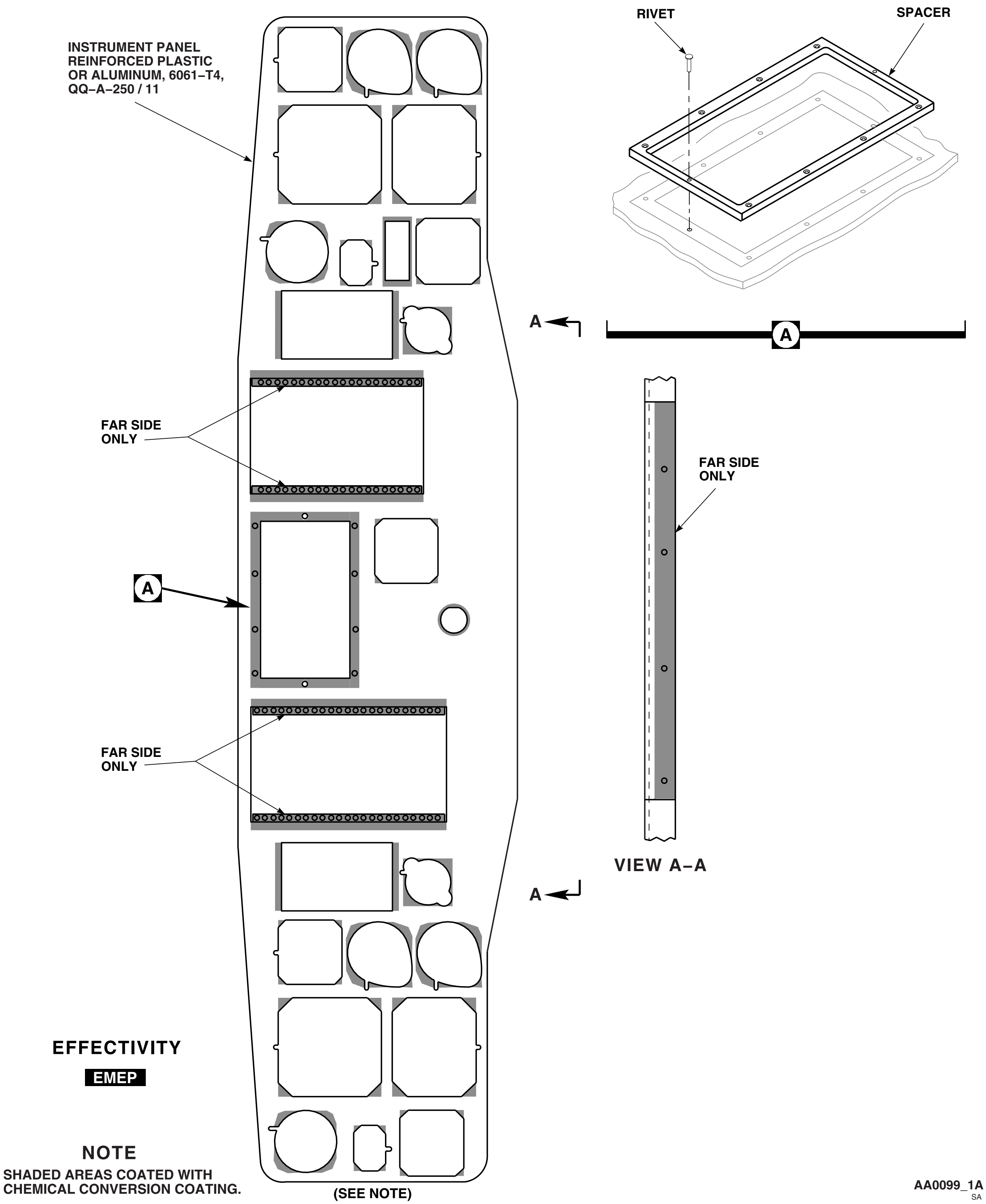


Figure 2. Repair Instrument Panel. (Sheet 1 of 2)

INSTALL - Continued

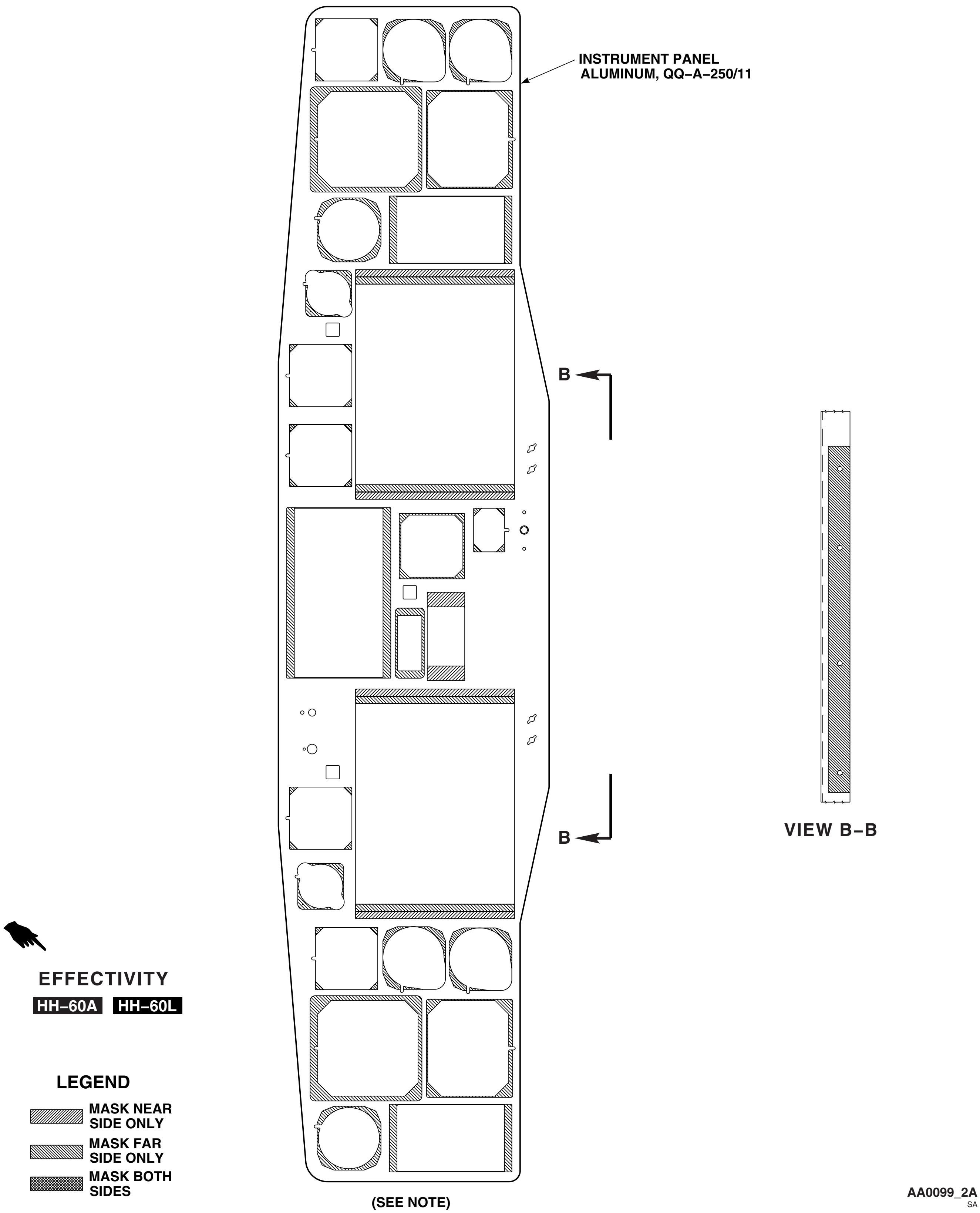


Figure 2. Repair Instrument Panel. (Sheet 2 of 2)

INSTALL - Continued

b. Position instrument panel and install bolts, washers, and bonding straps at RBL 1.5 and LBL 4.0 (Figure 1, Sheet 2, Detail B). Do not tighten bolts now.

c. **QA** Install remaining bolts and washers securing instrument panel to end supports, top supports, and lower console. **TORQUE ALL BOLTS TO 45 - 55 INCH-POUNDS.**

NOTE

Resistance measurements must be taken at least three times.

d. Using digital low-resistance ohmmeter, check resistance reading from instrument panel to lower console right/left end supports, and top support using digital low-resistance ohmmeter. **RESISTANCE SHALL NOT BE MORE THAN 5.0 MILLIOHMS**.

e. If resistance is greater than 5.0 milliohms, remove instrument panel and replace chemical conversion coating on instrument panel and lower console (WP 0381 00 and TM 55-1500-345-23).

4. Install these instruments in instrument panel:

a. Airspeed Indicators (WP 0771 00).

b. Barometric Altimeters (WP 0772 00).

c. Vertical Speed Indicators (WP 0770 00).

d. Pilot's/Copilot's Display Units (WP 0782 00).

e. Clocks (WP 0794 00).

f. Caution/Advisory Panel (WP 0785 00).

g. Central Display Unit (WP 0776 00).

h. Blade Deice Control Panel (WP 1223 00).

i. Blade Deice Test Panel (WP 1240 00).

j. Icing Rate Meter (WP 1221 00).

k. Keylock Ignition Switch (WP 0923 00).

l. **EH-60A** Flares Switch (TM 11-1520-249-23).

m. **EH-60A** Bearing Distance Heading Indicator (WP 0767 00).

n. **HH-60A HH-60L** Indicator Light Switch (WP 0800 00).

o. **HH-60A HH-60L** Lower Console Auxiliary Dimming Control (WP 0935 00).

p. **HH-60A HH-60L** Master Warning and Radar Altimeter Dimming Control (WP 0934 00).

q. **HH-60A HH-60L** Multifunction Display (WP 0799 00).

r. **MWO 50-74** Auxiliary Fuel Management Panel (WP 1448 00).

5. Install these instruments in instrument panel (TM 11-1520-237-23):

a. Stabilator Indicator.

b. HSI/VSI Mode Selectors.

c. Vertical Situation Indicators.

d. Horizontal Situation Indicators.

e. Radar Altimeter Receiver/Transmitter Indicator.

INSTALL - Continued

f. Radar Signals Indicator.

g. Height Indicator (Radar Altimeter).

h. **EH-60A** Systems Select Panel (TM 11-1520-249-23).

i. **EH-60A** ECM Antenna Switch (TM 11-1520-249-23).

j. **EH-60A** ASE Status Panel (TM 11-1520-249-23).

k. **HH-60A HH-60L** Stormscope.

l. **HH-60A HH-60L** Remote Display Unit.

m. **HH-60A HH-60L** BRG 1/DIST Switch.

6. **QA** **UH-60A 77-22714 - 78-22968** Install right and left kick panels, with screws and washers, on bottom of instrument panel and clips attached to structure (Figure 1, Sheet 2, Detail C).

7. Install glare shield (WP 0215 00).

8. **EMEP HH-60A HH-60L** Using digital low-resistance ohmmeter, do EME operational check between bottom edge of instrument panel and lower console (Figure 1, Sheet 2, Detail B). **RESISTANCE MUST NOT EXCEED 2.5 MILLIOHMS.**

9. **UH-60A UH-60L** Do operational checks (TM 11-1520-237-23).

10. **EH-60A** Do operational checks (TM 11-1520-249-23).

REPAIR

1. Repair structure of instrument panel (WP 0378 00).
2. Inspect areas with chemical conversion coating (Figure 2, Sheet 1 and Figure 2, Sheet 2).
3. Apply corrosion resistant coating, Item 111, WP 1803 00, as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE SECTION AVIONIC COMPARTMENT COVER

This WP supersedes WP 0217 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Fastener Tape, Hook, Item 124, WP 1803 00
Fastener Tape, Pile, Item 125, WP 1803 00
Towel, Machinery Wiping Item 344, WP 1803 00

Personnel Required

Airframe Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Pull avionic compartment cover from bottom, tip it down from top, and remove it from hook fastener tape on structure (Figure 1, Detail A).

REPAIR

1. Remove rivets and remove hinge and door from avionic compartment cover (TM 1-1500-204-23) (Figure 1, Detail A).
2. Remove rivets and remove hinge from door.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Pull damaged pile fastener tape from cover or door. Using a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, wipe surface clean of remaining adhesive.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Pull damaged hook fastener tape from structure. Using a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, wipe surface clean of remaining adhesive.

REPAIR - Continued

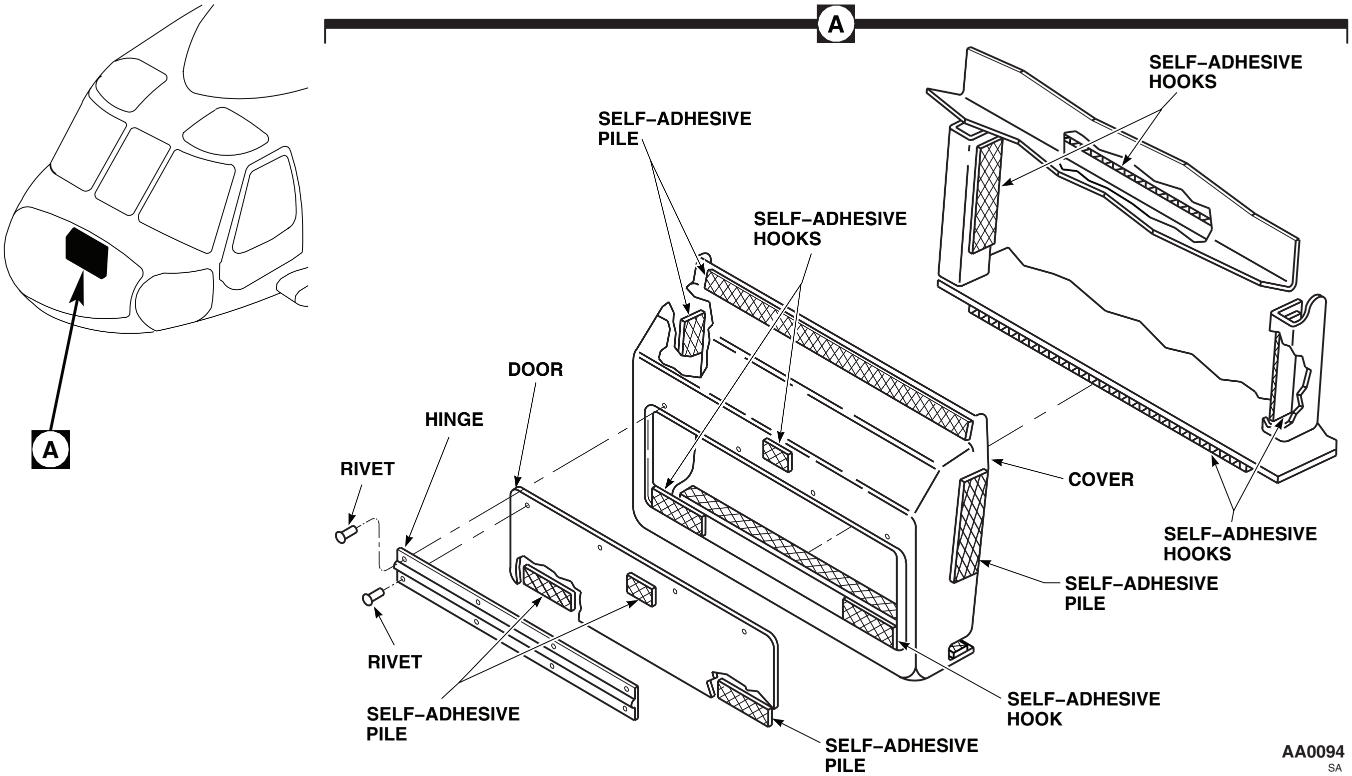


Figure 1. Replace Nose Section Avionic Compartment Cover.

5. Install hook fastener tape, Item 124, WP 1803 00, to structure and front of cover.
6. Install pile fastener tape, Item 125, WP 1803 00, to sides and back ends of cover.
7. Install pile fastener tape, Item 125, WP 1803 00, to door.
8. Rivet hinge to door and cover (TM 1-1500-204-23).

INSTALL

1. Turn off all electrical power.
2. **QA** Install cover in avionics compartment. Put top of cover behind upper structure flange and push in bottom and sides. Engage self-adhesive hook and pile (Figure 1, Detail A).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE SECTION THERMAL BARRIERS

This WP supersedes WP 0218 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 7, WP 1803 00
Adhesive, Item 31, WP 1803 00
Fastener Tape, Hook, Item 124, WP 1803 00
Fastener Tape, Pile, Item 125, WP 1803 00
Primer, Adhesive, Item 239, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Airframe Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0379 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and washers, and pull thermal barriers from supports and hook fastener tape (Figure 1, Detail A or Detail B).

REPAIR

1. Repair thermal barrier structure (WP 0379 00).
2. Pull damaged pile fastener tape from barrier and hook fastener tape from structure (Figure 1, Detail A or Detail B).
3. Pull damaged grommet from barrier.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Using a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, clean remaining adhesive from barrier and structure.
5. Coat mating surface of grommet strip and top of barrier with adhesive, Item 7, WP 1803 00, and bond grommet to barrier. Trim grommet to edge of barrier (TM 1-1500-204-23).
6. Apply a light coat of adhesive primer, Item 239, WP 1803 00, to thermal barrier and structure. Allow primer to dry for 10 minutes or until tacky.

REPAIR - Continued

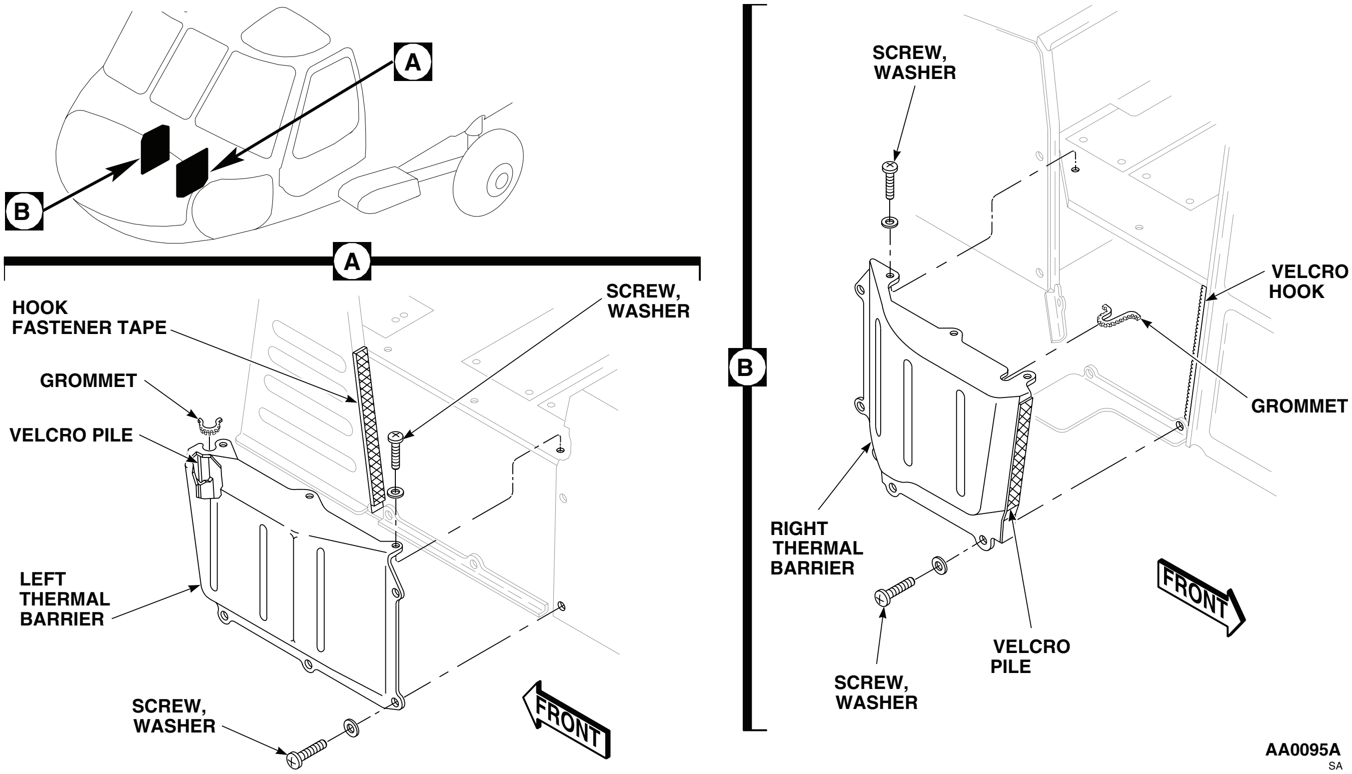


Figure 1. Replace Thermal Barriers.

7. Apply a light coat of adhesive, Item 31, WP 1803 00, to thermal barrier, structure, pile fastener tape, Item 125, WP 1803 00, and hook fastener tape, Item 124, WP 1803 00. Allow adhesive to become tacky.
8. Bond pile fastener tape, Item 125, WP 1803 00, to barrier and hook fastener tape, Item 124, WP 1803 00, to structure (TM 1-1500-204-23).

INSTALL

1. Turn off all electrical power.
2. **QA** Place thermal barrier against hook fastener tape, Item 124, WP 1803 00, and supports, guide wire bundle through grommet, and secure barrier with screws and washers (Figure 1, Detail A or Detail B).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE VIBRATION ABSORBER (SPRING TYPE)

This WP supersedes WP 0219 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Dial Indicator, MIL-I-18422
Gloves
Goggles/Face Shield
Micrometer, 0 to 1 Inch
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Bolt, NAS1305-24DH
Bolt, NAS1305-27D
Bolt, NAS1305-68D
Cloth, Abrasive, Item 75, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Grease, General Purpose, Item 136, WP 1803 00
Primer, Sealing Compound, Item 247, WP 1803 00
Sealing Compound, Item 282, WP 1803 00
Tags
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-265-23
TM 1-6625-724-13&P
TM 11-1520-237-23
TM 55-1500-345-23
WP 0356 00
WP 0381 00
WP 0384 00
WP 1803 00

REPLACE NOSE VIBRATION ABSORBER (SPRING TYPE)

Remove

1. Turn off all electrical power.
2. Remove glide slope antenna and support panel (TM 11-1520-237-23).
3. Remove nuts and washers from bolts attaching left fittings of absorber to structural fittings (Figure 1, Detail A). Do not remove bolts now.
4. Remove nut and washer from bolt attaching right lower fitting of absorber to structural fitting. Do not remove bolt now.

WARNING

Weight of vibration absorber (about 75 pounds) can cause personal injury. Do not remove attaching hardware unless absorber is supported.

5. With assistant supporting absorber, remove bolts attaching left fittings of absorber to structural fittings.
6. Lower left side of absorber onto structure.
7. With assistant supporting absorber, remove bolt supporting right lower fitting to structural fitting.
8. Lower right side, but still support absorber.

REPLACE NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued

9. Raise left side and move to left until right side clears structure. Continue to lower right side until absorber is removed from structure.

Install

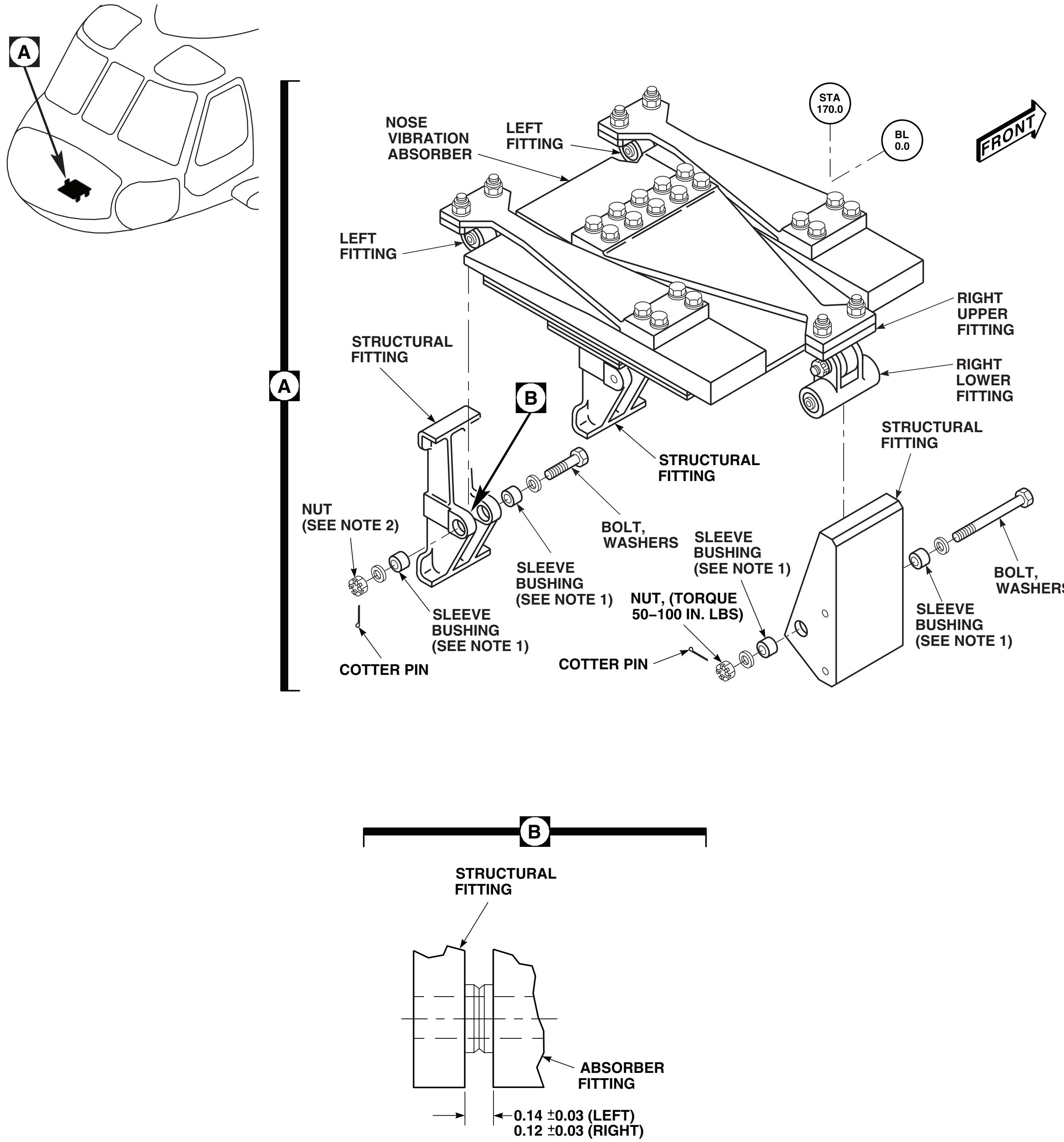
1. Turn off all electrical power.

NOTE

When installing a new vibration absorber, use same amount of weights as old absorber.

2. With aid of assistant, lift vibration absorber into nose structure and raise into place, keeping left side of absorber higher than right side.
3. Move absorber to left as far as possible to give clearance on right side.
4. **UH-60A 83-23924 - SUBQ UH-60L EH-60A** Raise right side so that right lower fitting engages structural fitting. Install sleeve bushings lubricated with general purpose grease, Item 136, WP 1803 00. Install bolt and washer with bolthead toward front (Figure 1, Detail A).
5. **UH-60A 83-23924 - SUBQ UH-60L EH-60A** Engage left absorber fittings with structural fittings, install sleeve bushings lubricated with general purpose grease, Item 136, WP 1803 00. Install bolts and washers with boltheads toward front.
6. Install washers and nuts on right and left fitting bolts.
7. Torque structural fitting bolts as follows:
   a. **TORQUE FITTING BOLT TO 80 - 95 INCH-POUNDS.** Loosen torque back to zero inch-pounds.
   b. **QA TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin.
8. **UH-60A 83-23924 - SUBQ UH-60L EH-60A** Check spring fitting alignment with left rear and right structural fittings as follows:
   a. Rotate left front fitting bolt two full turns.
   b. Rotate left rear and right fitting bolts two full turns.
   c. Measure gap between spring fitting face and structural fitting face (Figure 1, Detail A).
   d. **GAPS MUST BE 0.09 to 0.15-INCH FOR RIGHT FITTING AND 0.11 to 0.17-INCH FOR LEFT FITTING.**
   e. Adjust clearance as required (step 10. and step 11.).
9. **UH-60A 83-23924 - SUBQ UH-60L EH-60A** Check for absorber binding as follows:
   a. Grasp right side of absorber body. Rock absorber back and forth. Make sure the sleeve bushings slide from side to side in structural fitting.
   b. If right sleeve bushings do not slide in structural fitting, check bolt torque and/or slip fit of sleeve bushings.
   c. Check breakout torque on the left rear and right structural fitting bolt. **BREAKOUT TORQUE SHALL NOT EXCEED 10 INCH-POUNDS THROUGH FULL 360 DEGREE TURN.** If breakout torque exceeds 10 inch-pounds, check bolt torque and slip-fit of sleeve bushings.
   d. Check breakout torque on left forward structural fitting bolt. **BREAKOUT TORQUE SHALL EXCEED 50 INCH-POUNDS.** If breakout torque is less than 50 inch-pounds, check buildup and bolt torque.

REPLACE NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued



NOTES:

1. **UH60A 83–23924 – SUBQ** **UH60L** **EH60A**
2. TORQUE LEFT FRONT AND REAR FITTING NUTS TO 50–100 IN. LBS.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Nose Vibration Absorber (Spring Type).

REPLACE NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued

NOTE

Front side spring fitting locates against flanged bushing in left front structural fitting, no adjustment is required.

10. Adjust center spring fitting in right structural fitting, as follows:
    a. Make sure vibration absorber is installed properly.
    b. Loosen bolt attaching fitting to spring.
    c. **ADJUST FITTING TO GAP OF 0.09 TO 0.15-INCH EACH SIDE (Figure 1, Detail B).**
    d. **QA TORQUE NUTS TO 171 - 190 INCH-POUNDS.**
    e. Rotate right structural fitting bolt 2 full turns.
    f. Recheck gaps between fitting faces. If limits are not met, repeat step b. through step f.
11. Adjust rear side spring fitting in left rear structural fitting, as follows:
    a. Make sure vibration absorber is installed properly.
    b. Loosen bolts attaching fitting to spring.
    c. **ADJUST FITTING TO GAP OF 0.11 TO 0.17-INCH EACH SIDE (Figure 1, Detail B).**
    d. **QA TORQUE NUTS TO 171 - 190 INCH-POUNDS.**
    e. Rotate left rear structural fitting bolt 2 full turns.
    f. Recheck gaps between fitting faces. If limits are not met, repeat step b. through step f.
12. Install glide slope antenna and support panel (TM 11-1520-237-23).
13. Perform vibration check of absorber (TM 1-6625-724-13&P).

REPAIR NOSE VIBRATION ABSORBER (SPRING TYPE)

Disassemble

1. Remove right upper and lower fittings as follows:
    a. Remove bolt, washer, and nut attaching upper fitting to lower fitting (Figure 2, Sheet 1). Keep hardware for installation.
    b. Remove bolts, washers, and nuts attaching upper fitting to center spring. Keep hardware for installation of replacement fitting.
2. Remove bolts and nuts attaching left fittings to side spring. Keep hardware for installation of replacement fitting.

NOTE

Tag and record each weight for reinstallation.

3. Remove bolts, washers and nuts attaching small weights to body weight.
4. Remove bolts, washers and nuts attaching center weight to body weight.
5. Remove bolts, washers and nuts attaching center spring and liner to body weight.
6. Remove bolts, washers, and nuts attaching side spring and liners to body weight.

REPAIR NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued

Inspect/Repair

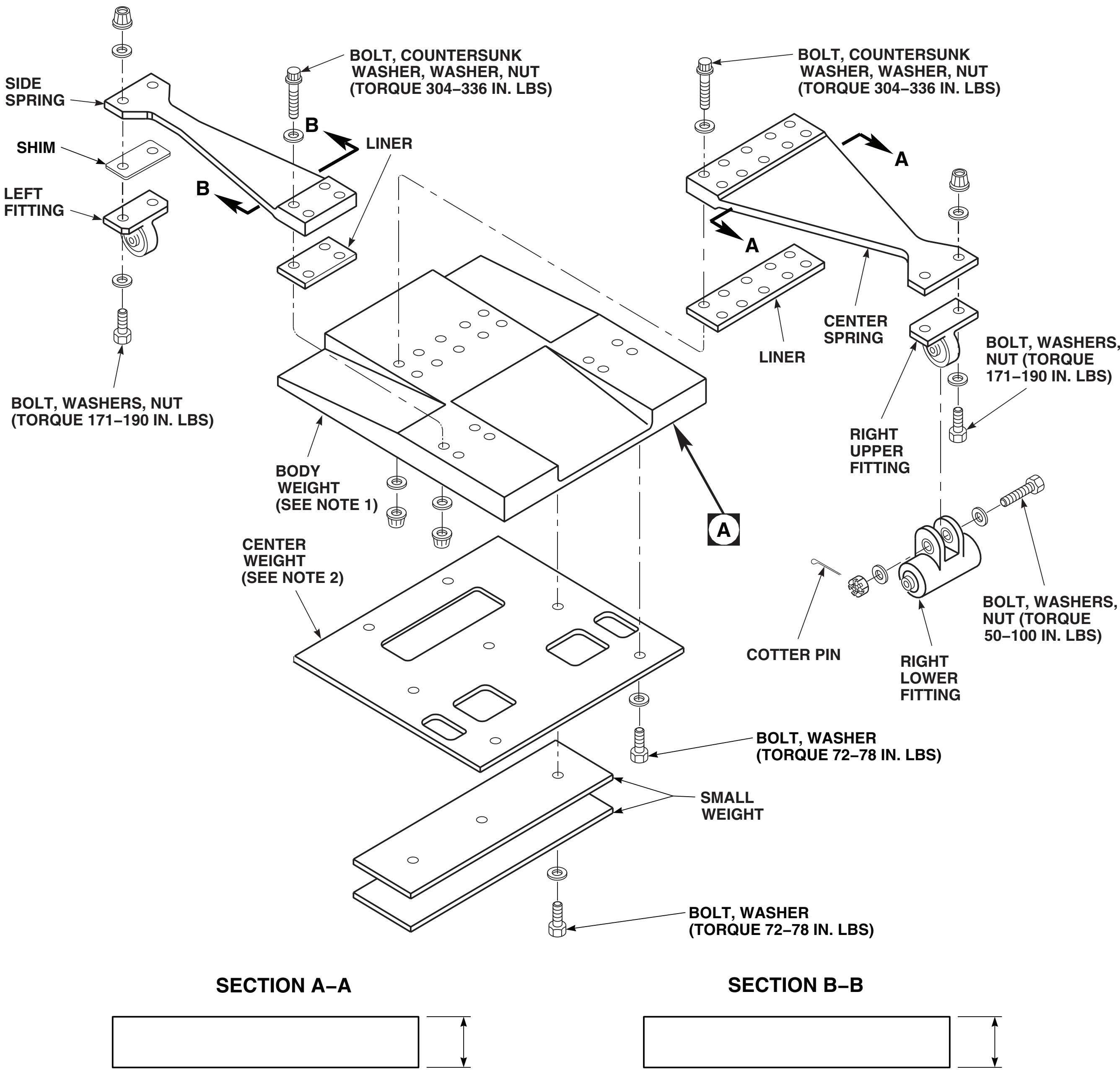
1. Check fitting bearings for wear as follows:
   a. Set up dial indicator with stylus tracking bearing inner race radial movement.
   b. **MAXIMUM RADIAL PLAY IS 0.005-INCH.** Replace bearings having more than 0.005-inch radial play (WP 0356 00).
2. Check bearing breakout torque as follows:
   a. Install a new NAS1305-27D bolt in bearing with enough washers on the bolt to seat nut. Install and tighten nut so that washers do not turn.
   b. Rotate bearing inner race four complete turns.
   c. Using torque wrench, turn bearing. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH A FULL 360 DEGREE TURN.**
3. Check all absorber attachment bushings for wear as follows:
   a. Install all absorber attachment bushings in respective airframe fitting bushing bores.
   b. Install a new NAS1305-27D bolt through bushings in left forward airframe fitting.
   c. Set up dial indicator with indicator stylus touching bolt.
   d. Take measurements as close to bushings as possible.
   e. **MAXIMUM RADIAL PLAY ALLOWED IS 0.002-INCH.** Replace bushings having more than 0.002-inch radial play (WP 0356 00). Repeat procedure.
   f. If replacing bushing does not correct problem, replace fitting.
   g. Install a new NAS1305-27D bolt through bushings in left aft airframe fitting.
   h. Repeat step 3. c. through step 3. f.
   i. Install a new NAS1305-68D bolt through bushings in right airframe fitting.
   j. Repeat step 3. c. through step 3. f.
4. **UH-60A 83-23924 - SUBQ UH-60L EH-60A** Check structural fitting slip-fit sleeve bushings as follows:
   a. Coat outer surface of sleeve bushing with general purpose grease, Item 136, WP 1803 00.
   b. Install sleeve bushing in sleeve bushing hole in structural fitting.
   c. Grasp sleeve bushing ends between thumb and forefinger and slide the bushing from back and forth in the sleeve bushing hole.
   d. Replace sleeve bushings that do not move freely in sleeve bushing hole. Repeat procedure.
   e. If replacing sleeve bushing does not correct problem, inspect sleeve bushing holes for burrs. Remove burrs using abrasive cloth, Item 75, WP 1803 00. Repeat procedure.
   f. If replacing the sleeve bushings and deburring sleeve bushing hole does not correct the problem, replace fitting.
5. Check springs for corrosion, gouges, nicks, and cracks as follows:

NOTE

Damage rework will remove shotpeen coverage and is considered acceptable in order to extend component service life.

   a. **SPRINGS 70219-02106-110 AND 70219-02106-111 DAMAGE MUST NOT EXCEED 0.315 - INCH**

REPAIR NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued



70219–02106–111 = 0.315 INCH MINIMUM THICKNESS
70219–02106–115 = 0.300 INCH MINIMUM THICKNESS

70219–02106–110 = 0.315 INCH MINIMUM THICKNESS
70219–02106–114 = 0.300 INCH MINIMUM THICKNESS

NOTES:

1. EFFECTIVITY: BODY WEIGHT, 70219–02107–103, ON VIBRATION ABSORBERS, 70219–02107–141, –042, AND –044; BODY WEIGHT, 70219–02107–109 ON VIBRATION ABSORBERS, 70219–02107–043 AND –046.
2. EFFECTIVITY: CENTER WEIGHT ON VIBRATION ABSORBERS 70219–02107–141, –042 (2 USED), –044 (2 USED), –045, AND –046.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 2. Nose Vibration Absorber. (Sheet 1 of 2)

REPAIR NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued

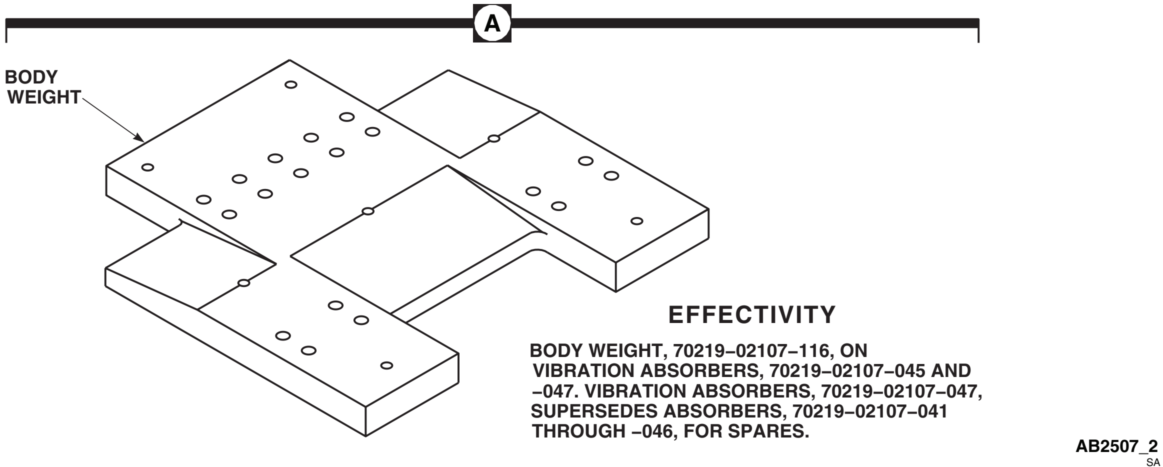


Figure 2. Nose Vibration Absorber. (Sheet 2 of 2)

MINIMUM THICKNESS. Blend damage using abrasive cloth, Item 82, WP 1803 00. **REPLACE SPRING IF DAMAGE EXCEEDS LIMITS (Figure 2, Sheet 1, Section AA or Figure 2, Sheet 1, Section BB).**

b. **SPRINGS 70219-02106-114 AND 70219-02106-115 DAMAGE MUST NOT EXCEED 0.300 - INCH MINIMUM THICKNESS.** Blend damage using abrasive cloth, Item 82, WP 1803 00. **REPLACE SPRING IF DAMAGE EXCEEDS LIMITS (Figure 2, Sheet 1, Section AA or Figure 2, Sheet 1, Section BB).**

c. **DAMAGE MUST NOT EXCEED 25% OF PERIMETER OF ANY CROSS SECTIONAL AREA.**

d. **NO DAMAGE ALLOWED WITHIN A RADIUS EQUAL TO ONE DIAMETER AT ANY ATTACHMENT HOLE.**

e. **IF DAMAGE IS FOUND IN OR AROUND ATTACHMENT HOLES CONTACT ENGINEERING AUTHORITY FOR DAMAGE ANALYSIS AND RECOMMENDED REPAIR OR REPLACEMENT.**

f. Touch up finish (WP 0381 00 and TM 55-1500-345-23).

6. Check attaching bolts and hardware for corrosion and wear. **MAXIMUM WEAR IS 0.002-INCH.** Replace worn or corroded hardware.

7. Check structural fittings for cracks and corrosion (TM 1-1520-265-23 and TM 1-1500-344-23). Replace or repair cracked or corroded fittings (WP 0384 00).

8. Check surrounding structure for cracks, dents, nicks or other obvious damage. Repair damage (WP 0384 00).

Assemble

1. Place center weight on body weight with boltholes lined up (Figure 2, Sheet 2).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean threads of all bolts and nuts with technical acetone, Item 5, WP 1803 00. Allow to air dry.

3. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts. Let dry for 5 to 15 minutes.

REPAIR NOSE VIBRATION ABSORBER (SPRING TYPE) - Continued

4. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts.
5. **QA** Install bolts and washers in center weight and body weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
6. Place liners and side spring on body weight with boltholes lined up.
7. Install countersunk washers on bolts with countersunk side facing boltheads.
8. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts and nuts. Let dry for 5 to 15 minutes.
9. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts and nuts.
10. **QA** Install bolts, with boltheads seated on spring side, countersunk washers, washers, and nuts. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**
11. Place liner and center spring on body weight with boltholes lined up.
12. Install countersunk washers on bolts with countersunk side facing boltheads.
13. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts and nuts. Let dry for 5 to 15 minutes.
14. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts and nuts.
15. **QA** Install bolts, with boltheads seated on spring side, countersunk washers, washers, and nuts. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**
16. Remove tags from small weights and position weights on center weight, with boltholes lined up.
17. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts. Let dry for 5 to 15 minutes.
18. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts.
19. **QA** Install bolts and washers in weights and center weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.** Remove tags.
20. Install single swivel bearing left fittings as follows:
    a. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts and nuts.
    b. **QA** Place fitting on side spring. Use any required combination of thick and thin washers. Install bolts, washers, and nuts. **TORQUE NUTS TO 171 - 190 INCH-POUNDS.**
21. Install right upper and lower fittings as follows:
    a. Install single swivel bearing upper fitting on center spring. Install bolt, washers, and nuts. Do not tighten nuts at this time.

NOTE

Lower fitting front end is on the side of flanged bushing.

    b. Install lower fitting on upper fitting with new NAS1305-24DH bolt. Make sure bolthead washer is against flanged bushing. Use any required combination of thick and thin washers.
    c. **QA** Install washer and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**
    d. Make sure upper fitting bearing is sandwiched between the flange and sleeve bushings and does not move from side to side. Install cotter pin.

ADJUST NOSE VIBRATION TUNING WEIGHTS

1. Determine weight change (TM 1-6625-724-13&P).

ADJUST NOSE VIBRATION TUNING WEIGHTS - Continued

2. Remove ILS glide slope antenna and support panel (TM 11-1520-237-23).
3. Remove bolts, washers, and nuts attaching tuning weights to vibration absorber.
4. Tune vibration absorber (TM 1-6625-724-13&P).
5. Place bolt through tuning weight holes. Check how many threads are visible.
6. If less than five threads are visible use longer bolt. If unthreaded portion of bolt is visible use shorter bolt.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Clean threads of tuning weight bolts with machinery towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Allow to air dry.
8. Apply sealing compound primer, Item 247, WP 1803 00. to threads of bolts. Let dry for 5 to 15 minutes.
9. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts.
10. **QA** Place tuning weights on absorber body, aligning boltholes. Install bolts, washers, and nuts. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
11. Install glide slope antenna and support panel (TM 11-1520-237-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE VIBRATION ABSORBER (BOX FRAME) UH-60L 95-26610, 95-26621-SUBQ

This WP supersedes WP 0220 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit SC 5180-99-B01
Airframe Repairer's Toolkit SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Sealing Compound, Item 258, WP 1803 00
Tags
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-265-23
TM 1-6625-724-13&P
TM 11-1520-237-23
WP 0384 00
WP 1803 00

REPLACE NOSE VIBRATION ABSORBER (BOX FRAME)

Remove

1. Turn off all electrical power.
2. Remove glide slope antenna and support panel (TM 11-1520-237-23).

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when removing attaching hardware. Box frame vibration absorber weighs approximately 50 pounds. Do not remove attaching hardware unless absorber is supported.

3. With assistant supporting absorber, remove bolts attaching absorber to left and right structural fittings (Figure 1, Detail A).
4. With aid of assistant, lower absorber until clear of structure.
5. Remove barrel-nuts and retainers from fittings.

Install

1. Turn off all electrical power.

REPLACE NOSE VIBRATION ABSORBER (BOX FRAME) - Continued

CAUTION

Damage to barrel-nut fitting will result if barrel-nut is incorrectly installed. To prevent damage to fitting make sure nut is installed with barrel (round side) towards bolthead.

NOTE

Make sure that only square barrel-nuts are used.

2. **QA** Install barrel-nuts and retainers into structural fittings (Figure 1, Detail A).
3. Apply sealing compound, Item 258, WP 1803 00, to mating surfaces of fittings and absorber assembly.

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when installing attaching hardware. Box frame vibration absorber weighs approximately 50 pounds.

4. With aid of assistant, lift vibration absorber into nose structure and place into position.
5. Make sure bolt holes line up and serrated plates on fittings and absorber are engaged.
6. Apply sealing compound, Item 258, WP 1803 00, to bolts.
7. **QA** Install bolts and washers. **TORQUE BOLTS TO 385 - 425 INCH-POUNDS**. Using nonelectrical wire, Item 354, WP 1803 00, lockwire bolts.
8. Install glide slope antenna and support panel (TM 11-1520-237-23).
9. Check cockpit main rotor 4/rev vibration per TM 1-6625-724-13&P, WP 0069 00, Task 7, Procedure 5. If vibration is within limits no absorber tuning is required. If vibration is not within limits and absorber tuning is required contact a Sikorsky Field Service Representative, or Sikorsky Aircraft for assistance.

REPAIR NOSE VIBRATION ABSORBER (BOX FRAME)

Disassemble

WARNING

Injury to personnel and damage to equipment will occur if housing attach bolts are loosened or removed. Box frame vibration absorber springs are always in compression. Make sure that housing attach bolts are not loosened or removed.

NOTE

- Tag and record each tuning weight for reinstallation.
- Disassembly of absorber is limited to removal of tuning weights and serrated plates.

1. Remove bolts and washers attaching short tuning weights to absorber mass (Figure 1, Detail A).
2. Remove bolts and washers attaching long tuning weights to absorber mass.
3. Remove screws attaching serrated plate to absorber assembly. Remove serrated plate.

Inspect

1. Check absorber upper and lower housings for cracks and corrosion, using fluorescent penetrant inspection kit (TM 1-1520-265-23 and TM 1-1500-344-23). **NO CRACKS ALLOWED.** If cracks are found, replace absorber.

REPAIR NOSE VIBRATION ABSORBER (BOX FRAME) - Continued

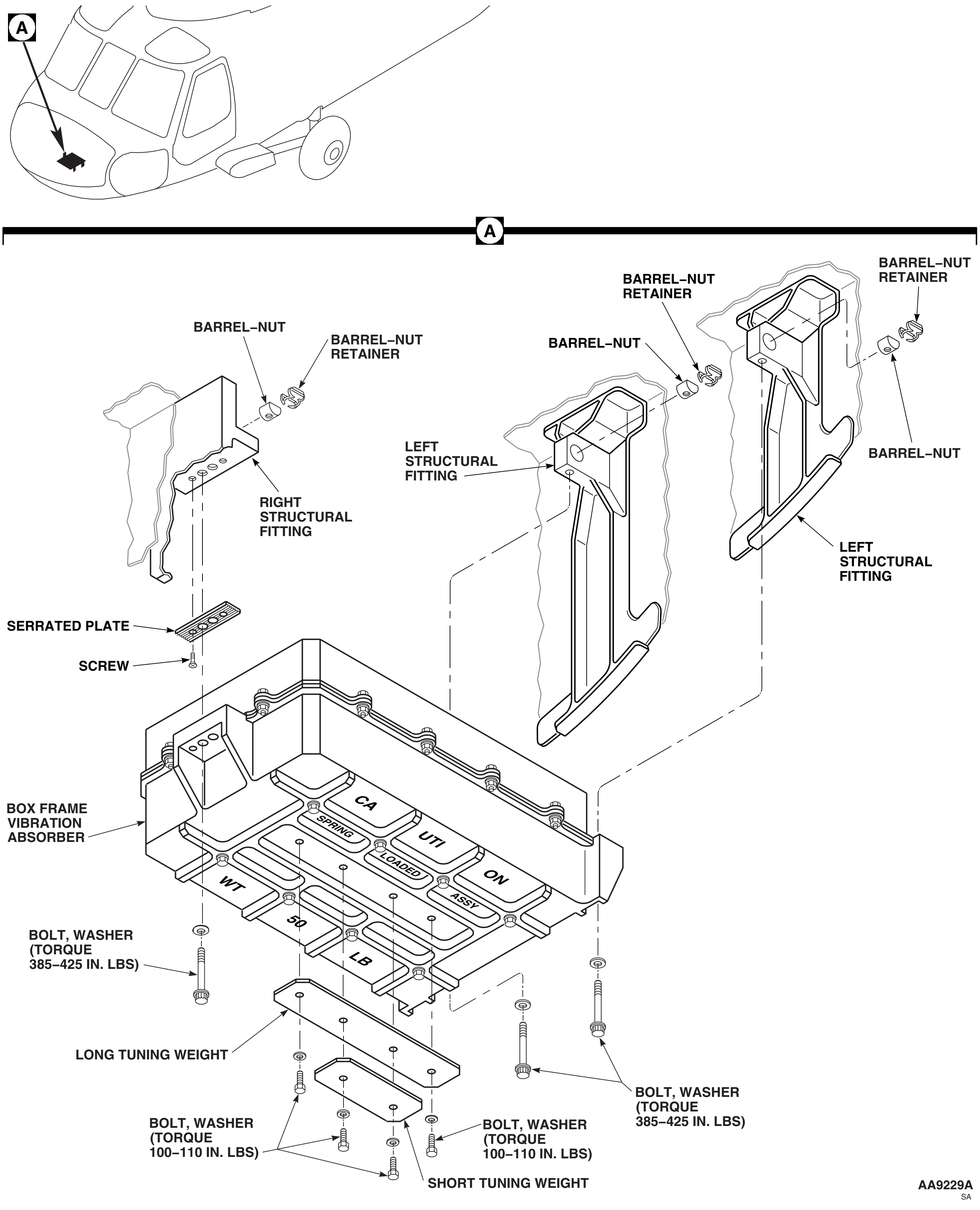


Figure 1. Nose Box Frame Vibration Absorber.

REPAIR NOSE VIBRATION ABSORBER (BOX FRAME) - Continued

2. Check serrated plate for damage to serrations.
3. Check airframe structural fittings for cracks and corrosion, using fluorescent penetrant inspection kit (TM 1-1520-265-23 and TM 1-1500-344-23). Replace and repair cracked and corroded fittings (WP 0384 00).
4. Check barrel-nuts and retainers for damage and corrosion. Replace damaged and corroded barrel-nuts and retainers.
5. Test barrel-nuts as follows:
   a. Place barrel-nut in soft jaw vise.
   b. Install bolt in each nut until two threads protrude from nut.
   c. Using a torque wrench, back out each bolt. **BREAKAWAY TORQUE SHALL BE AT LEAST 10 INCH-POUNDS.**
   d. If breakaway torque is less than 10 inch-pounds, replace barrel-nut.
6. Check surrounding structure for cracks, dents, nicks or other obvious damage (WP 0384 00).

Assemble

1. Apply sealing compound, Item 258, WP 1803 00, to threads of screws.
2. **QA** Install serrated plate to absorber with screws (Figure 1, Detail A).
3. Remove tags from short and long tuning weights. Position weights on absorber mass with bolt holes lined up.
4. Apply sealing compound, Item 258, WP 1803 00, to threads of bolts.
5. **QA** Install bolts and washers in tuning weights and absorber mass. **TORQUE BOLTS TO 100 - 110 INCH-POUNDS.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN VIBRATION ABSORBER (SPRING TYPE)

This WP supersedes WP 0221 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
C-Clamp, GGG-C-105
Dial Indicator, MIL-I-18422
Gloves
Goggles/Face Shield
Fluorescent Penetrant Inspection Kit, XMA101
Micrometer, 0 to 1 Inch
Torque Wrench, 0 - 16 in. oz.
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Wooden Tuning Wedge, Locally-Made, Figure, H-159, WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Bolt, NAS1305-24D
Bolt, NAS1305-27D
Bolt, NAS1305-68D
Bolt, NAS1306-38D
Bolt, NAS1306-380
Bolt, NAS1306-39D
Cloth, Abrasive, Item 75, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Grease, General Purpose, Item 136, WP 1803 00
Primer, Sealing Compound, Item 247, WP 1803 00
Sealing Compound, Item 282, WP 1803 00
Tags

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-265-23
TM 1-6625-724-13&P
TM 55-1500-345-23
WP 0258 00
WP 0357 00
WP 0358 00
WP 0381 00
WP 0387 00
WP 1803 00
WP 1805 00

REPLACE CABIN VIBRATION ABSORBER (SPRING TYPE)

Remove

1. Turn off all electrical power.
2. Remove cabin soundproofing (WP 0258 00).
3. Remove nuts and washers from bolts attaching left fittings of absorber to structural fittings (Figure 1, Sheet 1, Detail A or Figure 1, Sheet 2, Detail A). **DO NOT REMOVE BOLTS NOW.**
4. Remove nut and washer from bolt attaching right fitting of absorber to structural fitting. **DO NOT REMOVE BOLT NOW.**
5. Remove bolts, washers, and nuts attaching stop to right structural fitting.

REPLACE CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

WARNING

Weight of vibration absorber is about 75 pounds can cause personal injury. Do not remove attaching hardware unless absorber is supported.

6. With assistant supporting absorber, remove bolts attaching left side of absorber to structural fittings.
7. Lower left side of absorber onto structure.
8. With assistant supporting absorber, remove bolt attaching right side to structural fitting.
9. Lower right side but still support absorber.
10. Raise left side and move to left until right side clears structure, continue to lower right side until absorber is clear of structure.

Install

1. **UH-60A 77-22714 - 80-23491** Install vibration absorber as follows:
   a. Turn off all electrical power.
   b. With aid of assistant, place vibration absorber in overhead structure. Raise into place with left side of absorber higher than right side.
   c. Raise left side of absorber and move it left as far as possible give clearance on right side.
   d. Raise right side so that right lower fitting engages structural fitting (Figure 1, Sheet 1, Detail A).
   e. With assistant supporting absorber, install bolt with bolthead facing front.
   f. Engage left absorber fittings with structural fittings. Install bolts, with boltheads facing front.
   g. Install washers and nuts on left and right fitting bolts.
   h. Torque left front fitting bolt as follows:
      (1) **TORQUE BOLT TO 80 - 95 INCH-POUNDS.** Loosen torque back to zero inch-pounds.
      (2) **QA TORQUE BOLT TO 50 - 100 INCH-POUNDS.** Install cotter pin.
   i. Torque left rear and right fitting bolts as follows:
      (1) **TORQUE BOLTS TO 80 - 95 INCH-POUNDS.** Loosen torque back to zero inch-pounds.
      (2) **QA TORQUE BOLTS TO 47 - 53 INCH-POUNDS.** Install cotter pin.
   j. **QA** Install stop to right structural fitting with bolts, washers and nuts. **TORQUE TOP NUT TO 47 - 53 INCH-POUNDS. TORQUE BOTTOM NUT TO 95 - 100 INCH-POUNDS.**
   k. **UH-60A 77-22714 - 80-23491 UH-60L EH-60A** Check spring alignment at left and right airframe fittings as follows:
      (1) Rotate left front fitting bolt two full turns.
      (2) Rotate left rear and right fitting bolts two full turns.
      (3) Measure gap between structural fittings and washers. **MINIMUM GAP ALLOWED IS 0.020-INCH.**
      (4) If clearance is less than 0.020-inch at any measurement location, realign spring.
   l. Perform vibration check (TM 1-6625-724-13&P).
2. **UH-60A 80-23492-95-26623 UH-60L EH-60A** Install vibration absorber as follows:
   a. Turn off all electrical power.

REPLACE CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

NOTE

When installing a new vibration absorber, use same amount of weights as on old absorber.

b. With aid of assistant, position replacement absorber in overhead structure. Raise into place with left side higher than right side.

c. Raise left side of absorber and move it left as far as possible to give clearance on right side.

d. **UH-60A 80-23492-95-26623 UH-60L EH-60A** Raise right side so right lower fitting engages structural fitting. Install sleeve bushings lubricated with general purpose grease, Item 136, WP 1803 00. Install bolt with bolthead toward front (Figure 1, Sheet 2, Detail A).

e. **UH-60A 80-23492-95-26623 UH-60L EH-60A** Engage left absorber spring yokes with fittings, install sleeve bushings lubricated with general purp[ose grease, Item 136, WP 1803 00. Install bolts with boltheads toward front.

f. Torque fitting bolts as follows:

(1) **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.** Loosen torque back to 0 inch-pounds.

(2) **QA TORQUE RIGHT FITTING BOLT TO 50 - 100 INCH-POUNDS.** Install cotter pin.

(3) **QA TORQUE LEFT FITTING BOLTS 50 - 100 INCH-POUNDS.** Install cotter pin.

g. **QA** Install stop to right structural fitting with bolts, washers and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

h. Check spring alignment for left and right fittings as follows:

(1) Rotate left fitting bolts two full turns.

(2) Measure gap between outer side of spring yoke and inner side of bolt attachment washers. **MINIMUM GAP ALLOWED IS 0.020-INCH.**

(3) If clearance is less than 0.020-inch at any measurement location, realign spring.

i. **UH-60A 80-23492-95-26623 UH-60L EH-60A** Check absorber attachment bolts for breakout torque as follows:

(1) Rotate left side attachment bolts using torque wrench. **BREAKOUT TORQUE SHALL NOT EXCEED 10 INCH-POUNDS THROUGH FULL 360 DEGREE TURN.**

(2) If breakout torque exceeds 10 inch-pounds, check torque on bolts and slip-fit of sleeve bushings.

(3) Rotate right side attachment bolt using torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 50 INCH-POUNDS**. If breakout torque is more than 50 inch-pounds check buildup and bolt torque.

j. Perform vibration check (TM 1-6625-724-13&P).

NOTE

- Center spring (right structural fitting) is resting on a flanged bushing and cannot be out of alignment.
- Alignment procedure is the same for front and rear spring.

k. **UH-60A 80-23492 - SUBQ UH-60L EH-60A** Align front or rear spring as follows:

(1) Make sure absorber is correctly installed.

REPLACE CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

(2) Loosen four bolts holding spring to absorber body. Loosen bolts enough to allow bolt and nut to turn together with moderate wrench force on nut.

(3) Push or pull on spring to get 0.020-inch clearance between both sides of spring, yoke and washers (Figure 1, Sheet 2, View A-A).

(4) When spring alignment is correct, tighten two spring bolts.

(5) Check for 0.020-inch clearance between both sides of spring yoke and washers. If spring alignment is incorrect, loosen bolts and reset clearance.

(6) **TORQUE SPRING BOLTS TO 304 - 336 INCH-POUNDS.**

l. **UH-60A 83-23924 - SUBQ UH-60L EH-60A** Check absorber attachment bolts for breakout torque as follows:

(1) Rotate left side attachment bolts using torque wrench. **BREAKOUT TORQUE SHALL NOT EXCEED 10 INCH-POUNDS THROUGH FULL 360 DEGREE TURN.**

(2) If breakout torque exceeds 10 inch-pounds, check torque on bolts and slip-fit of sleeve bushings.

(3) Rotate right side attachment bolt using torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 50 INCH-POUNDS.** If breakout torque is more than 50 inch-pounds check buildup and bolt torque.

3. Install cabin soundproofing (WP 0258 00).

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE)

Disassemble

1. **UH-60A 77-22714 - 80-23491** Remove fittings as follows:

   a. Remove bolt, washer, and nut attaching right lower fitting to right upper fitting (Figure 1, Sheet 3, Detail B). Keep hardware for installation.

   b. Remove bolts, washers, and nuts attaching upper fitting to center spring. Keep hardware for installation of fitting.

   c. Remove bolts, washers, and nuts attaching left fittings to side springs. Keep hardware for installation of fittings.

2. **UH-60A 80-23492 - SUBQ UH-60L EH-60A** Remove bolt, washer, and nut securing right lower fitting to absorber (Figure 1, Sheet 1, Detail A).

NOTE

Tag and record each weight for reinstallation.

3. Remove bolts, washers, and nuts attaching small weights to body weight.
4. Remove bolts, washers, and nuts attaching center weight to body weight.
5. Remove bolts, washers, and nuts attaching center spring and liner to body weight.
6. Remove bolts, washers, and nuts attaching side springs and liners to body weight.

Inspect/Repair

1. **UH-60A 77-22714 - 80-23491** Inspect vibration absorber as follows:

   a. Check all swivel bearings and lower bearings for wear as follows:

   (1) Set up dial indicator with stylus tracking bearing inner race radial movement.

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

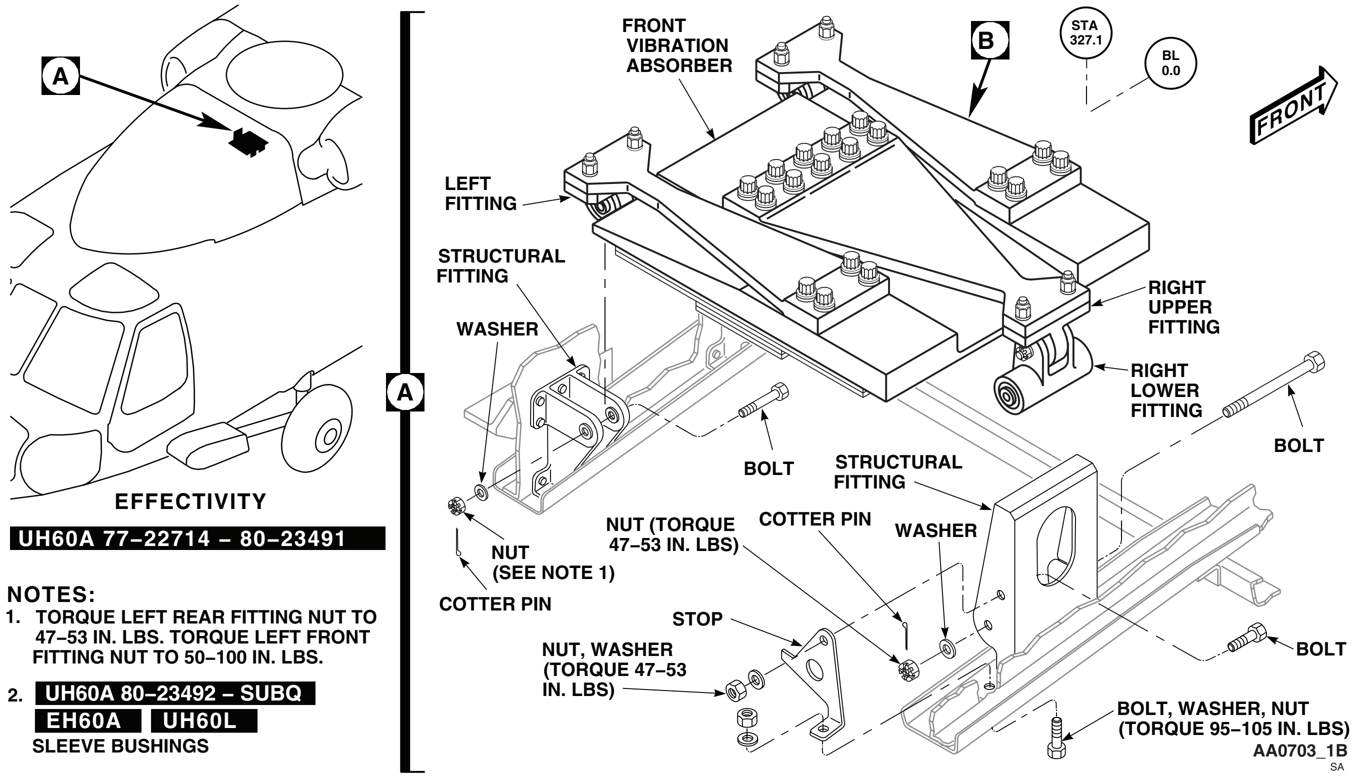


Figure 1. Cabin Vibration Absorber (Spring Type). (Sheet 1 of 4)

(2) **MAXIMUM RADIAL PLAY ALLOWED IS 0.005-INCH.** Replace bearings having more than 0.005-inch radial play (WP 0358 00).

b. Check bearing breakout torque on left and right swivel bearing and right lower bearing fitting as follows:

(1) Install new NAS1305-68D bolt in right lower fitting bearing with enough washers on bolt to seat nut. Install and tighten nut so that washers do not turn.

(2) Rotate bearing inner race four complete turns.

(3) Turn bearing inner race with torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

(4) Install NAS1305-27D bolt in right upper and left single swivel bearings with enough washers on bolt to seat nut.

(5) Install and tighten nut so that washers do not turn. Repeat step b. (2). and step b (3). **REPLACE ALL BEARINGS WITH BREAKOUT TORQUE GREATER THAN 10 INCH-OUNCES.**

c. Visually check structural fitting bushings for cracks. Replace cracked bushings.

d. Check structural fitting bushings for wear as follows:

(1) Install a new NAS1305-27D bolt through bushings in structural fittings.

(2) Set up dial indicator with indicator stylus touching bolt. Take measurement as close to bushing as possible.

(3) **MAXIMUM RADIAL PLAY ALLOWED IS 0.002-INCH.** Replace bushings having more than 0.002-inch radial play (WP 0357 00). Repeat procedure. If replacing bushing does not correct problem, replace structural fitting.

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

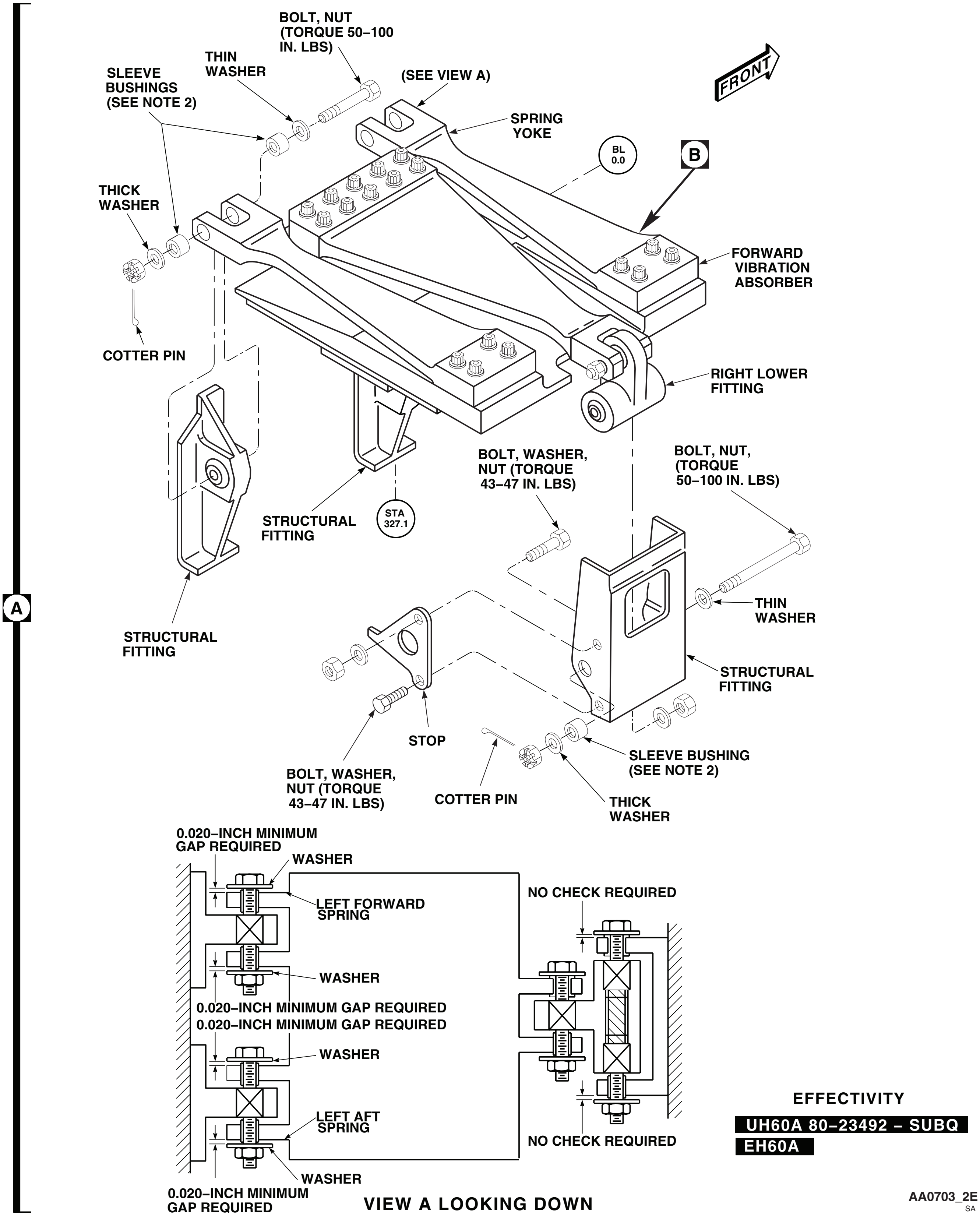


Figure 1. Cabin Vibration Absorber (Spring Type). (Sheet 2 of 4)

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

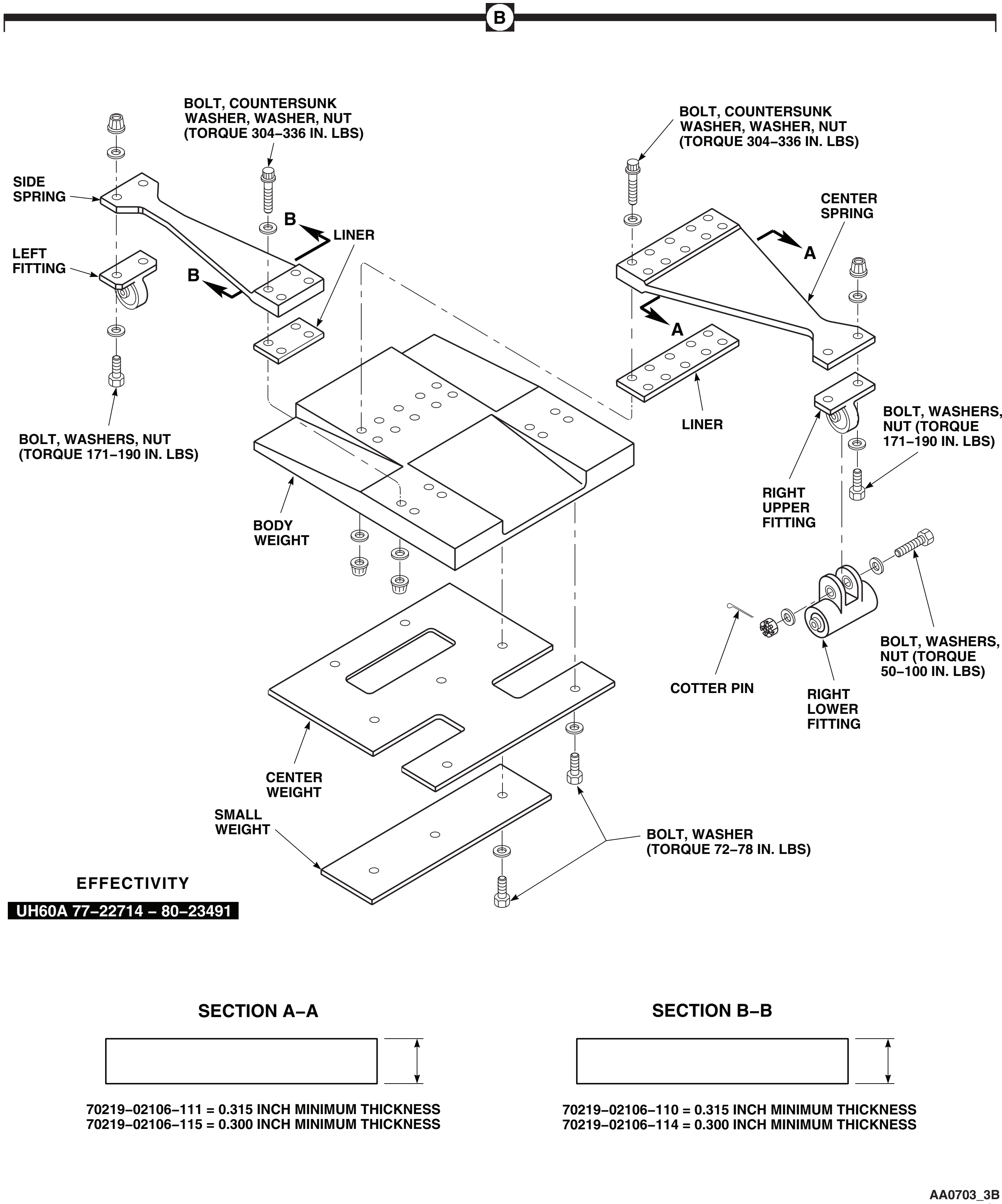


Figure 1. Cabin Vibration Absorber (Spring Type). (Sheet 3 of 4)

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

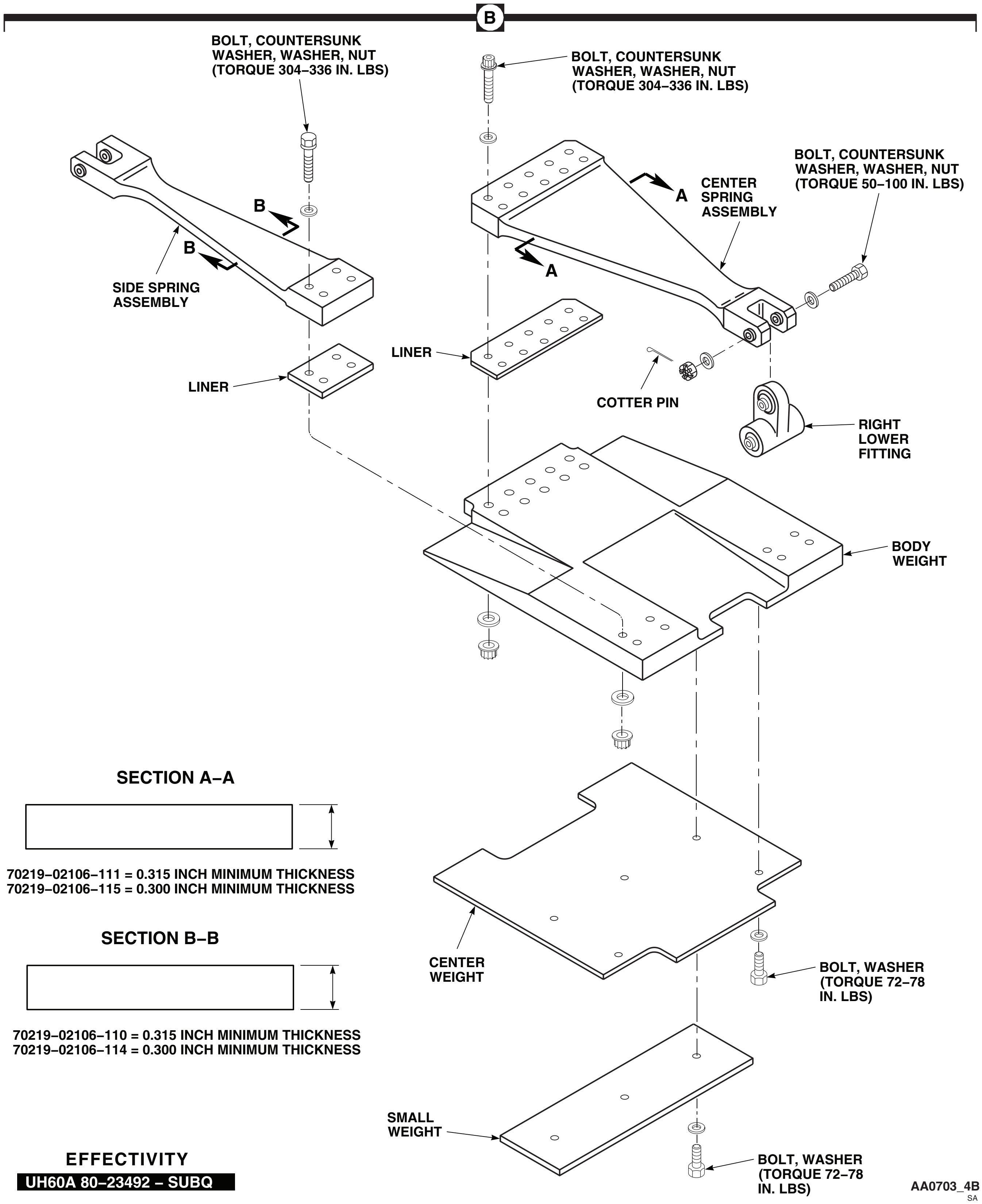


Figure 1. Cabin Vibration Absorber (Spring Type). (Sheet 4 of 4)

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

e. Check springs for corrosion, gouges, nicks, and cracks as follows:

NOTE

Damage rework will remove shotpeen coverage and is considered acceptable in order to extend component service life.

(1) **SPRINGS 70219-02106-110 AND 70219-02106-111 DAMAGE MUST NOT EXCEED 0.315-INCH MINIMUM THICKNESS.** Blend damage using abrasive cloth, Item 82, WP 1803 00. **REPLACE SPRING IF DAMAGE EXCEEDS LIMITS (Figure 1, Sheet 3, Section A-A or Section B-B).**

(2) **SPRINGS 70219-02106-114 AND 70219-02106-115 DAMAGE MUST NOT EXCEED 0.300 - INCH MINIMUM THICKNESS.** Blend damage using abrasive cloth, Item 82, WP 1803 00. **REPLACE SPRING IF DAMAGE EXCEEDS LIMITS (Figure 1, Sheet 3, Section A-A or Section B-B).**

(3) **DAMAGE MUST NOT EXCEED 25% OF PERIMETER OF ANY CROSS SECTIONAL AREA.**

(4) **NO DAMAGE ALLOWED WITHIN A RADIUS EQUAL TO ONE DIAMETER AT ANY ATTACHMENT HOLE.**

(5) **IF DAMAGE IS FOUND IN OR AROUND ATTACHMENT HOLES CONTACT ENGINEERING AUTHORITY FOR DAMAGE ANALYSIS AND RECOMMENDED REPAIR OR REPLACEMENT.**

(6) Touch up finish (WP 0381 00 and TM 55-1500-345-23).

f. Check structural fittings for cracks and corrosion, using fluorescent penetrant inspection kit (TM 1-1520-265-23 and TM 1-1500-344-23). Replace or repair damaged fittings (WP 0387 00).

g. Check attaching bolts and hardware for corrosion and wear. **MAXIMUM WEAR ALLOWED IS 0.002-INCH.** Replace worn or corroded hardware.

h. Check surrounding structure for cracks, dents, nicks, or other obvious damage. Repair damage (WP 0387 00).

2. **UH-60A 80-23492 - SUBQ UH-60L EH-60A** Inspect vibration absorber as follows:

a. Check lower fitting bearings and left structural fitting bearings for wear as follows:

(1) Set up dial indicator with stylus tracking bearing inner race radial movement.

(2) **MAXIMUM RADIAL PLAY ALLOWED IS 0.005-INCH.** Replace bearings having more than 0.005-inch radial play (WP 0358 00).

b. Check lower fitting bearings and left structural fitting bearings for breakout torque as follows:

(1) Install new NAS1306-39D bolt in left structural fitting bearing with enough washers on bolt to seat nut. Install and tighten nut so that washers do not turn.

(2) Rotate bearing inner race four complete turns.

(3) Turn bearing inner race with torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH A FULL 360 DEGREES.**

c. Check pressed-in bushings for wear as follows:

(1) Install a new NAS1306-380 bolt or NAS1305-68D bolt in structural fitting, lower fitting, or spring.

(2) Set up dial indicator with indicator stylus touching bolt. Take measurement as close to bushing as possible.

(3) **MAXIMUM RADIAL PLAY ALLOWED IS 0.002-INCH.** Replace bushings having more than 0.002-inch radial play (WP 0357 00). Repeat procedure. If replacing bushing does not correct problem, replace structural fitting, lower fitting or spring (WP 0357 00).

d. **UH-60A 80-23492 - SUBQ UH-60L EH-60A** Check slip-fit sleeve bushings as follows:

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

(1) Coat all sleeve bushings with general purpose grease, Item 136, WP 1803 00. Install bushings in spring and airframe fitting.

(2) Grasp sleeve bushing ends between thumb and forefinger and slide bushing side to side in bushing hole. Replace sleeve bushings that do not move freely. Repeat procedure.

(3) If replacing bushing does not correct problem, deburr bushing holes in spring and structural fitting with abrasive cloth, Item 75, WP 1803 00. Repeat procedure.

(4) If deburring sleeve bushing hole(s) does not correct problem, replace spring or airframe fitting.

e. Check springs for corrosion, gouges, nicks, and cracks as follows:

NOTE

Damage rework will remove shotpeen coverage and is considered acceptable in order to extend component service life.

(1) **SPRINGS 70219-02106-110 AND 70219-02106-111 DAMAGE MUST NOT EXCEED 0.315-INCH MINIMUM THICKNESS.** Blend damage using abrasive cloth, Item 82, WP 1803 00. **REPLACE SPRING IF DAMAGE EXCEEDS LIMITS (Figure 1, Sheet 3, Section A-A or Section B-B).**

(2) **SPRINGS 70219-02106-114 AND 70219-02106-115 DAMAGE MUST NOT EXCEED 0.300-INCH MINIMUM THICKNESS.** Blend damage using abrasive cloth, Item 82, WP 1803 00. **REPLACE SPRING IF DAMAGE EXCEEDS LIMITS (Figure 1, Sheet 3, Section A-A or Section B-B).**

(3) **DAMAGE MUST NOT EXCEED 25% OF PERIMETER OF ANY CROSS SECTIONAL AREA.**

(4) **NO DAMAGE ALLOWED WITHIN A RADIUS EQUAL TO ONE DIAMETER AT ANY ATTACHMENT HOLE.**

(5) **IF DAMAGE IS FOUND IN OR AROUND ATTACHMENT HOLES CONTACT ENGINEERING AUTHORITY FOR DAMAGE ANALYSIS AND RECOMMENDED REPAIR OR REPLACEMENT.**

(6) Touch up finish (WP 0381 00 and TM 55-1500-345-23).

f. Check attaching bolts and hardware for corrosion and wear. **MAXIMUM WEAR ALLOWED IS 0.002-INCH.** Replace worn or corroded hardware.

g. Check structural fittings for cracks and corrosion, using fluorescent penetrant inspection kit (TM 1-1520-265-23 and TM 1-1500-344-23). Replace or repair cracked or corroded fittings (WP 0357 00 or WP 0358 00).

h. Check surrounding structure for cracks, dents, nicks, or other obvious damage. **NO CRACKS ALLOWED.** Repair damage (WP 0387 00).

Assemble

1. Place center weight on body weight with boltholes lined up (Figure 1, Sheet 3, Detail Band Figure 1, Sheet 4, Detail B).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean threads of all bolts and nuts with technical acetone, Item 5, WP 1803 00. Allow to air dry.

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

3. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts. Let dry for 5 to 15 minutes.
4. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts.
5. **QA** Install bolts and washers in center weight and body weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
6. Place liners and side springs on body weight, with boltholes lined up.
7. Install countersunk washers on bolts with countersunk side facing bolthead.
8. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts and nuts.
9. **QA** Install bolts, with boltheads seated on spring side, countersunk washers, washers, and nuts. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**
10. Place liner and center spring on body weight with boltholes lined up.
11. Install countersunk washers on bolts with countersunk side facing boltheads.
12. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts. Let dry 5 to 15 minutes.
13. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts and nuts.
14. **QA** Install bolts, with boltheads seated on spring side, countersunk washers, washers, and nuts. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**
15. Remove tags from small weights and place weights on center weights, with boltholes lined up.
16. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts. Let dry for 5 to 15 minutes.
17. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts.
18. **QA** Install bolts and washers in weights and center weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
19. **UH-60A 77-22714 - 80-23491** Install left fittings as follows (Figure 1, Sheet 3, Detail B):
    a. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts and nuts.

NOTE

Thin or thick washer may be installed under bolthead or nut.

    b. **QA** Place fitting on side spring. Use any required combination of thick and thin washers. Install bolts, washers, and nuts. **TORQUE NUTS TO 171 - 190 INCH-POUNDS.**

20. **UH-60A 77-22714 - 80-23491** Install right fittings as follows:
    a. **QA** Place upper fitting on center spring. Install bolts, washers, and nuts. **TORQUE NUTS TO 171 - 190 INCH-POUNDS (Figure 1, Sheet 3, Detail B).**

NOTE

Front of lower fitting ends with flanged bushing.

    b. Install lower fitting on upper fitting with new NAS1305-24DH bolt and washer required (any combination of thick and thin). Make sure bolthead washer is against flanged bushing.
    c. **QA** Install nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**
    d. Install cotter pin.

21. **UH-60A 80-23492 - SUBQ UH-60L EH-60A** Install right lower fitting on center spring as follows (Figure 1, Sheet 4, Detail B):
    a. Install new NAS1306-38D bolt, and washer. Make sure bolthead washer is against flanged bushing.

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

NOTE

On right lower fittings with inspection holes, position fitting with inspection hole facing outboard.

b. **QA** Install washer and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**

c. Make sure fitting bearing is sandwiched between flanged bushing and sleeve bushing, and does not move side to side in yoke. Install cotter pin.

Adjust

NOTE

- Large center weight is about 5 pounds. Thick weight is about 1 pound. Thin weight is about 1/2 pound.
- Cabin tuning weight adjustment requires removal of interior soundproofing.
- Locally make wooden tuning wedge (Figure 159, WP 1805 00) before inflight tuning weight adjustments are made.
- Torquing and sealing compound application to bolts for in flight adjustments should be done after landing. Weights must not move on absorber body after tightening bolts.
- Bolts used during in flight adjustments with less than five threads showing must be changed after landing.
- If unthreaded portion of bolt is visible, and shorter bolts are unavailable, return to original weight and make adjustment on the ground.

1. If absorber has less than 3 pounds of tuning weights and a 5-pound center weight, remove center weight and install 5 pounds of tuning weight.
2. Locally make wooden tuning wedge (Figure 159, WP 1805 00) if vibration absorber tuning is to be performed in flight.
3. Insert locally made wooden tuning wedge (Figure 159, WP 1805 00) into absorber to reduce absorber vibrations while making inflight tuning weight changes.
4. Tune vibration absorber (TM 1-6625-724-13&P).
5. Place bolt through tuning weight holes. Check how many threads are visible.
6. If less than five threads are visible use longer bolt. If unthreaded portion of bolt is visible use shorter bolt.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Clean threads of tuning weight bolts with technical acetone, Item 5, WP 1803 00. Allow to air dry.
8. Apply sealing compound primer, Item 247, WP 1803 00, to threads of bolts. Let dry for 5 to 15 minutes.
9. Apply sealing compound, Item 282, WP 1803 00, to threads of bolts.

REPAIR CABIN VIBRATION ABSORBER (SPRING TYPE) - Continued

10. **QA** Place tuning weights on absorber body, lining up boltholes. Install bolts and washers. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN VIBRATION ABSORBER (BOX FRAME) UH-60L 95-26610, 95-26621-SUBQ

This WP supersedes WP 0222 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Vibration Absorber Support Assembly, Locally-Made, Figure 220, WP 1805 00

Material/Parts

Sealing Compound, Item 258, WP 1803 00
Tags
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-265-23
TM 1-6625-724-13&P
WP 0258 00
WP 0387 00
WP 1803 00
WP 1805 00

REPLACE CABIN VIBRATION ABSORBER (BOX FRAME)

Remove

1. Turn off all electrical power.
2. Remove cabin ceiling soundproofing (WP 0258 00).
3. Locally-make vibration absorber support assembly (Figure 220, WP 1805 00).

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when removing attaching hardware. Vibration absorber weighs 90 pounds. Do not remove attaching hardware unless absorber is supported with locally-made vibration absorber support assembly (Figure 220, WP 1805 00), and two assistants.

4. Have assistants support vibration absorber with locally-made vibration absorber support assembly (Figure 220, WP 1805 00).
5. Remove bolts and washers, attaching vibration absorber to structure fittings (Figure 1, Detail A).
6. With two assistants supporting vibration absorber, remove locally-made vibration absorber support assembly (Figure 220, WP 1805 00) from helicopter.
7. Remove barrel-nuts and retainers from fittings.

Install

1. Turn off all electrical power.

REPLACE CABIN VIBRATION ABSORBER (BOX FRAME) - Continued

CAUTION

Damage to barrel-nut fitting will result if barrel-nut is incorrectly installed. To prevent damage to fitting, make sure nut is installed with barrel (round side) towards bolthead.

NOTE

Make sure that only square barrel-nuts are used.

2. **QA** Install barrel-nuts and retainers into structural fittings (Figure 1, Detail A).
3. Apply sealing compound, Item 258, WP 1803 00, to mating surfaces of fittings and absorber.

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when installing. Use at least two assistants to lift absorber and support with locally-made vibration absorber support assembly (Figure 220, WP 1805 00). Box frame vibration absorber weighs 90 pounds.

4. With two assistants supporting vibration absorber, raise into position on cabin structure and support with locally-made absorber installation tool. Make sure bolt holes line up and serrated plate on fitting and absorber are engaged.
5. Apply sealing compound, Item 258, WP 1803 00, to bolts.
6. **QA** Install bolts and washers, **TORQUE BOLTS TO 385 - 425 INCH-POUNDS**. Using nonelectrical wire, Item 354, WP 1803 00, lockwire bolts
7. Install cabin ceiling soundproofing (WP 0258 00).
8. Check cockpit main rotor 4/rev vibration per TM 1-6625-724-13&P, WP 0069 00 , Task 7, Procedure 5. If vibration is within limits no absorber tuning is required. If vibration is not within limits and absorber tuning is required contact a Sikorsky Field Service Representative, or Sikorsky Aircraft for assistance.

REPAIR CABIN VIBRATION ABSORBER (BOX FRAME)

Disassemble

WARNING

Injury to personnel and damage to equipment will occur if housing attach bolts are loosened or removed. Box frame vibration absorber springs are always in compression. Make sure that housing attach bolts are not loosened or removed.

NOTE

- Tag and record each tuning weight for reinstallation.
- Disassembly of absorber is limited to removal of tuning weights and serrated plates.

1. Remove bolts and washers attaching short tuning weights to absorber mass (Figure 1, Detail A).
2. Remove bolts and washers attaching long tuning weights to absorber mass.
3. Remove screws attaching serrated plate to absorber assembly. Remove serrated plate.

Inspect

1. Check absorber upper and lower housings for cracks and corrosion, using fluorescent penetrant kit (TM 1-1520-265-23 and TM 1-1500-344-23). **NO CRACKS ALLOWED.** If cracks are found, replace absorber.

REPAIR CABIN VIBRATION ABSORBER (BOX FRAME) - Continued

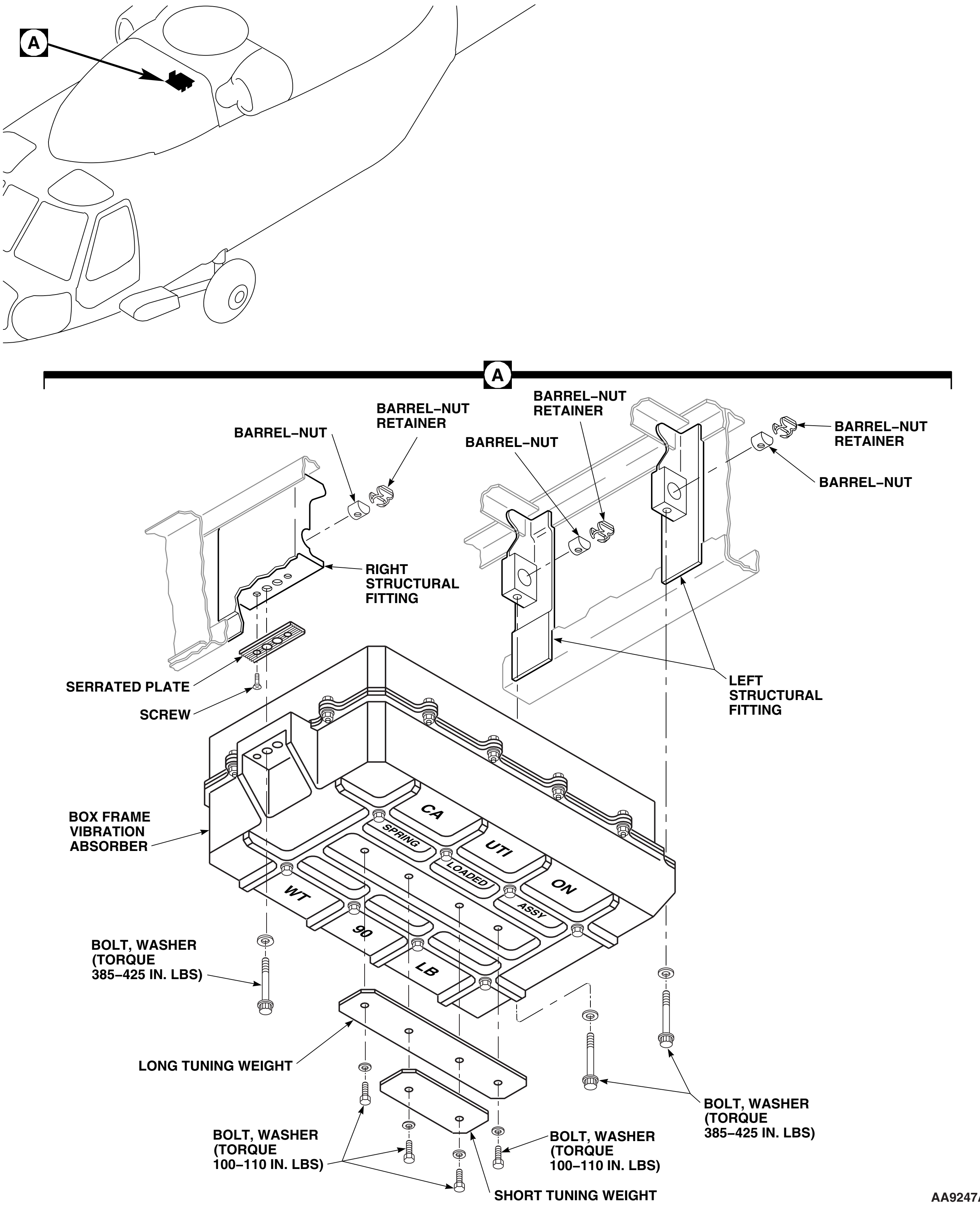


Figure 1. Cabin Vibration Absorber (Box Frame).

REPAIR CABIN VIBRATION ABSORBER (BOX FRAME) - Continued

2. Check serrated plate for damage to serrations. **NO DAMAGE ALLOWED.** If damage is found, replace serrated plated.
3. Check structural fittings for cracks and corrosion, using fluorescent penetrant inspection kit (TM 1-1520-265-23 and TM 1-1500-344-23). Replace or repair cracked and corroded fittings (WP 0387 00).
4. Check barrel-nuts and retainers for damage and corrosion. **NO DAMAGE OR CORROSION ALLOWED.** Replace damaged and corroded barrel-nuts and retainers.
5. Test barrel-nuts as follows:
   a. Place barrel-nut in soft jaw vise.
   b. Install bolt in each nut until two threads protrude from nut.
   c. Using a torque wrench, back out each bolt. **BREAKAWAY TORQUE SHALL BE AT LEAST 10 INCH-POUNDS.**
   d. If breakaway torque is less than 10 inch-pounds, replace barrel-nut.
6. Check surrounding structure for cracks, dents, nicks or other obvious damage (WP 0387 00).

Assemble

1. Apply sealing compound, Item 258, WP 1803 00, to threads of screws.
2. Install serrated plate to absorber with screws (Figure 1, Detail A).
3. Remove tags from short and long tuning weights. Position weights on absorber mass with bolt holes lined up.
4. Apply sealing compound, Item 258, WP 1803 00, to threads of bolts.
5. **QA** Install bolts and washers in tuning weights and absorber mass. **TORQUE BOLTS TO 100 - 110 INCH-POUNDS.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ROLL VIBRATION ABSORBER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
TM 1-1520-265-23
WP 0286 00
WP 0293 00
WP 0378 00
WP 0388 00
WP 1803 00

REMOVE

NOTE

EH60A UH60L Roll vibration absorbers are installed on the helicopters.

1. Turn off all electrical power.
2. Remove drag beam support fairing (WP 0286 00) and cockpit step (WP 0293 00).
3. Remove top and bottom bolts and washers from roll vibration absorber. Remove absorber (Figure 1, Detail A).
4. Remove barrel-nuts and retainers from fitting.

INSPECT

1. Inspect roll vibration absorber for corrosion, gouges, nicks, and cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
2. Inspect attachment bolts, washers, and barrel nuts for corrosion. Replace corroded hardware.

NOTE

Make sure that only square barrel-nuts are used.

3. Test attachment bolts and barrel-nuts as follows:
   a. Place barrel-nut in soft-jawed vise.
   b. Thread bolt into barrel-nut until two threads are showing beyond nut.
   c. Back off bolt using torque wrench. **BREAKAWAY TORQUE SHALL NOT BE LESS THAN 18 INCH-POUNDS.**
   d. If breakaway torque is less than 18 inch-pounds, replace bolt. Repeat step 3. a. through step 3. c.

INSPECT - Continued

e. If breakaway torque is still less than 18 inch-pounds, replace barrel-nut. Repeat step 3. a. through step 3. c.

4. Inspect surrounding structure for cracks, dents, nicks, or other obvious damage, using fluorescent penetrant kit (TM 1-1520-265-23 and TM 1-1500-344-23). **NO CRACKS ALLOWED.** Repair damage (WP 0378 00 and WP 0388 00).

INSTALL

1. Turn off all electrical power.

CAUTION

Damage to recess in fitting will occur if barrel-nut is installed incorrectly. Make sure nut is installed with rounded side toward bolthead.

2. **QA** Install barrel-nuts and retainers in fitting with rounded sides toward boltheads, if required (Figure 1, Detail A).
3. **QA** Install roll vibration absorber with bolt and washer in top and bottom hole. **TORQUE BOTTOM BOLT TO 845 - 935 INCH-POUNDS.** Remove bolt and washer from top hole.
4. Install drag beam support fairing (WP 0286 00). Do not install cockpit step now.
5. **QA** Insert cockpit step through hole in drag beam support fairing and position step between drag beam support and vibration absorber fitting. Install bolt, washer, and barrel nut. **TORQUE BOLT TO 845 - 935 INCH-POUNDS.**
6. **QA** Install screws and washers attaching cockpit step to drag beam support fairing.
7. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.

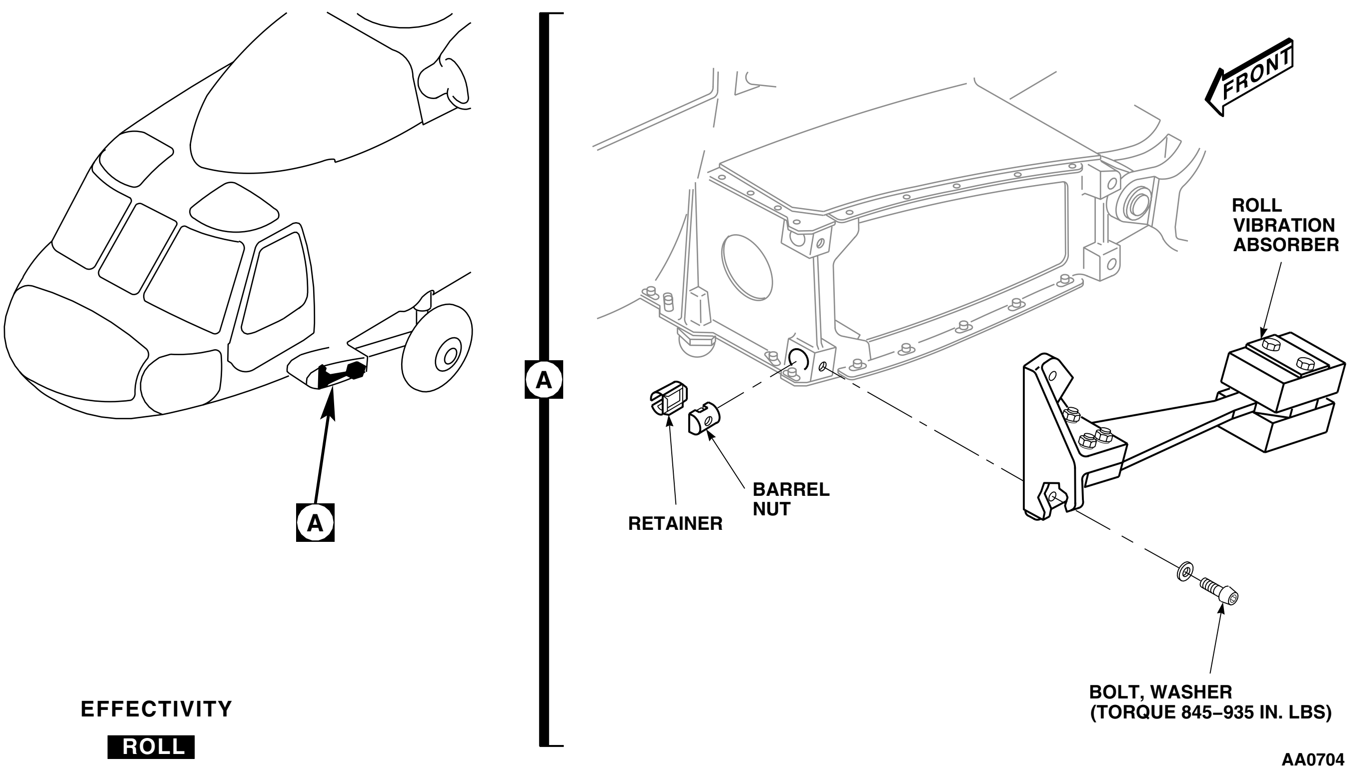


Figure 1. Replace Roll Vibration Absorber.

INSTALL - Continued

8. **QA** Install cover on drag beam support fairing with screws and washers.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

UPPER WINDOWS

This WP supersedes WP 0224 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Remover, Paint, Item 250, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 328, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 329, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

WP 0229 00
WP 0796 00
WP 1803 00

REMOVE

1. **EH-60A 85-24441 - SUBQ UH-60L EH-60A HCW** Remove free-air thermometer (WP 0796 00).
2. **UH-60A 80-23416 - SUBQ UH-60L EH-60A** Remove screws and carefully remove upper window and rain gutter (if installed) (Figure 1, Detail A).
3. Remove foam seal tape from flanges.

CAUTION

Do not allow paint remover to contact plastic windows, damage to plastic windows will result.

4. Clean flanges with paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean paint remover residue with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean machinery towel.

REMOVE - Continued

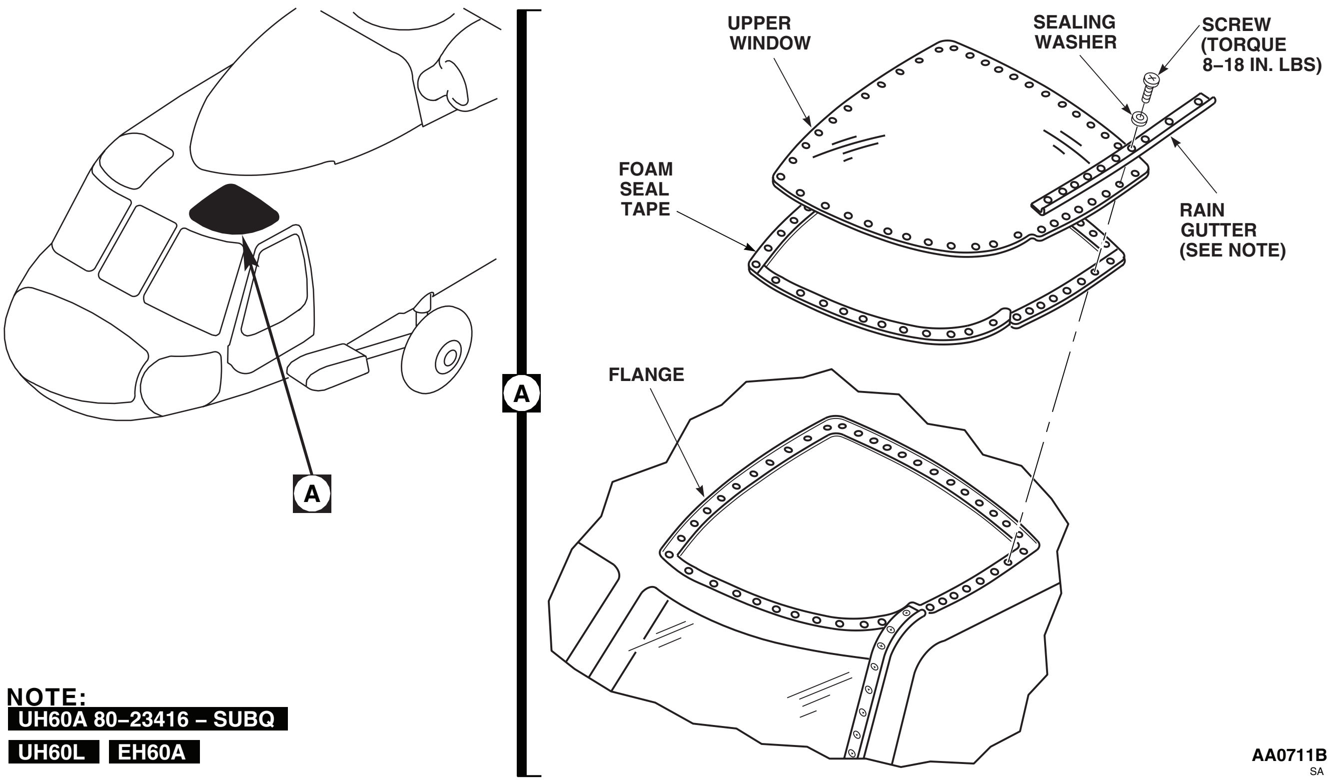


Figure 1. Replace Upper Window.

INSTALL

NOTE

For a better fit, cut and join tape at sharp corners.

1. Apply pressure sensitive adhesive tape, Item 328, WP 1803 00, to flanges of frame with sticky surface against flange. For proper window installation, apply additional layer of pressure sensitive adhesive tape, Item 329, WP 1803 00, in areas of joggle. Trim tape as necessary.

NOTE

- UH-60A 80-23416 - SUBQ UH-60L Rain gutters are installed.
- Replace sealing washers that have missing or unserviceable seals.

2. Coat fastening screws with petrolatum, Item 216, WP 1803 00, to prevent damaging or folding over sealing washers' rubber seals.

INSTALL - Continued

NOTE

The window installation screws should be gradually tightened in an alternating pattern as required to provide for uniform squeeze out of sealant about 0.100-inch wide. Torque should be at least 8 inch-pounds, but not more than 18 inch-pounds because of the wide range of breakaway torque possible for the anchor nuts. The torquing of all screws to the same value is not recommended as this can result in a waviness of the window edge.

3. **QA** Place window and rain gutter against flanges and install screws and sealing washers (Figure 1, Detail A). **TORQUE SCREWS TO 8 - 18 INCH-POUNDS.**
4. Clean the area along the window edges, to be sealed, using a nonmetallic scraper.
5. Fill all gaps around installed window with sealing compound, Item 273, WP 1803 00, to make sure surface is smooth and flush.
6. **EH-60A 85-24441 - SUBQ UH-60L EH-60A HCW** Install free-air thermometer on upper window (WP 0796 00).
7. Check window for water leaks (WP 0229 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

LOWER WINDOWS

This WP supersedes WP 0225 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Foil Tape, Locally-Made, Figure 170, WP 1805 00
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 70, WP 1803 00
Remover, Paint, Item 250, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 287, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 328, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 329, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
WP 0229 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove screws and carefully remove lower window from frame flanges (Figure 1 Detail A).
2. Remove vinyl/foam tape from flanges.

CAUTION

Do not allow paint remover to contact plastic windows, damage to plastic windows plastic windows will result.,

3. Clean flanges with paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean paint remover residue with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean machinery towel.

REPAIR FOIL TAPE

Repair cuts in foil tape by painting over cut area with sealing compound, Item 287, WP 1803 00.

REPAIR FOIL TAPE - Continued

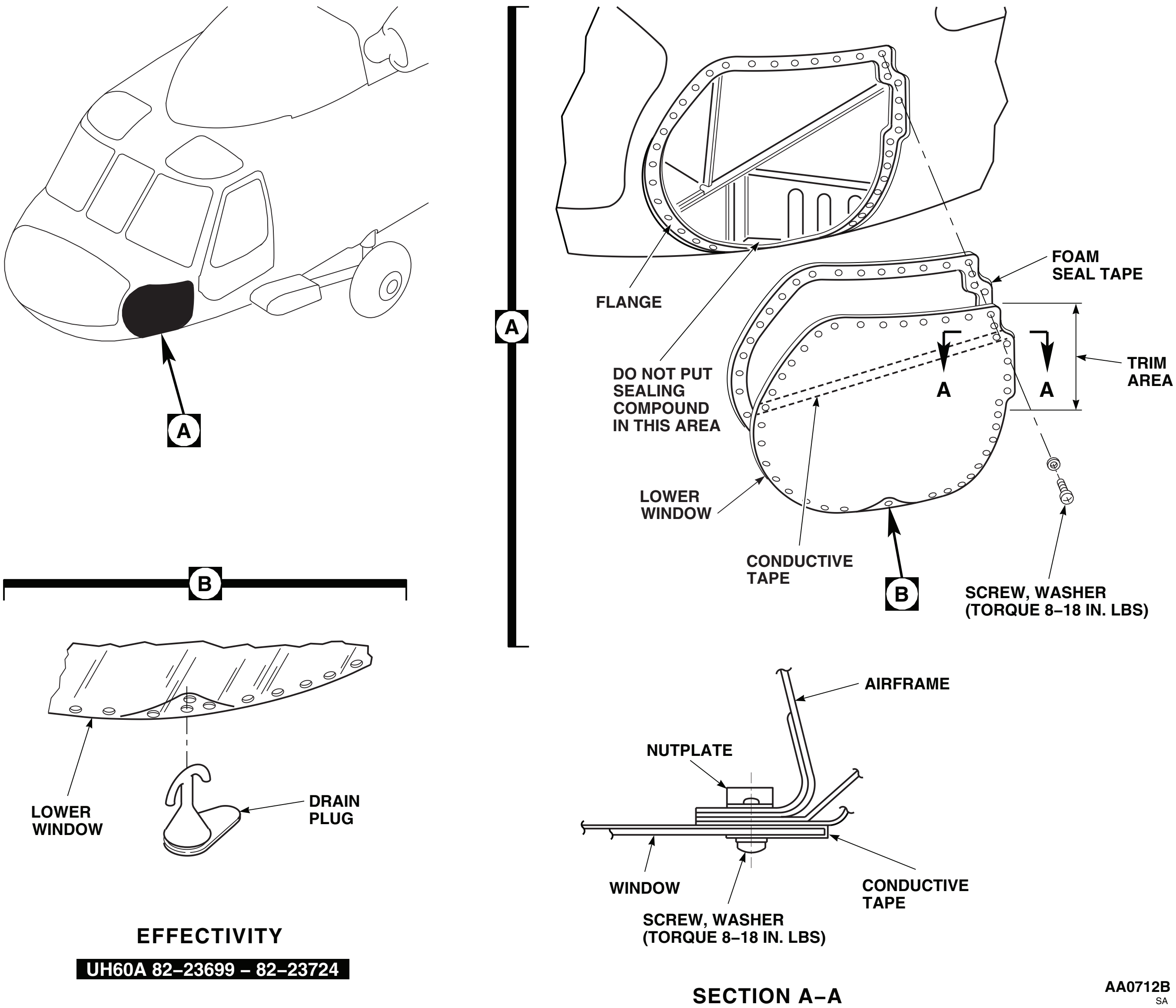


Figure 1. Replace Lower Window.

REPLACE FOIL TAPE

CAUTION

Use extreme care when using tools to remove tape to prevent damage to window.

1. Remove tape, using nonmetallic scraper.
2. Clean tape location with machinery wiping towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 70, WP 1803 00. Wipe dry with clean machinery wiping towel, Item 344, WP 1803 00.
3. Locally make foil tape, (WP 1805 00).
4. Peel backing from new foil tape. Apply foil tape to inside of window, overlapping outside about 1.5-inches centered on mounting holes. Press firmly to make sure there is a good bond.

INSTALL

1. Trial fit replacement window. Trim window edges to fit opening if needed, using procedures in TM 1-1500-204-23.

NOTE

For a better fit, cut and join tape at sharp corners.

2. Apply pressure sensitive adhesive tape, Item 328, WP 1803 00, to frame flanges with sticky surface against flange. Trim tape as necessary.
3. Apply an additional layer of pressure sensitive adhesive tape, Item 329, WP 1803 00, in areas of joggles to insure proper window installation.
4. Place window against flanges. Wrap foil tape around window edge so washer can make good contact with tape.

NOTE

The window installation screws should be gradually tightened in an alternating pattern as required to provide for uniform squeezeout of sealant about 0.100-inch wide. Torque should be at least 8 inch-pounds, but not more than 18 inch-pounds because of the wide range of breakaway torque possible for the anchor nuts. The torquing of all screws to the same value is not recommended as this can result in a waviness of the window edge.

5. **QA** Install screws and washers (Figure 1, Detail A). **TORQUE SCREWS TO 8 - 18 INCH-POUNDS.**
6. Clean the area along the window edges, to be sealed, using a nonmetallic scraper.
7. Fill all gaps around installed window with sealing compound, Item 273, WP 1803 00, making sure surface is smooth and flush.
8. **UH-60A 82-23699 - 82-23724** Install drain plug in hole at bottom of window (Figure 1, Detail B).
9. Check for continuity between foil tape and canopy structure.
10. Check window for water leaks (WP 0229 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

OUTBOARD WINDSHIELDS

This WP supersedes WP 0226 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.
Windshield Shim Kit, 70206-01003-043

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Remover, Paint, Item 250, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 263, WP 1803 00
Tags
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)

References
WP 0144 00
WP 0215 00
WP 0229 00
WP 0379 00
WP 1213 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

NOTE

You do not have to remove glare shield on helicopters with nutplates installed.

2. If necessary, remove glare shield (WP 0215 00).
3. Disconnect upper power/sensor wiring from windshield. Tag wires for proper identification.
4. Pull wiper arm away from windshield. Insert cotter pin in hole in arm to lock blade away from windshield surface. Remove wiper arm, if necessary (WP 1213 00).

NOTE

Screws used to attach the windshield are not all the same length or the same alloy steel. Tag or identify the screws as they are removed.

5. **UH-60A 77-22714 - 80-23415** Remove screws, washers, nuts (if applicable), packings, and windshield from airframe flanges (Figure 1, Detail A). Discard packing.
6. Disconnect lower power/sensor wiring from windshield. Tag wires for proper identification.
7. **UH-60A 80-23416 - SUBQ UH-60L EH-60A** Remove screws, washers, packings, and rain gutter from windshield. Discard packing.
8. **WSPS** Remove rain gutter as follows:
   a. Remove screws, washers, nuts, and wire-deflector insert from channel (Figure 1, Detail C).

REMOVE - Continued

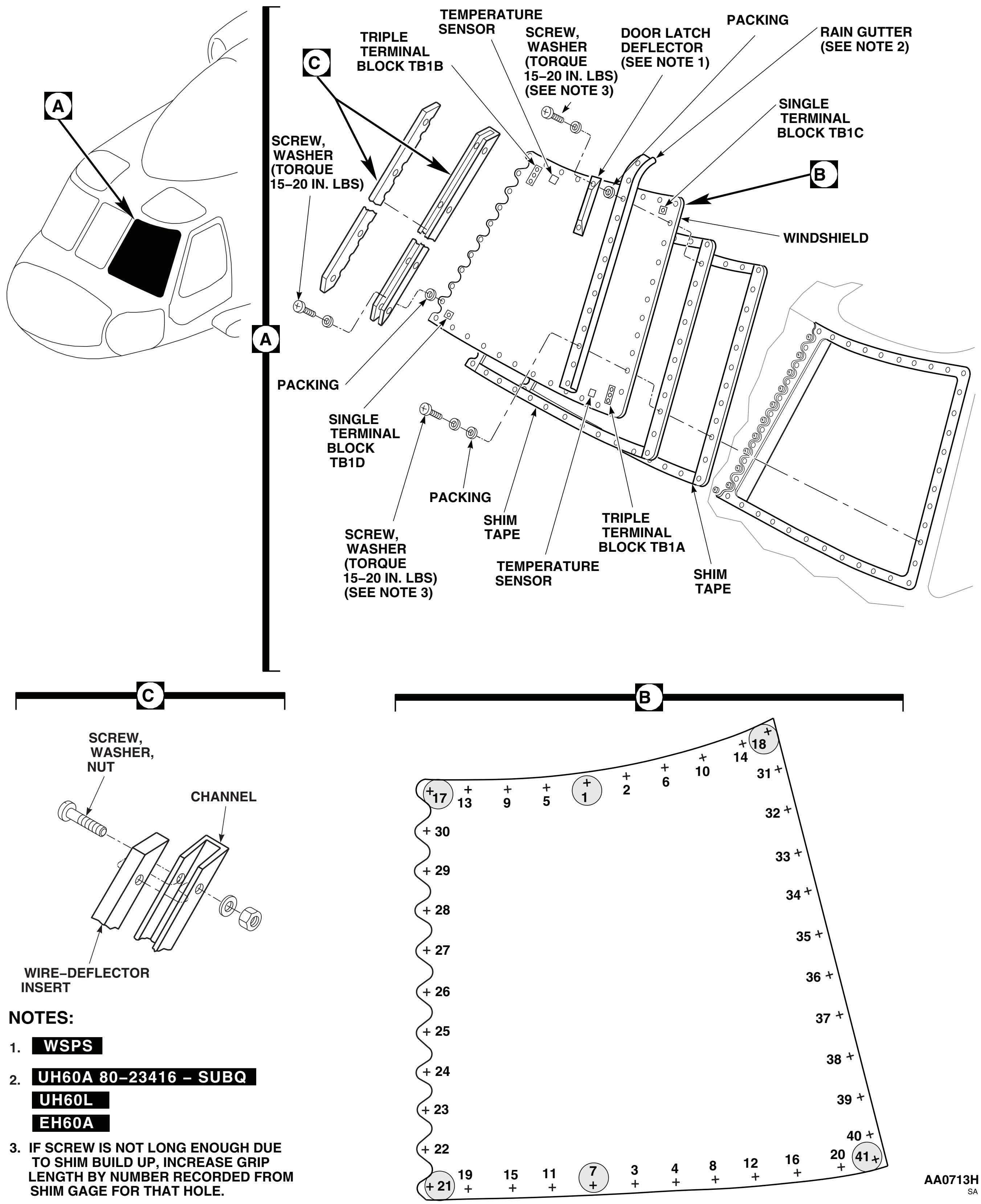


Figure 1. Replace Outboard Windshield.

REMOVE - Continued

b. Remove screws, washers, packings, channel, door-latch deflector, and rain gutter from windshield (Figure 1, Detail A). Discard packing.

c. Lift windshield from frame. Disconnect lower power/sensor wiring from windshield. Tag wires for proper identification.

d. Remove foam seal tape from flanges.

CAUTION

Do not allow paint remover to contact glass or damage to windshield may result.

9. Clean flanges with paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

10. Clean paint remover residue with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean machinery towel.

INSTALL

1. Turn off all electrical power.
2. Trial-fit replacement window in canopy.
3. If holes in window and canopy do not line up, do this:
   a. Mark holes in canopy that do not line up.
   b. Fill holes (WP 0379 00).
   c. Mark and drill new holes, using replacement window as template.
4. Place new windshield on cleaned window opening. Line up windshield with mounting holes (Figure 1, Detail A).

NOTE

Do not install screws in hole 18 and hole 41 at this time.

5. Loosely install screws in hole 1, hole 7, hole 17, and hole 21 (Figure 1, Detail B).
6. Push windshield down on inboard edge at hole 22 through hole 30 until windshield rests flush on canopy.

NOTE

If gage reading is on a line, use next higher number as gap reading.

7. Measure gap at each windshield mounting hole along outboard and bottom edges of windshield using shim gage provided in Windshield Shim Kit. Remove and reinstall screws at hole 21 and hole 7 for measurement.
8. Record number from gage on windshield shim record sheet provided in Windshield Shim Kit.
9. Release windshield and move screws from hole 17 and hole 21 to hole 18 and hole 41.
10. Push windshield down on outboard edge at hole 31 and hole 40 until windshield rests on canopy.

INSTALL - Continued

NOTE

If gage reading is on a line, use next higher number as gap reading.

11. Measure gap at each windshield mounting hole along top edge and center post section of windshield using shim gage provided in Windshield Shim Kit. Remove and reinstall screws at hole 1 and hole 18 for measurement.
12. Record number from gage on windshield shim record sheet provided in Windshield Shim Kit.
13. Release windshield, remove screws and remove windshield from helicopter.

NOTE

- If center windshield is installed, trim shim tape to fit openings at center post.
- 1/8-inch shim tape may be substituted for 1/16-inch shim tape by using 1 layer of 1/8-inch for each 2 layers of 1/16-inch shim tape.
- Use 3 inch hole spacing shim tape for top edge of windshield.
- Use 3.125-inch hole spacing shim tape for other edges of windshield.
- Stretch, compress, or cut shim tape as required to line up holes.

14. Install shim tape as follows:
    a. Install one layer of shim tape around canopy opening.

NOTE

Number recorded from gage readings equals number of layers of shim tape including first layer.

    b. Using shim tape, build up layers equal to number on shim record sheet recorded during step 8. and step 12.
15. **QA** Position windshield on canopy. Install screws, washers, and packings loosely to position windshield.

NOTE

- Do not install rain gutter or wire strike at this time.
- If any screw is not long enough due to shim tape build up, increase grip length by number recorded from shim gage for that hole.

16. Using sequence numbers shown, tighten all screws, washers, and packings in 5 inch-pound increments, **NOT TO EXCEED 20 INCH-POUNDS**, until even contact between windshield and shim tape is achieved (Figure 1, Detail B).
17. **QA** After even contact between windshield and shim tape is achieved for the entire pattern using same sequence, **RETORQUE ALL SCREWS TO 15-20 INCH-POUNDS.**
18. Fill all exterior gaps and edges around windshield with sealing compound, Item 273, WP 1803 00 making sure surface is smooth and flush.
19. Allow sealing compound to dry.
20. Carefully remove fasteners required to install rain gutter and wire strike deflectors.
21. **QA** Install heater wire, H4302A14C (pilot's windshield), or H4307A14C (copilot's windshield), on lower center terminal board, TB1D-6, with screw, lockwasher, and washer.

INSTALL - Continued

22. Check insulating boots on windshield deice wire ends for cracks or tears. **NONE ALLOWED.** Replace boots if cracked or torn.

NOTE

- **UH-60A 77-22714 - 80-23415 UH-60L** Rain gutters installed.
- **WSPS** Wire deflectors installed.

23. If rain gutter and wire deflectors are being installed, do this:
    a. Brush mating surfaces of rain gutter, door latch deflector, and wire deflector channel with sealing compound, Item 273, WP 1803 00 (Figure 1, Detail C).
    b. Install door latch deflector and rain gutter on outboard side of window with screws, washers, and packings. Place packings between channel and windshield (Figure 1, Detail A).
    c. **QA** Install wire deflector channel on inboard side of windshield with screws, washers, and packings.
24. Brush mating surface of wire deflector insert with sealing compound, Item 273, WP 1803 00. Install insert in channel with screws, washers, and nuts (Figure 1, Detail C).
25. Fill all gaps around installed windshield with additional sealing compound, Item 263, WP 1803 00, making sure surface is smooth and flush.
26. If removed, install glare shield (WP 0215 00).
27. Remove cotter pin from wiper arm or if removed install wiper arm (WP 1213 00).
28. Check windshield for water leaks (WP 0229 00).
29. Connect power/sensor wiring to pilot's windshield terminals as follows:
    a. **QA** H4301B18B to TB1A-1
    b. **QA** H4305B20 to TB1A-4
    c. **QA** H4304B20 to TB1A-3
    d. **QA** H4300A14A to TB1C-2
    e. **QA** H4300A14B to TB1B-5
    f. **QA** **UH-60A 77-22714 - 77-22719** Wire No. H305A24 is connected to TB1B-7, Wire No. H4304A24 is connected to TB1B-8.
    g. **QA** H4305B20 to TB1B-7
    h. **QA** H4304A20 to TB1B-8
30. Connect power/sensor wiring to copilot's windshield terminals as follows:
    a. **QA** H4308B18B to TB1A-1
    b. **QA** H4310B20 to TB1A-4
    c. **QA** H4311B20 to TB1A-3
    d. **QA** H4309A14C to TB1C-2
    e. **QA** H4308A14B to TB1B-5
    f. **QA** **UH-60A 77-22714 - 77-22719** Wire No. H4310A24 is connected to TB1B-7 and Wire No. H4311A24 is connected to TB1B-8.

INSTALL - Continued

g. **QA** H4310A20 to TB1B-7

h. **QA** H4310B20 to TB1B-7

i. **QA** **UH-60A 77-22714 - 77-22719** Wire No. H4310A24 is connected to TB1B-7 and Wire No. H4311A24 is connected to TB1B-8.

j. **QA** H4311A20 to TB1B-8

31. Do an operational check of windshield anti-ice (WP 0144 00).

32. Remove wire location tags.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CENTER WINDSHIELD (GLASS)

This WP supersedes WP 0227 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Remover, Paint, Item 250, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 263, WP 1803 00
Tags
Tape, Pressure Sensitive, Adhesive, Item 328, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)

References
WP 0144 00
WP 0215 00
WP 0229 00
WP 0796 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

NOTE

The plastic center windshield may be replaced by heated glass center windshield.

2. **W/O HCW** To replace plastic center windshield do this:
   a. Remove free-air thermometer from windshield (WP 0796 00).

   NOTE

   It is not necessary to remove glare shield on helicopters with nutplates installed.

   b. If necessary, remove glare shield (WP 0215 00).
   c. Remove screws, washers, packings (if installed), nut, and bonding jumper from windshield (Figure 1, Detail A). Discard packing.
3. **HCW** To replace heated glass center windshield do this:
   a. Turn off all electrical power.
   b. Disconnect upper power/sensor wiring from windshield. Tag wires for proper identification.
   c. Remove screws, washers, and packings from windshield. Discard packing.
4. **WSPS** Remove screws, washers, nuts, packings, wire deflector inserts, and channels from windshield. Discard packing.
5. Lift windshield from frame.
6. Disconnect lower power/sensor wiring from windshield.

REMOVE - Continued

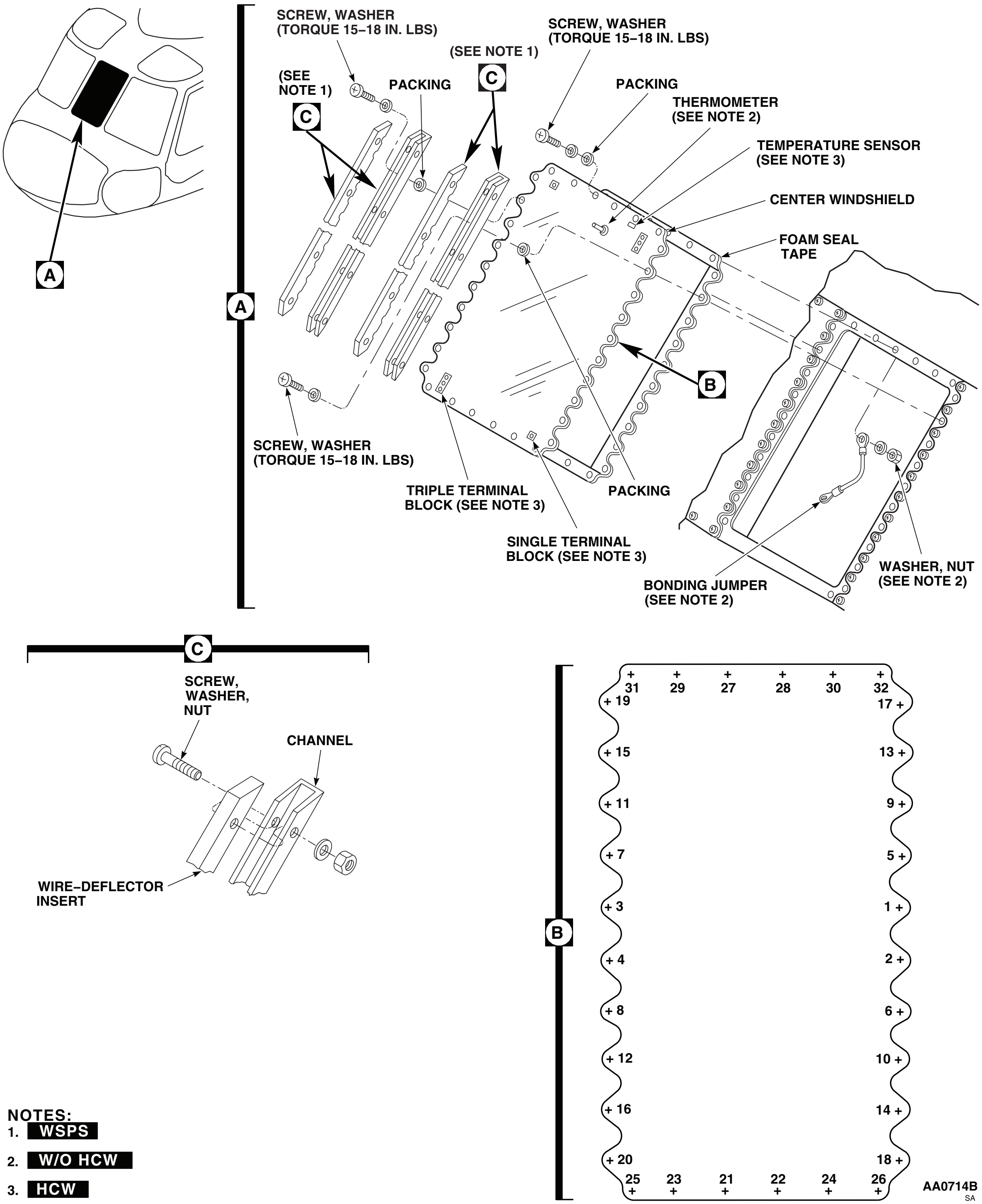


Figure 1. Replace Center Windshield.

REMOVE - Continued

7. Remove foam seal tape from flanges.

CAUTION

Do not allow paint remover to contact glass or damage to windshield will result.

8. Clean flanges with paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. Clean paint remover residue with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean machinery towel.

INSTALL

1. Turn off all electrical power.

NOTE

For a better fit, cut and join tape at sharp corners.

2. Apply pressure sensitive adhesive tape, Item 328, WP 1803 00, to frame flanges with sticky surface against flange.
3. Place windshield up to frame.

NOTE

Windshield attaching screws should be installed in the sequence shown in (Figure 1, Detail B).

4. **W/O HCW** Install windshield as follows:
   a. Install windshield with screws, washers, and packings. Do not tighten screws.
   b. **QA** In position 27 use screw, washer, packing, and nut. Install bonding jumper under nut.
5. **HCW** Install windshield as follows:
   a. **QA** Connect lower power/sensor wiring to windshield.
   b. Install windshield with screws, washers, and packings. Do not tighten screws.
6. **WSPS** Install windshield as follows:
   a. Apply light coat of sealing compound, Item 273, WP 1803 00, to mating edges of wire-deflector channels.
   b. Install channels on left and right edges of windshield with screws, washers, and packings. Install packings between washers and channels. Do not tighten screws.
   c. Loosely install remaining screws, washers and packings. Do not tighten screws.
7. Using sequence numbers shown in Figure 1, Detail B, tighten all screws just enough for even contact between windshield and sealer.
8. **QA USING SAME SEQUENCE, TORQUE ALL SCREWS TO 15 - 18 INCH-POUNDS.**

INSTALL - Continued

9. Fill all gaps around installed windshield with additional sealing compound, Item 263, WP 1803 00, making sure surface is smooth and flush.
10. **W/O HCW** Install free-air thermometer (WP 0796 00).
11. **QA** **UH-60A 80-23416 - SUBQ** Install wire-deflector inserts in channels using screws, washers, and nuts.
12. Check windshield for water leaks (WP 0229 00).
13. **HCW** Connect wires as follows:
    a. **QA** Connect upper power/sensor wiring to windshield. Remove wire locations tags.
    b. Do an operational check of windshield anti-ice (WP 0144 00).
14. If removed, install glare shield (WP 0215 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CENTER WINDSHIELD (PLASTIC)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Low-Resistance Ohmmeter, DLRO247000

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
WP 0227 00
WP 0796 00

NOTE

First time installation of glass center windshield requires installation of one free-air thermometer in each upper window (WP 0796 00).

REMOVE

NOTE

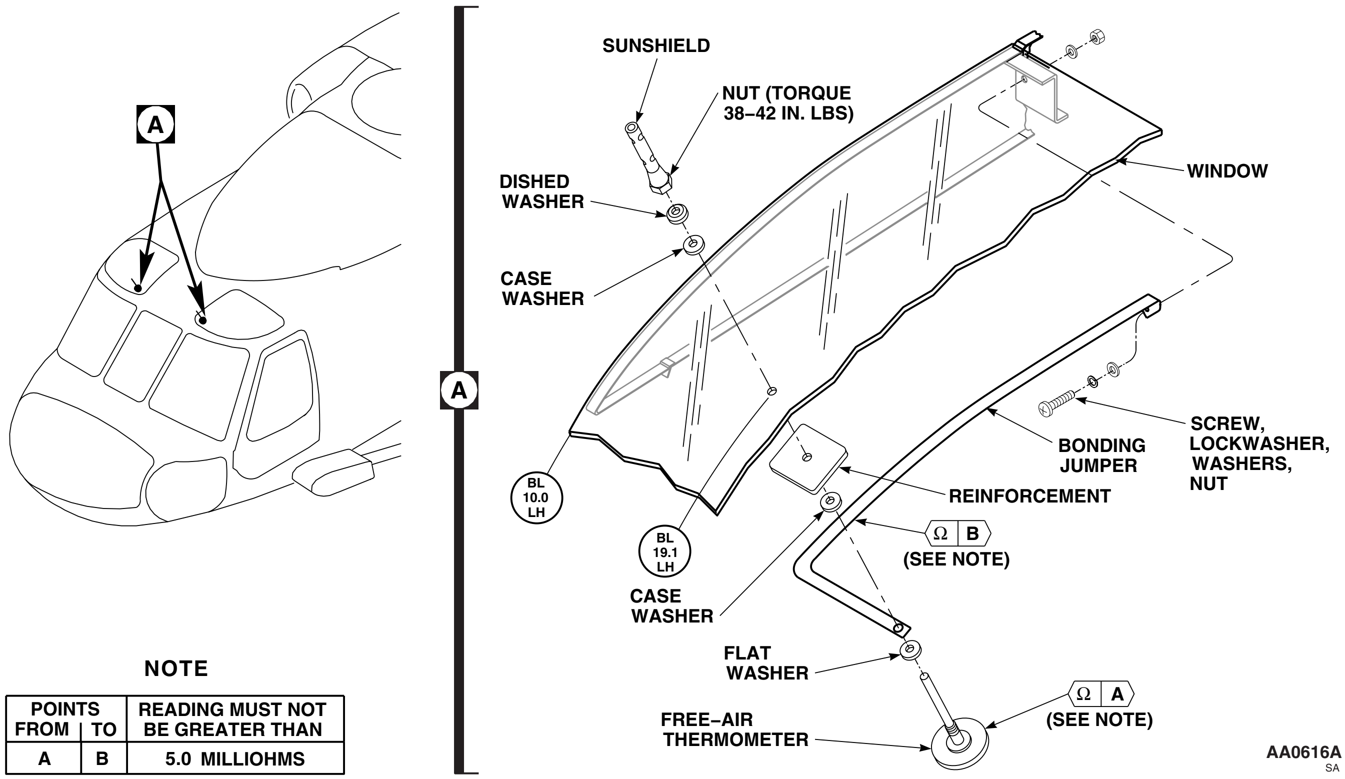
The plastic center windshield may be replaced by heated glass center windshield.

1. Turn off all electrical power.
2. Remove free-air thermometer from center windshield (WP 0796 00). Retain thermometer and hardware for reinstallation (Figure 1, Detail A).
3. Remove center windshield (WP 0227 00).

INSTALL

1. Turn off all electrical power.
2. Install heated center windshield (WP 0227 00).
3. **QA** Drill 0.455 to 0.460-inch diameter hole in each upper window, centered 3 inches from inboard side and front window edge (TM 1-1500-204-23).
4. **QA** Install bonding jumper by drilling out existing rivets and riveting end of jumper to canopy (WP 0796 00) (Figure 1, Detail A).
5. Install free-air thermometer (WP 0796 00).

INSTALL - Continued



NOTE

POINTS FROM	POINTS TO	READING MUST NOT BE GREATER THAN
A	B	5.0 MILLIOHMS

Figure 1. Replace Plastic Center Windshield.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WINDSHIELD AND WINDOW LEAK TEST

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02
Garden Hose, L-H-520
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Sealing Compound, Item 245 , WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 328, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)

References
WP 0224 00
WP 0225 00
WP 0226 00
WP 0227 00
WP 0228 00
WP 1803 00

WATER LEAK TEST AND CORRECTIVE ACTION

1. Turn off all electrical power.
2. Spray water from garden hose on cockpit exterior windows and windshields. Concentrate spray on sealed edges. Have assistant in cockpit look for leakage. Should leakage be found, do this:
   a. Check screw torque in leakage area. **TORQUE SCREWS TO 15 - 18 INCH-POUNDS.**
   b. Spray area where leakage was found. If leakage continues, dry area and apply sealing compound, Item 273, WP 1803 00, to leaky area.
3. Spray area again. If leakage still continues, remove window panel or windshield. Inspect frame for damage. If frame is not damaged, replace pressure sensitive adhesive tape, Item 328, WP 1803 00, and sealing compound, Item 273, WP 1803 00, and reinstall window (WP 0224 00, WP 0225 00, WP 0226 00, WP 0227 00, or WP 0228 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT FLOOR

This WP supersedes WP 0230 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 31, WP 1803 00
Fastener Tape, Hook, Item 124, WP 1803 00
Fastener Tape, Pile, Item 125, WP 1803 00
Thread, Item 340, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0388 00
WP 1803 00

REPLACE/REPAIR FLOOR COVER

Remove

NOTE

Repair/replace procedures are the same for both pilot's and copilot's floor covers.

1. Start at sides of floor cover and gently pull up on cover until pile fastener tape is free of hook fastener tape.
2. Remove cover.

Repair

1. Remove damaged hook fastener tape from floor and front cover, using a nonmetallic scraper (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Using a machinery wiping towel, Item 344, WP 1803 00, dampened in technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper, remove remaining adhesive from floor and front cover.
3. Within 15 minutes after cleaning, apply coat of adhesive, Item 31, WP 1803 00, to hook fastener tape, Item 124, WP 1803 00, floor, and front cover with putty knife.
4. After adhesive becomes tacky, press hook fastener tape, Item 124, WP 1803 00, to floor and front cover. Allow adhesive to cure for 16 hours before installing floor cover.
5. Repair or install pile fastener tape, Item 125, WP 1803 00, on floor cover by sewing with thread, Item 340, WP 1803 00.

REPLACE/REPAIR FLOOR COVER - Continued

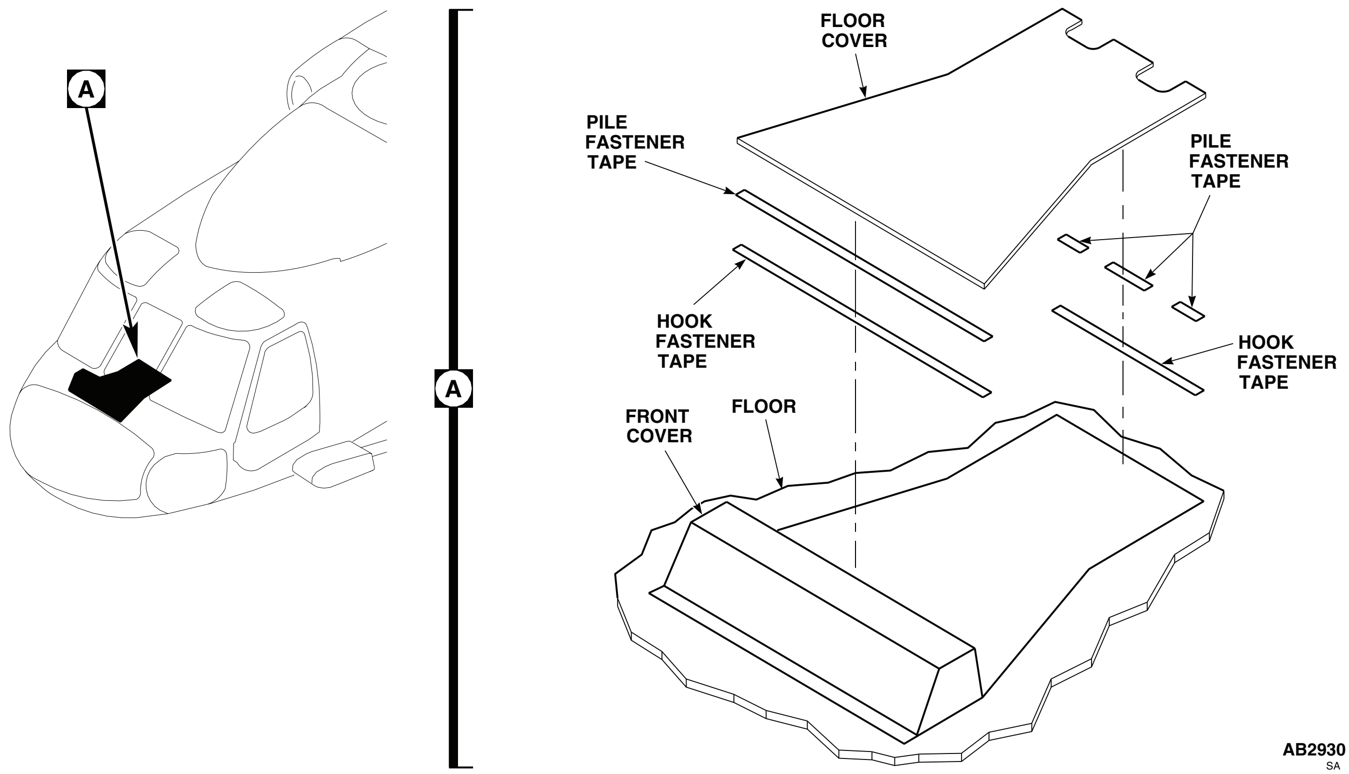


Figure 1. Replace/Repair Floor Cover.

Install

1. Position pile fastener tape, Item 125, WP 1803 00, over hook fastener tape, Item 124, WP 1803 00, on front cover and gently press down on cover (Figure 1, Detail A). Make sure pile fastener tape makes good contact with hook fastener tape.
2. Repeat procedure for hook fastener tape on floor and pile fastener tape on floor cover.

REPAIR COCKPIT FLOOR

See WP 0388 00 for repair instructions.

REPLACE/REPAIR ACCESS PANEL

Remove

NOTE

Repair/replace procedures are the same for both pilot's and copilot's access panels.

1. Remove screws and washers securing access panel to cockpit floor (Figure 2, Detail A).
2. Remove access panel.

Repair

Inspect panels for corrosion, if required treat for corrosion.

REPLACE/REPAIR ACCESS PANEL - Continued

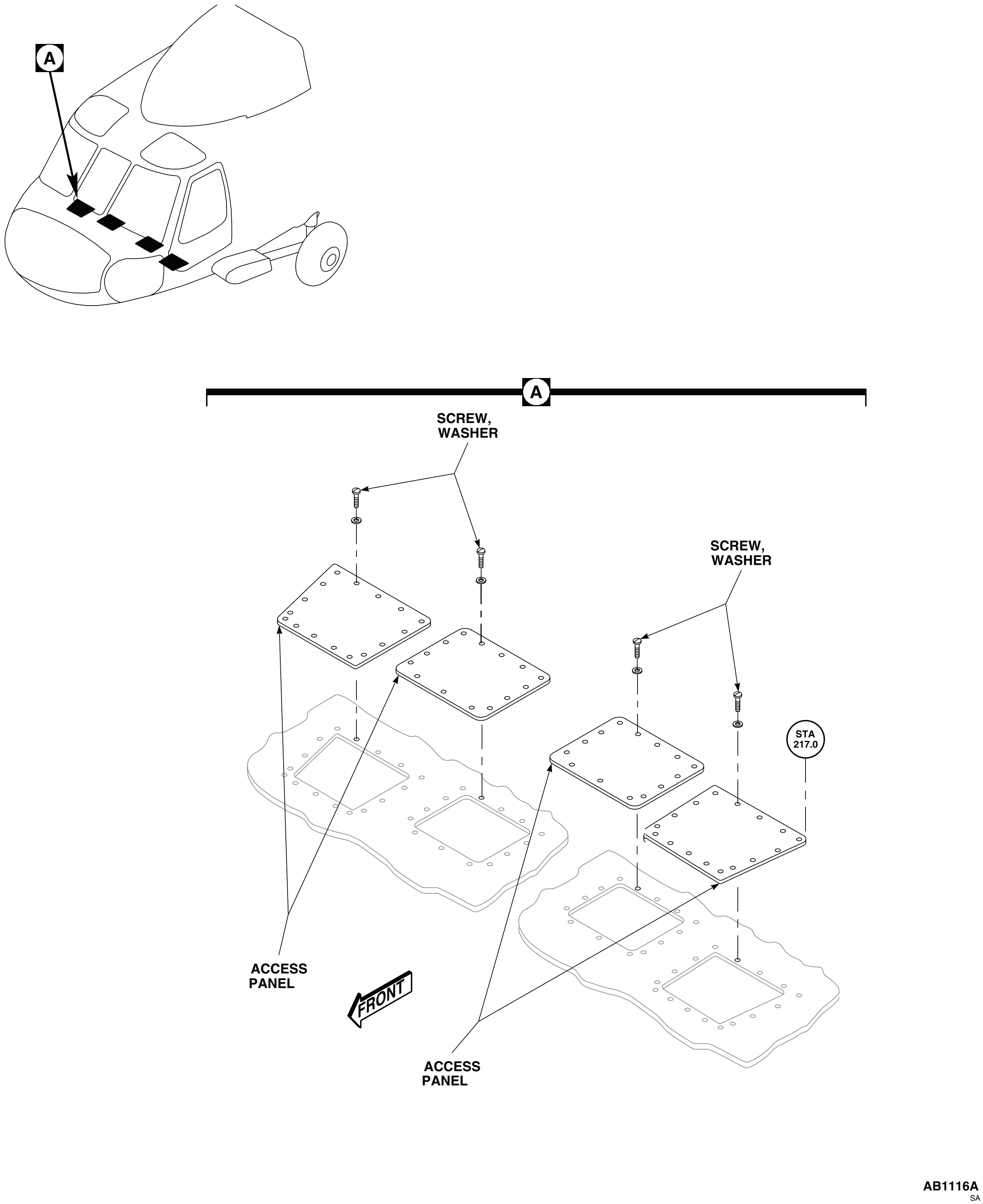


Figure 2. Replace/Repair Access Panels.

REPLACE/REPAIR ACCESS PANEL - Continued

Install

1. Position access panel on cockpit floor.
2. **QA** Secure access panel to cockpit floor using screws and washers (Figure 2, Detail A).
3. Inspect component security and make sure area is clean and free of foreign objects.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

PILOT'S/COPILOT'S SEATS, RA 30525-1, RA 100900-1, RA 100900-3, AND RA 100900-5

This WP supersedes WP 0231 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Die and Tap Set, GGG-T-330
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Heat Gun, MIL-H-45193
Nonmetallic Scraper
Spring Scale, AAA-S-133
Stabilizer Strut, Locally-Made, Figure 182, WP 1805 00
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 15, WP 1803 00
Adhesive, Item 16, WP 1803 00
Adhesive, Item 30, WP 1803 00
Adhesive, Item 45, WP 1803 00
Adhesive, Item 47, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cleaning Compound, Solvent, Item 72, WP 1803 00
Etching Solution, Item 121, WP 1803 00
Fastener Tape, Pile, Item 125, WP 1803 00
Paper, Abrasive, Item 210, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 283, WP 1803 00
Sealing Compound, Item 285, WP 1803 00
Sealing Compound, Item 291, WP 1803 00
Strap, Tiedown, Electrical, Item 302, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 11-1520-237-23
TM 55-1500-322-24
TM 55-1500-345-23
WP 0171 00
WP 0233 00
WP 0378 00
WP 0379 00
WP 0381 00
WP 1589 00
WP 1703 00
WP 1803 00
WP 1805 00

REPLACE INERTIA REEL

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up and full front.

REPLACE INERTIA REEL - Continued

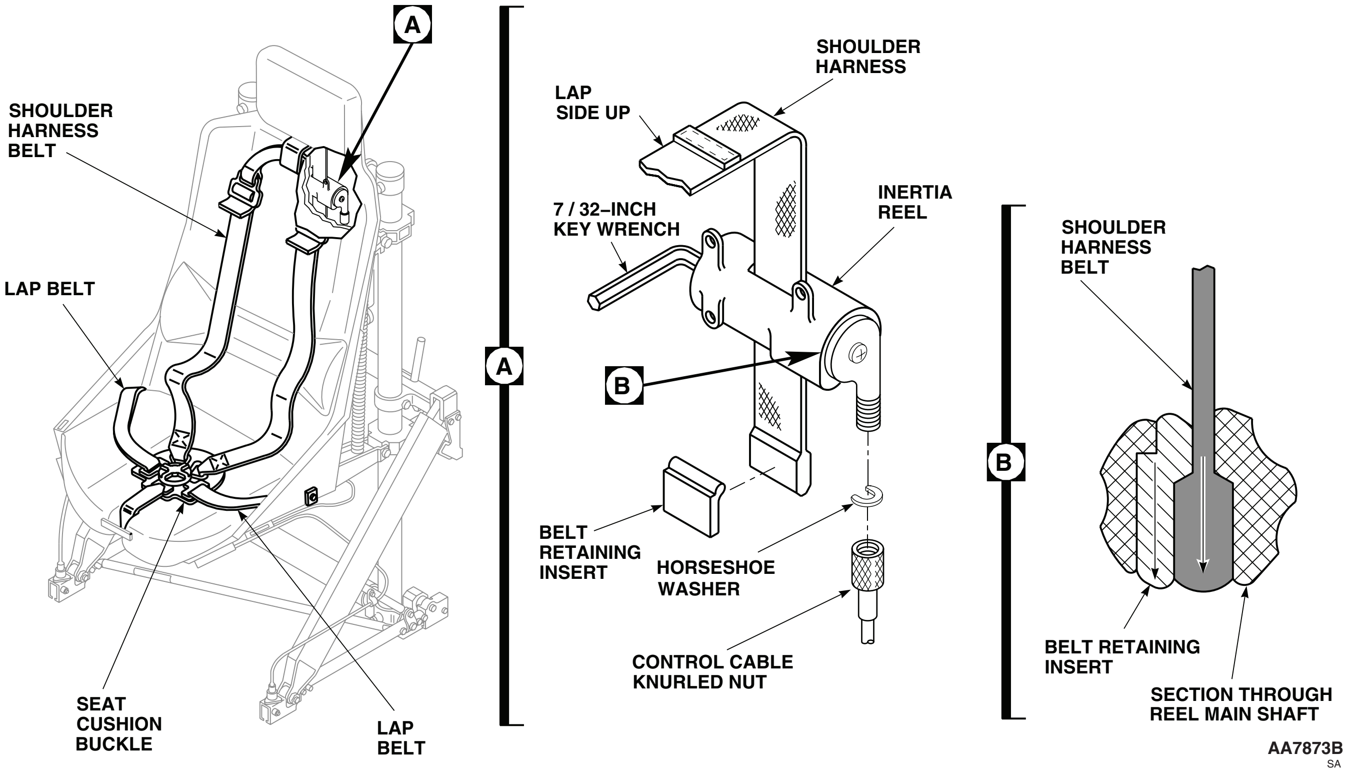


Figure 1. Replace Inertia Reel.

2. Remove seat back cushions to get to inertia reel attaching screws. Refer to, REPLACE SEAT CUSHIONS, in this work package.
3. Set inertia reel control handle in AUTO-LOCK (rear) (Figure 1, Detail A).
4. Remove screws, washers, and nuts, and pull inertia reel from back of seat.

CAUTION

To prevent damage to inertia reel due to uncontrolled release of spring tension, do not allow key wrench to become disengaged from reel shaft until spring has been unwound.

5. Hold inertia reel and pull shoulder harness belt fully out. Place 7/32-inch hex wrench in hex hole on end of reel housing, and hold reel shaft from turning.
6. Disconnect knurled nut from inertia reel.
7. Set inertia reel control handle in MANUAL LOCK (front).
8. Remove control handle cable from inertia reel.
9. Push belt retaining insert and belt, down and out of reel shaft and lower slot.
10. Remove belt retaining insert and pull belt from reel shaft.
11. Reinstall belt retaining insert 90 degrees from normal installed position. Slowly back off key wrench until power spring holds insert.

REPLACE INERTIA REEL - Continued

Install

CAUTION

Replacement reel is spring-wound. The spring is pre-wound by manufacturer. The belt retaining insert is installed 90 degrees from normal installation for shipping purposes only.

1. Install key in hex hole on end of replacement inertia reel housing. Rotate key wrench counterclockwise just enough to remove belt retaining insert (Figure 1, Detail A).

NOTE

Be sure lap belt seam of shoulder harness belt is up before installing.

2. While securing inertia reel, slip end of shoulder harness belt through slot in shaft and through bottom opening in housing.
3. Install belt retaining insert on end of belt and pull up into reel shaft. Keep tension on belt and remove key wrench from shaft. Allow belt to slowly wind on shaft until fully retracted.
4. **QA** Install inertia reel on seat back and secure with screws, washers, and nuts. **TORQUE NUTS TO 20 - 25 INCH-POUNDS.**
5. **QA** Connect control cable and knurled nut to inertia reel (Figure 1, Detail A) .
6. Replace seat back cushion. Refer to, REPLACE SEAT CUSHIONS, in this work package.

REPLACE INERTIA REEL CONTROL

Remove

1. Disconnect cable from inertia reel control by loosening cable knurled nut (Figure 2, Detail A).
2. Remove screws and washers securing inertia reel control. Remove inertia reel control from seat bucket (Figure 2, Detail A).
3. Remove horseshoe washer securing cable to inertia reel control.
4. Remove cable from inertia reel control.

Install

1. **QA** Position replacement inertia reel control on bracket (Figure 2, Detail A). Put two washers, as spacers, between control housing and bracket at each mounting hole. Attach control to bracket with screws and washers.
2. **QA** Connect control cable to inertia reel control with knurled nut (Figure 3, Detail A).
   a. Install cable to inertia reel control.
   b. Install horseshoe washer securing cable to inertia reel control.
3. Check operation of control cable and inertia reel control.

REPLACE INERTIA REEL CONTROL - Continued

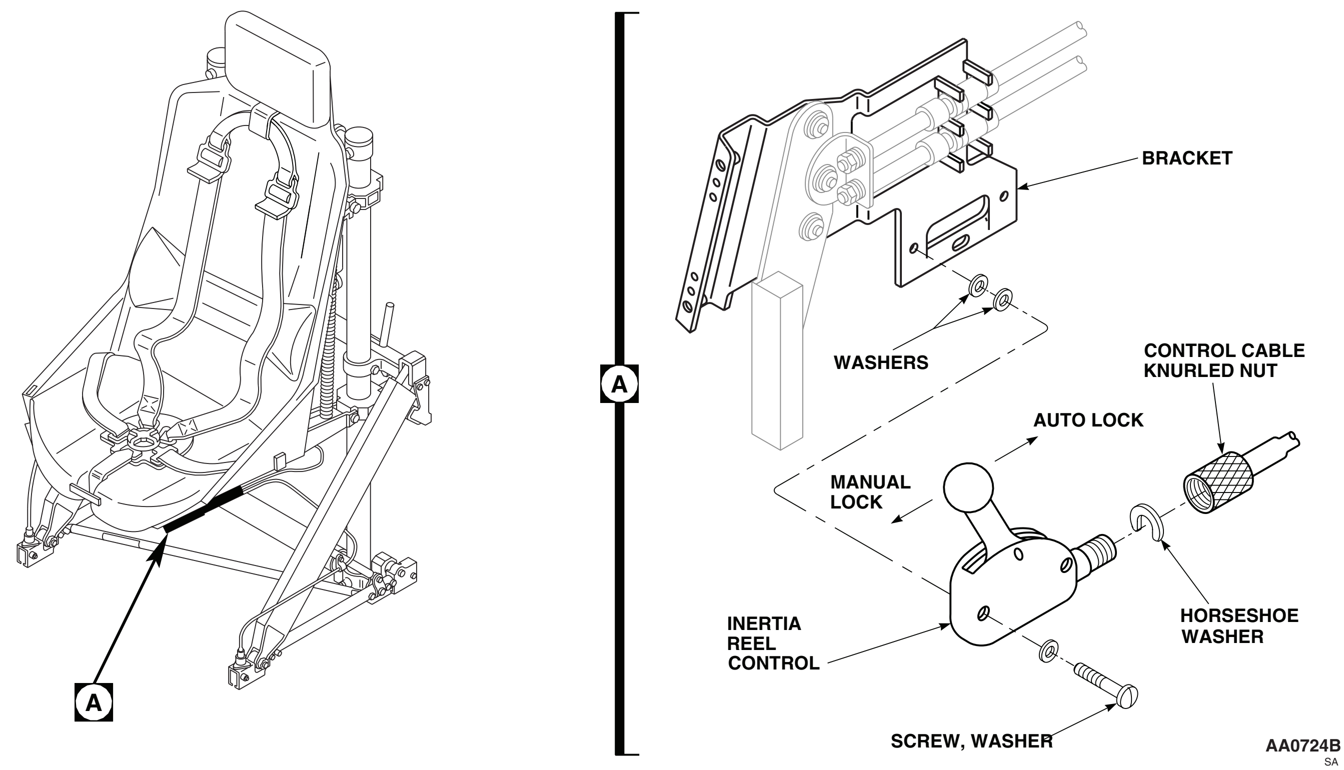


Figure 2. Replace Inertia Reel Control.

REPLACE SHOULDER HARNESS

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

Remove

1. Release shoulder harness from rotary buckle.
2. Set inertia reel control in AUTO-LOCK (rear).

CAUTION

To prevent damage to inertia reel due to uncontrolled release of spring tension, do not allow key wrench to become disengaged from reel shaft until harness is replaced.

3. Pull shoulder harness belt all the way out. Place 7/32-inch hex wrench in hex hole on end of reel housing, and hold reel shaft from turning (Figure 3, Detail A).

REPLACE SHOULDER HARNESS - Continued

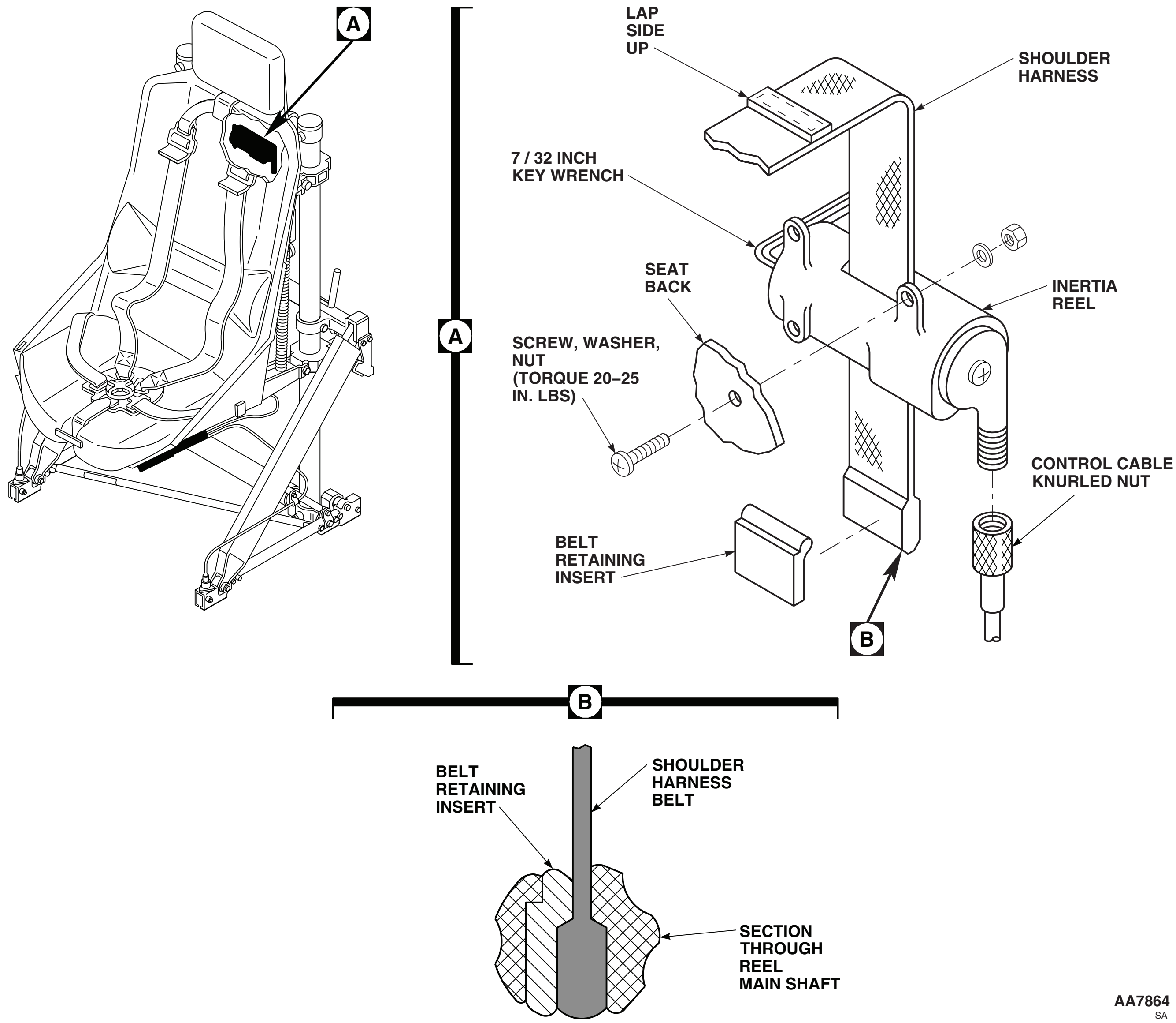


Figure 3. Replace Shoulder Harness.

NOTE

Newer styles of inertia reels do not have lower slots. In these cases inertia reel must be completely removed from seat back.

4. Push belt retaining insert and belt down and out of reel shaft and lower slot (Figure 3, Detail B).
5. Remove belt retaining insert, pull belt from reel shaft and through belt slot in seat back.

Install

1. Insert replacement shoulder harness through belt slot in seat back with overlapping side of belt up.

REPLACE SHOULDER HARNESS - Continued

NOTE

On ARA seat P/N D-3801 belt is to be routed through head rest guide.

2. Insert belt end through upper slot in reel housing and through slot in shaft until belt end sticks out through lower slot.
3. Install belt retaining insert on belt end. Pull belt up to seat insert and belt in reel main shaft. Maintain tension on belt.
4. Remove hex wrench from reel housing. Release belt tension and let belt rewind on main shaft.
5. **QA** Check that shoulder harness end fittings lock when engaged in rotary buckle.
6. Check operation of inertia reel.

REPLACE LAP BELT

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

Remove

1. Release lap belts from rotary buckle.
2. Lift seat cushion to get to attaching hardware. Remove flathead bolt, washers, and nut and remove lap belts from seat (Figure 4, Detail A).

Install

WARNING

Pilot and copilot seat lap belts must be compatible with the restraint buckles. Mismatched lap belt fittings and restraint buckles may degrade the integrity of the belt fittings attachment into the buckle. If installing a lap belt, confirm the lap belt fitting is correct for the restraint buckle.

1. **QA** Install replacement lap belt fittings on inside of seat and fasten with flathead screws, washers, and nuts (Figure 4, Detail A). Install screws from outside of seat, and a washer on each side of fittings.
2. **QA** Tighten nut, then back off one turn to allow free rotation of belt fitting.
3. Install seat cushion.
4. Engage lap belt in rotary buckle.

REPLACE LAP BELT - Continued

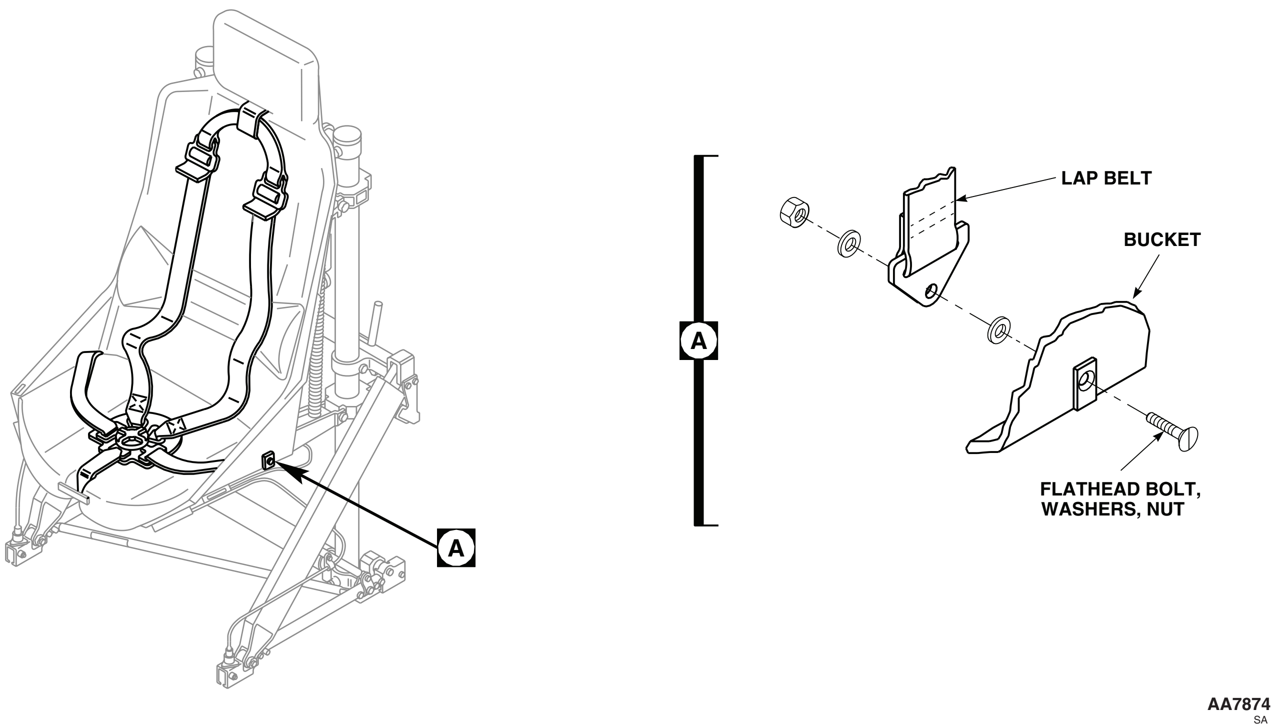


Figure 4. Replace Lap Belt.

REPLACE CROTCH BELT, BUCKLE, AND PAD

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

Remove

1. Release shoulder harness and lap belts from rotary buckle.
2. Remove screws, washers, and nuts securing crotch belt fitting to front of seat. Slide belt up through loop on seat cushion and remove crotch belt and buckle (Figure 5, Detail A).
3. Open buckle pad and remove screws and pad plate from rotary buckle (Figure 5, Detail B).

REPLACE CROTCH BELT, BUCKLE, AND PAD - Continued

Install

WARNING

Pilot and copilot seat restraint buckles must have same type of release mechanism. Different types of release mechanism could hinder an emergency egression. If installing a replacement buckle, make sure that both pilot and copilot seat restraint buckles have same type of release mechanism.

1. **QA** Place pad on replacement rotary buckle, install pad plate inside pad, and secure to rotary buckle with screws (Figure 5, Detail B).
2. **QA** Slide crotch belt down through loop on seat cushion and fasten belt fitting to front of seat with screws, washers, and nuts (Figure 5, Detail A). **TORQUE NUTS TO 20 - 25 INCH-POUNDS.**
3. Engage shoulder harness and lap belts in rotary buckle and check rotary buckle for correct latching and unlatching operation. Refer to, REPLACE CROTCH BELT, BUCKLE, AND PAD, in this work package.

REPLACE SEAT CUSHIONS

Remove

NOTE

Both seat, back and headrest cushions are removed and installed in the same manner.

Starting at corner of cushion, pull cushion from seat bucket.

Clean/Repair

1. Clean and/or repair seat cushions (TM 1-1500-204-23). Replace or reglue fastener tape, as required.
2. Install 2 inch wide pile fastener tape, Item 125, WP 1803 00, on seat bucket as required using adhesive, Item 30, WP 1803 00 (Figure 6).
3. Cut straps attaching lumbar support to seat back cushion.
4. Sear ends of straps to prevent unraveling.
5. Repair lumbar support (TM 1-1500-204-23).
6. Secure lumbar support to fastener tape strips on seat cushion.

Install

Install cushions in seat bucket by lining up and pressing down on cushion in area of fastener tape.

REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust full up.

REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE - Continued

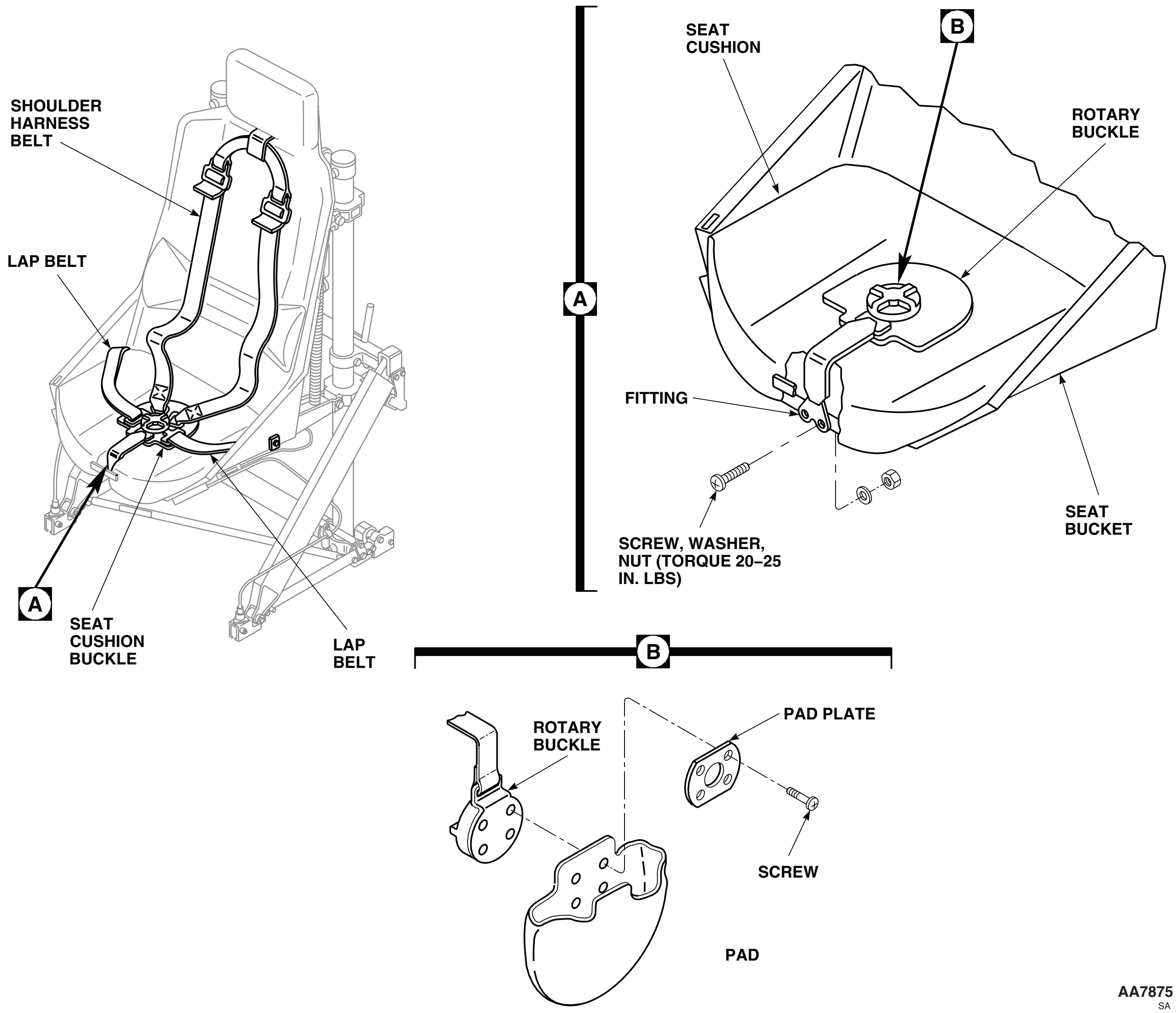


Figure 5. Replace Crotch Belt, Buckle, and Pad.

2. Remove inertia reel control from adjust baseplate. Refer to, REPLACE INERTIA REEL CONTROL, in this work package.
3. Remove bolts attaching lockpin housing to forward track fitting (Figure 7, Sheet 1, Detail A).
4. Remove screw, washer, and nut attaching cable and clamp to diagonal strut. Remove clamp and install screw, washer, and nut until ready to replace cable and clamp.
5. On seat, remove nuts and washers attaching release cables to bracket on release handle (Figure 7, Sheet 1, Detail B).
6. Remove screw, shaft clamp, and release cables from adjust handle baseplate. Remove cable and lockpin housing from helicopter.
7. Remove seat cushion.

REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE - Continued

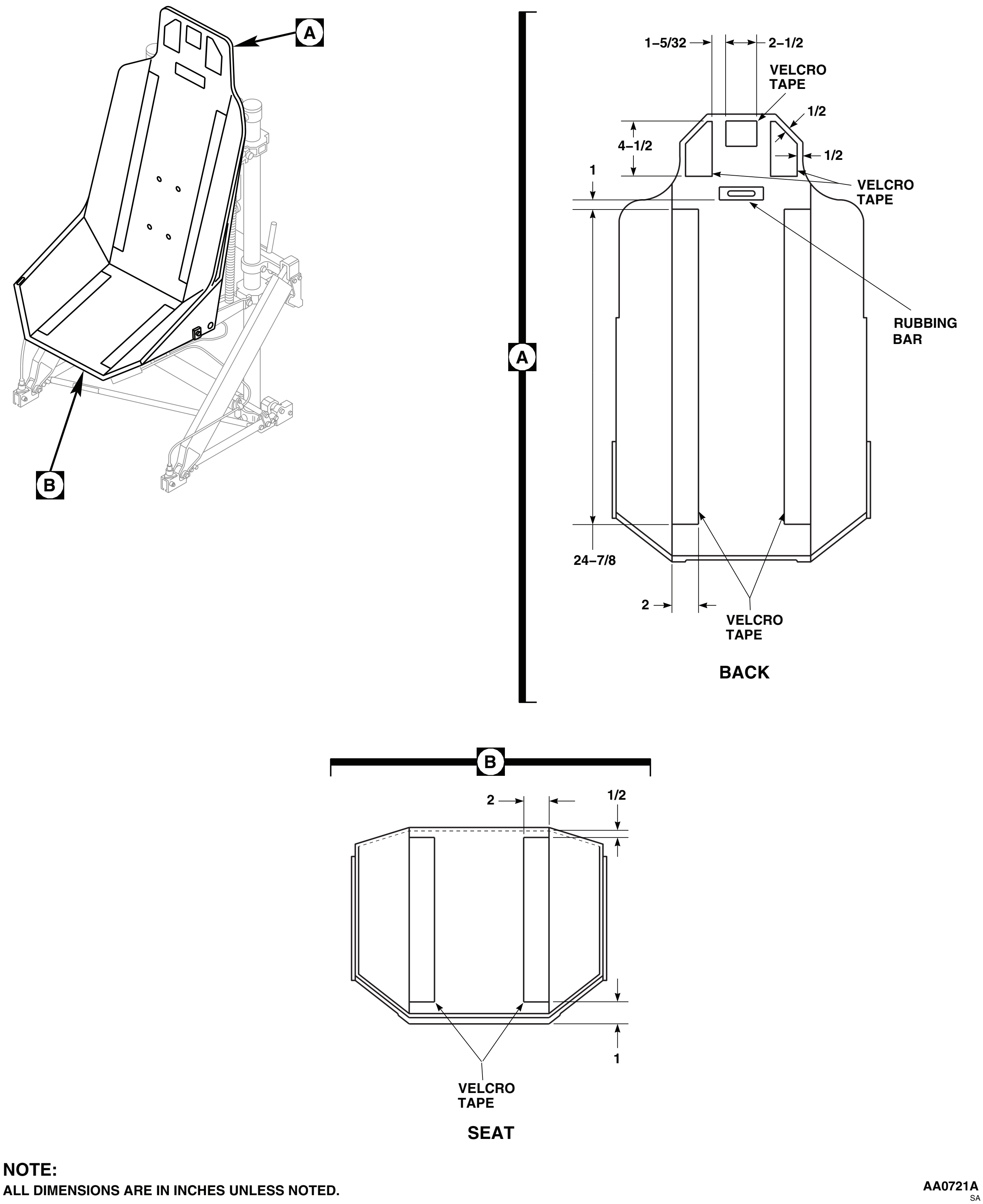


Figure 6. Seat Cushions.

REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE - Continued

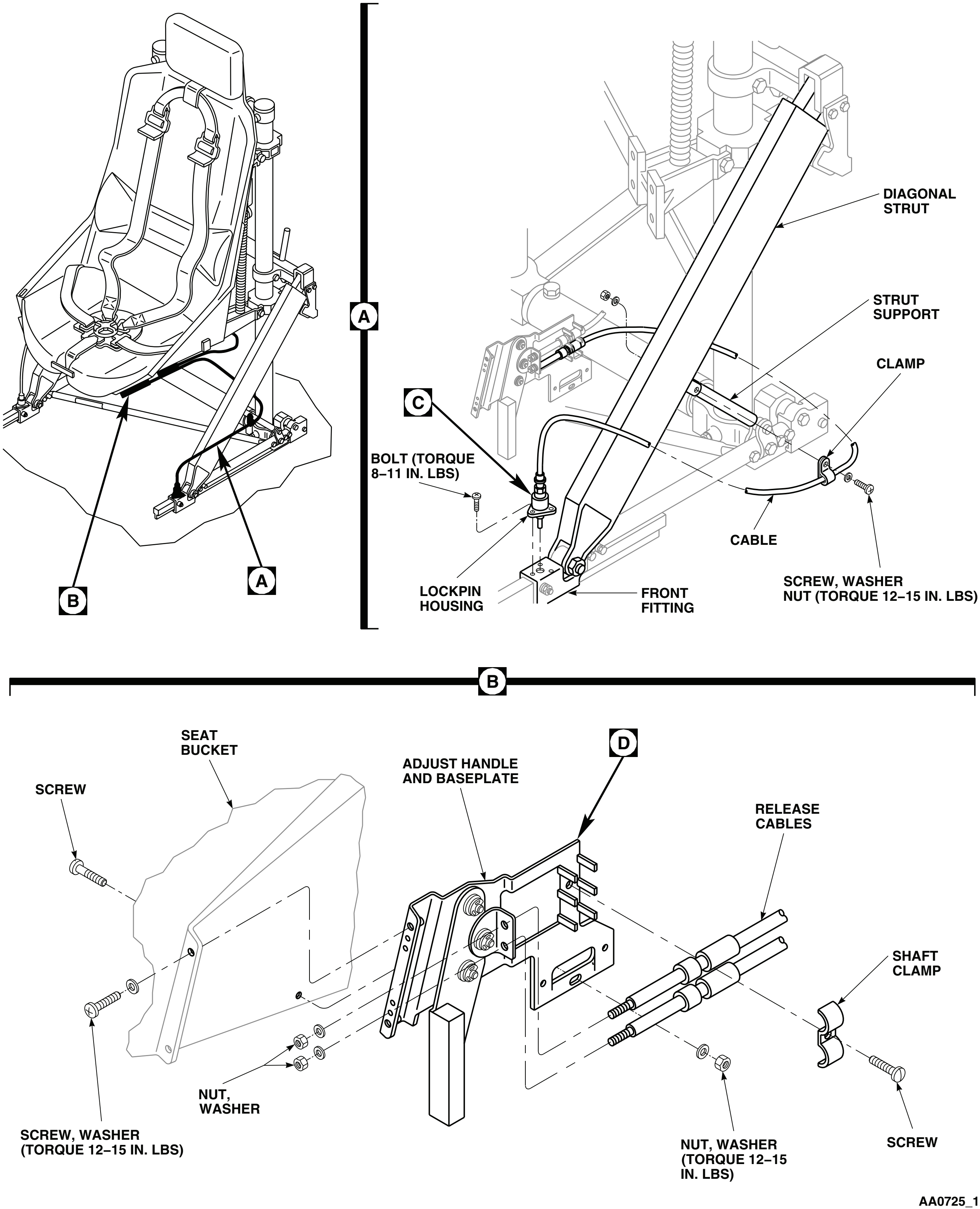


Figure 7. Adjust Handle and Baseplate. (Sheet 1 of 2)

REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE - Continued

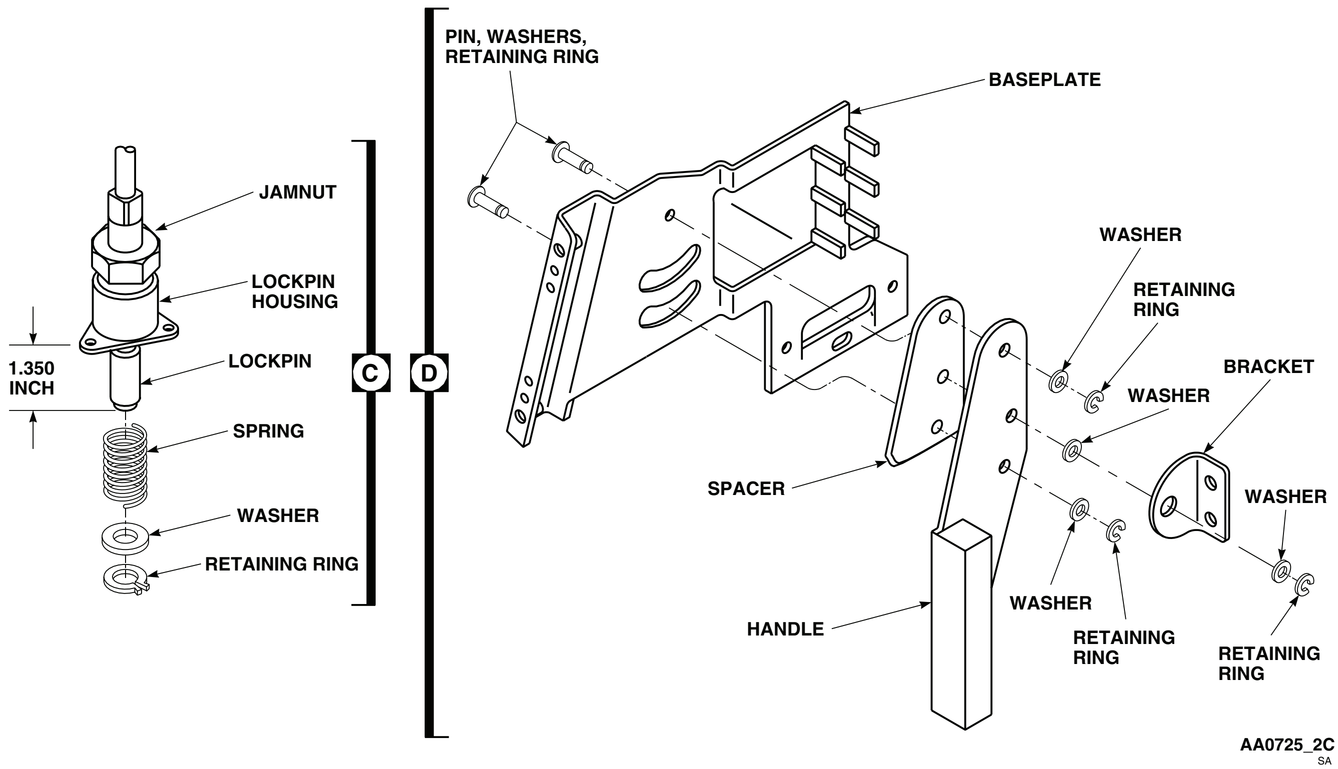


Figure 7. Adjust Handle and Baseplate. (Sheet 2 of 2)

8. Remove screw, washer, nut, and screws and washers attaching adjust handle and baseplate to seat bucket. Remove baseplate.

Repair Adjust Cable

1. Loosen jamnut on lockpin housing and turn housing up on cable until all tension is released on spring (Figure 7, Sheet 2, Detail C).
2. Remove retaining ring from lockpin and slip off washer and spring.
3. Clean housing and lockpin using machinery wiping towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry with clean machinery towel.
4. Slip replacement spring on lockpin. Install washer and retaining ring on lockpin.
5. Turn housing down on lockpin until dimension from bottom face of housing to tip of lockpin is 1.350-inches. Tighten jamnut until a sharp increase in torque is felt; then tighten 1/6 to 1/3 turn more.
6. Check operation of cable and pin.
7. Check clearance between lockpin and rail, while securing handle full forward. **CLEARANCE MUST BE BETWEEN 0.010 AND 0.050 - INCH.**
8. Adjust clearance as required by shimming between lockpin housing and front fitting.

Repair Adjust Handle Baseplate

1. Inspect parts for cracks, bends, corrosion, or hole elongation. Replace as necessary.
2. Remove retaining rings, washers, and pins, and remove adjust handle from baseplate (Figure 7, Sheet 2, Detail D).

REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE - Continued

3. Remove retaining ring, washers, and pin, and remove bracket from adjust handle.
4. Install bracket on adjust handle with pin, washers and retaining ring, with pin head against adjust handle. Install washer under pin head, one between adjust handle and bracket and one under retaining ring.
5. Install adjust handle and spacer on baseplate with pins, washers, and retaining ring, with pin head against baseplate, a washer between adjust handle and bracket, and a washer under retaining ring.

Install

1. **QA** Place adjust handle and baseplate on side of seat and attach to front of seat with screws and washers (Figure 7, Sheet 2, Detail D). Attach rear of baseplate to seat with screw, washer, and nut, with nut and washer against baseplate. **TORQUE FRONT SCREWS AND NUT TO 12 - 15 INCH-POUNDS.**
2. Install seat cushion. Refer to, REPLACE SEAT CUSHIONS, in this work package.
3. **QA** Install lockpin housing in front track fitting and attach with bolts. **TORQUE BOLTS TO 8 - 11 INCH-POUNDS.**
4. Route cable around outside of diagonal strut, and install end of cable in adjust handle bracket and cable shaft in baseplate clips. Clamp cable to base with shaft clamp and screw.
5. **QA** Install nuts and washers on cable ends, with a washer each side of bracket.
6. **QA** Remove screw, washer, and nut from diagonal strut and strut support. Clamp cable to diagonal strut with clamp, screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
7. Check operation of cable and stop.
8. Install inertia reel control on adjust baseplate. Refer to, REPLACE INERTIA REEL CONTROL, in this work package.

REPLACE UP-AND-DOWN EXTENSION SPRING AND RETAINER

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. While securing spring retainer up, pull down on spring and unhook spring from retainer. Remove retainer from bottom seat support (Figure 8, Detail A).
3. Unhook spring from attenuator bracket and remove spring.

Install

1. Hook replacement spring on attenuator bracket (Figure 8, Detail A).
2. Push retainer up through hole in bottom seat bracket, with loop on retainer leaning to rear.
3. **QA** Pull down on spring and hook it to retainer.
4. Make sure area is clean and free of foreign material.

REPLACE UP-AND-DOWN EXTENSION SPRING AND RETAINER - Continued

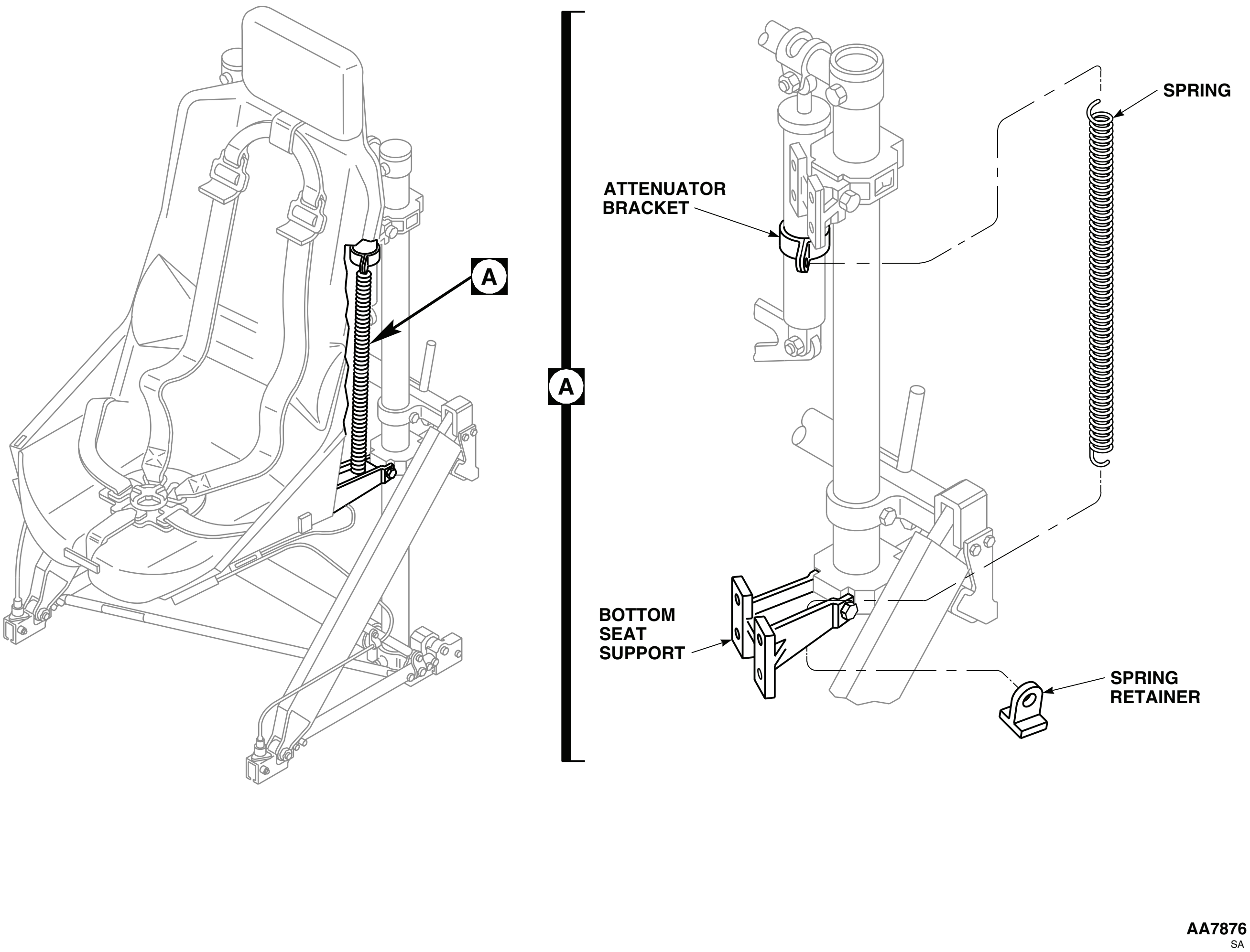


Figure 8. Replace Up-and-Down Extension Spring and Retainer.

REPLACE UP-AND-DOWN ATTENUATOR

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up. Remove first aid kit. Refer to, REPLACE PILOT'S/COPILOT'S SEAT, in this work package.
2. Remove tiedown straps and ICS cord from right attenuator (Figure 9, Detail A).
3. Provide a support for seat bucket by placing a block of wood or equivalent across seat rails in front of diagonal strut supports.
4. Unhook springs from attenuators.
5. Support seat bucket and remove bolt and nut securing attenuators to yoke. Lower seat to block.
6. Remove bolts and nuts securing attenuator to upper mounts and remove attenuators.

Install

1. **QA** Install replacement attenuators in mounts on upper support and install bolts and nuts (Figure 9, Detail A). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
2. **QA** Loosen jamnuts on attenuator clevis. Face spring brackets to front. Lift seat and line up clevis holes with yoke holes. Install bolts and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS. HOLD CLEVIS AND TIGHTEN JAMNUT UNTIL SHARP RISE IN TORQUE IS FELT, THEN TIGHTEN 1/6 TO 1/3 TURN MORE.**
3. Hook vertical springs to brackets on attenuators.
4. Remove support block from seat rails.
5. Place ICS cord next to right attenuator and secure to attenuator with electrical tiedown straps, Item 302, WP 1803 00.
6. Install first aid kit. Refer to, REPLACE PILOT'S/COPILOT'S SEAT, in this work package.
7. Check operation of up-and-down adjustment.
8. Make sure area is clean and free of foreign material.

REPLACE UP-AND-DOWN ATTENUATOR - Continued

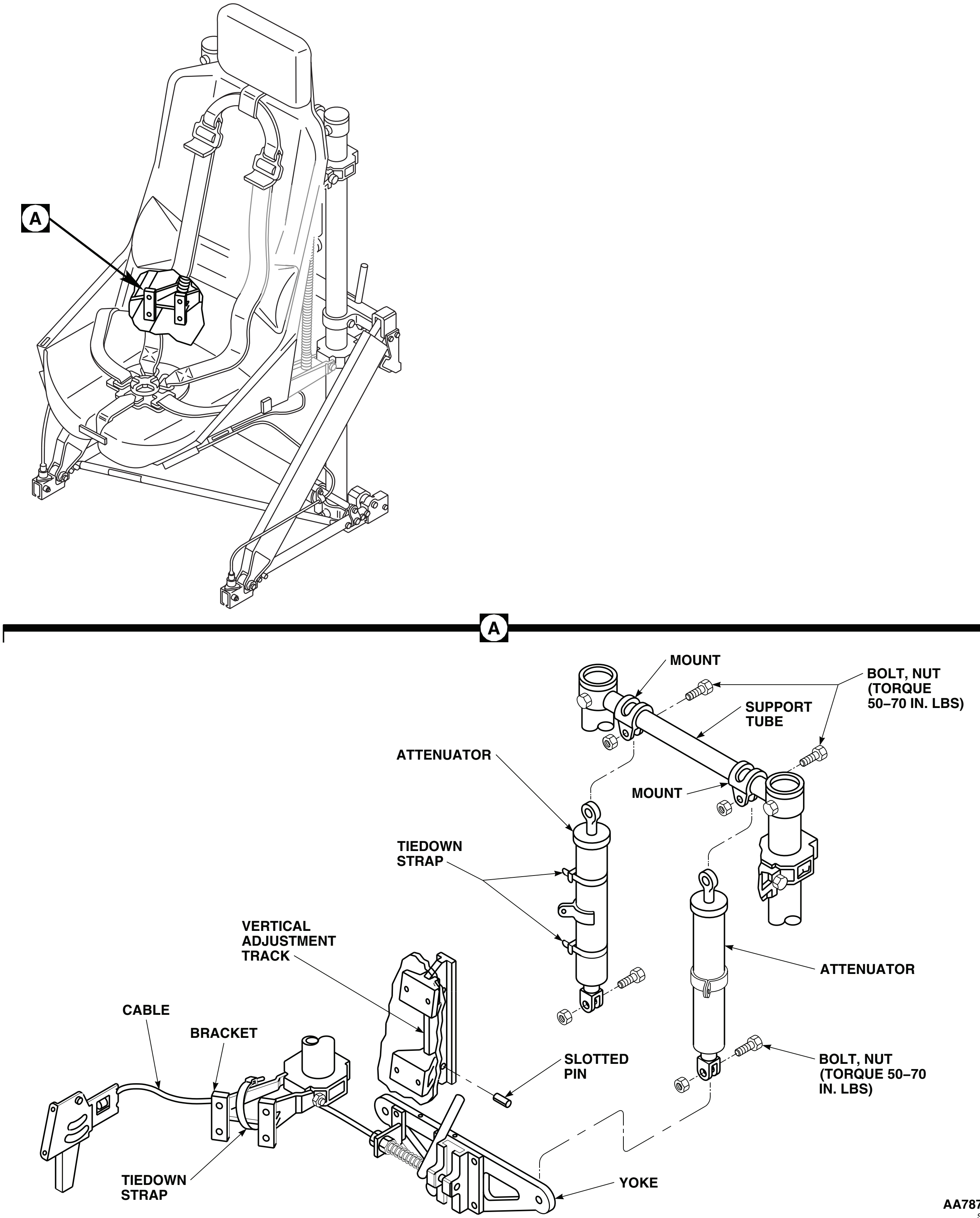


Figure 9. Replace Up-and-Down Attenuator.

REPLACE UP-AND-DOWN YOKE AND CABLE

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. Provide a support for seat bucket by placing a block of wood or equivalent across seat rails in front of diagonal strut supports.
3. Unhook springs from attenuators.
4. Support seat bucket and remove nuts and bolts securing attenuators to yoke. Lower seat and rest it on wood block (Figure 10, Sheet 1, Detail A).
5. Remove slotted pin from vertical adjustment track. Pull release handle and pull yoke down and off track.
6. Lift seat cushion on right side. Hold screw in seat bucket and remove nut and washer attaching clamp and baseplate to seat. Remove clamp (Figure 10, Sheet 1, Detail B).
7. Remove nut and washer and remove cable end from release handle bracket. Remove washer from cable end. Slide cable through tiedown strap on seat support. Remove yoke and cable.

Disassemble

1. Remove screws and washers attaching guide plate to yoke and bearing assembly (Figure 10, Sheet 2, Detail C).
2. Remove spring pin attaching release handle to lockpin and coupling. Remove insert from yoke and bearing assembly.
3. Remove screw and washer attaching release handle to insert on yoke, and remove handle. Remove guide plate from handle.
4. Hold cable, thread lockpin and coupling from cable. Remove spring. Remove lockpin and coupling from guide. Remove guide from yoke and bearing assembly.
5. Loosen jamnut and unthread cable from bracket on yoke.

Inspect

1. Check yoke and track guide for deformation, deep pitting corrosion damage to release handle pivot, and elongation of mounting holes.
2. If damage is found, replace yoke and bearing assembly.
3. Check cable bracket for cracks, corrosion, damaged threads, and loose rivets. If damage is found, replace bracket.
4. Check lockpin bushings in track guide for hole elongation or grooved wear that would prevent free operation.
5. Check lockpin and coupling for condition of threads, bends, and grooved wear that would prevent free operation and elongation of spring pin hole.
6. Check spring for deformation, corrosion, or compressed set that would prevent complete return of lockpin.
7. Check cable for kinks, corrosion, condition of threads, and free operation.

REPLACE UP-AND-DOWN YOKE AND CABLE - Continued

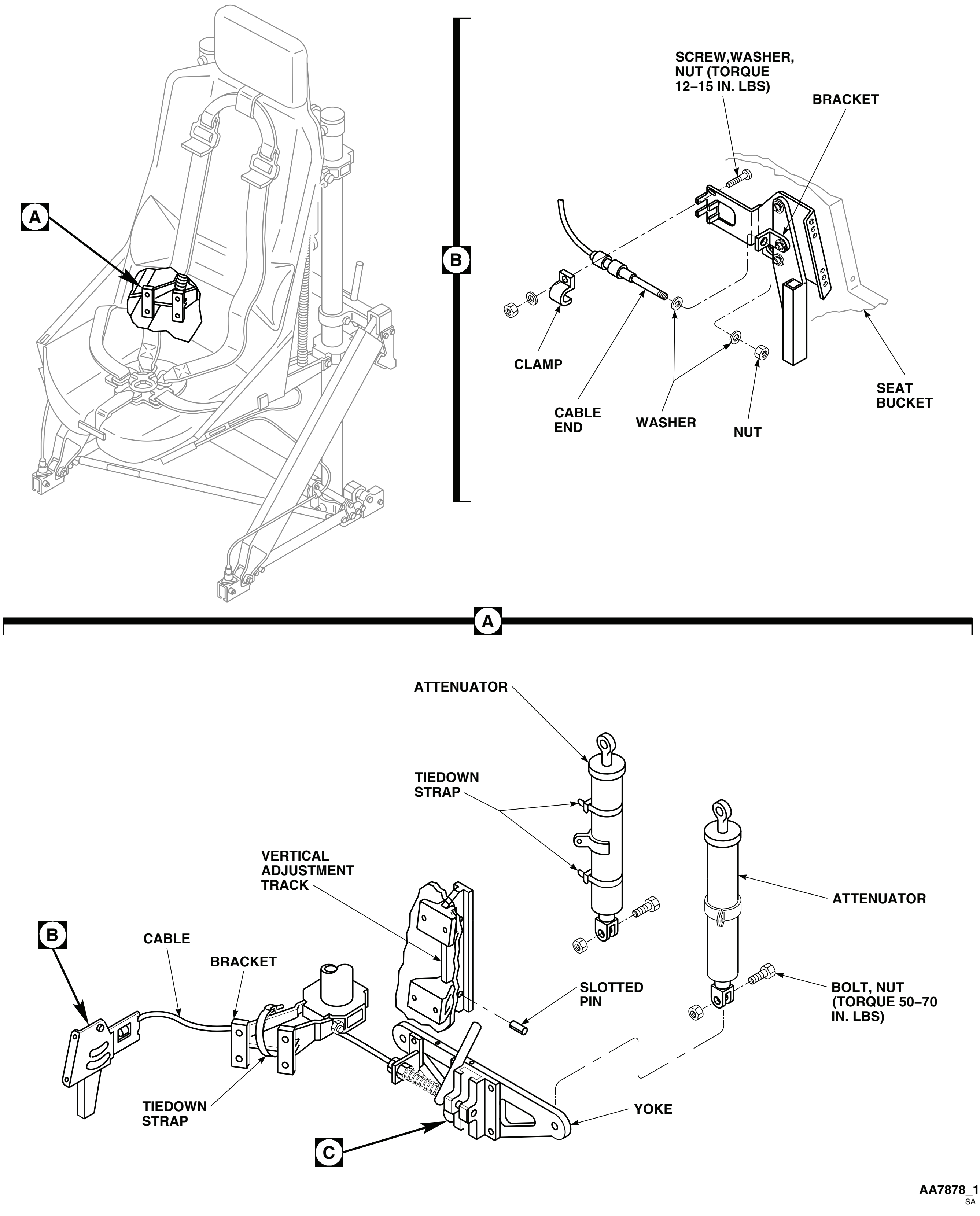


Figure 10. Replace Up-and-Down Yoke and Cable. (Sheet 1 of 2)

REPLACE UP-AND-DOWN YOKE AND CABLE - Continued

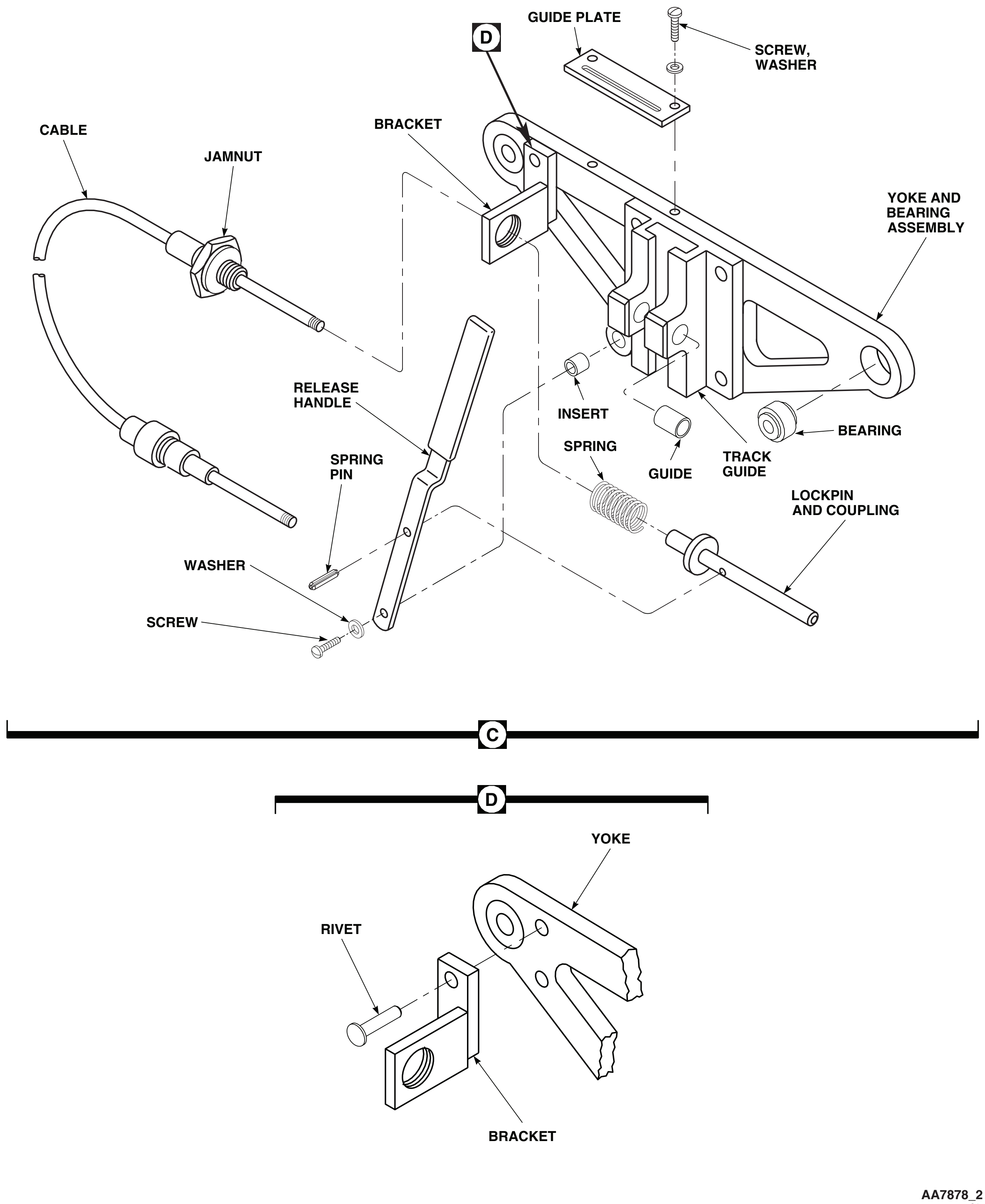


Figure 10. Replace Up-and-Down Yoke and Cable. (Sheet 2 of 2)

REPLACE UP-AND-DOWN YOKE AND CABLE - Continued

8. Check handle for deformation, deep pitting corrosion, elongation of mounting holes, and cracked or torn grip. Replace as necessary.
9. Check guide plate for deformation, deep pitting corrosion, and elongation of mounting holes.
10. Check bearing for damage. **NO DAMAGE ALLOWED.** Replace bearing if required.

Repair

1. If required drill out rivets and remove bracket. Place replacement bracket on yoke and install solid rivets (Figure 10, Sheet 2, Detail D) (TM 1-1500-204-23).
2. Remove minor surface corrosion, treat and touch up paint (WP 0381 00 and TM 55-1500-345-23).
3. If lockpin, coupling, and/or cable are not to be replaced, remove retaining compound from threads with 10-32 tap and die.
4. Replace bearing in yoke assembly if required (TM 55-1500-322-24).

Assemble

1. **QA** Install jamnut on cable housing (Figure 10, Sheet 2, Detail C). Thread cable housing into bracket on yoke. **TIGHTEN JAMNUT UNTIL A SHARP RISE IN TORQUE IS FELT. TORQUE 1/6 TO 1/3 TURN MORE.**
2. **QA** Install lockpin and coupling in yoke and bearing assembly.
3. Coat threads of cable with sealing compound, Item 285, WP 1803 00. Slip spring over cable, and thread lockpin and coupling on cable end until it bottoms.
4. Put guide plate on release handle so mounting holes are under grip.
5. **QA** Install release handle on yoke, with release handle grip over yoke, attach to yoke with washer and screw. Line up spring pin holes in lockpin and handle, and install spring pin.
6. Attach guide plate to yoke with screws and washers.
7. Check that lockpin slides in and out of track guide fitting freely, and that spring returns lockpin to fully locked position when handle is released.

Install

1. Place replacement yoke behind seat bucket. Feed cable up through tiedown strap on seat support.
2. **QA** Install cable end in release handle bracket with washers and nut. Make sure cable end is seated on clips on baseplate. Install clamp on cable, with screw, washer and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.** Push seat cushion down in place (Figure 10, Sheet 1, Detail B).
3. Pull release handle and install yoke on vertical adjustment track. Release handle and allow locking pin to engage lowest locking hole.
4. **QA** Install slotted pin in vertical adjustment track.
5. **QA** Lift seat and install attenuator fittings on yoke with bolts and nuts (Figure 10, Sheet 1, Detail A). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
6. Hook extension springs on attenuator brackets.
7. Remove support from seat rails.
8. Check operation of yoke and cable. Lock in each stop hole.
9. Make sure area is clean and free of foreign material.

REPLACE UP-AND-DOWN RELEASE HANDLE AND BASEPLATE

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. Remove seat cushion.
3. Remove screw, washer, nut, attaching cable clamp and baseplate to seat bucket. Remove clamp (Figure 11, Detail A).
4. Remove nut and washer from end release handle bracket. Pull cable from baseplate and bracket.
5. Remove screws and washers attaching base to front of seat bucket, and remove base.
6. Remove retaining rings, washers, spacer and pins attaching release handle to baseplate (Figure 11, Detail B).
7. Remove retaining ring, washers, and pin attaching bracket to handle.

Inspect

Inspect parts for cracks, bends, corrosion, or hole elongation. **NONE IS ALLOWED.** Replace parts as necessary.

Install

1. **QA** Assemble bracket to release handle with pin, washers, and retaining ring (Figure 11, Detail B).
2. **QA** Assemble handle and spacer to baseplate with pins, washers, and retaining rings.
3. **QA** Attach baseplate to front of seat bucket with screws and washers (Figure 11, Detail A). **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
4. **QA** Install cable end in handle bracket and cable shaft in baseplate with nuts and washers.
5. **QA** Install clamp on cable, baseplate and seat bucket with screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
6. Replace cushion and press down on fastener tape.
7. Check operation of up-and-down adjustment.
8. Make sure area is clean and free of foreign material.

REPLACE UP-AND-DOWN RELEASE HANDLE AND BASEPLATE - Continued

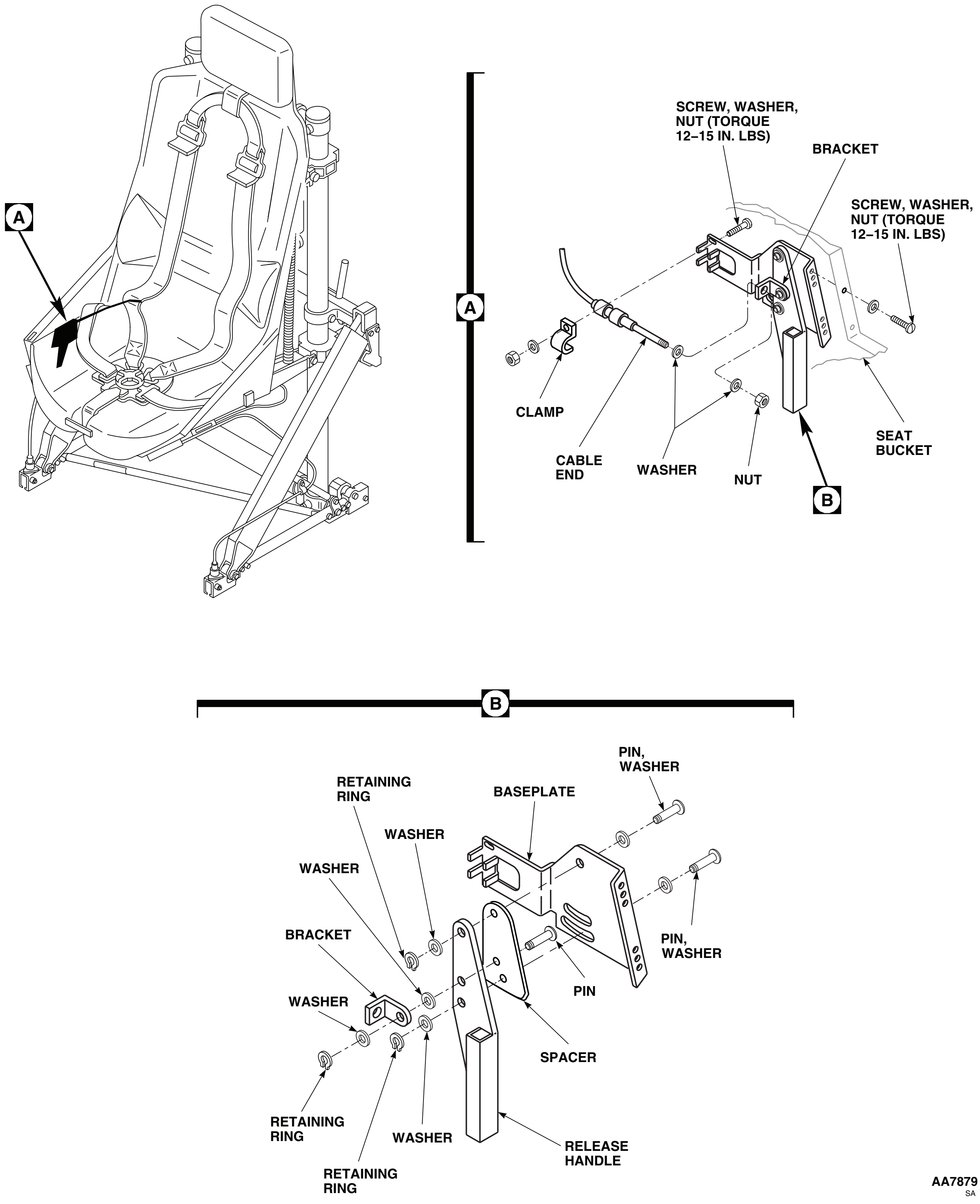


Figure 11. Replace Up-and-Down Release Handle and Baseplate.

REPLACE UP-AND-DOWN VERTICAL ADJUSTMENT TRACK

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. Provide a support for seat bucket by placing a block of wood or equivalent across seat rails in front of diagonal strut supports.
3. Unhook spring from attenuators (Figure 12, Detail A).
4. Support seat and remove bolts and nuts securing attenuators to yoke. Lower seat and rest it on wood block.
5. Remove slotted pin from vertical adjustment track. Pull release lever and pull yoke down and off track (Figure 12, Detail A).
6. Remove seat back cushion.
7. Remove screws, washers, and nuts attaching vertical adjustment track to seat. Remove track.

Install

1. Remove slotted pin from bottom of replacement vertical adjustment track.
2. **QA** Install vertical adjustment track on back of seat with screws, washers, and nuts. Install screws from front of seat back (Figure 12). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
3. Pull release handle and install yoke on vertical adjustment track, release handle and allow locking pin to engage lowest locking hole.
4. **QA** Install slotted pin in track.
5. **QA** Lift seat and install attenuator fittings on yoke with bolts and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
6. Hook extension springs on attenuator brackets.
7. Remove support from seat rails.
8. Check operation of seat up and down. Lock in each stop hole.
9. Make sure area is clean and free of foreign material.

REPLACE UP-AND-DOWN VERTICAL ADJUSTMENT TRACK - Continued

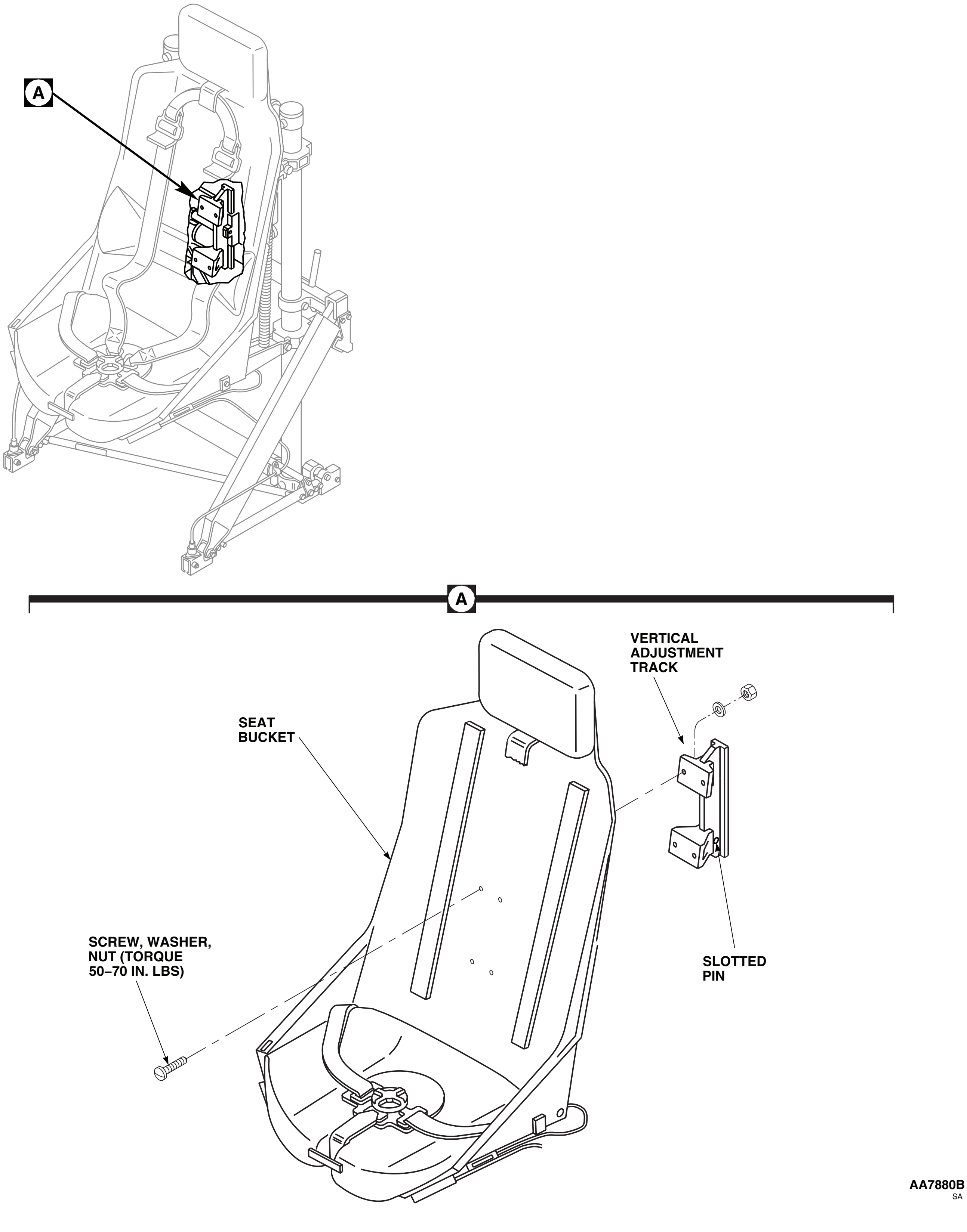


Figure 12. Replace Up-and-Down Vertical Adjustment Track.

REPLACE SEAT BUCKET

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. Remove shoulder harness, lap belts, and rotary buckle. Refer to, REPLACE SHOULDER HARNESS, REPLACE LAP BELT, and REPLACE CROTCH BELT, BUCKLE AND PAD, in this work package.
3. Remove seat cushions. Refer to, REPLACE SEAT CUSHIONS, in this work package.
4. Remove inertia reel control. Refer to, REPLACE INERTIA REEL CONTROL, in this work package.
5. Remove inertia reel. Refer to, REPLACE INERTIA REEL, in this work package.
6. Remove front-to-rear release handle and base. Refer to, REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE, in this work package.
7. Remove vertical adjustment release handle and base. Refer to, REPLACE UP-AND-DOWN RELEASE HANDLE AND BASEPLATE, in this work package.
8. Remove screws, washers, and nuts securing seat bucket to vertical adjustment track (Figure 13).
9. Support seat bucket and remove bolts and washers securing seat belt to supports on frame. Remove seat bucket.

Install

1. **QA** Install replacement seat bucket in frame and attach to supports with bolts and washers (Figure 13). **TORQUE BOLTS TO 100 - 140 INCH-POUNDS.**
2. **QA** Attach vertical adjustment track to bucket with bolts, washers, nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
3. Install vertical adjustment release handle and base. Refer to, REPLACE UP-AND-DOWN RELEASE HANDLE AND BASEPLATE, in this work package.
4. Install front-to-rear release handle and base. Refer to, REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE, in this work package.
5. Install inertia reel. Refer to, REPLACE INERTIA REEL, in this work package.
6. Install inertia reel control. Refer to, REPLACE INERTIA REEL CONTROL, in this work package.
7. Install seat cushions. Refer to, REPLACE SEAT CUSHIONS, in this work package.
8. Install shoulder harness, lap belts, and rotary buckle. Refer to, REPLACE SHOULDER HARNESS, REPLACE LAP BELT, and REPLACE CROTCH BELT, BUCKLE AND PAD, in this work package.
9. Check operation of seat and inertia reel.
10. Make sure area is clean and free of foreign material.

REPLACE SEAT BUCKET - Continued

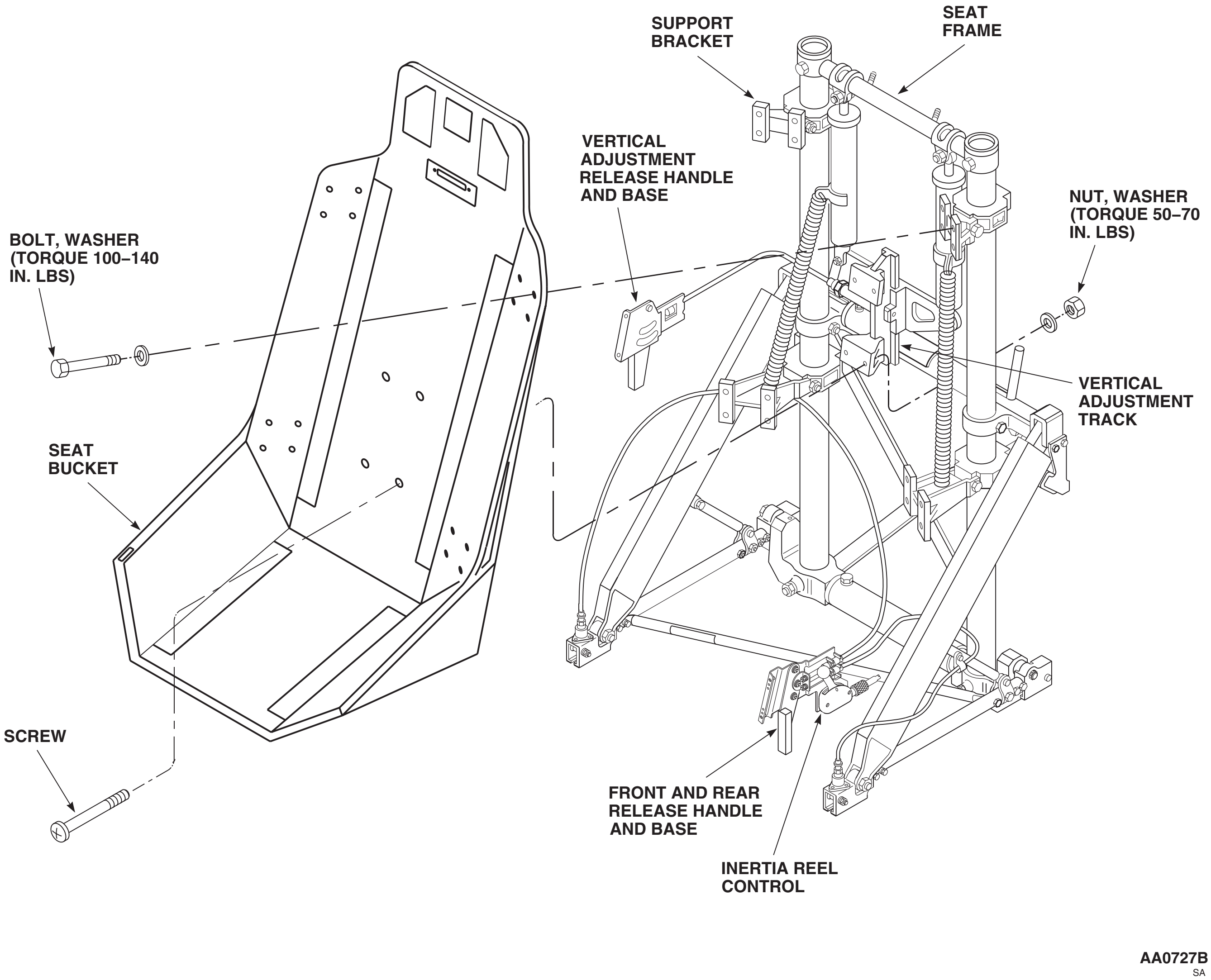


Figure 13. Replace Seat Bucket.

REPLACE SHOULDER HARNESS RUBBING BAR

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust full up.

REPLACE SHOULDER HARNESS RUBBING BAR - Continued

2. Remove headrest and seat back cushions. Refer to, REPLACE SEAT CUSHIONS, in this work package.
3. Remove shoulder harness. Refer to, REPLACE SHOULDER HARNESS, in this work package.
4. Remove screws and rubbing bar halves from seat (Figure 14, Detail A).

Install

NOTE

Flanged piece of rubbing bar is inserted from rear of armored seat bucket with thick side down. Flat piece of rubbing bar is positioned on front of armored seat bucket with thick side down.

1. Check fit of flat piece of rubbing bar with the seat bucket. File flat piece of rubbing bar as necessary to eliminate any interference with seat bucket.
2. **QA** Apply sealing compound, Item 291, WP 1803 00, to threads of screws. Install screws from the rear of seat bucket through the flanged piece into the flat piece of rubbing bar (Figure 14, Detail A).
3. Install shoulder harness. Refer to, REPLACE SHOULDER HARNESS, in this work package.
4. Install seat back and headrest cushions. Refer to, REPLACE SEAT CUSHIONS in this work package.
5. Make sure area is clean and free of foreign material.

REPLACE SEAT FRAME UPPER TUBE AND SUPPORT

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. Unhook extension springs from attenuator bracket. Refer to, REPLACE UP-AND-DOWN EXTENSION SPRING RETAINER, in this work package.
3. Unclamp ICS cord from upper tube. Cut tiedown straps securing ICS cord to right attenuator and move ICS cord to one side.
4. Provide a support for seat bucket by placing a wood block or equivalent across seat rails in front of diagonal strut supports.
5. Support seat and remove attenuator upper bolts, and nuts. Lower seat to rest on support.
6. Pry plugs from top of guide tubes.
7. Remove bolts, washers, and nuts attaching supports to top of guide tubes (Figure 15, Detail A).
8. Remove supports from guide tubes by tapping up one side, then other, with a mallet.

Install

1. **QA** Install replacement upper tube and supports on guide tubes by tapping down one side, then other, with a mallet, until attachment holes line up (Figure 15, Detail A). Face studs, on tube, toward rear.

REPLACE SEAT FRAME UPPER TUBE AND SUPPORT - Continued

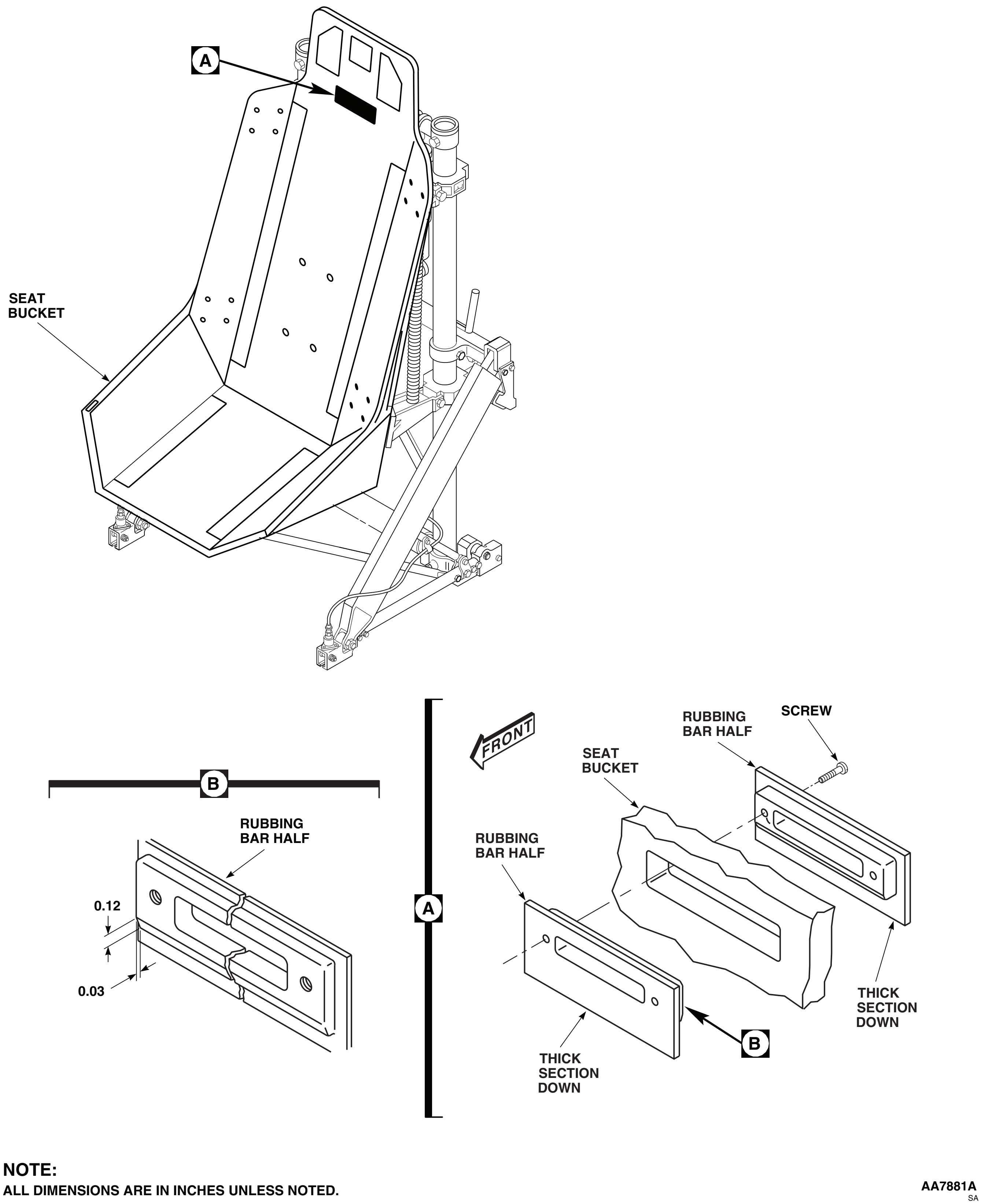


Figure 14. Replace Shoulder Harness Rubbing Bar.

REPLACE SEAT FRAME UPPER TUBE AND SUPPORT - Continued

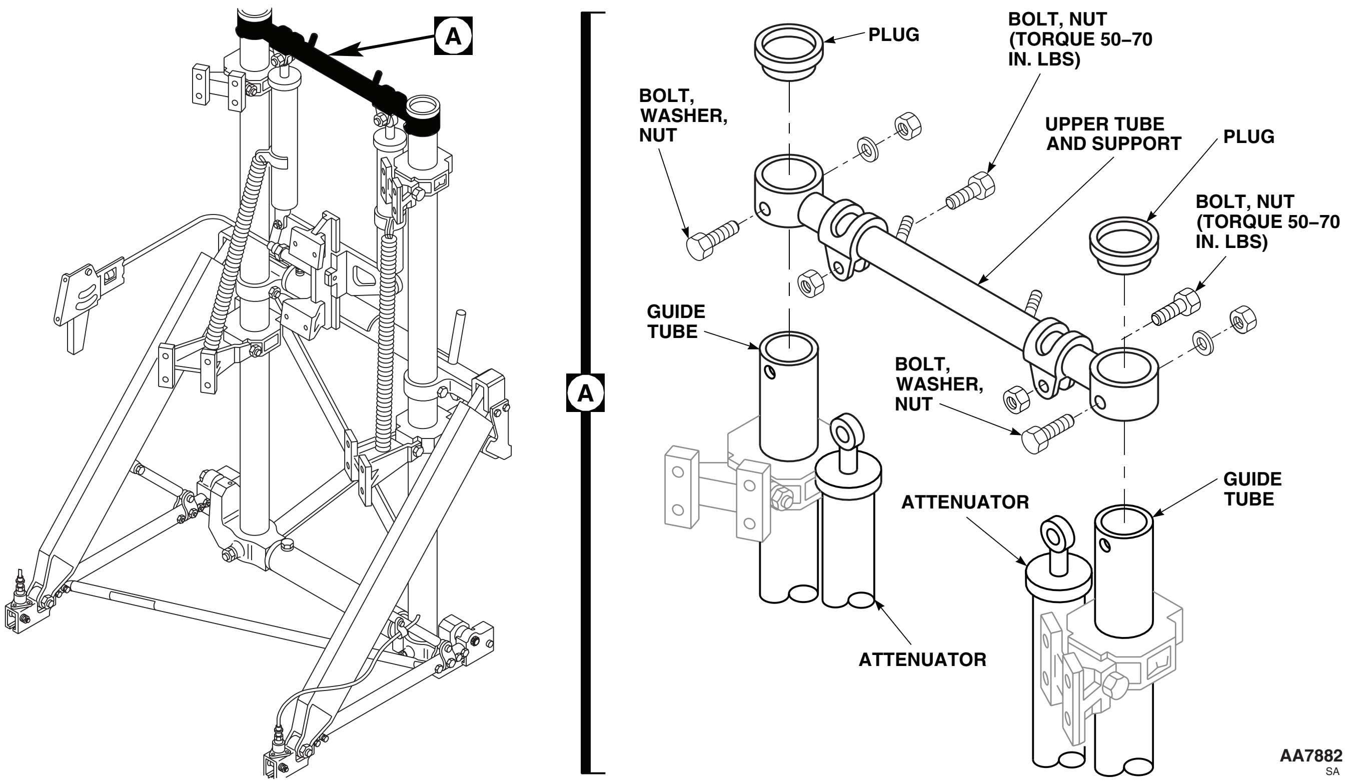


Figure 15. Replace Seat Frame Upper Tube and Supports.

2. Install bolts, washers, and nuts with boltheads toward front. **TIGHTEN NUTS UNTIL THERE IS NO PLAY; THEN BACK OFF UNTIL YOU GET 0.004 TO 0.010-INCH ENDPLAY.** Install plugs in top of guide tubes.
3. **QA** Lift seat and install attenuator rod ends in upper tube straps with bolts and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
4. Hook extension springs on attenuator brackets. Refer to, REPLACE UP-AND-DOWN EXTENSION SPRING RETAINER, in this work package.
5. Clamp ICS cord to upper tube. Tie ICS cord to right attenuator with two electrical tiedown straps, Item 302, WP 1803 00, one at each end of attenuator.
6. Remove support from seat rails.
7. Make sure area is clean and free of foreign material.

REPLACE SEAT FRAME UPPER TUBE AND SUPPORT - Continued

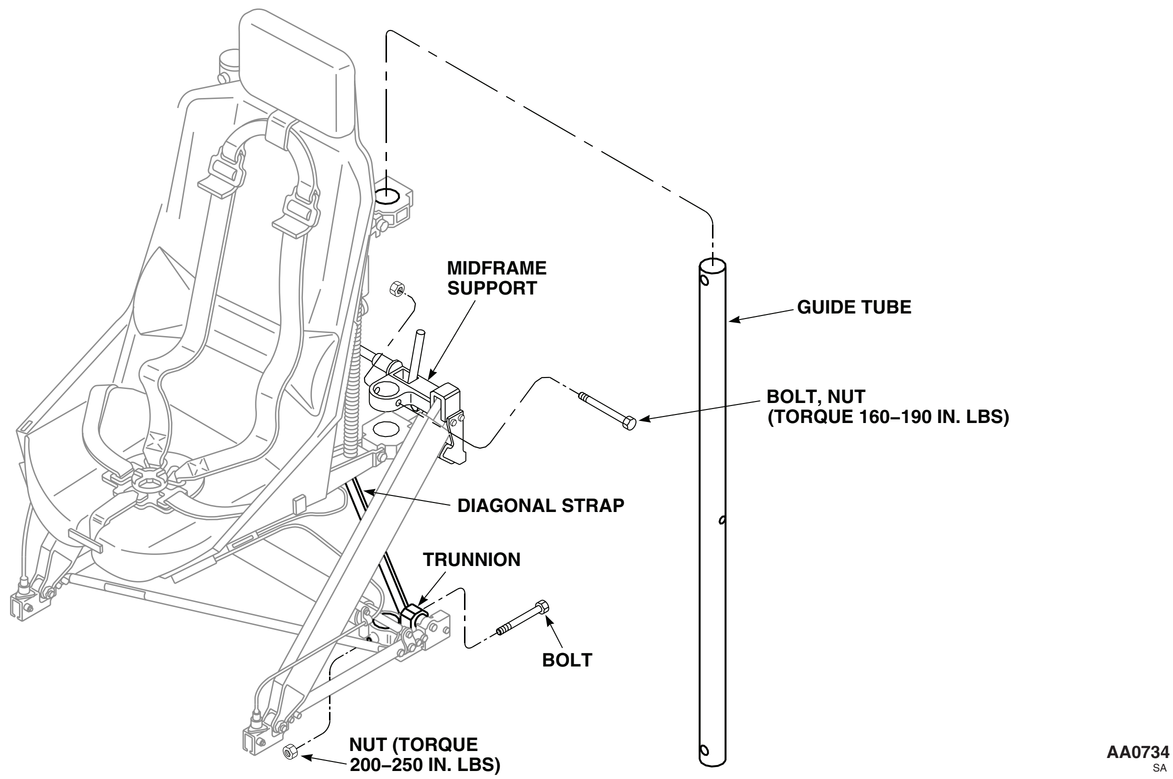


Figure 16. Replace Up-and-Down Guide Tube.

REPLACE UP-AND-DOWN GUIDE TUBE

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. If seat can be operated up, sit in seat and adjust it fully up. If not, be careful not to touch release handle on side of seat or emergency handle at rear of seat, before removing extension springs.
2. Remove upper tube and support brackets. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.
3. Remove midframe bolt and nut, and trunnion bolt and nut (Figure 16).
4. Twist, tap, and pull guide tube up from trunnion and through midframe support.

Install

1. Install tube through trunnion and midframe supports, line up butt holes (Figure 16).
2. **QA** Install trunnion bolt through diagonal strap and trunnion. Install nut. **TORQUE NUT TO 200 - 250 INCH-POUNDS.**

REPLACE UP-AND-DOWN GUIDE TUBE - Continued

3. **QA** Install midframe bolt and nut. **TORQUE NUT TO 160 - 190 INCH-POUNDS.**
4. Install upper tube and support brackets. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.
5. Check operation of up-and-down seat adjust.
6. Make sure area is clean and free of foreign material.

REPLACE MIDFRAME SUPPORT AND TUBE

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust full up.
2. Remove stowage panel from seat. Refer to, REPLACE PILOT'S STOWAGE BOX, PANEL, AND BAG, in this work package.
3. Remove up-and-down guide tube from side on which support is to be replaced. Refer to, REPLACE UP-AND-DOWN GUIDE TUBE, in this work package.
4. Remove stowage panel support bracket from midframe support (Figure 17).
5. Remove diagonal strap from midframe support.
6. Remove cable clamp and strut support from diagonal strut.
7. Remove midframe support from tube, disengage tilt release pin from diagonal strut, turn midframe support to clear diagonal strut. Lift diagonal strut from support. Remove support from tube.
8. Remove tube from other midframe support. Slide tube from support.

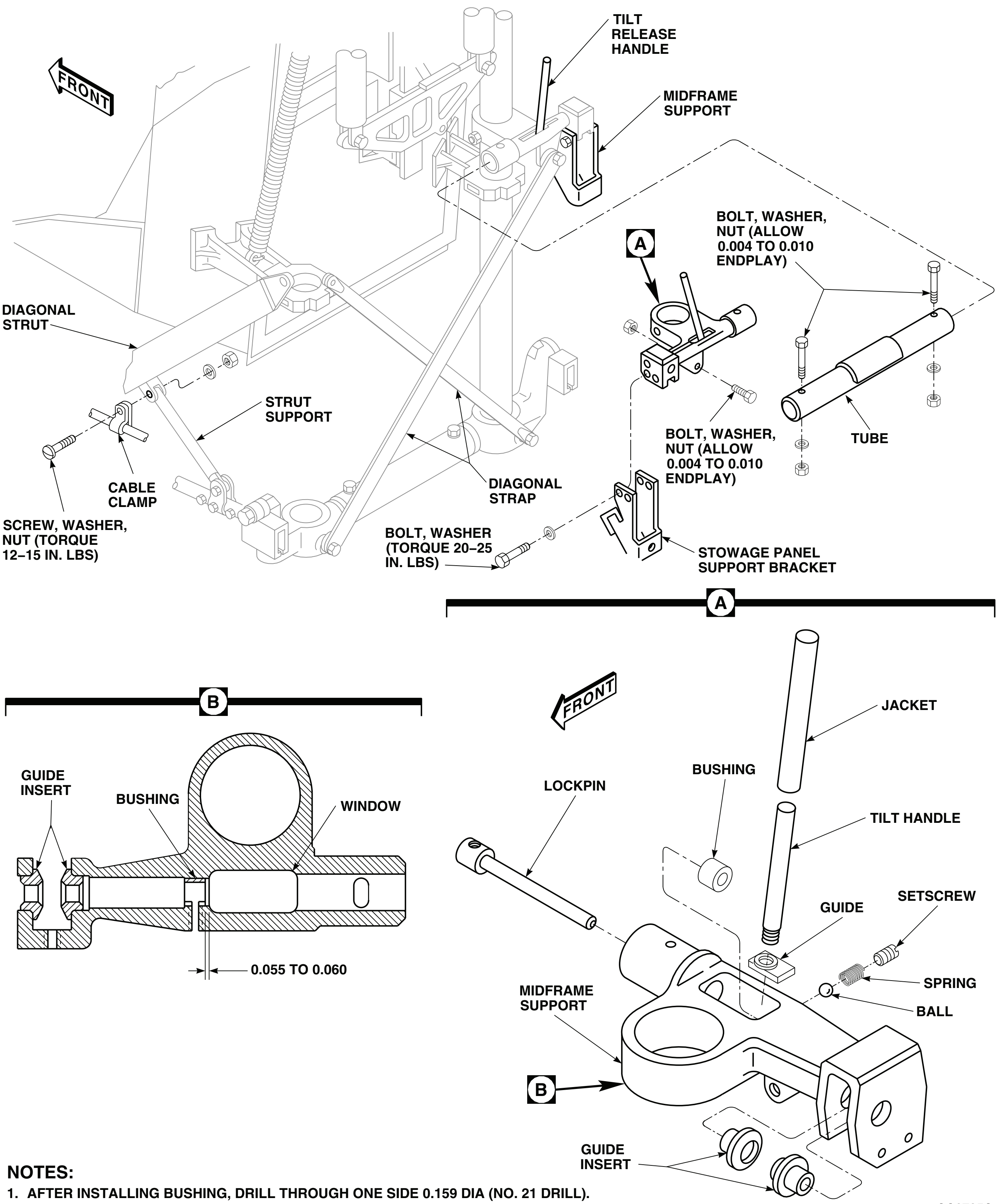
Disassemble

1. Unscrew tilt release handle from lockpin, remove guide (Figure 17, Detail A).
2. Remove setscrew, spring, and steel ball from support housing, and slide lockpin from housing.

Inspect

1. Inspect housing for cracks, deformation, corrosion, surface blemishes, and condition of paint. **REJECT HOUSING IF CRACKED, DEFORMED, OR HAS DEEP CORROSION PITTING.** Check all threaded holes for damaged threads.
2. Inspect bushing and guide inserts for grooved wear that would prevent free operation of lockpin. Replace if necessary.
3. Inspect setscrew, spring, and ball for wear, corrosion, deformed or compressed spring, and damaged threads. Replace if necessary.
4. Inspect lockpin and guide for bends, wear, grooves that would prevent lockpin from sliding freely in bushing or guides, corrosion, or damaged threads. Replace for any damage except damaged threads. Threads can be repaired by retapping. Replace lockpin with stripped threads.

REPLACE MIDFRAME SUPPORT AND TUBE - Continued



NOTES:

1. AFTER INSTALLING BUSHING, DRILL THROUGH ONE SIDE 0.159 DIA (NO. 21 DRILL).
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 17. Replace Midframe Support and Tube.

REPLACE MIDFRAME SUPPORT AND TUBE - Continued

NOTE

Normal wear along length of lockpin is acceptable if grooves or ridges are not seen.

5. Inspect handle for bends, corrosion, and damaged threads. Inspect jacket for cracking, peeling, or security. Replace as necessary. Damaged threads can be repaired. Replace handle with stripped threads.

Repair

1. Treat minor corrosion and touch up nicks, scratches, and paint (WP 0381 00 and TM 55-1500-345-23).
2. Use a 5/16 - 18 UNC tap in lockpin to clean out old adhesive and repair any damaged threads.
3. Use 5/16 - 18 UNC die to clean old adhesive and repair damaged threads on lockpin handle.
4. Use 10-32 tap to retap threaded holes in support.
5. Press out guide inserts and bushing from support.
6. Coat replacement bushing with primer coating, Item 243, WP 1803 00, and press it into support while primer is wet. End of bushing to be 0.055 to 0.060 - inch from window end of bore (Figure 17, Detail B).
7. Using No. 21 drill in setscrew hole, drill through one side of bushing. Deburr hole in bushing. Flush out chips with cleaning compound solvent, Item 71, WP 1803 00, and blow dry.
8. Coat outside diameters and face of flange on replacement guide inserts with primer coating, Item 243, WP 1803 00 and press into support while primer is wet.

Assemble

1. **QA** Slide replacement lockpin in support and into bushing and guide inserts (Figure 17, Detail A). Install guide on lockpin.
2. Coat threads of handle with sealing compound, Item 285, WP 1803 00. Thread handle into lockpin until it bottoms.
3. Check that lockpin moves freely in and out.

CAUTION

To prevent damage to spring, do not tighten setscrew beyond first feel of solid contact.

4. Install ball, spring, and setscrew in support. Tighten setscrew until spring bottoms (fully compressed), then back off 1 1/2 to 2 turns.
5. Check that lockpin moves freely except for extra force needed to unseat ball when pulling lockpin from insert guides.

Install

1. Slide replacement tube on midframe support on seat (Figure 17). Install bolt, washer, and nut.
2. Slide replacement support in tube, turn support and installed diagonal strut in support. Engage tilt release lockpin in strut.
3. Install bolts, washers, and nuts in tube and support. **TIGHTEN BOTH NUTS UNTIL THERE IS NO ENDPLAY, THEN BACK OFF TO ALLOW 0.004 TO 0.010-INCH ENDPLAY.**
4. **QA** Put strut support in diagonal strut clevis, put cable and clamp on clevis, and install screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
5. Bolt diagonal strap to midframe support. **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF TO ALLOW 0.004 TO 0.010-INCH ENDPLAY.**

REPLACE MIDFRAME SUPPORT AND TUBE - Continued

6. **QA** Bolt stowage panel support bracket to midframe support. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
7. Install up-and-down guide tube. Refer to, REPLACE UP-AND-DOWN GUIDE TUBE, in this work package.
8. Install stowage panel on midframe supports. Refer to, REPLACE PILOT'S STOWAGE BOX, PANEL, AND BAG, in this work package.
9. Check operation of tilt release and up-and-down adjust system.
10. Make sure area is clean and free of foreign material.

REPLACE TRUNNION AND TUBE

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it fully up.
2. Remove upper tube and supports. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.
3. Remove both midframe support bolts and nuts (Figure 18, Detail A).
4. Remove bolts and nuts from diagonal straps, trunnions, and guide tubes.
5. Twist and lift both guide tubes up far enough to clear trunnions.
6. Remove bolts attaching trunnions to track fitting bearings. Remove trunnion spacer and shims. Record number of shims removed from each side. Remove tube and trunnions.
7. Remove bolts, washers, and nuts, and remove trunnions from tube.

Install

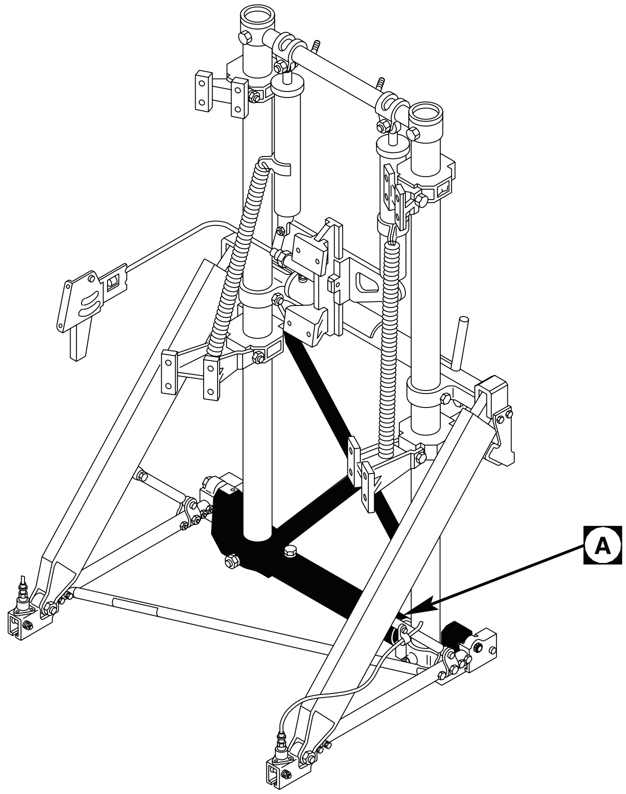
1. **QA** Install replacement tube on replacement trunnions, and install bolts, washers, and nuts. Install one washer under boltheads, and two washers under each nut (Figure 18, Detail A). **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF NUT UNTIL YOU HAVE 0.004 TO 0.010-INCH ENDPLAY.**
2. Install replacement tube and trunnions between seat track fittings at spherical bearing (resilient mounts). Install same number of shims on each side as removed, between bearings and trunnions. Install bolts. **TIGHTEN BOLT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF UNTIL THERE IS 0.004 TO 0.010-INCH ENDPLAY.**
3. Clean threaded end of trunnion bolt with cleaning compound solvent, Item 71, WP 1803 00 and allow time to air dry.

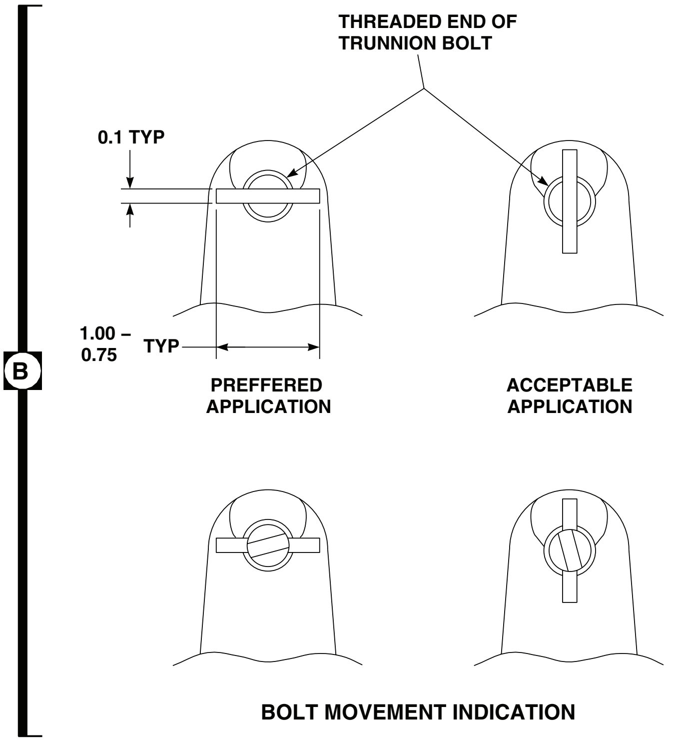
NOTE

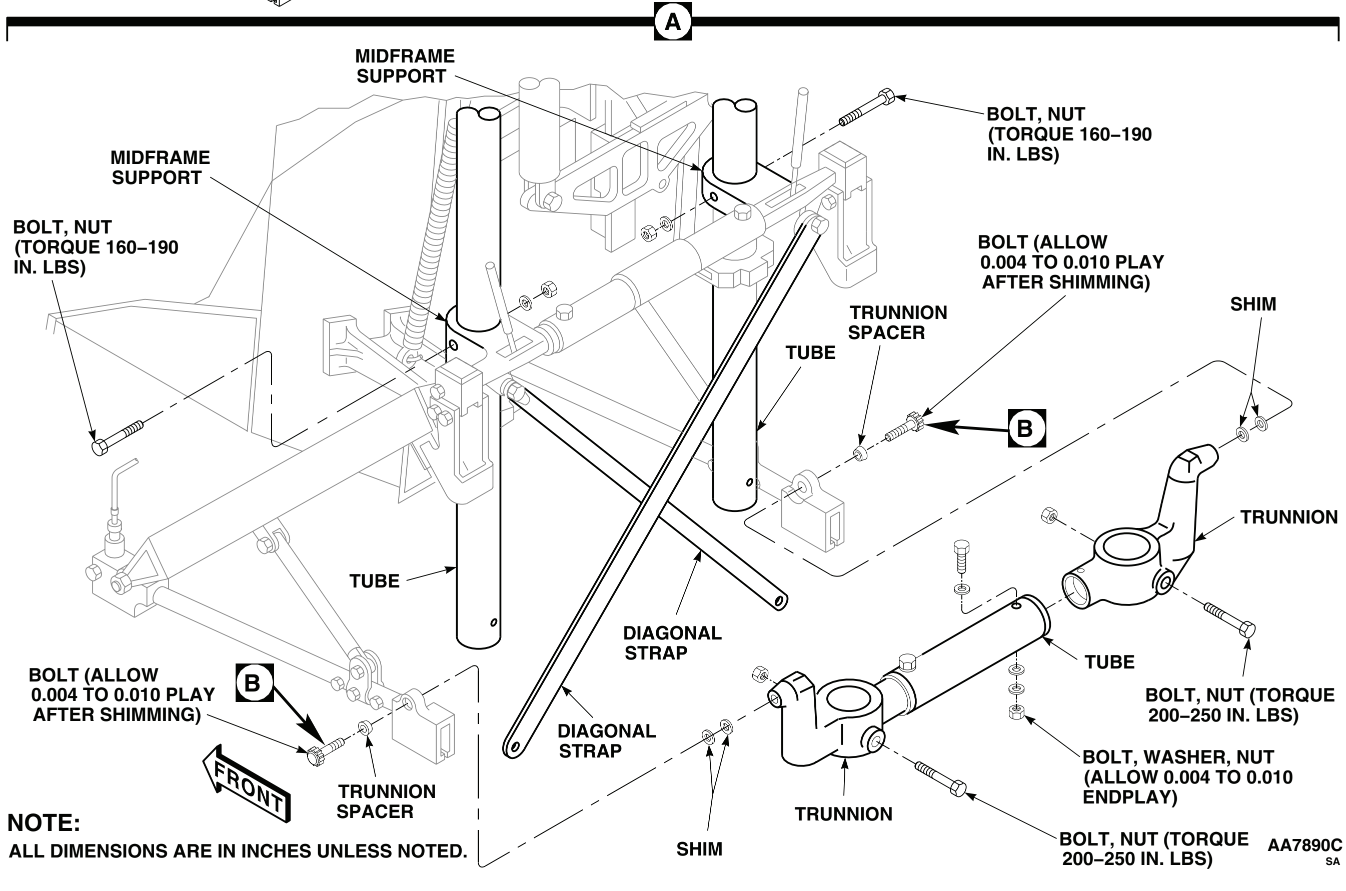
Torque stripe width shall be approximately 1/2 of fastner diameter, not to exceed 3/16 of an inch. The stripe shall extend a minimum of 1/2 inch past the edge of the threaded area onto the base part (Figure 18, Detail B)

4. Apply torque stripe (Figure 18, Detail B), (WP 0381 00). Color is optional, but must contrast with black.

REPLACE TRUNNION AND TUBE - Continued







NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 18. Replace Trunnion and Tube.

REPLACE TRUNNION AND TUBE - Continued

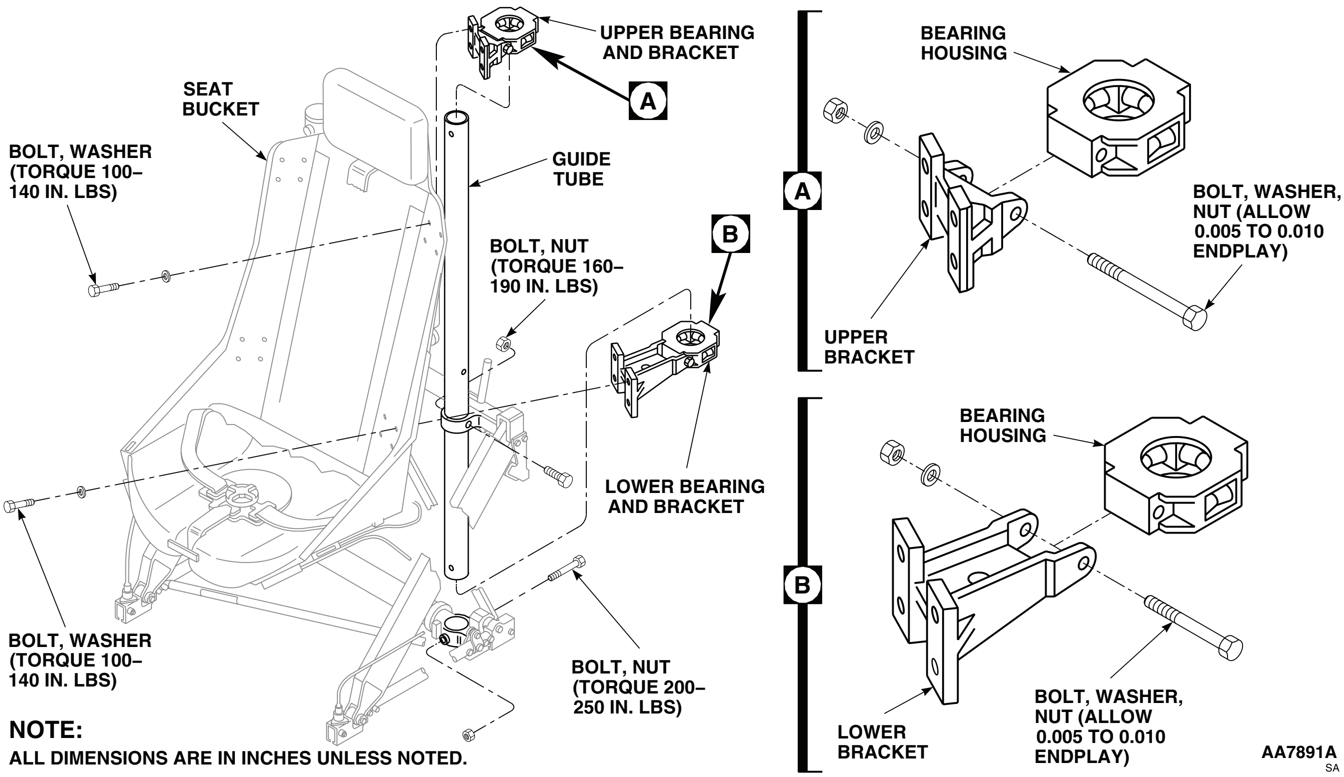


Figure 19. Replace Bearings and Support Brackets.

NOTE

If you do not know how many shims to install, install those you have, with the same number of shims on each side. Final adjustment will be done when seat is installed in helicopter.

5. Tap guide tubes down into trunnions and line up attaching boltholes in trunnions, midframe supports, and guide tubes.
6. **QA** Install diagonal straps, trunnions, and guide tubes with bolts and nuts. **TORQUE NUTS TO 200 - 250 INCH-POUNDS.**
7. **QA** Install bolts and nuts in midframe supports. **TORQUE NUTS TO 160 - 190 INCH-POUNDS.**
8. Install upper tube and supports. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.
9. Make sure area is clean and free of foreign material.

REPLACE BEARINGS AND SUPPORT BRACKETS

Remove

1. To remove upper bearing and support bracket do this (Figure 19, Detail A):

REPLACE BEARINGS AND SUPPORT BRACKETS - Continued

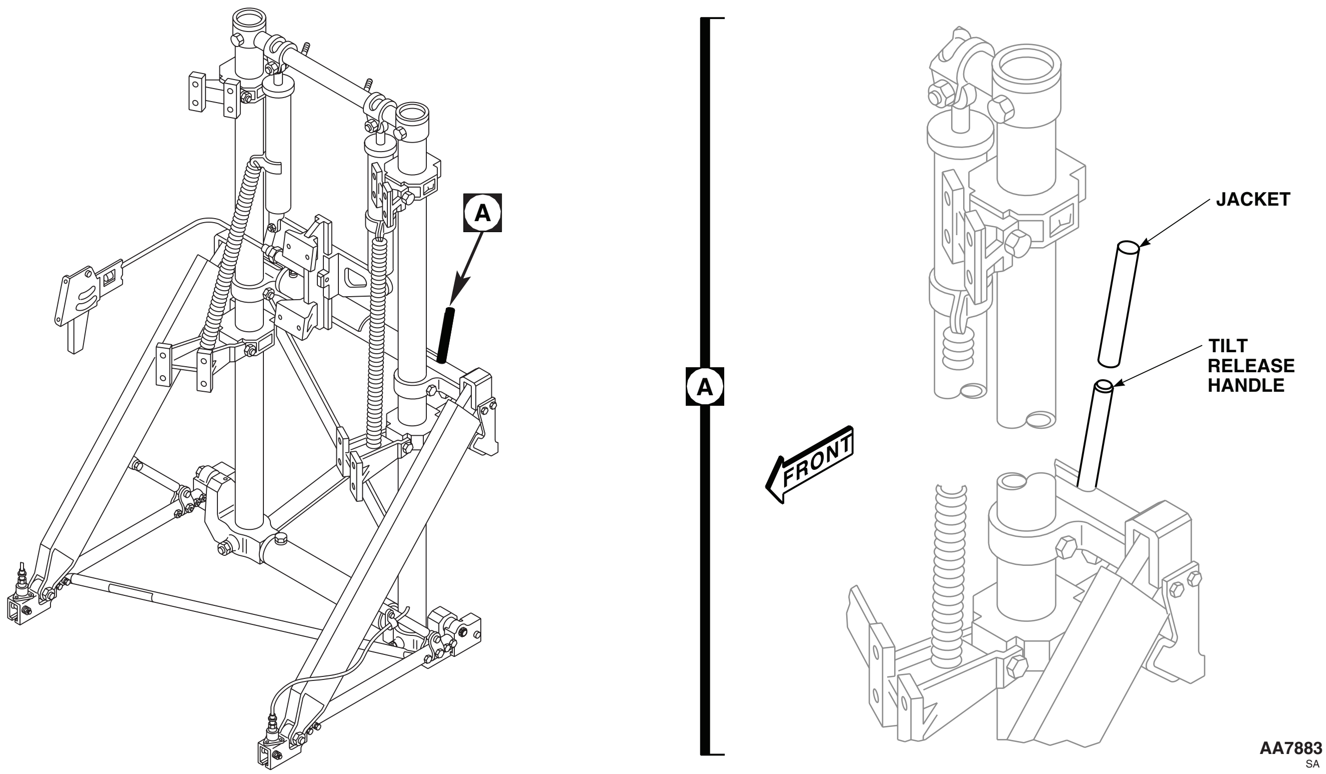


Figure 20. Replace Seat Frame Tilt Handle Jacket.

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

a. Sit in seat and adjust it fully up.

b. Remove upper tube and supports. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.

c. Remove seat back cushion.

d. Remove bolts and washers attaching seat bucket to support bracket, and slide bearing and bracket from guide tube.

e. Remove bolt, washer, nut, and bearing housing from upper bracket.

2. To remove lower bearing and support bracket do this (Figure 19, Detail B):

a. Remove upper tube and supports. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.

b. Remove bolts and washers from midframe support and trunnion on side from which support is to be removed.

c. Pull and twist guide tube up and out of trunnion 8 inches.

REPLACE BEARINGS AND SUPPORT BRACKETS - Continued

d. Remove seat back cushion.

e. Remove bolts and washers attaching seat bracket to support bracket. Slide bearing and bracket down from guide tube.

NOTE

On right support, remove tiedown strap holding up-and-down adjust cable.

f. Remove bolt, washers, and nut, and remove bearing housing from lower bracket.

Install

1. To install upper bearing and support bracket do this (Figure 19, Detail A):

a. **QA** Install replacement bearing housing in replacement upper bracket, and attach with bolt, washer, and nut.

b. **QA** Slide bearing and support bracket over guide tube. Install bolts and washers through seat bucket and attach support to seat bucket.

c. Install seat back cushion.

d. Install upper tube and supports. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.

e. Make sure area is clean and free of foreign material.

2. To install lower bearing and support bracket do this (Figure 19, Detail B):

a. **QA** Install bearing housing in lower bracket with bolt, washer, and nut.

b. **QA** Slide bearing and support bracket up on guide tube. Install support on seat bracket with bolts and washers through seat bucket. **TORQUE BOLTS TO 100 - 140 INCH-POUNDS.**

c. **QA** Tap guide tube down into trunnion, line up boltholes, and install bolt and nut through diagonal strap and trunnion. **TORQUE NUT TO 200 - 250 INCH-POUNDS.**

d. **QA** Install bolt and nut in midframe support. **TORQUE NUT TO 160 - 190 INCH-POUNDS.**

e. Install seat back cushion.

f. Install upper tube and support. Refer to, REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS, in this work package.

g. Make sure area is clean and free of foreign material.

REPLACE SEAT FRAME TILT HANDLE JACKET

Remove

1. Cut handle jacket and peel from tilt release handle (Figure 20, Detail A).

REPLACE SEAT FRAME TILT HANDLE JACKET - Continued

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves, goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Remove old adhesive from handle with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper.

Install

1. Apply a thin coat of adhesive, Item 45, WP 1803 00, to entire surface of handle. Allow to air dry for 10 minutes.
2. **QA** Slide replacement jacket over handle and press down firmly (Figure 20, Detail A).

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves, goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Remove any extra adhesive and clean handle with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

REPLACE SEAT FRAME DIAGONAL STRAP

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust to 3 to 4 stops below full up position.
2. Remove stowage box and panel.
3. Remove clamps securing ICS cord to diagonal strap.
4. Remove bolt, washers, and nut attaching upper end of strap to midframe support (Figure 21, Detail A).

NOTE

Remove diagonal straps one at a time from lower mounts to prevent guide tubes from slipping down into trunnions.

5. Remove bolt, washer, and nut attaching lower end of strap to trunnion mount. Remove strap. If new strap is not going to be installed now, install bolt to keep guide tube and trunnion mount in place.

Repair

If necessary, replace wear strip on diagonal strap using heat shrink tubing.

REPLACE SEAT FRAME DIAGONAL STRAP - Continued

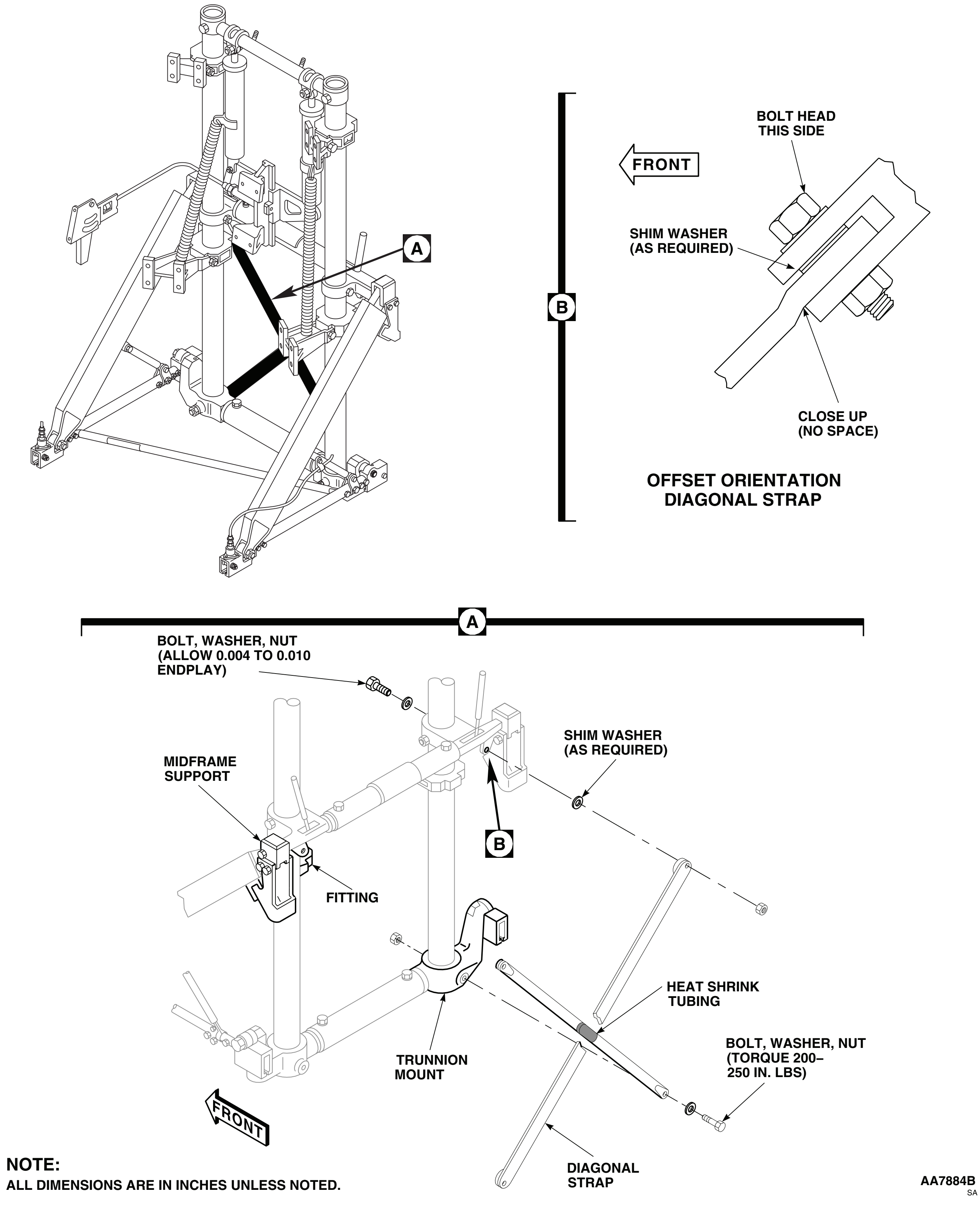


Figure 21. Replace Seat Frame Diagonal Strap.

REPLACE SEAT FRAME DIAGONAL STRAP - Continued

Install

WARNING

Injury to personnel and damage to equipment will result if diagonal straps are installed improperly. Diagonal straps offset portion must face towards seat bucket. Make sure diagonal straps are installed correctly.

NOTE

Replacement diagonal straps, P/N 104460-2, will be manufactured as a one piece design. When used as replacement for previous straps, P/N 100018-001 with welded doubler attachment ends, they will require replacement as pairs (no intermixing of strap configurations). Straps, P/N 104460-2, have an ″offset″ on upper attachment and must be installed on seat attachment point with protruding portion of ″offset″ positioned facing towards seat bucket.

1. **QA** Install bottom end of replacement strap on trunnion mount with bolt, washer, and nut (Figure 21, Detail A). **TORQUE NUT TO 200 - 250 INCH-POUNDS.** Make sure rubber spacer faces other diagonal strap.
2. Install upper end of replacement strap in clevis of midframe support with bolt, washers, shim washers (as required), and nut to eliminate play between strap and clevis. **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF TO GET 0.004 TO 0.010-INCH ENDPLAY.**
3. Clamp ICS cord to diagonal strap.
4. Install stowage box and panel.
5. Make sure area is clean and free of foreign material.

REPLACE SEAT FRAME DIAGONAL STRUT AND SUPPORT

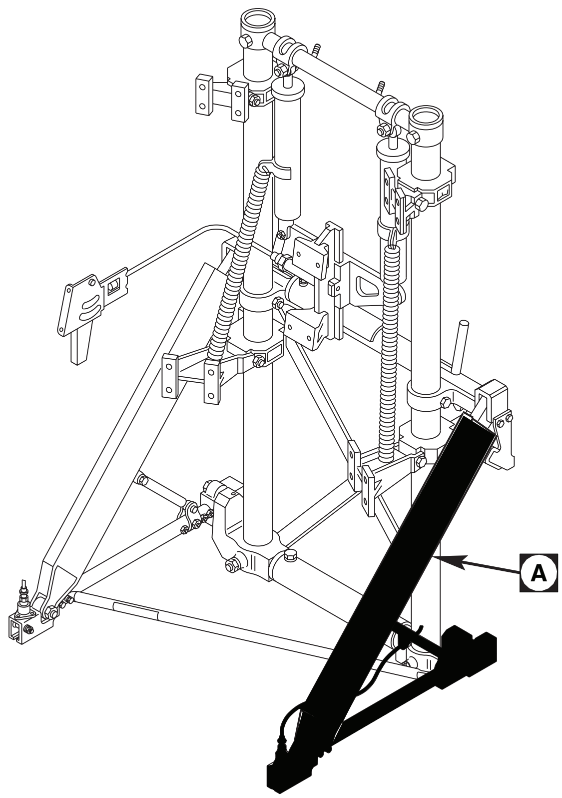
Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Remove diagonal strut support as follows:
   a. Sit in seat and adjust it full up.
   b. Remove screw, washer, and nut securing strut support and clamp to diagonal strut (Figure 22, Detail A).
   c. Remove screw and nut attaching strut support to track rail strap, and remove strut.
2. Remove diagonal strut as follows:
   a. Remove screw, washer, and nut attaching release cable, clamp, and strut support to diagonal strut.
   b. Remove bolt and nut attaching lower end of diagonal strut to front rail fitting. Use a hex wrench in bolt shank; hold bolt while removing nut.
   c. Pull tilt release handle and disengage upper end of diagonal strut from midframe support. Remove strut from track fitting. Remove bearing spacers.

REPLACE SEAT FRAME DIAGONAL STRUT AND SUPPORT - Continued



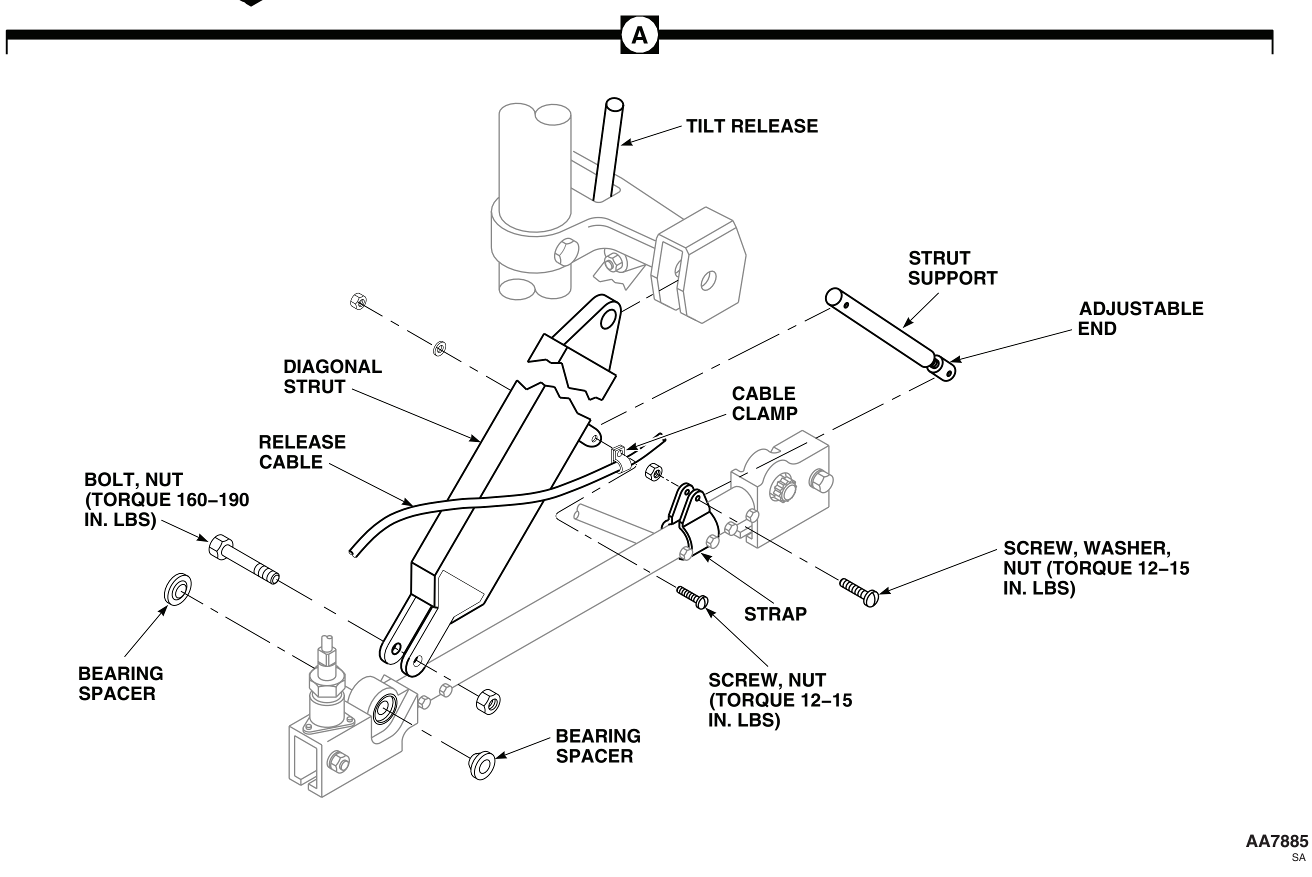


Figure 22. Replace Seat Frame Diagonal Strut and Support.

REPLACE SEAT FRAME DIAGONAL STRUT AND SUPPORT - Continued

d. Make sure area is clean and free of foreign material.

Install

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Install diagonal strut as follows:

   a. Install bearing spacers in seat track fitting. Slide clevis end of diagonal strut over spacers. Make sure support fitting faces down (Figure 22, Detail A).

   b. Pull tilt release handle and install upper end of diagonal strut in midframe support. Engage tilt release pin.

   c. **QA** Install strut clevis with special bolt and nut. Hold bolt shank with key wrench. **TORQUE NUT TO 160 - 190 INCH-POUNDS.**

   d. **QA** Install strut support in diagonal strut fitting. Install clamp and release cable on outboard side of fitting with screw, washer and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**

2. Install diagonal strut support as follows:

   a. **QA** Install replacement support strut between track rail strap and diagonal strut fitting, with adjustable end to track rail strap. Install clamp and release cable on outboard side of diagonal strut fitting. Attach clamp and support to diagonal strut fitting with screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**

   b. If necessary loosen jamnut at support adjustable end, and adjust end until attaching holes line up.

   c. **QA** Attach support to track rail strap with screw and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS. TIGHTEN JAMNUT UNTIL A SHARP RISE IN TORQUE IS FELT, THEN TIGHTEN 1/6 TO 1/3 TURNS MORE.**

   d. Make sure area is clean and free of foreign material.

REPLACE SEAT FRAME FRANGIBLE STRUT

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.

2. Remove bolts, washers, and nuts from both ends of strut, and remove strut (Figure 23, Detail A).

REPLACE SEAT FRAME FRANGIBLE STRUT - Continued

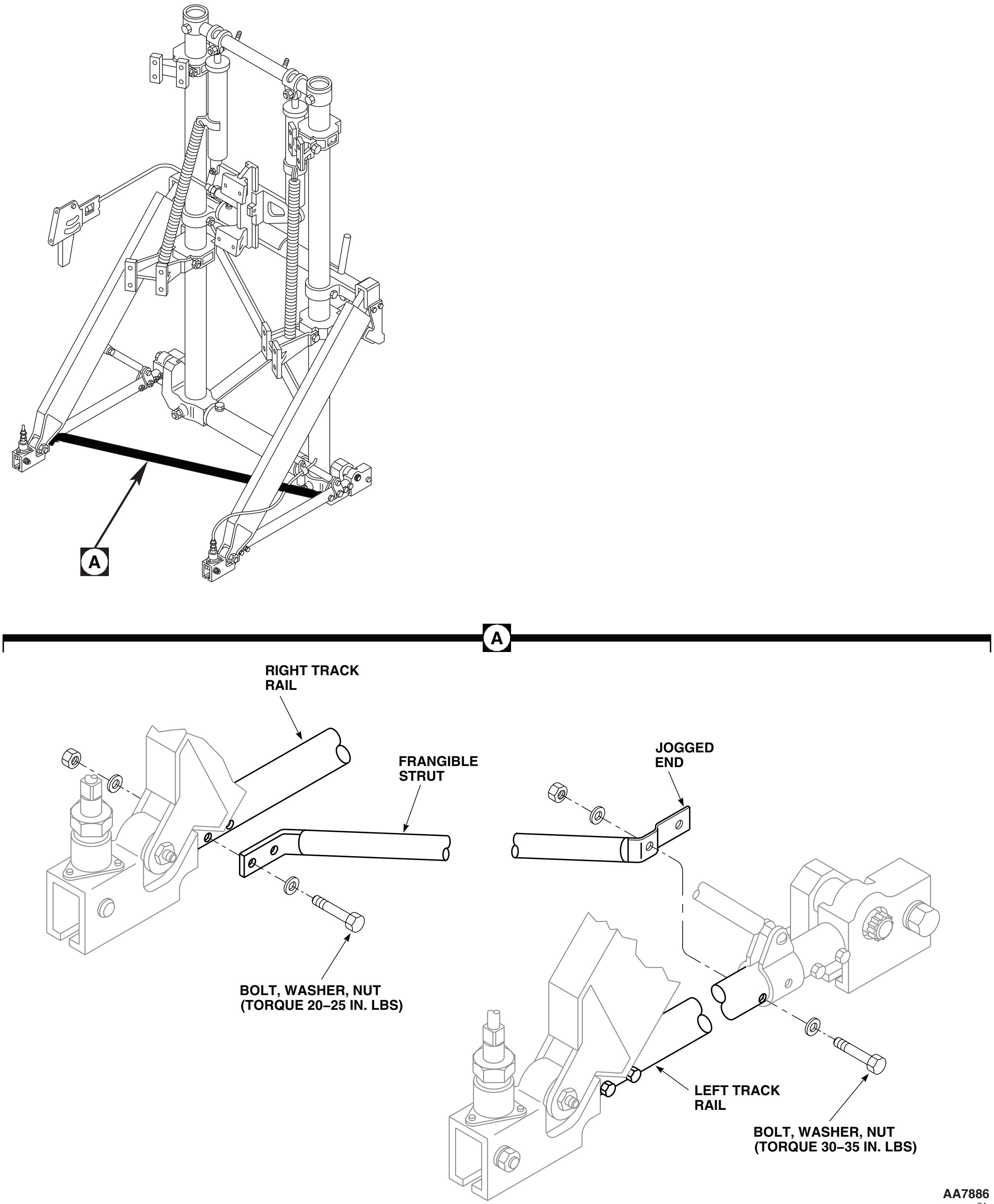


Figure 23. Replace Seat Frame Frangible Strut.

REPLACE SEAT FRAME FRANGIBLE STRUT - Continued

Install

1. **QA** Install replacement frangible strut between track rails, with joggled end fitting on left track rail, using bolts, washers, and nuts (Figure 23, Detail A). **TORQUE RIGHT SIDE NUTS TO 20 - 25 INCH-POUNDS. TORQUE LEFT SIDE NUTS TO 30 - 35 INCH-POUNDS.**

NOTE

All bolts are installed with heads toward left side of seat frame.

2. Make sure area is clean and free of foreign material.

REPLACE SEAT FRAME TRACK ROLLERS AND BEARINGS

Remove

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

1. Sit in seat and adjust it full up.
2. Remove stowage box and panel.
3. Remove seat from helicopter.

CAUTION

Padded support blocks must be placed at midframe supports to prevent damage to tilt handles or seat frame.

4. Provide block of wood with padding to place under midframe support. Carefully lay seat on its back on support blocks.
5. Remove bolts and pin-housing from front track fitting (Figure 24, Detail A).
6. Remove screw and nut attaching strut support to track strap. Remove screws and nuts attaching frangible strut to track.
7. Remove bolt, washer, and nut attaching diagonal strut to front fitting. Remove spacers.

NOTE

Record number of shims removed.

8. Remove bolt attaching rear track fitting to trunnion. Remove spacer and shims. Remove track.

Disassemble

1. Install key wrench in front bolt shank, hold bolt and remove nut and washer. Tap bolt from front fitting. Remove roller with eccentric bushing from inside fitting. Push sliding bushing from nut side of fitting (Figure 24, Detail B).
2. Repeat step 1 for rear fitting.
3. Remove retainers, seals, and eccentric bushings from rollers.
4. Remove mounts as follows:

REPLACE SEAT FRAME TRACK ROLLERS AND BEARINGS - Continued

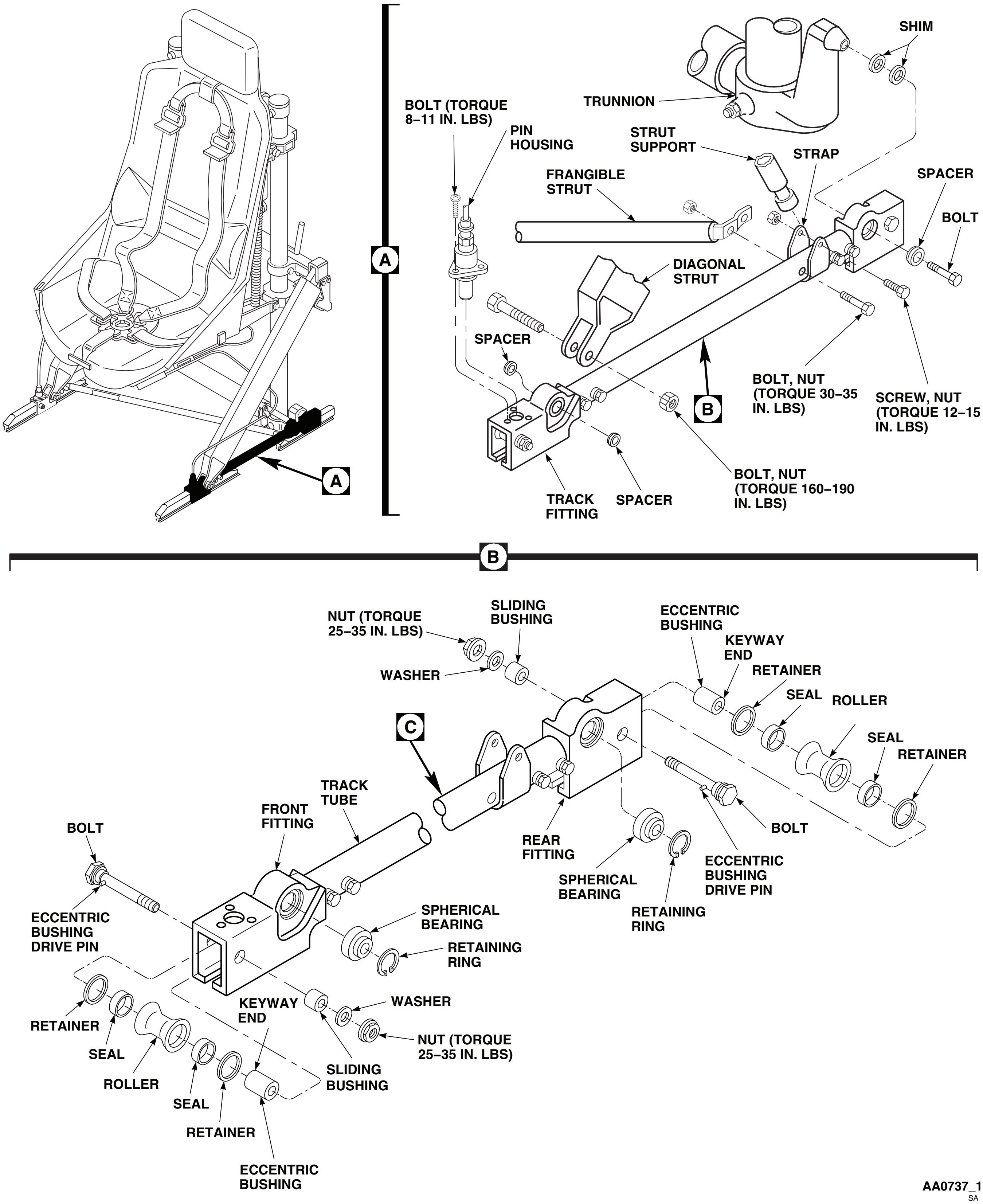


Figure 24. Replace Seat Frame Track Rollers and Bearings. (Sheet 1 of 2)

REPLACE SEAT FRAME TRACK ROLLERS AND BEARINGS - Continued

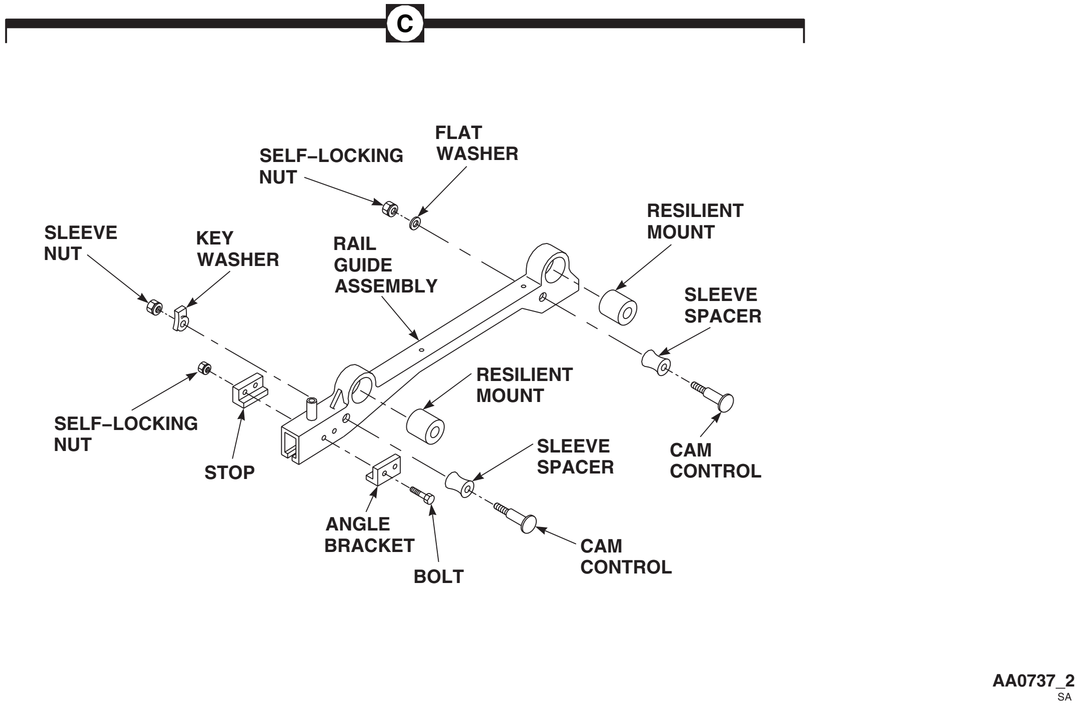


Figure 24. Replace Seat Frame Track Rollers and Bearings. (Sheet 2 of 2)

a. For spherical mounts, remove retaining rings and press spherical bearings from fittings.

b. For resilient mounts, press front isolator from front rail guide ring. Press rear isolator, which is bonded to rear ring of rail guide, or scrape out from rear rail guide ring.

Inspect

1. Clean all parts using machinery towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry with clean machinery towel.
2. Inspect track tube and fittings for distortion, cracks, deep pitting corrosion, deep gouges, and hole elongations. Replace track assembly if damage is found.
3. Inspect rollers, eccentric bushings, sliding bushings, and bolts for distortion, cracks, deep pitting corrosion, deep gouges, and hole elongations. Replace damaged parts as needed.
4. Inspect front and rear bolts for damaged threads, worn bushing, bent or broken eccentric bushing drive pin. Replace as necessary.
5. Inspect threaded holes in front fitting for thread damage.
6. Inspect spherical bearing (resilient mounts) (WP 0232 00). Inspect needle bearings in rollers (TM 55-1500-322-24).

Repair

1. Treat minor corrosion and touch up paint nicks, and scratches (WP 0381 00 and TM 55-1500-345-23).
2. Use 6-32 UNC-2B tap and retap three screwholes in front track, if threads are damaged. If threads are stripped, replace track.

REPLACE SEAT FRAME TRACK ROLLERS AND BEARINGS - Continued

Assemble

1. Install mounts as follows:

   a. For spherical mounts, press replacement spherical bearings into fittings and install retaining ring (Figure 24, Detail B).

   NOTE

   Do not apply sealing compound to front isolator or front ring on rail guide.

   b. For resilient mounts, press front isolator into front ring of rail guide (Figure 24, Detail C. Apply sealing compound, Item 283, WP 1803 00, to contacting surface of rear isolator and rear ring of rail guide. Press isolator into rear ring of rail guide.

2. **QA** Install seals and retainers in roller. Install eccentric bushing in roller.
3. Install roller with eccentric bushing in front fitting, with keyway end of eccentric bushing facing side opposite to spherical bearing (resilient mounts) retaining ring. Face high lobe of eccentric bushing toward front. Install bolt through fitting and roller, and engage drive pin with keyway in eccentric bushing.
4. **QA** Install sliding bushing in fitting. Install washer and nut on bolt.

   NOTE

   Rollers will be adjusted when seat is installed in helicopter.

5. Install roller with eccentric bushing in rear fitting, with keyway end of eccentric bushing facing same side as spherical bearing (resilient mounts) retaining ring. Face high lobe of eccentric bushing toward rear. Install bolt through fitting and roller, and engage drive pin with keyway in eccentric bushing.
6. **QA** Install sliding bushing in fitting. Install washer and nut on bolt.

Install

1. Clean track fittings using machinery towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry using clean machinery towel Item 344, WP 1803 00.
2. **QA** Install rear fitting bearing against trunnion (Figure 24, Detail A). Put same number of shim washers, between trunnion and bearing, as removed. Install spacer against bearing and attach fitting to trunnion with bolt. **TIGHTEN BOLT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF UNTIL THERE IS 0.004 TO 0.010-INCH TOTAL PLAY BETWEEN TRUNNION AND BOLTHEAD (Figure 24, Detail B).**
3. **QA** Install spacers on each side of spherical bearing (resilient mounts) in front fitting with smaller diameter toward bearing. Install fitting in diagonal strut with bolt and nut. **TORQUE NUT TO 160 - 190 INCH-POUNDS.**
4. **QA** Install diagonal strut support in track bracket with screw and nut. **TORQUE NUT 12 - 15 INCH-POUNDS.**
5. **QA** Install end of frangible strut on track tube with screws and nuts. **TORQUE NUTS TO 30 - 35 INCH-POUNDS.**
6. **QA** Install pin-housing in front fitting with bolts. **TORQUE BOLTS TO 8 - 11 INCH-POUNDS.**
7. Set seat upright.
8. Install stowage panel and box.
9. **QA** Install seat in helicopter.
10. Make sure area is clean and free of foreign material.

REPLACE PILOT'S STOWAGE BOX, PANEL, AND BAG

Remove

1. Turn off all electrical power.
2. Remove screws, washers, and nuts fastening stowage bag to stowage panel (Figure 25, Detail A).
3. **UH-60A 85-24462 - SUBQ** Disconnect electrical connector, J455, from cabin dome lights dimmer switch.
4. Loosen fasteners and lift ICS control panel enough to get to electrical connector. Disconnect connector from control panel and remove control panel. Pull connector through bottom of stowage box (Figure 25, Detail B).
5. On right side of stowage panel, remove screws and washers securing stowage panel and ICS cord and clamps to midframe support (Figure 25, Detail A).
6. On left side of stowage panel, remove screws and washers securing stowage box and panel to midframe support. Remove panel.
7. Remove screws, washers, and nuts and remove stowage box from panel.
8. Remove bolts and washers and remove stowage panel support brackets from midframe supports.

Repair

1. Turn off all electrical power.
2. If necessary, repair stowage box and stowage panel (WP 0379 00 and TM 1-1500-204-23).
3. Remove screws, washers, and nuts securing tube and cover to cord terminals. Remove cord from cover. If clamps are damaged, remove them from tube (Figure 25, Detail C).
4. Install replacement cord in cover. Connect tube and cover to cord terminals with screws, washers, and nuts. Install replacement clamps on tube.
5. Repair minor rips or tears in covers by stitching (TM 1-1500-204-23).

Install

1. Turn off all electrical power.
2. **QA** Install stowage panel support brackets on midframe supports with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS (Figure 25, Detail A).**
3. Install stowage box on panel with screws, washers, and nuts.
4. **QA** Install stowage panel on midframe supports with screws and washers. On right side, install ICS cord and clamps. **TORQUE SCREWS TO 20 - 25 INCH-POUNDS.**
5. **UH-60A 85-24462 - SUBQ** Connect electrical connector, J455, to cabin dome lights dimmer switch.
6. **QA** Pull electrical connector up through stowage box and connect electrical connector to ICS control panel. Install control panel in stowage box and fasten with fasteners (Figure 25, Detail B).
7. Install stowage bag on replacement stowage panel with screws, washers, and nuts.
8. Check operation of ICS.

REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG

Remove

1. Turn off all electrical power.
2. Pull first-aid kit snap fasteners from mounting studs and remove kit from stowage panel (Figure 26, Detail A).

REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG - Continued

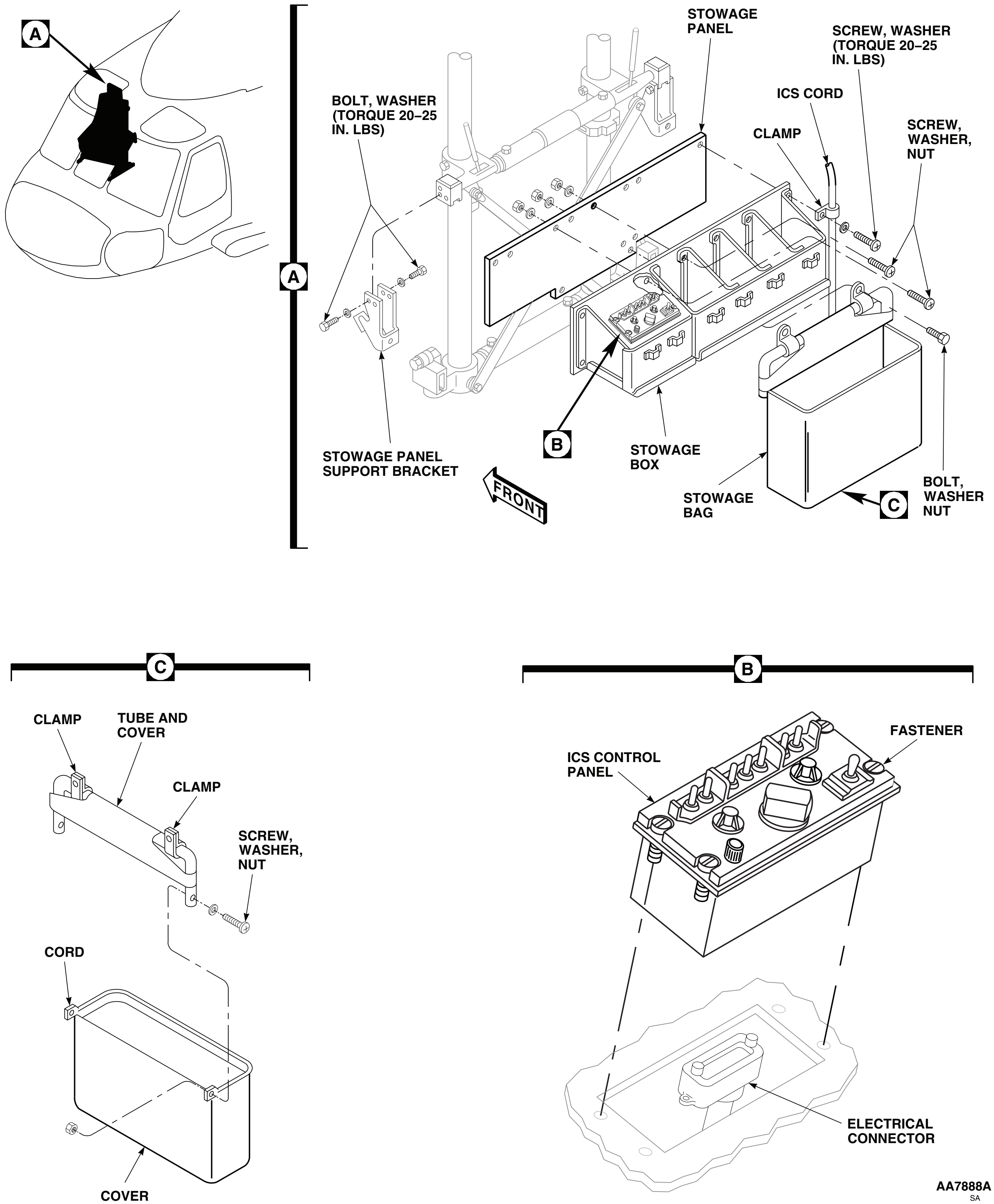


Figure 25. Replace Pilot's Stowage Box, Panel, and Bag.

REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG - Continued

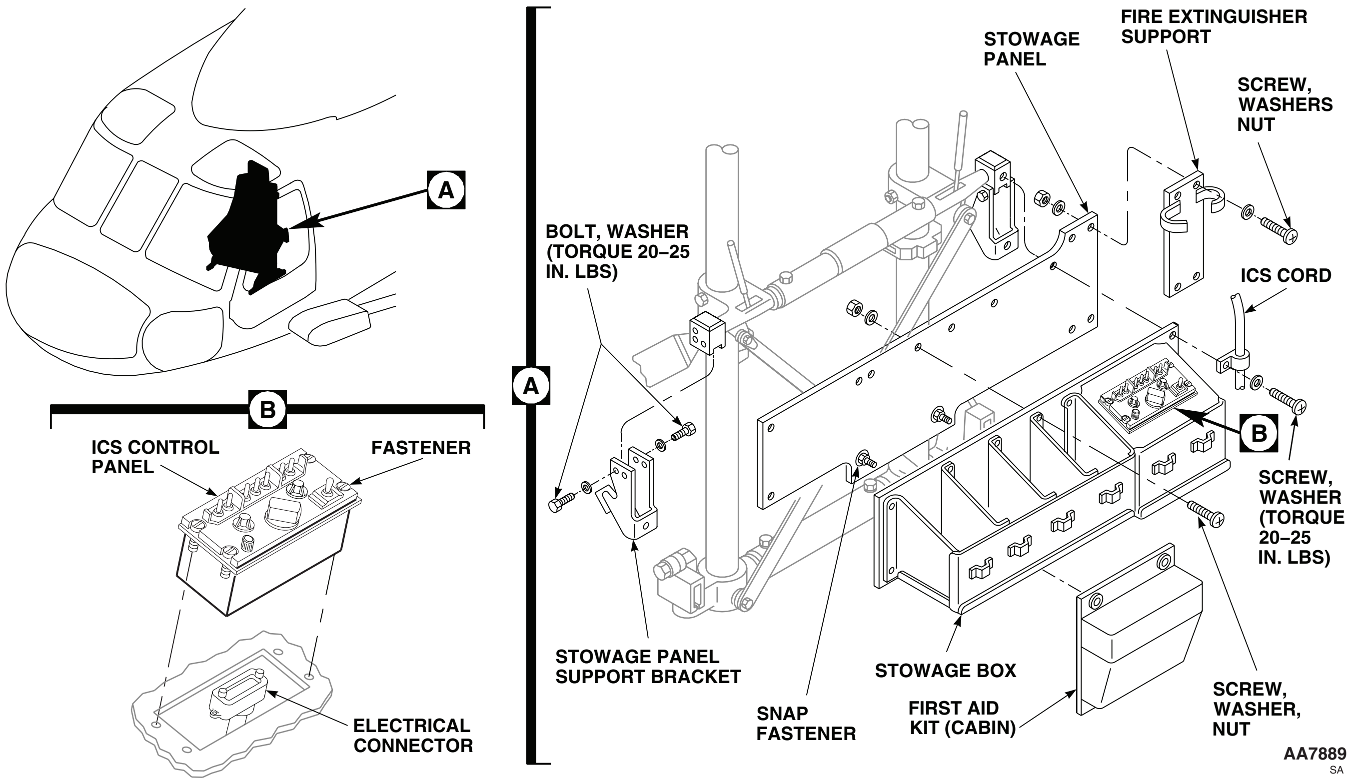


Figure 26. Replace Copilot's Stowage Box, Panel, and Bag.

3. Remove fire extinguisher from support and unscrew support from stowage panel.
4. **UH-60A 85-24462 - SUBQ** Disconnect electrical wires from utility light terminal block. ■
5. Loosen fasteners and lift ICS control panel enough to get to electrical connector. Disconnect connector from control panel and remove control panel. Pull connector through bottom of stowage box (Figure 26, Detail B).
6. On right side of stowage panel, remove screws and washers securing stowage panel and ICS cord and clamps to midframe support (Figure 26, Detail A).
7. On left side of panel, remove screws and washers securing stowage box and panel to midframe support. Remove box and panel.
8. Remove screws, washers, and nuts and remove stowage box assembly from panel.
9. Remove bolts and washers and remove stowage panel support brackets from midframe supports.

Repair

1. Turn off all electrical power.
2. If necessary, repair stowage box and stowage panel (WP 0379 00 and TM 1-1500-204-23). Replace snap fasteners in stowage panel.

Install

1. Turn off all electrical power.
2. **QA** Install stowage panel support brackets on midframe supports with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS (Figure 26, Detail A).**

REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG - Continued

3. Install stowage box assembly on panel with screws, washers, and nuts.
4. **QA** Install replacement stowage panel on midframe supports with screws and washers. On right side, install ICS cord and clamps. **TORQUE SCREWS TO 20 - 25 INCH-POUNDS**.
5. **QA** **UH-60A 85-24462 - SUBQ** Connect electrical wires to utility light terminal block. Cover terminal ends with rubber boots.
6. Pull electrical connector up through stowage box and connect electrical connector to ICS control panel. Install control panel in stowage box and fasten with fasteners (Figure 26, Detail B).
7. Install fire extinguisher support on stowage panel and secure with screws, washers, and nuts. Replace fire extinguisher in support (Figure 26, Detail A).
8. Reinstall first-aid kit on mounting studs of stowage panel.
9. Check operation of ICS.

REPLACE ARMORED WING

Remove

1. Turn off all electrical power.
2. If installed, disconnect pilot's and copilot's lateral air bag modules electrical connector, P5 (pilot), and electrical connector, P7 (copilot), (Figure 27, Sheet 1, Detail A).
3. Remove pilot's and copilot's lateral air bag modules (WP 1589 00).
4. Remove pin and washer from upper cable end. Loosen jamnut, remove upper cable end from lever support (Figure 27, Sheet 1, Detail B).
5. Remove cotter pin, jamnut from lower end of cable. Pull cable out of fitting. Remove cable from helicopter (Figure 27, Sheet 1, Detail C).
6. Remove spring and washer from bracket.
7. Remove spring pins from upper and lower slide tubes.
8. Slide moveable wing to front, remove from helicopter.
9. Remove screw, washers, and nuts attaching stationary wing to fuselage. Remove stationary wing (Figure 27, Sheet 1, Detail C and Figure 27, Sheet 1, Detail D).

Inspect

1. Check release cable lever and support for damage and elongated holes. Replace as necessary.
2. Check tubes for dents. Replace dented tubes if they bind in bearings.
3. Check Teflon bearings for wear. Replace if worn to point where tube binds in bearing (Figure 27, Sheet 2, Detail F).
4. Check cable for binding. Replace if it will not slide freely in sleeve.
5. Check brackets for damage. Repair or replace as necessary (WP 0378 00).
6. Check slide tubes and bushings for general cleanliness. Clean as necessary using Machenery Towel, Item 344, WP 1803 00, moistened with Cleaning Compound, Item 72, WP 1803 00.

Repair

1. Remove bolt, washer, nut, and lever from moveable wing (Figure 27, Sheet 1, Detail B).
2. Remove screw, washers, nut, and rivets attaching support lever to moveable wing.

REPLACE ARMORED WING - Continued

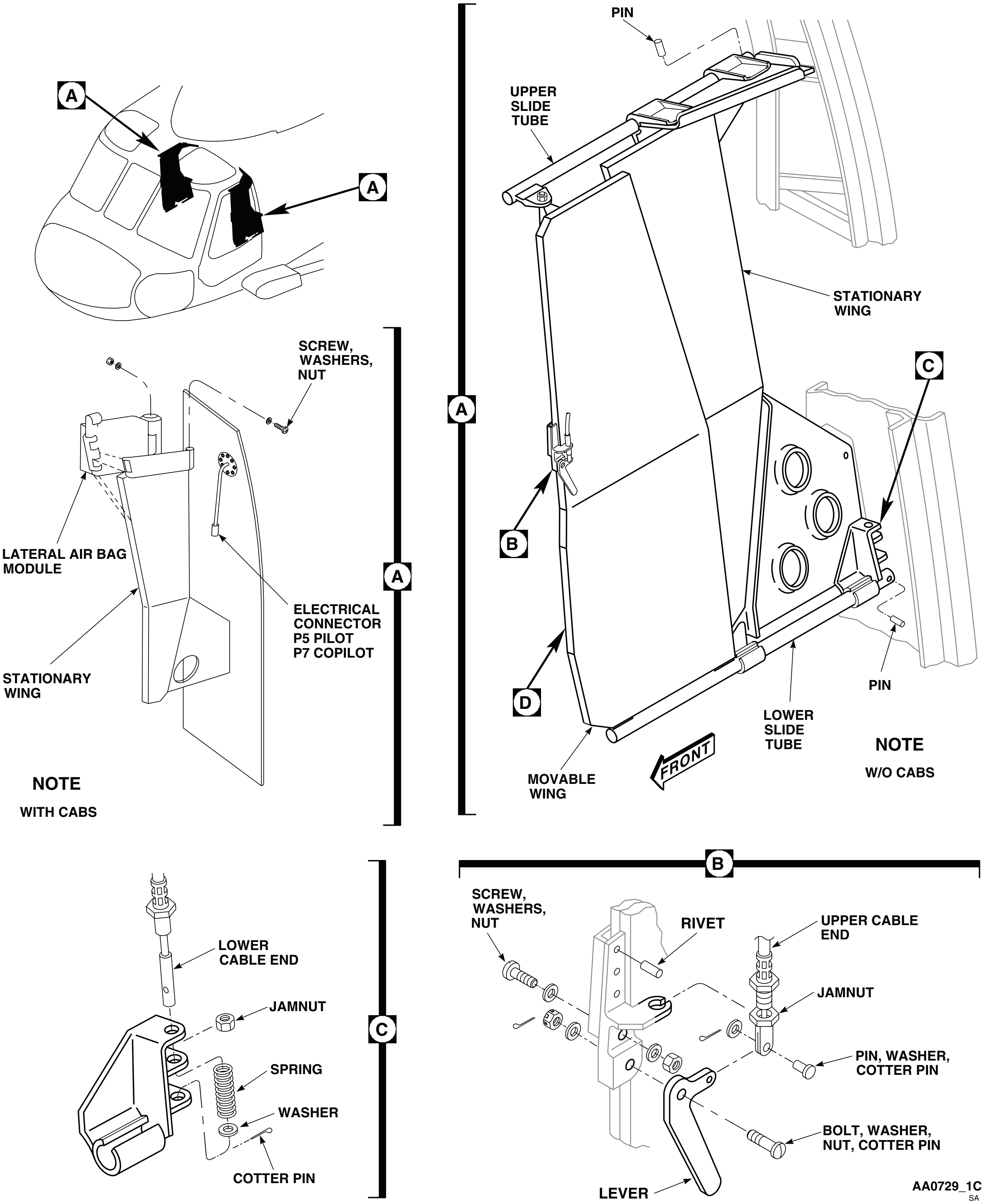


Figure 27. Replace Armored Wing. (Sheet 1 of 2)

REPLACE ARMORED WING - Continued

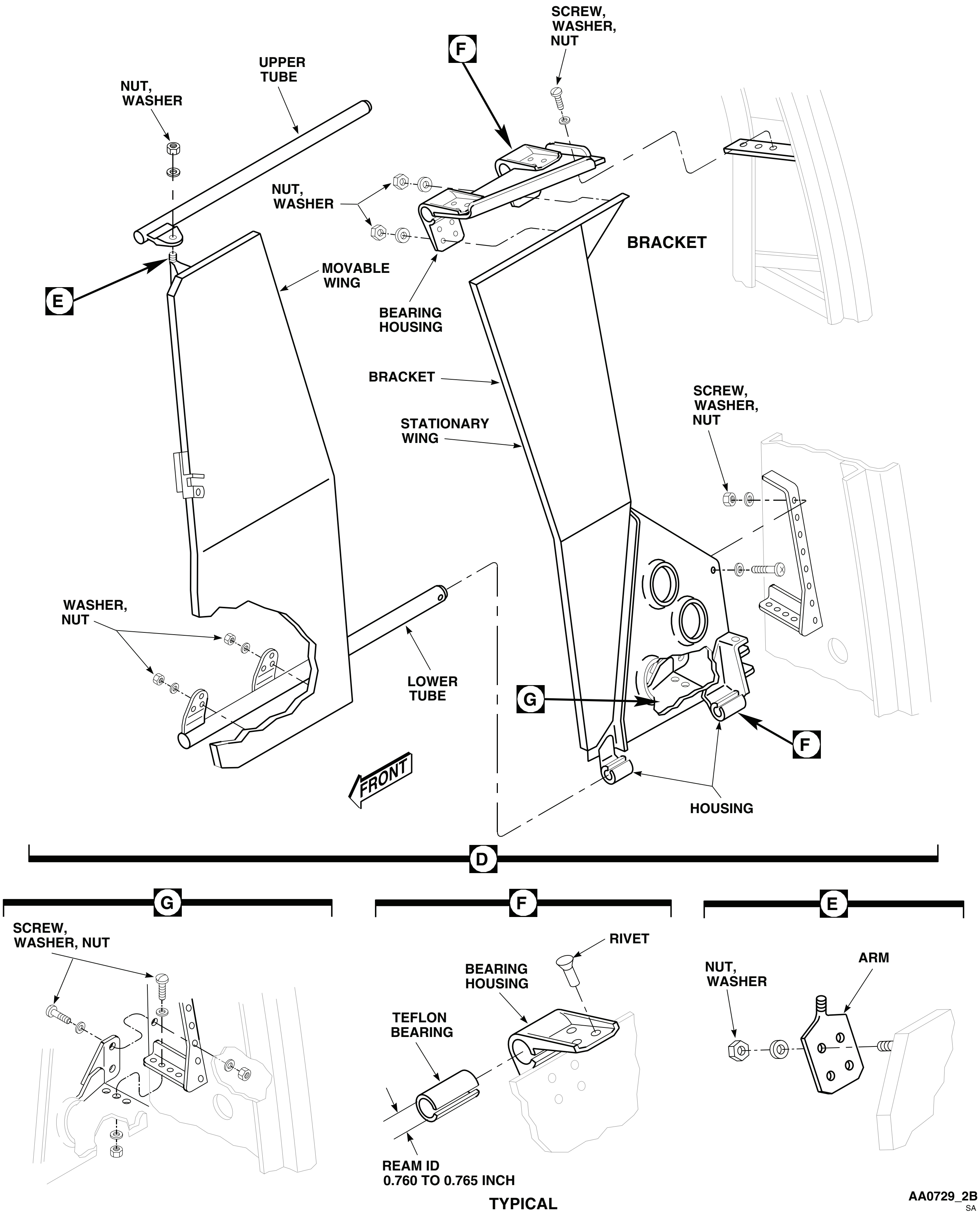


Figure 27. Replace Armored Wing. (Sheet 2 of 2)

REPLACE ARMORED WING - Continued

3. Remove nut, washers, and upper tube from moveable wing (Figure 27, Sheet 2, Detail D).
4. Remove nuts, washers, and arm from moveable wing (Figure 27, Sheet 2, Detail E).
5. Remove nuts, washers, and lower tube from moveable wing (Figure 27, Sheet 2, Detail D).
6. Remove nuts, washers, and rivets, and remove bracket and bearing housings from stationary wing.
7. Remove rivets and bearing housings from brackets (Figure 27, Sheet 2, Detail F).

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

8. Replace Teflon bearings in bearing housings as follows:
   a. Cut damaged bearing from housing.
   b. Use cleaning compound solvent, Item 72, WP 1803 00, and remove old adhesive.

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves, goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   c. Clean housing with machinery towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry using clean machinery towel.
   d. Before bonding, etch outside of bearing with etching solution, Item 121, WP 1803 00, following manufacturer's instructions.
   e. Apply a coat of adhesive, Item 15, WP 1803 00, to bearing housing.
   f. Press Teflon bearing into housing and let cure for 12 hours at 70°F(21°C) to 85°F(29°C).
   g. If adhesive, Item 15, WP 1803 00, is not available, apply a coat of adhesive, Item 16, WP 1803 00, to bearing housing.
   h. Press Teflon bearing into housing and, using heat gun, apply heat 235°F(113°C) to 265°F(129°C) to bearing and housing and allow adhesive to cure for 60 minutes.
   i. After completing bonding, ream Teflon bearing ID to 0.760 to 0.765-inch.
   j. Cut slot in Teflon bearing in bearing housings that are slotted.
9. Install lower bearing housings on brackets with solid rivets (TM 1-1500-204-23).
10. Install upper bearing housings and bracket on stationary wing with solid rivets and nuts and washers (TM 1-1500-204-23).
11. Install lower tube on moveable wing with nuts and washers (Figure 27, Sheet 2, Detail D).
12. Install arm with nuts and washers on moveable wing (Figure 27, Sheet 2, Detail E).

REPLACE ARMORED WING - Continued

13. Install upper tube on moveable wing with nuts and washers (Figure 27, Sheet 2, Detail D).
14. Install lever support on moveable wing with screw, washer, nut and solid rivets (TM 1-1500-204-23) (Figure 27, Sheet 1, Detail A).
15. Install lever on moveable wing with bolt, washers, nut, and cotter pin.

Repair Wing Panels

NOTE

It is not necessary to remove stationary panel for repair. Moveable panels should be removed to facilitate workspace for stationary panel.

1. Remove moveable panel. Refer to, REPLACE ARMORED WING, in this work package.
2. Using putty knife, peel back canvas cover from panel.
3. For center bond separations do the following:

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves, goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   a. Clean area with abrasive paper, Item 210, WP 1803 00, and machinery towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
   b. Gently spread open disbonded area and inject with adhesive, Item 15, WP 1803 00.
   c. Clamp area closed with clamps or bracing, allow 24 hours for adhesive to cure.
4. For edge separations, inject with adhesive, Item 47, WP 1803 00, allow 24 hours for adhesive to cure.
5. Reattach canvas cover using adhesive, Item 14, WP 1803 00, and reinstall panel. Refer to, REPLACE ARMORED WING, in this work package.
6. Touch up paint as necessary.

Install

1. **QA** Install stationary wing to fuselage with screws, washers, and nuts (Figure 27, Sheet 2, Detail D and Figure 27, Sheet 2, Detail G).
2. Slide upper and lower tubes into bearings on stationary wing and install moveable wing (Figure 27, Sheet 1, Detail A).
3. Press spring pins in upper and lower tubes.
4. **QA** Install upper end of cable in lever with pin and cotter pin (Figure 27, Sheet 1, Detail B).
5. **QA** Slide cable end in lever support. Tighten jamnut and lockwire using nonelectrical wire, Item 357, WP 1803 00.
6. **QA** Install spring with washer on bottom in bracket on stationary wing (Figure 27, Sheet 1, Detail C).
7. **QA** Slide lower end of cable through bracket, jamnut, spring, and washer. Compress spring; install cotter pin in end of cable. Tighten jamnut.
8. Turn off all electrical power, disconnect battery and pull CABS circuit breaker.

REPLACE ARMORED WING - Continued

9. Install lateral air bag module on stationary wing and fasten with screws, washers, and nuts (WP 1589 00).
10. Inspect and treat electrical connectors (WP 1703 00).
11. **QA** Connect electrical connector P5 (pilot), and electrical connector P7 (copilot) (Figure 27, Sheet 1, Detail A).
12. Do an operational check of air bag system (WP 0171 00).
13. Check armored wing for smooth operation.

REPLACE PILOT'S/COPILOT'S SEAT

Remove

1. Turn off all electrical power.

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

2. Sit in seat. Under right front of seat, release VERT adjust lever and adjust seat to full up. Under left front of seat, pull LONG adjust lever and slide seat fully to rear.
3. Remove gunner's seat and two troop seats next to cargo door on side through which seat will be removed (WP 0233 00).
4. Remove stowage box and panel from seat.
5. Pull first aid kit snap fasteners from mounting studs on back of pilot's or copilot's seat, and remove kit.
6. On copilot's seat, remove fire extinguisher and mounting bracket. On both seats, remove stowage box and panel.
7. Remove screws, washers, and nuts, and remove clamps and spacer securing ICS control cable to seat top support tube. Remove tiedown straps holding ICS cable to right attenuator (Figure 28, Sheet 1, Detail A, Detail B, Detail C, and Detail D).

CAUTION

Move ICS cable out of way when removing seat to prevent damage to cable.

8. Remove screws, washers, and nuts, and remove clamps and ICS cable from cross brace (Figure 28, Sheet 2, Detail E and Detail F).
9. Move ICS cable out of way so it will not be damaged when removing and installing seat. Keep clamps, screws, washers, and nuts with ICS cable.
10. Remove screws securing raised floor panel at front of seat tracks, and remove panel.
11. Remove bolts and stops from front of front seat tracks.
12. If necessary, locally make stabilizer strut (Figure 182, WP 1805 00).

REPLACE PILOT'S/COPILOT'S SEAT - Continued

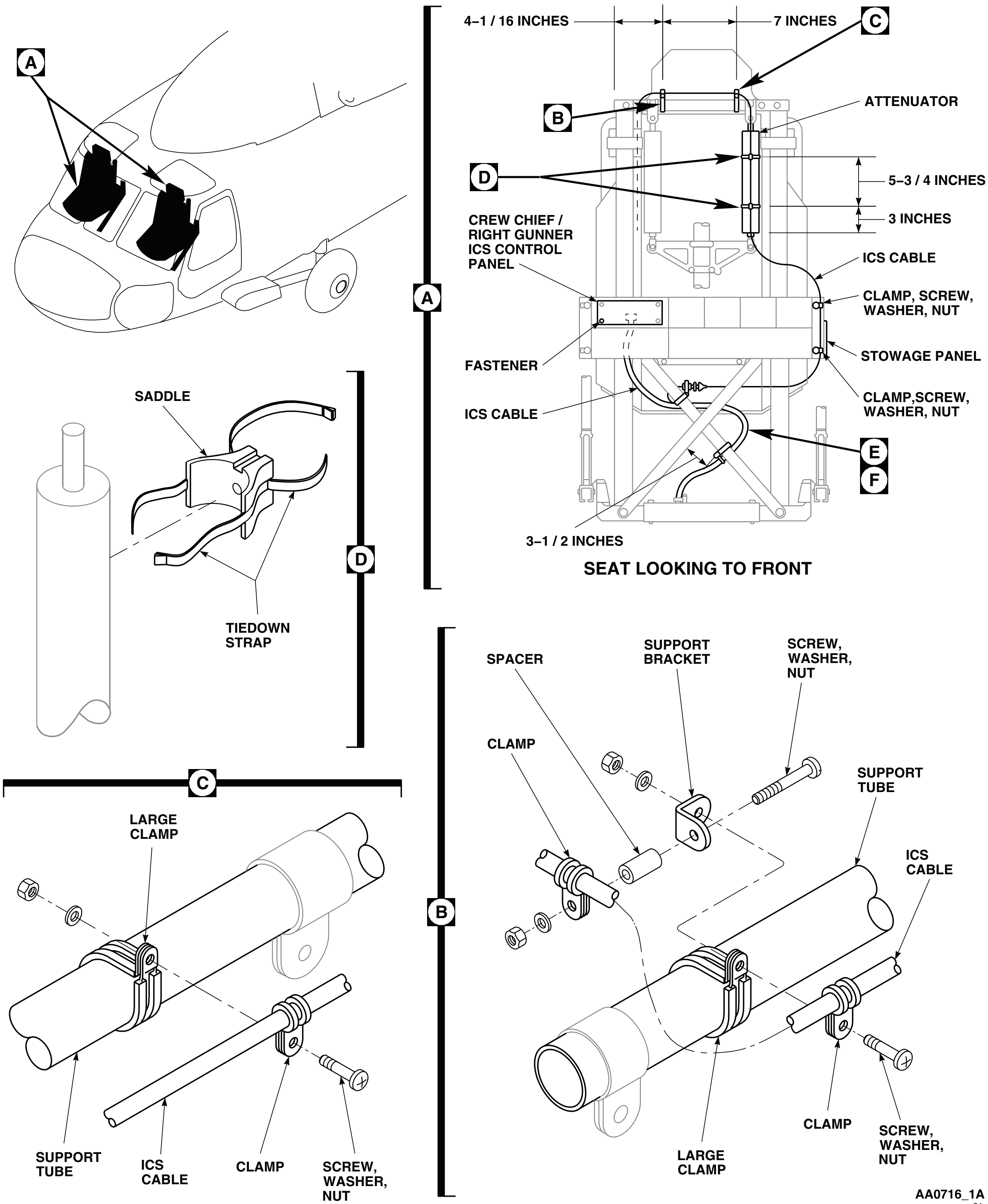
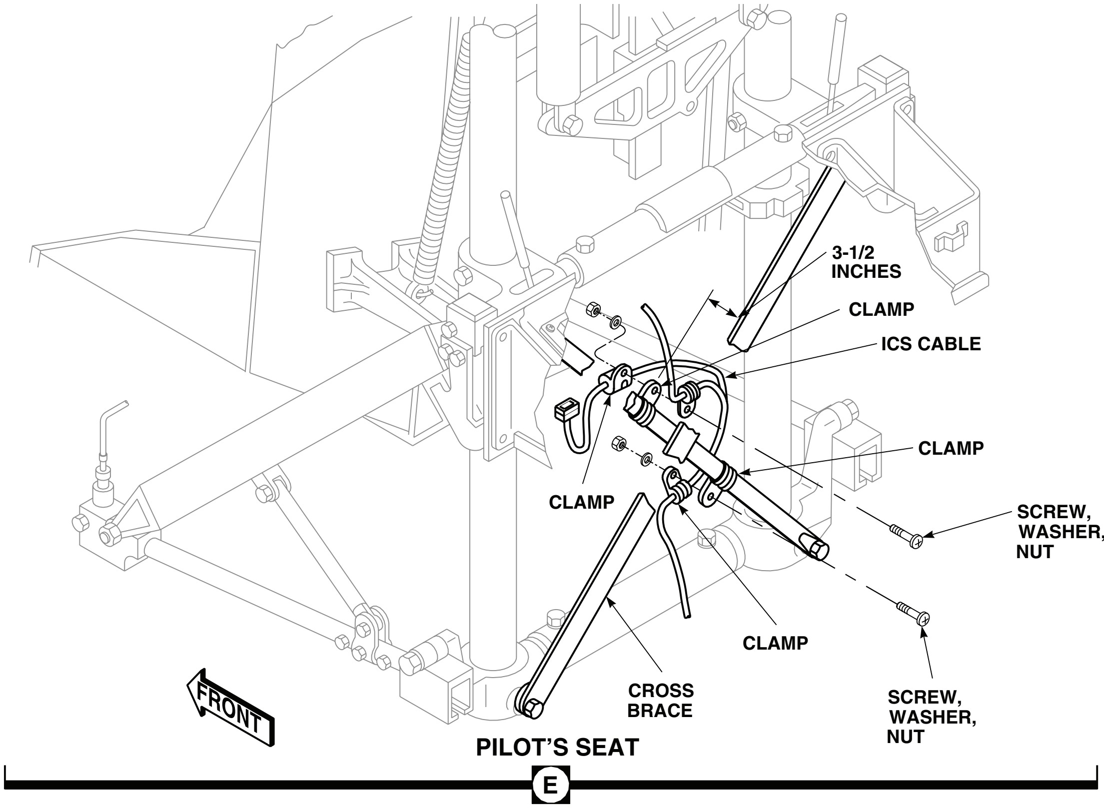


Figure 28. Replace Pilot's/Copilot's Seat. (Sheet 1 of 2)

REPLACE PILOT'S/COPILOT'S SEAT - Continued



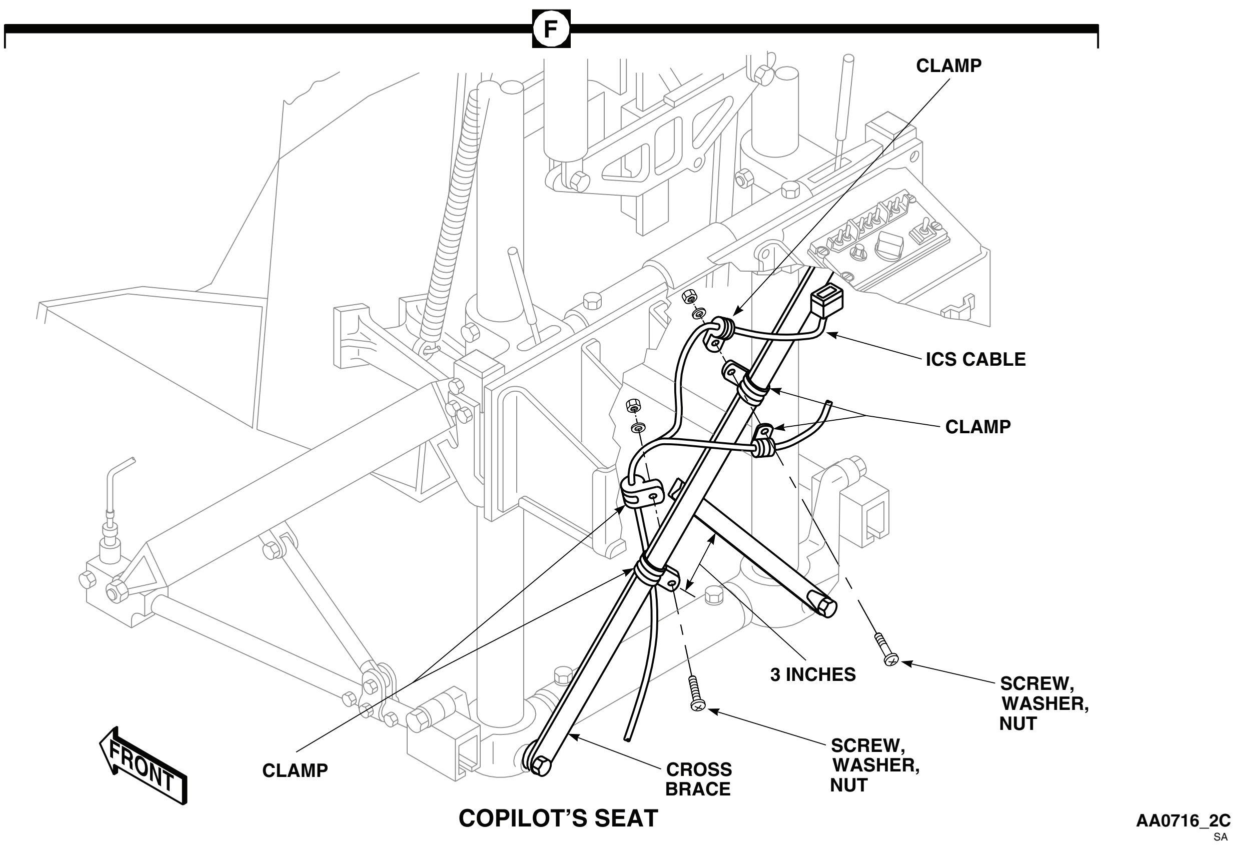


Figure 28. Replace Pilot's/Copilot's Seat. (Sheet 2 of 2)

REPLACE PILOT'S/COPILOT'S SEAT - Continued

CAUTION

Do not remove seat from tracks unless red stabilizer strut is installed to prevent damage to seat.

13. Install red stabilizer strut on front seat rail fittings with screws.

CAUTION

- Do not lift seat by frangible strut or track fittings or components can be bent or otherwise damaged by careless handling.
- Do not bump, push on, or apply twisting loads to strut or damage to diagonal strut will result when tilting seat backward.
- Abrasive damage can be caused by carelessness in moving seat. Carry seat to new location. Do not drag it across ground or floor on track fittings, lower cross frame fittings, seat sides, or seat back.

14. Under front left side of seat bucket, pull LONG adjust lever and roll seat forward until track fittings clear tracks. Lift seat by guide tubes, middle or lower cross members in back of seat, and forward edge of bucket. If necessary for clearance, tilt seat back into cabin by using TILT RELEASE levers. With aid of assistant, lift seat through cabin and out cargo door.
15. Place seat on pallet for transporting.

Install

1. Turn off all electrical power.

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

2. Sit in replacement seat. Under right front of seat, release VERT adjust lever and adjust seat to full up position.

CAUTION

Track fittings can be damaged through improper engagement with seat tracks if width between fittings is incorrect. Do not install seat unless red stabilizer strut is in place.

3. If necessary, install red stabilizer strut on front seat track fittings with screws.

REPLACE PILOT'S/COPILOT'S SEAT - Continued

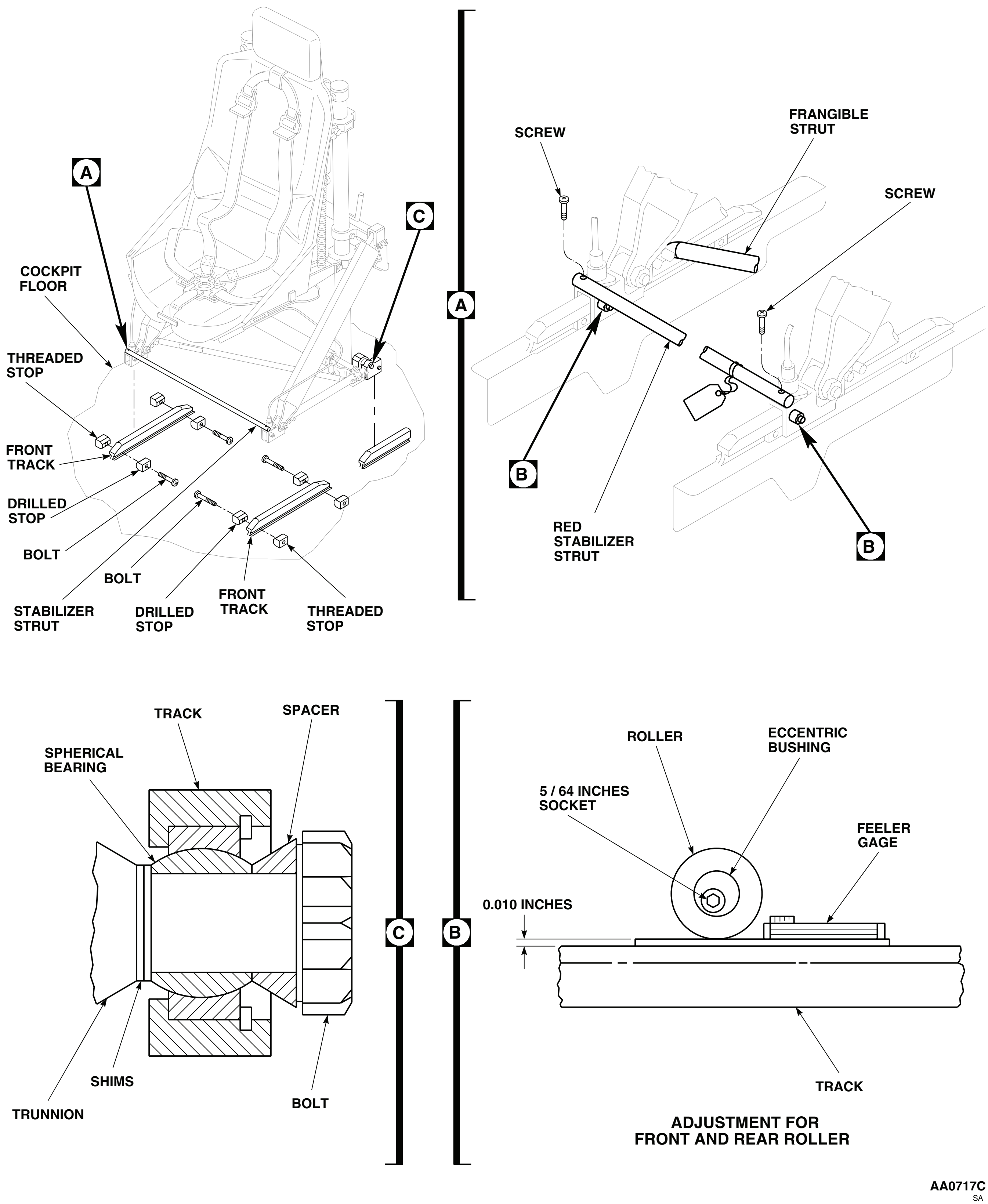


Figure 29. Adjust Pilot's/Copilot's Seat.

REPLACE PILOT'S/COPILOT'S SEAT - Continued

WARNING

Pilot and copilot seat restraint buckles must have same type of release mechanism. Different types of release mechanism could hinder an emergency egression. If installing a replacement seat, make sure that both pilot and copilot seat restraint buckles have same type of release mechanism.

CAUTION

- Seat components can be bent or otherwise damaged by careless handling. Do not lift seat by frangible strut or track fittings.
- Avoid damage to diagonal strut when tilting seat backward. Do not bump, push on, or apply twisting loads to strut.
- Abrasive damage can be caused by carelessness in moving seat. Carry seat to new location. Do not drag it across ground or floor on track fittings, lower cross frame fittings, seat sides, or seat back.

4. Lift replacement seat by guide tubes, middle or lower cross members in back of seat, and forward edge of bucket. Install seat through cargo door, place in cockpit with track fittings in front of and in line with tracks on cockpit floor.
5. Pull LONG adjust lever, roll seat onto tracks against rear stops.
6. If seat rear track fittings do not line up with floor tracks, or if they bind when seat is moved to rear, adjust shims between trunnions and spherical bearing (resilient mounts) in track fitting as follows (Figure 29, Detail C):
   a. Do one side at a time and try to have same number of shims on each side. Keep adjusting until seat rolls freely for entire length of track.
   b. Remove bolt and spacer and add or remove shims as necessary to free track fitting from binding on track.
   c. Install spacer and bolt. Tighten bolt until there is no endplay then back off until you have 0.004 to 0.010-inch play between parts.

NOTE

If replacing seat stop, a maximum of 0.062-inch may be removed from stop length, if required.

7. **QA** Install seat stops (chamfered side against track) on front track with bolts.
8. Rear stops must face properly. On seats, RA 100900-1 and RA 100900-5, face red end of stop to front, on seats, RA 30525-1, face red end of stop to rear.
9. Install raised floor panel in front of seat. Install screws.
10. Pull LONG adjust lever and move seat forward against stops.
11. Route ICS cables and clamp to cross braces (Figure 28, Sheet 1, Detail A and Figure 28, Sheet 2, Detail E and Detail F). Allow at least 1 1/2-inch slack in ICS cable between first clamp on cross brace, and place where cable comes out of floor at STA 247.0.
12. Install stowage panel and stowage box. At same time attach ICS cable to stowage panel right side with clamps, bolts, washers, and nuts. Install washers under bolthead and nuts.

REPLACE PILOT'S/COPILOT'S SEAT - Continued

WARNING

Injury to personnel and damage to euipment will occur if improper tiedown straps are used to attach ICS cable to attenuator. Attenuator movement will be restricked if incorrect tiedown straps are used. Use only electrical tiedown straps, Item 302, WP 1803 00 to attach ICS cable to attenuator.

13. **QA** Route ICS cable up alone right attenuator, continue across seat top support tube, to left side of seat back cushion. Tie ICS cord to attenuator with two electrical tiedown straps, Item 302, WP 1803 00 (Figure 28, Sheet 1, Detail A).
14. **QA** Attach two saddle assemblies to right attenuator using electrical tiedown straps, Item 302, WP 1803 00 (Figure 28, Sheet 1, Detail D).
15. **QA** Attach ICS cable to saddle assemblies using electrical tiedown straps, Item 302, WP 1803 00 (Figure 28, Sheet 1, Detail D).
16. **QA** Clamp ICS cord to right side of seat top support with clamps, screw, washer, and nut (Figure 28, Sheet 1, Detail C).

NOTE

The ICS cable should be 16 inches in length from final attachment clamp.

17. **QA** Clamp ICS cord to left side of seat top support tube with clamps, support bracket, spacer, screw, washer, and nut. Position so ICS cable goes over left side of seat back cushion without touching headrest (Figure 28, Sheet 1, Detail B).
18. **QA** Connect ICS cable electrical connector to gunner's ICS control panel and install panel in stowage box. Lock fasteners.
19. On copilot's seat, install fire extinguisher mounting bracket on right end of stowage panel with screws, washers, and nuts.
20. Install fire extinguisher in mounting bracket and connect latch.
21. Install first aid kit on studs on seat top support tube and snap fasteners closed.
22. Inspect for excessive grease and dirt buildup on tracks and rollers. Clean as follows:
    a. Remove front stops from seat tracks. Move seat to front of tracks.
    b. Tilt seat to rear to expose bottom of seat rail guides and horizontal lockpins. Clean using machinery wiping towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry with clean machinery towel.
    c. Reinstall seat to track.
    d. Reinstall front stops.
23. Adjust track rollers as follows:
    a. Loosen nut on track roller bolt.
    b. Insert a 0.010-inch feeler gage between roller and track rail (Figure 29, Detail B).

REPLACE PILOT'S/COPILOT'S SEAT - Continued

CAUTION

To prevent damage to roller, fitting, or bolt, do not try to turn adjustment bolt 360 degrees. Roller will jam against fitting and damage cross pin in bolt.

c. Using 5/64-inch key wrench in nut end of bolt, turn eccentric bushing until track fitting touches underside of track rail.

d. **QA TORQUE NUT TO 25 - 35 INCH-POUNDS, WHILE MAINTAINING ADJUSTMENT WITH KEY WRENCH.**

e. Remove feeler gage. Remove key wrench.

24. Pull LONG adjust lever and roll seat stop to stop. Check that seat rolls freely on track.
25. Install gunner's and troop seats (WP 0233 00).
26. Check operation of ICS (TM 11-1520-237-23).
27. Make sure area is clean and free of foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

PILOT'S/COPILOT'S SEATS, D-3801-1 AND D-3177

This WP supersedes WP 0232 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drag Link Adjusting Tool, Locally-Made, Figure Figure 162, WP 1805 00
Drag Link Tool, GGG-B-1222
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Cleaning Compound, Solvent, Item 71, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 11-1520-237-23
WP 0231 00
WP 0233 00
WP 1803 00
WP 1805 00

REPLACE INERTIA REEL

See WP 0231 00 for maintenance instructions.

REPLACE INERTIA REEL CONTROL

See WP 0231 00 for maintenance instructions.

REPLACE SHOULDER HARNESS

See WP 0231 00 for maintenance instructions.

REPLACE LAP BELT

See WP 0231 00 for maintenance instructions.

REPLACE CROTCH BELT, BUCKLE, AND PAD

See WP 0231 00 for maintenance instructions.

REPLACE SEAT CUSHIONS

See WP 0231 00 for maintenance instructions.

REPLACE ARMORED WING

See WP 0231 00 for maintenance instructions.

FRONT-TO-REAR ADJUST SYSTEM

Inspect/Adjust Tracks and Rollers

1. Inspect for excessive grease and dirt build-up on tracks and rollers. Clean as follows:

FRONT-TO-REAR ADJUST SYSTEM - Continued

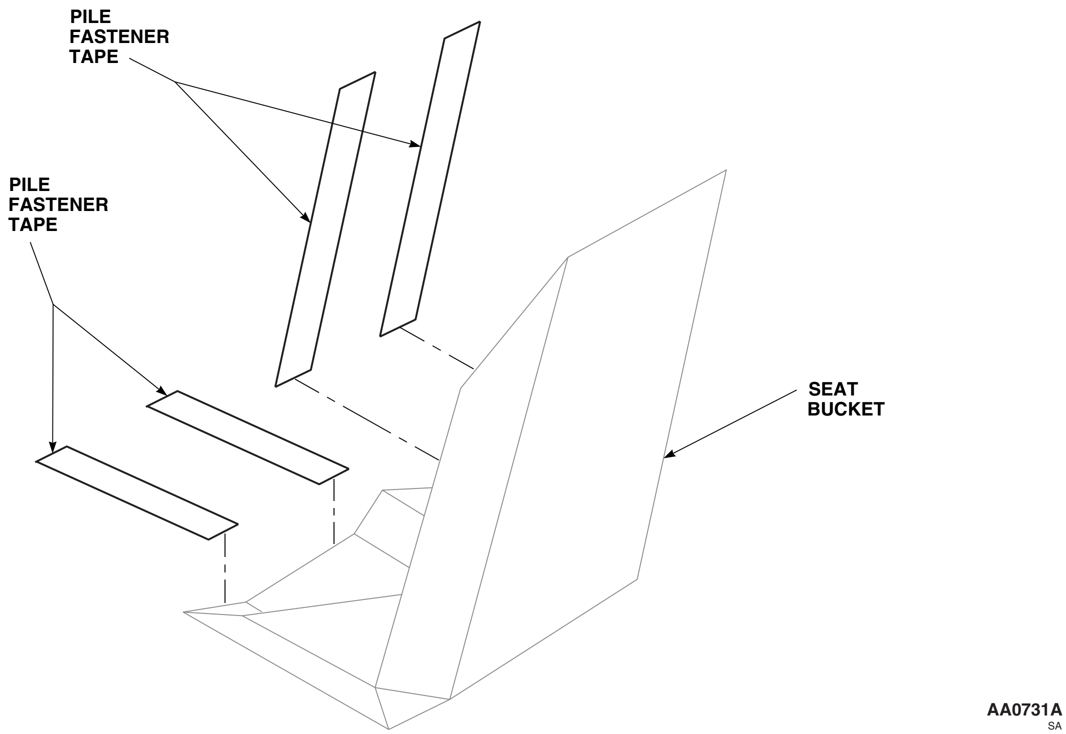


Figure 1. Replace Seat Cushions.

a. Remove front stops from seat tracks. Move seat to front off tracks. Refer to, REPLACE PILOT'S/COPILOT'S SEAT, in this work package.

b. Tilt seat to rear to expose bottom of seat rail guides and horizontal lockpins. Clean using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry using clean machinery towel.

c. Inspect guide rails for cracks. Replace guide rail if large cracks in material are found. Rails determined to have hairline cracks may remain in service until next scheduled PMS (40 hour) inspection, and will be inspected at each Daily Inspection for condition and accelerated crack growth.

d. Inspect front and rear fittings for distortion and cracks (Figure 3, Detail A). Distortion of fittings will facilitate looseness of seat assembly when installed, and may lead to failure of seat rail guide. Replace rail guide if distorted, cracked, or if seat assembly rocks from side to side and cannot be adjusted as outlined in step 3.

e. Reinstall seat to track and install stops. Refer to, REPLACE PILOT'S/COPILOT'S SEAT, in this work package.

2. Check for excessive clearance between seat rail guides and tracks. **CLEARANCE SHALL NOT EXCEED 0.010-INCH BETWEEN UNDERSIDE OF TRACK RAIL AND SEAT GUIDE.**

3. If seat rail guides to track clearance is excessive, reduce clearance by adjusting rollers on each side of seat (two rollers per side). Loosen nuts and turn front and rear cams to achieve smooth front to rear seat travel with minimum rocking of seat.

4. Inspect spherical bearing or resilient mounts for cracks (Figure 3). Inspect for rippling of rubber material. Rippling occurs before tearing and failure of resilient mount. Once torn, damage to rail guide will occur due to metal-to-metal contact of support mount bolts. **REPLACE SPHERICAL BEARING OR RESILIENT MOUNTS IMMEDIATELY IF CRACKS ARE FOUND OR IF NOT POSSIBLE, REPLACEMENT(S) MAY BE MADE AT**

FRONT-TO-REAR ADJUST SYSTEM - Continued

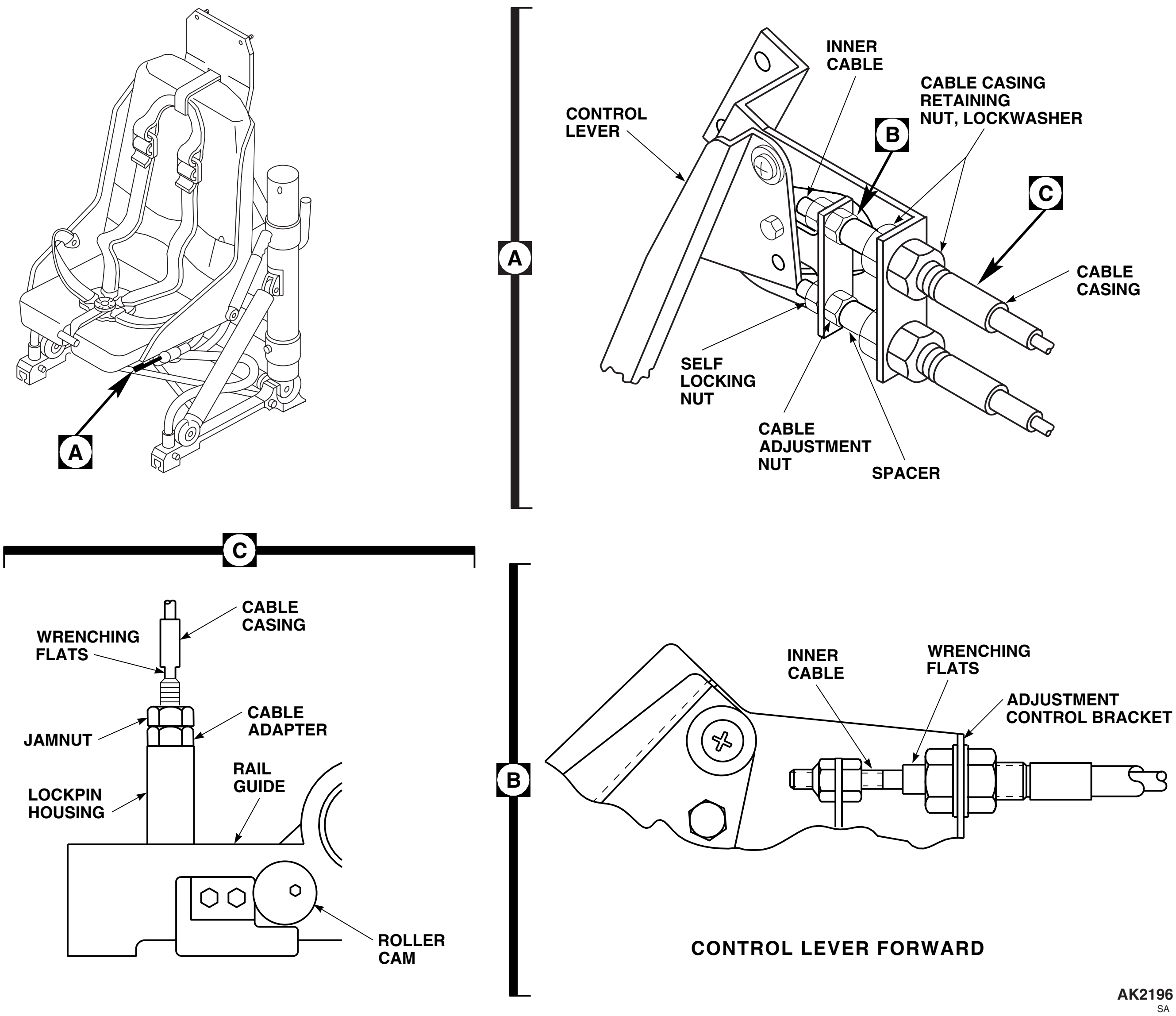


Figure 2. Adjust Rollers and Lockpins.

NEXT PREVENTIVE MAINTENANCE SERVICES (40 HOUR) PMS INSPECTION INTERVAL BUT IN NO CASE LATER THAN PREVENTIVE MAINTENANCE SERVICES (40 HOUR) PMS INTERVAL FROM RECORDED TIME/DATE OF ORIGINAL INSPECTION DISCOVERY OF CRACKED SPERICAL BEARING (RESILIENT MOUNTS).

Adjust Horizontal Lockpins

1. Remove seat, tilt to rear. Refer to, REPLACE PILOT'S/COPILOT'S SEAT, in this work package.
2. Pull pin release lever located at front left side of seat bucket and hold.
3. Slide back spacer on top end of cable to expose wrenching flats on inner cable near nut (Figure 2, Detail A and Detail B).
4. Hold cable and loosen jamnut at end of cable. Do not remove nut from cable.

FRONT-TO-REAR ADJUST SYSTEM - Continued

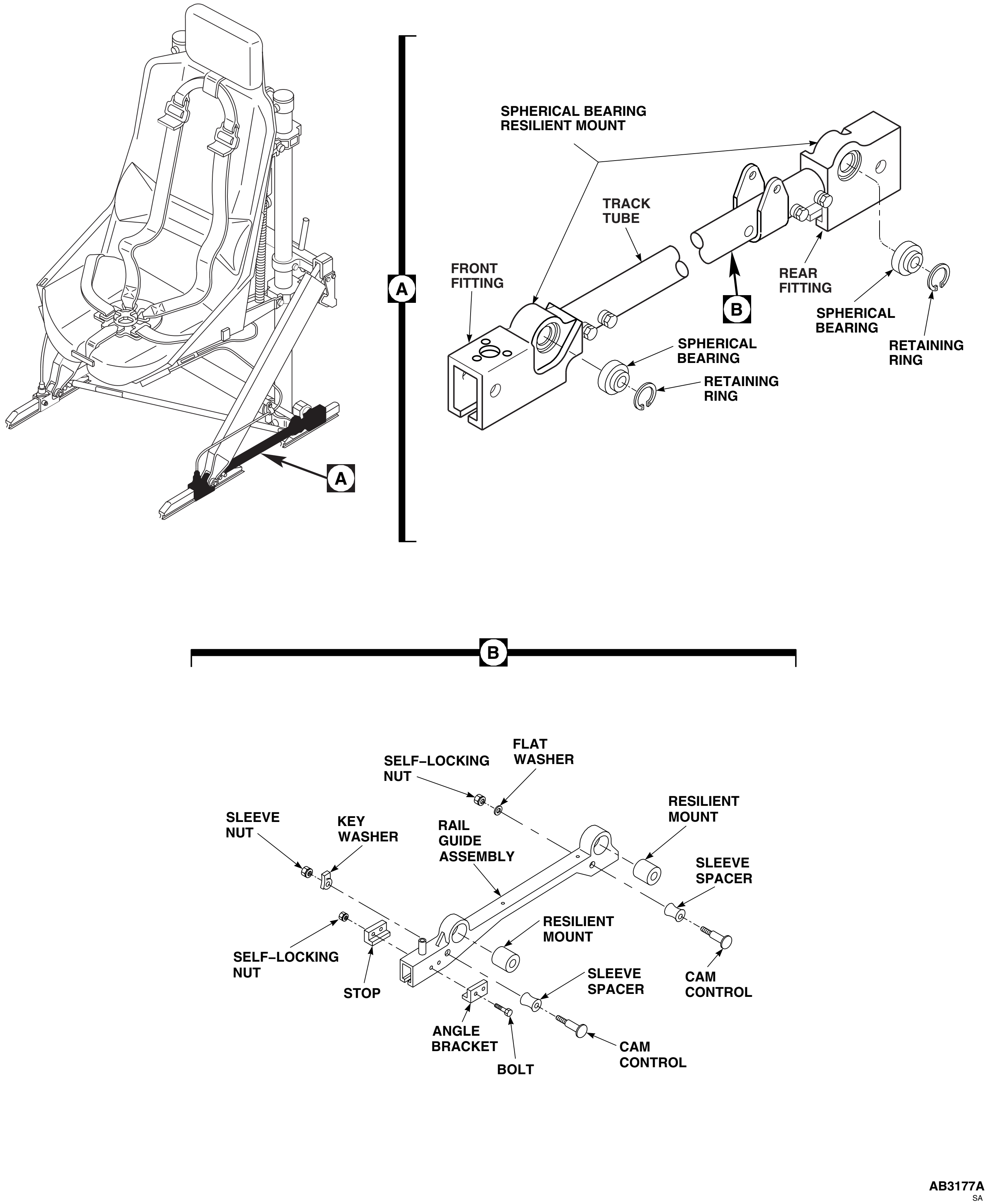


Figure 3. Inspect Spherical Bearing or Resilient Mounts.

FRONT-TO-REAR ADJUST SYSTEM - Continued

5. Loosen nuts retaining cable housing to lever bracket.
6. Hold cable outer cover at wrenching flats and back jamnut off cable adapter at lockpin-housing. Continue until nut reaches bottom of cable cover threads.
7. Unscrew cable adapter from lockpin-housing while holding cable cover, until it bottoms against jamnut.
8. Back off jamnut and remove adapter from lockpin housing.
9. Turn cable housing and adapter clockwise. Reinstall cable into lockpin-housing. Care should be taken when turning cable through its bends and through mounting holes in brackets and lever.
10. **QA** Snug adapter to lockpin housing and tighten until a sharp rise in torque is felt. **TIGHTEN 1/6 TO 1/3 TURN MORE.**
11. **QA** Hold adapter and tighten jamnut until a sharp rise in torque is felt. **TIGHTEN 1/6 TO 1/3 TURN MORE.**
12. Check that lockpin extends 3/8 to 13/32-inch beyond end of lockpin housing.
13. **QA** Tighten nuts retaining cable housing to lever bracket until a sharp rise in torque is felt. **TIGHTEN 1/6 TO 1/3 TURN MORE.**
14. **QA** Adjust cable adjustment nut until minimal play exists between lever brackets and cable spacer. Spacer should rotate freely. Tighten locknut until a sharp rise in torque is felt. **TIGHTEN 1/6 TO 1/3 TURN MORE.**
15. Check operation of lever movement and that the lockpins stroke at the same time.
16. **QA** Reinstall seat on track and install front stops. Refer to, REPLACE PILOT'S/COPILOT'S SEAT, in this work package.
17. Check clearance between lockpin and rail, while holding control lever full forward. **CLEARANCE MUST BE BETWEEN 0.010 AND 0.050-INCH.**
18. Adjust clearance as required by shimming or equivalent means.

REPLACE PILOT'S STOWAGE BOX, PANEL AND BAG

Remove

1. **UH-60A 85-24462 - SUBQ** Disconnect electrical connector, J455, from cabin dome lights dimmer switch.
2. Disconnect ICS cable from ICS control panel and remove ICS control panel from stowage box (Figure 4, Detail B).
3. Remove clamps attaching ICS cable to stowage panel.
4. Remove screws, washers, and nuts attaching stowage panel to pilot's seat.
5. Remove screws, washers, and nuts fastening box assembly to panel.
6. Remove bolts, washers, and nuts attaching stowage bag to stowage panel.
7. If necessary, remove stowage panel support brackets from seat frame.

Repair

1. Repair stowage box and stowage panel (TM 1-1500-204-23).
2. Remove screws, washers, and nuts securing tube and cover to cord terminals.
3. Remove cord from cover. Remove clamps from tube (Figure 4, Detail C).
4. Install replacement cord in cover.
5. Connect tube and cover to cord terminals with screws, washers, and nuts.

REPLACE PILOT'S STOWAGE BOX, PANEL AND BAG - Continued

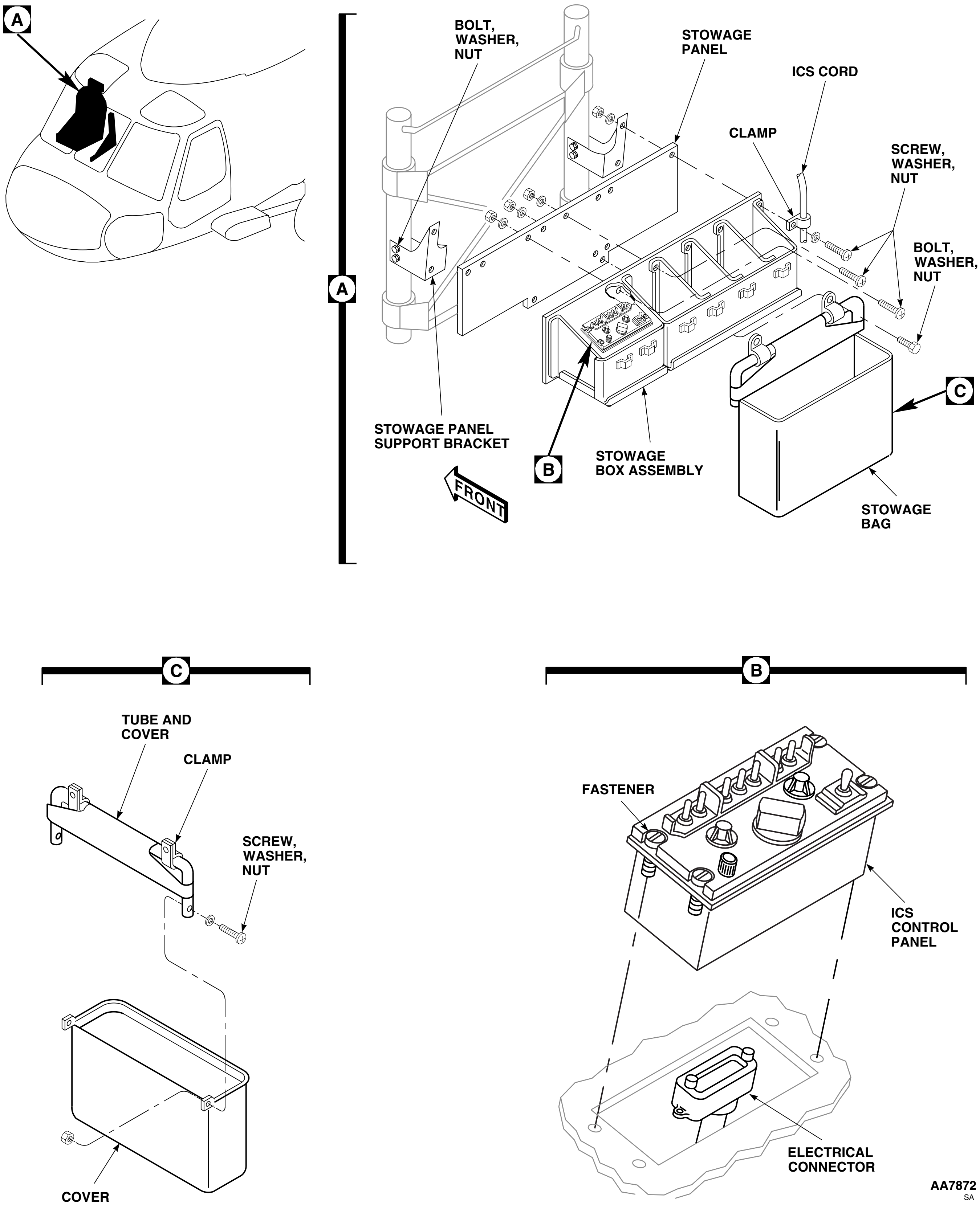


Figure 4. Replace Pilot's Stowage Box, Panel and Bag.

REPLACE PILOT'S STOWAGE BOX, PANEL AND BAG - Continued

6. Install replacement clamps on tube.
7. Repair minor rips or tears in covers (TM 1-1500-204-23).

Install

1. If stowage panel support brackets were removed, install them on seat frame with bolts, washers, and nuts (Figure 4, Detail A).
2. Install stowage box assembly on stowage panel with screws, washers, and nuts.
3. Attach stowage bag to panel with bolts, washers, and nuts.
4. Install stowage panel on seat frame with screws, washers, and nuts.
5. Clamp ICS cable to stowage panel.
6. Install ICS control panel in stowage box (Figure 4, Detail B).
7. **UH-60A 85-24462 - SUBQ** Connect electrical connector, J455, to cabin dome lights dimmer switch.
8. Connect ICS cable connector to ICS control panel.
9. Check operation of ICS (TM 11-1520-237-23).

REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG

Remove

1. Turn off all electrical power.
2. Pull first-aid kit snap fasteners from mounting studs and remove kit from stowage panel.
3. Remove fire extinguisher from support. Remove screws, washers, and nuts securing support to stowage panel (Figure 5, Detail A).
4. **UH-60A 85-24462 - SUBQ** Disconnect all electrical wires from utility light terminal block.
5. Disconnect ICS cable from ICS control panel. Remove ICS control panel from stowage box (Figure 5, Detail B).
6. Remove clamps attaching ICS cable to stowage panel.
7. Remove screws, washers, and nuts attaching stowage panel to copilot's seat.
8. Remove screws, washers, and nuts fastening box assembly to panel.
9. If necessary, remove screws, washers, and nuts securing stowage panel support brackets to seat frame.

Repair

1. Turn off all electrical power.
2. Repair stowage box and stowage panel. Replace snap fasteners in stowage panel (TM 1-1500-204-23).

Install

1. Turn off all electrical power.
2. If removed, install stowage panel support brackets on seat frame with screws, washers, and nuts.
3. Install stowage box assembly on stowage panel with screws, washers, and nuts (Figure 5, Detail A).
4. Install stowage panel on seat frame with screws, washers, and nuts.
5. Clamp ICS cable to stowage panel.
6. Install ICS control panel in stowage box (Figure 5, Detail B).

REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG - Continued

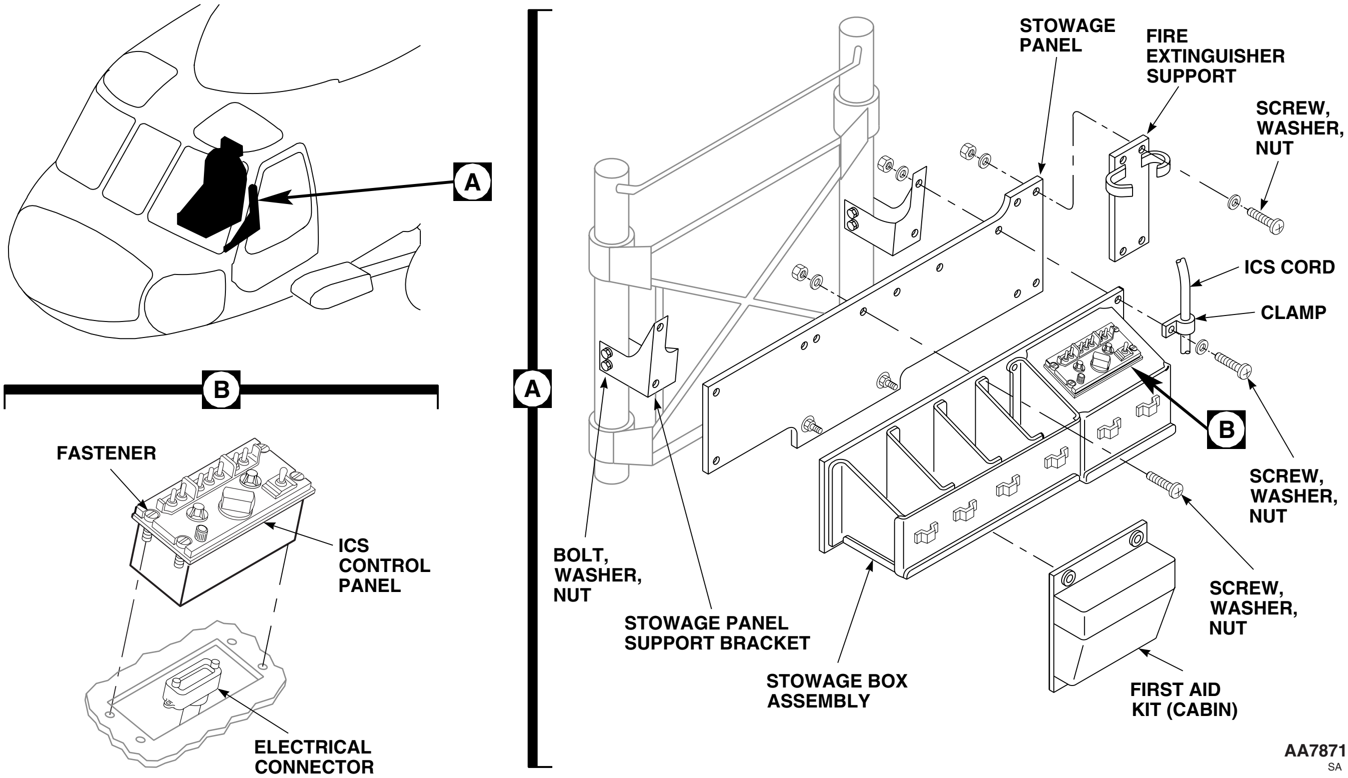


Figure 5. Replace Copilot's Stowage Box and Panel.

7. **UH-60A 85-24462 - SUBQ** Connect electrical wires to utility light terminal block. Cover terminal ends with rubber boots.
8. Connect ICS cable connector to ICS control panel.
9. Check operation of ICS (TM 11-1520-237-23).
10. Attach fire extinguisher support to stowage panel with screws, washers, and nuts.
11. Install fire extinguisher in support.
12. Attach first aid kit to mounting studs on stowage panel.

SHOCK ABSORBER CLEVIS

Repair

1. Modify locally-made drag link adjusting tool, if necessary, by grinding down bit thickness to fit retaining nut recesses (WP 1805 00). Use this tool in following procedures, using a 9/16-inch open end wrench to prevent clevis bolt from turning.
2. Before tightening nut on loose clevis bolt, back off nut and measure running torque, using torque wrench. **RUNNING TORQUE MUST BE AT LEAST 20 INCH-POUNDS.**
3. If running torque is less than 20 inch-pounds, replace clevis bolt (Figure 6, Detail A).
4. **QA** If running torque is 20 inch-pounds or more, **TORQUE RETAINING NUT TO 65 - 75 INCH-POUNDS PLUS RUNNING TORQUE READING.**
5. Inspect again for looseness.

SHOCK ABSORBER CLEVIS - Continued

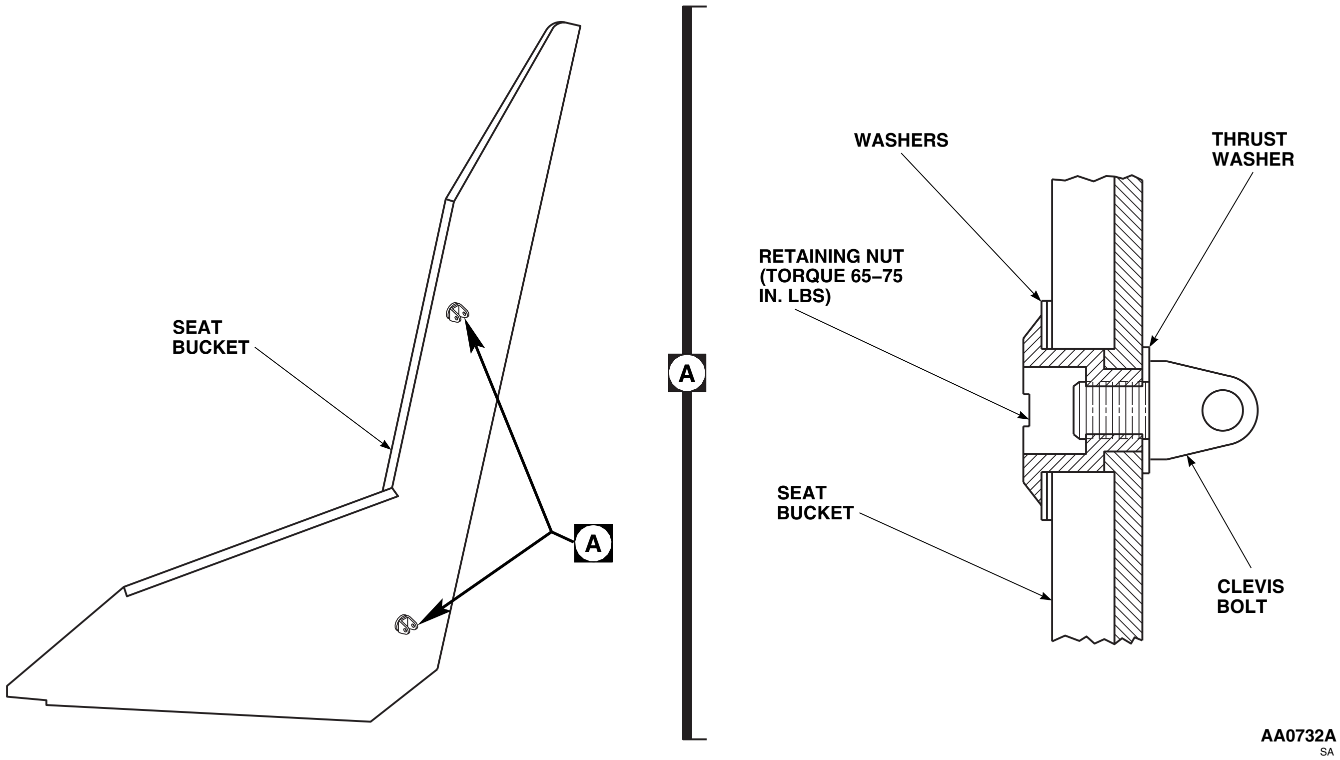


Figure 6. Shock Absorber Clevis.

6. If looseness continues, reinstall clevis using up to 3 washers, in any combination, under retaining nut.
7. Make sure that clevis bolt extends through retaining nut beyond bolt chamfer and one thread.

Replace

1. Remove shoulder bolt, washers, and nut fastening shock absorber to clevis.
2. Remove clevis bolt, retaining nut, and washers.
3. Install clevis bolt with thrust washer, and retaining nut with required stackup of washers. Position clevis to engage shock absorber.
4. **QA** Attach shock absorber to clevis with shoulder bolt, washers, and nut. **TORQUE CLEVIS RETAINING NUT TO 65 - 75 INCH-POUNDS PLUS MEASURED RUNNING TORQUE.**
5. Install back cushion. Refer to, WP 0231 00, in this work package.

ARMORED WING

See WP 0231 00 for maintenance instructions.

REPLACE PILOT'S/COPILOT'S SEAT

Remove

1. Turn off all electrical power.

REPLACE PILOT'S/COPILOT'S SEAT - Continued

WARNING

To prevent injury to personnel, do not pull forward on control lever unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and control lever is pulled forward, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes will be seriously injured.

2. Remove first aid kit and/or stowage bag.
3. Remove clamp and disconnect Intercommunication System (ICS) cable. Disconnect connector from ICS control panel. Unclamp ICS cable from stowage panel (Figure 7, Detail A and B).
4. Remove fire extinguisher and mounting bracket from copilot seat. Remove stowage panel from back of seats.
5. Remove bolts and front stops from front seat tracks (Figure 8, Detail A).
6. Pull forward on the horizontal adjustment control lever (front left side). Move seat to track openings.
7. Remove gunner and troop seats next to cargo door (WP 0233 00).

CAUTION

ICS cable will be damaged if seat contacts cable during seat removal. Move ICS cable out of way before removing seat to prevent damage.

8. With the aid of an assistant, lift seat from tracks and remove from helicopter.

Install

1. Turn off all electrical power.

WARNING

To prevent injury to personnel, do not pull forward on control lever unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and control lever is pulled forward, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes will be seriously injured.

2. Adjust seat to full down position by pulling forward on the control lever on the front right side and pushing bucket downward.

WARNING

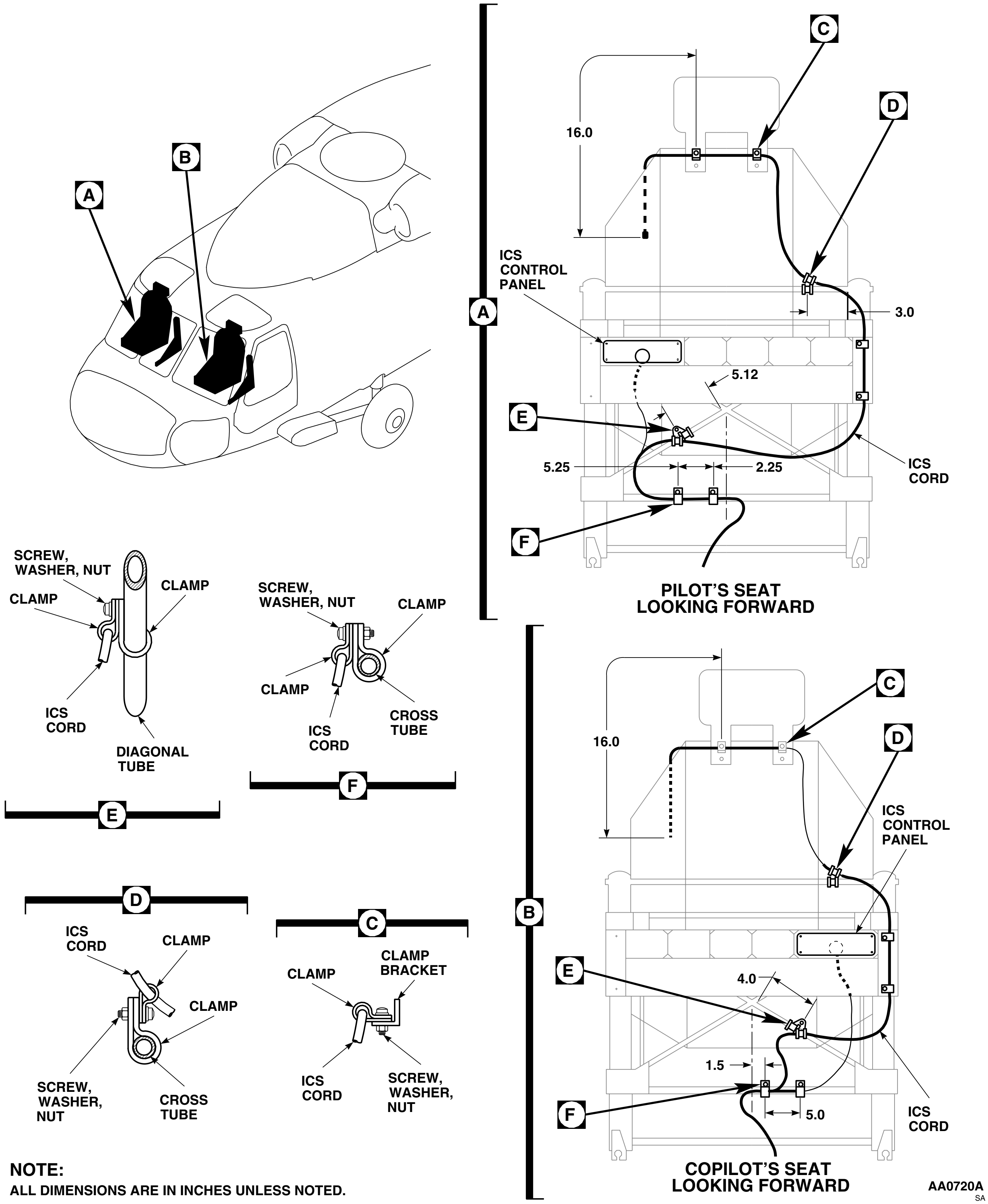
Pilot and copilot seat restraint buckles must have same type of release mechanism. Different types of release mechanism could hinder an emergency egression. If installing a replacement seat, make sure that both pilot and copilot seat restraint buckles have same type of release mechanism.

CAUTION

Move ICS cable out of way when installing seat to prevent damaging cable.

3. With aid of assistant, position replacement seat in helicopter and lift into track openings.
4. Pull forward on horizontal adjustment control lever and move seat to full backward position.

REPLACE PILOT'S/COPILOT'S SEAT - Continued



NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 7. Replace Pilot's/Copilot's Seat.

REPLACE PILOT'S/COPILOT'S SEAT - Continued

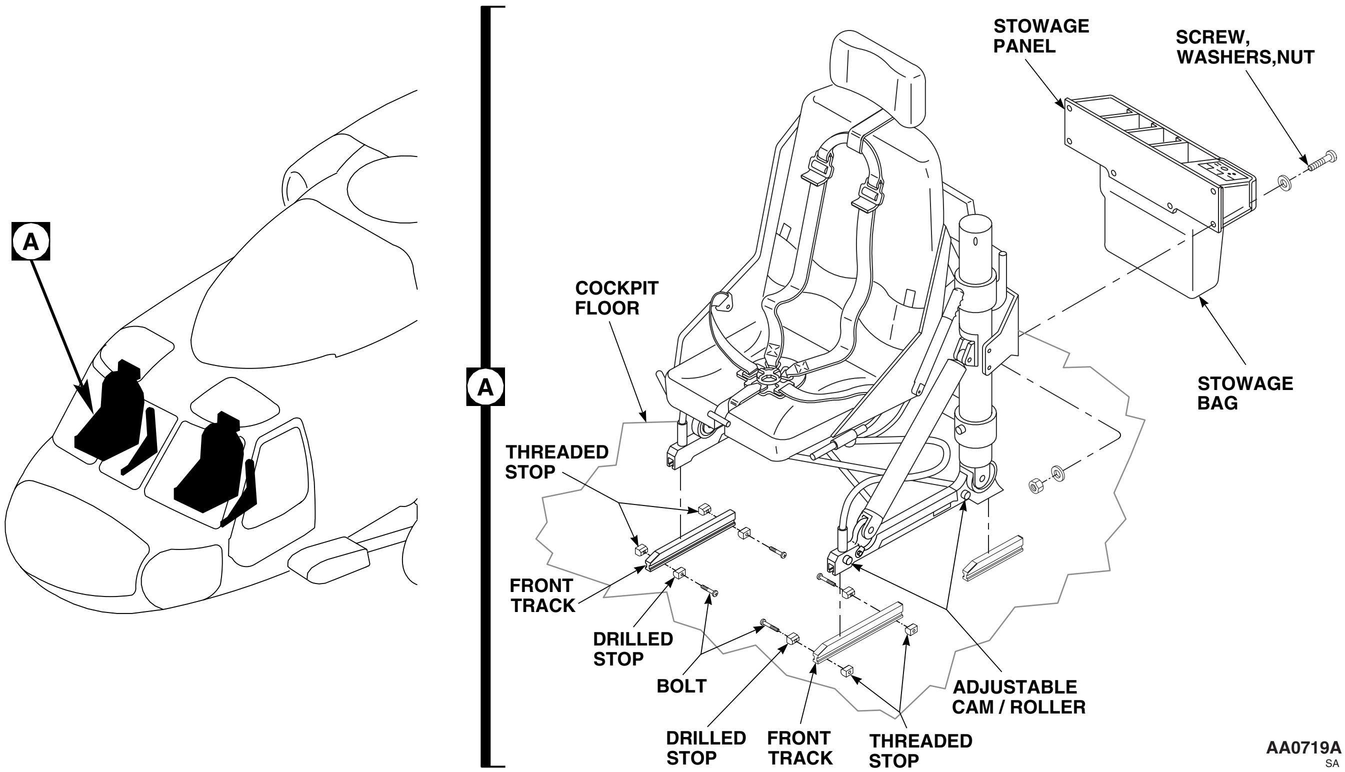


Figure 8. Replace Pilot's/Copilot's Seat Stowage Panel.

NOTE

If replacing seat stops, a maximum of 0.062-inch may be removed from stop length, if required.

5. **QA** Install front seat stops (chamfered side against track) on front track with screws (Figure 8, Detail A).
6. **QA** **UH-60A 88-26084 - SUBQ UH-60L 89-26149 - 92-26442** Install rear stops (chamfered side against track) with bolts, red ends of stops to rear when using ARA or Norton seats and yellow end of stops to rear when using Simula seats.

NOTE

It is acceptable to use stops with yellow portion forward when using Simula seats.

7. **QA** **UH-60L 92-26443 - SUBQ** Install rear stops (chamfered side against track) with bolts.
8. Adjust two rollers on each side of seat by loosening nuts and turning front and rear cams with 1/4-inch Allen wrench for smoothest front to rear seat travel.
9. Install stowage panel on back of seat with screws, nuts and washers. Attach fire extinguisher and mounting bracket to copilot's seat.
10. Install first aid kit and/or stowage bag.
11. Clamp ICS cable to stowage panel and connect to ICS control panel (Figure 7, Detail A and Detail B).

REPLACE PILOT'S/COPILOT'S SEAT - Continued

WARNING

To prevent injury to personnel, do not pull forward on control lever unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and control lever is pulled forward, the seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes will be seriously injured.

12. Check vertical operation of seat. (Control lever on front right side). Cycle seat up and down.

NOTE

To be sure of proper longitudinal operation, occupy seat during check.

13. Check longitudinal operation of seat. (Control lever on front left side). Move seat forward and backward. Readjust roller cams as required.
14. Check operation of ICS System.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/GUNNER'S SEAT

This WP supersedes WP 0233 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Riveter, Hand, G-740-ELM
Gloves
Goggles/Face Shield
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 15, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 55-1500-322-24
TM 55-1500-345-23
WP 0234 00
WP 0235 00
WP 0258 00
WP 0377 00
WP 0378 00
WP 0380 00
WP 0381 00
WP 1803 00

REPLACE TROOP/GUNNER'S SEAT HEAD REST

Remove

1. Remove screws and washers securing brackets and head rest to seat frame (Figure 1).
2. Remove head rest from seat.

Install

1. Place replacement head rest in position on seat.
2. Fasten head rest and brackets to seat with screws and washers (Figure 1, Detail A).

REPLACE LOWER ENERGY ATTENUATOR

Remove

1. Disengage disconnect fitting from stud in cabin floor (Figure 2, Detail A).
2. Remove bolt, washers, and nuts attaching attenuator to seat frame (Figure 2, Detail B).
3. Remove cotter pin, washer, and headed pin attaching lever, attenuator, spring, shim washers, and washer to disconnect fitting. Keep shim washers for reinstallation (Figure 2, Detail A).

Install

1. **ADJUST ATTENUATOR LENGTH TO 21.43 TO 21.49-INCHES MEASURED BETWEEN END FITTING HOLE CENTERS (Figure 2, Detail C).**

REPLACE LOWER ENERGY ATTENUATOR - Continued

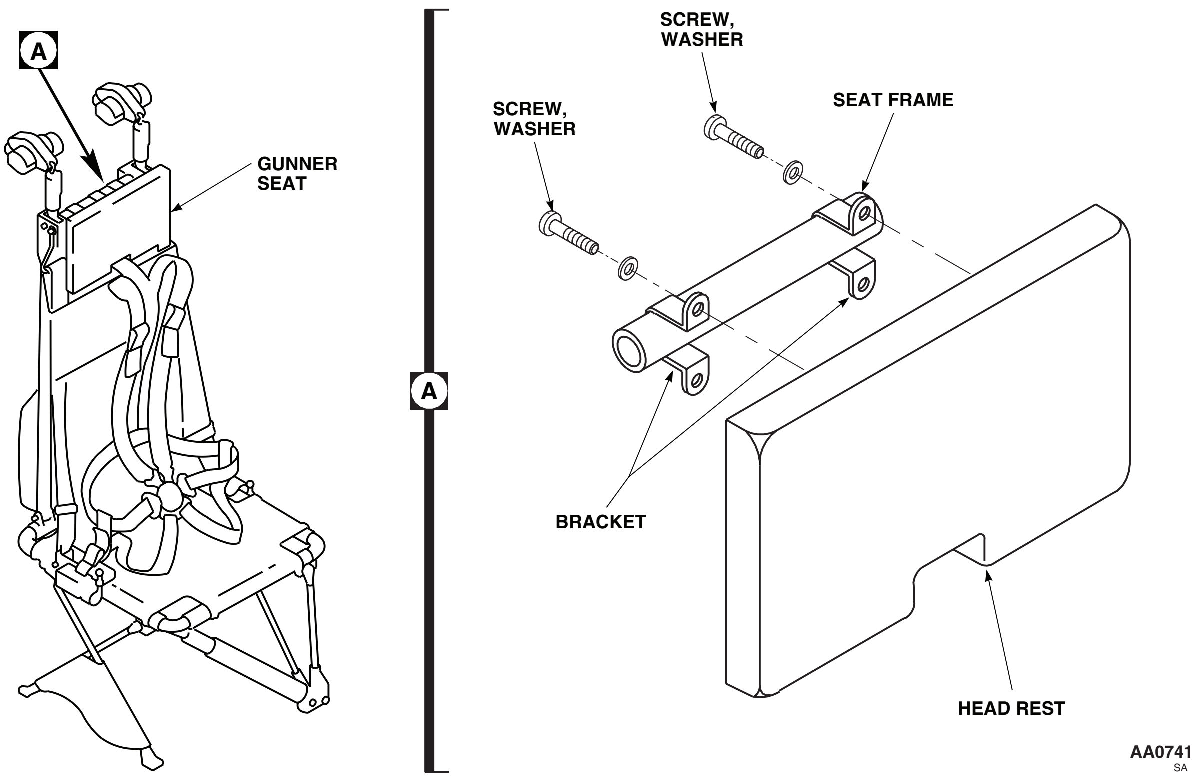


Figure 1. Replace Gunner's Seat Head Rest.

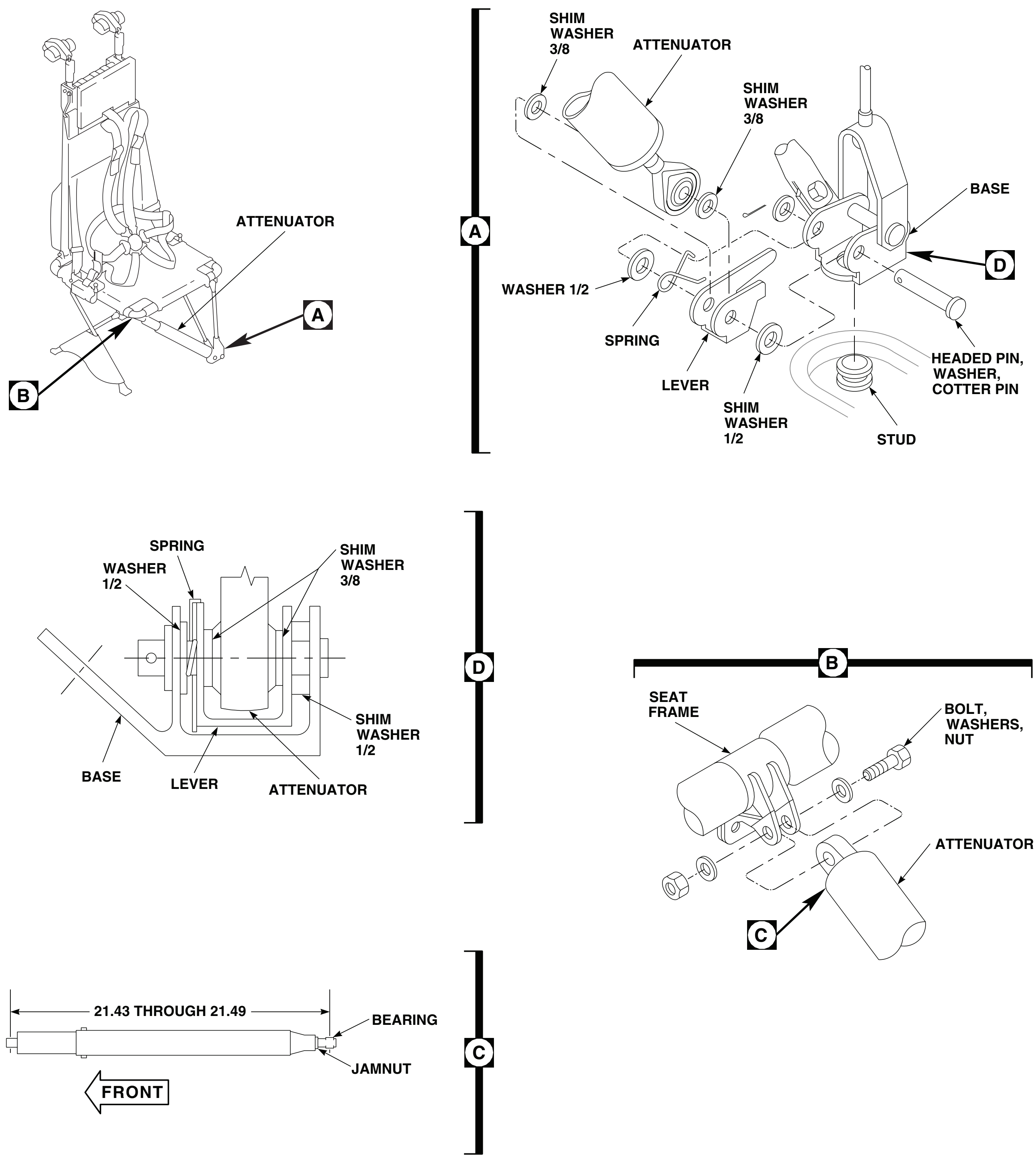
2. Center rod end of replacement attenuator in lever. Install a 3/8-inch shim washer on each side of rod end.
3. Install a 1/2-inch shim washer between lever and fitting and partially insert headed pin. Install spring and 1/2-inch plain washer. Fully insert headed pin (Figure 2, Detail A and Figure 2, Detail D).
4. **CHECK FOR 0.005 TO 0.010-INCH ENDPLAY BETWEEN LEVER AND ATTENUATOR ROD END. ALSO 0.005 TO 0.010-INCH ENDPLAY BETWEEN LEVER AND FITTING.** Adjust shim washers as required to get endplay.
5. Hook spring onto lever while under tension.
6. After adjusting shims, install washer and cotter pin (Figure 2, Detail A).
7. Bolt attenuator to seat frame with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY, ATTENUATOR MUST BE FREE TO PIVOT (Figure 2, Detail B).**
8. Engage disconnect fitting with stud on cabin floor (Figure 2, Detail A).

REPLACE UPPER ENERGY ATTENUATOR

Remove

1. Remove latch.
2. On rear-facing troop seats, release anti-sag device cable tension.
3. Remove bolts, washers, and nuts from upper ring. On rear-facing troop seats, retain upper attach link on roller retaining bolt (Figure 3, Detail A).
4. Withdraw hanger wire with rollers, Teflon washers, and washers.

REPLACE UPPER ENERGY ATTENUATOR - Continued



NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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SA

Figure 2. Replace Lower Energy Attenuators.

REPLACE UPPER ENERGY ATTENUATOR - Continued

Inspect

1. Inspect upper energy attenuator rollers and hanger wires for corrosion (TM 1-1500-344-23).
2. Remove minor corrosion (TM 1-1500-344-23).
3. Inspect upper energy attenuator wires to make sure there is no visible indication of metal abrasion on the attenuator wire +/- 0.5 inch from the flat bottom edge of the crescent bushing. Replace upper energy attenuator wires as required.
4. Only a maximum of three nicks and scratches 0.5 inch below the crescent bushing is allowed. Replace upper energy attenuator wires as required.
5. A maximum of .005 inch deep and 1/8 inch long for each nick and scratch. Replace upper energy attenuator wires as required.
6. Replace all crescent bushings that have a bent and/or cracked edge.

Install

WARNING

Injury to personnel and damage to equipment will result if retention bolt nuts are excessively tightened. Crushing of support tube and roller binding will result. Make sure shimming and tightening procedures are followed.

1. Determine roller assembly end play as follows:

NOTE

Roller end play shall be calculated without hanger wires installed.

   a. Place two washers along with roller(s) into support tube. Temporarily install retaining bolt through washers and each roller. Slide rollers and washers to one side of support tube.
   b. Measure distance between support tube and washer/roller stackup using a feeler gage.
   c. Using dimension obtained from step b, add washers until clearance does not exceed 0.030-inch.
   d. Place washers equally and adjacent to each respective roller into support tube and install retaining bolt.
   e. Make sure rollers are free to move and clearance is between 0.010-inch and 0.030-inch (Figure 3, Detail A, View A-A).
   f. Remove roller assemblies from support tube.

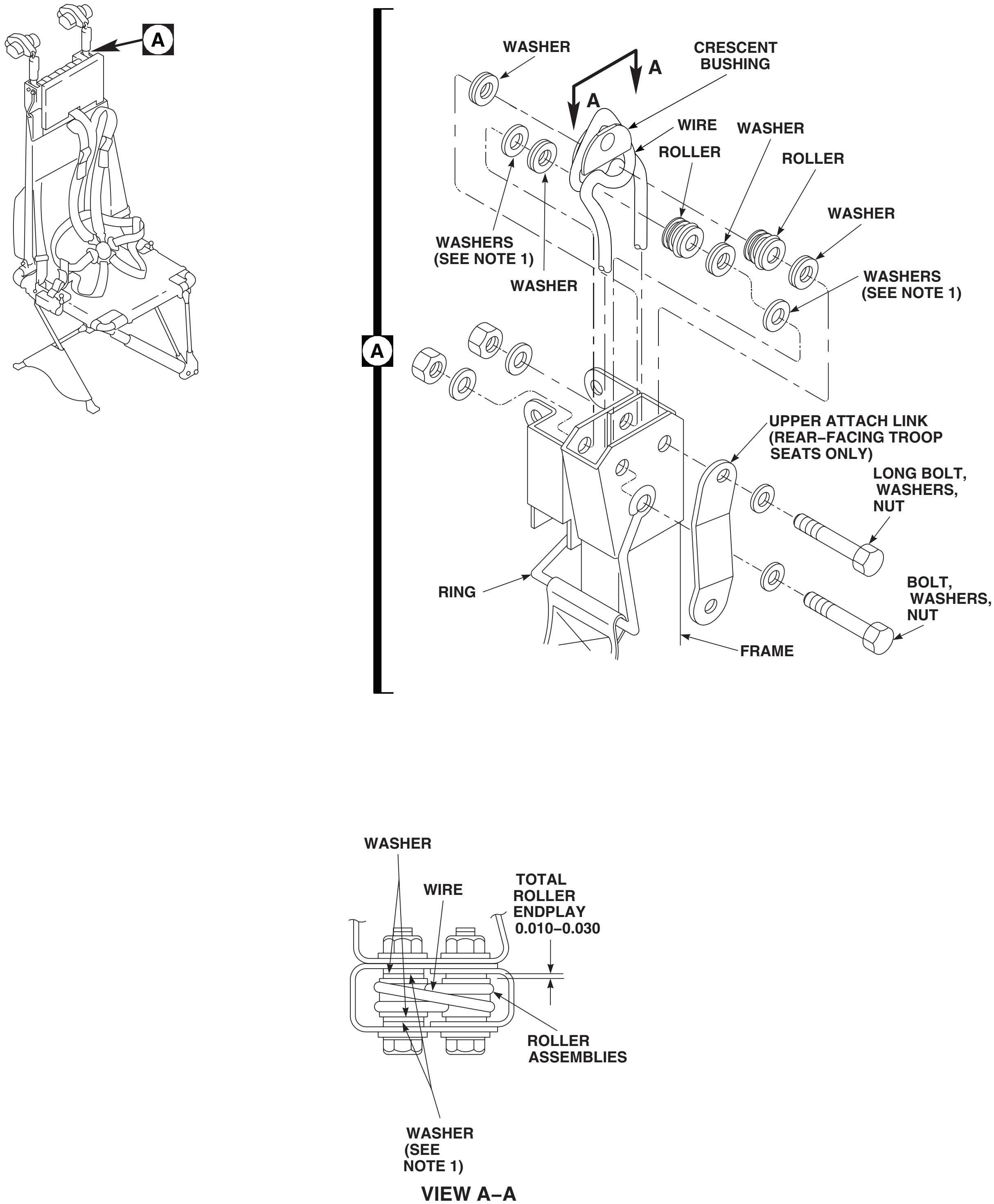
2. Install hanger wire and roller assembly into support tube as follows:
   a. Place roller through each wire loop.

NOTE

- Washers are located adjacent to roller.
- All troop/gunner seats shall have crescent bushing installed under upper attenuation wire radius (flat edge down) at re-assembly.

   b. Assemble each roller assembly with previously determined hardware stackup.
   c. Install roller assemblies into frame as follows:
      (1) Install bushing, front roller assembly, ring, bolt, washer, and nut. **DO NOT TIGHTEN NUT AT THIS TIME.**
      (2) Install rear roller assembly, bolt, washer, and nut. **DO NOT TIGHTEN NUT AT THIS TIME.**
      (3) On rear facing troop seats, check that upper attach link is installed on rear roller retaining bolt.

REPLACE UPPER ENERGY ATTENUATOR - Continued



NOTES:

1. ADD WASHERS AS REQUIRED TO OBTAIN ROLLER END PLAY.
2. CRESCENT BUSHING REMOVED FOR CLARITY.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 3. Replace Upper Energy Attenuators.

REPLACE UPPER ENERGY ATTENUATOR - Continued

NOTE

Washers adjacent to rollers shall be able to rotate with minimum pressure.

d. Make sure washers rotate with minimum pressure.

CAUTION

Damage to support tube and binding of roller assemblies will occur if roller retaining bolt nuts are over tightened. Do not over tighten roller retaining bolt nuts.

e. Tighten nuts so washers rotate and roller assemblies shall have a 0.010-inch to 0.030-inch end play (Figure 3, Detail A, View AA).

f. Install latch.

g. On rear-facing troop seats, adjust turnbuckle tension to remove slack on cable assembly and install locking clips.

REPLACE CABLE

Remove

1. Remove bolt, washers, and nut attaching cable to seat support (Figure 4, Detail A).
2. On cables with end clips, disengage clip from stud in cabin floor. Remove bolt, washers, and nut attaching cable to end clip. Keep clip and spring for reinstallation (Figure 4, Detail C).
3. On cables that attach to disconnect fittings, do this:
   a. Disengage disconnect fitting from stud in cabin floor (Figure 4, Detail B).
   b. Remove headed pin, washer, and cotter pin attaching vertical cable to disconnect fitting.
   c. Unbolt diagonal cables from fitting.

Install

1. On cables that attach to disconnect fitting, place replacement cable assembly on disconnect fitting. Install headed pin, washer, and cotter pin (Figure 4, Detail B).
2. Connect diagonal cable to disconnect fitting with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY, CABLE MUST BE FREE TO PIVOT.**
3. Engage disconnect fitting with stud in cabin floor.
4. On cables with end clips, attach end clip and spring to replacement cable with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY, CABLE MUST BE FREE TO PIVOT (Figure 4, Detail C).**
5. Engage end clip with stud in cabin floor.
6. Fasten cable to seat support with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY, CABLE MUST BE FREE TO PIVOT (Figure 4, Detail A).**

REPLACE ANTI-SAG DEVICE

Remove

1. Remove turnbuckle locking clips and release tension on cable (Figure 5, Detail B).
2. Release locking lever on seat support latch.

REPLACE ANTI-SAG DEVICE - Continued

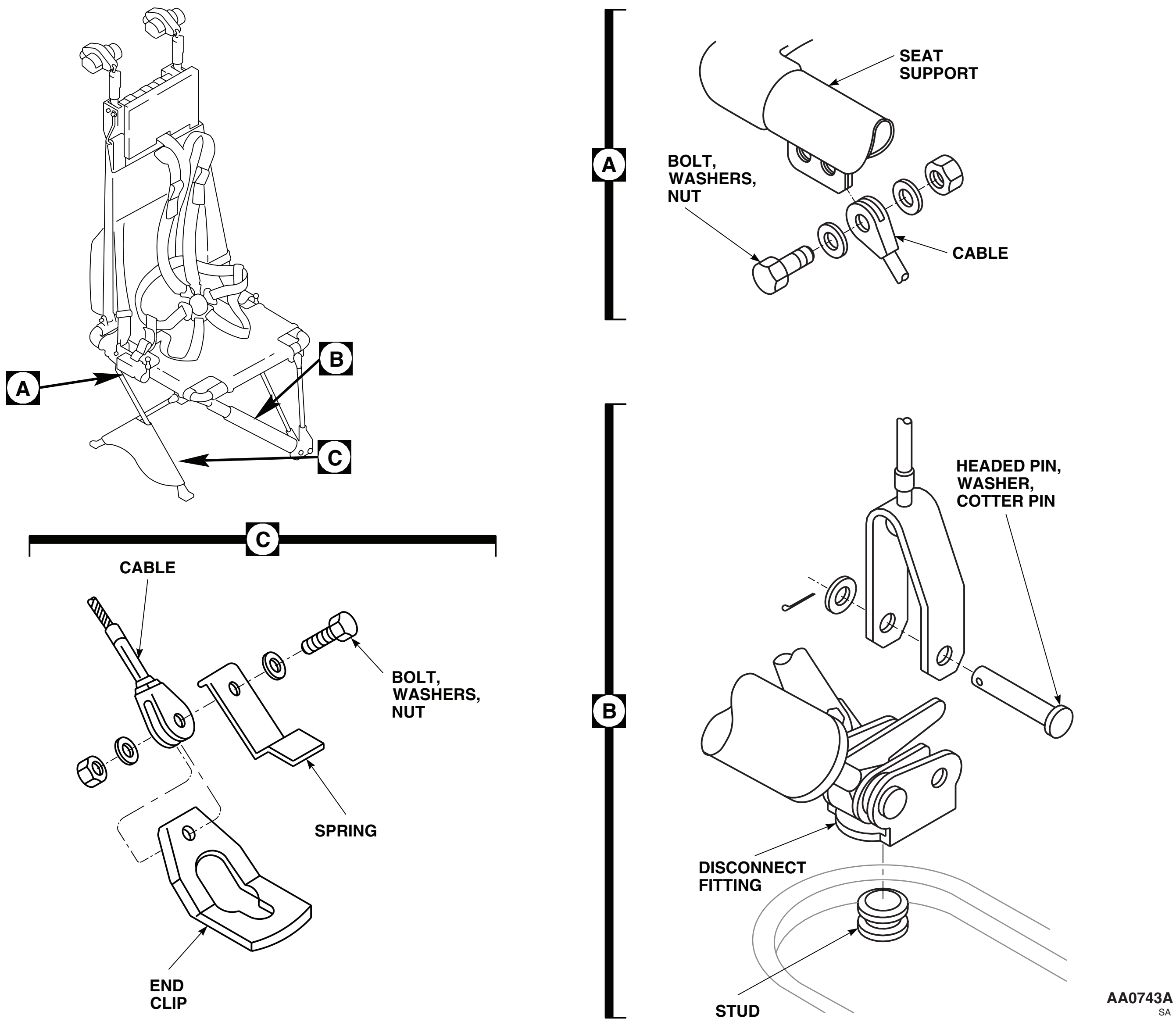


Figure 4. Replace Cable.

WARNING

Injury to personnel and damage to equipment will result if rollers are released when removing roller retaining bolt. Make sure that roller assembly remains in place when removing bolt.

3. Remove roller retaining bolt, washers, and nut (Figure 5, Detail A).
4. Remove upper attach link from roller retaining bolt. Temporarily reinstall roller retaining bolt in seat frame.
5. Remove bolt, washer, and nut from lower web retainer, lower attach link, and seat support frame.
6. If required, disconnect cable from upper and lower attach links by removing pins, washers, and cotter pin.

REPLACE ANTI-SAG DEVICE - Continued

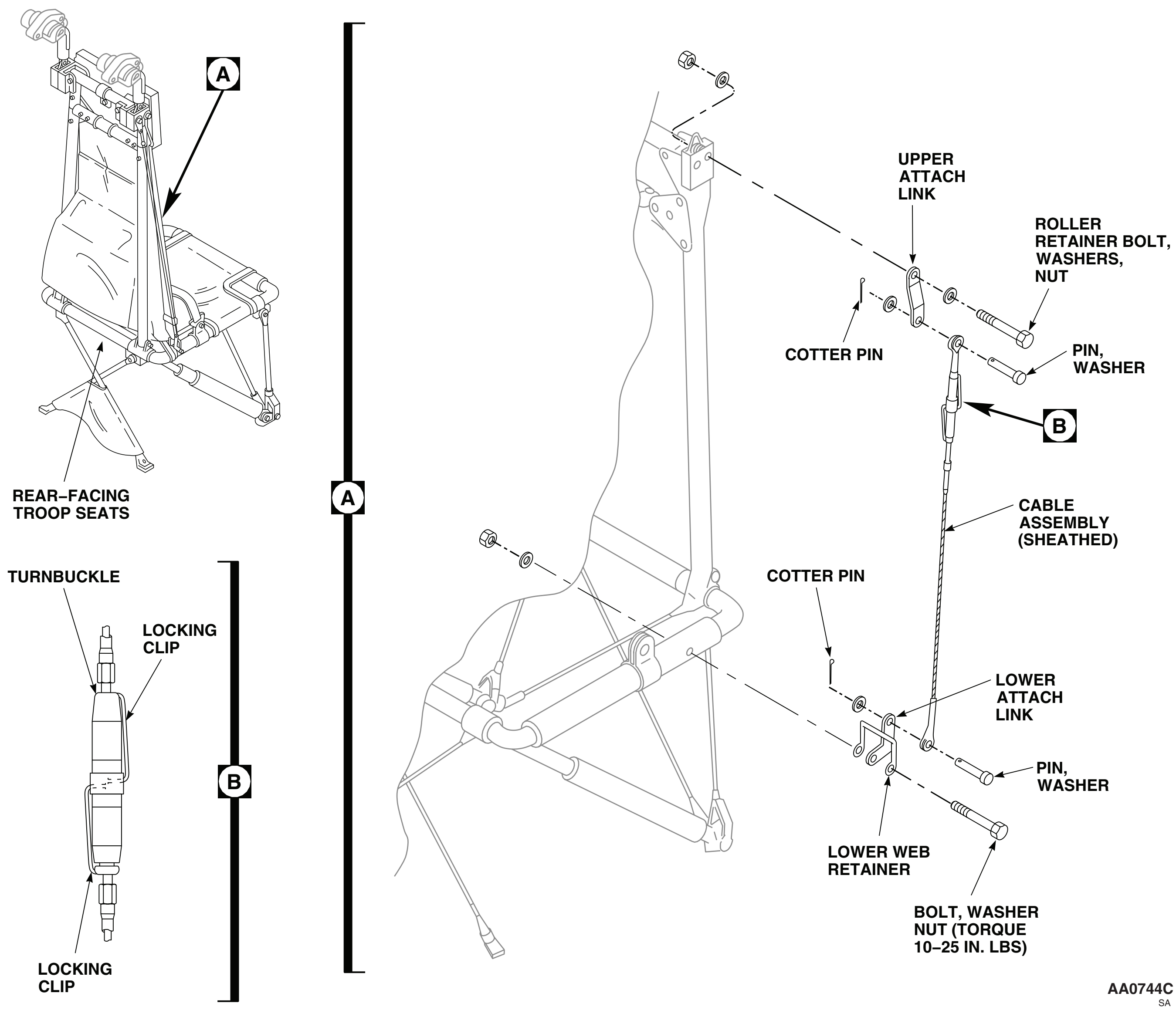


Figure 5. Replace Anti-Sag Device.

Install

1. Remove temporarily installed roller retaining bolt. Install upper attach link on roller retainer bolt and reinstall bolt, washers, and nut (Figure 5, Detail A).
2. If cable was disconnected from upper attach link, connect it using pin, washer, and cotter pin.

NOTE

Ensure lower attach link is installed on outside of seat frame.

3. Install lower attach link and lower web retainer on seat support frame with bolt, washer, and nut. **TORQUE NUT TO 10 - 25 INCH-POUNDS.**
4. If cable was disconnected from lower attach link, connect it using pin, washer, and cotter pin.

REPLACE ANTI-SAG DEVICE - Continued

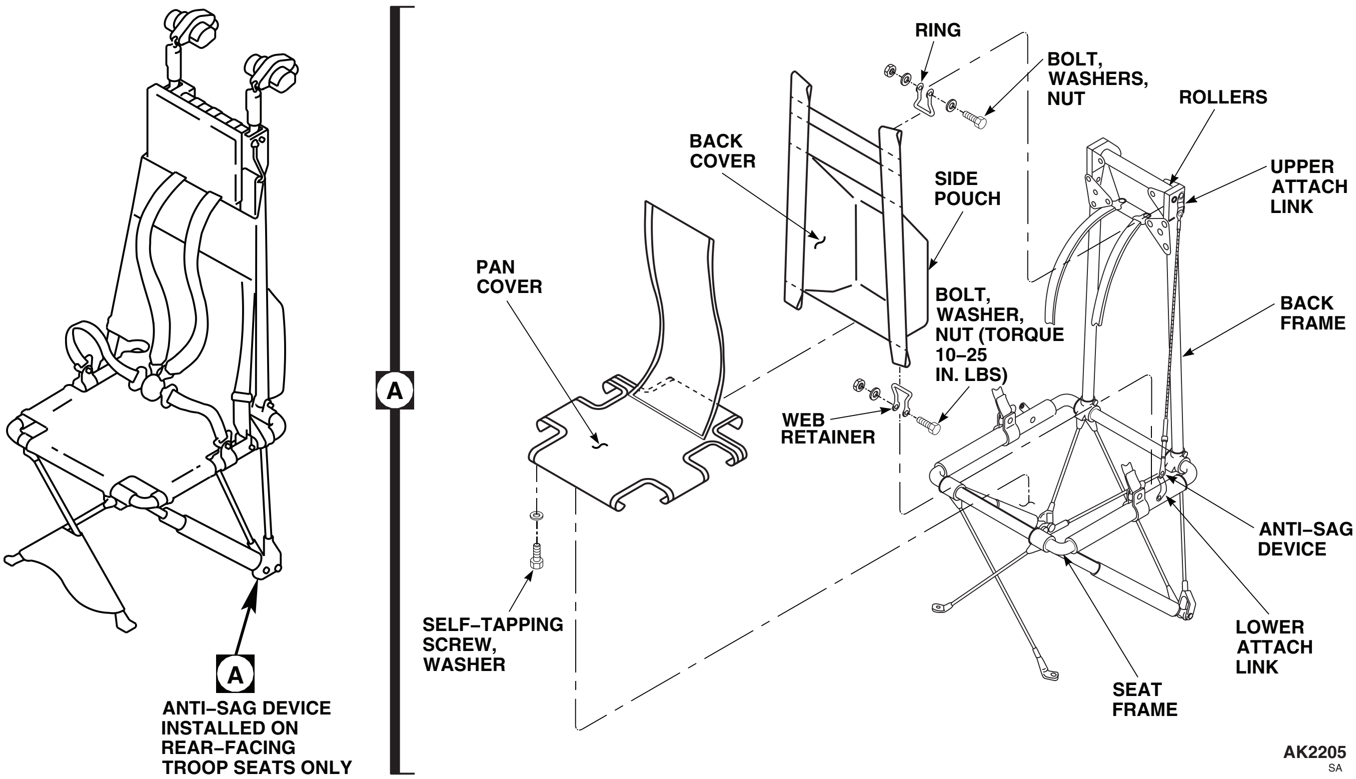


Figure 6. Replace Covers.

5. Lock locking lever on seat latch.
6. Tighten turnbuckle until slack is removed from cable.
7. Insert turnbuckle locking clips (Figure 5, Detail B).

REPLACE TROOP/GUNNER'S SEAT COVERS

Remove

1. Remove troop seat cover as follows:
   a. Remove self-tapping screws and washers attaching pan cover to seat frame. Remove cover (Figure 6, Detail A).
   b. Remove bolts, washers, and nuts attaching web retainers to seat frame. On rear-facing troop seats, remove lower attach link.
   c. Remove bolts, washers, and nuts attaching rings to back frame.
   d. Remove back cover from frame.
   e. Remove all web retainers and rings from cover.
2. Remove gunner's seat cover as follows:
   a. Remove self-tapping screws and washers attaching pan cover to seat frame. Remove cover.
   b. Remove bolts, washers, and nuts attaching rings to seat frame.
   c. Remove headed pin, washers, and cotter pin attaching web retainer and belt reel yoke to seat frame.

REPLACE TROOP/GUNNER'S SEAT COVERS - Continued

d. Remove back cover from frame.

e. Remove all web retainers and rings from cover.

Install

NOTE

All fasteners in seat frames covered by seat cover fabric must be smooth and free of burrs prior to installation of seat cover.

1. Install troop seat cover as follows:

   a. Slip retainers and rings through ends of straps of replacement back cover (Figure 6, Detail A).

   b. Place rings on back frame and install bolts, washers, and nuts. **MAKE SURE A 0.010 TO 0.030-INCH CLEARANCE EXISTS BETWEEN ROLLERS AND FRAME.**

   c. Place web retainers and lower part of back cover on seat frame. On rear-facing troop seats, position lower attach link between seat frame and retainer. Install bolt, washers, and nut.

   d. Place pan cover around tubes of seat frame. Pull cover to slightly tighten material. Install self-tapping screws and washers. If holes for self-tapping screws become oversized, replace self-tapping screws with machine screws and rivnuts (TM 1-1500-204-23).

2. Install gunner seat cover as follows:

   a. Slip retainers and rings through ends of straps of replacement back cover.

   b. Place rings on upper back frame and install bolts, washers, and nuts. **MAKE SURE A 0.010 TO 0.030-INCH CLEARANCE EXISTS BETWEEN ROLLERS AND FRAME.**

   c. Place web retainers, belt reel yoke, and lower part of back cover on seat frame. Install headed pin, washers, and cotter pin.

   d. Place pan cover around tubes of seat frame. Pull cover to slightly tighten material. Install self-tapping screws and washers. If holes for self-tapping screws become oversized, replace self-tapping screws with machine screws and rivnuts (TM 1-1500-204-23).

REPLACE LATCH

Remove

WARNING

Tension in anti-sag device cable can cause personal injury when releasing locking lever. Relieve cable tension before releasing locking lever.

1. Release locking levers to relieve seat tension.

2. Unhook latch from support bracket in cabin overhead (Figure 7, Detail A).

3. Remove bolt, washers, and nut, and remove latch and bushing from seat hanger wire.

Install

1. Place bushing (flat edge down) in seat hanger wire. Put replacement latch in place and install bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY, LATCH MUST BE FREE TO PIVOT (Figure 7, Detail A).**

2. Hook latch onto support bracket.

3. Add tension to seat by locking seat levers.

REPLACE LATCH - Continued

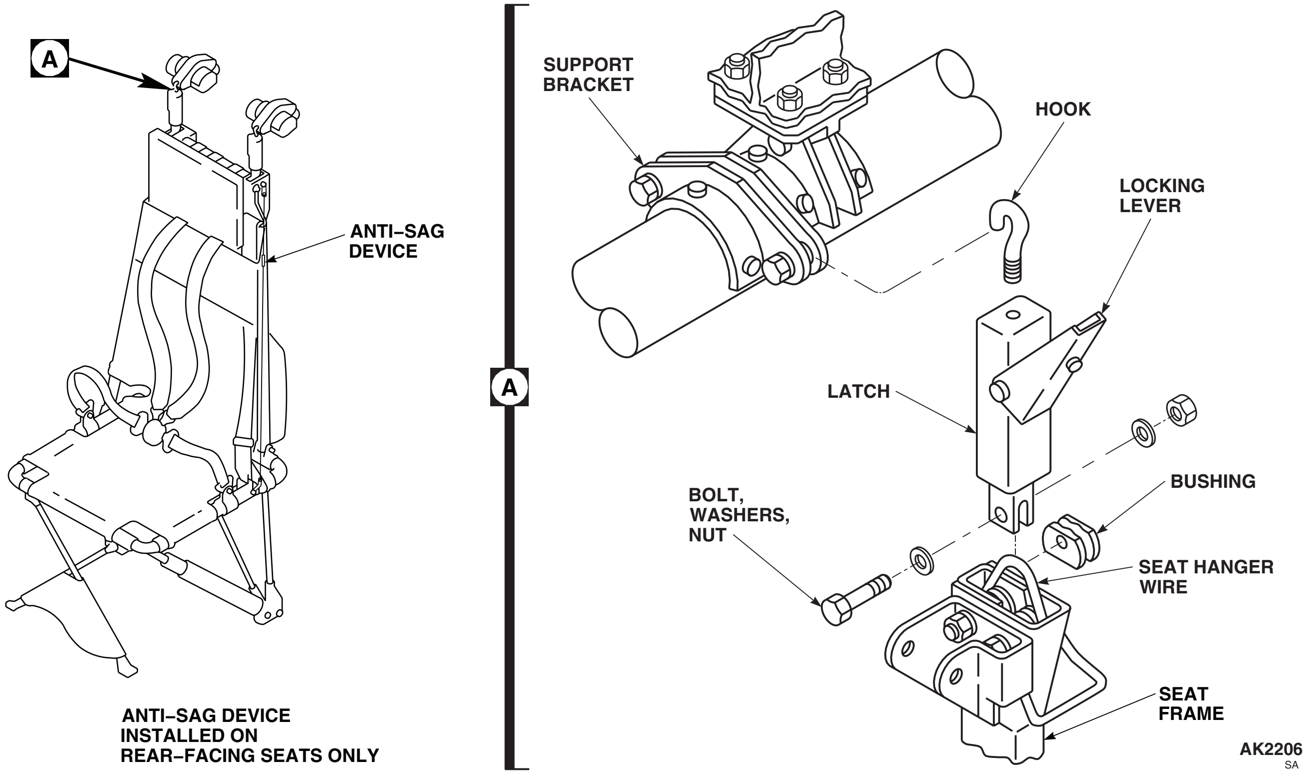


Figure 7. Replace Latch.

REPLACE DISCONNECT FITTING

Remove

1. Release lock lever and disengage disconnect fitting from stud in cabin floor (Figure 8, Detail A).
2. Remove attenuator from disconnect fitting.
3. Remove bolt, washers, and nut securing diagonal cable to disconnect fitting.
4. Remove headed pin, washer, and cotter pin attaching vertical cable to disconnect fitting.

Install

1. Place vertical cable on replacement disconnect fitting. Install headed pin, washer, and cotter pin (Figure 8, Detail A).
2. Connect diagonal cable to fitting, with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY, CABLE MUST BE FREE TO PIVOT.**
3. Install attenuator in disconnect fitting.
4. Engage disconnect fitting with stud in cabin floor.

REPLACE DISCONNECT FITTING - Continued

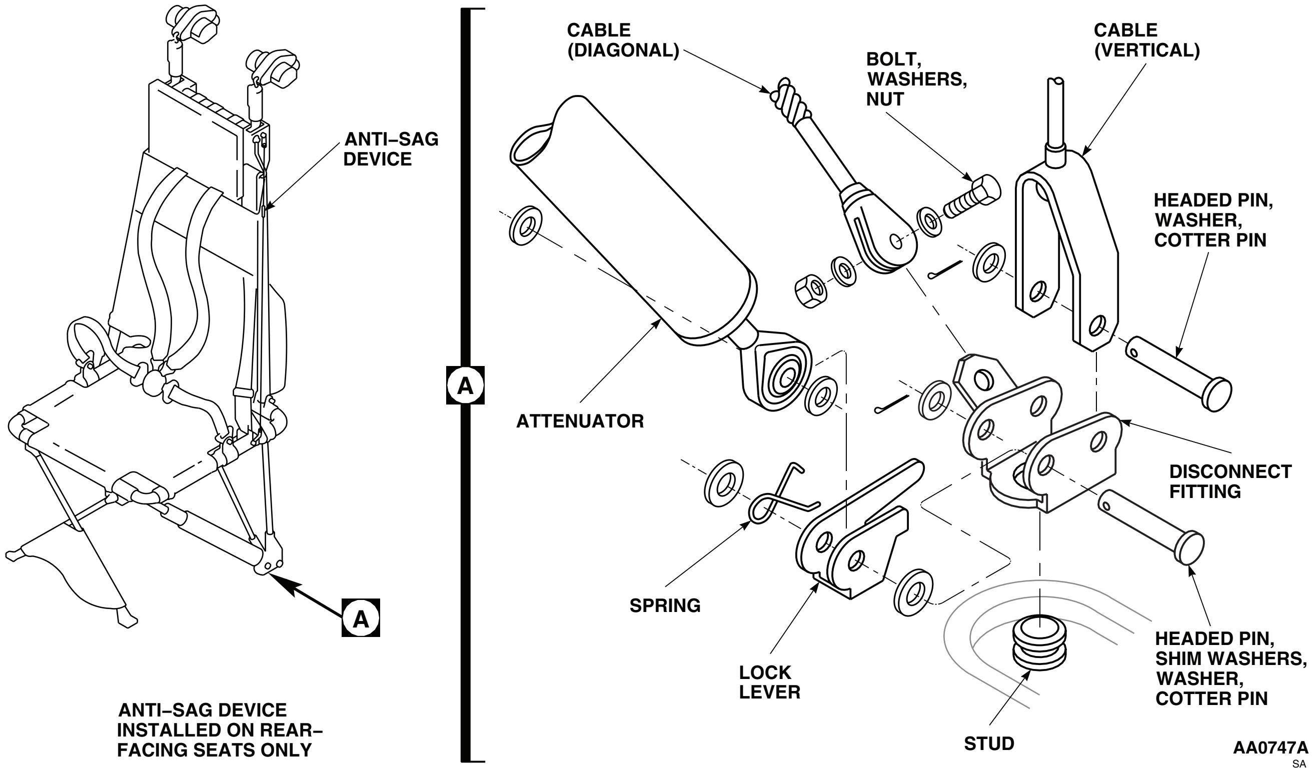


Figure 8. Replace Disconnect Fitting.

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORTS

Remove

CAUTION

Damage to upper support and support fittings will result if support is allowed to hang from end support fitting. Upper support will bend and damage support fitting. Upper support must be completely removed.

1. To remove upper support for gunner's and troop commander's seats do this:
   a. Remove gunner's and troop commander's seats.
   b. Disconnect ICS cords at disconnects.
   c. Remove covers from tube ends.
   d. Remove bolt, washers, and nut from upper support and remove spacers from fittings (Figure 9, Detail A).
   e. If replacing upper support, unbolt and remove clamps and ICS cords from support tubes.
2. To remove troop seat upper support, do this:
   a. Remove center troop seats.
   b. Remove bolt, washers, and nut securing upper support to fittings.

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORTS - Continued

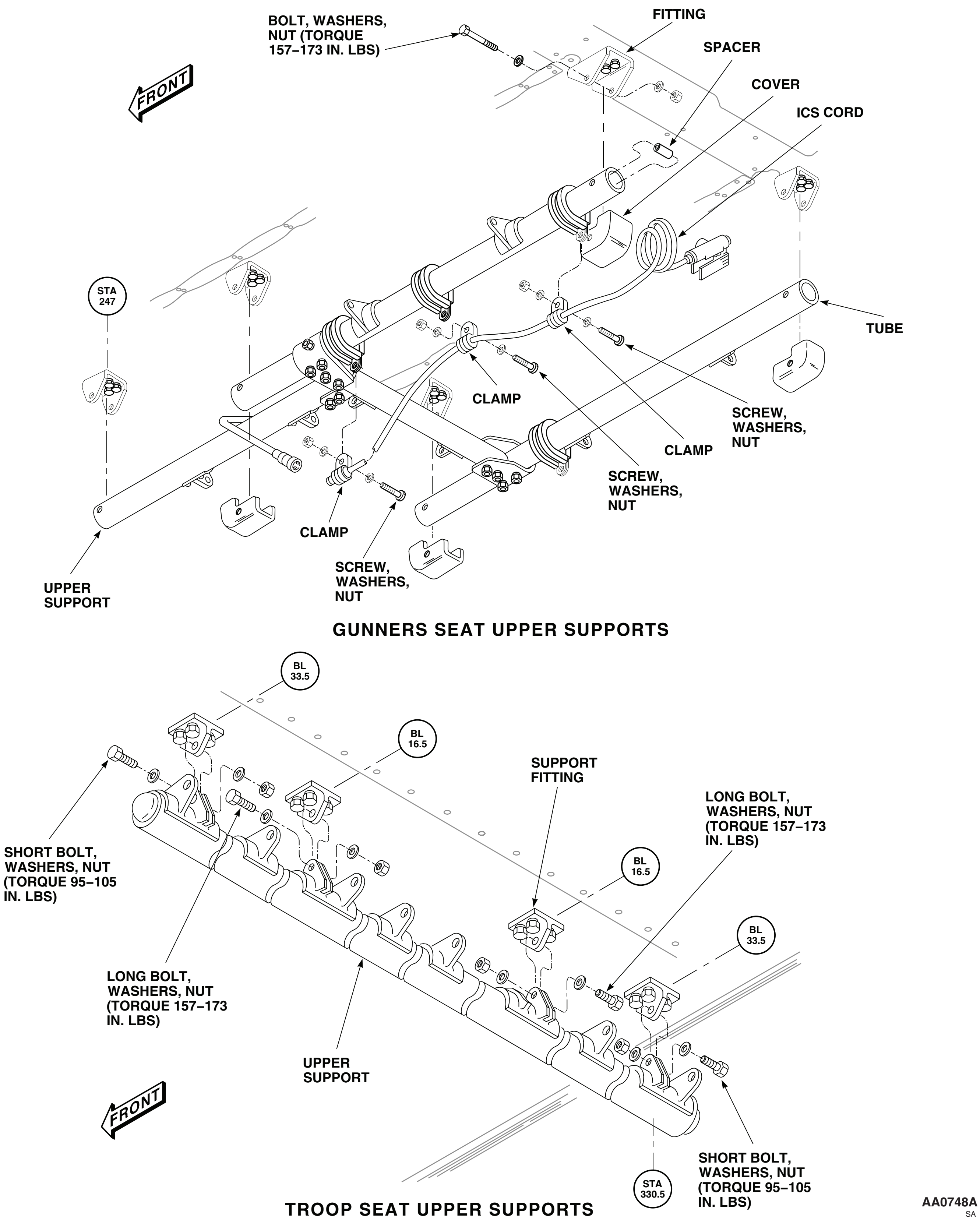


Figure 9. Replace Seat Upper Supports.

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORTS - Continued

Install

1. To install upper support for gunner's and troop commander's seat, do this:
   a. If ICS cords on upper support were removed, install them using clamps, screws, washers, and nuts (Figure 9, Detail A).
   b. Position upper support in fittings, and install spacers in ends of tubes.
   c. Line up holes in fittings, spacers, and tubes. Install bolts, washers, and nuts. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

   d. Clean mating surfaces of cover and tube using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
   e. Apply thin coat of adhesive, Item 15, WP 1803 00, to mating surfaces inside tube and outside cover.
   f. Install cover in tube and make 0.050-inch fillet of adhesive squeeze out.
   g. Allow 12 hours at 70°F(21°C) to 85°F(29°C) for adhesive cure.
   h. Connect ICS cords at disconnects.
   i. Install gunner's and troop commander's seats.
2. To install troop seat upper support, do this:
   a. Position upper support in fittings and install short bolts, washers, and nuts in fittings near ends of support (BL 33.50). **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
   b. Install long bolts, washers, and nuts in fittings near center of support (BL 16.50). **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**
   c. Install center troop seats.

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORT FITTINGS

Remove

1. Remove troop/gunner's seats.
2. Remove troop/gunner's seat upper supports.
3. Open soundproofing panels as required (WP 0258 00).
4. Remove bolts, washers, radius blocks, and nuts attaching gunner's seat support fittings at STA 265.5, STA 295.0, LBL 10.0, and RBL 10.0 (Figure 10, Sheet 1, Detail A).
5. Remove bolts, washers, and nuts attaching support fittings at STA 330.5, LBL 16.5 and RBL 16.5.
6. Remove bolts and washers attaching troop seat support fitting at STA 330.5, LBL 33.5.
7. Remove blind rivets attaching support fitting at STA 330.5, RBL 33.5 (TM 1-1500-204-23).
8. Remove bolts, washers, nuts, angle bracket, and retaining ring attaching seat support fittings at STA 398.0 (Figure 10, Sheet 2, Detail C).

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORT FITTINGS - Continued

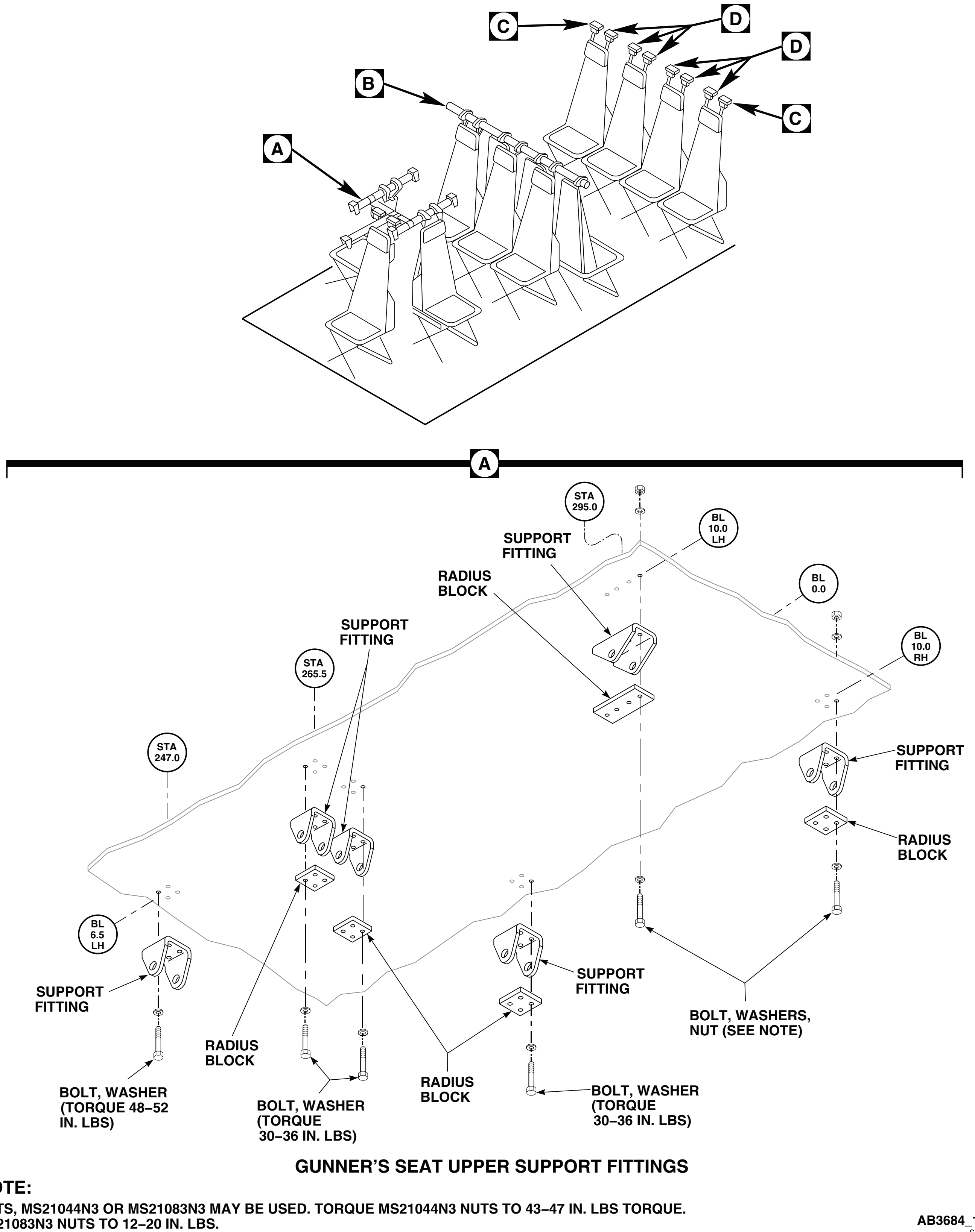


Figure 10. Replace Seat Upper Support Fittings. (Sheet 1 of 2)

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORT FITTINGS - Continued

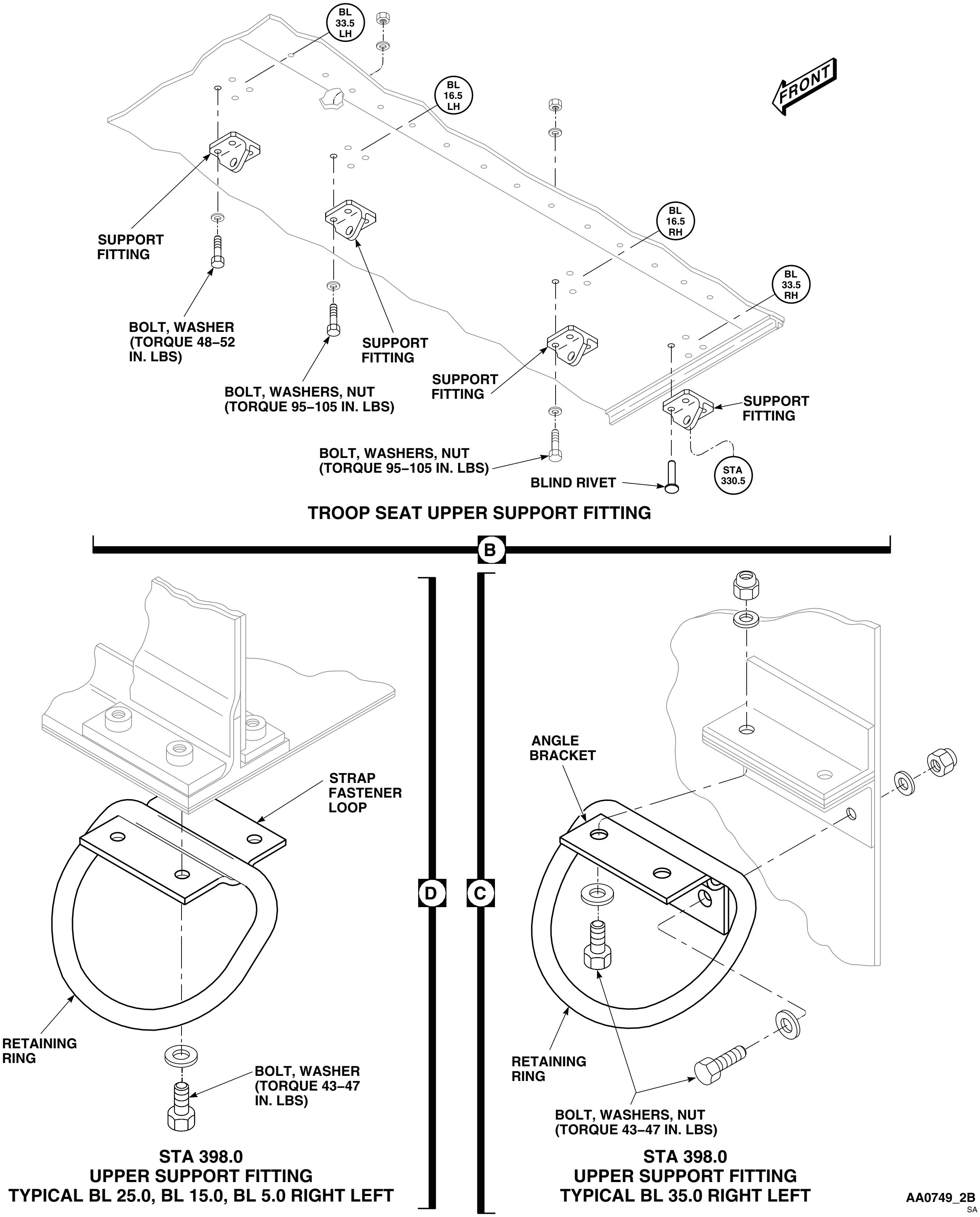


Figure 10. Replace Seat Upper Support Fittings. (Sheet 2 of 2)

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORT FITTINGS - Continued

9. Remove bolts, washers, strap fastener loop, and retaining ring attaching seat support fittings at STA 398.0 (Figure 10, Sheet 2, Detail D).

Install

WARNING

- Injury to personnel and damage to equipment will result if improper length bolts are used to install support fittings. Bolts can bottom out on nutplates or nuts before proper torque is reached. Make sure proper length bolts are used.
- A minimum of 1.5 threads must show through nutplates or nuts. Up to three washers may be used under boltheads to obtain proper thread engagement.

CAUTION

Damage to mounting hardware, fittings and radius blocks will occur if radius blocks are not properly installed. Make sure radius blocks are installed so that contours on radius block match radius of inside corners of support fittings.

1. Check self-locking feature of nutplates or nuts (TM 1-1500-204-23).
2. Install troop /gunner seat support fitting at STA 247.0, LBL 6.5 (Figure 10, Sheet 1, Detail A) and STA 330.5, LBL 33.5 (Figure 10, Sheet 2, Detail B) with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS**.
3. Install support fitting at STA 330.5, RBL 33.5 with blind rivets (TM 1-1500-204-23).
4. Install support fittings at STA 330.5, LBL 16.5 and RBL 16.5 with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
5. Install gunner's seat support fittings at STA 265.5 and STA 295.0, LBL 10.0 and RBL 10.0 with bolts, washers, radius blocks, and nuts. **TORQUE NUTS TO 30 - 36 INCH-POUNDS.**
6. Install seat support fittings at STA 398.0 with bolts, washers, nuts, angle bracket, and retaining ring (Figure 10, Sheet 2, Detail C). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
7. Install seat support fittings at STA 398.0 with bolts, washers, strap fastener loop, and retaining ring (Figure 10, Sheet 2, Detail D). **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**
8. Close soundproofing panels (WP 0258 00).
9. Install troop/gunner's seat upper support.
10. Install troop/gunner's seats.

Repair

1. Troop/gunner seats are tubular aluminum alloy framework covered with fabric and held rigid by diagonal cable assemblies. Quick-disconnect floor fittings, upper support tubes, and support fittings are formed or extruded aluminum alloy. Restraint fittings and upper latch attachment fittings are 4130 steel welded assemblies (Figure 11, Sheet 1, Figure 11, Sheet 2, and Figure 11, Sheet 3).
2. Evaluate damage to aluminum parts (WP 0377 00).
3. Use WP 0378 00 and WP 0380 00 to repair or replace aluminum parts.
4. Evaluate damage or wear on steel welded assemblies using WP 0377 00 limits. Blend out minor damage and restore cadmium-plating (TM 1-1500-344-23). Replace welded assemblies damaged or worn beyond negligible damage limits.
5. Use this paragraph and TM 1-1500-204-23 for damage limits or repair on straps, webbing, and covers.

REPLACE TROOP/GUNNER'S SEAT UPPER SUPPORT FITTINGS - Continued

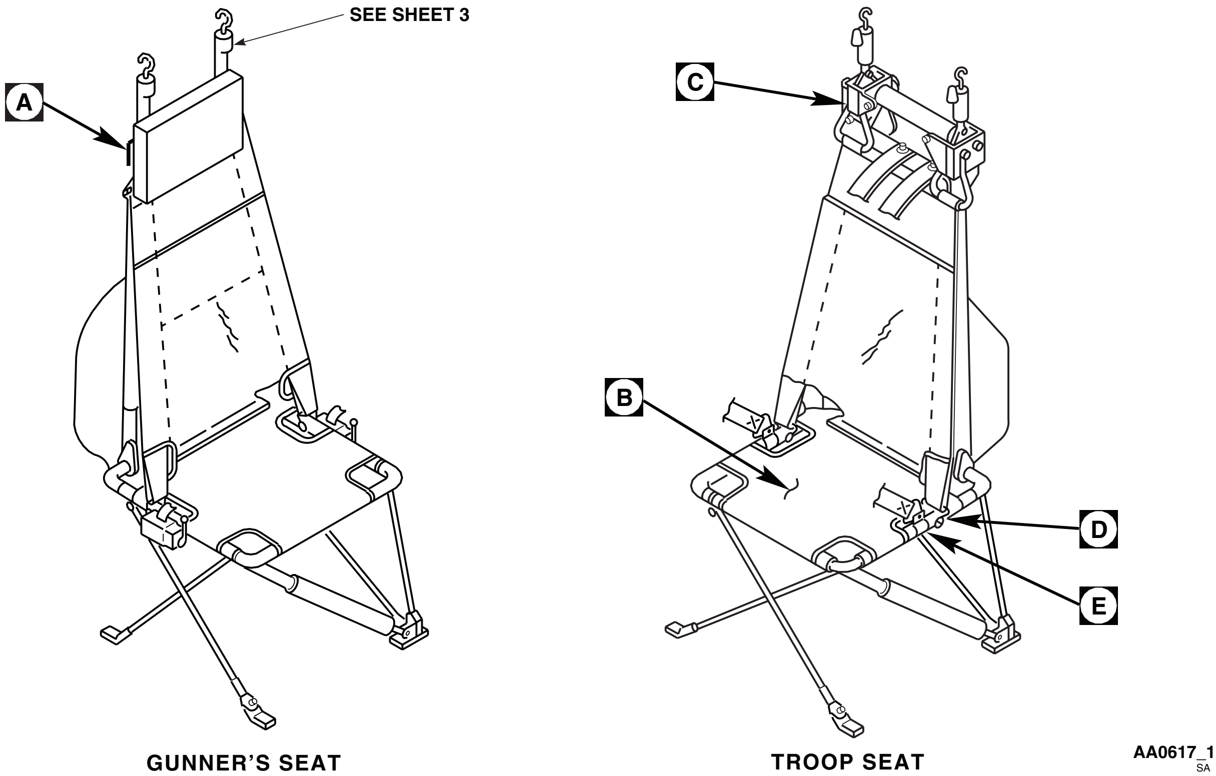


Figure 11. Repair Troop/Gunner's Seat. (Sheet 1 of 3)

6. To repair troop and gunner's seat pan covers, damaged by rivets on seat frame, sew gray webbing, 2 inches wide, to inner surface of pan cover (Figure 11, Sheet 2, Detail B). Stitch webbing (TM 1-1500-204-23).
7. To reduce wear on seat cover, replace protruding head blind rivet with flush head rivet, in two places (Figure 11, Sheet 2, Detail A).
8. Use TM 1-1500-204-23 for tubing repair.
9. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPLACE TROOP/GUNNER'S SEAT

NOTE

- This paragraph also applies to troop commander's seat.
- Do not place tool boxes, tools, or oily rags on seat covers while performing maintenance.

Remove

1. Turn off all electrical power.
2. Release seat latch locking levers (Figure 12, Sheet 1, Detail D).
3. Disengage disconnect fittings and cable clips from floor attachment studs (Figure 12, Sheet 1, Detail E).

REPLACE TROOP/GUNNER'S SEAT - Continued

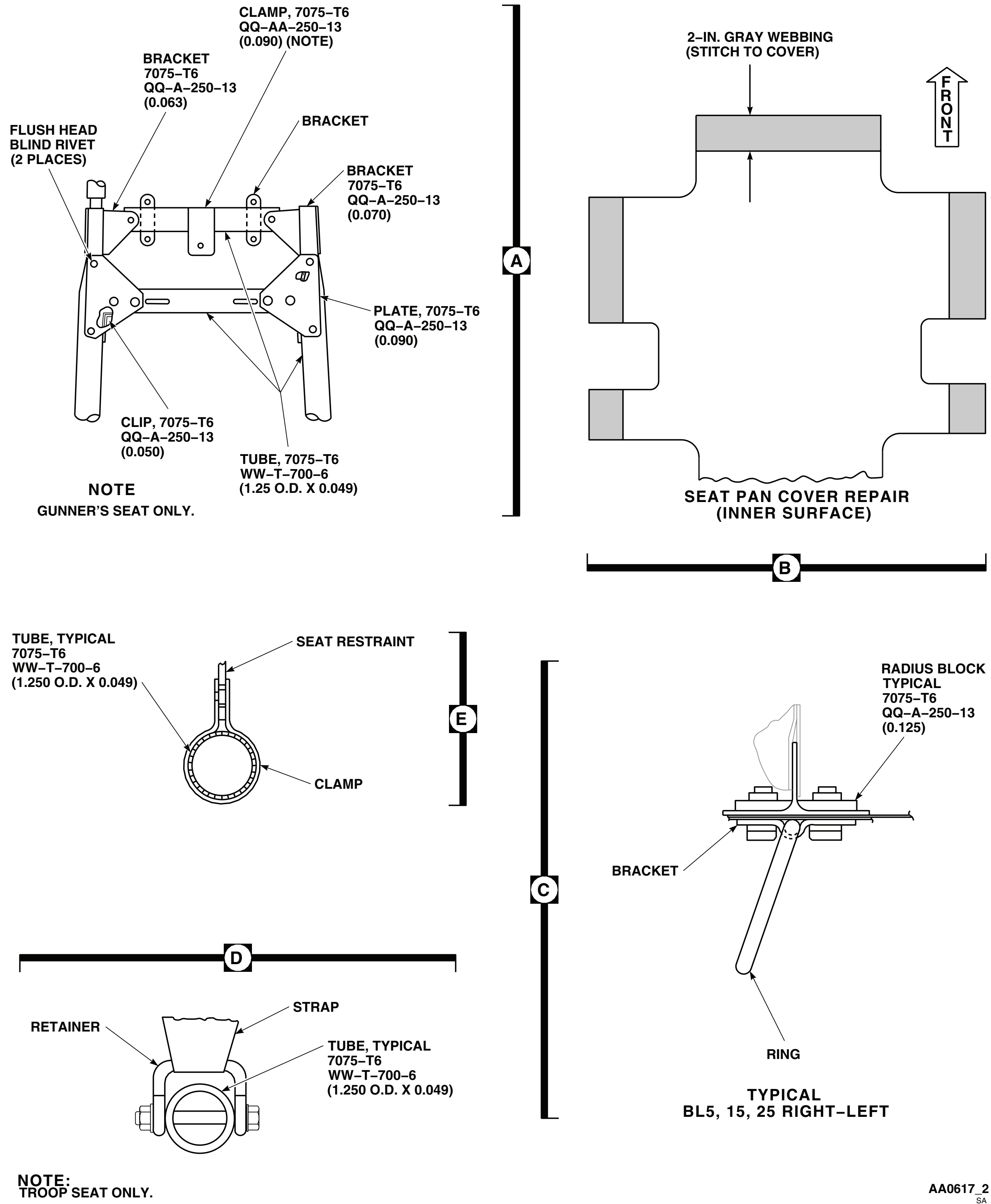


Figure 11. Repair Troop/Gunner's Seat. (Sheet 2 of 3)

REPLACE TROOP/GUNNER'S SEAT - Continued

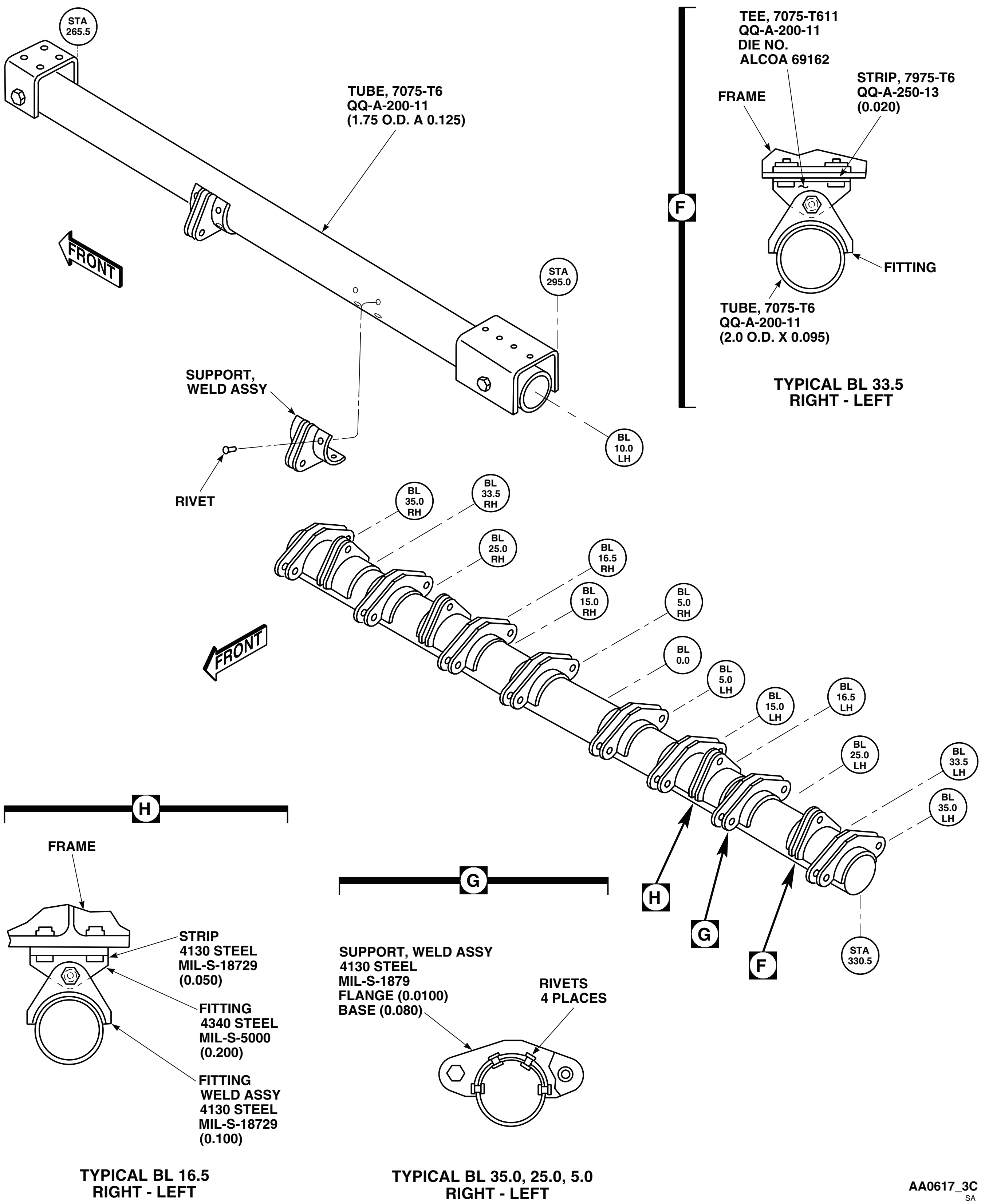


Figure 11. Repair Troop/Gunner's Seat. (Sheet 3 of 3)

REPLACE TROOP/GUNNER'S SEAT - Continued

CAUTION

Damage to attenuator wire will occur if troop/gunner's seat is not properly handled and stored. Mishandling will cause wire to bend, twist and not be centered on roller depression. Make sure to use care when handling or storing troop/gunner's seat.

4. Unhook seat latches from upper support brackets (Figure 12, Sheet 1, Detail C). Remove seat.

Inspect

1. Check upper latches and latch locking mechanism for cracks and broken springs. **NO CRACKS ALLOWED.** Replace cracked latches.
2. Check attenuator wire for wear, cracks, flatness (wire diameter), deformation, or other damage. **NO CRACKS ALLOWED. WIRE DAMAGE SHALL BE LIMITED TO A MINIMUM DIAMETER OF 0.090-INCH.** Replace damaged attenuator wire. There shall be no wires installed that are deformed so that they are uncentered on roller depressions. Replace any unserviceable attenuator wire. Refer to REPLACE CABLE, in this work package.
3. Check seat back, pan frames, and frame fittings for cracks, loose, sheared, and missing fasteners. **NO CRACKS ALLOWED.** Repair or replace damaged frames.
4. Check anti-sag device cable assembly on rear-facing troop seats for broken or loose strands, cable sheathing for tears or damage, and attaching hardware for cracks and security. Replace damaged and worn parts.
5. Check fabric for tears, holes, and loose or missing stitching. Check seat pan for loose or missing fabric retaining screws. Replace missing screws. Repair or replace damaged seat coverings (TM 1-1500-204-23).
6. Check area between upper seat cover ring and seat frame for cracks and wear. Replace damaged ring. Replace rivets (TM 1-1500-204-23).
7. Check attenuator tubing for misalignment, dents, cracks, and torque stripes for tube extension. **NO BENDS, CRACKS, OR TUBE EXTENSION ALLOWED.** Replace bent, cracked, or extended attenuators.
8. Check attenuator tube end fittings for loose, sheared, or missing fasteners. Check bearing end for free movement. Repair loose, sheared, or missing fasteners. Replace damaged bearings (TM 55-1500-322-24).
9. Check disconnect hardware for cracks, wear, and missing or broken parts. **NO CRACKS ALLOWED.** Replace broken disconnect parts.
10. Check cables for broken strands. If three or more strands are broken, replace cable.
11. Check cables for kinks. If cables cannot be straightened by hand, replace damaged cables.

NOTE

Check seat configuration to make sure that anti-sag device cable assembly is installed on rear facing troop seats.

12. Check cable attaching hardware for cracks and security. **NO CRACKS ALLOWED.** Replace damaged hardware.
13. Inspect troopseat restraint system (WP 0234 00) or gunner's seat restraint system (WP 0235 00).

Install

1. Turn off all electrical power.

REPLACE TROOP/GUNNER'S SEAT - Continued

WARNING

- To reduce personal injury in a hard landing, make sure attenuators are installed with small ends pointing up and toward nose of aircraft.
- Crashworthiness of the aft outboard, forward facing troop seats will be decreased if the auxiliary cabin heater duct installations at STA 398.0 are present. The duct and diffuser installation near the floor protrudes beneath the troop seat allowing the seat to strike the diffuser and stroke envelope, potential decrease the energy attenuation provided by the seat stroke action. **DO NOT INSTALL AFT OUTBOARD, FORWARD FACING TROOP SETS OF HEATER DUCTS ARE INSTALLED IN THE AFT BULKHEAD AT STA 398.0.**

2. Hook seat latches on overhead support brackets (Figure 12, Sheet 1, Detail C). Do not lock lever.
3. Install attenuator and cable clip fittings on floor attachment studs (Figure 12, Sheet 1, Detail D and Detail E).
4. Screw in upper seat latch hooks to take up cable slack (Figure 12, Sheet 1, Detail F).
5. Add tension to seat by locking upper latch lever.

INSTALL FORWARD FACING CENTER SEAT

1. Turn off all electrical power.

CAUTION

Damage to upper support and support fittings will result if support is allowed to hang from end support fitting, Upper support will bend and damage support fitting. Upper support must be completely removed.

2. Remove center section seat support tube assembly at STA 272.0.
3. Install center section seat support tube assembly making sure seat attachment fittings are facing forward.
4. Install seat assembly.

INSTALL FORWARD FACING CENTER SEAT - Continued

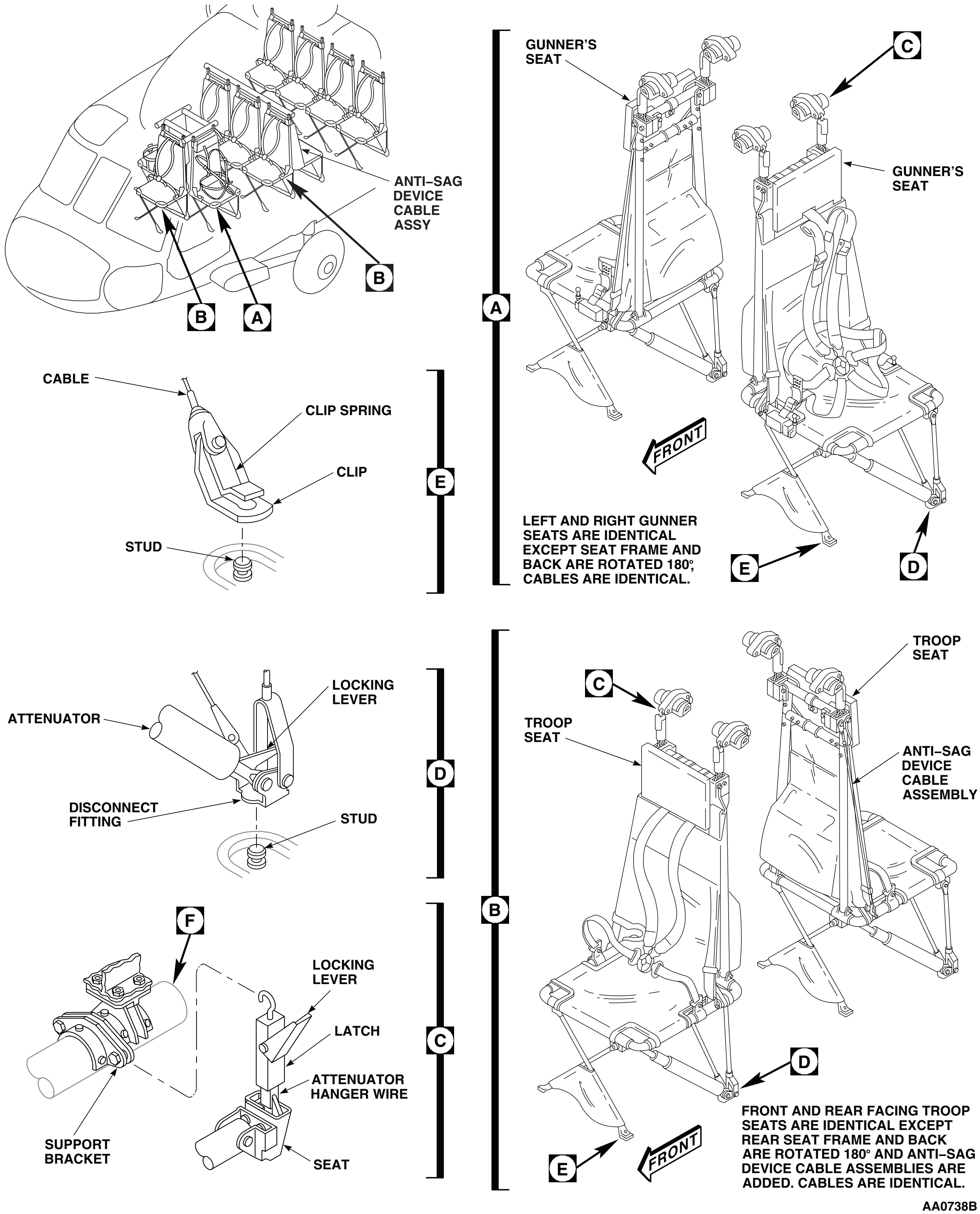


Figure 12. Troop/Gunner's Seat. (Sheet 1 of 2)

INSTALL FORWARD FACING CENTER SEAT - Continued

F

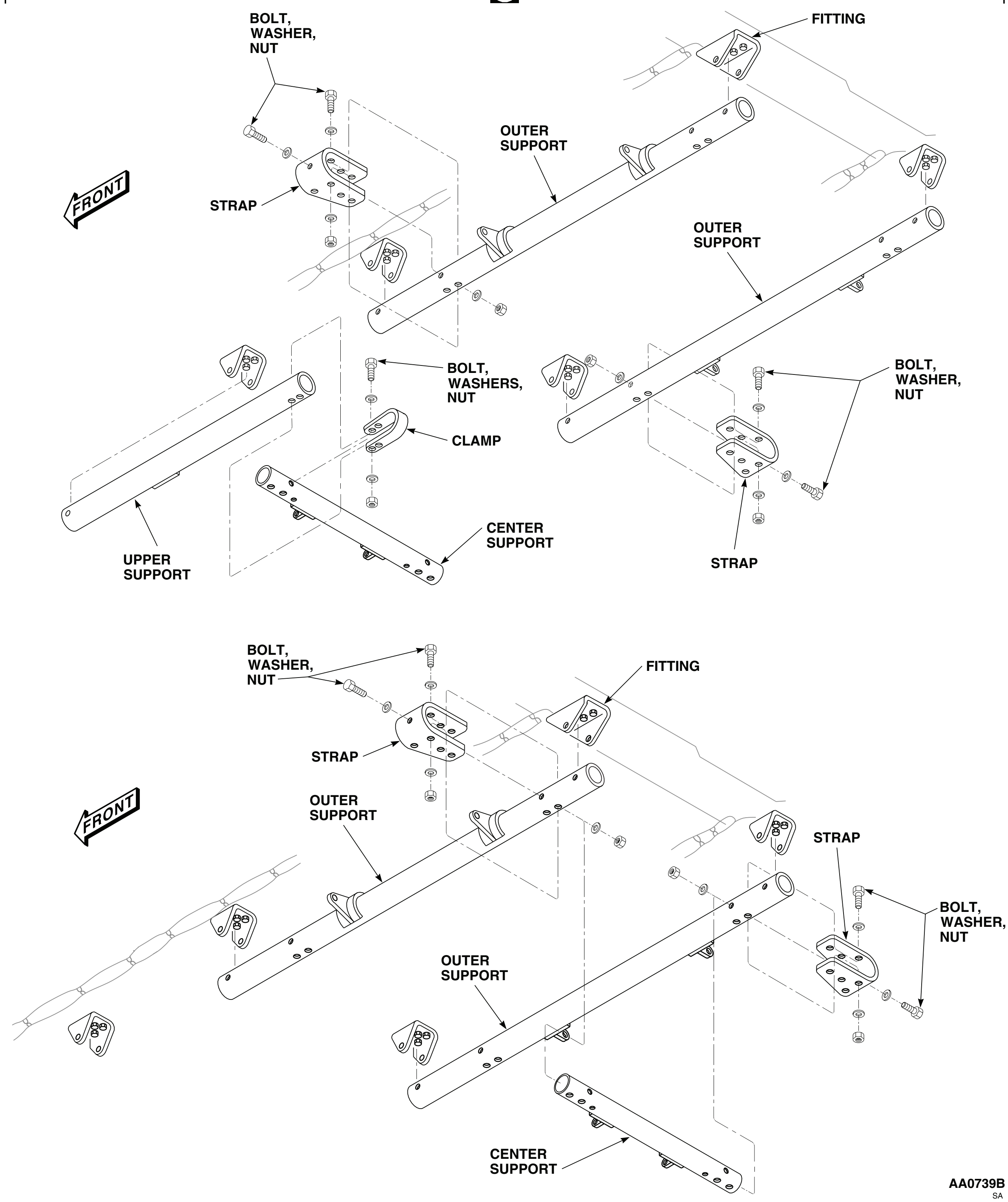


Figure 12. Troop/Gunner's Seat. (Sheet 2 of 2)

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP SEAT RESTRAINT SYSTEM

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23

NOTE

Maintenance procedures for troop commander's seat restraint are the same as for troop seat restraint.

REMOVE

1. Turn off all electrical power.
2. Remove bolt, washers, and nut securing lap belt to clamp on frame on each side of seat (Figure 1, Detail B).
3. Remove bolts, washers, nuts, spacers, and clamps anchoring shoulder straps to rear of seat frame (Figure 1, Detail A).
4. Remove restraint system from seat.

INSTALL

1. Turn off all electrical power.
2. Place restraint system on seat.
3. **QA** Attach shoulder strap fittings to clamps on rear of seat frame with spacers, bolts, washers, and nuts. **TORQUE NUTS TO 90 - 100 INCH-POUNDS (Figure 1, Detail A).**
4. **QA** Bolt lap belt to clamps on seat frame with bolts, washers, and nuts. **BELT MUST BE FREE TO PIVOT (Figure 1, Detail B).**

TEST

1. Sit in troop seat. Attach lap belt and shoulder harness to buckle.
2. Adjust lap belt and shoulder harness to maximum tightness. Lean hard against restraints. **ADJUSTMENTS MUST HOLD.** Harnesses which loosen must be replaced or repaired (TM 1-1500-204-23).
3. Lean into restraints while operating buckle release. **RESTRAINT RELEASE SHALL REQUIRE RELEASE PRESSURE NO GREATER THAN CAN BE APPLIED WITH ONE FINGER.** Replace buckles failing release test.

TEST - Continued

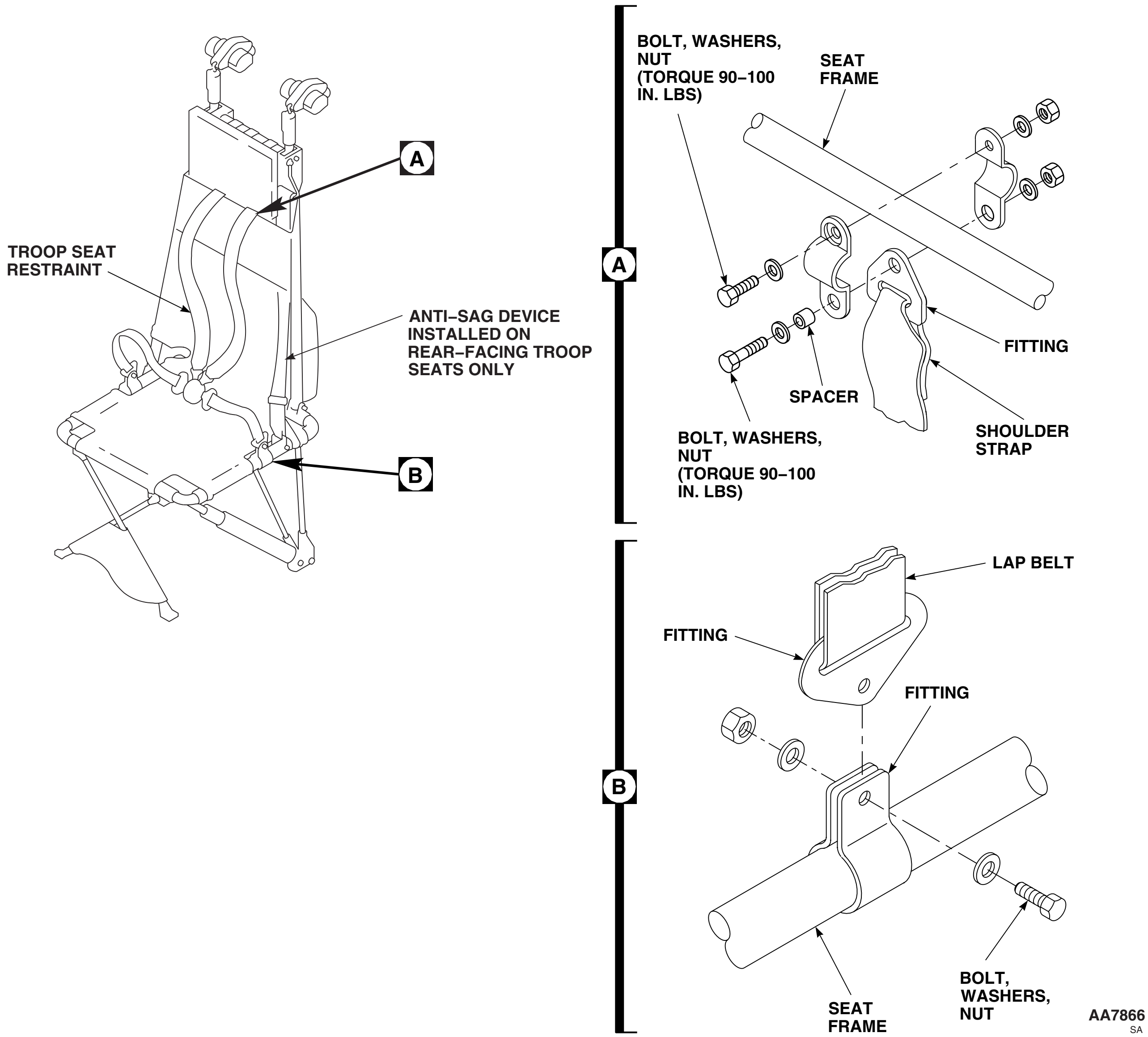


Figure 1. Troop Seat Restraint System.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S SEAT RESTRAINT SYSTEM

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0233 00

REMOVE

1. Remove head rest (WP 0233 00).
2. Remove bolts, washers, and nuts securing lap belt reels to seat frame (Figure 1, Detail A).
3. Remove screw, washers and nut securing shoulder strap reel to rear of seat frame (Figure 1, Detail B).
4. Remove restraint from seat.

WARNING

Accidental roller release can cause personal injury and reel damage. Use care to prevent accidental release. Replace a sprung reel.

NOTE

Shoulder harness is sewn onto inertial reel roller. Remove and replace shoulder harness and reel as a unit.

5. To remove and install lap belt on inertia reel roller, do this:
   a. Remove reel cover.
   b. Fully extend belt.
   c. Insert a 7/32 hex wrench in reel. Hold roller in position (Figure 1, Detail C).
   d. Push belt and insert out of roller.
   e. Withdraw belt. Do not release roller.
6. To replace belt on roller, do this:
   a. Carefully wind spring until tight. Back off 1/2 to 1 1/2 turns.
   b. Install belt and insert with insert closest to single roller groove (Figure 1, Detail C).

INSTALL

1. Place restraint system on seat.
2. **QA** Bolt shoulder strap and reel to rear of seat frame (Figure 1, Detail B).

INSTALL - Continued

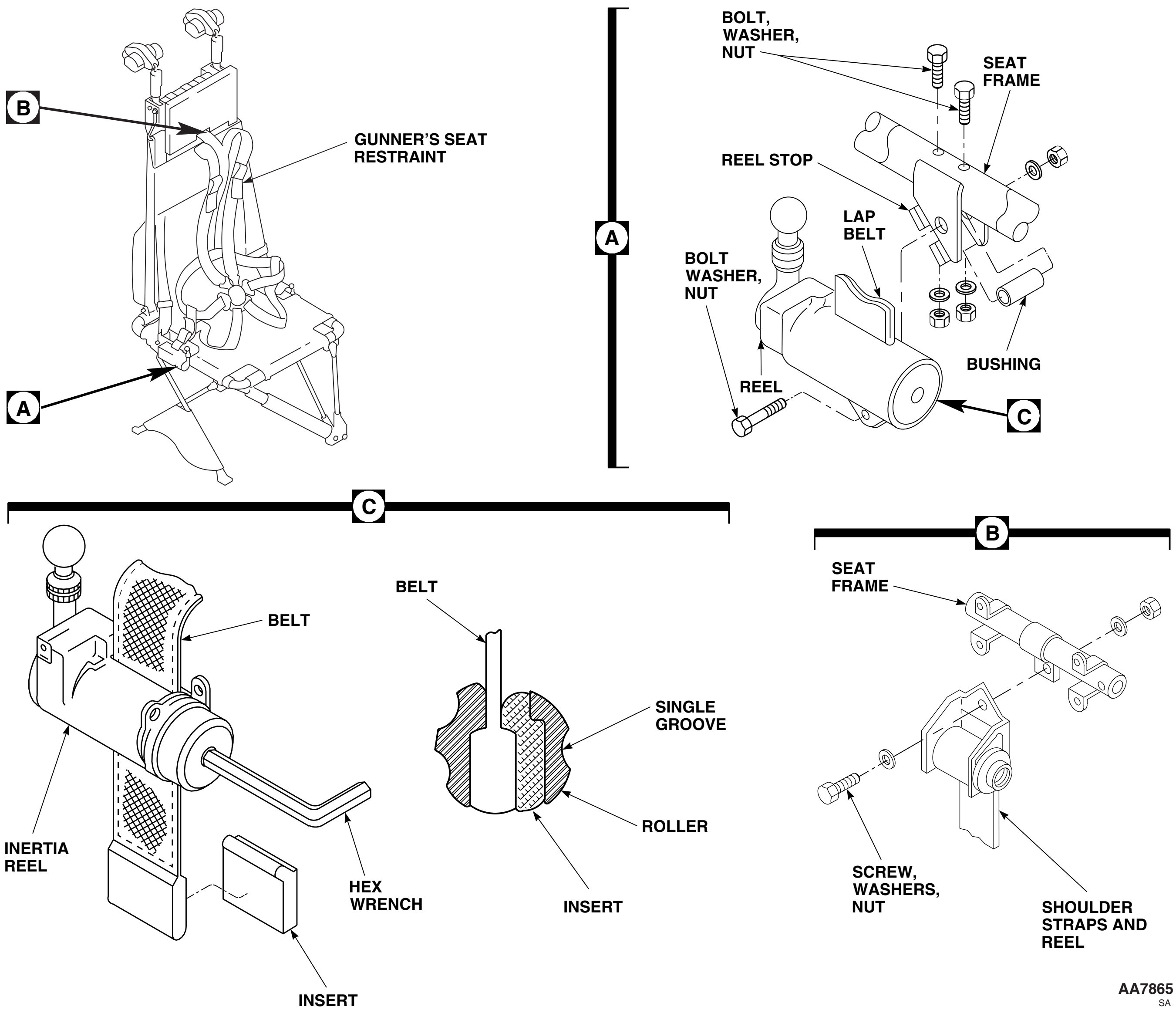


Figure 1. Replace Gunner's Seat Restraint System.

3. Install lap belt reels to seat frame. **REELS MUST BE FREE TO PIVOT (Figure 1, Detail A).**
4. Install head rest (WP 0233 00).

TEST

1. Sit in gunner's seat. Attach lap belt and shoulder harness to buckle.
2. Adjust lap belt and shoulder harness to maximum tightness.
3. Put lap belt inertia reels in MANUAL-LOCK.
4. Lean hard into restraint system. **HARNESS ADJUSTMENTS MUST HOLD.** Harnesses which loosen must be replaced or repaired (TM 1-1500-204-23). Some webbing will come out of inertia reels as webbing tightens on rollers.

TEST - Continued

5. Lean into restraints while operating buckle release. **RESTRAINT RELEASE SHALL REQUIRE PRESSURE NO GREATER THAN CAN BE APPLIED WITH ONE FINGER.** Replace buckles failing release test.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MISSION OPERATOR'S SEAT EH60A

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REPLACE MISSION OPERATOR'S SEAT

Remove

1. Turn off all electrical power.

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Injury to personnel could result.

2. Sit in seat. Under right front of seat, release VERT adjust lever and adjust seat to full up. Under left front of seat, pull HORIZ adjust lever and slide seat to front to gain access to stop (Figure 1).
3. On DF operator seat track/frame assembly, remove two screws securing stop to each track. Refer to, REPLACE DF TRACK/FRAME ASSEMBLY, in this work package.
4. On ECM operator seat track/frame assembly, remove screw securing stop to each track and remove screws securing bridges to each track. Refer to, REPLACE ECM TRACK/FRAME ASSEMBLY, in this work package.

NOTE

Make sure controls of ECM equipment do not get engaged due to forward travel of the seat.

5. Release HORIZ adjust lever and slide seat to the front until track fittings reach bridge/stop gaps of track/frame assembly.
6. Lift seat out of track/frame assembly (Figure 1).
7. Remove seat from helicopter.
8. Reinstall track/frame assembly bridges and stops.

Install

1. Turn off all electrical power.

REPLACE MISSION OPERATOR'S SEAT - Continued

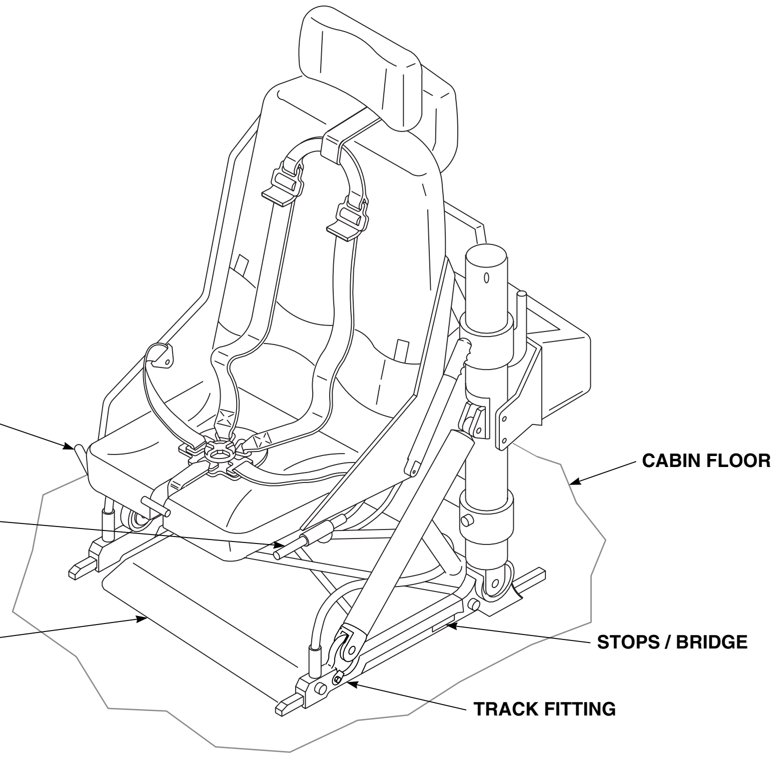


Figure 1. Replace Mission Operator's Seat.

WARNING

To prevent injury to personnel, do not release either the normal or emergency vertical adjust levers unless someone is sitting in seat. The extension springs are under load at all times. With seat at lowest position, the vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, the seat will be snapped to highest stop. Injury to personnel could result.

2. Sit in seat. Under right front of seat, release VERT adjust lever and adjust seat to full up position (Figure 1).
3. Remove track/frame assembly bridges and stops.
4. Position seat over track/frame assembly with track fittings in gaps left by removal of bridges and/or stops.
5. Release HORIZ adjust lever and slide seat slightly to the rear on tracks.

NOTE

On DF operator's seat make sure stop is installed with notch down and aft.

6. **QA** Install stop on each track DF seat or install bridge and stops on each track for ECM seat. Refer to REPLACE DF TRACK/FRAME ASSEMBLY and REPLACE ECM TRACK/FRAME ASSEMBLY, in this work package.
7. Release HORIZ adjust lever and slide seat to the rear to check bridge/stops installation.

REPLACE DF TRACK/FRAME ASSEMBLY

Remove

1. Turn off all electrical power.
2. Remove DF operator’s seat. Refer to REPLACE MISSION OPERATOR’S SEAT, in this work package.

NOTE

Washers and crushable cores may be loose in frame insert. Take care not to lose when removing assembly.

3. Remove bolts securing track/frame assembly to cabin floor (Figure 2, Detail A).
4. Remove track/frame assembly from helicopter.

Disassemble

1. Remove washers and crushable cores from frame assembly. Inspect for physical damage or other signs of deterioration. Replace as required (Figure 2, Detail B).
2. Remove screws securing stop to each track.
3. Remove bolts and washers securing track on each side of frame assembly. Remove track.

Assemble

NOTE

Washers are only installed on outboard track.

1. **QA** Install each track on frame assembly with washers and bolts. **TORQUE TO 50 - 70 INCH-POUNDS (Figure 2, Detail B).**
2. **QA** Install stop on each track with screws.
3. Install crushable cores and washers in frame assembly.

Install

1. Turn off all electrical power.
2. **QA** Position track/frame assembly over mounting holes in cabin floor (Figure 2, Detail A).
3. Make sure washers and crushable cores are present in frame inserts.
4. **QA** Install bolts to secure track frame assembly to cabin floor. **TORQUE TO 260 - 280 INCH-POUNDS.**
5. Install DF operator’s seat. Refer to REPLACE MISSION OPERATOR’S SEAT, in this work package.

REPLACE ECM TRACK/FRAME ASSEMBLY

Remove

1. Turn off all electrical power.
2. Remove ECM operator’s seat. Refer to REPLACE MISSION OPERATOR’S SEAT, in this work package.

REPLACE ECM TRACK/FRAME ASSEMBLY - Continued

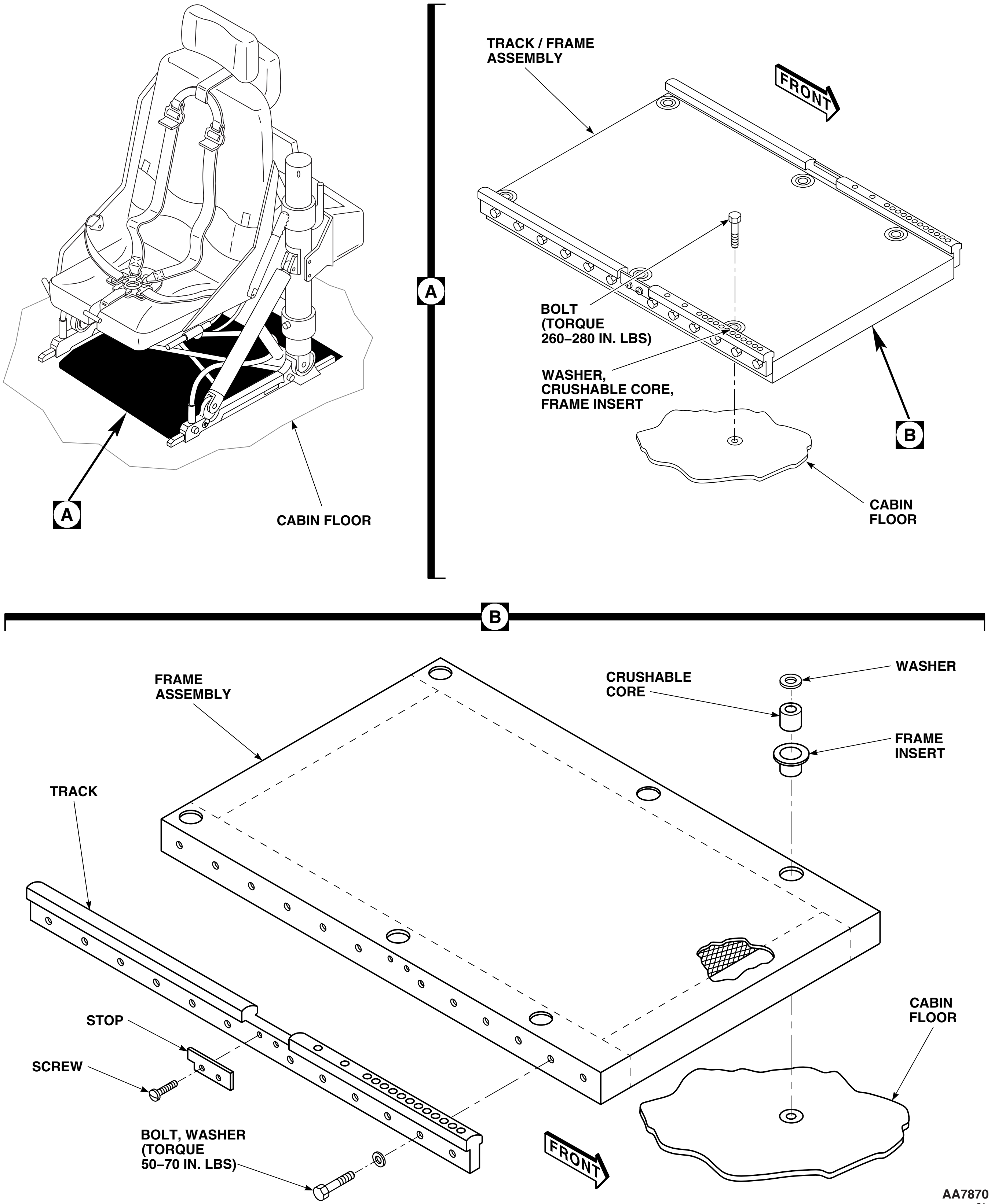


Figure 2. Replace DF Track/Frame Assembly.

REPLACE ECM TRACK/FRAME ASSEMBLY - Continued

NOTE

Washers and crushable cores may be loose in insert. Take care not to lose when removing assembly.

3. Remove bolts securing track/frame assembly to cabin floor (Figure 3, Detail A).
4. Remove track/frame assembly from aircraft.

Disassemble

1. Remove washers and crushable cores from frame assembly. Inspect for physical damage or other signs of deterioration. Replace as required (Figure 3, Detail B).
2. Remove screws securing bridge to each track. Remove bridge
3. Remove screws securing stop to each track. Remove stops.
4. Remove bolts and washers securing track on each side of frame assembly. Remove track.

Assemble

1. **QA** Install each track on frame assembly with washers and bolts (Figure 3, Detail B). **TORQUE TO 50 - 70 INCH-POUNDS.**
2. **QA** Install stop on each track with screws.
3. Install crushable cores and washers in frame assembly.

Install

1. Turn off all electrical power.
2. Position track/frame assembly over mounting holes in cabin floor (Figure 3, Detail A).
3. Make sure washers and crushable cores are present in frame inserts.
4. **QA** Install bolts to secure track/frame assembly to cabin floor. **TORQUE TO 260 - 280 INCH-POUNDS.**
5. Install ECM operator’s seat. Refer to REPLACE MISSION OPERATOR’S SEAT, in this work package.

REPLACE ECM TRACK/FRAME ASSEMBLY - Continued

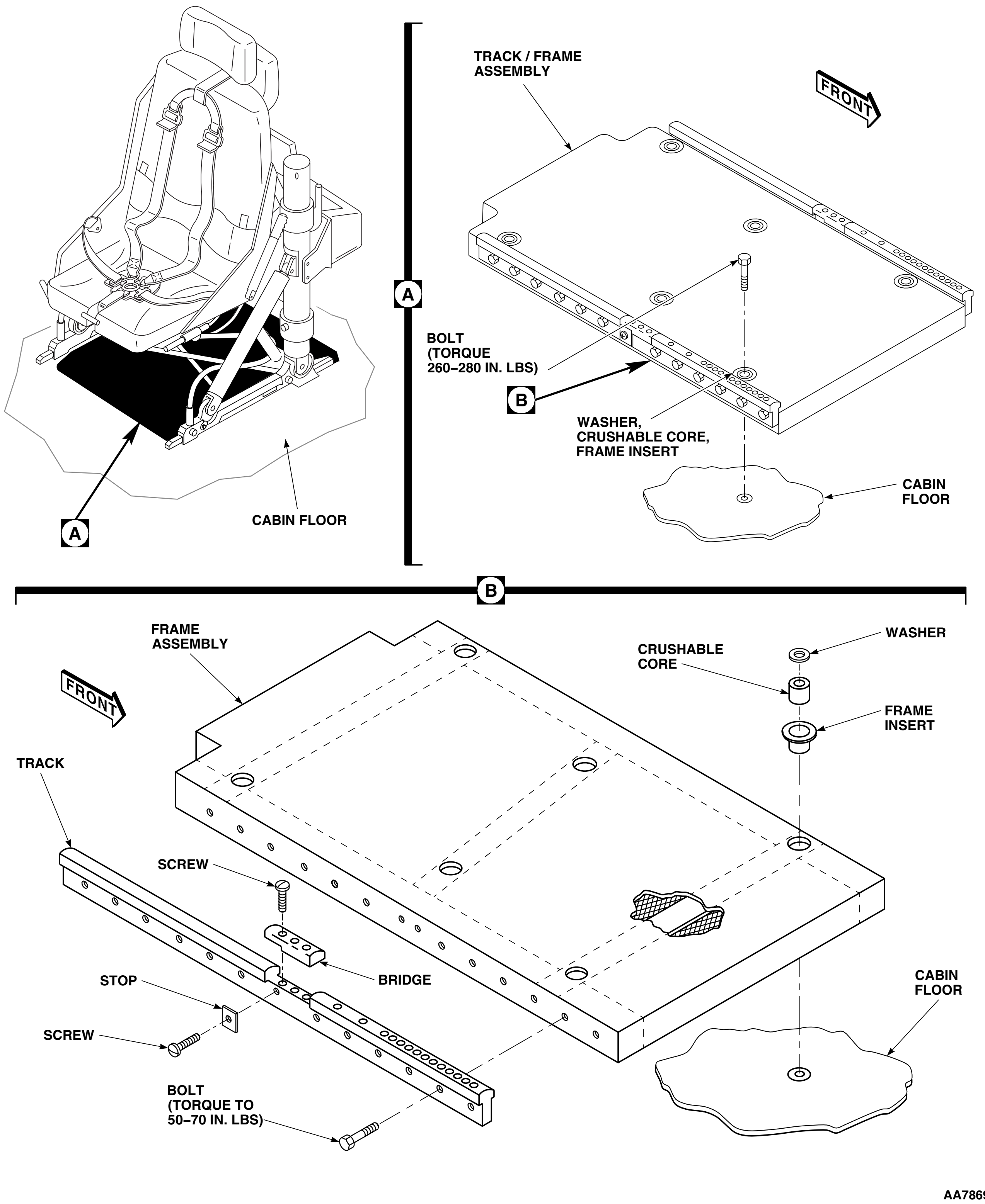


Figure 3. Replace ECM Track/Frame Assembly.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Sealing Compound, Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 11-1520-237-23
WP 0238 00
WP 0399 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove VHF/FM homing antenna from helicopter (Figure 1, Detail A) (TM 11-1520-237-23).
3. Tie string to antenna lead and push lead back into helicopter.
4. Remove fairing cap covering upper window tracks (WP 0399 00).

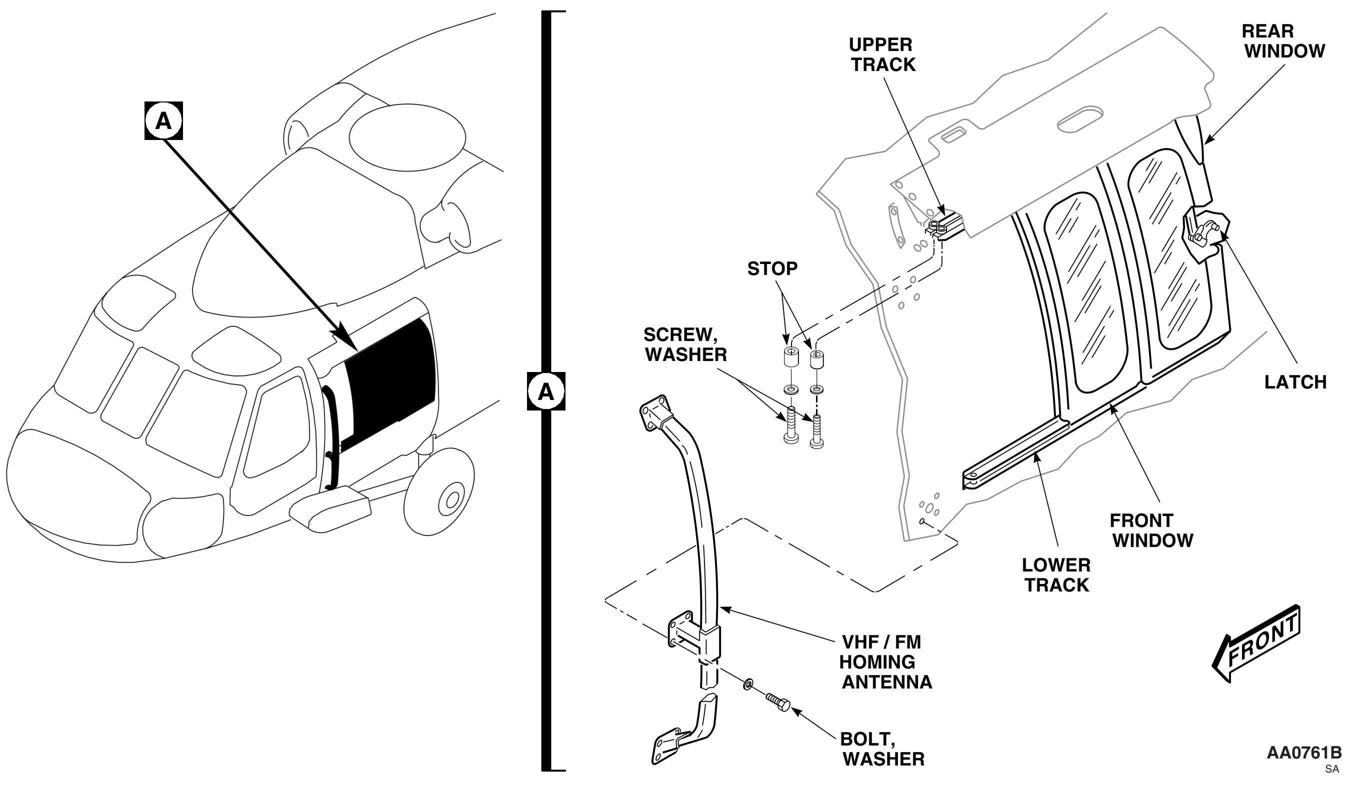


Figure 1. Replace Window.

REMOVE - Continued

5. Remove screws, washers, and stops from upper tracks.
6. Unlatch rear window. Slide both windows out of tracks and remove from helicopter.

INSTALL

1. Turn off all electrical power.
2. Line up rear window with tracks and start window on track about 6 inches. Start front window on tracks and slide both windows to closed position (Figure 1, Detail A).
3. Check clearance between window assemblies and fuselage to avoid chafing of outer skin. **MINIMUM CLEARANCE ALLOWED IS 0.040-INCH.**
4. If rear window frame binds against forward window, trim rear frame (TM 1-1500-204-23).
5. Latch rear window.
6. Check that rear seal is tight against frame with no gaps.
7. If necessary, on helicopters with windows, 70000-02400-041 and 70000-02400-042, adjust pin block (WP 0238 00).
8. If necessary, on helicopters with windows, 70216-02402-047 through 70216-02402-050, adjust pivot plate (WP 0238 00).
9. **QA** Install screws, washers, and stops in upper tracks.
10. Install fairing cap over upper window tracks (WP 0399 00).
11. Seal fairing cap with sealing compound, Item 273, WP 1803 00.
12. Retrieve antenna lead from inside helicopter by pulling on string.
13. Install VHF/FM homing antenna on helicopter (TM 11-1520-237-23). Connect antenna lead and remove string.
14. Make sure area is clean and free of foreign material.
15. Do a functional check of VHF/FM Homing System (TM 11-1520-237-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW LATCH ASSEMBLY

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Cleaning Compound, Solvent, Item 71, WP 1803 00
Primer, Adhesive, Item 240, WP 1803 00
Tape, Insulation, Electrical, Item 319, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0400 00
WP 1803 00

REMOVE

1. Open rear gunner's window.
2. Disconnect latch spring from latch (Figure 1, Sheet 1, Detail A or Figure 1, Sheet 2, Detail A).
3. Remove weldment (or pivot plate and spacer plate), handle, and latch by removing screws (or bolts) and washers.
4. If necessary, on helicopters with windows, 70000-02400-041 and 70000-02400-042, remove nuts, washers, and bolts securing pin block and plate to window frame(s). Remove pin block and plate Figure 1.
5. If necessary, on helicopters with windows, 70216-02402-041 through 70216-02402-044, and 70216-02402-047 through 70216-02402-050, remove cotter pin, headed pin, and washers from window frame(s).
6. If necessary, on helicopters with windows 70216-02402-049 and 70216-02402-050, remove catch (Figure 2).

INSPECT

1. Check latch parts, mounting hardware and framework for cracks, misalignment, binding, interference, and loose or missing parts.
2. Repair or replace loose, missing, crooked, bent, or broken parts (WP 0400 00).

REPAIR

1. If required, remove tape covering from latch handle using a nonmetallic scraper and cleaning compound solvent, Item 71, WP 1803 00. Wipe surface with clean machinery wiping towel, Item 344, WP 1803 00.
2. Spray on one coat of adhesive primer, Item 240, WP 1803 00, and install new electrical insulation tape, Item 319, WP 1803 00. Trim tape to shape of handle.

INSTALL

1. **QA** Assemble handle and latch to weldment (or pivot plate), and attach to window(s) with screws (or bolts) and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 1, Sheet 1, Detail A or Figure 1, Sheet 2, Detail A).
2. Connect latch spring to latch.

INSTALL - Continued

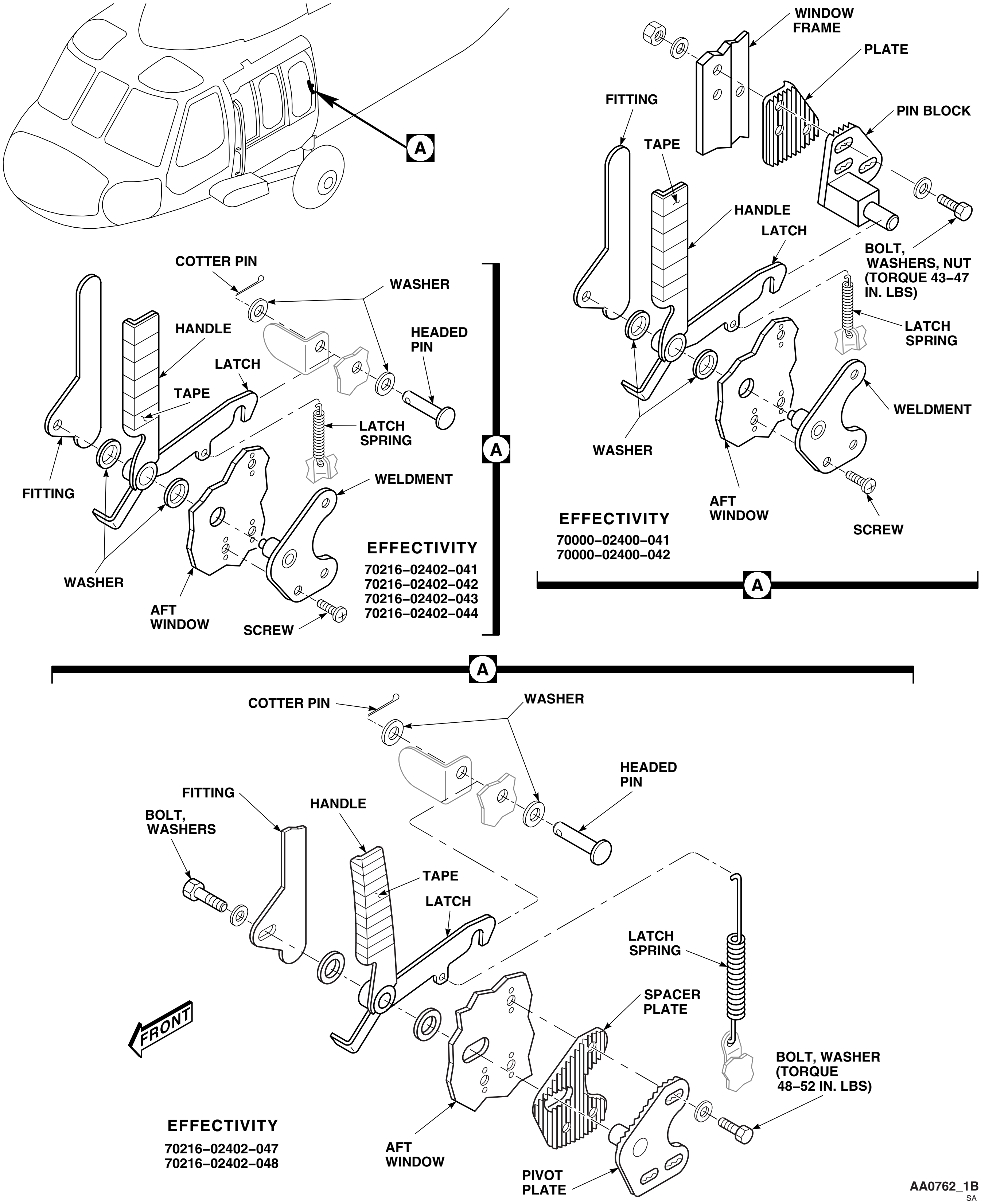


Figure 1. Replace Gunner’s Window Latch Assembly. (Sheet 1 of 2)

INSTALL - Continued

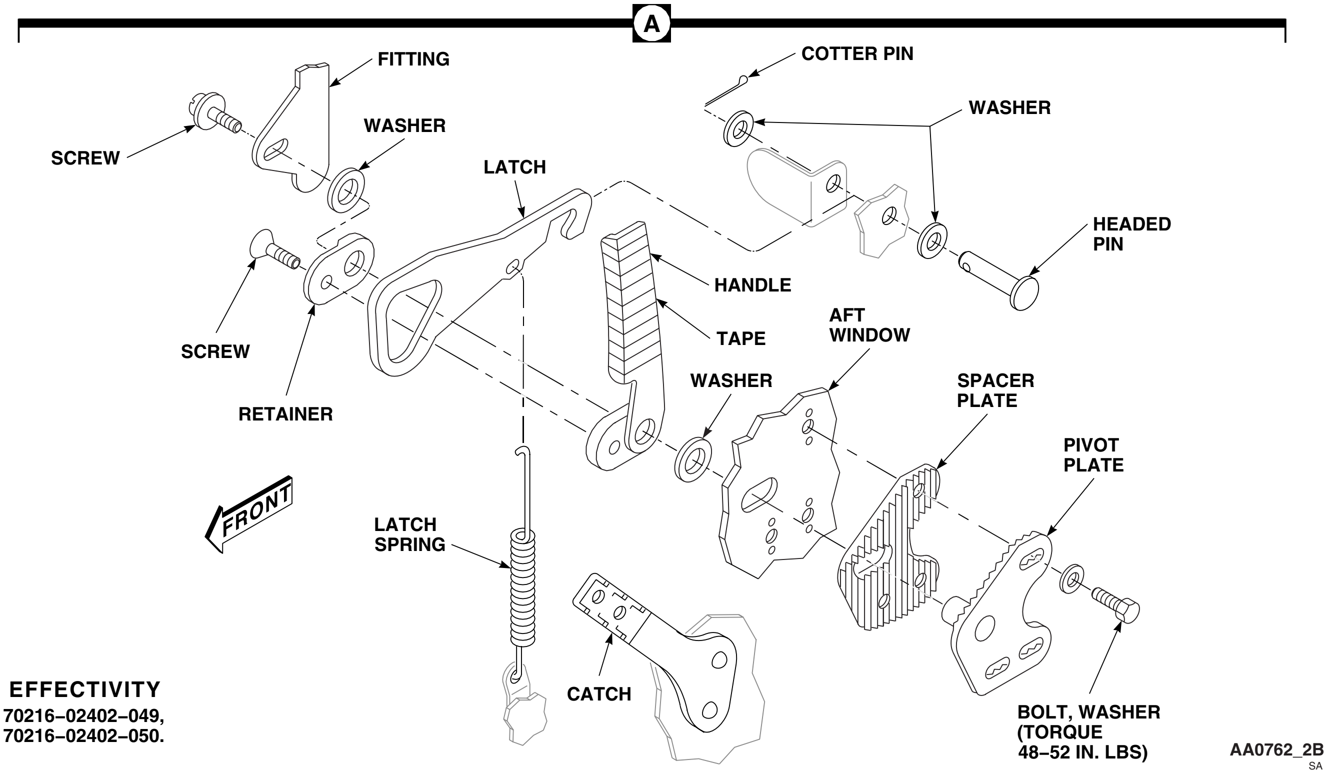


Figure 1. Replace Gunner's Window Latch Assembly. (Sheet 2 of 2)

3. **QA** Where required, install pin block and plate on window frame with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
4. If required, on window frame, install headed pin with light washer under head, another washer, and cotter pin.
5. If required, install catch with rivets (TM 1-1500-204-23).
6. Open and close window a few times to check for correct latch operation. Check that rear seal is tight against window frame with no gaps.
7. If necessary, on helicopters with windows 70000-02400-041 and 70000-02400-042, adjust pin block by loosening nuts and bolts and moving pin block one serration at a time across serrated plate, until seal is compressed against window frame with no obvious gaps when window is closed.
8. **QA** Tighten nuts and bolts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
9. **QA** If necessary, on helicopters with windows 70216-02402-041 and 70216-02402-047 through 70216-02402-050, adjust latching action. Unlatch window, loosen three bolts securing pivot plate to window, move pivot plate forward or back as needed, and retighten bolts. Adjust until seal is compressed against window frame with no gaps, when window is latched. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW SLIDE

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Remove screws attaching slide assembly to window (Figure 1, Detail A).
2. Remove slide assembly.

INSTALL

1. Place replacement slide assembly on window.
2. **QA** Install screws and washers (Figure 1, Detail A).

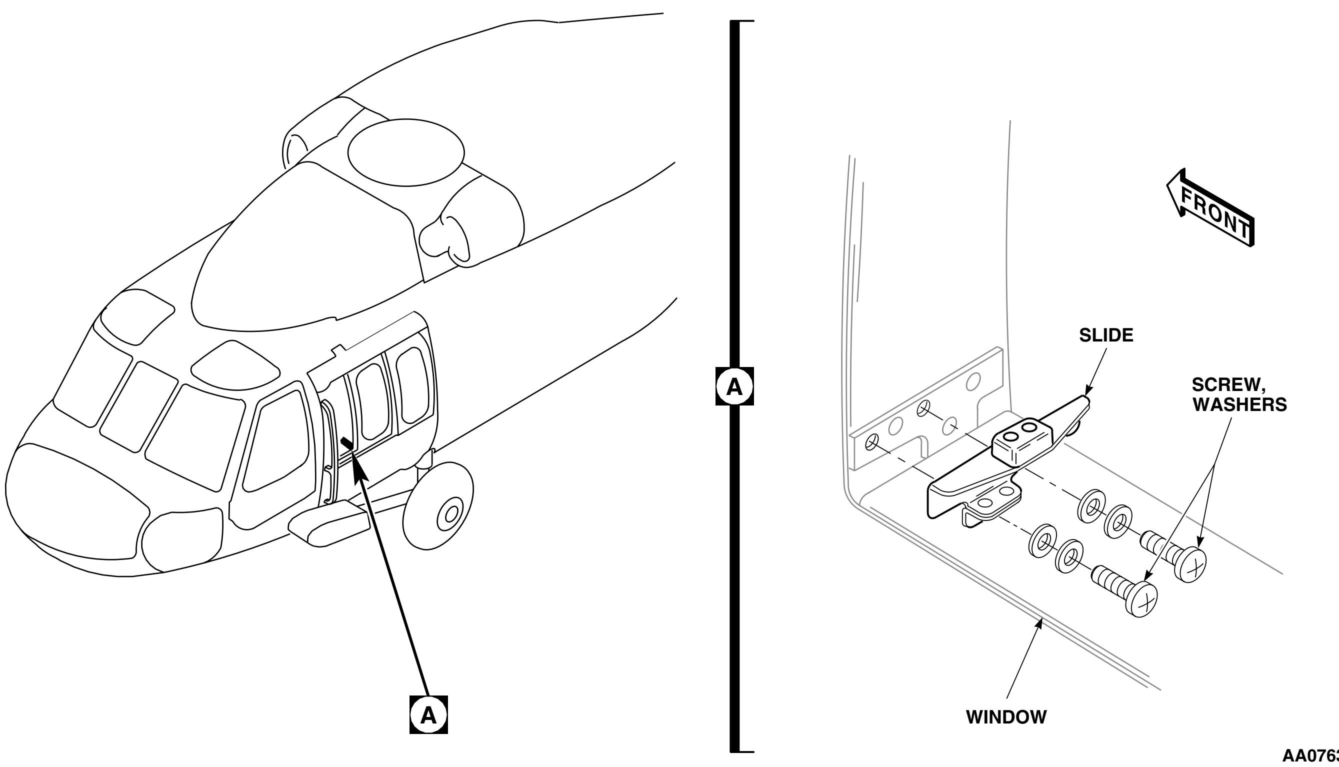


Figure 1. Replace Gunners Window Slide.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW SEALS

This WP supersedes WP 0240 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 40, WP 1803 00
Gunner's Window Brush Seal, Locally-Made, Figure 183, WP 1805 00
Petrolatum, Item 216, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0237 00
WP 0381 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove window(s) (WP 0237 00).

NOTE

A rubber seal bonded to a vertical angle bears against rear surface of front window forward seal with window closed. Remove and install this seal in same way as seals bonded to windows.

2. Remove damaged rear seal(s) from window using a nonmetallic scraper.
3. Remove front seal and retainer by drilling out rivets (Figure 1, Detail A).
4. **UH-60A 83-23837 - SUBQ UH-60L** Drill out rivets and remove clip (Figure 1, Detail B).
5. Remove damaged seal plug from lower window track using nonmetallic scraper.
6. Remove damaged brush seals from lower track and rear window by pushing out of retainer grooves, using nonmetallic scraper (Figure 1, Detail C).

REMOVE - Continued

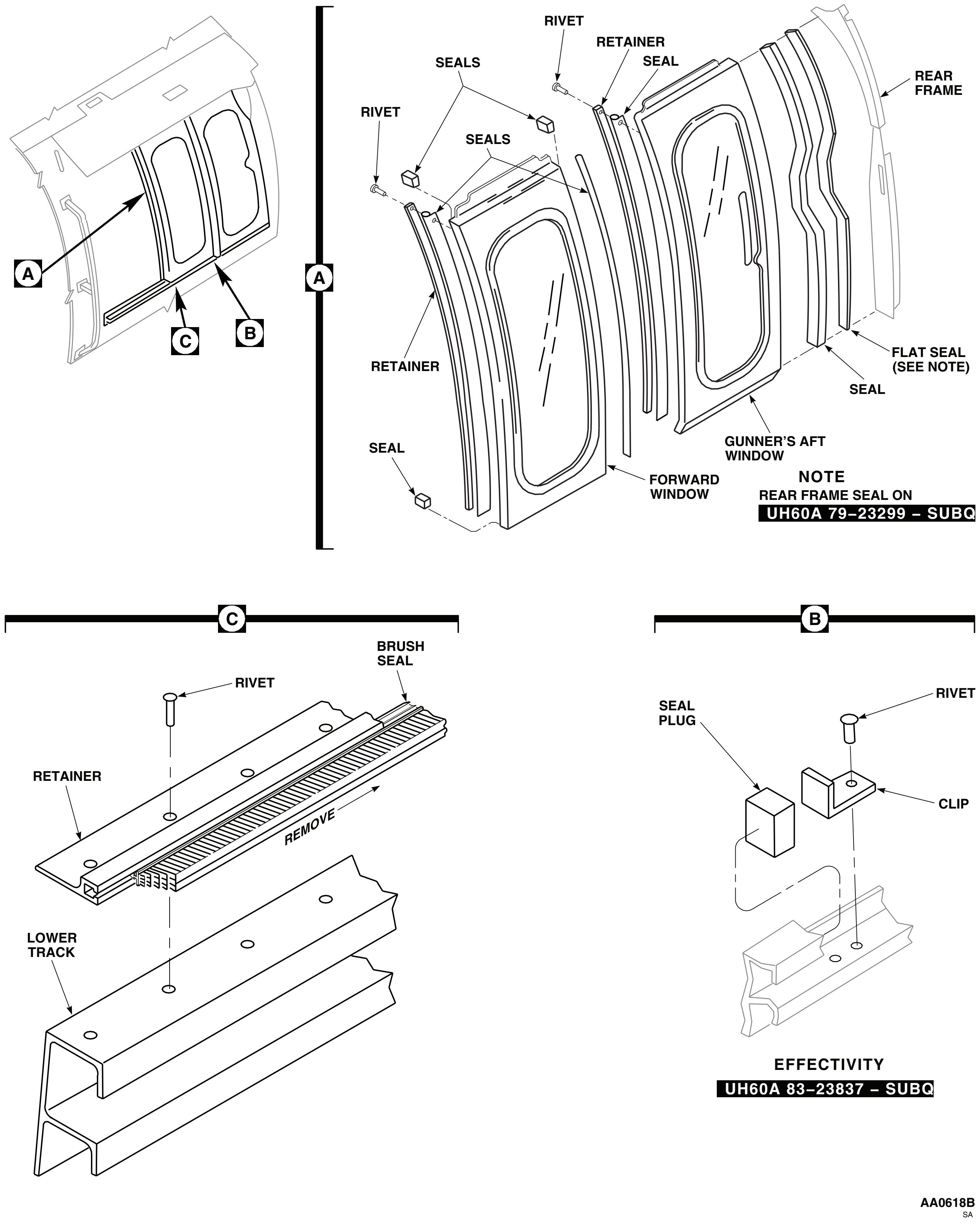


Figure 1. Replace Gunner's Window Seals.

INSTALL

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to plastic windows will result if technical acetone contacts them. Do not allow technical acetone to contact plastic windows.

1. Clean adhesive from window seal bonding surfaces with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper. Wipe dry with a clean machinery wiping towel.
2. Place replacement seal(s) on window, rear frame, and track (Figure 1, Detail A). Bond seals in position with adhesive, Item 14, WP 1803 00. Fill any gaps between seal and window with adhesive.
3. Bond vertical seal to angle, attached to rear of front door with adhesive, Item 40, WP 1803 00.
4. Bond vertical seals to angles attached to fuselage at STA 265.0, as follows:
   a. **UH-60A UH-60L 89-26149, 89-26154 UH-60L 89-26179 - 90-26232 EH-60A** Use adhesive, Item 40, WP 1803 00.
   b. **UH-60A 90-26233 - SUBQ** Use adhesive, Item 14, WP 1803 00.
5. Install front seal and retainer with rivets (Figure 1, Detail C).
6. **UH-60A 83-23837 - SUBQ UH-60L EH-60A** Install clip with rivets (Figure 1, Detail B).
7. Locally make gunner's window brush seal (WP 1805 00).
8. Slide replacement seal into retainer groove, using petrolatum, Item 216, WP 1803 00, for lubrication if required.
9. Install window(s) (WP 0237 00).
10. Spray water on window and check for leaks.

NOTE

All seals are to be free of primer and paint overspray.

11. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW UPPER TRACK SEALS

This WP supersedes WP 0241 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Gunner's Window Brush Seal, Locally-Made, Figure 183, WP 1805 00
Petrolatum, Item 216, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Washer, 70500-02056-102

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 1803 00
WP 1805 00

REMOVE

NOTE

Seals are replaced same way on both sides of helicopter.

1. Slide both windows fully open.
2. **UH-60A 77-22714 - 83-23842** Remove screws, washers, and nuts attaching seal assembly and rear gunners strap to airframe structure. Remove seal assembly (Figure 1, Detail A).
3. **UH-60A 83-23843 - SUBQ UH-60L EH-60A** Remove screws and washers attaching seal assembly and rear gunners strap to airframe structure. Remove seal assembly (Figure 1, Detail A).
4. Remove damaged brush seals from retainer assembly by pushing out of retainer grooves, using nonmetallic scraper (Figure 1, Detail D).
5. **UH-60A 85-24439 - SUBQ UH-60L EH-60A** Replace seal plug as follows:
   a. Remove seal plug from bracket using nonmetallic scraper (Figure 1, Detail C).

REMOVE - Continued

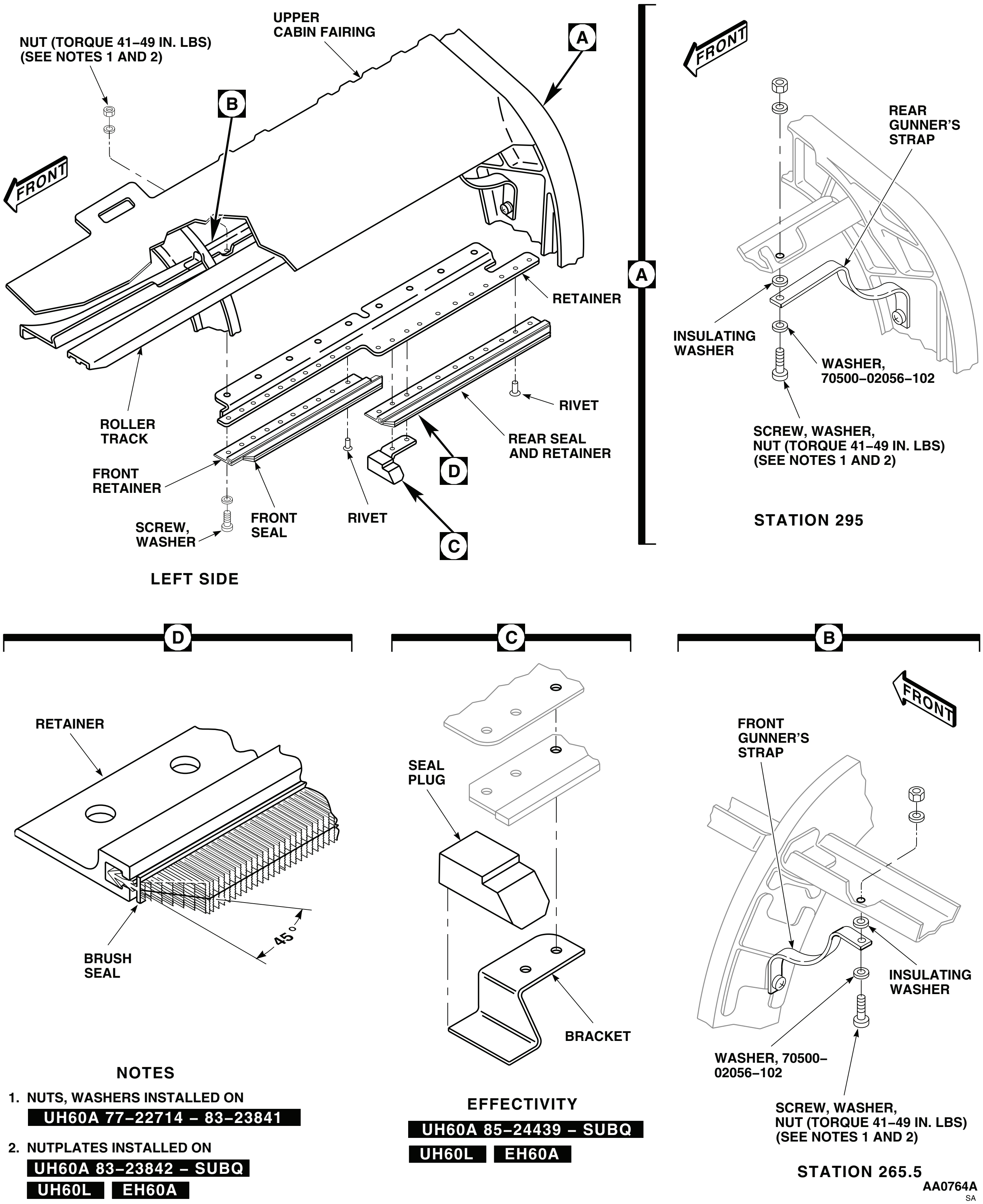


Figure 1. Replace Gunner's Window Upper Track Seals.

REMOVE - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to plastic windows will result if technical acetone contacts them. Do not allow technical acetone to contact plastic windows.

b. Remove adhesive residue with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper. Wipe dry with a clean machinery wiping towel.

INSTALL

1. Apply adhesive, Item 14, WP 1803 00, to mating surfaces of seal plug and bracket. Let dry 10 minutes.
2. Press seal firmly into position on bracket.
3. Trim seal plug to 0.10-inch minimum interference with rear end of front door.
4. Locally make gunner's window brush seal (WP 1805 00).
5. Slide replacement seal into retainer groove using petrolatum, Item 216, WP 1803 00, if required (Figure 1, Detail D).

NOTE

- Long screws are installed through gunner's straps.
- Large washer, 70500-02056-102 is installed with round edge against straps.
- Insulating washer is installed between strap and structure.

6. **UH-60A 77-22714 - 83-23842** Loosely attach seal assembly to helicopter structure with screws, washers, and nuts (Figure 1, Detail A). Make sure rear gunner's strap is attached to second screw from rear and front strap is attached to second screw from front.
7. **UH-60A 83-23843 - SUBQ UH-60L EH-60A** Loosely attach seal assembly to helicopter structure with screws and washers. Make sure rear gunner's strap is attached to second screw from rear and front strap is attached to second screw from front.
8. Slide windows closed, and adjust seals to make contact with windows. Tighten screws from inside cabin.
9. **UH-60A 85-24403 - SUBQ UH-60L EH-60A** Remove rubber seal plug between upper track and fuselage, near forward end of track.

INSTALL - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to plastic windows will result if technical acetone contacts them. Do not allow technical acetone to contact plastic windows.

10. Remove old seal with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper. Wipe dry with clean machinery wiping towel.
11. Coat contacting surfaces of new seal, track, and fuselage with adhesive, Item 14, WP 1803 00. Let dry for about 10 minutes and press seal firmly into place.
12. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW UPPER AND LOWER ROLLERS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Spring Scale, AAA-S-133

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0237 00

REPLACE ROLLER ASSEMBLY

Remove

NOTE

Replacement of all roller assemblies is the same.

1. If necessary, trim angle on upper seal to provide additional clearance for roller bracket.
2. Remove window(s) (WP 0237 00).
3. On helicopters with windows, 70216-02401-047, 70216-02401-048, 70216-02401-051, and 70216-02401-052; 70216-02402-043, 70216-02402-044, and 70216-02402-047 through 70216-02402-050, remove screws, washers, and nuts attaching roller to window(s). Remove roller (Figure 1).
4. On helicopters with window, 70216-02402-041, 70216-02402-042 through 70216-02402-044, remove screws attaching roller to window. Remove roller.

Install

1. **QA** On helicopters with windows, 70216-02401-047, 70216-02401-048, 70216-02401-051, and 70216-02401-052; 70216-02402-043, 70216-02402-044, and 70216-02402-047 through 70216-02402-050, place replacement roller on window. Fasten roller to window with screws, washers, and nuts (Figure 1).
2. **QA** On helicopters with window, 70216-02402-041, 70216-02402-042 through 70216-02402-044, place replacement roller on window. Fasten roller to window with screws.
3. Install window(s) (WP 0237 00).
4. Slide window to front and rear, and check for correct operation.
5. Using spring scale, close window. **MAXIMUM FORCE TO CLOSE WINDOW NOT TO EXCEED 15 LBS.**
6. **QA** On helicopters with windows, 70216-02401-043 through 70216-02401-048, 70216-02401-051, and 70216-02401-052, adjust window(s) as necessary. To do this, loosen roller screws and nuts, move window in or out, and tighten screws and nuts.

REPLACE ROLLER ASSEMBLY - Continued

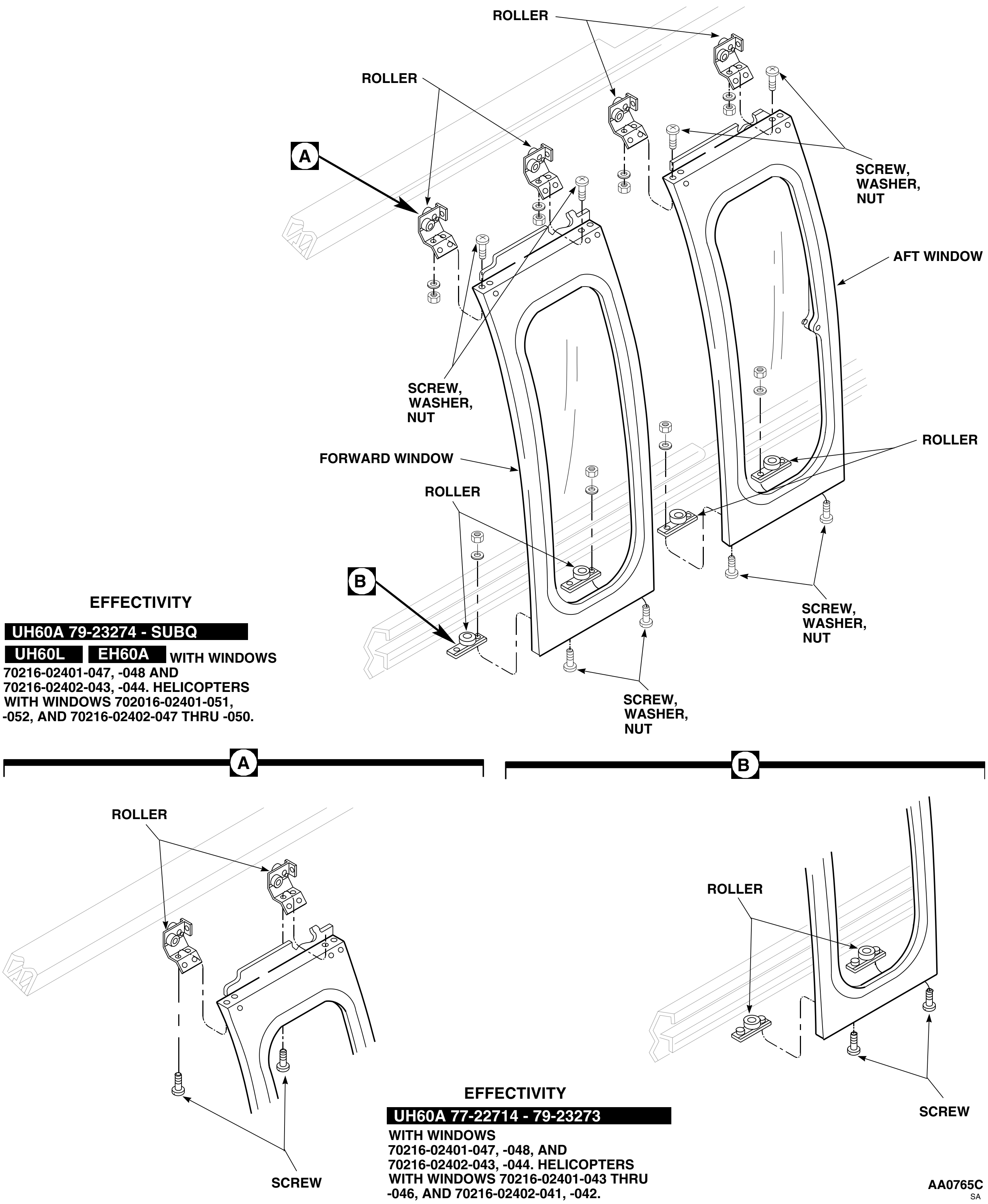


Figure 1. Replace Upper/Lower Rollers.

REPLACE LOWER ROLLERS

1. On helicopters with windows, 70216-02401-043 through 70216-02401-048, 70216-02401-051, 70216-02401-052; 70216-02402-049 and 70216-02402-050, remove lower roller assembly from window. Refer to, REPLACE ROLLER ASSEMBLY, in this work package.
2. Remove pin-rivet and collar from lower roller assembly (TM 1-1500-204-23). Remove roller and washer (Figure 2, Sheet 1).
3. On helicopters with windows 70216-02402-049, and 70216-02402-050, remove rivet retaining nutplate to plate, if required (TM 1-1500-204-23).
4. Install roller, washer, to plate with pin-rivet and collar (TM 1-1500-204-23).
5. On helicopters with windows 70216-02402-049, and 70216-02402-050, install nutplates with rivets (TM 1-1500-204-23).
6. Install roller assemblies on windows. Refer to, REPLACE ROLLER ASSEMBLY, in this work package.

REPLACE UPPER ROLLERS

1. Remove upper roller assembly from window. Refer to, REPLACE ROLLER ASSEMBLY, in this work package.
2. On roller assemblies, 70216-02407-045 through 70216-02407-050 and 70216-02407-054, remove pin-rivets and collars retaining rollers washer, and spacer to collar support (TM 1-1500-204-23) (Figure 2, Sheet 2).
3. On roller assemblies, 70216-02407-052 and 70216-02407-053, remove pin-rivets and collars retaining rollers, washers and tee to roller support (TM 1-1500-204-23) (Figure 2, Sheet 2).
4. Remove rivets retaining seal and plate to tee if required (TM 1-1500-204-23).

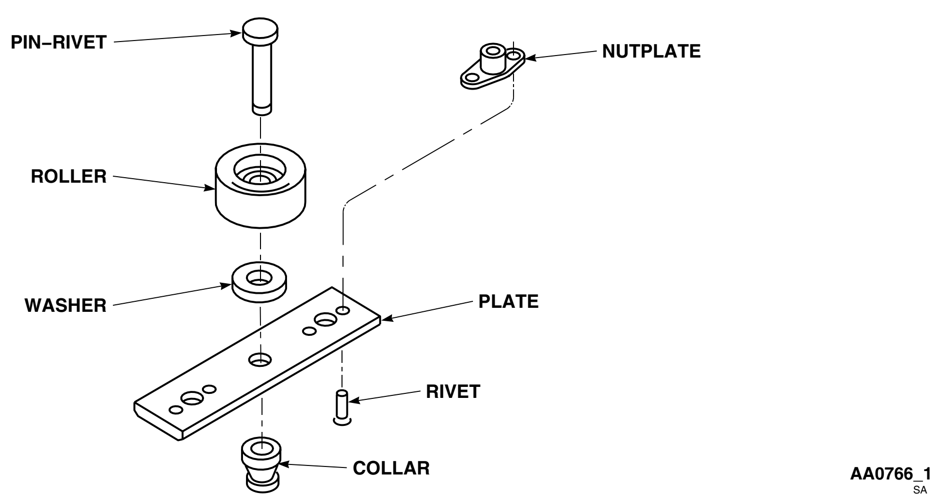


Figure 2. Repair Rollers. (Sheet 1 of 2)

REPLACE UPPER ROLLERS - Continued

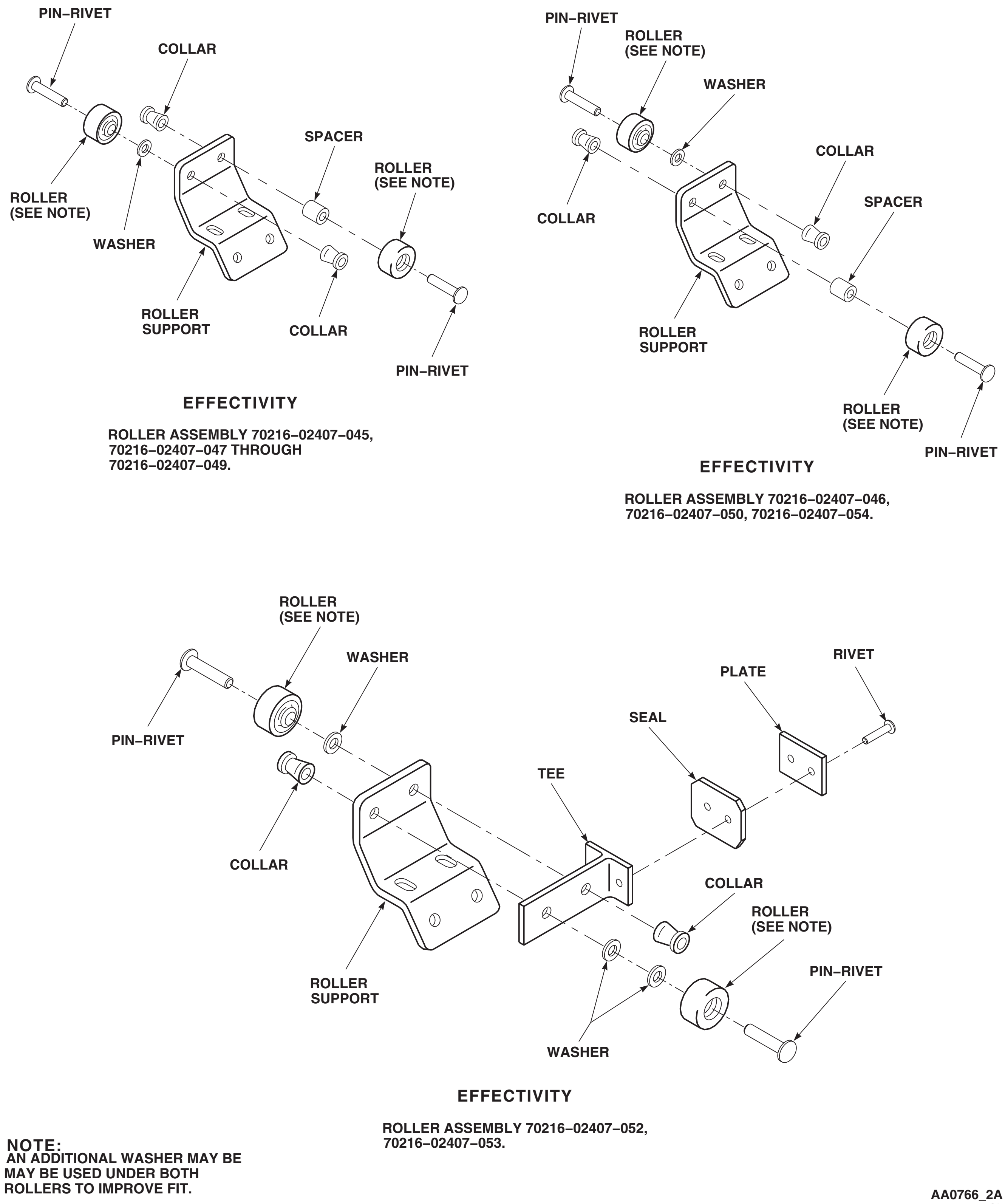


Figure 2. Repair Rollers. (Sheet 2 of 2)

REPLACE UPPER ROLLERS - Continued

NOTE

On roller assemblies, 70216-02407-045 through 70216-02407-050 and 70216-02407-054, an additional washer, may be used to improve fit of roller.

5. On roller assemblies, 70216-02407-045 through70216-02407-050 and 70216-02407-054, install rollers, spacer, washer, additional washer if required, to roller support with pin-rivet and collar (TM 1-1500-204-23) (Figure 2, Sheet 2). **MAXIMUM DISTANCE BETWEEN TOP OF ROLLERS SHALL NOT BE MORE THAN 1.08-INCHES.**
6. On roller assemblies, 70216-02407-052 and 70216-02407-053, install seal and plate to tee with rivets, if required (Figure 2, Sheet 2).

NOTE

On roller assemblies, 70216-02407-052 and 70216-02407-053, an additional washer may be used to improve fit of roller.

7. On roller assemblies, 70216-02407-052 and 70216-02407-053, install rollers, washers, and tee to roller support with pin-rivet and collar (TM 1-1500-204-23). **MAXIMUM DISTANCE BETWEEN TOP OF ROLLERS SHALL NOT BE MORE THAN 1.09-INCHES.**
8. Install roller assemblies on windows. Refer to, REPLACE ROLLER ASSEMBLY, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

GUNNER'S WINDOW SECURITY DEVICE

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0288 00

NOTE

UH60A 84-23938 - SUBQ UH60L EH60A Replacement of right and left security devices is the same.

REMOVE

1. Remove landing gear upper fairing (WP 0288 00).
2. Open rear gunner's window.
3. Remove screw, washers, and nut attaching cable assembly to frame (Figure 1, Detail A).
4. Remove cable assembly.
5. Remove nuts, washers, and pin from frame.

INSTALL

1. Install nut and washer on replacement pin and install pin into frame with remaining nut and washer (Figure 1, Detail A).
2. Attach cable assembly to frame with screw, washers, and nut.
3. Install landing gear upper fairing (WP 0288 00).
4. Close rear gunner's window.

INSTALL - Continued

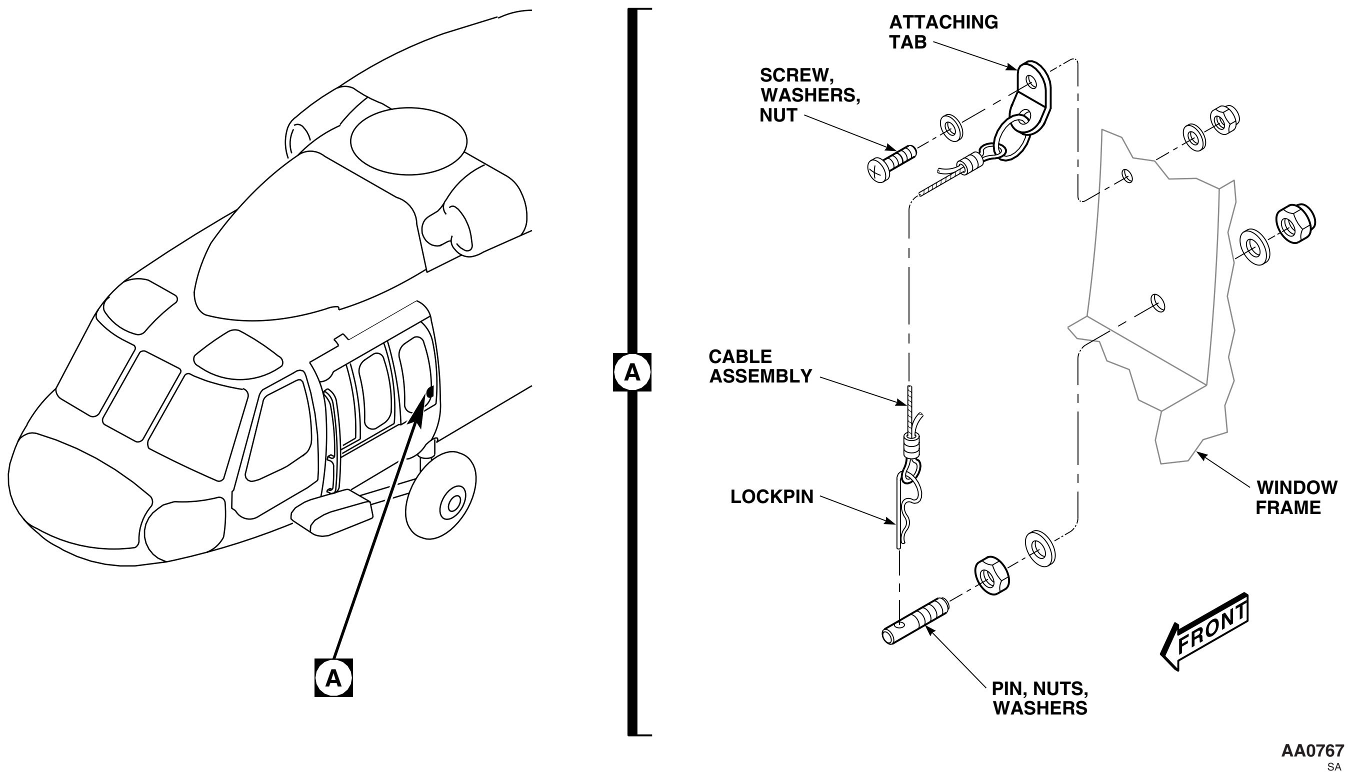


Figure 1. Replace Gunner's Window Security Device.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Molybdenum Disulfide, Item 197, WP 1803 00
Petrolatum, Item 216, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0179 00
WP 0247 00
WP 0255 00
WP 0495 00
WP 1803 00

REMOVE

1. Remove stop(s) from upper track by removing bolt(s), washers, and nuts (Figure 1, Detail A).
2. Remove stop from lower track by removing screw (Figure 1, Detail C).
3. HIRSS Remove HIRSS extender outboard strut (WP 0495 00).
4. Open door. As door reaches full open, manually keep open stop from engaging by holding door stop open.
5. With aid of assistant, slide door rearward, out of upper and lower tracks, and remove it from helicopter.
6. If door is being replaced, remove door lock (WP 0255 00).

INSPECT

Inspect upper and lower roller support assemblies for wear. If wear is found. Refer to WP 0245 00.

INSTALL

1. On a replacement door, install door lock removed from old door (WP 0255 00).

CAUTION

Damage to latching mechanism may result if door is closed with handle in the locked position.

NOTE

Up to 35 pounds of pressure may be needed to roll door forward on tracks.

2. Lightly apply molybdenum disulfide, Item 197, WP 1803 00, to upper and lower tracks to ease installation. Apply molybdenum disulfide only to surface of tracks where rollers on door make contact.
3. Apply a thin coat of petrolatum, Item 216, WP 1803 00, to upper and lower track seals.

INSTALL - Continued

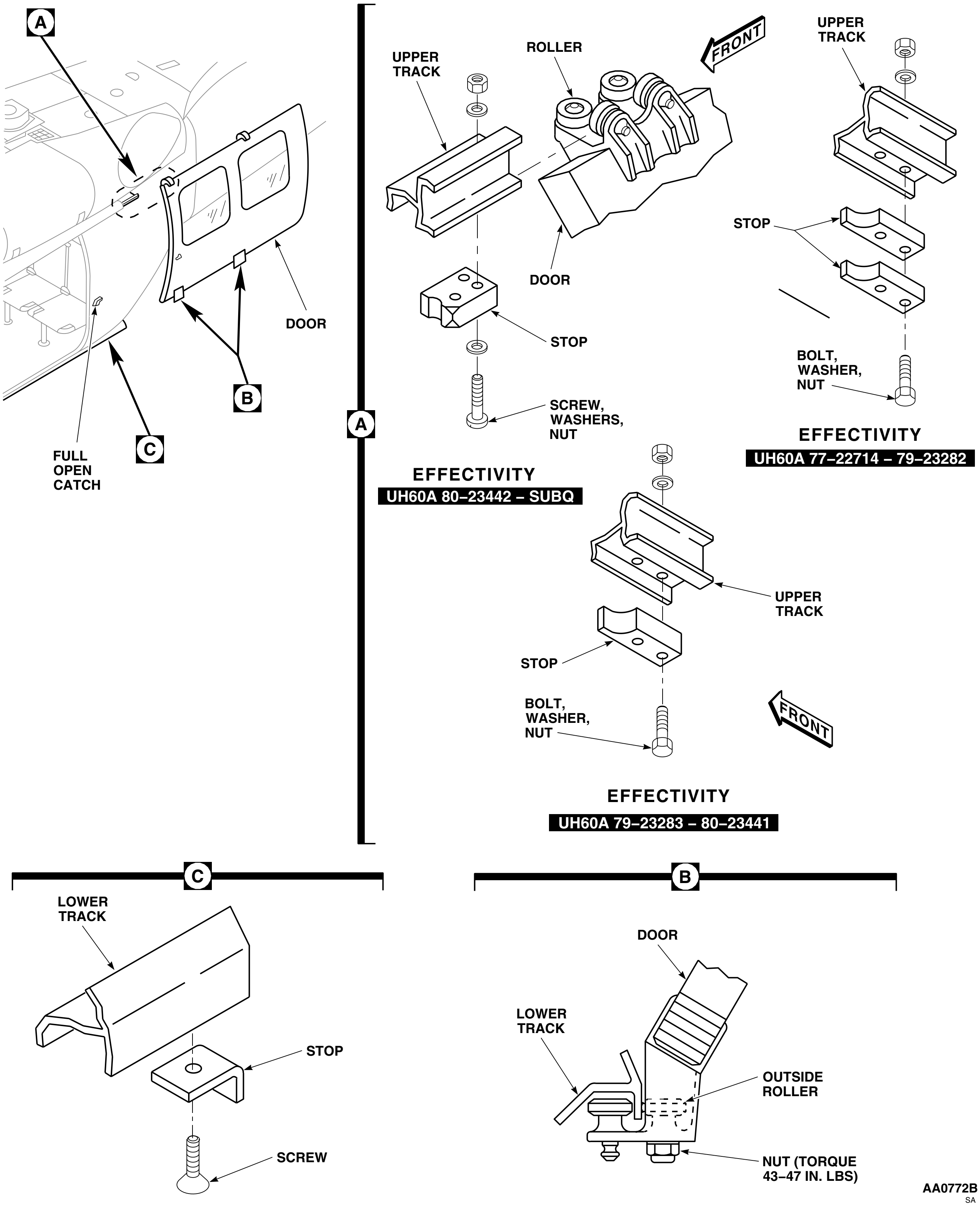


Figure 1. Replace Troop/Cargo Door.

INSTALL - Continued

4. With aid of assistant, line up front rollers on door with upper and lower tracks. Guide front seal flap into inboard bottom angle of upper track. Guide rear seal flap under inboard bottom edge of upper track. Open door latch and carefully slide door forward until all rollers are in tracks, and close door.

NOTE

Thick stop replaces thin stops. Do not interchange stops.

5. **QA** Install stop(s) in upper track and secure with bolt(s), washers, and nuts (Figure 1, Detail A).
6. **QA** Install stop on lower track, with stop leg toward rear, and secure with screw (Figure 1, Detail C).
7. Open and close door a few times to check operation.
8. Perform cargo door window jettison check (WP 0177 00).
9. **QA** If necessary, adjust outside roller in lower roller assemblies by loosening nut and moving roller against track to prevent door rattle. Move roller away from track to prevent binding. **TORQUE NUT TO 43 - 47 INCH-POUNDS (Figure 1, Detail B).**
10. **HIRSS** Install HIRSS extender outboard strut (WP 0495 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR WINDOW/VENT

This WP supersedes WP 0245 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00
Wire, Nonelectrical, Item 351, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TB 1-1520-237-20-273
TM 55-1500-345-23
WP 0179 00
WP 0381 00
WP 1803 00

REMOVE WINDOW

1. Close troop/cargo door.

CAUTION

Windows will be damaged from jettison if emergency release lever is turned. Do not turn emergency release lever until instructed to.

2. If only one window need be removed, secure other window in place with pressure sensitive adhesive tape, Item 334, WP 1803 00.
3. To gain access to emergency release lever, pull guard from cover (if installed) or pull handle strap.
4. With an assistant supporting window(s) from outside, turn emergency release lever in door to OPEN POSITION.
5. Push out on bottom of window and remove from helicopter (Figure 1, Detail A).

INSTALL WINDOW

1. From outside helicopter, have assistant lift window(s) into position.
2. Line up tabs at top of window with slots in door frame (Figure 1, Detail A).
3. Have assistant push in at bottom of window from outside helicopter.
4. Turn emergency release lever to LOCKED POSITION.
5. Perform cargo door window jettison check (WP 0179 00).
6. **QA** Using nonelectrical wire, Item 351, WP 1803 00, lockwire jettison handle.
7. **QA** If guard was removed, install it on cover by pressing on pile and hook fastener tape or position handle strap over emergency release lever.
8. If necessary, identify window as emergency exit (WP 0381 00 and TM 55-1500-345-23).

INSTALL WINDOW - Continued

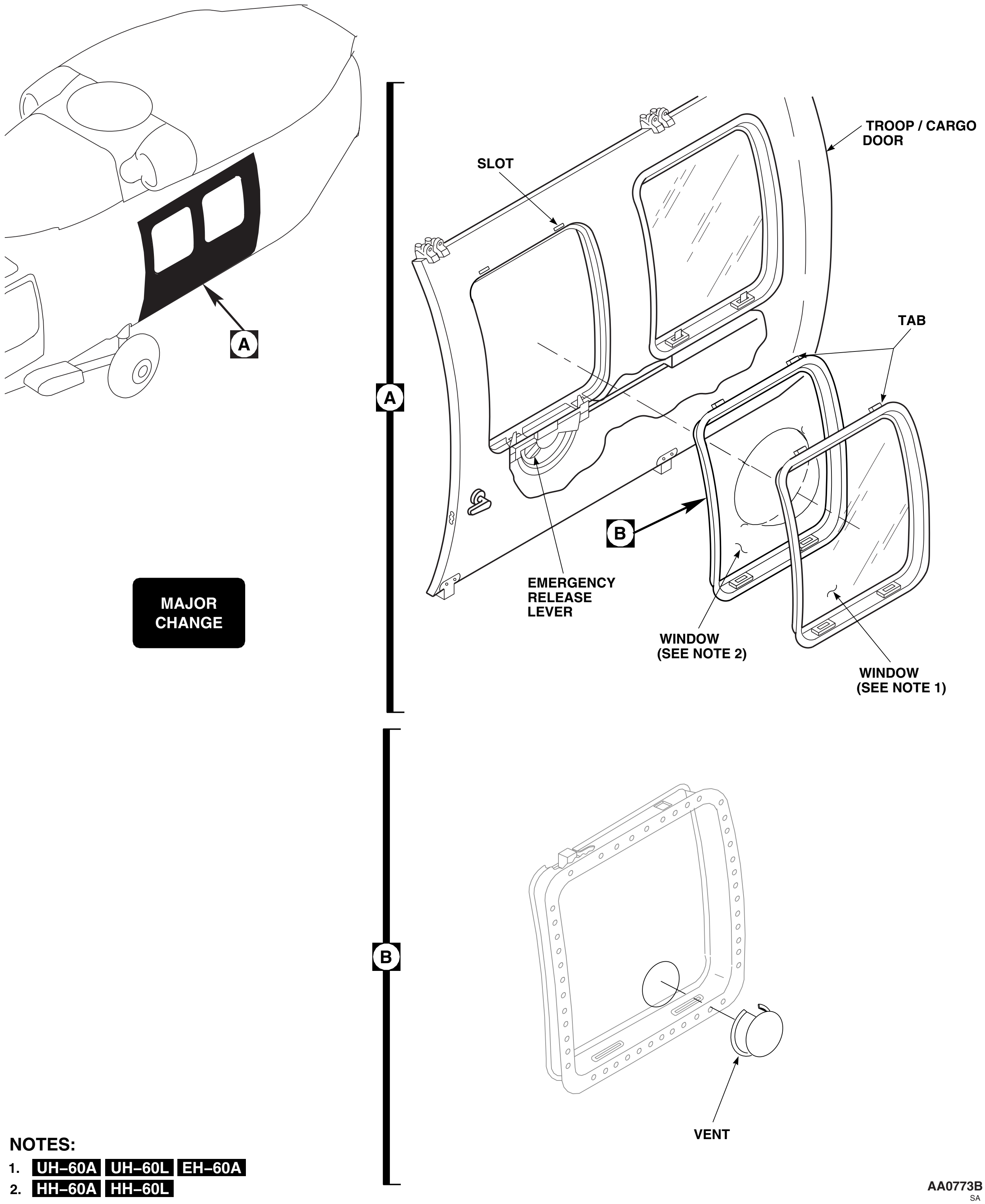


Figure 1. Replace Troop/Cargo Door Window.

INSTALL WINDOW - Continued

9. If necessary, remove masking tape from window.

REMOVE VENT

NOTE

This vent is an option installed per TB 1–1520–237–20–273.

1. Remove metal clip.
2. Open Vent.
3. Compress forward inboard portion of vent while gently pulling towards aft (Figure 1, Detail B).

INSTALL VENT

NOTE

Applying a non-petroleum lubrication to the outside mating surface may assist in installation.

1. Place aft portion of vent into window pane vent hole.
2. Gently compress forward portion of vent and press completely into window pane (Figure 1, Detail B).
3. Close vent.
4. Install metal clip.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR LOWER TRACK

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0244 00
WP 0287 00
WP 0381 00
WP 1803 00

REMOVE

1. Remove troop/cargo door (WP 0244 00).
2. Remove main landing gear lower fairing (WP 0287 00).
3. Remove screws and washers attaching lower track to fairing (Figure1, Detail A).
4. Remove screws and washers attaching lower track to fuselage. Remove spacers.
5. Remove lower track.

INSTALL

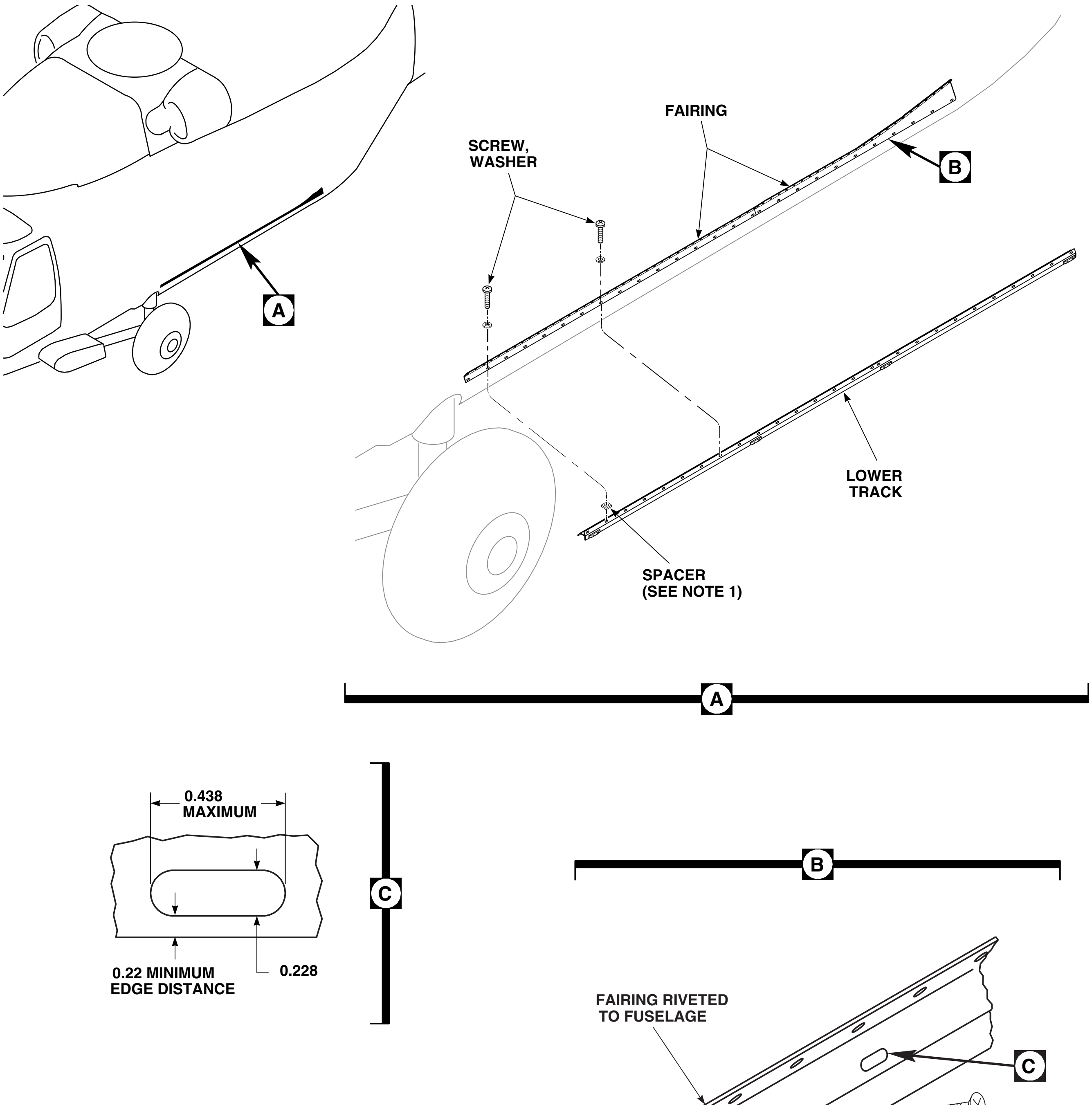
1. Place replacement lower track on fuselage.
2. Check alignment of fairing holes with lower track nutplates. If any holes do not line up, do this:
   a. Mark fairing holes that do not line up.
   b. Slot marked holes in fairing to dimensions shown in Figure1, Detail C. Slot may be made in any direction provided minimum edge distance is maintained.
   c. Touch up paint in slotted area (WP 0381 00 and TM 55-1500-345-23).
3. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.

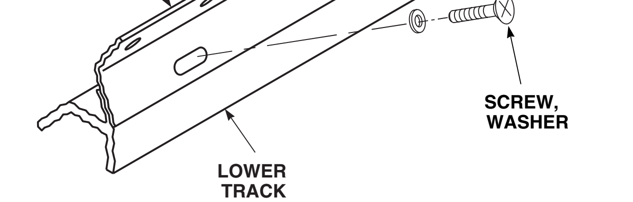
NOTE

Track spacers vary in size. Correct spacer location is given in Figure1.

4. **QA** Place spacers between track and fuselage and install screws and washers (Figure1, Detail A).
5. Install main landing gear lower fairing (WP 0287 00).
6. Install troop/cargo door (WP 0244 00).

INSTALL - Continued





NOTES:

1. SPACER

LOCATION	SIZE
STA 308.0	0.125 BY 1.00 BY 2.00
STA 325.0	0.125 BY 1.00 BY 1.60
STA 343.0	0.125 BY 1.00 BY 1.60
STA 353.0	0.125 BY 1.00 BY 0.75
STA 363.0	0.125 BY 1.00 BY 1.60
STA 379.0	0.070 BY 1.00 BY 2.13
STA 398.0	0.125 BY 1.00 BY 1.31

2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB1601A
SA

Figure 1. Replace Troop/Cargo Lower Track.

INSTALL - Continued

7. Open and close door a few times to check operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR SEALS

This WP supersedes WP 0247 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Riveter, Blind, G-740-LM

Material/Parts
Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 10, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 40, WP 1803 00
Cargo Door Seal, Locally-Made, Figure 76, WP 1805 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 55-1500-345-23
WP 0244 00
WP 0245 00
WP 0377 00
WP 0381 00
WP 1803 00
WP 1805 00

REPLACE TROOP/CARGO WINDOW SEALS

Remove

1. Remove window(s) from troop/cargo door (WP 0245 00).
2. Remove damaged bulb seal from flange of window opening in door, using nonmetallic scraper (Figure 1).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean remaining adhesive from window opening flange using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

NOTE

UH-60A 77-22714 - 83-23900 Have a seal on the window flange at top of window only.
UH-60A 83-23901 - SUBQ Have additional, seals for front and rear window flanges.

4. Remove damaged seal(s) from window flange using a nonmetallic scraper.

REPLACE TROOP/CARGO WINDOW SEALS - Continued

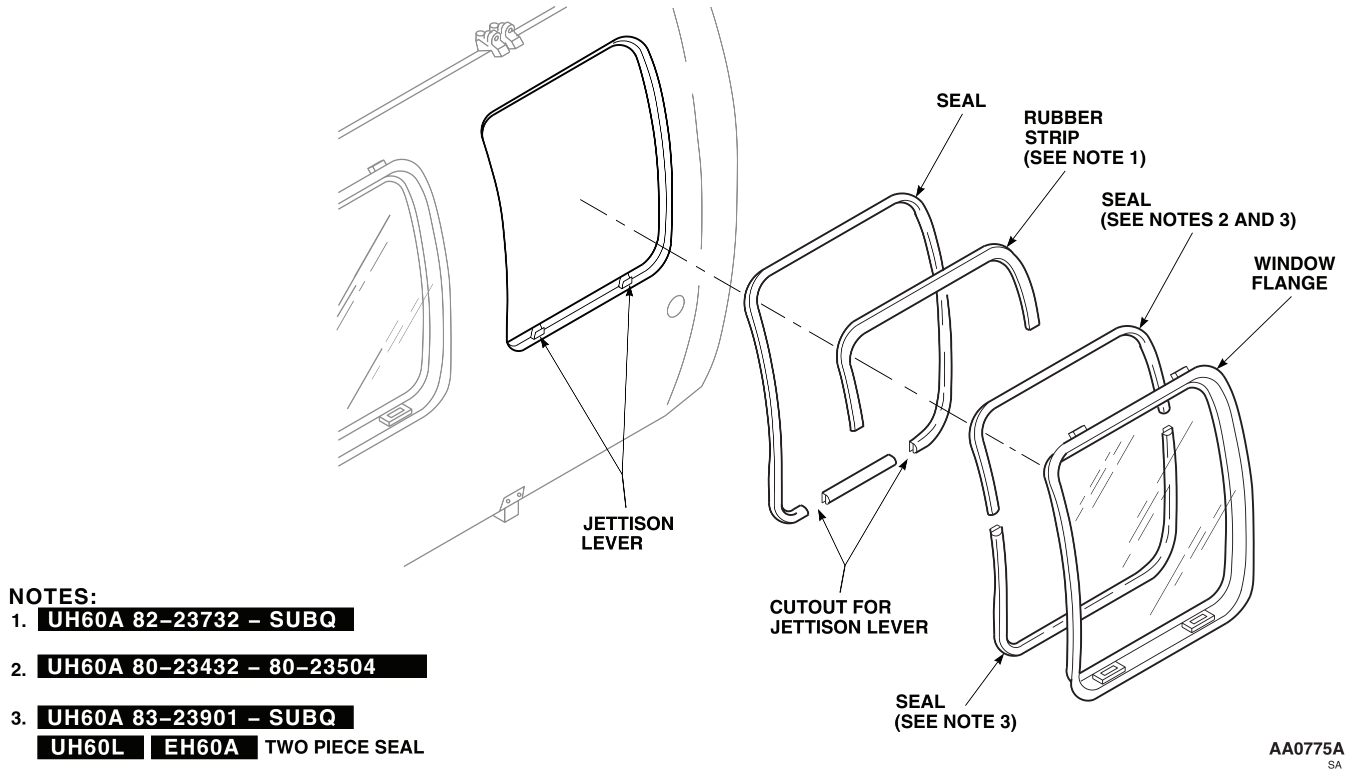


Figure 1. Replace Window Seals.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean remaining adhesive from window flange using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

Install

1. Place replacement seal on flange inside window opening in door, bulb of seal facing outboard. Trim length to fit opening (Figure 1).
2. Mark location of window jettison levers and cut seal to fit. Remove seal from door.
3. Put on a thin even coat of adhesive, Item 14, WP 1803 00, to window opening flange and mating surfaces of replacement seal (TM 1-1500-204-23).
4. Starting at bottom of window frame, install bulb seal on flange. Do not stretch seal around curved parts of frame.
5. UH-60A 77-22714 - 83-23900 Place replacement seal around top of window flange and cut seal to size of old seal.

REPLACE TROOP/CARGO WINDOW SEALS - Continued

WARNING

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6. Lightly roughen mating surface of seal using abrasive cloth assortment, Item 1, WP 1803 00. Clean roughened surface of seal using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
7. Apply a light coat of adhesive, Item 10, WP 1803 00, on window flange and roughened surface of seal. Allow adhesive to dry until tacky.
8. Install seal on window flange. Do not stretch seal around corners of window. Do not let seal lose contact with window flange.
9. Allow seal installation to cure at room temperature for 24 hours.

WARNING

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10. **UH-60A 83-23901 - SUBQ UH-60L EH-60A** Lightly roughen mating surface of replacement seal(s) using abrasive cloth assortment, Item 1, WP 1803 00. Clean roughened surface of seal using a clean cloth dampened with technical acetone, Item 5, WP 1803 00.
11. Apply a light coat of adhesive, Item 10, WP 1803 00, to roughened surface of seal(s) and window flange. Allow to air dry until tacky.
12. Install seal(s) on window flange. Do not stretch top seal around corners of window flange. Do not let seal(s) lose contact with window flange.
13. Allow seal installation to cure at room temperature for 24 hours.
14. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
15. Install window(s) in troop/cargo door (WP 0245 00).
16. Spray water on window and check for leaks.

REPLACE TROOP/CARGO DOOR FRONT SEALS

Remove

1. Cut away old bulb seals to seal retainers (Figure 2, Sheet 2, Detail F).
2. Remove retainers by drilling out rivets (TM 1-1500-204-23).
3. Pull damaged seal(s) from door frame.

REPLACE TROOP/CARGO DOOR FRONT SEALS - Continued

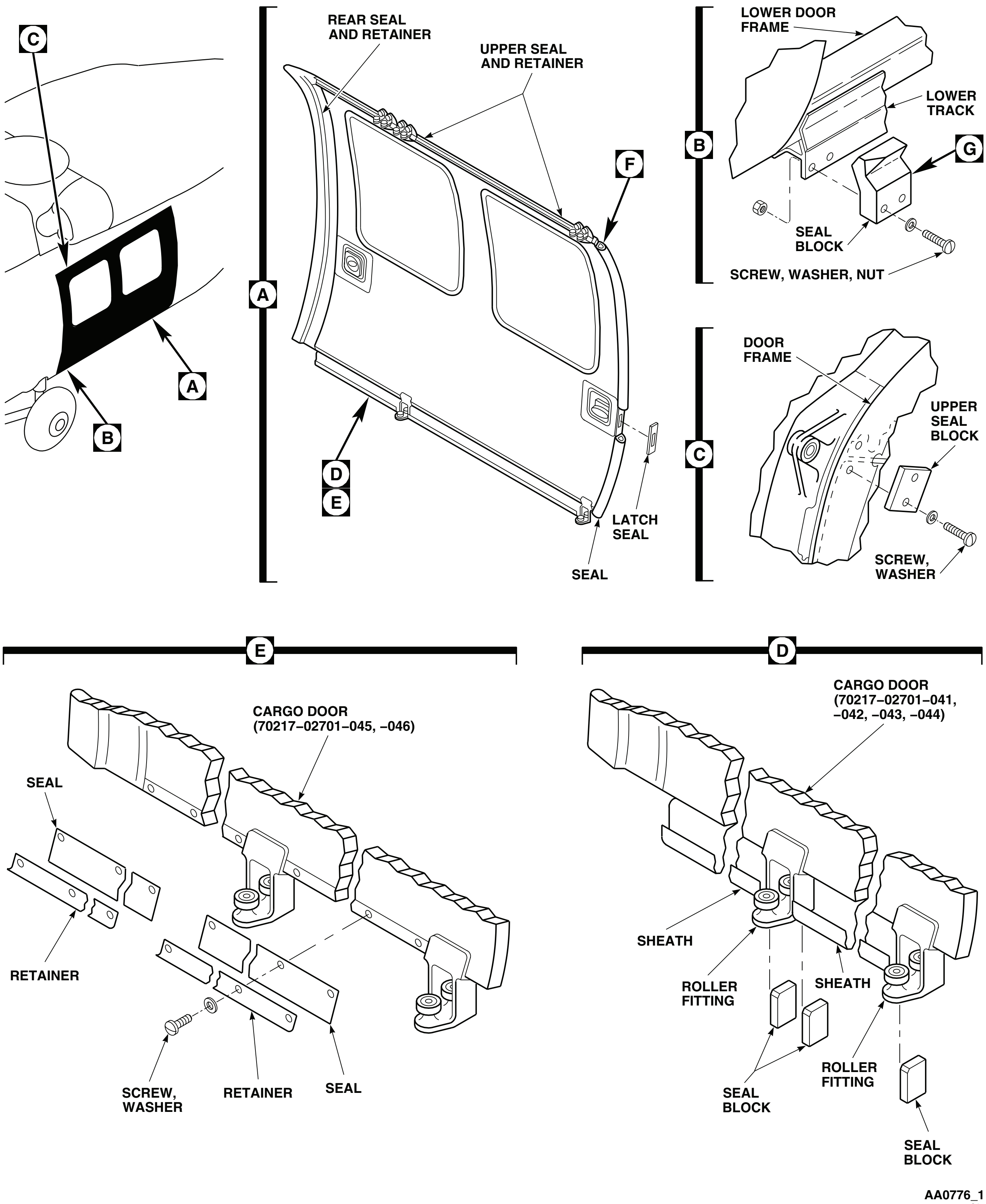

Figure 2. Replace Front and Lower Seals. (Sheet 1 of 2)

REPLACE TROOP/CARGO DOOR FRONT SEALS - Continued

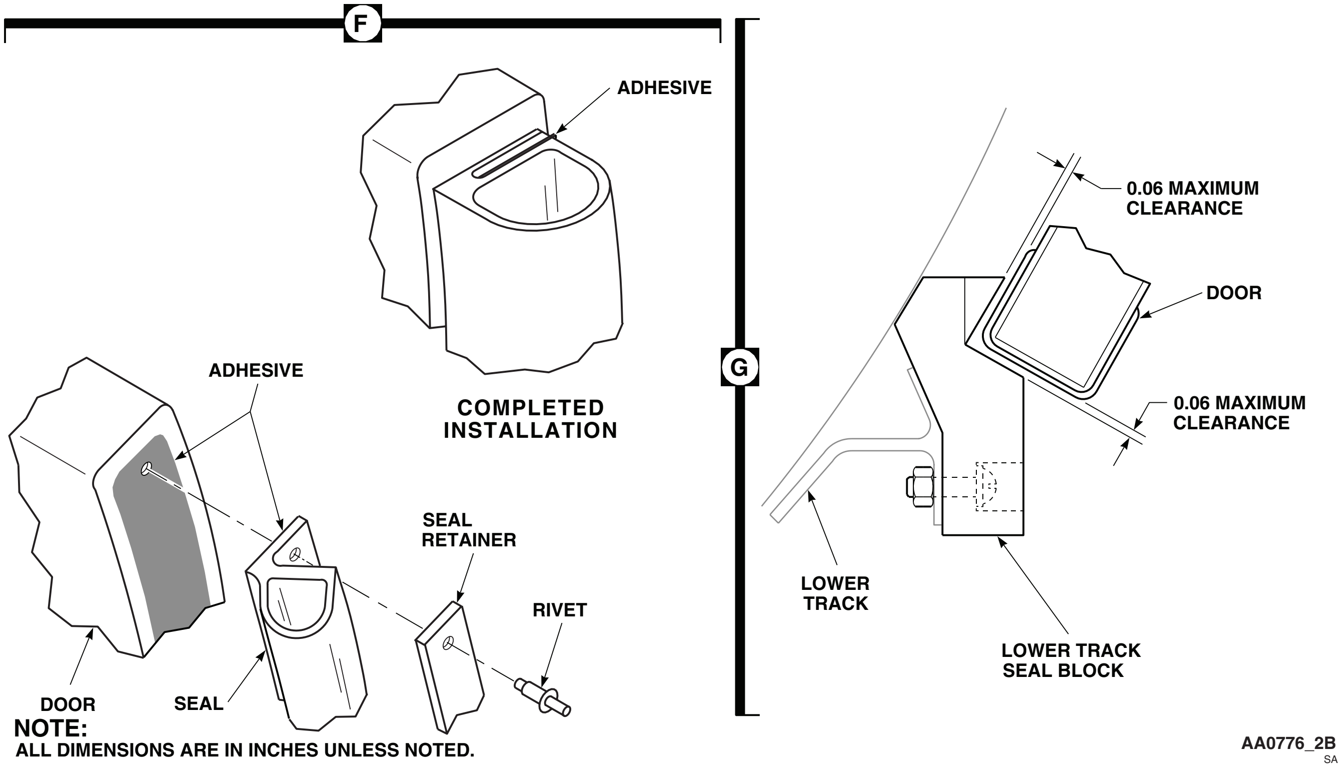

Figure 2. Replace Front and Lower Seals. (Sheet 2 of 2)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Remove seal and adhesive debris with nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

Install

1. Place replacement bulb seal(s) on door frame and trim to fit (Figure 2, Sheet 1, Detail A). Trim latch seal to fit, if necessary.
2. On helicopters with doors, 70217-02701-043 through 70217-02701-046, do this:
   a. Attach seal and retainer to door using clecos. Drill rivet holes and countersink as required.
   b. Remove seal and retainer. Deburr as required.

REPLACE TROOP/CARGO DOOR FRONT SEALS - Continued

WARNING

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c. Wipe door, seal, and retainer clean with cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

3. Put a thin even coat of adhesive, Item 10, WP 1803 00, or adhesive, Item 40, WP 1803 00, on door frame and mating surfaces of seals.
4. On helicopters with doors, 70217-02701-043 through 70217-02701-046, hold seal retainer and seal assemblies together with cleco fasteners.
5. Line up latch seal with latch opening and press seal in place.
6. Lineup and press bulb seal on door frame starting at latch seal and moving to end of door. Do not stretch seal while installing it. Wipe excess sealant off installation.
7. On helicopters with doors, 70217-02701-043 through 70217-02701-046, install standard or oversize rivets (Figure 2, Sheet 2, Detail F) (TM 1-1500-204-23).
8. Apply an even coat of adhesive, Item 40, WP 1803 00, over seal retainer and seal joint face.
9. Close and press seal over retainer.

WARNING

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10. Remove excess adhesive using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
11. Close cargo door and latch.

REPLACE TROOP/CARGO UPPER DOOR FRAME SEAL BLOCK

Remove

1. At fuselage upper door frame, remove screws and washers, and remove upper door frame seal block from door frame (Figure 2, Sheet 1, Detail C).
2. Scrape remaining adhesive and parts of seal block from frame using a nonmetallic scraper.

REPLACE TROOP/CARGO UPPER DOOR FRAME SEAL BLOCK - Continued

WARNING

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3. Using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, clean remaining adhesive from door frame. Wipe dry.

Install

1. Put a thin even coat of adhesive, Item 14, WP 1803 00, on mating surface of door frames and largest flat area of seal blocks (TM 1-1500-204-23).
2. Face angled side of seal block outboard.
3. Press seal block on door frame and secure with screws and washers (Figure 2, Sheet 1, Detail C).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, clean extra adhesive from door frame.

REPLACE TROOP/CARGO DOOR BOTTOM SEAL OR SEAL BLOCKS/SHEATH SEAL

Remove

1. On doors, 70217-02701-045 and 70217-02701-046, remove bottom seal as follows:
   a. Remove door (WP 0244 00).
   b. Remove screws, washers, retainer, and seal from inside edge of door (Figure 2, Sheet 1, Detail E).
2. On doors, 70217-02701-041 through 70217-02701-044, remove bottom seal blocks as follows:
   a. Pull off damaged seal block(s) (Figure 2, Sheet 1, Detail D).
   b. Scrape off remaining parts of seal block, using nonmetallic scraper.

WARNING

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   c. Using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, clean remaining adhesive from cargo door and roller fittings.
3. On doors 70217-02701-041 through 70217-02701-044, remove sheath seal as follows:

REPLACE TROOP/CARGO DOOR BOTTOM SEAL OR SEAL BLOCKS/SHEATH SEAL - Continued

a. Remove fasteners holding sheath seal section(s). Remove old seal (Figure 2, Sheet 1, Detail D).

WARNING

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b. Remove old sealant, as required using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper.

Install

1. On doors, 70217-02701-045 and 70217-02701-046, install bottom seal as follows:

   a. Locally-make cargo door seal (Figure 76, WP 1805 00).

   b. Place replacement seal on inside edge of door.

   c. Place retainer on seal, with flange toward bottom of door, and install screws, washers (Figure 2, Sheet 1, Detail E).

   d. Apply a bead of sealing compound, Item 273, WP 1803 00, along outside edge of door where it touches seal.

   e. Install door (WP 0244 00).

2. On doors, 70217-02701-041 through 70217-02701-044, install bottom seal blocks as follows:

   a. Put a thin even coat of adhesive, Item 14, WP 1803 00, on mating area of seal blocks and door and roller fittings (TM 1-1500-204-23).

   b. Press seal block on door and roller fitting with chamfered edge toward inside corner of sheath seal (Figure 2, Sheet 1, Detail D).

3. On doors, 70217-02701-041 through 70217-02701-044, install sheath seal as follows:

   a. Line up holes in new sheath seal with existing holes in door. Use hole holders to hold in position. Drill holes as required. Remove sheath seal.

WARNING

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   b. Clean door and seal joining surfaces using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

   c. Apply a thin even coat of sealing compound, Item 273, WP 1803 00, to joining surfaces of door and sheath seal.

   d. Use clecos to hold seal in place on door. Install rivets using blind riveter per TM 1-1500-204-23.

REPLACE TROOP/CARGO LOWER TRACK SEAL BLOCK

Remove

1. Remove screws, washers, nuts, and seal block from lower track (Figure 2, Sheet 1, Detail B).
2. Scrape remaining sealing compound from fuselage and lower track with nonmetallic scraper.

WARNING

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3. Using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, clean remaining sealing compound from lower track. Wipe dry.

Install

1. Place replacement seal on lower track. Face angled side of seal block outboard and secure with screws, washers, and nuts (Figure 2, Sheet 1, Detail B).
2. Close door over seal, measure, and if necessary, trim seal to allow no more than 0.060-inch clearance between door and seal.
3. Apply bead of sealing compound, Item 273, WP 1803 00, along seal edges contacting track and fuselage.

REPLACE TROOP/CARGO UPPER DOOR SEAL

Remove

1. Remove door from helicopter (WP 0244 00). Place door on padded saw horses or sandbags.
2. Drill out rivets holding seal and retainer to door (TM 1-1500-204-23) (Figure 3, Detail D). Remove screws and washers if installed.
3. Drill and install rivnuts at fastener locations (TM 1-1500-204-23) (Figure 3, Detail B).
4. Replace seal retainer if damaged beyond limits (WP 0377 00).

WARNING

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5. Remove old seal with a nonmetallic scraper, using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, to remove remaining sealing compound.

Install

1. Clamp old retainer as a template over replacement retainer and seal. Drill 5/32-inch diameter rivet holes as required. Deburr retainer (Figure 3, Detail D).

REPLACE TROOP/CARGO UPPER DOOR SEAL - Continued

WARNING

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2. Clean parts and door bonding area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
3. Apply an even coat of adhesive, Item 14, WP 1803 00, to door.
4. Place retainer on seal and install screws and washers (TM 1-1500-204-23).
5. Apply bead of sealing compound, Item 273, WP 1803 00, around all edges of installation (Figure 3, Detail A).

WARNING

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6. Remove excess sealing compound with technical acetone, Item 5, WP 1803 00.
7. Install door on helicopter (WP 0244 00).

REPLACE TROOP/CARGO DOOR FAIRING SEAL

Remove

1. Remove door from helicopter (WP 0244 00). Place on padded saw horses or sandbags.
2. Drill out rivets holding seal and seal retainer to door fairing (Figure 3, Detail D).
3. Drive rivets out of holes with a small pin punch.
4. Strip retainer and seal from fairing.

WARNING

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5. Remove old adhesive using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper.

Install

1. Line up new seal and seal retainer over existing holes in door fairing. Use clecos to hold in position (Figure 3, Detail D).
2. Drill holes as required (TM 1-1500-204-23).
3. Remove seal and retainer.

REPLACE TROOP/CARGO DOOR FAIRING SEAL - Continued

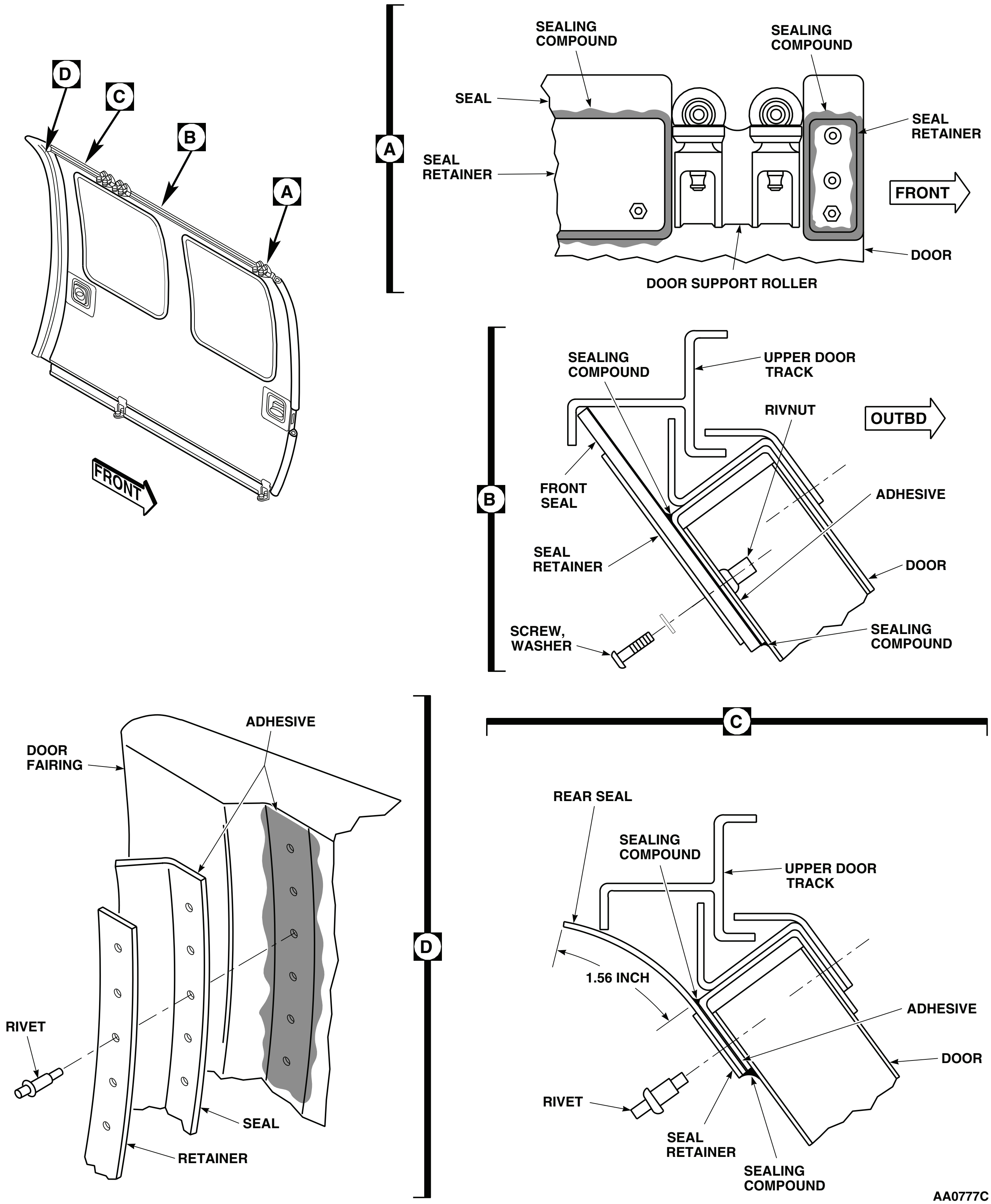

Figure 3. Replace Rear and Upper Seals.

REPLACE TROOP/CARGO DOOR FAIRING SEAL - Continued

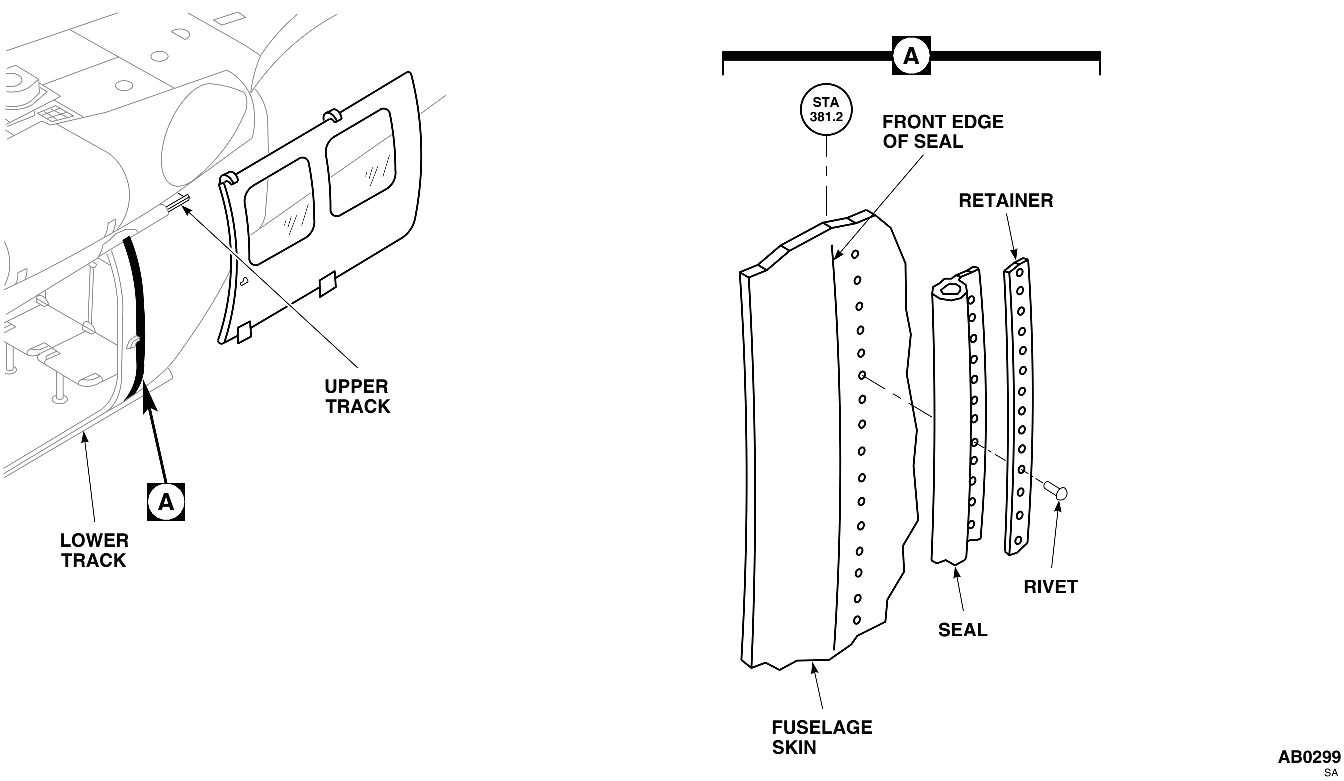

Figure 4. Replace Fuselage Seal.

WARNING

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4. Roughen seal and fairing joint surfaces with abrasive cloth assortment, Item 1, WP 1803 00, and wipe with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, removing all grit.
5. Apply a coat of adhesive, Item 14, WP 1803 00, to the door and seal joining surfaces. Let adhesive dry for at least 10 minutes.
6. Fasten seal and seal retainer together and install on door fairing with 1/8-inch diameter cleco in every third hole.
7. Rivet seal and retainer to door fairing (TM 1-1500-204-23).

REPLACE FUSELAGE REAR DOOR SEAL

Remove

1. Remove troop/cargo door (WP 0244 00).
2. Carefully drill out and remove blind rivets securing retainer and seal to fuselage skin (Figure 4, Detail A).
3. Clean up rivet debris.

REPLACE FUSELAGE REAR DOOR SEAL - Continued

4. Inspect retainer for damage and replace if beyond negligible damage limits (WP 0377 00).

Install

1. Position new seal and retainer and temporarily hold in place with 1/8-inch diameter clecos in every third hole (Figure 4, Detail A).
2. Install blind rivets using a blind riveter.
3. Install troop/cargo door (WP 0244 00).
4. Check open/close function of troop/cargo door (WP 0244 00).
5. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR FAIRING SEAL

This WP supersedes WP 0248 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0244 00
WP 1803 00

REMOVE

1. Remove troop/cargo door (WP 0244 00).
2. Drill out and remove rivets holding upper door track fairing seal and seal retainer to fairing (Figure 1, Detail A).
3. Remove retainer and seal from fairing.

WARNING

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4. Remove all old sealing compound from fairing using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

INSTALL

1. Locate and line up pilot holes in retainer with existing rivet holes in fairing (Figure 1, Detail A).
2. Hold retainer and seal in position using clecos.
3. Open rivet holes to correct rivet size using fairing rivet holes as a pattern.
4. Remove seal and retainer from fairing.
5. Roughen new seal and fairing bonding surfaces with abrasive cloth assortment, Item 1, WP 1803 00.

INSTALL - Continued

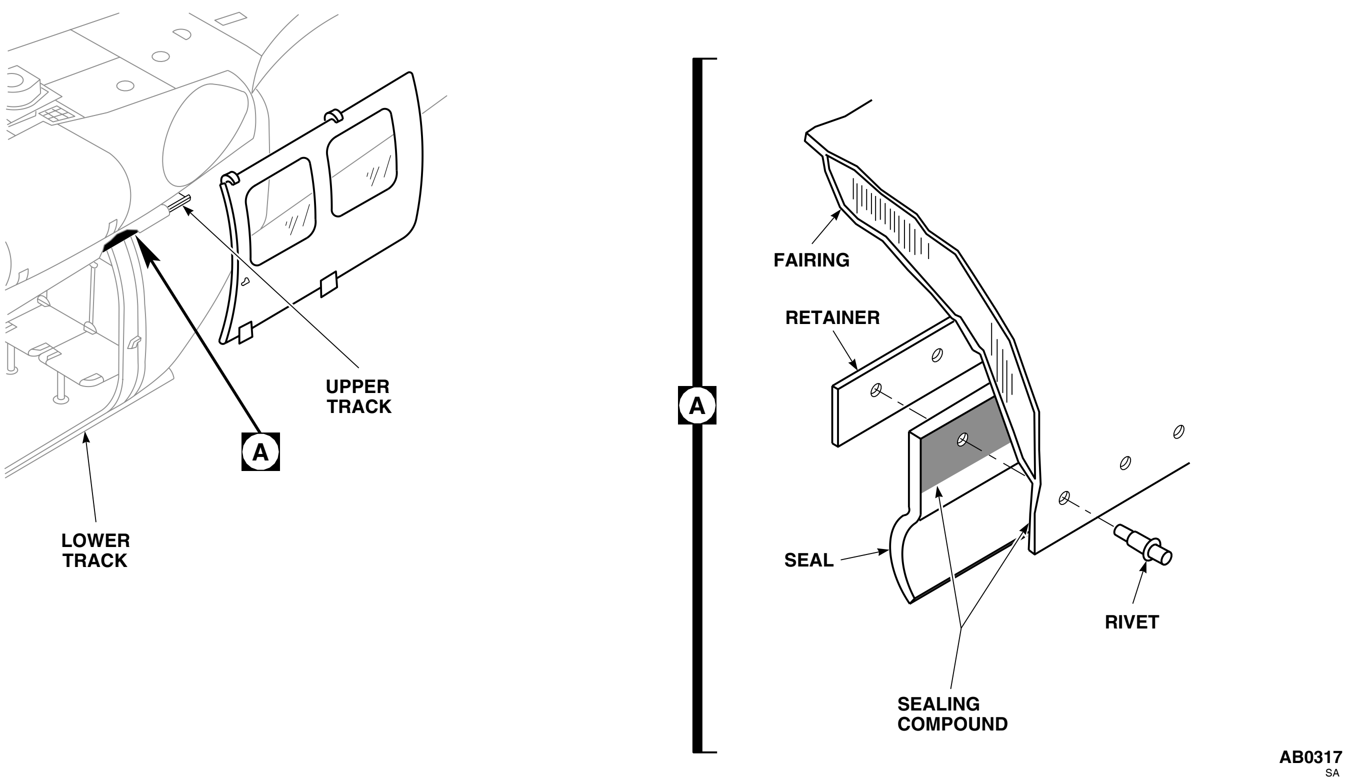

Figure 1. Replace Troop/Cargo Door Track Fairing Seal.

WARNING

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6. Wipe clean with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth cloth.
7. Fasten seal and retainer together with 1/8-inch diameter clecos.
8. Apply coat of sealing compound, Item 273, WP 1803 00, on seal and fairing bonding surfaces.
9. Rivet seal to fairing while sealing compound is wet (TM 1-1500-204-23).
10. Install troop/cargo door (WP 0244 00).
11. Open and close door to check operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR HANDLES

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Sealing Compound, Item 281, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0255 00
WP 1803 00

REMOVE

1. Remove two screws attaching outside handle to door (Figure 1, Detail A).

2. Remove handle.

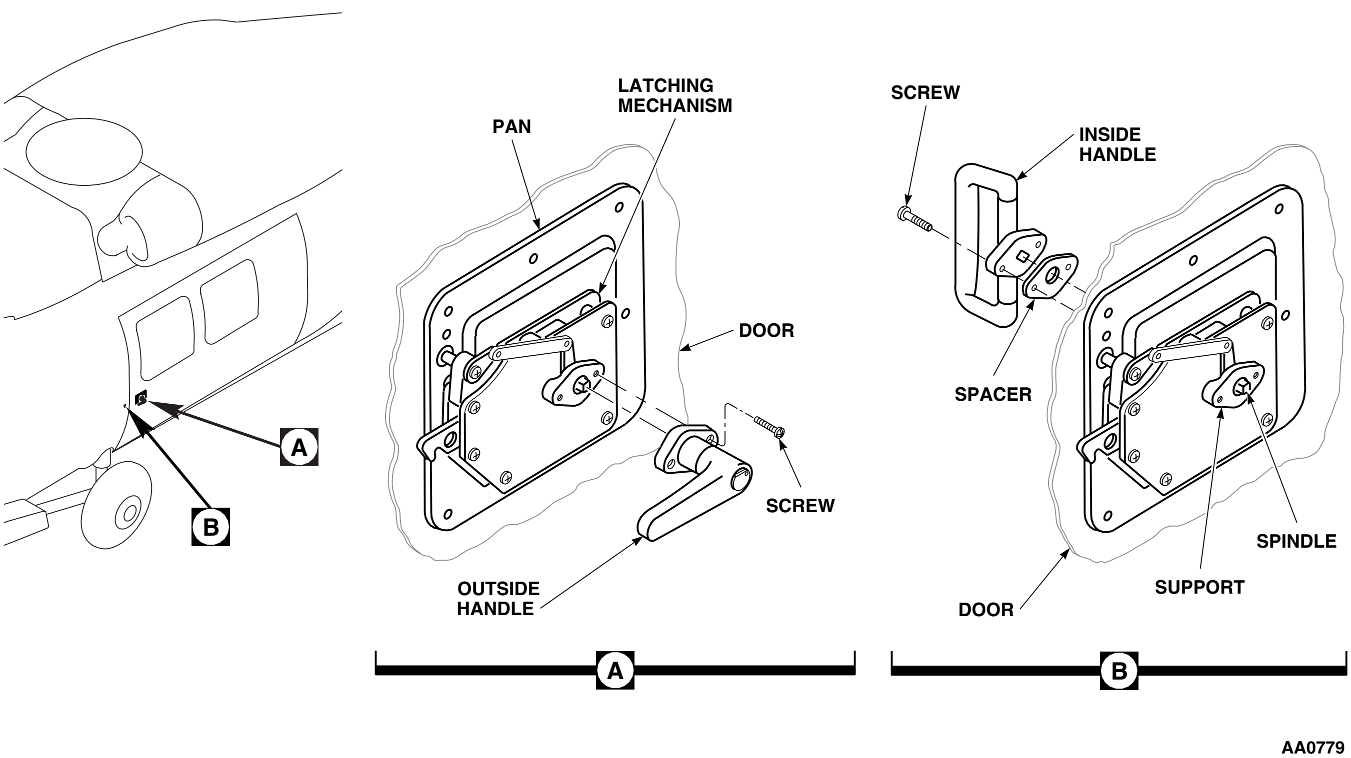


Figure 1. Replace Troop/Cargo Door Handles.

REMOVE - Continued

3. Remove two screws attaching inside handle to door (Figure 1, Detail B).
4. Remove handle.

NOTE

Do not pull spindle out of latch, as internal parts will come loose.

5. If handle is being replaced, remove door lock (WP 0255 00).

INSTALL

1. **QA** Place inside handle on door. Line up handle with spindle (Figure 1, Detail B).
2. Coat threads of screws with sealing compound, Item 281, WP 1803 00, and install.
3. On a replacement outside handle, install door lock (WP 0255 00).
4. **QA** Place outside handle on door (Figure 1, Detail A).
5. Coat threads of screws with sealing compound, Item 281, WP 1803 00, and install.
6. Open and close door using both inside and outside handles to check operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR LATCH ASSEMBLY

This WP supersedes WP 0250 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing Installation Kit, GGG-P-643
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Lubricant, Solid Film, Item 180, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 281, WP 1803 00
Sealing Compound, Item 282, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
WP 0249 00
WP 0252 00
WP 1803 00

REMOVE

1. Open troop/cargo door enough to get at both sides of latch (about halfway open).
2. Block door to keep it from sliding open or closed while removing latch.
3. Remove inside and outside handles, leaving spindle in latch (WP 0249 00).
4. Remove screws and washers attaching latch pan to inside surface of door (Figure 1, Sheet 1, Detail A).
5. Remove latch assembly.

NOTE

Do not pull spindle out of latch, as internal parts will come loose.

6. Remove screw and washer, and remove catch from support.
7. If replacing sleeve bearing, do this:
   a. Remove old sealing compound from edges of sleeve bearing using a nonmetallic scraper.
   b. Press sleeve bearing from pan, using arbor press and tools from bushing installation kit.

REMOVE - Continued

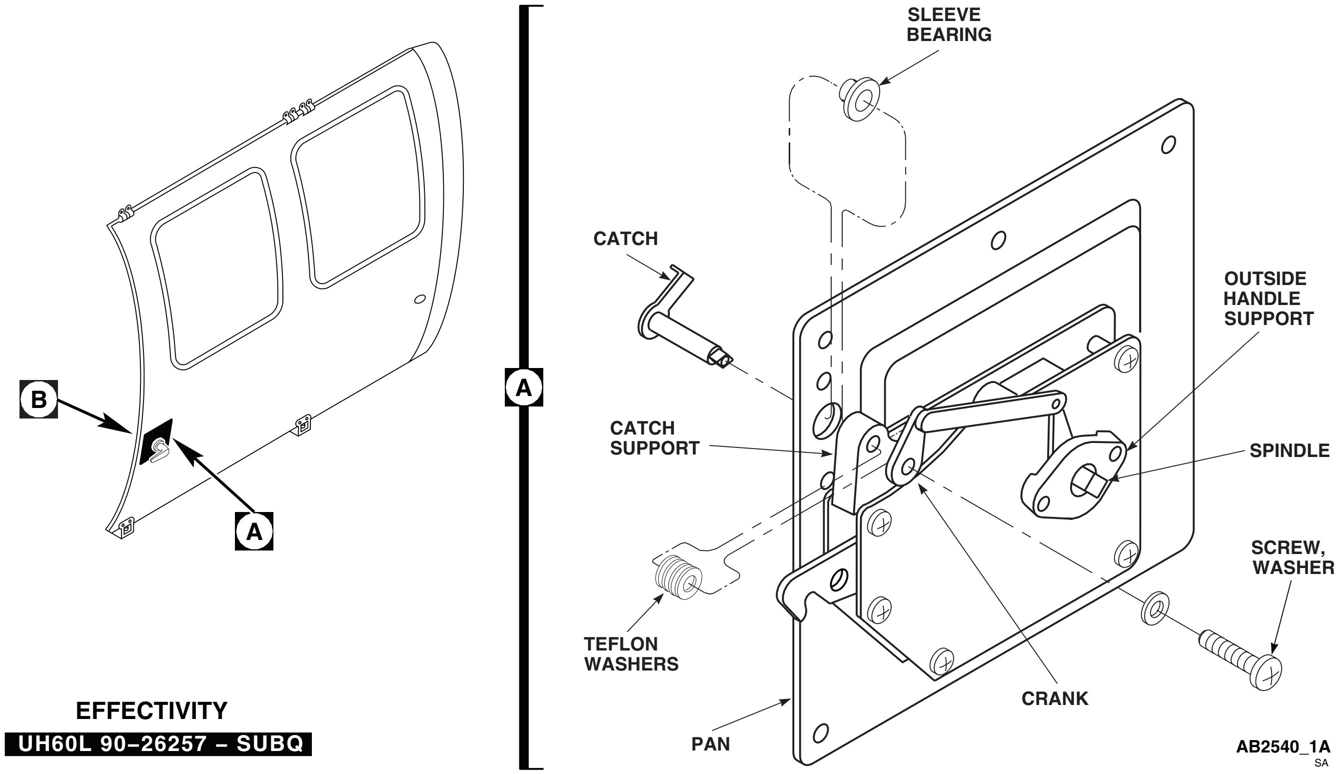


Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 1 of 7)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Clean remaining sealing compound from pan using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

d. Press new sleeve bearing in pan, using arbor press and tools from bushing installation kit.

e. Apply a bead of sealing compound, Item 273, WP 1803 00, around edge of flange of bearing and pan, and around joint of pan and sleeve of bearing.

NOTE

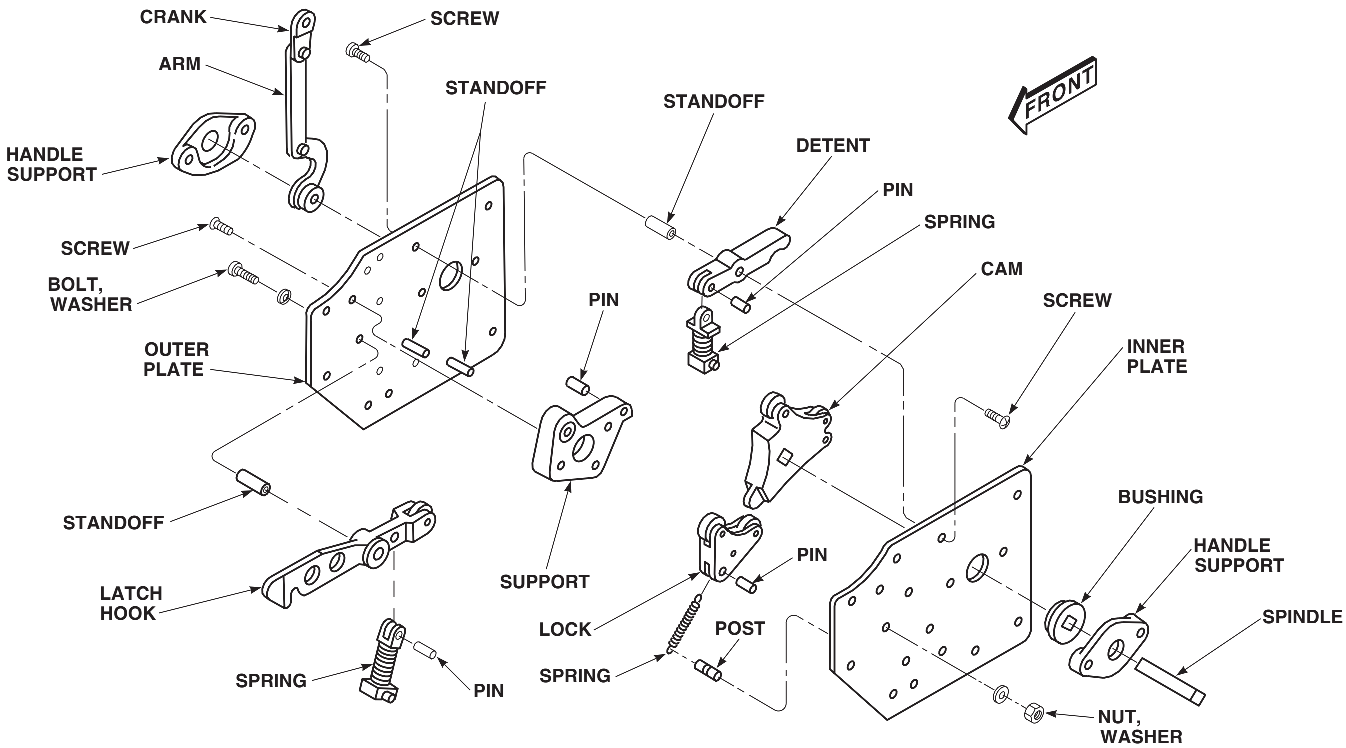
UH-60L 90-26257 - SUBQ Have a laminated shim that is bonded to the latch pan (L/H 70217–02718–131, R/H 70217–02718–132) to center the latch in the cargo door.

8. **UH-60L 90-26257 - SUBQ** Peel laminated shim as required to eliminate interference between latch hook and hook opening in cargo door.

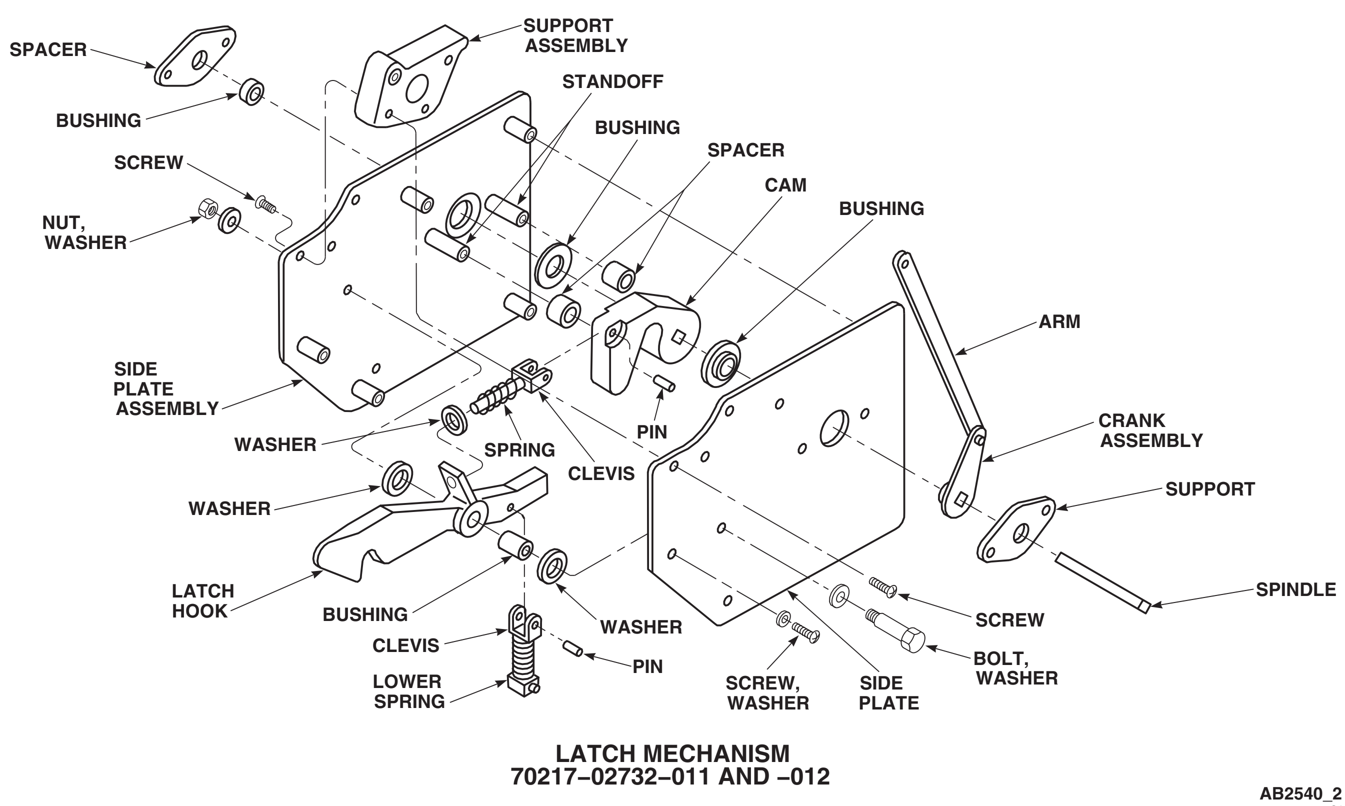
DISASSEMBLE

1. Disassemble latch assembly, 70217-02718-011 or 70217-02718-012, as follows:

DISASSEMBLE - Continued



LATCH MECHANISM
70217–02718–011 AND –012



LATCH MECHANISM
70217–02732–011 AND –012

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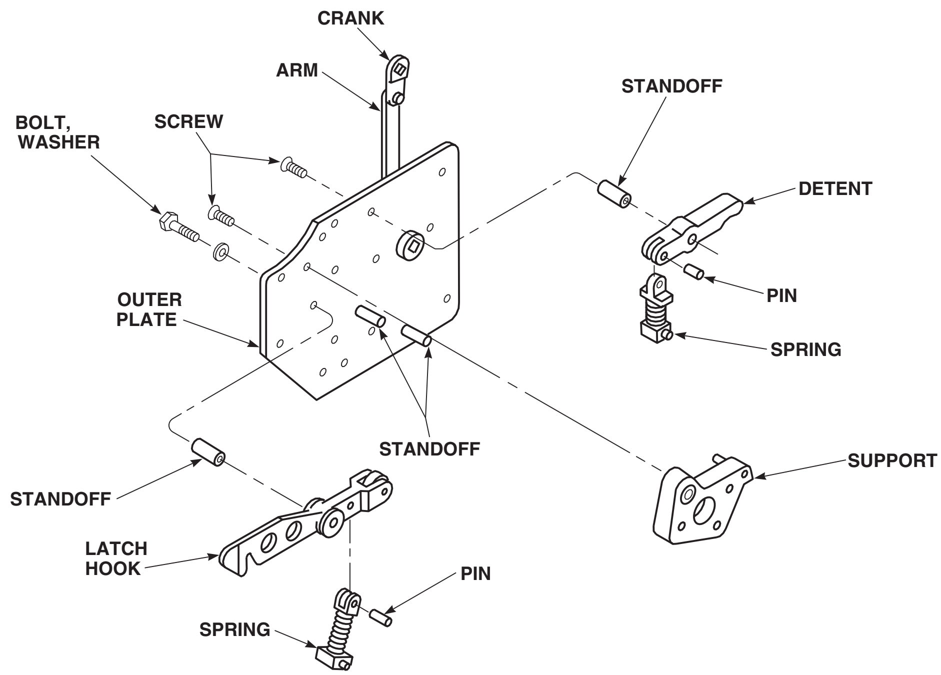
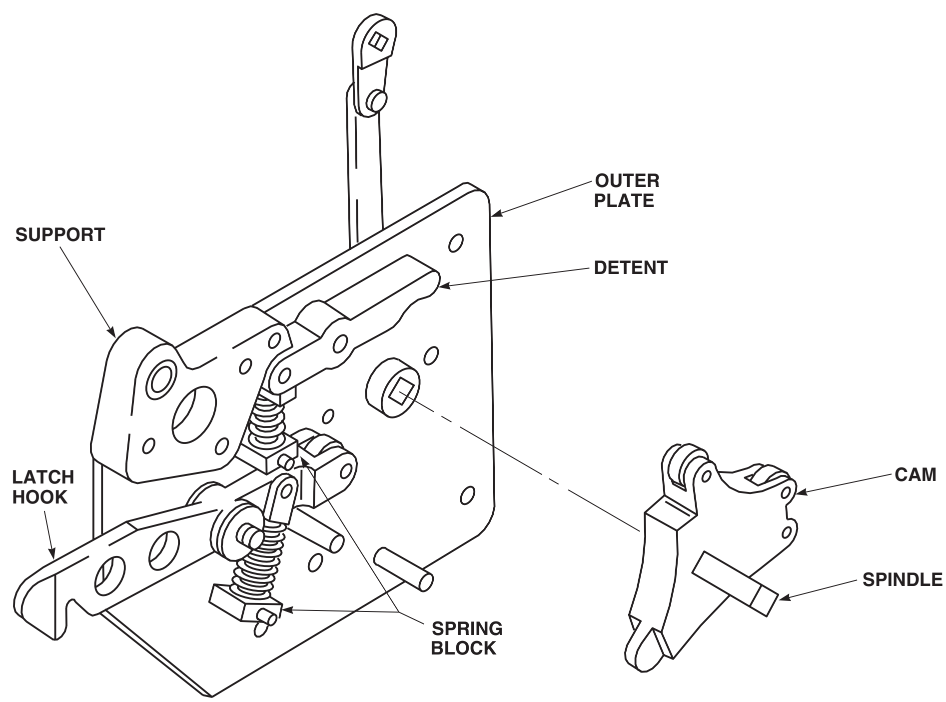
SA

Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 2 of 7)

DISASSEMBLE - Continued



LATCH MECHANISM
70217–02718–011 AND 012

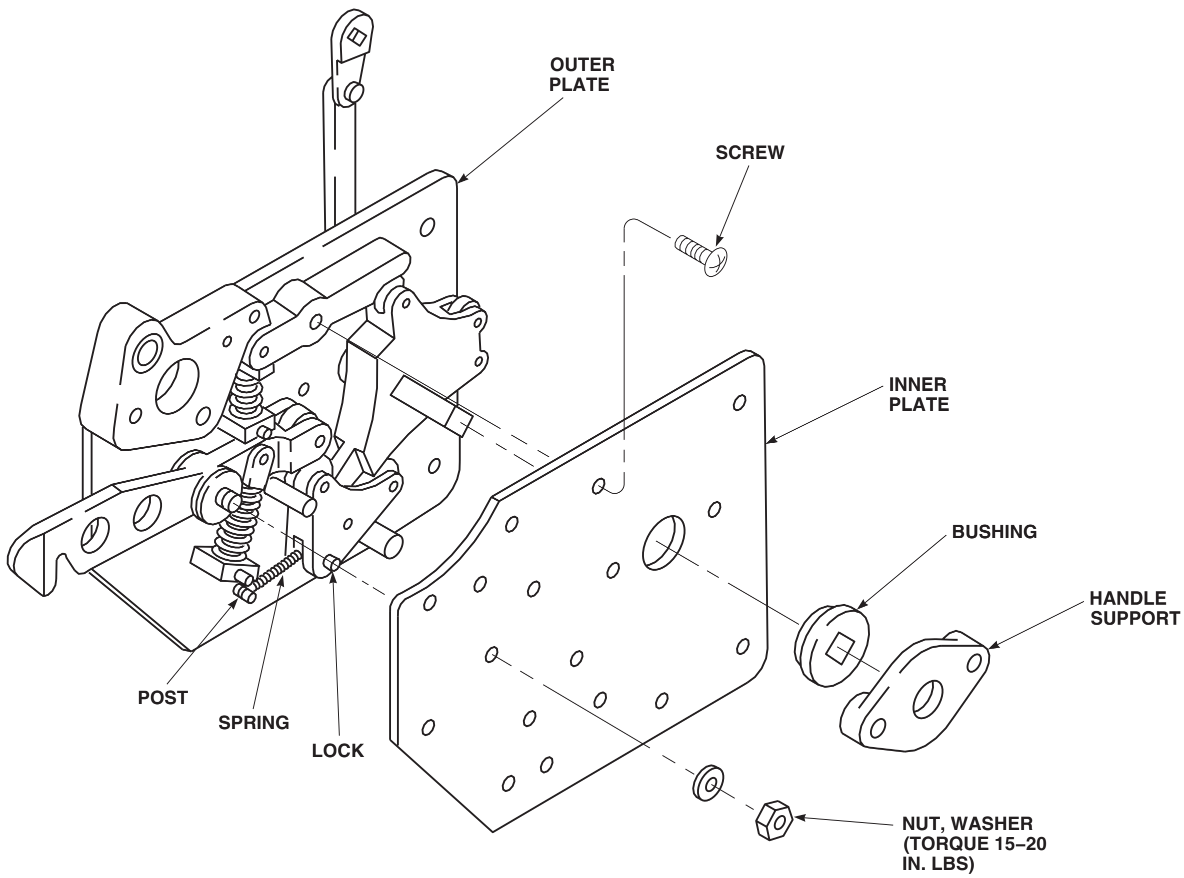


LATCH MECHANISM
70217–02718–011 AND –012

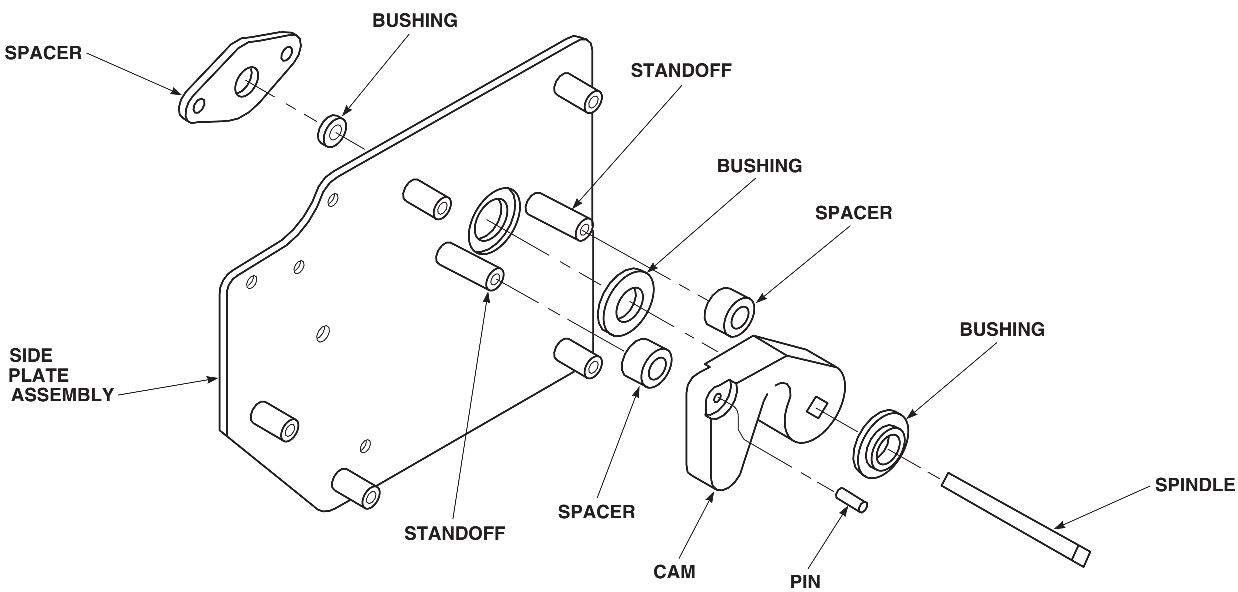
AB2540_3
SA

Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 3 of 7)

DISASSEMBLE - Continued



LATCH MECHANISM
70217–02718–011 AND –012

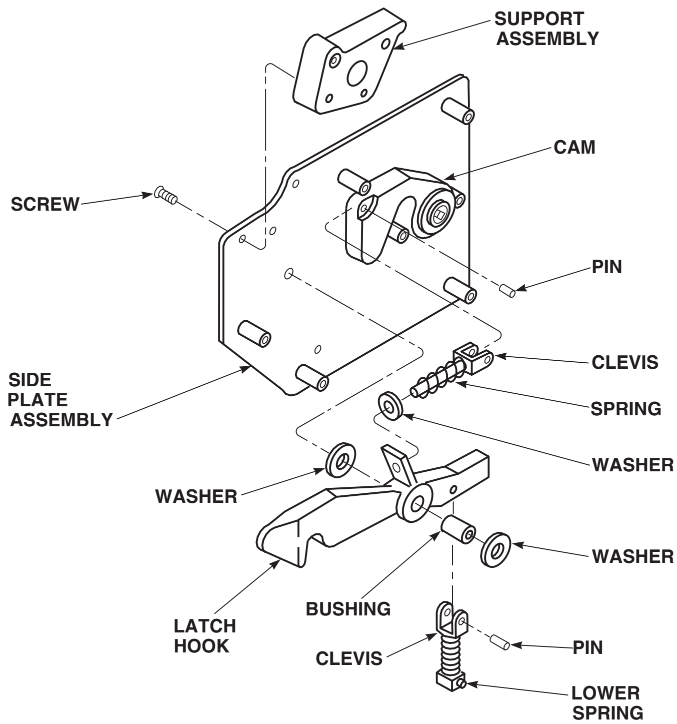


LATCH MECHANISM
70217–02732–011 AND –012

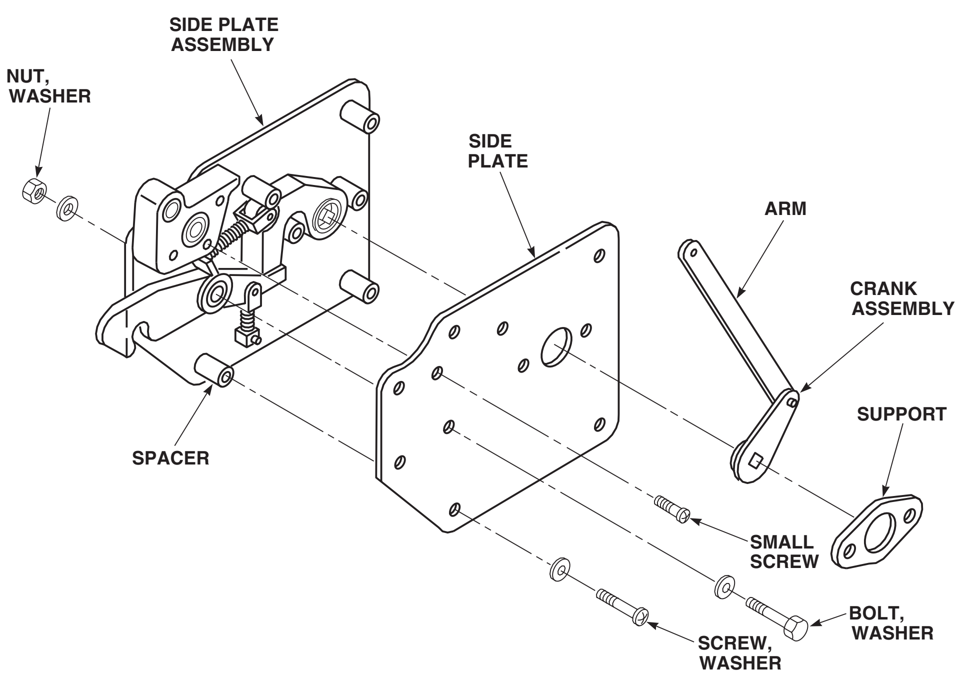
AB2540_4
SA

Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 4 of 7)

DISASSEMBLE - Continued



LATCH MECHANISM
70217–02732–011 AND –012

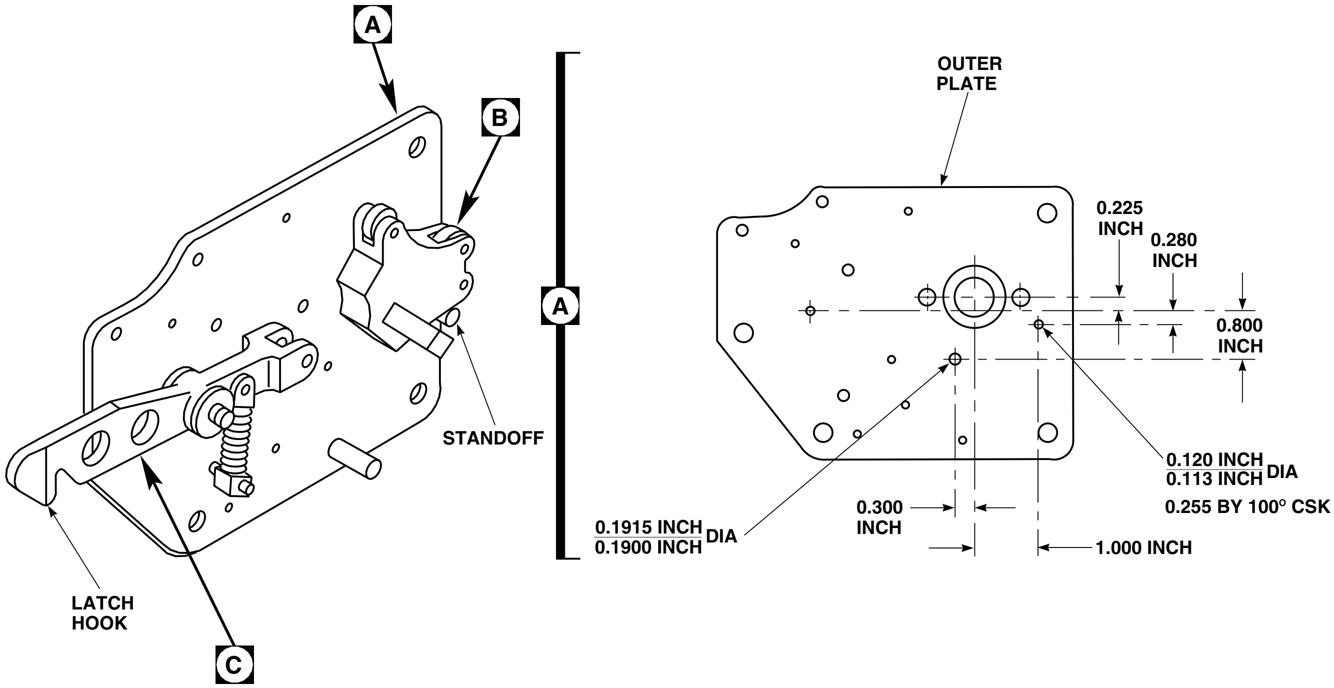


LATCH MECHANISM
70217–02732–011 AND –012

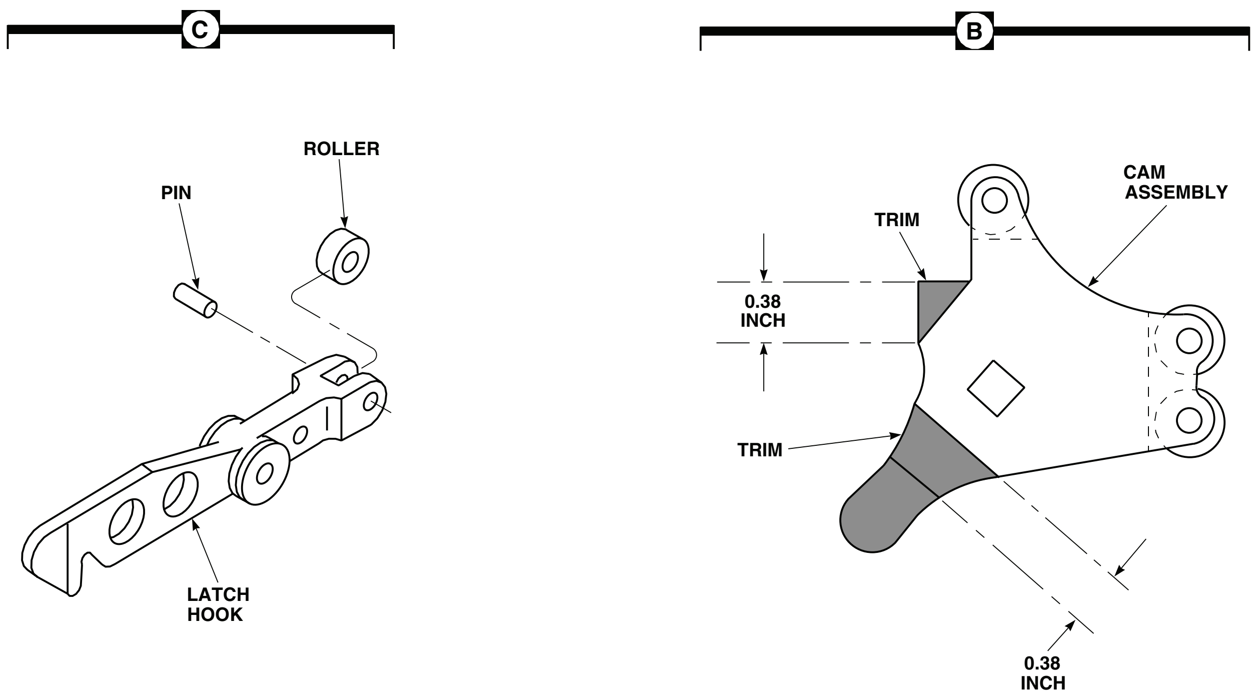
AB2540_5
SA

Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 5 of 7)

DISASSEMBLE - Continued



LATCH MECHANISM
70217-02718-011 AND -012



LATCH MECHANISM
70217-02718-011 AND -012

AB2540_6A
SA

Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 6 of 7)

DISASSEMBLE - Continued

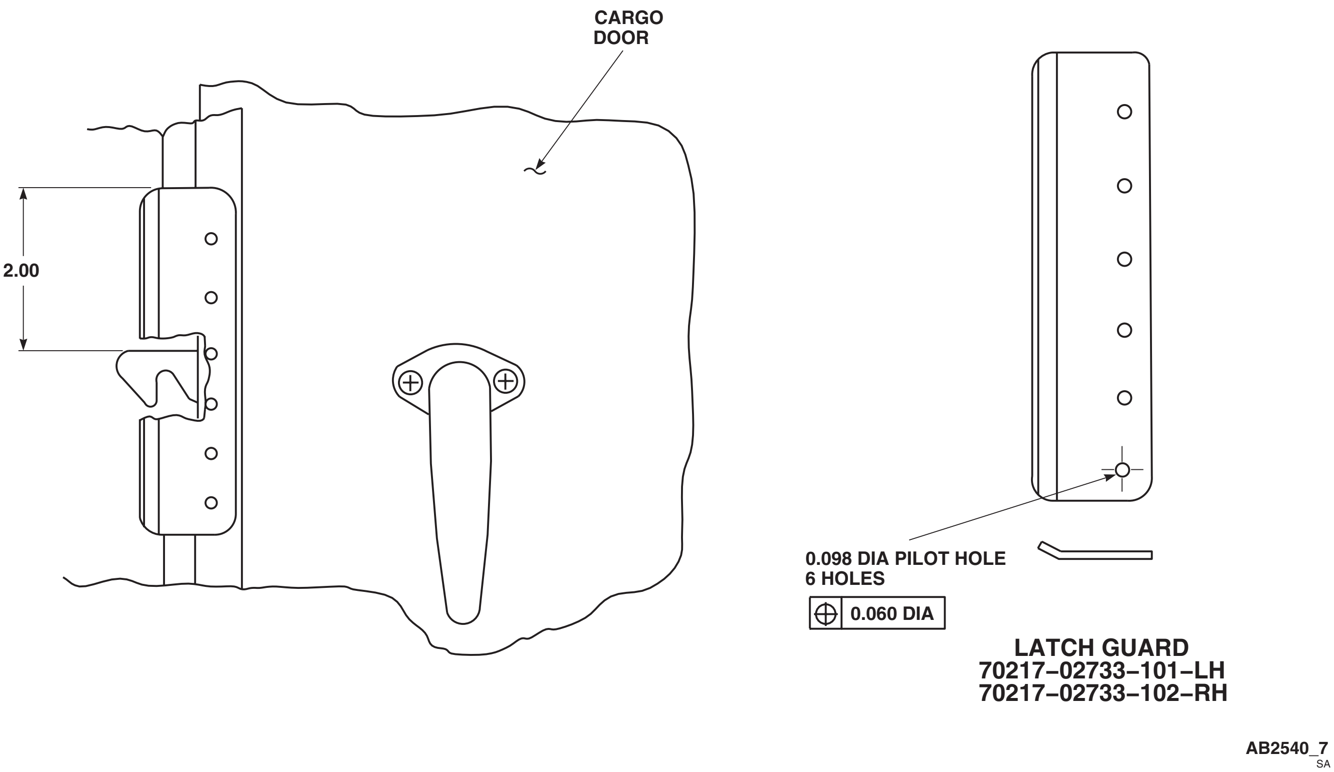


Figure 1. Replace Troop/Cargo Door Latch Assembly. (Sheet 7 of 7)

a. Lift outside handle support off standoffs (Figure 1, Sheet 2).

b. Slide arm off spindle.

c. Remove nut and washer from latch hook bolt.

d. Place latch on bench, outer plate down.

e. Lift inside handle support off standoffs.

f. Remove all screws in inner plate. Lift inner plate off latch.

g. Remove bushing from inner plate.

h. Remove spindle and cam.

i. Lift lock from plate. Drive out pin, remove spring from lock.

j. Lift detent off standoff.

k. Remove latch hook and spring.

l. Remove pin and spring from detent.

m. Remove pin and spring from latch hook.

n. Turn latch over, remove screw and support from plate.

o. Remove remaining bolt, screws, and standoffs from outer plate.

p. If inner or outer plate bushings are worn, press out and press in new bushings.

DISASSEMBLE - Continued

2. Disassemble latch assembly, 70217-02732-011 or 70217-02732-012, as follows:
   a. Remove outside handle support from side plate (Figure 1, Sheet 2).
   b. Slide arm and crank assembly off spindle.
   c. Remove nut and washer from latch hook bolt and remove bolt.
   d. Place latch on bench with screwheads facing up.
   e. Remove screws from sideplate and remove sideplate from latch.
   f. Remove pin from latch hook and clevis end of lower spring.
   g. Remove lower spring.
   h. Remove pin from spring clevis and cam.
   i. Remove spring and washer from latch hook.
   j. Remove latch hook, washers, and bushing from sideplate assembly.
   k. Remove cam from latch assembly.
   l. Remove spacers from standoffs.
   m. Remove spindle from latch assembly.
   n. Remove support from latch assembly.
   o. Remove spacer from sideplate assembly.
   p. If sideplate or spacer bushings are worn, replace them.
   q. Inspect all parts for signs of binding, too much wear, loose bushings, cracks.
   r. Replace all worn, broken, or cracked parts.

REPAIR

NOTE

- This repair procedure applies only to helicopters which have latches, 70217-02718-011 and 70217-02718-012 installed.
- Dispose of parts not being reused per local policies.
- When disassembling latch, remove latch hook and spring as an assembly.

1. Disassemble latch (WP 0252 00).
2. Remove post from outer plate and discard post (Figure 1, Sheet 4).
3. Using a No. 32 drill bit, drill hole (0.113 to 0.120-inch diameter) through both inner and outer side plates. Countersink outside of holes 100 degrees at 0.225-inch. See Figure 1, Sheet 6, for hole locations.
4. Using a No. 11 drill bit, drill a hole (0.1900 to 0.1915-inch diameter) through both sideplates.
5. Remove standoff from center of outer plate and press into new hole.
6. Remove pin from latch hook and remove roller. Discard pin and roller.
7. Remove shaded areas of cam.
8. To touch up trimmed areas of cam, do this:

REPAIR - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Clean trimmed areas of cam using technical acetone, Item 5, WP 1803 00. Allow area to dry completely.

b. Apply corrosion resistant coating, Item 111, WP 1803 00, to areas per TM 1-1500-344-23.

NOTE

Unused holes in plates may be left unplugged.

9. Reusing pin, install spring, 70217-02718-051, on end of latch hook.

NOTE

Lock and spring will not be reused when reassembling latch.

10. Reassemble latch (WP 0252 00).

11. **UH-60A 80-23492 - SUBQ** If required, replace latch guard as follows:

a. Remove rivets from latch guard (TM 1-1500-204-23).

b. Remove latch guard.

c. Position replacement latch guard vertically on troop/cargo door so that upper end of guard in 2 inches above latch hook, and mark rivet holes (Figure 1, Sheet 7).

d. Using a No. 30 drill bit, drill premarked holes in troop/cargo door.

e. Position latch guard (with bend facing outward) on troop/cargo door, and install rivets (TM 1-1500-204-23).

ASSEMBLE

1. Assemble latch assembly, 70217-02718-011 or 70217-02718-012 as follows:

a. Install standoffs on outer plate with screws (Figure 1, Sheet 3).

b. Install support on outer plate with screw. Make sure pin in support is in hole in outer plate.

NOTE

If installing new support, drill hole and install new pin. Use old support as pattern.

c. Install arm on outer plate bushing.

d. Place bolt, washer for latch hook through hole in outer plate.

e. Install springs in detent and latch hook with pins.

f. Place latch hook on bolt.

g. Place detent on standoff.

h. Place detent spring block and latch hook spring block in holes on plate (Figure 1, Sheet 3).

i. Place lockspring on post, install spring in lock with pin. Place lock on standoff on plate (Figure 1, Sheet 4).

j. Hold cam in position on plate, push spindle through cam and outer plate (Figure 1, Sheet 3).

ASSEMBLE - Continued

k. Place post in hole on outer plate. Hold in position with needlenose pliers (Figure 1, Sheet 4).

l. Place inner plate on built-up latch, making sure post and spring blocks are seated in holes in inner plate. Fasten inner plate to latch with screws.

m. Install nut, washer on latch hook bolt. **TORQUE NUT TO 15 - 20 INCH-POUNDS.**

n. Slide bushing over spindle until it seats in inner plate bushing.

o. Place both door handle supports over standoffs.

2. Assemble latch assembly, 70217-02732-011 or 70217-02732-012 as follows:

a. Lubricate all parts with solid film lubricant, Item 180, WP 1803 00, before assembly.

WARNING

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b. Clean outside diameter of bushings and inner bore of sideplates using cheesecloth cloth, Item 85, WP 1803 00, and technical acetone, Item 5, WP 1803 00.

c. Apply a 1/16-inch bead of sealing compound, Item 282, WP 1803 00, to leading edge of bushing outside diameter and edge of bore in sideplates.

d. Carefully press bushings in sideplate (Figure 1, Sheet 4).

WARNING

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e. Wipe excess sealing compound from inner bore of bushings using cheesecloth cloth, Item 85, WP 1803 00, and technical acetone, Item 5, WP 1803 00.

f. Press bushing into spacer plate.

g. Install spacer plate on sideplate assembly.

h. Install spindle through bushing in spacer plate.

i. Install spacers on standoffs.

j. Install clevis rod on cam with pin.

k. Install cam on spindle and sideplate assembly (Figure 1, Sheet 4).

l. Position support assembly on sideplate (Figure 1, Sheet 5). Coat threads of screws with sealing compound, Item 282, WP 1803 00, and install screws.

m. Install upper spring and washer on clevis rod.

n. Connect clevis end of lower spring assembly to latch hook with pin.

o. Position washers and latch hook on sideplate assembly, engaging end of clevis rod in latch hook tab hole.

ASSEMBLE - Continued

p. Install bushing in center hole of latch hook.

q. Place end of lower spring assembly in hole in sideplate assembly (Figure 1, Sheet 5).

r. Position sideplate on spacers, engaging lower spring assembly in holes.

s. Coat threads of screws with sealing compound, Item 282, WP 1803 00, and install screws and washers.

t. Install bolt and washer through sideplates and latch hook. Install washer and nut on bolt.

u. Install arm and crank on spindle.

v. Install handle support on sideplate.

w. Coat threads of small screws with sealing compound, Item 282, WP 1803 00, and install screws in sideplate and handle support.

INSTALL

1. **QA** Insert catch shaft through sleeve bearing, catch support, and Teflon washers. Coat threads of screw with sealing compound, Item 281, WP 1803 00. Install square hole of crank onto catch shaft with washer and screw (Figure 1, Sheet 1, Detail A).
2. **QA** Coat threads of screws with sealing compound, Item 281, WP 1803 00, and fasten latch assembly to inside surface of door with screws and washers.
3. **QA** Coat threads of screws with sealing compound, Item 281, WP 1803 00. Install inside handle on door with screws.
4. **QA** Put outside handle on door. Coat threads of screws with sealing compound, Item 281, WP 1803 00, and install.
5. Open and close door using both inside and outside handles to check operation.
6. Apply bead of sealing compound, Item 273, WP 1803 00, in angle of outside handle base and door. Fair to fillet at least 1/8-inch wide.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR RECEIVER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic’s Toolkit, SC 5180-99-B01
Airframe Repairer’s Toolkit, SC 5180-99-B02

Material/Parts
Shim, 70217-02700-109

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Open troop/cargo door.
2. **UH60A 77-22714 - 80-23491** Remove countersunk screws attaching receiver to door frame (Figure 1, Detail A).
3. **UH60A 80-23492 - SUBQ EH60A UH60L** Remove countersunk screws, washers, and nuts, and remove adjuster plate, shim, and receiver (Figure 1, Detail A).

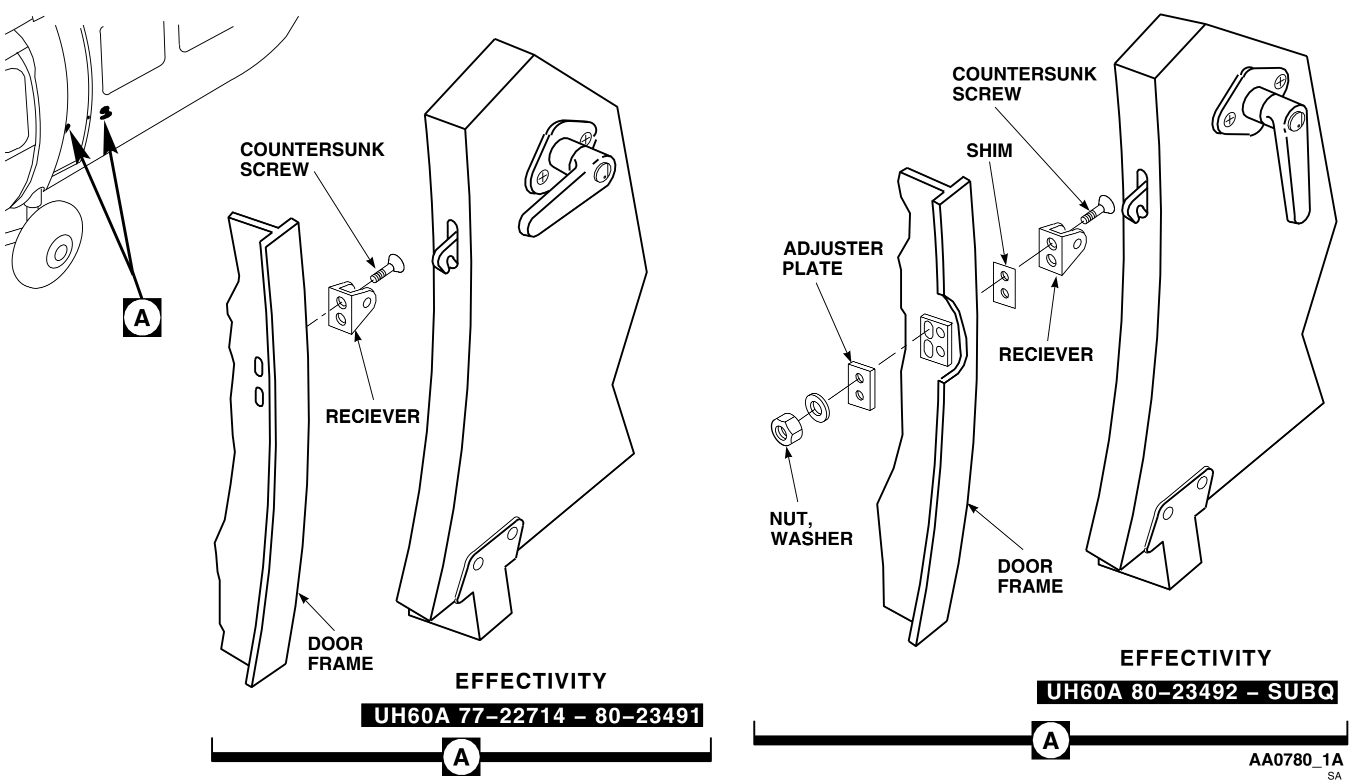

Figure 1. Replace Troop/Cargo Receiver.

INSTALL

1. **QA** **UH60A 77-22714 - 80-23491** Place receiver on door frame and install countersunk screws (Figure 1).
2. **UH60A 80-23492 - SUBQ EH60A UH60L** Install shim and receiver on door jamb and install all countersunk screws (Figure 1).
3. On other side of door jamb, install adjuster plate on screws, with serrated side against serrated backup plate.
4. **QA** Install nuts and washers.
5. Close door and check position of latch in receiver and compression of seal.
6. If necessary, adjust receiver as follows:
   a. Loosen nuts to allow movement of receiver.
   b. Lower receiver to its lowest limit.
   c. With latch horizontal, raise receiver until pin touches inside of latch hook without raising latch.
   d. Tighten nuts while checking that latch does not move.
   e. If latch moves, drop adjuster plate one or two serrations.
   f. **QA** Tighten nuts.
7. With door fully closed, door seal should not be compressed more than 1/4-inch.
8. If necessary, add shim, 70217-02700-109, under receiver to keep seal compression to 1/4-inch with door fully closed.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR STOP MECHANISM

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0244 00

REMOVE

1. Remove troop/cargo door (WP 0244 00).
2. Remove pin, screw, and washers attaching stop to frame at STA 379.0 (Figure 1, Detail A).
3. Remove stop, spacers, spring, and plastic washers.

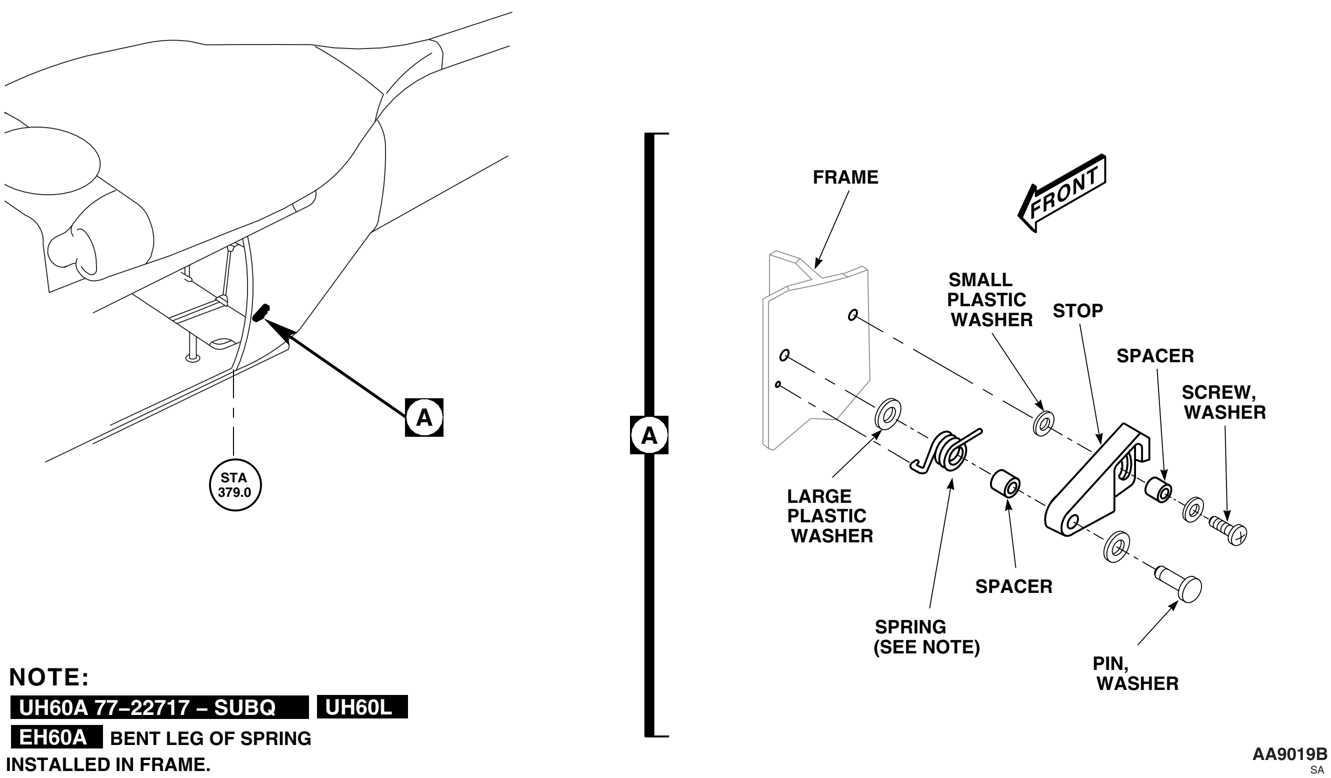

Figure 1. Replace Troop/Cargo Stop Mechanism.

INSTALL

1. **QA** Place new plastic washers between stop and frame at STA 379.0 (Figure 1, Detail A).

NOTE

UH60A 77-22717 - SUBQ UH60L EH60A Bent leg of spring should be inserted into hole in frame.

2. Place stop in position and insert spacers and spring.
3. **QA** Install attaching pin, screw, and washers.
4. Install troop/cargo door (WP 0244 00).
5. Open door fully and check stop operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR ROLLER SUPPORTS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Depth Micrometer, 0 to 1 inch
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Abrasive Cloth, Assortment, Item 1, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Washer, AN960KD10
Washer, AN960KD10LL

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 55-1500-322-24
WP 0244 00
WP 1803 00

REMOVE

1. Remove troop/cargo door from helicopter (WP 0244 00).
2. Remove screws, washers, and nuts attaching roller supports to door (Figure 1, Detail A).
3. Remove roller support from door. Tag each roller support for location upon removal. Keep spacer for installation of replacement roller support.

DISASSEMBLE

1. Remove pins, washers, and collars and remove rollers from upper roller support fitting (Figure 1, Detail B).
2. Remove pins, washers, collar, and nut, and remove rollers from lower roller support (Figure 1, Detail E).

INSPECT

CAUTION

Bearings will be damaged by loss of lubricant if soaked in solvent. Keep solvent away from bearings when cleaning parts.

1. Clean all parts with clean cloth wet with cleaning compound solvent, Item 71, WP 1803 00, or equivalent, to avoid saturation of bearings. Inspect bearings (TM 1-1500-204-23).
2. On helicopters with doors, 70217-02701-041 through 70217-02701-046, inspect roller support fittings, 70217-02717-041 and 70217-02717-042, as follows:
   a. Inspect upper front roller support fitting for groove on angled surface using depth micrometer. **MAXIMUM DAMAGE IS 0.030-INCH.**
   b. Inspect upper rear roller support fitting for groove on angled surface using depth micrometer. **NO WEAR ALLOWED.**
   c. Inspect lower front roller support fitting for groove between rollers using depth micrometer. **MAXIMUM DAMAGE IS 0.060-INCH.**

INSPECT - Continued

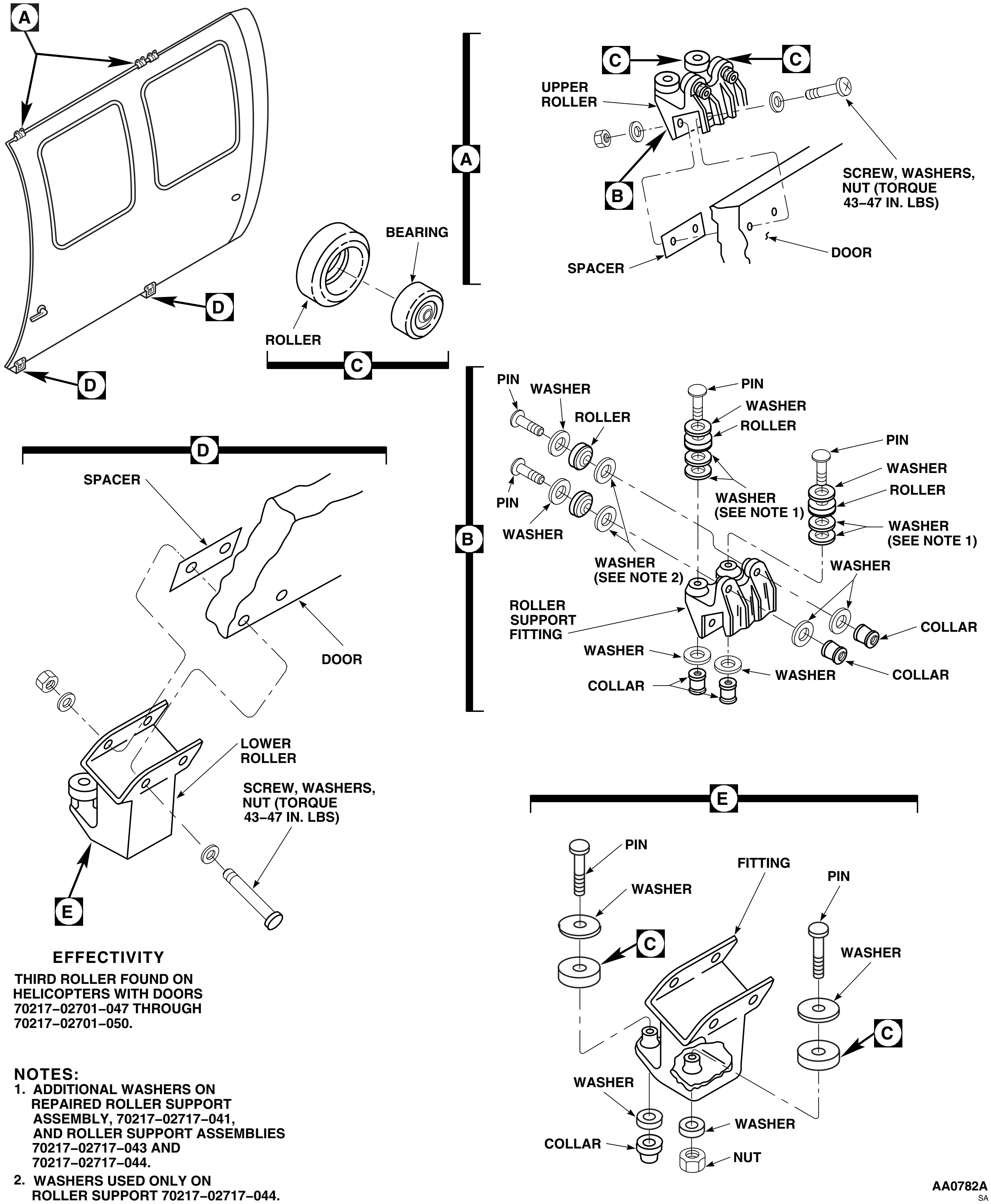

Figure 1. Replace Troop/Cargo Door Rollers and Supports.

INSPECT - Continued

d. Inspect lower rear roller support fitting for groove between rollers using depth micrometer. **MAXIMUM DAMAGE IS 0.025-INCH.**

3. On helicopter with doors, 70217-02701-047 and 70217-02701-048, inspect roller support fittings, 70217-02717-041 and 70217-02717-042, as follows:

a. Inspect upper front roller support fitting for groove on angled surface using depth micrometer. **MAXIMUM WEAR IS 0.030-INCH.**

b. Inspect middle and rear upper roller support fitting for groove on angled surface using depth micrometer. **MAXIMUM WEAR IS 0.015-INCH.**

c. Inspect lower front roller support fitting for groove between rollers using depth micrometer. **MAXIMUM WEAR IS 0.060-INCH.**

d. Inspect lower rear roller support fitting for groove between rollers using depth micrometer. **MAXIMUM WEAR IS 0.060-INCH.**

e. If wear limits are exceeded, replace roller support fitting.

f. Repair damage as follows:

(1) Blend to remove groove using abrasive cloth assortment, Item 1, WP 1803 00.

WARNING

Injury to personnel and damage to equipment will result if all upper roller assemblies do not have the same part number or additional washers under the horizontal rollers. Door could separate from helicopter. All upper roller assemblies must have the same part number or be reworked to add additional washers under the horizontal rollers.

(2) Install horizontal rollers on repaired roller support, 70217-02717-041, with washers, AN960KD10 and AN960KD10LL, between mounting boss and chamfered edge of roller. Install pins, washers and collars with washer under pin head and collar (Figure 1, Detail B).

(3) Install vertical rollers on upper support with chamfered edge toward mounting boss. Install pins, washers, and collars with washer under pin head and collar.

4. Inspect support fittings for cracks, distortion, and broken or missing pieces. Replace damaged fittings with fittings having same part number.

5. Check roller bearings for smooth operation. A slight rolling resistance caused by lubricant and seal drag is normal. Replace dry, loose, rough, seized, or sticking bearings and bearings having loose, open, missing, bent, or dented seals (TM 55-1500-322-24).

6. Check bearing radial wear. **REPLACE BEARINGS HAVING RADIAL WEAR OF OVER 0.0025-INCH.**

7. Check rollers for cracks, flatspots, and missing pieces. Replace damaged rollers.

ASSEMBLE

1. On nonreworked upper roller support, 70217-02717-041, install rollers with roller chamfered edge toward mounting boss. Install pins, washers, and collars, with a washer under pin head and collar (Figure 1, Detail B).

2. On repaired roller support, 70217-02717-041, and roller supports, 70217-02717-043 and 70217-02717-044, install horizontal rollers with washers, AN960KD10 and AN960KD10LL, between mounting boss and chamfered edge of roller. Install pins, washers, and collars with washer under pin head and collar (Figure 1, Detail B).

3. Install vertical rollers on upper support with rollers chamfered edge toward mounting boss. Install pins, washers, and collars with washer under pin head and collar.

ASSEMBLE - Continued

4. Install rollers on lower roller support, with roller chamfered edge toward mounting boss. Install pin, washers, and collar in inboard roller with a washer under pin head and collar. Install pin, washers, and nut in outboard roller with washer under pin head and nuts (Figure 1, Detail E).

INSTALL

WARNING

Injury to personnel and damage to equipment will result if all upper roller assemblies do not have the same part number or additional washers under the horizontal rollers. Door could separate from helicopter. All upper roller assemblies must have the same part number or be reworked to add additional washers under the horizontal rollers.

1. Place roller support on door in same location as removed (Figure 1, Detail A and Figure 1 Detail D).
2. Slide spacer in between support and inside of door.
3. **QA** Install screws, washers, and nuts attaching roller support to door. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
4. Wipe door tracks clean using cheesecloth cloth, Item 85, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry with clean cheesecloth cloth.
5. Install troop/cargo door on helicopter (WP 0244 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR UPPER FAIRINGS

This WP supersedes WP 0254 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 107, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0244 00
WP 0313 00
WP 0379 00
WP 0381 00
WP 1803 00

REMOVE

1. Remove screws, washers, and upper door track fairing between STA 420.0 and STA 428.0 (Figure 1, Detail A).
2. Remove troop/cargo door (WP 0244 00).
3. Remove engine cowling (WP 0313 00).
4. Remove bolt, screws, and washers securing inlet shroud fairing. Remove fairing (Figure 1, Detail B).
5. Remove screws and washers from engine fairing and upper fuselage fairing between STA 308.0 and STA 398.0. Remove fairings with rubber seal attached.
6. If necessary, remove rubber seal from fairings.
7. On engine fairing, if necessary, replace block seal (Figure 1, Detail C).
8. Remove seal from clip with nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. Clean off old adhesive using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

REMOVE - Continued

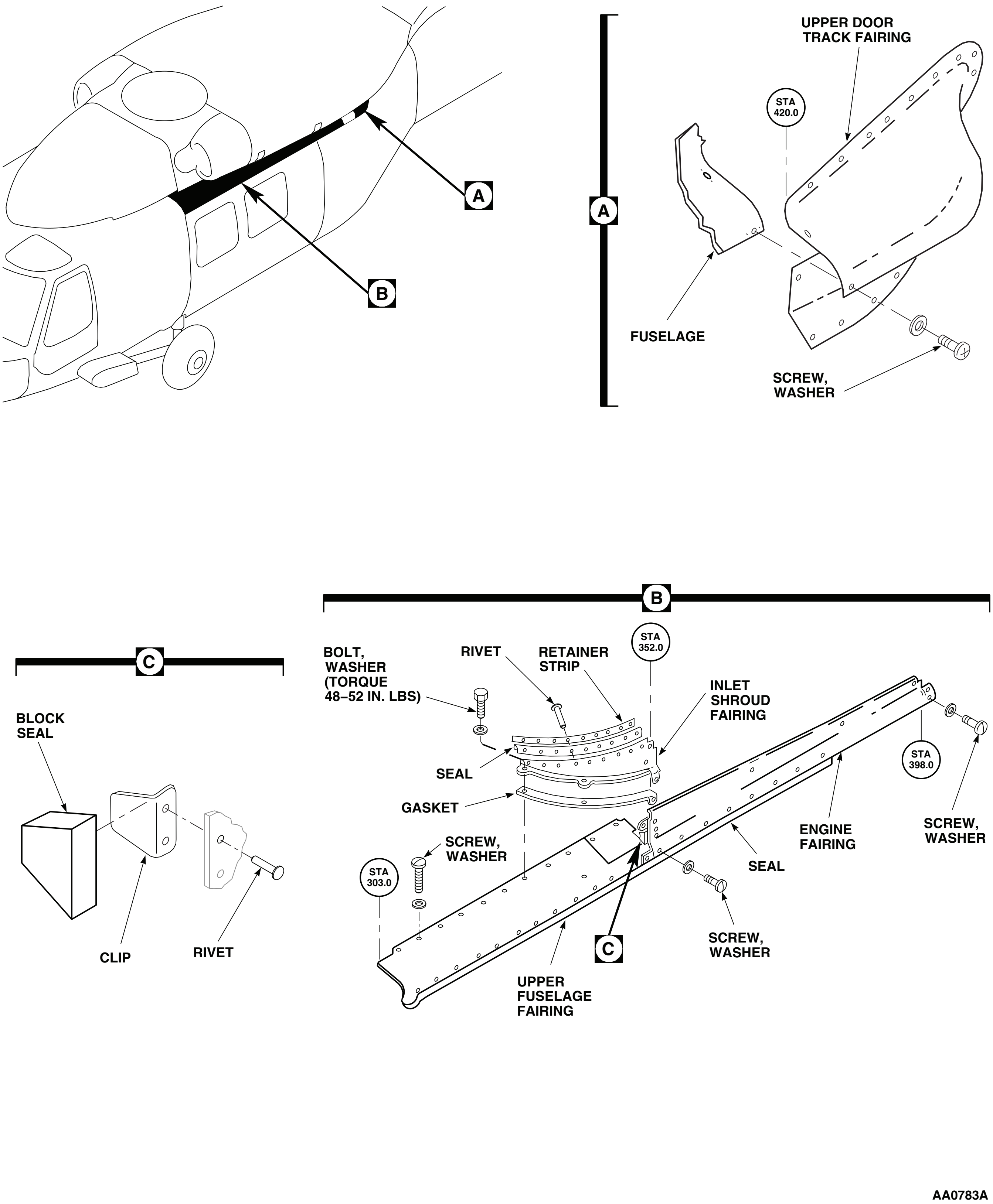

Figure 1. Replace Troop/Cargo Door Upper Track Fairings.

REMOVE - Continued

10. Check fairings for damage and repair as necessary (WP 0379 00).

REPAIR

1. On inlet shroud fairing, drill out rivets that secure retaining strip. Remove strip and seal (Figure 1, Detail B).
2. Remove gasket from fairing flange using a nonmetallic scraper.

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3. Clean old adhesive from fairing using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
4. Drill rivet holes through new seal, using retaining strip as a template.
5. Make sure fairing surface that contacts seal is wiped clean. Fasten seal and retaining strip in place with rivets.
6. Apply a thin, even coat of adhesive, Item 14, WP 1803 00, to joining surfaces of fairing flange and gasket. Let adhesive dry for 10 minutes.
7. Bond gasket to fairing flange.

INSTALL

1. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
2. **QA** Install engine fairing with screws and washers (Figure 1, Detail B).
3. If necessary, install block seal as follows:
   a. Apply adhesive, Item 14, WP 1803 00, to mating surfaces of block seal and clip. Let dry for 10 minutes.
   b. Position seal and bond to clip (Figure 1, Detail C).
   c. Seal front side of seal with sealing compound, Item 273, WP 1803 00.
4. Sand upper fuselage fairing and fuselage joining surfaces with abrasive cloth assortment, Item 1, WP 1803 00.

WARNING

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5. Wipe joining surfaces using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Before solvent dries, wipe surfaces with clean cheesecloth cloth, turning cloth with each wipe.
6. Apply sealing compound, Item 273, WP 1803 00, to fairing and fuselage joining surfaces.
7. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
8. **QA** Install upper fuselage fairing and fasten with screws and washers (Figure 1, Detail B).

INSTALL - Continued

9. Apply sealing compound, Item 273, WP 1803 00, around top of fairing, all edges, fasteners, and bottom where fairing meets fuselage frame.
10. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
11. **QA** Install inlet shroud fairing with screws, washers, and bolt. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**
12. If removed, install rubber seal.
13. Install engine cowling (WP 0313 00).
14. Check that engine cowling does not bind against fairing. If it does, do this:
    a. Remove engine cowling (WP 0313 00).
    b. Slit fairing lengthwise and remove piece of fairing to shorten distance between top of fairing and mounting screws (WP 0379 00).
    c. Repaint fairing (WP 0381 00and TM 55-1500-345-23).
    d. Install engine cowling (WP 0313 00).
15. Install troop/cargo door (WP 0244 00).
16. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
17. **QA** Install upper door track fairing between STA 420.0 and STA 428.0 with screws and washers (Figure 1, Detail A).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR LOCK

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Sealing Compound, Item 280, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

NOTE

Lock cylinders are available only in sets. When replacing troop/cargo door lock cylinder, replace lock cylinders on all doors.

REMOVE

1. Unlock troop/cargo door.
2. Remove screws attaching outside handle to door. Remove handle (Figure 1).
3. Remove snapring and washers fastening handle to escutcheon (Figure 1, Detail A). Remove escutcheon from handle.
4. Gently tap lock case with rubber mallet to remove it from handle.
5. Remove two pins holding shell in lock case. Remove shell from lock case.
6. Using thin bladed screwdriver, depress retaining ring into cylinder groove and remove cylinder from shell.

INSTALL

1. Compress retaining ring, insert cylinder into shell, and snap into place (Figure 1, Detail A).
2. Insert shell into lock case and install two pins. Pins must be flush with surface of lock case.
3. Install lock case in handle.
4. Install washer on handle and handle in escutcheon.
5. **QA** Install two washers and snapring on handle.
6. Place handle on door. Coat threads of screws with sealing compound, Item 280, WP 1803 00, and install (Figure 1).
7. Open and close door a few times to check operation.

INSTALL - Continued

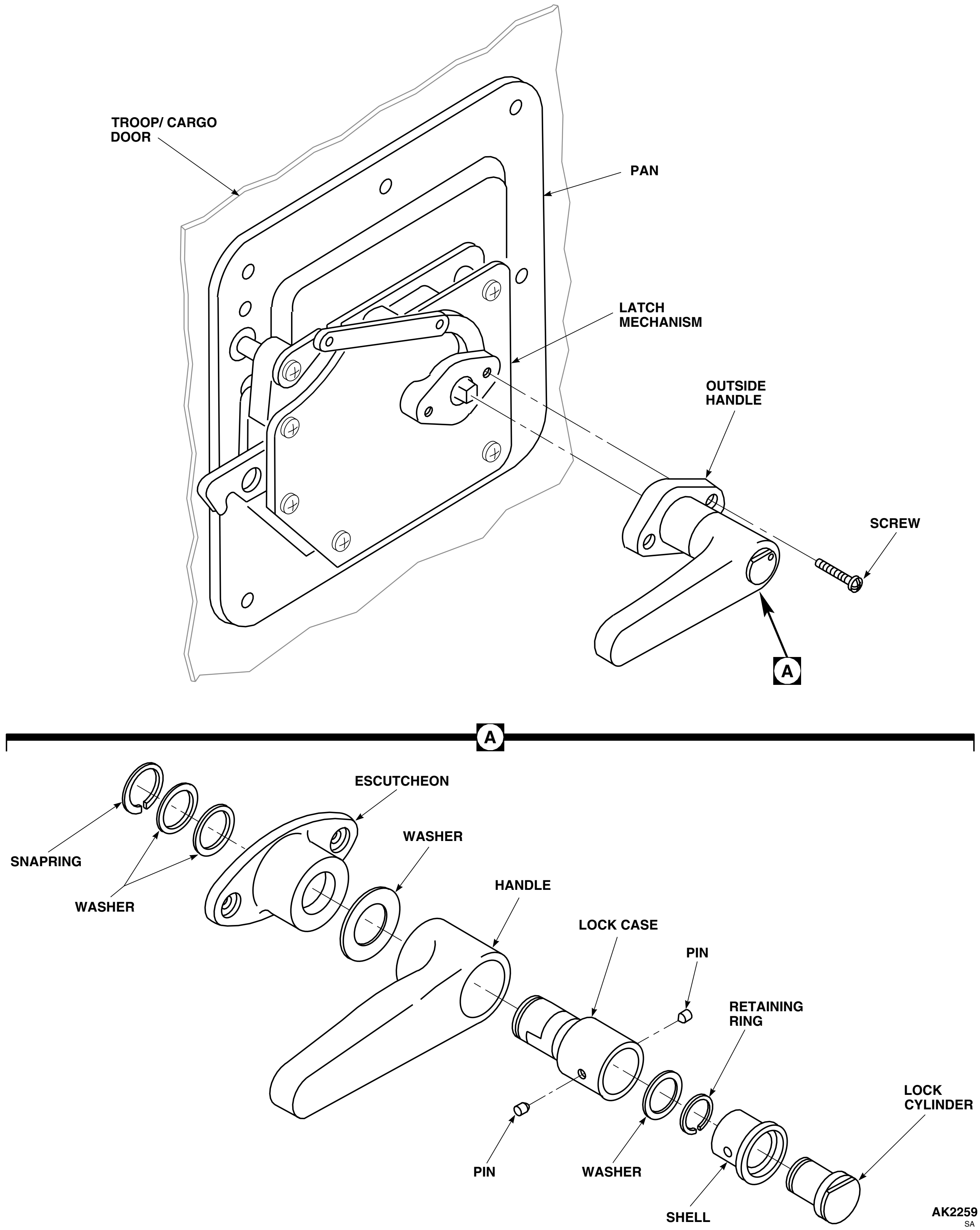

Figure 1. Replace Troop/Cargo Door Lock.

INSTALL - Continued

8. Close and lock troop/cargo door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR WINDOW JETTISON MECHANISM

This WP supersedes WP 0256 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0245 00
WP 1803 00

DISASSEMBLE MECHANISM

1. If windows are installed, remove them from troop/cargo door (WP 0245 00).
2. Remove screws and washers and remove front and rear access covers from inside of troop/cargo door (Figure 1, Sheet 1, Detail A).
3. Disconnect long arm and link from front lever in rear window by removing pin.
4. Disconnect link from rear lever in rear window by removing pin.
5. Disconnect long arm and link from rear lever in front window by removing pin and remove arm.
6. Disconnect link from front lever in front window by removing pin.
7. Remove arm from troop/cargo door by removing screws.
8. Remove rivets from front window lever supports and remove levers.
9. Remove rivets from rear window lever supports and remove levers.

NOTE

- Disassembly of all levers is the same.
- When removing pin from lever, Teflon washers will fall from pin.

10. Remove cotter pin and slide pin from supports, washers, and lever (Figure 1, Sheet 2, Detail E and Detail F).
11. Keep Teflon washers for reassembly.

DISASSEMBLE ARM

1. Remove rivets from spring supports and arm and remove supports (Figure 1, Sheet 2, Detail B).

DISASSEMBLE ARM - Continued

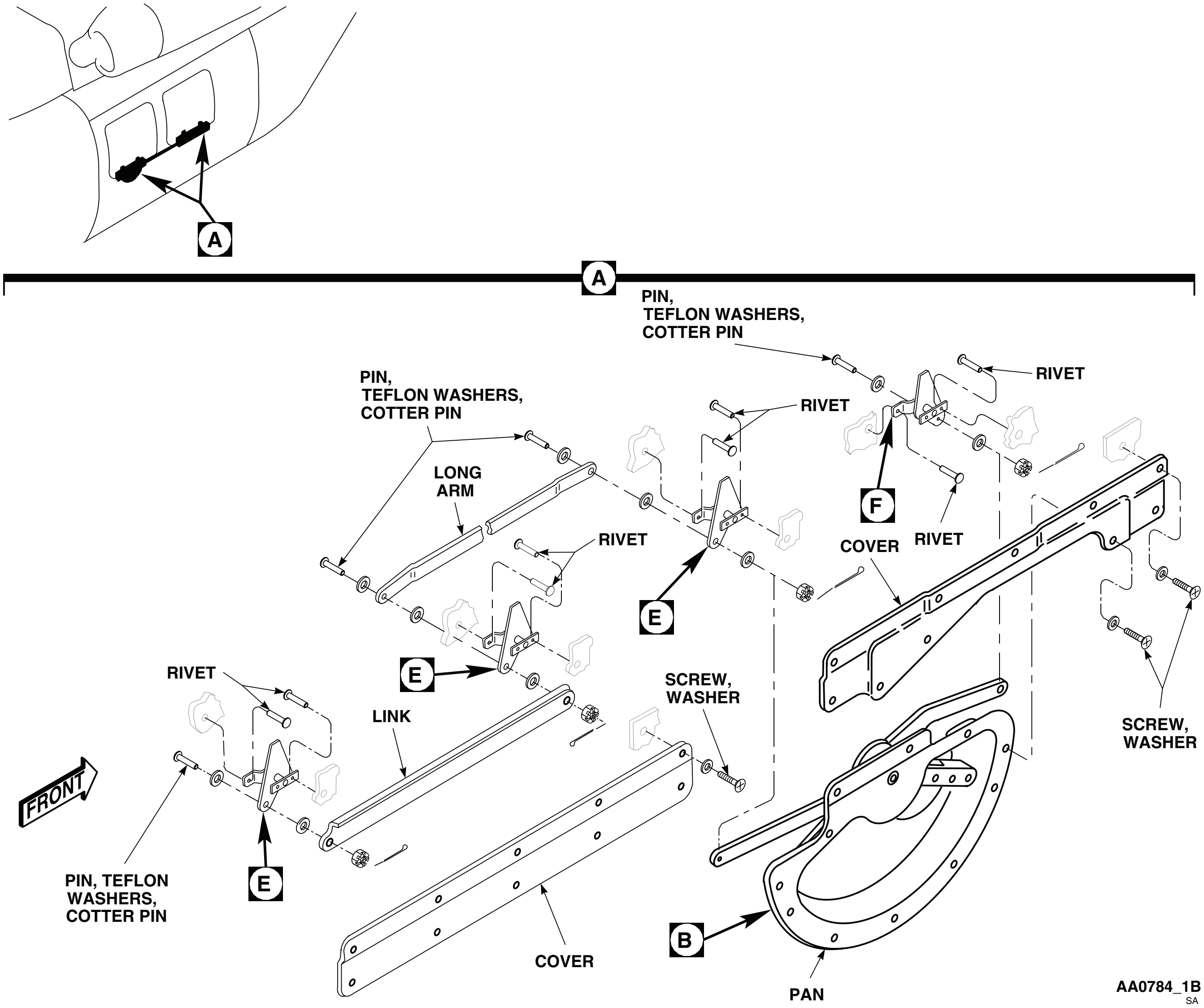

Figure 1. Troop/Cargo Door Window Jettison Mechanism. (Sheet 1 of 2)

NOTE

When removing screw from center weld, plastic and Teflon washers will fall from screw. Do not lose washers.

2. To remove center weld from arm, 70217-02708-049, do this (Figure 1, Sheet 2):
   a. Remove rivets from center plate and arm.
   b. Remove nut from screw and remove center plate from center weld.
   c. Remove screw and spacer from center plate.
3. Remove center weld from arm, 70217-02708-046, by removing screw from center weld and support.
4. Remove screw and spacer from support.

DISASSEMBLE ARM - Continued

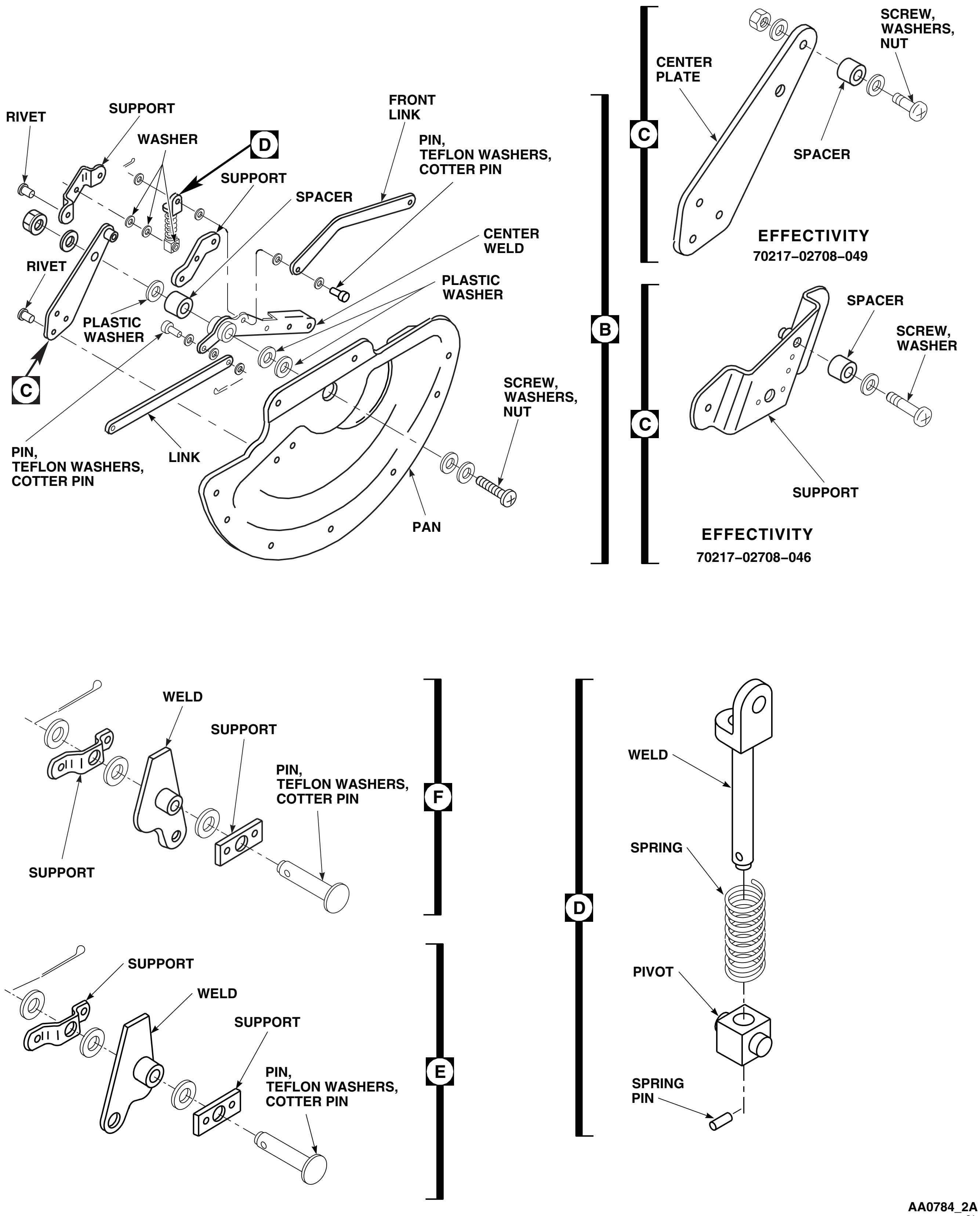

Figure 1. Troop/Cargo Door Window Jettison Mechanism. (Sheet 2 of 2)

DISASSEMBLE ARM - Continued

5. Disconnect spring and rear link from center weld.
6. Disconnect front link from center weld.
7. Remove center weld from arm.
8. Remove spring pin from weld and remove spring and pivot.

ASSEMBLE ARM

1. **QA** Install spring and pivot on weld and install spring pin in bottom of weld (Figure 1, Sheet 2, Detail D).
2. Slide screw through washers, arm, jettison handle, and spacer (Figure 1, Sheet 2, Detail B).
3. Install front link on jettison handle as follows:
   a. Install Teflon washer under head of pin.
   b. Install pin (head of pin outboard from arm) in jettison handle.
   c. **QA** Install Teflon washer, rear link, and Teflon washer on pin. Install cotter pin.
4. Install front link and spring on jettison handle as follows:
   a. Install Teflon washer under head of pin and install pin in link.
   b. Install Teflon washer on pin, and install pin with link (head of pin toward arm) in jettison handle.
   c. **QA** Install Teflon washer, spring assembly, and Teflon washer on pin. Install cotter pin.
5. Install jettison handle on arm as follows:

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   a. If washer became unbonded from arm, clean old adhesive from arm, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper.
   b. Bond washer to arm using adhesive, Item 11, WP 1803 00.
   c. Install washer under head of screw and install screw in arm.
   d. Install plastic washers on screw.
   e. Install jettison handle, spacer, and plastic washer on screw.
   f. **QA** Install screw, washers, spacer, and nut on center plate (Figure 1, Sheet 2, Detail C).
   g. Install center plate (with screwhead facing arm) on jettison handle.
   h. Line up rivets holes in center plate and arm and install rivets (TM 1-1500-204-23).
6. Install washer and nut on screw.

ASSEMBLE MECHANISM

1. Assemble two levers for front window and one lever for rear window as follows (Figure 1, Sheet 1, Detail A):
   a. Slide pin through flat surface and install Teflon washer on pin.

ASSEMBLE MECHANISM - Continued

b. Install weld on pin with cam facing left.

c. **QA** Install Teflon washer, support, and Teflon washer on pin. Install cotter pin.

2. Assemble one lever for rear window as follows:

a. Slide pin through flat support and install Teflon washer.

b. Install weld on pin with cam facing right.

c. **QA** Install Teflon washer, support, and Teflon washer on pin. Install cotter pin.

3. Install levers in rear window of troop/cargo door as follows:

a. Position lever in rear slot of rear window, with cam facing rear.

b. Install round head rivets from outside of door and flush head rivets from inside of door (TM 1-1500-204-23).

c. Position lever in front slot of rear window, with cam facing front.

d. **QA** Install round head rivets from outside of door, and flush head rivets from inside of door (TM 1-1500-204-23).

4. To install levers in front window of troop/cargo door do this:

a. Position levers in front and rear slots of front window, with cams facing front.

b. Install round head rivets from outside of door, and flush head rivets from inside of door.

5. Install arm on door using screws and washers.

6. Connect link to front lever in front window with pin (head facing out), Teflon washers, and cotter pin.

7. Slide long arm in center of troop/cargo door.

8. Connect long arm and link to rear lever in front window as follows:

a. Install Teflon washer under head of pin.

b. Install pin (head facing out) through arm, Teflon washer, and lever.

c. **QA** Install Teflon washer, link, and Teflon washer. Install cotter pin.

9. Connect link to rear lever in rear window as follows:

a. Install Teflon washer under head of pin and install pin (head facing out) through lever.

b. Install Teflon washer, link, Teflon washer, and cotter pin.

10. Connect long arm and link to front lever in rear window as follows:

a. Install Teflon washer under head of pin and install pin (head facing out) through long arm.

b. Install Teflon washer on pin and install pin through lever.

c. **QA** Install Teflon washer, link, Teflon washer, and cotter pin.

11. **QA** Install front and rear access covers, using screws and washers.

12. If windows were removed, install them (WP 0245 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TROOP/CARGO DOOR WINDOW JETTISON HANDLE GUARD/WINDOW JETTISON HANDLE STRAP

This WP supersedes WP 0257 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 31, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Fastener Tape, Hook, Item 124, WP 1803 00
Fastener Tape, Pile, Item 125, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REPLACE HANDLE GUARD

Remove

Pull handle guard from door (Figure 1, Detail A).

Repair

1. Peel hook fastener tape from guard and pile fastener tape from jettison arm and access cover, using a nonmetallic scraper (Figure 1, Detail A).

WARNING

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2. Remove all old adhesive with cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Measure and cut hook fastener tape, Item 124, WP 1803 00, and pile fastener tape, Item 125, WP 1803 00, to length (Figure 1, Detail A). Trim tapes to match upper and front edges of handle guard.
4. Brush a thin even coat of adhesive, Item 31, WP 1803 00, on arm/cover and pile fastener tape bonding surfaces. Let adhesive air-dry for at least 10 minutes.
5. Carefully press pile fastener tape into position on arm and cover. Allow 16 hours for adhesive to dry before handling.
6. Brush a thin even coat of adhesive, Item 31, WP 1803 00, on handle guard and hook fastener tape bonding surfaces. Let adhesive dry for at least 10 minutes.
7. Carefully press hook fastener tape into position on guard. Allow 16 hours for adhesive to dry before handling.
8. Cut 0.5-inch diameter holes in both hook and pile fastener tape to match holes in handle guard.

REPLACE HANDLE GUARD - Continued

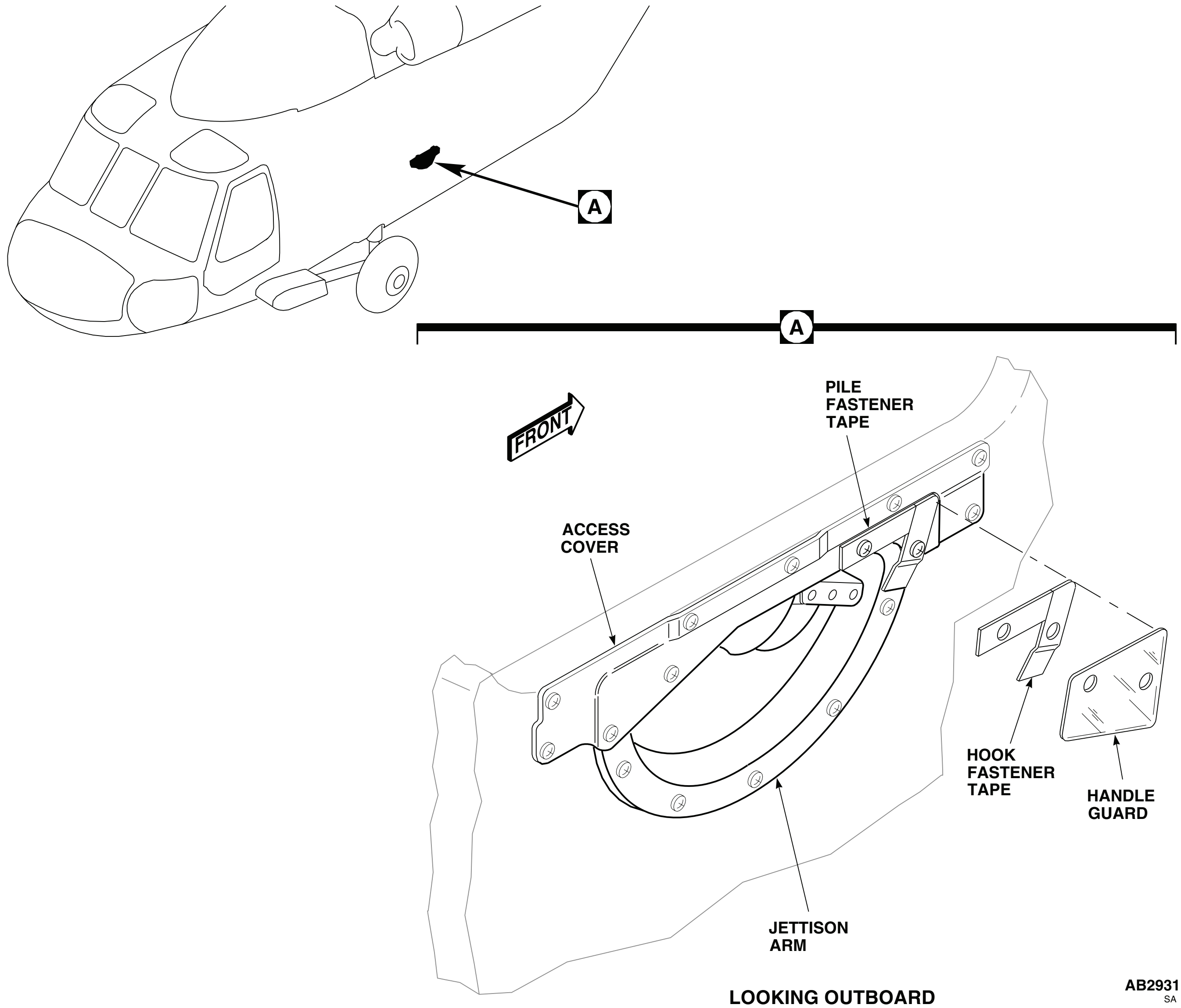

Figure 1. Replace Troop/Cargo Door Window Jettison Handle Guard.

Install

Install handle guard by pressing in place, engaging hook and pile fastening tape.

REPLACE HANDLE STRAP

Remove

Pull handle strap to free emergency release lever (Figure 2, Detail A).

Repair

1. Peel hook fastener tape and handle strap from jettison arm, using a nonmetallic scraper (Figure 2, Detail A).

REPLACE HANDLE STRAP - Continued

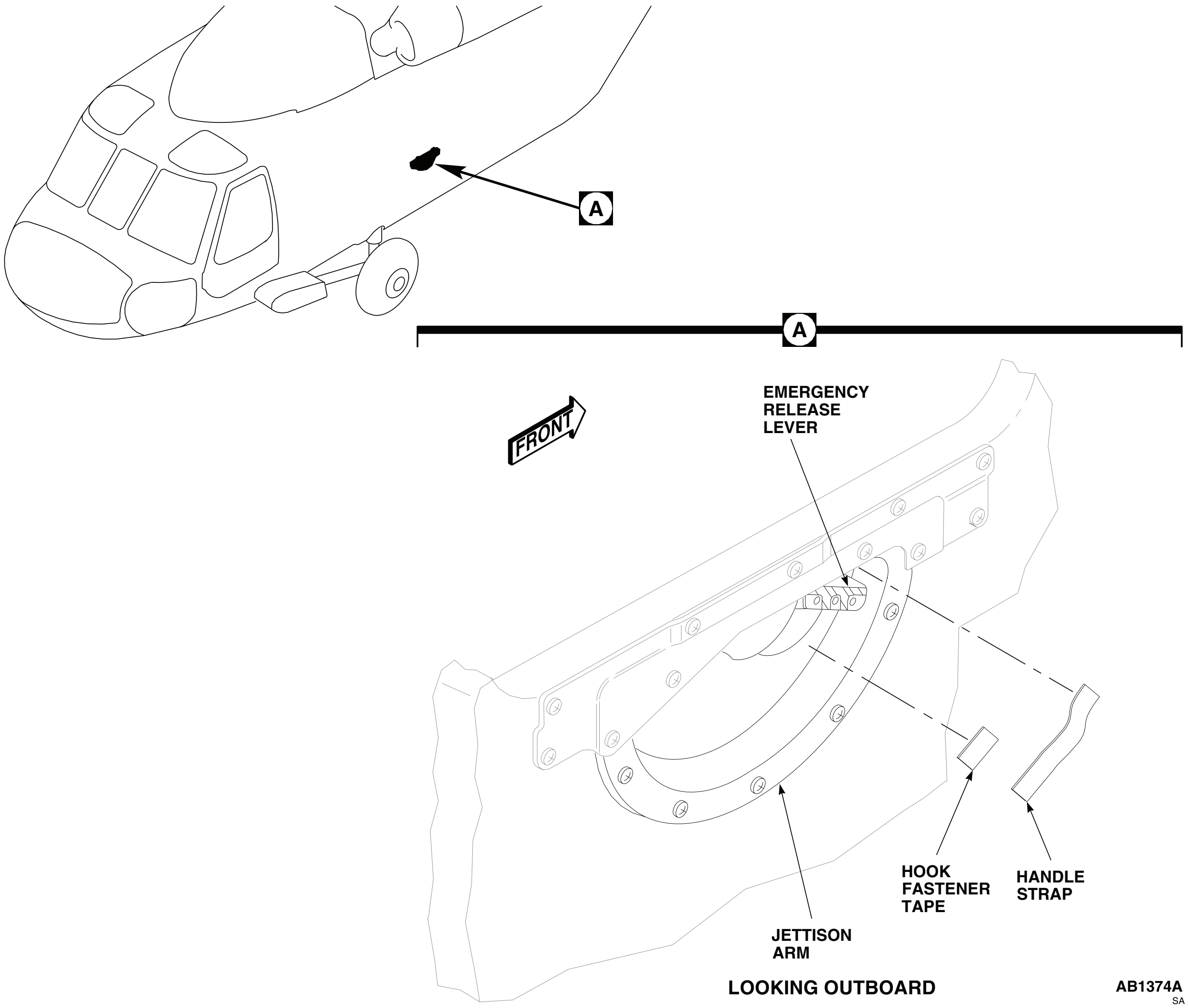

Figure 2. Replace Troop/Cargo Door Window Jettison Handle Strap.

WARNING

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2. Remove all old adhesive with cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

3. Measure and cut hook fastener tape, Item 118, WP 1803 00, to length (Figure 2, Detail A). Trim tape to match length of pile fastener tape of handle strap.

REPLACE HANDLE STRAP - Continued

4. Brush a thin even coat of adhesive, Item 31, WP 1803 00, on hook fastener tape bonding surface. Let adhesive air-dry for at least 10 minutes.
5. Carefully press hook fastener tape into position on jettison arm. Allow 16 hours for adhesive to dry before handling.
6. Brush a thin even coat of adhesive, Item 31, WP 1803 00, on handle strap bonding surfaces on jettison arm. Let adhesive dry for at least 10 minutes.
7. Carefully press handle strap on jettison arm. Allow 16 hours for adhesive to dry before handling.

Install

Position handle strap over emergency release lever and press in place, engaging hook and pile fastener tape (Figure 2, Detail A).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

SOUNDPROOFING PANELS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0233 00

REMOVE

1. Remove troop/gunner's seat upper support fittings, if required (WP 0233 00).

NOTE

Removal of some soundproofing panels may require removal of various cabin furnishings/equipment. Refer to applicable task for removal of these particular items.

2. Remove soundproofing panels from cabin, ceiling, rear bulkhead, and sidewalls by removing attaching hardware and loosening 1/4 turn fasteners.

INSTALL

CAUTION

To avoid possible interference between flight control rods and soundproofing panels, make sure panel over vertical flight control rods overlaps cover (STA 247.0, WL 206.75 through WL 215.0).

1. Install soundproofing panels in cabin by putting panel in place and fastening in place with attaching hardware and 1/4 turn fasteners.
2. Install troop/gunner's seat upper support fittings, if required (WP 0233 00).
3. Reinstall any removed furnishings/equipment.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN TRANSMISSION DRIP PAN EH-60A

This WP supersedes WP 0259 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 10, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 29, WP 1803 00
Brush, Acid Swabbing, Item 57, WP 1803 00
Cloth, Abrasive, Item 79, WP 1803 00
Ethyl Alcohol, Absolute, ACS, Item 122, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0233 00
WP 0258 00
WP 0261 01
WP 0288 00
WP 0290 00
WP 0379 00
WP 1803 00

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES

Remove

NOTE

For DP3-5-01 Drip Pan, refer to WP 0261 01.

1. Remove soundproofing panels at STA 308.0, left and right side, and at STA 383.0 left side (WP 0258 00).
2. UH-60A 79-23318 - 82-23747 Remove support clamps, screws, washers and nuts holding drain tube(s) to overhead panels (Figure 1, Sheet 1, Detail A).
3. Unclamp tube(s) from drip pan and bulkhead fitting(s). Remove tube(s). Remove hose clamps from tube(s).
4. UH-60A 84-23970 - SUBQ Remove main landing gear strut fairings (WP 0288 00 or WP 0290 00).
5. Working inside strut well, remove support clamps, screws, washers, spacers, and nuts holding drain tubes to STA 308.0 frame.
6. Remove hose clamps and tube(s) from bulkhead fittings. Remove drain tubes.
7. Remove clamps from drain tube(s) (Figure 1, Sheet 1 and Figure 1, Sheet 2).

Install

1. UH-60A 79-23318 - 82-23747 Slip support clamps onto replacement tubes.
2. Install hose clamps on each end and install tubes on bulkhead fittings. Tighten hose clamps.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

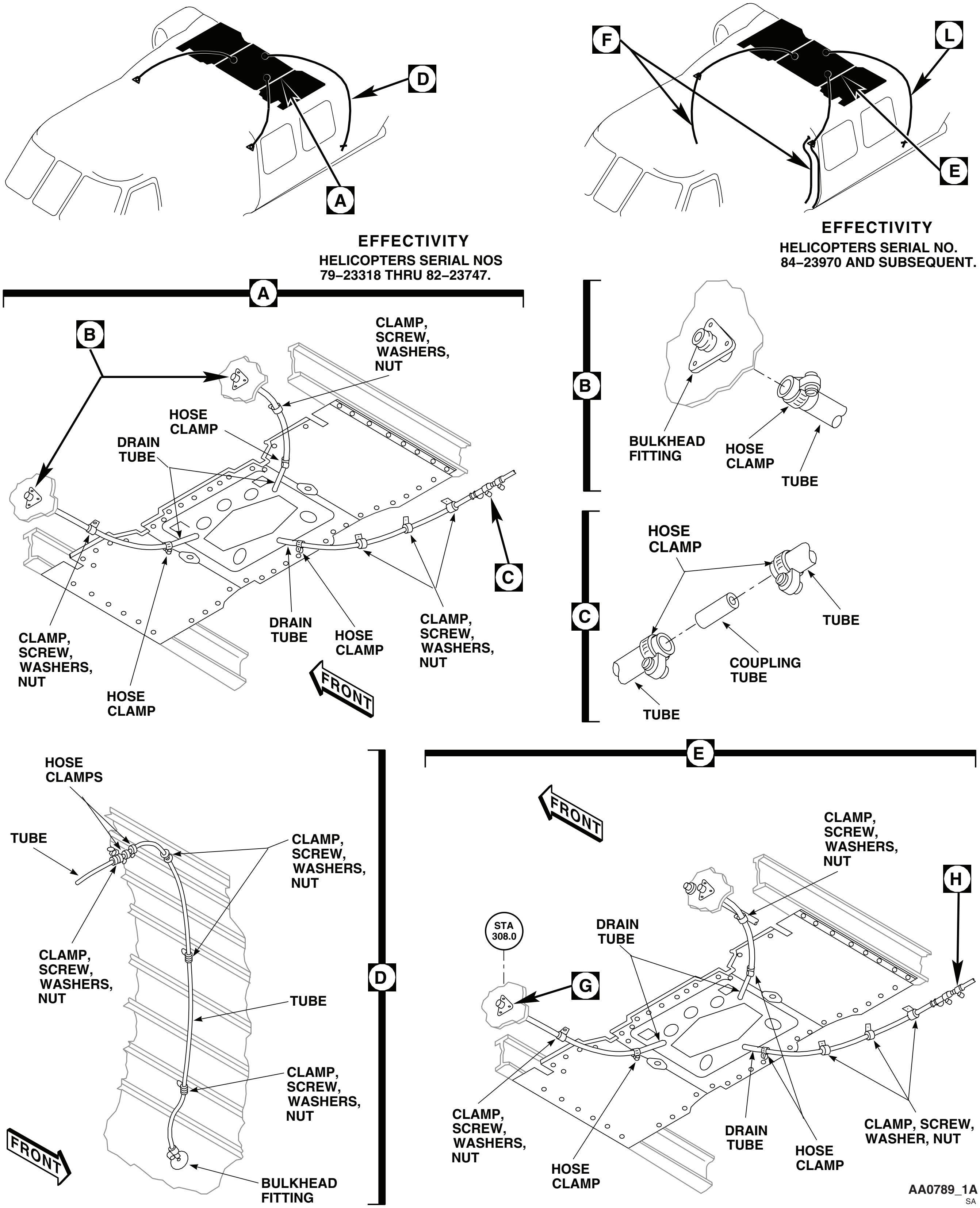

Figure 1. Replace Main Transmission Drip Pan Drain Tubes. (Sheet 1 of 2)

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

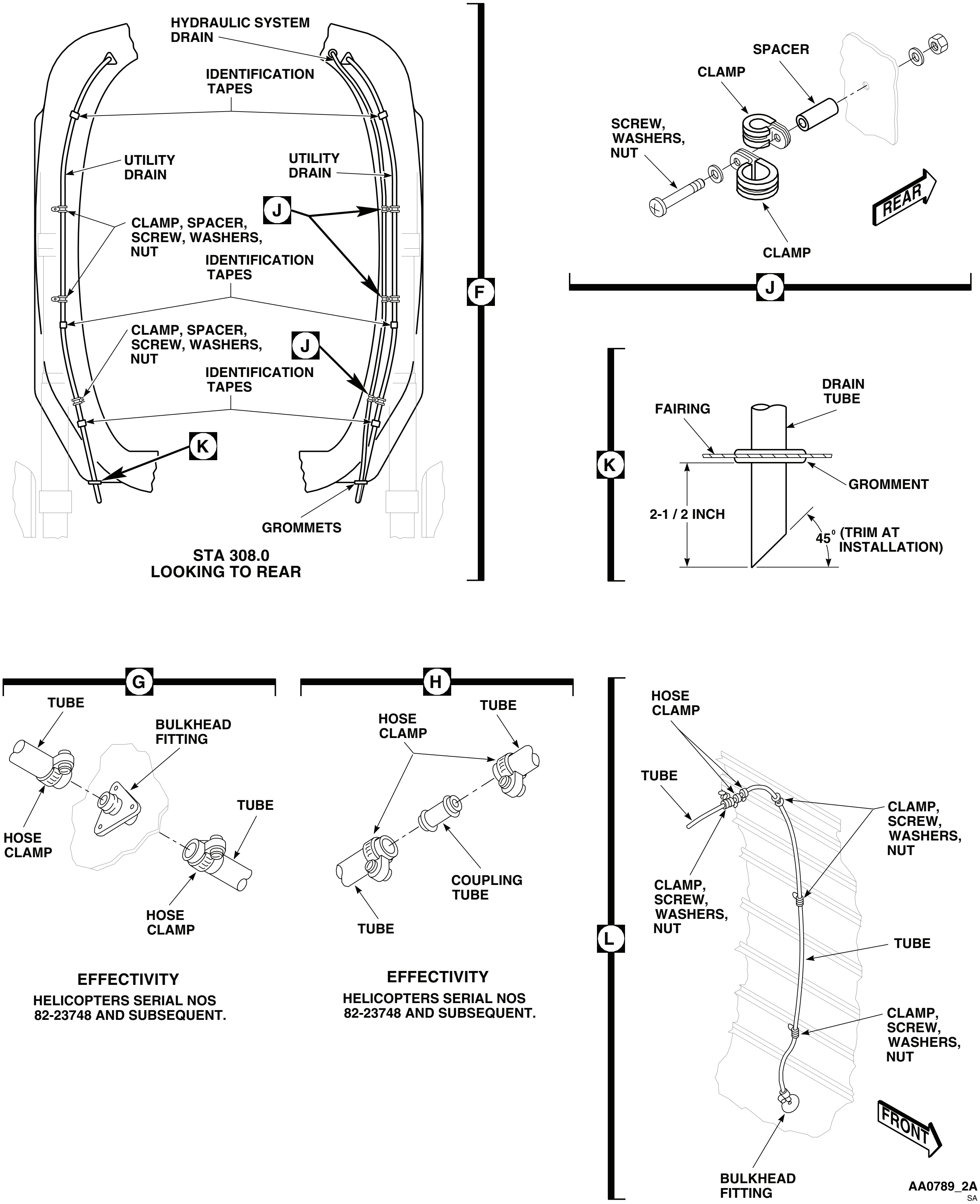

Figure 1. Replace Main Transmission Drip Pan Drain Tubes. (Sheet 2 of 2)

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

3. **UH-60A 84-23970 - SUBQ** Work inside strut well and install clamps on STA 308.0 frame with screws, washers, spacers, and nuts (Figure 1, Sheet 2, Detail J).
4. Install main landing gear strut fairings (WP 0288 00 or WP 0290 00). Slip tubing through grommet.
5. Leave 2 1/2-inches of tubing extended below fairing. Cut away extra length to make a 45 degree opening facing rear (Figure 1, Sheet 2, Detail K).
6. Slip support clamps onto overhead tubes.
7. Install hose clamps on each end. Install tubes on drip pan and bulkhead fittings. Tighten hose clamps.
8. Fasten support clamps to overhead with screws and washers. Install washers under clamp as necessary to keep drain tube sloping from drip pan fitting to bulkhead fitting.
9. Apply utility drain identification tape (Figure 1, Sheet 2, Detail F) (TM 1-1500-204-23).
10. Make sure area is clean and free of foreign material before closing up.
11. Install soundproofing (WP 0258 00).

REPLACE MAIN TRANSMISSION DRIP PAN AND SUPPORT

Remove

NOTE

To allow removal of drip pan without removing upper seat support a slot at left/right corners, 1.60-inches in from outboard edge, 2.20-inches wide by 5.20-inches deep may be cut. Repair edges of slot per procedures in WP 0379 00.

1. Remove troop/gunner's seat upper supports (WP 0233 00).
2. Loosen hose clamps and remove drain tubes from drip pan drain fittings. Refer to REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES, in this work package.
3. Support drip pan. Turn studs fastening drip pan to support (Figure 2). Remove drip pan.
4. If required, remove rivets and remove support from beams. Scrape sealing compound from beams using nonmetallic scraper.

Install

1. Seal all overlapping surfaces between beam flanges and support with sealing compound Item 273, WP 1803 00.
2. If required, install support on beams and fasten with rivets (Figure 2).
3. Line up stud holes of replacement drip pan with receptacles in support. Fasten drip pan to support.
4. Install drain tubes on drip pan drain fittings and tighten hose clamps. Refer to REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES, in this work package.
5. Install troop/gunner's seat upper support (WP 0233 00).

REPLACE MAIN TRANSMISSION DRIP PAN SEALS

Remove

1. Remove drip pan from support. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN AND SUPPORT, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS - Continued

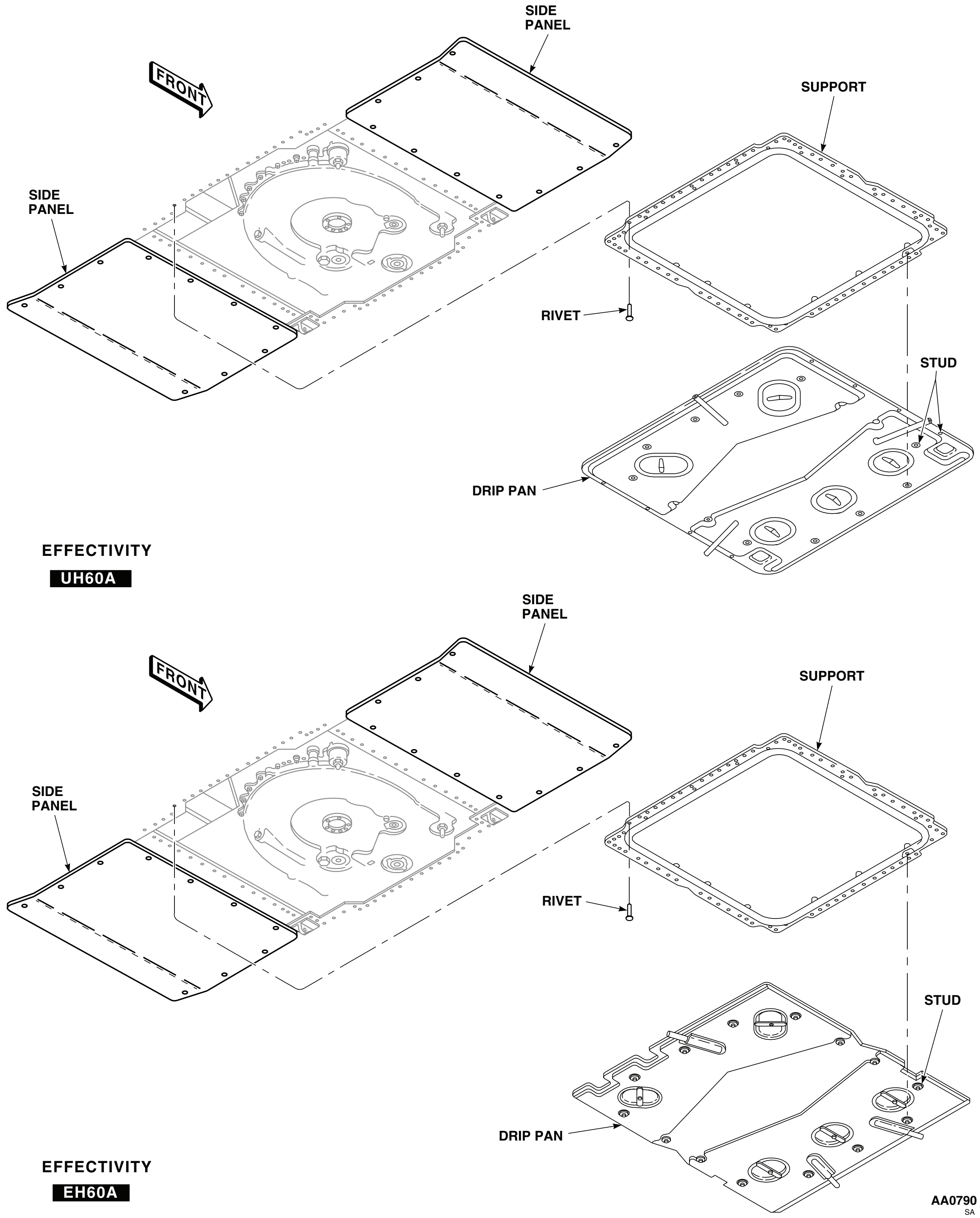

Figure 2. Replace Main Transmission Drip Pan Drip Pan and Support.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS - Continued

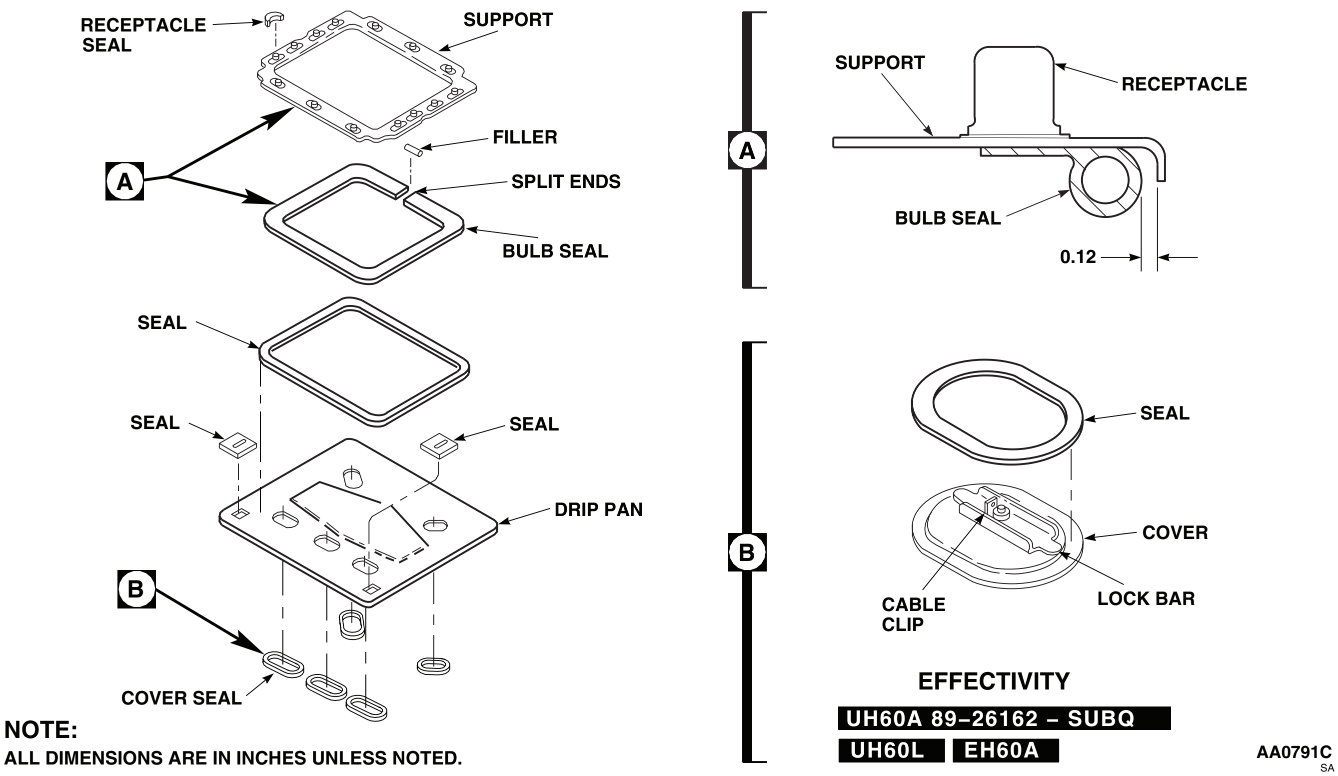

Figure 3. Replace Main Transmission Drip Pan Seals.

2. Scrape off old seals from drip pan using a nonmetallic scraper (Figure 3).

WARNING

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3. Clean off any remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

Install

1. Clean replacement seals using machinery wiping towel, Item 344, WP 1803 00, moistened with ACS absolute ethyl alcohol, Item 122, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

2. Thoroughly scuff up bonding surface of seals and drip pan with abrasive cloth, Item 79, WP 1803 00. Wipe off all dust with a clean machinery wiping towel, Item 344, WP 1803 00.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS - Continued

WARNING

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3. Do a final cleaning of surfaces to be bonded with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Within 15 minutes after cleaning surfaces to be bonded, brush one coat of adhesive, Item 14, WP 1803 00, on drip pan and one coat on replacement seal.
5. Allow adhesive to air-dry for 10 to 15 minutes.
6. Install replacement seal(s) on drip pan. Join together firmly while making sure entire mating surfaces are in direct contact. Do not move or disturb bonded surfaces after mating (Figure 3).

WARNING

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7. Wipe off extra adhesive with a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
8. Cure time for adhesive is at least 12 hours, after which drip pan may be reinstalled. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN AND SUPPORT, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN COVER SEALS

Remove

NOTE

UH-60A 89-26162 - SUBQ UH-60L EH-60A Cover seals are installed on the covers instead of on the drip pan.

1. Remove cover from drip pan and disconnect retaining cable (Figure 3, Detail B).
2. Scrape off old seal from cover, using nonmetallic scraper.

WARNING

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3. Clean off any remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean machinery wiping towel, Item 344, WP 1803 00.

REPLACE MAIN TRANSMISSION DRIP PAN COVER SEALS - Continued

Install

1. Scuff up bonding surface of seal with abrasive cloth, Item 79, WP 1803 00.
2. Wipe off all dust with clean machinery wiping towel, Item 344, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. Clean seal bonding surface with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Let air dry.

NOTE

W/O MWO 50-54 UH-60L 89-26149, 89-26154 UH-60L 89-26179 - 92-26420 EH-60A Use adhesive, Item 10, WP 1803 00. **UH-60L 92-26421 - SUBQ** Use adhesive, Item 29, WP 1803 00.

4. Apply even coat of adhesive, Item 10 or adhesive Item 29, WP 1803 00, to bonding surfaces of both seal and cover, using clean acid swabbing brush, Item 57, WP 1803 00.
5. Install seal on cover and apply moderate pressure to ensure firm contact at all points (Figure 3, Detail B).

WARNING

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6. Wipe off excess adhesive with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. Cure adhesive for 24 hours at room temperature. For adhesive, Item 10, WP 1803 00, alternate curing time is 1 hour at 140°F(60°C) to 160°F(71°C).
8. Connect retaining cable to clip and install cover on drip pan.

REPLACE MAIN TRANSMISSION DRIP PAN SUPPORT SEAL

Remove

1. Remove drip pan and support from ceiling panels. Refer to REPLACE MAIN TRANSMISSION DRIP PAN AND SUPPORT, in this work package.
2. Scrape off old seal from support using a nonmetallic scraper (Figure 3, Detail A).

REPLACE MAIN TRANSMISSION DRIP PAN SUPPORT SEAL - Continued

WARNING

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3. Clean off any remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

Install

1. Clean replacement seal using machinery wiping towel, Item 344, WP 1803 00, moistened with ACS absolute ethyl alcohol, Item 122, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.
2. Thoroughly scuff up surface of seal and support to be bonded with abrasive cloth, Item 79, WP 1803 00. Wipe off all dust with a clean machinery wiping towel, Item 344, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. Do a final cleaning of surfaces to be bonded with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Before coating seal or support with adhesive, do this:
   a. Position bulb seal on support (Figure 3, Detail A).
   b. Place ends of seal at rear end of support half way between center of support and right side. Install filler, an equal distance in each end of bulb seal.
   c. Slit flat part of seal as necessary to allow seal to bend at corners.
   d. Punch or drill four 0.098-inch diameter holes through only outside surface of seal. A hole should be at about center of each side of support to allow seal to compress.
5. Coat surfaces of seal and support to be bonded with adhesive, Item 10, WP 1803 00.
6. Allow adhesive to air dry for about 1/2 to 1 hour, or until sticky.
7. Install seal on support. Join ends together firmly, while making sure entire mating surfaces are in direct contact. Do not move or disturb bonded surfaces after mating.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

8. Wipe off extra adhesive with a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
9. Air-cure bonded seal for at least 24 hours. For greatest strength, cure about 7 days.

REPLACE MAIN TRANSMISSION DRIP PAN SUPPORT SEAL - Continued

10. Cure time can be cut short by applying heat at 110°F(43°C) to 130°F(54°C) for 16 hours.
11. Install support. Refer to REPLACE MAIN TRANSMISSION DRIP PAN AND SUPPORT, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN TRANSMISSION DRIP PAN UH-60A 77-22714-79-23317SUBQ

This WP supersedes WP 0260 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit SC 5180-99-B01
Airframe Repairer's Toolkit SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Abrasive, Item 79, WP 1803 00
Ethyl Alcohol, Absolute, ACS, Item 122, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0258 00
WP 0261 01
WP 1803 00

REPAIR OR REPLACEMENT

NOTE

For DP3-5-01 Drip Pan, refer to WP 0261 01.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES

Remove

1. Remove soundproofing panels at STA 308.0, left and right side, and at STA 383.0, left side (WP 0258 00).
2. Between drip pan fittings and side bulkhead fittings, remove support clamps, screws, and washers holding drain tube(s) to overhead panels (Figure 1, Detail A).

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

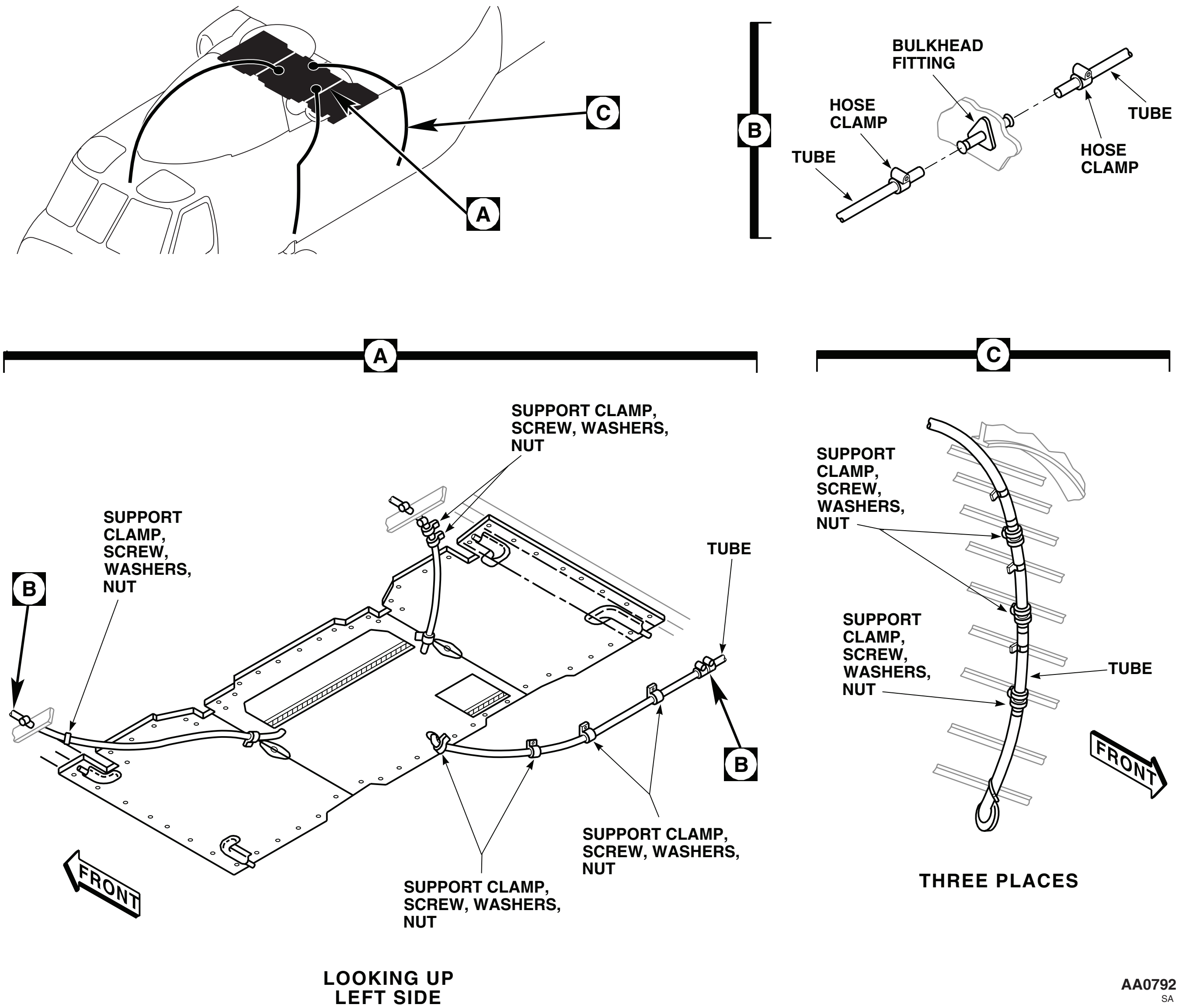

Figure 1. Replace Main Transmission Drip Pan Drain Tubes.

3. Unclamp tube(s) from drip pan fitting(s) and bulkhead fitting(s) (Figure 1, Detail B). Remove tube(s) from fittings. Remove hose clamps from tube(s).
4. Remove support clamps, screws, washers, nuts holding drain tubes to side frame (Figure 1, Detail C). Unclamp tubes from upper and lower bulkhead fittings. Remove drain tubes from fittings. Remove clamps from drain tubes.

Install

1. **QA** Install hose clamps on each end of replacement side drain tube(s). Install tube(s) on upper and lower bulkhead fittings (Figure 1, Detail B). Tighten hose clamps.
2. Install support clamps on drain tube(s) and install clamps on brackets with screws, washers, and nuts (Figure 1, Detail C).

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

3. Install hose clamps on each end of replacement overhead drain tube(s). Install tube(s) on drip pan and upper bulkhead fitting. Tighten hose clamps.
4. Install support clamps on drain tube and attach to overhead with screws and washers (Figure 1, Detail A). Install washers under clamps, as necessary, so that drain tube slopes from drip pan fitting to bulkhead fitting.
5. Make sure area is clean and free of foreign material before closing up.
6. Install soundproofing (WP 0258 00).

REPLACE MAIN TRANSMISSION DRIP PAN

Remove

1. Loosen hose clamps and remove drain tubes from drip pan drain fittings. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES, in this work package.

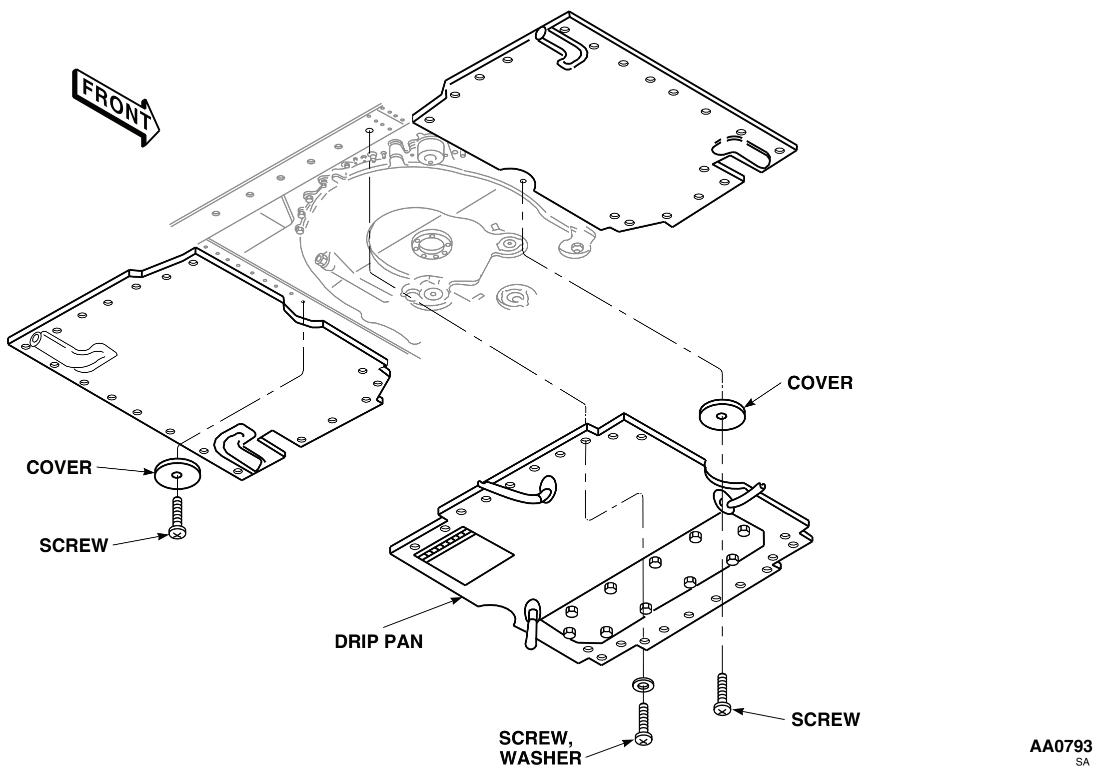

Figure 2. Replace Main Transmission Drip Pan.

2. Remove screws and remove covers from each side of drip pan (Figure 2).
3. Support drip pan and remove screws and washers attaching drip pan to overhead supports. Remove drip pan.

Install

1. Make sure area is clean and free of foreign material before installing drip pan.
2. **QA** Place drip pan up against supports, and line up screwholes with receptacles in supports (Figure 2). Install screws and washers.
3. **QA** Install covers on each side of drip pan and install screws.

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

4. Install drain tubes on drip pan drain fittings and tighten hose clamps. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS

Remove

1. Remove drip pan (door) from transmission support. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Scrape off old seal(s) from drip pan and mating surface of transmission support, using a nonmetallic scraper (Figure 3).

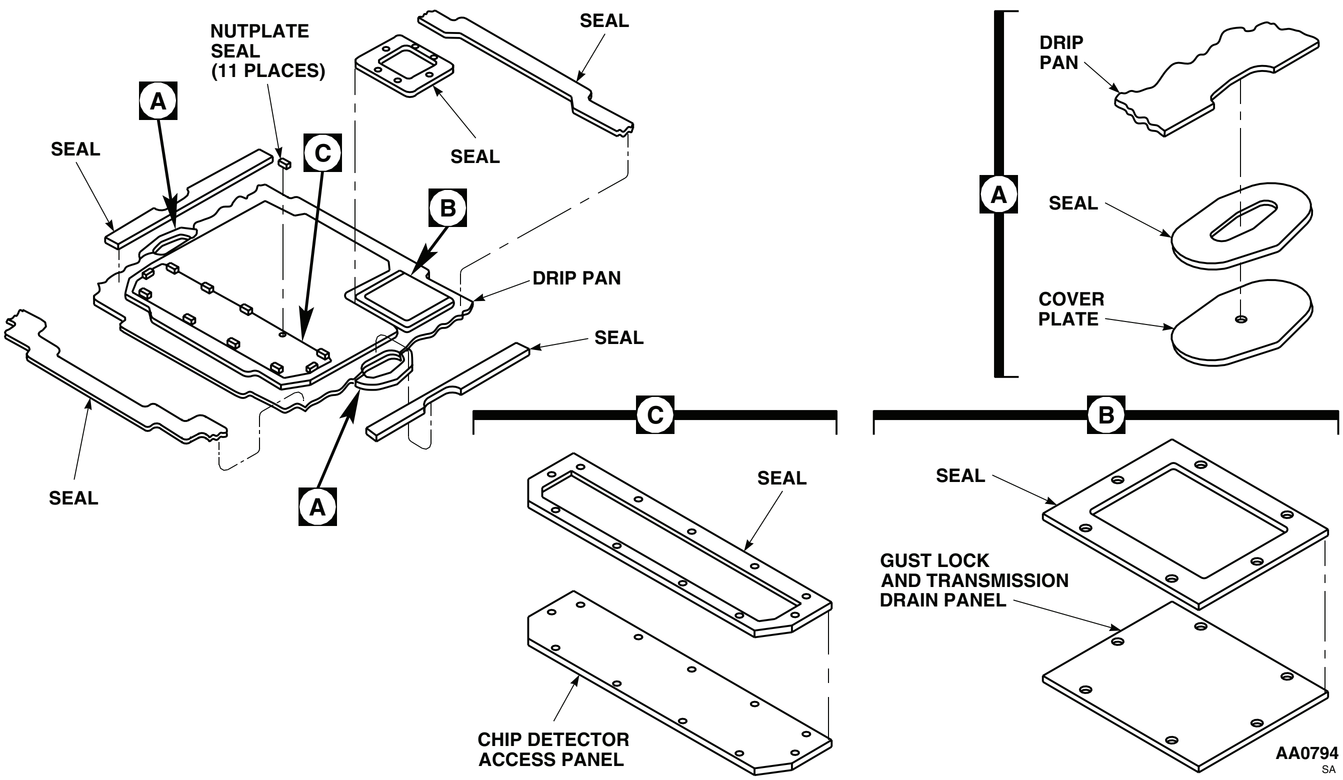

Figure 3. Replace Main Transmission Drip Pan Seals.

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean off any remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS - Continued

Install

1. Clean replacement seal(s) using machinery wiping towel, Item 344, WP 1803 00, moistened with ACS absolute ethyl alcohol, Item 122, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.
2. Thoroughly scuff up surface of seal(s) and drip pan to be bonded with abrasive cloth, Item 79, WP 1803 00. Wipe off all dust with a clean machinery wiping towel, Item 344, WP 1803 00.

WARNING

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3. Do a final cleaning of surfaces to be bonded with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Within 15 minutes after cleaning surfaces to be bonded, brush one coat of adhesive, Item 14, WP 1803 00, on drip pan, and one coat on replacement seal.
5. Allow adhesive to air-dry about 10 to 15 minutes.
6. Install replacement seal(s) on drip pan. Join together firmly while making sure entire mating surfaces are in direct contact (Figure 3). Do not move or disturb bonded surfaces after mating.

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Wipe off extra adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
8. Cure time for bonded seal(s) is at least 12 hours, after which drip pan may be reinstalled. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN TRANSMISSION DRIP PAN UH-60A 77-22722

This WP supersedes WP 0261 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/face shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 79, WP 1803 00
Ethyl Alcohol, Absolute, ACS, Item 122, WP 1803 00
Primer, Adhesive, Item 237, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0258 00
WP 0261 01
WP 1803 00

NOTE

For DP3-5-01 Drip Pan, refer to WP 0261 01.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES

Remove

1. Remove soundproofing panels from STA 308.0 and STA 385.0, left and right sides (WP 0258 00).
2. At two front corners and right rear corner of drip pan, remove screws and washers from drain tube covers on ceiling soundproofing panels (Figure 1, Detail B). Remove covers.
3. Loosen hose clamps at drip pan and bulkhead fittings, and remove tube(s).
4. At left rear corner of drip pan, loosen hose clamps at drip pan and bulkhead fittings (Figure 1, Detail C). Remove screws and washers and remove panel with tube from supports. Remove hose clamps.
5. If necessary to replace drain tube, remove rivets, remove supports from panel, and remove drain tube.
6. On side of drain tubes, remove support clamps holding drain tubes to side frame (Figure 1, Detail D).
7. At upper and lower fitting, loosen hose clamps and remove drain tubes. Remove hose clamps from tubes.

Install

1. Install hose clamp on each end of replacement side drain tube and install tubes on upper and lower bulkhead fittings. Tighten hose clamps (Figure 1, Detail D).
2. Install support clamps on drain tubes and install clamps on brackets with screws, washers, and nuts.
3. If removed, install replacement drain tube on panel (Figure 1, Detail C). Install supports over drain tube and on panel. Install rivets and washers in supports and panel with rivet head against support (TM 1-1500-204-23).

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

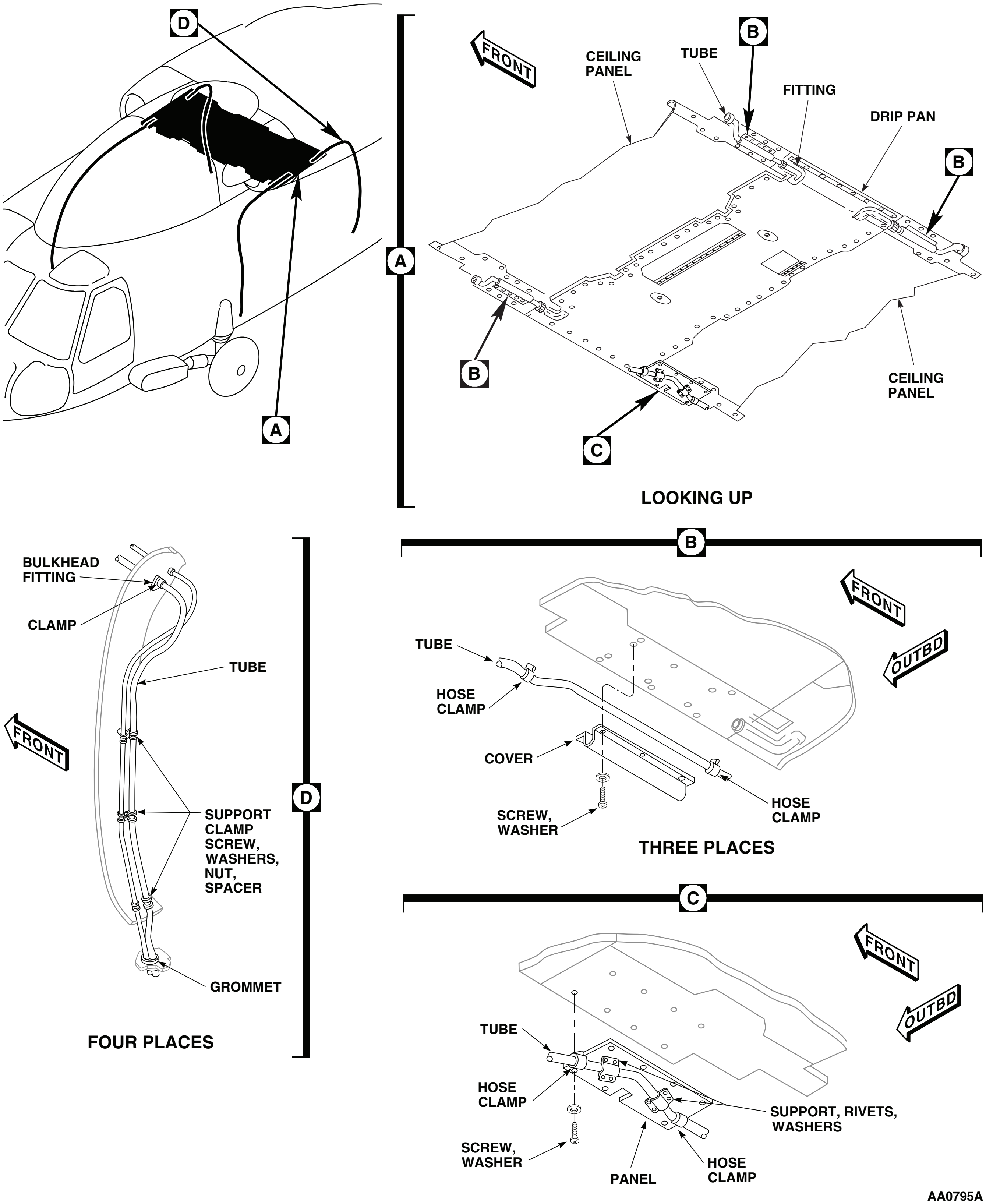

Figure 1. Replace Main Transmission Drip Pan Drain Tubes.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBES - Continued

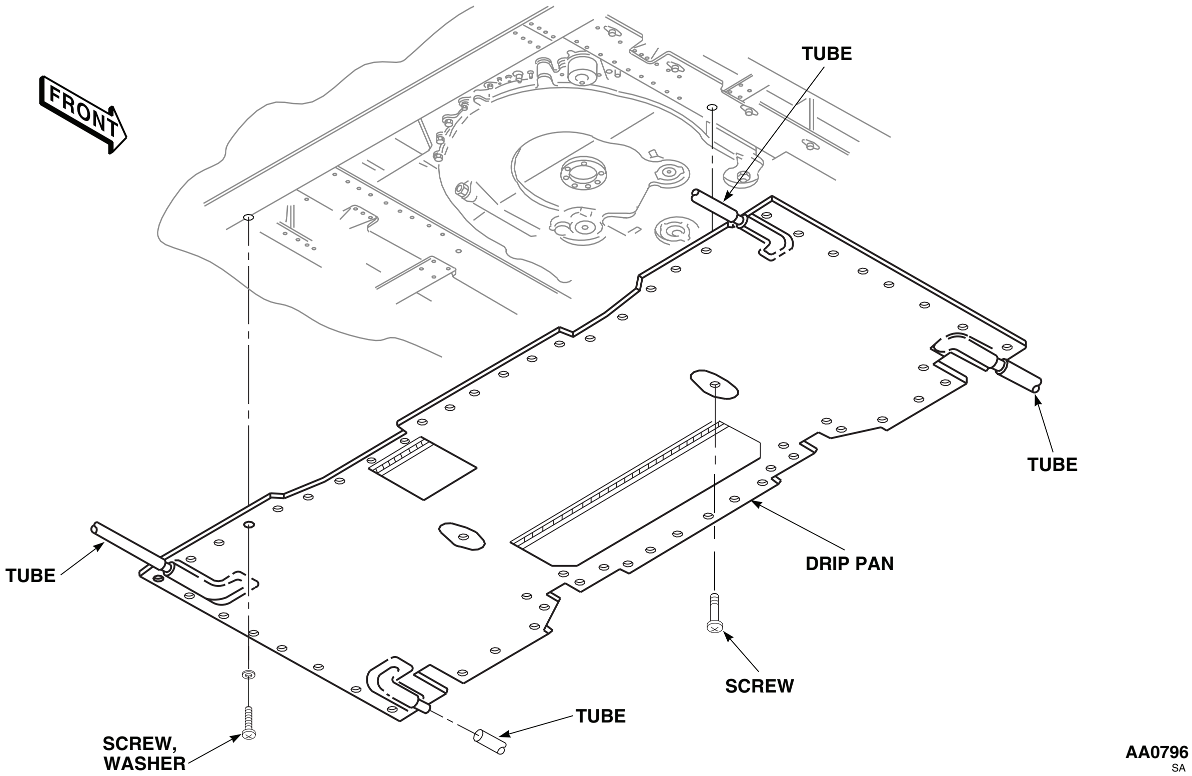

Figure 2. Replace Main Transmission Drip Pan.

4. Install hose clamps on drain tube. At left rear corner of drip pan, install panel to overhead supports with screws and washers. Install drain tube on drip pan and bulkhead fittings. Tighten hose clamps.
5. At two front corners and right rear corner of drip pan, install hose clamps on each end of replacement drain tube(s). Install drain tube(s) on drip pan fitting(s) and bulkhead fitting(s). Tighten hose clamps (Figure 1, Detail B).
6. Install cover(s) over drain tube(s) and secure with screws and washers.
7. Make sure area is clean and free of foreign material before installing soundproofing.
8. Install soundproofing panels (WP 0258 00).

REPLACE MAIN TRANSMISSION DRIP PAN

Remove

1. Remove screws and washers from drain tube covers on soundproofing (Figure 2).
2. Remove drain tubes from end fittings.
3. Remove screws and washers attaching drip pan to helicopter cabin overhead between STA 360.0 and STA 398.0. Remove drip pan.

Install

1. **QA** Line up screw holes of replacement drip pan with mating holes in transmission support. Install screws and washers (Figure 2).
2. Attach drain tubes to drip pan.

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

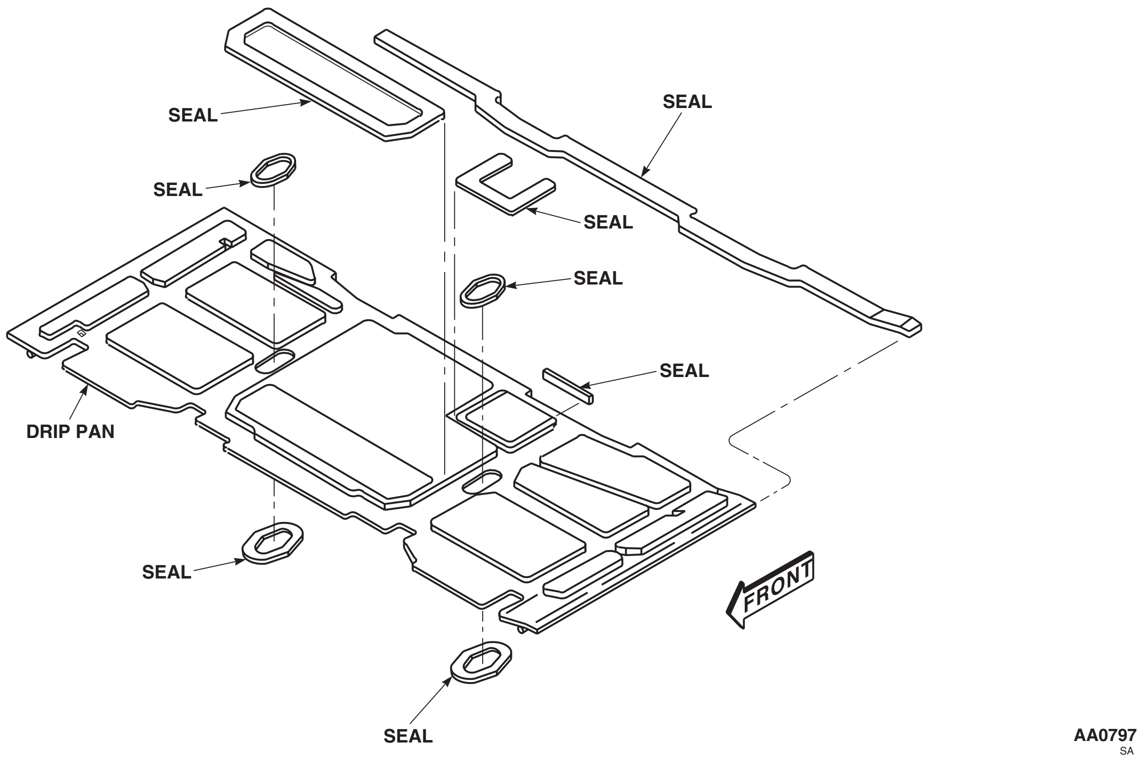

Figure 3. Replace Main Transmission Drip Pan Seals.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS

Remove

1. Remove drip pan from transmission support. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Scrape off old seal(s) from drip pan and mating surface of transmission support using a nonmetallic scraper (Figure 3).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. Clean off any remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

Install

1. Clean replacement seal(s) using machinery wiping towel, Item 344, WP 1803 00, moistened with ACS absolute ethyl alcohol, Item 122, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

REPLACE MAIN TRANSMISSION DRIP PAN SEALS - Continued

2. Thoroughly scuff up surface of seal(s) and drip pan to be bonded with abrasive cloth, Item 79, WP 1803 00. Wipe off all dust with a clean machinery wiping towel, Item 344, WP 1803 00.

WARNING

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3. Clean off any remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Coat replacement seal(s) on drip pan with adhesive primer, Item 237, WP 1803 00. Allow adhesive to air dry about 15 minutes.
5. Install replacement seal(s) on drip pan (Figure 3). Join together firmly while making sure entire mating surfaces are in direct contact. Do not move or disturb bonded surfaces after mating.
6. Wipe off extra adhesive using a machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. Cure time for bonded seal(s) is about 12 hours, after which drip pan may be reinstalled. Refer to REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN TRANSMISSION DRIP PAN DP3-5-01

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Nonmetallic Scraper

Material/Parts

Corrosion Resistant Coating, Item 110, WP 1803 00
Isopropyl Alcohol, Technical, Item 170, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Sealing Compound, Item 257, WP 1803 00
Tape, Pressure Sensitive Item 329, WP 1803 00
Towel Machinery, Wiping, Item 344, WP 1803 00
Tubing Nonmetallic, Tygothane, Item 345, WP 1803 00

Personnel Required

Aircraft Structural Repairer, MOS 15G (1)
UH-60 Helicopter Repairer, MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0258 00
WP 0259 00
WP 0260 00
WP 0261 00
WP 0381 00
WP 1803 00

REPLACE MAIN TRANSMISSION DRIP PAN

Remove

NOTE

If installing drip pan, DP3–5–01, for the first time, follow instructions for initial drip pan removal (WP 0261 00, WP 0259 00, or WP 0260 00)

1. Remove drain tube. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE, in this work package.
2. Disengage pan clips securing drip pan to drip pan frame (Figure 1).
3. Remove drip pan from drip pan frame.

Prepare to Install

NOTE

- Drip pans 70500-02060-019, 70500-02070-041, and 70500-02070-066 must be removed prior to installation of drip pan, DP3-5-01.(WP 0259 00, WP 0260 00, or WP 0261 00).
- If sheet metal cracks or working rivets are not repaired, leakage can occur.

1. Inspect area inside of transmission well for presence and security of sealant covering screw receptacles, ensuring all sheet metal joints and rivets are sealed to prevent fluid leakage bypassing drip pan assembly.
2. Repair any sheet metal cracks (TM 1-1500-204-23).

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

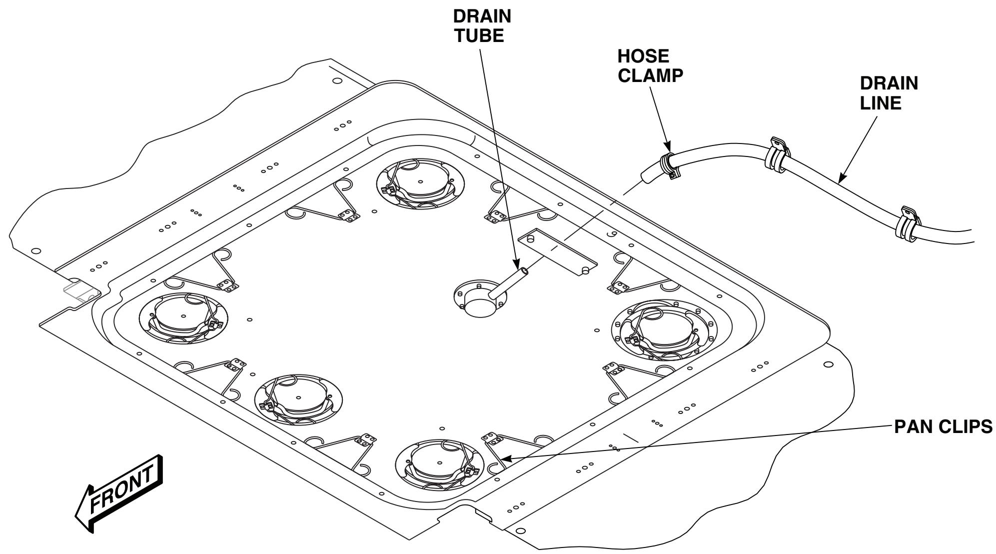


Figure 1. Replace Main Transmission Drip Pan.

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

NOTE

Sealant can release from mating surface of transmission well if mating surface is not properly prepared.

3. Remove old sealant and adhesive from transmission well.
4. Define burnish area by placing drip pan frame around transmission well and use a marker to trace perimeter of drip pan frame (Figure 2, Detail A).
5. Burnish mating surface and treat burnished metal with corrosion resistant coating, (Item 110, WP 1803 00) (TM 1-1500-204-23).

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

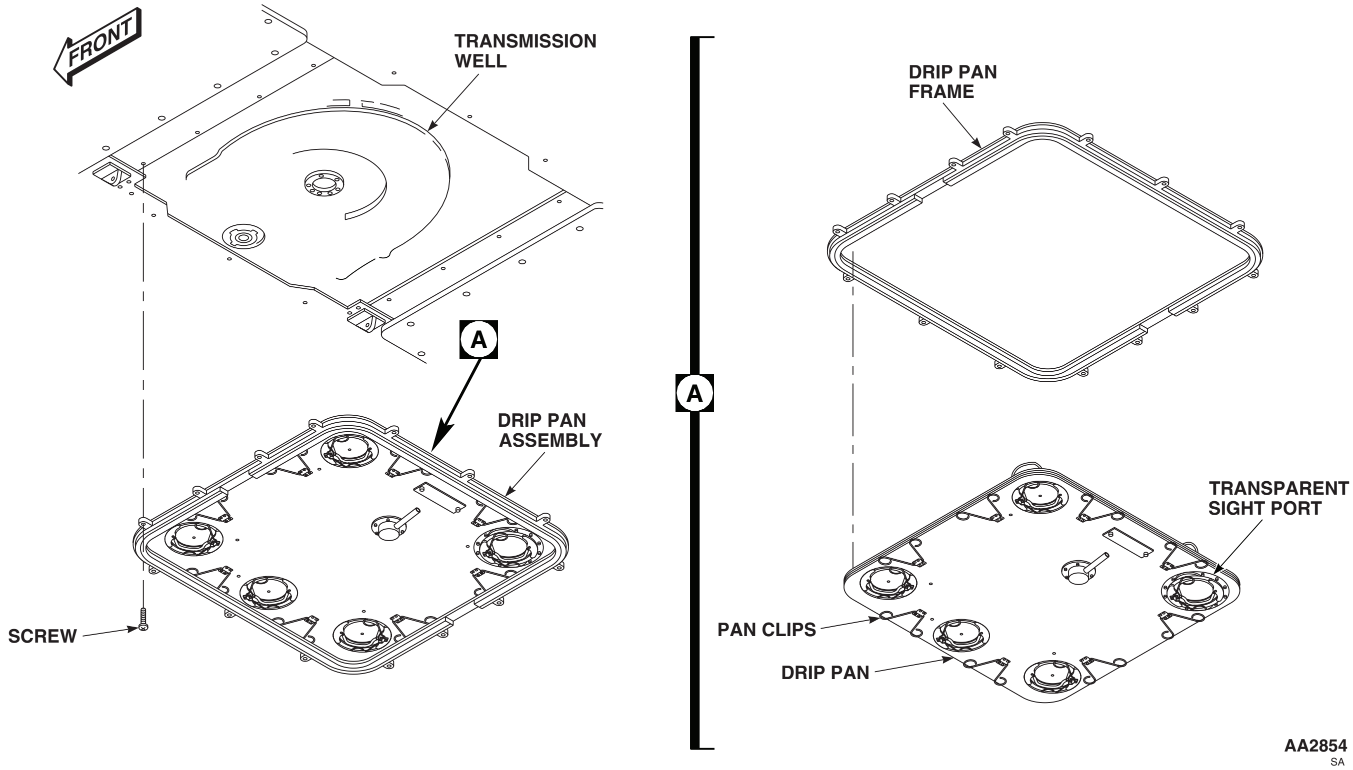


Figure 2. Replace Main Transmission Drip Pan.

NOTE

Verify gaskets are lubricated before drip pan assembly installation.

6. Install drip pan into drip pan frame. Ensure transparent sight port is properly located (Figure 2, Detail A).
7. Fasten pan clips by engaging clips into slots in drip pan frame.

CAUTION

Do not install drip pan frame without drip pan installed. Frame will warp and may cause leakage or difficulty with installation.

NOTE

Do not use sealant during test fitting of drip pan assembly.

8. Install drip pan assembly tightly to allow for fit. Drip pan is designed to allow for adjustment.
9. Test fit drip pan assembly by placing it over the transmission well. Ensure the transparent sight port is properly located.
10. Tighten all screws in a criss-cross pattern until snug.
11. Check drip pan frame for good fit around transmission well.
12. Remove access cover and check for contact between main rotor de-ice harness and supports and drip pan assembly. If contact is observed, adjust harness and supports as needed. If contact is not observed, but pan still fails to install easily, remove pan from frame and adjust wiring harnesses as needed.

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

13. Ensure access to main transmission oil filter bowl is not impeded.
14. Remove drip pan assembly from transmission well.
15. Without removing drip pan from drip pan frame, mask upper surface of drip pan where it contacts drip pan frame with pressure sensitive tape, Item 329, WP 1803 00. This will prevent excess sealant from adhering to drip pan.

NOTE

Proper application of sealant is indicated by a small amount squeezing from edges of drip pan assembly when installed in transmission well.

16. Apply no less than a 1/4 inch bead of sealing compound, Item 257, WP 1803 00, to mating surface of drip pan frame.

CAUTION

Sealant must be allowed to cure properly IAW manufacturer's instruction or leakage can occur.

NOTE

If drip pan frame is installed without drip pan installed, frame can bend or warp when fastened to transmission well. Ensure drip pan and drip pan frame are installed as an assembly.

17. Carefully place drip pan assembly over transmission well. Tighten screws in a criss-cross fashion until snug.
18. Smooth all excessive sealant so that a fillet is formed around the entire inner and outer edge of drip pan frame, filling any voids. Allow sealant to cure IAW manufacturer's instruction.

Drip Pan Leak Test

NOTE

- Water hose should be maneuvered down into main transmission well from top of aircraft to drip pan sight port window to prevent leakage around peripheral structures.
- If leak is observed outside perimeter of drip pan assembly, refer to appropriate technical manual for repair.

1. Cap drain tube and fill transmission well with water to a one-inch depth and check for leaks.
2. If leakage occurs during leak test, remove drain cap, drain transmission well, and repeat drip pan installation procedures.
3. If no leakage occurs, proceed with trim ring installation.

Initial Trim Ring Installation

1. Remove drip pan from drip pan frame (Figure 2, Detail A).
2. Touch-up and prime perimeter around drip pan frame as needed WP 0381 00 and TM 55-1500-345-23 .

NOTE

- Install trim ring raised lip side to transmission well.
- If helicopter is equipped with a hoist, trim ring will be installed under soundproofing.

3. Fit trim ring around drip pan frame, ensuring proper placement for snug fit (Figure 3).

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

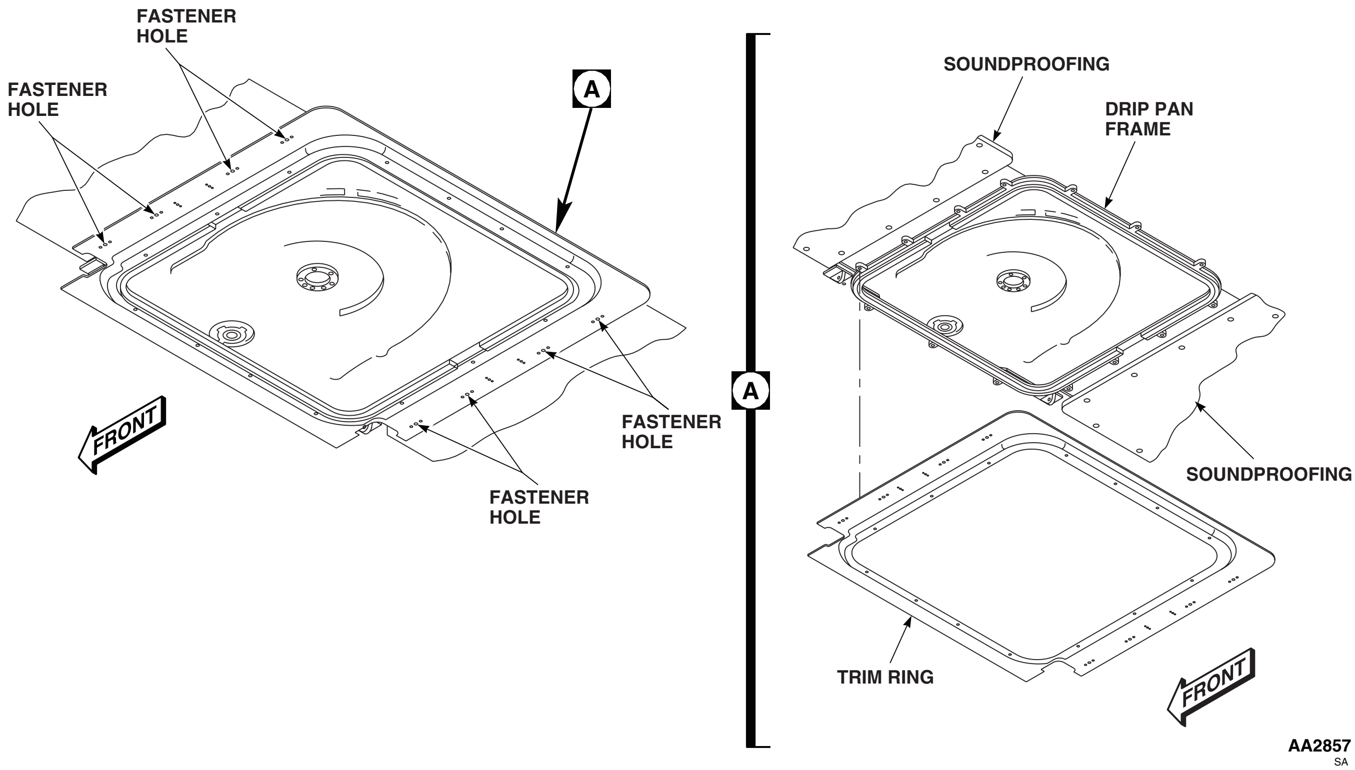


Figure 3. Initial Trim Ring Installation.

4. Position trim ring onto drip pan frame, aligning with soundproofing fastener holes for match drilling.

NOTE

- On UH-60A/L with MEDEVAC mission kit installed models, check fit with openings for carousel before installing trim ring.
- On UH-60A/L with MEDEVAC mission kit installed models, use obsolete drip pan as a template for cut-outs for carousel mount.
- For installation with carousel and FRIES Bar, modify trim ring as needed.
- For HH-60 models, when fastening trim ring to soundproofing on right side, use fasteners present in soundproofing.

5. Place trim ring back into position around drip pan frame and install fasteners to secure trim ring to soundproofing.
6. Connect drain tube. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE, in this work package.
7. Remove and discard two forward drain tubes on RH and LH sides and plug drain tube holes at the point of drain line removal.

REPLACE MAIN TRANSMISSION DRIP PAN - Continued

Install

NOTE

Verify seal is lubricated and installed before drip pan installation If seal is not installed or needs lubrication refer to, REPLACE MAIN TRANSMISSION DRIP PAN in this work package.

1. Install drip pan into drip pan frame.
2. Engage pan clips to secure drip pan to drip pan frame.
3. Connect drain tube. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN TRIM RING

Remove

NOTE

If installing drip pan for first time, follow instructions for initial trim ring installation. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

1. Remove drain tube. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE, in this work package.
2. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
3. Loosen fasteners securing soundproofing to trim ring (Figure 4).
4. Remove screws securing drip pan frame to transmission well.
5. Remove trim ring.

Installation

1. Install trim ring to drip pan frame.
2. Tighten fasteners securing soundproofing to trim ring.
3. Install drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
4. Install drain tube. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN SEAL

Remove

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove damaged seal from seal channel (Figure 4).

Install

1. Using a clean machinery wiping towel, Item 344, WP 1803 00, apply a small amount of petrolatum, Item 216, WP 1803 00, to replacement seal and spread evenly over entire seal.

REPLACE MAIN TRANSMISSION DRIP PAN SEAL - Continued

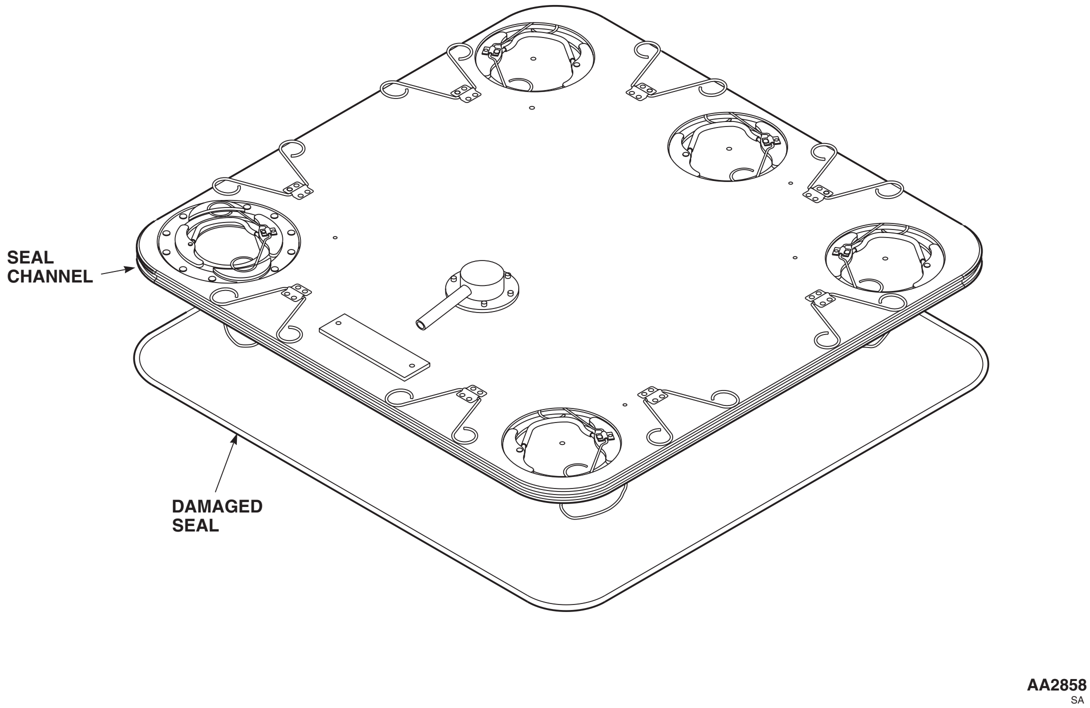


Figure 4. Replace Main Transmission Drip Pan Seal.

NOTE

Avoid excessive stretching of replacement seal. Excessive stretching may cause damage to seal or cause seal to leak.

2. Install seal into drip pan channel (Figure 4).
3. Equalize tension on seal by gently lifting a portion of seal out of channel and running a finger around entire pan, between drip pan and seal. Ensure seal resets into channel properly.
4. Install drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN PORT SEALS

Remove

NOTE

When drip pan is removed check lubrication on pan seal lubricated with petrolatum.

1. Unfasten port cover with damaged seal by releasing cover retaining clips and using pull handle to remove cover from port.
2. Remove damaged seal from port cover seal channel (Figure 5).

Install

1. Using a clean machinery wiping towel, Item 344, WP 1803 00, apply a small amount of petrolatum, Item 216, WP 1803 00, to replacement seal and spread evenly over entire seal.

REPLACE MAIN TRANSMISSION DRIP PAN PORT SEALS - Continued

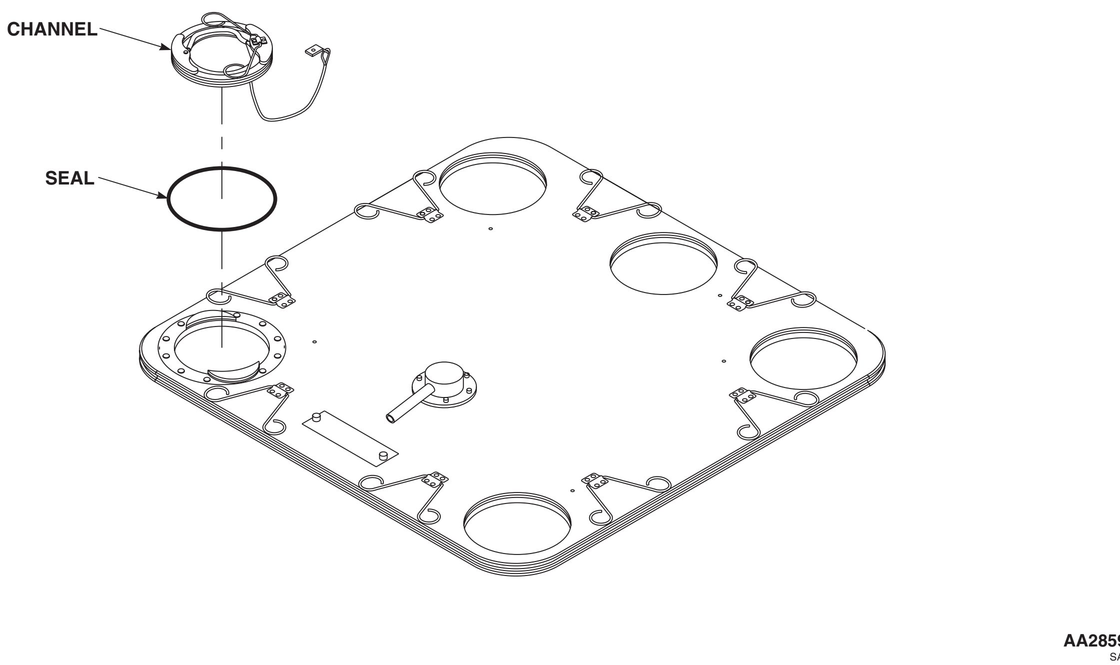


Figure 5. Replace Main Transmission Drip Pan Port Seals.

CAUTION

Avoid excessive stretching of replacement seal. Excessive stretching may cause damage to seal or cause seal to leak.

2. Install seal into port channel (Figure 4).
3. Equalize tension on seal by gently lifting a portion of seal out of channel and running a finger around entire port, between port and seal. Ensure seal resets into channel properly.
4. Secure port cover into drip pan.

REPLACE MAIN TRANSMISSION DRIP PAN CLIP

Remove

CAUTION

When removing rivets and clip use caution to avoid scratching anodized coating on drip pan.

NOTE

Bent clips can usually be easily bent back into shape with no detrimental effects. Clips only need replacement if broken or bent to the extent they will not return to a functional shape.

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN CLIP - Continued

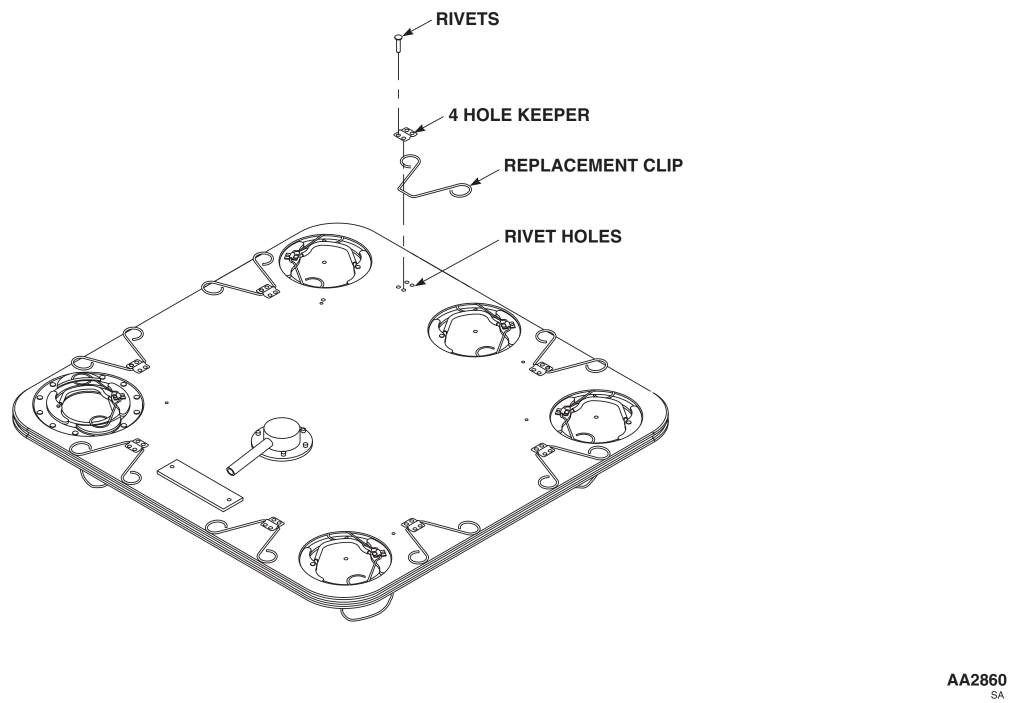


Figure 6. Replace Main Transmission Drip Pan Clip.

2. Remove rivets holding keeper (Figure 6) (TM 1-1500-204-23).

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

3. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

Install

1. Place replacement clip in keeper.
2. Fasten keeper to drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN KEEPERS, in this work package.
3. Apply a small amount of sealing compound, Item 257, WP 1803 00, to the back of rivet holes to prevent leakage. Allow sealant to cure IAW manufacturers instructions.
4. Install drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN CLIP - Continued

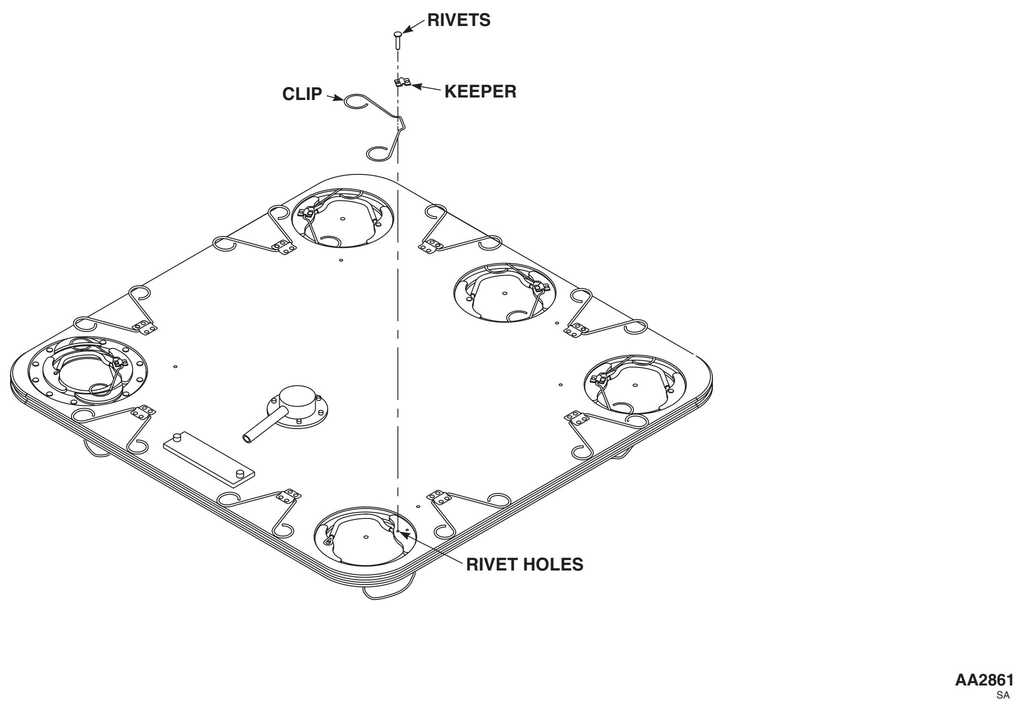


Figure 7. Replace Main Transmission Drip Pan Port Clip.

REPLACE MAIN TRANSMISSION DRIP PAN PORT CLIP

Remove

CAUTION

When removing rivets and clip use caution to avoid scratching anodized coating on drip pan.

NOTE

Bent clips can usually be easily bent back into shape with no detrimental effects. Clips only need replacement if broken or bent to the extent they will not return to a functional shape.

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove rivets (Figure 7) (TM 1-1500-204-23).

REPLACE MAIN TRANSMISSION DRIP PAN PORT CLIP - Continued

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

3. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00

Install

1. Place clip in keeper (Figure 7).
2. Fasten keeper to drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN KEEPERS, in this work package. Ensure pull handle is in place.
3. Apply a small amount of sealing compound, Item 257, WP 1803 00, to back of rivet holes to prevent leakage. Allow sealant to cure IAW manufactures instructions.
4. Install drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN KEEPERS

Remove

CAUTION

When removing rivets and clip use caution to avoid scratching anodized coating on drip pan.

NOTE

Removal for two and four hole clip keepers is the same except as noted.

1. Remove drip pan from drip pan frame. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove rivets from keeper (Figure 7) (TM 1-1500-204-23).

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

3. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

REPLACE MAIN TRANSMISSION DRIP PAN KEEPERS - Continued

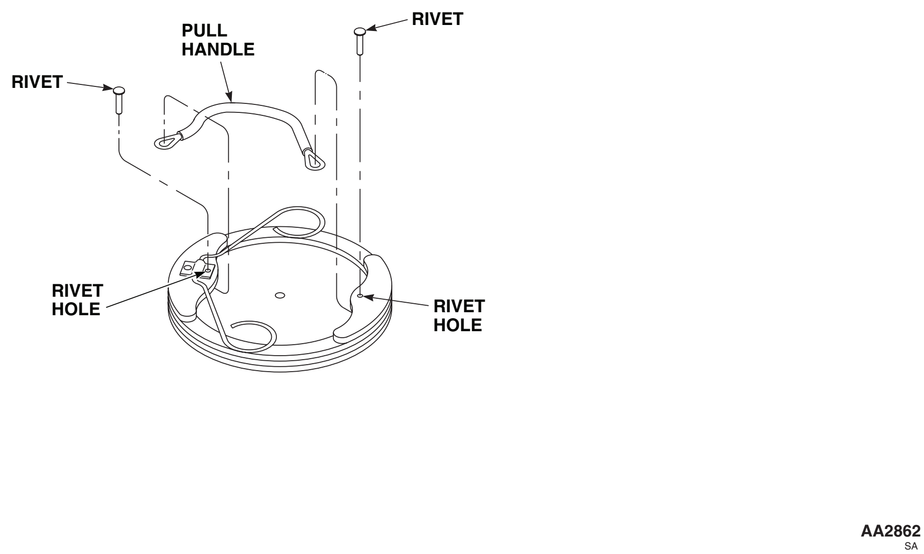


Figure 8. Replace Main Transmission Drip Pan Port Pull Handle.

Install

1. Place clip in keeper.
2. Fasten keeper to drip pan with blind rivets (TM 1-1500-204-23).
3. Apply a small amount of sealing compound, Item 257, WP 1803 00, to back of rivet holes to prevent leakage. Allow sealant to cure IAW manufactures instructions.
4. Install drip pan to drip pan frame. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN PORT PULL HANDLE

Remove

CAUTION

When removing rivets use caution to avoid scratching anodized coating on drip pan.

1. Remove drip pan from drip pan frame. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove rivets from damaged pull handle (Figure 8) (TM 1-1500-204-23).
3. Remove pull handle.

REPLACE MAIN TRANSMISSION DRIP PAN PORT PULL HANDLE - Continued

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

4. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

Install

1. Position pull handle end splices in the appropriate recesses, aligning with rivet holes.
2. Install rivets, making sure shaft of rivet passes through correct splice Figure 8 (TM 1-1500-204-23).
3. Apply a small amount of sealing compound, Item 257, WP 1803 00, to back of rivet holes to prevent leaks. Allow sealant to cure IAW manufactures instructions.
4. Install drip pan to drip pan frame. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN PORT LANYARD

Remove

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove rivets connecting lanyard to port and drip pan (Figure 9) (TM 1-1500-204-23).
3. Remove lanyard.

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

4. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

Install

1. Apply a small amount of sealing compound, Item 257, WP 1803 00 to area around rivet holes to prevent leakage from under mounting holes (Figure 9, Detail A).
2. Install lanyard and fasten with rivets (TM 1-1500-204-23).
3. Apply a small amount of sealing compound, Item 257, WP 1803 00, to back of rivets. Allow sealant to cure IAW manufactures instructions.
4. **QA** Install drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN PORT LANYARD - Continued

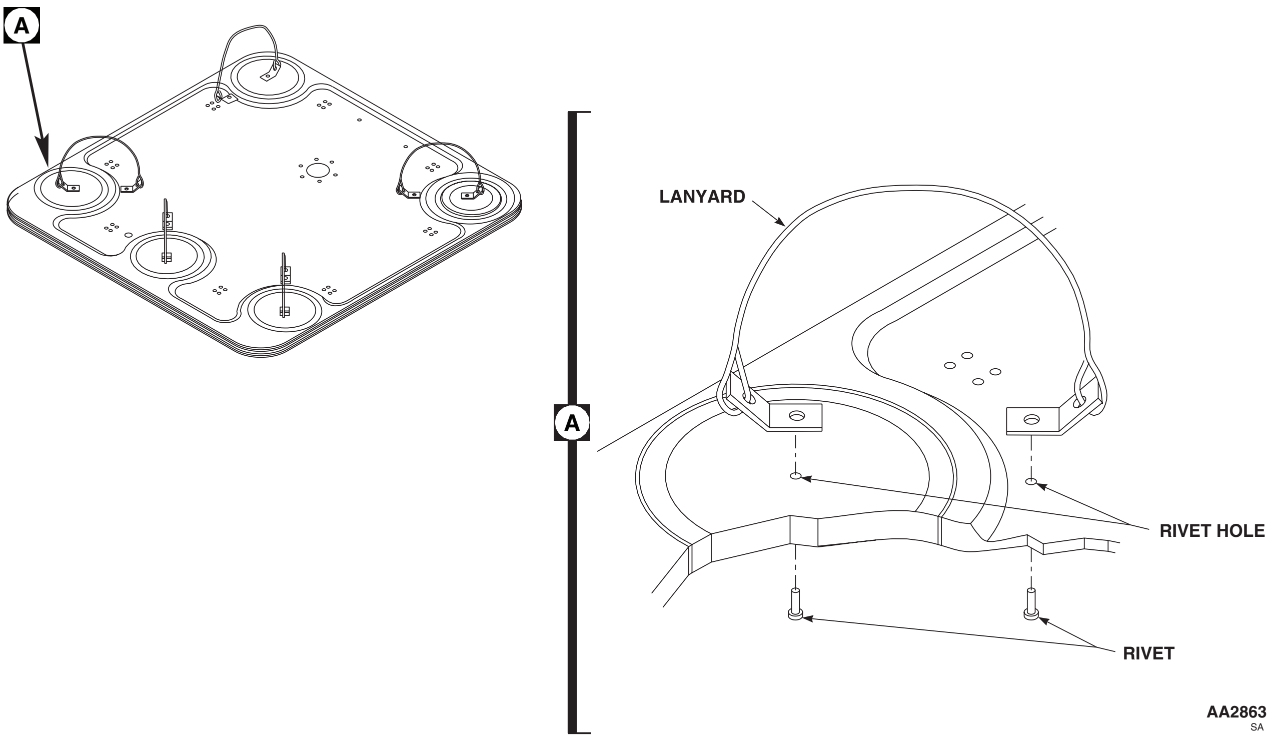


Figure 9. Replace Main Transmission Drip Pan Port Lanyard.

REPLACE MAIN TRANSMISSION DRIP PAN PORT COVER

Remove

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove sealant from rivets attaching lanyard to port cover.
3. Remove lanyard. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN PORT LANYARD, in this work package.
4. Release clips securing port cover and remove port cover from drip pan.

Install

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

1. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

REPLACE MAIN TRANSMISSION DRIP PAN PORT COVER - Continued

2. Wet install rivets attaching lanyard to drip pan (TM 1-1500-204-23).
3. Apply a small amount of sealing compound, Item 257, WP 1803 00, to back of rivet holes to prevent leakage. Allow sealant to cure IAW manufactures instructions.
4. Install port cover to drip pan with clips.
5. Install drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE AND DRAIN BASKET

Remove

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.
2. Remove rivets securing drain basket assembly (Figure 10) (TM 1-1500-204-23).
3. Remove drain basket assembly.

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush them with clean water (15 minutes) and get immediate medical help.

4. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

Install

1. Install replacement drain tube or drain basket, as necessary (Figure 10).
2. Apply sealing compound, Item 257, WP 1803 00, to flange surface of drain basket and onto threaded portion of drain tube.
3. Screw drain tube into drain basket, ensuring sealant completely seals joint.

NOTE

Ensure drain tube is installed to drip pan to allow for connection to rear drain line.

4. Install drain tube and drain basket to drip pan with rivets ensuring offset holes match so drain tube is properly oriented (TM 1-1500-204-23).
5. Apply sealing compound, Item 257, WP 1803 00, to flange surface of drain basket and onto threaded portion of drain tube. Allow sealant to cure IAW manufactures instructions.
6. Install drip pan to drip pan frame. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE AND DRAIN BASKET - Continued

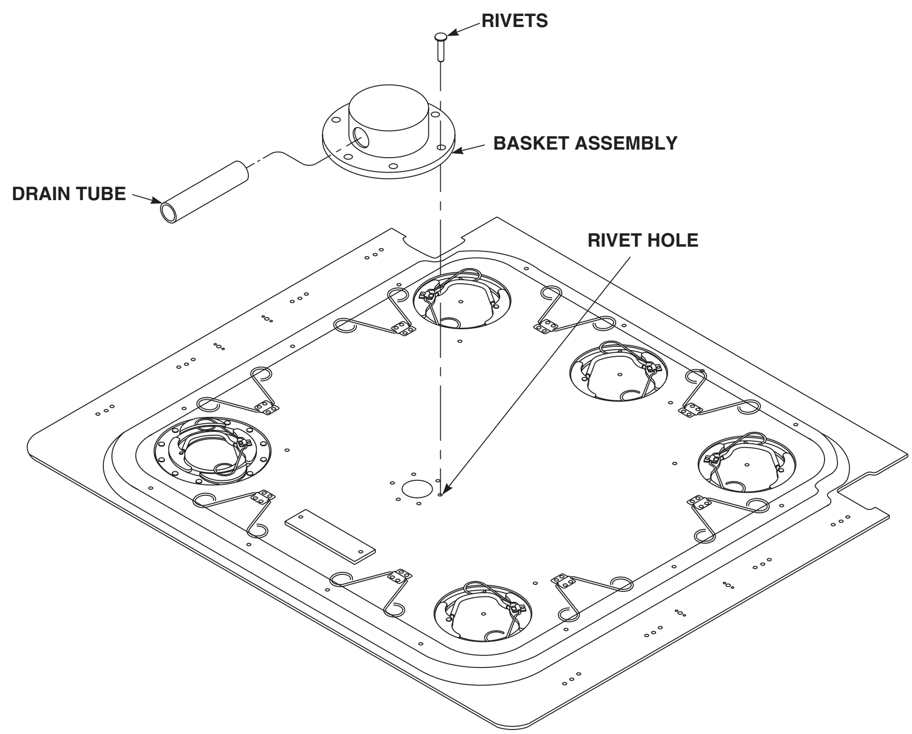


Figure 10. Replace Main Transmission Drip Pan Drain Tube and Drain Basket.

REPLACE MAIN TRANSMISSION DRIP PAN DRAIN TUBE

Remove

NOTE

Drip pan DP3–5–01 uses only rear drain line. When drip pans, 70500–02060–019, 70500–02070–041 and 70500–02070–066 are replaced with drip DP3–5–01, rear drain line tubing from drip pan to drain line will require replacement because drain line is to short.

1. Remove support clamps, screws, washers and nuts holding drain line to overhead panels. (Figure 11).
2. Loosen drain line hose clamps from drip pan drain tube and coupling tube. Remove drain line. Remove hose clamps from both ends of drain line.

Install

1. Cut a length of nonmetallic tubing, Item 345, WP 1803 00, 54 inches long.
2. Slip nose clamps onto each end of drain line.
3. Install hose clamps onto each end of drain line and install drain line onto overhead panels.

REPLACE MAIN TRANSMISSION DRIP PAN SIGHT PORT

Remove Stationary Sight Port

1. Remove drip pan. Refer to, REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

REPLACE MAIN TRANSMISSION DRIP PAN SIGHT PORT - Continued

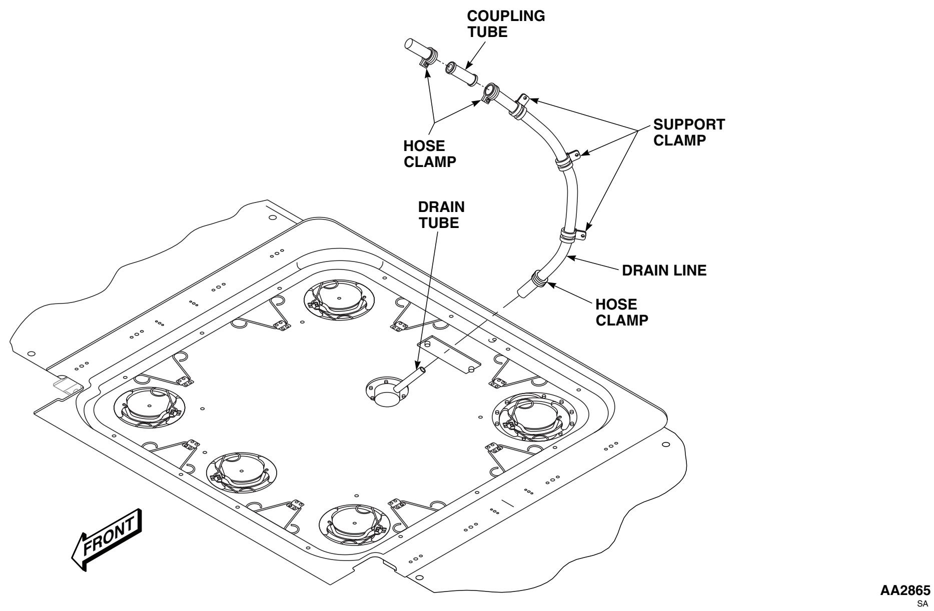


Figure 11. Replace Main Transmission Drip Pan Drain Tube.

2. Remove rivets securing the stationary sight port window and doubler plate (Figure 12) (TM 1-1500-204-23).

CAUTION

When prying off doubler plate, sight port window may break.

NOTE

If sight port window and doubler plate will not release, use prybar to remove doubler plate. Place bar between doubler plate and sight glass and pry off doubler plate.

3. Remove doubler plate and sight port window (Figure 12).

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush eyes with clean water (15 minutes) and get immediate medical help.

4. Remove cured sealant from rivet holes and clean area with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.

Install Removable Sight Port

1. Apply sealing compound, Item 257, WP 1803 00, to mounting surface (Figure 13).

REPLACE MAIN TRANSMISSION DRIP PAN SIGHT PORT - Continued

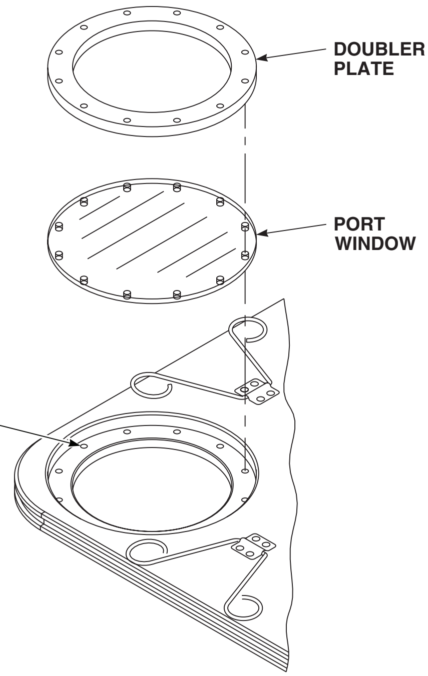


Figure 12. Remove Main Transmission Drip Pan Stationary Sight Port.

NOTE

- When adapter ring is inserted, ensure clip receptacles cover the corresponding rivet holes.
- Ensure alignment of lanyard is 45° from outer edge of drip pan.

2. Remove sight port from adapter ring.
3. Insert adapter ring into sight port opening.
4. Confirm orientation and ensure adjacent clips clear adapter ring ears, and that clips will lock (Figure 13).
5. Ensure other rivet holes align, then apply firm pressure to seat adapter ring.
6. Install adapter ring by wet install rivets with sealing compound, Item 257, WP 1803 00 (TM 1-1500-204-23). Allow sealant to cure IAW manufacturer's instructions.

REPLACE MAIN TRANSMISSION DRIP PAN SIGHT PORT - Continued

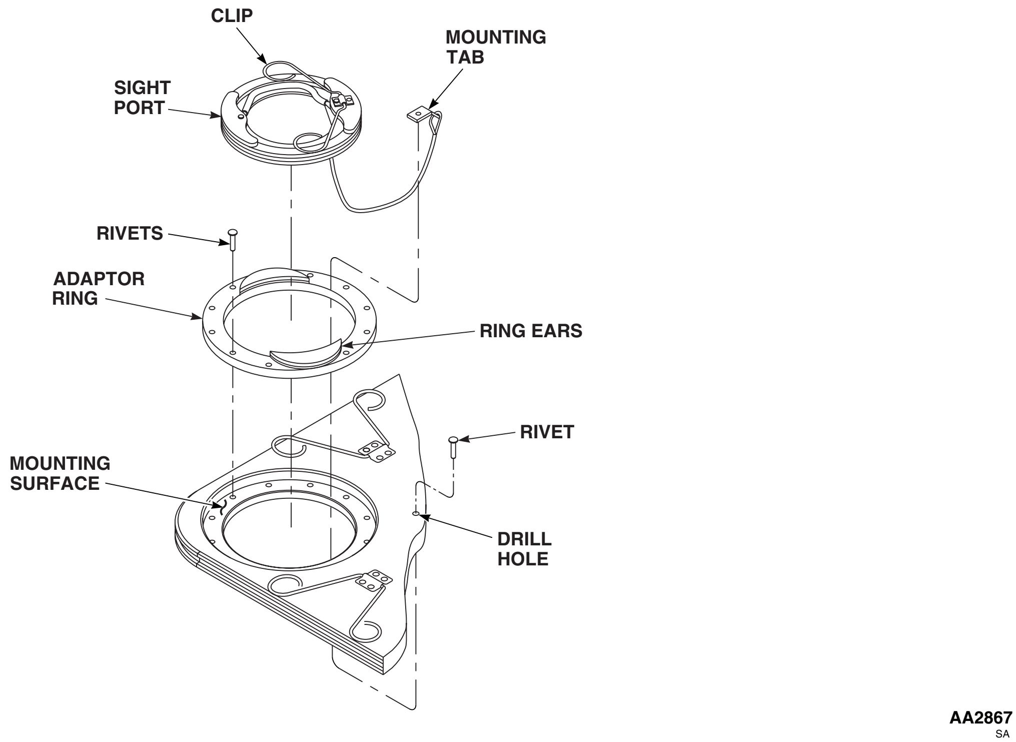


Figure 13. Install Main Transmission Drip Pan Removable Sight Port.

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush them with clean water (15 minutes) and get immediate medical help.

7. Clean excess sealant with machinery wiping towel, Item 344, WP 1803 00 moistened with isopropyl alcohol, Item 170, WP 1803 00.

CAUTION

Solvents will damage sight port window . Do not allow solvents to come in contact with sight port window.

8. Place sight port into sight port opening, orienting sight port so lanyard is opposite the corner of drip pan.
9. Engage port clips into clip receptacles.

REPLACE MAIN TRANSMISSION DRIP PAN SIGHT PORT - Continued

NOTE

Exact placement of mounting tab hole is not critical. The location should be approximately 2 inches from sight port opening.

10. Drill hole through drip pan using a 0.128 inch (No. 30) drill bit.
11. Apply a small amount of sealing compound, Item 257, WP 1803 00, to hole on side lanyard will be mounted.

NOTE

- Orient mounting tab away from port to alleviate binding.
- Install rivet head from interior side of drip pan.

12. Rivet lanyard mounting tab.(TM 1-1500-204-23).
13. Remove old dataplate by removing rivets holding dataplate to drip pan (TM 1-1500-204-23).

NOTE

When removing old sealant use caution to avoid scratching anodized coating on drip pan.

14. Remove old sealant from drip pan. using nonmetallic scraper.

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible.
- Get fresh air and immediate medical attention. Wash contacted skin with clean water for 15 minutes. If solvent contacts eyes, flush them with clean water (15 minutes) and get immediate medical help.

15. Clean area thoroughly with machinery wiping towel, Item 344, WP 1803 00, moistened with isopropyl alcohol, Item 170, WP 1803 00.
16. Align new dataplate then match drill to existing rivet holes.
17. Wet install rivets on dataplate with sealing compound, Item 257, WP 1803 00 (TM 1-1500-204-23). Allow sealant to cure IAW manufacturer's instructions.
18. Dome seal lanyard mounting tab to dataplate, rivets (TM 1-1500-204-23) that secure adapter ring, and two obsolete holes in adapter ring using sealing compound, Item 257, WP 1803 00. Allow sealant to cure IAW manufacturer's instructions.
19. Install drip pan. REPLACE MAIN TRANSMISSION DRIP PAN, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN FLOOR

This WP supersedes WP 0262 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive Primer, Item 23, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Isopropyl Alcohol, Technical, Item 170, WP 1803 00
Kit, Packing Material, Item 200, WP 1803 00
Packing Material, Item 201, WP 1803 00
Packing Material, Item 202, WP 1803 00
Packing Material, Item 203, WP 1803 00
Packing Material, Item 204, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Shim, 70214-02555-107
Shim, 70214-02555-109
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
WP 0233 00
WP 0379 00
WP 0402 00
WP 0857 00
WP 1803 00

NOTE

Replacement instructions for any section of cabin floor is the same, except as noted.

REMOVE

1. Turn off all electrical power.
2. If removing section of cabin floor from STA 247.0 to STA 288.0, remove battery (WP 0857 00).
3. Remove troop seats (WP 0233 00).
4. Remove screws from floor panels and attachment fitting floor tiedowns (Figure 1). Remove floor panels from helicopter.

NOTE

Packing material does not have to be replaced when floorboards are removed unless damage to packing material occurs or maintenance is performed in areas where packing material is applied.

5. Remove old packing material and clean areas that packing material will be applied using machinery wiping towel, Item 344, WP 1803 00, dampened with technical isopropyl alcohol, Item 170, WP 1803 00.

INSPECT

1. Visually inspect fasteners, skin under stringers and around bulkheads, and floor attachment fittings for corrosion.
2. Inspect tub section for complete coverage of corrosion protection. If there is complete coverage of corrosion protection, do not clean. If any portion of tub section has no corrosion protection, clean unprotected portion with machinery wiping towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00. Wipe cleaned area with machinery wiping towel, Item 344, WP 1803 00, moistened with technical isopropyl alcohol, Item 170, WP 1803 00. Allow to dry completely. Apply one coat of epoxy primer coating kit, Item 120, WP 1803 00.

INSTALL

NOTE

Two shims, 70214-02555-109, replace one shim, 70214-02555-107.

1. Turn off all electrical power.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. If shims have become disbonded, do this:
   a. Wipe bottom of pan and solid side of shim using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
   b. Check for corrosion on pan and shim. Remove any corrosion and repair area as required (WP 0402 00).
   c. Remove any remaining adhesive from pan and shim using a nonmetallic scraper. Clean area using a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
   d. Prime solid side of shim using adhesive primer, Item 23, WP 1803 00. Allow primer to dry at room temperature for at least 1 hour.
   e. Apply a light coat of adhesive, Item 11, WP 1803 00, to solid side of shim.
   f. **QA** Place shim on bottom of pan, with holes in shim and pan lined up.
   g. Allow adhesive, Item 11, WP 1803 00, to cure at room temperature for at least 1 hour.
3. If shim is lost, do this:
   a. Peel replacement shim so that there is a maximum gap of 0.010-inch between bottom of floor panel and top of shim. When using solid shim, machine to obtain gap.
   b. Bond new shim to bottom of pan using bonding procedure in step 2.
4. Apply two coats of primer coating, Item 243, WP 1803 00, to structure attachment area flanges.

INSTALL - Continued

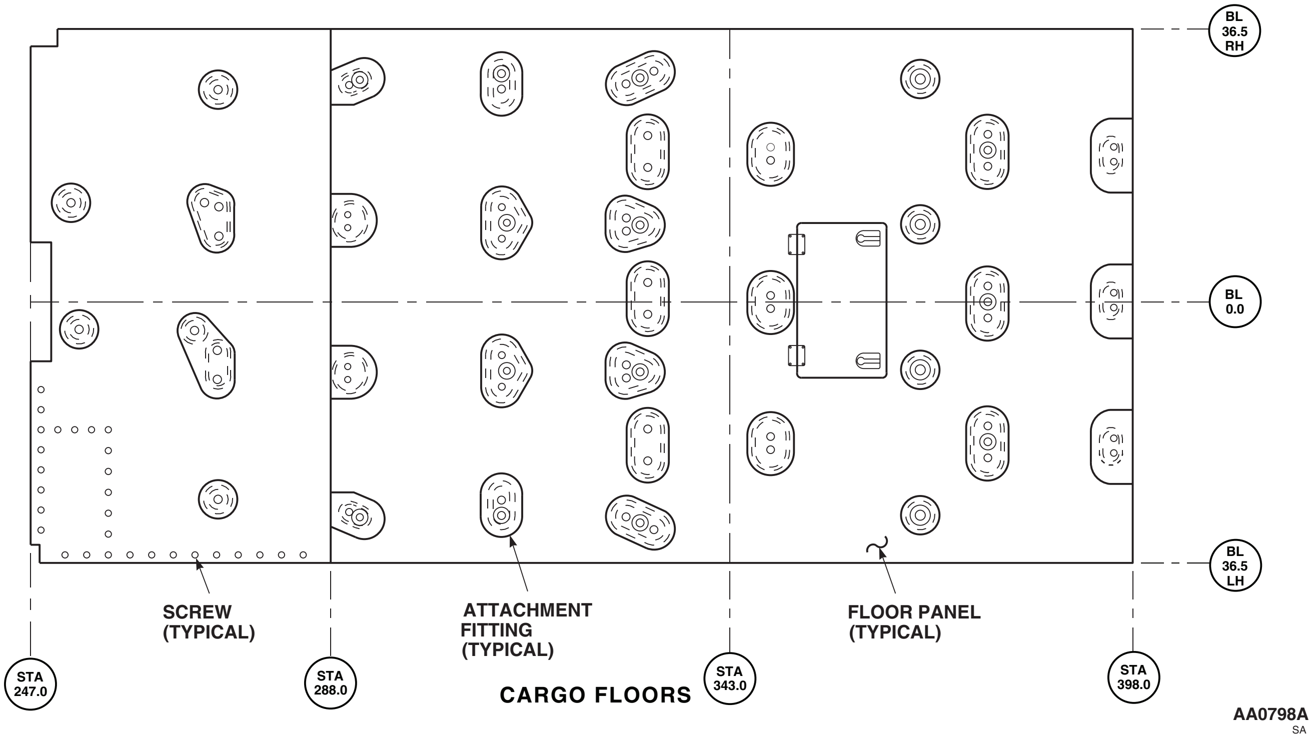


Figure 1. Replace Cabin Floor.

NOTE

Packing material is self adhesive with peel away release film on both sides. When using packing material to seal floor panels, release film should remain in place on top side of packing material. Only remove film on bottom side of packing material. T-caps require complete coverage.

5. Apply packing material kit, Item 200, WP 1803 00, as follows:
   a. Cut packing material to length. After primer has dried, apply packing material, Item 201, WP 1803 00, Item 202, WP 1803 00, Item 206, or Item 204, WP 1803 00 as required, to structure with adhesive side of packing material adhering to structure. Overlap packing material in corners by 1/4-inch for most effective sealing. If overlapping is not complete leakage may occur.
   b. On all other floor interface structures, center and apply 1.50-inch long strips of packing material, Item 202, WP 1803 00 over screw holes.
6. If screw holes do not line up with nutplates, do this:
   a. Mark holes that do not line up, remove floor panel.
   b. Fill out-of-line holes (WP 0379 00).
   c. Mark, drill, and countersink new holes in floor panels (TM 1-1500-204-23).
7. Make sure area is clean and free of foreign material before closing up.
8. **QA** Place floor panels on structure flanges (Figure 1).

INSTALL - Continued

9. **QA** Line up screw holes with nutplates and install screws. **TORQUE SCREWS NO MORE THAN 10 INCH-POUNDS GREATER THAN TORQUE NEEDED TO SEAT SCREWS.**
10. Install troop seats (WP 0233 00).
11. Attach troop seat quick-disconnects to floor panel fittings (WP 0233 00).
12. If removed, install battery and cover (WP 0857 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO HOOK DOOR STRAP SEAL

This WP supersedes WP 0263 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cargo Hook Access Seal, Locally-Made, Figure 60, WP 1805 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0266 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove cargo hook door (WP 0266 00).
2. **UH-60A 80-23453 - SUBQ** Remove seal on strap as follows:
   a. Remove damaged seal from strap using a nonmetallic scraper (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   b. Clean any remaining adhesive from strap using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

INSTALL

1. **UH-60A 80-23453 - SUBQ** Install seal on strap as follows:
   a. Locally make new cargo hook access seal (WP 1805 00).
   b. Roughen side of new seal using abrasive cloth assortment, Item 1, WP 1803 00. Clean roughened surface of seal using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
   c. Apply a light coat of adhesive, Item 14, WP 1803 00, to roughened side of seal and edge of strap. Allow adhesive to dry for about 10 minutes or until it is tacky.
   d. Install seal on strap, making sure all mating surfaces are in contact. Allow adhesive to dry at room temperature for 24 hours.

INSTALL - Continued

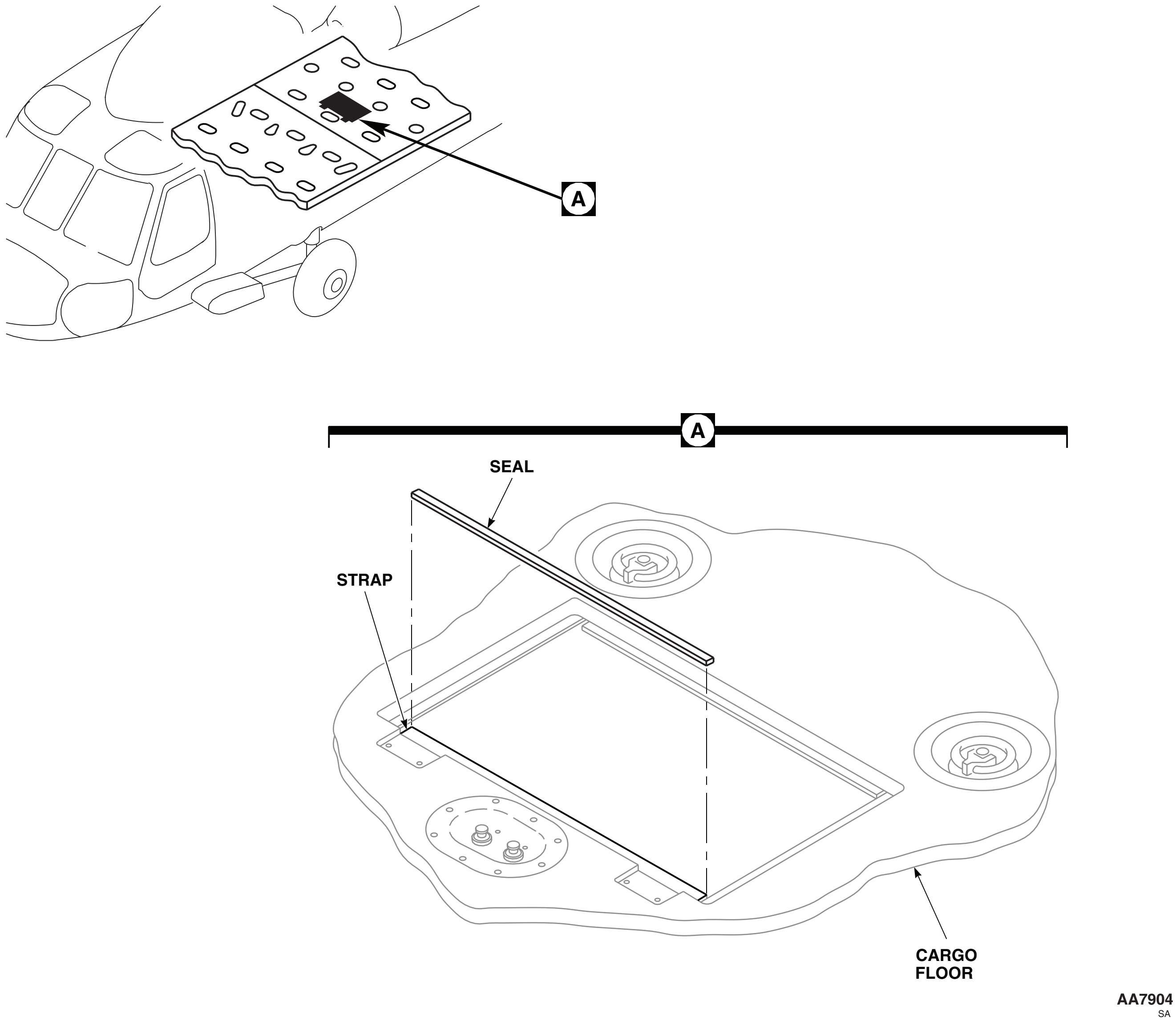


Figure 1. Cargo Hook Door Strap Seal.

2. Install cargo hook door (WP 0266 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN FLOOR SEAT STUD

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Sealing Compound, Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0262 00
WP 1803 00

REMOVE

1. Remove cabin floor (WP 0262 00).
2. Remove sealing compound from one keeper rivet. Drill out rivet (Figure 1, Detail B).
3. Remove retaining ring.
4. Turn keeper to one side so stud can be removed.
5. If washer was installed, remove it from stud.

INSTALL

1. If washer was removed, place washer on replacement stud and place stud in correct floor stud hole (Figure 1, Detail B).
2. Place retaining ring on stud.
3. Line up hole in keeper with mating hole in cabin floor. Install rivet.
4. Apply sealing compound, Item 273, WP 1803 00, to rivet.
5. Install cabin floor (WP 0262 00).

INSTALL - Continued

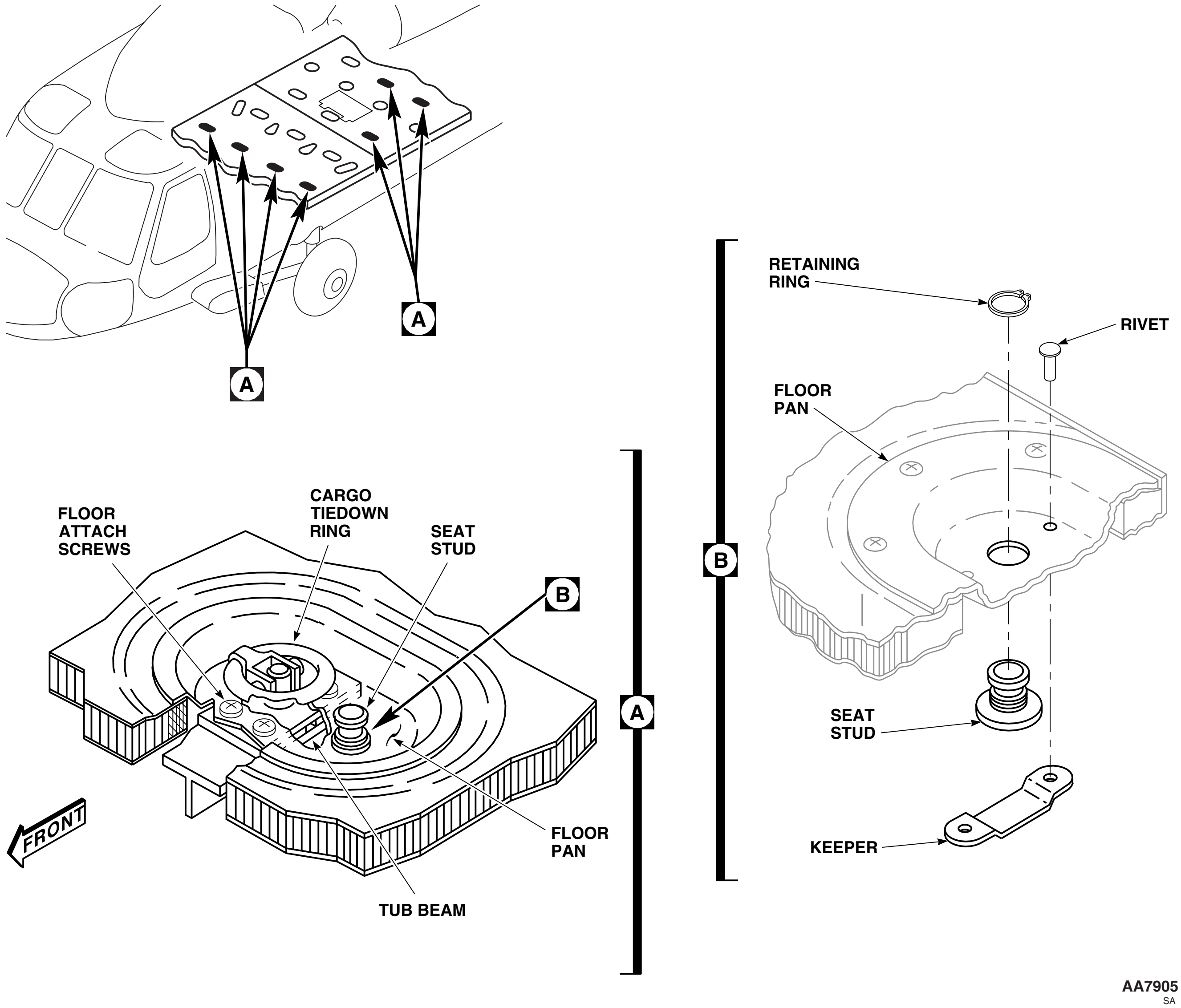


Figure 1. Replace Cabin Floor Seat Stud.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN FLOOR LEVELING PLATE

This WP supersedes WP 0265 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bubble Protractor
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Plumb Bob, 5720-10
Spirit Level, GGG-L-211, Type VII

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0258 00
WP 0381 00
WP 1675 00
WP 1803 00

REMOVE

Remove old leveling plate using nonmetallic scraper and carbon dioxide, Item 64, WP 1803 00.

INSTALL

1. Open engine cowling work platform.
2. Remove soundproofing panel below main transmission and directly above floor panel leveling location (WP 0258 00).
3. Jack helicopter (WP 1675 00).
4. Hang plumb bob from slot in fitting in cabin overhead beam at STA 309.62 and LBL 36.0 (Figure 1, Detail A).
5. Place spirit level or bubble protractor on transmission support beam near transmission.
6. While observing spirit level or protractor, adjust jacks to bring helicopter to level attitude both lengthwise and laterally.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Clean leveling plate and floor panel plate using cheesecloth cloth,Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
8. Coat underside of leveling plate with adhesive, Item 11, WP 1803 00.

INSTALL - Continued

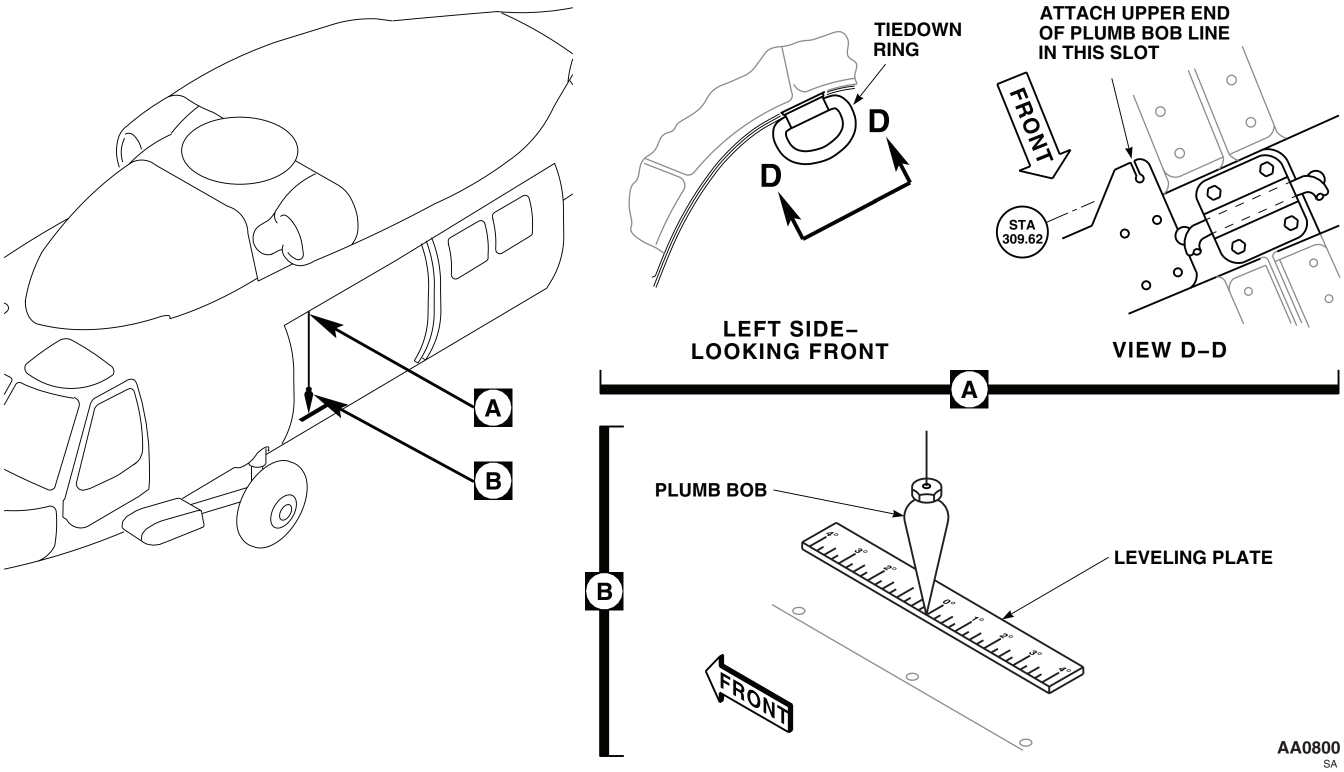


Figure 1. Replace Cabin Floor Leveling Plate.

9. Position plate on floor with zero mark at edge exactly under point of plumb bob, and with long scale line parallel to centerline of helicopter (Figure 1, Detail B).
10. Remove plumb bob and install soundproofing panel (WP 0258 00).
11. Remove level or protractor from transmission support beam and close engine cowling work platform.
12. Lower helicopter and remove jacks (WP 1675 00).
13. Touch up paint around leveling plate (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO HOOK DOOR

This WP supersedes WP 0266 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0233 00
WP 0379 00
WP 1803 00

REMOVE

1. Remove troop seats (WP 0233 00).
2. Open cargo hook door and support in open position.
3. Remove screws, washers, and nuts securing cargo hook door to hinges (Figure 1, Detail A). Remove door.

REPAIR

Refer to WP 0379 00 to repair any skin or structural damage to door.

INSTALL

1. Install seals on replacement door as follows (Figure 1, Detail A):
   a. Measure and cut new seals to fit front, rear, and side of door.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   b. Clean edges of replacement door using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
   c. Roughen side of new seal using abrasive cloth, Item 82, WP 1803 00. Clean roughened surface of seal, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
   d. Apply a light coat of adhesive, Item 14, WP 1803 00, to roughened side of seal and edge of door. Allow adhesive to dry for about 10 minutes or until it is tacky.
   e. Install seals on edges of door, making sure all mating surfaces are in contact.

INSTALL - Continued

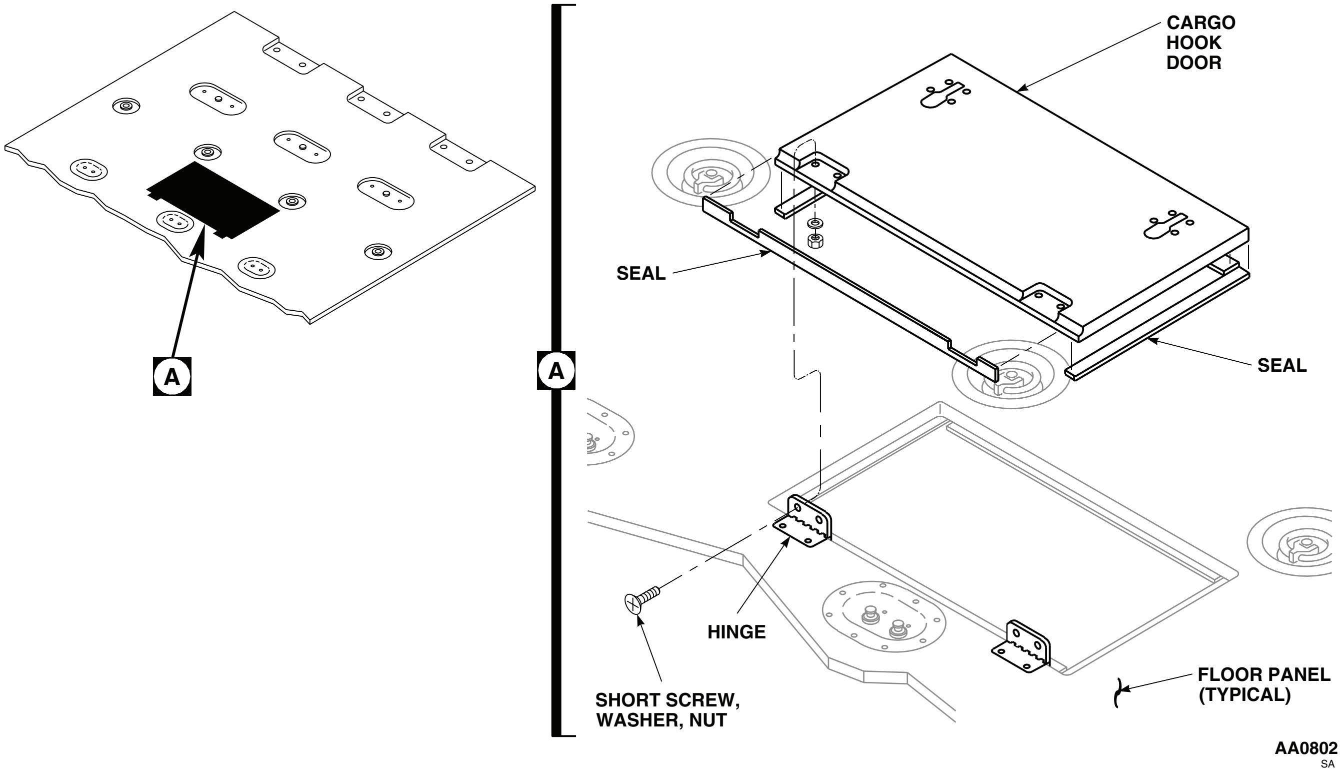


Figure 1. Replace Cargo Hook Door.

f. Allow adhesive to dry at room temperature for 24 hours.

2. **QA** Install new door on hinges using screws, washers, and nuts.
3. Close and latch cargo hook door.
4. Install troop seats (WP 0233 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO HOOK DOOR LATCH

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Rivet

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Carefully drill out and remove four rivets securing damaged latch to cargo hook door using a No. 40 drill and 3/32-inch drive pin punch (Figure 1, Detail A).
2. Remove latch and make sure area is clean and free of foreign material.

INSTALL

1. If required, open pilot holes using No. 40 drill. Deburr holes.
2. Position latch in cargo door and hold in place using three 3/32-inch diameter clecos.
3. Check operation of door latch.
4. Install one rivet in empty hole (Figure 1, Detail A).
5. Remove one cleco from diagonal corner hole and install rivet.
6. **QA** Remove remaining two clecos and install rivets.

INSTALL - Continued

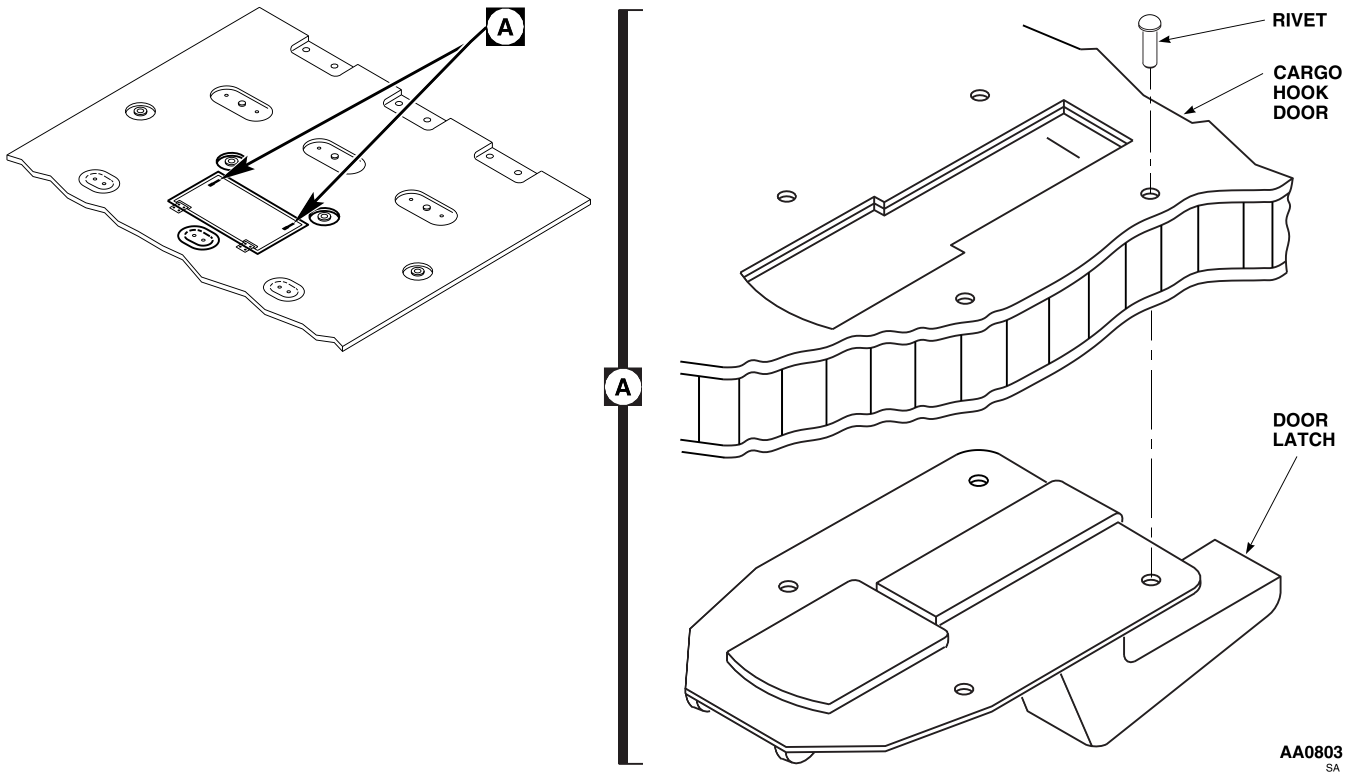


Figure 1. Replace Cargo Hook Door Latch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO HOOK DOOR SEAL

This WP supersedes WP 0268 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical Item 5, WP 1803 00
Adhesive Item 14, WP 1803 00
Cloth, Abrasive Item 82, WP 1803 00
Cloth, Cheesecloth Item 85, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0266 00
WP 1803 00

REMOVE

1. Remove cargo hook door (WP 0266 00).
2. Remove damaged seal using a nonmetallic scraper, being careful not to damage other seals (Figure 1).

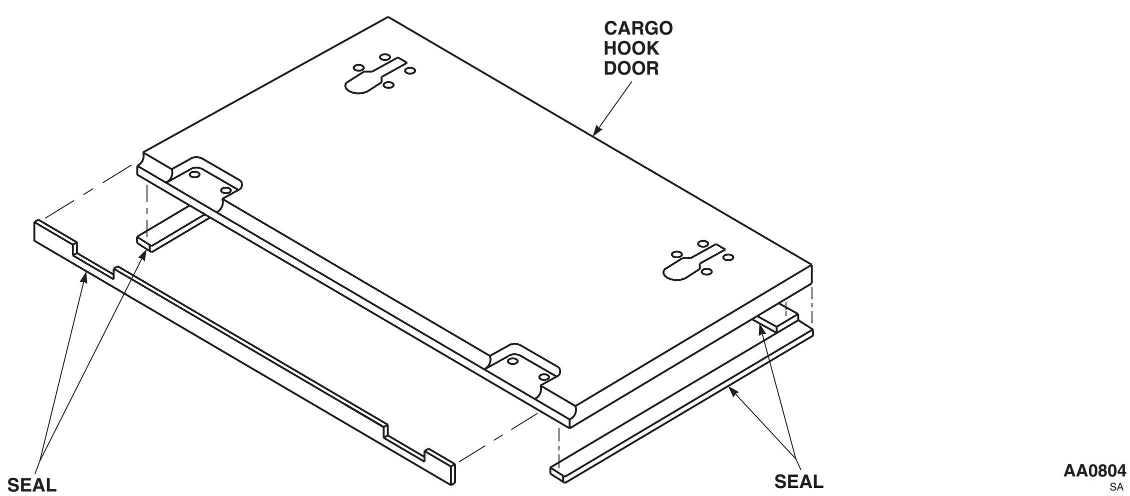


Figure 1. Replace Cargo Hook Door Seal.

REMOVE - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean any remaining adhesive from cargo hook door using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

INSTALL

1. Measure and cut new seal.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Roughen side of new seal using abrasive cloth, Item 82, WP 1803 00. Clean roughened surface of seal using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
3. Apply a light coat of adhesive, Item 14, WP 1803 00, to roughened side of seal and edge of door. Allow adhesive to dry for about 10 minutes or until tacky.
4. Install seal on edge of door making sure all mating surfaces are completely touching. Allow adhesive to dry at room temperature for 24 hours (Figure 1).
5. Install cargo hook door (WP 0266 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO HOOK DOOR HINGES

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0233 00

REMOVE

1. Remove troop seats (WP 0233 00).
2. Open cargo hook door.
3. Remove nut, washers, and screws and remove hinge from cargo floor and cargo hook door (Figure 1).

INSTALL

1. Position hinge on cargo floor and install long bolts, washers, and nuts (Figure 1).
2. **QA** Install end of hinge on cargo hook door using short screws, washers, and nuts.
3. Close and latch cargo hook door.
4. Install troop seats (WP 0233 00).

INSTALL - Continued

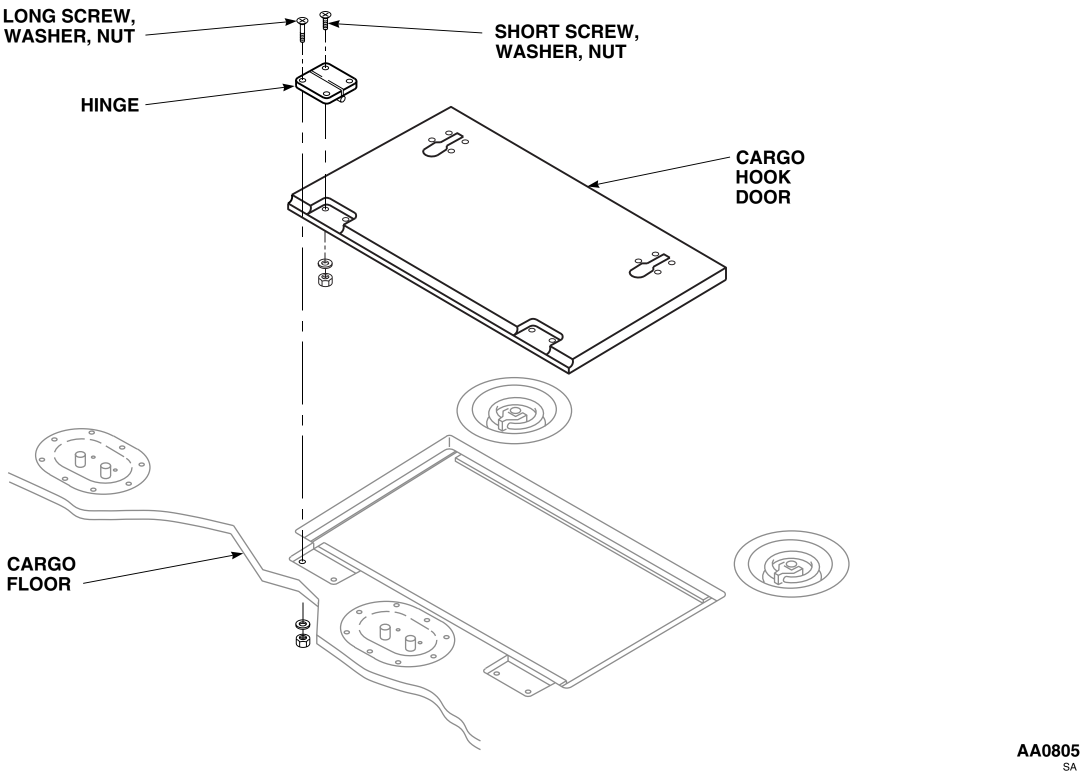


Figure 1. Replace Cargo Hook Door Hinges.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO HOOK DOOR REPAIR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0379 00

REPAIR CARGO HOOK DOOR

Refer to WP 0379 00 to repair any skin or structural damage to door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE UPPER CUTTER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper,
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

1. Open main rotor pylon sliding cover.
2. Remove bolt, washers, and nut securing upper cutter struts to upper cutter (Figure 1, Detail A).
3. Remove bolt, washers, and nut securing upper cutter struts to main rotor pylon sliding cover (Figure 1, Detail B).
4. Remove bolt, washers, and nut securing upper cutter to main rotor pylon sliding cover.
5. Remove sealing compound from main rotor pylon sliding cover, struts, and cutter with nonmetallic scraper.

INSTALL

1. Install upper cutter onto main rotor pylon sliding cover with bolts and washers. Do not torque bolts now (Figure 1, Detail A).
2. Apply sealing compound, Item 273, WP 1803 00, around upper cutter strut mounting holes, and to shank of bolt.
3. Install upper cutter struts onto main rotor pylon sliding cover with screws, washers, shims, and nuts. Do not torque nuts now (Figure 1, Detail B).
4. **QA** Secure upper cutter struts to upper cutter with bolt, washer, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.** Apply sealing compound, Item 273, WP 1803 00, around bolthead and nut (Figure 1, Detail A).
5. Tighten nuts on upper cutter strut screws.
6. **QA TORQUE UPPER CUTTER MOUNT BOLTS TO 48 - 52 INCH-POUNDS.**
7. Seal cutter struts to main rotor pylon sliding cover with sealing compound, Item 273, WP 1803 00.
8. Close main rotor pylon sliding cover.

INSTALL - Continued

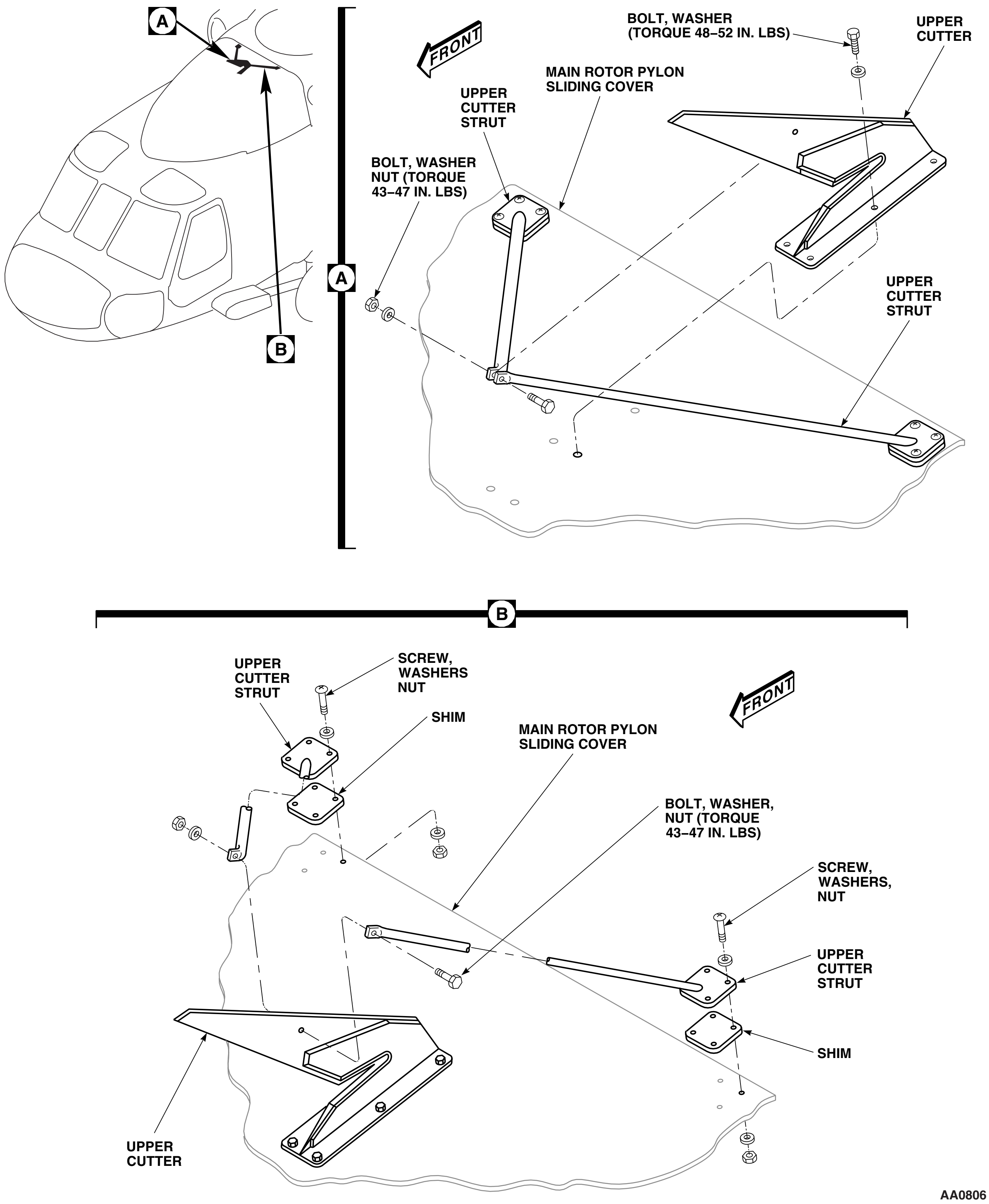


Figure 1. Replace Wire Strike Upper Cutter.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE PITOT CUTTER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

1. Remove screws and washers securing pitot cutter to main rotor pylon sliding cover (Figure 1, Detail A). Remove pitot cutter.
2. Remove sealing compound from main rotor pylon sliding cover with nonmetallic scraper.

INSTALL

1. Install pitot cutter onto main rotor pylon sliding cover as follows: (Figure 1, Detail A).
   a. **QA** Install screws and washers in two forward holes.
   b. **QA** Install screws and washers in four middle holes. **MAXIMUM PROTRUSION OF SCREWS BELOW NUTPLATES IS 0.09-INCH.** Adjust, as required, by adding washers under screw heads.
   c. **QA** Install screws, washers, lockwashers, and nuts in two aft holes with screwhead down.
2. Seal screwheads, washers, and nuts with sealing compound, Item 273, WP 1803 00.
3. Seal pitot cutter to main rotor pylon sliding cover with sealing compound, Item 273, WP 1803 00.

INSTALL - Continued

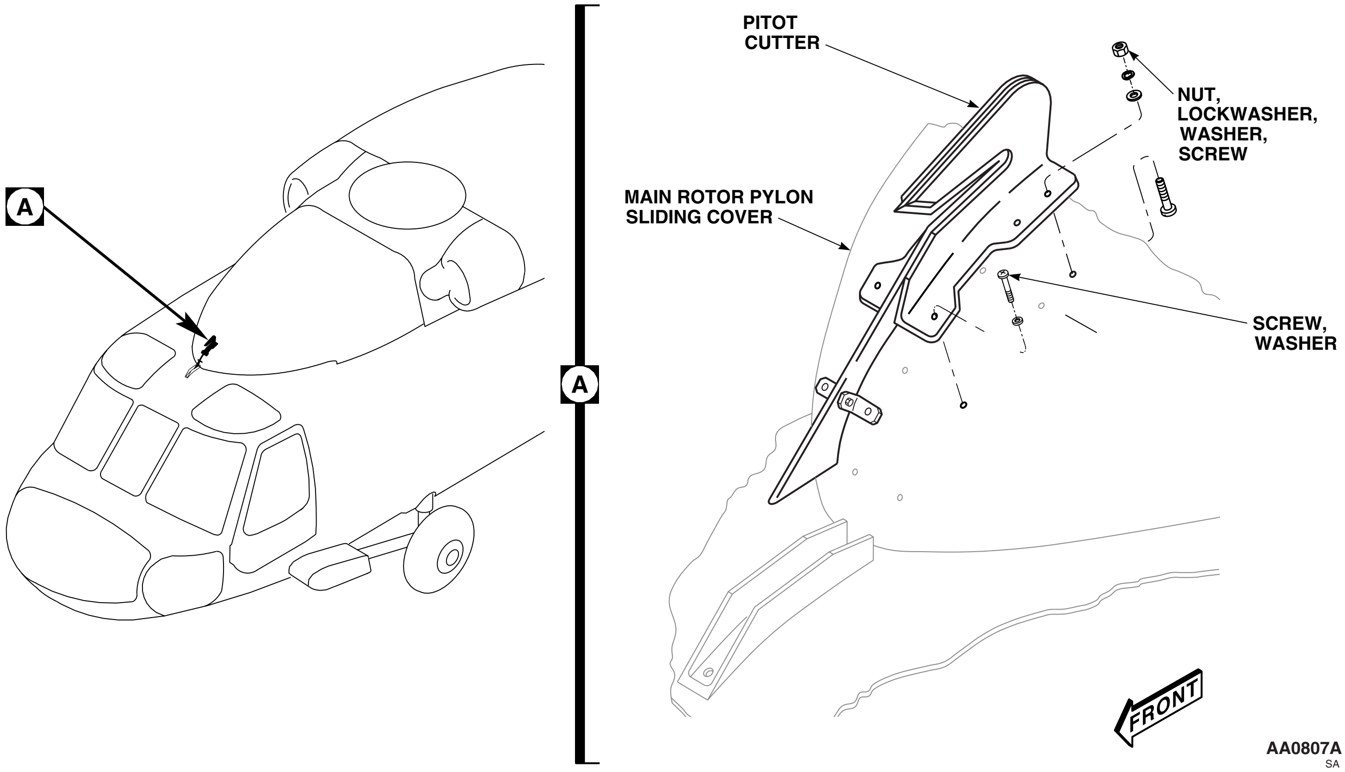


Figure 1. Replace Wire Strike Pitot Cutter.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE PITOT CUTTER GUIDE CHANNEL

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0272 00
WP 1803 00

REMOVE

1. Remove pitot cutter (WP 0272 00).
2. Remove screws and remove pitot cutter guide channel from upper canopy (Figure 1, Detail A).
3. Remove sealing compound from upper canopy with nonmetallic scraper.

INSTALL

1. **QA** Install pitot cutter guide channel onto upper canopy with screws (Figure 1, Detail A).
2. Seal screwheads with sealing compound, Item 273, WP 1803 00.
3. Seal guide channel to canopy with sealing compound, Item 273, WP 1803 00.
4. Install pitot cutter (WP 0272 00).

INSTALL - Continued

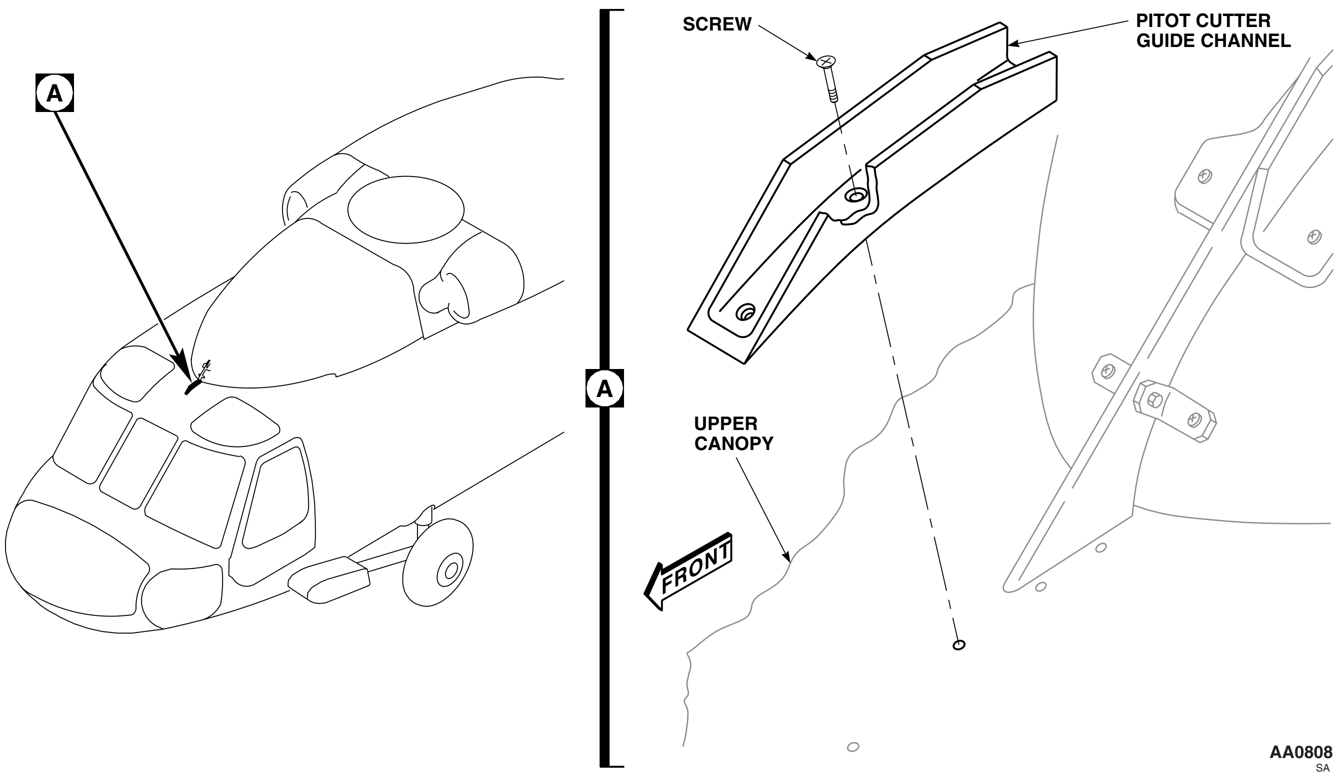


Figure 1. Replace Wire Strike Pitot Cutter Guide Channel.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE WINDSHIELD WIPER DRIVE POST DEFLECTOR

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper

Material/Parts

Sealing Compound Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1803 00

NOTE

The following procedures apply to both pilot's and copilot's windshield wiper drive post deflectors.

REMOVE

1. Remove screws and washers and remove windshield wiper drive post deflector from nose structure (Figure 1, Detail A).
2. Remove sealing compound from nose structure and deflector with nonmetallic scraper.

INSTALL

1. **QA** Install windshield wiper drive post deflector onto nose structure with screws and washers (Figure 1, Detail A).
2. Seal screwheads and washers with sealing compound, Item 273, WP 1803 00.
3. Seal deflector to nose structure with sealing compound, Item 273, WP 1803 00.

INSTALL - Continued

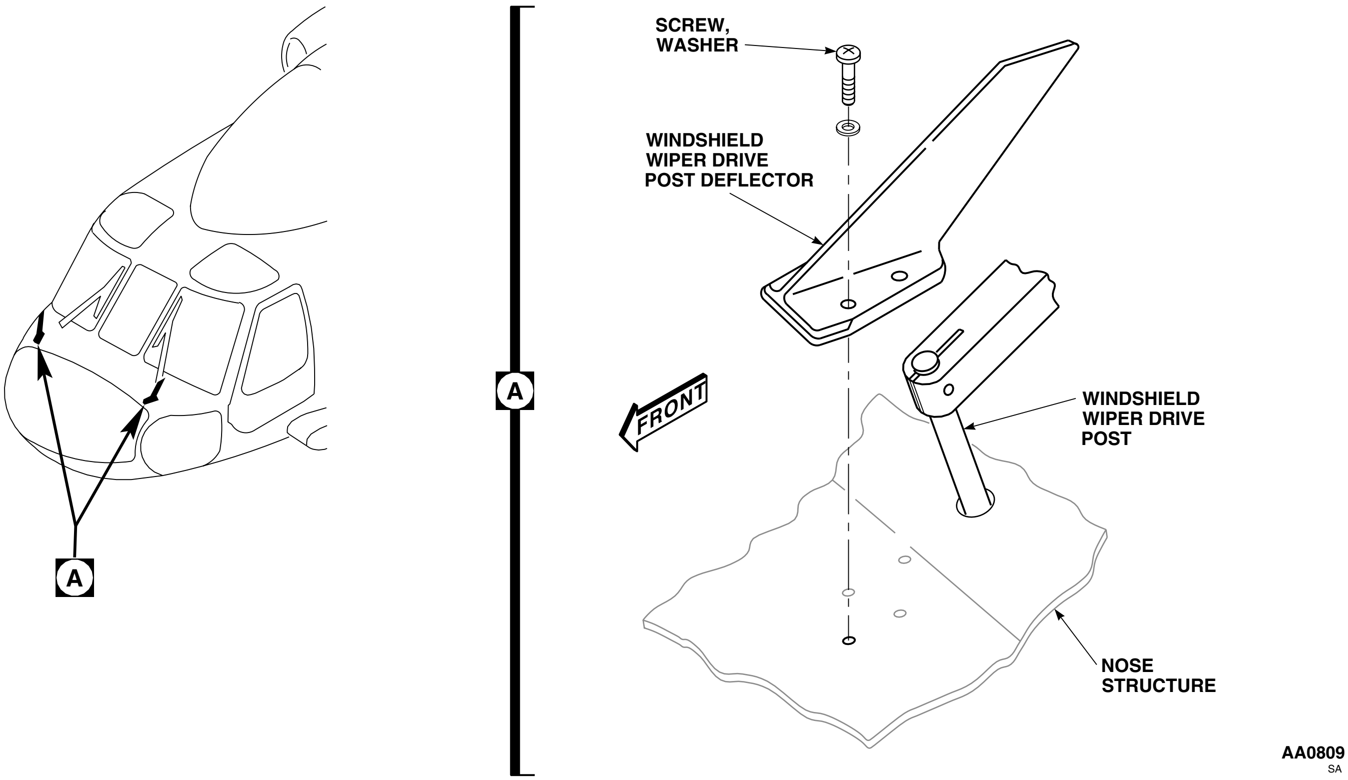


Figure 1. Replace Wire Strike Windshield Wiper Deflector.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE MAIN LANDING GEAR CUTTER

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both left and right main landing gear cutters.

REMOVE

1. Remove breakaway tip from main landing gear cutter by removing bolts, washers, and nuts (Figure 1, Detail A).
2. Remove bolt, washers, and nut and remove main landing gear cutter from clamps.

INSTALL

1. **QA** Position main landing gear cutter onto drag beam while aligning holes in main landing gear cutter with holes in clamps, and install bolts, washers, and nuts. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS (Figure 1, Detail A).**
2. **QA** Install breakaway tip on main landing gear cutter with bolts, washers, and nuts. **TORQUE BOLTS TO 85 - 95 INCH-POUNDS.**

INSTALL - Continued

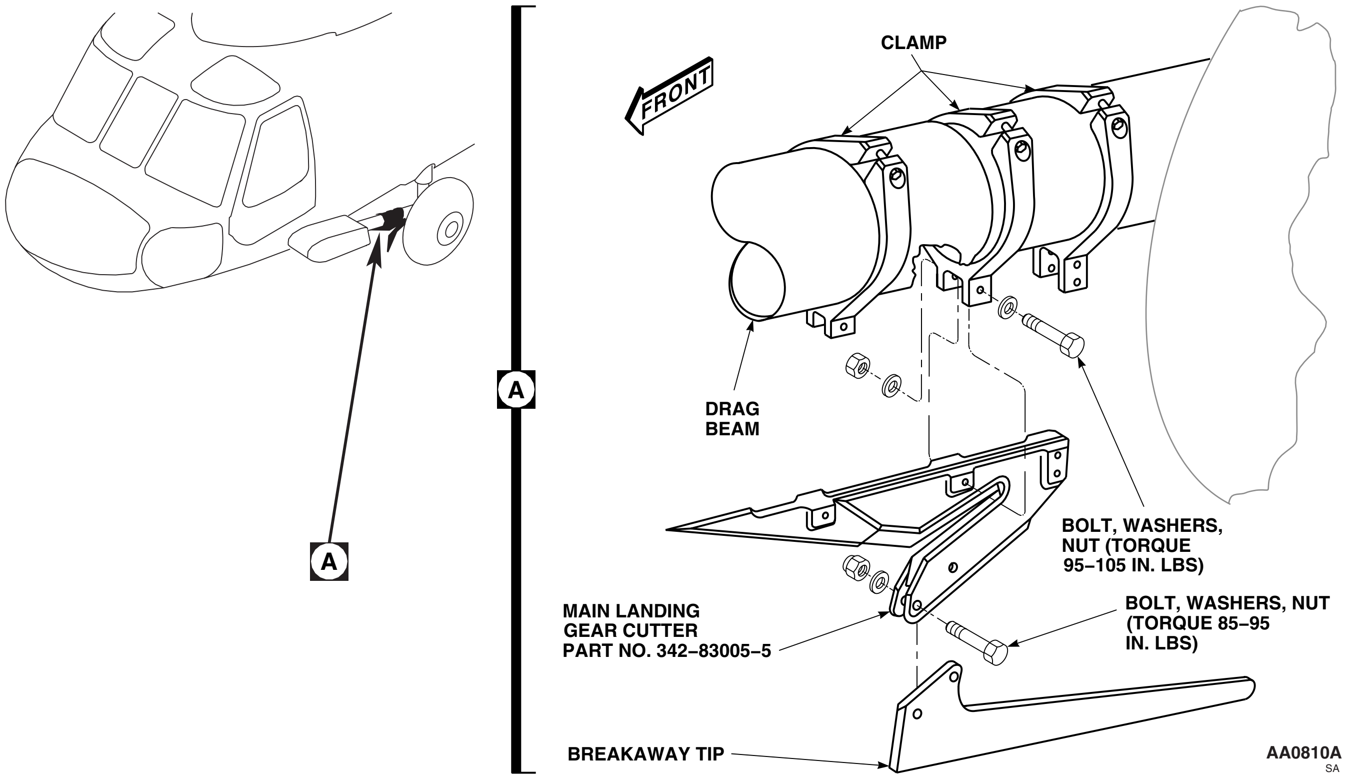


Figure 1. Replace Wire Strike Main Landing Gear Cutter.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE MAIN LANDING GEAR CUTTER CLAMP

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0275 00
WP 1803 00

NOTE

The following procedures apply to all main landing gear cutter clamps on both left and right main landing gears.

REMOVE

1. If removing forward, middle, or all three main landing gear cutter clamps, remove main landing gear cutter (WP 0275 00) (Figure 1, Detail A or Detail B).
2. If removing main landing gear cutter rear clamp only, remove bolts, washers, and nut that secure main landing gear cutter to clamp.
3. Remove bolt, washers, and nut and remove main landing gear cutter clamp from drag beam.

INSTALL

1. Apply sealing compound, Item 273, WP 1803 00, to mating surfaces of cutter clamps and drag beam.
2. **QA** Position main landing gear cutter clamps onto drag beam, and install washers and nuts handtight (Figure 1, Detail A or Detail B).
3. If installing forward, middle, or all three main landing gear cutter clamps, install main landing gear cutter (WP 0275 00).
4. **QA** If installing main landing gear cutter rear clamp only, install bolts, washers, and nuts that secure main landing gear cutter to clamp. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS.**

NOTE

Remove and replace one clamp at a time to insure proper alignment. Make sure cutter tip is aligned to vertical centerline of drag beam if all three clamps are removed.

5. If all three clamps were removed, adjust cutter by doing this:
   a. Bolt cutter to clamps.
   b. Install clamps loosely on drag beam.
   c. Position cutter tip to within 0.030 to 0.060-inch of drag beam radius (Figure 1).

INSTALL - Continued

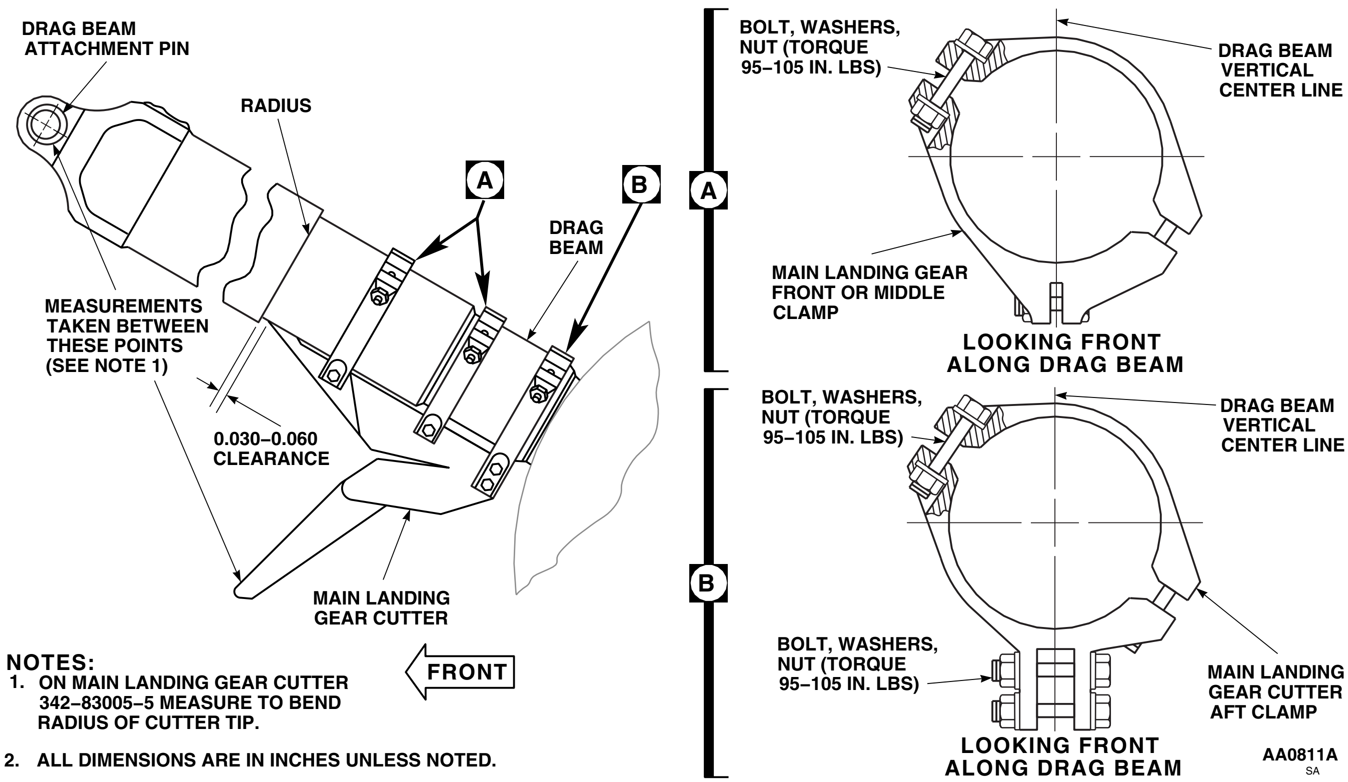


Figure 1. Replace Wire Strike Main Landing Gear Cutter Clamp.

d. When installing cutter, 342-83005-5, measure distance from edge of drag beam attachment pin to tip or bend radius of cutter. Take measurements from both inboard and outboard sides of drag beam. **MEASUREMENTS MUST BE WITHIN 1/8-INCH OF EACH OTHER.**

e. When installing cutter, 342-83005-5, visually align deflector with center of tiedown ring located between deflector and wheel hub.

6. **QA TORQUE MAIN LANDING GEAR CUTTER CLAMP BOLTS TO 95 - 105 INCH-POUNDS.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE MAIN LANDING GEAR JOINT DEFLECTOR

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0286 00

NOTE

The following procedures apply to both left and right main landing gear deflectors.

REMOVE

1. Remove main landing gear support fairing (WP 0286 00).
2. Remove bolt, washer, and main landing gear joint deflector from fuselage (Figure 1, Detail A).
3. Remove barrel-nuts and retainers from fitting.

INSTALL

NOTE

Make sure that only square barrel-nuts are used.

1. Test attachment bolts and barrel-nuts as follows:
   a. Place barrel-nut in soft-jawed vise.
   b. Thread bolt into barrel-nut until two threads are showing beyond nut.
   c. Back off bolt using torque wrench. **BREAK AWAY TORQUE SHALL NOT BE LESS THAN 18 INCH-POUNDS.**
   d. If breakaway torque is less than 18 inch-pounds, replace bolt. Repeat step a through step c.
   e. If breakaway torque is less than 18 inch-pounds, replace barrel-nut. Repeat step a through step c.

CAUTION

Damage to fitting will result if barrel-nut is incorrectly installed. To prevent damage make sure barrel-nut is installed with barrel (round side) towards bolthead.

2. **QA** Install barrel-nuts and retainers in fitting with barrel sides towards boltheads (Figure 1, Detail A).
3. **QA** Install main landing gear joint deflector onto fuselage with bolt and washer. **TORQUE BOLTS TO 732 - 808 INCH-POUNDS.**

INSTALL - Continued

4. Install main landing gear support fairing (WP 0286 00).

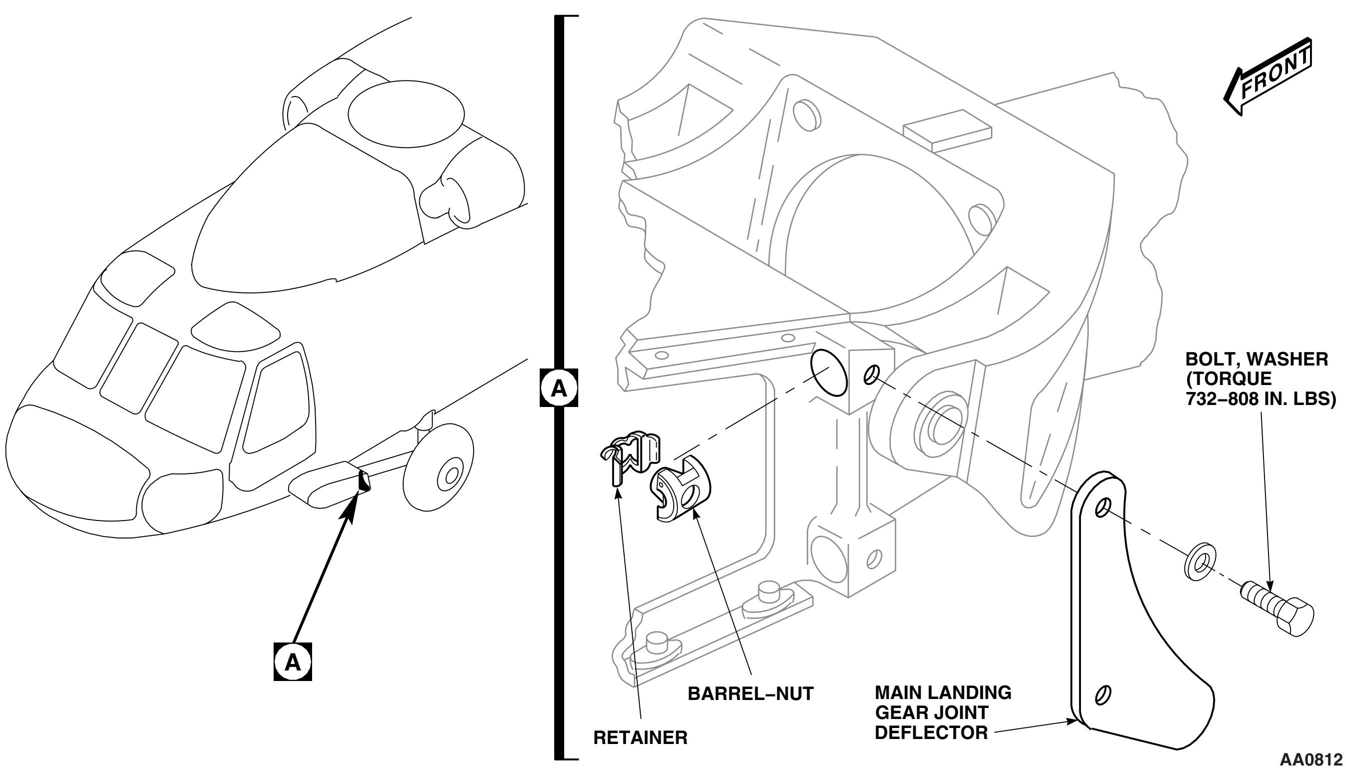


Figure 1. Replace Wire Strike Main Landing Gear Joint Deflector.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE TAIL LANDING GEAR DEFLECTOR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

Remove bolts, washers and nut securing tail landing gear deflector to clamp (Figure 1). Remove deflector.

INSTALL

1. Position tail landing gear deflector onto yoke while aligning holes in tail landing gear deflector with holes in clamp, and install bolts, washers, and nuts. Maintain minimum possible clearance between deflector and yoke (Figure 1).
2. Position tail landing gear deflector so that there is 0.070-inch clearance between deflector and tail landing gear fork.
3. **QA TORQUE TAIL LANDING GEAR DEFLECTOR CLAMP BOLTS TO 53 - 57 INCH-POUNDS.**

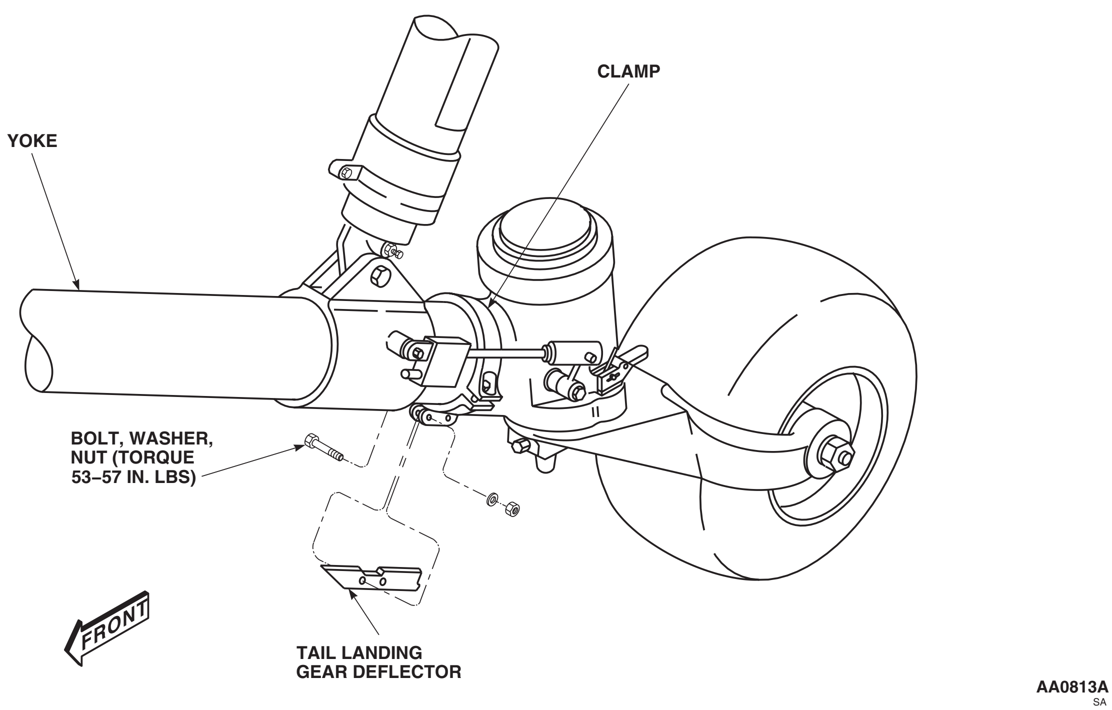


Figure 1. Replace Wire Strike Tail Landing Gear Deflector.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE TAIL LANDING GEAR DEFLECTOR CLAMP

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

References
WP 0278 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Remove tail landing gear deflector (WP 0278 00).
2. Unbolt tail landing gear deflector clamp from yoke (Figure 1, Detail B).

INSTALL

1. Position tail landing gear deflector clamp onto yoke, and install bolts, washers, and nuts handtight (Figure 1, Detail A).
2. Install tail landing gear deflector (WP 0278 00).
3. Align deflector with vertical centerline of yoke.
4. Move defector clamp along yoke to obtain 0.070-inch clearance between deflector and tail landing gear fork (Figure 1, Detail B).
5. **QA TORQUE TAIL LANDING GEAR DEFLECTOR CLAMP BOLTS TO 119 - 131 INCH-POUNDS.**

INSTALL - Continued

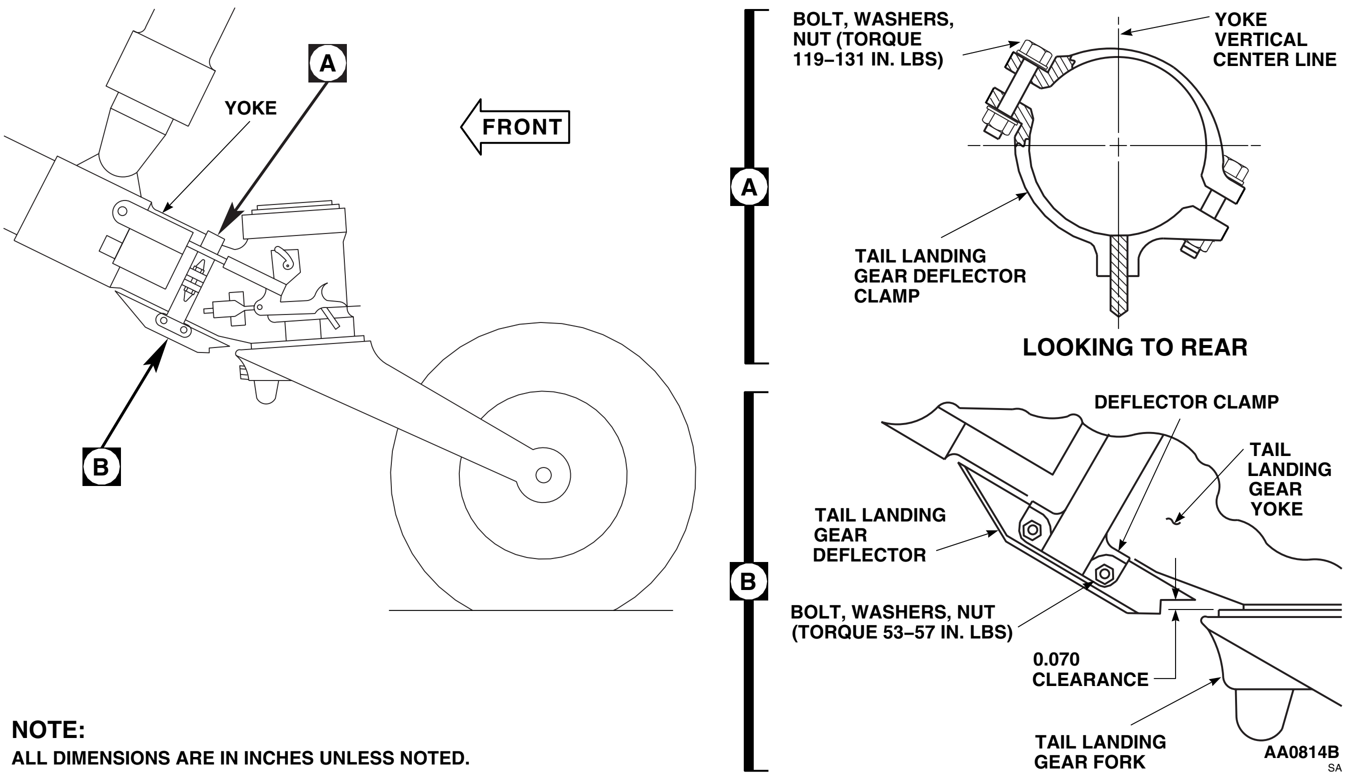


Figure 1. Replace Wire Strike Tail Landing Gear Deflector Clamp.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE STEP DEFLECTOR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

NOTE

The following procedures apply to left step deflector and to right step deflector when fuel probe is not installed.

REMOVE

1. Remove screws and washers and remove step deflector from fuselage (Figure 1, Detail A).
2. Remove sealing compound from fuselage with nonmetallic scraper.

INSTALL

1. **QA** Install step deflector onto fuselage with screws and washers (Figure 1, Deatil A).
2. Seal screwheads and washers with sealing compound, Item 273, WP 1803 00.
3. Seal step deflector to fuselage with sealing compound, Item 273, WP 1803 00.

INSTALL - Continued

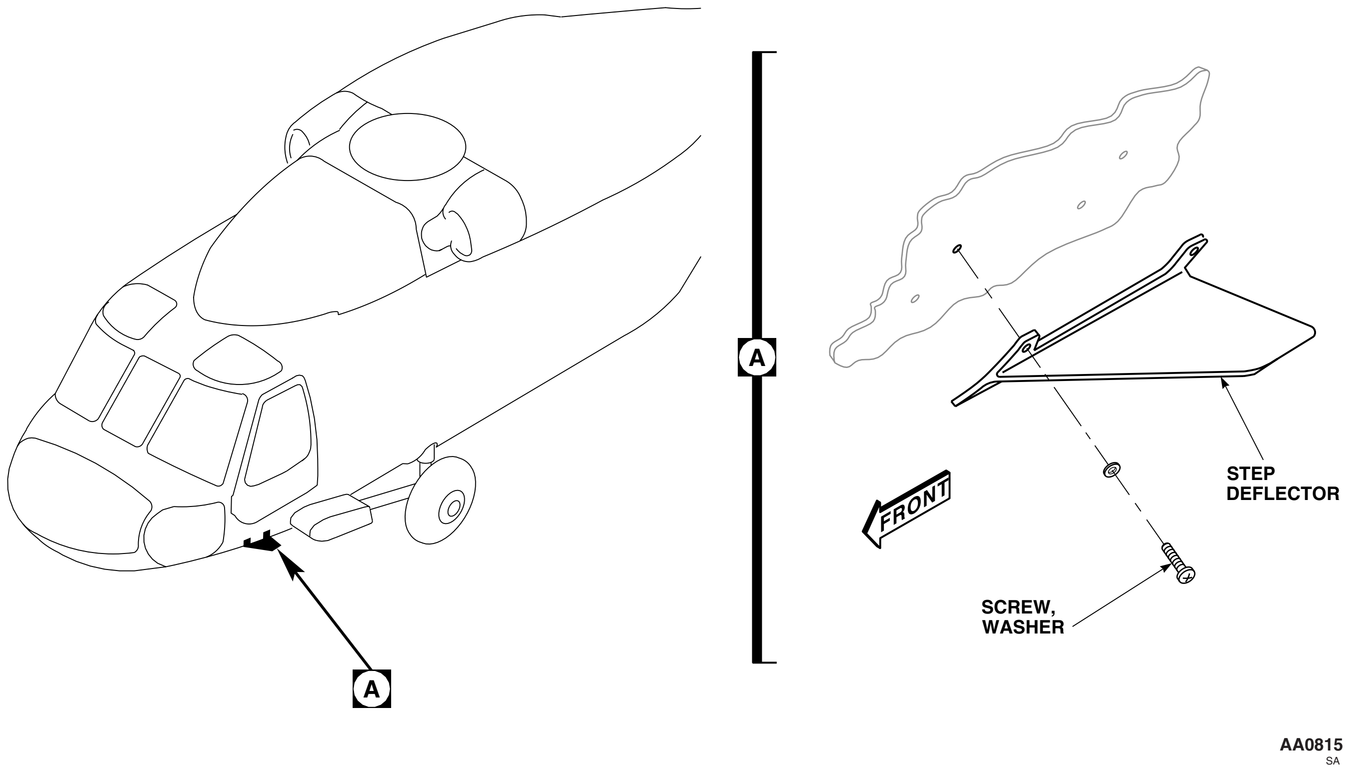


Figure 1. Replace Wire Strike Step Deflector.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE STEP EXTENSION

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both left step extension and to right step extension when fuel probe is not installed.

REMOVE

1. Remove plate from step extension by removing screws, lock washers and washers (Figure 1, Detail A).
2. Remove step extension from step by removing nuts, washers, and screws.

INSTALL

1. **QA** Install step extension onto step with screws, washers, and nuts (Figure 1, Detail A).
2. **QA** Install plate onto step extension with screws, lockwashers, and washers.

INSTALL - Continued

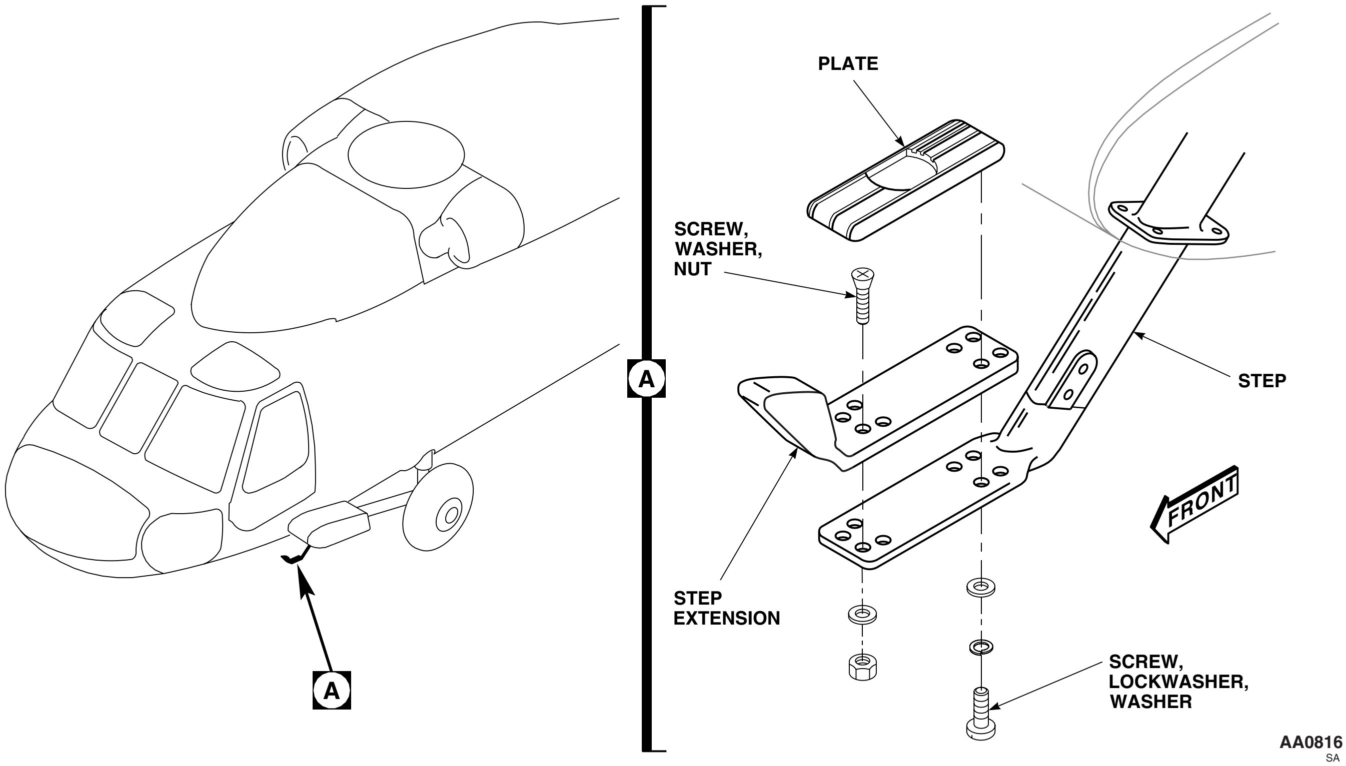


Figure 1. Replace Wire Strike Step Extension.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE COCKPIT DOOR LATCH DEFLECTOR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

NOTE

The following procedures apply to both left and right door latch deflectors.

REMOVE

1. Remove door latch deflector from windshield frame by removing screws, washers, and packings (Figure 1, Detail A). Discard packing.
2. Remove sealing compound from frame with nonmetallic scraper.

INSTALL

1. **QA** Install door latch deflector onto windshield frame with screws, washers, and packings. **TORQUE SCREWS 15 - 18 INCH-POUNDS (Figure 1, Detail A)**.
2. Seal screwheads and washers with sealing compound, Item 273, WP 1803 00.
3. Seal door latch deflector to frame with sealing compound, Item 273, WP 1803 00.

INSTALL - Continued

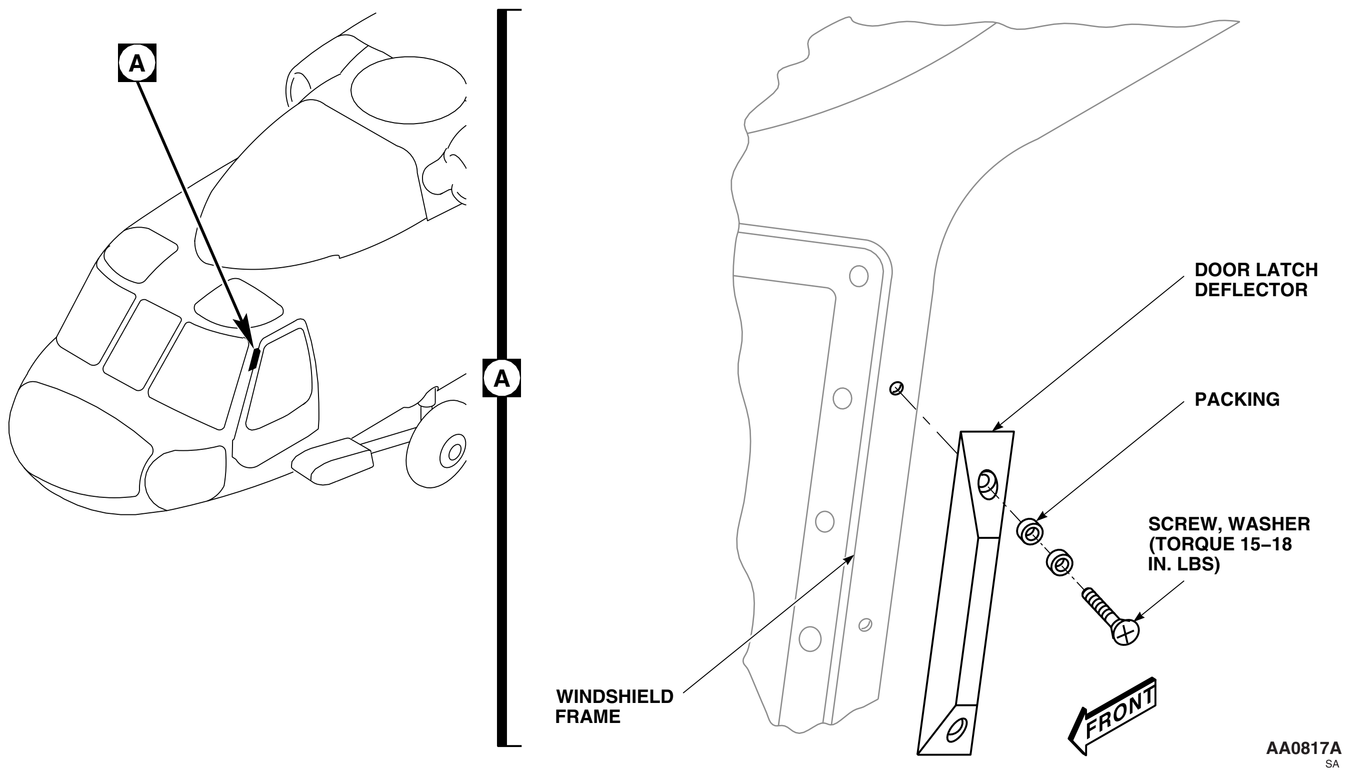


Figure 1. Replace Wire Strike Door Latch Defector.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE WINDSHIELD CENTER POST DEFLECTOR INSERT

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both pilot's and copilot's windshield center post deflector inserts.

REMOVE

Remove windshield center post deflector insert from channel by removing nuts, washers, and screws (Figure 1, Detail A).

INSTALL

Install windshield center post deflector insert into channel with screws, washers, and nuts (Figure 1, Detail A). **QA check**

INSTALL - Continued

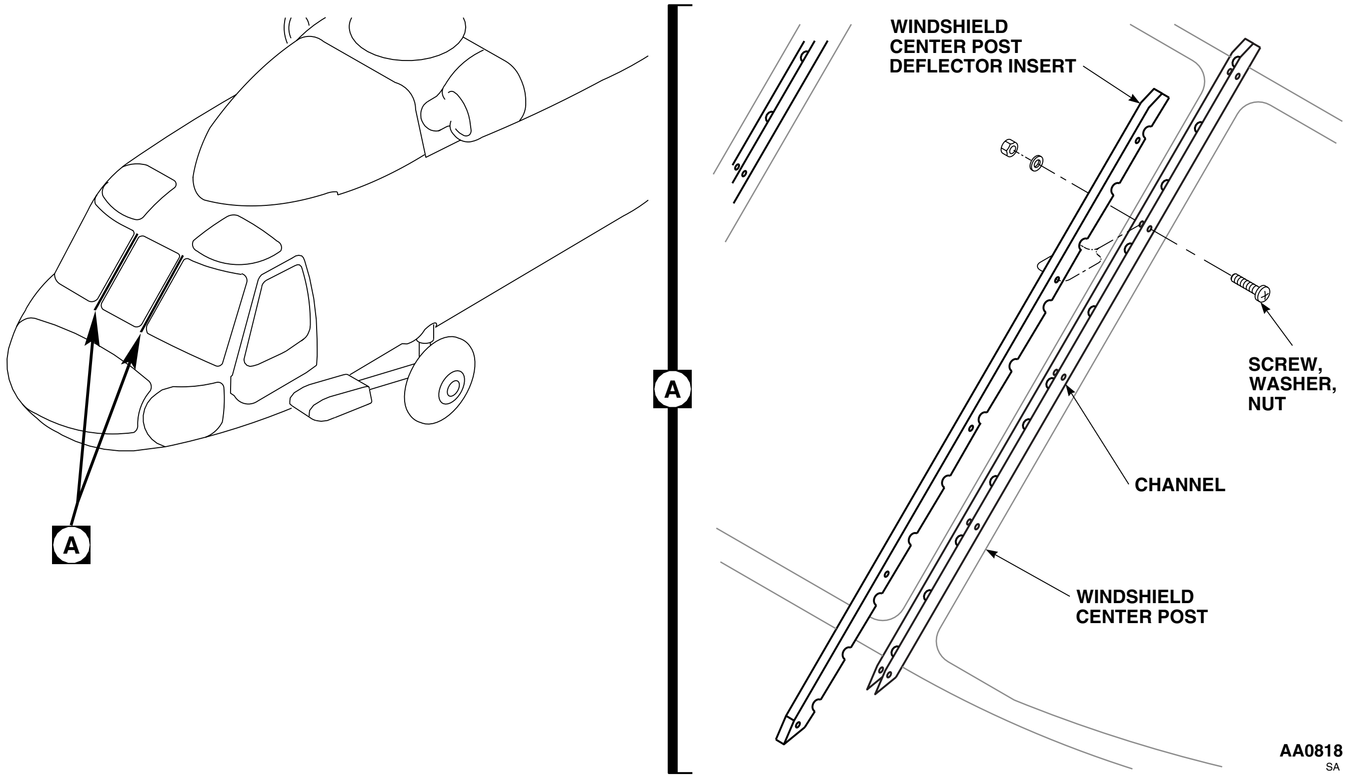


Figure 1. Replace Wire Strike Windshield Center Post Deflector Insert.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

WIRE STRIKE WINDSHIELD CENTER POST DEFLECTOR CHANNEL

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
No. 11 (0.191-Inch Diameter) Drill
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Sealing Compound Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0283 00
WP 1803 00

NOTE

The following procedures apply to both pilot's and copilot's windshield center post deflector channels.

REMOVE

1. Remove windshield center post deflector insert (WP 0283 00).
2. Remove windshield center post deflector channel from windshield center post by removing screws, washers, and packings (Figure 1, Detail A). Discard packing.
3. Remove old sealing compound from windshield center post deflector channel and windshield with a nonmetallic scraper.

INSTALL

NOTE

Center post deflector channel may need to have screwholes enlarged.

1. Enlarge screwholes in deflector channel into transverse slots, using No. 11 drill. **LENGTH OF SLOT SHALL NOT EXCEED INSIDE WIDTH OF CHANNEL.**
2. Apply a 1/8-inch wide bead of sealing compound, Item 273, WP 1803 00, around edges of deflector channel, and position deflector on windshield over center post, with packings, in position between channel and windshield (Figure 1, Detail A).
3. **QA** Apply sealing compound, Item 273, WP 1803 00, to threads of screws, and install screws and washers. **TORQUE SCREWS TO 15 - 18 INCH-POUNDS.**
4. Remove excess sealant from edges of deflector and screw heads.
5. Install windshield center post deflector (WP 0283 00).

INSTALL - Continued

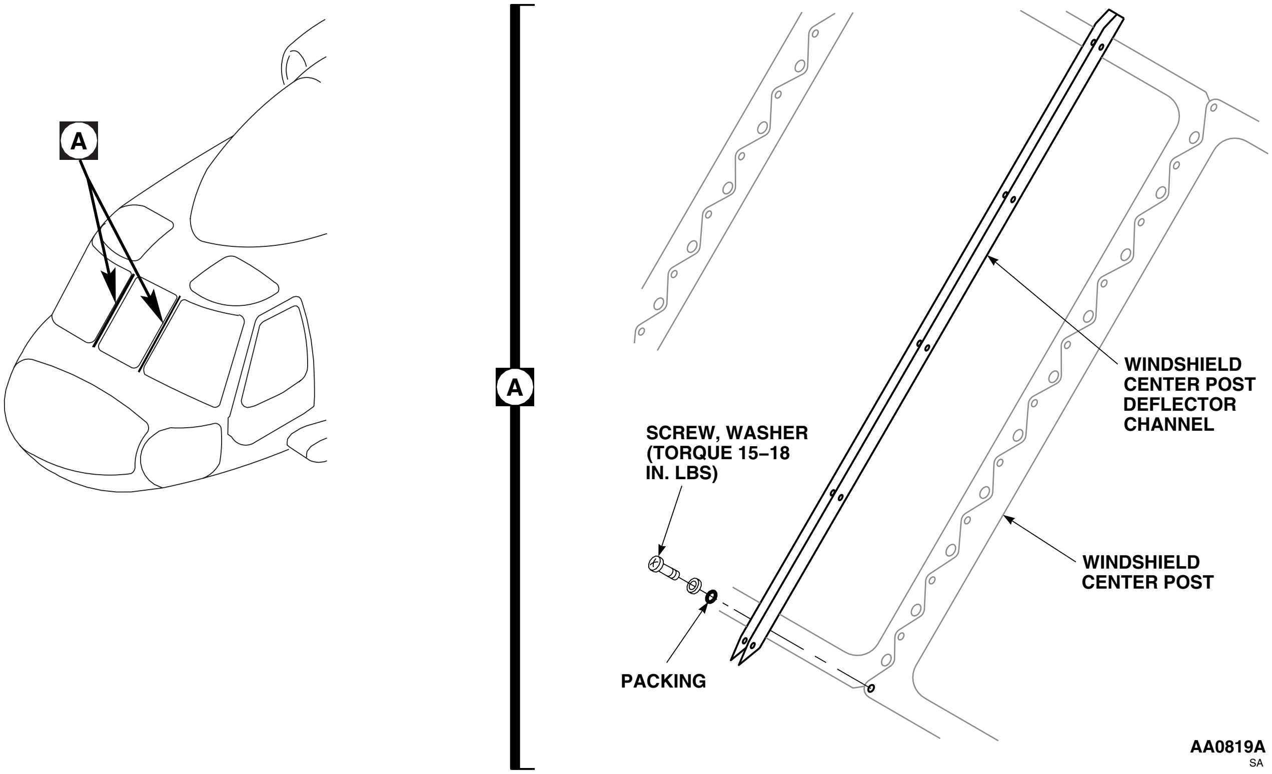


Figure 1. Replace Wire Strike Windshield Center Post Deflector Channel.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

FUSELAGE JACKPADS

This WP supersedes WP 0285 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 30-200 in. lbs.

Material/Parts
Acetone, Technical Item 5, WP 1803 00
Antiseize Compound Item 49, WP 1803 00
Cleaning Compound, Solvent Item 71, WP 1803 00
Cloth, Cheesecloth Item 85, WP 1803 00
Corrosion Preventive Compound Item 107, WP 1803 00
Sealing Compound Item 273, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 1803 00

REPLACE DRAG BEAM SUPPORT JACKPAD

Remove

1. Remove jackpad (Figure 1, Detail A).
2. Wipe stud and insert threads clean using cheesecloth cloth, Item 85, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

Inspect

1. Check jackpad stud for misalignment and thread damage.
   a. Replace pads having bent studs.
   b. Repair damaged threads (TM 1-1500-204-23).
2. Check pad threaded insert for tightness. Repair or replace loose or damaged inserts (TM 1-1500-204-23).

Install

1. Coat jackpad stud threads with antiseize compound, Item 49, WP 1803 00.
2. **QA** Install jackpad (Figure 1, Detail A). **TORQUE JACKPAD TO 176 - 194 INCH-POUNDS.**

REPLACE TAIL CONE JACKPAD

Remove

1. Remove screws and washers from front section of tail landing gear fairing (Figure 2, Detail A). Remove fairing.
2. Remove jackpad attaching hardware. Discard used hardware (TM 1-1500-204-23).
3. Using a nonmetallic scraper, separate bond between jackpad and structure. Remove jackpad.

REPLACE TAIL CONE JACKPAD - Continued

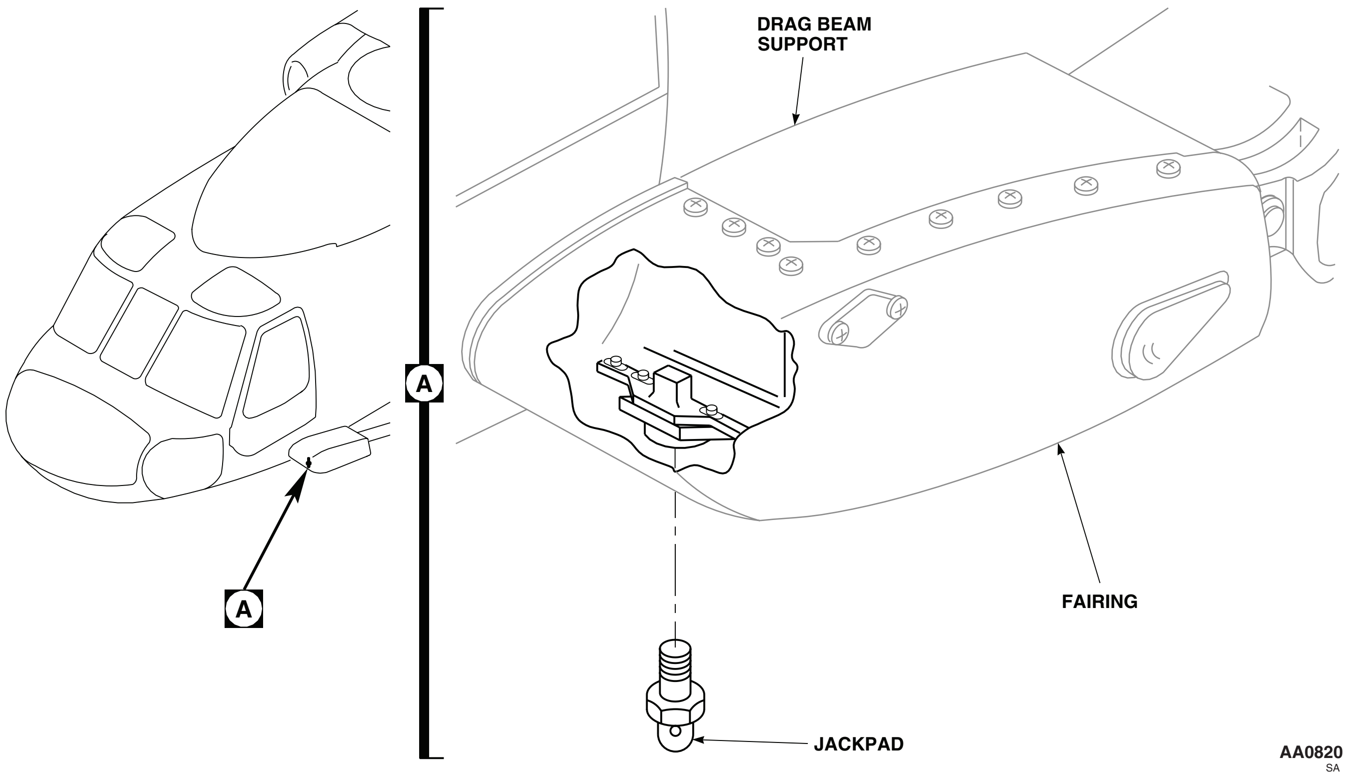


Figure 1. Replace Drag Beam Support Jackpad.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Remove sealing compound debris using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00 and a nonmetallic scraper.

Install

1. Using clean cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, wipe jackpad, fuselage mating surfaces, fastener holes, and fasteners, until clean.
2. Coat new fastener pins, jackpad, and fuselage joint surfaces with sealing compound, Item 273, WP 1803 00 (Figure 2, Detail A).
3. Install jackpad with pins and collars while sealing compound is wet (TM 1-1500-204-23).
4. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
5. **QA** Install fairing with screws and washers.

REPLACE TAIL CONE JACKPAD - Continued

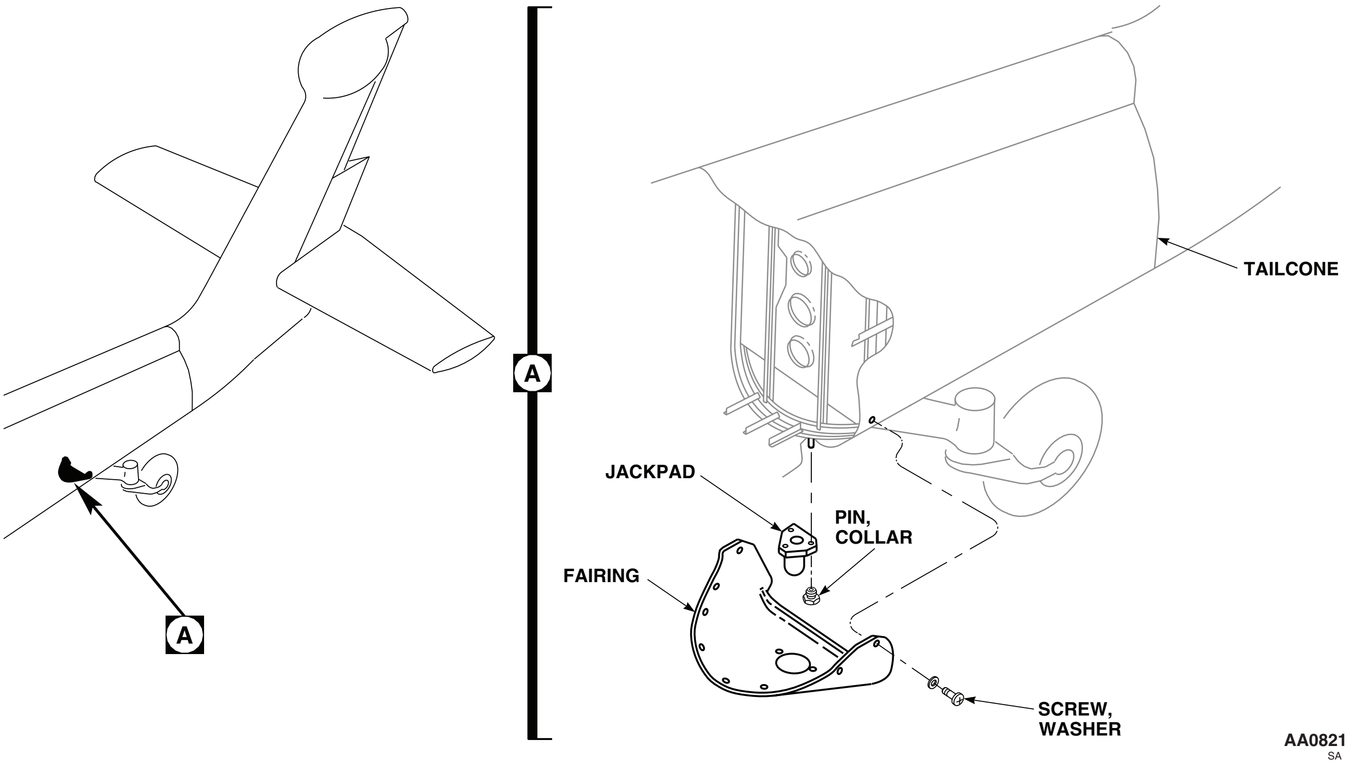


Figure 2. Replace Tail Cone Jackpad.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

DRAG BEAM SUPPORT FAIRING

This WP supersedes WP 0286 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Abrasive Cloth, Assortment Item 1, WP 1803 00
Acetone, Technical Item 5, WP 1803 00
Adhesive Item 14, WP 1803 00
Cloth, Cheesecloth Item 85, WP 1803 00
Corrosion Preventive Compound Item 107, WP 1803 00
Sealing Compound Item 273, WP 1803 00
Tape, Lacing and Tying Item 321, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0113 00
WP 0293 00
WP 0379 00
WP 0905 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove cockpit step (WP 0293 00).
3. Remove screws and washers, and separate fairing from fuselage enough to get at electrical connector for position light (Figure 1, Detail A).
4. Disconnect electrical connector, P254 (left) or P253 (right), from position light.
5. Remove fairing.
6. If fairing is to be replaced, remove position light (WP 0905 00).

REPAIR

1. Repair support fairing (WP 0379 00).
2. To replace fairing seal, do this:
   a. Remove damaged seal, seal debris, and adhesive using a nonmetallic scraper.
   b. Roughen seal and fairing bonding surfaces using abrasive cloth assortment, Item 1, WP 1803 00.

REPAIR - Continued

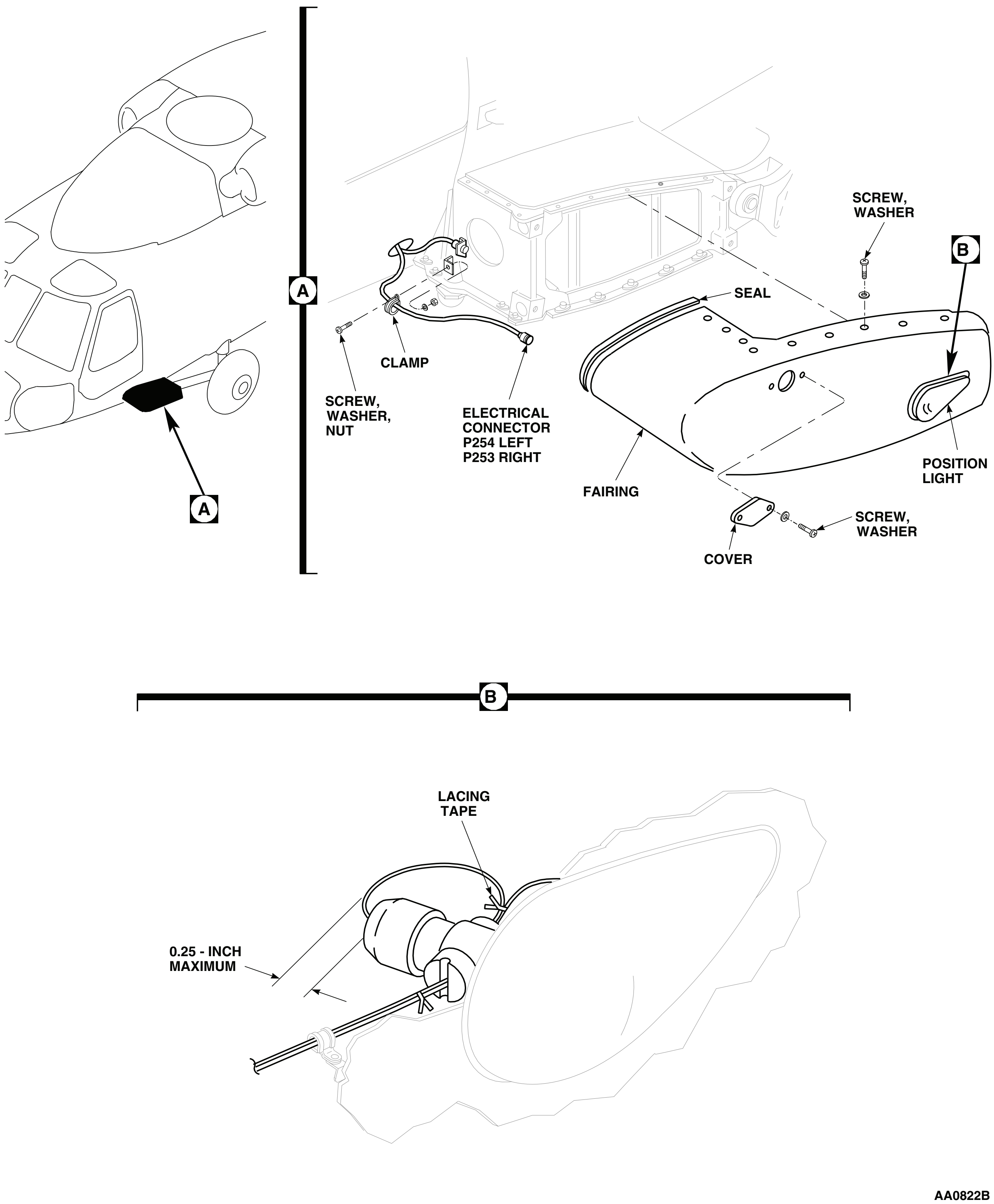


Figure 1. Replace Drag Beam Support Fairing.

REPAIR - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Wipe bonding surfaces clean with cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.

d. Apply a thin even coat of adhesive, Item 14, WP 1803 00, to seal and fairing bonding surfaces. Allow adhesive to become tacky.

e. Bond seal to fairing. Work out wrinkles or blisters. Check seal installation for complete contact.

f. Remove extra adhesive.

INSTALL

1. Turn off all electrical power.
2. If a replacement fairing is being installed, install position light on fairing (WP 0905 00).
3. Inspect and treat electrical connector (WP 1703 00).
4. **QA** Connect electrical connector, P254 (left) or P253 (right), to position light (Figure 1, Detail A).
5. Check position light wire harness for clearance (Figure 1, Detail B). Adjust as necessary and tie wires together with lacing and tying tape, Item 321, WP 1803 00.
6. Place fairing into position against helicopter.
7. Make sure area is clean and free of foreign material before closing up.
8. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
9. **QA** Position fairing to line up with screwholes in fuselage and install attachment screws and washers.
10. Seal joining area between fairing and fuselage with sealing compound, Item 273, WP 1803 00.
11. Install step (WP 0293 00).
12. Check operation of position light (WP 0113 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

LOWER MAIN LANDING GEAR FAIRING W/O ESSS

This WP supersedes WP 0287 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive Item 10, WP 1803 00
Cloth, Abrasive Item 77, WP 1803 00
Cloth, Cheesecloth Item 85, WP 1803 00
Corrosion Preventive Compound Item 107, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0379 00
WP 1803 00

REMOVE FAIRING

NOTE

Procedures are the same for both left and right side.

1. Remove screws, washers, and lower fairing cover from lower fairing (Figure 1, Detail A).
2. Remove screws, washers, and lower fairing from fuselage.

INSTALL FAIRING

1. If installing new fairing, do this:
   a. Trial-fit replacement fairing on fuselage.
   b. Mark holes that do not line up with nutplates.
   c. Fill marked holes using procedures in WP 0379 00.
   d. Locate and drill new holes in fairing, using fuselage as template.
2. Make sure area is clean and free of foreign material before closing up.
3. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
4. **QA** Fasten fairing to fuselage with screws and washers (Figure 1, Detail A).
5. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
6. **QA** Install cover on fairing and fasten in place with screws and washers.

REPLACE FAIRING RUBBER SEAL

1. Remove damaged rubber seal using nonmetallic scraper (Figure 1, Detail A).

REPLACE FAIRING RUBBER SEAL - Continued

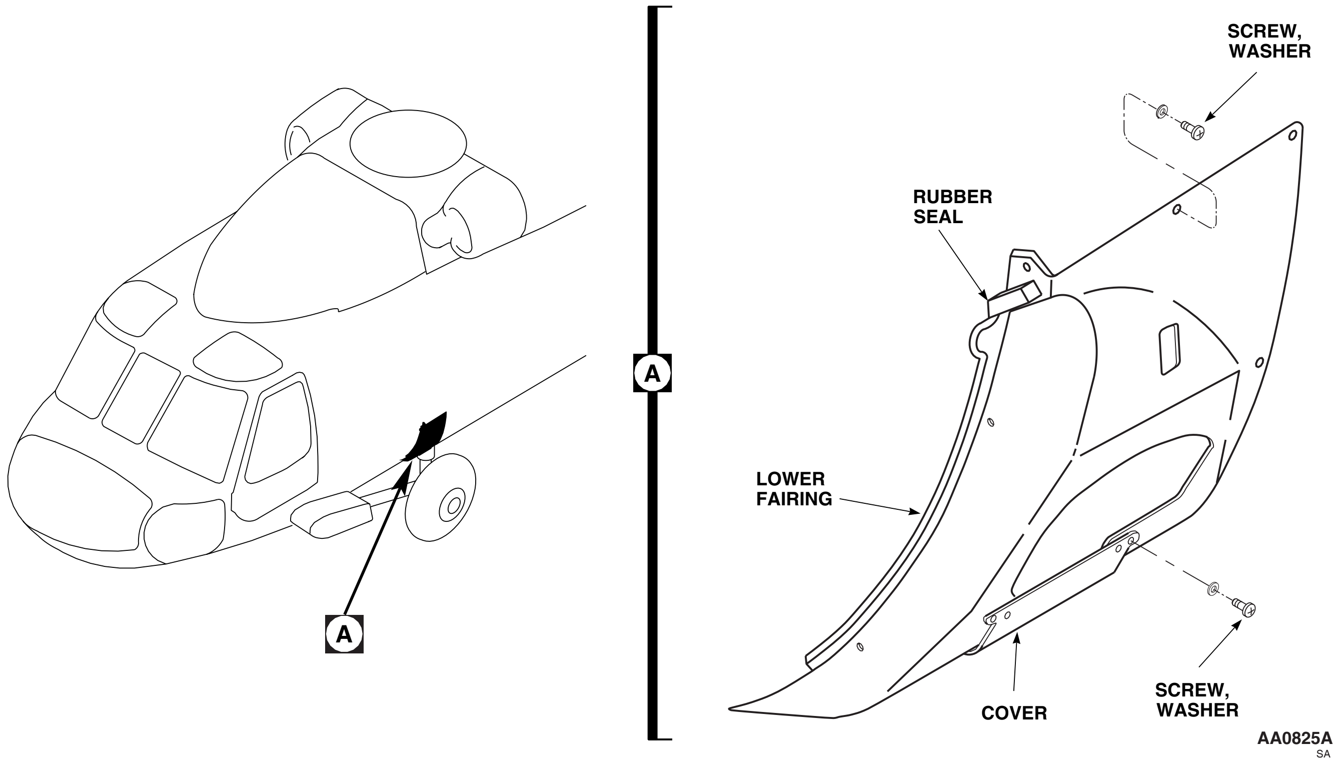


Figure 1. Replace Lower Main Landing Gear Fairing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean old adhesive from fairing using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Lightly abrade both rubber seal and fairing bonding area surface with abrasive cloth, Item 77, WP 1803 00.
4. Wipe bonding surfaces clean with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, to remove any sanding residue.
5. Apply a thin coat of adhesive, Item 10, WP 1803 00, to rubber seal and fairing bonding surfaces. Allow sufficient time for solvent to evaporate completely so that very little tack remains. Assembly before evaporation of solvent will result in failure of bond.
6. Position rubber seal on fairing surface, making sure entire mating surfaces are in direct contact. Clamp rubber seal in place.
7. Allow adhesive to dry for 24 hours at room temperature before removing clamp.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

UPPER MAIN LANDING GEAR FAIRING W/O ESSS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0379 00
WP 1803 00

REMOVE

NOTE

Procedures are the same for both left and right side.

1. Remove screws and washers and unfasten upper part of fairing from fuselage (Figure 1, Detail A).
2. Remove upper fairing.
3. Check fairing door hinge to make sure pin is properly retained by peening of both ends.

INSTALL

1. If installing new fairing, do this:
   a. Trial-fit fairing on fuselage.
   b. Mark holes or studs that do not line up with nutplates or receptacles.
   c. Fill marked holes using procedures in WP 0379 00.
   d. Mark and drill new holes using fuselage as template.
   e. If removed, replace stud.
2. Make sure area is clean and free of foreign material before closing up.
3. Place fairing into position on fuselage (Figure 1, Detail A).
4. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
5. **QA** Attach fairing to fuselage with screws and washers.
6. Fasten upper part of fairing to fuselage.

INSTALL - Continued

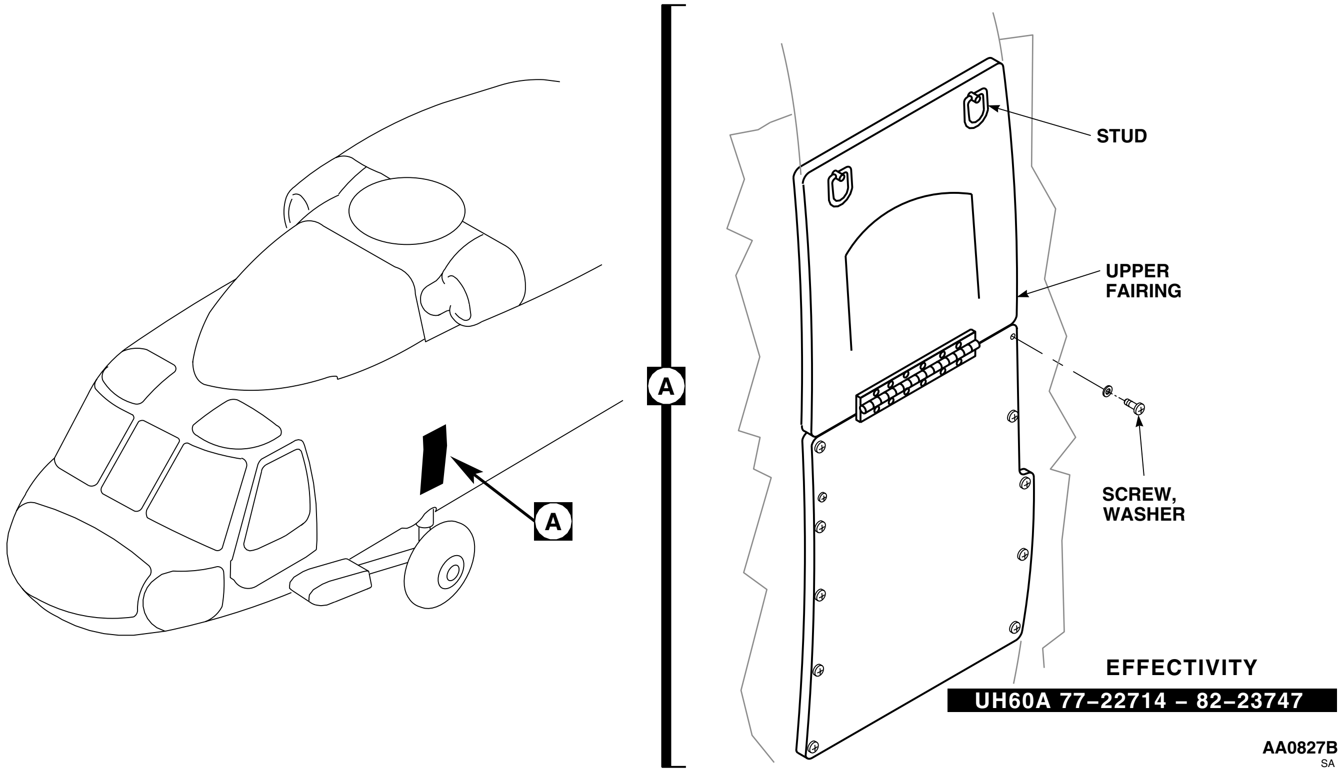


Figure 1. Replace Upper Main Landing Gear Fairing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

LOWER MAIN LANDING GEAR FAIRING ESSS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Preventive Compound Item 107, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0379 00
WP 1803 00

REMOVE

NOTE

Procedures are the same for both left and right side.

1. Unlock studs, and remove lower cap fairings from ESSS fittings (Figure 1, Detail A).
2. Remove screws, washers, and lower fairing cover from lower fairing.
3. Remove screws, washers, and lower fairing from fuselage.

INSPECT

1. Check drain tube exit holes for missing grommets. Replace grommets as necessary (Figure 1, Detail A).
2. For left fairing, cut grommet to correct length and install. **GAP BETWEEN CUT ENDS MUST NOT BE OVER 0.025-INCH.**
3. For right fairing, split round grommet and insert in hole.

INSTALL

1. If installing new fairings, do this:
   a. Trial-fit replacement fairing on fuselage.
   b. Mark holes that do not line up with nutplates.
   c. Fill marked holes using procedures in WP 0379 00.
   d. Locate and drill new holes in fairing, using fuselage as template.
2. Make sure area is clean and free of foreign material before closing up.
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. **QA** Fasten fairing to fuselage with screws and washers (Figure 1, Detail A).
5. Insert drain tubes through fairing exit holes.
6. Apply corrosion preventive compound, Item 105, WP 1803 00, to fairing attachment hardware.
7. **QA** Install cover on fairing and fasten in place with screws and washers.

INSTALL - Continued

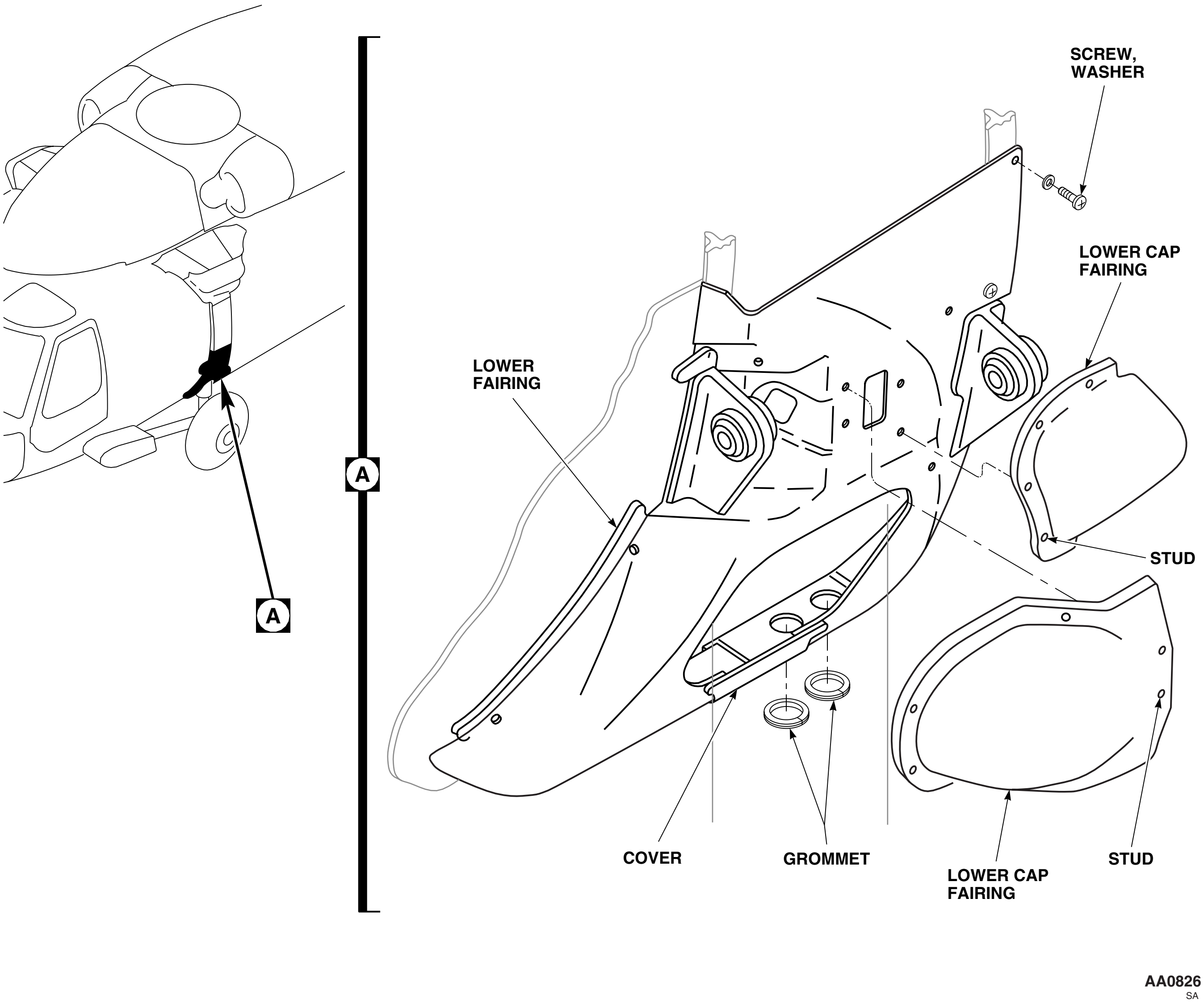


Figure 1. Replace Lower Main Landing Gear Fairing.

8. **QA** Install lower cap fairings over fittings with studs.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

UPPER MAIN LANDING GEAR FAIRING ESSS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Preventive Compound Item 107, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

NOTE

Procedures are the same for both left and right side.

1. Remove screws and washers and unfasten door of fairing from fuselage (Figure 1, Detail A).
2. Remove upper fairing.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. Place fairing into position on fuselage (Figure 1, Detail A).
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. **QA** Attach fairing to fuselage with screws and washers.
5. **QA** Fasten upper part of fairing to fuselage.

INSTALL - Continued

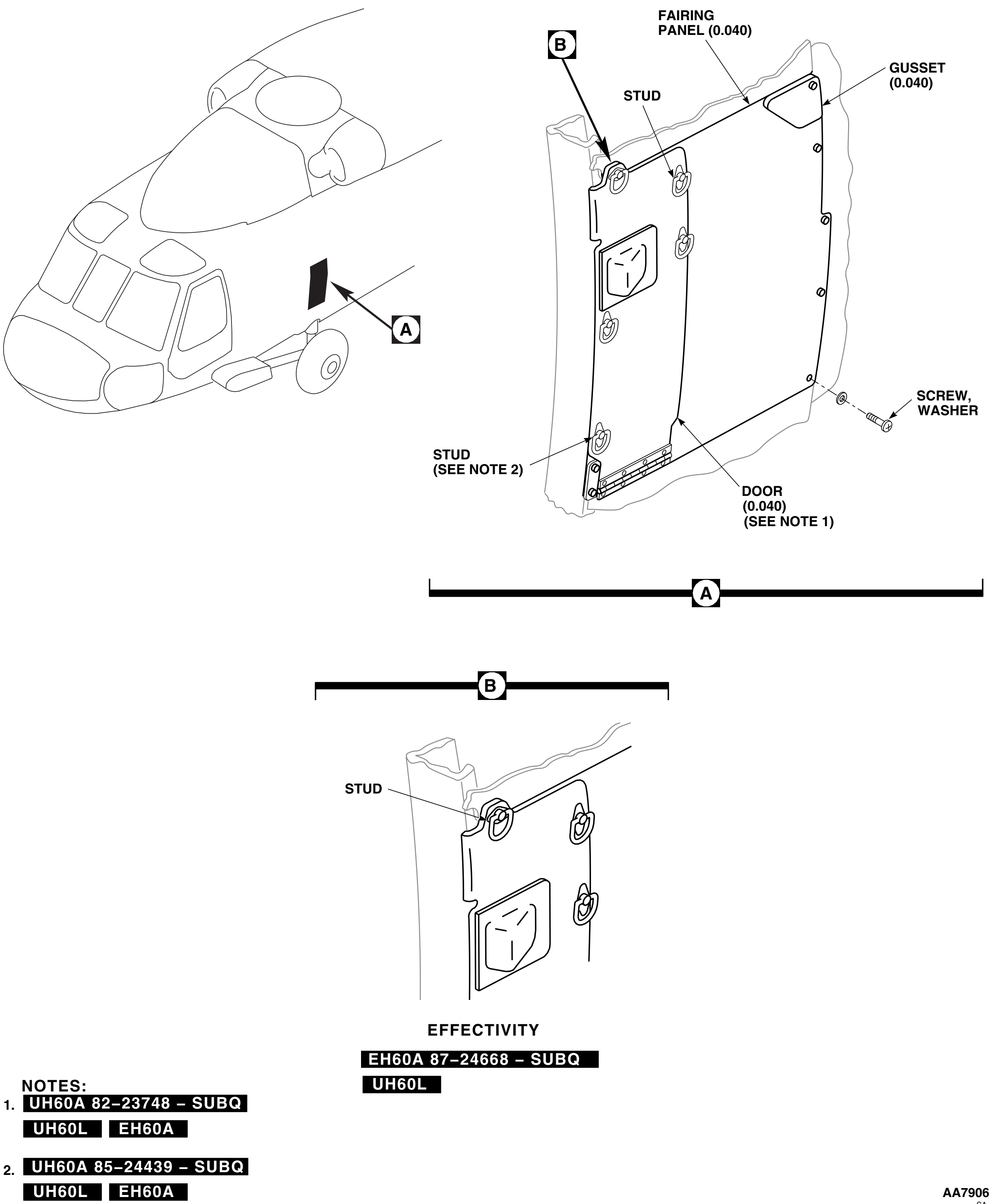


Figure 1. Replace Upper Main Landing Gear Fairing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL LANDING GEAR FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Preventive Compound Item 107, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0378 00
WP 0379 00
WP 1803 00

REMOVE

1. Remove screws and washers attaching section of fairing covering jack point forward of yoke (Figure 1, Detail A). Remove fairing from helicopter.
2. Remove screws and washers attaching front plate to fairing. Remove plate from fairing.
3. Remove screws and washers attaching rear plate to fairing. Remove plate from fairing.
4. Remove screws and washers attaching left and right fairings to helicopter. Remove fairings from helicopter.
5. Check condition of removed fairing sections and plates. If damaged, repair using instructions in WP 0378 00 and WP 0379 00.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
3. **QA** Place left and right fairings in position on helicopter fuselage. Install screws and washers (Figure 1, Detail A).
4. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
5. **QA** Place rear plate in position on fairing. Fasten plate to fairings with screws and washers.
6. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
7. **QA** Place front plate in position on fairing. Fasten plate to fairings with screws and washers.
8. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
9. **QA** Position front section of fairing, covering jack point forward of yoke, on fuselage. Attach with screws and washers.

INSTALL - Continued

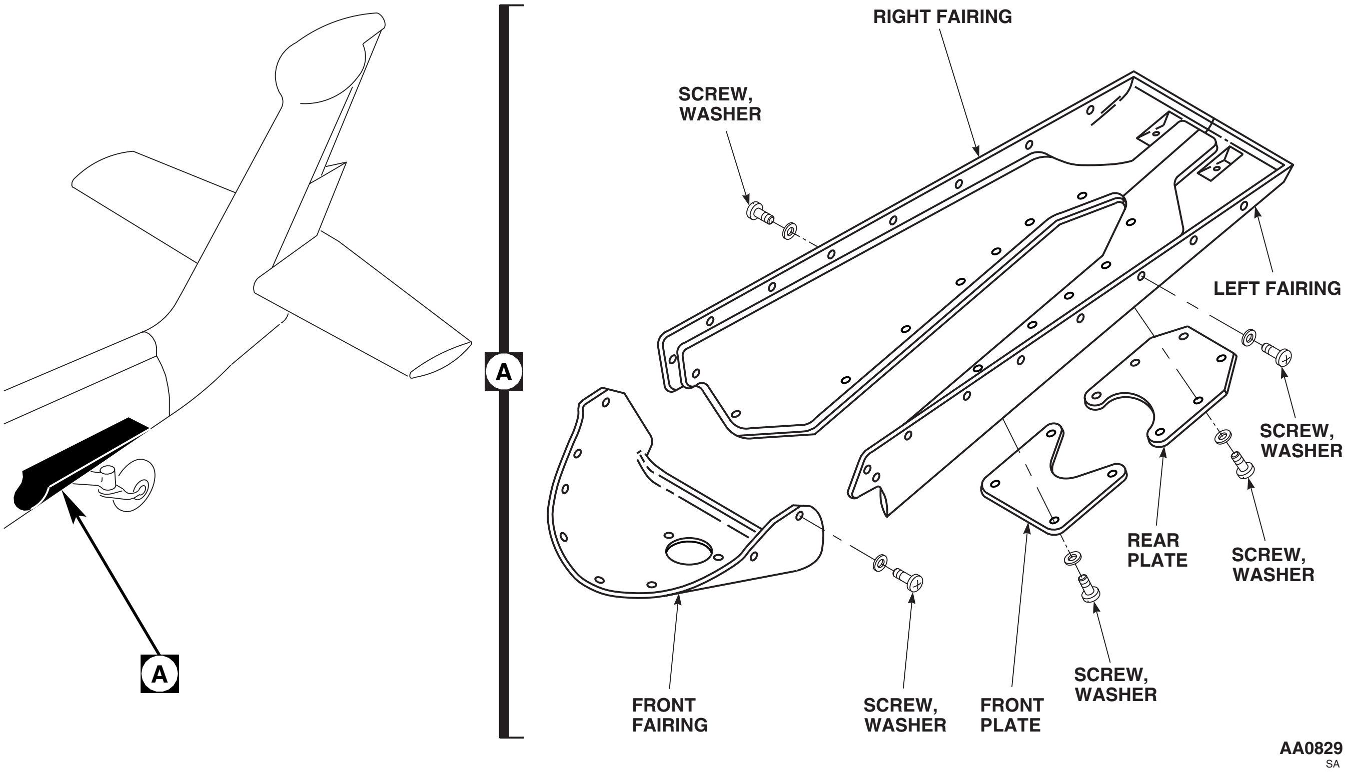


Figure 1. Replace Tail Landing Gear Fairing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

UPPER ESSS FITTING FAIRING AND PLATFORM ESSS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Corrosion Preventive Compound Item 107, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

NOTE

Procedures are the same for left and right side.

1. Unfasten studs and remove ESSS cap fairing (Figure 1, Detail A).
2. Unfasten studs and screws holding front ESSS fairing section. Remove fairing section.
3. Unfasten all studs and screws holding center platform and hinged rear ESSS fairing section. Remove fairing.

INSTALL

1. Line up ESSS center and rear fairing panels over fastener holes.
2. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
3. **QA** Install screws and washers in center fairing platform followed by rear hinged panel studs. See Figure 1, Detail B for fastener size and location.
4. Line up front ESSS fairing panel over fastener holes.
5. **QA** Install front fairing panel with studs (Figure 1, Detail A).

INSTALL - Continued

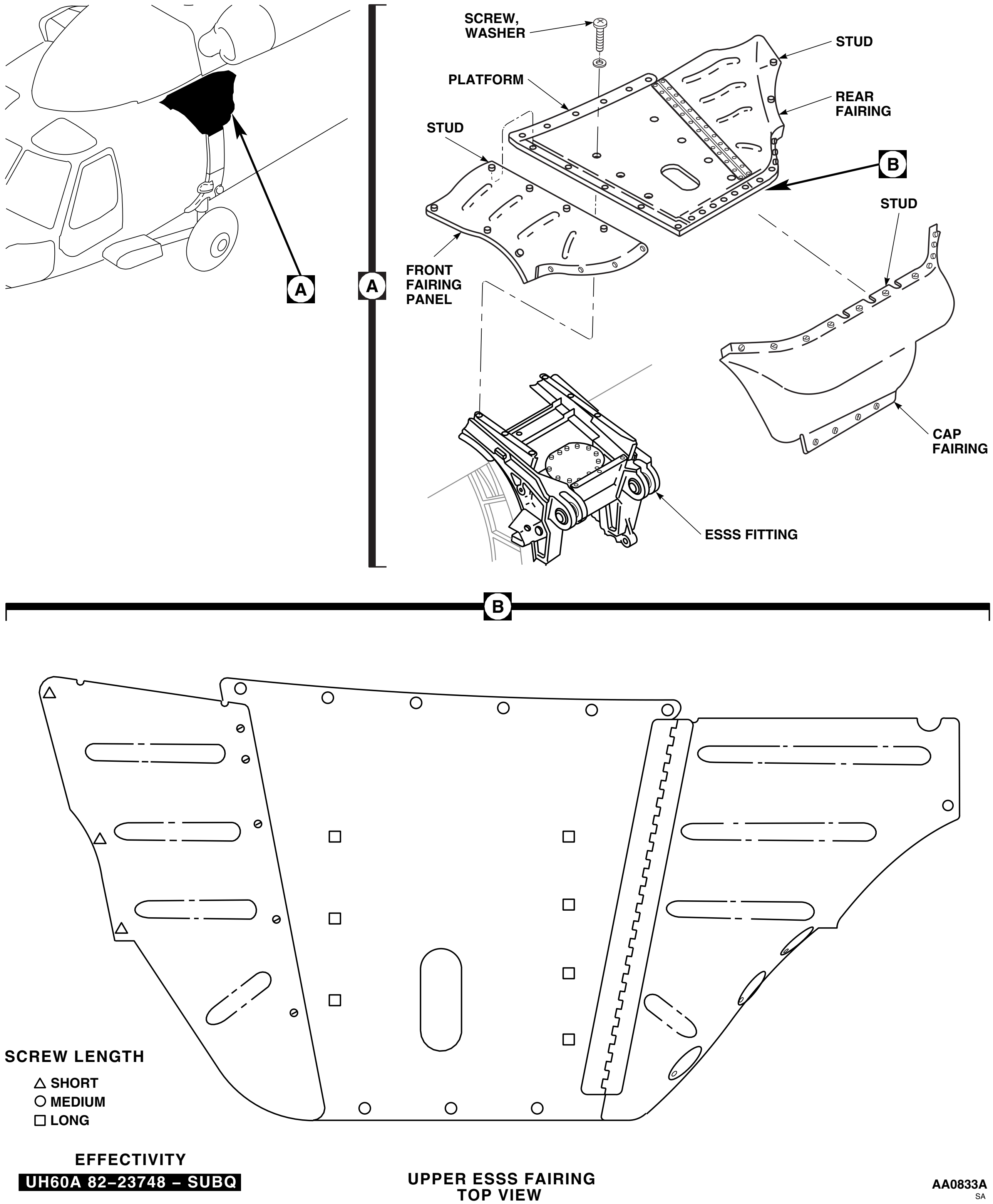


Figure 1. Replace ESSS Fitting Fairing And Platform.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT STEPS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0281 00
WP 0286 00
WP 1803 00

REMOVE

1. WSPS Remove plate and step extension from step (WP 0281 00).
2. Remove support fairing cover to get to step upper attachment bolt and washer (Figure 1, Detail A).
3. Remove screws and washers attaching step.

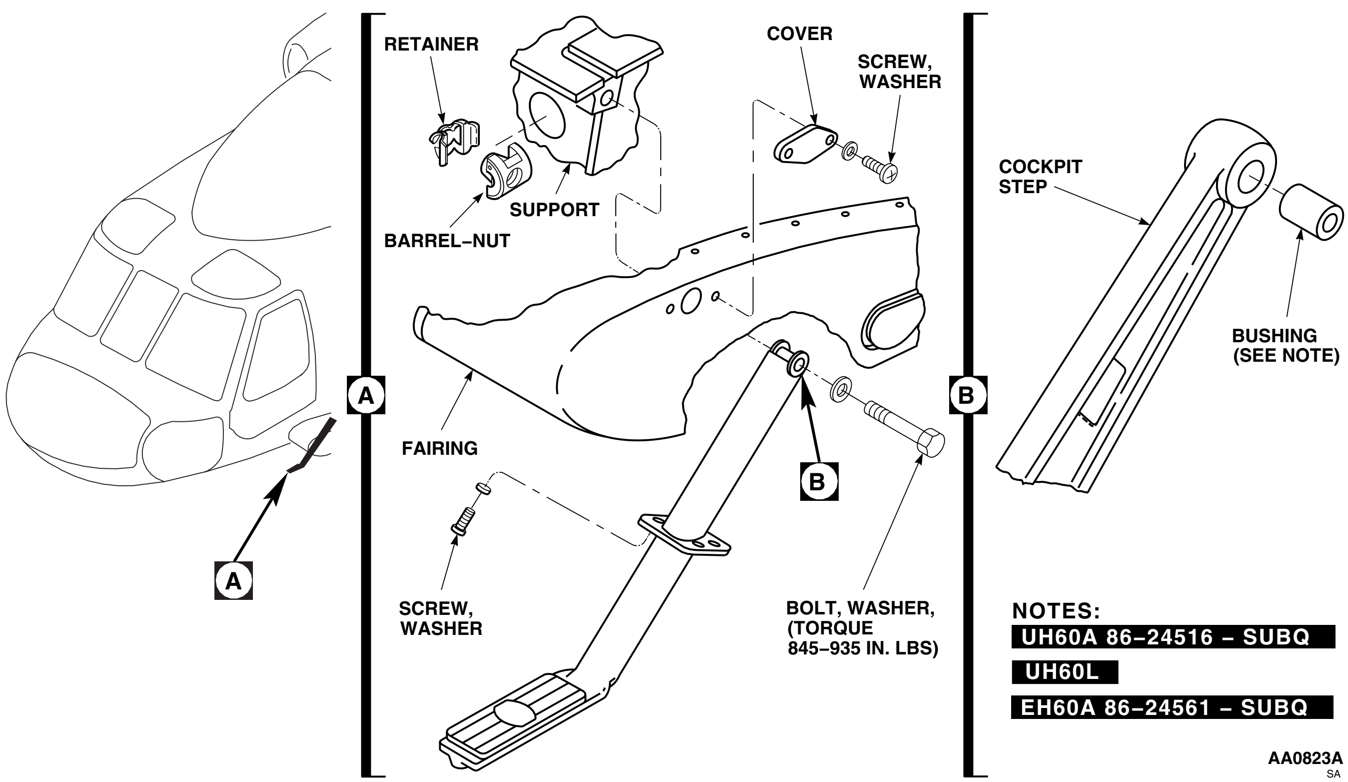


Figure 1. Replace Cockpit Steps and Tow Plates.

REMOVE - Continued

4. Remove bolt and washer attaching step support. Remove fairing.
5. Remove main landing gear support fairing (WP 0286 00).
6. Remove barrel-nut and retainer from fitting.

REPAIR

WSPS Using arbor press and drift, press bushing from cockpit step (Figure 1, Detail B).

INSTALL

NOTE

Make sure that only square barrel-nut is used.

1. Test attachment bolt and barrel-nut as follows:
   a. Place barrel-nut in soft-jawed vise.
   b. Thread bolt into barrel-nut until two threads are showing beyond nut.
   c. Back off bolt using torque wrench. **BREAKAWAY TORQUE SHALL NOT BE LESS THAN 18 INCH-POUNDS.**
   d. If breakaway torque is less than 18 inch-pounds, replace bolt. Repeat step 1. (a) through step 1. (c).
   e. If breakaway torque is still less than 18 inch-pounds, replace barrel-nut. Repeat step 1. (a) through step 1. (c).

CAUTION

Damage to fitting will result if barrel-nut is incorrectly installed. To prevent damage make sure barrel-nut is installed with barrel (round side) towards bolthead.

2. **QA** Install barrel-nut and retainer in fitting with barrel side toward bolthead (Figure 1, Detail A).
3. **QA** Install main landing gear support fairing (WP 0286 00).
4. **QA** Attach step to support with bolt and washer. **TORQUE BOLT TO 845 - 935 INCH-POUNDS.**
5. **QA** Install screws attaching step to support.
6. Make sure area is clean and free of foreign material before closing up.
7. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
8. **QA** Install support fairing cover with screws and washers.
9. **WSPS** Install step extension (WP 0281 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN STEPS W/O ESSS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

NOTE

Procedures are the same for left and right side.

1. Open upper shock strut fairing.
2. Remove attaching bolts, washers, and nuts (Figure 1, Detail A).
3. Remove step from fuselage.

INSTALL

1. Place step on fuselage (Figure 1, Detail A).
2. **QA** Install bolts, washers, and nuts with both boltheads facing to front. Make sure one washer is installed under bolthead. **TIGHTEN NUTS FINGERTIGHT.** Install cotter pins.

NOTE

Do not torque nuts. Step must be free to turn.

3. Make sure area is clean and free of foreign material before closing up.
4. Close upper shock strut fairing.

INSTALL - Continued

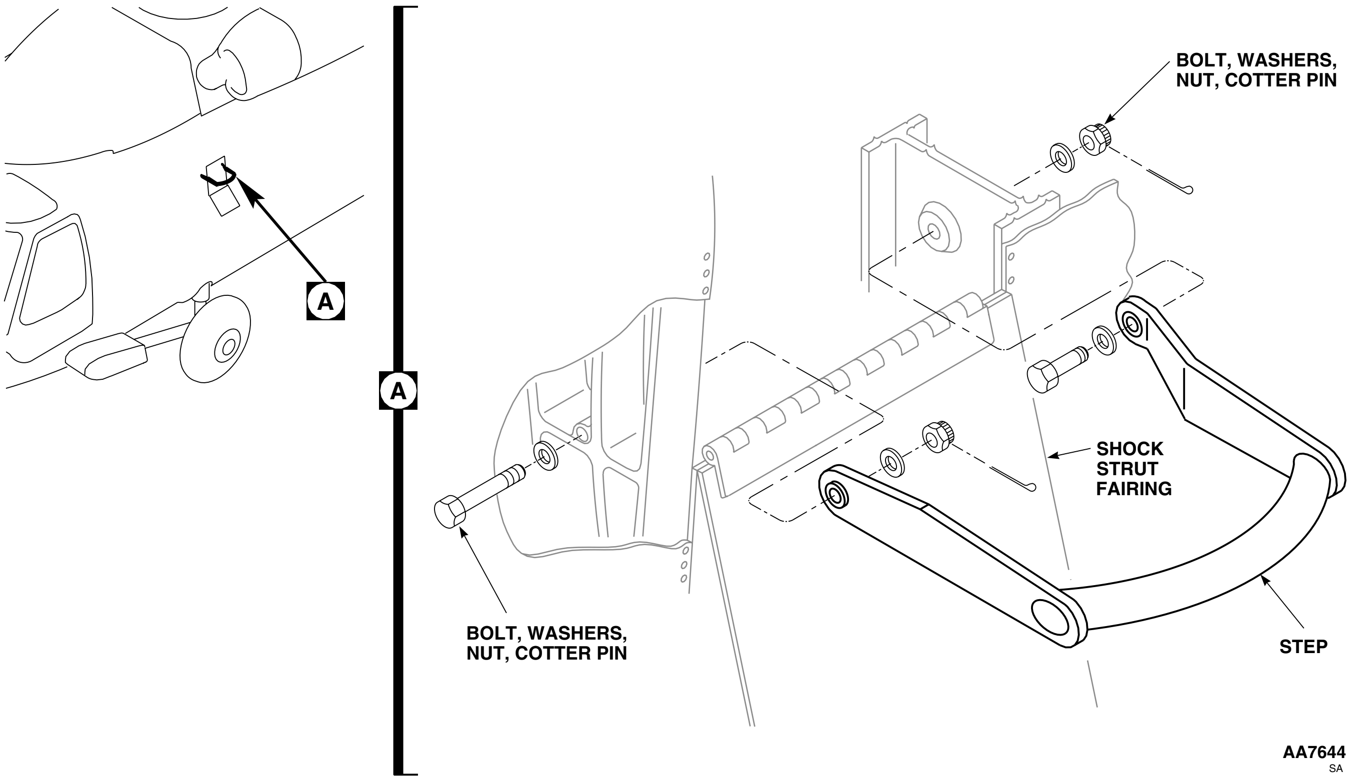


Figure 1. Replace Cabin Steps.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN STEPS ESSS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Adhesive, Item 14, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

NOTE

Removal and installation procedures are the same for left and right side.

1. Open step fairing.
2. Remove bolts, washers, nuts, laminated shims, and angle from bulkhead (Figure 1, Detail A).

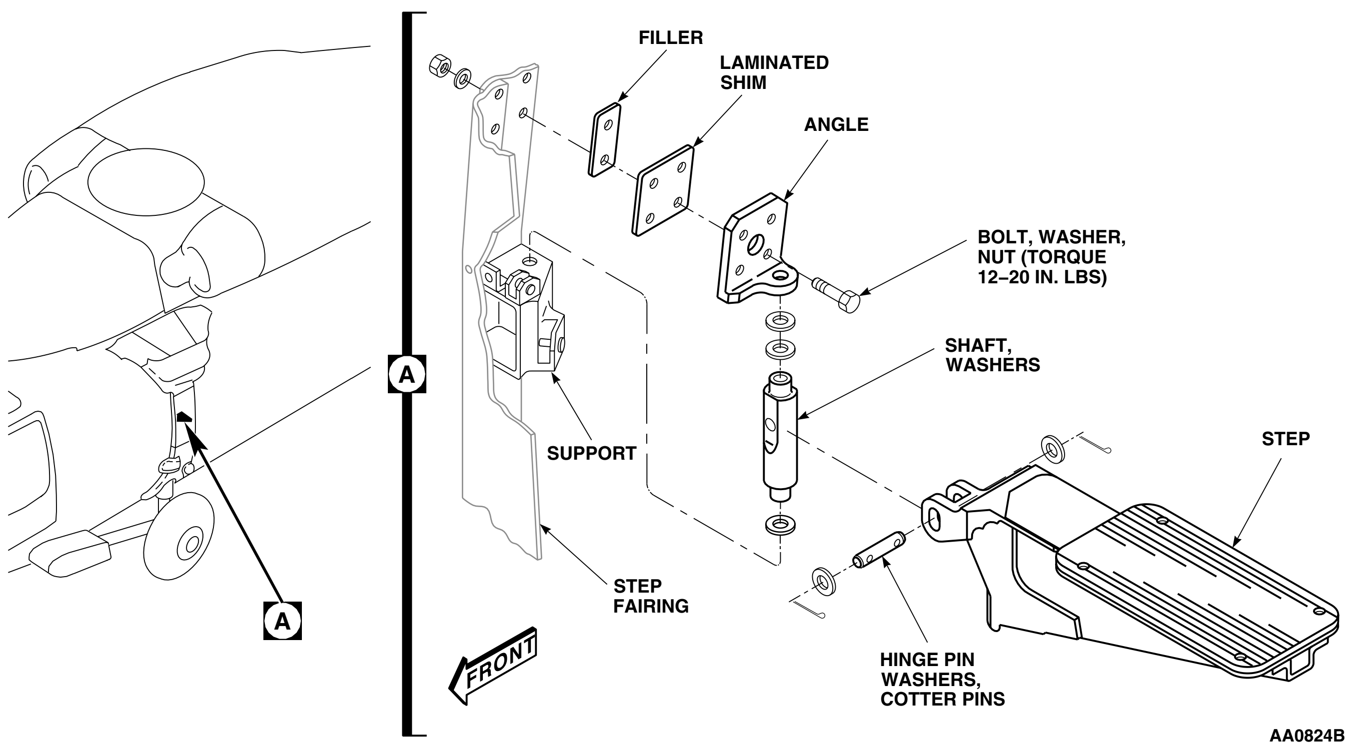


Figure 1. Replace Cabin Steps.

REMOVE - Continued

3. Remove step from helicopter.
4. If necessary, remove hinge pin and step from shaft.

INSTALL

1. **QA** If step was removed, install on shaft with hinge pin, washers, and cotter pins (Figure 1, Detail A).
2. Place shaft and washer in lower support.
3. Install washers on top of shaft.
4. Install filler and laminated shim on bulkhead using adhesive, Item 14, WP 1803 00. Drill fastener holes.
5. Install shaft with washers in angle, and fasten to bulkhead with bolts, washers, and nuts.
6. Measure clearance between washers on shaft and angle. **GAP MUST BE 0.020 TO 0.060-INCH.**
7. **QA** Add or subtract washers as necessary to get proper gap. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**
8. Fold step.
9. Close and fasten fairing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON SLIDING COVER

This WP supersedes WP 0296 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Caliper, Vernier, GGG-C-111
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Micrometer, Outside, 0 to 1 inch
Nonmetallic Scraper
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive Primer, Item 23, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Grommet, Nonmetallic, Item 139, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 0271 00
WP 0272 00
WP 0297 00
WP 0298 00
WP 0299 00
WP 0377 00
WP 0379 00
WP 1803 00

WARNING

To prevent injury to personnel, keep hands away from sharp edges of main rotor pylon sliding cover near track and mating edge, and ventilation blower inlet hole, when opening and closing cover. Use care to avoid injury.

REMOVE

1. WSPS Remove pitot wire cutter (WP 0272 00).
2. Unlatch and slide sliding cover forward. Slide cover against stops (Figure 1, Detail A and Detail B).
3. WSPS Remove upper wire cutter (WP 0271 00).
4. With aid of an assistant, pull up on rings of quick-release stop pins, slide cover forward, and remove.

INSPECT

1. Visually inspect latch for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. Remove corrosion from latch.
2. Raise latch, loosen jamnut and inspect fit of latch hook and jamnut on threaded stud. **IF LATCH HOOK OR JAMNUT IS LOOSE OR WOBBLES ON THREAD, REPLACE LATCH.**
3. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

INSPECT - Continued

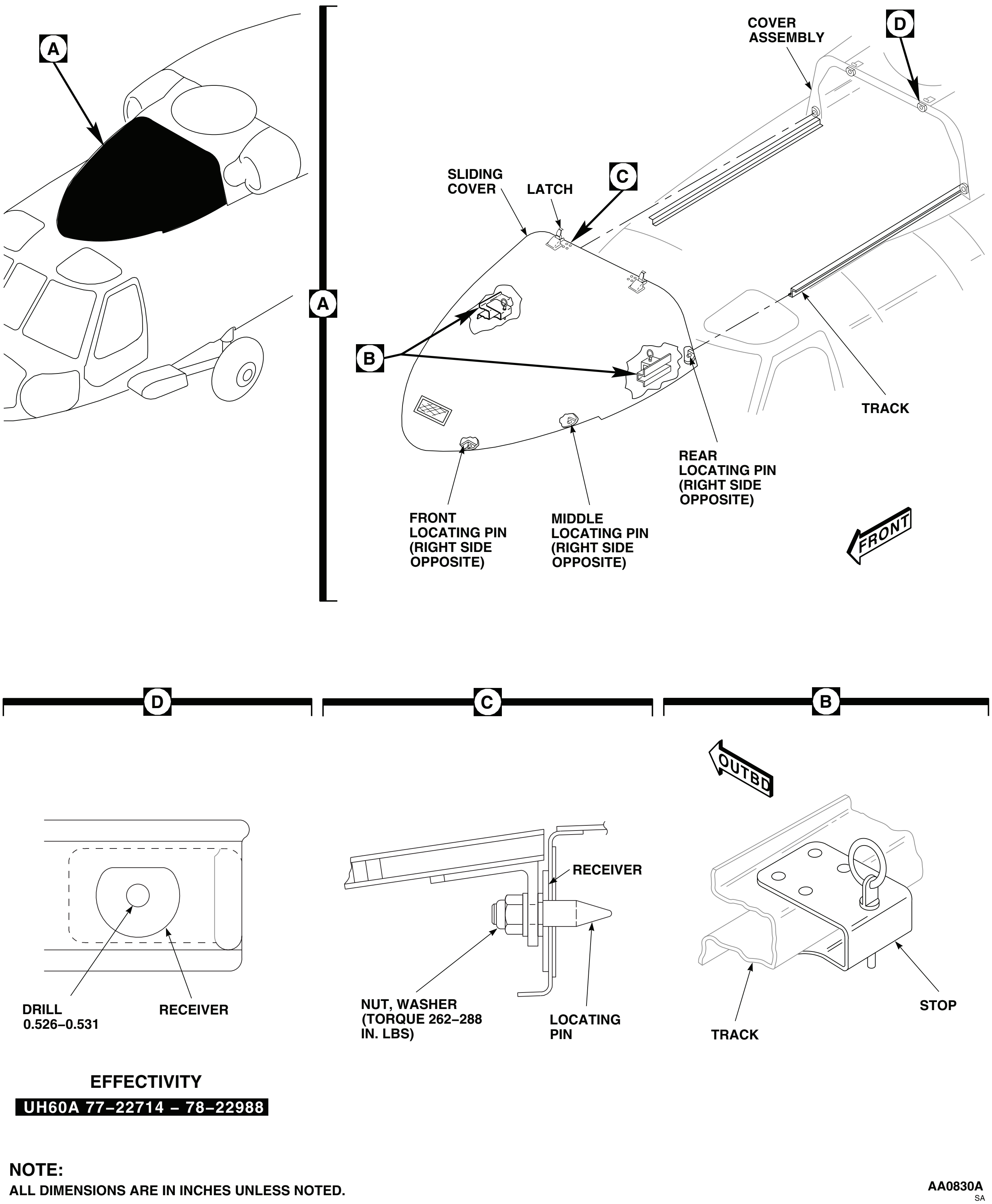


Figure 1. Replace Main Rotor Pylon Sliding Cover.

INSPECT - Continued

4. Visually inspect roller assemblies for disbonding of nylon rollers from bushings.
5. Replace damaged sliding cover rollers (WP 0298 00 or WP 0299 00).
6. Inspect pin receivers and support structure for cracks. **NO CRACKS ALLOWED.** Repair cracks (WP 0377 00).
7. Inspect pin receiver holes for wear using vernier caliper and outside micrometer caliper from caliper set. **WEAR SHALL NOT EXCEED 0.60-INCH.**
8. **UH-60A 86-24516 - SUBQ UH-60L EH-60A 86-24561 - SUBQ** Inspect forward locating pin support for wear or damage to nylon edging grommet.
9. Replace pin support edging grommet as follows:
   a. Remove damage edging grommet using nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   b. Remove adhesive residue from pin support and clean bonding surfaces thoroughly using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   c. Apply thin coat of adhesive primer, Item 23, WP 1803 00, to bonding surfaces of pin support.
   d. Let primer cure for at least 1 hour at room temperature.
   e. Cut piece of nonmetallic grommet, Item 139, WP 1803 00, to fit edge of pin support.
   f. Mix adhesive, Item 11, WP 1803 00, and apply to bonding surfaces of pin support and nylon grommet.
   g. Immediately install grommet on edge of pin support and press in place.

INSTALL

WARNING

Injury to personnel and damage to equipment will result if rollers are damaged. Care must be taken to insure that rollers properly engage tracks during installation of main rotor pylon sliding cover. Damaged rollers will result in separation of sliding cover from helicopter.

1. With aid of an assistant, position cover on tracks. Pull up on rings of quick-release stop pins and slide cover rearward (Figure 1, Detail A). Release pins.
2. Loosen nuts attaching six locating pins on sliding cover. Tighten fingertight (Figure 1, Detail C).
3. Slide cover open and closed a few times to check for proper operation.
4. **QA** After checking for proper line up and operation, **TORQUE LOCATOR PIN NUT TO 262 - 288 INCH-POUNDS.**
5. With cover latches fully engaged, check gap between locator pin shoulder and socket surface to these limits:
   a. Upper Rear: **NO GAP ALLOWED.**
   b. Lower Rear: **MAXIMUM GAP ALLOWED IS 0.120-INCH.**
   c. Middle and Front: **MAXIMUM GAP ALLOWED IS 0.250-INCH.**

INSTALL - Continued

NOTE

UH-60A 77-22714 - 78-22988 Replacement covers may bind due to lack of clearance between locating pins and their receivers.

6. If receiver holes are binding, enlarge receiver holes to 0.526 to 0.531-inch (Figure 1, Detail D).
7. If there is too much gap between closed cover and pylon, build up rear of cover (WP 0379 00). **MAXIMUM GAP ALLOWED IS 0.187-INCH.**
8. **WSPS** Install upper wire cutter (WP 0271 00).
9. Make sure area is clean and free of foreign material before closing up.
10. Close and fasten latches, adjusting if necessary (WP 0297 00).
11. Install pitot cutter (WP 0272 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON SLIDING COVER LATCH

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Polyurethane, Coating, Item 227, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REPLACE

1. Unlatch and slide sliding cover forward a little.
2. Remove screws, washers, and nuts attaching latch to sliding cover (Figure 1, Detail A). Remove latch.
3. Place replacement latch in position on sliding cover.
4. Install screws, washers, and nuts.
5. Adjust latch if it does not lock firmly.
6. Make sure area is clean and free of foreign material before closing up.
7. Slide cover rearward and fasten latches.

ADJUST

1. Loosen jamnut on threaded stud (Figure 1, Detail B).
2. Screw latch hook on or off stud until latch locks firmly.
3. **TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

REPAIR

1. To remove corrosion from latch, do this:
   a. Remove latch from sliding cover.
   b. Loosen jamnut and remove latch hook from stud. Remove jamnut from threaded stud (Figure 1, Detail B).
   c. Remove corrosion manually from latch, using scouring pad, Item 205, WP 1803 00. **PITS ON NON-THREADED AREAS OF LATCH SHALL NOT EXCEED 0.005-INCH AFTER REWORK.**
   d. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
   e. Apply compound solution to corroded areas of latch. Rinse thoroughly with clean water.
   f. Touch up paint with two coats of epoxy primer coating kit, Item 120, WP 1803 00, and top coat with polyurethane coating, Item 227, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

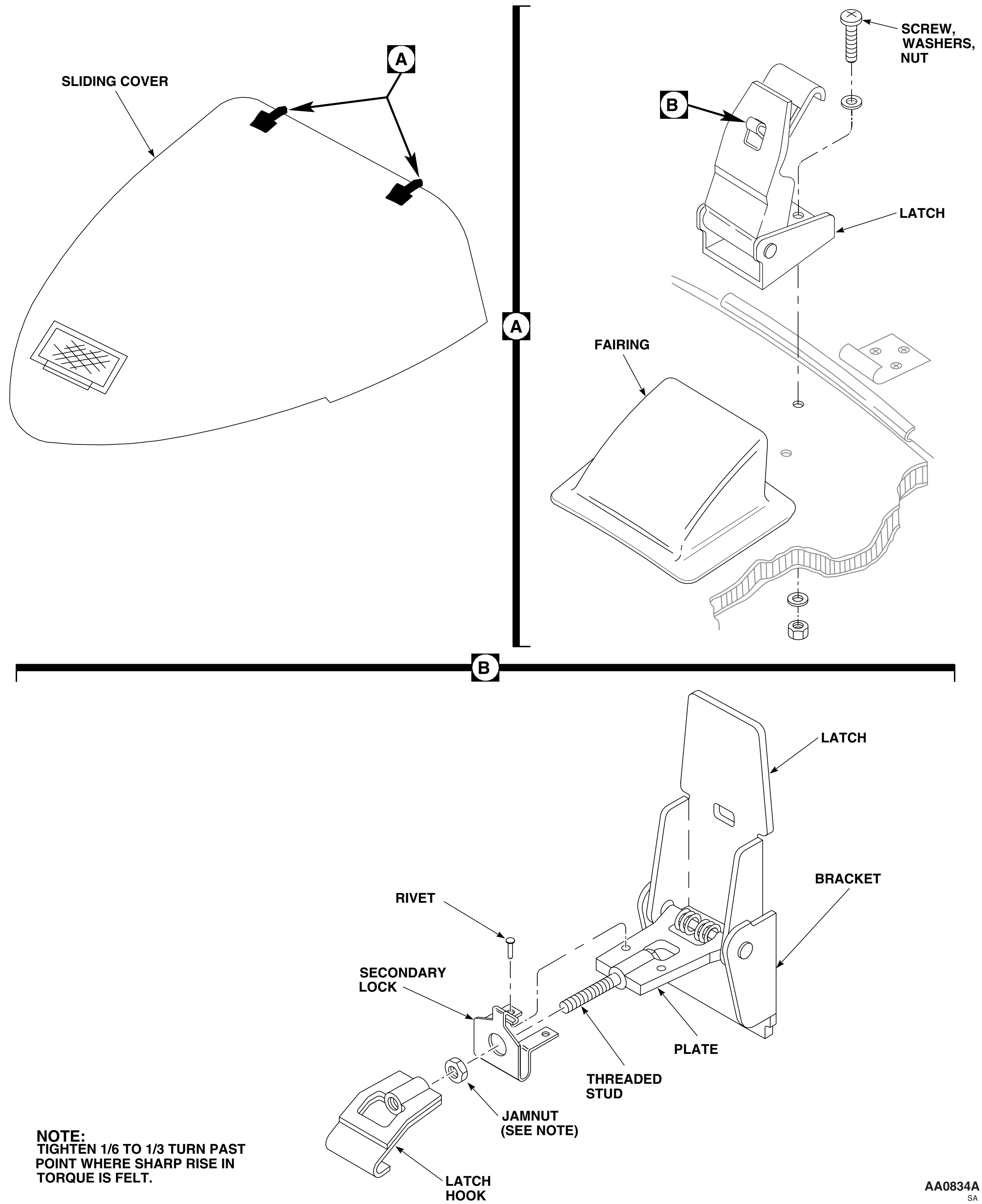


Figure 1. Replace Main Rotor Pylon Sliding Cover Latch.

REPAIR - Continued

g. Install jamnut on threaded stud. Install latch hook on threaded stud.

h. Install and adjust latch on sliding cover.

2. To replace secondary lock, do this:

a. Remove latch from door.

b. Loosen jamnut and remove latch hook from threaded stud. Remove jamnut from stud.

c. Remove rivets attaching secondary lock to plate. Remove lock.

d. Position replacement secondary lock on plate and attach with rivets.

e. Install jamnut on threaded stud. Install latch hook on threaded stud.

f. Install and adjust latch on door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON SLIDING COVER VERTICAL ROLLERS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0296 00

REMOVE

NOTE

Replacement instructions for any vertical roller are the same.

1. Remove sliding cover (WP 0296 00).
2. Place cover on a padded surface.
3. Remove bolt, washers, and bushing attaching roller to web (Figure 1, Detail B). Remove roller.

INSTALL

1. Insert bushing in roller (Figure 1, Detail B).
2. **QA** Install bolt, small washer, bushing, roller, and large washer on web. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.**
3. Install sliding cover (WP 0296 00).
4. Make sure area is clean and free of foreign material.
5. Slide cover rearwards and fasten latches.

INSTALL - Continued

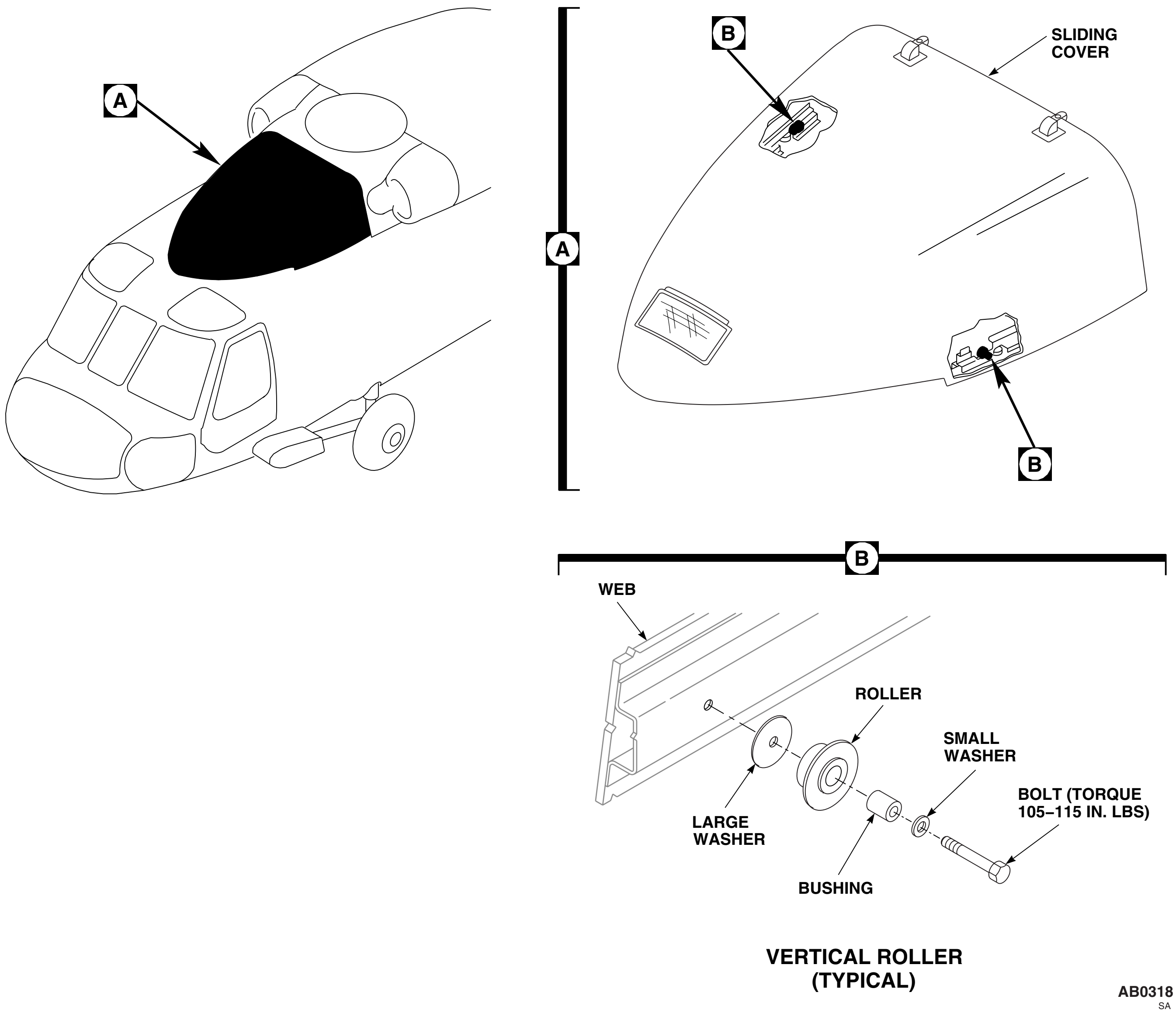


Figure 1. Replace Main Rotor Pylon Sliding Cover Vertical Rollers.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON SLIDING COVER HORIZONTAL ROLLERS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0296 00
WP 0379 00
WP 0381 00

REMOVE

1. If interference occurs between rivets, MS20470AD5-6, and forward horizontal main pylon roller, replace two rivets, MS20470AD5-6, with countersunk rivets, NAS1097AD5, as follows:
   a. Remove two rivets, MS2040AD5-6.
   b. Remove paint from the location of the original rivets (WP 0381 00 and TM 55-1500-345-23) and inspect area for possible damage to composite
   c. If damage is found, repair composite per WP 0379 00.
   d. Install countersunk rivets, NAS1097AD5, in addition to 1.5 inch x 0.75 inch doubler underneath the bucked tails of the rivets.

NOTE

Doubler should be 0.040 inch thick with rounded corners. Adjust doubler as needed to maintain edge and pitch distance.

   e. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

NOTE

Replacement instructions for any horizontal roller are the same.

2. Remove sliding cover (WP 0296 00).
3. Place cover on a padded surface.
4. Remove bolt, washers, and bushing attaching roller to web (Figure 1, Detail B). Remove roller.

INSTALL

1. Insert bushing in roller (Figure 1, Detail B).
2. **QA** Install bolt, small washer, bushing, roller, and large washer on web. **TORQUE NUT TO 105 - 115 INCH-POUNDS.**
3. Install sliding cover (WP 0296 00).
4. Make sure area is clean and free of foreign material.

INSTALL - Continued

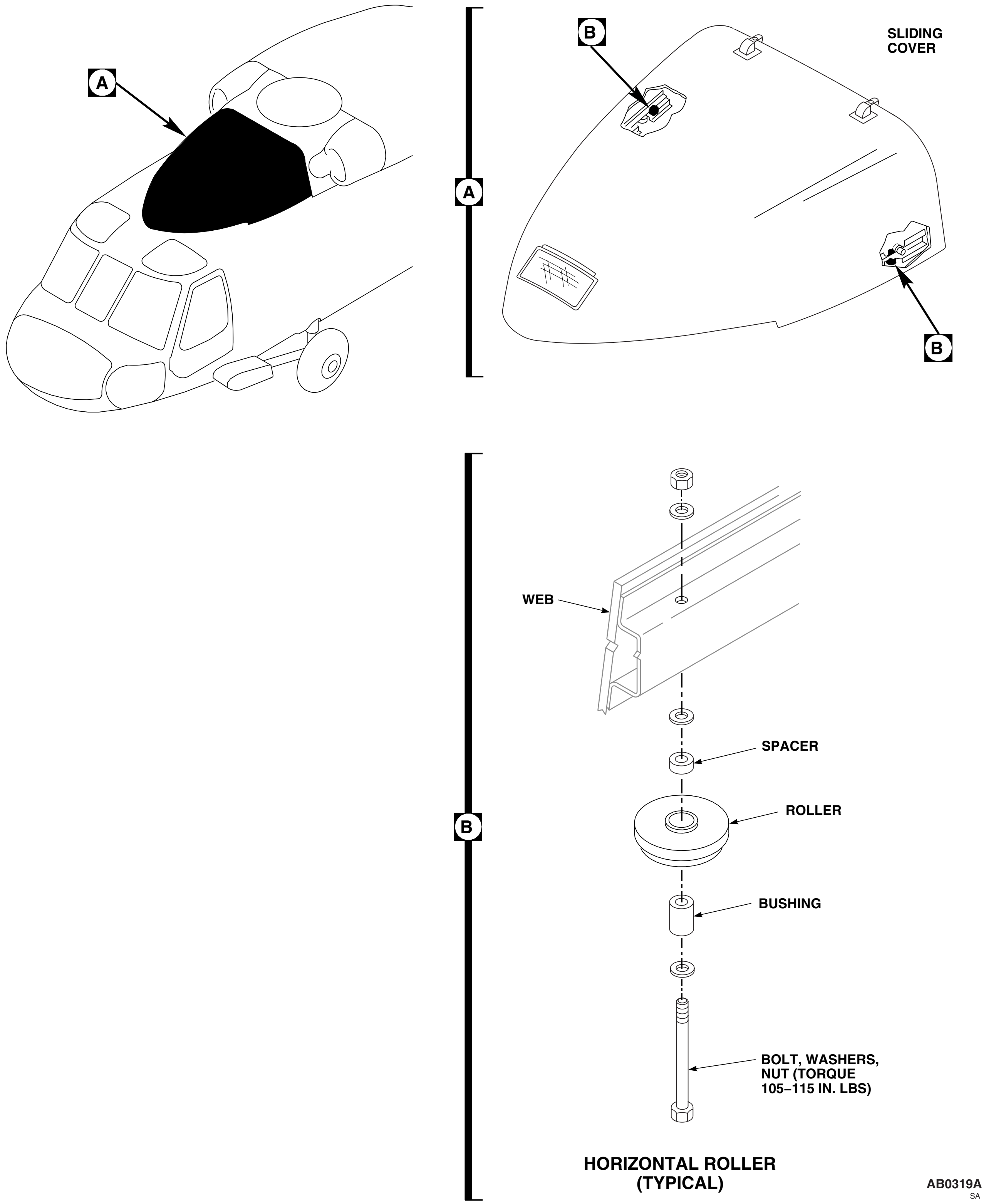


Figure 1. Replace Main Rotor Pylon Sliding Cover Horizontal Rollers.

INSTALL - Continued

5. Slide cover rearwards and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON WORK PLATFORM

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Unlatch and open sliding cover.
2. Remove quick-release pin from aft stop, place stop in remove position and install pin.
3. Push handle down and disengage arms from tracks.
4. Slide work platform forward and remove from lower track (Figure 1, Sheet 1, Detail A).
5. Remove bolts and washers attaching left and right tracks to platform (Figure 1, Sheet 1, Detail B). Remove tracks.
6. Remove bolt, washers, and nut attaching aft stop to bracket on platform (Figure 1, Sheet 2, Detail C).
7. Remove rivets securing aft stop bracket to platform. Remove bracket.

DISASSEMBLE

1. Remove rivets securing handle and arm assembly brackets to lower tracks. Remove handle and arm assembly (Figure 1, Sheet 1, Detail B).
2. Remove springs from arms.

ASSEMBLE

1. Install springs on arms (Figure 1, Sheet 1, Detail B).
2. Secure handle and arm assembly brackets on lower tracks with rivets.

INSTALL

1. Install aft stop bracket on platform with rivets (Figure 1, Sheet 2, Detail C).
2. Install aft stop on bracket with bolt, washers, and nut. Install cotter pin. Place stop and quick-release pin in remove position
3. Install bolts and washers attaching left and right tracks to platform. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS (Figure 1, Sheet 1, Detail B).**
4. Place work platform on front end of lower track (Figure 1, Sheet 1, Detail A).
5. Push handle down and slide platform back.
6. Release handle to allow arms to engage in slots in tracks.
7. Remove quick-release pin from remove position and install in stop position.

INSTALL - Continued

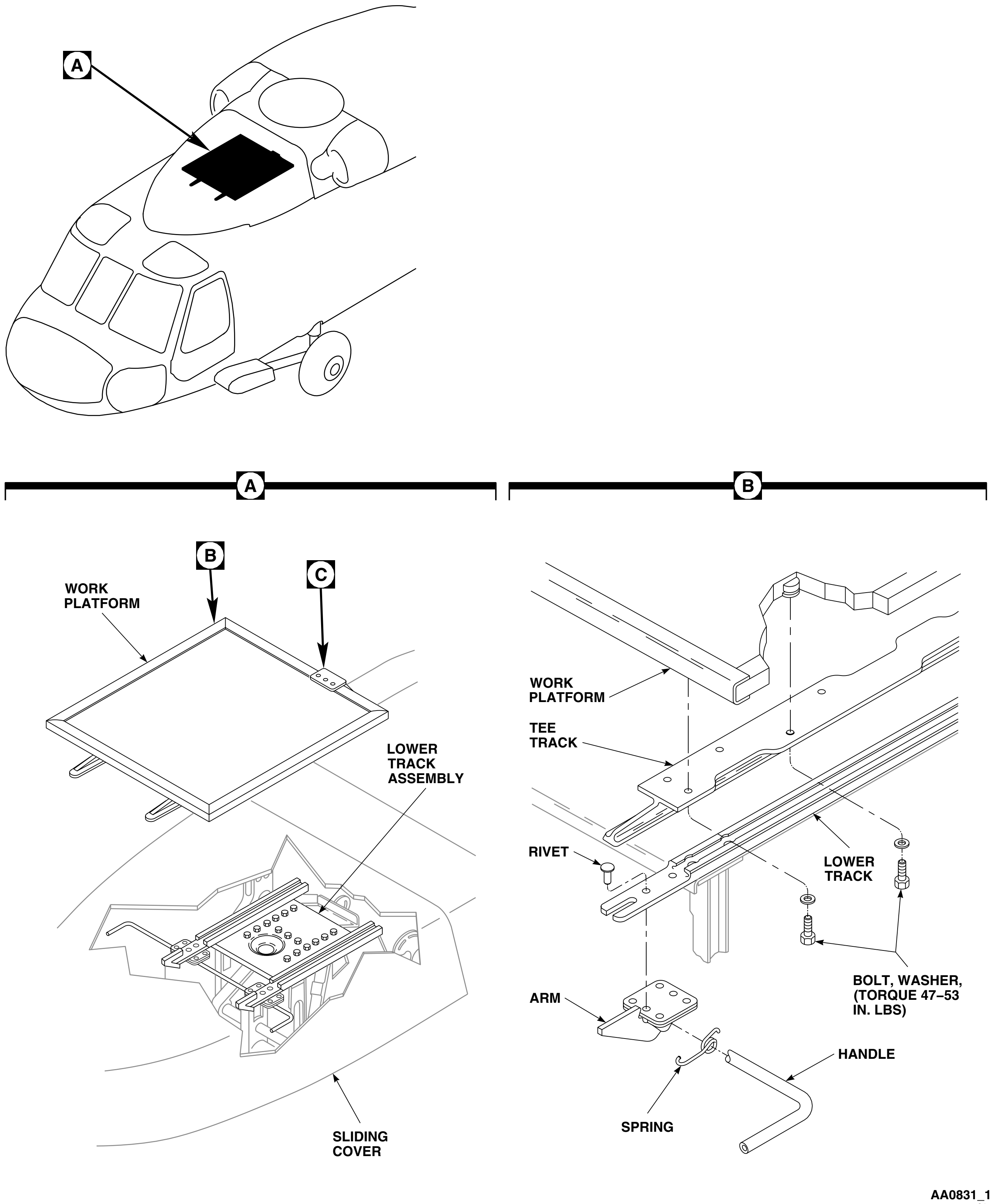


Figure 1. Replace Work Platform. (Sheet 1 of 2)

INSTALL - Continued

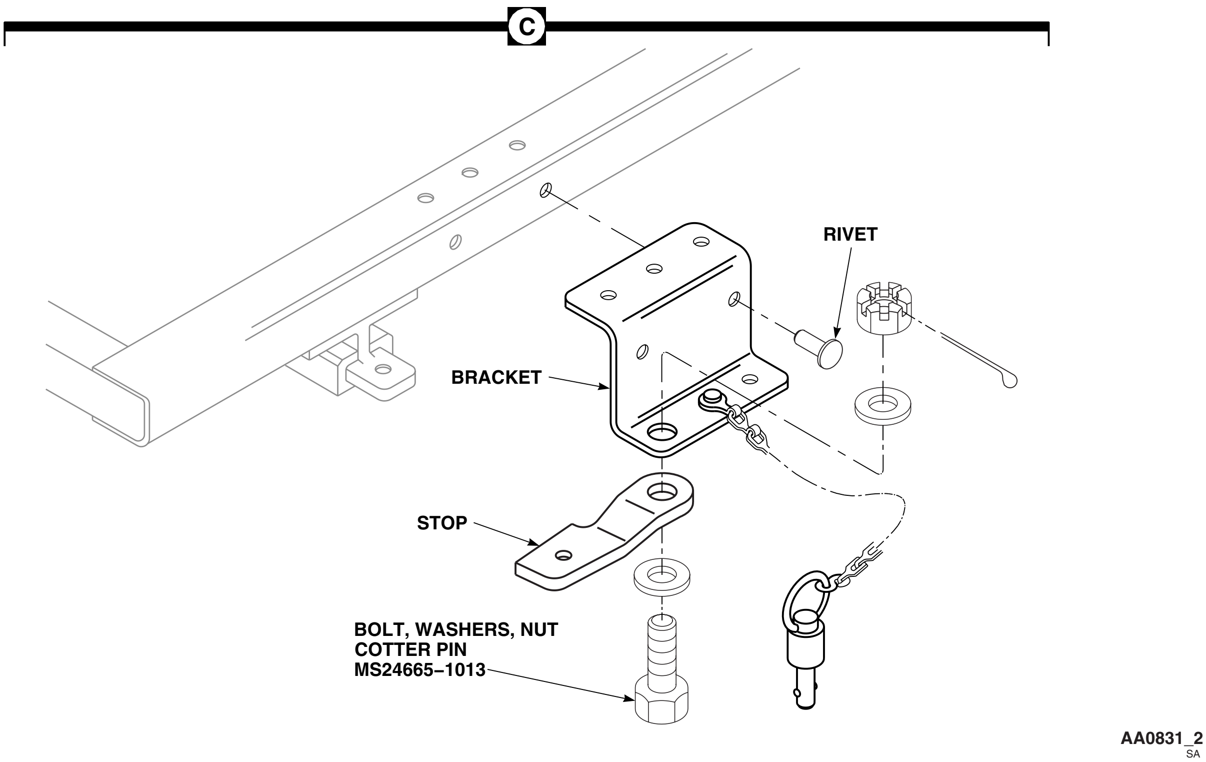


Figure 1. Replace Work Platform. (Sheet 2 of 2)

8. Make sure area is clean and free of foreign material before closing sliding cover.
9. Close sliding cover and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON AIR INLET FAIRINGS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00
Wire, Nonelectrical, Item 353, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0484 00
WP 0999 00
WP 1803 00

REMOVE

1. Unlatch sliding cover and slide cover to the front.
2. Remove air inlet from helicopter (WP 0484 00).
3. **ESSS** If installed, remove fuel system fixed components (WP 0999 00).
4. Remove screws, washers, and inlet shroud fairing from helicopter (Figure 1, Detail A).
5. Remove bolt, washers, and nut securing support tube clevis to fairing beam fitting.
6. Remove screws and washers holding edge of fairing to deck support and cover assembly.
7. **UH60A 77-22714 - 80-23452** Remove screws holding fairing to brackets attached to gimbal on input gear box housing.
8. **UH60A 80-23453 - 83-23888** Remove bolts and large washers holding fairing to bracket attached to gimbal on input gear box housing (Figure 1, Detail B).
9. **UH60A 83-23889 - SUBQ** Remove bolts, washers, and nuts holding fairing to bracket attached to gimbal on input gear box housing.
10. Remove screws holding fairing frame and beam ends to deck bracket. Remove fairing from helicopter (Figure 1, Detail A).

INSTALL

1. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
2. **QA** Place fairing frame against deck bracket and install washers and screws (Figure 1, Detail A).
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. **QA** **UH60A 77-22714 - 80-23452** Fasten fairing to brackets on gimbal with washers and screws.
5. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
6. **QA** **UH60A 80-23453 - 83-23888** Fasten fairing to brackets on gimbal with large washers and bolts. Tighten bolts only until grommet starts to compress (Figure 1, Detail B). Using nonelectrical wire, Item 353, WP 1803 00, lockwire bolts.
7. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.

INSTALL - Continued

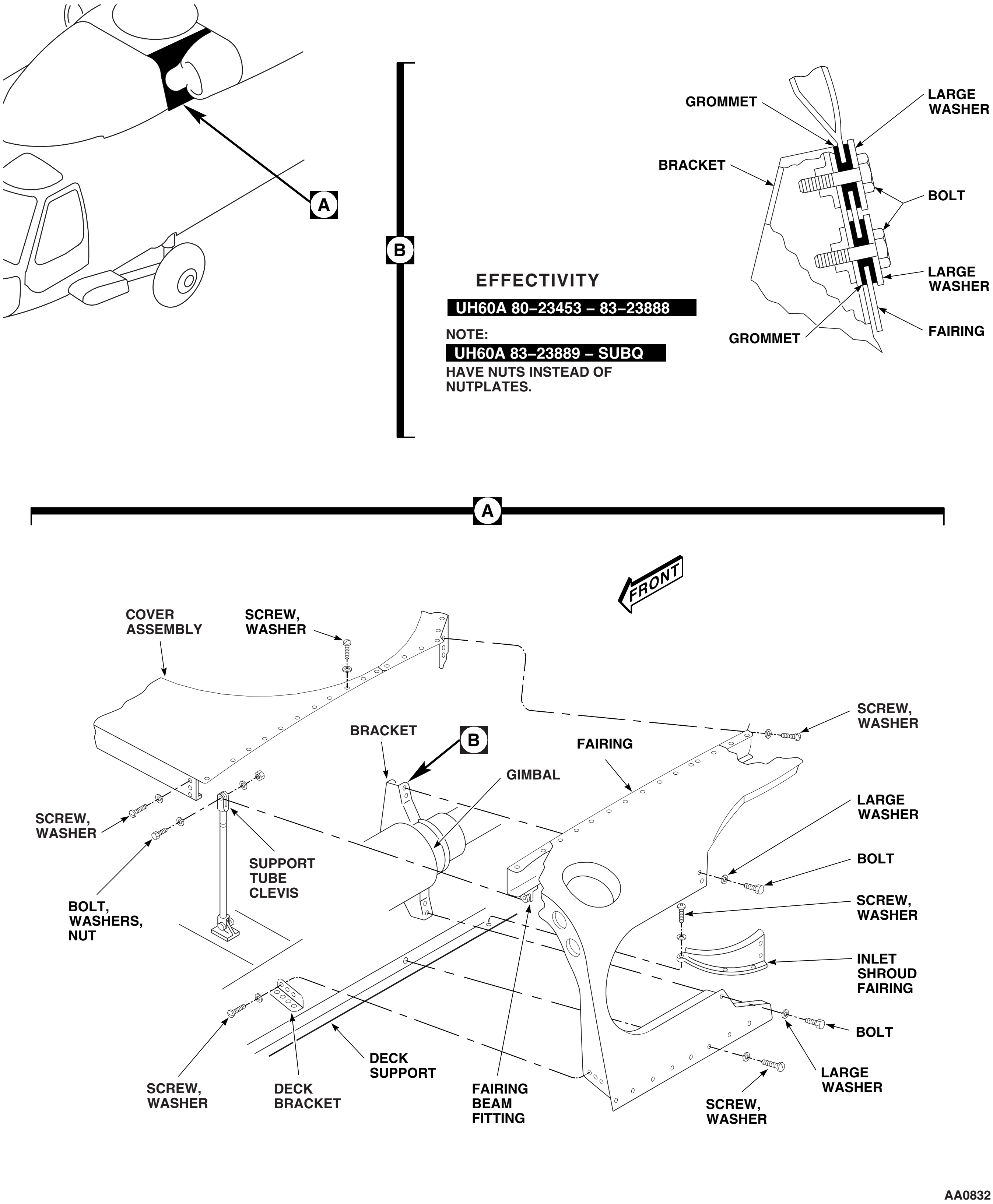


Figure 1. Replace Main Rotor Pylon Air Inlet Fairings.

INSTALL - Continued

8. **UH60A 83-23889 - SUBQ** Fasten fairing to brackets with large washers, bolts, and nuts. Tighten bolts until grommet starts to compress. Using nonelectrical wire, Item 353, WP 1803 00, lockwire bolts.
9. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
10. **QA** Fasten fairing edges to cover assembly and deck support with washers and screws (Figure 1, Detail A).
11. **QA** Bolt support tube clevis to fairing beam fitting.
12. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
13. **QA** Install fixed fairing with washers and screws.
14. Install air inlet on helicopter (WP 0484 00).
15. **ESSS** If removed, install fuel system fixed components (WP 0999 00).
16. Make sure area is clean and free of foreign material before closing up.
17. Move sliding cover to closed position. Fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON TRANSMISSION COVER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0297 00

REMOVE

1. Unlatch and open sliding cover (Figure 1, Detail A).

NOTE

Transmission cover is made up of four separate covers. All are replaced in the same way.

2. Remove screws and washers attaching cover to fuselage. Remove cover.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. **QA** Position replacement cover on fuselage and install attaching screws and washers (Figure 1, Detail A).
3. If forward cover was replaced, check sliding cover latches for correct operation. Adjust latches if necessary (WP 0297 00).
4. Close and latch sliding cover.

INSTALL - Continued

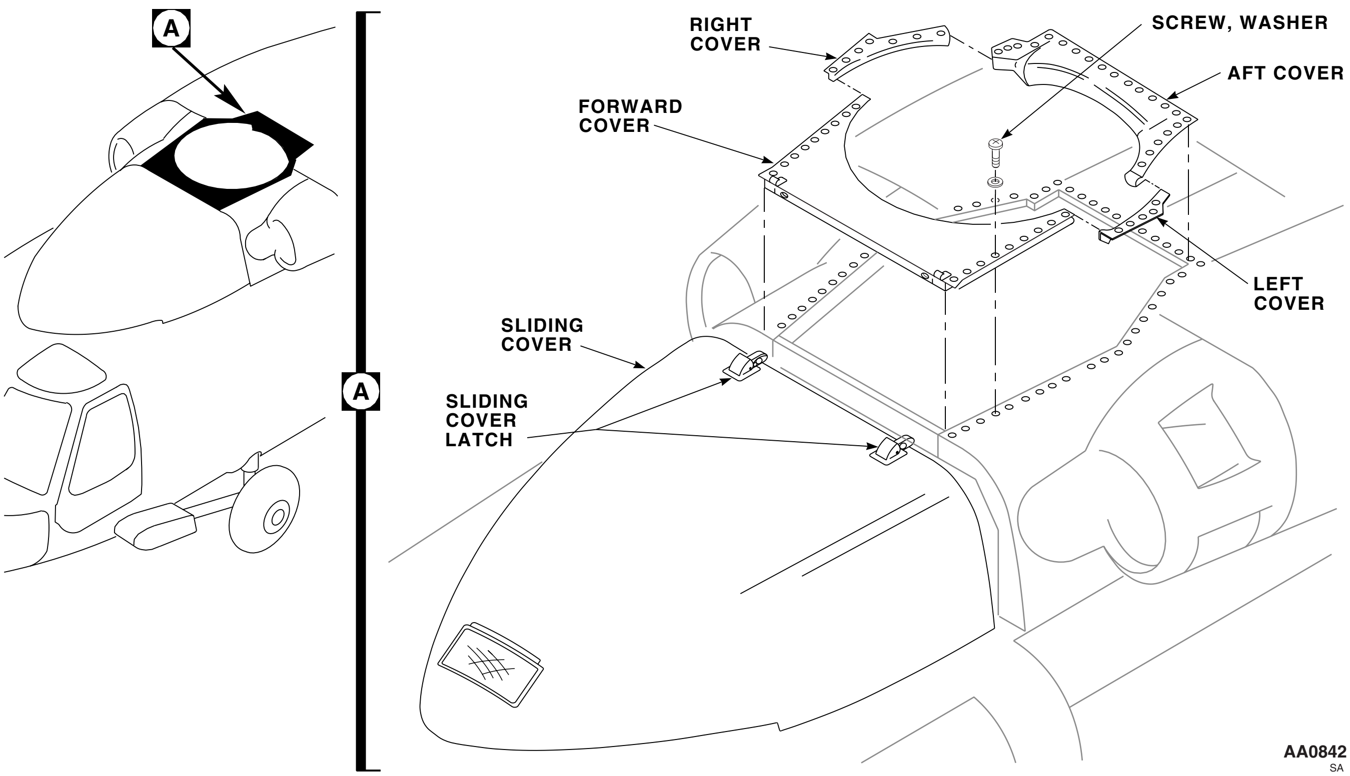


Figure 1. Replace Main Rotor Pylon Transmission Cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

APU ACCESS DOOR

This WP supersedes WP 0303 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Chafing Strip, Locally-Made, Figure 57, WP 1805 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Pylon Door Chafing Strip, Locally-Made, Figure 28, WP 1805 00
Rubber Sheet, Solid, Item 256, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0379 00
WP 0381 00
WP 1803 00
WP 1805 00

REMOVE

1. Unlatch APU compartment access door.
2. Squeeze hinge-release pins together at each hinge (Figure 1, Detail A). Remove access door from helicopter.

INSPECT/REPAIR

1. Visually inspect hinges, catches, and catch fittings for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked item or door assembly, as required.
2. Remove corrosion and blend out wear, using scouring pad, Item 205, WP 1803 00. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. If chafing strip(s) require replacement do this:
   a. Remove chafing strip and old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 1, Detail B).
   b. Locally make pylon door chafing strip, (WP 1805 00), and locally make chafing strip, (WP 1805 00), from solid rubber sheet, Item 256, WP 1803 00.
   c. Lightly abrade bonding surface of strip, using scouring pad, Item 205, WP 1803 00.

INSPECT/REPAIR - Continued

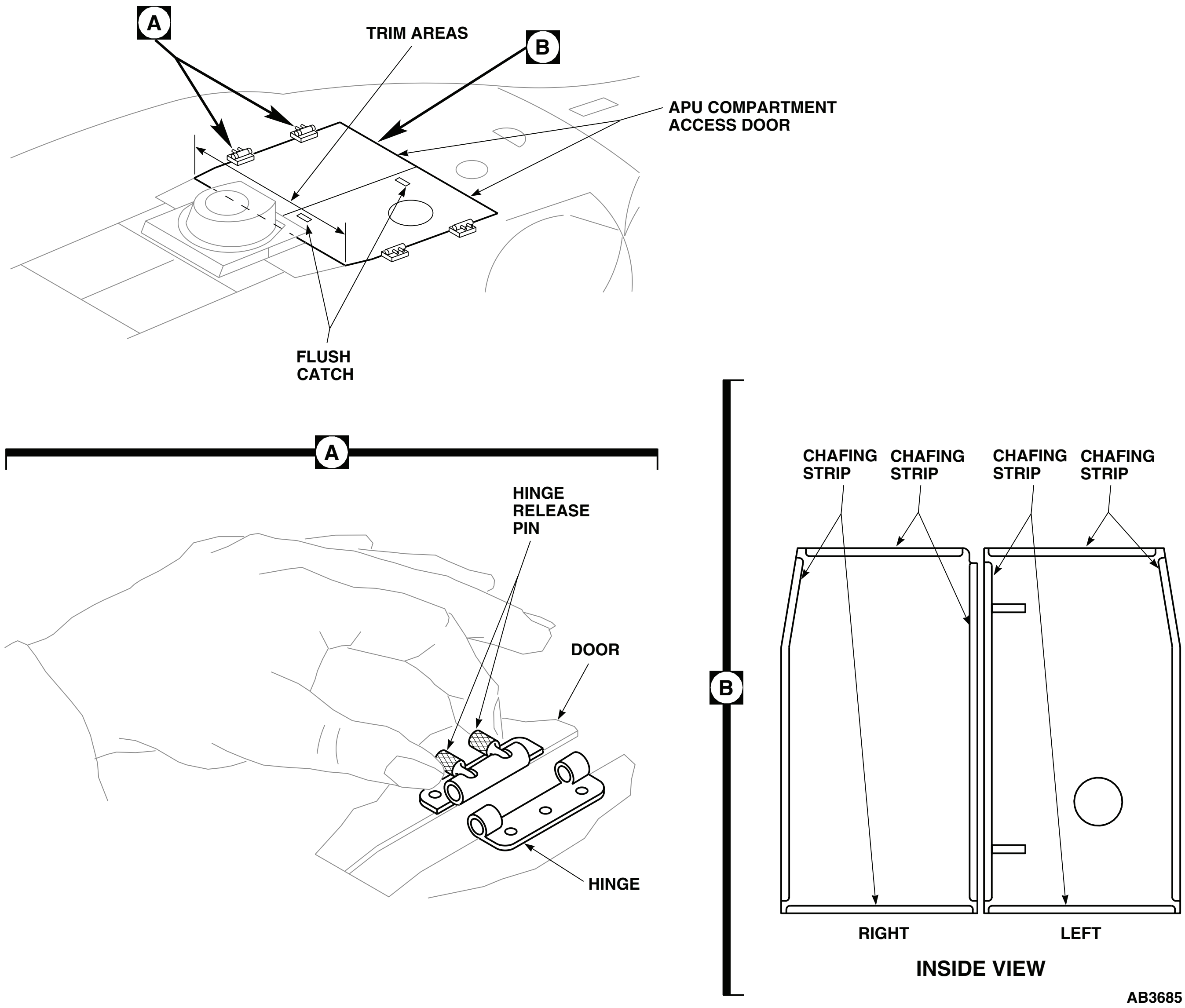


Figure 1. Replace APU Access Door.

d. Thoroughly clean bonding surfaces of chafing strap and door, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

e. Apply even coat of adhesive, Item 14, WP 1803 00, to both bonding surfaces.

f. Let adhesive dry for about 10 minutes until tacky.

g. Install chafing strip on door and press firmly in place.

4. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Place door over compartment. Squeeze hinge-release pins together at each hinge, put hinge halves together, and release pins (Figure 1, Detail A).

INSTALL - Continued

2. Close right door.
3. Check IR jammer wire bundle for chafing by door. If wire bundle is chafed by door, trim contact area on door using medium-cut file to a maximum of 1/4-inch.
4. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
5. If trimmed area extends into honeycomb core, repair core (WP 0379 00).
6. Check door for fit. If door edge binds against frames, trim area (Figure 1) (WP 0379 00).
7. Make sure area is clean and free of foreign material before closing up.

NOTE

Installed access doors may be warped 0.75-inch above surface of surrounding pylon. Hand pressure up to 16 pounds may be used to hold door closed for fastening.

8. Close and fasten APU compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

APU SCREENS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Remove screws and washers attaching APU screen to fuselage (Figure 1, Detail A).
2. Remove screen.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. Place replacement APU screen in position (Figure 1, Detail A).
3. **QA** Install screws and washers attaching screen to fuselage.

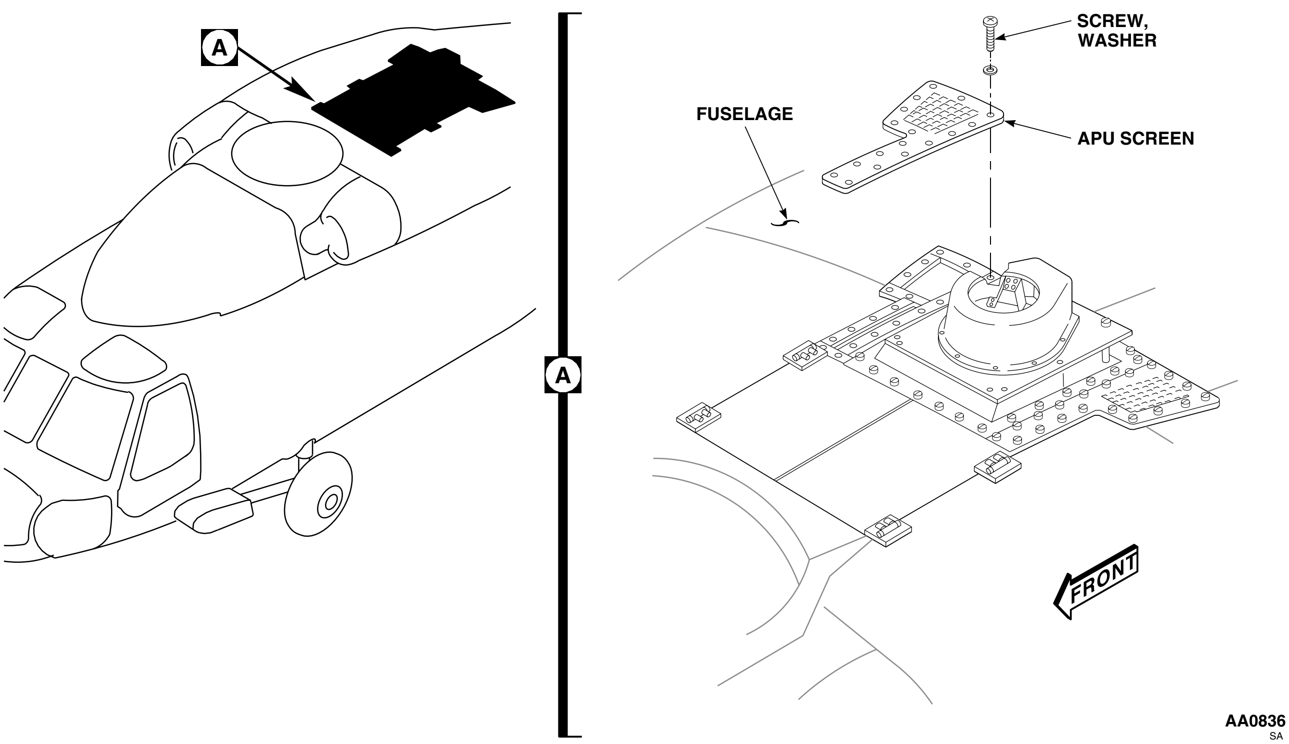


Figure 1. Replace APU Screens.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

APU ACCESS DOOR BEAM

This WP supersedes WP 305 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Primer, Sealing Compound, Item 245, WP 1803 00
Sealing Compound, Item 284, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0303 00
WP 1803 00

REMOVE

1. Remove APU access doors (WP 0303 00).
2. Remove screws and washers securing APU access door beam to support structure and remove beam (Figure 1, Detail A).

INSTALL

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean beam attaching screws and support structure nutplate threads, using machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
2. Apply sealing compound primer, Item 245, WP 1803 00, to beam attaching screws and nutplates. Allow primer to dry for five minutes.
3. Apply sealing compound, Item 284, WP 1803 00, to beam attaching screws and nutplates.
4. **QA** Secure beam to support structure with attaching screws and washers (Figure 1, Detail A).
5. Install APU access doors (WP 0303 00).

INSTALL - Continued

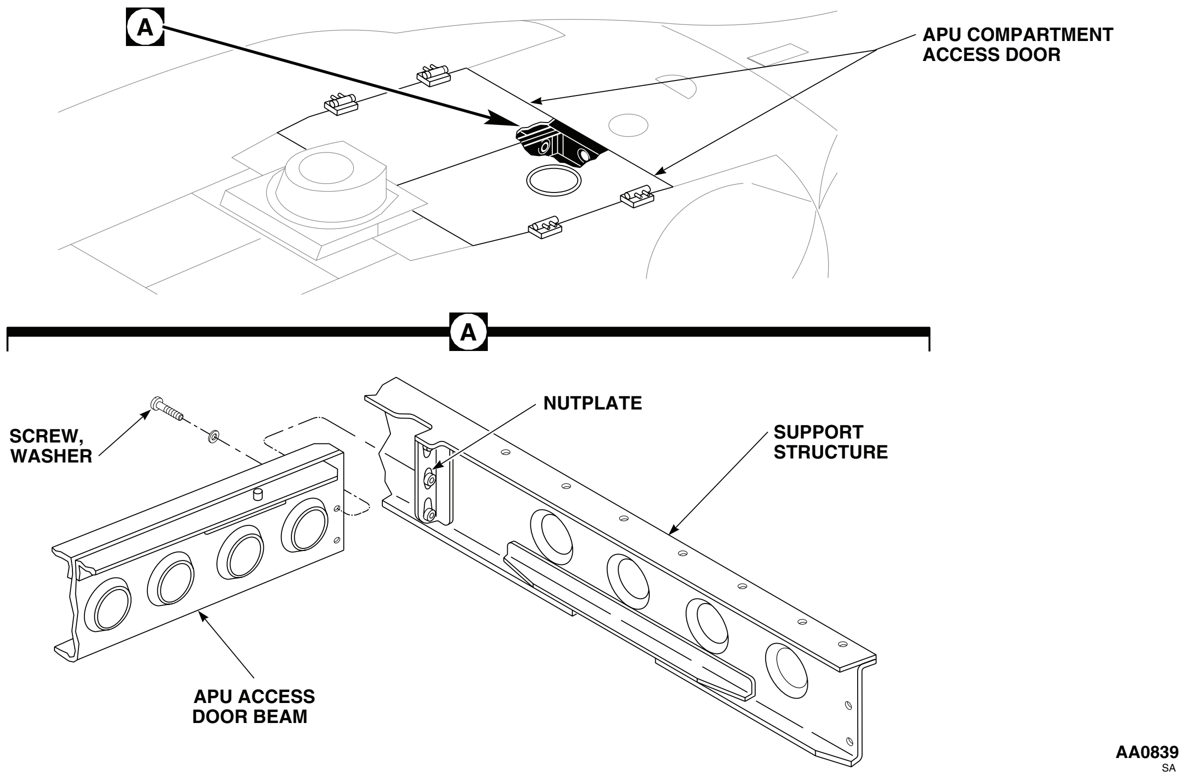


Figure 1. Replace APU Access Door Beam.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

APU EXHAUST FAIRING

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0379 00
WP 0381 00
WP 1803 00

REMOVE

1. Remove APU exhaust fairing attachment screws and washers (Figure 1, Detail A).
2. Remove APU exhaust fairing.

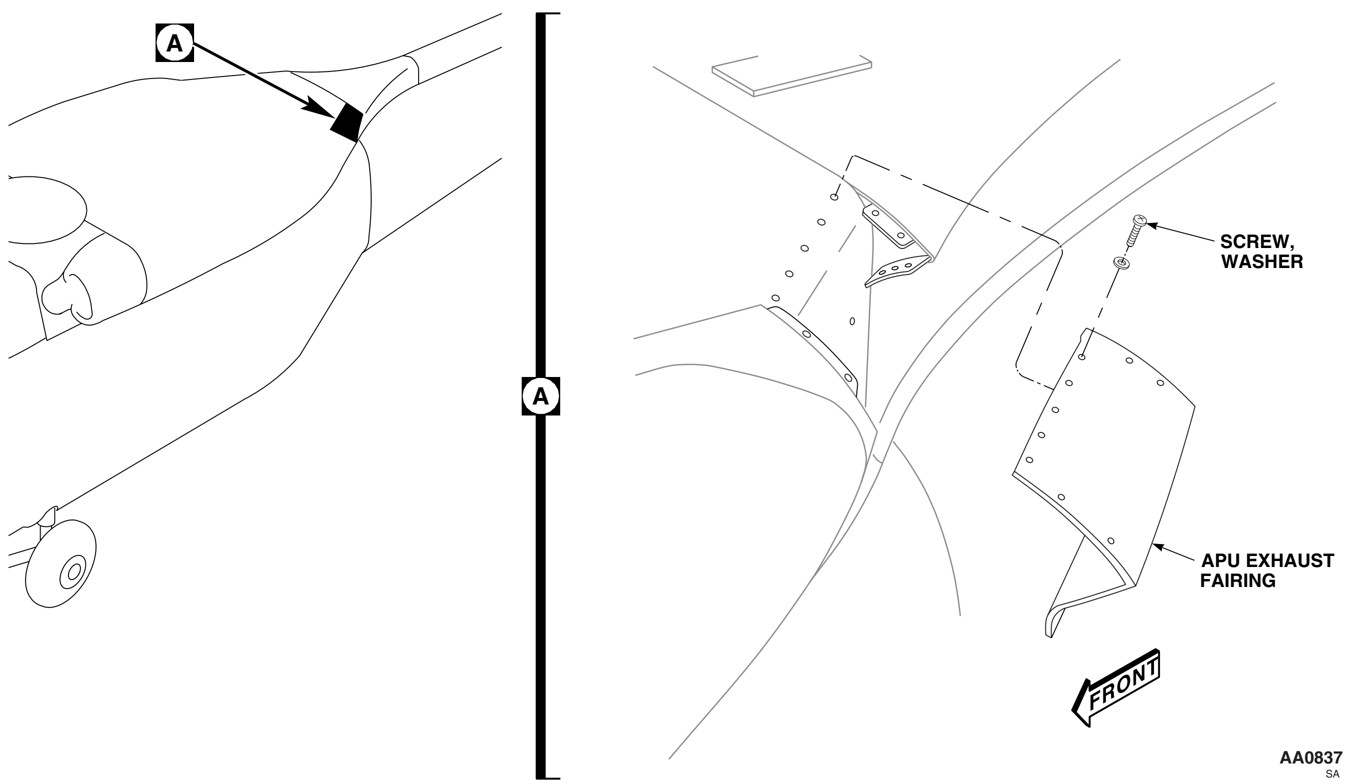


Figure 1. Replace APU Exhaust Fairing.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. Position replacement fairing and check that holes line up (Figure 1, Detail A).
3. If holes do not line up, do this:
   a. Mark out-of-line holes.
   b. Remove fairing, fill holes (WP 0379 00) and touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
   c. Replace fairing. Mark and drill new holes.
4. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
5. **QA** Install APU exhaust fairing with screws and washers.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

APU PANEL

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 11-1520-237-23
WP 0303 00
WP 0306 00
WP 0311 00
WP 0379 00
WP 1360 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove APU exhaust fairings (WP 0306 00).
3. Remove screws, washers, and fairing supports (Figure 1, Detail A).
4. Remove APU access doors (WP 0303 00).
5. Remove top IFF antenna (TM 11-1520-237-23).
6. Remove rear pylon fairing (WP 0311 00).
7. Loosen nut, remove formation light electrical connector from bracket. Remove clamps, free electrical harness from APU panel.
8. **UH60A 80-23509 - SUBQ** Remove screws and washers holding panel to fire bottle support.
9. **UH60A 77-22714 - 80-23508** Remove rivets, screws, and washers holding panel to fire bottle support.
10. Remove APU exhaust duct (WP 1360 00).
11. Remove clamps from top IFF antenna cables.
12. Remove remaining screws and washers holding panel to fuselage.
13. Remove panel.

INSTALL

1. To install new panel, do this:
   a. Install APU access doors (WP 0303 00).
   b. Trial fit APU panel. Trim edges that bind against firewall or APU access doors (WP 0379 00).
   c. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.

INSTALL - Continued

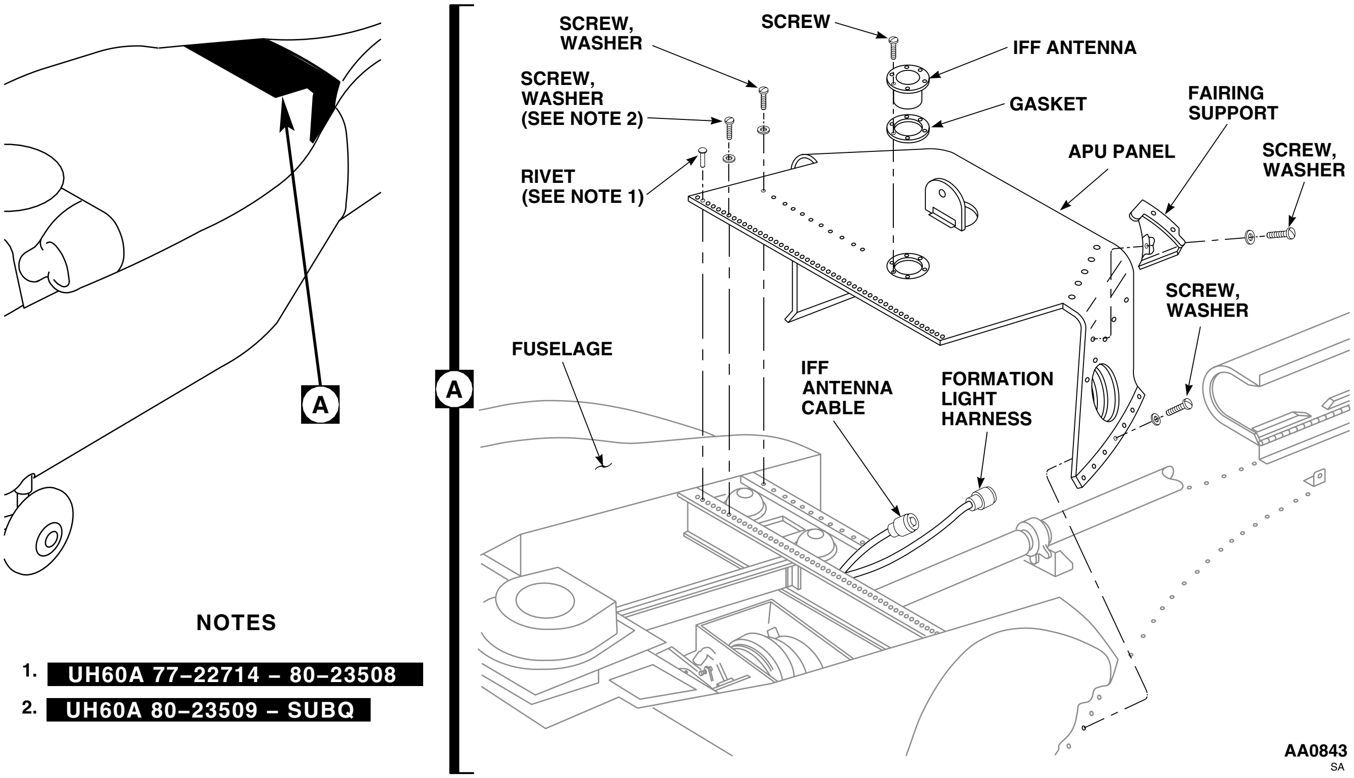


Figure 1. Replace APU Panel.

d. Install rear pylon fairing with two screws (WP 0311 00).

e. Mark and trim APU panel if it binds against rear pylon fairing.

f. Mark all holes that do not line up with attachment holes in fuselage or fire bottle support.

g. Remove fairing and panel from helicopter, and fill in marked holes (WP 0379 00).

h. Replace APU panel on helicopter. Mark and pilot-drill fastener holes where required (TM 1-1500-204-23).

i. Remove fairing. Enlarge pilot holes for fastening screws and rivets (TM 1-1500-204-23).

2. **QA** UH60A 80-23509 - SUBQ Install panel on helicopter and attach with screws and washers (Figure 1, Detail A).

3. **QA** UH60A 77-22714 - 80-23508 Install panel on helicopter with screws and washers and rivet panel to fire extinguisher support.

4. Make sure area is clean and free of foreign material before closing up.

5. Install APU exhaust duct (WP 1360 00).

6. **QA** Clamp formation light electrical harness to panel. Install connector in bracket with nut and washer.

7. Install rear pylon fairing (WP 0311 00).

8. **QA** Attach top IFF antenna cable to cover with screws, washers, nuts, and clamps.

9. Install top IFF antenna (TM 11-1520-237-23).

10. Install APU access doors (WP 0303 00).

INSTALL - Continued

11. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
12. **QA** Install fairing supports with screws, and washers.
13. Install APU exhaust fairings (WP 0306 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

OIL COOLER COMPARTMENT ACCESS DOOR

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Pad, Scouring, Item 205, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0303 00
WP 0379 00
WP 0381 00
WP 1803 00

REMOVE

1. Unfasten oil cooler compartment access door.
2. Squeeze hinge-release pins together at each hinge (Figure 1, Detail B and Detail C). Remove access door from helicopter.

INSPECT/REPAIR

1. Visually inspect hinges, catches, and catch fittings for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked item or door assembly, as required.
2. Remove corrosion and blend out wear, using scouring pad, Item 205, WP 1803 00. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.
3. If chafing strap(s) require replacement, follow procedures in WP 0303 00, Figure 1, Detail B.
4. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Place door over compartment. Squeeze hinge-release pins together at each hinge, put hinge halves together, and release pins (Figure 1, Detail B and Detail C).
2. **UH60A 79-23288 - SUBQ EH60A UH60L** Fit doors between firewalls. If necessary, loosen firewall hinge half attaching screws and nuts. Adjust doors, then tighten screws and nuts (Figure 1, Detail D).

NOTE

UH60A 77-22714 - 79-23287 Hinges are riveted.

3. Check door for fit. If door edge binds against frame, trim area shown in Figure 1, Detail A. Trim edge using procedures in WP 0379 00.
4. Make sure area is clean and free of foreign material before closing up.

INSTALL - Continued

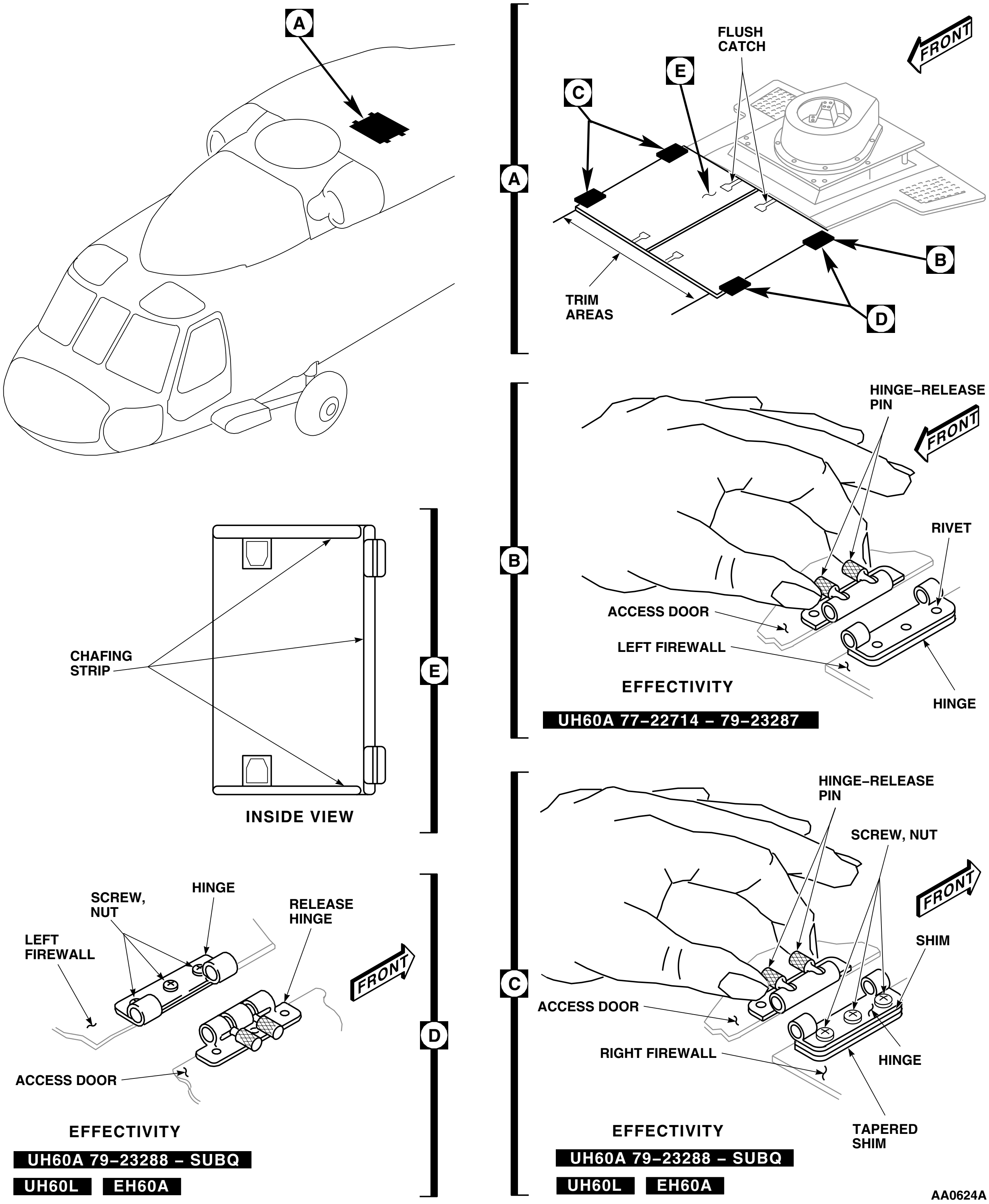


Figure 1. Replace Oil Cooler Compartment Access Door.

INSTALL - Continued

NOTE

Installed access doors may be warped 0.30-inch above surface of surrounding pylon. Hand pressure up to 16 pounds may be used to hold door closed for fastening.

5. Close and fasten oil cooler compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

IR JAMMER FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and washers attaching IR jammer fairing to IR jammer panel (Figure 1, Detail A).
3. Carefully remove IR jammer fairing. Make sure IR jammer transmitter wiring is clear of fairing when removing.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. **QA** Place replacement fairing over IR jammer transmitter. Make sure transmitter wiring is clear of fairing and is through slot in fairing (Figure 1, Detail A).
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. **QA** Install screws and washers attaching fairing to IR jammer panel.

INSTALL - Continued

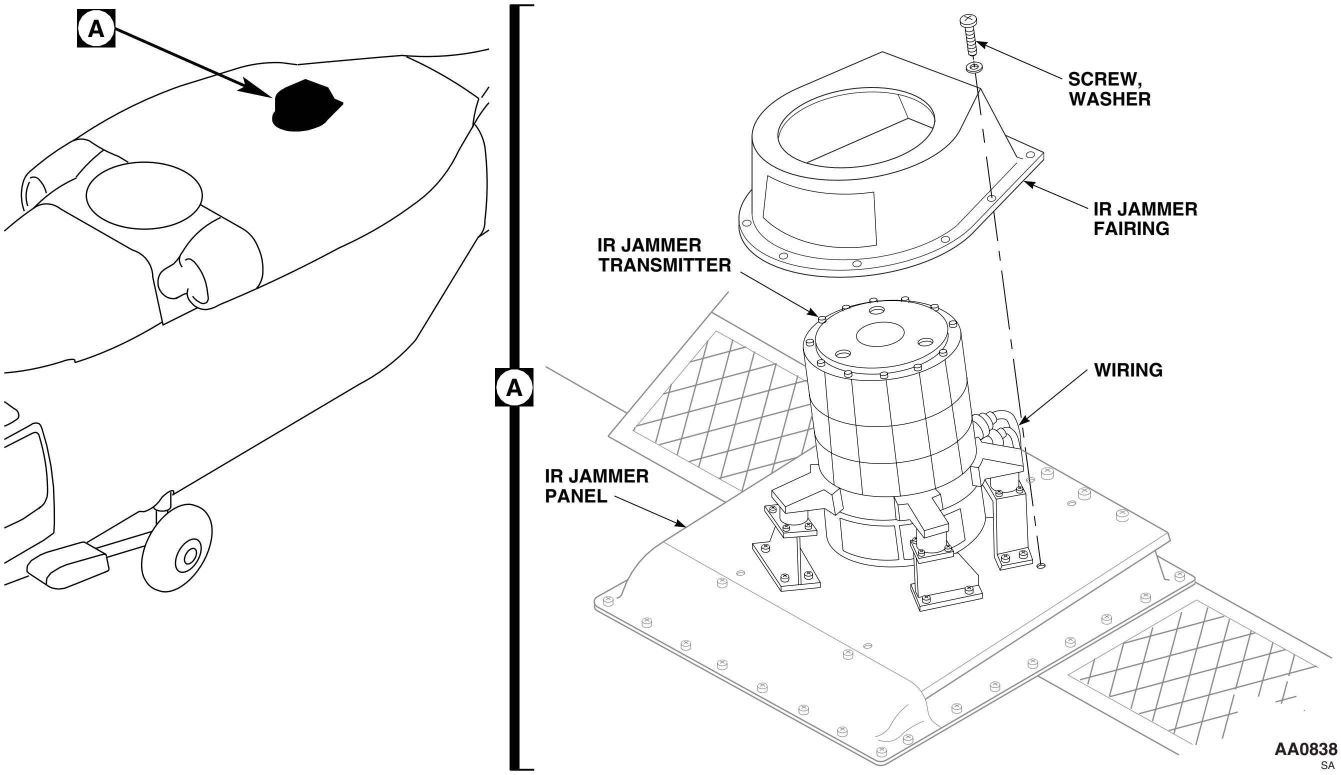


Figure 1. Replace IR Jammer Fairing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

IR JAMMER PANEL

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Wire, Nonelectrical, Item 352, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 11-1520-237-23
WP 0309 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove IR jammer fairing (WP 0309 00).
3. Disconnect electrical connectors from rear of IR jammer transmitter (TM 11-1520-237-23).
4. Remove screws, washers, and spacers attaching IR jammer panel to oil cooler support. Remove panel (Figure 1, Detail A).
5. Remove screws, washers, and nuts attaching mount support brackets to panel.

INSTALL

1. **QA** Install mount support brackets to IR jammer panel with screws, washers, and nuts (Figure 1, Detail A).
2. **QA** Place panel on oil cooler support. Position spacers, with a washer between spacer and panel, and a washer between spacer and oil cooler support, under aft end of panel. Install screws and washers securing panel and spacers to oil cooler support.
3. **QA** Install remaining screws and washers attaching panel to support.
4. Install IR jammer transmitter (TM 11-1520-237-23).
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Connect electrical connectors to receptacles on IR jammer transmitter. Lockwire, using nonelectrical wire, Item 352, WP 1803 00.
7. Make sure area is clean and free of foreign material before closing up.
8. Install IR jammer fairing (WP 0309 00).

INSTALL - Continued

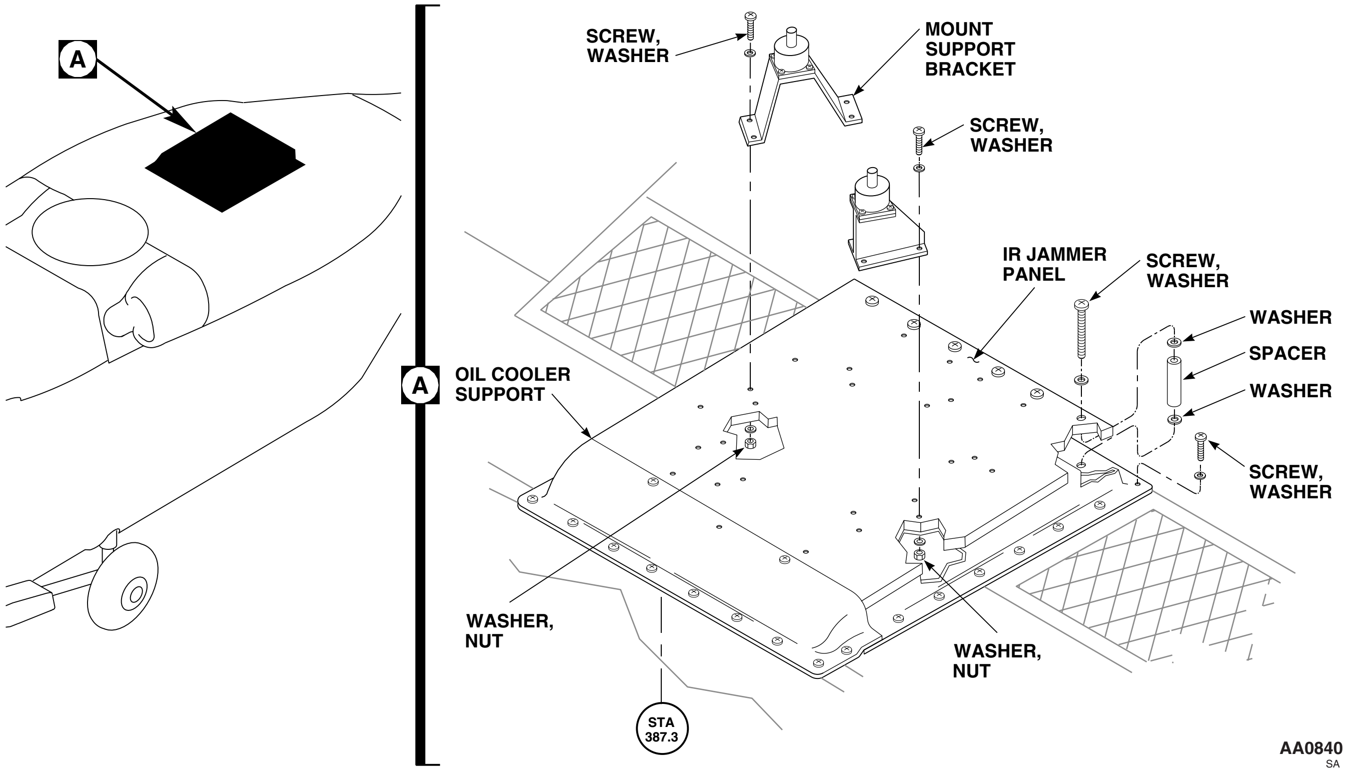


Figure 1. Replace IR Jammer Panel.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN ROTOR PYLON REAR FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 55-1500-345-23
WP 0111 00
WP 0379 00
WP 0381 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open tail rotor drive shaft cover.
3. Open access panel on APU panel and disconnect formation light electrical connectors, P402 (Figure 1, Detail A).
4. Remove screws and washers attaching rear pylon fairing to fuselage and APU panel.
5. Remove fairing.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. Place replacement fairing in position (Figure 1, Detail A).
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. **QA** Install screws and washers attaching fairing to fuselage and APU panel.
5. If installing new fairing, do this:
   a. Trial fit fairing on fuselage.
   b. **UH60A 77-22714 - 79-23286** If nutplates in straps do not line up with holes in APU panel, drill out rivets and remove strap from fairing.
   c. **UH60A 79-23287 - SUBQ** Remove screws and washers and remove straps from fairing.
   d. Mark any other holes in fairing that do not line up with nutplates in fuselage.
   e. Remove fairing. Fill and touch up paint-marked holes and empty rivet holes using procedures in WP 0379 00, WP 0381 00, and TM 55-1500-345-23.
   f. Install loose straps on APU panel with screws and washers. If necessary, fill, mark, and redrill holes in APU panel to fit strap.
   g. Install fairing. Mark new holes using procedures in TM 1-1500-204-23.

INSTALL - Continued

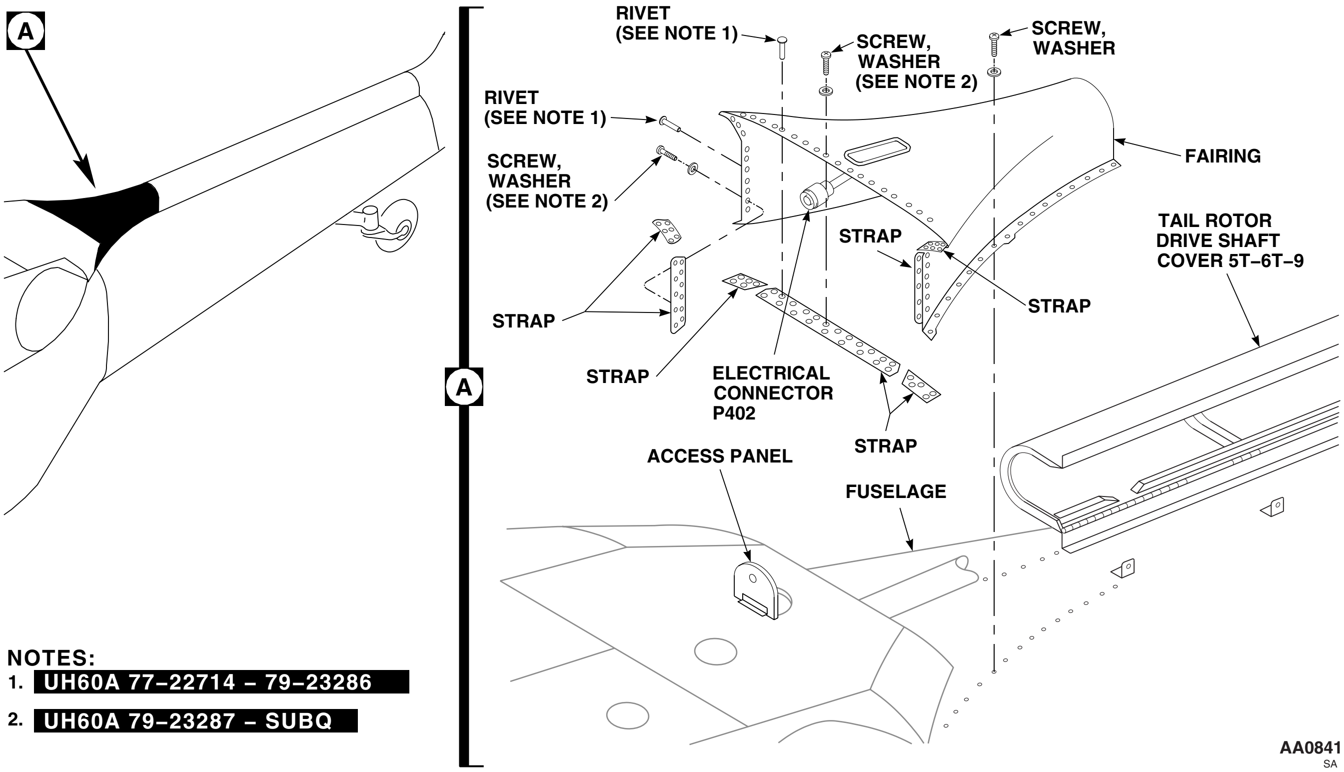


Figure 1. Replace Main Rotor Pylon Rear Fairing.

h. Remove fairing and loose straps. Drill marked holes.

i. **UH60A 77-22714 - 79-23286** Install loose straps on fairing with rivets.

j. **UH60A 79-23287 - SUBQ** Install loose straps on fairing with screws and washers. Tighten screws.

6. Inspect and treat electrical connector (WP 1703 00).
7. **QA** Connect electrical connectors, P402, to formation light electrical harness.
8. Close and fasten access panel on APU panel.
9. Close tail rotor drive shaft cover.
10. Do an operational check of formation light (WP 0111 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

YAW PUSH ROD BOOT AND SHROUD

This WP supersedes WP 0312 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0300 00
WP 0723 00
WP 1123 00
WP 1803 00

REMOVE

1. Disconnect all electrical and hydraulic power.
2. Open main rotor pylon sliding cover.
3. Remove work platform (WP 0300 00).
4. Remove yaw push rod (WP 1123 00).
5. Remove transfer module manifold (WP 0723 00).
6. Remove boot from shroud using a nonmetallic scraper (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. If replacing boot only, remove remaining adhesive from shroud using nonmetallic scraper and machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
8. Remove rivets from shroud (TM 1-1500-204-23) and remove shroud.
9. Remove remaining sealing compound from helicopter fuselage using nonmetallic scraper and machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

INSTALL

1. Lightly abrade inner bonding surface of boot using abrasive cloth, Item 78, WP 1803 00.

INSTALL - Continued

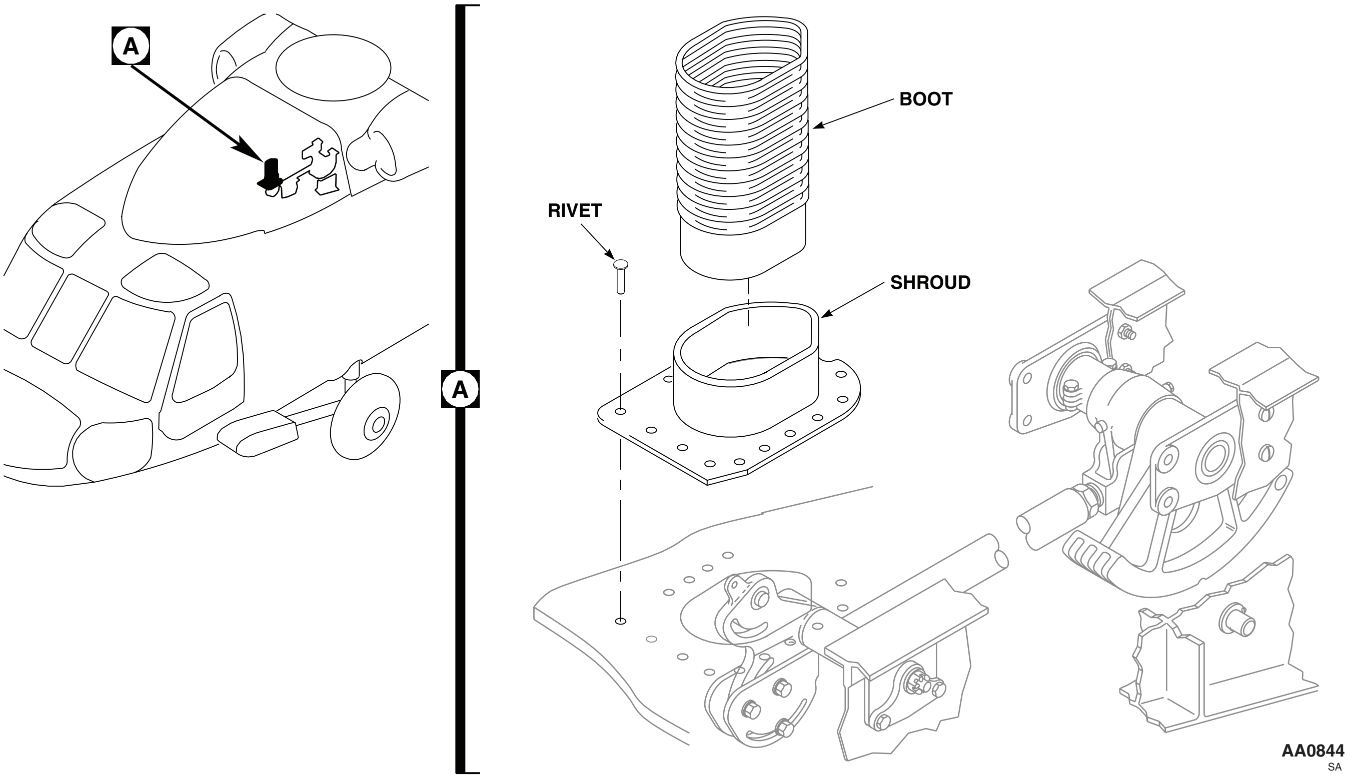


Figure 1. Replace Yaw Push Rod Boot and Shroud.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Wipe abraded area using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Wipe bonding surface of shroud with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
4. Mix 100 parts by weight of part A to 22 parts by weight of part B of adhesive, Item 11, WP 1803 00.
5. Apply a light coat of adhesive, Item 11, WP 1803 00, to bonding surface of boot and shroud.
6. **QA** Install boot on shroud making sure of good contact between bonding surfaces (Figure 1, Detail A).
7. Allow adhesive, Item 11, WP 1803 00, to cure at room temperature for a minimum of 12 hours.
8. Apply a light coat of sealing compound, Item 273, WP 1803 00, to bottom of shroud.
9. **QA** Position boot and shroud on helicopter fuselage.
10. Install rivets wet with sealing compound, Item 273, WP 1803 00 (TM 1-1500-204-23).
11. Install transfer module manifold (WP 0723 00).

INSTALL - Continued

12. Install yaw push rod to mixer lever (WP 1123 00).
13. Make sure area is clean and free of foreign material.
14. Install work platform (WP 0300 00).
15. Close and latch main rotor pylon sliding cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Cloth, Abrasive, Item 75, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-344-23
WP 0379 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

CAUTION

To prevent damage to helicopter, do not let engine cowling drop to open position.

2. Open engine cowling.

CAUTION

To prevent cowling from pivoting down causing possible damage to helicopter, be ready to support cowling when unbolting front and rear supports.

3. With aid of assistant, remove bolt, washer, and nut securing front and rear supports to cowling (Figure 1, Sheet 1).
4. With aid of assistant, remove bolt, washer, and nut securing cowling to forward and aft fittings and remove cowling from helicopter.

INSPECT/REPAIR

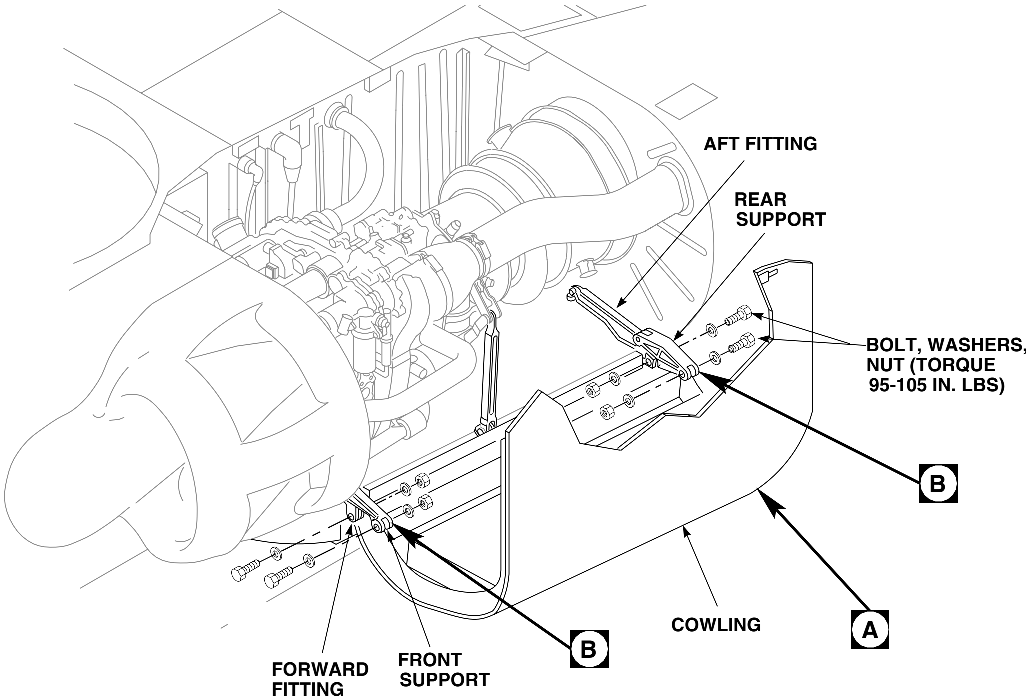
1. Visually inspect hinge bolts for corrosion (TM 1-1500-344-23). Corrosion is indicated by pitting.
2. Replace corroded bolts or remove corrosion.

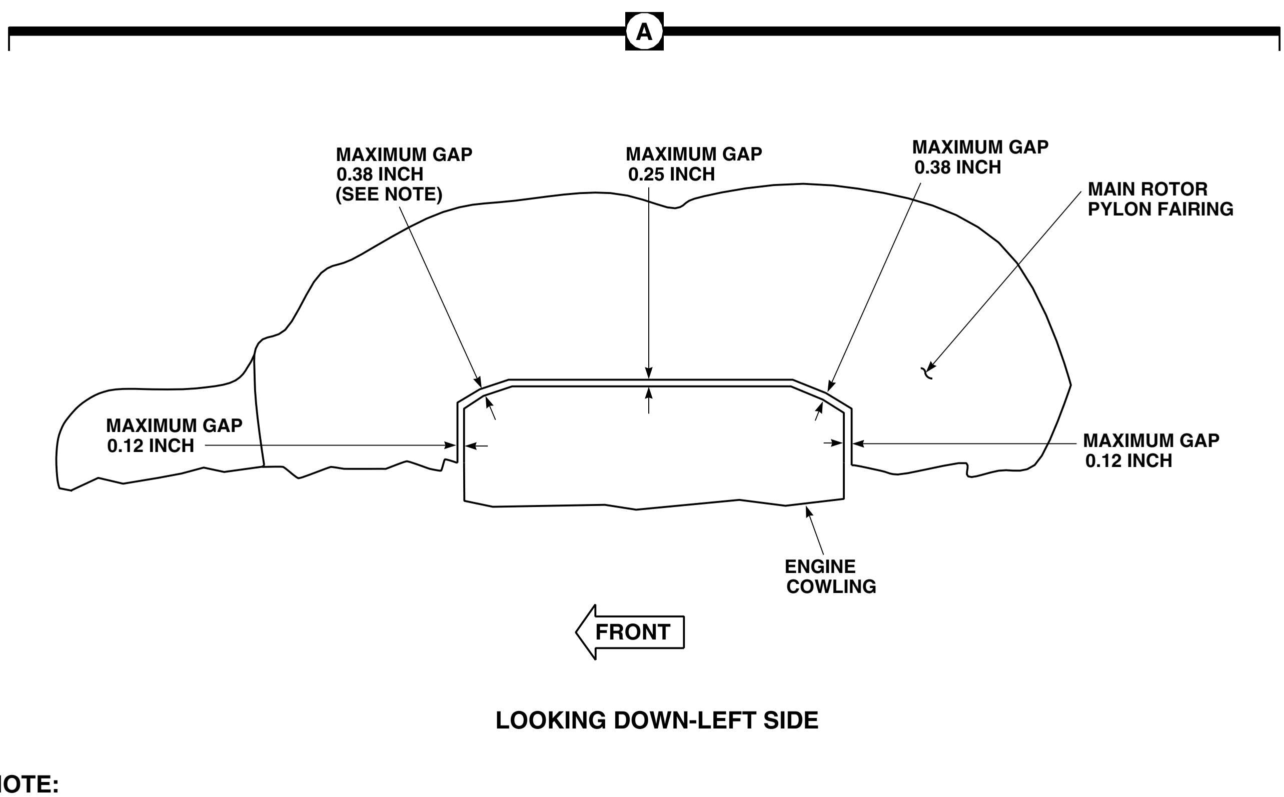
NOTE

If possible, replace corroded bolts. Cadmium plating will be removed during corrosion removal.

3. Remove corrosion manually, using scouring pad, Item 205, WP 1803 00. **NO PITS ALLOWED AFTER CLEAN-UP.**
4. Inspect front and rear supports for wear or damage. **NO DAMAGE BEYOND LIMITS ALLOWED (Figure 1, Sheet 2).** Replace front or rear support if worn or damaged beyond limits.

INSPECT/REPAIR - Continued





NOTE:
1.00 MAX GAP RIGHT SIDE OF COWLING.

AA0846_1D
SA

Figure 1. Replace Engine Cowling. (Sheet 1 of 2)

INSPECT/REPAIR - Continued

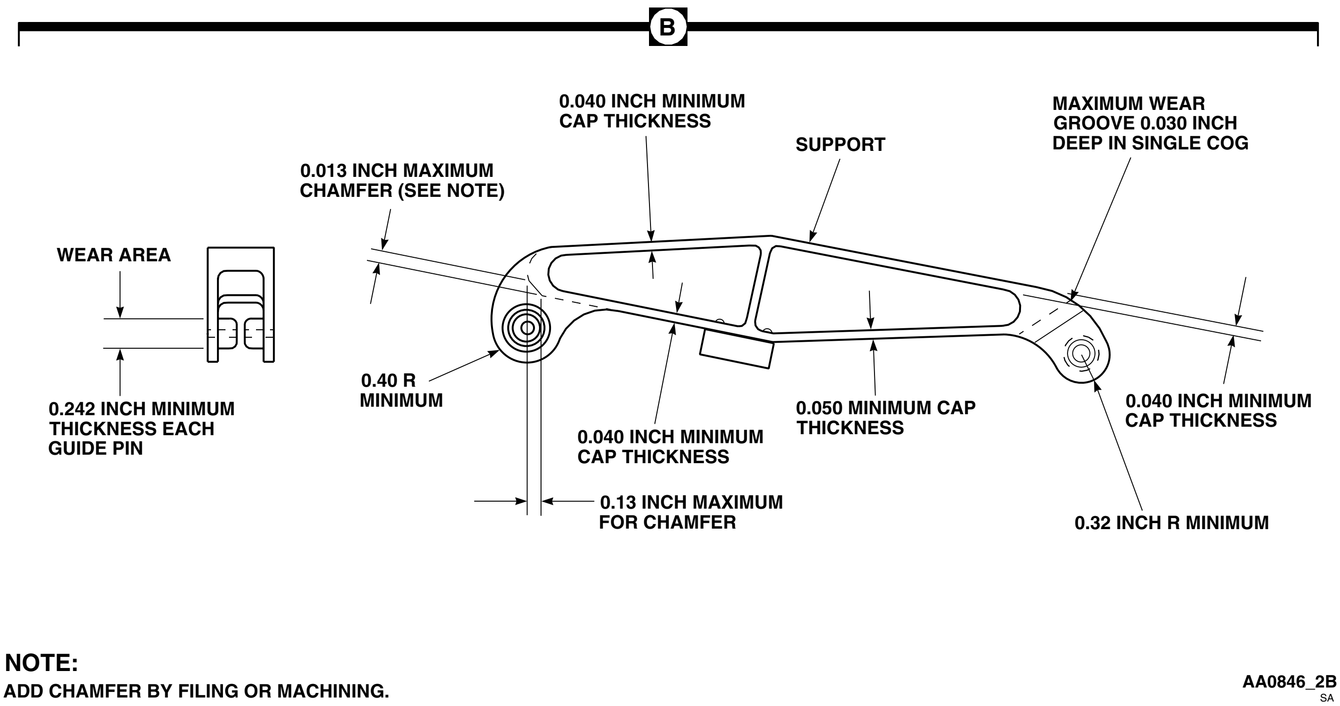


Figure 1. Replace Engine Cowling. (Sheet 2 of 2)

5. Inspect front and rear support guide pin for wear. **GUIDE PIN DIAMETER SHALL NOT BE LESS THAN 0.242-INCH.** Replace front or rear support if diameter is below limit.
6. Blend wear and damage to front or rear supports using abrasive cloth, Item 75, WP 1803 00. **DO NOT BLEND TO DEPTH BEYOND DAMAGE.**

INSTALL

1. Turn off all electrical power.
2. **QA** With aid of assistant, place cowling on forward and aft fittings and install bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS (Figure 1, Sheet 1).**
3. **QA** Bolt front and rear support to cowling. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
4. Check fit of cowling. If necessary, trim edges of cowling (WP 0379 00). Gap tolerances are given in Figure 1, Sheet 1, Detail A. Gaps are same for left and right engine cowling except as noted.
5. Make sure area is clean and free of foreign material before closing up.
6. Close and latch engine cowling.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING LOWER FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0313 00

REPLACE

Refer to WP 0313 00 for removal and installation procedures.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING BLANKET INSULATION

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Open engine cowling.
2. Remove screws, washers, and remove blanket insulation from cowling (Figure 1, Detail A).

INSTALL

1. **QA** Place blanket insulation on cowling and install washers and screws (Figure 1, Detail A).
2. Make sure area is clean and free of foreign material before closing up.
3. Close engine cowling and latch.

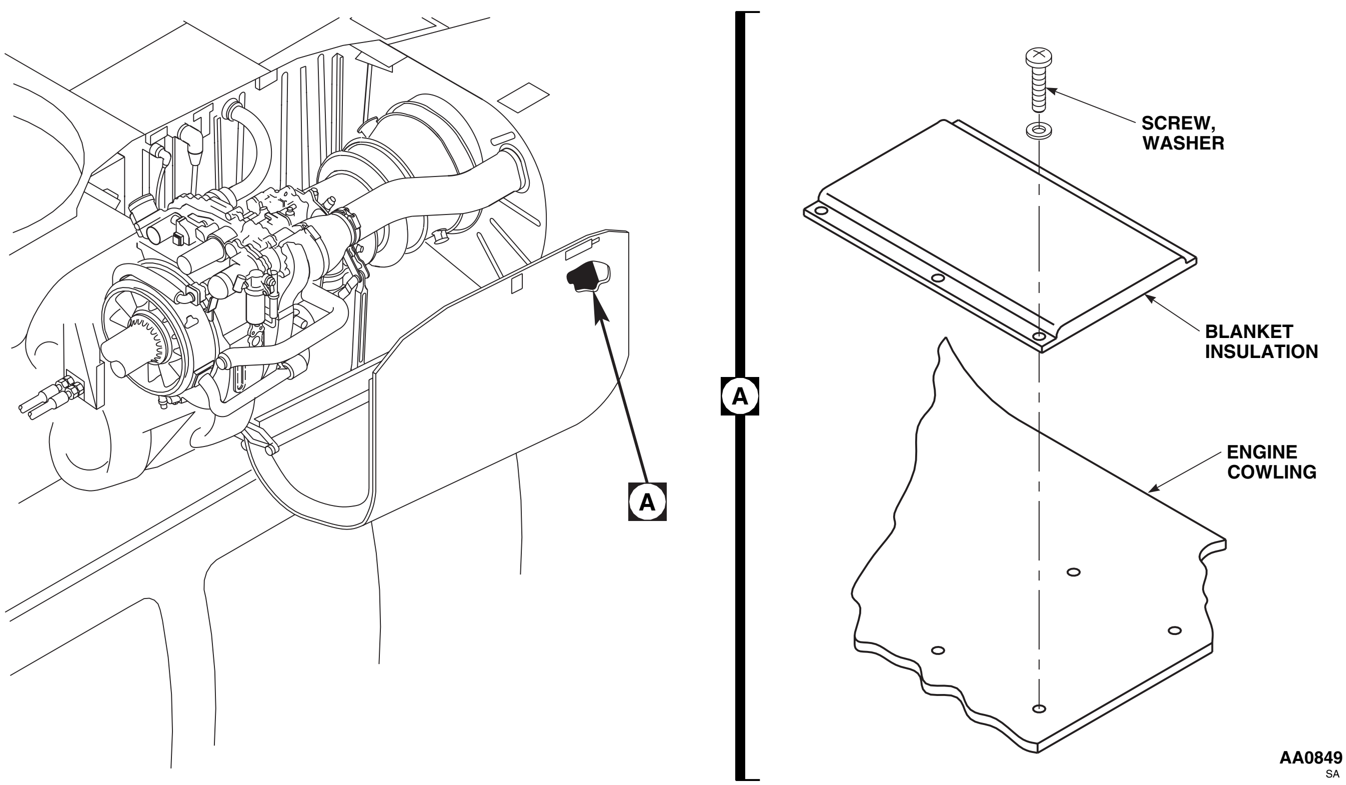


Figure 1. Replace Engine Cowling Blanket Insulation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING LATCH

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Polyurethane, Coating, Item 227, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. Open engine cowling.
2. Remove clevis bolts, washers, bushings, and nuts attaching lockpin tubes to latch handle assembly. Remove lockpin tubes (Figure 1, Sheet 1, Sheet A).
3. Remove screws, washers, and nuts attaching latch handle release to engine cowling. Remove latch handle release.
4. Remove screws, washers, and nuts attaching latch handle assembly to engine cowling. Remove latch handle assembly.
5. Remove screws, washers, and nuts attaching lockpin fitting to cowling. Remove fitting.

DISASSEMBLE

1. Disassemble latch handle assembly as follows:
   a. Remove clevis pin, washer, and cotter pin attaching latch handle to handle support (Figure 1, Sheet 1).
   b. Remove latch handle.
2. Disassemble lockpin tube as follows:
   a. Remove clevis bolt, washer, nut, and cotter pin attaching lockpin to lockpin tube (Figure 1, Sheet 1).
   b. Remove lockpin.

REPAIR

1. Remove corrosion manually, using scouring pad, Item 205, WP 1803 00.
2. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
3. Apply compound solution to corroded areas of latch handle. Rinse thoroughly with clean water.
4. Touch up paint nicks, scratches, and corroded areas, with epoxy primer coating kit, Item 120, WP 1803 00, and coating polyurethane Item 227, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

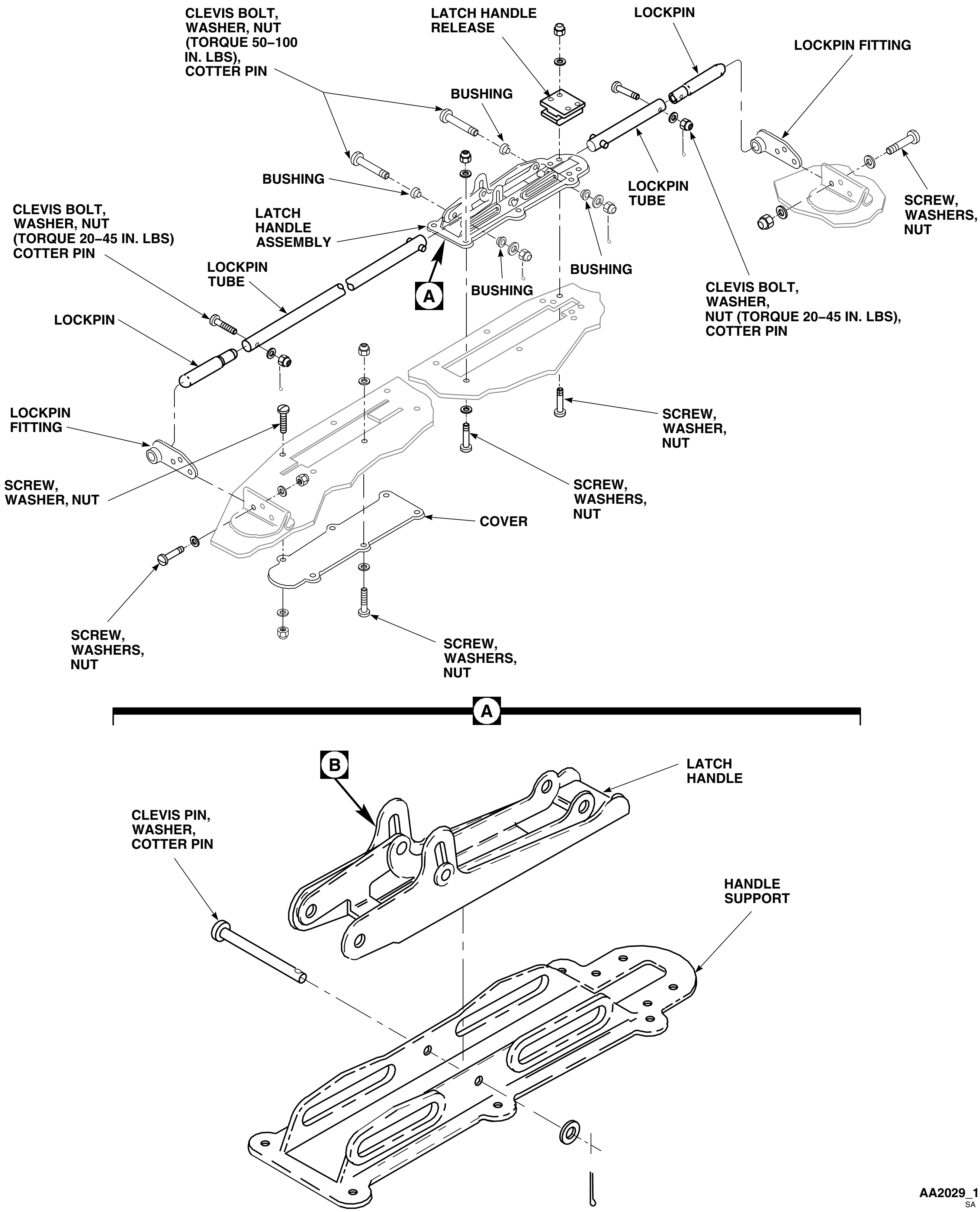


Figure 1. Replace Engine Cowling Latch. (Sheet 1 of 2)

REPAIR - Continued

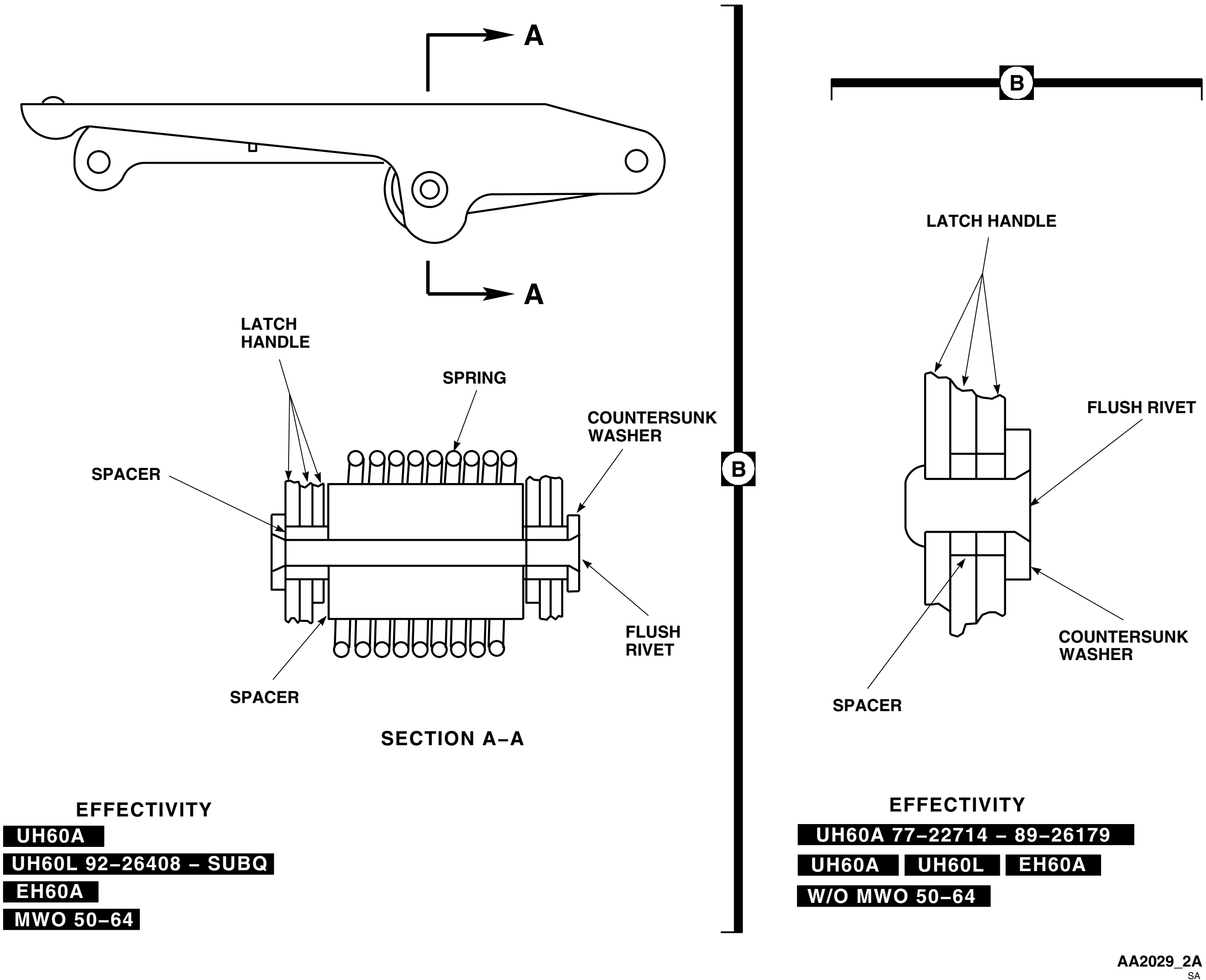


Figure 1. Replace Engine Cowling Latch. (Sheet 2 of 2)

ASSEMBLE

1. Assemble latch handle as follows:

   a. Position latch handle in handle support (Figure 1, Sheet 1).

   b. **QA** Install clevis pin, washer, and cotter pin.

2. Assemble lockpin tube as follows:

   a. **QA** Install lockpin in lockpin tube using clevis bolt, washer, and nut (Figure 1, Sheet 1). **TORQUE NUT TO 20 - 45 INCH-POUNDS.**

   b. **QA** Install cotter pin.

INSTALL

1. Install latch handle assembly on engine cowling with screws, washers, and nuts (Figure 1, Sheet 1).
2. Install latch handle release on engine cowling using screws, washers, and nuts.
3. Install lockpin fittings on engine cowling using screws, washers, and nuts.
4. Insert lockpin tube in fitting and other end in latch handle assembly.

NOTE

Free play of bushings is required for proper latch actuation.

5. **QA** Install clevis bolt, bushings, and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**
6. **QA** Install cotter pin.
7. Check action of latch handle and lockpins for smooth operation.
8. Make sure area is clean and free of foreign material.
9. Close and latch engine cowling several times to be sure of proper operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING FITTING

This WP supersedes WP 0317 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 32, WP 1803 00
Cloth, Abrasive, Item 77, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Wear Strip, Locally-Made, Figure 201, WP 1805 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0313 00
WP 0381 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove engine cowling from helicopter (WP 0313 00).
2. Remove pin and collar attaching forward fitting to upper door track (Figure 1, Sheet 1, Detail C).
3. Remove bolt, washer, and nut attaching aft fitting to upper door track (Figure 1, Sheet 1, Detail D).
4. Remove bolts, washers, and nuts attaching forward or aft fittings to fittings on engine deck (Figure 1, Detail A or Detail B).
5. Remove fitting.

INSPECT

1. Inspect forward fitting for wear. **CAP THICKNESS SHALL NOT BE LESS THAN 0.040-INCH (Figure 1, Sheet 2, Detail E)**. Replace or repair fitting as required.
2. Inspect aft fitting for wear. **CAP THICKNESS SHALL NOT BE LESS THAN 0.055-INCH IN ZONE A AND 0.065-INCH IN ZONE B (Figure 1, Sheet 2, Detail F)**. Replace or repair fitting as required.

REPAIR

1. Blend wear to fittings using abrasive cloth, Item 77, WP 1803 00. **DO NOT BLEND TO DEPTH BEYOND DAMAGE.**
2. Bond wear strip to aft fitting as follows:
   a. Locally make wear strip (WP 1805 00).

REPAIR - Continued

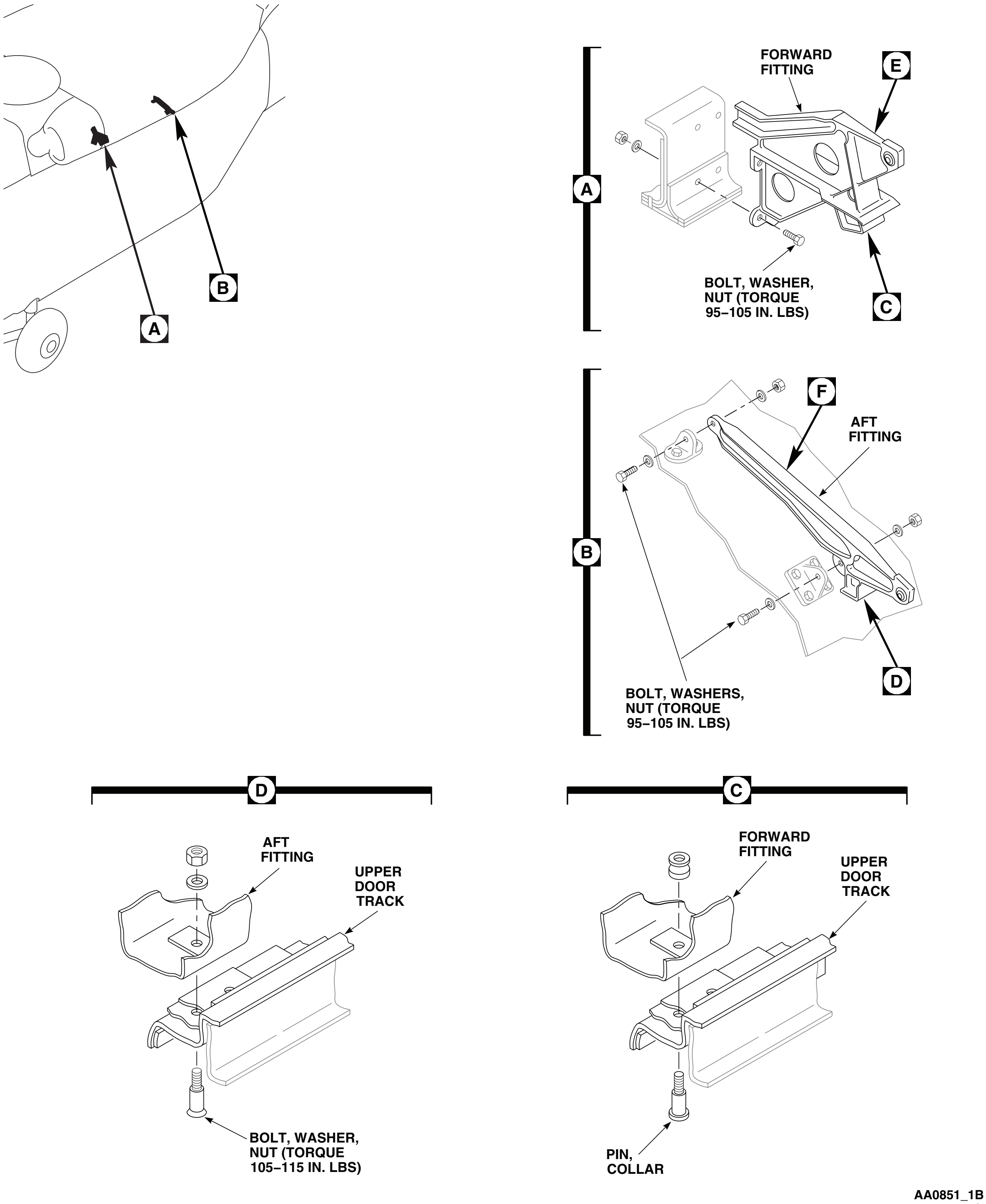
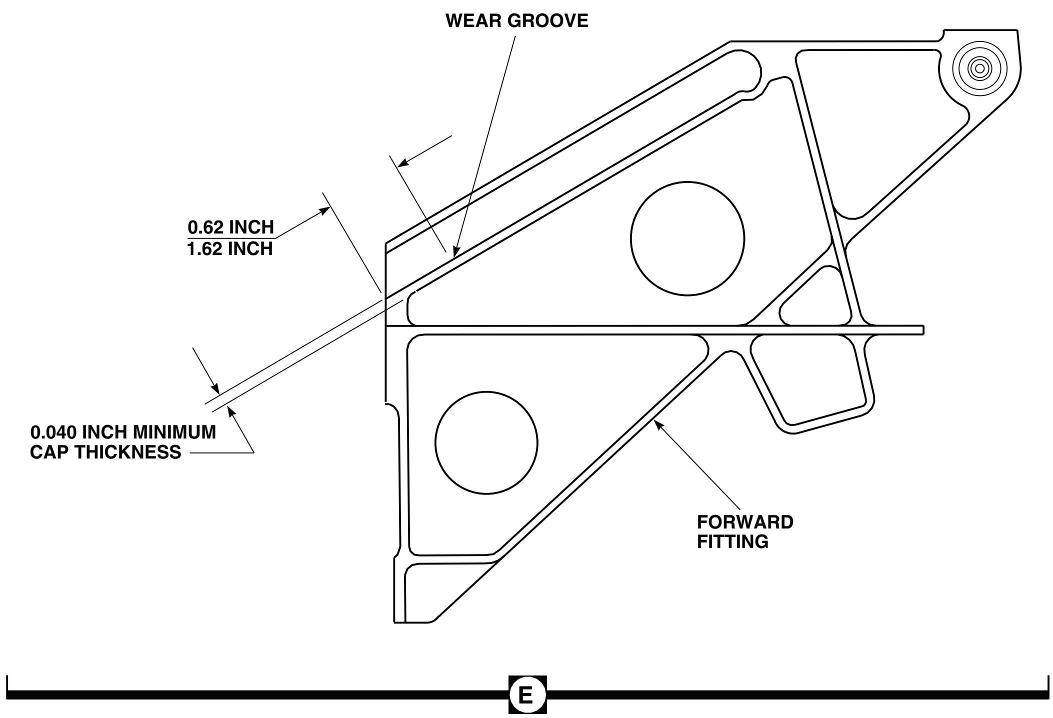


Figure 1. Replace Engine Cowling Fittings. (Sheet 1 of 2)

REPAIR - Continued



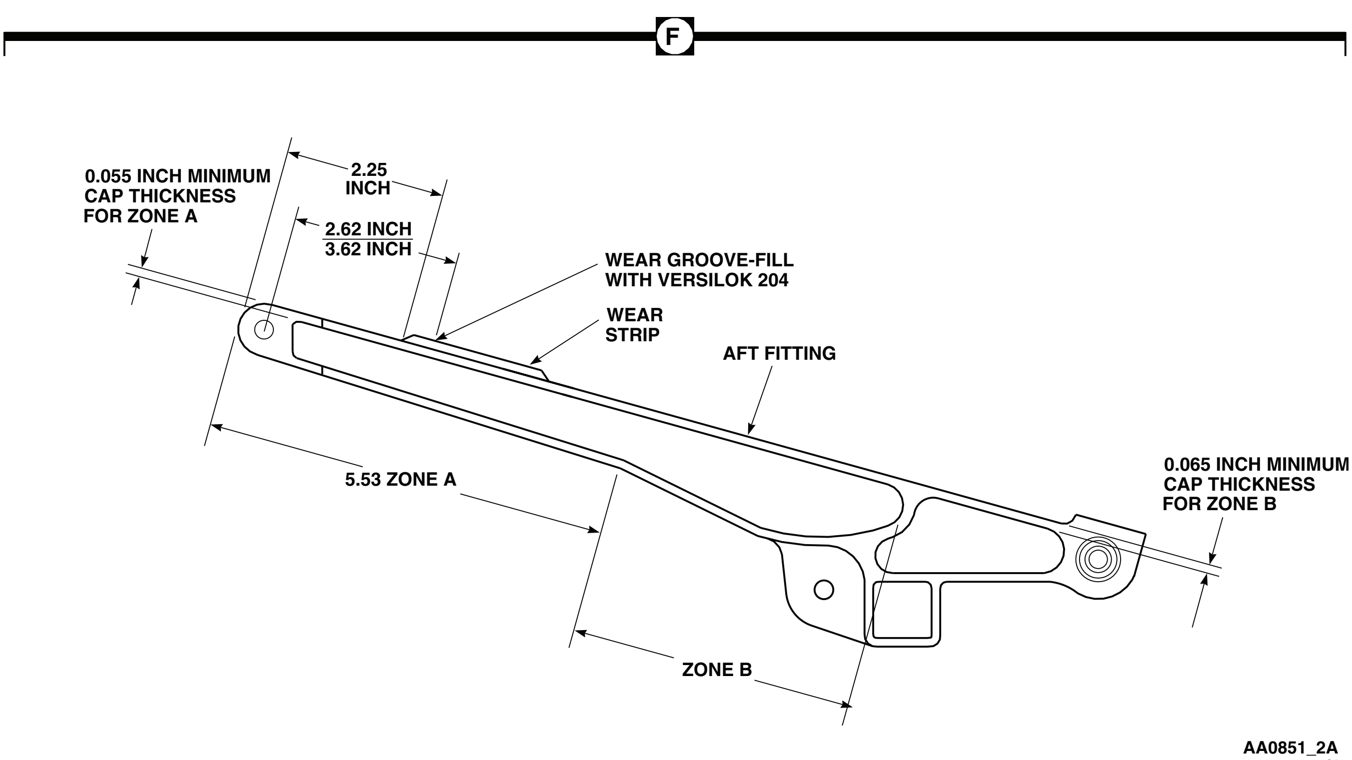


Figure 1. Replace Engine Cowling Fittings. (Sheet 2 of 2)

REPAIR - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Clean bonding surface of aft fitting and wear strip with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply adhesive, Item 32, WP 1803 00, to each surface to be bonded.

d. Allow adhesive to air dry at room temperature for a minimum of 3 minutes.

e. Apply a bead of adhesive, Item 32, WP 1803 00, to aft fitting.

f. Position wear strip on aft fitting (Figure 1, Sheet 2, Detail F).

g. Press wear strip firmly in place.

h. Allow to cure at room temperature for 24 hours.

i. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Place replacement fitting on engine deck (Figure 1).
2. **QA** Install bolt, washer, and nut attaching aft fitting to upper door track. **TORQUE NUT TO 105 - 115 INCH-POUNDS (Figure 1)**.
3. Install pin and collar attaching forward fitting to upper door track (Figure 1).
4. **QA** Install bolts, washers, and nuts attaching fittings to engine deck. **TORQUE NUTS TO 95 - 105 INCH-POUNDS Figure 1, Sheet 1, Detail A and Figure 1, Sheet 1, Detail B).**
5. Install engine cowling (WP 0313 00).
6. Make sure area is clean and free of foreign material before closing up.
7. Close and latch cowling.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING SUPPORT FITTING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

Replacement instructions are the same for both front and rear fittings.

REMOVE

1. Remove screws, nuts, and washers securing support fitting to engine cowling (Figure 1).
2. Remove support fitting and laminated shims.

INSTALL

1. Install support fitting and laminated shims, aligning fastener holes (Figure 1).
2. **QA** Install screws, washers, and nuts.

INSTALL - Continued

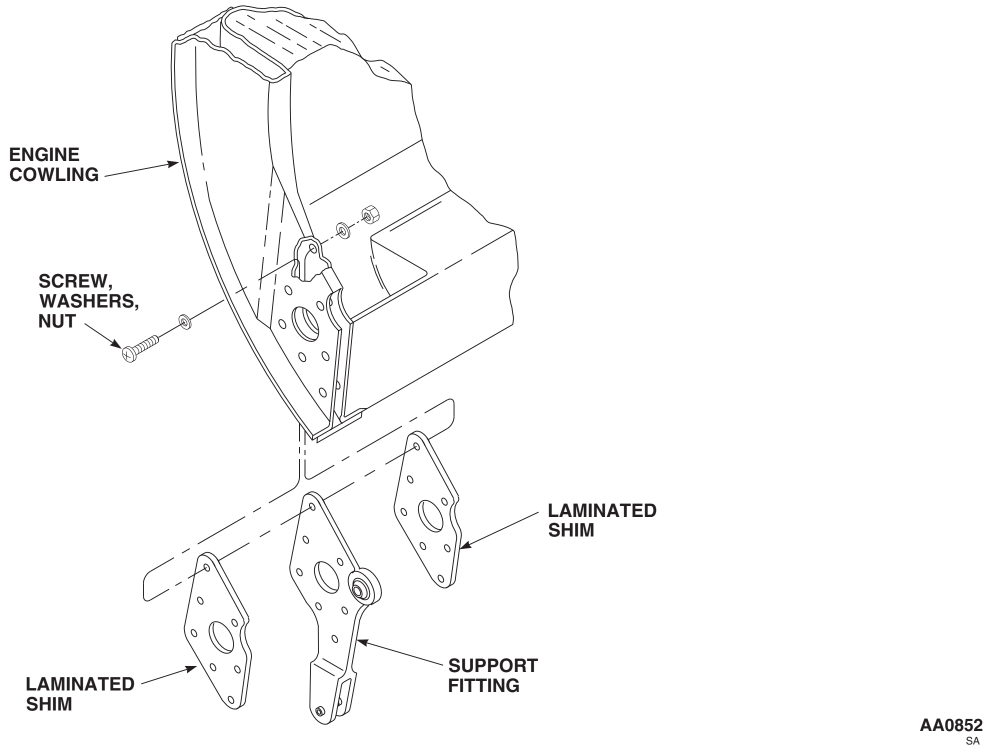


Figure 1. Replace Engine Cowling Support Fitting.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COMPARTMENT HEAT SHIELD

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Open engine cowling.
2. Disconnect engine drain tubes and elbows on rear engine compartment deck (Figure 1, Detail A).
3. Remove screws and washers from heat shield.
4. Slightly bend rear end of heat shield up enough to clear elbows in deck. Remove heat shield, spacers, washers, and packings from rear engine compartment deck. Discard packings.

INSTALL

1. Slightly bend rear end of heat shield up enough to clear elbows in deck. Place heat shield over elbows on rear engine compartment deck (Figure 1, Detail A).
2. **QA** Position spacers, washers, and packing beneath heat shield at each attachment hole, install screws and washers.
3. Connect engine drain tubes and hose to elbows and union on rear engine compartment deck.
4. Make sure area is clean and free of foreign material before closing up.
5. Close engine cowling and latch.

INSTALL - Continued

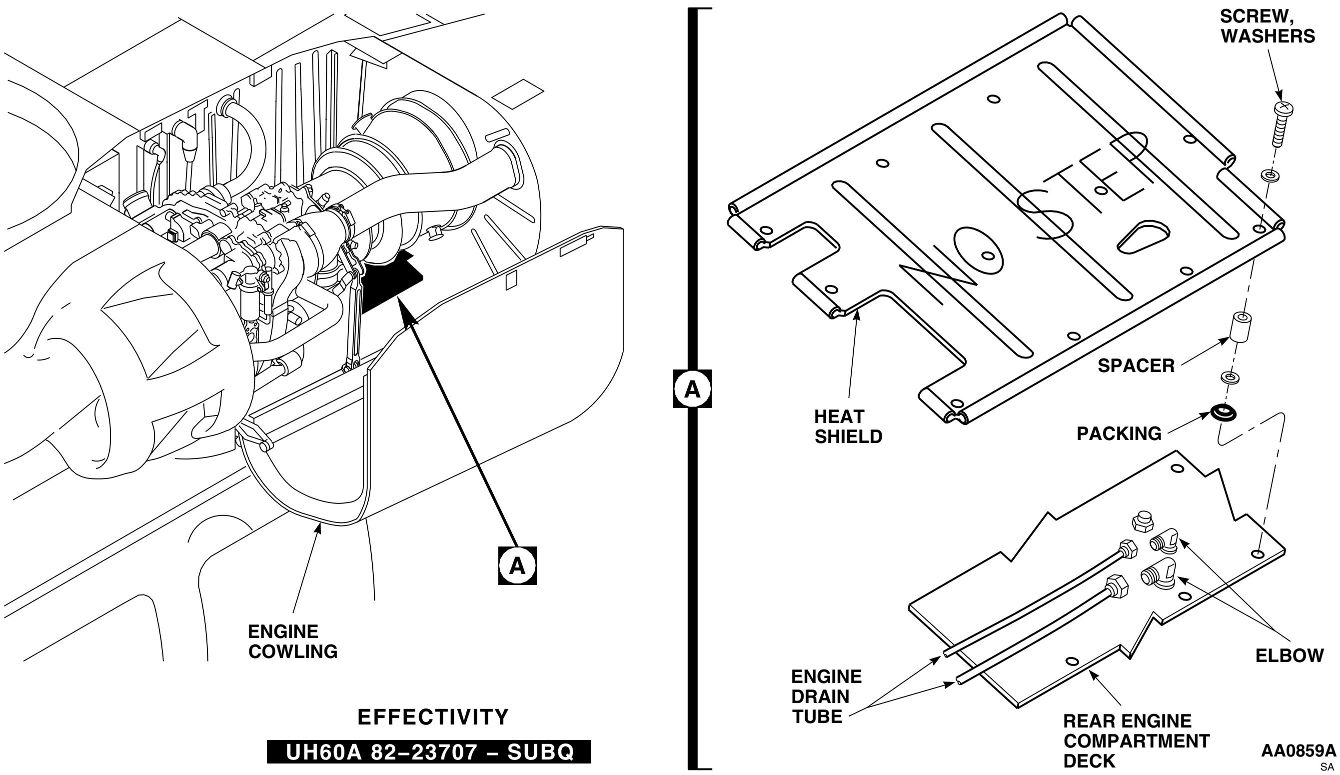


Figure 1. Replace Engine Compartment Heat Shield.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR EH60A

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door (Figure 1, Detail A).
3. Support door in open position while removing outboard end of strut from door.
4. Disengage outboard end of strut from support bracket on avionics door.
5. Lower door to closed position.
6. Remove clevis pins, holding hinge halves together. Remove door.

INSTALL

1. Turn off all electrical power.
2. Place access door in open position on helicopter, engaging hinge halves (Figure 1, Detail A).
3. **QA** Line up hinge bores and install clevis pin and cotter pin, in each hinge.
4. Support door in open position.
5. Attach outboard end of strut to support bracket on avionics compartment door.
6. Close door and check if door is centered within door opening.
7. If centering is required, adjust door by loosening screws and nuts that hold door hinge fittings and reposition door as necessary. Retighten screws.
8. Make sure area is clean and free of foreign material before closing up.
9. Close door and fasten latches.

INSTALL - Continued

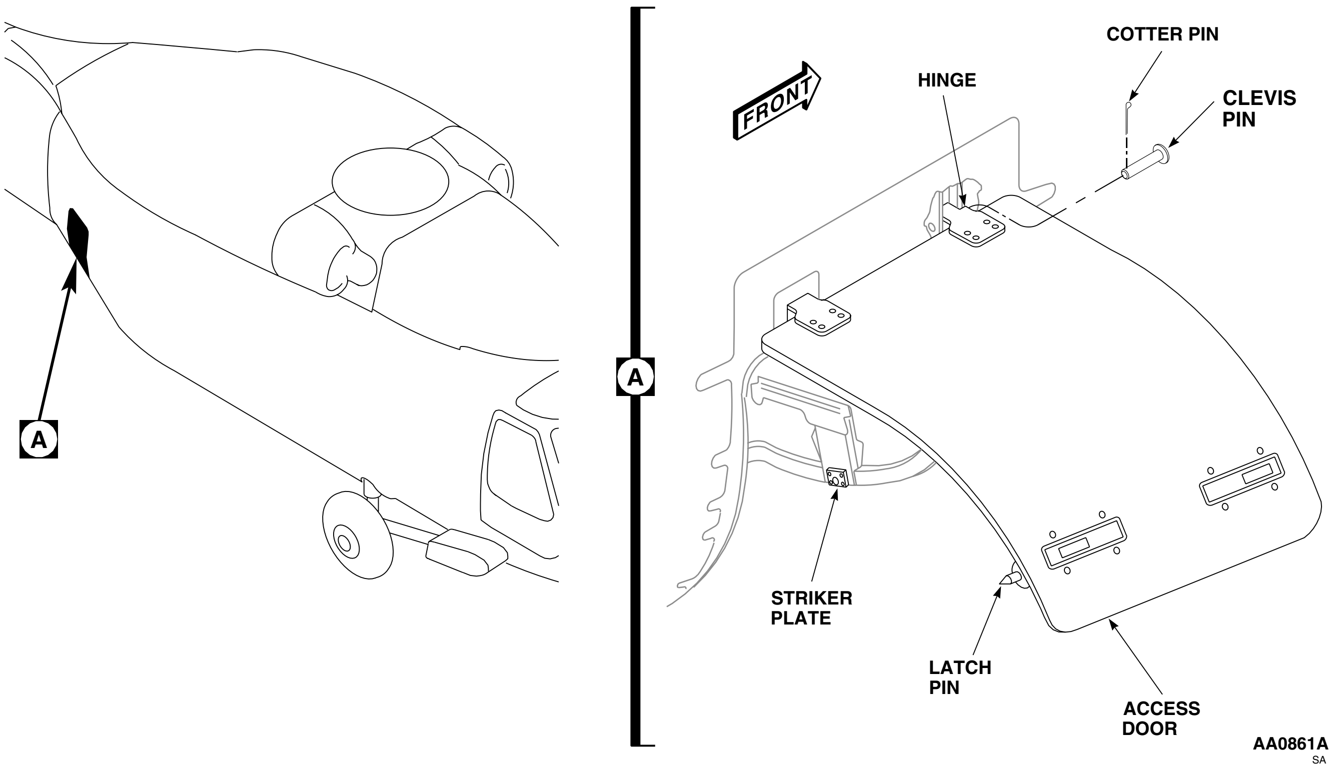


Figure 1. Replace Transition Section Avionics Door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR STRUT EH60A

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Support door in open position while removing outboard end of strut from support bracket.
4. Disengage outboard end of strut from support bracket on avionics door (Figure 1, Detail A).
5. Unpin inboard end of strut from support fitting on compartment bulkhead. Remove strut from compartment.

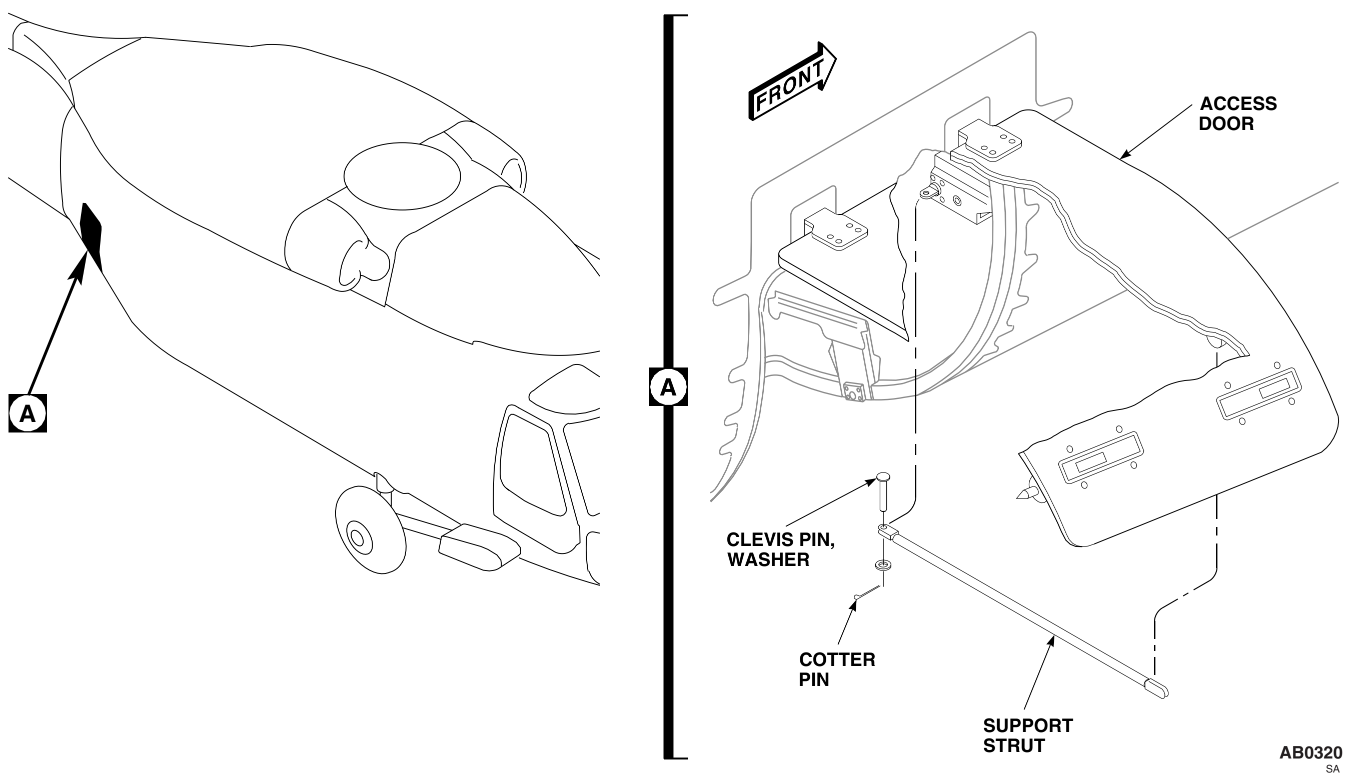


Figure 1. Replace Transition Section Avionics Door Strut.

INSTALL

1. **QA** Install clevis end of strut on inboard support fitting with pin. Install washer and cotter pin (Figure 1, Detail A).
2. **QA** Attach outboard end of strut to support bracket on avionics door.
3. Remove temporary avionics compartment door support.
4. Make sure area is clean and free of foreign material before closing up.
5. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR EYEBOLT AND SUPPORT EH60A

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0321 00

REMOVE

1. Remove avionics door strut (WP 0321 00).
2. Drill out rivets holding eyebolt support to bulkhead (Figure 1, Detail B) (TM 1-1500-204-23). Remove support with eyebolt.
3. Remove nut, washers, and eyebolt from support.

INSTALL

CAUTION

Equipment can be damaged if eyebolt nut is too tight. Do not overtighten nut. Eyebolt should turn freely after cotter pin is installed.

1. **QA** Install eyebolt, washers, and nut on support. Tighten nut fingertight. If cotter pin hole in eyebolt does not line up with slots in nut, loosen nut to next slot to install cotter pin. Install cotter pin (Figure 1, Detail B).
2. **QA** Install eyebolt support on compartment bulkhead and rivet in place (TM 1-1500-204-23).
3. Install avionics door strut (WP 0321 00).

INSTALL - Continued

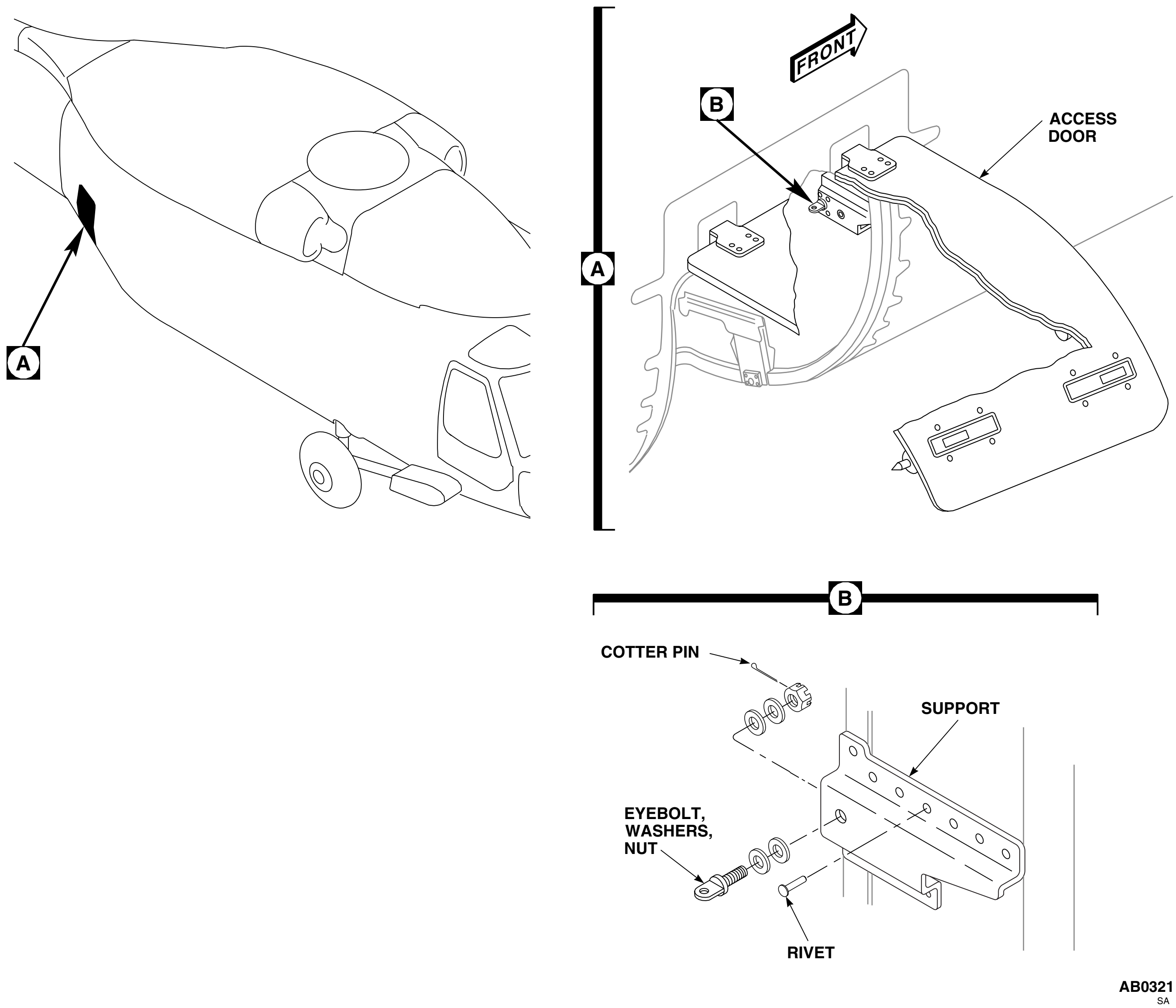


Figure 1. Replace Transition Section Avionics Door Eyebolt and Support.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR STRUT LOCK FITTING EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

Remove bolts, washers, and strut lock fitting from access door (Figure 1, Detail B).

INSTALL

1. Install strut lock fitting on access door with bolts and washers (Figure 1, Detail B).
2. **QA TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Using nonelectrical wire, Item 354, WP 1803 00, lock-wire boltheads together.

INSTALL - Continued

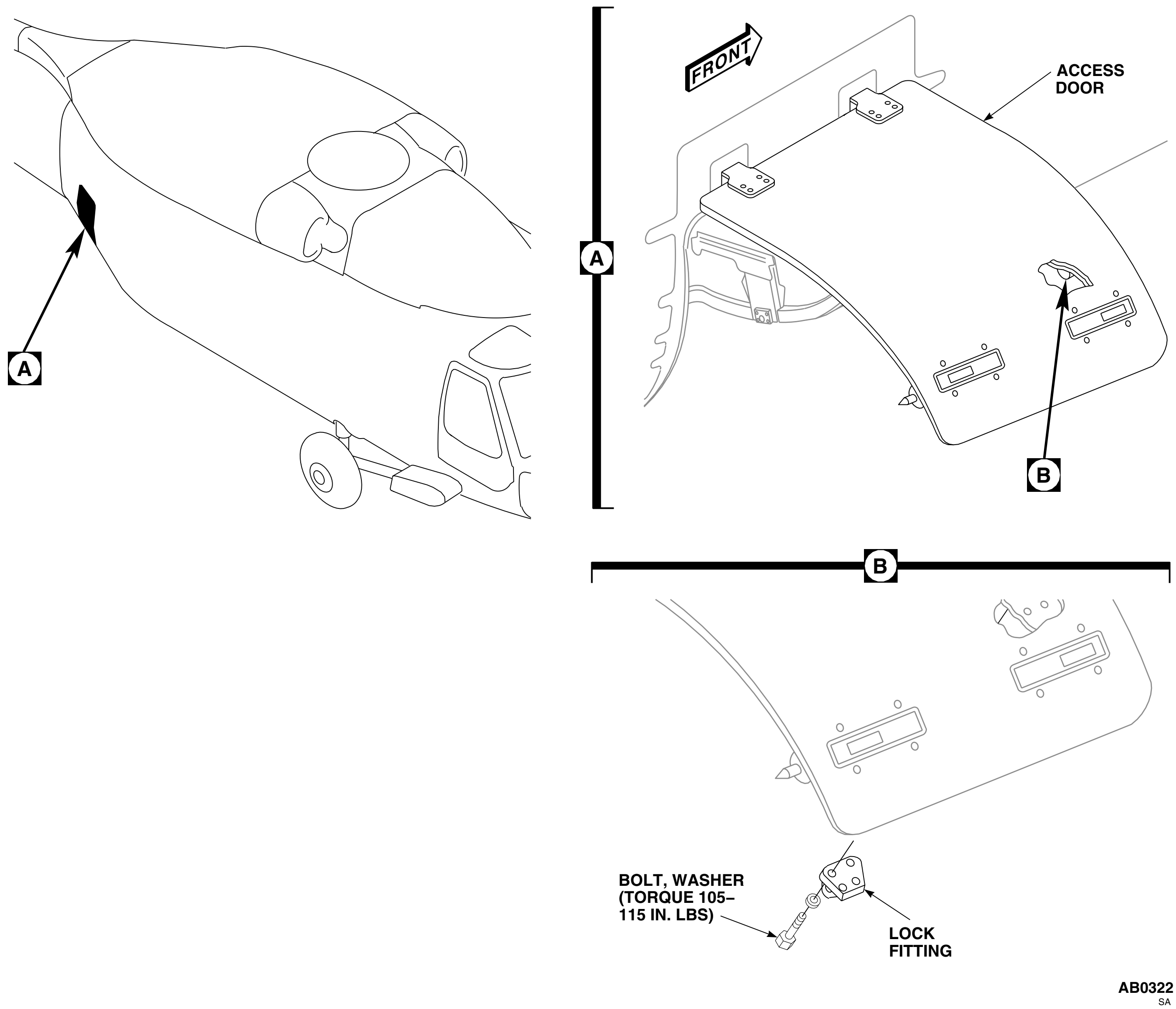


Figure 1. Replace Transition Section Avionics Door Strut Lock Fitting.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR HINGE EH-60A

This WP supersedes WP 0324 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 15, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Support door in open position while replacing hinge.
4. Remove clevis pins holding hinge parts together.
5. Remove screws, washers and nuts holding hinge to door (Figure 1, Detail A). Remove hinge from door.

INSTALL

1. Turn off all electrical power.
2. **QA** Assemble hinge parts together with clevis pins and cotter pins.
3. Position door on aft transition section of helicopter.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Wipe hinge clean with a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
5. Apply sealing compound, Item 273, WP 1803 00, to door hinge screw holes and screws.
6. Apply adhesive, Item 15, WP 1803 00, to raised pad area on avionics door.
7. **QA** Immediately, install hinge on door with screws, washers, and nuts (Figure 1, Detail A).
8. Remove door support and close door. See if door is centered within door opening.

INSTALL - Continued

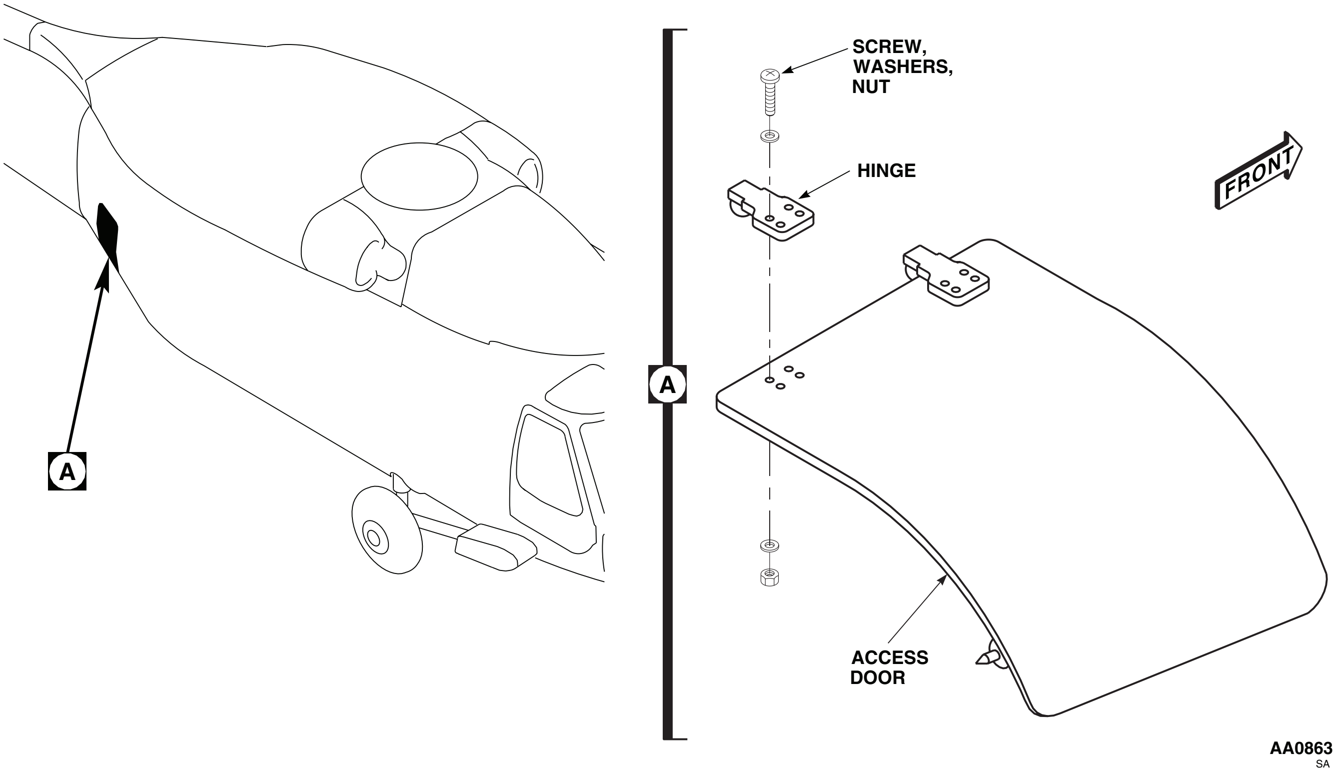


Figure 1. Replace Transition Section Avionics Door Hinge.

9. If centering is required, adjust door by loosening screws and nuts that hold door hinge and reposition door as necessary. Retighten screws.
10. Wipe off extra sealing compound.
11. Make sure area is clean and free of foreign material before closing up.
12. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR SEALS EH-60A

This WP supersedes WP 0325 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 9, WP 1803 00
Adhesive, Item 14, WP 1803 00
Plastic, Sheet, Item 220, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Stirring Stick, Beverage, Item 300, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0320 00
WP 1803 00

REMOVE

1. Remove avionics compartment access door (WP 0320 00).
2. Remove old seals from door and jamb using nonmetallic scraper (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Remove remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

INSTALL

1. Fit new seal to door with spliced joint at bottom center. Locate and remove 30 to 45 degree sectional cuts to keep seal from bunching or stretching around corners (Figure 1, Detail A).
2. Split door seal to prevent disbonding in bolt recess area.
3. Fit new seal to jamb with spliced joints at top and bottom center.
4. Roughen all adhesive contact surfaces of new seals with abrasive cloth assortment, Item 1, WP 1803 00.

INSTALL - Continued

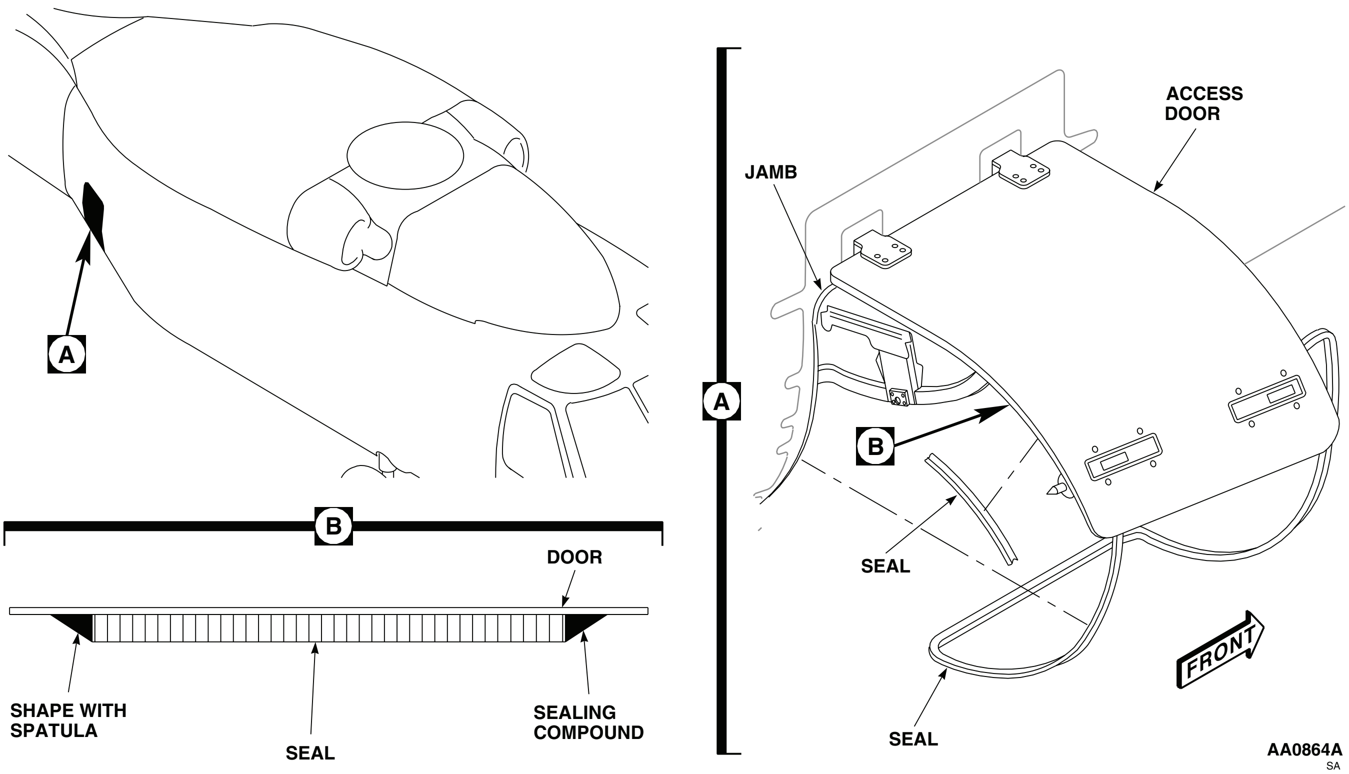


Figure 1. Replace Transition Section Avionics Door Seals.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Wipe all adhesive contact areas with a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
6. Line up jamb seal joints on a flat surface covered with plastic sheet, Item 220, WP 1803 00.
7. Wipe areas with a machinery towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
8. Apply adhesive, Item 9, WP 1803 00, to seal splice.
9. Apply an even coat of adhesive, Item 14, WP 1803 00, to door, jamb and seal joining surfaces. Allow adhesive to air dry for 30 minutes or until tacky.
10. Attach seal to door and to jamb. Work out gaps, trapped air, and strained areas all the way around seal installation. Remove extra adhesive with a machinery towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

INSTALL - Continued

11. Mix and apply a bead of sealing compound, Item 273, WP 1803 00, around the inner and outer edges of the door seal installation. Shape seal bead as shown with a beverage stirring stick, Item 300, WP 1803 00. Allow to cure at least 24 hours (Figure 1, Detail B).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TRANSITION SECTION AVIONICS DOOR LATCH EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Grease Pencil

Material/Parts
Grease, Aircraft, Item 132, WP 1803 00
Sealing Compound, Item 284, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

NOTE

All instructions are the same for both latches.

REMOVE

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Remove screws and washers, and nuts from door and latch assembly. Remove cover, retainer, and latch as an assembly (Figure 1, Sheet 1, Detail B).

REPAIR

1. Remove bolt and washer from end of cover. Slide cover off of retainer and end of latch bolt (Figure 1, Sheet 2, Detail E).
2. Remove screws, washers, and nuts and separate retainer and latch.
3. Lubricate moving parts in latch with aircraft grease, Item 132, WP 1803 00. Place latch in retainer and install with screws, washers and nuts.

NOTE

Latch bolt should be installed over closed end of retainer; it should also be on same side of retainer as mounting flanges.

4. Install cover by passing flat end over latch bolt and set on mounting flanges of retainer. Install bolt and washer on end of cover.

INSTALL

1. Turn off all electrical power.
2. **QA** Install latch assembly on door with screws, washers, and nuts (Figure 1, Sheet 1, Detail B).
3. Apply fillet of sealing compound, Item 284, WP 1803 00, to edge of latch and panel of door.

INSTALL - Continued

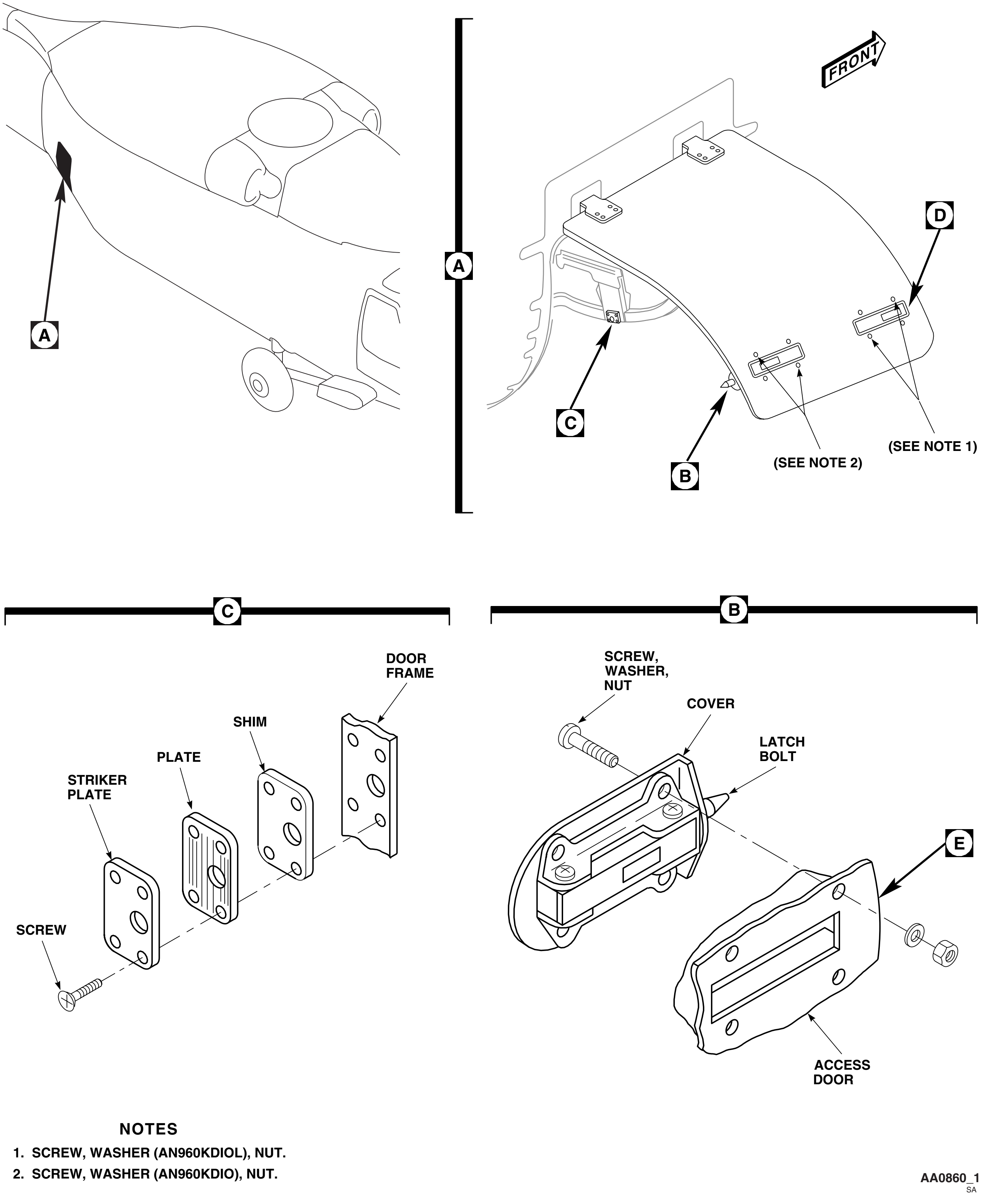


Figure 1. Replace/Repair Transition Section Avionics Door Latch. (Sheet 1 of 2)

INSTALL - Continued

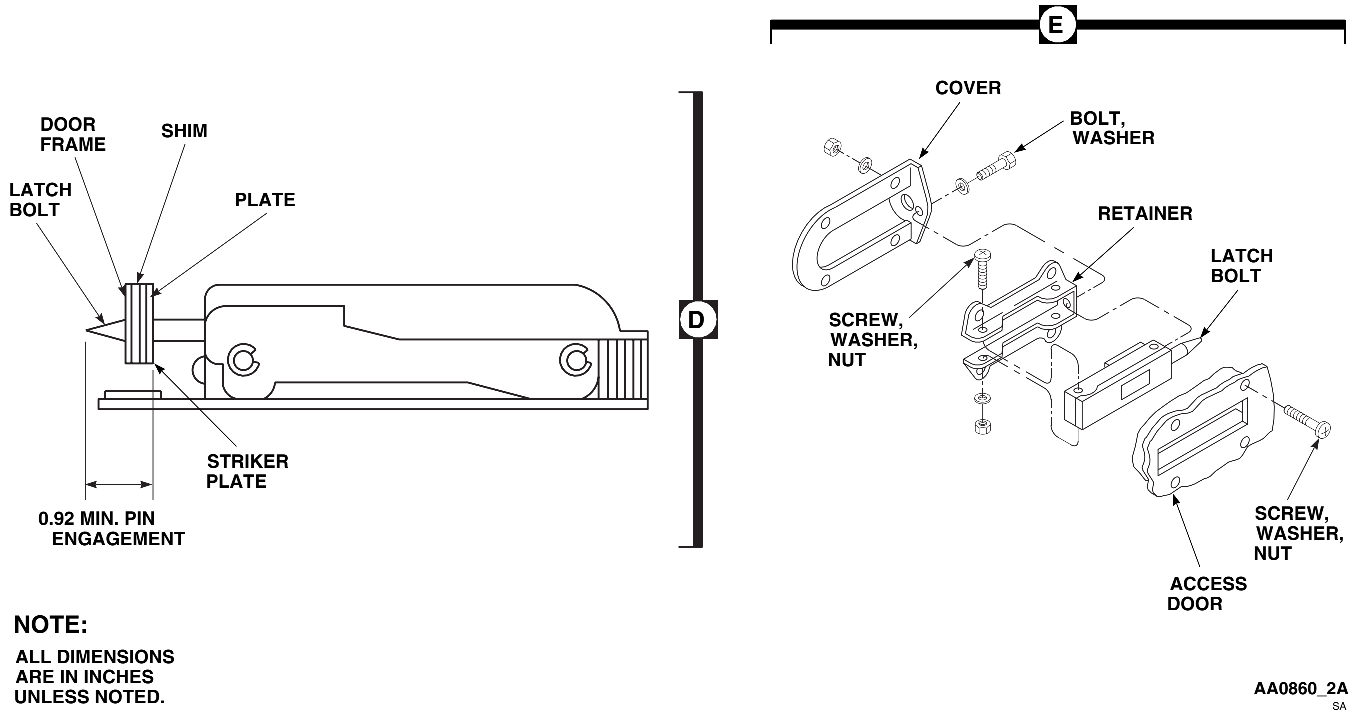


Figure 1. Replace/Repair Transition Section Avionics Door Latch. (Sheet 2 of 2)

ADJUST

1. Coat latch bolt with grease pencil to provide mark.
2. Close door and fully engage latch. Reopen door. **MEASURE MARK ON LATCH BOLT. AT LEAST 0.92-INCH OF LATCH BOLT MUST EXTEND INTO HOLE ON STRIKER PLATE ON DOOR FRAME** (Figure 1, Sheet 2, Detail D).
3. If limit is not met, adjust position of striker plate and baseplate by doing this:
   a. Determine difference between actual engagement of latch bolt and required minimum engagement.
   b. Remove screws holding striker plate and baseplate to door frame. Remove plates and shim if installed under baseplate (Figure 1, Sheet 1, Detail C).

NOTE

A new shim is 0.125-inch thick and consists of 0.002-inch laminations.

   c. Peel new shim so that latch bolt will extend 0.92-inch or more beyond striker plate surface.
   d. Install shim, baseplate, and striker plate with screws.

ADJUST - Continued

NOTE

Serrated faces of plates should face each other. Baseplate has largest latch bolt hole.

e. Close and latch access door. Recheck depth of latch bolt engagement.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR DRIVE SHAFT COVER

This WP supersedes WP 0327 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lubricating Oil, Molybdenum Disulfide, Item 189, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Polyurethane Coating, Item 227, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 55-1500-345-23
WP 0379 00
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Unfasten studs, open tail rotor drive shaft cover and remove nut, washer, and screw securing stop (Figure 1, Sheet 1, Detail A). Remove stop.
3. On rear tail rotor drive shaft cover, disconnect formation light electrical connector, P506 (Figure 1, Sheet 2, Detail B).
4. Remove hinge pin and tail rotor drive shaft cover (Figure 1, Sheet 1, Detail A).
5. Remove nuts, washers, screws, and formation light from rear tail rotor drive shaft cover (Figure 1, Sheet 2, Detail B).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Clean light using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

REPAIR

1. Remove corrosion manually, using scouring pad, Item 205, WP 1803 00 (TM 1-1500-344-23). **PITS ON HINGES SHALL NOT EXCEED 0.005-INCH DEEP AFTER CLEANUP.**
2. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, and 1 part water.

REPAIR - Continued

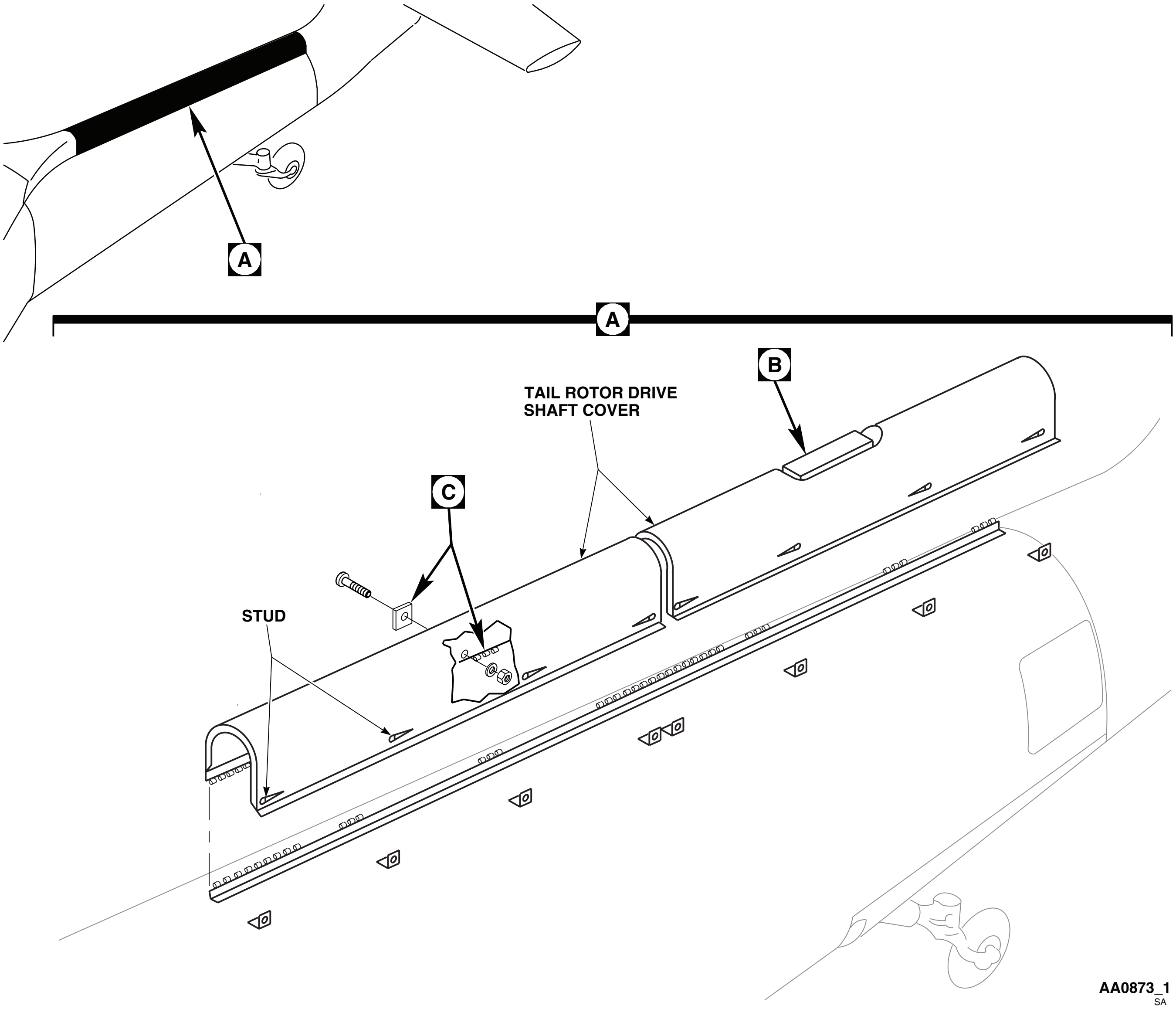


Figure 1. Replace Tail Rotor Drive Shaft Cover. (Sheet 1 of 2)

3. Apply compound solution to corroded areas of fasteners and hinges.
4. Rinse thoroughly with clean water.

CAUTION

Paint on bearing surfaces can cause improper operation of hinges and fasteners. Keep paint off of bearing surfaces.

5. Touch up paint with epoxy primer coating kit, Item 120, WP 1803 00, and polyurethane coating, Item 227, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
6. Coat hinge pins with molybdenum disulfide lubricating oil, Item 189, WP 1803 00.

REPAIR - Continued

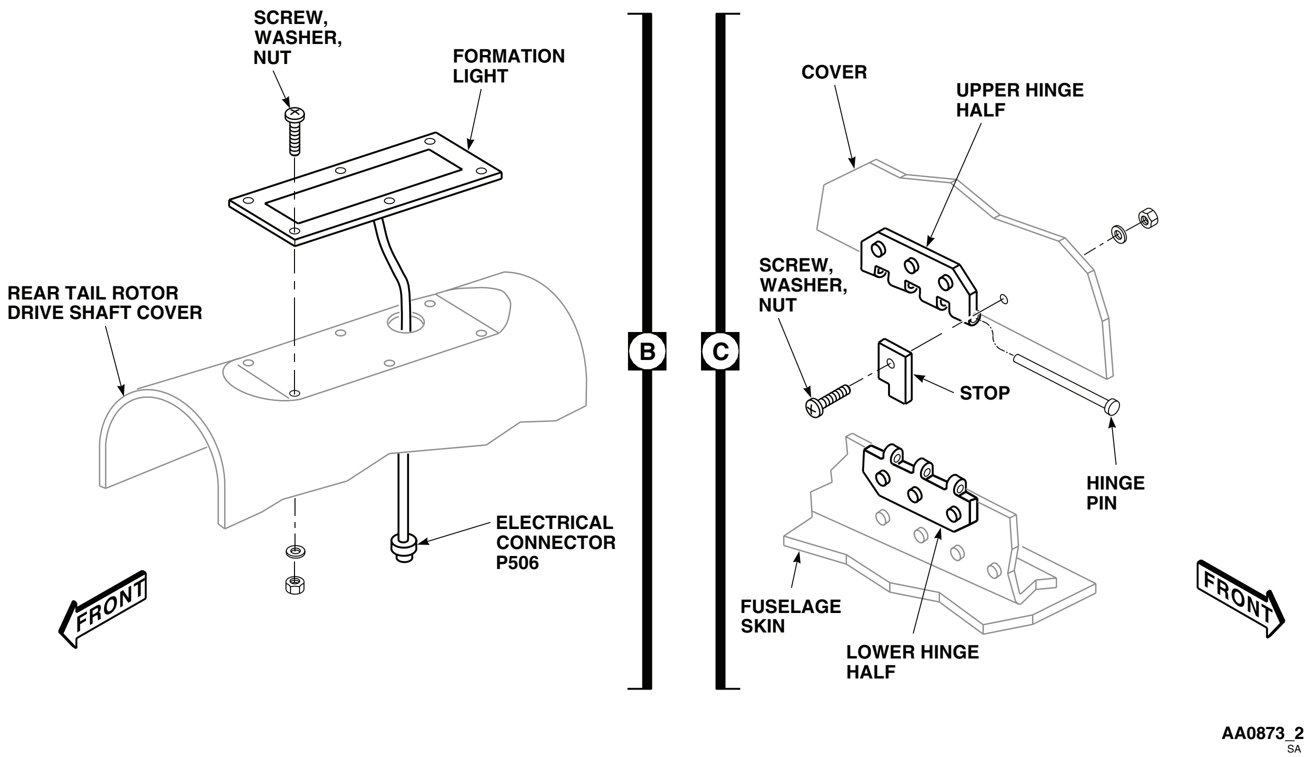


Figure 1. Replace Tail Rotor Drive Shaft Cover. (Sheet 2 of 2)

INSTALL

1. **QA** Fasten formation light to rear tail rotor drive shaft cover with screws, washers, and nuts (Figure 1, Sheet 2, Detail B).
2. Apply sealing compound, Item 273, WP 1803 00, around edge of light.
3. **QA** Line up hinge on tail rotor drive shaft cover with lower hinge half and install hinge pin with swaged end on pin stop side (Figure 1, Sheet 1, Detail A).
4. Close tail rotor drive shaft cover, check that studs line up with their receptacles. If they do not, do this:
   a. Drill out rivets holding ejector spring to tail rotor drive shaft cover.
   b. Remove tail rotor drive shaft cover from tail cone.
   c. Fill in rivet holes (WP 0379 00).
   d. Install tail rotor drive shaft cover, fasten all studs.
   e. Mark and drill new holes using ejector spring as template. Rivet ejector spring to tail rotor drive shaft cover (TM 1-1500-204-23).
5. **QA** Connect formation light electrical connector, P506.
6. **QA** Attach stop with screw, washer, and nut.
7. Check open/close function of tail rotor drive shaft cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Cargo Straps, MS 3367-1-0
Fluorescent Penetrant Inspection Kit, XMA101
Padding
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistants - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-MTF
TM 1-1520-265-23
TM 55-1500-345-23
WP 0329 00
WP 0330 00
WP 0331 00
WP 0332 00
WP 0333 00
WP 0335 00
WP 0336 00
WP 0337 00
WP 0338 00
WP 0381 00
WP 0417 00
WP 0585 00
WP 0659 00
WP 0668 00
WP 0681 00
WP 1145 00
WP 1703 00
WP 1746 00

REMOVE

1. Turn off all electrical power.
2. Remove all pylon fairings, covers/antennas from pylon (WP 0329 00 , WP 0330 00, WP 0331 00, WP 0332 00, WP 0333 00 and WP 0335 00).
3. Remove stabilator (WP 0338 00).
4. Remove tail rotor assembly (WP 0585 00).
5. Remove tail rotor drive shaft (WP 0668 00).
6. Remove tail gear box (WP 0681 00).
7. Remove intermediate gear box (WP 0659 00).
8. Remove tail rotor control cables from pylon (WP 1145 00).
9. Disconnect electrical wiring, coaxial cables, and bonding jumpers at pylon to tail cone disconnects.
10. Attach cargo straps to pylon (Figure 1).

REMOVE - Continued

WARNING

- Hoist or hoist cable may break, causing injury to personnel or damage to equipment, if an improper capacity hoist is used. Pylon weighs 176 pounds. Select a hoist capable of supporting tail rotor pylon structure.
- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

11. Position overhead hoist over pylon and attach hoist to cargo straps.
12. Raise up on hoist to provide tension on straps taking weight of pylon off attachment bolts.
13. Remove bolts, washers, and nuts in pylon hinges (Figure 1, Detail A).
14. Back off tapered pins from pylon, from inside tail cone.

NOTE

Should condition of taper pins require replacement, taper pins must be removed and installed from tail pylon side of tail cone bulkhead.

15. With one assistant operating overhead hoist, and another to guide pylon, remove pylon-to-tail cone attachment bolts (Figure 1, Detail B).

NOTE

Adjust strap tension as required to balance pylon.

16. Carefully separate pylon from tail cone, making sure all electrical and hydraulic connections have been disconnected.
17. Slowly hoist pylon away from tail cone. Place pylon on padded saw horses or equivalent.
18. Disconnect straps from hoist, and unfasten straps from pylon.
19. If pylon is being replaced, remove tapered pins from tail cone (Figure 1, Detail A).
20. Remove steps (WP 0336 00 and WP 0337 00).
21. If pylon is being replaced, remove barrel-nuts and retainers from pylon.

INSPECT

NOTE

If inspection facilities are not available, use all new hardware.

1. Inspect tapered pins, pylon attachment bolts, washers, barrel-nuts, retainers, and pylon hinge bolts, washers, and nuts, using fluorescent penetrant inspection kit (TM 1-1520-265-23).

INSPECT - Continued

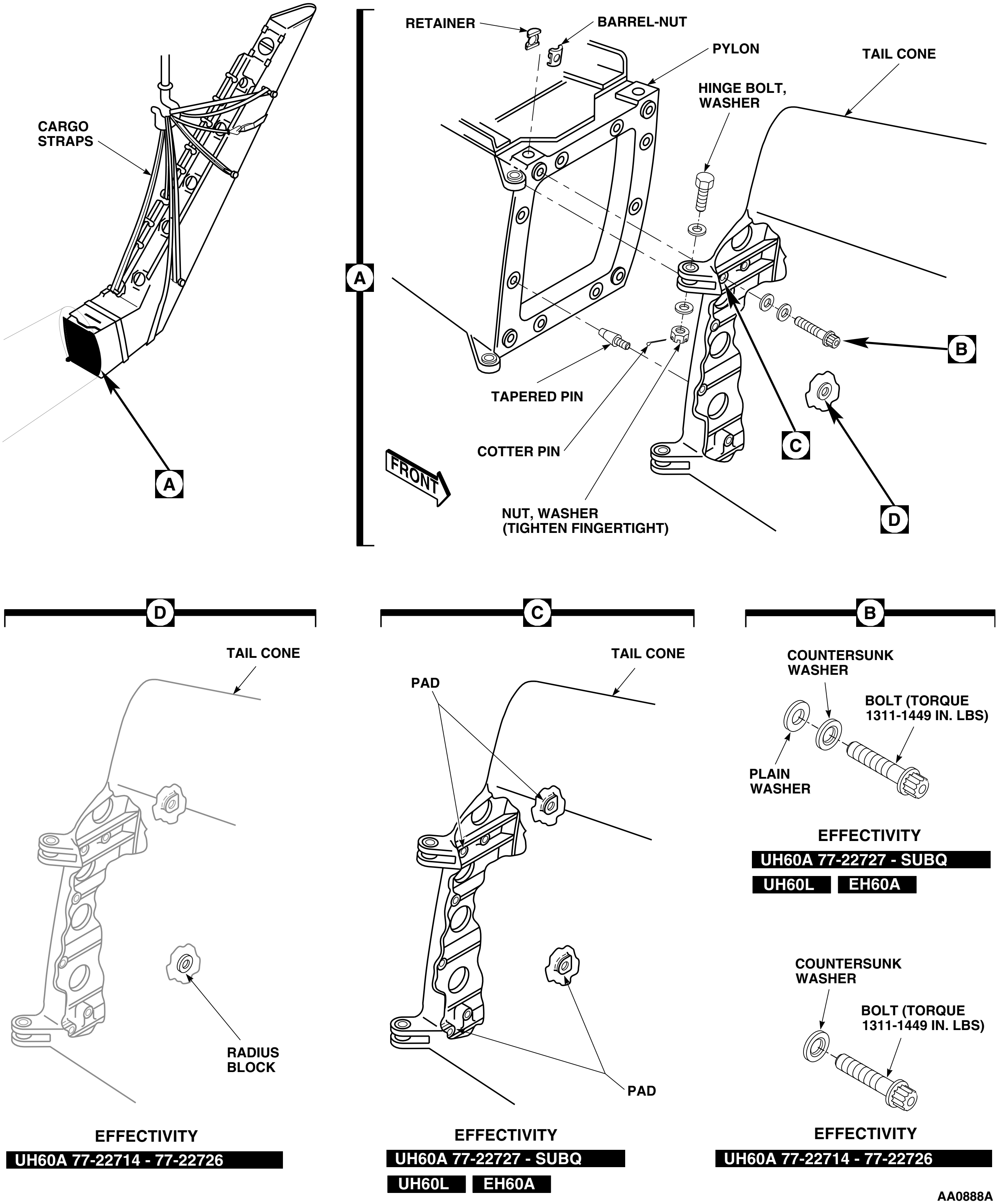


Figure 1. Replace Tail Rotor Pylon.

INSPECT - Continued

WARNING

CSI

Installation of radius pads is a critical characteristic. Make sure radius pads are bonded securely to tail cone.

2. **UH60A 77-22727 - SUBQ** Check tail cone attachment points for four installed radius pads (Figure 1, Detail C). Replace missing radius pads (WP 0417 00).

WARNING

CSI

Installation of radius block is a critical characteristic. Make sure radius block is bonded securely to tail cone.

3. **UH60A 77-22714 - 77-22726** Check for one installed radius block at lower left tail cone attach point (Figure 1, Detail D). Replace missing radius block (WP 0417 00).
4. Repair or replace fittings as required (WP 0417 00). Use a fluorescent penetrant inspection kit to inspect tail cone/ pylon fitting for cracks (TM 1-1520-265-23).

INSTALL

1. Turn off all electrical power.

WARNING

CSI

Installation of bolts, washers, barrel-nuts, and retainers, and torque of tension bolts are critical characteristics. Verification of correct installation and torque is required.

NOTE

- Make sure fluorescent penetrant inspection of all attaching hardware has been done. If inspection facilities are not available, use all new hardware.
- Make sure only square barrel nuts are used.

2. Test attachment bolts and barrel nuts as follows:
   a. Put nut in a vise.
   b. Thread bolt into nut until two threads stick out past nut.
   c. Using torque wrench, check breaking torque needed to loosen bolt. **IF TORQUE NEEDED IS NO LESS THAN 24 INCH-POUNDS, BOLT AND NUT ARE GOOD TO USE. IF TORQUE NEEDED IS LESS THAN 24 INCH-POUNDS, REPLACE NUT AND RECHECK TORQUE. IF TORQUE IS STILL LESS THAN 24 INCH-POUNDS, REPLACE BOLT.**

CAUTION

Barrel-nut must be properly installed to prevent serious damage to barrel-nut recess when bolt is torqued.

3. **QA** Install barrel nuts and retainers in replacement pylon. Barrel side of nut must be toward bolthead.

INSTALL - Continued

4. **QA** Install tapered pins in tail cone. Back off tapered pins so they will not be in the way when pylon is positioned against tail cone.
5. Reinstall steps (WP 0336 00 and WP 0337 00).
6. Attach cargo straps to pylon (Figure 1).
7. Position overhead hoist over pylon and attach hoist to straps.

NOTE

Adjust strap tension as required to balance pylon.

8. With one assistant operating hoist, and another to guide pylon, carefully position pylon against tail cone.

WARNING

CSI

Installation of bolts, washers, barrel nuts and retainers, and torque of tension bolts are critical characteristics. Verification of correct installation and torque is required.

9. **UH60A 77-22714 - 77-22726** Use one countersunk washer on each attachment bolt (Figure 1, Detail B).
10. **UH60A 77-22727 - SUBQ UH60L EH60A** Use one countersunk and one plain washer on each attachment bolt (Figure 1, Detail B).
11. Install four bolts with washers. Run bolts down until pylon evenly contacts tail cone. Do not torque bolts now.
12. Turn all tapered pins into mating bushings in pylon.
13. **QA TORQUE PYLON ATTACHMENT BOLTS TO 1311 - 1449 INCH-POUNDS.** Apply torque stripes (WP 0381 00 and TM 55-1500-345-23).
14. **QA** Install pylon hinge bolts, washers, and nuts. **TIGHTEN NUTS FINGERTIGHT.** Install cotter pin (Figure 1, Detail A).

NOTE

Hinge bolts must be free to rotate when attachment bolts are installed.

15. Disconnect straps from hoist and unfasten straps from pylon.
16. Paint and stencil pylon as required (WP 0381 00 and TM 55-1500-345-23).
17. Install tail rotor control cables in pylon (WP 1145 00).
18. Install intermediate gear box (WP 0659 00).
19. Install tail gear box (WP 0681 00).
20. Install tail rotor drive shaft (WP 0668 00).
21. Install tail rotor assembly (WP 0585 00).
22. Install stabilator (WP 0338 00).
23. Inspect and treat electrical connector (WP 1703 00).
24. **QA** Connect all electrical connectors, coaxial connectors, and bonding jumpers.
25. Make sure area is clean and free of foreign material before closing up.

INSTALL - Continued

26. Install all pylon fairings, covers/antennas on pylon (WP 0329 00, WP 0330 00, WP 0331 00, WP 0332 00, WP 0333 00 and WP 0335 00).
27. Check DA Form 2408-13-1 to make sure torque checks are shown as due 40 flight hours after installation.
28. Do a weight and balance check (WP 1746 00).
29. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON LOWER FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 11-1520-237-23
WP 0379 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn on electrical power.
2. Place MAN SLEW switch, on STABILATOR CONTROL panel, to neutral position, or 0 degrees.
3. Turn off all electrical power.
4. Loosen studs and open hinge cover in bottom of fairing (Figure 1, Detail A).
5. Disconnect bonding jumper connected to radar warning system receiver, APR-39.
6. Disconnect coaxial connectors from receptacles, J3, J4, and J5, of receiver.
7. Remove screws, bolts, and washers attaching fairing to pylon.
8. Remove fairing.
9. If fairing is being replaced, remove radar warning system receiver, APR-39, if installed and antennas (TM 11-1520-237-23).

INSTALL

1. Make sure stabilator is in neutral position or 0 degrees.
2. Turn off all electrical power.
3. If new fairing is being installed, do this:
   a. Trial-fit new fairing on pylon.
   b. Trim fairing as necessary on both sides to fit pylon. **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
   c. Mark any holes in fairing that do not line up with nutplates in pylon.
   d. Fill marked holes using procedures in WP 0379 00.
   e. Locate and drill new holes in fairing, using pylon as template.

INSTALL - Continued

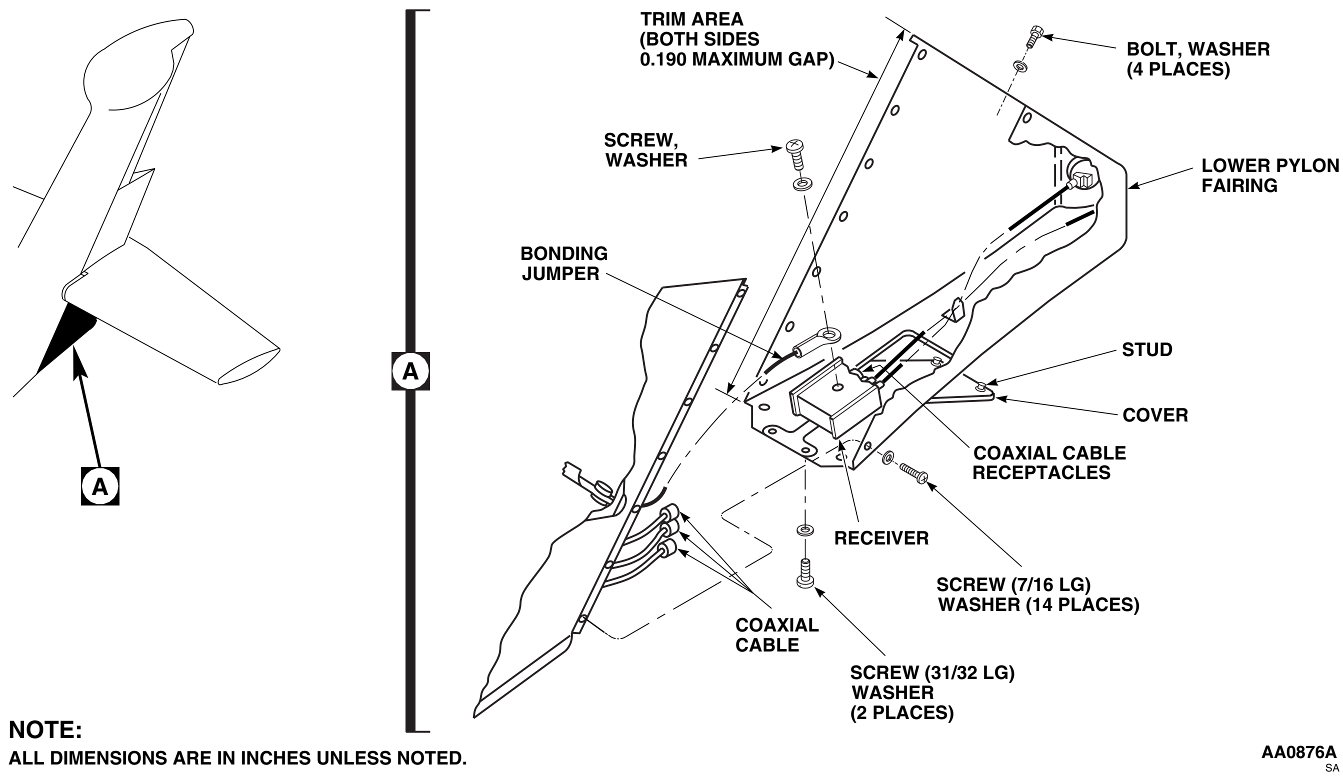


Figure 1. Replace Tail Rotor Pylon Lower Fairing.

f. Install radar warning system antennas, ARP-39, and receiver on fairing (TM 11-1520-237-23).

4. Make sure area is clean and free of foreign material before closing up.
5. Place fairing in position on pylon. Line up screw holes in fairing with pylon.
6. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
7. **QA** Install long screws and washers in bottom front holes in fairing (Figure 1, Detail A).
8. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
9. **QA** Install short screws and washers in side holes in fairing.
10. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
11. **QA** Install bolts and washers in rear top holes in fairing.
12. **QA** Connect bonding jumper to radar warning receiver, APR-39.
13. Inspect and treat coaxial connector (WP 1703 00).
14. **QA** Connect coaxial connectors to receptacles, J3, J4, and J5, of receiver.
15. Close cover in bottom of fairing and tighten studs.
16. Check operation of radar warning system, APR-39 (TM 11-1520-237-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON STABILATOR-TO-PYLON FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

NOTE

Replacement of left or right stabilator/pylon fairing is the same.

1. Remove screws attaching fairing to pylon (Figure 1, Detail A).
2. Remove fairing.

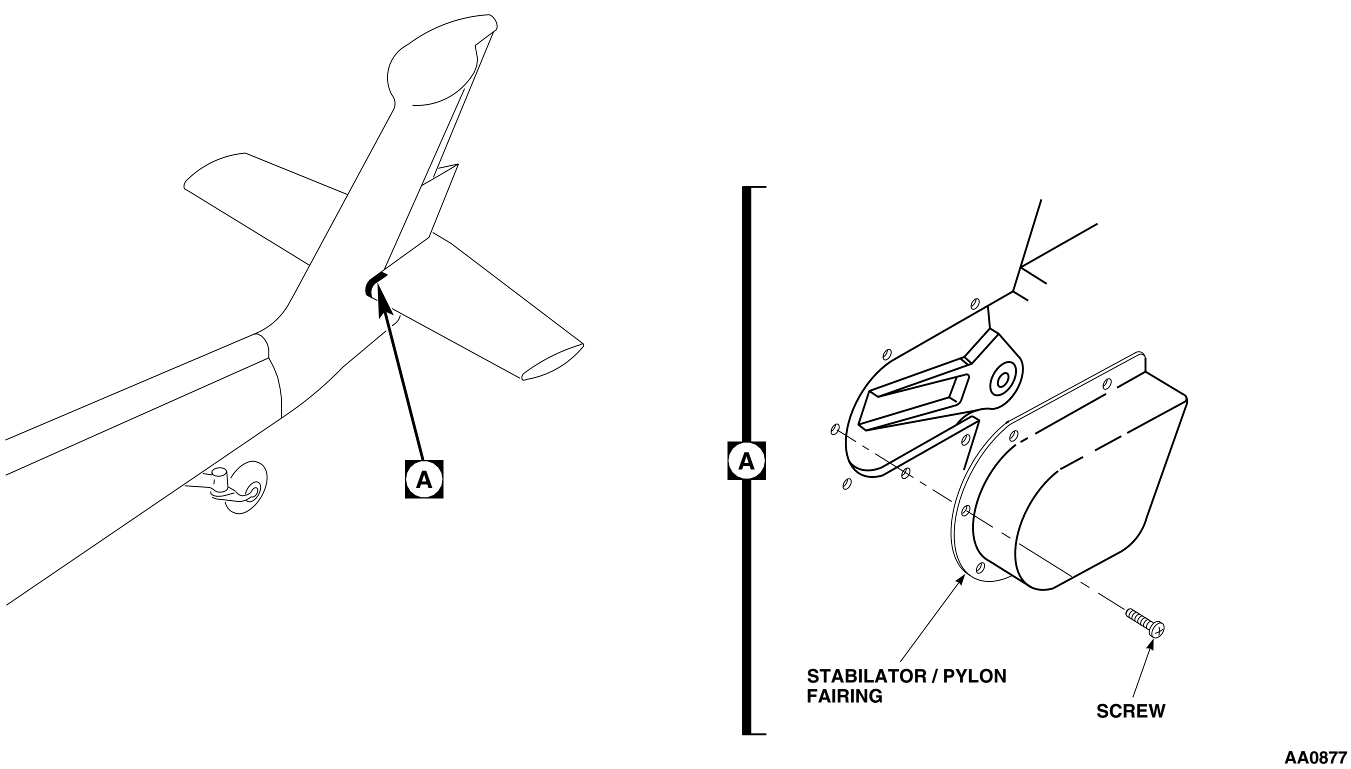


Figure 1. Replace Tail Rotor Pylon Stabilator-to-Pylon Fairing.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.
2. Place replacement fairing in position on pylon (Figure 1, Detail A).
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. **QA** Install attachment screws.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON CAMBERED FAIRING

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Corrosion Preventive Compound, Item 107, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0379 00
WP 1803 00

REMOVE

1. Remove bolts attaching bottom of fairing to pylon (Figure 1, Detail A).

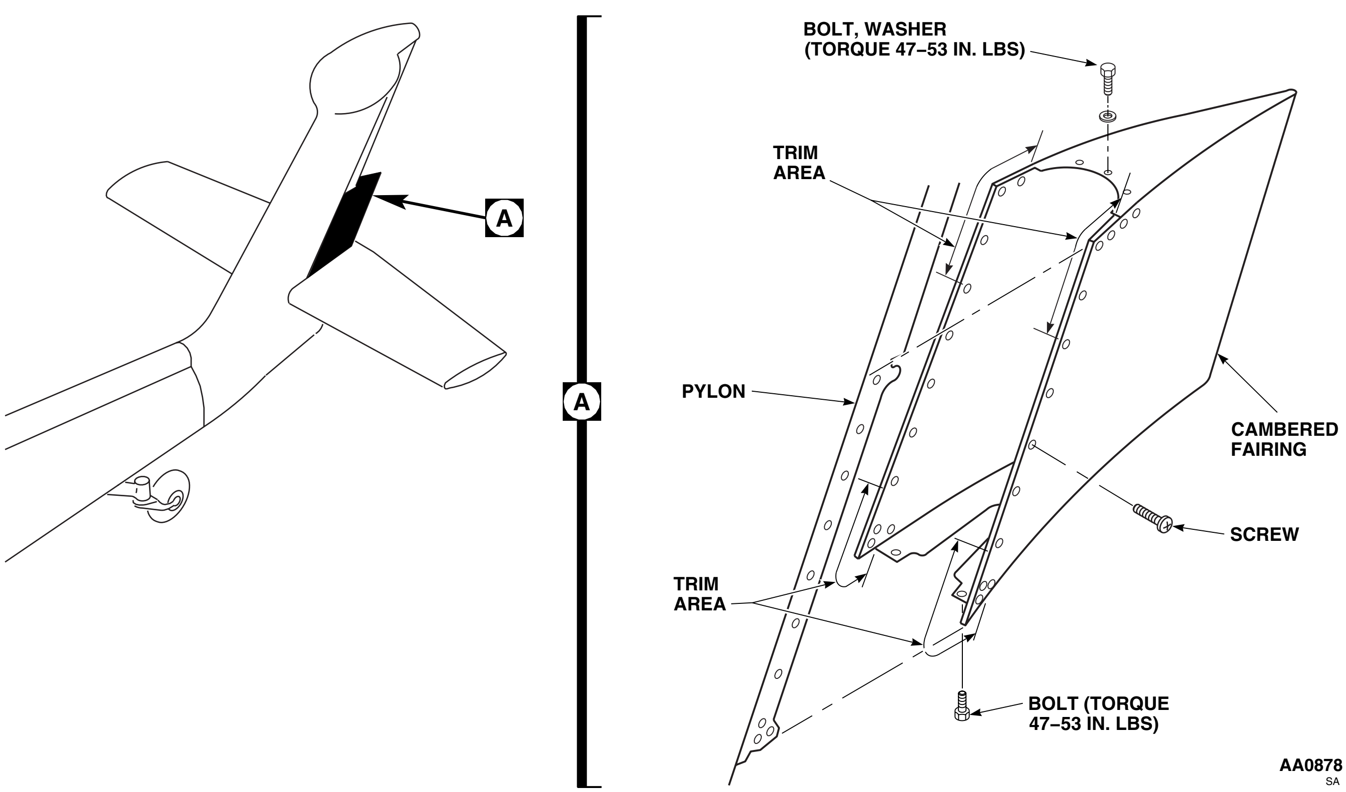


Figure 1. Replace Tail Rotor Pylon Cambered Fairing.

REMOVE - Continued

2. Working from pylon steps or a maintenance work platform, remove bolts and washers attaching top of fairing to pylon.
3. Remove screws from both sides of fairing.
4. Remove fairing.

INSTALL

1. Make sure area is clean and free of foreign material before closing up.

NOTE

Trim fairing, mark and drill holes using procedures in TM 1-1500-204-23.

2. If installing new fairing, do this:
   a. Trial-fit fairing with bolts.
   b. If needed, trim areas shown in Figure 1, Detail A (WP 0379 00). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
   c. Mark screwholes.
   d. Remove fairing, drill screwholes.
3. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
4. Place fairing on pylon and install two screws on top and bottom of each side to hold fairing in position (Figure 1, Detail A).
5. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
6. **QA** Install bolts and washers attaching top of fairing to pylon. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
7. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
8. **QA** Install bolts attaching bottom of fairing to pylon. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
9. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
10. **QA** Install remaining screws on both sides of fairing.
11. Apply sealing compound, Item 273, WP 1803 00, to all joining of fairing, pylon, antennas, and fastening hardware.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON TRAILING EDGE FAIRING/ANTENNA

This WP supersedes WP 0332 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Cleaning Solvent, DS 108, Item 368, WP 1803 00
Cloth, Cheesecloth, CCC-C-440, Item 85, WP 1803 00
Cloth, Cleaning, Miracle Wipe L001, Item 86, WP 1803 00
Connector Wrap, Dual Tape, Item 259, WP 1803 00
Corrosion Resistant Coating, MIL-C-81706 Class 3, Item 111, WP 1803 00
Gasket, SG876000-01
Sealing Compound, CA 1000, Item 258, WP 1803 00
Sealing Compound, PR-1775 B-2, Item 271, WP 1803 00
Sealing Compound, AMS 3265 B-1/2, Item 367, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 11-1520-237-23
WP 0378 00
WP 0379 00
WP 0381 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Working from a maintenance work platform, remove sealant from screwheads. Remove screws and washers attaching fairing to pylon (Figure 1, Detail A).
3. Separate antenna from pylon enough to get at coaxial connector.
4. Disconnect coaxial connector from antenna receptacle.
5. Remove fairing/antenna.
6. Remove gasket from antenna.

INSTALL

1. Turn off all electrical power.

NOTE

Antenna and aircraft mating surfaces shall be free of dust, oil, grease, fingerprints, and other contamination prior to antenna installation.

2. Clean fairing/antenna and helicopter mating surfaces using clean cleaning cloth, Item 86, WP 1803 00, or clean cheesecloth, Item 85, WP 1803 00, moistened with Cleaning Solvent, DS108, Item 368, WP 1803 00.
3. Inspect chemical coated surfaces, remove any corrosion,and restore electrically conductive coating with coating conforming to MIL-C-81706, Class 3, (Item 111, WP 1803 00).

INSTALL - Continued

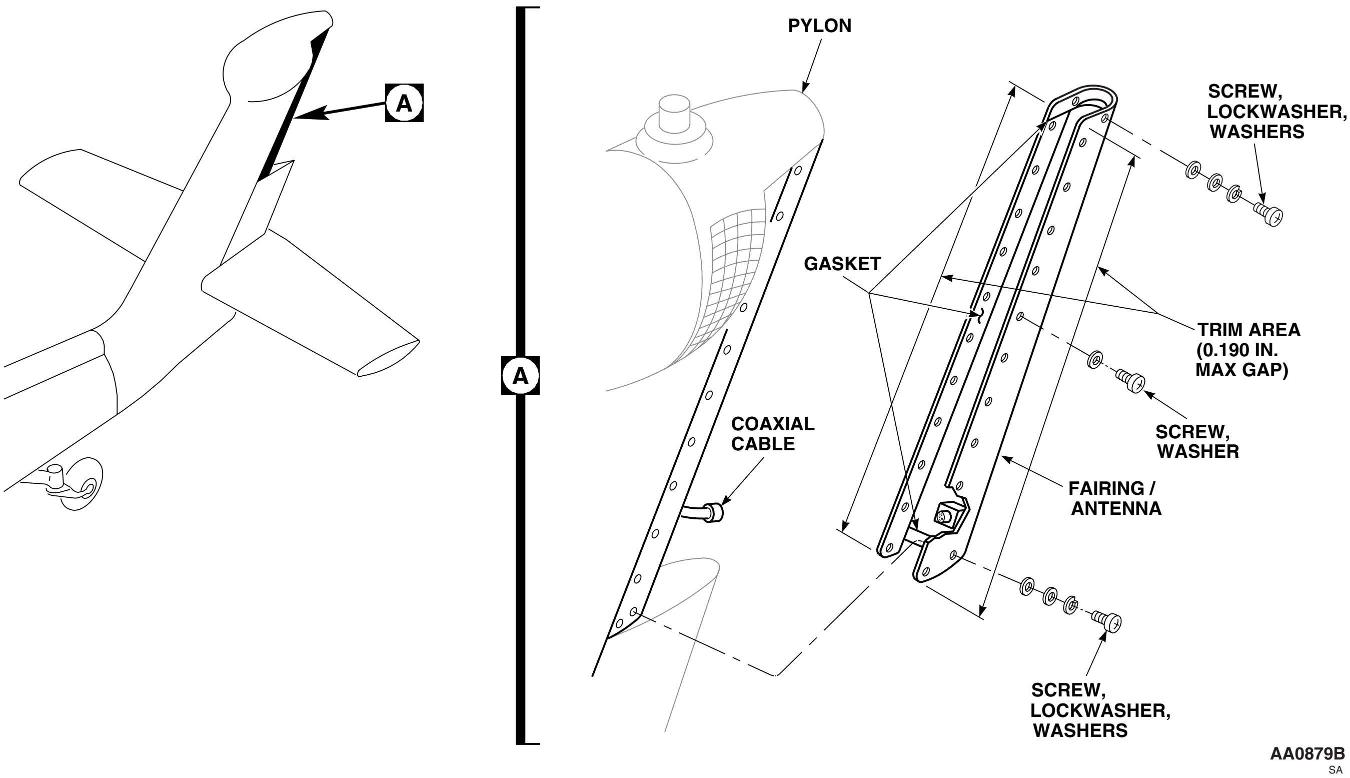


Figure 1. Replace Tail Rotor Pylon Trailing Edge Fairing/Antenna.

4. Inspect chemical coated surfaces and coat any bare aluminum skin with coating conforming to MIL-C-81706, Class 3, (Item 111, WP 1803 00).
5. If installing new fairing, do this:
   a. Trial fit fairing to pylon. Trim bottom edge if it binds against fairing.
   b. If needed, trim areas shown in Figure 1, Detail A (WP 0379 00). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
   c. Mark holes in fairing that do not line up with nutplates in pylon.
   d. Remove fairing, fill in marked holes using procedures in WP 0379 00.
   e. Place fairing on pylon. Mark and drill new holes using procedures in TM 1-1500-204-23.
6. Working from a maintenance work platform, place fairing in position on pylon.
7. Inspect and treat coaxial connector (WP 1703 00).

INSTALL - Continued

CAUTION

Gasket is supplied with protective release film on both sides of gasket. Leave release film in place until ready to install gasket. Do not fold or bend gasket when removing from packaging.

NOTE

Verify that fastener hole cutouts in upper and lower halves of gasket align with antenna when positioned for installation.

8. Trim perimeter of gasket flush with antenna if needed.
9. Remove release film from side of gasket marked ″antenna side″ and position gasket over antenna.

NOTE

Release film should remain on exposed ″aircraft side″ of gasket until immediately prior to antenna installation.

10. **QA** Connect coaxial connectors to antenna receptacle.

NOTE

- The Connector Wrap, Dual Tape Kit, Item 259, WP 1803 00, contains both the Stretch Seal Tape and Self-Fusing Tape.
- A tight wrap is necessary to ensure a proper seal. Coverage shall be from base of antenna to at least 1/2 inch beyond connector and on top of insulation jacket of coaxial cable.

11. Apply Streach Seal Tape, Item 259, WP 1803 00, to connector. Remove release film from stickey side of tape making sure tape remains free of dirt or other contaminants. Wrap stickey side around mated connector with 50% overlap while stretching tape 25% to 50% to ensure a tight wrap.
12. Apply Self-Fusing Tape, Item 259, WP 1803 00, to connector over top of the Stretch Seal Tape. Remove small amount of release film, making sure tape remains free of dirt or other contaminants. Begin wrapping Self-Fusing Tape around connector behind the coupling ring (nut). After a full wrap is completed, continue wrapping with a 50% overlap using the line in middle of tape as a guide. Stretching tape 50% to 300% to ensure a tight fit will aid in fusing process. Continue wrapping until tape completes at least one wrap past the end of Streach Seal tape. Make one full wrap over top of the previous wrap at first and last wrap.
13. Remove release film from gasket on exposed ″aircraft side″. Carefully position fairing/antenna on helicopter pylon.
14. Apply sealing compound, CA 1000, Item 258, WP 1803 00, to mounting screw threads immediately prior to installation. Wipe away sealant squeezedout after fasteners are installed.

NOTE

Left and right side of antenna gasket does not have cutout holes for mounting screws. Carefully press screws through gasket material.

15. Install screws and washers in remaining holes on each side of fairing.
16. In the top three holes, and bottom two holes of each side of fairing install screws, lockwashers and washers (Figure 1, Detail A).
17. Clean antenna, surrounding aircraft surfaces, and screw heads with clean cleaning cloth, Item 86, WP 1803 00, moistened with cleaning solvent, DS-108, Item 368, WP 1803 00.

INSTALL - Continued

18. Apply sealing compound, PR-1775 B-2, Item 271, WP 1803 00, or sealing compound, AMS 3265 B-1/2, Item 277, WP 1803 00, around mounting screw heads , into any unused mounting holes, and around perimeter of antenna base to form a water-tight seal with the airframe.
19. Check operation of troop commander's antenna and missile warning system (TM 11-1520-237-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR GEAR BOX FAIRING

This WP supersedes WP 0333 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Abrasive, Item 80, WP 1803 00
Cloth, Cheesecloth Item 85, WP 1803 00
Corrosion Preventive Compound, Item 107, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Primer, Adhesive, Item 240, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 330, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0115 00
WP 0379 00
WP 0903 00
WP 0910 00
WP 0911 00
WP 0913 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Working from a maintenance work platform, unfasten spring-type fasteners attaching front fairing to pylon (Figure 1).
3. Remove fairing.
4. Disconnect anticollision light electrical connections (WP 0913 00).
5. Disconnect incandescent position light electrical connections (WP 0910 00).
6. Disconnect infrared position light electrical connections (WP 0911 00).
7. Make sure bonding jumpers are disconnected.
8. Remove screws and washers attaching rear fairing to pylon.
9. Remove fairing.

INSPECT/REPAIR

1. Inspect tail gear box cover for missing or damaged seals.
2. If seal(s) are missing, or damaged so that they do not prevent chafing between gear box and fairing, do this:
   a. Remove damaged seal(s) using a nonmetallic scraper.

INSPECT/REPAIR - Continued

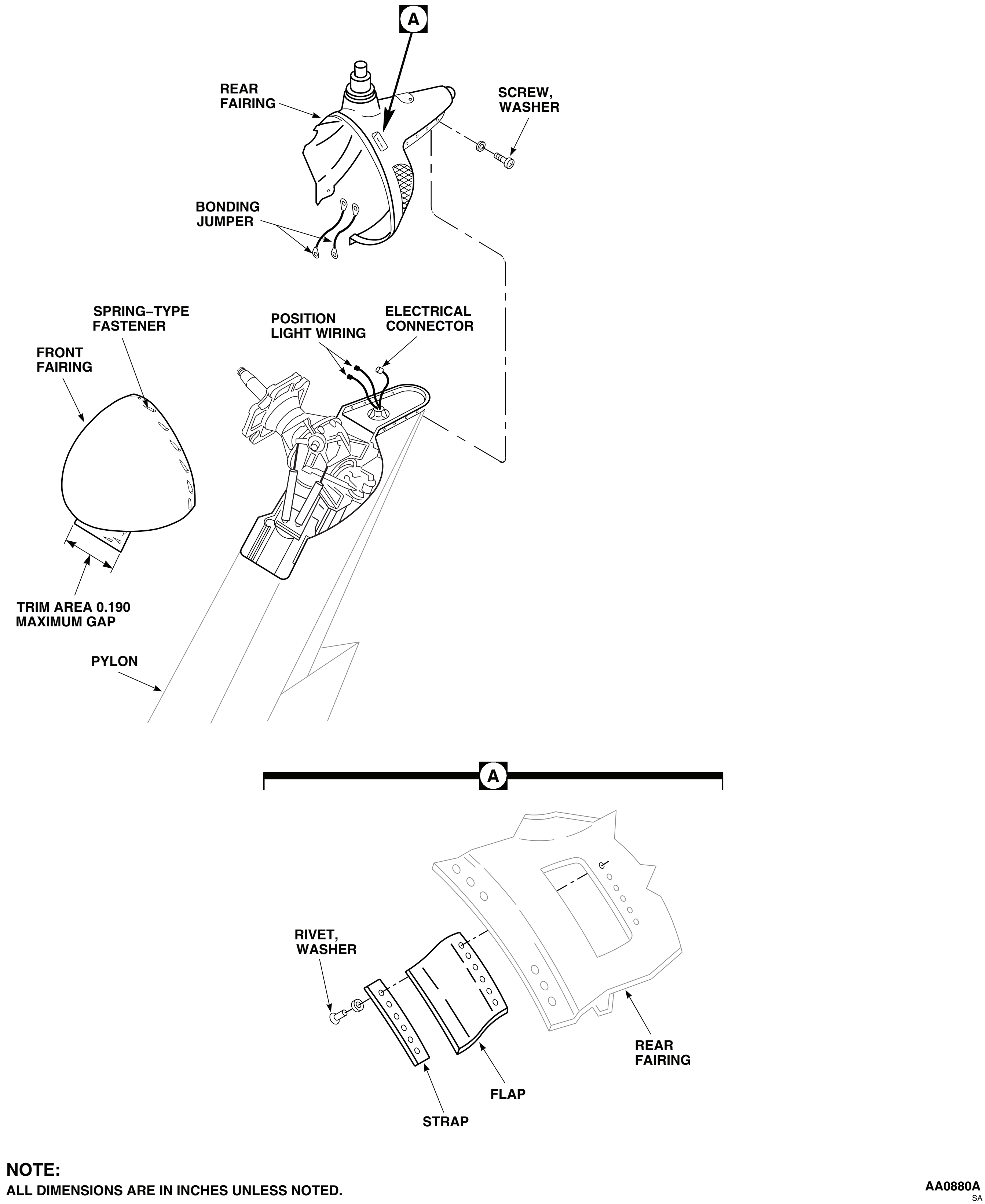


Figure 1. Replace Tail Rotor Gear Box Fairing.

INSPECT/REPAIR - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Clean any remaining adhesive on edge of fairing using technical acetone, Item 5, WP 1803 00.

c. Roughen inside edges of new seal(s) using abrasive cloth, Item 80, WP 1803 00. Clean roughened edges of seal(s) using technical acetone, Item 5, WP 1803 00.

d. Apply a light coat of adhesive, Item 14, WP 1803 00, to roughened edges of seal(s) and edge of tail gear box. Allow adhesive to dry for about 10 minutes or until it is tacky.

3. Install seal(s) on edge of fairing, making sure all mating surfaces are in contact. Allow adhesive to dry for 24 hours.

4. **UH-60A 83-23843 - SUBQ** Inspect rear fairing handhold flap and strap for damage (Figure 1).

5. If flap and strap are damaged and need replacement, do this:

a. Drill out rivets holding flap and strap to fairing.

b. Remove flap and strap from fairing.

c. Position replacement flap and strap on fairing and install with washers and rivets (TM 1-1500-204-23).

6. Inspect gear box fairing from missing or worn chafing tape.

7. If tape is missing or worn so that chafing is not prevented, do this:

a. Remove worn tape using a nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Clean dirt and grease from fairing using cheescloth cloth, Item 85, WP 1803 00 and technical acetone, Item 5, WP 1803 00.

c. Lightly roughen surface with scouring pad, Item 205, WP 1803 00.

d. Clean roughened surface of fairing using cheescloth cloth, Item 85, WP 1803 00 and technical acetone, Item 5, WP 1803 00.

e. Measure and cut new tape.

f. Spray a light coat of adhesive primer, Item 240, WP 1803 00, on area where tape is to be installed.

g. Install adhesive pressure sensitive tape, Item 330, WP 1803 00, on fairing applying pressure to be sure of good bond.

8. If cover is being replaced, remove anticollision light (WP 0913 00) and position light (WP 0903 00) from fairing.

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

2. If not installed, install anticollision and position light (WP 0913 00) and position light (WP 0903 00) from fairing.
3. Working from a maintenance work platform, place rear cover on pylon (Figure 1, Detail A).
4. Inspect and treat electrical connector (WP 1703 00).
5. Connect incandescent position light electrical connector (WP 0910 00).
6. Inspect and treat electrical connector (WP 1703 00).
7. Connect anticollision light electrical connector (WP 0913 00).
8. **QA** Make sure bonding jumpers are connected to pylon structure.
9. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
10. **QA** Install screws and washers attaching rear fairing to pylon.
11. Make sure area is clean and free of foreign material before closing up.
12. **QA** Install front fairing and fasten spring-type fasteners.
13. If installing new fairing, do this:
    a. Trial fit fairing on pylon. Trim edges as necessary (WP 0379 00). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
    b. Mark any holes in fairing that do not line up.
    c. Remove cover, fill marked holes and touch up paint using procedures in WP 0379 00.
    d. Install cover on pylon, mark and drill new holes, using pylon as template.
    e. If spring fastener does not line up, remove and move receptacle using procedures in WP 0379 00.
14. Check operation of anticollision and position lights (WP 0115 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON LEADING EDGE COVER/ANTENNA

This WP supersedes WP 334 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Bumper Pad, 10-267-114
Cloth, Abrasive, Item 83, WP 1803 00
Cloth, Cheese Item 85, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Epoxy Coating Kit, Item 118, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 327, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 11-1520-237-23
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Working from pylon steps or a maintenance work platform, unfasten spring-type fasteners attaching cover/antenna to pylon.
3. Hinge pylon drive shaft cover open. Disconnect coaxial connector from antenna receptacle (Figure 1, Detail A).
4. Lift up on cover/antenna to disengage hinges and remove cover/antenna from pylon.

REPAIR

NOTE

Replacement of upper or lower leading edge cover/antenna bumper pads is the same.

1. Mask area around bumper pad with pressure sensitive adhesive tape, Item 327, WP 1803 00 (Figure 1, Detail B).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Using a nonmetallic scraper and cheescloth cloth, Item 85, WP 1803 00 moistened with technical acetone, Item 5, WP 1803 00, carefully remove damaged bumper pad.
3. Apply a light coat of adhesive, Item 11, WP 1803 00, to large flat surface of bumper pad, 10-267-114.
4. Install bumper pad on leading edge cover/antenna.

REPAIR - Continued

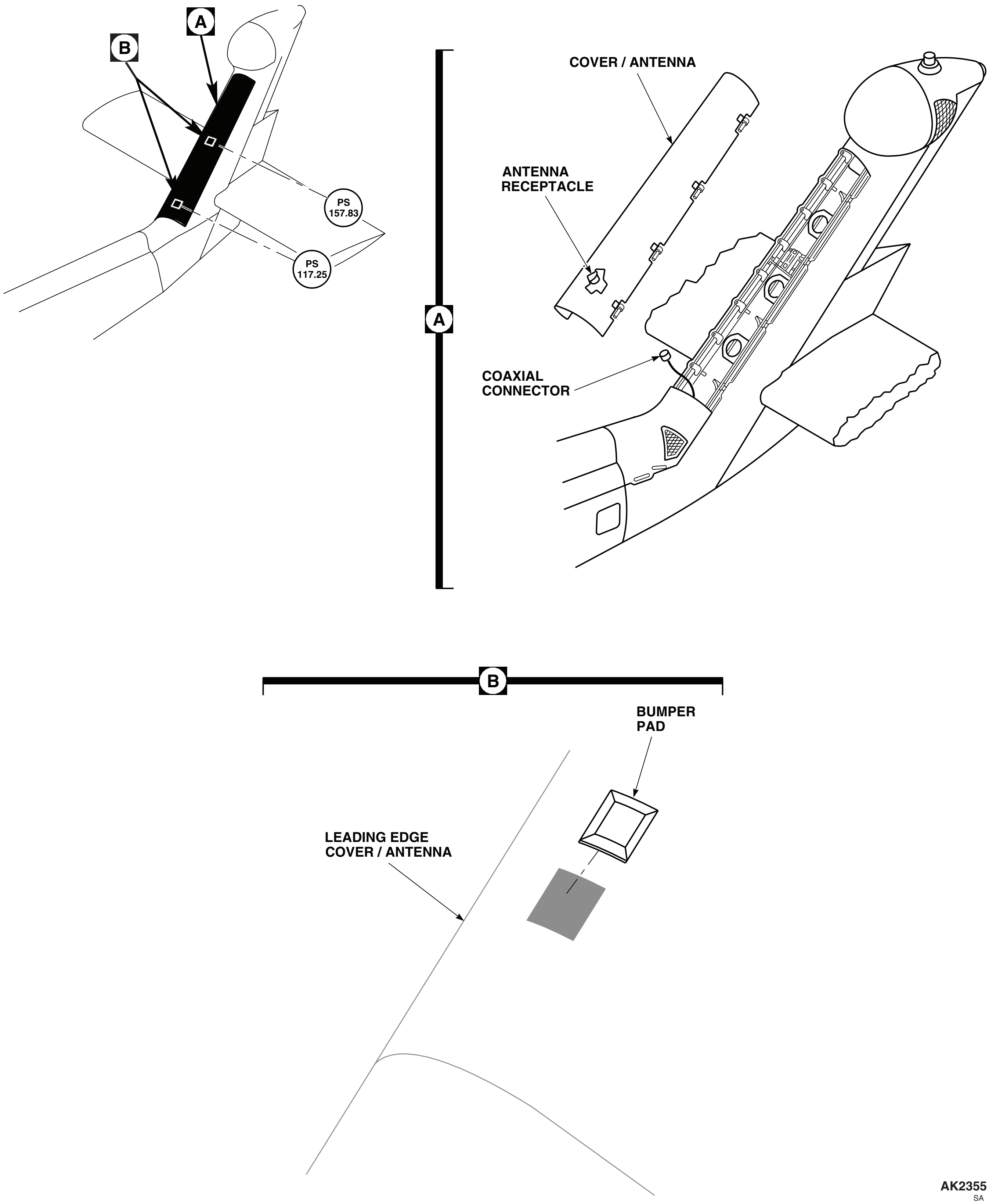


Figure 1. Maintain Tail Rotor Pylon Leading Edge Cover/Antenna.

REPAIR - Continued

5. Turn off all electrical power.

NOTE

Spot painting may be done with antenna installed on helicopter. Remove antenna, if antenna requires total repainting.

6. Remove antenna, if required.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Clean outer surface with cleaning cloth, Item 86, WP 1803 00, soaked with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to antenna elements will occur if antenna surface is sanded excessively. Use extreme care when sanding antenna surface.

8. Lightly hand sand antenna with abrasive cloth, Item 83, WP 1803 00.
9. Reclean surface with cleaning cloth, Item 86, WP 1803 00, soaked with technical acetone, Item 5, WP 1803 00.
10. Apply two coats of epoxy coating kit, Item 118, WP 1803 00.

NOTE

Allow at least 1/2 hour between coats.

11. Install antenna, if removed.
12. Perform operational check of antenna (TM 11-1520-237-23).

INSTALL

1. Turn off all electrical power.
2. Working from pylon steps or a maintenance work platform, insert cover/antenna hinge pins in mating hinge fittings on pylon (Figure 1, Detail A).
3. Inspect and treat coaxial connector (WP 1703 00).
4. **QA** Connect coaxial connector to antenna receptacle.
5. Make sure area is clean and free of foreign material before closing up.
6. Hinge cover/antenna closed.
7. **QA** Fasten cover to pylon with spring-type fasteners.
8. Perform operational check of antenna (TM 11-1520-237-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

INTERMEDIATE GEAR BOX FAIRING

This WP supersedes WP 0335 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheese, Item 85 WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Primer, Adhesive, Item 240, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 330, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Unfasten spring-type fasteners attaching intermediate gear box fairing to pylon (Figure 1, Detail A).
3. Remove fairing from helicopter.
4. If gearbox fairing, 70205-06051-043, is chafing pylon, replace worn or missing tape.
5. If tape is missing or worn, so that chafing is not prevented, do this:
   a. Remove worn tape using a nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   b. Clean dirt and grease from fairing using cheescloth cloth, Item 85, WP 1803 00 and technical acetone, Item 5, WP 1803 00.
   c. Measure and cut new tape.
   d. Spray a light coat of adhesive primer, Item 240, WP 1803 00, on area where tape is to be installed.
   e. Install pressure sensitive adhesive tape, Item 330, WP 1803 00, on fairing.

REMOVE - Continued

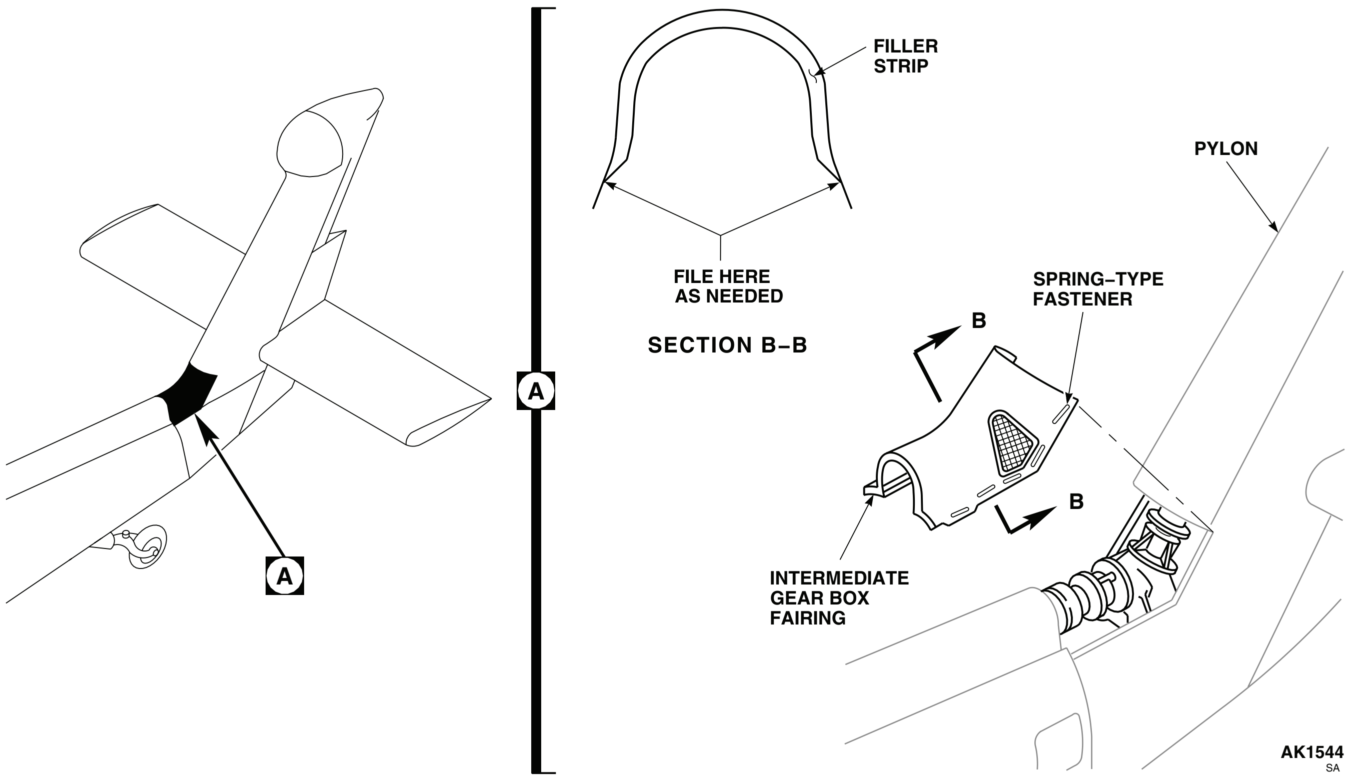


Figure 1. Replace Intermediate Gear Box Fairing.

INSTALL

NOTE

On some replacement fairings the tapered ends of filler strips may contact the pylon skin, making installation difficult. If this happens, file tapered ends as needed to let fairing seat properly. Touch up filed area with paint using procedures in WP 0381 00 and TM 55-1500-345-23.

1. Make sure area is clean and free of foreign material before closing up.
2. Place replacement fairing in position on pylon.
3. **QA** Fasten fairing with spring-type fasteners (Figure 1, Detail A).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON STEPS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Spring Tester

Material/Parts
Primer Coating, Item 243, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0337 00
WP 1803 00

NOTE

Replacement instructions for all steps except the lower are the same. To replace lower step, refer to WP 0337 00.

REMOVE

1. Turn off all electrical power.
2. Remove screws and washers attaching step to pylon (Figure 1, Sheet 1, Detail A).
3. Remove step from pylon.

DISASSEMBLE

WARNING

Tread is installed with approximately 16 pounds of spring pressure: personal injury may occur when removing pin from tread and housing. Make sure step is fully extended before removing pin.

1. Extend step fully and remove pin from tread and housing (Figure 1, Sheet 2, Detail B).
2. Remove tread from housing.
3. Remove rivets from tread and ring (TM 1-1500-204-23). Remove ring from tread.
4. Remove rivets from cap and tread (TM 1-1500-204-23) (Figure 1, Sheet 2, Detail C). Remove cap from tread.
5. Remove spring from housing.
6. To check spring tension, do this:
   a. Using spring tester, apply 16 to 17 pounds of pressure to compress spring.
   b. Measure compressed length of spring. **LENGTH OF SPRING MUST BE 1.79-INCHES.**

DISASSEMBLE - Continued

c. If length of spring is less than 1.79-inches, replace spring.

ASSEMBLE

1. Install spring in housing (Figure 1, Sheet 2, Detail B).
2. Install cap in tread and line up holes (Figure 1, Sheet 2, Detail C).
3. Install rivets and grind heads flush (TM 1-1500-204-23).
4. Install ring in tread and line up holes in tread and ring.
5. Install rivets and grind heads flush (TM 1-1500-204-23).
6. Install tread in housing. Line up hole in tread with slot in housing.
7. **QA** Press pin into tread. Stake pin 2 places on each end.
8. Check for smooth operation by alternately extending and pushing in tread.

INSTALL

1. Turn off all electrical power.
2. Prime pylon-to-step contact surfaces with primer coating, Item 243, WP 1803 00.
3. **QA** Install step in pylon and fasten with screws and washers (Figure 1, Detail A).

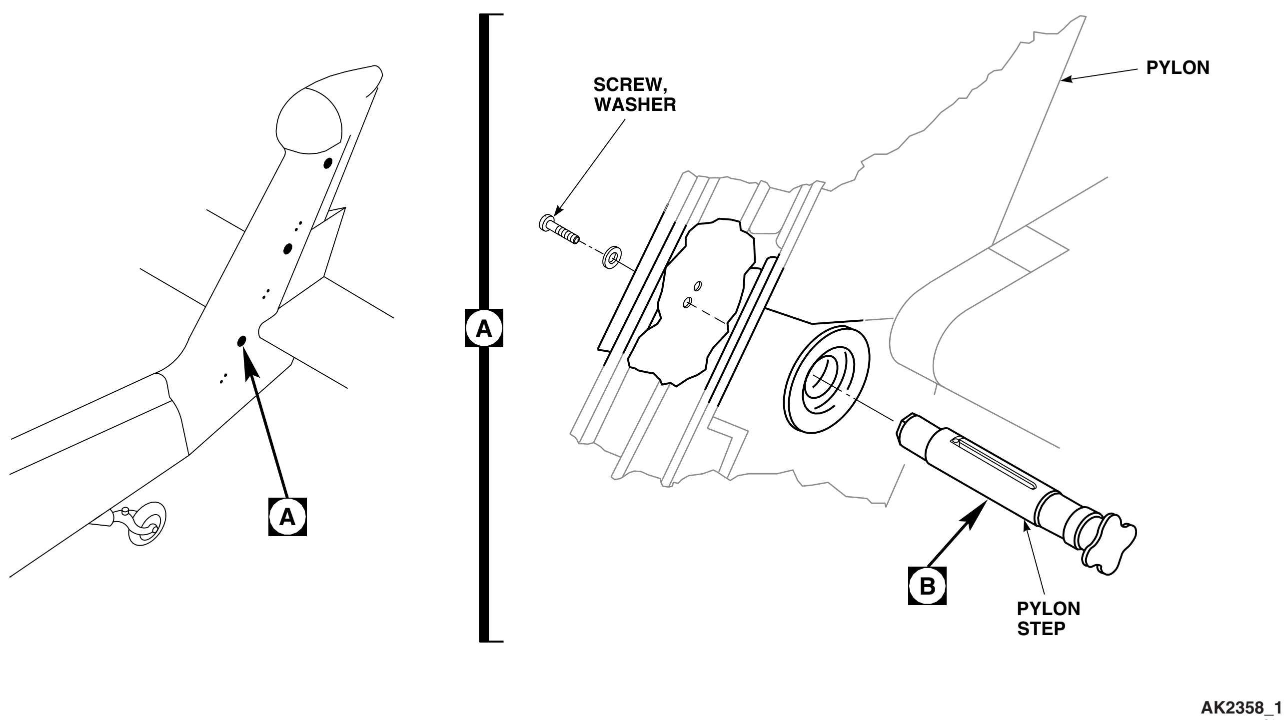


Figure 1. Replace Tail Rotor Pylon Steps. (Sheet 1 of 2)

INSTALL - Continued

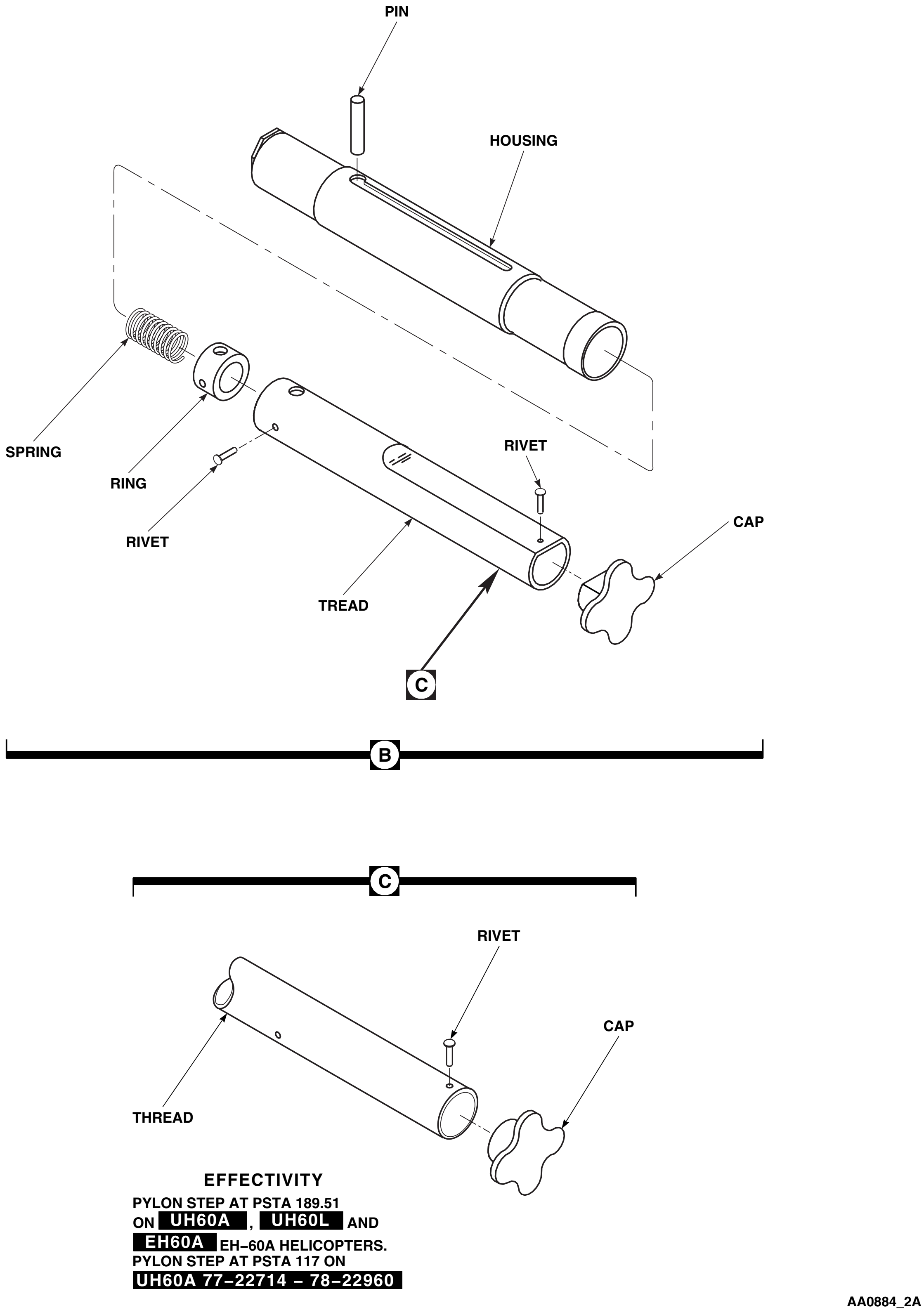


Figure 1. Replace Tail Rotor Pylon Steps. (Sheet 2 of 2)

INSTALL - Continued

4. Extend and push in tread a few times to check operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR PYLON LOWER STEP

This WP supersedes WP 0337 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1681 00

REMOVE

1. Turn off all electrical power.
2. Fold pylon 90 degrees (WP 1681 00).
3. Push up and turn step towards right side of helicopter and lower step from pylon (Figure 1, Detail A).

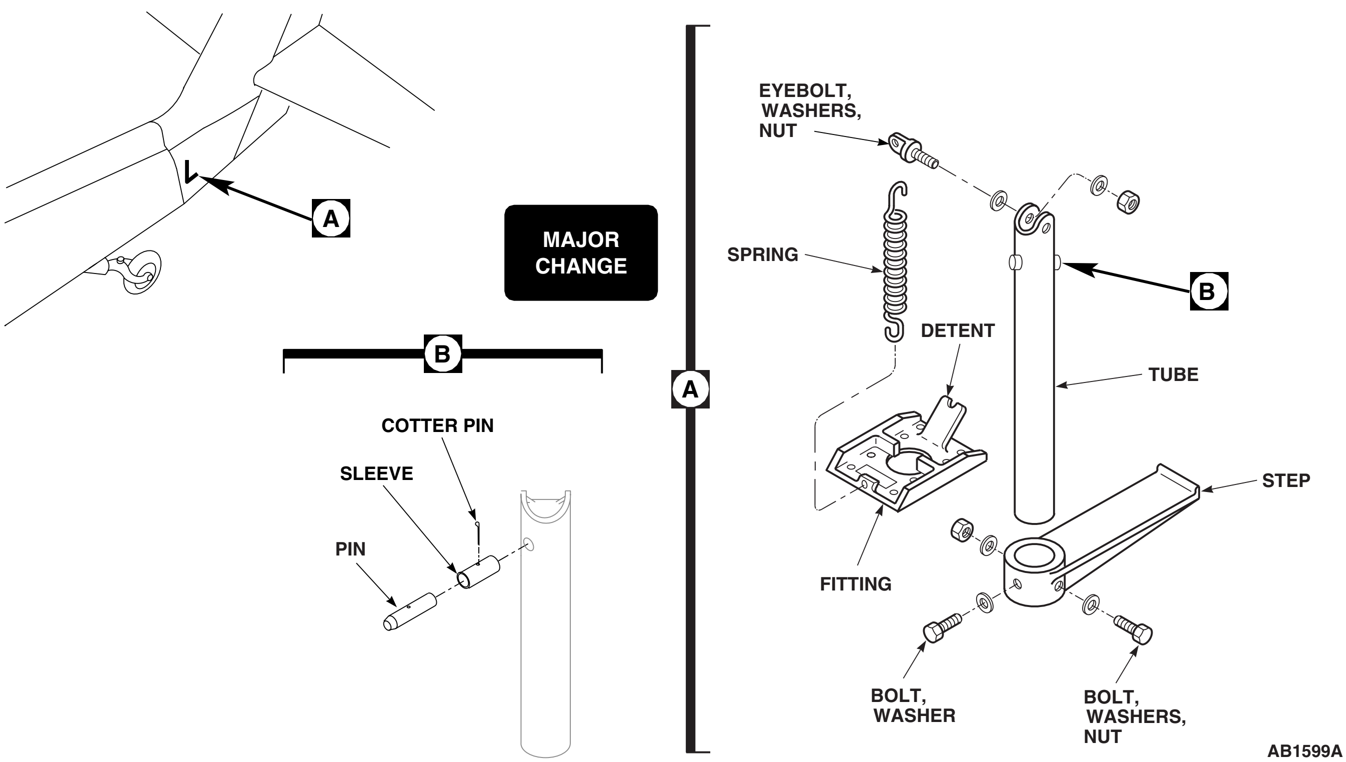


Figure 1. Replace Tail Rotor Pylon Lower Step.

REMOVE - Continued

4. Remove bolts, washers, and nuts attaching step to tube. Remove step.
5. Disconnect spring from tube.
6. Pull tube up and out of pylon.
7. If tube will be reinstalled, check condition of eyebolt and pin in tube. Replace eyebolt and pin if damaged or worn.
8. To remove pin from tube, do this:
   a. From inside of tube, remove cotter pin from pin and sleeve spacer. Remove pin and sleve spacer from tube.
   b. Check condition of sleeve spacer. Replace sleeve spacer if damaged or worn.

NOTE

Replacement sleeve spacers may not have a hole for the cotter pin. If a new sleeve spacer is required, a hole for the cotter pin may need to be drilled.

   c. If replacement sleeve spacer does not have a hole for the cotter pin, drill a 0.078-inch diameter hole with a No. 47 drill through part. Deburr holes in sleeve spacer (Figure 1, Detail B).

INSTALL

1. If eyebolt is not installed, install it, washers, and nut in tube (Figure 1, Detail A).
2. If pin is replaced or not installed, place sleeve spacer in tube and slide the pin through the sleeve spacer and both sides of the tube. Allow pin to protrude out both sides of tube. Align holes in pin and sleeve spacer and secure with cotter pin (Figure 1, Detail B).
3. From inside of pylon, insert tube through pylon into position. Make sure pin of tube is in pylon fitting (detent position).
4. Slide tube down to its lowered position.
5. **QA** Connect spring to eyebolt.
6. **QA** Slide step on tube and install bolts, washers, and nuts.
7. Check operation of step assembly a few times.
8. Push step up and turn to its stowed position.
9. Unfold pylon (WP 1681 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR

This WP supersedes WP 0338 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Cradle Assembly, 70700-20435-041
Gloves
Goggles/Face Shield
Stabilator Bolt Removal Tool, Locally-Made, Figure 169, WP 1805 00
Stabilator Drift Pin, Locally-Made, Figure 254, WP 1805 00
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Antiseize Compound, Item 51, WP 1803 00
Antiseize Compound, Item 52, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Corrosion Preventive Compound, Item 105, WP 1803 00
Corrosion Preventive Compound, Item 107, WP 1803 00
Nutplate MS21075-4
Sealing Compound, Item 271, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-MTF
TM 55-1500-345-23
WP 0329 00
WP 0330 00
WP 0381 00
WP 0428 00
WP 1685 00
WP 1703 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove accessible screws securing right and left stabilator-pylon fairings (Figure 1, Sheet 1, Detail A).
2. Connect external electrical power (WP 1685 00).
3. Place EXT PWR switch, on upper console, ON.
4. With assistant watching stabilator, place MAN SLEW to -8 degrees (trailing edge up).
5. Turn off all electrical power.
6. Pull these circuit breakers:
   a. STAB CONTR NO. 2 PRIMARY AC
   b. STAB CONTR AC ESNTL BUS
   c. STAB PWR NO. 2 PRIMARY DC
   d. STAB PWR DC ESNTL BUS
7. Remove remaining screws and right and left fairings.
8. Remove lower pylon fairing (WP 0329 00).
9. Disconnect formation lights electrical connectors from receptacles on pylon.

REMOVE - Continued

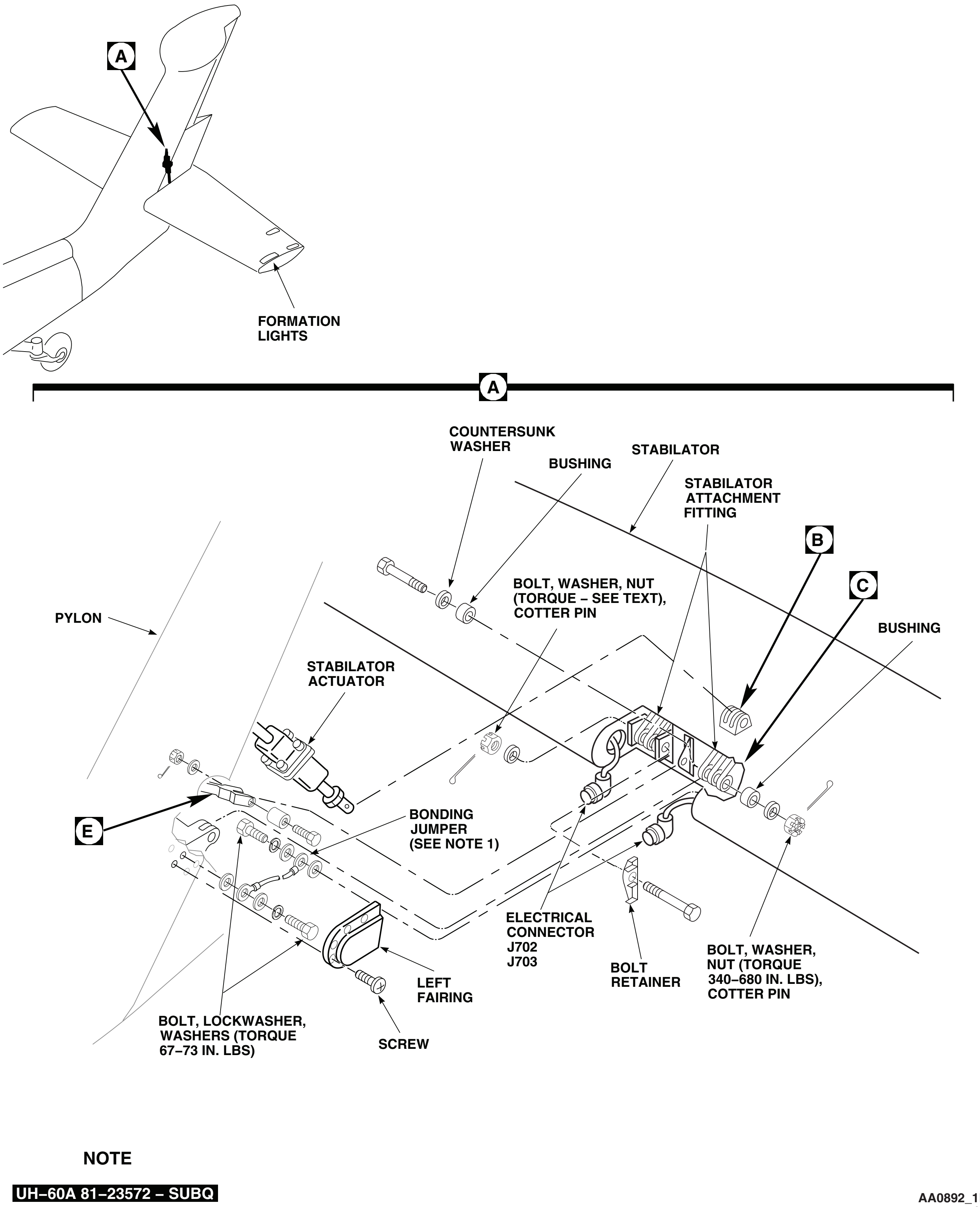
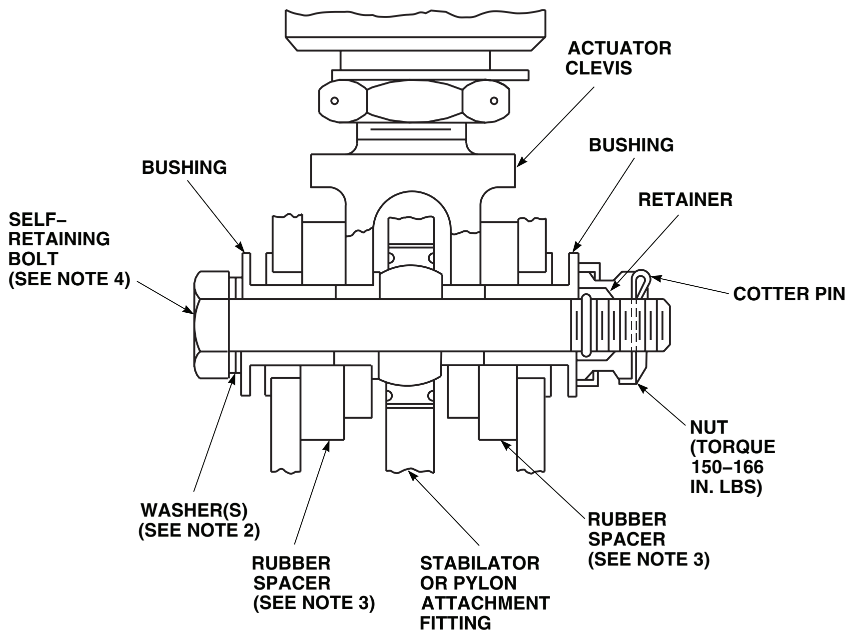


Figure 1. Replace Stabilator. (Sheet 1 of 4)

REMOVE - Continued



EFFECTIVITY

UH–60A 79–23274 – SUBQ

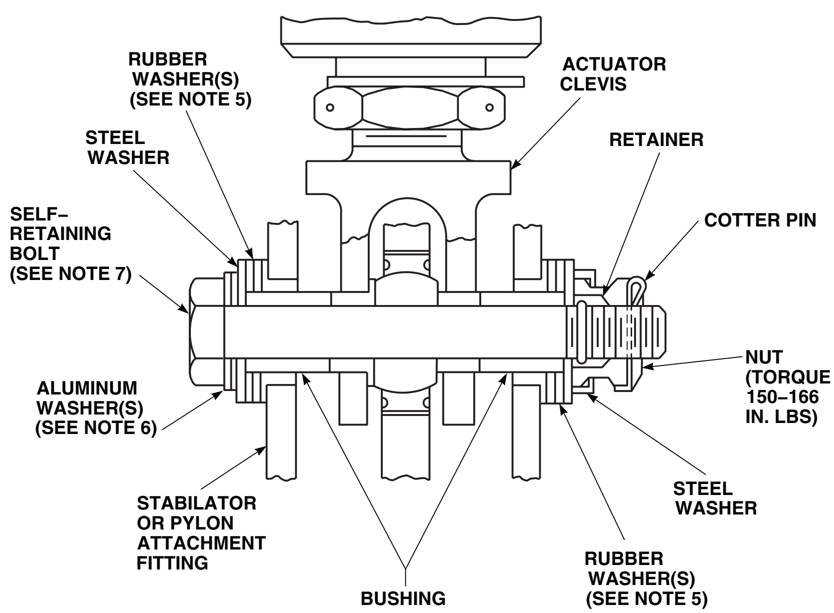
UH–60L EH–60A

NOTES

2. INSTALL WASHERS UNDER BOLT HEAD (3 MAXIMUM), TO MAKE SURE NUT BOTTOMS AGAINST BUSHING AND NOT ON RETAINER.
3. APPLY A LIGHT COAT OF PETROLATUM TO FACE OF RUBBER SPACERS FOR EASE OF INSTALLATION.
4. BOLTHEADS MAY BE INSTALLED IN EITHER DIRECTION.

B

B



EFFECTIVITY

UH–60A 77–22714 – 79–23273

NOTES

5. SHIM WITH RUBBER WASHERS AS REQUIRED TO ELIMINATE ENDPLAY. RUBBER WASHERS TO BE INSTALLED AT BOTH ENDS. MINIMUM OF ONE WASHER AT EACH END WITH MAXIMUM COMBINED TOTAL OF FOUR.
6. INSTALL ALUMINUM WASHERS UNDER BOLTHEAD AS REQUIRES TO MAKE SURE NUT BOTTOMS AGAINST STEEL WASHERS AND NOT RETAINER.
7. BOLTHEADS MAY BE INSTALLED IN EITHER DIRECTION.

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Figure 1. Replace Stabilator. (Sheet 2 of 4)

REMOVE - Continued

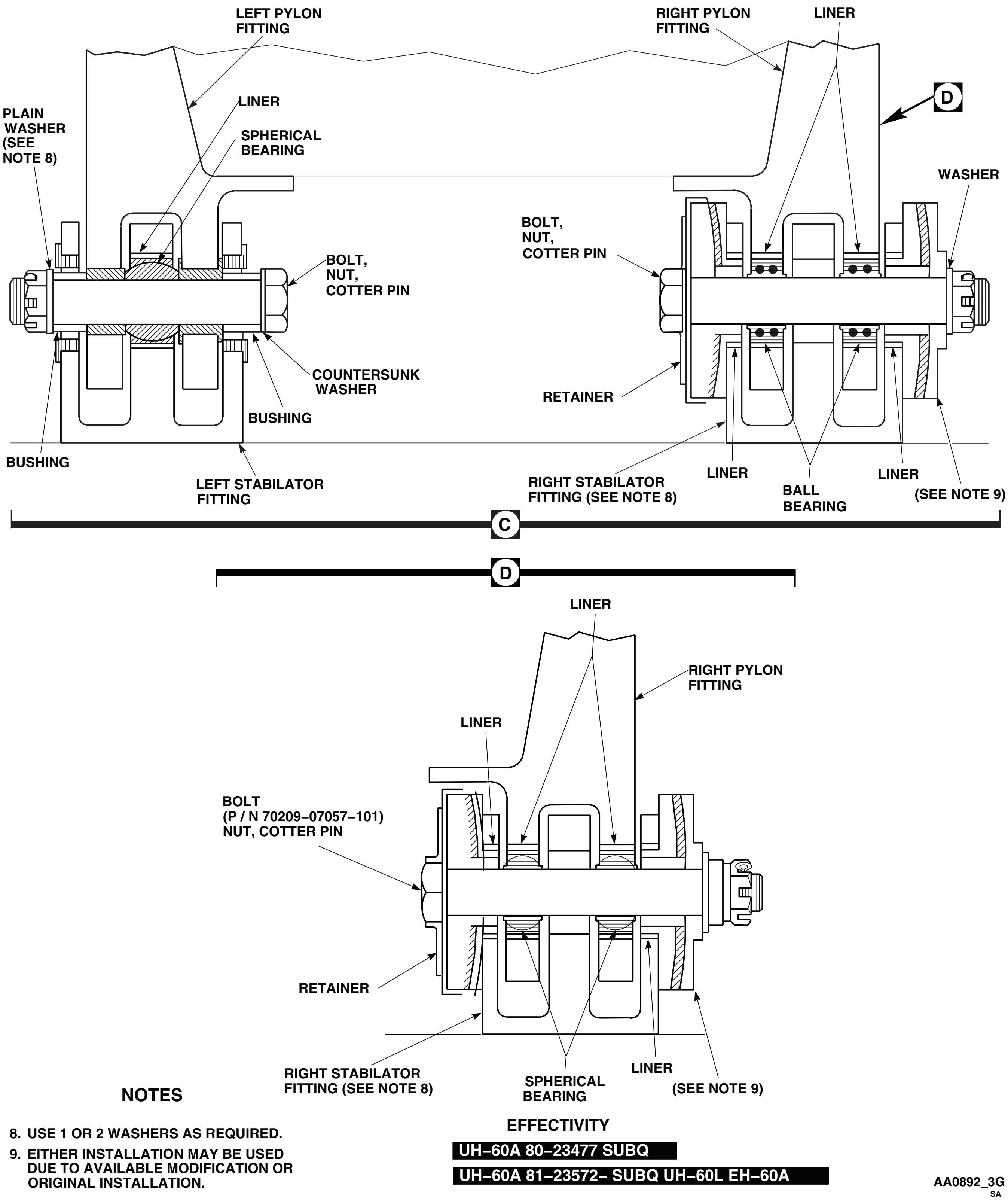


Figure 1. Replace Stabilator. (Sheet 3 of 4)

REMOVE - Continued

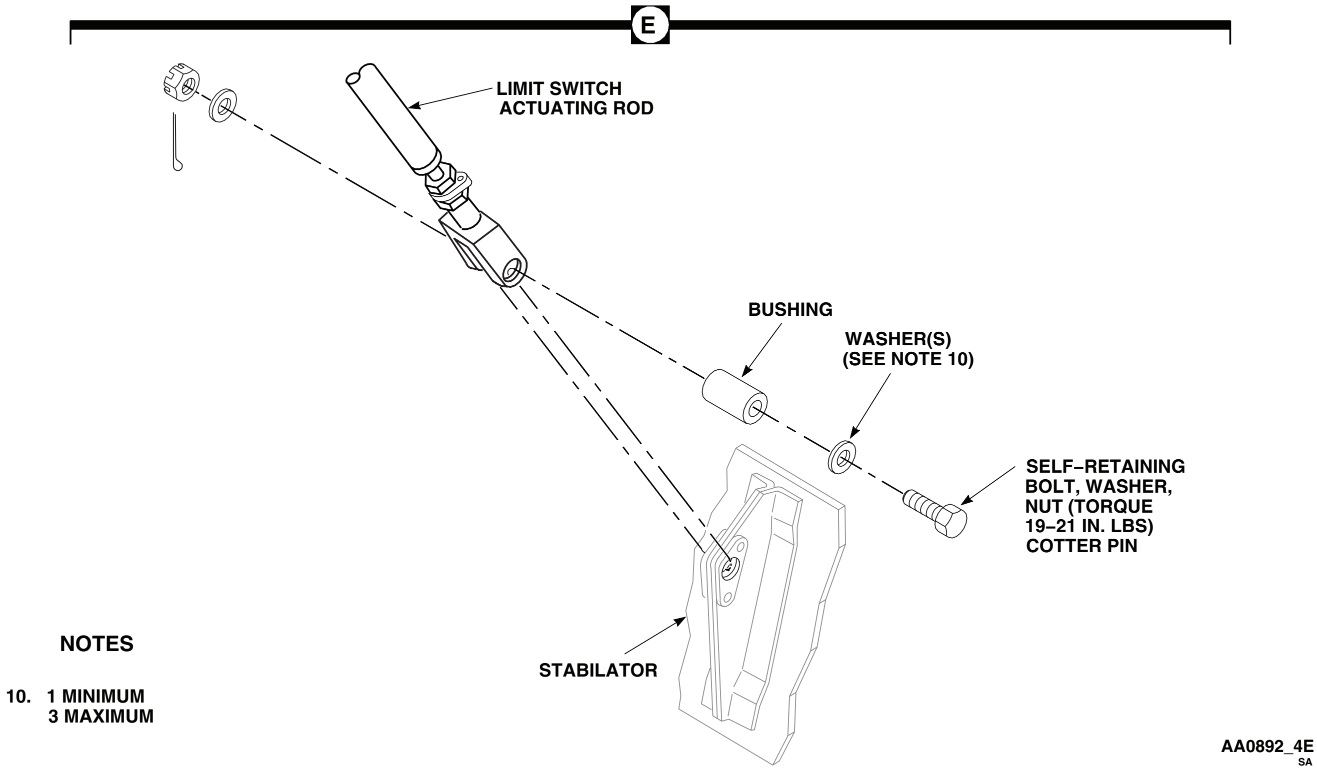


Figure 1. Replace Stabilator. (Sheet 4 of 4)

10. Remove cotter pin, nut, and washer from bolt securing limit switch actuator rod to stabilator (Figure 1, Sheet 4, Detail E). Do not remove bolt now.
11. Remove cotter pins, nuts, and washers from stabilator attach bolts. Do not remove bolts and bushings now.
12. Turn on electrical power.
13. Push in these circuit breakers:
    a. STAB PWR NO. 2 PRIMARY DC
    b. STAB PWR DC ESNTL BUS

NOTE

STABILATOR AUTO CONTROL RESET will not light.

14. With assistant watching stabilator, place MAN SLEW to 0 degrees (level position) to get to stabilator actuator attaching bolt.
15. Pull out both dc circuit breakers.
16. Turn off all electrical power.

CAUTION

Forcible removal of self-retaining bolt will damage equipment. Use tool to press release pin in center of bolthead while removing bolt.

17. Remove bolt and bushing holding limit switch actuator rod to stabilator. Disengage rod from stabilator fitting.

REMOVE - Continued

WARNING

CSI

The outside diameter and bolthead shoulder of stabilator actuator attaching bolts and stabilator actuator attach fitting lug areas are critical surfaces which must not be damaged during and after removal of actuator assembly. Use extreme caution when removing and provide protective covering after removal of actuator assembly.

18. With two assistants supporting stabilator, remove bolt, washers, bushings, and nut holding stabilator actuator to stabilator.
19. **UH-60A 79-23274 - SUBQ UH-60L EH-60A** Remove bolt, washers, bushings, spacers, and nut holding stabilator actuator to stabilator.
20. **UH-60A 81-23572 - SUBQ UH-60L EH-60A** Remove bolt and disconnect bonding jumpers from stabilator.
21. Remove left bolt, washers, and bushing holding stabilator to pylon.

NOTE

Stabilator bolt removal tool (locally made) may be used on right and left stabilator attach bolts.

22. If corroded or frozen bolt cannot be removed by hand, locally make stabilator bolt removal tool (WP 1805 00) and remove as follows:
    a. Select hex nut of proper length from tool and run it onto stabilator bolt.
    b. Place steel plate of tool against inside of stabilator rib, opposite bolt.
    c. Hold bolt head with wrench and back off tool hexnut with second wrench, with nut bearing against steel plate of tool.
    d. When bolt has broken free, remove hex nut and steel plate, and remove bolt.

NOTE

Right stabilator attach bolt, 70209-07057-101, may become discolored (reddish-brown) while in service. Surface discoloration to this bolt is not a cause for replacement.

23. Gently rocking stabilator, to aid in removal, remove right bolt and bolt retainer or countersunk washer holding stabilator to pylon.
24. Remove stabilator and install in cradle assembly, 70700-20435-041.
25. Inspect bonding jumper and hardware for corrosion. Replace bonding jumper and attaching hardware if necessary. Inspect bonding jumper airframe riv-nuts for condition of threads and security. If damaged riv-nut is found, replace with floating nutplate, MS21075-4 and **TORQUE BOLT SECURING ″JUMPER″ TO 67 - 73 INCH-POUNDS.**

INSPECT

1. Inspect top stabilator surface at front spar, both left and right sides, for skin cracks and/or loose rivets. Repair cracks (WP 0428 00) or replace stabilator.
2. **UH-60A 84-23935 - SUBQ UH-60L EH-60A** Remove screws, washers, and leading edge inspection plate cover.
3. **UH-60A 84-23935 - SUBQ UH-60L EH-60A** Using a flashlight and inspection mirror, check top internal spar cap and web for cracks and/or loose rivets. Replace stabilator with cracks, or loose rivets.

REMOVE RECAP TAIL ROTOR PYLON LEFT STABILATOR ATTACHMENT FITTING WASHER

NOTE

The following procedure applies to UH-60A's that have been modified by the A to A Recapitalization Program.

1. Remove steel washer from fitting with a sharp nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Using a nonmetallic scraper and technical acetone, Item 5, WP 1803 00, remove old sealing compound from fitting.

INSTALL RECAP TAIL ROTOR PYLON LEFT STABILATOR ATTACHMENT FITTING WASHER

NOTE

Some helicopters may have a steel and gasket washer combination bonded together which has been replaced by the procedures below.

1. Clean bonding surface of fitting using machinery towel, Item 344, WP 1803 00.
2. Bond steel washer (70070-20588-101) to outboard face of lug on left hand fitting assembly with a film of sealing compound, Item 271, WP 1803 00. Position steel washer to lug face utilizing attachment bolt to ensure that washer is installed concentric to bushing and parallel with lug face.
3. Initially apply a thicker layer of sealant (0.060) and squeeze washer against lug face leaving a thin (approximately 0.020 thick) layer of sealant between washer and lug face. Verify 100% squeeze - out of sealant along inboard diameter and outside diameter of steel washer.

CAUTION

Allow sealing compound to cure for a minimum of 24 hours prior to installing stabilator assembly on pylon.

4. Remove all sealant squeeze - out from inboard diameter. Utilize the squeeze - out of steel washer to create a thin layer of sealant along the outside diameter of washer at interface with lug face.

INSTALL

WARNING

CSI

Stabilator fittings on pylon, stabilator attach bolts, actuator fitting on stabilator and attach bolt have critical areas that must not be damaged during installation. Installation of stabilator attach bolts and actuator attach bolt have critical characteristics. Verification of proper installation and torque limits must be accomplished.

1. Turn off all electrical power.

INSTALL - Continued

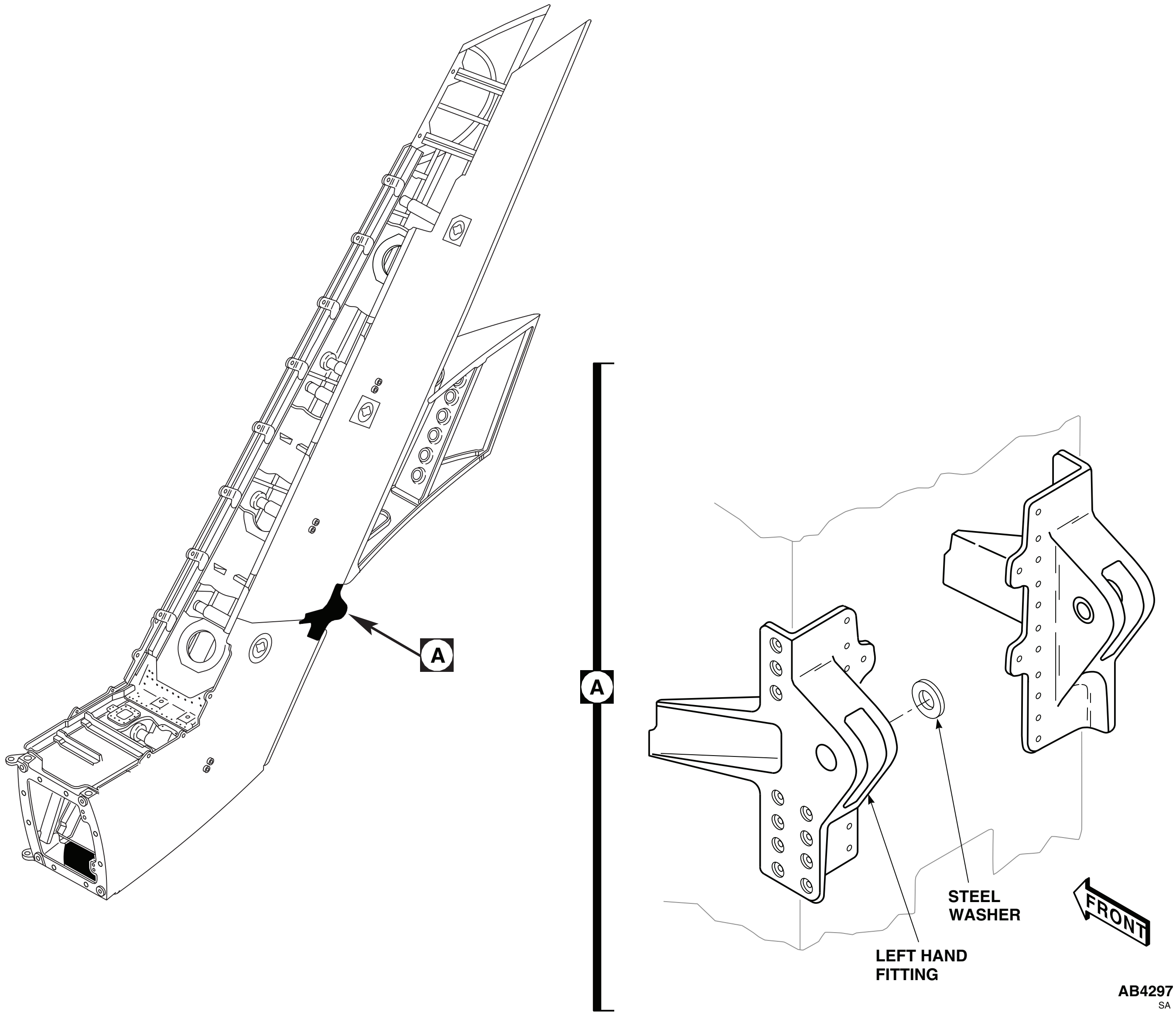


Figure 2. Left Hand Tail Pylon Stabilator Fitting.

2. Pull inner sleeve bushing out a little from left pylon fitting, to allow better clearance for positioning of attaching parts.
3. Clean area where bonding jumper is attached using a clean cloth and cleaning compound solvent, Item 71, WP 1803 00.
4. Have assistants place stabilator on helicopter, being careful not to hit limit switch actuator rod.
5. Coat attachment fittings with antiseize compound, Item 52, WP 1803 00.

NOTE

Use a locally-made drift to help line up parts, before installing attaching parts.

6. Locally make stabilator drift pin (WP 1805 00). Push drift through inboard side of right attachment fittings.

INSTALL - Continued

7. Coat shank of right bolt with antiseize compound, Item 51, WP 1803 00

CAUTION

Damage may result if bolts are forced through fittings. Ensure fittings are properly lined up properly before inserting bolts.

NOTE

UH-60A 77-22714 - 80-23476 Misalignment interference of 0.050-inch is permissible between inboard and outboard elastomeric bearings when installing bolt in right fitting.

8. If bolt retainer was removed, reinstall. Push bolt through retainer and inboard bearing, pushing drift out with bolt. Have assistants rock stabilator to aid in installation. Do not install nut now.
9. Coat bolt shank with antiseize compound, Item 51, WP 1803 00.
10. Push left bolt, countersunk washer, and bushing through left stabilator fittings. Do not force bolt. Do not install nut. Have assistants gently rock stabilator to aid installation (Figure 1, Sheet 3, Detail C).
11. **QA** **UH-60A 81-23572 - SUBQ UH-60L EH-60A** Connect bonding jumpers to stabilator using bolts, lockwashers, and washers. **TORQUE BOLTS TO 67- 73 INCH-POUNDS.**

WARNING

CSI

The outside diameter and bolthead shoulder of stabilator actuator attaching bolts and stabilator actuator attach fitting lug areas are critical surfaces which must not be damaged during installation of actuator assembly. After removal of protective covering, use extreme caution when installing actuator assembly.

NOTE

If any attaching hardware for upper or lower clevis is lost, on **UH-60A 77-22714 - 79-23273** , replace all attaching hardware, with hardware used on **UH-60A 79-23274 - SUBQ** . Attaching hardware cannot be intermixed. Use same stack up for both upper and lower clevises.

12. Place stabilator actuator in attachment fitting.

WARNING

CSI

Proper hardware installation, torque, proof torque, and safetying of self-retaining bolts are critical characteristics and must be verified.

CAUTION

Do not use force while inserting bolts through fittings, as damage may result from possible out-of-line condition.

13. Install self retaining bolt and washers (under bolthead).
14. Install nut on bolt.
15. **QA TORQUE STABILATOR ACTUATOR ATTACH NUT TO 150 - 166 INCH-POUNDS.** Install cotter pin.

INSTALL - Continued

16. **PROOF TORQUE BOLTHEAD TO 45 - 47 INCH-POUNDS.** Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
17. Coat actuator limit switch actuator rod bolt and bushing with corrosion preventive compound, Item 105, WP 1803 00. Do not apply compound to bolt threads.

WARNING

CSI

Proper hardware installation, torque, proof torque, and safetying of self-retaining bolt are critical characteristics and must be verified.

CAUTION

To prevent damage to equipment, do not attach limit switch actuator rod to stabilator unless stabilator actuator is in place.

NOTE

- Install up to three washers under bolthead to maintain proper grip length of bolt.
- Before installing nut, be sure that no portion of self-retaining element of bolt (ring or balls) is inside bolt hole.

18. **QA** Position actuator limit switch actuator rod with stabilator fitting (Figure 1, Sheet 4, Detail E). Install bushing, self-retaining bolt, washer, and nut. **TORQUE TO 19 - 21 INCH-POUNDS.** Install cotter pin.
19. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
20. Turn on all electrical power.
21. Push in these circuit breakers:
    a. STAB PWR NO. 2 PRIMARY DC
    b. STAB PWR ESNTL BUS
22. With assistant watching stabilator, place MAN SLEW to -8 degrees (trailing edge up).
23. Turn off all electrical power.
24. Pull out these circuit breakers:
    a. STAB PWR NO. 2 PRIMARY DC
    b. STAB PWR ESNTL BUS
25. Install washer and nut on right stabilator attach bolt (Figure 1, Sheet 3, Detail C and Detail D).

INSTALL - Continued

WARNING

CSI

The right stabilator attaching bolt installation has critical characteristics. Verification of nut tightening procedure and cotter pin installation are critical characteristics.

CAUTION

To prevent damage to fitting, make sure retainer is properly aligned with elastomeric bearing and bolthead before torquing nut. Hold bolthead securely while torquing nut.

26. **UH-60A 77-22714 - 80-23476** To torque right nut, do this: ■

   a. **TIGHTEN NUT UNTIL BOLTHEAD, WASHER, AND NUT ARE FIRMLY SEATED. BACK OFF NUT UNTIL WASHER UNDER NUT IS LOOSE. TIGHTEN NUT AGAIN ONLY UNTIL WASHER IS SEATED AND CANNOT BE TURNED BY FINGER PRESSURE.**

   b. **QA IF SLOT IN NUT IS NOT LINED UP WITH COTTER PIN HOLE IN BOLT, KEEP TIGHTENING NUT UNTIL HOLE AND SLOT ARE LINED UP.** Install cotter pin. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

   c. **QA IF SLOT IN NUT IS LINED UP WITH COTTER PIN HOLE IN BOLT, KEEP TIGHTENING NUT UNTIL IT TURNS ONLY ONE MORE SLOT.** Install cotter pin. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

WARNING

CSI

The right stabilator attaching bolt installation has critical characteristics. Verification of nut tightening procedure and cotter pin installation are critical characteristics.

CAUTION

To prevent damage to fitting, make sure retainer is properly aligned with elastomeric bearing and bolthead before torquing nut. Hold bolthead securely while torquing nut.

27. **UH-60A 80-23477 - SUBQ** To torque right nut, do this: ■

   a. **TIGHTEN NUT TO 25 - 50 INCH-POUNDS.**

   b. **QA IF SLOT IN NUT IS LINED UP WITH COTTER PIN HOLE IN BOLT.** Install cotter pin. Apply torque stripe (WP 0381 00).

   c. **QA IF SLOT IN NUT IS NOT LINED UP WITH COTTER PIN HOLE IN BOLT, KEEP TIGHTENING NUT UNTIL HOLE AND SLOT ARE LINED UP. DO NOT EXCEED 50 INCH-POUNDS.** Install cotter pin. Apply torque stripe (WP 0381 00).

WARNING

CSI

The left stabilator attach bolt installation has critical characteristics. Verification of nut torque and cotter pin installation are critical characteristics.

28. **QA** Install bushings, washers, and nut on left stabilator attach bolt. **TORQUE TO 340 - 680 INCH-POUNDS.** Install cotter pin (Figure 1, Sheet 3, Detail C). Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

29. Inspect and treat electrical connector (WP 1703 00).
30. **QA** Connect formation lights electrical connectors to receptacles on pylon.
31. Install lower pylon fairing (WP 0329 00).
32. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
33. Attach right and left pylon fairings with accessible screws (WP 0330 00).
34. Turn on all electrical power.
35. Push in these circuit breakers:
    a. STAB PWR NO. 2 PRIMARY DC
    b. STAB PWR ESNTL BUS
36. With assistant watching stabilator, check operation by manually slewing through full travel. Leave stabilator fully down.
37. Push in these circuit breakers:
    a. STAB CONTR NO. 2 PRIMARY AC
    b. STAB CONTR AC ESNTL BUS
38. Remove external power from helicopter (WP 1685 00).
39. Apply corrosion preventive compound, Item 107, WP 1803 00, to fairing attachment hardware.
40. **QA** Install remaining screws in stabilator pylon fairings.

NOTE

If stabilator has been removed from and reinstalled on same serial numbered helicopter, without repair or adjustment of stabilator, no maintenance test flight is required.

41. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ELECTROSTATIC DISCHARGER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Heat Gun, LFT500

Material/Parts
Insulation Sleeving, Electrical, Item 163, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

Remove screws, washers, and electrostatic discharger from stabilator (Figure 1).

REPAIR

1. Repair cracks in electrostatic discharger rubber coating as follows:
   a. Cut a piece of electrical insulation sleeving, <Item 163, WP 1803 00, 1 1/2-inches long.
   b. Slide insulation sleeving over electrostatic discharger. Make sure insulation sleeving extends 3/4-inch beyond both sides of crack.
   c. Shrink insulation sleeving around electrostatic discharger using heat gun.

INSTALL

Install electrostatic discharger on stabilator with washers and screws (Figure 1).

INSTALL - Continued

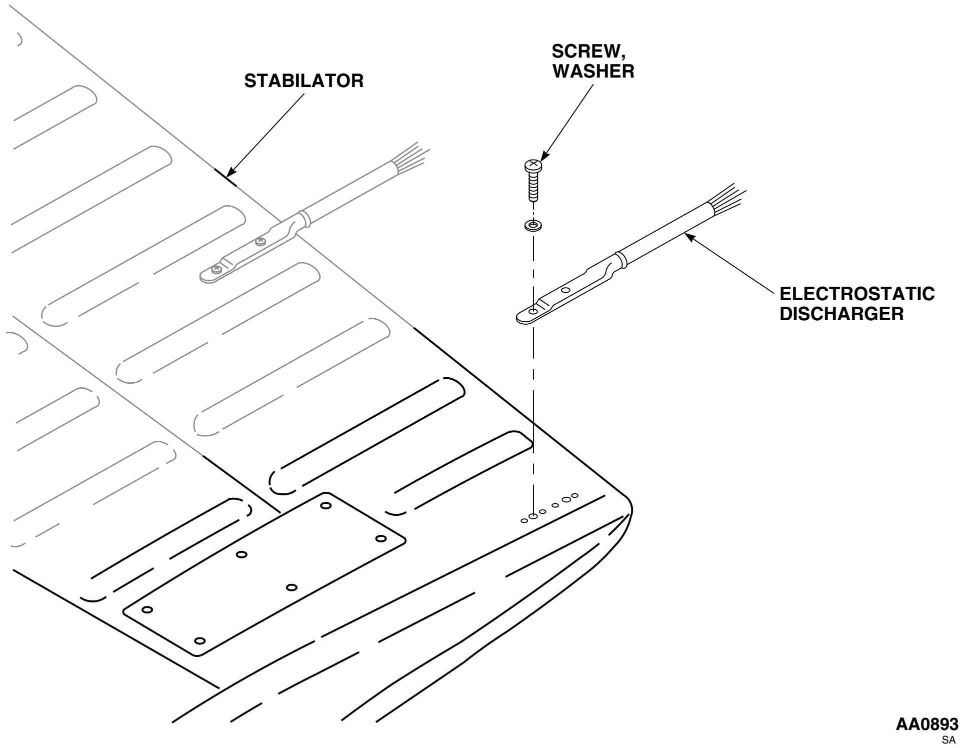


Figure 1. Replace Stabilator Electrostatic Discharger.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ATTACH FITTINGS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Polyurethane Coating, Item 227, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 55-1500-345-23
WP 0338 00
WP 0381 00
WP 1696 00
WP 1803 00

REMOVE

1. Remove stabilator (WP 0338 00).
2. Remove access covers on underside of stabilator.
3. Remove bolts, washers, square washers, and nuts. Remove fitting (Figure 1, Detail A).

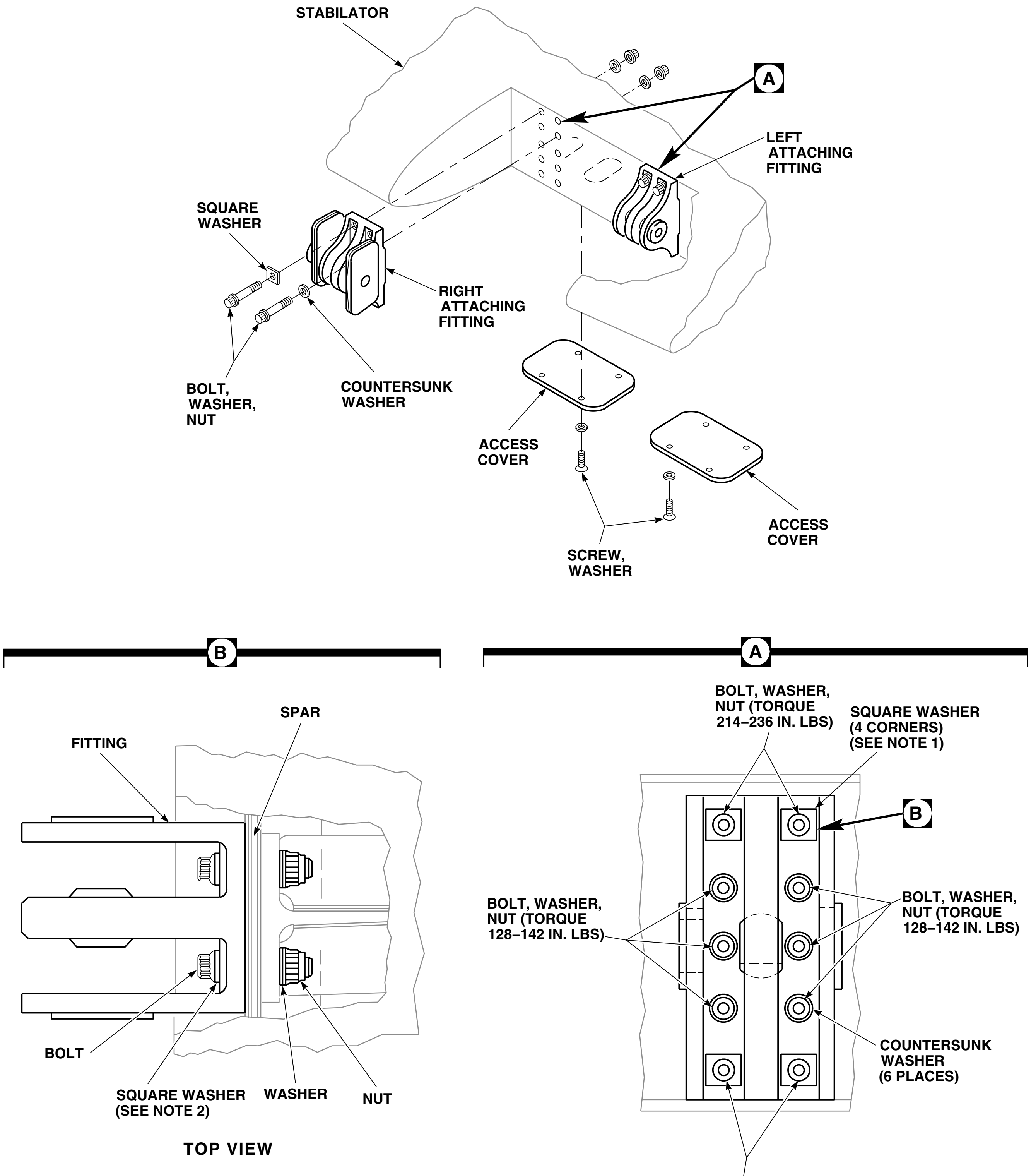
INSPECT

Visually inspect right side fitting for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. **PITS SHALL NOT EXCEED 0.050-INCH IN DIAMETER.** Remove corrosion from fitting.

REPAIR

1. Remove corrosion manually using scouring pad, Item 205, WP 1803 00.
2. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
3. Apply compound solution to corroded areas of fitting. Rinse thoroughly with clean water.
4. Touch up paint with epoxy primer coating kit, Item 120, WP 1803 00, and aircraft green polyurethane coating, Item 227, WP 1803 00 (WP 0381 00 and TM 1-1520-345-23).

REPAIR - Continued



NOTES:
1. EITHER 70201–07056–119 OR 70201–07056–120 MAYBE USED.
2. MATE RADIUS OF SQUARE WASHERS (RADIUS BLOCKS) WITH RADIUS OF FITTING.

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Figure 1. Replace Stabilator Attach Fittings.

INSTALL

WARNING

CSI

- Stabilator actuator attach fittings have critical areas that must not be damaged during installation. **NICKS, DENTS SCRATCHES AND TOOL OR SCUFF MARKS WITHIN CRITICAL CHARACTERISTIC AREA SHALL NOT EXCEED ACCEPTANCE LIMITS.**
- Damage to equipment could result if locator pin in right stabilator attach fitting is missing. Make sure pin has not fallen out and that the pin mates properly with locator hole in stabilator.

1. Position fitting on stabilator.
2. Install four corner bolts, square washers, washers, and nuts. Put square washers under boltheads, with radius sides facing left to right and toward fitting (Figure 1, Detail B).
3. Install middle bolts, countersunk washers, washers, and nuts. Install countersunk washers under bolthead, with countersunk side toward bolthead.
4. **QA TORQUE TOP AND BOTTOM CORNER BOLTS IN A DIAGONAL PATTERN (TOP LEFT CORNER, BOTTOM RIGHT CORNER, TOP RIGHT CORNER, BOTTOM LEFT CORNER) 214 - 236 INCH-POUNDS (Figure 1, Detail A).**
5. Check DA Form 2408-13-1 to make sure top and bottom corner bolts torque check is shown due 40 flight hours after installation (WP 1696 00).
6. **QA TORQUE CENTER BOLTS IN A DIAGONAL PATTERN (TOP LEFT CENTER, BOTTOM RIGHT CENTER, TOP RIGHT CENTER, BOTTOM LEFT CENTER; THEN TWO MIDDLE BOLTS) 128 - 142 INCH-POUNDS.**
7. **QA** Fasten access cover to stabilator with screws and washers.
8. Install stabilator (WP 0338 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR TIP CAP

This WP supersedes WP 0341 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts
Acetone, Technical, Item 5, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)

References
TM 1-1500-204-23
TM 55-1500-345-23
WP 0377 00
WP 0381 00
WP 1803 00

REMOVE

1. Carefully drill out and remove 34 rivets securing tip cap to stabilator using 1/8-inch diameter drill and 3/32-inch diameter drive pin punch (Figure 1).

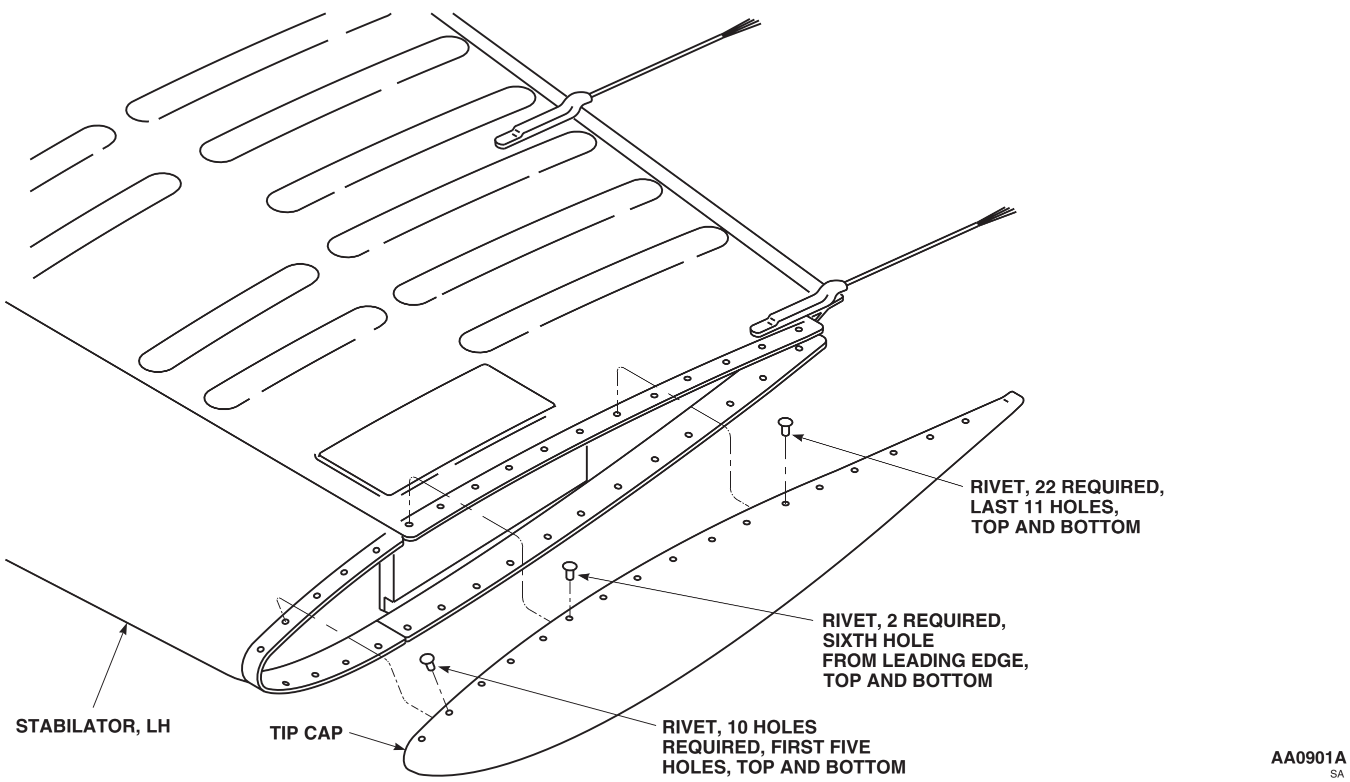


Figure 1. Replace Stabilator Tip Cap.

REMOVE - Continued

2. Remove tip cap, using a nonmetallic scraper remove old sealing compound.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean up rivet debris and wipe area clean with a cloth moistened with technical acetone, Item 5, WP 1803 00. Make sure area is clean and free of foreign material.

INSTALL

1. Position tip cap on stabilator and temporarily hold in place using 1/8-inch diameter clecos.
2. Install rivets as follows:
    a. 10 blind rivets, in first five holes, top and bottom, in leading edge (Figure 1).
    b. 2 blind rivets, in sixth hole from leading edge, top and bottom, attaching tip cap to forward spar.
    c. 22 blind rivets, in remaining 11 holes, top and bottom, from forward spar to trailing edge.
3. Install one rivet of required length between each cleco, top and bottom.
4. Install second rivet of required length between each cleco, top and bottom.
5. **QA** Remove clecos and install remaining rivets.
6. Inspect rivets for acceptable condition (TM 1-1500-204-23).
7. For corrosion protection and watertightness, seal tip cap to stabilator (WP 0377 00).
8. Touch up paint finish as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ACTUATOR

This WP supersedes WP 0342 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Digital Low-Resistance Multimeter, DLRO247000
Electrical Repairer's Toolkit, SC 5180-99-B06
Fluorescent Penetrant Inspection Kit, XMA101
Padded Support for Stabilator
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Corrosion Removing Compound, Item 108, WP 1803 00
Lubricant, Solid Film, Item 180, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

MIL-STD-1686C
TM 1-1500-204-23
TM 1-1520-265-23
TM 11-1520-237-23
TM 55-1520-345-23
WP 0187 00
WP 0331 00
WP 0334 00
WP 0338 00
WP 0344 00
WP 0345 00
WP 0381 00
WP 0424 00
WP 1703 00
WP 1803 00

REMOVE

1. Pull STAB CONTR ac circuit breakers on pilot's circuit breaker panel. Make sure STAB PWR dc circuit breaker and STAB IND ac circuit breaker on pilot's circuit breaker panel and STAB PWR dc circuit breaker on upper console are pushed in.
2. Turn on all electrical power.
3. Establish ICS communication between cockpit and maintenance personnel at stabilator. Shut off beeping tone in headsets by pressing and then releasing pilot's MASTER CAUTION PRESS TO RESET capsule.
4. Using MAN SLEW switch on STABILATOR CONTROLS panel, position stabilator as required to get to lower stabilator actuator attachment hardware (Figure 1, Sheet 1, Detail A).
5. Turn off all electrical power.
6. Open pylon leading edge cover/antenna. Remove cover/antenna from pylon only if it interferes with actuator assembly removal (WP 0334 00).

REMOVE - Continued

NOTE

EMEP Pin filtered adapters are considered part of harness assembly. Do not remove pin filtered adapters from electrical connectors, J606 and J607.

7. W/O EMEP Remove screws and lift connector bracket from under pylon drive shaft (Figure 1, Sheet 1, Detail A).
8. EMEP Unfasten screws and remove both connector brackets (Figure 1, Sheet 1, Detail C).
9. Disconnect electrical connectors, P606 and P607, from electrical connectors, J606 and J607.
10. Remove cambered fairing, if it will interfere with actuator assembly removal (WP 0331 00).
11. MWO 50-54 Remove nut and washers securing grounding strap to grounding bracket on actuator assembly (Figure 1, Sheet 2, Detail D). Remove strap from bracket.

CAUTION

To prevent damage to limit switches connected to stabilator, position sensor actuator rod, stabilator, or lower fairing; be sure support is high enough to prevent stabilator from going further than 35 degrees trailing edge down. To prevent damage to stabilator trailing edge, be sure support is well padded.

12. Place padded support below stabilator trailing edge.

WARNING

CSI

The outside diameter and bolthead shoulder of the actuator attachment bolts, and stabilator actuator attachment fitting lug areas are critical surfaces which must not be damaged during and after removal of actuator assembly. Use extreme caution when removing actuator assembly and provide protective covering after removal.

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

13. Remove bolt, washer(s), spacer(s), bushing(s), and nut from actuator assembly lower stabilator actuator attach fitting. Rest stabilator on padded support (Figure 1, Sheet 2, Detail E).
14. Install protective sleeve on attach bolt.

WARNING

Injury to personnel and damage to equipment will result if moving parts of actuator come in contact with personnel or helicopter. Make sure actuator shafts do not come in contact with personnel or any part of helicopter when positioning shafts.

15. Position actuator to ease removal as follows:
    a. Temporarily connect electrical connectors, P606 and P607, to electrical connectors, J606 and J607.
    b. Turn on electrical power.
    c. Place MAN SLEW switch UP or DN to position actuator as required.

REMOVE - Continued

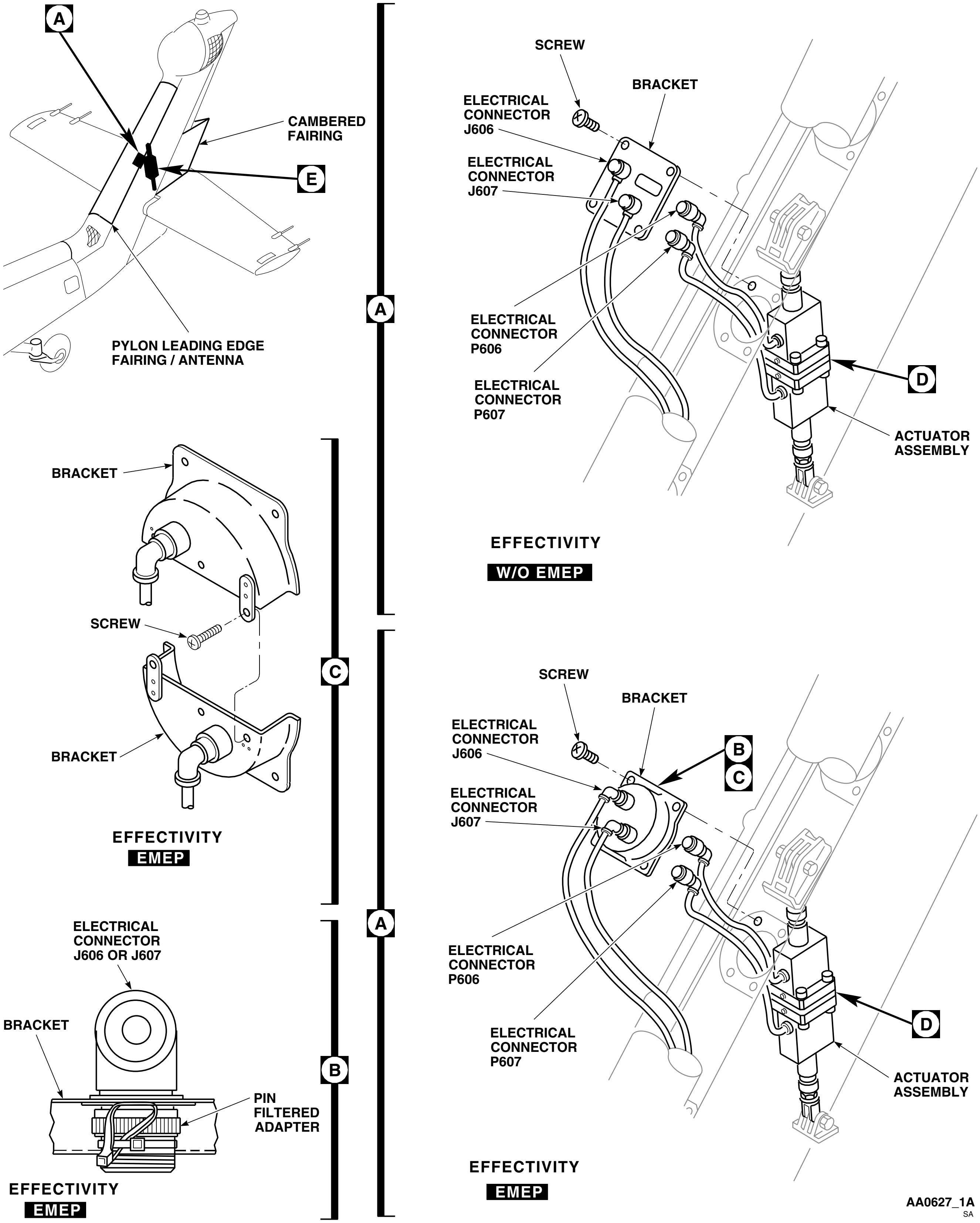


Figure 1. Replace Stabilator Actuator. (Sheet 1 of 3)

REMOVE - Continued

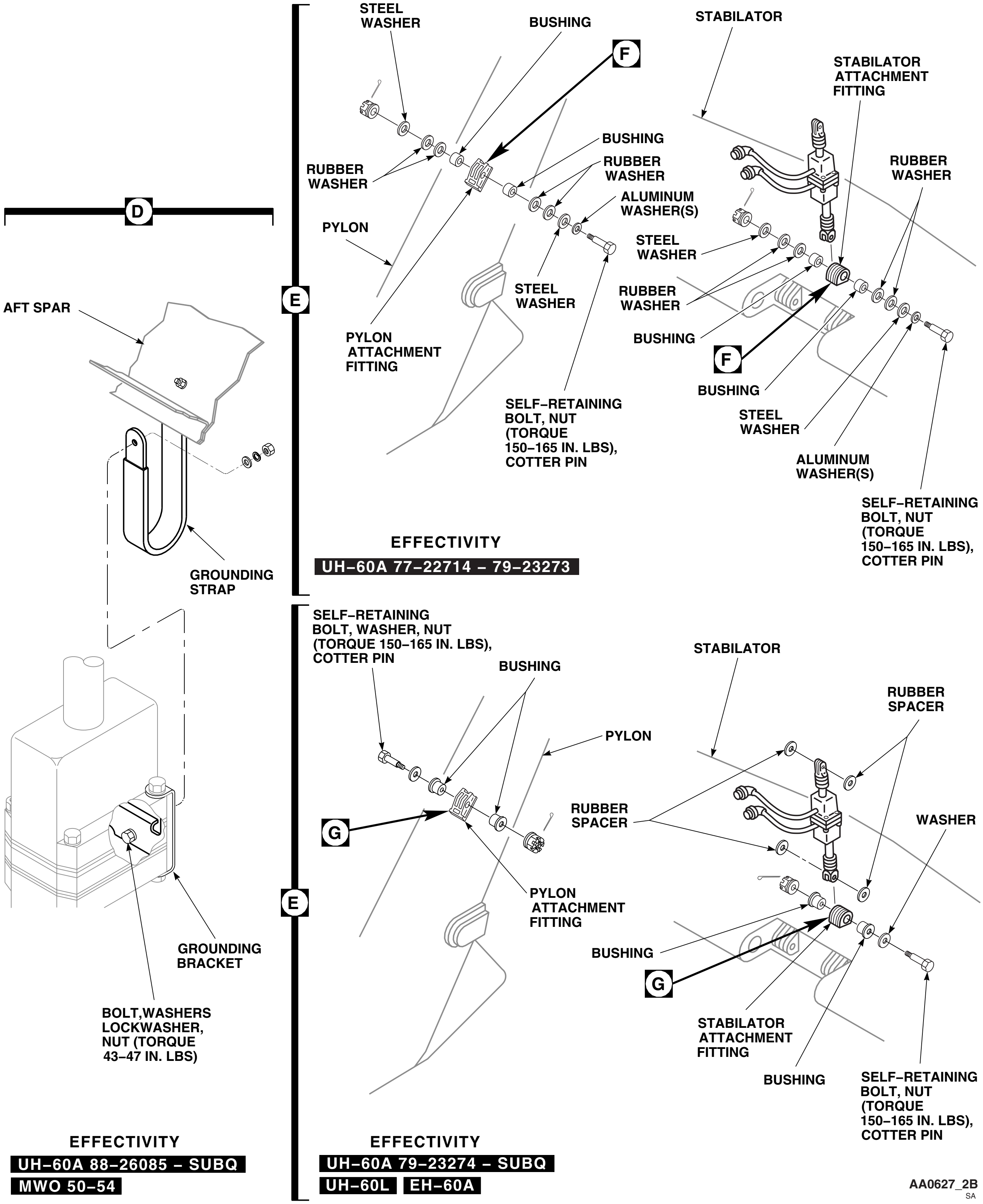


Figure 1. Replace Stabilator Actuator. (Sheet 2 of 3)

REMOVE - Continued

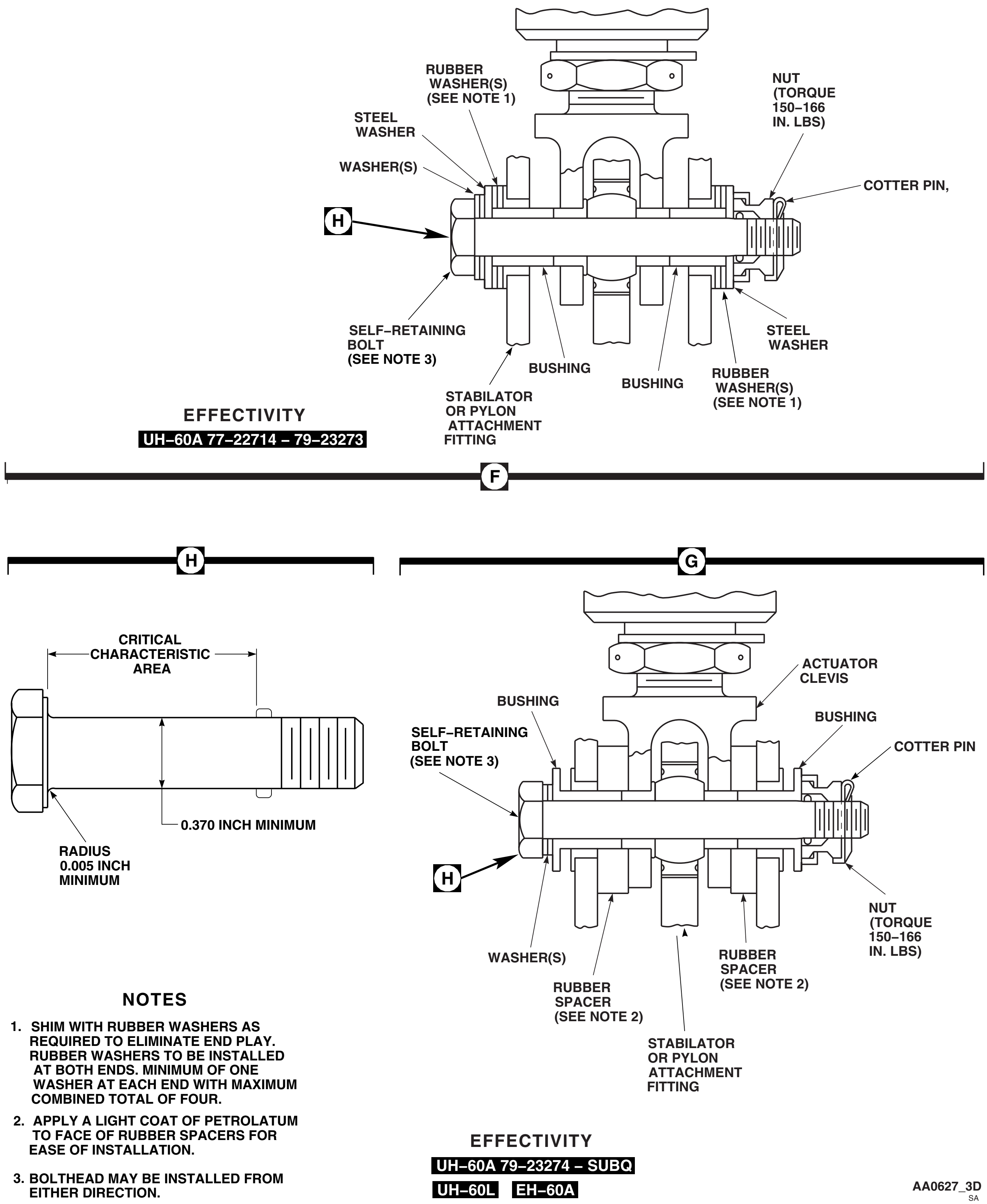


Figure 1. Replace Stabilator Actuator. (Sheet 3 of 3)

REMOVE - Continued

d. Turn off electrical power.

e. Disconnect electrical connectors, J606 and J607, from electrical connectors, P606 and P607.

16. Remove bolt, washer(s), spacer(s), bushing(s), and nut from actuator assembly upper pylon attach fitting. Remove actuator. Install protective sleeve on bolt.

INSPECT

1. Check actuator cables and electrical connectors, P606 and P607, for damage. **NO DAMAGE ALLOWED.**

WARNING

CSI

- The outside diameter and bolthead shoulder of stabilator actuator attachment bolts and stabilator actuator attachment fitting lug areas are critical surfaces which must not be damaged during installation of actuator assembly.
- Proper bonding of liner in center lug of actuator attach fitting is a critical characteristic and must be verified.

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

2. Inspect stabilator actuator attachment fitting for liner edge separation (WP 0424 00 and WP 0187 00).
3. Inspect stabilator actuator attachment fitting lug areas on stabilator (WP 0424 00 and WP 0187 00).
4. Check actuator housing for damage and cracks (TM 1-1500-204-23). **NO DAMAGE OR CRACKS ALLOWED.** Minor scratches are acceptable.
5. Inspect attaching hardware for damage and scratches. Replace parts as needed. **NO SCRATCHES, NICKS, BURRS, FRETTING, OR SCORING ALLOWED ON BOLTS OR BUSHINGS.** Visually inspect outside diameter and shoulder of bolts for nicks, dents and scratches. **SCRATCHES NOT THROUGH CADMIUM PLATING ARE ALLOWED.** Replace bolts, if damage is found (WP 0338 00).

NOTE

Repair actuator attach bolt immediately upon discovery of damage and replace bolt at earliest possible opportunity.

6. If damage to actuator attach bolt has penetrated into parent metal, and new bolt is not available for replacement, make temporary repair as follows:

WARNING

CSI

Minimum shank diameter and head-to-shank radius, as given in Figure 1, Sheet 3, Detail H, must be retained.

a. Repair damaged area, using abrasive cloth, Item 83, WP 1803 00, or abrasive cloth, Item 82, WP 1803 00, or abrasive mat, Item 2, WP 1803 00.

INSPECT - Continued

b. **BLEND DAMAGED AREA TO MAXIMUM DEPTH OF 0.003-INCH.** Blend reworked area to smooth transition into adjacent areas.

c. Inspect bolt for cracks, using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.**

d. Apply corrosion removing compound, Item 108, WP 1803 00. Rinse thoroughly with water and let dry.

e. If process is available, brush cadmium plate repaired area of bolt. As an alternate, apply solid film lubricant, Item 180, WP 1803 00.

7. Inspect actuator upper and lower rubber spacers. Replace damaged, deteriorated, or missing spacers.

8. Remove surface corrosion from actuator shafts with abrasive mat, Item 2, WP 1803 00. Wipe shafts clean with machinery wiping towel, Item 344, WP 1803 00.

NOTE

Stabilator actuator shafts must be fully extended for complete inspection of shafts.

9. If shafts are not fully extended, extend shafts as follows:

WARNING

Injury to personnel and damage to equipment will result if moving parts of actuator come in contact with personnel or helicopter. Make sure actuator shafts do not come in contact with personnel or any part of helicopter when extending shafts.

a. Temporarily connect electrical connectors P606 and P607 to electrical connectors J606 and J607.

b. Turn on electrical power.

c. Place MAN SLEW switch to DN position to extend shafts.

d. Turn off electrical power.

e. Disconnect electrical connectors, P606 and P607, from electrical connectors, J606 and J607.

10. Check stabilator actuator assembly output shafts for scratches, nicks, and burrs. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP. NICKS AND BURRS SHALL NOT EXCEED 0.250-INCH LONG, 0.050-INCH WIDE, AND 0.010-INCH DEEP.** Blend out all rough and raised edges with abrasive cloth, Item 82, WP 1803 00.

11. **UH-60A 88-26085 - SUBQ UH-60L MWO 50-54** Inspect stabilator actuator grounding strap as follows (Figure 1, Sheet 2, Detail D):

a. Check stabilator actuator grounding strap for security, cracks, or tears in shrink tubing. **NO CRACKS OR TEARS ALLOWED.** Replace strap if damage is found (WP 0345 00).

b. Check for disbonding of shrink tubing at strap end fittings. **NO DISBONDING ALLOWED.** Replace strap if disbonding is found (WP 0345 00).

c. Inspect chemical conversion coating on mating surface of strap end and grounding bracket (WP 0381 00 and TM 55-1520-345-23).

INSTALL

WARNING

CSI

- The outside diameter and bolthead shoulder of the actuator attachment bolts, and stabilator actuator attachment fitting lug areas on stabilator are critical surfaces which must not be damaged during actuator assembly installation. After removal of protective covering, use extreme caution when installing.
- Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

NOTE

- **UH-60A 77-22714 - 79-23273** If any attaching hardware for upper or lower clevis is lost, replace all attaching hardware, with hardware used on **UH-60A 79-23274 - SUBQ** . Attaching hardware cannot be intermixed. Use same stackup for both upper and lower clevises.
- Actuator adjustment is required when one or both actuators have been replaced.

1. Adjust both actuators, if required (WP 0344 00).
2. Position upper clevis (long clevis) in pylon attachment fitting.
3. Inspect and treat electrical connector (WP 1703 00).
4. Connect upper actuator electrical connector, P606, to electrical connector, J606.
5. Inspect and treat electrical connector (WP 1703 00).
6. Connect lower actuator electrical connector, P607, to electrical connector, J607.
7. **EMEP** Do EME operational check of pin filtered adapters as follows:

NOTE

Resistance check applies to stabilator actuator 70400-06641-117 only. Resistance measurements must be taken at least three times.

   a. Using a digital low-resistance multimeter, measure resistance between shell of each pin filtered adapter and surface of actuator housing. **RESISTANCE SHALL NOT BE MORE THAN 5.0 MILLIOHMS.**

   b. If resistance is more than 5.0 milliohms, clean surface between pin filtered adapters and actuator assembly connectors.

8. Fasten connector bracket to tail rotor pylon with screws (Figure 1, Sheet 1, Detail A).
9. **UH-60L 90-26272 UH-60L 90-26293 - SUBQ** Fasten connector brackets to tail rotor pylon and then to each other with screws (Figure 1, Sheet 1, Detail C).

INSTALL - Continued

10. Check that STAB CONTR circuit breaker is still out. Turn on electrical power. Establish ICS communication between cockpit and maintenance personnel at stabilator. Shut off beeping tone in headsets by pressing and then releasing pilot's MASTER CAUTION PRESS TO RESET capsule.

WARNING

Injury to personnel and damage to equipment will result if moving parts of actuator come in contact with personnel or helicopter. Make sure actuator shafts do not come in contact with personnel or any part of helicopter when extending shafts.

CAUTION

Damage to limit switch will occur if stabilator is allowed to go further than 35 degrees trailing edge down. To prevent damage to limit switch, do not allow stabilator to exceed 35 degrees trailing edge down.

NOTE

If stabilator actuator installation attachment bolts are replaced, update the 2408–18 inspection A-475.

11. Have assistant support trailing edge of stabilator and help line up boltholes in actuator lower clevis and attachment fitting. Using MAN SLEW switch on STABILATOR CONTROLS panel, position stabilator actuator to line up boltholes.
12. **UH-60A 77-22714 - 79-23273** Refer to Figure 1, Sheet 3, Detail F, for installing self-retaining bolts and attaching hardware.
13. Install self-retaining bolt and washers (under bolthead) (WP 0187 00).
14. Install nut on bolt. **TORQUE NUT TO 150 - 165 INCH-POUNDS.** Install cotter pin.
15. **PROOF TORQUE BOLTHEAD TO 45 - 47 INCH-POUNDS.** Apply torque stripe (WP 0381 00 and TM 55-1520-345-23).
16. **UH-60A 79-23274 - SUBQ UH-60L EH-60A** Refer to Figure 1, Sheet 3, Detail G, for installing self-retaining bolts and attaching hardware.
17. Install self-retaining bolt and washers (under bolthead) (WP 0187 00).
18. Install nut on bolt. **TORQUE NUT TO 150 - 165 INCH-POUNDS.** Install cotter pin.
19. **PROOF TORQUE BOLTHEAD TO 45 - 47 INCH-POUNDS.** Apply torque stripe (WP 0381 00 and TM 55-1520-345-23).
20. Remove support from under stabilator.
21. **MWO 50-54** Install stabilator grounding strap as follows:
    a. Inspect chemical conversion coated mating surfaces of grounding bracket and grounding strap (WP 0381 00).
    b. Install grounding strap on actuator assembly grounding bracket and secure with washers and nut (Figure 1, Sheet 2, Detail D). **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
    c. Do EME operational check of grounding strap as follows:

INSTALL - Continued

NOTE

Resistance measurements must be taken at least three times.

(1) Measure resistance between shell of each actuator housing and rear spar of tail rotor pylon. **RESISTANCE SHALL NOT BE MORE THAN 7.5 MILLIOHMS.**

(2) If resistance is more than 7.5 milliohms, remove grounding strap and replace chemical conversion coating on grounding bracket and grounding strap (WP 0381 00 and TM 55-1520-345-23).

(3) Repeat step 21. b. and step 21. c. (1). If resistance is still more than 7.5 milliohms, replace actuator assembly.

d. Cover bolt, washers, and nut securing strap to bracket with sealing compound, Item 273, WP 1803 00.

22. Make sure area is clean and free of foreign material.

23. Install cambered fairing (WP 0331 00).

24. **QA** Secure pylon leading edge fairing/antenna.

25. Adjust stabilator position transmitter assembly (TM 11-1520-237-23). Leave test set hooked up to do next step.

26. Adjust stabilator amplifier bias (TM 11-1520-237-23).

27. Check stabilator auto mode as follows:

NOTE

- Make sure all personnel are clear of stabilator.
- Stabilator auto mode check does not require hook up of test set.

a. On STABILATOR CONTROLS panel, press and release AUTO CONTROL RESET switch.

(1) STAB POS indicators should move to 34 degrees to 42 degrees DN.

(2) STABILATOR and MASTER CAUTION capsules should be off.

(3) Beeping tone in headsets should be off.

(4) Visually verify stabilator position.

b. Note position of stabilator as indicated on both STAB POS indicators. Press and hold TEST switch for about 3 seconds.

(1) STAB POS indicators should move 5 degrees to 12 degrees up from noted position.

(2) AUTO CONTROL RESET switch ON legend should go off.

(3) STABILATOR and MASTER CAUTION capsules should be on.

(4) Beeping tone in headsets should be audible.

c. Press MASTER CAUTION PRESS TO RESET capsule.

(1) Beeping tone should go off.

(2) Both MASTER CAUTION capsules should go off.

d. Press and release AUTO CONTROL RESET switch.

(1) AUTO CONTROL RESET switch ON legend should go on.

(2) STABILATOR capsule should be off.

(3) STAB POS indicators should move to 34 degrees to 42 degrees DN.

INSTALL - Continued

(4) Visually verify stabilator position.

e. Place MAN SLEW switch to UP. Hold until stabilator stops moving.

(1) STAB POS indicators should move to 6 degrees to 10 degrees up within about 7 seconds.

(2) STABILATOR and MASTER CAUTION capsules should go on.

(3) AUTO CONTROL RESET switch ON legend should go off.

(4) Beeping tone should be audible.

(5) Visually verify stabilator position.

f. Place MAN SLEW switch to DN. Hold until STAB POS indicators read 0 degrees.

g. Visually verify stabilator position.

h. Press and release AUTO CONTROL RESET switch.

(1) AUTO CONTROL RESET switch ON legend should go on.

(2) STABILATOR and MASTER CAUTION capsules should go off.

(3) STAB POS indicators should move to 34 degrees to 42 degrees DN.

(4) Beeping tone in headsets should go off.

(5) Visually verify stabilator position.

i. If any part of stabilator checks fail, go to (TM 11-1520-237-23).

28. Turn off all electrical power.

29. Make sure area is clean and free of foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ACTUATOR REPAIR

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Digital Low-Resistance Multimeter, DLRO247000
Electrical Repairer's Toolkit, SC 5180-99-B06
Heat Gun, LFT500
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Corrosion Preventive Compound, Item 106, WP 1803 00
Corrosion Removing Compound, Item 109, WP 1803 00
Corrosion Removing Compound, Item 108, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

MIL-STD-1686C
TM 1-1500-204-23
TM 1-1500-344-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0344 00
WP 0375 00
WP 0381 00
WP 1803 00

DISASSEMBLE

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

NOTE

Disassemble stabilator actuator assembly only to the extent necessary to replace damaged parts.

1. Back off jamnuts from each clevis (Figure 1, Sheet 1, Detail A).
2. Note number of exposed threads on each clevis. Remove upper and lower clevises from actuator assembly. Also note and record number of turns required to unscrew each clevis. Remove jamnuts and lockwashers.
3. **MWO 50-54** Remove bolts, washers, and nuts, from actuators, remove grounding bracket, separate upper and lower actuators. Remove boot (Figure 1, Sheet 2, Detail A).
4. **W/O MWO 50-54** Remove bolts, washers, and nuts, and separate upper and lower actuators. Remove boot (Figure 1, Sheet 2, Detail A).

INSPECT

1. Inspect 0.375-inch ID of threaded shank for application of epoxy primer. **PRIMER MUST BE EVIDENT.**

INSPECT - Continued

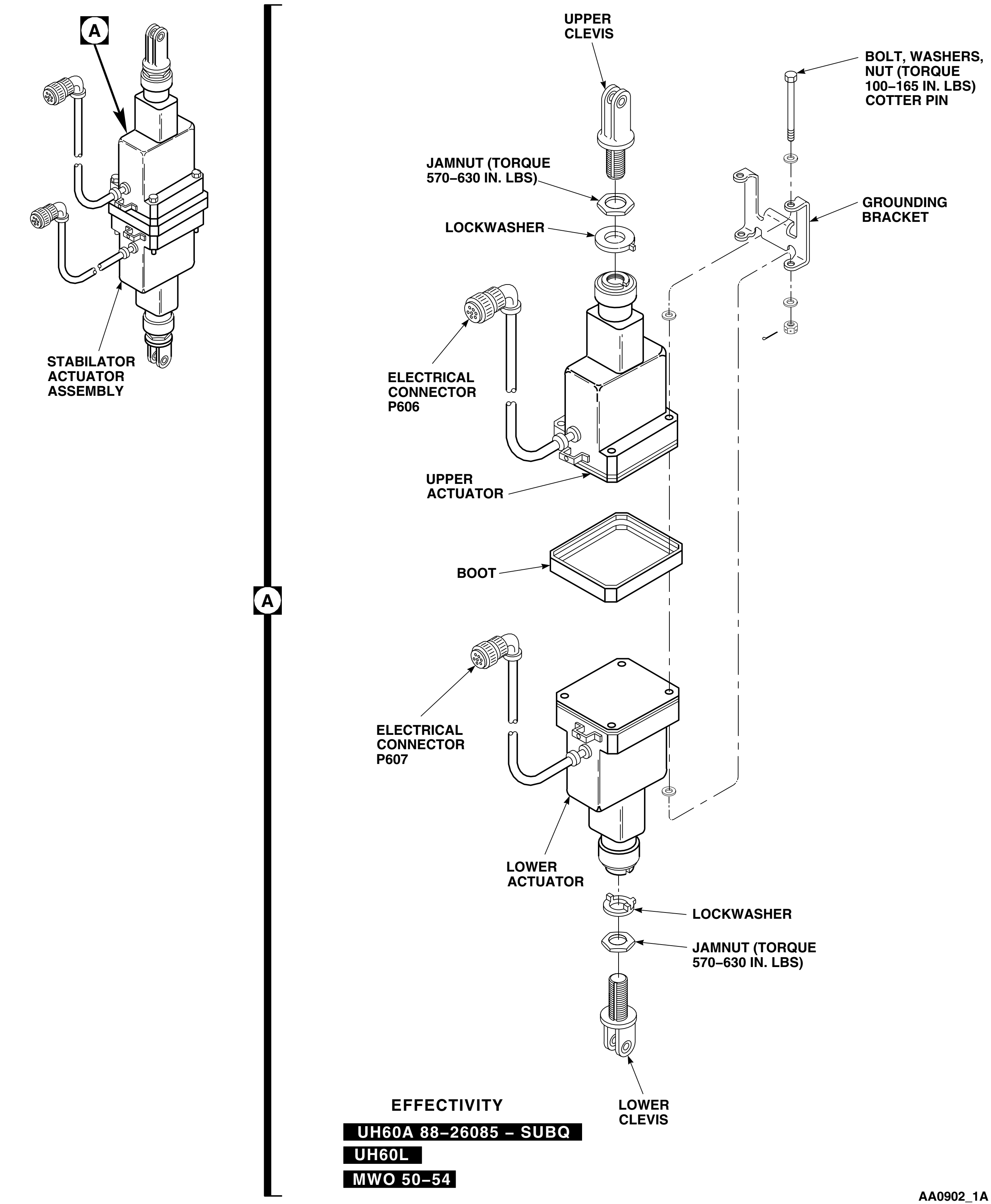


Figure 1. Stabilator Actuator Repair. (Sheet 1 of 2)

INSPECT - Continued

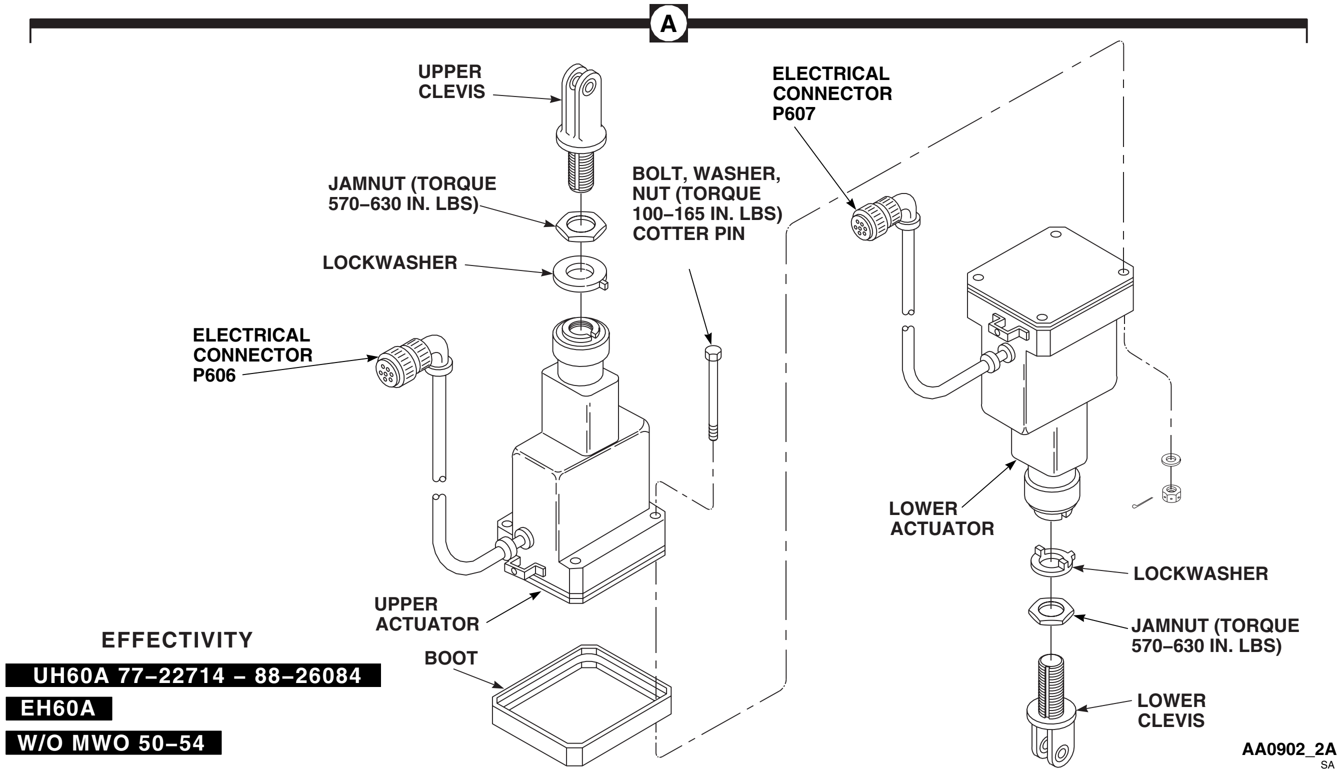


Figure 1. Stabilator Actuator Repair. (Sheet 2 of 2)

2. If primer is not evident proceed as follows:

   a. Apply corrosion removing compound, Item 109, WP 1803 00 or Item 108, WP 1803 00, per manufacturers instructions.

   b. Rinse thoroughly in clean (preferably hot) water.

   c. Allow to air dry completely prior to installation.

   d. Apply epoxy primer coating kit, Item 120, WP 1803 00, to inside surface of bore.

3. Measure ID of clevis bushings. If ID exceeds 0.3760-inch, replace bushings (WP 0375 00).

4. Inspect threaded portion of clevis. **NO NICKS, DENTS, OR CORROSION PITS ALLOWED. REPLACE CLEVIS IF DAMAGE IS FOUND.**

5. On all clevis surfaces, except thread portion and fillet radii, blend out nicks and scratches up to 0.010-inch deep using abrasive cloth, Item 82, WP 1803 00. Magnetic particle inspect clevis (TM 1-1520-265-23) and restore cadmium-plating (TM 1-1500-344-23). Touch up paint (WP 0381 00 and TM 55-1500-345-23).

ASSEMBLE

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

1. Coat threads of upper and lower clevis with corrosion preventive compound, Item 106, WP 1803 00 (Figure 1, Sheet 1, Detail A).
2. Install upper clevis, jamnut, and lockwasher on upper actuator as follows:
   a. Make sure tang on lockwasher is not broken.
   b. Thread upper clevis into actuator same number of turns recorded during disassembly. **MAXIMUM NUMBER OF EXPOSED THREADS ON CLEVIS SHALL NOT EXCEED FOUR.**

NOTE

Make sure correct type lock washers are used on clevis rod ends for the type of actuator that is installed.

   c. Seat lockwasher in key slot in actuator.

WARNING

CSI

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

   d. **QA TORQUE JAMNUT TO 570 - 630 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire jamnut.
   e. Apply torque stripe (WP 0381 00 and TM-55-1500-345-23).
3. Install lower clevis, jamnut, and lockwasher on lower actuator as follows:
   a. Make sure tang on lockwasher is not broken.
   b. Thread lower clevis into actuator same number of turns recorded during disassembly. **MAXIMUM NUMBER OF EXPOSED THREADS ON CLEVIS SHALL NOT EXCEED FOUR.**

NOTE

Make sure correct type lock washers are used on clevis rod ends for the type of actuator that is installed.

   c. Seat lockwasher in key slot in actuator.

WARNING

CSI

Verification of proper hardware installation, torque and safety installation are critical characteristics.

   d. **QA TORQUE JAMNUT TO 570 - 630 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire jamnut.

ASSEMBLE - Continued

e. Apply torque stripe (WP 0381 00 and TM-55-1500-345-23).

NOTE

The following step is required on actuator, 70400-06641-117, only.

4. **MWO 50-54** Inspect chemical conversion coated actuator mating surfaces, bolt hole spotfaces, and grounding bracket mating surfaces for compliance with electrical bonding requirements (WP 0381 00 and TM 55-1500-345-23).

5. **W/O MWO 50-54** Clean actuator mating surfaces with machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

NOTE

- Assemble actuators with electrical cables on same side.
- Boltheads must be on upper actuator.

6. **MWO 50-54** Assemble actuators as follows:

a. Do EME operational check of stabilator actuators as follows:

NOTE

- Resistance check is required on actuator 70400-06641-117 only.
- Resistance measurements must be taken at least three times.

(1) Measure resistance between connector shell and actuator housing on upper and lower actuators using digital low-resistance multimeter. **RESISTANCE SHALL NOT BE MORE THAN 10 MILLIOHMS.**

(2) If resistance is more than 10 milliohms, replace actuator.

b. Position actuator boot on upper actuator as required to leave mating surface exposed (Figure 1, Sheet 1, Detail A).

c. **QA** Position grounding bracket and washers across both actuators.

d. **QA** Secure actuators together with bolts, washers, and nuts. **TORQUE NUTS TO 100 - 165 INCH-POUNDS.** Install cotter pin.

e. Do EME operational check between both stabilator actuator housings as follows:

NOTE

- Resistance check is required only if stabilator actuator assembly is made up of two 70400-06641-117 actuators only.
- Resistance measurements must be taken at least three times.

(1) Measure resistance between both stabilator actuator housings using digital low-resistance multimeter. **RESISTANCE SHALL NOT BE MORE THAN 5 MILLIOHMS.**

(2) If resistance is more than 5 milliohms, disassemble actuators and replace chemical conversion coated surfaces (WP 0381 00 and TM 55-1500-345-23).

f. Do EME operational check between both connector shells as follows:

ASSEMBLE - Continued

NOTE

- Resistance check is required only if stabilator actuator assembly is made up of two 70400-06641-117 actuators only.
- Resistance measurements must be taken at least three times.

(1) Measure resistance between both connector shells using digital low-resistance multimeter. **RESISTANCES SHALL NOT BE MORE THAN 25 MILLIOHMS.**

(2) If resistance is more than 25 milliohms, disassemble actuators and repeat step 6. a. and step 6. e.

(3) If resistance is still more than 25 milliohms, replace actuator assembly.

g. Apply sealing compound, Item 273, WP 1803 00, over mounting hardware and to area where actuators meet. Allow to cure for 24 hours at room temperature, or 2 hours using heat gun at 150°F (65°C) to 180°F (82°C).

h. Position actuator boot on area where actuators meet after sealing compound cures.

7. **W/O MWO 50-54** Assemble actuators as follows:

a. **QA** Secure actuators together with bolts, washers, and nuts. **TORQUE NUTS TO 100 - 165 INCH-POUNDS.** Install cotter pin (Figure 1, Sheet 2, Detail A).

b. Apply sealing compound, Item 273, WP 1803 00, over mounting hardware and to area where actuators meet. Allow to cure for 24 hours at room temperature, or 2 hours using heat gun at 150°F(65°C) to 180°F(82°C).

c. Position actuator boot on area where actuators meet after sealing compound cures.

8. Check both connectors for identification tapes. If necessary, install new identification tapes (TM 1-1500-204-23). Upper actuator connector is marked P606 to J606 UPR STAB ACT. Lower actuator connector is marked P607 to J607 LWR STAB ACT.

CAUTION

On actuator, 70400-06641-117, damage to wiring may result if connector is rotated more than 90 degrees. On all other actuators, connector must not rotate more than 180 degrees.

9. Check actuator housings for minor surface scratches. Polish out any superficial scratches with abrasive cloth, Item 82, WP 1803 00.
10. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
11. Adjust actuators (WP 0344 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ACTUATOR ADJUSTMENT

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Protractor, F7710
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
MIL-STD-1686C
TM 55-1500-345-23
WP 0258 00
WP 0342 00
WP 0343 00
WP 0381 00
WP 1803 00

ADJUSTMENT

WARNING

Injury to personnel and damage to equipment will result due to loss of jamnut torque. Do not torque actuator upper or lower clevises while installed on aircraft. Adjust clevises and torque jamnuts only with actuator removed from aircraft.

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

NOTE

Adjust actuators when upper and lower clevises are not threaded equally into actuators, or when one or both actuators are being replaced.

1. Assemble actuators (WP 0343 00).
2. Turn on all electrical power.
3. Place actuator assembly close to pylon. Connect electrical connector P606, to J606 on tail pylon.
4. Place MAN SLEW switch to UP, until actuator stops. Measure and record length of exposed end of shaft. Add 1.786-inches to this dimension to get rigging dimension for upper actuator (Figure 1).
5. Place MAN SLEW switch to DOWN until exposed length of shaft is same as rigging dimension ± 0.010-inch.
6. Disconnect electrical connector P606 from J606.
7. **QA** Connect electrical connector P607 to J607 on tail pylon.

ADJUSTMENT - Continued

8. Place MAN SLEW switch to UP, until actuator stops. Measure and record exposed end of shaft. Add 1.786-inches to this dimension to get rigging dimension for lower actuator.
9. Place MAN SLEW switch to DOWN until exposed length of shaft is same as rigging dimension ±0.010-inch.
10. Disconnect electrical connector, P607 from J607 on tail pylon.
11. Loosen jamnuts and thread upper and lower clevis into their shafts.
12. Back both clevises out of their shafts four turns. **HANDTIGHTEN JAMNUTS.**
13. Install actuator assembly on stabilator and pylon (WP 0342 00). Do not torque attachment nuts now.
14. Place protractor fixture on top of stabilator between LBL 9.0 and RBL 9.0, with lip of fixture plate over forward edge of stabilator.
15. Pull back fixture hook and secure to trailing edge of stabilator so that lip of fixture fits snugly against forward edge of stabilator.
16. Remove soundproofing panels from cabin ceiling between STA 360.0 and STA 398.0 (WP 0258 00).
17. Place protractor against bottom side of right transmission beam and set protractor scale at 0 degrees.
18. Remove protractor from beam and place on protractor fixture or bar through fold fittings.
19. **ANGLE OF STABILATOR SHOULD BE 7.5 DEGREES TO 8.5 DEGREES (TRAILING EDGE UP).**
20. If stabilator is not at proper angle, remove actuator assembly and thread both upper and lower clevis in or out at same time to get proper reading. **MAXIMUM NUMBER OF EXPOSED THREADS ON EITHER CLEVIS SHALL NOT EXCEED FOUR.**

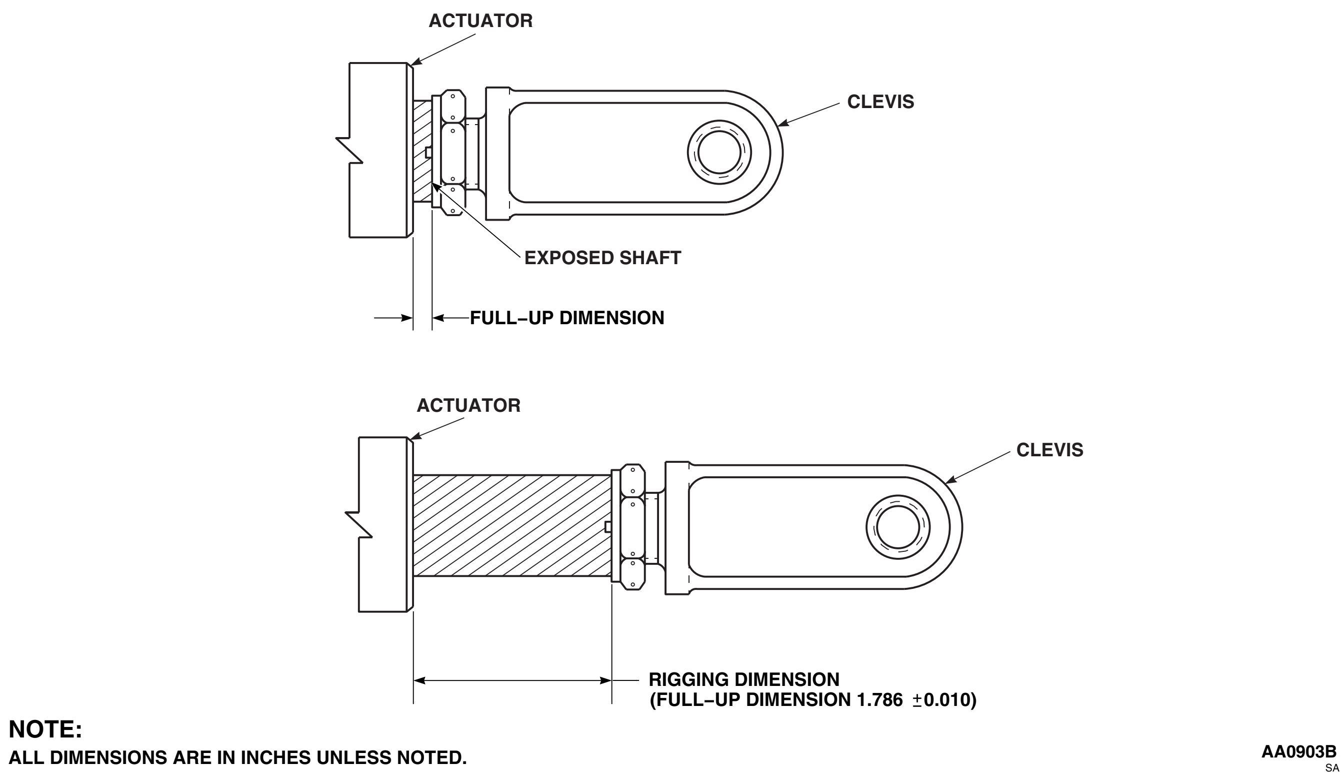


Figure 1. Stabilator Actuator Adjustment.

ADJUSTMENT - Continued

21. When stabilator is at 7.5 degrees to 8.5 degrees, remove actuator assembly. Perform final torque on upper and lower clevis as follows:

    a. **QA** Make sure lockwasher is seated in key slot in actuator.

WARNING

CSI

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

    b. **QA TORQUE JAMNUT TO 570 - 630 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire jamnut.

    c. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

22. Turn off all electrical power.
23. Install actuator assembly on helicopter (WP 0342 00).
24. Install soundproofing panels to cabin ceiling between STA 360.0 and STA 398.0 (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ACTUATOR GROUNDING STRAP UH60A 88-26085 - SUBQ UH60L MWO 50-54

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Low-Resistance Digital Multimeter, DLRO247000
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

MIL-STD-1686C
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

1. Remove nut and washers securing grounding strap to grounding bracket on actuator assembly (Figure 1, Detail A). Remove strap from bracket.
2. Remove bolt, washers, and nut securing grounding strap to rear spar. Remove strap from rear spar.

INSTALL

CAUTION

The 70400-06641-117 actuator contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when touching, removing or inserting (MIL-STD-1686C).

1. **QA** Secure 45 degree strap end fitting to rear spar with bolt, washers, lockwasher, and nut (Figure 1, Detail A). **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
2. **QA** Secure strap to grounding bracket on actuator assembly with bolt, washers, lockwasher, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
3. Do EME operational check as follows:
   a. Check resistance reading from stabilator actuator to airframe using digital multimeter. **RESISTANCES MUST NOT EXCEED 10 MILLIOHMS.**
   b. If resistance is greater than 10 milliohms, inspect chemical conversion coated surfaces (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

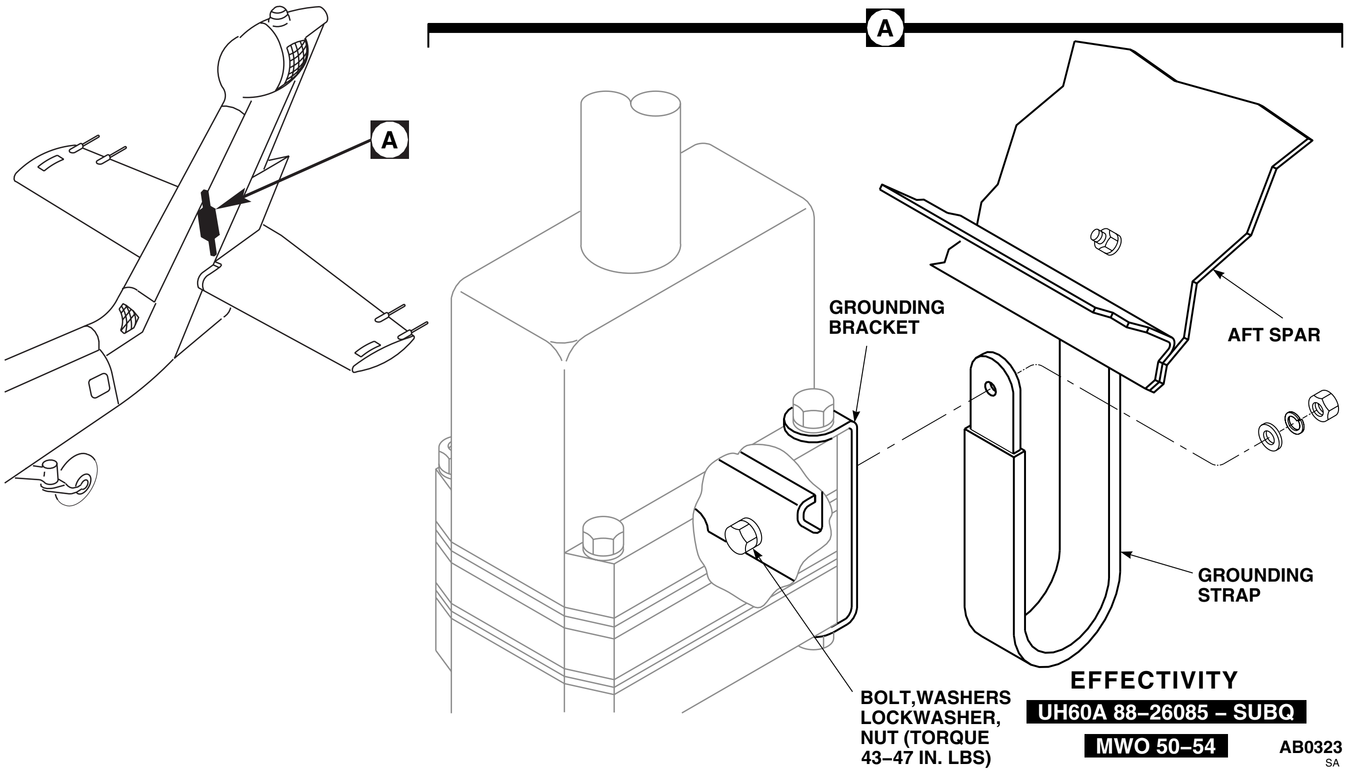


Figure 1. Replace Stabilator Actuator Grounding Strap.

4. Cover nuts, bolts, and washers securing strap to rear spar and bracket with sealing compound, Item 273, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

POSITION TRANSMITTER LIMIT SWITCH, 70400-06705

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 11-1520-237-23
WP 0329 00
WP 0330 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open pylon leading edge fairing.
3. Disconnect electrical connectors, P604 and P605, from helicopter receptacles (Figure 1, Detail A).
4. Remove left and right stabilator pylon fairings (WP 0330 00).
5. Remove lower pylon fairing (WP 0329 00).
6. Remove bolts and washers from mount assembly and remove cover.
7. Remove lockwire holding locknut to actuator rod.
8. Loosen and remove locknut from actuator rod.
9. Remove bolts and washers holding mount assembly to pylon.

CAUTION

Damage may occur to actuator rod if mount assembly is removed before actuator rod is disconnected.

10. Hold mount assembly so that sensor lever arm is in a fixed position (Figure 1, Detail B).
11. Turn lower end of actuator rod to unthread rod from sensor lever trunnion.
12. Remove mount assembly from helicopter.

INSTALL

1. Place mount assembly so that sensor lever trunnion is lined up with actuator rod.
2. Hold sensor lever centered between microswitches, and thread actuator rod into sensor lever trunnion until mount assembly is firm against pylon (Figure 1, Detail A and Detail B).
3. **QA** Install mount assembly to pylon with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

INSTALL - Continued

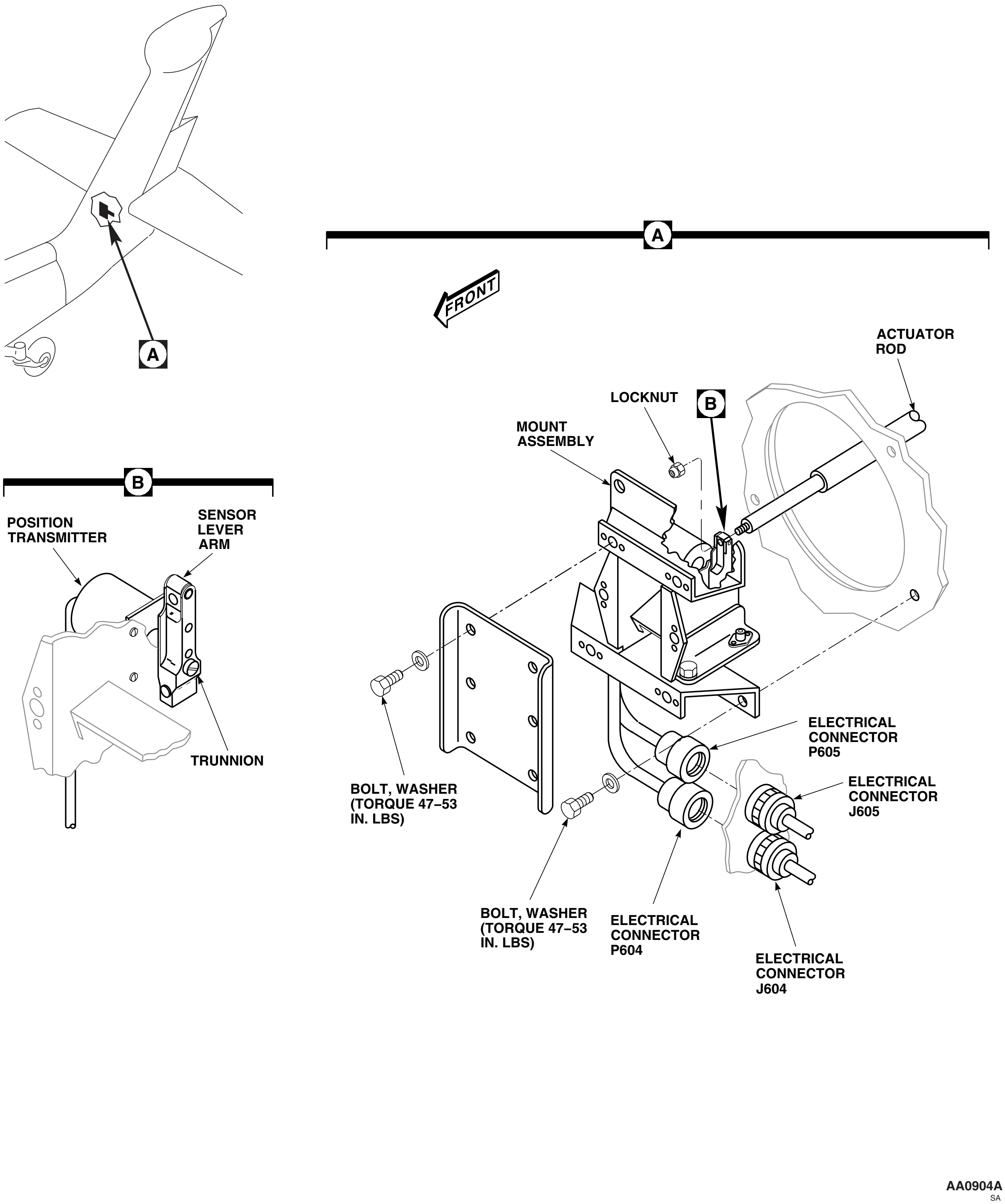


Figure 1. Replace Position Transmitter Limit Switch, 70400-06705.

INSTALL - Continued

4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect electrical connectors, P604 and P605, to helicopter receptacles, J604 and J605.
6. Install cover to mount assembly with bolts and washers.
7. Do a position transmitter adjustment (TM 11-1520-237-23).
8. Install and tighten locknut on actuator rod.
9. **QA** Using nonelectrical wire, Item 354, WP 1803 00, lockwire locknut.
10. Install left and right stabilator pylon fairings (WP 0330 00).
11. Install lower pylon fairing (WP 0329 00).
12. Do an operational check of stabilator (TM 11-1520-237-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

POSITION TRANSMITTER LIMIT SWITCH, 70400-06712

This WP supersedes WP 0347 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive Primer, Item 23, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 105, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 1-1520-237-MTF
TM 11-1520-237-23
TM 55-1500-345-23
WP 0069 00
WP 0329 00
WP 0330 00
WP 0377 00
WP 0381 00
WP 0669 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove left and right stabilator/pylon fairings (WP 0330 00).
3. Remove lower pylon fairing (WP 0329 00).
4. Remove bolt, bushing, washers, nut, and actuator rod end from stabilator fitting (Figure 1, Detail A).
5. Mark jamnut location on actuator rod and record number of free threads above nut.
6. Loosen jamnut. Remove rod end, locking device, and jamnut from actuator rod.
7. Remove section IV tail rotor drive shaft (WP 0669 00).

NOTE

If bolts are installed, replace them with screws.

8. Remove screws, washers, spacers, and cover from housing.
9. Disconnect electrical connectors, P604 and P605, from electrical connectors, J604 and J605 (Figure 1, Detail C).

REMOVE - Continued

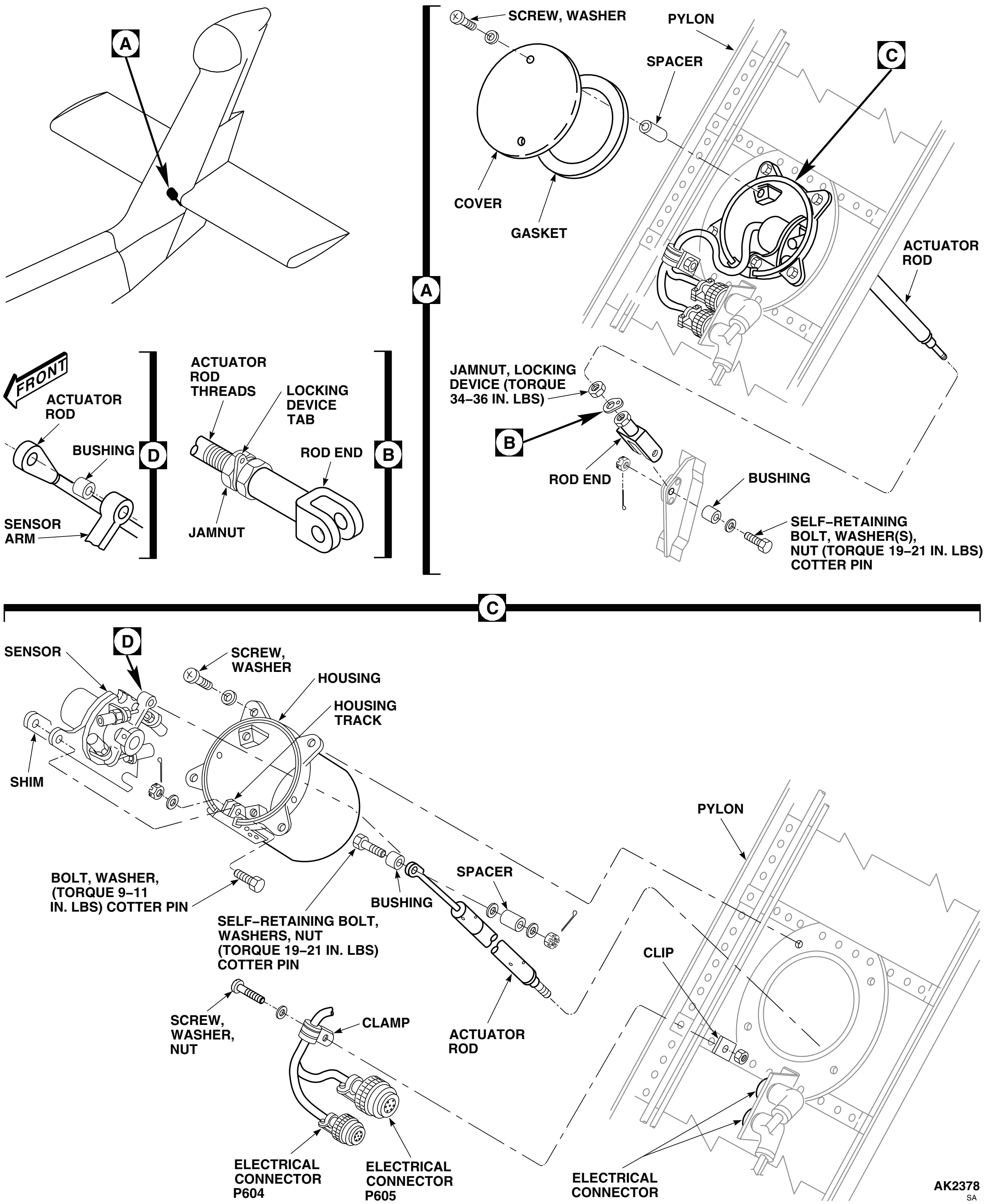


Figure 1. Replace Position Transmitter Limit Switch, 70400-06712.

REMOVE - Continued

10. Remove screw, washer, nut, and clamp holding electrical harness on pylon.
11. Remove bolt, washer, and nut holding sensor in housing. Slide sensor from housing.
12. Remove bolt, washers, nut, spacer, and bushing holding actuator rod on transmitter. Remove actuator rod.

INSPECT

1. Check housing for dents, cracks, corrosion, and broken welds (TM 1-1500-204-23 and TM 1-1500-344-23). Replace if damaged.
2. Check housing cover for loose or damaged gasket. Rebond gasket to cover, if necessary, with sealing compound, Item 273, WP 1803 00.
3. Visually check actuator rod for cracks, distortion, and thread damage. Replace if cracked, visibly bent, or has damaged threads.
4. Check bushings for corrosion and wear. Replace if visibly worn or corroded.

INSTALL

1. If damaged housing is being replaced, ensure good electrical bonding as follows:
   a. Remove any paint or other solid material from mating surfaces of housing lugs and pylon spar, using nonmetallic scraper or file to avoid abrasive residue.
   b. Thoroughly clean mating surfaces, using cheesecloth cloth, Item 85, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00.
   c. Wipe dry with clean cheesecloth.
   d. Apply protection against corrosion to all mating surfaces, using corrosion resistant coating, Item 110, WP 1803 00.
   e. Install housing on pylon spar and fasten with screws and washers (Figure 1, Detail C).
   f. Seal all edges of mating surfaces with sealing compound, Item 273, WP 1803 00 (WP 0377 00).

WARNING

CSI

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

2. Check that actuator rod lower end has mark for jamnut on it. If it does not, do this:
   a. Install jamnut, locking device, and lower rod end on rod.
   b. Thread lower rod end on rod so that distance between rod end holes is 15/64 to 23/64-inches.
   c. Note number of free threads, and mark location of jamnut.
   d. Remove lower rod end, jamnut, and locking device from rod end.
3. Adjust position sensor and limit switches (WP 0069 00).

NOTE

Actuator rod end is installed inboard of sensor arm.

4. **QA** Install actuator rod on sensor.

INSTALL - Continued

5. Coat actuator limit switch actuator rod bolt and bushing with corrosion preventive compound, Item 105, WP 1803 00. Do not apply compound to bolt threads.

WARNING

CSI

Proper hardware installation, torque, proof torque, and safetying of self-retaining bolt are critical characteristics and must be verified.

6. Install self-retaining bolt, washer(s), spacer, and bushing, as follows:

NOTE

Do not place more than three washers under nut. Do not place any washers under bolt-head.

a. Use washers under nut (not more than 3) to adjust position of bolt self-retaining element.

b. Place washers in any required combination between sensor arm and spacer, between spacer and nut, or some in both locations.

c. Bolt self-retaining element must ride in spacer grooves without being restrained or depressed. With selected washer(s) in place and nut not installed, perceptible axial play must be present.

7. **QA** Install nut on bolt. **TORQUE NUT TO 19 - 21 INCH-POUNDS.** Install cotter pin.

8. Slide sensor base and shim into track in housing. Check for sideplay. **SIDEPLAY MUST BE LESS THAN 0.003-INCH.** Peel shim as necessary to get clearance.

9. Bond shim to sensor base, as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Clean bond surfaces with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

NOTE

Either adhesive, Item 11, WP 1803 00, or adhesive, Item 14, WP 1803 00, may be used.

b. If adhesive, Item 14, WP 1803 00, is used, apply even coat to both bond surfaces.

(1) Let adhesive dry for 10 minutes until tacky.

(2) Install shim on sensor base, and allow adhesive to cure for at least 1 hour.

c. If adhesive, Item 11, WP 1803 00, is used, do this:

(1) Apply a thin, even coat of adhesive primer, Item 23, WP 1803 00, to both bond surfaces.

(2) Let primer dry for 1 hour.

(3) Apply even coat of adhesive, Item 11, WP 1803 00, to both bond surfaces.

INSTALL - Continued

(4) Immediately install shim on sensor base.

(5) Cure adhesive for 2 hours at 110°F(43°C) to 130°F(54°C).

10. **QA** Install bolt, washer, and nut in track. **TORQUE NUT TO 9 - 11 INCH-POUNDS.** Install cotter pin. Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolthead to sensor base.
11. Pass length of lockwire through holes in pylon that actuator rod will pass through. Attach wire to lower rod end. Have assistant stand behind pylon to help guide rod end through hole in pylon rear spar.
12. **QA** Install housing in pylon with screws and washers.
13. **QA** Install wiring harness on pylon with screw, washer, nut, and clamps.
14. Inspect and treat electrical connector (WP 1703 00).
15. **QA** Connect electrical connectors, P604 and P605, to electrical connectors, J604 and J605.
16. **QA** Install cover and gasket on housing with bolts, washers, and spacers.
17. Install jamnut on lower end of actuator rod to mark, leaving number of free threads above nut as recorded during removal.

WARNING

CSI

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

18. Install rod end, as follows:
    a. Install locking device against jamnut.
    b. Thread rod end onto actuator rod, engaging locking device lugs in rod end slots so that final tab position, when snugly against jamnut, is as shown (Figure 1, Detail B).
    c. **QA TORQUE JAMNUT TO 34 - 36 INCH-POUNDS.**
    d. Verify thread engagement by inserting nonelectrical wire, Item 354, WP 1803 00, into rod end inspection hole nearest jamnut (some rod ends have two inspection holes). If nonelectrical wire passes through, review rod end adjustment.
    e. **QA** Lockwire jamnut to locking device tab using nonelectrical wire, Item 354, WP 1803 00.
19. **QA** Install lower actuator rod end on stabilator.
20. Coat sensor/limit switch actuator rod bolt and bushing with corrosion preventive compound, Item 105, WP 1803 00. Do not apply compound to bolt threads.
21. Install self-retaining bolt, washer(s) under head, and bushing, as follows:
    a. Install bolt with washers under head (3 or less) to ensure correct grip length and cotter pin position.
    b. Make sure that bolt has noticeable axial motion and self-retaining element is not restrained or depressed. Change washer stackup for correct installation.
22. **QA** Install nut on bolt. **TORQUE NUT TO 19 - 21 INCH-POUNDS.** Install cotter pin.
23. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
24. Do stabilator operational check (TM 11-1520-237-23).

INSTALL - Continued

25. Make sure area is clean and free of foreign material.
26. Install section IV tail rotor drive shaft (WP 0669 00).
27. Install lower pylon fairing (WP 0329 00).
28. Install left and right stabilator/pylon fairings (WP 0330 00).
29. Check DA Form 2408-13-1 to make sure section IV tail rotor drive shaft installation torque check is shown as due 9 to 11 flight hours after installation.
30. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

BATTERY ACCESS DOOR HH-60L

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Pad, Scouring, Item 205, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0381 00
WP 0931 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

WARNING

Injury to personnel will result if battery access door is not carefully opened. Battery weighs approximately 35 pounds and is mounted to inside of battery access door. Battery access door must be supported while screws are being removed to prevent battery access door from abruptly swinging open.

2. With an assistant supporting battery access door, remove screws and washers securing forward end and both sides of battery access door only. Carefully allow battery access door swing open (Figure 1, Detail A).
3. Remove battery (WP 0931 00).
4. With an assistant supporting battery access door, remove screws securing cable support bracket to door, remove remaining screws and washers securing battery access door to hinge. Remove battery access door.

INSPECT/REPAIR

1. Visually inspect battery access door, battery support angles, and hinge, for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked item or door assembly, as required.
2. Remove corrosion and blend out wear, using scouring pad, Item 205, WP 1803 00. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.
3. If support angles require replacement do the following (Figure 1, Detail B):
   a. Remove bolts securing support angles to door. Remove support angles.
   b. Visually inspect battery access door for corrosion.
   c. Remove corrosion, using scouring pad, Item 205, WP 1803 00.
   d. Position new support angles on battery access door and secure using bolts.
4. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSPECT/REPAIR - Continued

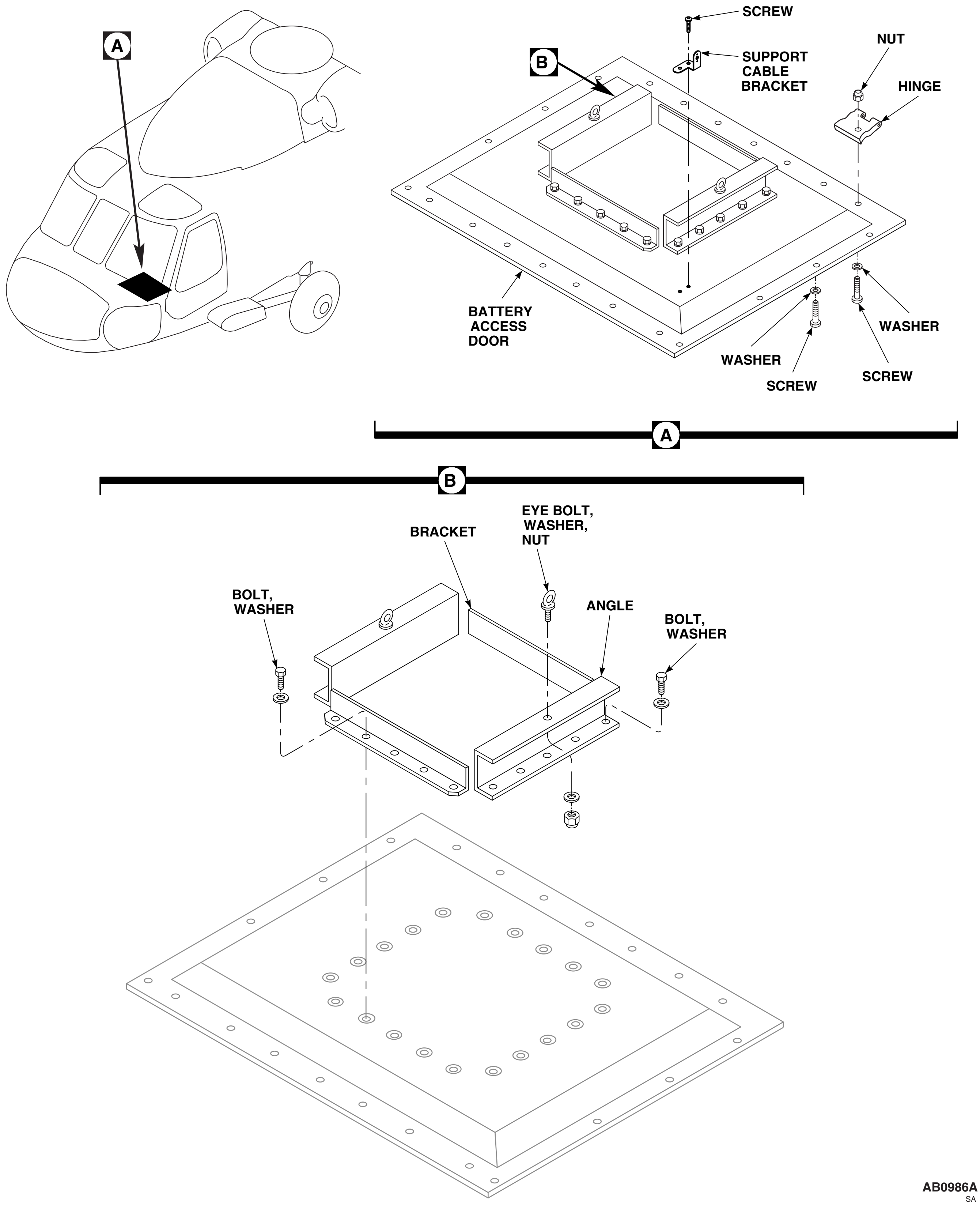


Figure 1. Replace Battery Access Door.

INSTALL

1. Turn off all electrical power.
2. **QA** With an assistant, position battery access door in open position on helicopter, securing to hinge using screws, washers, and nuts. Position cable support bracket on door and secure using screws (Figure 1, Detail A).
3. Install battery (WP 0931 00).
4. Make sure area is clean and free of foreign material before closing up.

WARNING

Injury to personnel will result if battery access door is not carefully supported. Battery weighs approximately 35 pounds and is mounted to inside of battery access door. Battery access door must be supported while screws are being installed to prevent battery access door from abruptly swinging open.

5. **QA** With an assistant supporting door in closed position, secure battery access door with remaining screws and washers.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

AVIONICS COMPARTMENT ACCESS DOOR HH-60L

This WP supersedes WP 0349 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Digital Low-Resistance Ohmmeter, Model DL-RO247000
Electrical Repairer's Toolkit, SC 5180-99-B06
Gloves
Goggles/Face Shield
Grease Pencil
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 15, WP 1803 00
Adhesive, Item 40, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Grease, Aircraft, Item 132, WP 1803 00
Isopropyl Alcohol, Technical, Item 170, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REPLACE AVIONICS COMPARTMENT ACCESS DOOR

Remove

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door (Figure 1, Detail A).
3. Support door in open position while disconnecting hinge halves.
4. Remove screws and retainers.
5. Remove cotter pins and pins holding hinge halves together. Remove door.

Install

1. Turn off all electrical power.
2. Place access door in open position on helicopter, engaging hinge halves.
3. **QA** Line up hinge bores and install pins and cotter pins, in each hinge (Figure 1, Detail A).
4. **QA** Install retainers with screws.
5. Close door and check if door is centered within door opening.
6. If centering is required, adjust door by loosening screws and nuts that hold door hinge fittings and reposition door as necessary. Retighten screws.
7. Make sure area is clean and free of foreign material before closing up.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR - Continued

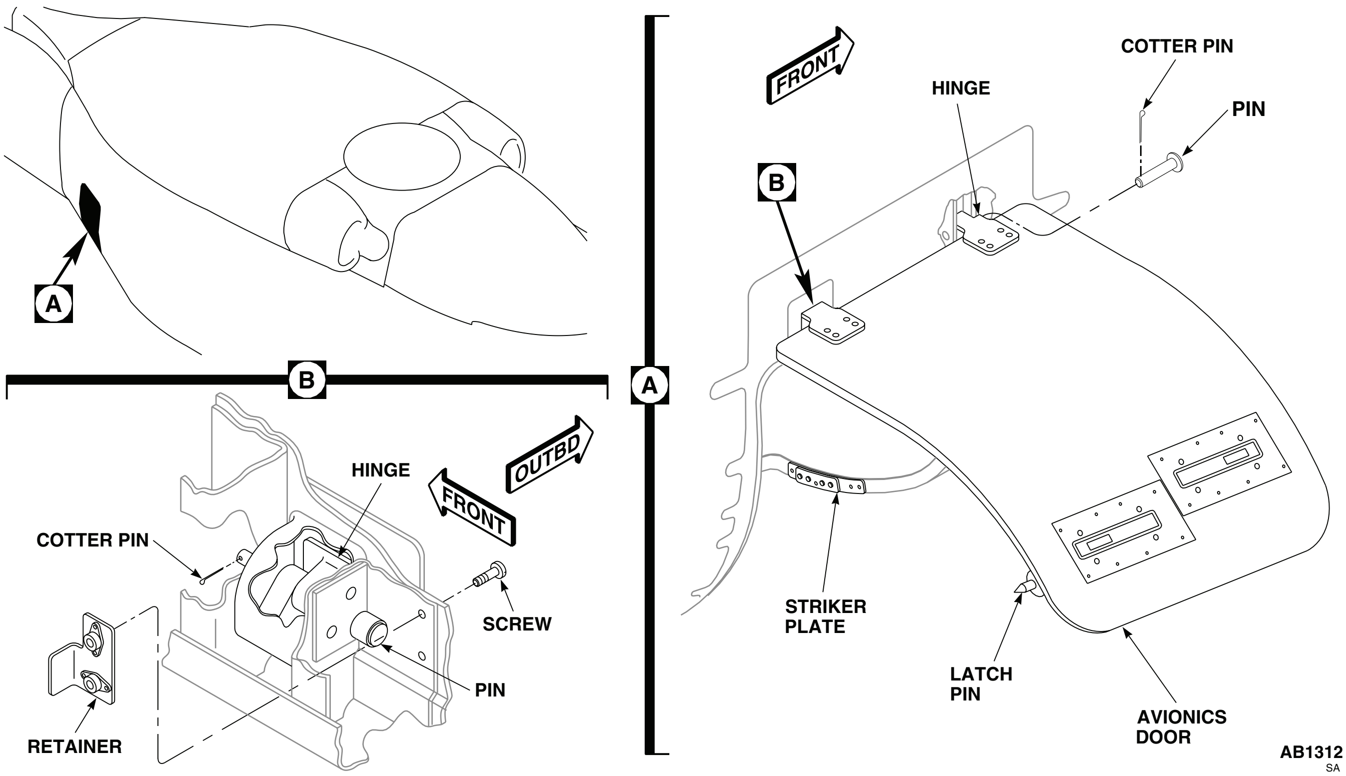


Figure 1. Replace Avionics Compartment Access Door.

8. Close door and fasten latches.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR STRUT

Remove

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Support door in open position with temporary support.
4. Remove lock end of strut from stowage clip on bulkhead (Figure 2, Detail D).
5. Remove cotter pin, washer, and pin securing strut clevis to eyebolt on compartment bulkhead. Remove strut (Figure 2, Detail A).

Install

1. Turn off all electrical power.
2. **QA** Install clevis end of strut on eyebolt with pin (Figure 2, Detail A). Install washer, and cotter pin.
3. Engage strut lock with fitting on door.
4. Remove temporary door support.
5. Make sure area is clean and free of foreign material before closing up.
6. Disengage strut lock from door fitting and stow strut end in spring clip on bulkhead.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR STRUT - Continued

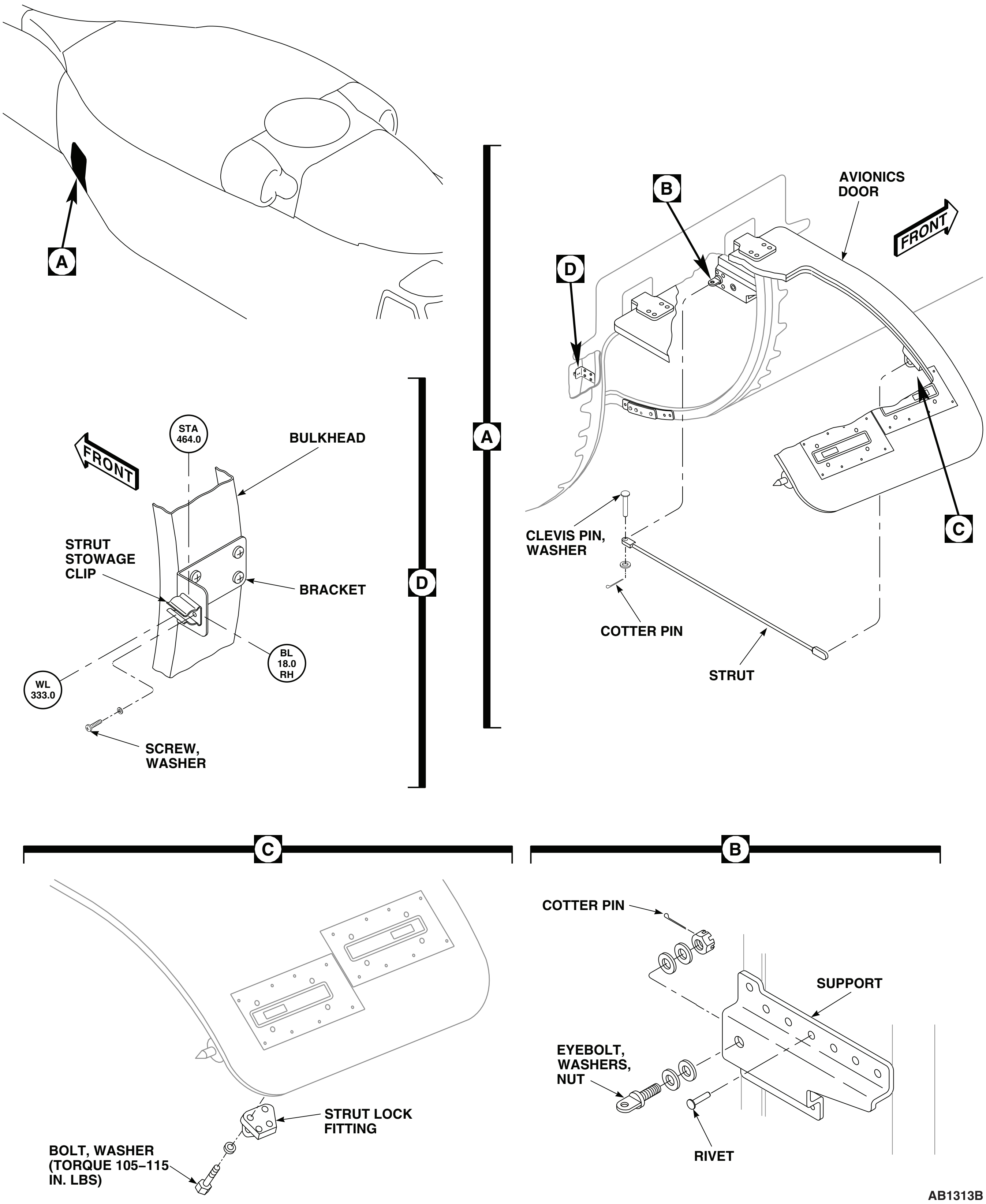


Figure 2. Replace Avionics Compartment Access Door Strut, Eyebolt, Lock Fitting.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR STRUT - Continued

7. Close door and fasten latches.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR EYEBOLT

Remove

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Support door in open position with temporary support.
4. Drill out rivets holding eyebolt support to bulkhead (TM 1-1500-204-23) (Figure 2, Detail B). Remove support with eyebolt.
5. Remove nut, washers, and eyebolt from support.

Install

1. Turn off all electrical power.

CAUTION

Equipment can be damaged if eyebolt nut is too tight. Do not overtighten nut. Eyebolt should turn freely after cotter pin is installed.

2. **QA** Install eyebolt, washers, and nut on support (Figure 2, Detail B). Tighten nut fingertight. If cotter pin hole in eyebolt does not line up with slots in nut, loosen nut to next slot to install cotter pin. Install cotter pin.
3. **QA** Install eyebolt support on compartment bulkhead and rivet in place (TM 1-1500-204-23).
4. Remove temporary door support.
5. Make sure area is clean and free of foreign material before closing up.
6. Close door and fasten latches.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR STRUT LOCK FITTING

Remove

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Support door in open position with temporary support.
4. Remove bolts, washers, and strut lock fitting from access door (Figure 2, Detail C).

Install

1. Turn off all electrical power.
2. Install strut lock fitting on access door with bolts and washers (Figure 2, Detail C).
3. **QA TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Using nonelectrical wire, Item 354, WP 1803 00, lockwire boltheads together.
4. Remove temporary door support.
5. Make sure area is clean and free of foreign material before closing up.
6. Close door and fasten latches.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR STRUT LOCK FITTING - Continued

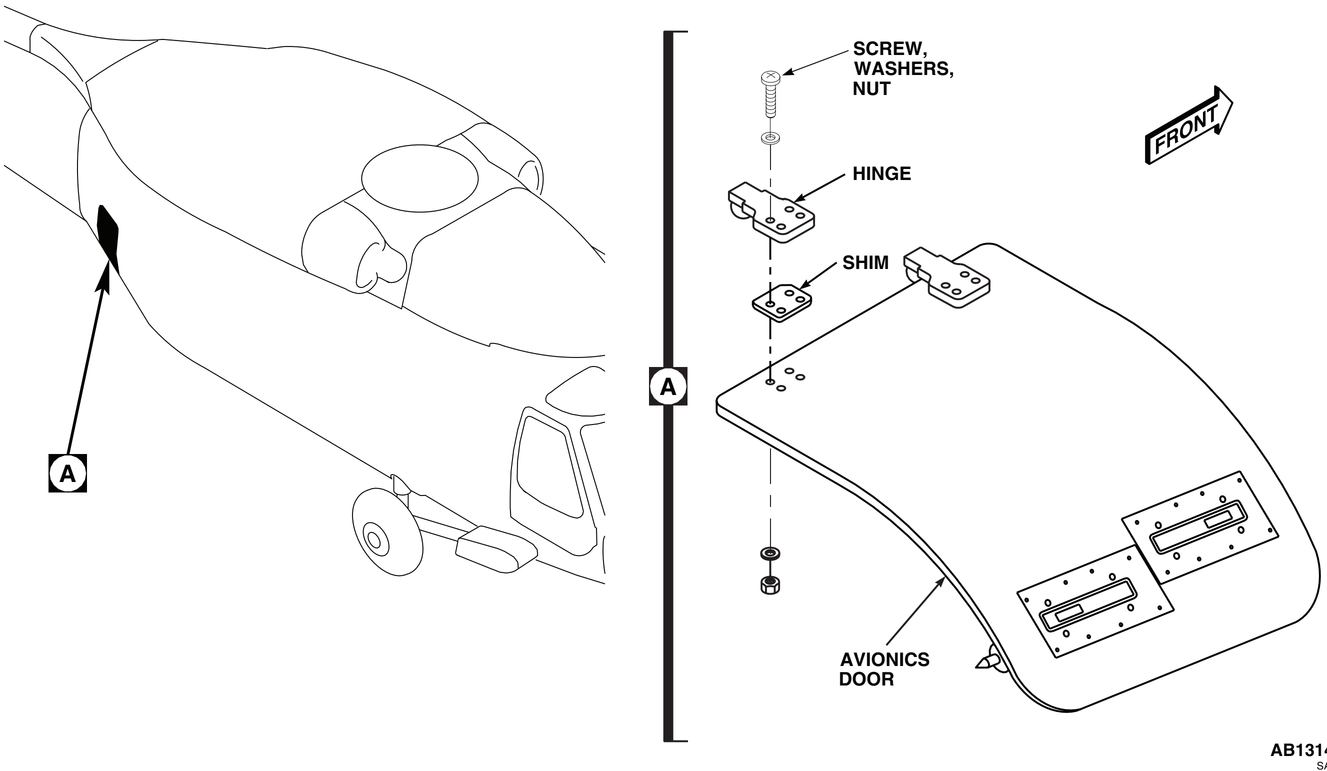


Figure 3. Replace Avionics Compartment Access Door Hinge.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR HINGE

Remove

1. Turn off all electrical power.
2. Remove avionics compartment access door. Refer to, REPLACE AVIONICS COMPARTMENT ACCESS DOOR, in this work package.
3. Remove screws, washers and nuts holding hinge and shim to door (Figure 3, Detail A). Remove hinge from door.

Install

1. Turn off all electrical power.
2. **QA** Assemble hinge parts together with pins and cotter pins. Install retainers with screws.
3. Position door on aft transition section of helicopter.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR HINGE - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Wipe hinge clean with a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
5. Apply sealing compound, Item 273, WP 1803 00, to door hinge screw holes and screws.
6. Apply adhesive, Item 15, WP 1803 00, to raised pad area on avionics door.
7. Immediately, install hinge and shim on door with screws, washers, and nuts (Figure 3, Detail A).
8. Close door. See if door is centered within door opening.
9. If centering is required, adjust door by loosening screws and nuts that hold door hinge and reposition door as necessary. Retighten screws.
10. Wipe off extra sealing compound.
11. Make sure area is clean and free of foreign material before closing up.
12. Close door and fasten latches.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR JAMB GASKET

Remove

1. Turn off all electrical power.
2. Remove avionics compartment access door. Refer to, REPLACE AVIONICS COMPARTMENT ACCESS DOOR, in this work package.
3. Remove adhesive fillets sealing gasket edges with nonmetallic scraper.
4. Pull gasket out of retainers and remove from jamb (Figure 4, Detail A).

Install

1. Turn off all electrical power.
2. Clean chemical conversion coated (mating) surfaces (around all mounting holes) with cheesecloth cloth, Item 85, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00. Dry surfaces with clean, dry cheesecloth cloth, Item 85, WP 1803 00.
3. Inspect chemical conversion coated (mating) surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (WP 0381 00 and TM 55-1500-345-23).
4. Starting at bottom center of jamb, insert gasket into retainers, continuing around door opening.
5. Cut off free end of gasket to meet other end in splice (Figure 4, Detail A).

CAUTION

Adhesive on contacting surfaces of gasket can adversely affect electrical bonding. All adhesive must be kept away from these surfaces.

6. Apply adhesive, Item 40, WP 1803 00, to each end of gasket, join, and insert gasket into bottom retainer.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR JAMB GASKET - Continued

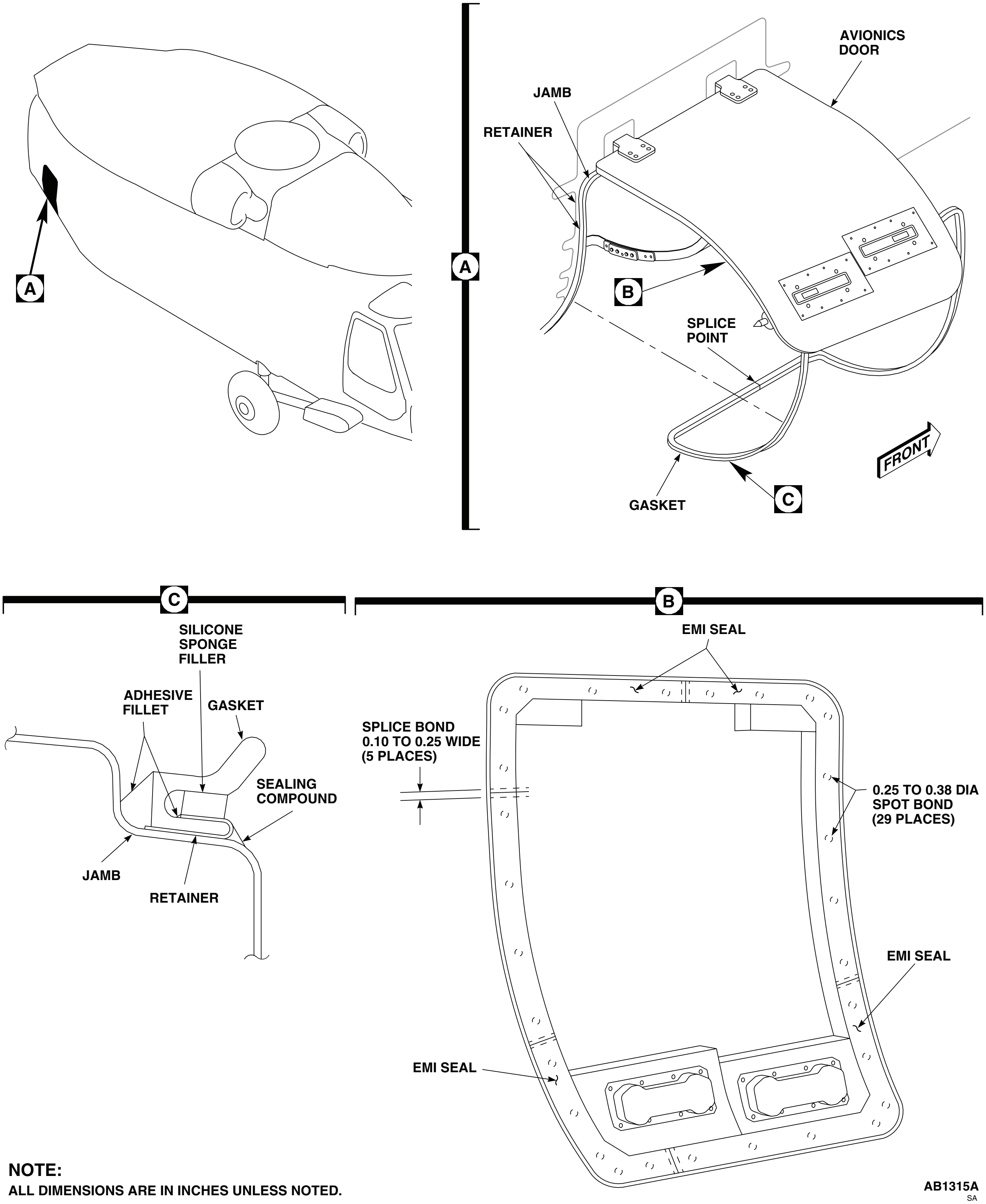


Figure 4. Replace Avionics Compartment Access Door Gasket Seals.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR JAMB GASKET - Continued

7. Install avionics compartment access door. Refer to, REPLACE AVIONICS COMPARTMENT ACCESS DOOR, in this work package.
8. As required for full contact between door and gasket, insert silicone sponge filler into gasket slot (Figure 4, Detail C). Fix filler in place with adhesive, Item 40, WP 1803 00.
9. Seal all edges of gasket and filler with fillet of adhesive, Item 40, WP 1803 00.

WARNING

Isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated areas. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If congestion occurs induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

10. Clean conductive seals and gaskets on door and jamb, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical isopropyl alcohol, Item 170, WP 1803 00.
11. Do EME operational check as follows:
    a. Check resistance from seals to door jamb using digital ohmmeter. **RESISTANCES MUST NOT EXCEED 100 MILLIOHMS.**
    b. If resistance is greater than 100 milliohms, remove gasket and apply chemical conversion coating (WP 0381 00, and TM 55-1500-345-23).
12. If required, replace or repair fillet of sealing compound, Item 273, WP 1803 00, at joining of retainers and jamb (Figure 1, Detail C).
13. Make sure area is clean and free of foreign material before closing up.
14. Close door and fasten latches.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR SEALS

Remove

1. Turn off all electrical power.
2. Remove avionics compartment access door. Refer to, REPLACE AVIONICS COMPARTMENT ACCESS DOOR, in this work package.
3. Remove old seals from door using nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Remove remaining adhesive using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

Install

1. Turn off all electrical power.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR SEALS - Continued

2. Wipe mating surfaces of seals and door with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Apply adhesive, Item 40, WP 1803 00, as spot bonding (Figure 4, Detail B), and install seals on door.
4. Fill spaces between adjacent seals, 0.10 to 0.25-inch wide, with adhesive, Item 40, WP 1803 00.
5. Apply fillet of adhesive, Item 40, WP 1803 00, along inner and outer edges of seals all around door.
6. Allow adhesive to cure for at least 24 hours.
7. Install avionics compartment access door. Refer to, REPLACE AVIONICS COMPARTMENT ACCESS DOOR, in this work package.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR LATCH

NOTE

All instructions are the same for both latches.

Remove

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Remove screws and washers, and remove cover (Figure 5, Sheet 1, Detail B).
4. Remove screws, washers, and nuts, and remove latch.

Repair

1. Remove screws, washers, radius block and nuts and separate support angles, laminated shims and latch (Figure 5, Sheet 2, Detail E).
2. Lubricate moving parts in latch with aircraft grease, Item 132, WP 1803 00.
3. Assemble latch, laminated shims and support angles with screws, washers, radius blocks and nuts.

Install

1. Turn off all electrical power.
2. Fasten latch support angles to door with screws, washers, and nuts (Figure 5, Sheet 1, Detail B).
3. **QA** Install cover with screws and washers.
4. Apply fillet of sealing compound, Item 273, WP 1803 00, all around joint of cover with access door.
5. Make sure area is clean and free of foreign material before closing up.
6. Close door and fasten latches.

Adjust

1. Turn off all electrical power.
2. Unlatch and open avionics compartment access door.
3. Coat latch bolt with grease pencil to provide mark.
4. Close door and fully engage latch. Reopen door. **MEASURE MARK ON LATCH BOLT. AT LEAST 1 INCH OF LATCH BOLT MUST EXTEND INTO HOLE ON STRIKER PLATE ON DOOR FRAME (Figure 5, Sheet 2, Detail D)**.

REPLACE AVIONICS COMPARTMENT ACCESS DOOR LATCH - Continued

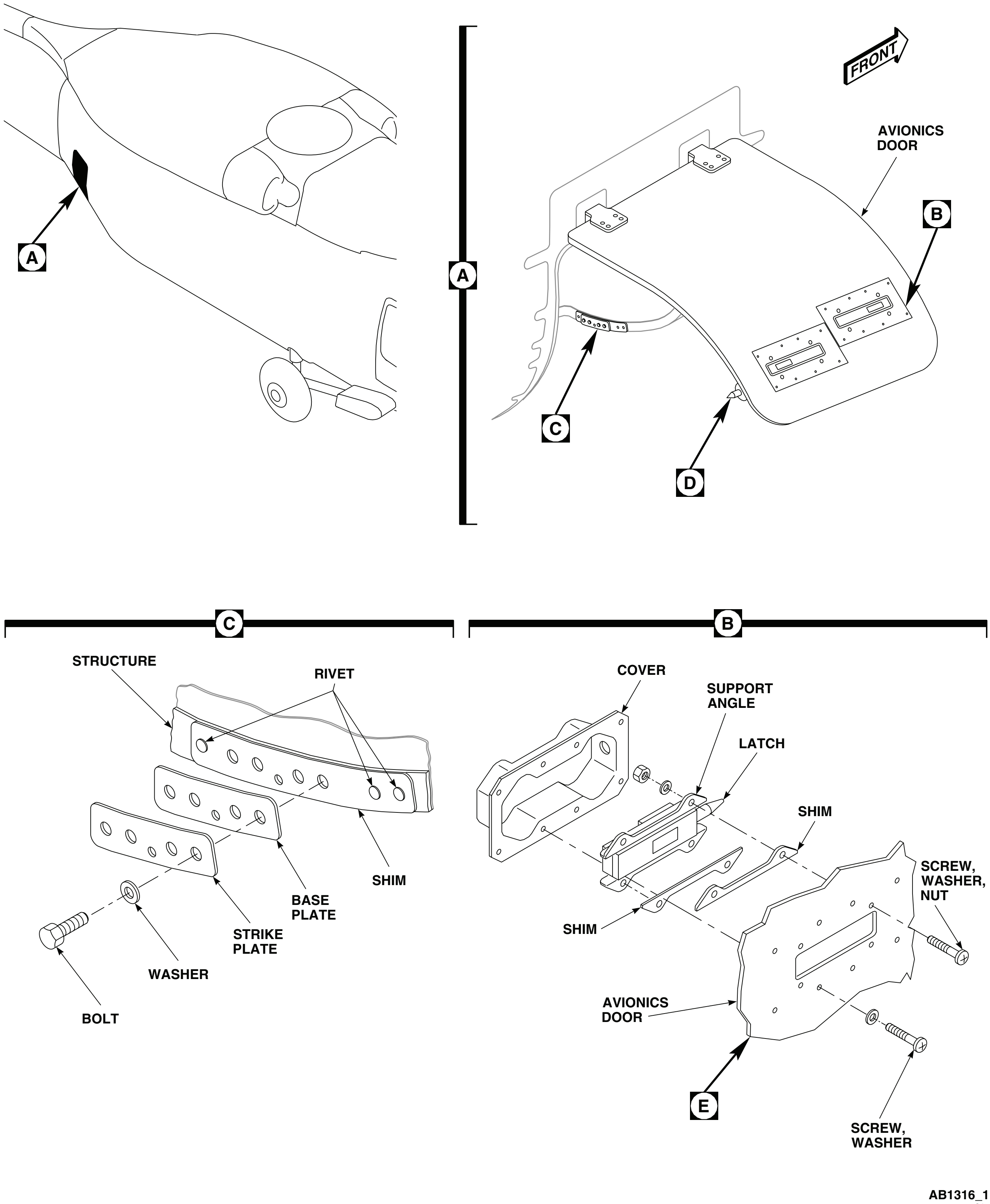


Figure 5. Replace/Maintain Avionics Compartment Access Door Latch. (Sheet 1 of 2)

REPLACE AVIONICS COMPARTMENT ACCESS DOOR LATCH - Continued

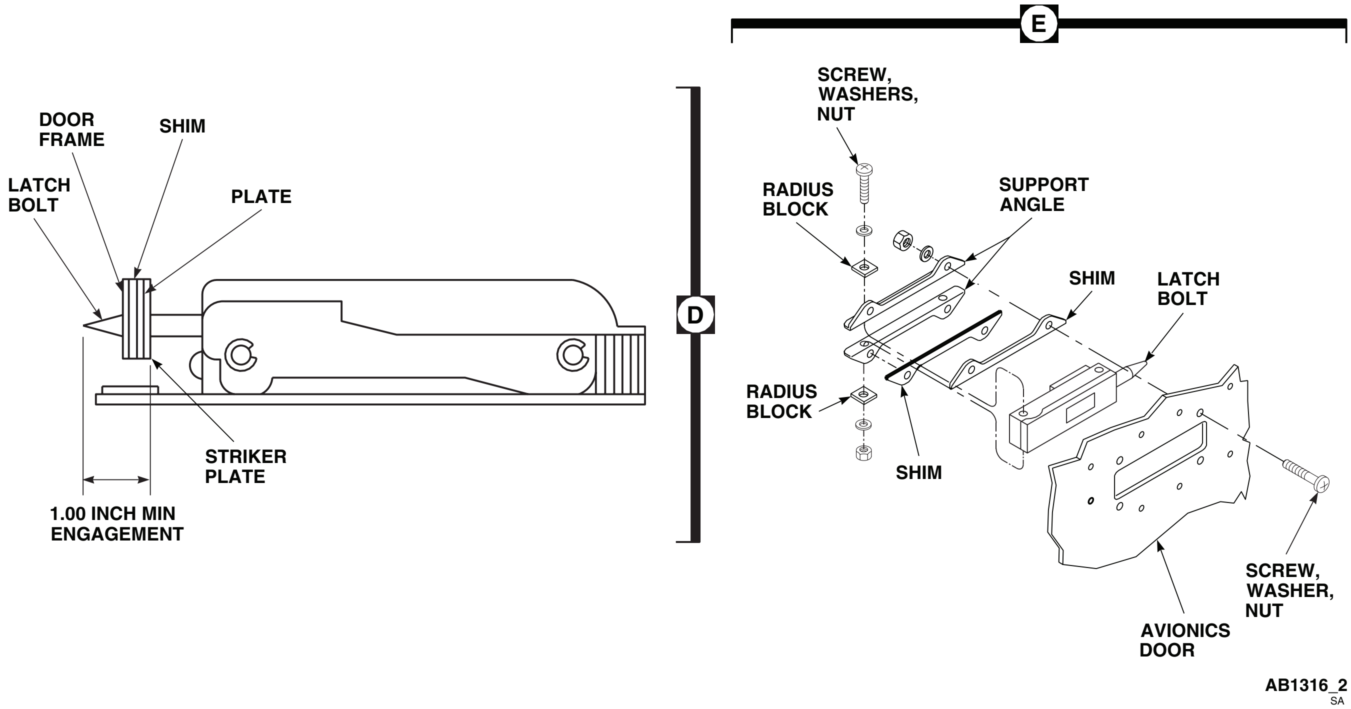


Figure 5. Replace/Maintain Avionics Compartment Access Door Latch. (Sheet 2 of 2)

5. If limit is not met, adjust position of striker plate and baseplate by doing this:
   a. Determine difference between actual engagement of latch bolt and required minimum engagement.
   b. Remove screws holding striker plate and baseplate to door frame. Remove shim if installed under baseplate.
   c. Peel new shim so that latch bolt will extend 1 inch or more beyond striker plate surface.

NOTE

A new shim is 0.125-inch thick and consists of 0.002-inch laminations.

   d. Install shim, baseplate, and striker plate with screws.

NOTE

Serrated faces of plates should face each other. Baseplate has largest latch bolt hole.

   e. Close and latch access door. Recheck depth of latch bolt engagement.
6. Make sure area is clean and free of foreign material before closing up.
7. Close door and fasten latches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

RESCUE HOIST CABLE BUMPER GUARD

This WP supersedes WP 0350 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Pad, Scouring, Item 205, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. Turn off all electrical power.

NOTE

Landing gear does not have to be removed from helicopter, only access to rescue hoist cable bumper guard screws is required.

2. Remove screws securing land gear fairing to helicopter.

WARNING

Injury to personnel will result if rescue hoist cable bumper guard is not supported correctly for removal. Rescue hoist cable bumper guard must be supported while screws are being removed to prevent rescue hoist cable bumper guard from falling.

3. With an assistant supporting rescue hoist cable bumper guard, remove screws and washers securing rescue hoist cable bumper guard to supports. Carefully lower bumper guard to the ground (Figure 1, Detail A).

INSPECT/REPAIR

1. Visually inspect rescue hoist cable bumper guard, and airframe supports, for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked items as required.
2. Remove corrosion and blend out wear, using scouring pad, Item 205, WP 1803 00. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.
3. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

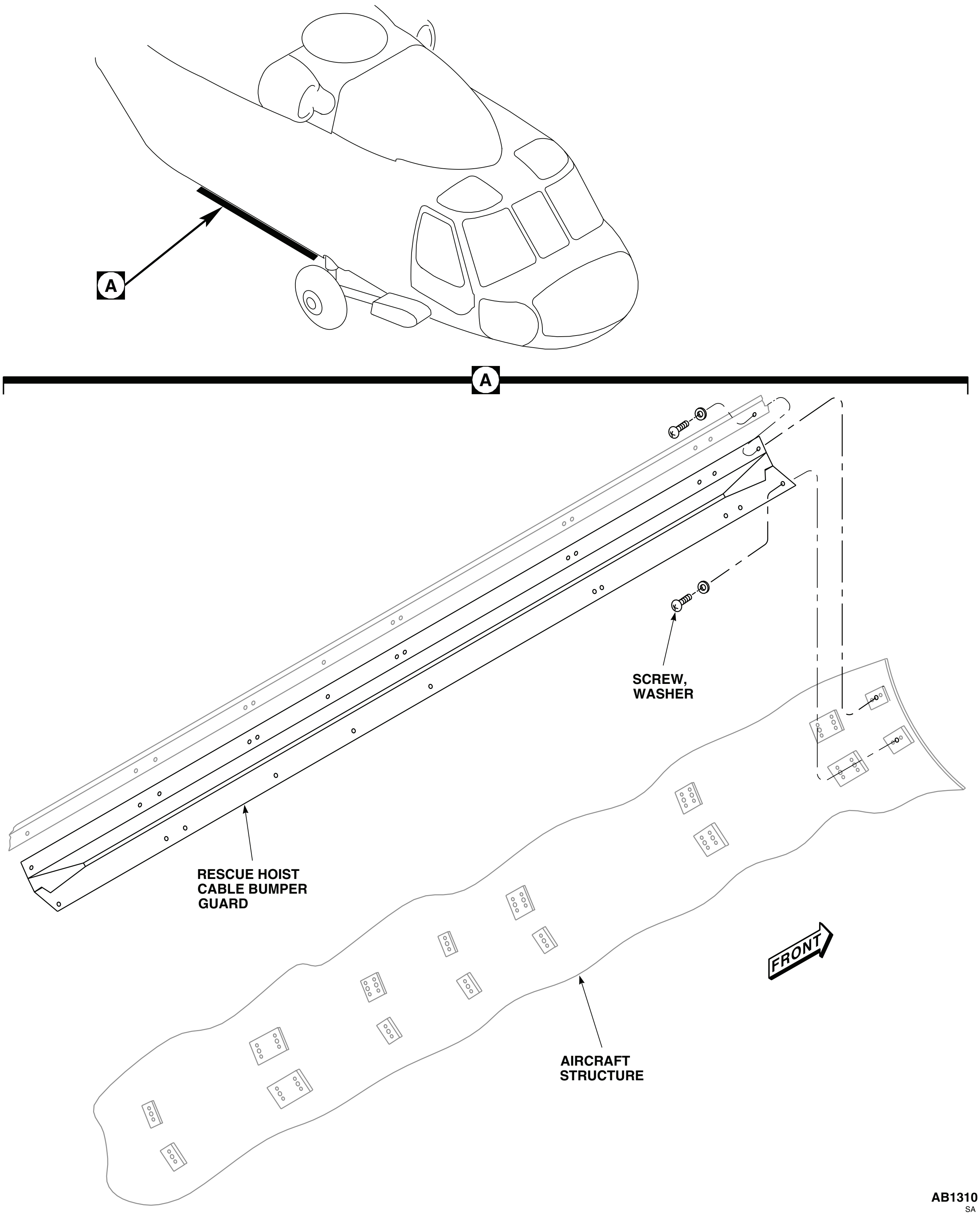


Figure 1. Replace Rescue Hoist Cable Bumper Guard.

INSTALL - Continued

WARNING

Injury to personnel will result if rescue hoist cable bumper guard is not supported correctly. Rescue hoist cable bumper guard must be supported while screws are being installed to prevent rescue hoist cable bumper guard from falling.

2. **QA** With an assistant, position rescue hoist cable bumper guard on helicopter, secure using screws and washer (Figure 1, Detail A).
3. **QA** Install and tighten screws securing landing gear fairing to helicopter.
4. Make sure area is clean and free of foreign objects.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

COCKPIT DOOR HINGE SUPPORT FITTING (AVIM)

This WP supersedes WP 0351 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drift
Gloves
Goggles/Face Shield
Hammer, Slide, CG46-2
Puller Set, GGG-P-781

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Applicator, Disposable, Item 53, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Primer, Sealing Compound, Item 248, WP 1803 00
Sealing Compound, Item 293, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0199 00
WP 0201 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove hinge assembly and gasket (WP 0199 00).
3. Remove screws and washers holding cover to inside of door, and remove cover (Figure 1).
4. Remove components of door jettison mechanism necessary to access hinge support fitting (WP 0201 00).

REPAIR

1. Replace bushings by doing this:
   a. Remove one bushing using puller from puller set and slide hammer (Figure 1, Detail A) (TM 1-1500-204-23).
   b. Using slide hammer and drift, remove second bushing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   c. Remove primer residue from outer surfaces of fitting and inside of bore with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

REPAIR - Continued

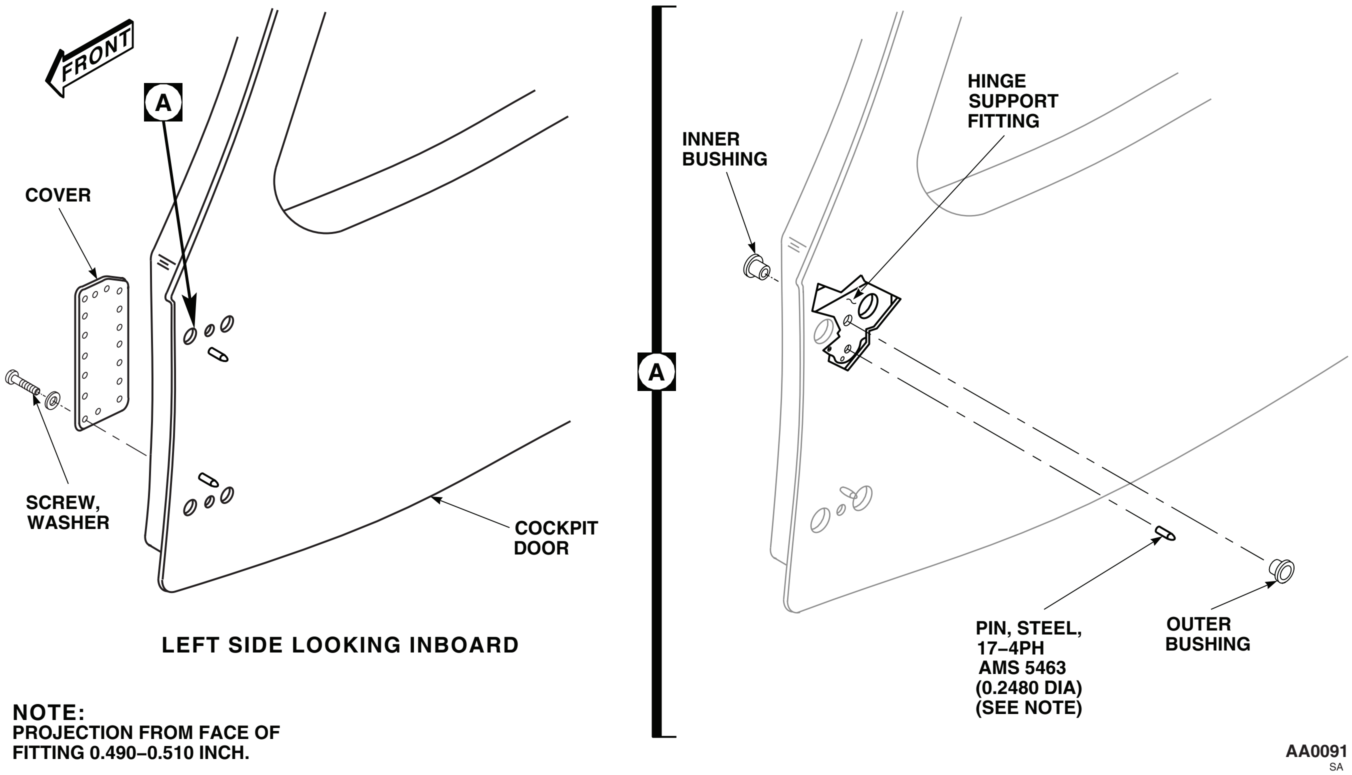


Figure 1. Repair Cockpit Door Hinge Support Fitting (AVIM).

WARNING

Personal injury will occur if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear protective gloves when handling carbon dioxide or chilled parts.

d. Put two new bushings in container of carbon dioxide, Item 64, WP 1803 00, for 30 minutes.

e. Remove one bushing from carbon dioxide, and apply thin coat of primer coating, Item 243, WP 1803 00, to all outer surfaces that will contact hinge support fitting.

f. **QA** While primer is still wet, install bushing in inner side of fitting, using puller from puller set and slide hammer (TM 1-1500-204-23).

g. Remove second bushing from carbon dioxide and apply thin coat of primer coating, Item 243, WP 1803 00, to all outer surfaces that will contact hinge support fitting.

h. **QA** While primer is still wet, install second bushing in outer side of fitting, using appropriate drift and slide hammer (TM 1-1500-204-23).

2. Replace pin by doing this:

a. Grip pin with pliers, twist to break bond, and remove from fitting.

REPAIR - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Remove adhesive residue from inside of pin socket with disposable applicator, Item 53, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.

c. Thoroughly clean inside of pin socket and surface of new pin with technical acetone, Item 5, WP 1803 00. Use disposable applicator, Item 53, WP 1803 00, for socket and cheesecloth cloth, Item 85, WP 1803 00, for pin surface.

d. Apply a thin coat of sealing compound primer, Item 248, WP 1803 00, on inside of socket and on mating pin surface to 0.50-inch from tip. Let dry for 5 minutes.

e. Coat inside of socket, and mating pin surface to 0.50-inch from tip, with sealing compound, Item 293, WP 1803 00.

f. Insert pin into socket until tip is 0.490 to 0.510-inch from fitting surface.

INSTALL

1. Turn off all electrical power.
2. Install components of door jettison mechanism (WP 0201 00).
3. **QA** Install cover on inside of door with screws and washers (Figure 1).
4. Install door hinge assembly and gasket (WP 0199 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

VIBRATION ABSORBER LOWER FITTING BUSHINGS (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing Installation Kit, GGG-P-643
Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Sealing Compound, Item 293, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

WARNING

Bushings, 70219-02150-101 through 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust and wear goggles while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

On fittings, 70219-02105-042 and 70219-02105-054, remove bushings per TM 1-1500-204-23 using bushing installation kit (Figure 1, Detail B).

INSTALL

1. Wipe bushings and fitting holes clean, using a machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Avoid getting solvent into bearing. Wipe dry.
2. Coat bushings and fitting holes with sealing compound, Item 293, WP 1803 00.

NOTE

Inspection hole identifies fitting front end (Figure 1, Detail B).

3. **QA** Using bushing installation kit and arbor press, install flanged bushing in front ear with flange inside.
4. **QA** Using bushing installation kit and arbor press, press sleeve bushing into fitting ear until it is 0.812 to 0.813-inch from flanged bushing (Figure 1, Detail B).
5. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

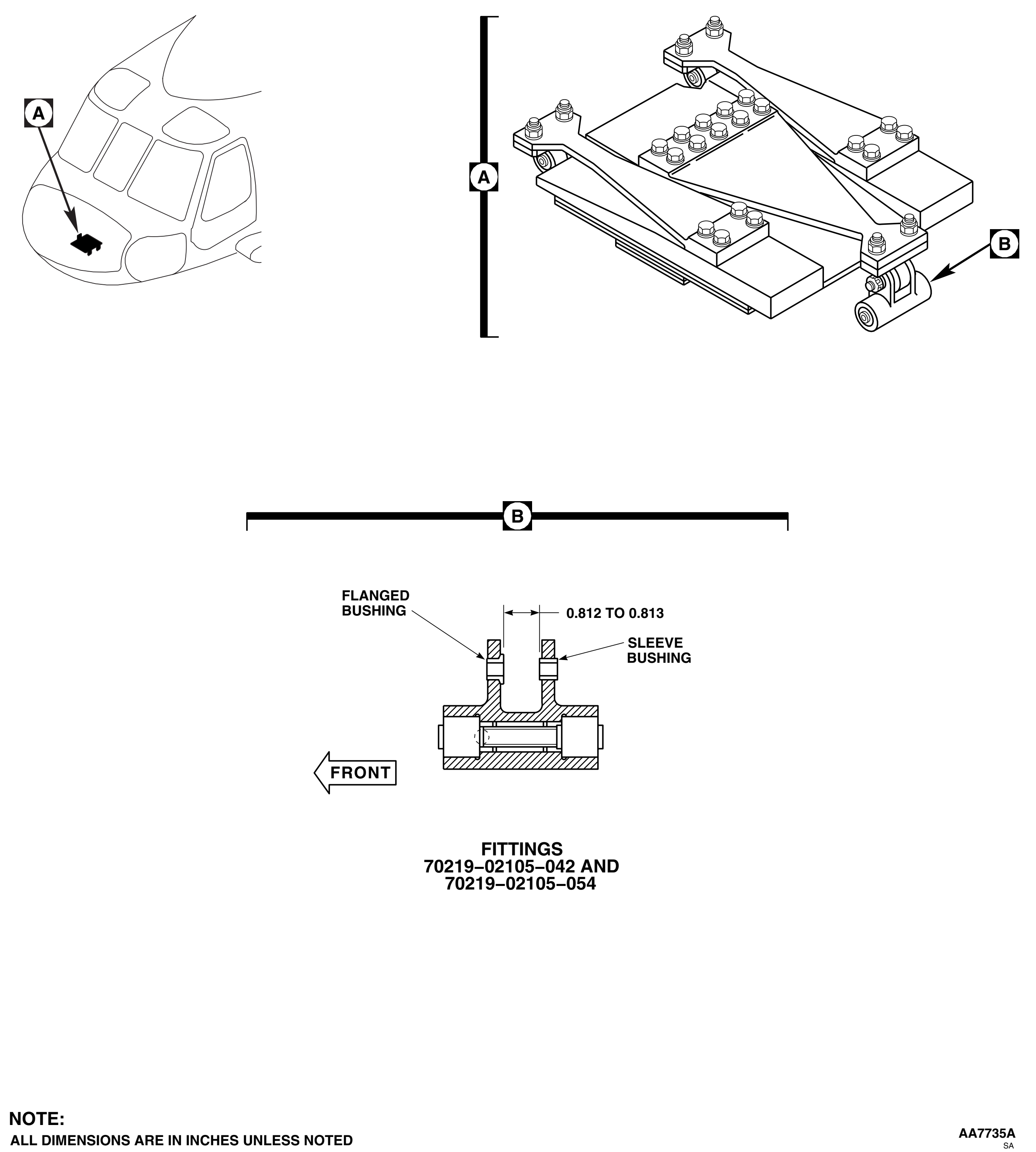


Figure 1. Replace Vibration Absorber Lower Fitting Bushings (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

NOSE VIBRATION ABSORBER FITTING BEARINGS (AVIM)

This WP supersedes WP 0353 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Arbor Press, 00-P-636
Bushing Installation Kit, GGG-P-643
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13
Staking Tools
Torque Wrench, 0 - 16 in. oz.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Applicator, Disposable, Item 53, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Grease, General Purpose, Item 136, WP 1803 00
Primer, Sealing Compound, Item 248, WP 1803 00
Sealing Compound, Item 294, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 55-1500-322-24
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE FITTING ASSEMBLY UPPER AND LOWER BEARINGS AND SPACER

1. Using bushing installation kit and arbor press, remove bearings and spacers from lower fittings 70219-02105-042, 70219-02105-054, and 70219-02127-041 (Figure 1, Detail C).
2. Using bushing installation kit and arbor press, remove bearings from upper fittings 70219-02105-041, 70219-02105-053, and 70219-02127-042 (Figure 1, Detail B).
3. On lower fittings, 70219-02105-042 and 70219-02105-054, clean fitting bore and spacer, using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry.
4. On lower fitting, 70219-02127-041, clean fitting bore and spacer as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Clean fitting bore using disposable applicator, Item 53, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

b. Remove retaining compound residue from fitting bore, using abrasive cloth Item 83, WP 1803 00.

REMOVE FITTING ASSEMBLY UPPER AND LOWER BEARINGS AND SPACER - Continued

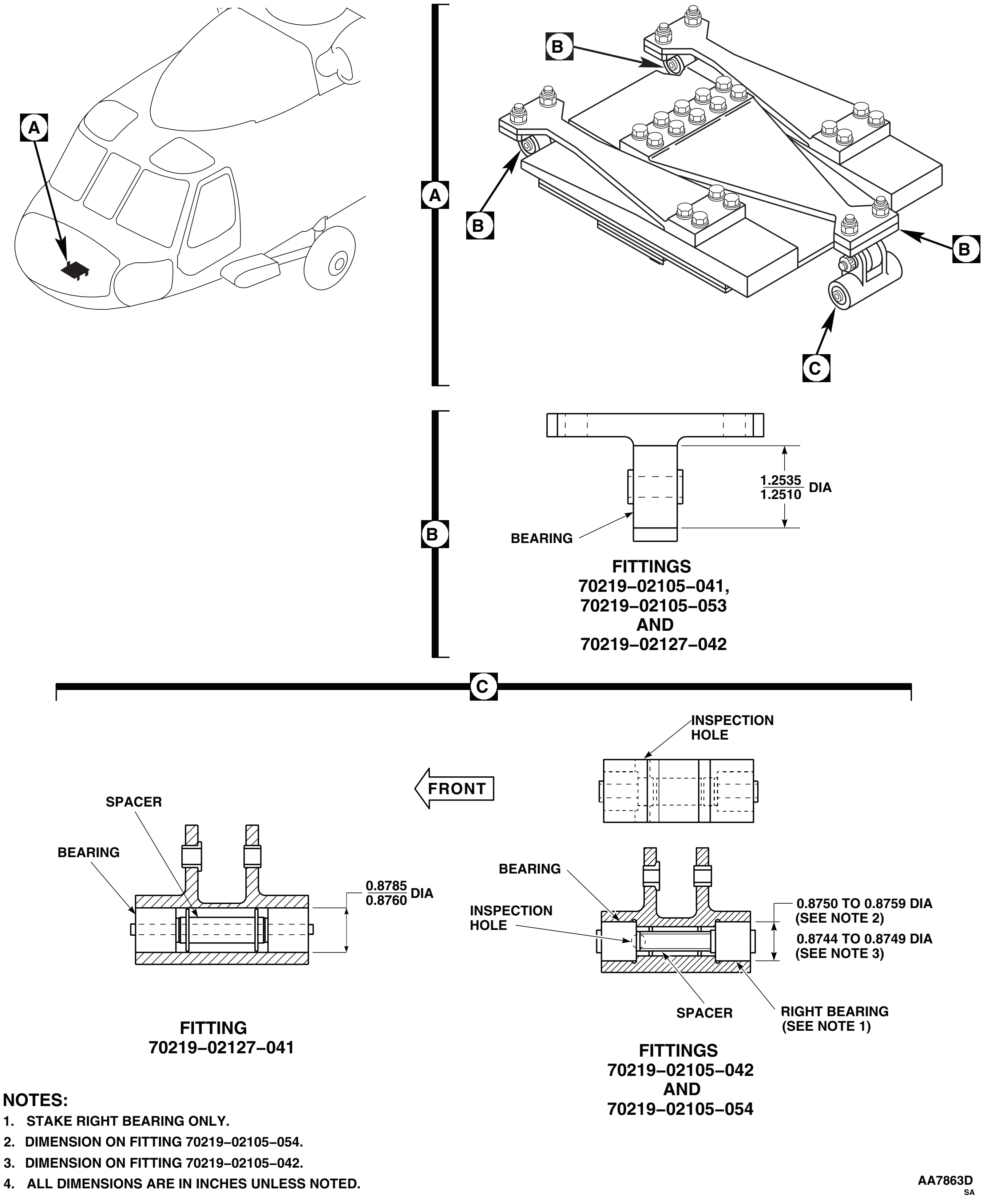


Figure 1. Replace Nose Vibration Absorber Fitting Bearings (AVIM).

REMOVE FITTING ASSEMBLY UPPER AND LOWER BEARINGS AND SPACER - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Remove abrasive residue from fitting bore using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

d. Clean spacer using machinery wiping towel, Item 344, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry with clean machinery wiping towel, Item 344, WP 1803 00.

5. On upper fittings, 70219-02105-041, 70219-02105-053, and 70219-02127-042, clean bore as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Clean bore, using disposable applicator, Item 53, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

b. Remove sealing compound residue, using abrasive cloth, Item 82, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Remove abrasive residue using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

INSTALL LOWER FITTING ASSEMBLY BEARINGS AND SPACER

1. Prior to installing bearings, perform breakout torque check, as follows:

a. Install bolt in bearings with enough washers on bolt to seat nut. Install and tighten nut so that washers do not rotate.

b. Rotate bearing inner race 4 complete turns.

c. Using torque wrench rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 4 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

d. Remove bolt, washers, and nut.

2. On fitting assembly, 70219-02105-042, install bearings and spacer as follows:

a. Apply general purpose grease, Item 136, WP 1803 00, to outside diameter of bearings.

INSTALL LOWER FITTING ASSEMBLY BEARINGS AND SPACER - Continued

NOTE

Use correct sizing tools and arbor press so inner and outer races of bearing contact at the same time.

b. Using bushing installation kit and arbor press, press right bearing into fitting until bearing seats against shoulder of housing (Figure 1, Detail C).

c. Install spacer.

CAUTION

Bearing preload will result in premature bearing wear. Do not attempt to bottom left bearing outer race against housing shoulder.

d. Using bushing installation kit and arbor press, press left bearing into fitting until bearing inner race just touches spacer. Use inspection hole to verify (Figure 1, Detail C).

e. Perform breakout torque check, as follows:

(1) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 95 - 105 INCH-POUNDS.**

(2) Rotate bearing inner race 4 complete turns.

(3) Using torque wrench, rotate bearing. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

(4) Remove bolt, washers, and nut.

f. Using staking tools, pressure stake right bearing only (TM 55-1500-322-24).

3. On fitting 70219-02105-054, install bearings and spacer as follows:

a. Apply general purpose grease, Item 136, WP 1803 00, to outside diameter of bearings.

NOTE

During bearing installation, if a press fit condition exists, do not attempt to install the bearing. The bearing must be a slip fit.

b. Install right bearing into fitting (Figure 1, Detail C).

c. Install spacer.

NOTE

During bearing installation, if a press fit condition exists, do not attempt to install the bearing. The bearing must be a slip fit.

d. Install left bearing into fitting (Figure 1, Detail C).

e. Perform breakout torque check, as follows:

(1) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 95 - 105 INCH-POUNDS.**

(2) Rotate bearing inner races 4 complete turns.

(3) Using torque wrench, rotate bearing. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

(4) Remove bolt, washers, and nut.

INSTALL LOWER FITTING ASSEMBLY BEARINGS AND SPACER - Continued

f. Using staking tools, pressure stake right bearing only (TM 55-1500-322-24).

4. On fitting, 70219-02127-041, install bearings and spacer as follows:

a. Prepare bonding surfaces of bearing and bearing bore as follows:

(1) Mask bearing interior against contamination using pressure sensitive adhesive tape, Item 334, WP 1803 00.

(2) Sand bonding surface of bearing and bearing bore lightly, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

(3) Clean off abrasive residue using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

(4) Remove pressure sensitive adhesive tape from bearing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Clean bonding surface of fitting and bearing using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply light coat of sealing compound primer, Item 248, WP 1803 00, to bearing outside diameter and inside diameter of fitting bores, using disposable applicator, Item 53, WP 1803 00.

CAUTION

Protect primed surfaces from contamination during drying. Do not touch primed surfaces with bare hands.

d. Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step 4.(b) through step 4.(d).

e. Install spacer in fitting bore. Make sure spacer is located approximately center of fitting bore.

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

f. Apply even coat of sealing compound, Item 294, WP 1803 00, to outside diameter of bearings and inside diameter of fitting bores, using disposable applicator, Item 53, WP 1803 00.

INSTALL LOWER FITTING ASSEMBLY BEARINGS AND SPACER - Continued

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If this time is exceeded, step 4. b. through step 4. f. must be repeated.

g. Immediately install bearings in fitting bore with slight twisting motion to distribute retaining compound.

NOTE

Adhesive sets in five minutes.

h. Prior to adhesive set, perform the following:

(1) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 95 - 105 INCH-POUNDS.**

(2) Adjust position by centering bearing and spacer within fitting bore.

i. Allow fitting assembly to cure for 24 hours minimum.

j. Perform breakout torque check as follows:

(1) Rotate bearing inner races 4 complete turns.

(2) Using torque wrench, rotate bearing. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

(3) Remove bolt, washers, and nut.

INSTALL UPPER FITTING ASSEMBLY BEARING

1. Prior to installing upper bearing, perform bearing breakout torque check, as follows:

a. Install bolt through bearing. Install enough washers on bolt to seat nut. Install and tighten nut so that washers do not rotate.

b. Rotate bearing inner race 4 complete turns.

c. Using torque wrench rotate bearing. **BREAKOUT TORQUE MUST NOT EXCEED 5 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

d. Remove nut, washers, and bolt from bearing.

2. Prepare bonding surface of bearing and bearing bore as follows:

a. Mask bearing interior against contamination using pressure sensitive adhesive tape, Item 334, WP 1803 00.

b. Sand bonding surface of bearing and bearing bore lightly, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Clean off abrasive residue using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

d. Remove masking tape from bearing.

INSTALL UPPER FITTING ASSEMBLY BEARING - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean bonding surfaces of fitting and bearing, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Apply light coat of sealing compound primer, Item 248, WP 1803 00, to bearing outside diameter and inside diameter of fitting bore, using disposable applicator, Item 53, WP 1803 00.

CAUTION

Protect primed surfaces from contamination during drying. Do not touch primed surfaces with bare hands.

5. Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step 3. through step 5.

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

6. Apply even coat of sealing compound, Item 294, WP 1803 00, to outside diameter of bearing and bonding surface of fitting, using disposable applicator, Item 53, WP 1803 00.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If this time is exceeded, step 3. through step 6. must be repeated.

7. **QA** Immediately install bearing in fitting with slight twisting motion to distribute retaining compound. Bearing must be centered in fitting bore within 0.020-inch. Observe squeeze out at bond line and wipe off excess immediately.
8. Allow fitting assembly to cure for 24 hours minimum.
9. On reworked fitting assemblies 70219-02105-041 and 70219-02105-053, perform the following:
   a. Apply corrosion resistant coating, Item 110, WP 1803 00, to any exposed reworked surfaces.
   b. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
10. Perform breakout torque check, as follows:
    a. **QA** Install bolt through bearing. Install enough washers on bolt to seat nut. Install and tighten nut so that washers do not rotate.
    b. Rotate bearing inner race 4 complete turns.
    c. Using torque wrench rotate bearing. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH FULL 360 DEGREE TURN WHEN MISALIGNING INNER AND OUTER RACES BY 5 DEGREES.**

INSTALL UPPER FITTING ASSEMBLY BEARING - Continued

d. Remove nut, washers, and bolt from bearing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CABIN VIBRATION ABSORBER FITTING BEARINGS (AVIM)

This WP supersedes WP 0354 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing Installation Kit, GGG-P-643
Fluorescent Penetrant Inspection Kit, XMA101
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13
Staking Tools
Torque Wrench, 0 - 16 in. oz.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Applicator, Disposable, Item 53, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Grease, General Purpose, Item 136, WP 1803 00
Primer, Sealing Compound, Item 248, WP 1803 00
Primer, Sealing Compound, Item 263, WP 1803 00
Sealing Compound, Item 294, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1520-265-23
TM 55-1500-322-24
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE FITTING ASSEMBLY BEARING

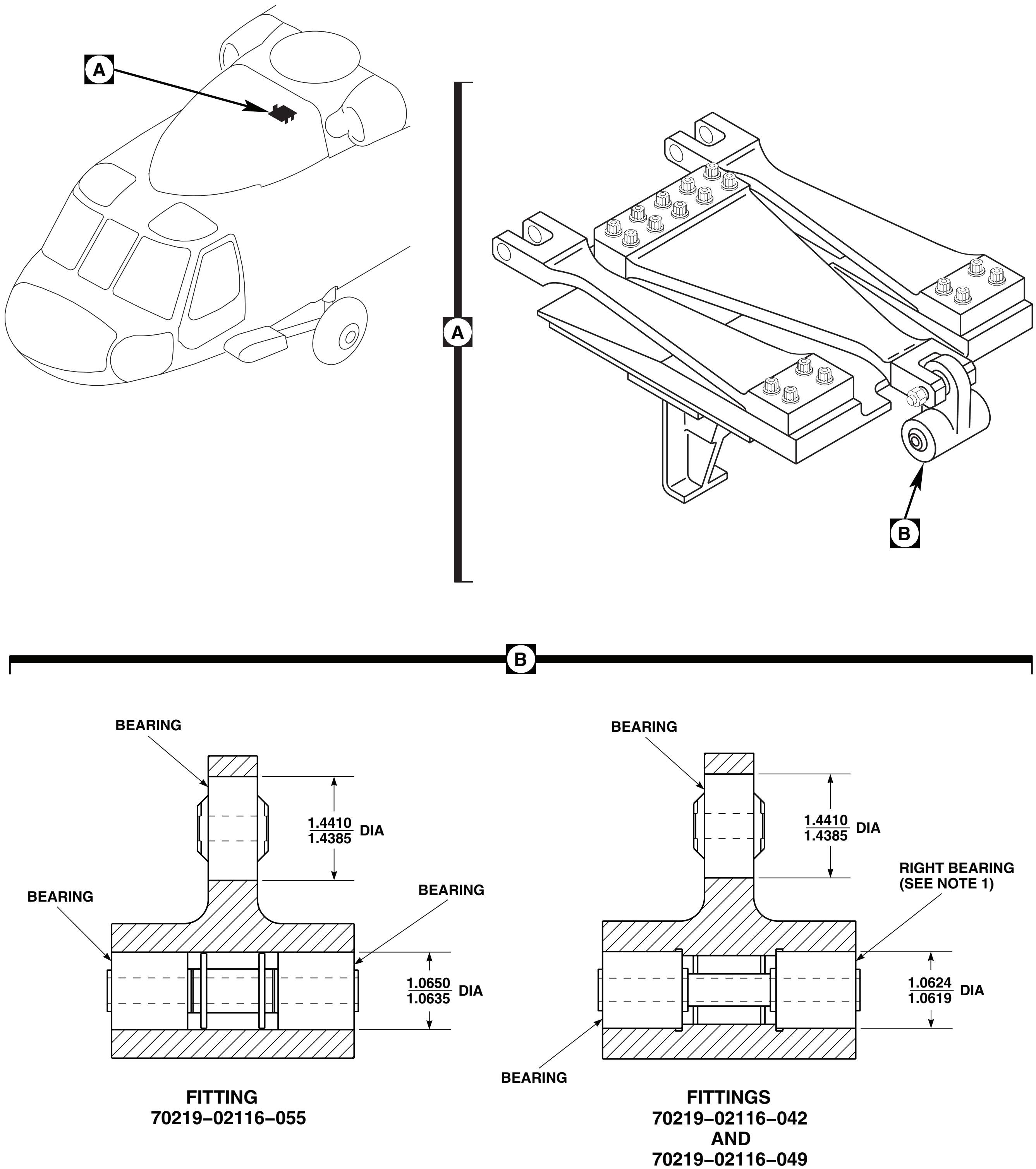
1. Using bushing installation kit and arbor press, remove upper bearing from fittings 70219-02116-042, 70219-02116-049, and 70219-02116-055 (Figure 1, Detail B) (TM 55-1500-345-23).
2. On fittings, 70219-02116-042 and 70219-02116-049, perform the following:
   a. Ream bearing bore 1.4385 to 1.4410-inch diameter.
   b. Blend out staking protrusions.
   c. Inspect reworked areas using fluorescent penetrant kit (TM 1-1520-265-23). **NO CRACKS ALLOWED**.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean fitting bore, using disposable applicator, Item 53, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Remove retaining compound residue, using abrasive cloth, Item 83, WP 1803 00.

REMOVE FITTING ASSEMBLY BEARING - Continued



NOTES:

1. ON FITTING 70219–02116–042 STAKE RIGHT BEARING ONLY.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Replace Cabin Vibration Absorber Fitting Bearings (AVIM).

REMOVE FITTING ASSEMBLY BEARING - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Remove abrasive residue, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

INSTALL FITTING ASSEMBLY BEARING

1. Prior to installing upper bearing, perform breakout torque check, as follows:
   a. Install bolt through bearing. Install enough washers on bolt to seat nut. Install and tighten nut so that washers do not rotate.
   b. Rotate bearing inner race 4 complete turns.
   c. Using torque wrench rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 5 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**
   d. Remove nut, washers, and bolt from bearing.
2. Prepare bonding surfaces of bearing and bearing bore as follows:
   a. Mask bearing interior against contamination using pressure sensitive adhesive tape, Item 334, WP 1803 00.
   b. Lightly sand bonding surface of bearing in circumferential direction, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   c. Clean off abrasive residue using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   d. Remove masking tape from bearing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean bonding surfaces of fitting and bearing using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Apply light coat of sealing compound primer, Item 263, WP 1803 00, to bearing outside diameter and inside diameter of fitting bore, using disposable applicator, Item 53, WP 1803 00.

INSTALL FITTING ASSEMBLY BEARING - Continued

NOTE

Protect primed surfaces from contamination during drying. Do not touch primed surfaces with bare hands.

5. Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step 3. through step 5.

NOTE

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of sealing compound will make it unusable.

6. Apply even coat of sealing compound, Item 248, WP 1803 00, to outside diameter of bearing and inside diameter of fitting bore, using disposable applicator, Item 53, WP 1803 00.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of sealing compound application. If this time is exceeded, step 3. through step 6. must be repeated.

7. **QA** Immediately install bearing in fitting with slight twisting motion to distribute sealing compound. Bearing must be centered in fitting bore within 0.020-inch. Observe squeeze out at bond line and wipe off excess immediately.
8. Allow fitting assembly to cure for 24 hours minimum.
9. On reworked assemblies, 70219-02116-042 and 70219-02116-049, perform the following:
   a. Apply corrosion resistant compound, Item 110, WP 1803 00, to any exposed reworked surfaces.
   b. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REMOVE FITTING ASSEMBLY BEARINGS AND SPACER

1. Using bushing installation kit and arbor press, remove lower bearings and spacer from fittings 70219-02116-042, 70219-02116-049, and 70219-02116-055 (Figure 1, Detail B) (TM 55-1500-345-23).
2. On fittings, 70219-02116-042 and 70219-02116-049, clean fitting bores and spacer using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry.
3. On fitting, 70219-02116-055, clean fitting bores and spacer, as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   a. Clean fitting bores, using disposable applicator, Item 53, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   b. Remove retaining compound residue from fitting bores, using abrasive cloth, Item 83, WP 1803 00.

REMOVE FITTING ASSEMBLY BEARINGS AND SPACER - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

c. Remove abrasive residue from fitting bores using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

d. Clean spacer using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry.

INSTALL FITTING ASSEMBLY BEARINGS AND SPACER

1. Prior to installing bearings, perform breakout torque check, as follows:

   a. **QA** Install bolt in bearings with enough washers on bolt to seat nut. Install and tighten nut so that washers do not rotate.

   b. Rotate bearing inner race 4 complete turns.

   c. Using torque wrench rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 4 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**

   d. Remove bolt, washers, and nut.

2. On fittings, 70219-02116-042 and 70219-02116-049, install bearings and spacer, as follows:

   a. Apply general purpose grease, Item 136, WP 1803 00, to outside diameter of bearings.

NOTE

Use correct sizing tools and arbor press so inner and outer races of bearing contact at the same time.

   b. **QA** Using bushing installation kit and arbor press, press right bearing into fittings, 70219-02116-042 and 70219-02116-049, until bearing seats against shoulder of housing (Figure 1, Detail B).

   c. Install spacer.

CAUTION

Bearing preload will result in premature bearing wear. Do not attempt to bottom left bearing outer race against housing shoulder.

   d. On fitting assembly, 70219-02116-042, use bushing installation kit and arbor press, press left bearing into fitting until bearing inner race just touches spacer. Use inspection hole to verify (Figure 1, Detail B).

NOTE

During bearing installation, if a press fit condition exists, do not attempt to install the bearing. The bearing must be a slip fit.

   e. On fitting assembly, 70219-02116-049, install left bearing into fitting (Figure 1, Detail B).

   f. Perform breakout torque check, as follows:

      (1) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**

INSTALL FITTING ASSEMBLY BEARINGS AND SPACER - Continued

(2) Rotate bearing inner races 4 complete turns.

(3) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360 DEGREE TURN MUST NOT EXCEED 10 INCH-OUNCES.**

(4) Remove bolt, washers, and nut.

g. On fitting assembly, 70219-02116-042, using locally made staking tools, pressure stake right bearing only (TM 55-1500-322-24).

3. On fitting, 70219-02116-055, install bearings and spacer, as follows:

a. Prepare bonding surfaces of bearing and bearing bore as follows:

(1) Mask bearing interior against contamination using pressure sensitive adhesive tape, Item 334, WP 1803 00.

(2) Lightly sand bonding surface of bearing in circumferential direction, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

(3) Clean off abrasive residue using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

(4) Remove masking tape from bearing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Clean bonding surfaces of fitting and bearing using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

c. Apply light coat of sealing compound primer, Item 248, WP 1803 00, to bearing outside diameter and inside diameter of fitting bores, using disposable applicator, Item 53, WP 1803 00.

NOTE

Protect primed surfaces from contamination during drying. Do not touch primed surfaces with bare hands.

d. Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step 3.b. through step 3.f.

e. **QA** Install spacer in fitting bore. Make sure spacer is located approximately center of fitting bore.

INSTALL FITTING ASSEMBLY BEARINGS AND SPACER - Continued

NOTE

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of sealing compound will make it unusable.

f. Apply even coat of sealing compound, Item 294, WP 1803 00, to outside diameter of bearings and inside diameter of fitting bores, using disposable applicator, Item 53, WP 1803 00.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of sealing compound application. If this time is exceeded, step 3.b. through step 3.f. must be repeated.

g. **QA** Immediately install bearings in fitting bores with slight twisting motion to distribute sealing compound.

NOTE

Adhesive sets in five minutes.

h. Prior to adhesive set, perform the following:

(1) **QA** Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**

(2) Adjust position by centering bearings and spacer within fitting bore.

i. Allow fitting assembly to cure for 24 hours minimum.

j. Perform breakout torque check, as follows:

(1) Rotate bearing inner races 4 complete turns.

(2) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360 DEGREE TURN MUST NOT EXCEED 10 INCH-OUNCES.**

(3) Remove bolt, washers, and nut.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

VIBRATION ABSORBER SPRING BUSHINGS (AVIM)

This WP supersedes WP 0355 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing Installation Kit, GGG-P-643
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Sealing Compound, Item 292, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References
WP 0219 00
WP 1803 00

REMOVE

WARNING

Bushings, 70219-02150-101 through 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust and wear goggles while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

Remove bushings using an extractor from bushing installation kit (Figure 1, Detail A or Detail B).

INSTALL

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Wipe replacement bushings and spring bushi ng holes clean using machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry.

NOTE

Use sealing compound, Item 292, WP 1803 00, on all fitting assemblies with shoulder bushings. Coat bushings and spring bushing holes.

2. If required, wet bushing and spring bushing holes with epoxy primer coating kit, Item 120, WP 1803 00.

INSTALL - Continued

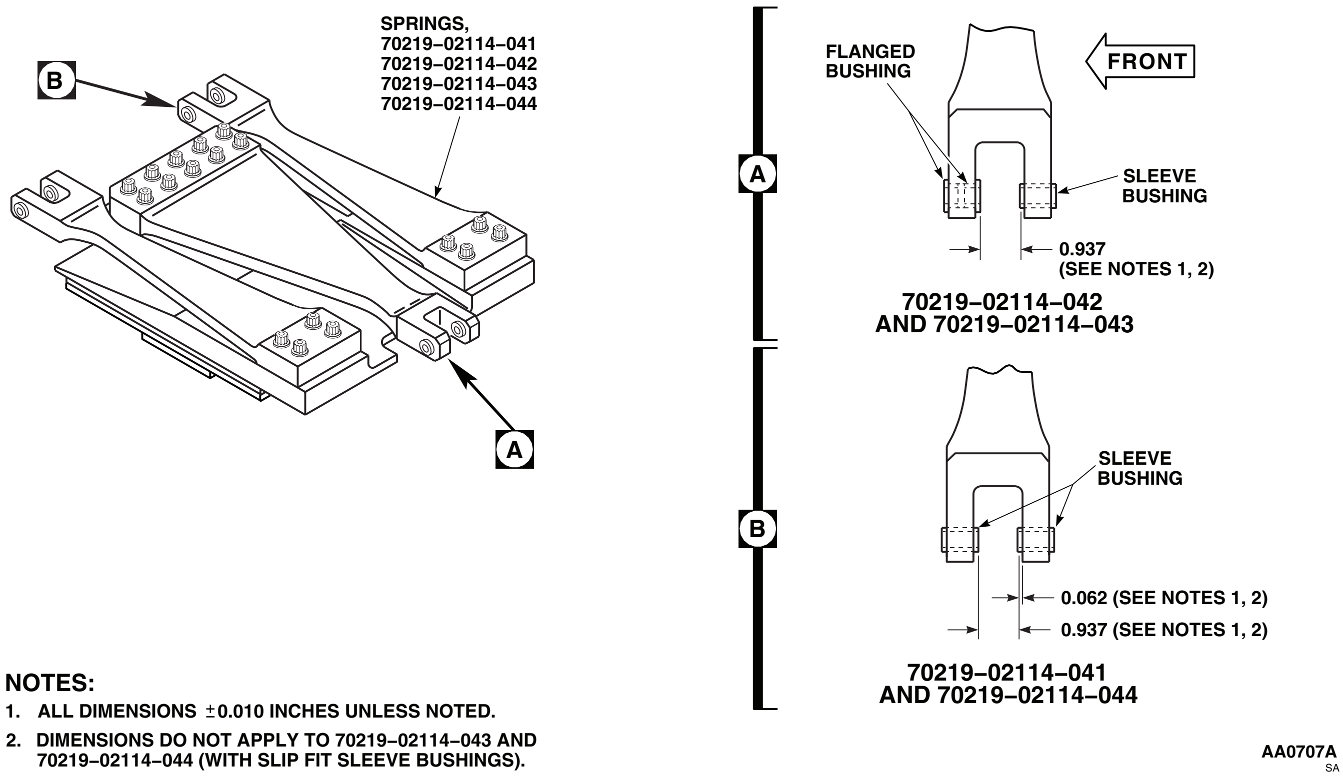


Figure 1. Replace Vibration Absorber Spring Bushings (AVIM).

NOTE

On springs, 70219-02114-043 and 70219-02114-044, loose sleeve bushings are installed at time of vibration absorber installation (WP 0219 00).

3. **QA** On spring, 70219-02114-041, press in front sleeve bushing to protrude 0.062-inch inward from yoke ear (Figure 1, Detail B).
4. **QA** On springs, 70219-02114-042 and 70219-02114-043, press flanged bushings into front yoke ear (Figure 1, Detail A).
5. **QA** Press sleeve bushing into opposite ear to dimensions shown in (Figure 1).
6. Leave squeezed out primer in place.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

VIBRATION ABSORBER AIRFRAME NOSE FITTING BUSHINGS (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing Installation Kit, GGG-P-643
Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Grease, Instrument, Item 137, WP 1803 00
Sealing Compound, Item 292, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

WP 1803 00

REMOVE

WARNING

Bushings, 70219-02150-101 through 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust and wear goggles while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

Remove bushings, using an extractor from bushing installation kit (Figure 1, Sheet 1, Detail A and Figure 1, Sheet 2, Detail B, Detail C, Detail D and Detail E). Discard old bushings.

INSTALL

1. Wipe new bushings and fitting holes clean using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent Item 71, WP 1803 00. Wipe dry.
2. **QA** **UH60A 83-23924 - SUBQ** Install all slip-fit sleeve bushings, 70219-02150-102 and 70219-02150-104, with outer surface coated with instrument grease, Item 137, WP 1803 00. Use no sealing compound on these bushings.
3. Use sealing compound, Item 292, WP 1803 00, on all flanged bushings and fitting holes.
4. **QA** Using bushing installation kit and arbor press, press flanged bushing into front ear with flange inside (Figure 1, Sheet 2, Detail B, Detail C and Detail D).
5. **QA** Using bushing installation kit and arbor press, on fittings with two sleeve bushings, press front bushing in fitting to dimension shown in Figure 1, Sheet 2, Detail E.
6. Using bushing installation kit and arbor press, press plain bushing in opposite fitting ear using sizing block.

INSTALL - Continued

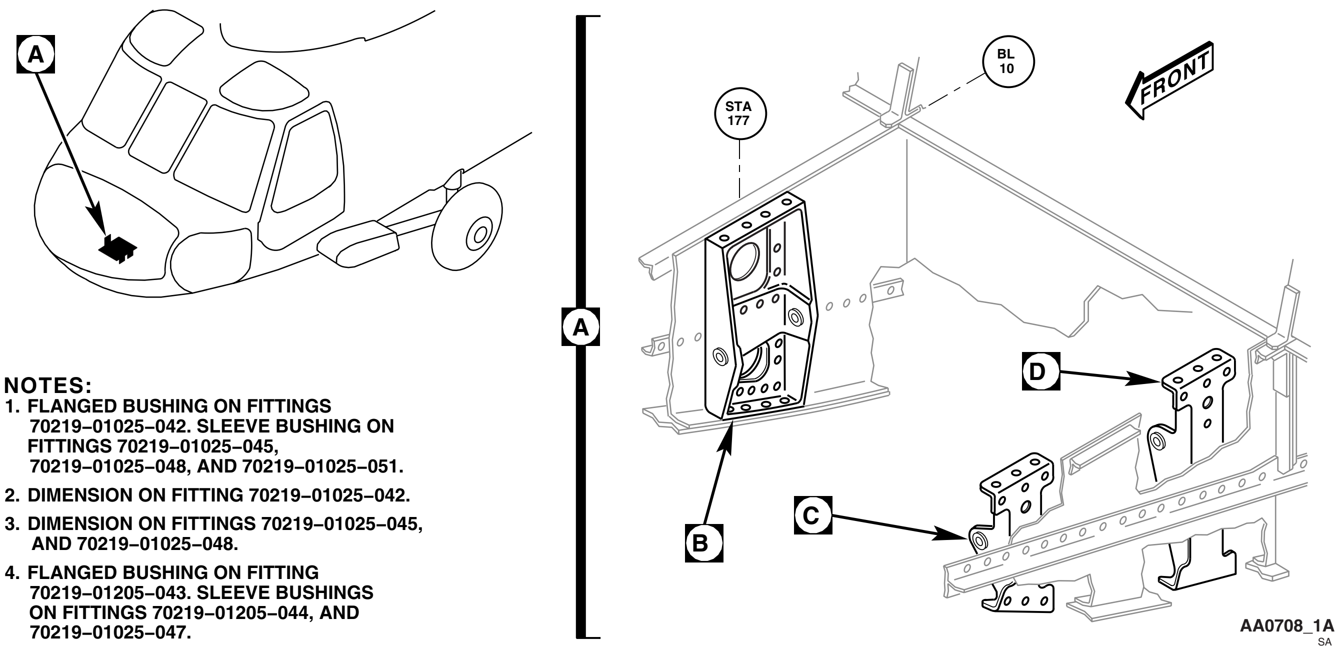


Figure 1. Replace Vibration Absorber Airframe Nose Fitting Bushings (AVIM). (Sheet 1 of 2)

INSTALL - Continued

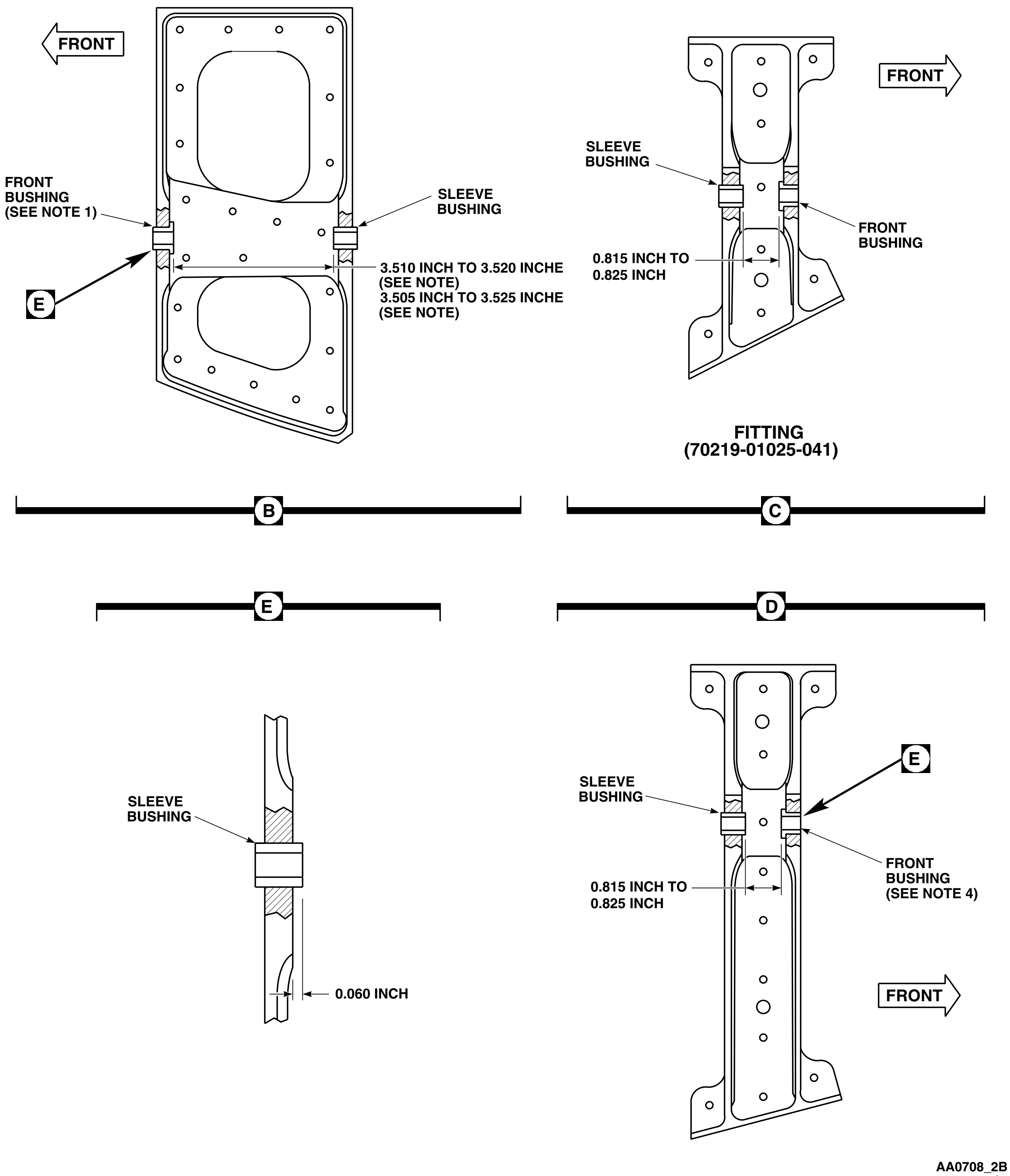


Figure 1. Replace Vibration Absorber Airframe Nose Fitting Bushings (AVIM). (Sheet 2 of 2)

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

VIBRATION ABSORBER AIRFRAME CABIN FITTING BUSHINGS (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing Installation Kit, GGG-P-643
Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Corrosion Preventive Compound, Item 103, WP 1803 00
Grease, Aircraft, Item 132, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

WP 1803 00

REMOVE

WARNING

Bushings, 70219-02150-101 through 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust and wear goggles while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

Remove bushings using an extractor from bushing installation kit (Figure 1, Sheet 1, Detail B and Detail C and Figure 1, Sheet 2, Detail D, Detail E and Detail F). Discard old bushings.

INSTALL

1. Wipe new bushings and fitting holes clean using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Wipe dry.
2. Wipe bushing outer surfaces and fitting holes with corrosion preventive compound, Item 103, WP 1803 00.
3. **UH60A 83-23924 - SUBQ** Install all slip-fit sleeve bushings, 70219-02150-101, with outer surface coated with aircraft grease, Item 132, WP 1803 00.
4. **QA** On fittings, 70219-02105-045, 70219-02105-046, and 70219-02105-047, press flanged bushings into fittings with flange inside, using bushing installation kit and arbor press (Figure 1, Sheet 1, Detail B and Detail C and Figure 1, Sheet 2, Detail D).
5. **QA** On fittings with two sleeve bushings, and using bushing installation kit and arbor press, press front bushing in fitting to dimension shown in (Figure 1, Sheet 1, Detail B and Detail C).
6. **QA** Using bushing installation kit and arbor press, press plain bushing into opposite fitting ear to dimension shown in (Figure 1, Sheet 2, Detail D, Detail E and Detail F).

INSTALL - Continued

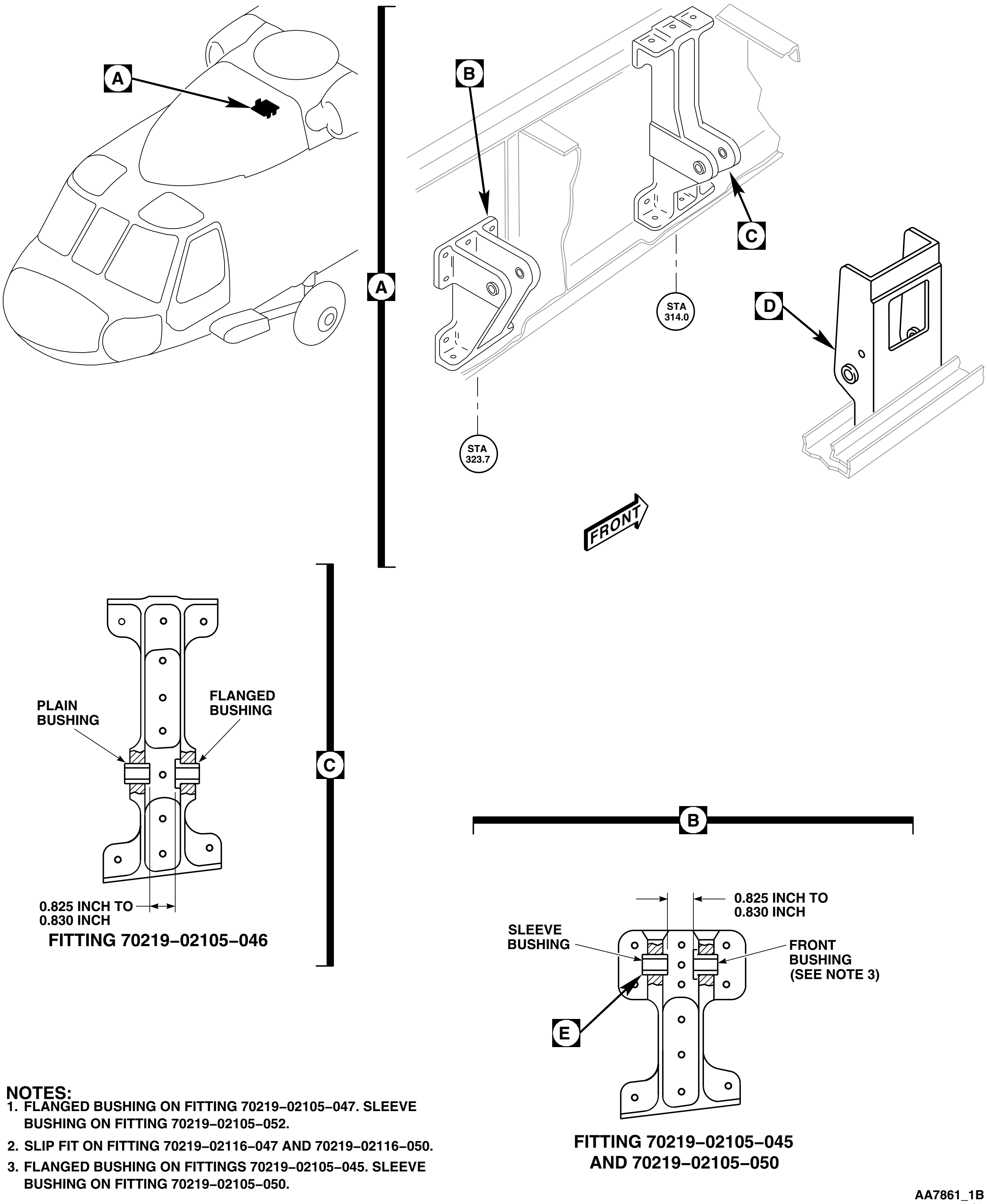


Figure 1. Replace Vibration Absorber Airframe Cabin Fitting Bushings (AVIM). (Sheet 1 of 2)

INSTALL - Continued

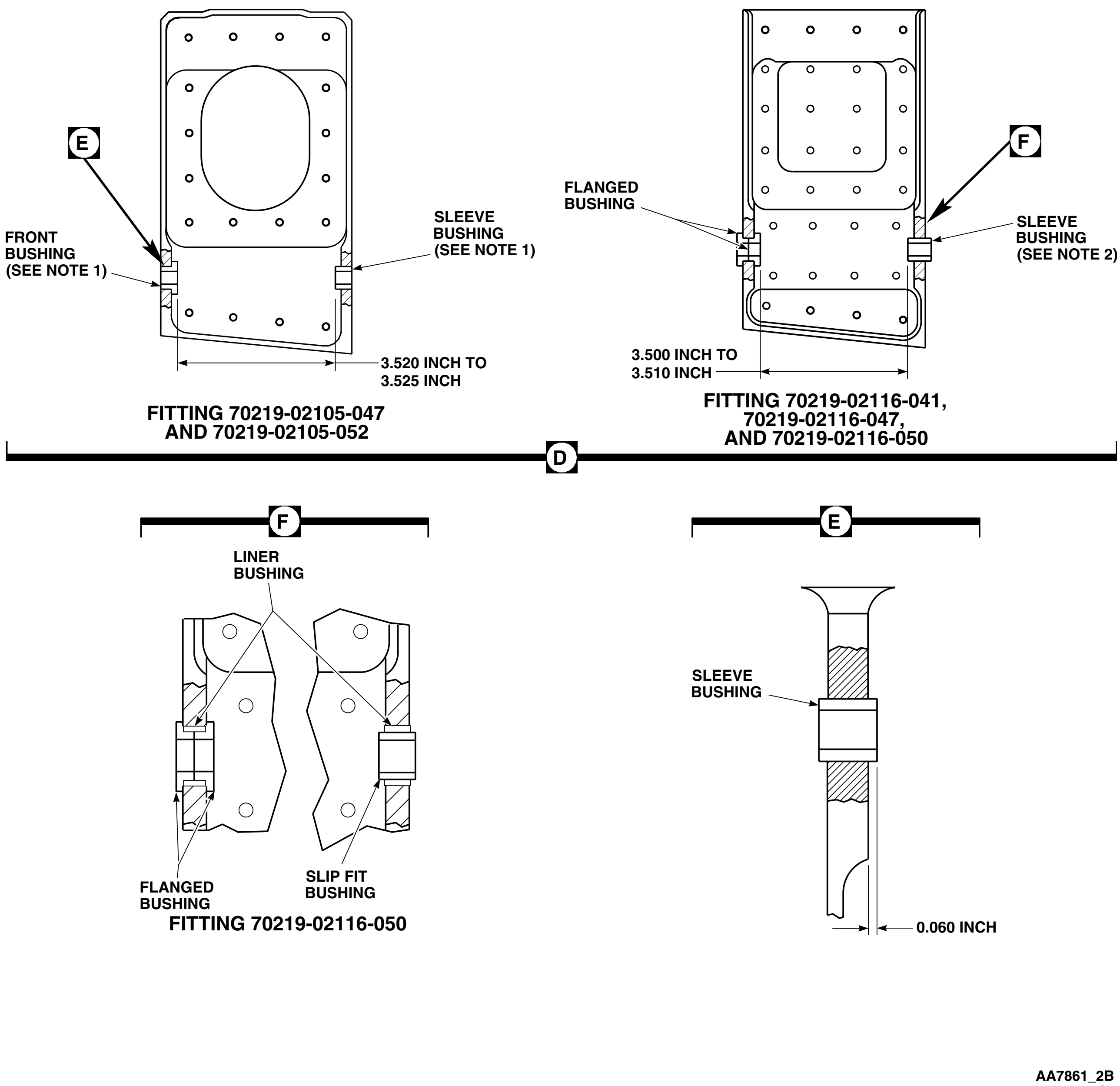


Figure 1. Replace Vibration Absorber Airframe Cabin Fitting Bushings (AVIM). (Sheet 2 of 2)

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

VIBRATION ABSORBER AIRFRAME CABIN FITTING BEARINGS (AVIM)

This WP supersedes WP 0358 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Arbor Press, 00-P-636
Bushing Installation Kit, GGG-P-643
Fluorescent Penetrant Inspection Kit, XMA101
Fly Cutter, 65700-40016-150
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13
Staking Tools
Torque Wrench, 0 - 16 in. oz.,

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Applicator, Disposable, Item 53, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Primer, Sealing Compound, Item 248, WP 1803 00
Sealing Compound, Item 294, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
TM 55-1500-322-24
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. Drill out rivets (TM 1-1500-204-23)and remove fitting from structure (Figure 1, Detail A).
2. On fittings, 70219-02116-045 and 70219-02116-046, remove staking using fly cutter and drill press (TM 1-1500-204-23).
3. Press bearing out of fitting, using arbor press and properly sized arbor and anvil (TM 55-1500-322-24) (Figure 1, Detail B and Detail C).
4. On fittings, 70219-02116-046 and 70219-02116-045, perform the following:
   a. Ream bearing bore 1.4385 to 1.4410-inch diameter.
   b. Blend out staking protrusions.
   c. Inspect reworked areas using fluorescent penetrant kit (TM 1-1520-265-23). **NO CRACKS ALLOWED**.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean fitting bore, using disposable applicator, Item 53, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

REMOVE - Continued

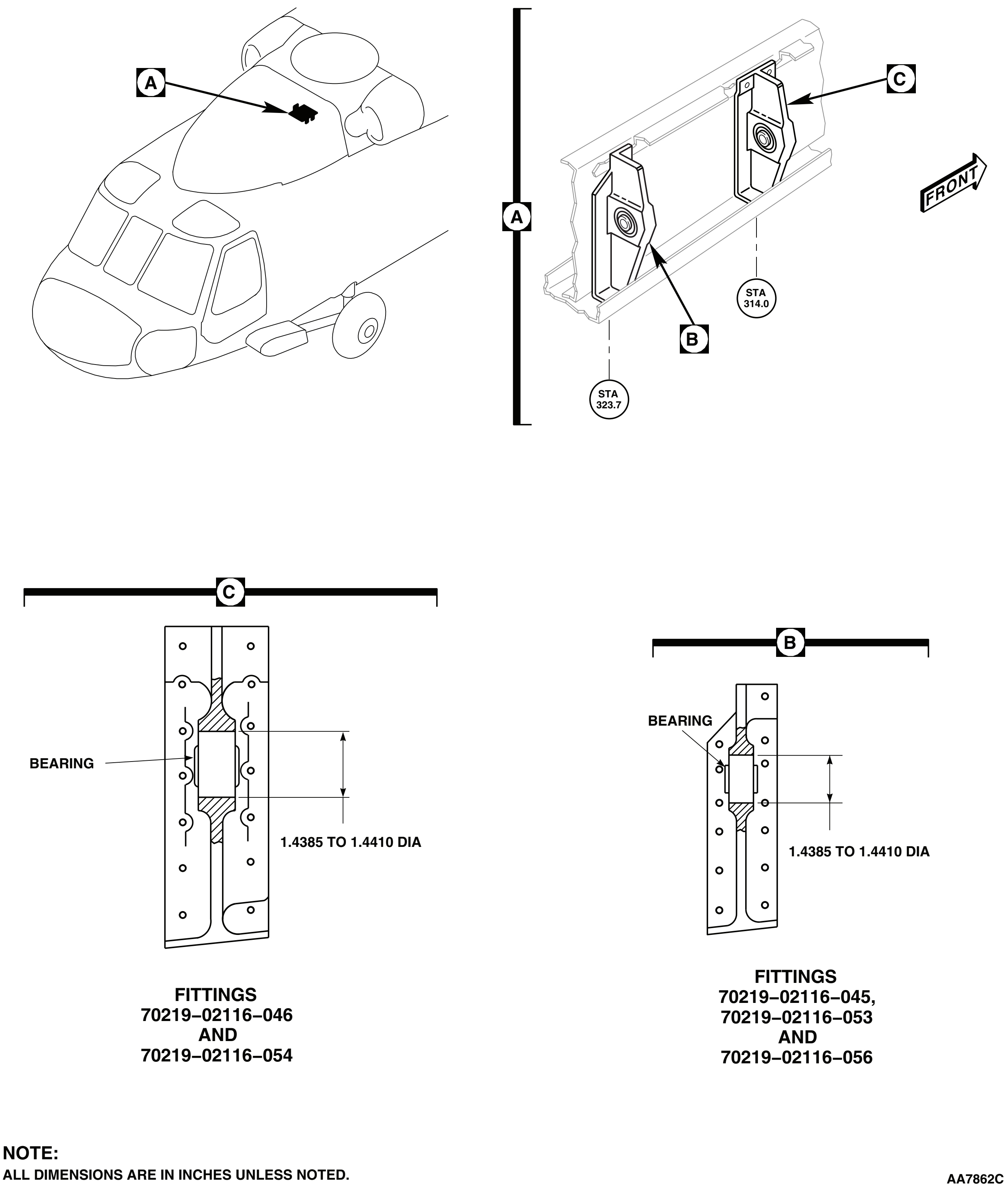


Figure 1. Replace Vibration Absorber Airframe Cabin Fitting Bearings (AVIM).

REMOVE - Continued

6. Remove sealing compound residue, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Remove abrasive residue, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

INSTALL

1. Prior to installing upper bearing, perform breakout torque check, as follows:
   a. Install bolt through bearing. Install enough washers on bolt to seat nut. Install and tighten nut so that washers do not rotate.
   b. Rotate bearing inner race 4 complete turns.
   c. Using torque wrench rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 5 INCH-OUNCES THROUGH FULL 360 DEGREE TURN.**
   d. Remove nut, washers, and bolt from bearing.
2. Prepare bonding surfaces of bearing and bearing bore as follows:
   a. Mask bearing interior against contamination using pressure sensitive adhesive tape, Item 334, WP 1803 00.
   b. Lightly sand bonding surface of bearing in circumferential direction, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   c. Clean off abrasive residue using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
   d. Remove masking tape from bearing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean bonding surfaces of fitting and bearing using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Apply light coat of sealing compound primer, Item 248, WP 1803 00, to bearing outside diameter and inside diameter of fitting bore, using disposable applicator, Item 53, WP 1803 00.

INSTALL - Continued

NOTE

Protect primed surfaces from contamination during drying. Do not touch primed surfaces with bare hands.

5. Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step 3. through step 5.

NOTE

If sealing compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of sealing compound will make it unusable.

6. Apply even coat of sealing compound, Item 294, WP 1803 00, to outside diameter of bearing and inside diameter of fitting bore, using disposable applicator, Item 53, WP 1803 00.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of sealing compound application. If this time is exceeded, step 3. through step 6. must be repeated.

7. **QA** Immediately install bearing in fitting with slight twisting motion to distribute sealing compound. Bearing must centered in fitting bore within 0.020-inch. Observe squeeze out at bond line and wipe off excess immediately (Figure 1, Detail B and Detail C).
8. Allow fitting assembly to cure for 24 hours minimum.
9. On reworked fitting assemblies, 70219-02116-045 and 70219-02116-046, perform the following:
   a. Apply corrosion resistant coating, Item 110, WP 1803 00, to any exposed reworked surfaces.
   b. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

MAIN LANDING GEAR FUSELAGE FITTING SPHERICAL BEARING (AVIM)

This WP supersedes WP 0359 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Arbor Prress, 00-P-636
Bearing Installation/Removal Tool, MLG, 707000-77533-043
Fluorescent Penetrant Inspection Kit, XMA101
Fly Cutter, 65700-40016-150
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13
Spinning Tool, STD-1850-458-011

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 106, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REPLACE MAIN LANDING GEAR FUSELAGE FITTING BEARING ASSEMBLY (REMOVED FROM HELICOPTER) (AVIM).

NOTE

The following procedure is for replacing the main landing gear fuselage fitting bearing assembly when assembly is removed from helicopter.

REMOVE

CAUTION

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

1. Clamp fitting assembly to baseplate of drill press, using fly cutter, machine off part of bearing outer race swaged over chamfer on fitting.
2. Press out bearing using arbor press and anvil and driver combination (TM 1-1500-204-23).

REPAIR

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean primer residue from bore and chamfer using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.
2. Check bearing bore lug for galling or scoring. **NO MORE THAN 0.010-INCH IF 25% OF CONTACT AREA IS AFFECTED.**
3. Blend out galling or scoring in bearing bore lug using abrasive cloth, Item 82, WP 1803 00.
4. Perform magnetic particle inspection of bearing bore lug per TM 1-1520-265-23.

INSTALL

1. Make sure fitting bore, chamfers, and adjacent surfaces are clean and dry.
2. Apply a light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and outer race of bearing.
3. **QA** Press bearing into fitting using arbor press and anvil and driver combination (TM 1-1500-204-23).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer, if any from bearing v-grooves using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and wipe dry.
5. Fix fitting with bearing on drill press base with proper spinning tool and anvil.

CAUTION

Do not allow petrolatum on ball bearing, or on exposed edge of the fabric liner. If petrolatum contacts these surfaces, remove petrolatum immediately.

6. Lubricate v-grooves of bearing and spinning tool with petrolatum, Item 216, WP 1803 00.

CAUTION

Do not allow spinning tool to make contact with ball side of groove as binding may result.

7. Swage v-groove lip over chamfer in fitting bore (TM 1-1500-204-23).
8. **QA** Turn fitting over and repeat swaging process on other side.
9. Clean petrolatum from v-groove using cheesecloth cloth, Item 85, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00.
10. Touch up bearing surface with corrosion preventive compound, Item 106, WP 1803 00.

INSTALL - Continued

11. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPLACE MAIN LANDING GEAR FUSELAGE FITTING BEARING ASSEMBLY (INSTALLED ON HELICOPTER) (AVIM).

NOTE

The following procedure is for replacing the main landing gear fuselage fitting bearing assembly when assembly is installed on the helicopter.

REMOVE

CAUTION

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

1. Clamp fitting assembly to baseplate of drill press, using fly cutter, machine off part of bearing outer race swaged over chamfer on fitting.
2. Following the instructions within the tool kit, carefully remove staking from the bearing.
3. Remove bearing using the MLG bearing installation/removal tool (707000-77533-043).

REPAIR

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean primer residue from bore and chamfer using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.
2. Check bearing bore lug for galling or scoring. **NO MORE THAN 0.010-INCH IF 25% OF CONTACT AREA IS AFFECTED.**
3. Blend out galling or scoring in bearing bore lug using abrasive cloth, Item 82, WP 1803 00.
4. Perform magnetic particle inspection of bearing bore lug per TM 1-1520-265-23.

INSTALL

1. Make sure fitting bore, chamfers, and adjacent surfaces are clean and dry.
2. Apply a light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and outer race of bearing.
3. Following the instructions within the tool kit, carefully install the new bearing.
4. Install bearing using the MLG bearing installation/removal tool (707000-77533-043).

INSTALL - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean primer, if any from bearing v-grooves using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and wipe dry.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO DOOR WINDOW (AVIM)

This WP supersedes WP 0360 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 10, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 293, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

REMOVE

1. If installed, remove seal from window with nonmetallic scraper (Figure 1).
2. Drill out rivets, remove strap and carefully work plexiglass window loose from frame, using nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to plexiglass windows will result if technical acetone comes in contact with it. Do not allow technical acetone to contact plexiglass windows.

3. Clean old sealing compound and adhesive from frame using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper. Wipe dry with a clean cheesecloth.

INSTALL

1. Position strap over replacement window (Figure 1).
2. Using strap as template, drill enough holes in plexiglass window to position plexiglass window and strap on frame.
3. Apply sealing compound, Item 293, WP 1803 00, to area on frame where window will be installed.
4. **QA** Install strap and glass on frame with button head rivets (TM 1-1500-204-23).

INSTALL - Continued

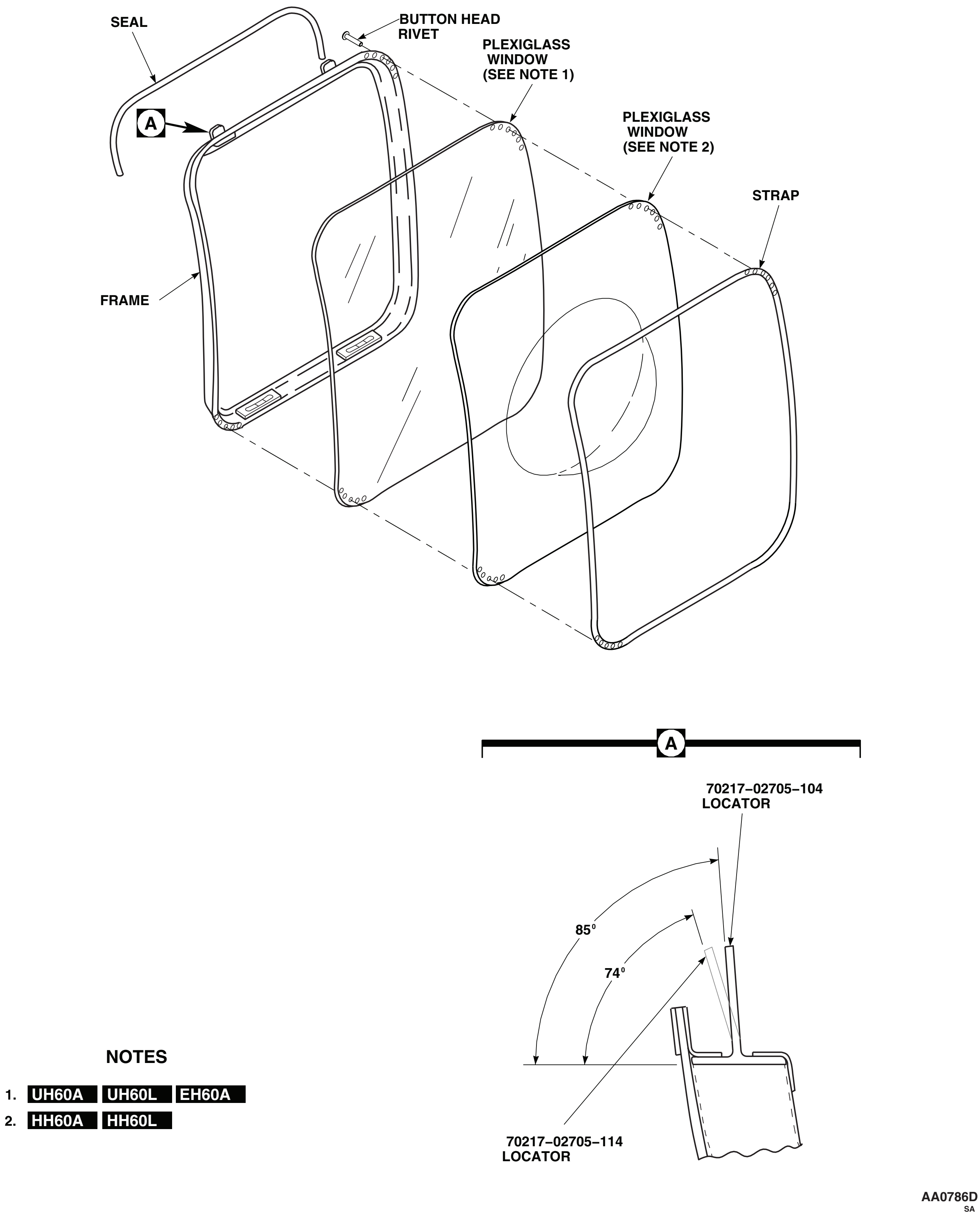


Figure 1. Replace Cargo Door Window (AVIM).

INSTALL - Continued

NOTE

Install seal on window with locator, 70217-02705-104 (bend angle 85 degrees) installed (Figure 1, Detail A). Do not install seal on windows with locator, 70217-02705-114 (bend angle 74 degrees), installed.

5. Roughen surface of new seal with abrasive cloth assortment, Item 1, WP 1803 00.
6. Bond new seal in place with adhesive, Item 10, WP 1803 00.
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

CARGO DOOR WINDOW FRAME (AVIM)

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)

References
TM 1-1500-204-23
TM 55-1500-345-23
WP 0378 00
WP 0381 00
WP 0401 00
WP 1408 00

GENERAL REPAIR

1. Replace blackout curtain, hook and pile fastener tape as required (WP 1408 00).

NOTE

Strap bonded to inside of frame opposite window may be cut in four places to get required fit on assembly.

2. Evaluate damage and repair per WP 0401 00 and WP 0378 00.

FRAME REPAIR

1. Carefully drill out rivets and remove locators from frame (TM 1-1500-204-23) (Figure 1, Detail A).
2. Carefully drill out rivets, remove cam and doubler from frame (TM 1-1500-204-23) (Figure 1).

NOTE

Upper and lower straps are found only on frame, 70217-02705-043.

3. Carefully drill out rivets and remove upper and lower straps from frame.
4. Install locators in frame with rivets (Figure 1, Detail A).

NOTE

Use blind rivets in locations noted (Figure 1, Detail A).

5. Install doublers and cams on bottom of frame with rivets.

NOTE

Install upper and lower straps only on frame, 70217-02705-043.

6. Install upper and lower straps on corners of frame with rivets.
7. Touch up paint (WP 0381 00 and TM 55-1500-345-23).

FRAME REPAIR - Continued

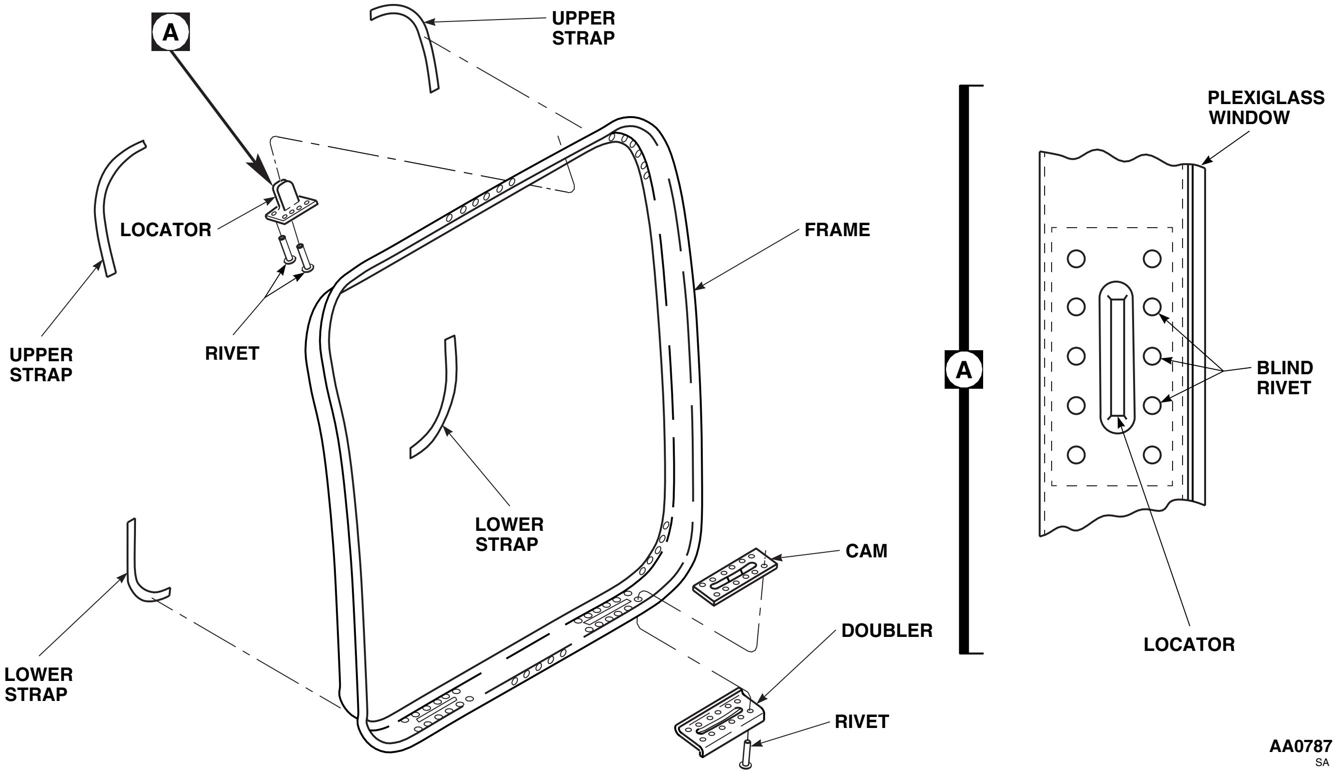


Figure 1. Repair Cargo Door Window Frame (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

SOUNDPROOFING PANEL (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Steam Iron, A-A-632

Material/Parts

Adhesive, Item 14, WP 1803 00
Adhesive, Item 30, WP 1803 00
Adhesive, Item 31, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
WP 0258 00
WP 0378 00
WP 0379 00
WP 1803 00

REPAIR

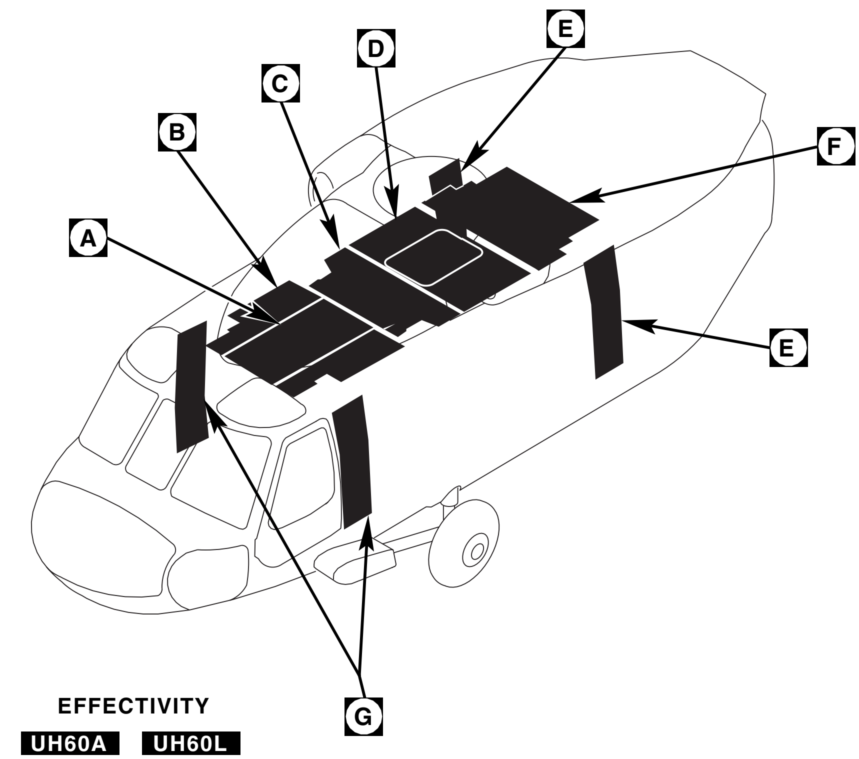
1. Soundproofing panels are bonded sandwich construction of Tedlar-coated Nomex cloth facing, foam, aluminum stiffening panel, backing foam, and Tedlar-coated Nomex cloth backing. Layers are stitched and bonded to form a soundproofing blanket. In some areas, Kevlar/epoxy laminates are used as panel material in place of aluminum.

NOTE

When applicable, two materials are shown on Figures for face cloth and back cloth. One is basic material with foam and the other is trim cloth only.

2. Tears and rips in the outer cloth are repaired by bonding Tedlar/Nomex facing or backing cloth over torn area using adhesive, Item 31, WP 1803 00. Use facing cloth for interior patching and backing cloth for patching backside of panel. Cut cloth patch 1 inch larger, on all sides, than tear dimensions. Lay film adhesive (1/8-inch larger than patch size) on repair area and center over tear. Cover with cloth patch and iron on patch with steam iron heated to 310°F(155°C). As alternate adhesives, use adhesive, Item 30, or adhesive, Item 14, WP 1803 00, to bond cloth patch to panel. Apply adhesive carefully to prevent discoloring panel outside repair area.
3. Fasteners pulling through stiffening panels are repaired per WP 0378 00 for aluminum and WP 0379 00 for Kevlar/epoxy laminates. Peel back blanket material to expose damaged area and apply patch 2 x 2 inches over hole. Rebond foam and cloth using noted adhesives, and install fasteners per TM 1-1500-204-23.
4. Tedlar coating on facing or backing cloth is oil-proof and may be cleaned using cheesecloth cloth, Item 85, WP 1803 00, moistened with cleaning compound solvent, Item 71, WP 1803 00.
5. Rebond loosened seals with adhesive, Item 14, WP 1803 00, and loosened fastener tape using adhesive, Item 31, WP 1803 00, per Notes, Figure 1, Sheet 13.
6. Repair honeycomb core damage per WP 0379 00.
7. Remove and reinstall soundproofing panels per WP 0258 00.

REPAIR - Continued



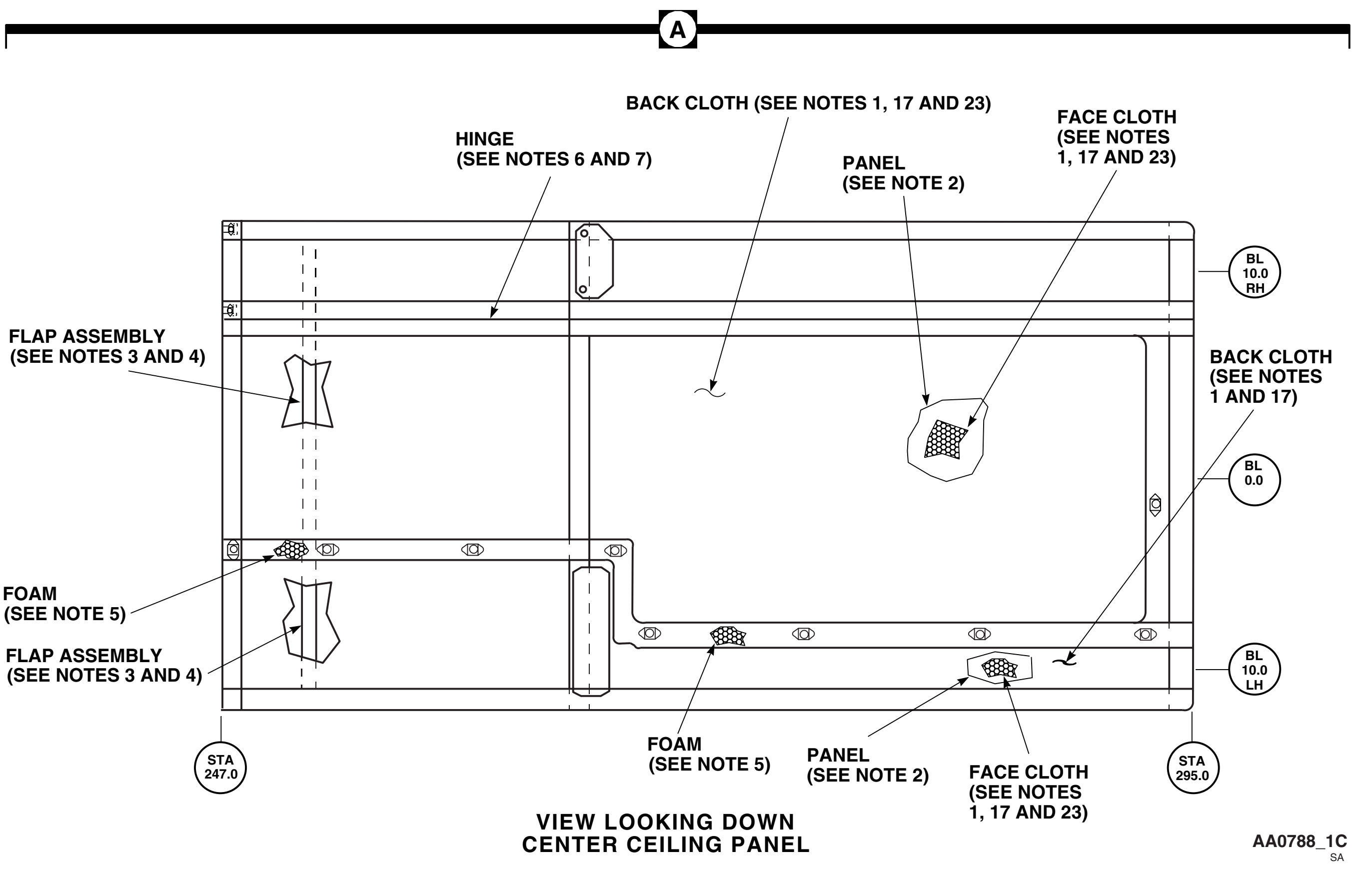
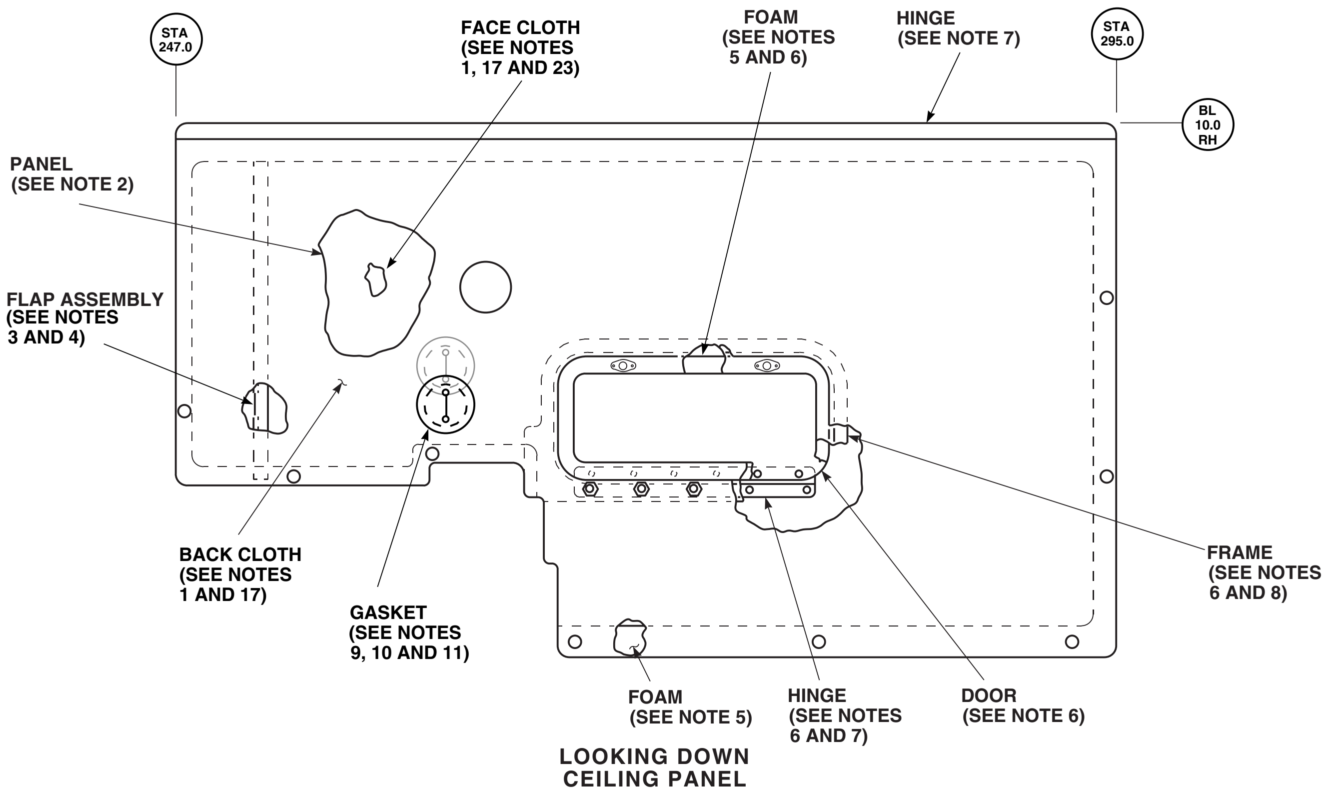


Figure 1. Soundproofing Panels (AVIM). (Sheet 1 of 13)

REPAIR - Continued



B

C

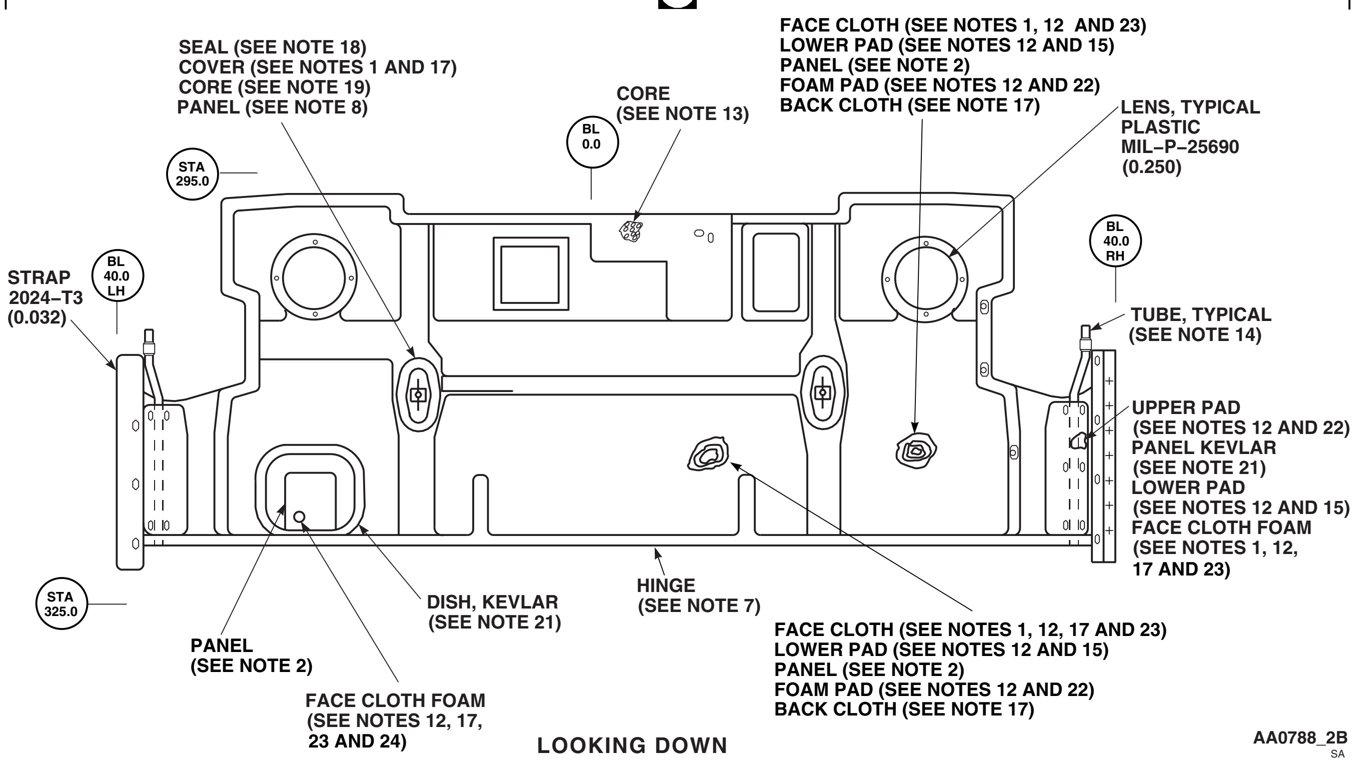
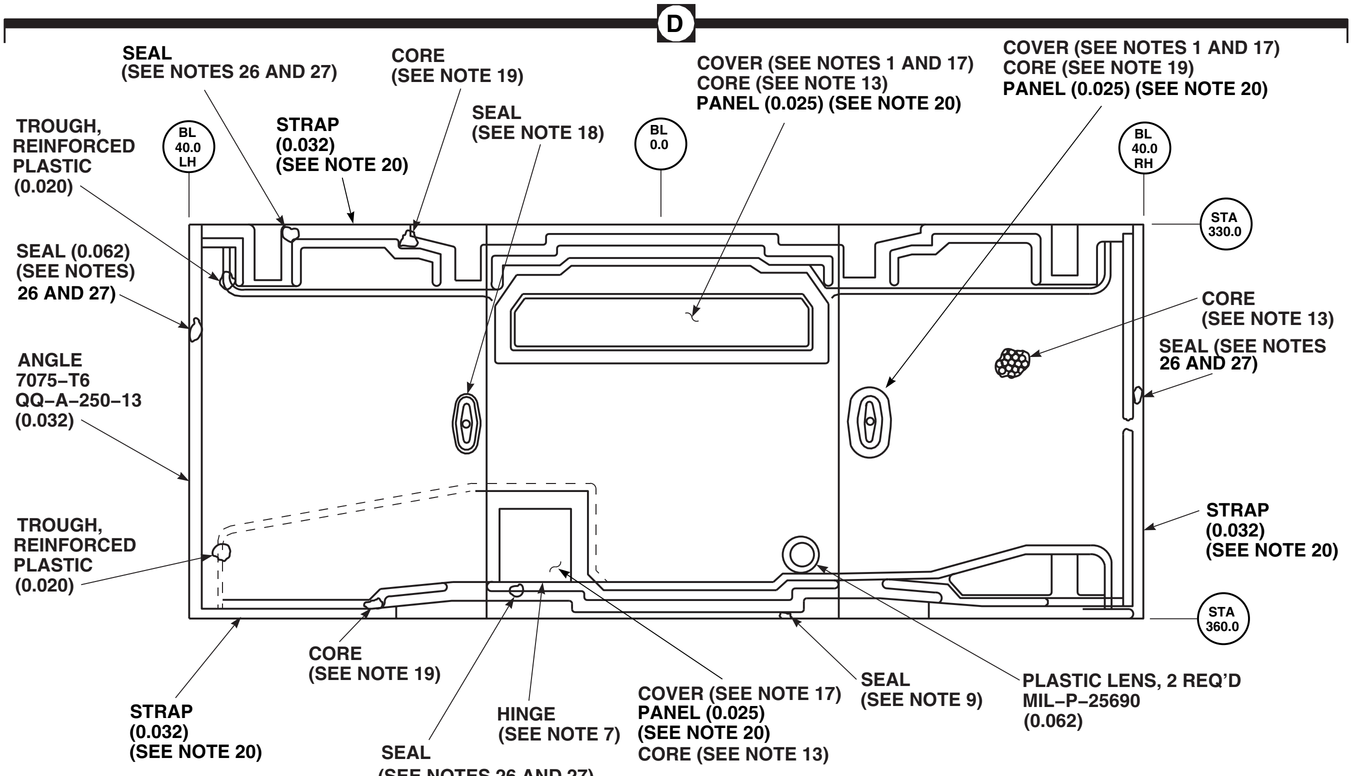


Figure 1. Soundproofing Panels (AVIM). (Sheet 2 of 13)

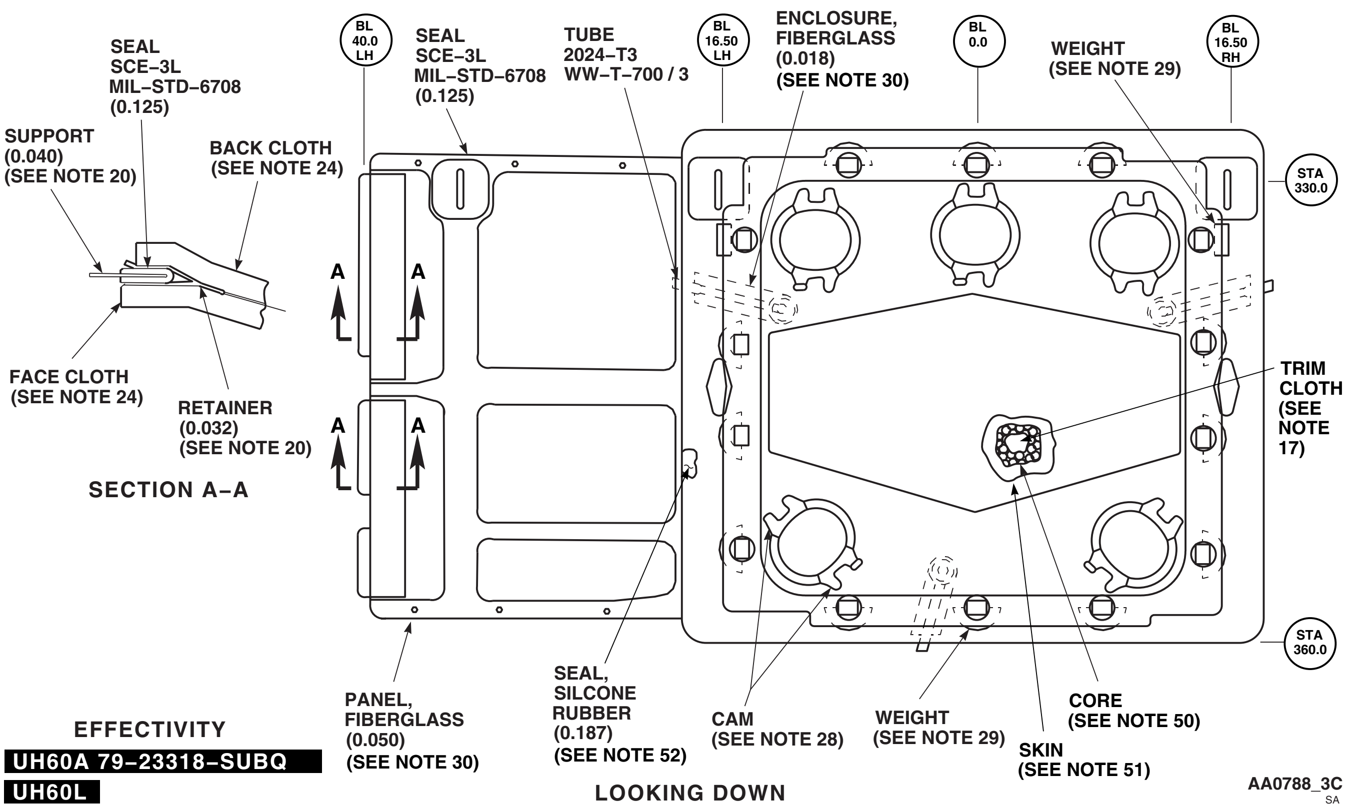
REPAIR - Continued



EFFECTIVITY

UH60A 77–22714–79–23317

LOOKING DOWN



EFFECTIVITY

UH60A 79–23318–SUBQ

UH60L

LOOKING DOWN

Figure 1. Soundproofing Panels (AVIM). (Sheet 3 of 13)

REPAIR - Continued

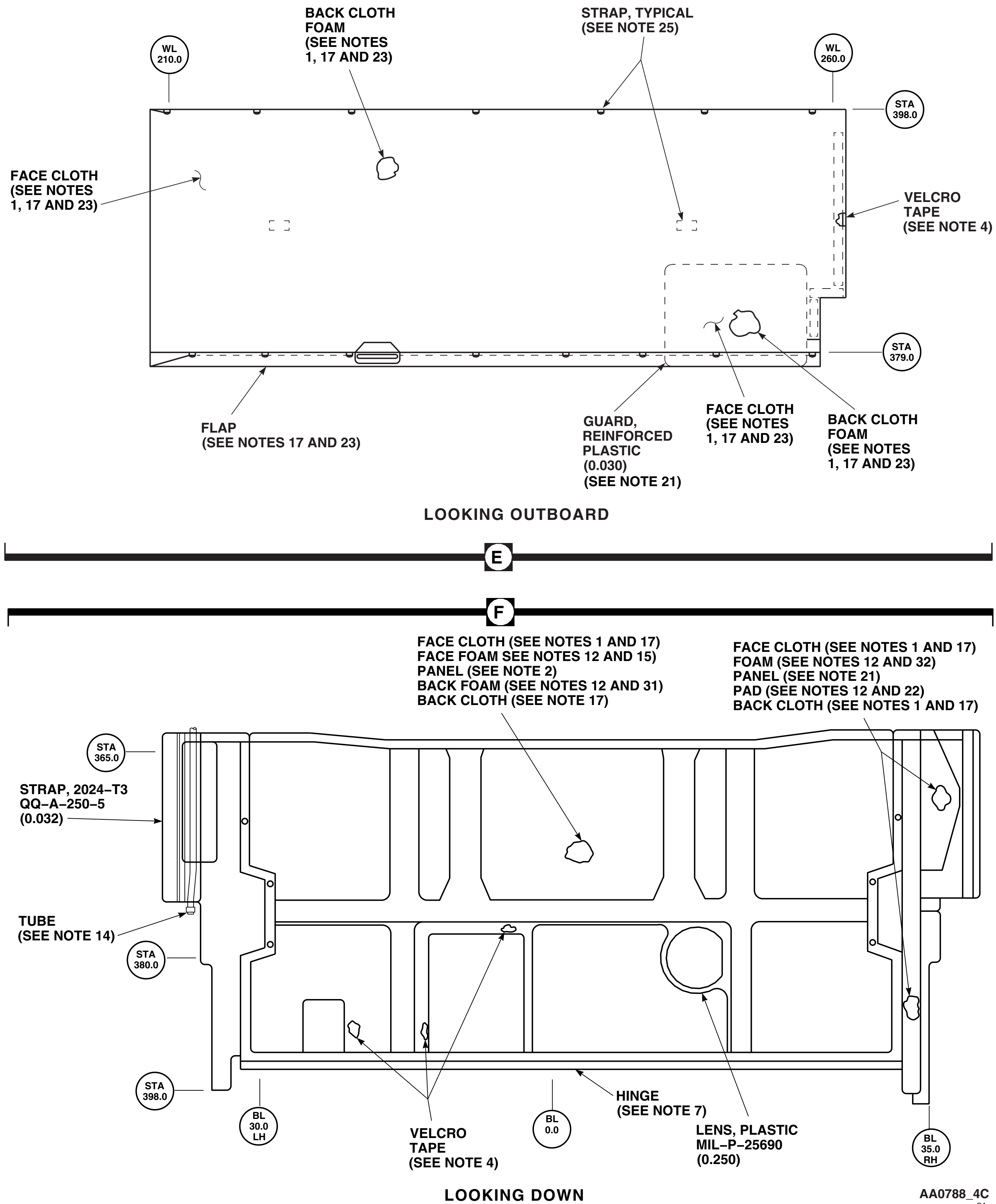
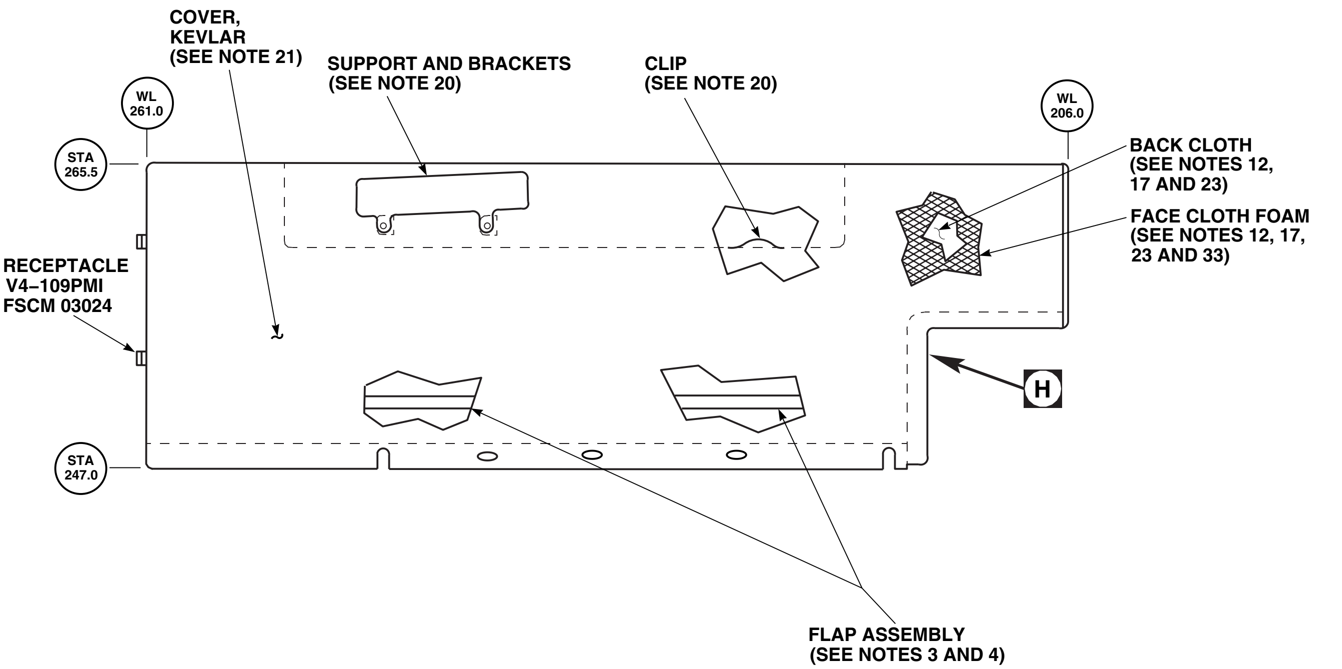


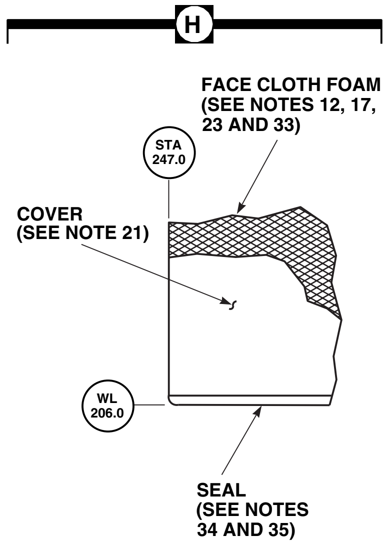
Figure 1. Soundproofing Panels (AVIM). (Sheet 4 of 13)

REPAIR - Continued



VIEW RH
LOOKING OUTBOARD

G



AA0788_5D
SA

Figure 1. Soundproofing Panels (AVIM). (Sheet 5 of 13)

REPAIR - Continued

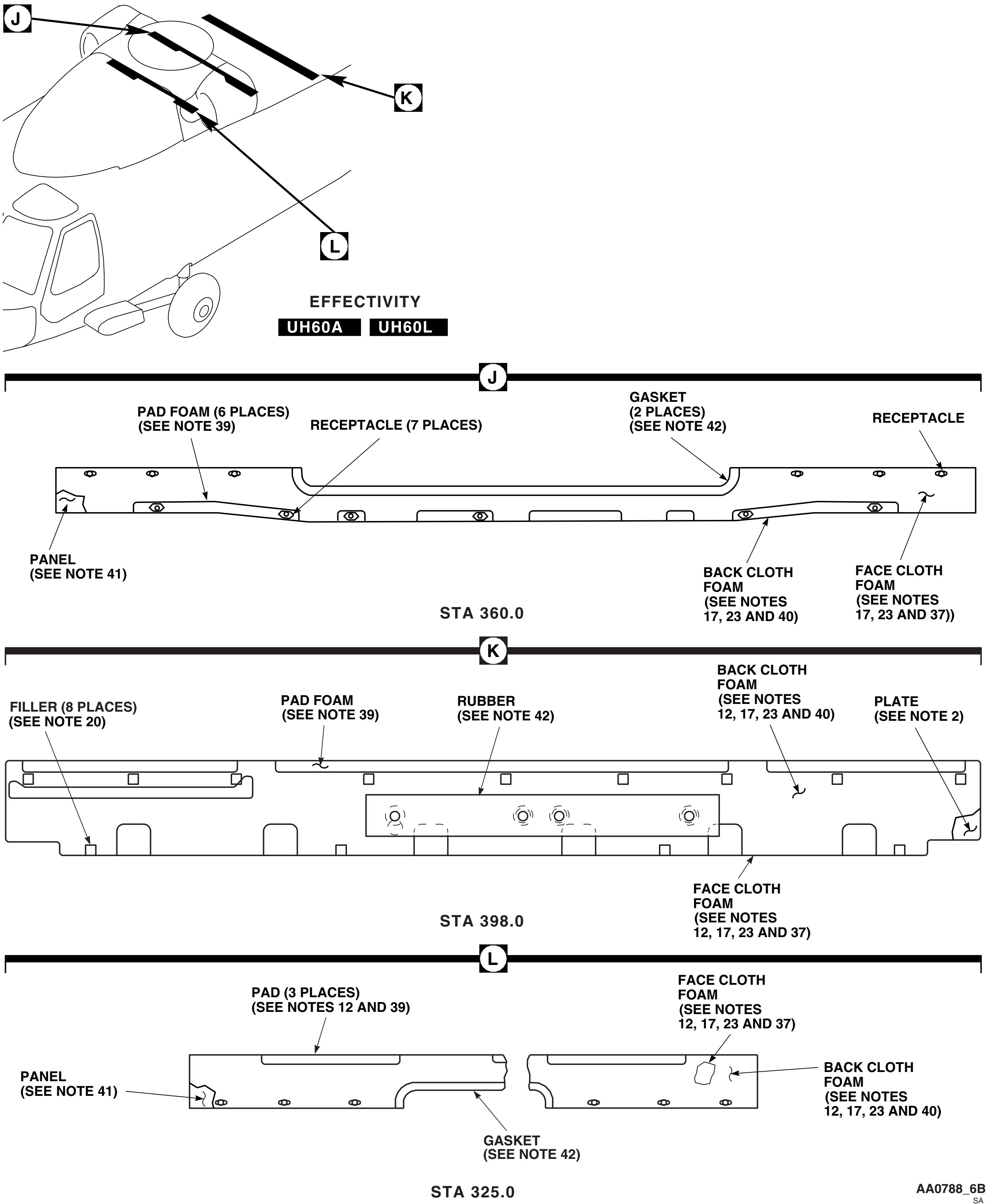
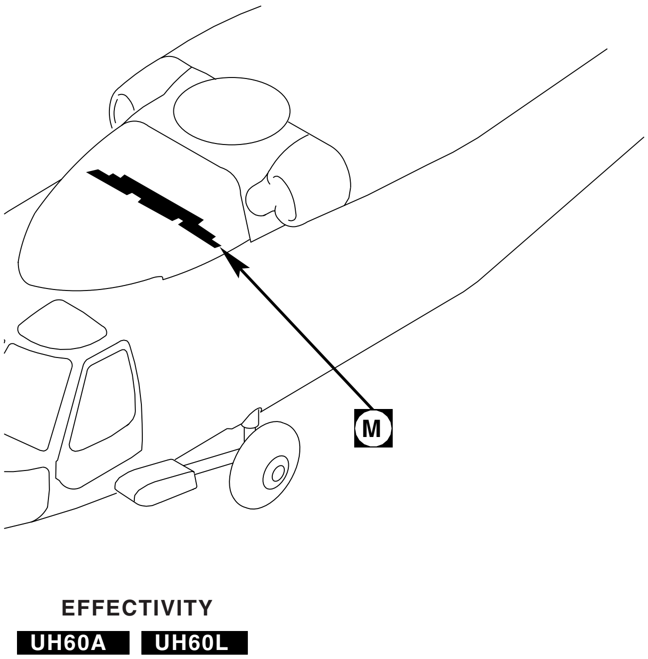


Figure 1. Soundproofing Panels (AVIM). (Sheet 6 of 13)

REPAIR - Continued



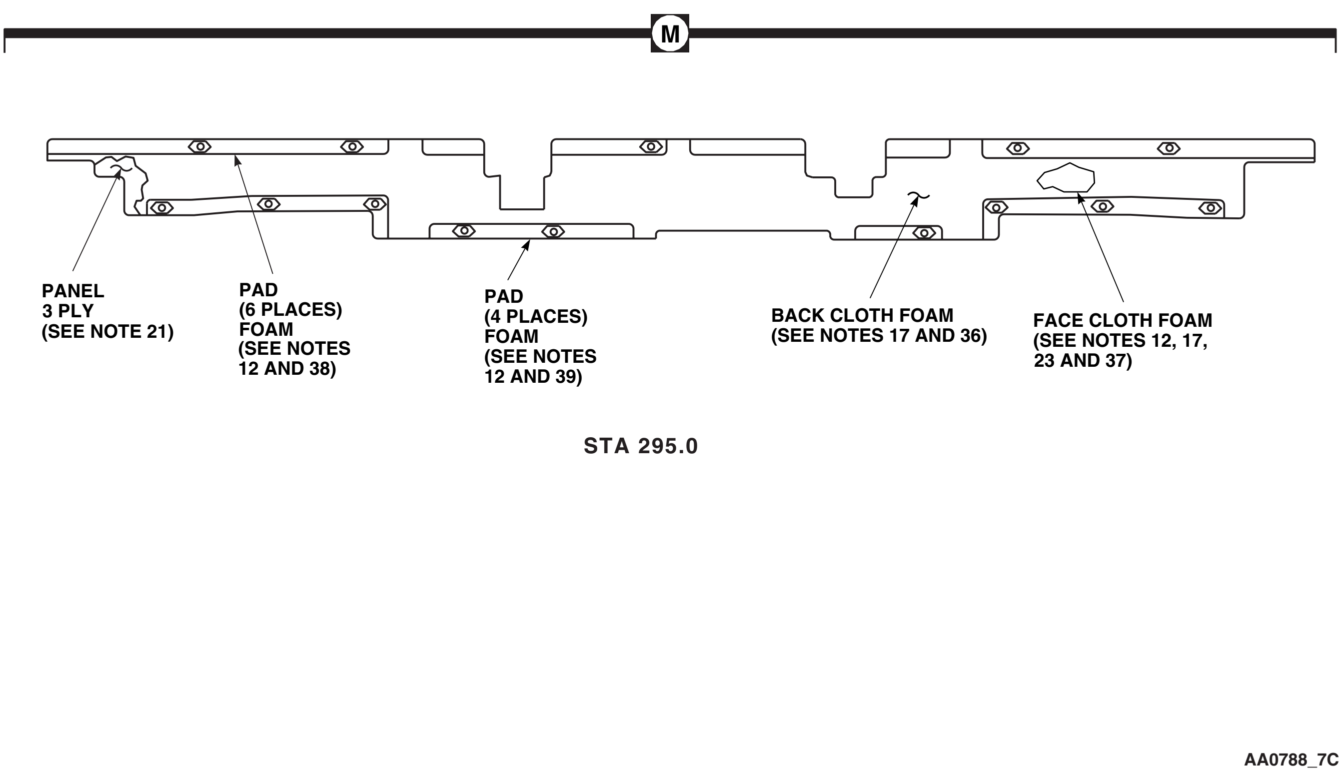


Figure 1. Soundproofing Panels (AVIM). (Sheet 7 of 13)

REPAIR - Continued

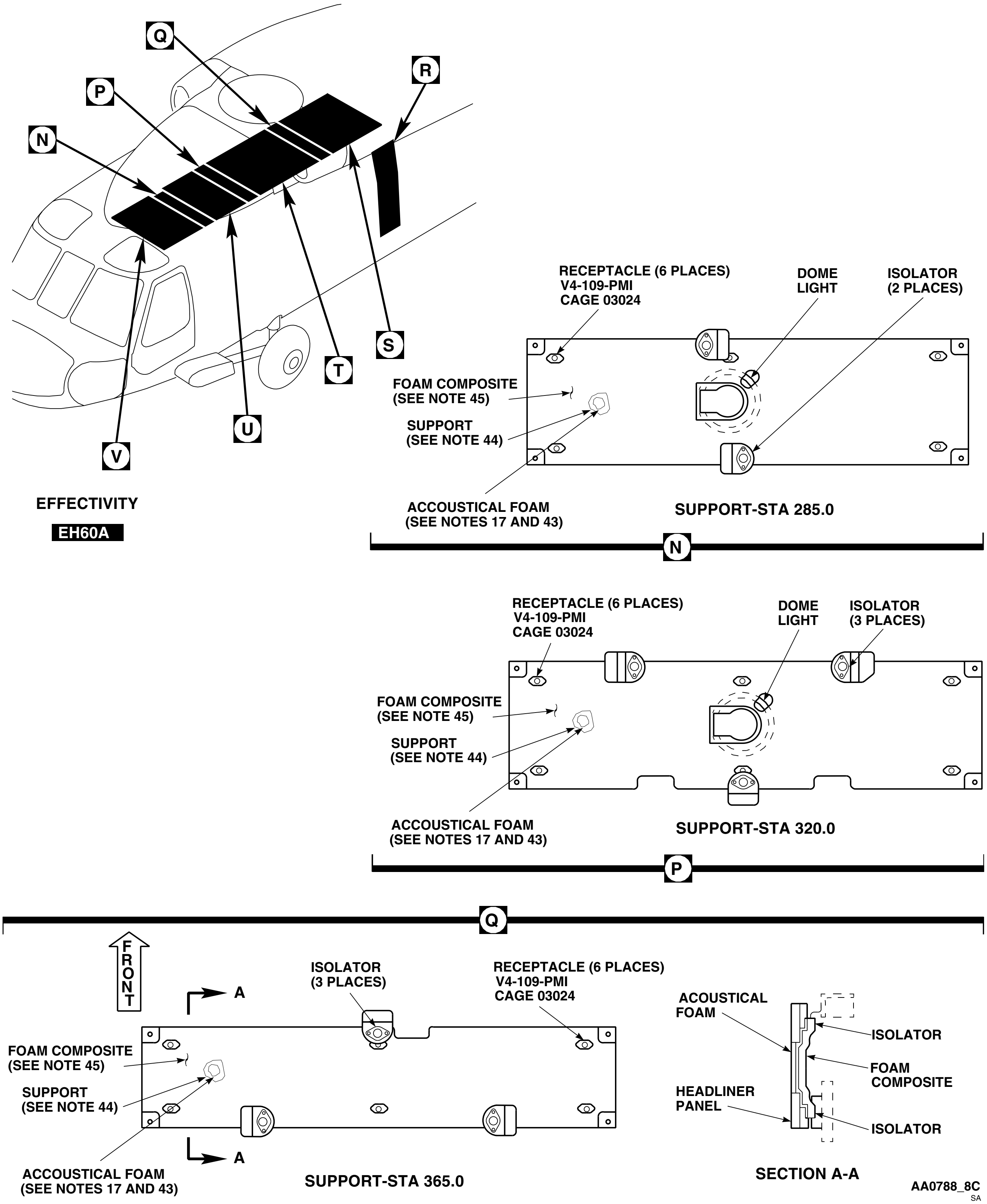
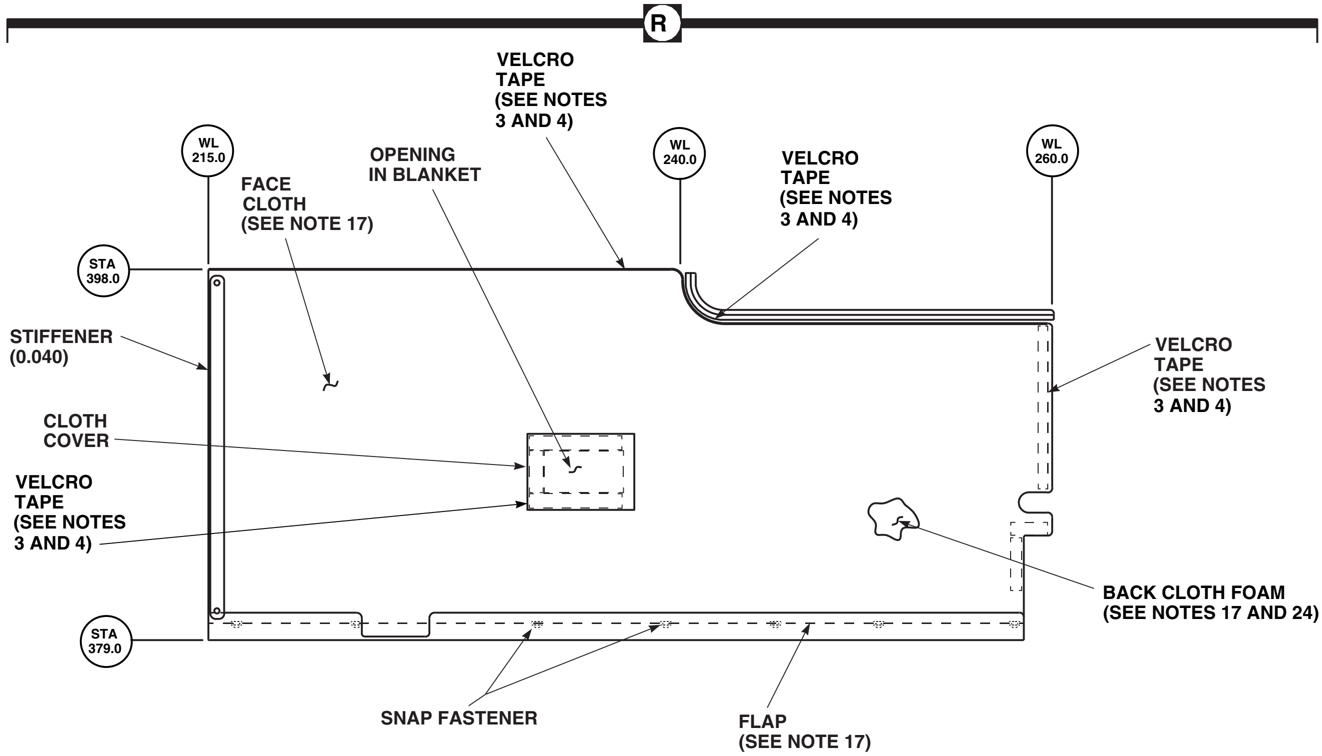
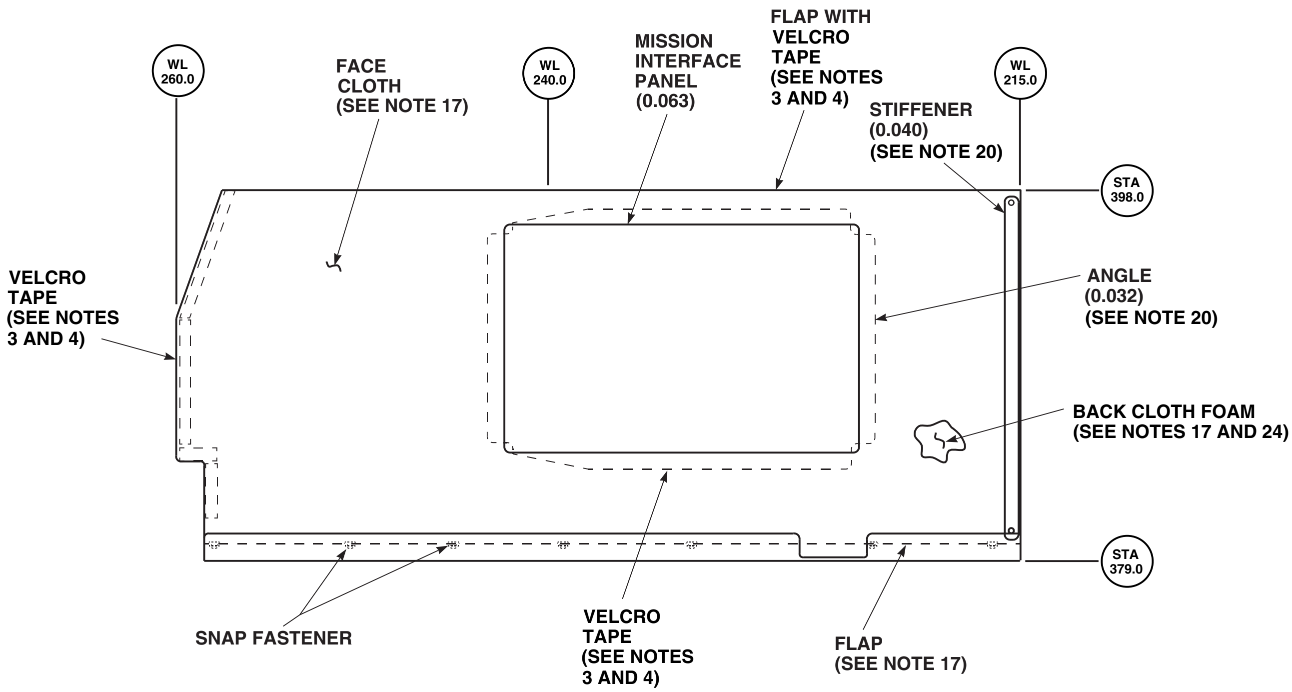


Figure 1. Soundproofing Panels (AVIM). (Sheet 8 of 13)

REPAIR - Continued



LEFT SIDE, LOOKING OUTBOARD



RIGHT SIDE, LOOKING OUTBOARD

AA0788_9B
SA

Figure 1. Soundproofing Panels (AVIM). (Sheet 9 of 13)

REPAIR - Continued

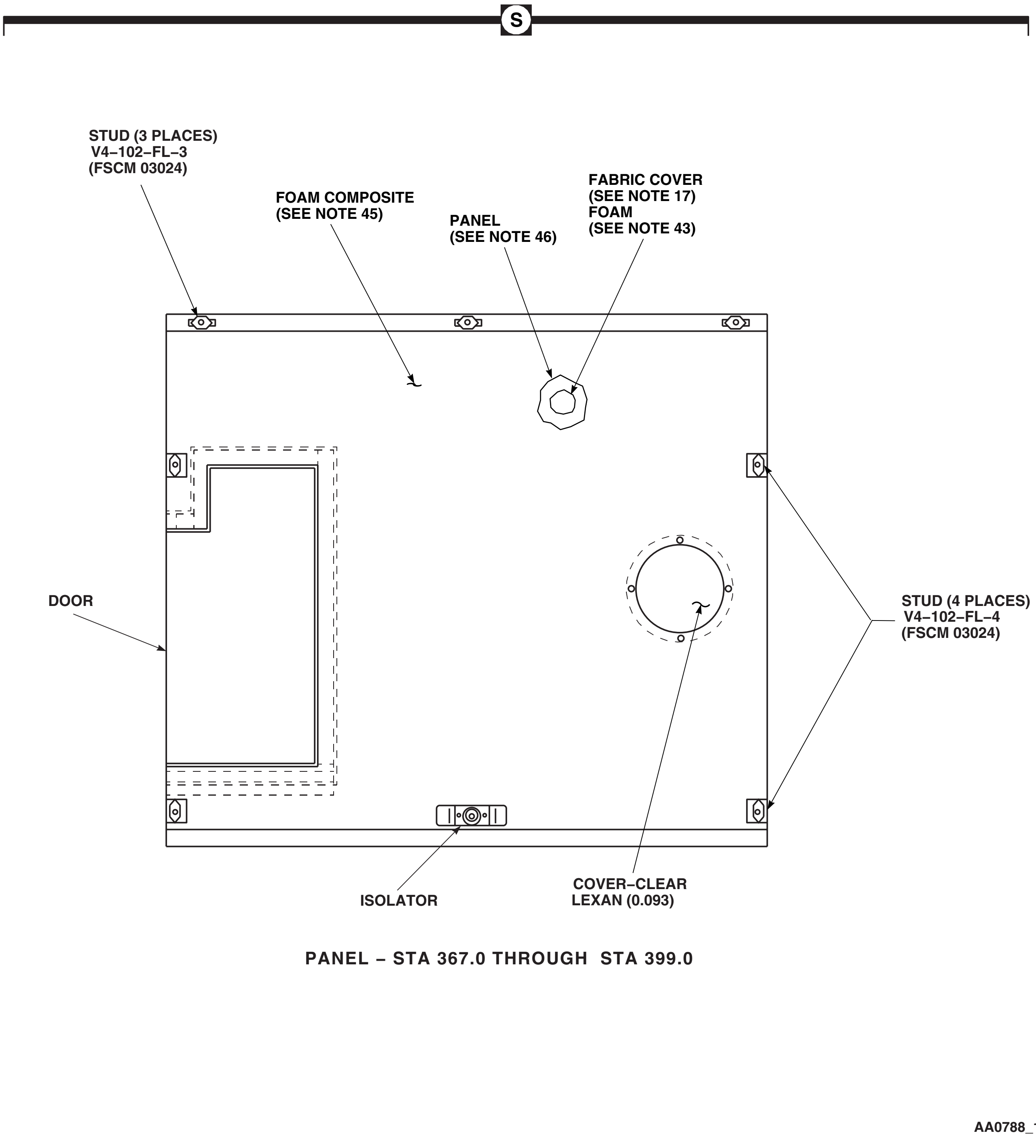
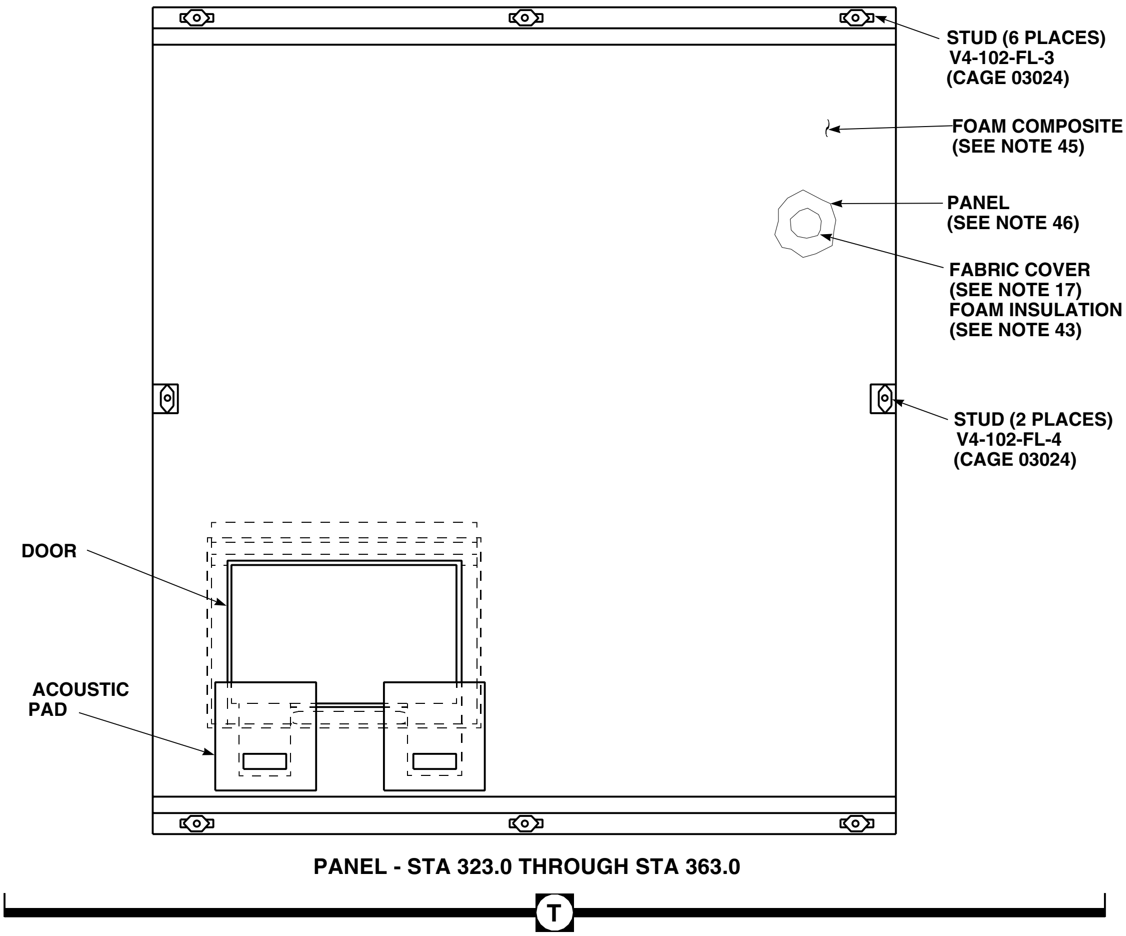


Figure 1. Soundproofing Panels (AVIM). (Sheet 10 of 13)

REPAIR - Continued



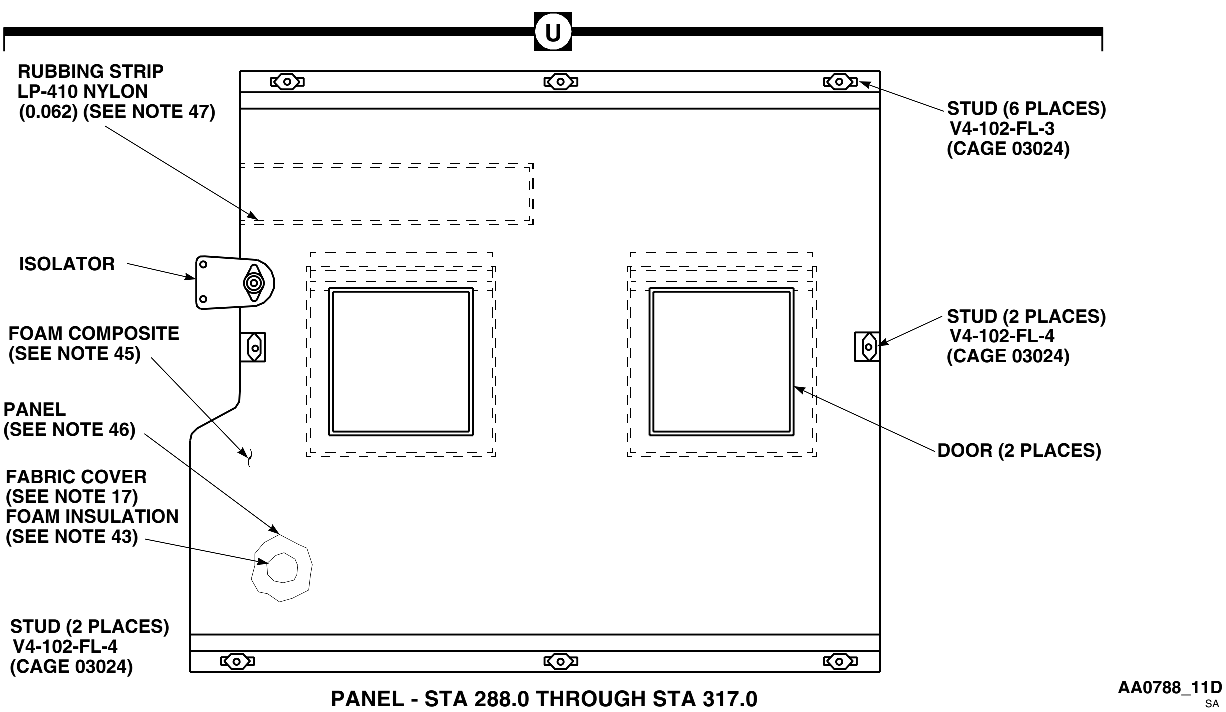


Figure 1. Soundproofing Panels (AVIM). (Sheet 11 of 13)

REPAIR - Continued

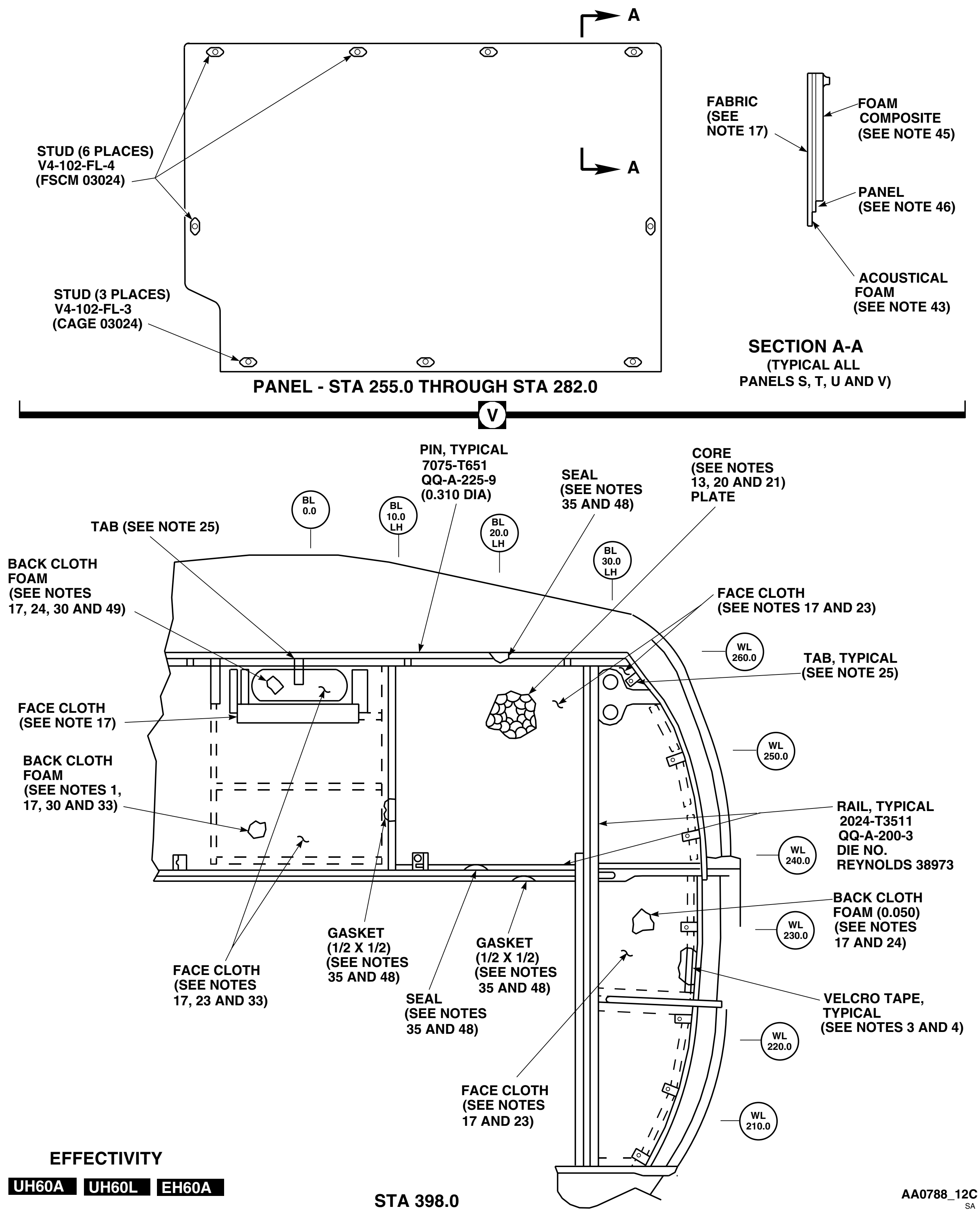


Figure 1. Soundproofing Panels (AVIM). (Sheet 12 of 13)

REPAIR - Continued

NOTES:

1. MAKE FROM SS9932–027 (TRIM CLOTH ON FOAM).
2. ALUMINUM ALLOY, 6061–T6, QQ–A–250–11 (0.016).
3. FLAP, MIL–C–20696, TYPE I, CL3, COLOR LIGHT GULL GREY,
   LOOP P/N SJ3418–200/1.00 VELCRO LOOP,
   HOOK P/N SJ 3419–200/1.00 VELCRO HOOK.
4. BOND TAPE LOOP TO STRUCTURE; BOND OR SEW TAPE HOOKS TO FLAP. USE ADHESIVE, ITEM 30, WP 1803 00.
5. MAKE FROM SS9232–030 (FOAM).
6. RIGHT SIDE ONLY.
7. POLYHINGE, COLOR GREY, 1.00 THK BY 1.50 WIDE.
8. ALUMINUM ALLOY, 6061–T6, QQ–A–250–11 (0.025).
9. RUBBER, 0.062 THK, MIL–R–6855, CL2, GR60.
10. UH60A 79–23343 – SUBQ UH60L EH60A
11. RIGHT SIDE GASKET LOCATION SHOWN IN PHANTOM.
12. BOND FOAM TO PANELS AND COATED NOMEX CLOTH TO FOAM WITH ADHESIVE, ITEM 13, APPENDIX D.
13. FIBERGLASS/HONEYCOMB/FLYSCREEN CORE, HRH–10 NOMEX, NO. 1507 (0.50).
14. POLYETHYLENE TUBE, TYPE 3 (0.62 OD BY 0.06 WALL).
15. MAKE FROM SS9232–035 (FOAM).
16. MAKE FROM SS9232–034 (FOAM).
17. MAKE FROM SS9232–024 (TRIM CLOTH).
18. PYRELL FOAM (0.12), CAGE 78112.
19. BALSA CORE, PLANK, MIL–S–7998.
20. ALUMINUM ALLOY, 7075–T6, QQ–A–250–13.
21. KEVLAR PREPREG, SS9612–008 (0.030).
22. MAKE FROM SS9232–032 (FOAM).
23. COLOR SAGE GREEN.
24. MAKE FROM SS9232–025 (TRIM CLOTH ON FOAM).
25. NYLON WEBBING, GRAY, MIL–W–4088, TYPE 1A, BUTTON–MS27980–IN, SOCKET–MS27980–6N.
26. NITRILE RUBBER, MIL–R–6855, CLASS 1, GRADE 60 (0.062).
27. BOND SEAL TO CLOTH WITH ADHESIVE, ITEM 13, WP 1803 00.
28. UH60A 79–23318 – 81–23570 QQ–A–250–5 (0.190).
    UH60A 81–23571 – SUBQ UH60L EH60A CASTING, ASM 4260, 356–T6.
29. UH60A 79–23318 – 79–23347 LEAD (0.250).
    UH60A 79–23348 – SUBQ UH60L EH60A STEEL 4130 COND A OR N.
30. E–GLASS PREPREG, SS9578–007.
31. MAKE FROM SS9232–033 (FOAM).
32. MAKE FROM SS9232–031 (FOAM).
33. MAKE FROM SS9232–026 (TRIM CLOTH ON FOAM).
34. RUBBER, ASTM D1056, 2B2L.
35. BOND SEAL TO COVER WITH ADHESIVE, ITEM 30, WP 1803 00.
36. MAKE FROM SS9232–014.
37. MAKE FROM SS9232–018 (TRIM CLOTH ON FOAM).
38. MAKE FROM SS9232–021 (FOAM).
39. MAKE FROM SS9232–022 (FOAM).
40. MAKE FROM SS9232–019 (TRIM CLOTH ON FOAM).
41. ALUMINUM ALLOY, 6061–TO, QQ–A–250–13 (0.032).
42. MAKE FROM MIL–STD–670, SCEIIL.
43. MAKE FROM SS9232–029 (FOAM).
44. F–600011, COLOR NO. 701, 0.125 THK LEXAN HAIRCELL TEXTURE, SS9507, TYPE II,
    ALTERNATE MATERIAL BOLTARON ABS 6800, SS9507, TYPE III, 0.125 THK HAIRCELL,
    COLOR NO. 37038, PER MIL–STD–595.
45. 635–227–635–OC, 635–454–635–OC, 635–227–635–RP.
46. 0.25 THK DIVINYCELL, TYPE HT70, LAMINATED WITH 2 PLIES PER SS9578–008 EACH SIDE.
47. EH60A 86–24562 – SUBQ
48. MAKE FROM MIL–STD–670, SCE3LCD.
49. BOND SS9578–007 TO CLOTH SIDE WITH ADHESIVE, ITEM 30, WP 1803 00.
50. ACOUSTICORE, HRH–10–1/4 (1.5) (WITH 0.6#/FT3 FIBERGLASS BATTTING, CORE TO BE 0.312 THK.
51. NO. 1507 FLYSCREEN (PREPREG FIBERGLASS), PER SS9231.
52. RUBBER, MIL–STD–670B.

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SA

Figure 1. Soundproofing Panels (AVIM). (Sheet 13 of 13)

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING SUPPORT FITTING BEARING (AVIM)

This WP supersedes WP 0363 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Arbor Press, 00-P-636
Flycutter, 65700-40016-150
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Spinning Tool, STD-1850-458-011

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 106, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-322-24
WP 1803 00

NOTE

Bearing replacement is the same for both front and rear support fittings.

REMOVE

CAUTION

Damage to fitting chamfer will occur if staked lip is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

1. Clamp fitting to drill press base plate.
2. Using flycutter, machine off the part of bearing outer race swaged over chamfer on fitting (Figure 1, Detail A).
3. Press out bearing using arbor press and driver and anvil combination (TM 55-1500-322-24).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer residue from bore and chamfer, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

INSTALL

1. Make sure that fitting bore, chamfers, and adjacent surfaces are clean and dry.

INSTALL - Continued

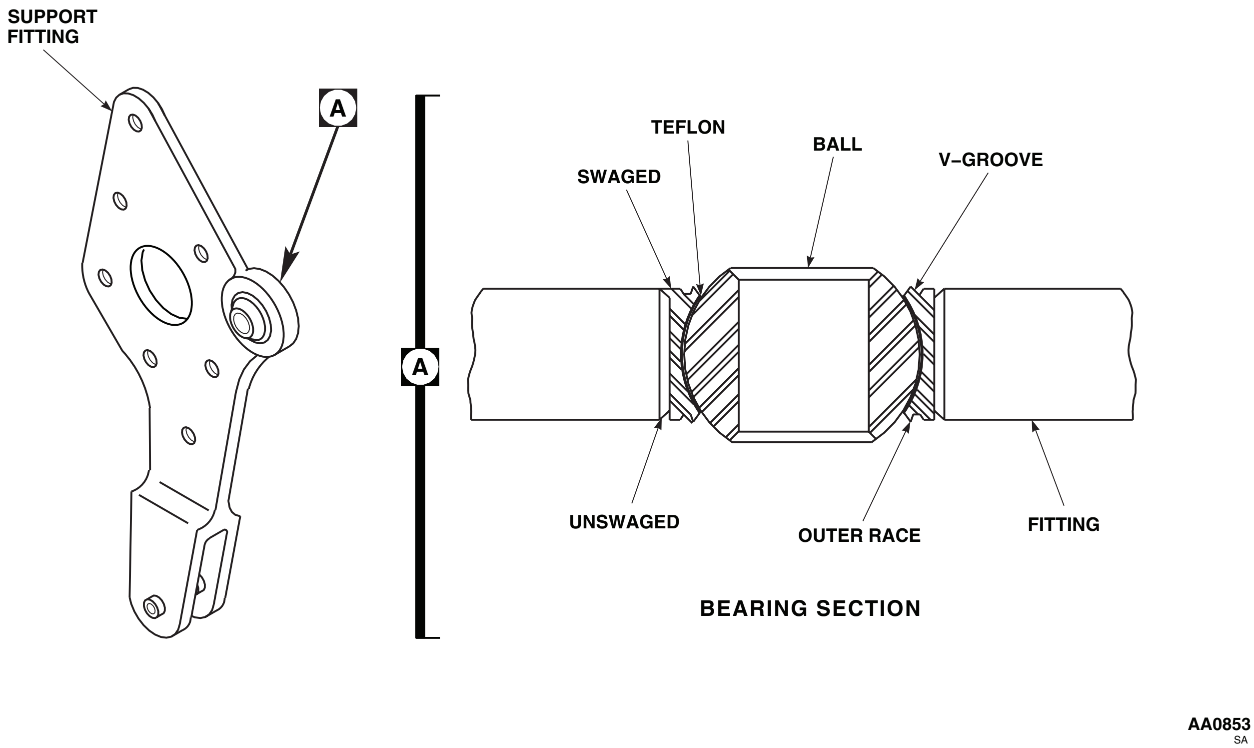


Figure 1. Replace Engine Cowling Support Fitting Bearing (AVIM).

2. Apply light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and outside surface of bearing outer race.
3. **QA** Press bearing into fitting using arbor press and driver and anvil combination (TM 55-1500-322-24).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer, if any, from bearing V-grooves using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.
5. Fix fitting on drill press baseplate, with proper spinning tool and anvil.

CAUTION

To prevent damage to Teflon, keep petrolatum from bearing ball and Teflon liner. Remove petrolatum immediately if it gets on these surfaces.

6. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 216, WP 1803 00.

INSTALL - Continued

CAUTION

Bearing may be damaged if spinning tool contacts ball side of V-groove. Deformation of outer race can cause binding of bearing.

7. Swage V-groove lip over chamfer in fitting bore (TM 55-1500-322-24) (Figure 1, Detail A).
8. **QA** Turn fitting over and repeat swaging process on other side.
9. Clean petrolatum, Item 216, WP 1803 00, from V-groove with cleaning compound solvent, Item 71, WP 1803 00, on cheesecloth cloth, Item 85, WP 1803 00.
10. Touch up bearing surfaces with corrosion preventive compound, Item 106, WP 1803 00.
11. Touch up fitting surfaces with primer coating, Item 243, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING SUPPORT FITTING BUSHINGS (AVIM)

This WP supersedes WP 0364 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Arbor Press, 00-P-636
Bushing Installation Kit, GGG-P-643
Gloves
Goggles/Face Shield

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

1. Press bushings out of fitting, using arbor press and bushing installation kit (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Remove primer residue from surfaces of fitting clevis and inside of bore using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

INSTALL

1. Make sure that inside of fitting clevis bores are clean and dry.
2. Apply thin coat of primer coating, Item 243, WP 1803 00, to bushing outer surface (on flanged bushing, coat under surface of flange).
3. Apply thin coat of primer coating, Item 243, WP 1803 00, to inside of clevis bore.
4. **QA** Press bushing into bore of fitting clevis using arbor press and bushing installation kit.
5. On rear fitting, install flanged bushing first and use appropriate spacer blocks for gap dimension when installing plain bushing (Figure 1, Detail A).
6. On front fitting, install front plain bushing to protrude 0.050-inch into clevis slot and use spacer blocks for gap dimension when installing rear plain bushing.

INSTALL - Continued

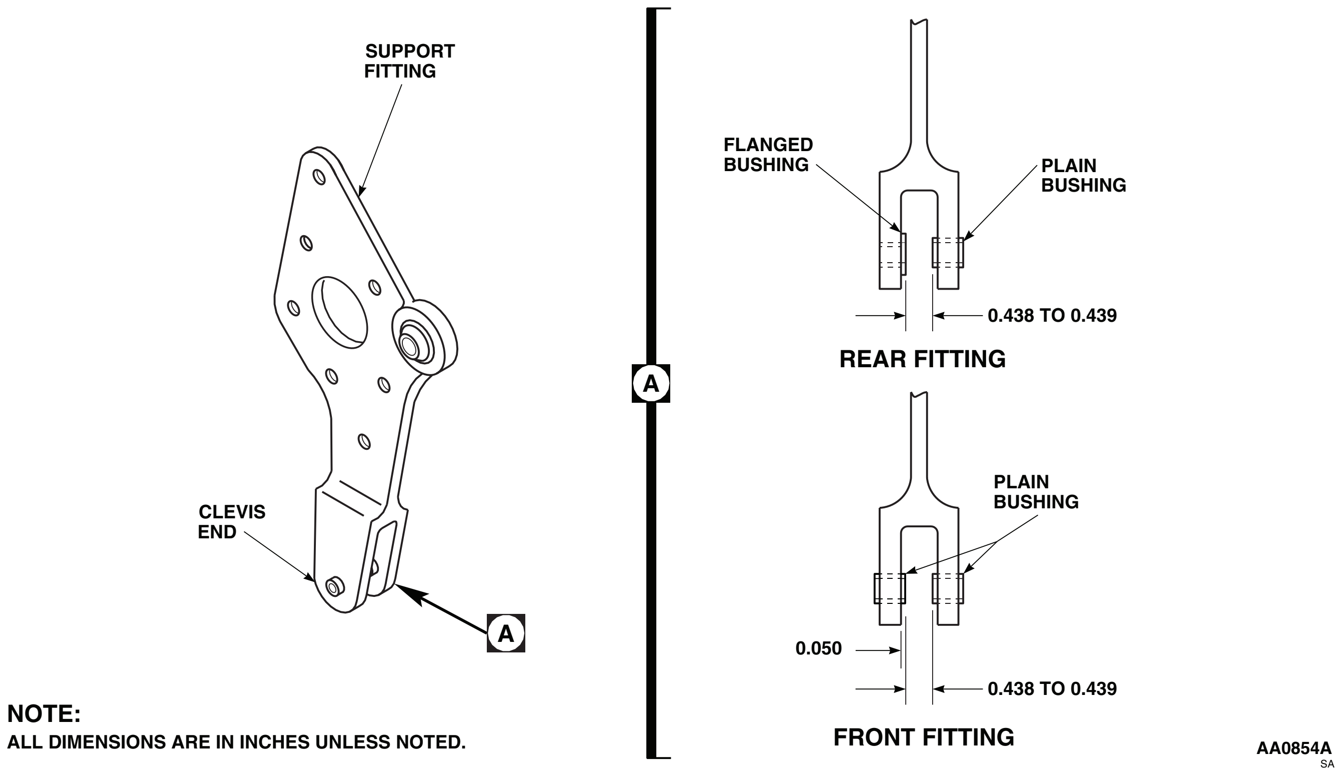


Figure 1. Replace Engine Cowling Support Fitting Bushings (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

ENGINE COWLING FITTING BEARING (AVIM)

This WP supersedes WP 0365 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Arbor Press, 00-P-636
Flycutter, 65700-40016-150
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Spinning Tool, STD-1850-458-011

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 106, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-322-24
WP 1803 00

REMOVE

CAUTION

Damage to fitting chamfer will occur if staked lip is improperly removed. Be careful not to damage chamfer of fitting when machining off staked lip of bearing.

NOTE

Bearing replacement is the same for both front and rear cowling fittings.

1. Clamp fitting to baseplate of drill press.
2. Using flycutter, machine off lip of bearing outer race that is swaged over chamfer on fitting (Figure 1, Detail A).
3. Press out bearing, using arbor press and driver and anvil combination (TM 55-1500-322-24).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer residue from fitting bore and chamfer, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

INSTALL

1. Make sure that fitting bore, chamfer, and adjacent surfaces are clean and dry.

INSTALL - Continued

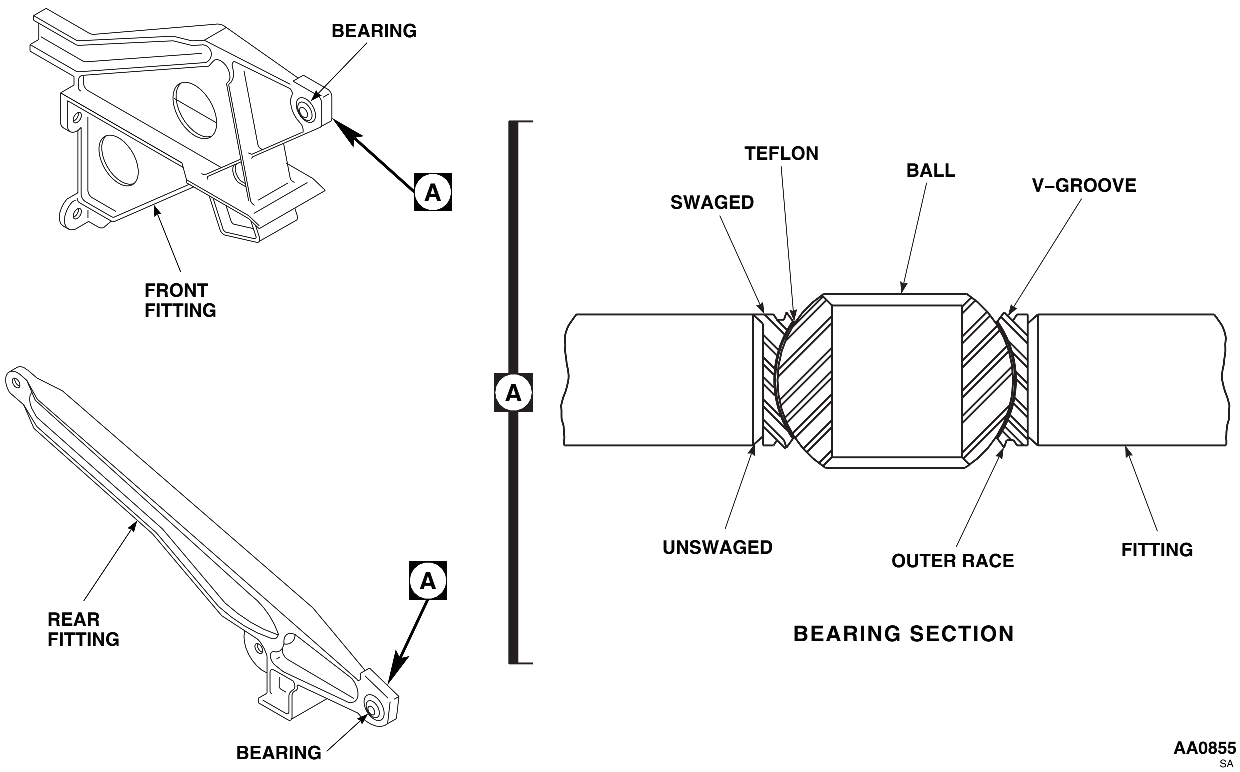


Figure 1. Replace Engine Cowling Fitting Bearing (AVIM).

2. Apply light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and to outside surfaces of bearing outer race.
3. **QA** Press bearing into fitting using arbor press and driver and anvil combination (TM 55-1500-322-24).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer, if any, from bearing V-grooves using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.
5. Fix fitting on drill press baseplate, with proper spinning tool and anvil.

CAUTION

To prevent damage to Teflon, keep petrolatum from bearing ball and Teflon liner. Remove petrolatum immediately if it gets on these surfaces.

6. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 216, WP 1803 00.

INSTALL - Continued

CAUTION

Deformation of outer race can cause binding of bearing. Do not let spinning tool contact ball side of V-groove.

7. Swage V-groove lip over chamfer in fitting bore (TM 55-1500-322-24) (Figure 1, Detail A).
8. **QA** Turn fitting over and repeat swaging process on other side.
9. Clean petrolatum, Item 216, WP 1803 00, from V-groove with cleaning compound solvent, Item 71, WP 1803 00, on cheesecloth cloth, Item 85, WP 1803 00.
10. Touch up bearing surfaces with corrosion preventive compound, Item 106, WP 1803 00.
11. Touch up fitting surfaces with primer coating, Item 243, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

OIL COOLER SUPPORT (AVIM)

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02
Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts
Sealing Compound, Item 273, WP 1803 00

Personnel Required
Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References
TM 1-1500-204-23
TM 55-1500-345-23
WP 0369 00
WP 0377 00
WP 0378 00
WP 0380 00
WP 0381 00
WP 1803 00

REPAIR

1. The oil cooler blower forward and rear supports include formed and extruded members made from 7075-T6, QQ-A-250-13 Alclad sheet stock and 7075-T6511, QQ-A-200-11 aluminum alloy extrusion (Figure 1). Support members are primary structure.
2. Evaluate damage to support members (WP 0377 00).
3. Use WP 0378 00 and WP 0380 00 for typical support member repair.
4. All repairs to oil cooler blower supports must maintain correct alignment with tail drive shaft (WP 0369 00).
5. **QA** Replace loose or damaged nutplates on forward support using rivets (TM 1-1500-204-23).
6. For corrosion protection and watertightness, seal all support parts which fasten to structure with sealing compound, Item 273, WP 1803 00 (WP 0377 00).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

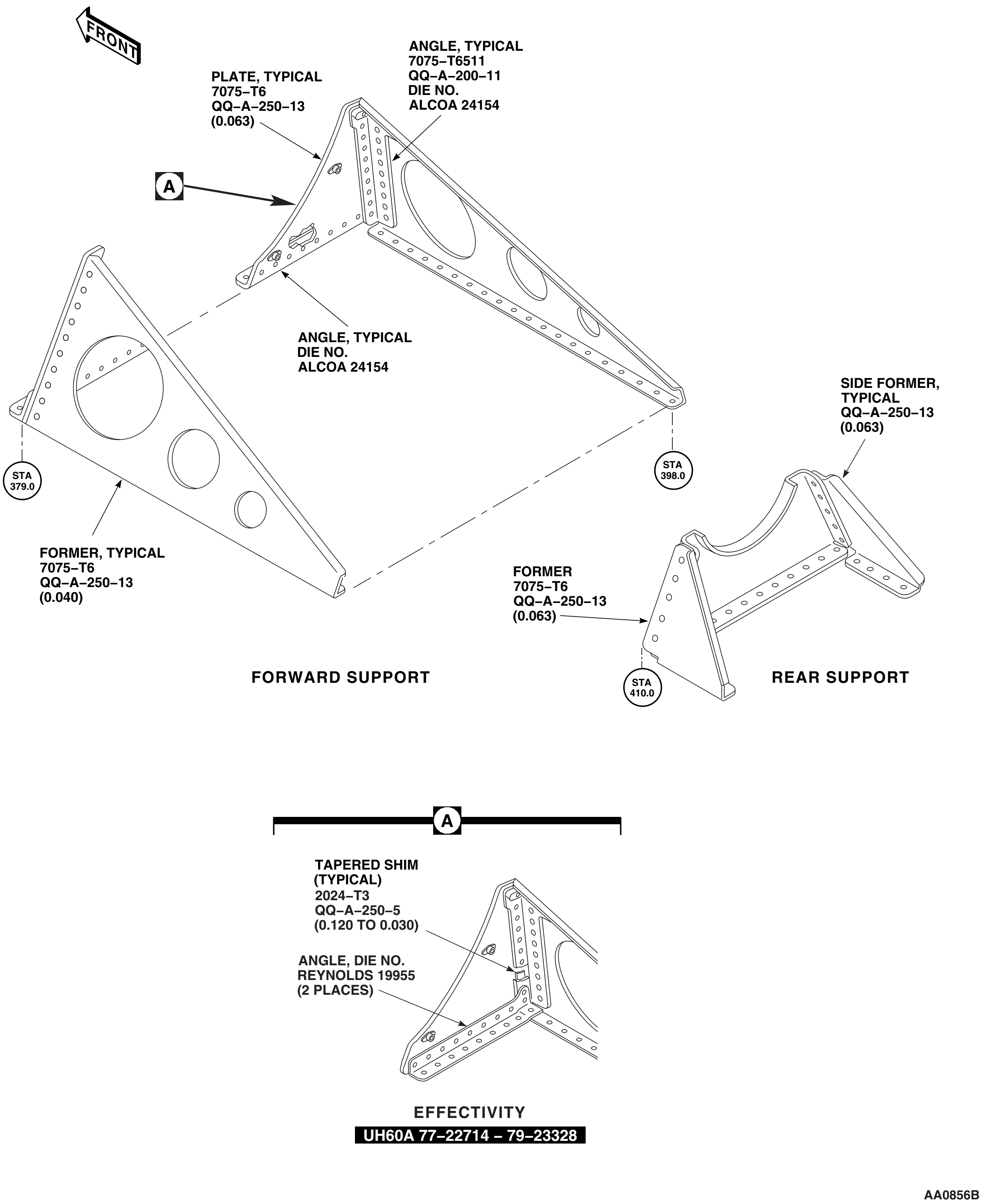


Figure 1. Repair Oil Cooler Support (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

OIL COOLER FORWARD SUPPORT (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Gun Kit, G-740-ELM
Clamp, A-A-429
Drill Set, GGG-D-751
Nonmetallic Mallet, 25714
Powertrain Repairer's Toolkit, SC 5180-99-B13
Tail Rotor Drive Shaft Alignment Fixture Set, 70700-20480-041

Material/Parts

Collar, NAS1080-5
Collar, NAS1080-6
Lockbolt, HL2086
Lockbolt, NAS1465-5
Lockbolt, NAS1465-6
Rivet, MS20470AD5
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 1-6625-724-13&P
TM 11-1520-237-23
TM 55-1500-345-23
WP 0233 00
WP 0258 00
WP 0308 00
WP 0381 00
WP 0666 00
WP 0667 00
WP 0668 00
WP 0678 00
WP 0689 00
WP 0690 00
WP 1615 00
WP 1616 00
WP 1803 00

REMOVE

CAUTION

- Remove left or right forward support assembly. Remaining forward support assembly will be used to reinstall forward support assembly.
- If left and right forward support assemblies must both be removed and one of the forward plates cannot be used as a reference point, contact depot maintenance.

NOTE

- The following instructions apply to both right or left forward support assembly.
- On helicopters with tail numbers prior to 79-23329, base angle (Figure 1, Sheet 2, Detail C) faces aft.

1. Defuel main fuel tanks (WP 1615 00 or WP 1616 00).
2. Remove rear troop seats (WP 0233 00).
3. Remove soundproofing panels from left and right stowage compartments and center structure (WP 0258 00) and unhook cargo netting.

REMOVE - Continued

4. Remove No. 1 and No. 2 stabilator amplifier (TM 11-1520-237-23).
5. Remove access doors and fairing (WP 0308 00).
6. Open tail rotor drive shaft cover.
7. Remove oil cooler, and tail rotor drive shaft Section II and Section III (WP 0667 00, WP 0668 00, and WP 0678 00).
8. Inspect support angles, formers, L/H and R/H plates, and upper plating. Inspect underside of support from inside of cabin.
9. Cut collars and remove lockbolts from lower support angle (TM 1-1500-204-23) (Figure 1, Sheet 1).
10. Carefully drill out and remove remaining rivets in lower support angle and rivets securing former to structure (TM 1-1500-204-23).
11. Remove forward support assembly.
12. Make sure area is clean and free of foreign material.

REPAIR

Remove and replace damaged parts as required (Figure 1, Sheet 1 and TM 1-1500-204-23). If plates are replaced, refer to **DRILL OIL COOLER INSTALLATION HOLES**.

INSTALL

CAUTION

Forward oil cooler support will be misaligned if target wire is not centered. Make sure target wire is centered in front adapter plate.

1. Install target wire in front adapter plate. Tighten all Allen head set screws in 1/2 turn increments following tightening sequence until wire cannot be moved by hand (Figure 1, Sheet 3, Detail F and Detail G).
2. Install front adapter plate to main gear tail takeoff flange using bolts and tapered knurled nuts supplied in alignment kit (Figure 1, Sheet 2 and Figure 1, Sheet 3, Detail H and Detail G).
3. Install rear adapter plate as follows:
   a. Prior to installation of the rear adapter plate, back off Allen head set screws to permit insertion of the alignment wire (Figure 1, Sheet 3, Detail F).
   b. Install rear adapter plate to input flange of intermediate gear box using bolts and tapered knurled nuts supplied in alignment kit (Figure 1, Sheet 3, Detail H).
   c. Turn hand knob on rear adapter plate counterclockwise until Allen head set screws in the locking mechanism are visible through holes in adapter plate barrel when the locking mechanism is pressed forward.

CAUTION

Target wire will break if over tightened. Use care when turning hand knob to tighten target wire.

   d. While maintaining the locking mechanism in the forward position, insert target wire through hand knob and through the locking mechanism to a point where the target wire is lightly tensioned over the length of the rear adapter plate.

INSTALL - Continued

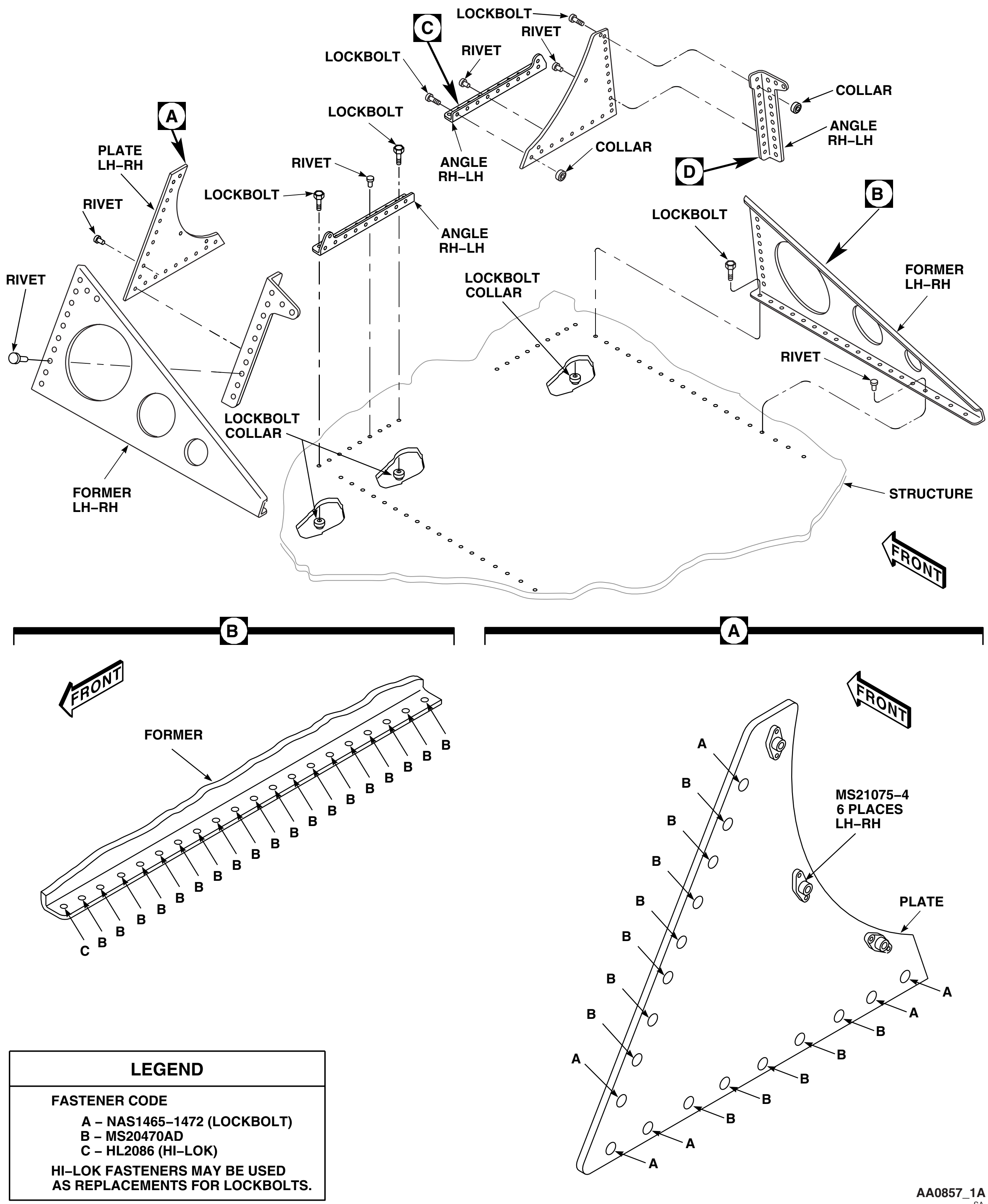


Figure 1. Replace Oil Cooler Forward Support (AVIM). (Sheet 1 of 3)

INSTALL - Continued

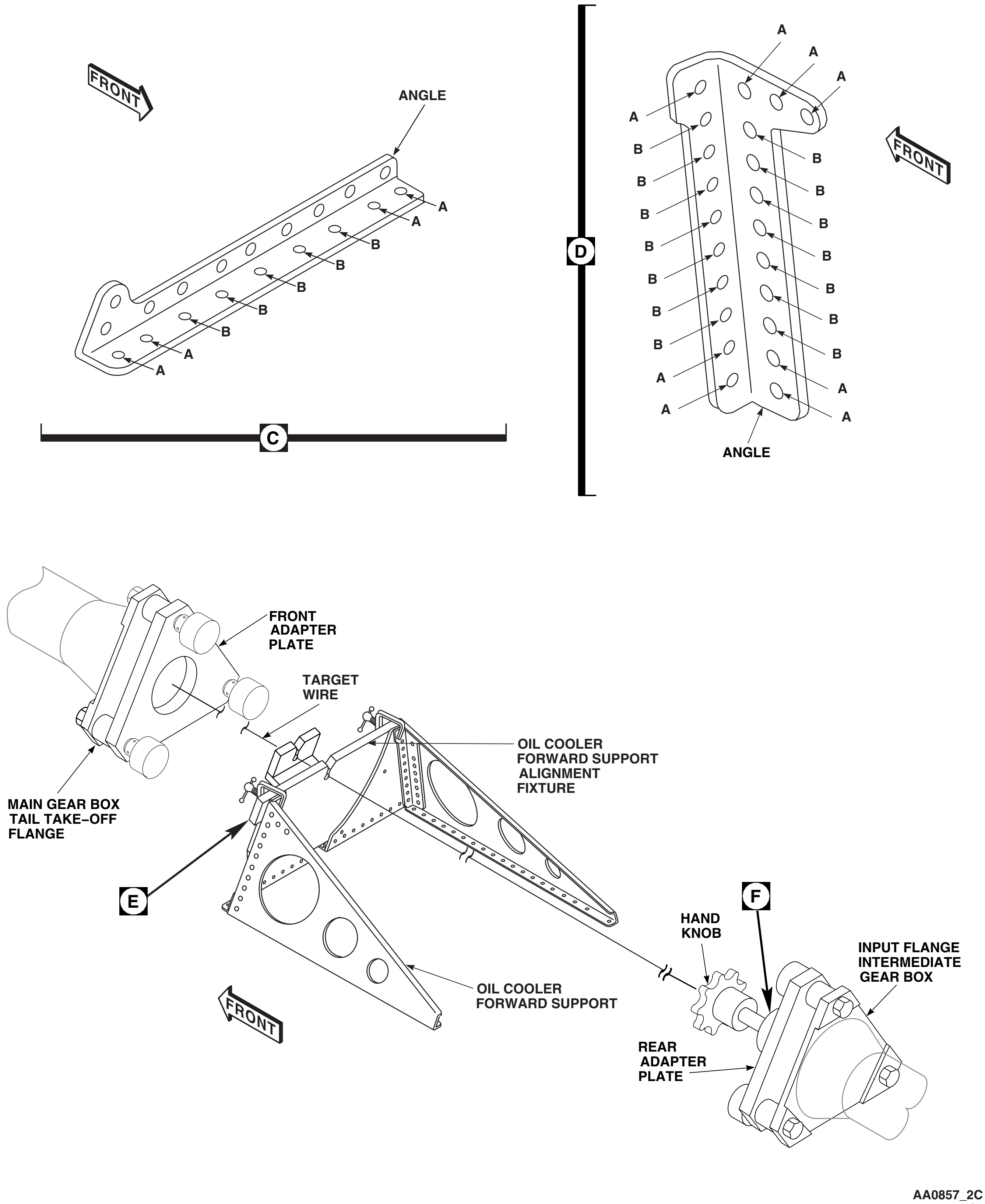


Figure 1. Replace Oil Cooler Forward Support (AVIM). (Sheet 2 of 3)

INSTALL - Continued

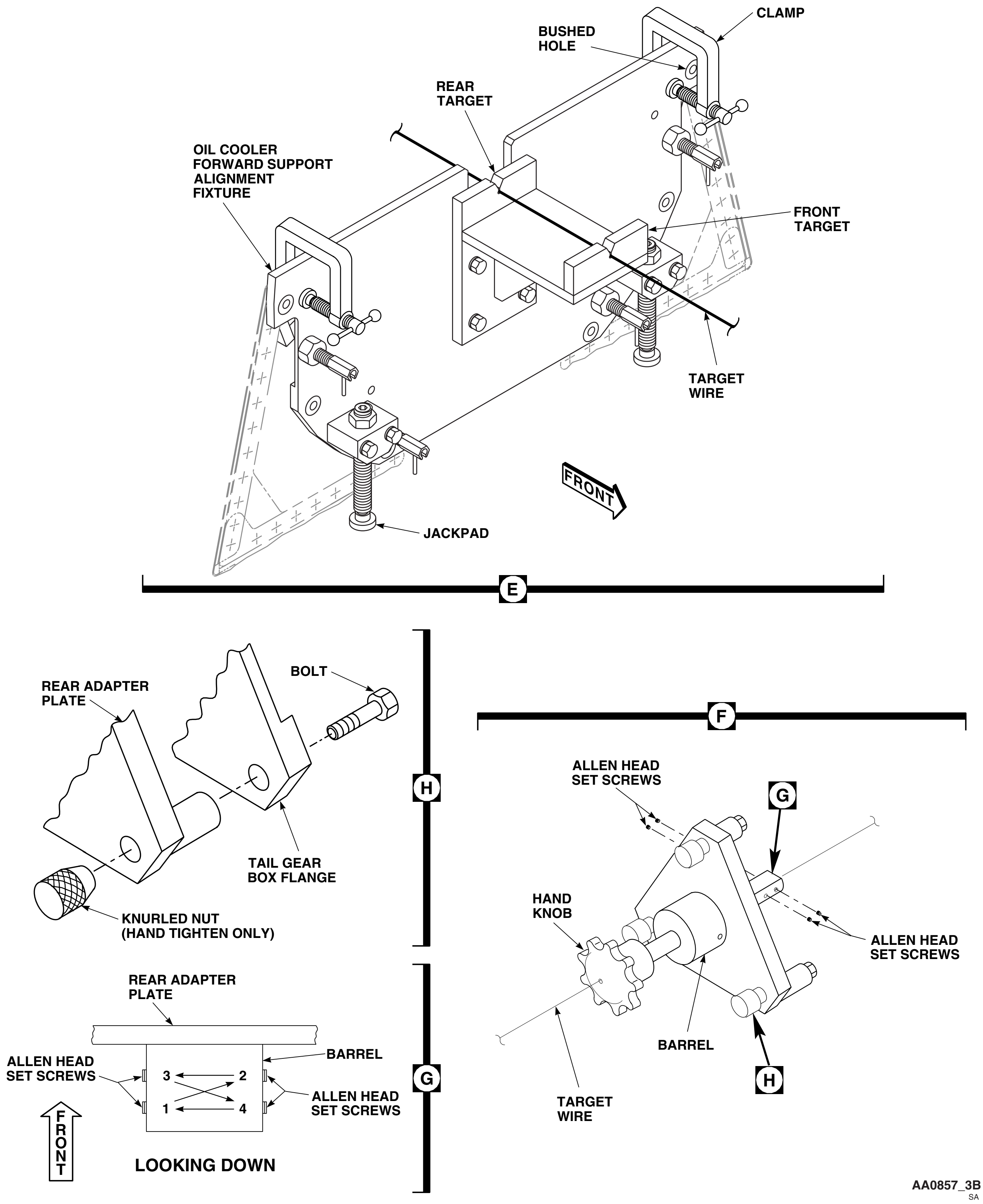


Figure 1. Replace Oil Cooler Forward Support (AVIM). (Sheet 3 of 3)

INSTALL - Continued

NOTE

When the target wire is lightly tensioned, it should be visible between the aft end of the rear adapter plate and the input flange of intermediate gear box.

e. Tighten all Allen head set screws (located inside the barrel of the rear adapter plate) in 1/2 turn increments following tightening sequence until wire cannot be moved by hand (Figure 1, Sheet 3, Detail G).

f. Remove rear adapter plate (with clamped target wire) from the input flange of intermediate gear box temporarily to complete clamping process of target wire.

g. To complete clamping, slide rear adapter plate forward on target wire until all Allen head set screw holes (2 holes through diameter) are visible and torque down all four Allen head set screws.

NOTE

Make sure rear adapter plate is completely seated (and centered) on input flange of intermediate gearbox.

h. Reinstall rear adapter plate on input flange of intermediate gear box.

CAUTION

Target wire will break if over tightened. Use care when turning hand knob to tighten target wire.

i. Turn hand knob clockwise to required target wire tension.

4. Seal for dissimilar metals insulation during installation, using sealing compound, Item 273, WP 1803 00 (TM 1-1500-344-23).
5. Position oil cooler forward support alignment fixture on oil cooler forward support assembly and secure with installation bolts.
6. Carefully position assembled alignment fixture and oil cooler forward support on structure at STA 379.0.
7. Carefully place target wire in hole of forward and rear target.
8. While making sure target wire remains centered in hole of forward and rear targets, temporarily secure oil cooler support to structure using clecos.
9. Insert required lockbolt pins in lower support angle between clecos and install collars from inside helicopter structure (TM 1-1500-204-23).
10. Install required rivets between clecos attaching former and lower support angle to structure (TM 1-1500-204-23).
11. Remove clecos and install remaining rivets and lockbolts (TM 1-1500-204-23).
12. Turn hand knob to relieve tension on target wire. Remove target wire.
13. Remove bolts securing front adapter plate to main gear box tail takeoff flange. Remove adapter plate.
14. Remove bolts securing rear adapter plate to intermediate gear box input flange. Remove adapter plate.
15. Remove bolts securing oil cooler forward support alignment fixture to oil cooler forward support assembly. Remove alignment fixture.
16. Remove tools, clean up repair area, and check parts for secure attachment.
17. Touch up paint as required (WP 0381 00 and TM 1-1500-345-23).
18. Apply a fillet of sealing compound, Item 273, WP 1803 00, around periphery of forward oil cooler support.

INSTALL - Continued

19. Install oil cooler, and tail rotor drive shaft Section II and Section III (WP 0667 00, WP 0668 00, and WP 0678 00).
20. Install access doors and fairing (WP 0308 00).
21. Install No. 1 and No. 2 stabilator amplifier (TM 11-1500-237-23).
22. Install soundproofing panels (WP 0258 00) and unhook cargo netting.
23. Do tail rotor drive shaft coupling shimming check (WP 0690 00).
24. Do tail rotor drive shaft coupling alignment check (WP 0689 00).
25. Close tail rotor drive shaft cover.
26. Do oil cooler vibration check (TM 1-6625-724-13&P).
27. Install rear troop seats (WP 0233 00).

DRILL OIL COOLER INSTALLATION HOLES

CAUTION

- Oil cooler will be misaligned if installation holes in oil cooler forward support are drilled improperly. Make sure oil cooler forward support alignment fixture is used to locate installation holes.
- Forward oil cooler support will be misaligned if target wire is not centered. Make sure target wire is centered in front adapter plate.

1. Install target wire in front adapter plate.
2. Install front adapter plate to main gear tail takeoff flange using bolts and tapered knurled nuts supplied in alignment kit (Figure 1, Sheet 2 and Figure 1, Sheet 3, Detail H and Detail G).
3. Install rear adapter plate as follows:
   a. Prior to installation of the rear adapter plate, back off Allen head set screws to permit insertion of the alignment wire (Figure 1, Sheet 3, Detail F).
   b. Install rear adapter plate to input flange of intermediate gear box using bolts and tapered knurled nuts supplied in alignment kit (Figure 1, Sheet 3, Detail H).
   c. Turn hand knob on rear adapter plate counterclockwise until Allen head set screws in the locking mechanism are visible through holes in adapter plate barrel when the locking mechanism is pressed forward.

CAUTION

Target wire will break if over tightened. Use care when turning hand knob to tighten target wire.

   d. While maintaining the locking mechanism in the forward position, insert target wire through hand knob and through the locking mechanism to a point where the target wire is lightly tensioned over the length of the rear adapter plate.

NOTE

When the target wire is lightly tensioned, it should be visible between the aft end of the rear adapter plate and the input flange of intermediate gear box.

   e. Tighten all Allen head set screws (located inside the barrel of the rear adapter plate) in 1/2 turn increments following tightening sequence until wire cannot be moved by hand (Figure 1, Sheet 3, Detail G).

DRILL OIL COOLER INSTALLATION HOLES - Continued

f. Remove rear adapter plate (with clamped target wire) from the input flange of intermediate gear box temporarily to complete clamping process of target wire.

g. To complete clamping, slide rear adapter plate forward on target wire until all Allen head set screw holes (2 holes through diameter) are visible and torque down all four Allen head set screws.

NOTE

Make sure rear adapter plate is completely seated (and centered) on input flange of intermediate gearbox.

h. Reinstall rear adapter plate on input flange of intermediate gear box.

CAUTION

Target wire will break if over tightened. Use care when turning hand knob to tighten target wire.

i. Turn hand knob clockwise to required target wire tension.

4. Carefully place oil cooler forward support alignment fixture against plates of oil cooler forward support and lightly clamp alignment fixture to plates (Figure 1, Sheet 3, Detail E).

5. If only one plate was replaced, do the following:

   a. Using a combination of jackpad adjustments, for vertical positioning, and gentle taps with a nonmetallic mallet, for horizontal positioning, align existing installation holes in plate with bushed holes in alignment fixture.

   b. While making sure target wire is centered in forward and rear target, secure alignment fixture to plate using installation bolts.

   c. Drill three holes in plate through bushed holes in alignment fixture using 17/64-inch diameter drill. **HOLE DIAMETER SHALL BE +0.004-INCH/-0.001-INCH FOR A 17/64-INCH DRILLED HOLE.**

   d. Remove bolts securing alignment fixture to plate. Remove alignment fixture.

   e. Using rivets, secure nutplates to plate (TM 1-1500-204-23).

6. If both plates were replaced, do the following:

   a. Using a combination of jackpad adjustments, for vertical positioning, and gentle taps with a nonmetallic mallet, for horizontal positioning, center target wire in hole of forward and rear targets. Tighten clamps.

   b. Drill three holes in each plate through bushed holes in alignment fixture using 17/64-inch diameter drill. **HOLE DIAMETER SHALL BE +0.004-INCH/-0.001-INCH FOR A 17/64-INCH DRILLED HOLE.**

   c. Remove clamps securing alignment fixture to oil cooler forward support assembly. Remove alignment fixture.

   d. Using rivets, secure nutplates to plate (TM 1-1500-204-23).

7. Turn hand knob to relieve tension on target wire. Remove target wire.

8. Remove bolts securing front adapter plate to main gear box tail takeoff flange. Remove adapter plate.

9. Remove bolts securing rear adapter plate to intermediate gear box input flange. Remove adapter plate.

10. Remove tools, clean up repair area, and check parts for secure attachment.

11. Touch up paint as required (WP 0381 00 and TM 1-1500-345-23).

12. Install oil cooler, and tail rotor drive shaft Section II and Section III (WP 0678 00, WP 0667 00, and WP 0668 00).

13. Install access doors and fairing (WP 0308 00).

DRILL OIL COOLER INSTALLATION HOLES - Continued

14. Do tail rotor drive shaft coupling shimming check (WP 0690 00).
15. Do tail rotor drive shaft coupling alignment check (WP 0689 00).
16. Close tail rotor drive shaft cover.
17. Do oil cooler vibration check (TM 1-6625-724-13&P).
18. Install soundproofing panels (WP 0258 00) and unhook cargo netting.
19. Install rear troop seats (WP 0233 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

OIL COOLER REAR SUPPORT (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Gun Kit, G-740-ELM
Drill Set, GGG-D-751
Powertrain Repairer's Toolkit, SC 5180-99-B13
Tail Rotor Drive Shaft Alignment Fixture Set, 70700-20480-041

Material/Parts

Epoxy Coating Kit, Item 118, WP 1803 00
Rivet, MS20470AD5
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 55-1500-345-23
WP 0233 00
WP 0258 00
WP 0308 00
WP 0381 00
WP 0666 00
WP 0667 00
WP 0675 00
WP 1803 00

REMOVE

1. Remove rear troop seats (WP 0233 00 00).
2. Remove soundproofing panel from left and right stowage compartment and center structure (WP 0258 00) and unhook cargo netting.
3. Remove access doors and fairing (WP 0308 00).
4. Remove oil cooler fan, radiator, and tail rotor drive shaft Section I and Section II (WP 0666 00, WP 0667 00, and WP 0675 00).
5. Unlock studs fastening left and right panels on top of fuel cells. Remove panels.
6. Remove screws and front and top panels from center structure.
7. From top side, carefully drill out and remove twenty-two rivets securing rear support assembly to upper structure, using No. 24 (0.152-inch diameter) drill and 1/8-inch diameter drive pin punch (TM 1-1500-204-23) (Figure 1). Remove support assembly.
8. Clean up rivet debris.

REPAIR

Replace damaged component parts as required (Figure 1).

REPAIR - Continued

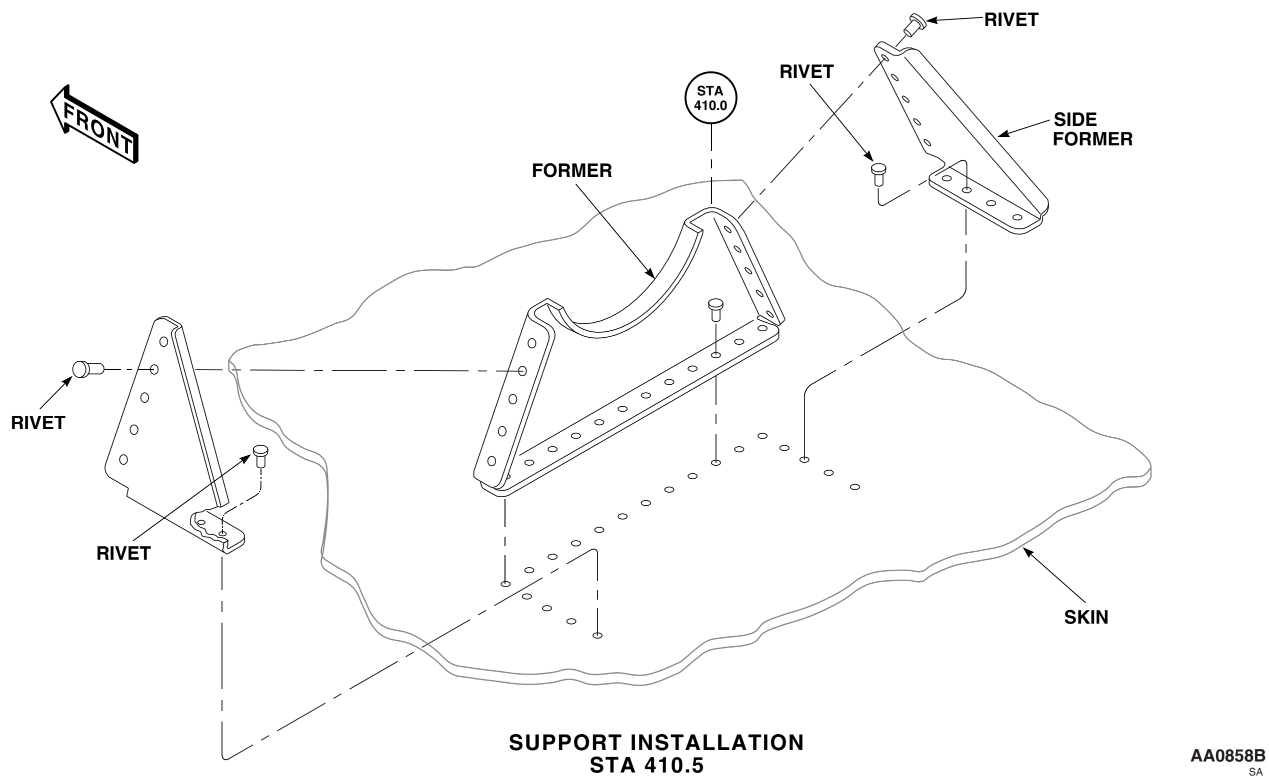


Figure 1. Replace Oil Cooler Rear Support (AVIM).

INSTALL

CAUTION

Do not drill four upper holes in support used to attach support housing until support is installed. To avoid damage to equipment through misalignment, these holes must be drilled using tail drive shaft support alignment fixture.

1. Seal for dissimilar metals insulation during installation, using epoxy coating kit, Item 118, WP 1803 00 (TM 1-1500-344-23).
2. Position rear support assembly on upper structure at STA 410.54 and hold in place with No. 40 clecos (Figure 1). Perform shaft support alignment and shimming for STA 410.54.
3. **QA** Install rivets in between clecos. Remove clecos and install remaining rivets ().
4. Set up tail drive shaft support alignment kit and drill four 17/64 (0.265)-inch diameter holes in support.
5. Remove tools and make sure that area is clean and free of foreign material.
6. Install center structure and stowage compartment soundproofing panels (WP 0258 00), and rear troop seats (WP 0233 00).
7. Remove and store alignment kit.
8. Make sure area is clean and free of foreign material.
9. Seal for watertightness, using sealing compound, Item 273, WP 1803 00.
10. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

11. Install oil cooler fan, radiator, and tail drive shaft Section I and Section II (WP 0666 00, WP 0667 00, and WP 0675 00).
12. Install access doors and fairing (WP 0308 00).
13. Make sure area is clean and free of foreign material.
14. Close and secure all access panels and fairings (WP 0308 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM)

This WP supersedes WP 0369 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill, Size F (0.257-Inch Diameter)
Tail Rotor Drive Shaft Alignment Fixture Set, 70700-20480-041

Material/Parts

Shim Kit, 70070-20232-001

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-6625-724-13&P
TM 55-1500-345-23
WP 0233 00
WP 0236 00
WP 0258 00
WP 0377 00
WP 0378 00
WP 0381 00
WP 0666 00
WP 0667 00
WP 0668 00
WP 1615 00
WP 1616 00

REPLACE TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM)

Remove

NOTE

- Only the two piece configuration of the drive shaft support shall be used to replace an existing configuration.
- The following kits will provide components for the two piece configuration: kit 70070-20488-012 for STA 410.54, assembly 70070-20488-022 for STA 470.98, assembly 70070-20266-013 for STA 531.43, and assembly 70070-20266-014 for STA 591.88.
- Removal procedures are the same for tail rotor drive shaft supports at STA 410.54, STA 470.98, STA 531.43, and STA 591.88 unless noted.

1. Open tail rotor drive shaft cover.

NOTE

There is no need to remove oil cooler except when performing procedures at STA 410.54 bracket. Target wire will route through center of shaft.

2. Remove tail rotor drive shaft Section I, Section II and Section III (WP 0666 00, WP 0667 00, and WP 0668 00).
3. Remove rivets attaching tail rotor drive shaft support to bracket (Figure 2, Sheet 1 and TM 1-1500-204-23).
4. Clean up debris.

REPLACE TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM) - Continued

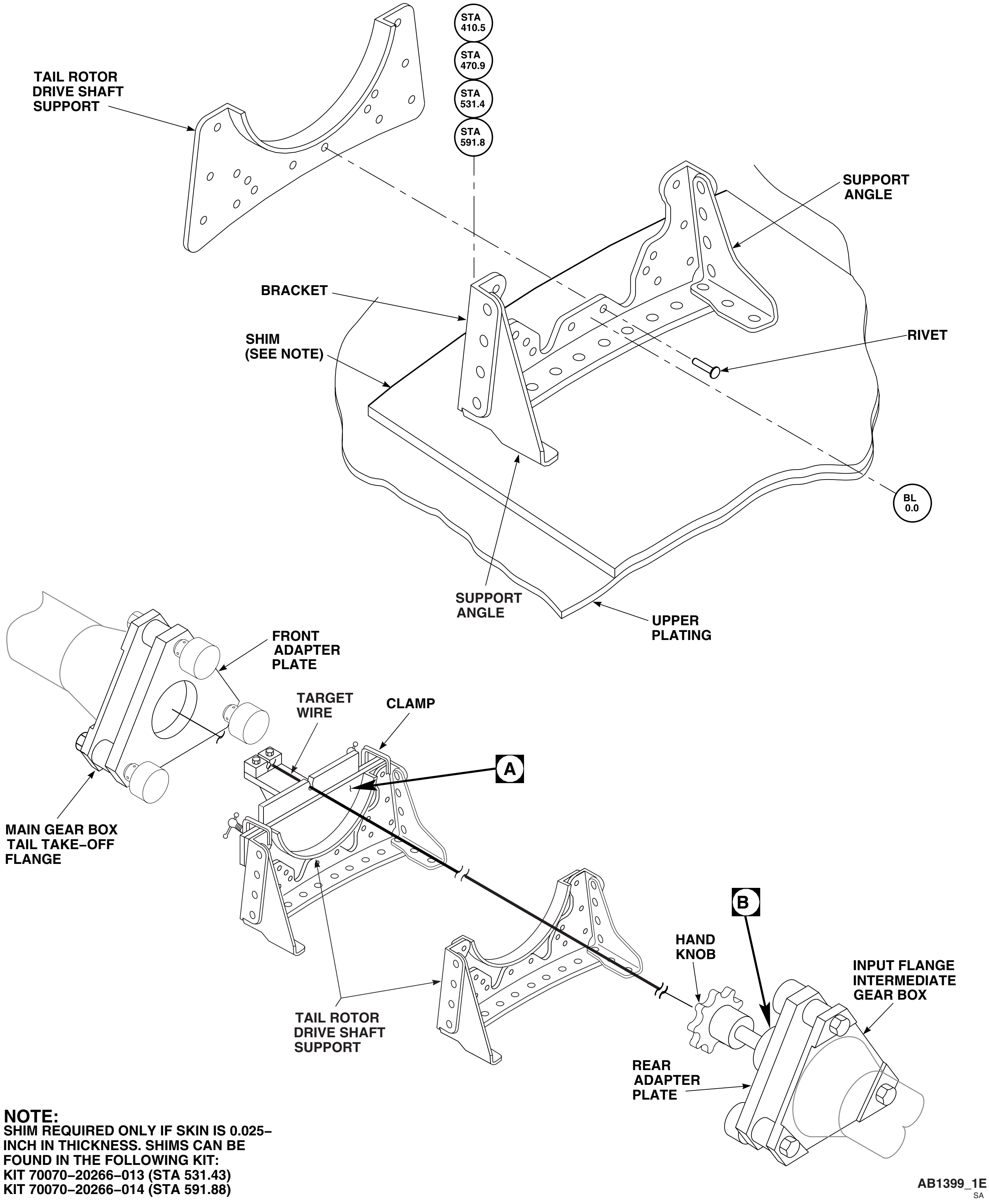


Figure 1. Tail Rotor Drive Shaft Supports (AVIM). (Sheet 1 of 2)

REPLACE TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM) - Continued

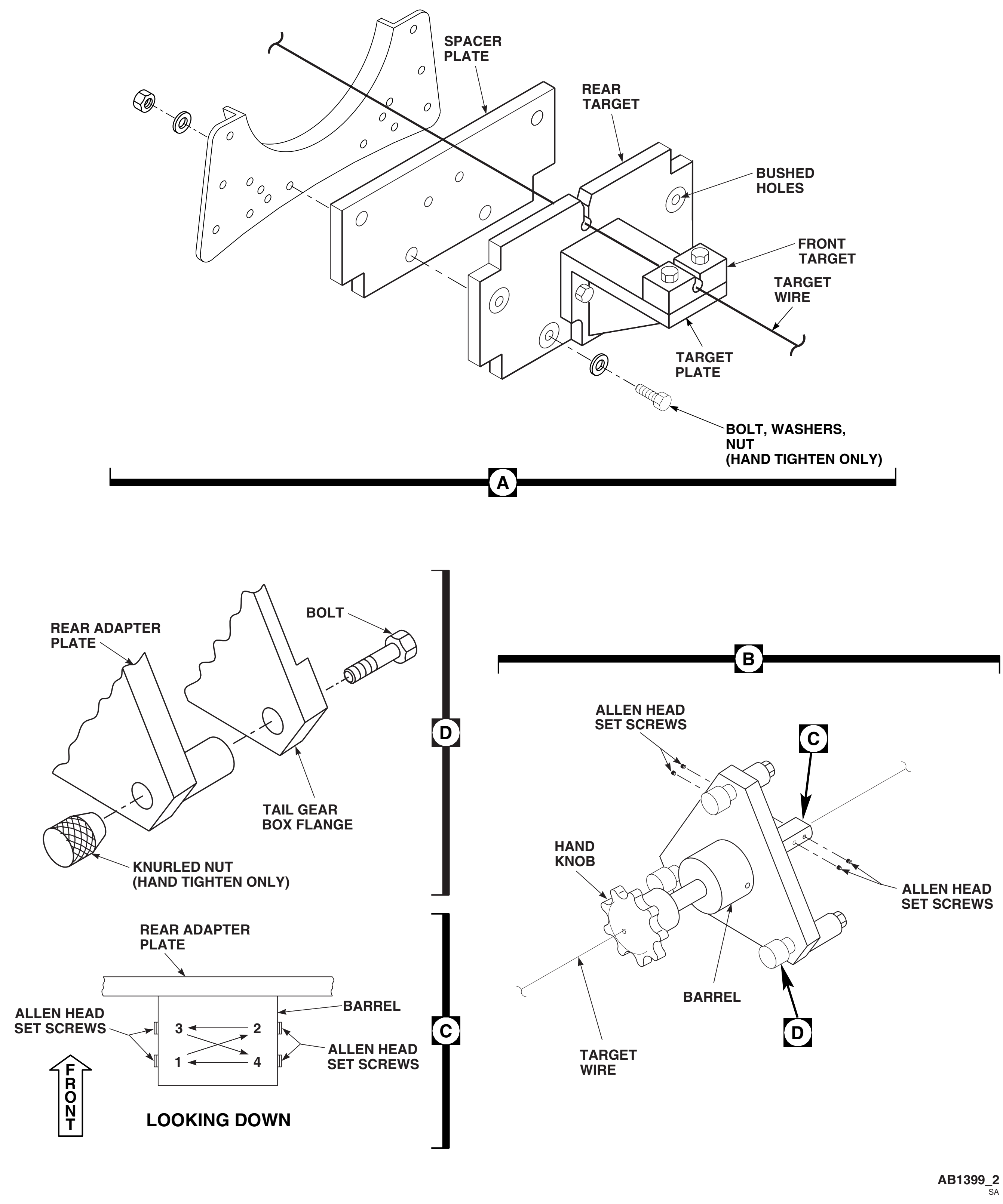


Figure 1. Tail Rotor Drive Shaft Supports (AVIM). (Sheet 2 of 2)

REPLACE TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM) - Continued

Inspect

NOTE

Make sure that underside inspection of tail rotor drive shaft support, is made from the inside of transition section or tailcone.

1. Inspect support angles, tail rotor drive shaft support, bracket, and upper plating for damage (WP 0377 00). Repair damaged support angles, tail rotor drive shaft support, brackets, and upper plating (WP 0378 00).
2. Inspect viscous damper bearing mount holes for elongation. **MAXIMUM HOLE DIAMETER SHALL NOT EXCEED 0.2696-INCH DIA.**

Install

NOTE

Install procedures are the same for tail rotor drive shaft supports at STA 410.54, STA 470.98, STA 531.43, and STA 591.88 unless noted.

1. Defuel main fuel tanks (WP 1615 00 or WP 1616 00).
2. If tail drive shaft support has been removed at STA 531.43 or STA 591.88, do the following:
   a. Measure helicopter skin thickness in area of removed support.
   b. **QA** If helicopter skin thickness is 0.025-inch install doubler 70070-20266-101 between STA 521.0 and STA 545.0, and install doubler 70070-20266-102 between STA 585.0 and STA 605.0.
   c. Additional kit applications are as follows:
      (1) Kit 70070-20266-011 will also provide longer side supports for the STA 531.43 installation.
      (2) Kit 70070-20266-012 will also provide longer side supports for the STA 591.88 installation.
      (3) Kit 70070-20266-013 will also provide the two piece configuration supports for the STA 531.43 installation.
      (4) Kit 70070-20266-014 will also provide the two piece configuration supports for the STA 591.88 installation.

NOTE

Make sure target wire is centered in front adapter plate. Forward oil cooler support will be misaligned if target wire is not centered.

3. Install target wire in front adapter plate. Tighten all Allen head set screws in 1/2 turn increments following tightening sequence until wire cannot be moved by hand (Figure 2, Sheet 2, Detail C).

CAUTION

Tail rotor drive shaft supports will be misaligned if installed improperly. Make sure tail rotor drive shaft support alignment tool is used to locate tail rotor drive shaft supports.

NOTE

Seal for dissimilar metals insulation during installation of support angles, tail rotor drive shaft support, bracket, and fasteners using procedures in WP 0377 00.

4. Install front adapter plate to main gear box tail takeoff flange using bolts and tapered knurled nuts supplied in alignment kit (Figure 2 and Figure 2, Sheet 2, Detail B).
5. Install rear adapter plate as follows:

REPLACE TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM) - Continued

a. Prior to installation of the rear adapter plate, back off Allen head set screws to permit insertion of the alignment wire (Figure 2, Sheet 2, Detail B).

b. Install rear adapter plate to input flange of intermediate gear box using bolts and tapered knurled nuts supplied in alignment kit (Figure 2, Sheet 1 and Figure 2, Sheet 2, Detail D).

c. Turn hand knob on rear adapter plate counterclockwise until Allen head set screws in the locking mechanism are visible through holes in adapter plate barrel when the locking mechanism is pressed forward.

CAUTION

Target wire will break if over tightened. Use care when turning hand knob to tighten target wire.

d. While maintaining the locking mechanism in the forward position, insert target wire through hand knob and through the locking mechanism to a point where the target wire is lightly tensioned over the length of the rear adapter plate.

NOTE

When the target wire is lightly tensioned, it should be visible between the aft end of the rear adapter plate and the input flange of intermediate gear box.

e. Tighten all Allen head set screws (located inside the barrel of the rear adapter plate) in 1/2 turn increments following tightening sequence until wire cannot be moved by hand (Figure 2, Sheet 2, Detail C).

f. Remove rear adapter plate (with clamped target wire) from input flange of intermediate gear box temporarily to complete clamping process of target wire.

g. To complete clamping, slide rear adapter plate forward on target wire until all Allen head set screw holes (2 holes through diameter) are visible. Torque down all four Allen head set screws.

NOTE

Make sure rear adapter plate is completely seated (and centered) on input flange of intermediate gearbox.

h. Reinstall rear adapter plate on input flange of intermediate gear box.

i. Turn hand knob clockwise to required target wire tension,

6. Bolt target plate and spacer plate to front side of tail rotor drive shaft support (Figure 2, Sheet 2, Detail A). **HANDTIGHTEN BOLTS UNTIL NO MOVEMENT OF TAIL ROTOR DRIVE SHAFT SUPPORT IS DETECTED.**

7. Position target plate with attached tail rotor drive shaft support as follows:

   a. Install target plate so that shoulders of tail rotor drive shaft support line up with top of bracket. Secure with clamps (Figure 2, Sheet 1).

   b. Position target wire in target plate hole.

   c. Check wire in target hole for proper position. **WIRE MUST NOT BE IN CONTACT WITH ANY SIDE OF FRONT AND/OR REAR TARGET HOLES.**

   d. If wire is **NOT** in proper position loosen clamps and position as necessary to obtain proper alignment. Retighten clamps.

REPLACE TAIL ROTOR DRIVE SHAFT SUPPORT (AVIM) - Continued

CAUTION

Damage to bracket and/or tail rotor drive shaft support may occur if tail rotor drive shaft support slips during drilling. Make sure tail rotor drive shaft support is securely clamped before drilling.

e. When wire is in proper position, drill four holes in tail rotor drive shaft support through drill bushings in target plate using ″F″ drill (0.257 inch diameter drill.) Maximum hole diameter shall not exceed 0.2696-inch dia.

f. Loosen clamps and remove target plate, spacer plate and tail rotor drive shaft support from bracket. Remove bolts, washers and nuts securing tail rotor drive shaft support to target plate and spacer plate. Remove tail rotor drive shaft support from spacer plate.

g. Remove bolts, washers and nuts securing tail rotor drive shaft support to target plate and spacer plate. Remove tail rotor drive shaft support from spacer plate.

8. Secure tail rotor drive shaft to bracket using rivets (TM 1-1500-204-23).
9. Reattach target plate to confirm alignment. If out of alignment repeat step 6. through step 9.
10. Turn hand knob to relieve tension on target wire.
11. On rear adapter plate loosen all Allen head set screws and remove target wire. Remove tapered knurled nuts and bolts securing rear adapter plate to intermediate gear box flanges. Remove rear adapter plate from helicopter.
12. On front adapter plate loosen all Allen head set screws and remove target wire. Remove tapered knurled nuts and bolts securing front adapter plate to main gear box tail take-off flanges. Remove front adapter plate from helicopter.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ELASTOMERIC BEARING (AVIM)

This WP supersedes WP 0370 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
- Arbor Press, 00-P-636
- C-Clamp, A-A-429
- Gloves
- Goggles/Face Shield
- Nonmetallic Drift, 25714
- Powertrain Repairer's Toolkit, SC 5180-99-B13
- Stabilator Fitting Support, Locally-Made, Figure 171, WP 1805 00

Material/Parts
- Acetone, Technical, Item 5, WP 1803 00
- Adhesive, Item 11, WP 1803 00
- Adhesive Primer, Item 23, WP 1803 00
- Cloth, Abrasive, Item 82, WP 1803 00
- Cloth, Abrasive, Item 83, WP 1803 00
- Cloth, Cheesecloth, Item 85, WP 1803 00
- Mat, Reinforcing, Fiberous Glass, Item 196, WP 1803 00
- Plastic Strip, Item 224, WP 1803 00
- Tape, Pressure Sensitive, Adhesive, Item 327, WP 1803 00

Personnel Required
- Aircraft Powertrain Repairer MOS 15D (1)

References
- TM 55-1500-345-23
- WP 0381 00
- WP 1803 00
- WP 1805 00

REMOVE

1. Place fitting assembly on workbench so that bearing assembly is on edge of bench.
2. Using a nonmetallic drift, alternately tap and twist ends of bearing assembly.
3. Remove bearing assembly from fitting (Figure 1, Detail B).

INSTALL

1. If rebonding existing elastomeric bearing assembly to fitting, do this:
   a. Cover exposed rubber on ends of assembly using pressure sensitive adhesive tape, Item 327, WP 1803 00.
   b. Using abrasive cloth, Item 83, WP 1803 00, carefully remove remaining adhesive from mating surface of fitting and bearing assembly.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

   c. Using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, clean mating surface of fitting and bearing assembly. Remove masking tape from ends of bearing assembly. Proceed to step 3.
2. If replacing elastomeric bearing assembly, do this:

INSTALL - Continued

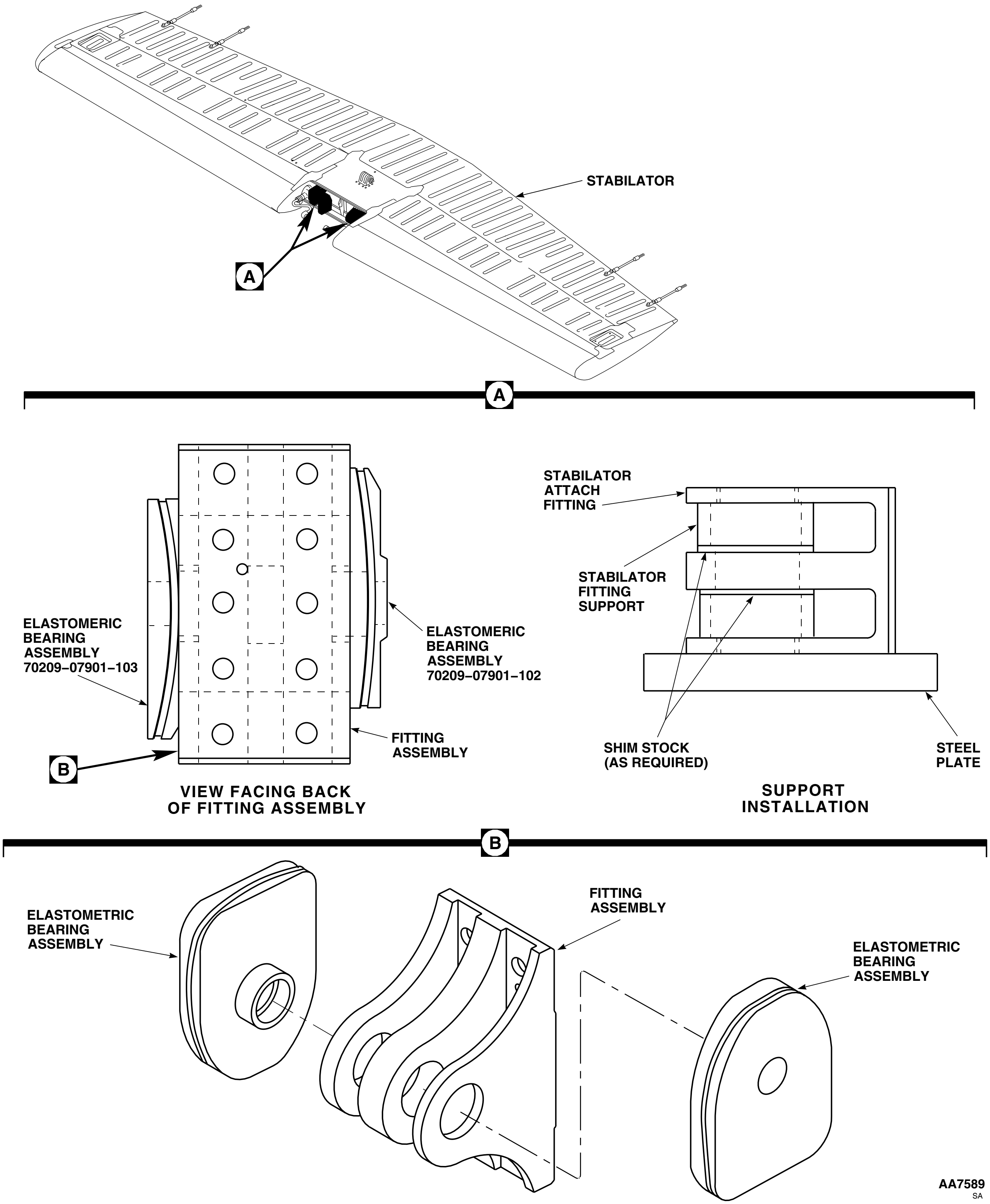


Figure 1. Replace Stabilator Elastomeric Bearing (AVIM).

INSTALL - Continued

a. Using abrasive cloth, Item 82, WP 1803 00, carefully remove adhesive from mating surface of fitting.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

b. Using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, clean mating surface of fitting.

3. Apply light coat of adhesive primer, Item 23, WP 1803 00, to mating surface of fitting and bearing assembly. Allow primer to cure at room temperature for at least 1 hour.
4. Locally make two stabilator fitting supports (Figure 171, WP 1805 00).

CAUTION

Damage to fitting legs will result if stabilator fitting supports are not tightly shimmed. Fitting legs will bend. Make sure supports are tightly shimmed before pressing in elastomeric bearing.

5. **QA** Install supports and plastic strip, Item 224, WP 1803 00, between center and outer legs of fitting (Figure 1, Detail A).
6. Mix adhesive, Item 11, WP 1803 00, per manufacturer's instructions (100 parts by weight of part A to 22 parts weight of part B) and apply a light coat of adhesive to mating surface of fitting and bearing assembly.
7. **QA** Install reinforcing fiberous glass mat, Item 196, WP 1803 00 to mating area of fitting. Cut out mat to diameter of fitting hole and trim to contour of fitting leg.

CAUTION

Damage to liners will occur if bearing is not properly seated in fitting. Liners will become unseated. Use care when pressing bearing in fitting.

NOTE

To properly position elastomeric bearing assemblies on fitting, place fitting on bench with lugs facing down and alignment pin towards top left side. Bearing assembly, 70209-07901-103, will be bonded to side of fitting nearest alignment pin with chamfered end of bearing down and facing rear. Refer to Figure 1, Detail A to correctly install bearing assemblies.

8. **QA** Position bearing assembly on fitting. Carefully press bearing assembly into fitting using an arbor press. Make sure liners do not become unseated.

INSTALL - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. Remove fitting assembly from arbor press and clean extra adhesive from around fitting and bearing assembly, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
10. Maintain pressure on fitting and bearing assembly by installing C-clamps on ends of bearing and ears of fitting. Allow adhesive to dry at least 24 hours at room temperature or 2 hours at 110°F(43°C) to 130°F(54°C).
11. **QA** When adhesive is dry, remove C-clamps, supports, and shims. Touch up unprotected areas (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

POSITION SENSOR/LIMIT SWITCHES (AVIM)

INITIAL SETUP:

Tools and Special Tools

Electrical Connector Maintenance Repair Kit, DMC 420
Electrical Repairer's Toolkit, SC 5180-99-B06
Heat Gun, LFT500
Insert/Extract Tool, MS3447-20

Material/Parts

Adhesive, Item 7, WP 1803 00
Insulation, Sleeving, Item 169, WP 1803 00
Insulation Sleeving, Electrical, Item 159, WP 1803 00
Strap, Tiedown, Electrical, Item 302, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required

Aircraft Electricial Repairer MOS 15F (1)

References

WP 0069 00
WP 1803 00

NOTE

Procedures for replacement of No. 1 and No. 2 down and No. 1 and No. 2 up limit switches is the same, except as noted.

REMOVE

1. Turn off all electrical power.
2. Remove nut, washers, screw, and small clamp attached to large clamp on No. 1 down switch. If No. 1 down switch is being replaced, remove large and small clamps (Figure 1, Sheet 1).
3. Remove locknut, lockwasher, and keyway-washer from switch.
4. Remove switch from assembly.
5. Remove cable tiedown straps from wire bundle.
6. Remove identification plate from wire bundle going to switch.
7. Unscrew backshell from connector.
8. Using insert/extract tool remove applicable wires and pins from connector (Table 1).
9. Remove switch and wire harness from assembly.

INSTALL

1. Turn off all electrical power.
2. Cut switch wires to length shown in Table 1 for applicable switch (No. 1 up, No. 1 down, No. 2 up, or No. 2 down).
3. Cut four switch wires not being used to length (Figure 1, Sheet 1).
4. Install sleeving insulation, Item 169, WP 1803 00, on four shorter switch wires, and using heat gun, heat-shrink end caps to switch wires.

INSTALL - Continued

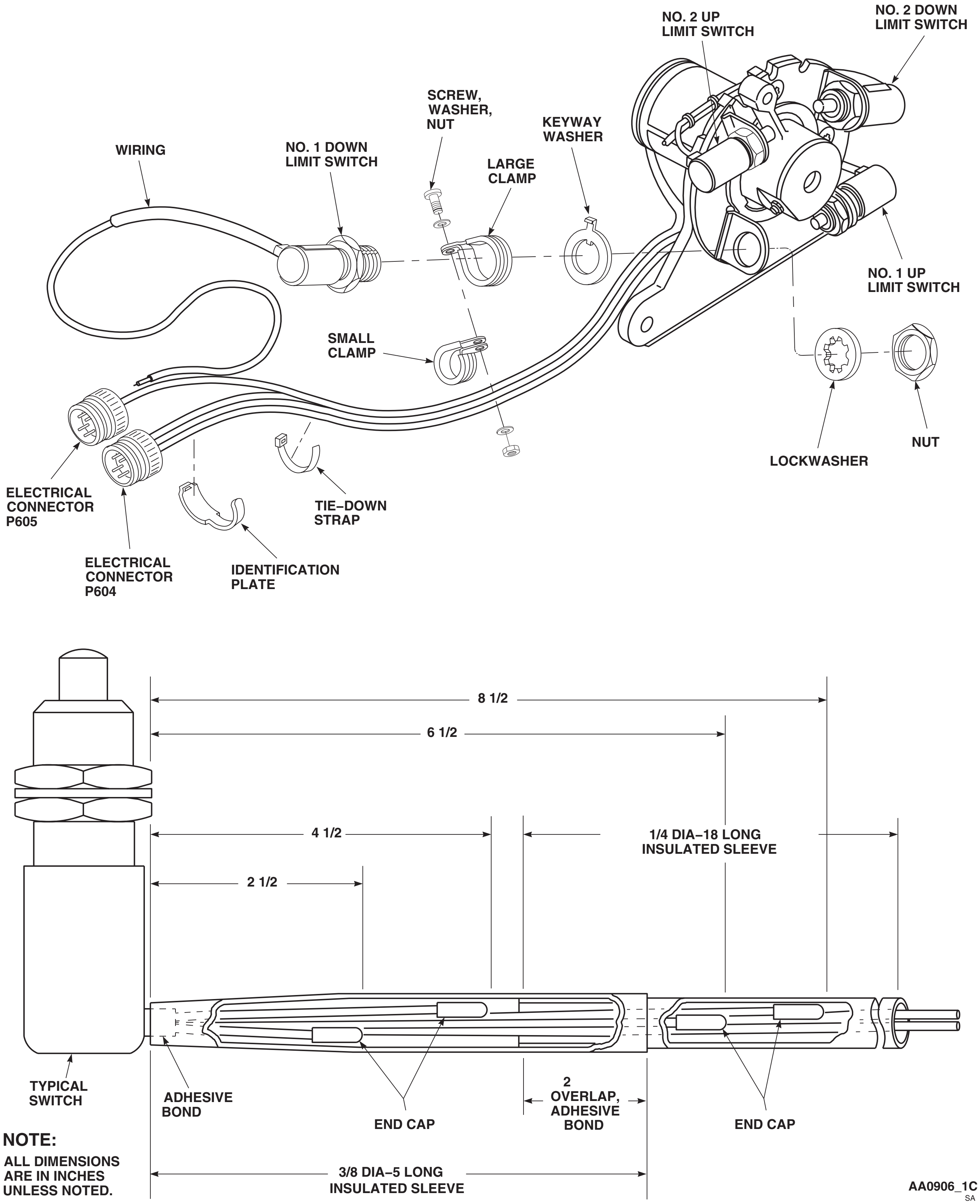


Figure 1. Replace Position Sensor/Limit Switches (AVIM). (Sheet 1 of 2)

INSTALL - Continued

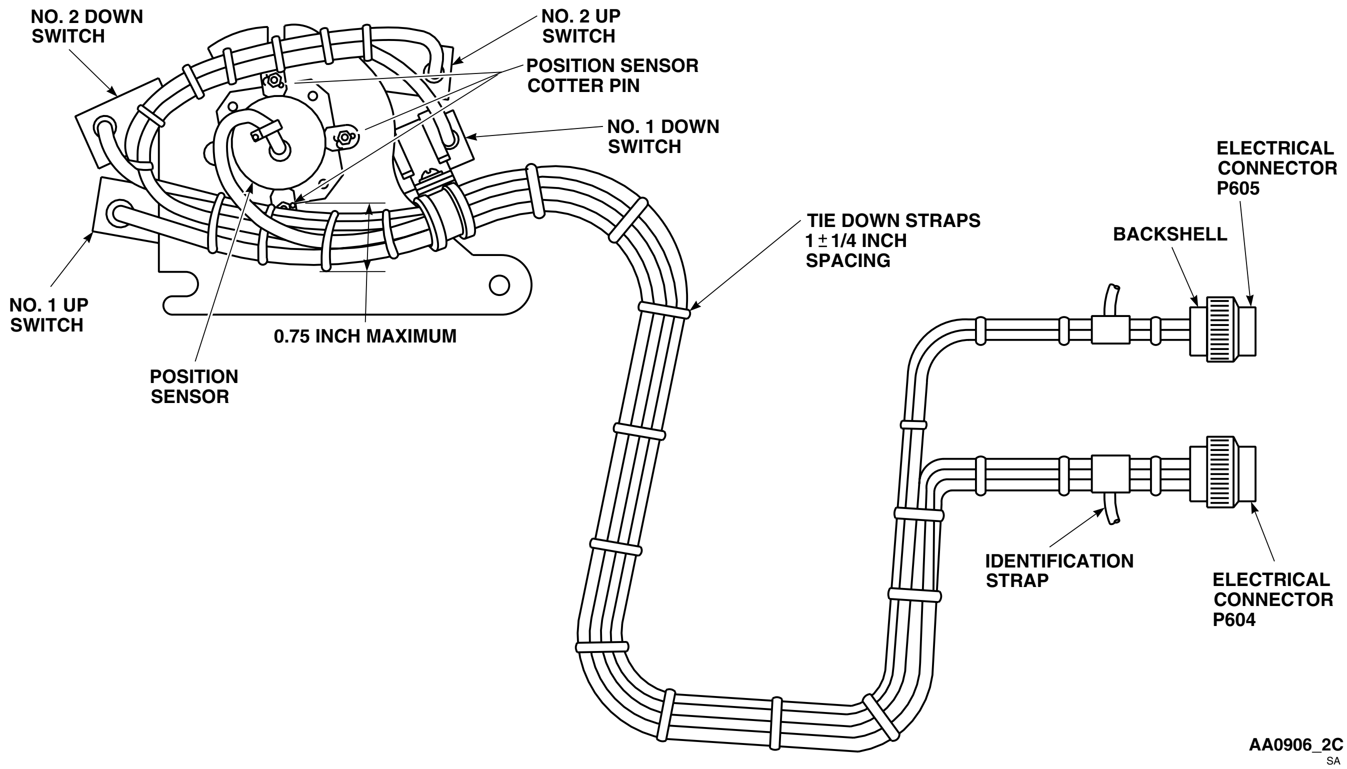


Figure 1. Replace Position Sensor/Limit Switches (AVIM). (Sheet 2 of 2)

5. Slide 3/8-inch diameter by 5 inches long electrical insulation sleeving, Item 159, WP 1803 00, over wires and bond sleeve to switch with adhesive, Item 7, WP 1803 00.
6. Slide 1/4-inch diameter electrical insulation sleeving, Item 159, WP 1803 00, over wires and into 3/8-inch diameter sleeve so that they overlap by 2 inches. Bond overlap with adhesive, Item 7, WP 1803 00.
7. Install new connector pins on two long wires from switch being replaced.
8. Install limit switch in stabilator position sensor/limit switch assembly with keyway-washer, lockwasher, and locknut. Do not lockwire until switch activation point is determined (WP 0069 00).
9. If No. 1 down limit switch is being installed, install large cable clamp over switch.
10. **QA** Route all switch wires in a bundle through small cable clamp attached to the large cable clamp with screw, washer, and nut (Figure 1, Sheet 1).
11. **QA** Slide switch wires through connector backshell, and using insert/extract tool install pins into applicable connector contacts shown in Table 1.
12. Thread backshell onto connector.
13. Do an operational check of stabilator position sensor/limit switch assembly (WP 0069 00).
14. Using heat gun, heat shrink tubing to limit switch.
15. **QA** Secure limit switch wire bundle with electrical tiedown straps, Item 302, WP 1803 00. Maintain at least 1/4-inch spacing between wire bundle and cotter pins on stabilator position sensor retaining nuts (Figure 1, Sheet 2).

INSTALL - Continued

Table 1. Wiring Requirements.

SWITCH	WIRE CODE	CONNECTOR NO. - PIN NO.	WIRE LENGTHS (INCHES)
NO. 1 DOWN	1 - 20	P605 - A	27
	3 - 20	P605 - B	27
NO. 1 UP	4 - 20	P605 - C	23
	6 - 20	P605 - D	23
NO. 2 DOWN	1 - 20	P604 - G	21
	3 - 20	P604 - H	21
NO. 2 UP	4 - 20	P604 - J	25
	6 - 20	P604 - K	25

16. Install identification straps to limit switch wire bundle (Figure 1, Sheet 2).
17. Do a stabilator position sensor/limit switch assembly operational check (WP 0069 00).
18. **QA** Lockwire stabilator position sensor/limit switch using nonelectrical wire Item 354, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR POSITION TRANSMITTER (AVIM)

This WP supersedes WP 0372 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Electrical Connector Maintenance Repair Kit, DMC 420
Electrical Repairer's Toolkit, SC 5180-99-B06
Gloves
Goggles/Face Shield
Heat Gun, LFT500
Insert/Extract Tool, MS3447-20
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Insulation Sleeving, Electrical, Item 159, WP 1803 00
Primer, Sealing Compound, Item 247, WP 1803 00
Sealing Compound, Item 282, WP 1803 00
Strap, Tiedown, Electrical, Item 302, WP 1803 00
Tags
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required

Aircraft Electricial Repairer MOS 15F (1)

References

WP 0069 00
WP 1803 00

NOTE

During disassembly, visually inspect mechanical parts (positioner, retainer, and base plate) and replace any part bent or distorted beyond fit or function.

REMOVE

1. Turn off all electrical power.
2. Loosen locknut and setscrew on pivot assembly (Figure 1).
3. Remove cotter pins, nuts, washers, synchro cleats, and bolts (and shim washers, if installed) from position sensor.
4. Pull position sensor away from stabilator position sensor/limit switch assembly.
5. Remove screw, nut, washer, and small cable clamp that is attached to large clamp on No. 1 down limit switch.
6. Remove backshell from connector, P604.
7. Tag wires, and, using insert/extract tool remove wires and pins from contact A through contact F on connector.
8. Remove identification strap and tiedown straps from wire bundle.
9. Remove position sensor from stabilator position sensor/limit switch assembly.

INSTALL

1. Turn off all electrical power.
2. Insert position sensor shaft through base, and orient zero indication mark on shaft collar with screw which is between No. 1 down limit switch and No. 2 up limit switch (Figure 1).

INSTALL - Continued

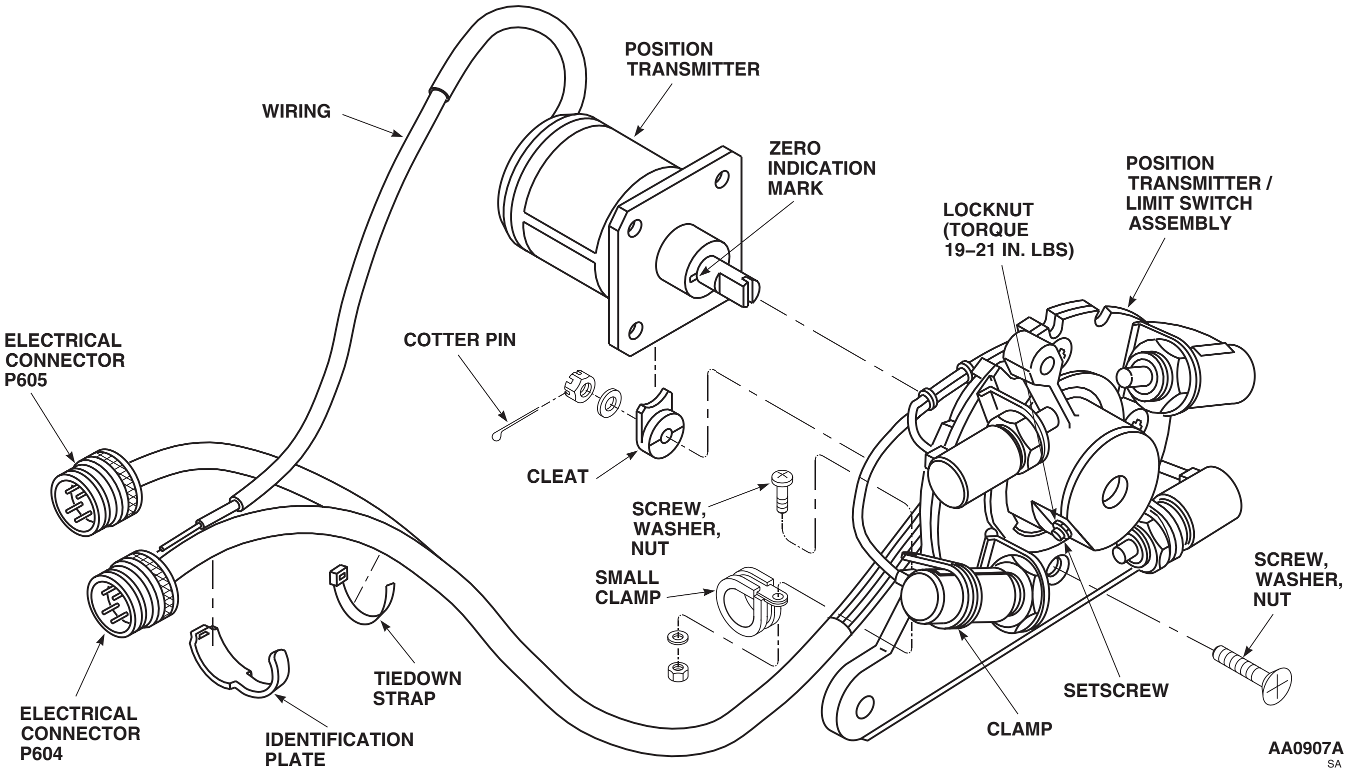


Figure 1. Replace Stabilator Position Transmitter (AVIM).

3. Secure position sensor to base with screws, synchro cleats, washers, and nuts. Install cotter pins.
4. Tighten nuts and check for binding in pivot assembly. If it binds, install shim washers on bolts between retainer and base. Use fewest shims permitting free play of pivot assembly. **SHIM THICKNESS SHALL NOT BE OVER 0.030-INCH.**
5. **QA** Tighten setscrew on pivot assembly. **TORQUE SETSCREW TO 19 - 21 INCH-POUNDS.**

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Clean inside threads of jamnut and threads of setscrew, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth.
7. Apply thin coat of sealing compound primer, Item 247, WP 1803 00, to threads of jamnut and setscrew. Let dry.
8. Apply thin coat of sealing compound, Item 282, WP 1803 00, to threads of jamnut and setscrew.
9. **QA** Install jamnut on setscrew. **TIGHTEN JAMNUT TO 1/6 TO 1/3 TURN BEYOND POINT WHERE SHARP RISE IN TORQUE IS FELT.**

INSTALL - Continued

10. Slide a 1/4-inch diameter, 3 inch long section of heat-shrink electrical insulation sleeving, Item 159, WP 1803 00, over end of position sensor cable. Leave 1 1/2-inches of wires exposed for installation of connector pins on wire ends.
11. Install pins on position sensor wires.
12. Slide 3 inch long section of heat-shrink electrical insulation sleeving, Item 159, WP 1803 00, over exposed wires leaving connector pins exposed. Using heat gun, heat shrink insulation tubing over cable harness.
13. Thread wires through connector backshell and using insert/extract tool insert pins into connector, P604, per Table 1.

Table 1. P604 Connector Wire Color/Pin Contact Chart.

WIRE COLOR	P604 CONTACT
BLUE	A
YELLOW	B
BLACK	C
RED/WHITE	D
BLK/WHITE	E
GREEN	F

14. **QA** Install backshell on connector, P604. Untag wires.
15. Adjust position sensor (WP 0069 00).
16. **QA** Lockwire, using nonelectrical wire Item 354, WP 1803 00, lockwire nut and setscrew to pivot assembly and install cotter pins through position sensor bolts after adjustments have been completed (WP 0069 00).
17. Install electrical tiedown straps, Item 302, WP 1803 00, and identification strap.
18. Install small cable clamp on harness and attach it to large cable clamp on No. 1 down limit switch with screw, nut, and washer.
19. Do a stabilator position sensor/limit switch assembly operational check (WP 0069 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

POSITION SENSOR/LIMIT SWITCH ELECTRICAL CONNECTOR (AVIM)

INITIAL SETUP:

Tools and Special Tools

Electrical Connector Maintenance Repair Kit, DMC 420
Electrical Repairer's Toolkit, SC 5180-99-B06
Insert/Extract Tool, MS3447-20

Material/Parts

Tags

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0069 00

NOTE

This procedure is the same for both electrical connectors, P604, and P605.

REMOVE

1. Turn off all electrical power.
2. Loosen backshell and slide it away from connector (Figure 1).
3. Tag wires and, using insert/extract tool, remove wires and pins from connector.
4. Remove connector.

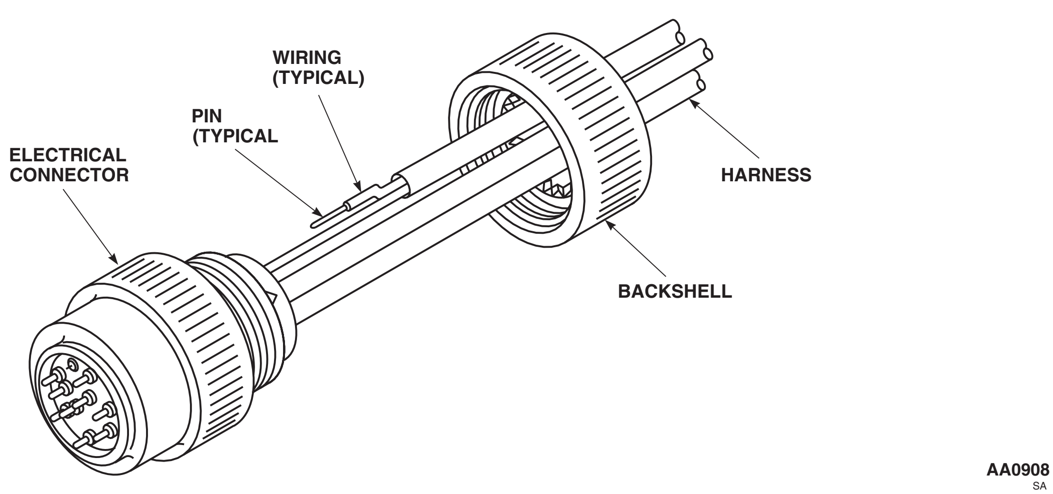


Figure 1. Replace Position Sensor/Limit Switch Electrical Connector (AVIM).

INSTALL

1. Turn off all electrical power.
2. Using insert/extract tool, insert pins in connector (Figure 1). Untag wires.
3. **QA** Slide backshell onto connector and tighten backshell.
4. Do a stabilator position sensor/limit switch assembly operational check (WP 0069 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR POSITION SENSOR (AVIM)

INITIAL SETUP:

Tools and Special Tools

Electrical Connector Maintenance Repair Kit, DMC 420
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0069 00

NOTE

Disassemble stabilator position sensor/limit switch assembly only as far as necessary to remove malfunctioning or damaged part.

REPAIR

1. Turn off all electrical power.

NOTE

To remove pivot assembly, any two diagonally opposite limit switches and their positioners must be removed.

2. Visually inspect the following mechanical parts. If part is damaged or corroded beyond fit or function, replace it (Figure 1).

- Pivot assembly
- Retainer
- Positioners
- Base
- Synchro cleats
- Clamps
- Hardware
- Shim

3. Do a stabilator position sensor/limit switch assembly operational check (WP 0069 00).

REPAIR - Continued

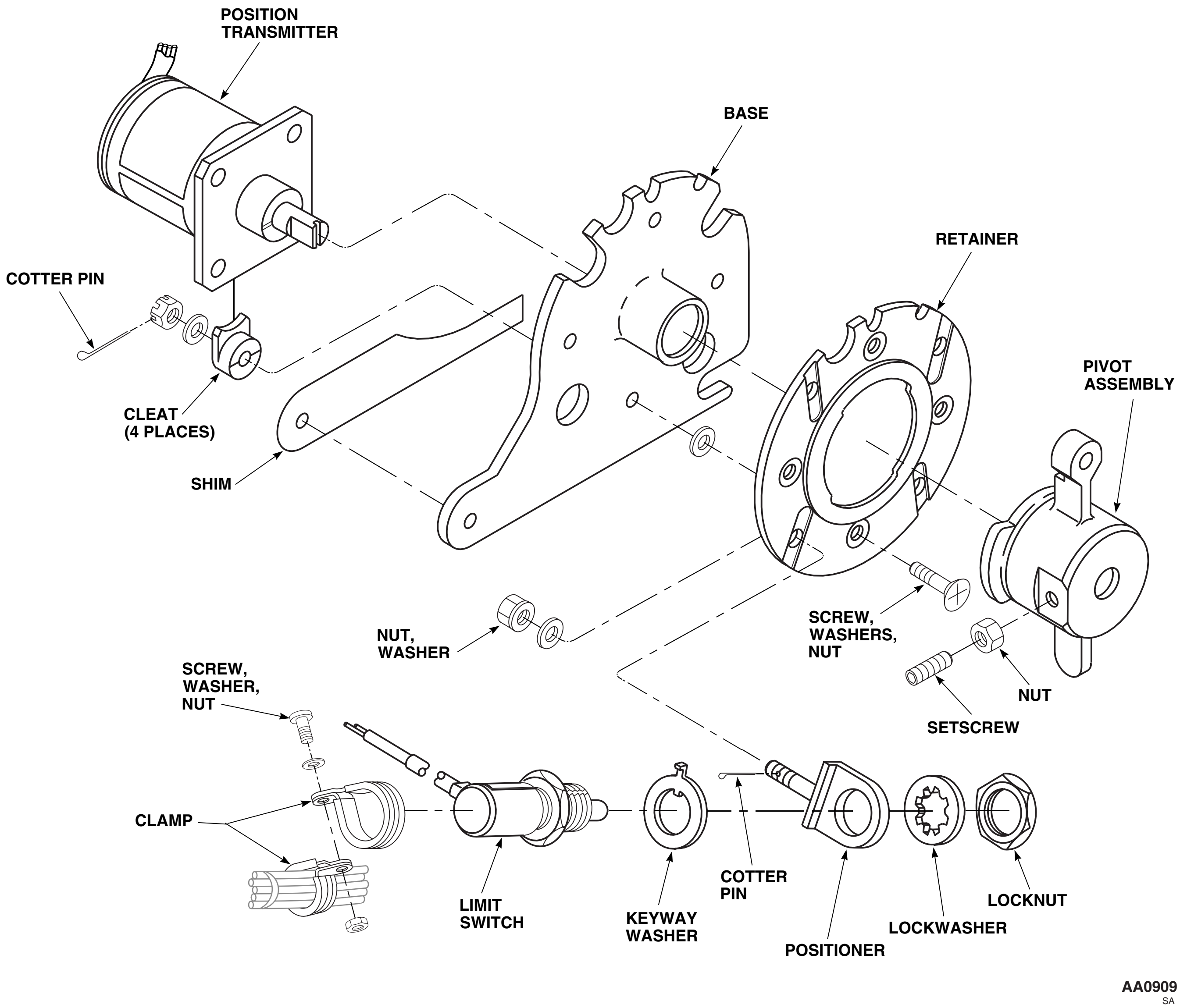


Figure 1. Replace Stabilator Position Sensor Mechanical Parts (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR ACTUATOR CLEVIS (AVIM)

This WP supersedes WP 0375 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanics Toolkit, SC 5180-99-B01
Arbor Press, 00-P-636
Drift
Gloves
Goggles/Face Shield
Reamer Set, GGG-R-00180

Material/Parts
Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0343 00
WP 1803 00

REPAIR

1. Remove stabilator actuator clevis from actuator (WP 0343 00).
2. Use arbor press and drift, press bushings from clevis (Figure 1).
3. Freeze replacement bushing in carbon dioxide, Item 64, WP 1803 00, for at least 30 minutes.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

.

4. Clean bushing holes in clevis with technical acetone, Item 5, WP 1803 00. Coat flanged bushing with epoxy primer coating kit, Item 120, WP 1803 00.
5. Using arbor press and drift, install flanged bushing in clevis with epoxy primer coating kit, Item 120, WP 1803 00. Bushing flange to be on inside of clevis.
6. **QA** Coat plain bushing with epoxy primer coating kit, Item 120, WP 1803 00. While still wet with epoxy primer, press plain bushing into clevis.
7. Set space between shoulder of flanged bushing and plain bushing at 0.407 to 0.406-inch (Figure 1).
8. Line ream bushings 0.3750 to 0.3755-inch.
9. Install stabilator actuator clevis (WP 0343 00).

REPAIR - Continued

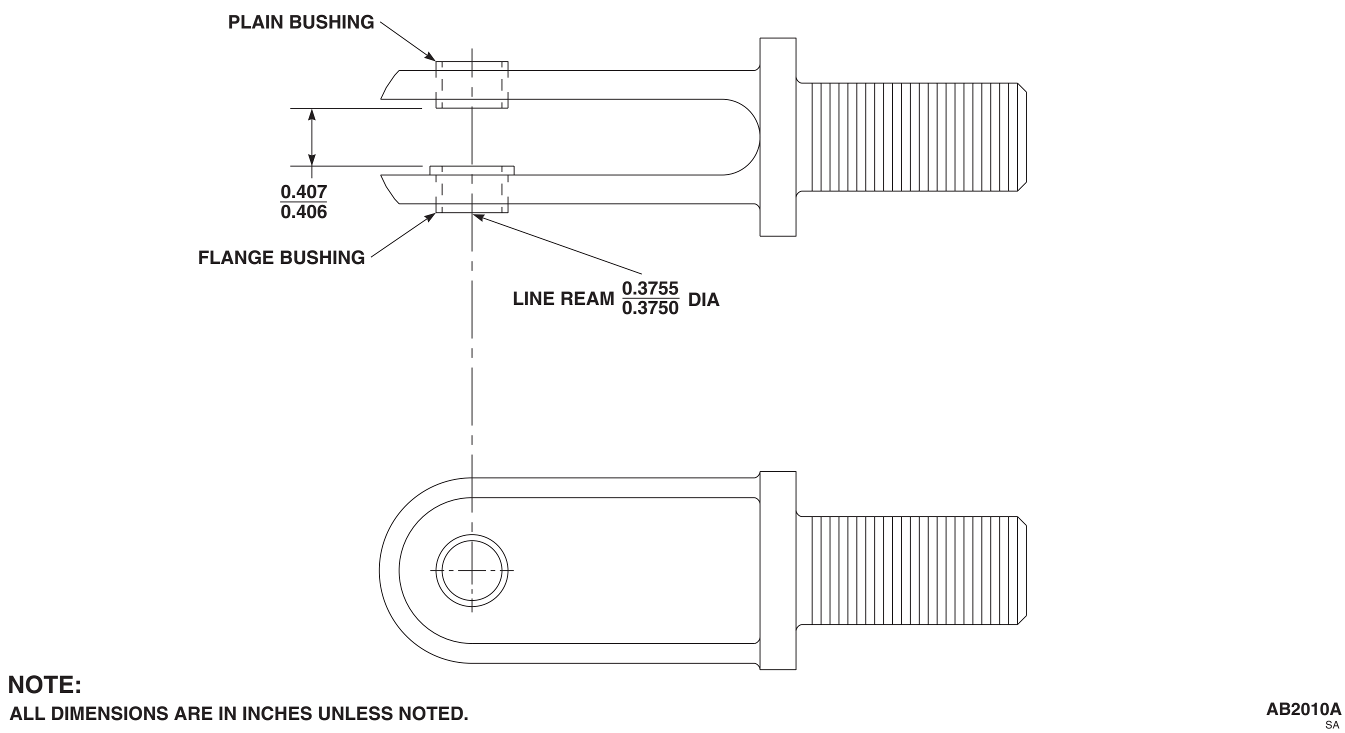


Figure 1. Repair Stabilator Actuator Clevis (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AIRFRAME

STABILATOR POSITION SENSOR BRACKET (AVIM)

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanics Toolkit, SC 5180-99-B01
Airframe Repairer Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Fluorescent Penetrant Inspection Kit, XMA101
Stabilator Position Sensor Bracket Locating Tool, Locally-Made, Figure H-263, WP 1805 00

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00
Isopropyl, Alcohol Technical, Item 170, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 1-1520-265-23
WP 0338 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove stabilator (WP 0338 00).
2. Carefully remove lockbolts and rivets securing stabilator position sensor bracket to stabilator (TM 1-1500-204-23). Remove position sensor bracket.
3. Clean position sensor bracket and mating surface on stabilator with technical isopropyl alcohol, Item 170, WP 1803 00.

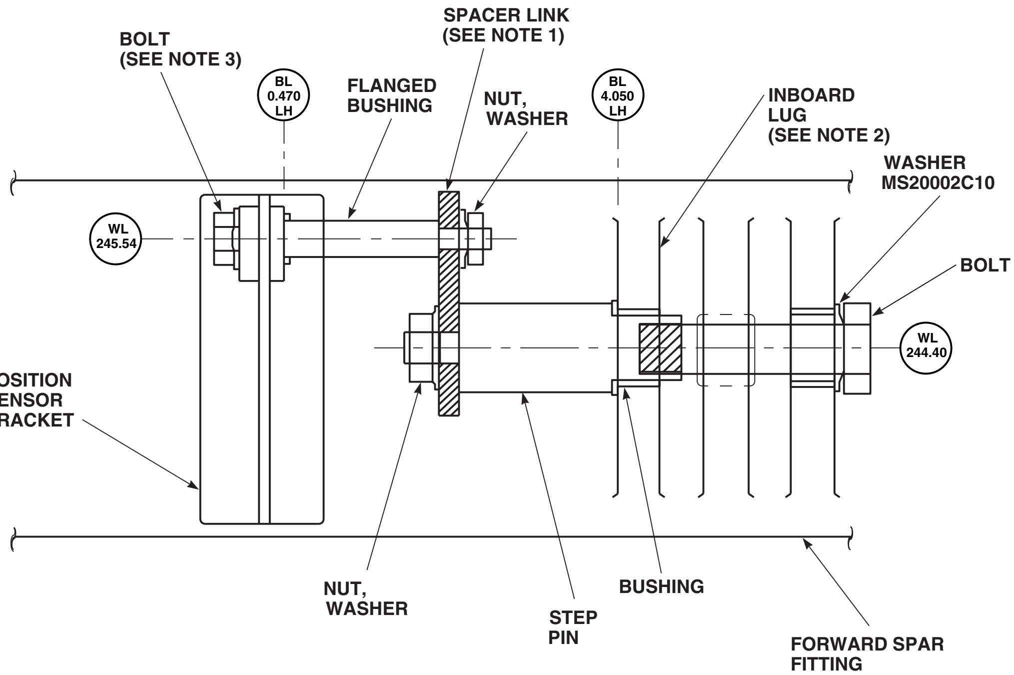
INSPECT

1. Using abrasive cloth, Item 82, WP 1803 00, repair nicks, scratches, and gouges in position sensor bracket and mating surface on front spar up to 0.010-inch deep, 0.040-inch wide, and 0.050-inch long. Using magnifying glass, visually inspect mating area of position sensor bracket on front spar for evidence of cracks or corrosion/pitting. Blend out damage to a maximum of 0.010-inch. **NO CRACKS ALLOWED.** If cracks are found, replace stabilator.
2. If cracks are suspected, perform nondestructive inspection using fluorescent penetrant inspection kit (TM 1-1520-265-23 and TM 1-1500-344-23). **NO CRACKS ALLOWED.** If cracks are found, replace stabilator.

INSTALL

1. Locally make position sensor bracket locating tool (WP 1805 00) and assemble on position sensor bracket and lug of forward spar (Figure 1). Rotate assembly until position sensor bracket is flush against mounting surface on front spar.
2. If previously installed position sensor bracket is being installed, do the following:
   a. Align mounting holes in position sensor bracket with holes in front spar.
   b. **QA** Using sealing compound, Item 273, WP 1803 00, wet install position sensor bracket on front spar and secure with lockbolts and rivets (TM 1-1500-204-23).
   c. Remove position sensor bracket locating tool.

INSTALL - Continued



NOTES:

1. INSTALL SPACER LINK ON THREADED END OF STEP PIN USING WASHER AND NUT. FINGER TIGHTEN ONLY AT THIS TIME.
2. INSERT LARGE DIAMETER END OF STEP PIN INTO INBOARD LUG OF FORWARD SPAR FITTING UNITL SHOULDER SEATS ON THE BUSHING. INSTALL BOLT WITH WASHER MS200002C10, UNDER MANUFACTURED HEAD. TIGHTEN ONLY UNTIL SHARP RISE IN TORQUE IS FELT.
3. INSTALL BOLT THROUGH POSITION SENSOR BRACKET, SLIDE FLANGED BUSHING OVER SHANK OF BOLT (FLANGE–END TOWARD BRACKET), INSTALL REMAINDER OF BOLT THROUGH THE SPACER LINK. INSTALL WASHER AND NUT. FINGER TIGHTEN.
4. ROTATE ASSEMBLY UNTIL POSITION SENSOR BRACKET IS FLUSH AGAINST MOUNTING SURFACE OF FORWARD SPAR FITTING.
5. TIGHTEN NUTS UNTIL NO RELATIVE MOVEMENT BETWEEN PARTS EXISTS.
6. MAINTAIN SUFFICIENT FORCE TO HOLD POSITION SENSOR BRACKET AGAINST FORWARD SPAR FITTING AND TRANSFER DRILL MOUNTING FASTENER HOLE LOCATIONS. DISASSEMBLE, DEBURR AND TOUCH–UP AS REQUIRED.

AB2503
SA

Figure 1. Replace Stabilator Position Sensor Bracket (AVIM).

INSTALL - Continued

d. Cover lockbolts and rivets with sealing compound, Item 273, WP 1803 00.

e. Apply a bead of sealing compound, Item 273, WP 1803 00, around perimeter of position sensor bracket.

3. If new position sensor bracket is being installed, do the following:

a. Back drill through front spar into position sensor bracket.

b. **QA** Using sealing compound, Item 273, WP 1803 00, wet install position sensor bracket on front spar and secure with lockbolts and rivets (TM 1-1500-204-23).

c. Remove position sensor bracket locating tool.

d. Cover lockbolts and rivets with sealing compound, Item 273, WP 1803 00.

e. Apply a bead of sealing compound, Item 273, WP 1803 00, around perimeter of position sensor bracket.

4. Install stabilator (WP 0338 00).

END OF WORK PACKAGE

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
General, United States Army
Chief of Staff

Official:

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0608616

DISTRIBUTION:
To be distributed in accordance with Initial Distribution Number (IDN) 311288, requirements for TM 1-1520-237-23-5.

These are the instructions for sending an electronic 2028

The following format must be used if submitting an electronic 2028. The subject line must be exactly the same and all fields must be included; however only the following fields are mandatory: 1, 3, 4, 5, 6, 7, 8, 9, 10, 13, 15, 16, 17, and 27.

From: “Whomever” <whomever@wherever.army.mil>
To: 2028@redstone.army.mil

Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT–93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123–123–1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

This is the text for the problem below line 27.

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.	Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE 8/30/02
TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, 35898	FROM: (Activity and location)(Include ZIP Code) MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565	

PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS

PUBLICATION/FORM NUMBER	DATE	TITLE
TM 9-1005-433-24	16 Sep 2002	Organizational, Direct Support, And General Support Maintenance Manual for Machine Gun, .50 Caliber M3P and M3P Machine Gun Electrical Test Set Used On Avenger Air Defense Weapon System

ITEM NO.	PAGE NO.	PARA-GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON
1	W PG 3		2			Test or Corrective Action column should identify a different WP number.

EXAMPLE

* Reference to line numbers within the paragraph or subparagraph.

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION	SIGNATURE
MSG, Jane Q. Doe, SFC	788--1234	

TO: (Forward direct to addressee listed in publication)	FROM: (Activity and location) (Include ZIP Code)	DATE
Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM–MMC–MA–NP Redstone Arsenal, 35898	MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565	8/30/02

PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS

PUBLICATION NUMBER	DATE	TITLE

PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION

PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)

EXAMPLE

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION	SIGNATURE
MSG, Jane Q. Doe, SFC	788–1234	

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.	Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE

TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898 - 5000	FROM: (Activity and location)(Include ZIP Code)

PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS

PUBLICATION/FORM NUMBER	DATE	TITLE

ITEM NO.	PAGE NO.	PARA– GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON

* Reference to line numbers within the paragraph or subparagraph.

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION	SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898	FROM: (Activity and location) (Include ZIP Code)	DATE

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TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION	SIGNATURE

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
1 decimeter = 10 centimeters = 3.94 inches
1 meter = 10 decimeters = 39.37 inches
1 dekameter = 10 meters = 32.8 feet
1 hectometer = 10 dekameters = 328.08 feet
1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
1 decigram = 10 centigrams = 1.54 grains
1 gram = 10 decigram = .035 ounce
1 decagram = 10 grams = .35 ounce
1 hectogram = 10 decagrams = 3.52 ounces
1 kilogram = 10 hectograms = 2.2 pounds
1 quintal = 100 kilograms = 220.46 pounds
1 metric ton = 10 quintals = 1.1 short tons

Liquid Measure

1 centiliter = 10 milliters = .34 fl. ounce
1 deciliter = 10 centiliters = 3.38 fl. ounces
1 liter = 10 deciliters = 33.81 fl. ounces
1 dekaliter = 10 liters = 2.64 gallons
1 hectoliter = 10 dekaliters = 26.42 gallons
1 kiloliter = 10 hectoliters = 264.18 gallons

Square Measure

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch
1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches
1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet
1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet
1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres
1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

Cubic Measure

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch
1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches
1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

Approximate Conversion Factors

To change	*To*	*Multiply by*	*To change*	*To*	*Multiply by*
inches	centimeters	2.540	ounce-inches	Newton-meters	.007062
feet	meters	.305	centimeters	inches	.394
yards	meters	.914	meters	feet	3.280
miles	kilometers	1.609	meters	yards	1.094
square inches	square centimeters	6.451	kilometers	miles	.621
square feet	square meters	.093	square centimeters	square inches	.155
square yards	square meters	.836	square meters	square feet	10.764
square miles	square kilometers	2.590	square meters	square yards	1.196
acres	square hectometers	.405	square kilometers	square miles	.386
cubic feet	cubic meters	.028	square hectometers	acres	2.471
cubic yards	cubic meters	.765	cubic meters	cubic feet	35.315
fluid ounces	milliliters	29,573	cubic meters	cubic yards	1.308
pints	liters	.473	milliliters	fluid ounces	.034
quarts	liters	.946	liters	pints	2.113
gallons	liters	3.785	liters	quarts	1.057
ounces	grams	28.349	liters	gallons	.264
pounds	kilograms	.454	grams	ounces	.035
short tons	metric tons	.907	kilograms	pounds	2.205
pound-feet	Newton-meters	1.356	metric tons	short tons	1.102
pound-inches	Newton-meters	.11296			

Temperature (Exact)

°F	Fahrenheit temperature	5/9 (after subtracting 32)	Celsius temperature	°C

PIN: 073175-000